

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS

CHAPTER 6 MAINTENANCE INSTRUCTIONS

*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19 and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

This information is furnished upon the condition that it will not be released to another nation without the specific authority of the Department of the Army of the United States, that it will be used for military purposes only, that individual or corporate rights originating in the information, whether patented or not, will be respected, that the recipient will report promptly to the United States, any known or suspected compromise, and that the information will be provided substantially the same degree of security afforded it by the Department of Defense of the United States. Also, regardless of any other markings on the document, it will not be downgraded or declassified without written approval of the originating United States agency.

DISTRIBUTION STATEMENT D - Distribution authorized to the DoD and DoD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5000.

WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C., Sec 2751, et. seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et. seq. Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

HEADQUARTERS, DEPARTMENT OF THE ARMY

17 April 2006

WARNING SUMMARY

Personnel performing operations, procedures, and practices which are included or implied in this technical manual shall observe the following warnings. To disregard these warnings and precautionary information can cause serious injury, death, or an aborted mission.

WARNING

FIRST AID - refer to FM 4-25-11.

AC POWER - before applying ac power to the helicopter, make sure that the area around the stabilator is clear of personnel and equipment. Should the stabilator be in any position other than trailing edge fully down, it may reposition automatically to fully down when ac power is applied.

CARBON MONOXIDE - during cold weather operations when preheating is necessary, all personnel shall become acquainted with the various types of heaters and where they are to be used. Be careful of accumulations of carbon monoxide and damage to heat-sensitive equipment.

CONSUMABLE MATERIALS - observe all cautions and warnings on the containers when using consumables. When applicable wear necessary protective gear during handling and use. If a consumable is flammable or explosive, MAKE SURE consumable and its vapors are kept away from heat, spark, and flame. MAKE SURE helicopter is properly grounded and firefighting equipment is readily available for use.

SAFETY PINS - make sure all safety pins are installed in ejector racks prior to performing any maintenance. To disregard this warning can cause serious injury, death, or an aborted mission.

ELECTRICAL GROUNDING OF THE HELICOPTER - army regulations require the grounding of the helicopter when doing all fueling and defueling operations. Do not operate helicopter electrical switches, except those essential for servicing during fueling and defueling. Do not smoke or use flame during fueling and defueling operations.

HIGH VOLTAGE - there are dangerous voltages in the helicopter. Use extreme care when working with equipment having these voltages.

HYDRAULIC SYSTEMS - there are dangerous high hydraulic system pressures in the helicopter. Use extreme care when working on systems having this pressure.

USE OF FIRE EXTINGUISHER - when fire extinguishers are used in a confined area, ventilate immediately. Serious injury or death could result if the area is not ventilated or the user does not use a self-contained breathing device.

MAIN ROTOR PYLON SLIDING COVER - when opening and closing main pylon sliding cover, keep hands away from sharp edges near track, mating end, and especially ventilation blower inlet hole.

TECHNICAL ACETONE - is extremely flammable and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

ALKALINE AQUEOUS CLEANER - alkaline cleaner is harmful to skin and eyes. Use adequate ventilation. Operators should wear protective gloves, aprons, and goggles. Upon contact with cleaning compound, wash affected area for at least 20 minutes with clean water. Seek medical attention.

CLEANING COMPOUND SOLVENT - cleaning Compound Solvent is combustible and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well-ventilated areas (or use approved respirator). Keep away from open flames or other sources of ignition.

ELECTRON - use electron in well ventilated area only. Avoid prolonged breathing of vapors. Avoid bodily contact. The use of chemical gloves and chemical splash goggles are required. Do not use near heat, spark or flame. This solvent is

reactive with acids and oxidizers; do not mix or cross-apply with other chemicals. Organic vapor respirator with dust and mist filter is recommended when solvent is spray applied. Keep containers closed between applications.

- Provide mechanical ventilation if used in confined spaces. Coordinate the use of this material with your supporting industrial hygiene and safety offices. Ensure you read and understand the material safety data sheet (MSDS) for electron prior to use.

ISOPROPYL ALCOHOL - isopropyl alcohol is a volatile and flammable liquid which must be kept away from flame and other ignition sources. It must be used only in well ventilated area. Personnel must wear protective gloves and eye protection. Excessive inhalation of vapors or contact with skin must be avoided. If contact occurs with eyes or skin, flush area thoroughly with water. If ingestion occurs, induce vomiting and call a doctor. Remove to fresh air if overexposed to vapor.

DS-108 - use in well ventilated area. Avoid breathing vapors. Organic vapor respirator with dust and mist filter is recommended. May be harmful by inhalation, ingestion, or skin absorption. Avoid bodily contact. Chemical safety glasses/goggles and chemical resistant gloves are required. Combustible, keep away from heat, spark, or flame. Vapors are heavier than air and may travel over a distance to an ignition source and then flash back. Do not mix or cross apply with other chemicals.

LOCKWIRE - in critical applications use lockwire instead of safety cable which is specifically prohibited from use in critical applications. The use of safety cable is restricted in these specific applications with a warning.

PURPOSE OF A WARNING - alerts personal to operation, procedure, practice, etc., which if not strictly observed, could result in personal injury or loss of life.

CHANGE
NO. 3

HEADQUARTERS DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 20 June 2008

TECHNICAL MANUAL
AVIATION UNIT AND INTERMEDIATE MAINTENANCE
FOR
ARMY MODELS
UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS
CHAPTER 6 MAINTENANCE INSTRUCTIONS

WARNING -This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C. Sec. 2751 et seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et seq. Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DISTRIBUTION STATEMENT D – Distribution authorized to the Department of Defense and U.S. DOD contractors only, for official use or for administrative or operational purposes. This determination was made on 15 June 2003. Other requests for this document must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5000.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

TM 1-1520-237-23-6, 17 April 2006, is updated as follows:

1. File this sheet in front of the manual for reference.
2. This change results from updated procedures for maintenance, checks and services, and/or updated lists of expendable/durable supplies and materials.
3. New or updated text is indicated by a vertical bar in the outer margin of the page.
4. Added illustrations are indicated by a vertical bar adjacent to the figure number. Changed illustrations are indicated by a miniature pointing hand adjacent to the updated area or a MAJOR CHANGE symbol and a vertical bar adjacent to the figure number.
5. Remove old pages and insert new pages as indicated below.

Remove Pages

a and b
A and B

Insert Pages

a and b
A and B

6. Replace the following work packages with their revised version.

Work Package Number

WP 0377 00
WP 0378 00
WP 0379 00
WP 0381 00
WP 0391 00

Work Package Number

WP 0393 00
WP 0395 00
WP 0396 00
WP 0397 00
WP 0398 00
WP 0399 00
WP 0400 00
WP 0403 00
WP 0405 00
WP 0409 00
WP 0413 00
WP 0415 00
WP 0417 00
WP 0421 00
WP 0424 00
WP 0425 00
WP 0426 00
WP 0430 00
WP 0431 00
WP 0438 00
WP 0440 00
WP 0441 00
WP 0441 01
WP 0442 00
WP 0462 00
WP 0463 00
WP 0464 00
WP 0467 00
WP 0468 00
WP 0469 00
WP 0473 00
WP 0474 00
WP 0485 00
WP 0486 00
WP 0508 00
WP 0509 00
WP 0523 00

CHANGE
NO. 3

TM 1-1520-237-23-6

By Order of the Secretary of the Army:

GEORGE W. CASEY, JR.
General, United States Army
Chief of Staff

Official:

A handwritten signature in black ink, reading "Joyce E. Morrow". The signature is fluid and cursive, with the first letters of each name being capitalized and prominent.

JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0815614

DISTRIBUTION:

To be distributed in accordance with the Initial Distribution Number (IDN) 311289, requirements for TM 1-1520-237-23-6.

CHANGE
NO. 2

HEADQUARTERS DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 15 December 2007

TECHNICAL MANUAL

**AVIATION UNIT AND INTERMEDIATE MAINTENANCE
FOR
ARMY MODELS
UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS**

WARNING -This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C. Sec. 2751 et seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et seq. Violation of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DISTRIBUTION STATEMENT D: Distribution authorized to the DOD and DOD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5230.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

TM 1-1520-237-23-6, 17 April 2006, is updated as follows:

1. File this sheet in front of the manual for reference.
2. This change results from updated procedures for maintenance, checks and services, and/or updated lists of expendable/durable supplies and materials.
3. New or updated text is indicated by a vertical bar in the outer margin of the page.
4. Added illustrations are indicated by a vertical bar adjacent to the figure number. Changed illustrations are indicated by a miniature pointing hand adjacent to the updated area or a MAJOR CHANGE symbol and a vertical bar adjacent to the figure number.
5. Remove old pages and insert new pages as indicated below.

Remove Pages

None
A and B

Insert Pages

a and b
A and B

6. Replace the following work packages with their revised version.

Work Package Number

WP 0385 00
WP 0403 00
WP 0411 00
WP 0417 00
WP 0428 00
WP 0431 00

Work Package Number

WP 0439 00
WP 0447 00
WP 0448 00
WP 0450 00
WP 0462 00
WP 0463 00
WP 0465 00
WP 0470 00
WP 0481 00
WP 0482 00
WP 0484 00
WP 0485 00
WP 0494 00
WP 0498 00
WP 0518 00

By Order of the Secretary of the Army:

Official:



JOYCE E. MORROW
*Administrative Assistant to the
Secretary of the Army*
0733120

GEORGE W. CASEY, JR.
*General, United States Army
Chief of Staff*

DISTRIBUTION:

To be distributed in accordance with Initial Distribution Number (IDN) 311289, requirements for TM 1-1520-237-23-6.

CHANGE
NO. 1

HEADQUARTERS
DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 21 March 2007

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A, AND HH-60L HELICOPTERS

WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act (Title 22, U.S.C. Sec. 2751 et seq.) or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 et seq. Violation of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DISTRIBUTION STATEMENT D. Distribution authorized to the DOD and DOD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5230.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

TM 1-1520-237-23-6, dated 17 April 2006, is changed as follows:

1. File these sheets in front of the manual for reference.
2. This change results from updated procedures for maintenance, checks and services, and/or updated lists of expendable/durable supplies and materials.
3. New or updated text is indicated by a vertical bar in the outer margin of the page.
4. Added illustrations are indicated by a vertical bar adjacent to the figure number. Changed illustrations are indicated by a miniature pointing hand adjacent to the updated area or a MAJOR CHANGE symbol and a vertical bar adjacent to the figure number.
5. Remove old pages and insert new pages as indicated below.

Remove pages

A and B
i through iv
v/vi Blank
Chapter 6 Title and Blank
Cover

Insert pages

A and B
i through iv
v/vi Blank
Chapter 6 Title and Blank
Cover

6. Replace the following work packages with their revised version.

Work Package Number

WP 0381 00
WP 0384 00
WP 0386 00
WP 0387 00
WP 0400 00

Work Package Number

WP 0401 00
WP 0402 00
WP 0405 00
WP 0407 00
WP 0415 00
WP 0431 00
WP 0438 00
WP 0439 00
WP 0442 00
WP 0451 00
WP 0462 00
WP 0464 00
WP 0465 00
WP 0470 00
WP 0473 00
WP 0478 00
WP 0481 00
WP 0484 00
WP 0486 00
WP 0489 00
WP 0494 00
WP 0495 00
WP 0501 00
WP 0502 00
WP 0523 00
WP 0524 00
WP 0528 00
WP 0529 00

7. Add the following new work packages.

Work Package Number

WP 0441 01

By Order of the Secretary of the Army:

PETER J. SCHOOMAKER
*General, United States
Army
Chief of Staff*

Official:



JOYCE E. MORROW
*Administrative Assistant to the
Secretary of the Army*
0704006

DISTRIBUTION:

To be distributed in accordance with Initial Distribution Number (IDN) 311289, requirements for TM 1-1520-237-23-6.

INSERT LATEST CHANGED PAGES/WORK PACKAGES. DESTROY SUPERSEDED DATA.

LIST OF EFFECTIVE PAGES/WORK PACKAGES

Dates of issue for original and changed pages / work packages are:

Original	0	17 April 2006
Change	1	21 March 2007
Change	2	15 December 2007
Change	3	20 June 2008

TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 12 AND TOTAL NUMBER OF WORK PACKAGES IS 158 CONSISTING OF THE FOLLOWING:

Page / WP No.	*Change No.	Page / WP No.	*Change No.	Page / WP No.	*Change No.
Volume 6 Title	1	WP 0400 00 (8 pgs)	3	WP 0429 00 (2 pgs)	0
a - b	3	WP 0401 00 (4 pgs)	1	WP 0430 00 (4 pgs)	3
A - B	3	WP 0402 00 (10 pgs)	1	WP 0431 00 (4 pgs)	3
i - v	1	WP 0403 00 (16 pgs)	3	WP 0432 00 (2 pgs)	0
vi Blank	1	WP 0404 00 (6 pgs)	0	WP 0433 00 (2 pgs)	0
Chapter 6 Title Page	1	WP 0405 00 (34 pgs)	3	WP 0434 00 (2 pgs)	0
WP 0377 00 (16 pgs)	3	WP 0406 00 (10 pgs)	0	WP 0435 00 (2 pgs)	1
WP 0378 00 (32 pgs)	3	WP 0407 00 (4 pgs)	1	WP 0436 00 (2 pgs)	0
WP 0379 00 (40 pgs)	3	WP 0408 00 (4 pgs)	0	WP 0437 00 (2 pgs)	0
WP 0380 00 (22 pgs)	0	WP 0409 00 (4 pgs)	3	WP 0438 00 (12 pgs)	3
WP 0381 00 (8 pgs)	3	WP 0410 00 (2 pgs)	0	WP 0439 00 (4 pgs)	2
WP 0382 00 (4 pgs)	0	WP 0411 00 (2 pgs)	2	WP 0440 00 (4 pgs)	3
WP 0383 00 (6 pgs)	0	WP 0412 00 (2 pgs)	0	WP 0441 00 (4 pgs)	3
WP 0384 00 (24 pgs)	1	WP 0413 00 (2 pgs)	3	WP 0441 01 (4 pgs)	3
WP 0385 00 (6 pgs)	2	WP 0414 00 (2 pgs)	0	WP 0442 00 (6 pgs)	3
WP 0386 00 (20 pgs)	1	WP 0415 00 (14 pgs)	3	WP 0443 00 (6 pgs)	0
WP 0387 00 (18 pgs)	1	WP 0416 00 (4 pgs)	0	WP 0444 00 (6 pgs)	0
WP 0388 00 (16 pgs)	0	WP 0417 00 (6 pgs)	3	WP 0445 00 (4 pgs)	0
WP 0389 00 (12 pgs)	0	WP 0418 00 (4 pgs)	0	WP 0446 00 (2 pgs)	0
WP 0390 00 (2 pgs)	0	WP 0419 00 (14 pgs)	0	WP 0447 00 (4 pgs)	2
WP 0391 00 (8 pgs)	3	WP 0420 00 (2 pgs)	0	WP 0448 00 (4 pgs)	2
WP 0392 00 (2 pgs)	0	WP 0421 00 (4 pgs)	3	WP 0449 00 (2 pgs)	0
WP 0393 00 (4 pgs)	3	WP 0422 00 (4 pgs)	0	WP 0450 00 (6 pgs)	2
WP 0394 00 (2 pgs)	0	WP 0423 00 (4 pgs)	0	WP 0451 00 (6 pgs)	1
WP 0395 00 (6 pgs)	3	WP 0424 00 (6 pgs)	3	WP 0452 00 (4 pgs)	0
WP 0396 00 (2 pgs)	3	WP 0425 00 (4 pgs)	3	WP 0453 00 (6 pgs)	0
WP 0397 00 (6 pgs)	3	WP 0426 00 (2 pgs)	3	WP 0454 00 (2 pgs)	0
WP 0398 00 (2 pgs)	3	WP 0427 00 (8 pgs)	0	WP 0455 00 (2 pgs)	0
WP 0399 00 (4 pgs)	3	WP 0428 00 (12 pgs)	2	WP 0456 00 (2 pgs)	0

* Zero in this column indicates an original page or work package.

LIST OF EFFECTIVE PAGES/WORK PACKAGES - CONTINUED

Page / WP No.	*Change No.	Page / WP No.	*Change No.	Page / WP No.	*Change No.
WP 0457 00 (8 pgs)	0	WP 0483 00 (2 pgs)	0	WP 0509 00 (4 pgs)	3
WP 0458 00 (4 pgs)	0	WP 0484 00 (10 pgs)	2	WP 0510 00 (2 pgs)	0
WP 0459 00 (4 pgs)	0	WP 0485 00 (10 pgs)	3	WP 0511 00 (2 pgs)	0
WP 0460 00 (4 pgs)	0	WP 0486 00 (16 pgs)	3	WP 0512 00 (6 pgs)	0
WP 0461 00 (2 pgs)	0	WP 0487 00 (4 pgs)	0	WP 0513 00 (4 pgs)	0
WP 0462 00 (14 pgs)	3	WP 0488 00 (2 pgs)	0	WP 0514 00 (2 pgs)	0
WP 0463 00 (10 pgs)	3	WP 0489 00 (14 pgs)	1	WP 0515 00 (4 pgs)	0
WP 0464 00 (6 pgs)	3	WP 0490 00 (4 pgs)	0	WP 0516 00 (2 pgs)	0
WP 0465 00 (8 pgs)	2	WP 0491 00 (4 pgs)	0	WP 0517 00 (2 pgs)	0
WP 0466 00 (14 pgs)	0	WP 0492 00 (4 pgs)	0	WP 0518 00 (2 pgs)	2
WP 0467 00 (4 pgs)	3	WP 0493 00 (6 pgs)	0	WP 0519 00 (2 pgs)	0
WP 0468 00 (8 pgs)	3	WP 0494 00 (14 pgs)	2	WP 0520 00 (2 pgs)	0
WP 0469 00 (6 pgs)	3	WP 0495 00 (8 pgs)	1	WP 0521 00 (2 pgs)	0
WP 0470 00 (6 pgs)	2	WP 0496 00 (2 pgs)	0	WP 0522 00 (2 pgs)	0
WP 0471 00 (8 pgs)	0	WP 0497 00 (2 pgs)	0	WP 0523 00 (6 pgs)	3
WP 0472 00 (6 pgs)	0	WP 0498 00 (4 pgs)	2	WP 0524 00 (4 pgs)	1
WP 0473 00 (14 pgs)	3	WP 0499 00 (2 pgs)	0	WP 0526 00 (2 pgs)	0
WP 0474 00 (4 pgs)	3	WP 0500 00 (2 pgs)	0	WP 0527 00 (2 pgs)	0
WP 0475 00 (2 pgs)	0	WP 0501 00 (4 pgs)	1	WP 0528 00 (4 pgs)	1
WP 0476 00 (6 pgs)	0	WP 0502 00 (2 pgs)	1	WP 0529 00 (4 pgs)	1
WP 0477 00 (2 pgs)	0	WP 0503 00 (2 pgs)	0	WP 0530 00 (4 pgs)	0
WP 0478 00 (2 pgs)	1	WP 0504 00 (2 pgs)	0	WP 0531 00 (2 pgs)	0
WP 0479 00 (2 pgs)	0	WP 0505 00 (2 pgs)	0	WP 0532 00 (2 pgs)	0
WP 0480 00 (2 pgs)	0	WP 0506 00 (6 pgs)	0	WP 0533 00 (2 pgs)	0
WP 0481 00 (22 pgs)	2	WP 0507 00 (4 pgs)	0		
WP 0482 00 (14 pgs)	2	WP 0508 00 (4 pgs)	3		

* Zero in this column indicates an original page or work package.

HEADQUARTERS,
DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 17 April 2006

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

Army Models UH-60A, UH-60L, EH-60A, HH-60A, and HH-60L Helicopters

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS.

You can help improve this manual. If you find any mistakes or if you know of a way to improve these procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms) located in the back of this manual directly to: Commander, U.S. Army Aviation and Missile Command, ATTN: AMSAM-MMC-MA-NP, Redstone Arsenal, AL 35898-5000. A reply will be furnished to you. You may also provide DA Form 2028 information to AMCOM via e-mail, fax, or the World Wide Web. Our fax number is: DSN 788-6546 or Commercial 256-842-6546. Our e-mail address is: 2028@redstone.army.mil. Instructions for sending an electronic 2028 may be found at the back of this manual immediately preceding the hard copy 2028. For the World Wide Web use: <https://amcom2028.redstone.army.mil>.

DISTRIBUTION STATEMENT D - Distribution authorized to the DoD and DoD contractors only due to Critical Technology effective as of 15 June 2003. Other requests must be referred to Commander, US Army Aviation and Missile Command, ATTN: SFAE-AV-UH/L, Redstone Arsenal, AL 35898-5000.

WARNING - This document contains technical data whose export is restricted by the Arms Export Control Act Title 22, U.S.C. Sec. 2751, *et. sec.* or the Export Administration Act of 1979, as amended, Title 50, U.S.C., App. 2401 *et seq.* Violations of these export laws are subject to severe criminal penalties. Disseminate in accordance with provisions of DoD Directive 5230.25.

DESTRUCTION NOTICE - Destroy by any method that will prevent disclosure of contents or reconstruction of the document.

TABLE OF CONTENTS

WP Sequence No.

CHAPTER 6 - MAINTENANCE INSTRUCTIONS

Structural Repair General Information	0377 00
Metal Structure Repair	0378 00
Composite Structure Repair	0379 00
Extrusion Charts	0380 00

*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19, and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

TABLE OF CONTENTS - Continued

	WP Sequence No.
Helicopter Painted Surfaces Touchup Repair	0381 00
Nose Door Repair	0382 00
Cockpit Door Repair	0383 00
Nose Section Framing Repair	0384 00
Nose Section Plating Repair	0385 00
Cabin Section Forward Framing Repair	0386 00
Cabin Section Upper Framing Repair	0387 00
Cabin Section Lower Framing Repair	0388 00
Cabin Section Plating Repair	0389 00
Cabin Section Beaded Inner Fuselage Plating STA 295.0 Through STA 308.0 Repair	0390 00
Cargo Door Upper Track Repair	0391 00
Cargo Door Lower Track Fairing Repair	0392 00
Cargo Door Lower Track Wear Strips Repair	0393 00
External Power Receptacle Cover	0394 00
External Stores Support System (ESSS) Fuselage Fitting Bearing	0395 00
Gunner's Window Lower Track	0396 00
Gunner's Window Track Repairs	0397 00
Gunner's Window Upper Track	0398 00
Gunner's Window Upper Track Fairing	0399 00
Gunner's Window Repair	0400 00
Troop/Cargo Door Repair	0401 00
Cabin Floor Repair	0402 00
Main Rotor Pylon Framing Repair	0403 00
Main Rotor Pylon Fairing Repair	0404 00
Transition Section Framing Repair	0405 00
Transition Section Plating Repair	0406 00
Transition Section Step Door Hinge WL 210.0 and WL 230.0 Repair	0407 00
Replace Transition Section Step Door Hinge WL 250.0	0408 00
Transition Section Fuel Sump Drain Door	0409 00
Transition Section Gravity Refuel Door	0410 00
Transition Section Pressure Refuel Door	0411 00
Transition Section Pneumatic Ground Start Door	0412 00
Transition Section APU Access Door Screws Repair	0413 00
Grounding Receptacle	0414 00
Tailcone Framing Repair	0415 00
Tailcone Plating Repair	0416 00
Tailcone Canted Hinge Bulkhead	0417 00
Tail Rotor Drive Shaft Covers Repair	0418 00
Tail Rotor Pylon Framing Repair	0419 00
Tail Rotor Pylon Plating Repair	0420 00
Tail Rotor Pylon Left Stabilator Attach Fitting	0421 00
Tail Rotor Pylon Right Stabilator Attach Fitting	0422 00
Tail Rotor Pylon Stabilator Actuator Attach Fitting	0423 00
Tail Rotor Pylon Stabilator Actuator Attach Fitting Bearings	0424 00

TABLE OF CONTENTS - Continued

	WP Sequence No.
Tail Rotor Pylon Cable Pulley Supports Repair	0425 00
Tail Rotor Pylon Canted Hinge Frame Repair	0426 00
Stabilator Framing Repair	0427 00
Stabilator Plating Repair	0428 00
Stabilator Tip Cap Repair	0429 00
Stabilator Actuator Attach Fitting Repair	0430 00
Main Landing Gear Drag Beam and Axle Inspections	0431 00
Main Landing Gear Inspection Shock Strut Quick Service Check	0432 00
Main Landing Gear Tire Inspection	0433 00
Main Landing Gear Wheel Brake Inspection	0434 00
Parking Brake Rod Inspection	0435 00
Tail Landing Gear Yoke Inspection	0436 00
Tail Landing Gear Tire Inspection	0437 00
Main Landing Gear Drag Beam	0438 00
Main Landing Gear Axle	0439 00
Brake Flange Spacers	0440 00
Drag Beam Support Fitting Bearings	0441 00
Main Landing Gear Drag Beam Bushings	0441 01
Main Landing Gear Drag Beam Jackpod and Tiedown Ring	0442 00
Main Landing Gear Drag Beam Switch	0443 00
Main Landing Gear Shock Strut	0444 00
Main Landing Gear Shock Strut Kneeling Valve	0445 00
Main Landing Gear Shock Strut Air Valve	0446 00
Main Landing Gear Wheel and Tire	0447 00
Main Landing Gear Wheel Valve	0448 00
Main Landing Gear Wheel Disk Drive Keys	0449 00
Main Landing Gear Tire	0450 00
Main Landing Gear Wheel Brake	0451 00
Master Brake Cylinder	0452 00
Main Landing Gear Wheel Brake Lines and Hoses	0453 00
Parking Brake Valve	0454 00
Parking Brake Valve Switch	0455 00
Slave Mixer Valve	0456 00
Bleed Main Landing Gear Wheel Brake System	0457 00
Parking Brake Handle and Mechanism	0458 00
Tail Landing Gear Shock Strut	0459 00
Tail Landing Gear Shock Strut Kneeling Valve	0460 00
Tail Landing Gear Shock Strut Air Valves	0461 00
Tail Landing Gear Yoke	0462 00
Tail Landing Gear Fork	0463 00
Tail Landing Gear Lock Mechanism	0464 00
Tail Landing Gear Axle, Wheel and Tire	0465 00
Main Landing Gear Shock Strut (AVIM)	0466 00
Main Landing Gear Shock Strut Lower Stage Spherical Bearing (AVIM)	0467 00

TABLE OF CONTENTS - Continued

	WP Sequence No.
Main Landing Gear Wheel (AVIM)	0468 00
Main Landing Gear Wheel Brake (AVIM).....	0469 00
Master Brake Cylinder (AVIM)	0470 00
Parking Brake Valve (AVIM).....	0471 00
Slave Mixer Valve (AVIM)	0472 00
Tail Landing Gear Shock Strut (AVIM).....	0473 00
Tail Landing Gear Yoke - Repair (AVIM)	0474 00
Tail Landing Gear Fork - Repair (AVIM).....	0475 00
Tail Landing Gear Wheel - Repair (AVIM)	0476 00
Engine Exhaust Module Inspection	0477 00
Engine Control Cable Inspection	0478 00
Engine Load-Demand Push/Pull Friction Check.....	0479 00
Inspect Pneumatic Tube	0480 00
Engine	0481 00
Demountable Power Package.....	0482 00
Engine LRU's	0483 00
Engine Air Inlet	0484 00
Aft Engine Mounts	0485 00
Forward Support Tube/Engine Output Shaft.....	0486 00
Engine Exhaust Module	0487 00
Engine Exhaust Module Ejector	0488 00
HIRSS	0489 00
HIRSS Rear Fairings and Panels	0490 00
HIRSS Suppressor Support Mounts.....	0491 00
HIRSS Nacelle Fairing Support Mounts	0492 00
HIRSS Nacelle Fairing.....	0493 00
HIRSS Suppressor Core and Baffle	0494 00
HIRSS Extender	0495 00
HIRSS Bulkhead Repair.....	0496 00
HIRSS Emissive Coating Repair	0497 00
Load-Demand Rotary Input	0498 00
Power-Available Rotary Input.....	0499 00
Engine Speed Control Potentiometer.....	0500 00
Engine Control Quadrant	0501 00
Engine Control Quadrant Canopy Plunger.....	0502 00
Engine Starter Switch Adjustment	0503 00
Engine Starter Abort Switch Adjustment.....	0504 00
Engine Starter Override Switch Adjustment	0505 00
Engine Load-Demand Push/Pull Cable	0506 00
Engine Power-Available Push/Pull Cable.....	0507 00
Fuselage Power-Available Push/Pull Cable.....	0508 00
Fuel Selector Valve Push/Pull Cable	0509 00
Fuel Selector Valve Control Box	0510 00
Engine Load-Demand Push/Pull Cable Supports.....	0511 00

TABLE OF CONTENTS - Continued

	WP Sequence No.
Engine Power-Available System Rigging and Adjustment.....	0512 00
Engine Mount Restraint Rings.....	0513 00
Engine Control Quadrant Front Cover Information Plate.....	0514 00
Aft Engine Mount Struts (AVIM).....	0515 00
Aft Engine Mount Fittings (AVIM).....	0516 00
Aft Engine Mount Links (AVIM).....	0517 00
Aft Engine Mount Supports (AVIM).....	0518 00
Engine Mount Crotch Assembly (AVIM).....	0519 00
Engine Exhaust Ejector and Fairings (AVIM)	0520 00
Engine Exhaust Module Duct Assembly (AVIM).....	0521 00
HIRSS Inlet Seal (AVIM)	0522 00
Rotary Input (AVIM).....	0523 00
Engine Control Quadrant Power Lever (AVIM).....	0524 00
Engine Control Quadrant T-Handle and Fuel Lever (AVIM).....	0525 00
Engine Control Quadrant Threaded Inserts (AVIM).....	0526 00
Engine Control Quadrant Engine Starter Switch (AVIM)	0527 00
Engine Control Quadrant Starter Abort Switch (AVIM)	0528 00
Engine Control Quadrant Starter Override Switch (AVIM)	0529 00
Engine Control Quadrant Fire Extinguisher Arming Switch (AVIM).....	0530 00
Engine Control Quadrant Left and Right Cover Information Plate (AVIM)	0531 00
Engine Control Quadrant Front Cover Information Plate Wiring Harness (AVIM).....	0532 00
Engine Load-Demand Cable Support Clip (AVIM).....	0533 00

CHAPTER 6
MAINTENANCE INSTRUCTIONS
FOR
ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A,
AND HH-60L
HELICOPTERS



UNIT LEVEL**AIRFRAME****STRUCTURAL REPAIR GENERAL INFORMATION**

This WP supersedes WP 0377 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
 Brush, Metal-Bound, Flat, H-B-695-1
 Drill Set, GGG-D-751
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Metal Tester, Model B
 Paint Roller, H-R-550
 Sealant Gun, 221830
 Vacuum Cleaner, MIL-V-21987

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cloth, Cheesecloth, Item 85, WP 1803 00
 Corrosion Preventive Compound, Item 102,
 WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Distilled Water, Reagent, Item 117, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Litmus Paper, ACS, Item 176, WP 1803 00

Sealing Compound, Item 266, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

FM 1-563
 MIL-C-46168
 TM 1-1500-204-23
 TM 1-1500-344-23
 TM 1-1520-237-PMI
 TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0378 00
 WP 0379 00
 WP 0380 00
 WP 0381 00
 WP 0576 00
 WP 0585 00
 WP 1803 00

GENERAL INFORMATION

This manual contains general instructions and information necessary for maintenance personnel to repair the primary and secondary structure. Responsibility for any specific repair may be determined by referring to the Maintenance Allocation Chart for the helicopter. The Fundamentals and Procedures of Airframe Maintenance (FM 1-563) shall be used in conjunction with the Maintenance Allocation Chart.

Refer to Aviation Unit Maintenance (AVUM) And Aviation Intermediate Maintenance (AVIM) Manual For General Aircraft Maintenance (TM 1-1500-204-23) for repair procedures not in this manual, and for information on fastener installation and repair materials.

Refer to, CORROSION PROTECTION AND WATER INTEGRITY SEALING, in this work package for specific corrosion protection and watertight sealing information.

Refer to Painting and Marking of Army Aircraft (TM 55-1500-345-23) for paint application.

Refer to WP 0381 00, WP 0576 00, and WP 0585 00 for specific paint requirements.

Stop-drilling cracks is a temporary repair. It requires a follow-up repair with another reinforcing repair at the next Periodic Inspection (PMI).

DAMAGE EVALUATION

1. Damage to the helicopter is caused by impact, wear, or parts malfunctioning. Impact damage will vary with the force and location of impact and the type of object causing the damage. Wear of parts is due to normal usage and/or misalignment of movable surfaces. Parts can be destroyed by fire, or may malfunction as a result of fatigue, corrosion, overstress, or previously undetected damage. Damage not caused by impact must be evaluated and corrected to prevent it from happening again. This may require reinforcement of the structure or additional corrosion prevention measures. Increased inspections and periodic maintenance may also be required. Each part in the damaged area must be inspected and the amount of damage determined and classified.
2. Structural repairs described in this section are intended for use in conjunction with (TM 1-1500-204-23).

NOTE

Repair at AVUM is limited to minor repair of sheet metal cracks, scratches, corrosion, and loose or missing hardware. These repairs can be accomplished using the airframe repairer's toolkit and portable hand tools. If any extensive damage occurs or major repair is required, repairs shall be accomplished by AVIM. Repair at AVIM is limited to repair of sheet metal cracks, scratches, corrosion, holes, and loose or missing hardware. If major damage occurs requiring jigs and fixtures, repairs shall be accomplished by next higher maintenance level.

3. The damage limits provided in WP 0379 00 on bonded panels are not intended to ground the aircraft. The limits are to provide guidance for scheduling repair or replacement at a schedule maintenance interval. When damage limits, particularly bond voids in bonded panels, are exceeded, the responsible unit maintenance authority will establish a recurring special inspection on the damaged area until the damage to the structure is corrected. If the damage is in the area that requires engineering authority for repair, engineering should be contacted in writing with a description of damage. If depot assistance will be required, unit should contact AMCOM, AMSAM-MMC-MA-NP, with your requirement.

CLASSIFICATION OF DAMAGE AND TYPES OF REPAIR

General

Airframe damage consists of bending, twisting, breaking, or reduction of material in the cross-section of a structural member. It may be a complete break, a hole, wrinkle, twist, nick, scratch, crack, or material eaten away by corrosion. In case of fire, it may be pitted, blistered, burned, or an eaten away area of the skin or skeleton. After examining damage, classify it as either negligible damage, reparable damage, or damage requiring replacement. This will aid in determining the type and size of the repair needed.

Negligible Damage

NOTE

Permissible damage limits for structural aluminum alloy members are explained and illustrated. Refer to, CLASSIFICATION OF DAMAGE AND TYPES OF REPAIR, in this work package. Permissible damage limits for reinforced plastic are described in WP 0379 00.

Negligible damage is damage which may be permitted to remain as is or which may be corrected by a simple repair procedure such as stop-drilling cracks, blending out nicks or scratches, or touching up protective finish on parts.

CLASSIFICATION OF DAMAGE AND TYPES OF REPAIR - Continued**Reparable Damage****NOTE**

Recommended typical repairs for aluminum alloy structural members are shown in WP 0378 00 and WP 0380 00. Recommended typical repairs for reinforced plastic components are shown in WP 0379 00. These repair examples may be used as shown, or used in any combination necessary to complete a repair.

1. Damage greater than negligible limits that requires the addition of a patch, splice, or reinforcement to restore the full load-carrying capacity of the structure is classified as reparable damage.
2. Cracks and scratches deeper than 10% of material thickness are stop-drilled using a 1/8-inch-diameter drill and patched. Nicks or gouges deeper than 10% of material thickness are drilled out and patched. Punctures, deep dents, and corroded areas of skin and webs are trimmed out and patched. Use same gage, type, and temper material as original part for patch material and use same type fasteners. Smooth dents should be tapped out using a soft mallet and the area reinforced with stiffening angles. Extensive damage to a long structural member is repaired by cutting away the damaged section and splicing in a piece identical in shape and material. Allow 0.030 to 0.060-inch clearance on each end of inserted member. Make splices long enough to include no less than five fasteners on each side of cut.

Damage Requiring Replacement

Damage which cannot be repaired by available equipment, or extensive damage which requires the replacement of an entire component, is damage requiring replacement. Small doublers, splices, clips, brackets, and short stiffening members, damaged beyond negligible limits, are usually replaced because repair of these members is impracticable. Some highly stressed structural members cannot be repaired without reducing the established margin of safety for the structure. Such members, when damaged beyond negligible limits, must be replaced. Sometimes, where damaged members have a shape which is difficult to repair, it may be better, in view of man-hours and cost, to replace the damaged member rather than attempt a repair. Damage to a removed structural component should be evaluated for possible repair at a higher level of maintenance. Replacement members shall be the same type material and temper as the original member and are installed using the same type fasteners. Information for the replacement or recommended substitution of extrusion is found in WP 0380 00, Extrusion Charts.

Cleaning Out Damaged Area**CAUTION**

If a part or section of any supporting structure is removed for replacement or repair, make sure that nearby structure is properly supported to prevent further distortion and damage.

NOTE

After a thorough investigation of damage, a course of action should be taken to restore damaged area to its original condition. Clean out damaged area first. Take care to prevent further damage during this operation. The following procedure for cleaning out a damaged area applies to all types of damage except rotor blade damage.

1. Restore all dented skin and supporting structure to its original shape and condition wherever possible.
2. Cut away all jagged edges and severely distorted material, removing enough material to eliminate any radiating cracks.
3. Leave as much radius as possible in corners of cutouts. Smooth out all edges.
4. Round out holes in flat or contoured sheets to form a smooth outline.

CLASSIFICATION OF DAMAGE AND TYPES OF REPAIR - Continued

5. If nicks or abrasions are greater than negligible limitations, they should be treated as cracks and prepared for patching.
6. Do not leave isolated cracks unattended even though area is considered non-stressed. Stop-drill crack using a 1/8-inch diameter drill.

INSPECTION OF DAMAGE**Preliminary External Inspection**

1. Clean up structure and make a thorough visual inspection of damaged members and nearby structure for local and secondary damage.
2. Check for secondary damage resulting from shock loads being transmitted to other structures.
3. Check nearby structure for dents, gouges, punctures, cracks, loose joints, and distortion.
4. Check all riveted or bolted joints for evidence of pulled fasteners or distorted holes. If extent of damage is in doubt, remove enough fasteners to remove parts, and inspect framing for condition and alignment.
5. Check all wrinkles in skin and webs to decide whether they are stress wrinkles. List all items to be removed or repaired. Note all suspect areas of conditions which will call for closer detailed inspection when actual repair procedures are started.

Detailed Inspection

Inspect each part in damaged area and classify all damage to determine whether parts can be repaired and reused, or because of damage limitations or extent of repair, if they should be replaced. For major damage to highly stressed areas of airframe or components, and following "sudden stoppage" of rotor blades, submit a detailed report to local engineering authority, or to contractor, for engineering analysis and specific repair recommendations. "Sudden stoppage" is defined as shock load to drive train or rotor system, with or without power applied. This shock load may result from blade strike, FOD, or compressor stall.

Fire Damage Evaluation**NOTE**

The first step in the evaluation of structural damage caused by fire is a thorough cleaning of the damaged area. Evaluation of the damage, which may be caused either by the high temperature or by subsequent corrosion, involves both visual and optical inspection, plus the use of test equipment and test procedures. Hardness tests will determine the approximate ultimate strength of a material, but it does not eliminate the possibility of hidden damage. When in doubt as to the serviceability of a part or component, replace it. '

Cleanup of Damaged Areas

1. Clean up as soon as possible after a fire, to reduce damage due to corrosion.
2. Shut off power source, remove batteries, fuel, loose equipment, and materials or interior accessories that are in the way or that may be damaged during cleanup and repair.
3. Remove access panels, drill out rivets and open seams if necessary, to prepare areas for complete flushing and draining.
4. Use preventive measures to protect areas not involved, using tape, barrier paper, and protective covers.
5. Remove all traces of foam fire extinguishing residue by flushing with clean, fresh water. Flushing of applicable area or part must be complete, internally and externally.

INSPECTION OF DAMAGE - Continued

6. Use an approved alkaline cleaner and remove all oil, soot, extinguishing products and fire residue from damaged area. Flush thoroughly with clean water and drain all areas completely. Wipe internal surfaces with clean cloths until dry. Use forced warm air to hasten drying.

Visual Inspection

1. Inspect area for visible fire damage, such as distortion, burning, or pitting.
2. Inspect parts near fire for blistered or discolored paint. These parts shall be inspected by nondestructive inspection methods before being returned to service.

Nondestructive Inspection

Hardness tests, using portable equipment, should be made on all materials suspected of exposure to high temperatures. Hardness tests are not entirely conclusive; however, they give a good indication of the reliability of the part. Check suspect areas for cracks, especially in the radius of formed parts or extruded members, using fluorescent penetrant inspection kit (TM 1-1520-265-23). If reliability of any part is doubtful, replace it.

Corrosion Damage**NOTE**

Extinguishing agents used to fight the fire may be corrosive when in contact with high temperatures and water. Materials consumed by fire may also release corrosive gases or by-products which are highly corrosive when combined with water. Check all areas that have been reached by smoke or fumes for evidence of corrosion.

1. Clean all affected areas with an approved cleaner.
2. Inspect affected area using reagent distilled water, Item 117, WP 1803 00, and ACS litmus paper, Item 176, WP 1803 00. Especially inspect skin laps and joining surfaces for evidence of corrosive acid residue.
3. Neutralize, with an alkaline wash, areas that turn blue litmus paper red.
4. For treatment of corrosion damage, refer to TM 1-1500-344-23.

NEGLECTIBLE DAMAGE LIMITATIONS**Damage Measurement**

1. Damage is measured by comparing the normal cross-section of a part with the depth and area affected because of holes, cracks, scratches, nicks, and corrosion. The amount of material loss is called damage reduction. Any damage reduction will affect the strength of the part.
2. Cross-section area of a part is found by adding the width of flanges and web, and multiplying this total by the material thickness. Use equation $(F + F + W) \times T = \text{Area of cross section}$ (Figure 1).
3. Damage reduction area is found by measuring the dimension of damage (C) along the cross-section and multiplying this dimension by total depth of the cleaned-up damage. Use equation $C \times T1 = \text{damage reduction area}$.
4. On high stressed structural members, if the damage reduction area is not more than 5% of cross-section area, damage is classified as negligible. On low stressed structural members, if damage reduction area is not more than 10% of the cross-section area, damage is classified as negligible. Exceptions to this rule are found in linear dimensions.
5. Negligible damage limit examples:
 - a. Dimension of cross-section: 1.0 inch flanges, 3.0 inch web, and 0.040-inch material thickness. Cross-section area would be 0.200 ($1.0 + 1.0 + 3.0 = 5.0$, then $5.0 \times 0.040 = 0.200$). Negligible damage limit for high stress frames (5% of cross-section area) would be 0.010 ($0.200 \times 0.05 = 0.010$). Negligible damage limit for low stress frames (10% of cross-section area) would be 0.020 ($0.200 \times 0.10 = 0.020$).

NEGLECTIBLE DAMAGE LIMITATIONS - Continued

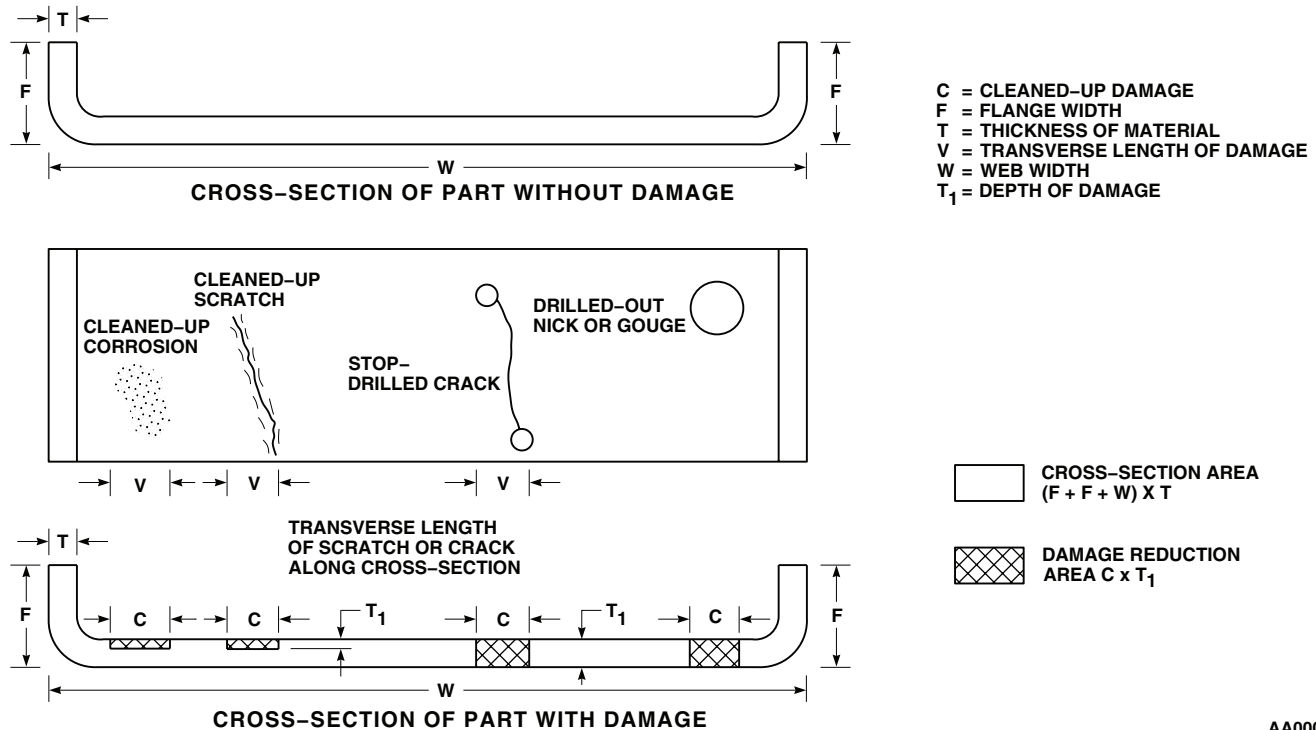
AA0001
SA

Figure 1. Damage Measurement.

- A scratch or corrosion, whose maximum depth after cleanup is 0.008-inch and transverse length (V) is 1.5-inches. Damage reduction area would be 0.012 (0.008 x 1.5 = 0.012). Damage would be within limits for low stress frames, but greater than allowable limits for high stress frames, which would need patching.
- A stop-drilled crack with transverse length (V) of 1.0 inch would have a damage reduction area of 0.040 (1.0 x 0.040 = 0.040). This would be greater than permitted negligible damage limits for both high and low stress frames, and each would need patching.
- A nick or gouge drilled out using a 1/4-inch diameter drill and polished would have a damage reduction area of 0.010 (0.250 x 0.040 = 0.010). This would be within permitted negligible damage limits for both high and low stress frames. Patching would be needed if a larger drill is used.

Negligible Damage Limits For Structural High Stress Frames

- Nicks, Scratches, or Corrosion. Do not remove more material than necessary. Determine cross-sectional area of part. $(F + F + W) \times T =$ area of cross-section. If damage reduction area clean up (Figure 2, Section A-A) is no more than 5% of cross section area, damage is negligible. Treat nicks and scratches as cracks if reduction area is more than 5% of cross section area.
- Nicks and Cracks. No nick or crack is allowed in radius. Stop-drill cracks, or nicks and scratches over damage limit, using 1/8-inch diameter drill. Cleaned-up damage to be not over damage reduction limits. Consider transverse length of damage when figuring damage reduction (Figure 1). If distance "E" is less than "D", consider damage goes to the edge of part. Nicks and gouges in flange edge, not over negligible limits, are blended out to not more than twice the depth of damage (Figure 2).

NEGLECTIBLE DAMAGE LIMITATIONS - Continued

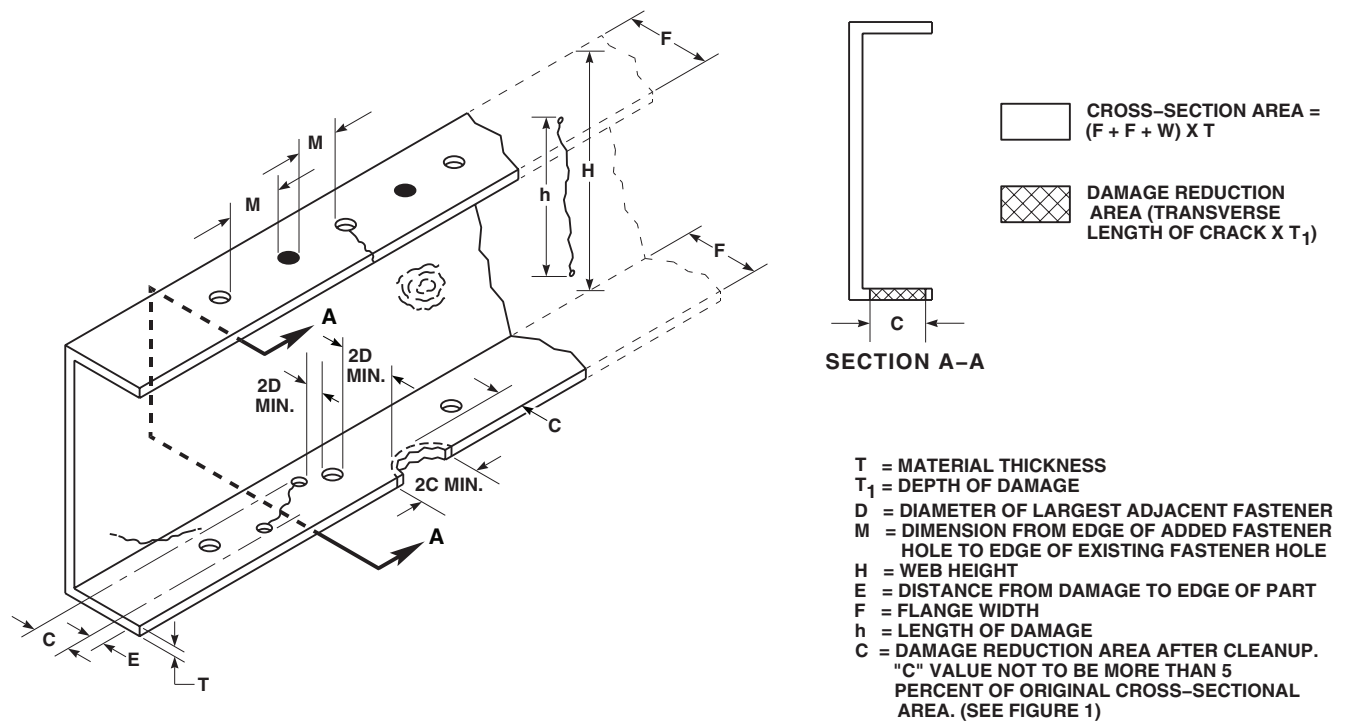
AA0002A
SA

Figure 2. Structural Frame Damage.

3. Dents. Smooth dents, free from sharp projections or abrasions, not over 1/8-inch in depth and 1 inch in diameter, are considered negligible. Treat as holes all dents over 1/8-inch in depth, or those with nicks or gouges. Trim out damage, sand to a smooth rounded hole, and patch as described in WP 0378 00.
4. Cracks in Flange Edge. When crack ends at a fastener hole, another fastener of the same type may be added on each side of damage hole, providing "M" dimension is not less than "2D" (Figure 2). If "M" dimension is less than "2D", repair flange (WP 0378 00).

Negligible Damage Limits For Extrusions

1. Cracks. Cracks are not allowed in extrusion radius (Figure 3). Stop-drill cracks using 1/8-inch diameter drill. Minimum distance allowable from adjacent damage or fastener hole is 2D. Consider transverse length of damage and cleanup when calculating damage reduction (Figure 1). If distance "E" is less than "D", consider damage goes to edge of part.
2. Nicks and Gouges. Refer to negligible damage limits for structural (high stress) frames. If nick or gouge is over negligible limits, use drill large enough to remove entire damage. Blend out nicks and gouges in flange edge, not greater than negligible limits, to not more than twice depth of damage.
3. Radius Damage. Nicks, scratches, and/or corrosion in extrusion radius not over 25% of "T" after cleanup. Blend damage to not more than twice damage depth.

Negligible Damage Limits For Primary Skin

1. Cracks Not Ending at Fastener Holes. Stop-drill using 1/8-inch diameter drill. Length of crack "L" not to be over 1/2-inch.

NEGLIGIBLE DAMAGE LIMITATIONS - Continued

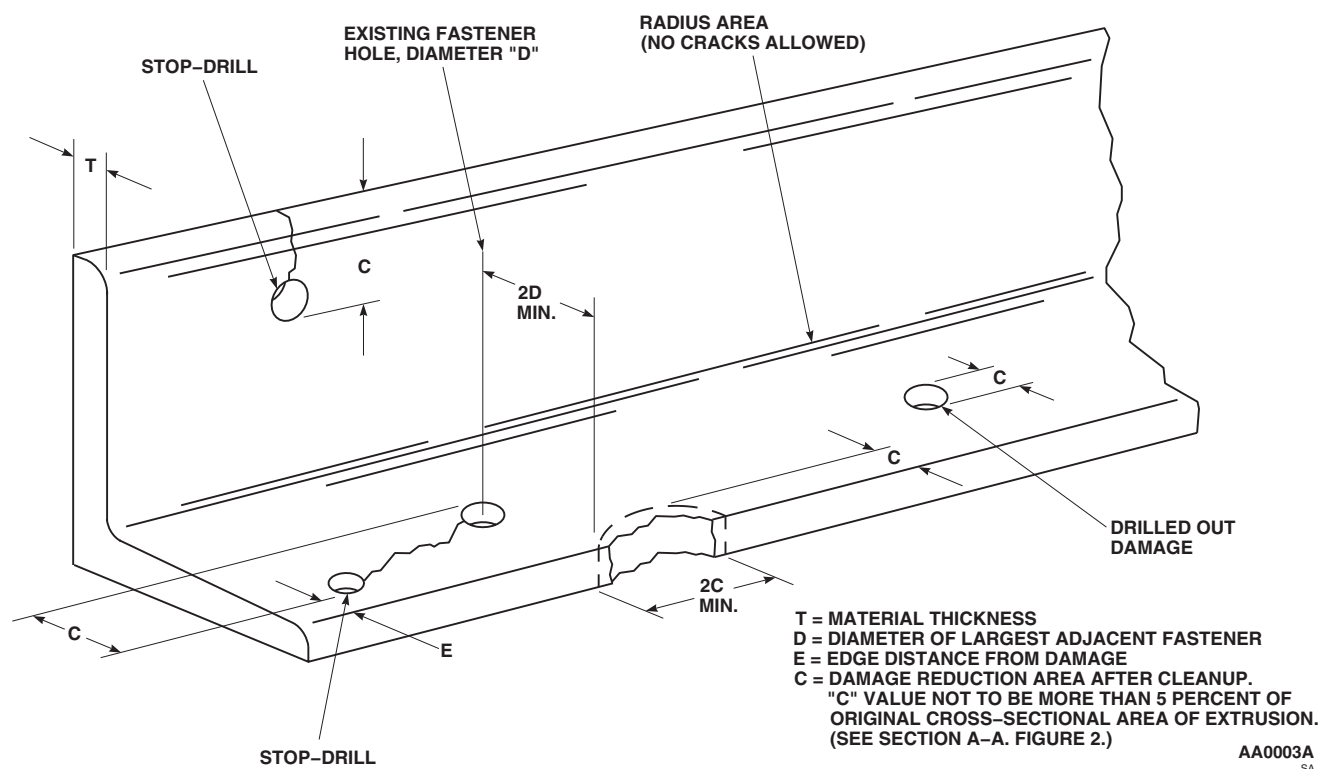


Figure 3. Extrusion Damage.

2. Edge Damage. Cleaned-up area not to be over 1/2-inch deep. Blend out damage to smooth rounded contour, not more than twice depth (Figure 4).
3. Nicks, Scratches, and Corrosion. Do not remove more material than necessary. Cleaned-up area (Figure 4, Section A-A) is not to be over these limits: Depth (T1) - 15% of T. Width (b) - 2 1/2 times material thickness T. Length (h) - 40% of H. Treat nicks and scratches as cracks if over limits T1, b, or h.
4. Dents. Smooth dents, free from sharp projections or abrasions, not over 1/4-inch deep. Treat as holes all dents over 1/4-inch deep, or those with nicks or gouges. Trim out damage, sand to a smooth rounded hole, and patch (WP 0378 00).

Negligible Damage Limits For Secondary Skin

1. Cracks Not Ending at Fastener Holes. Stop-drill using 1/8-inch diameter drill. Length of crack "L" not to be over 1 inch.
2. Edge Damage. Cleaned-up area to be not over 1 inch deep. Blend out damage to smooth rounded contour to not more than twice depth (Figure 4).
3. Nicks, Scratches, and Corrosion. Same as primary skin, except depth can be 35% of "T".
4. Dents. Same as primary skin.
5. Punctures and Cracks. Trim damage to a smooth rounded contour. Longest dimension not to be over 1 inch. Patch cleaned-up damage over 1 inch (WP 0378 00).

NEGLECTIBLE DAMAGE LIMITATIONS - Continued

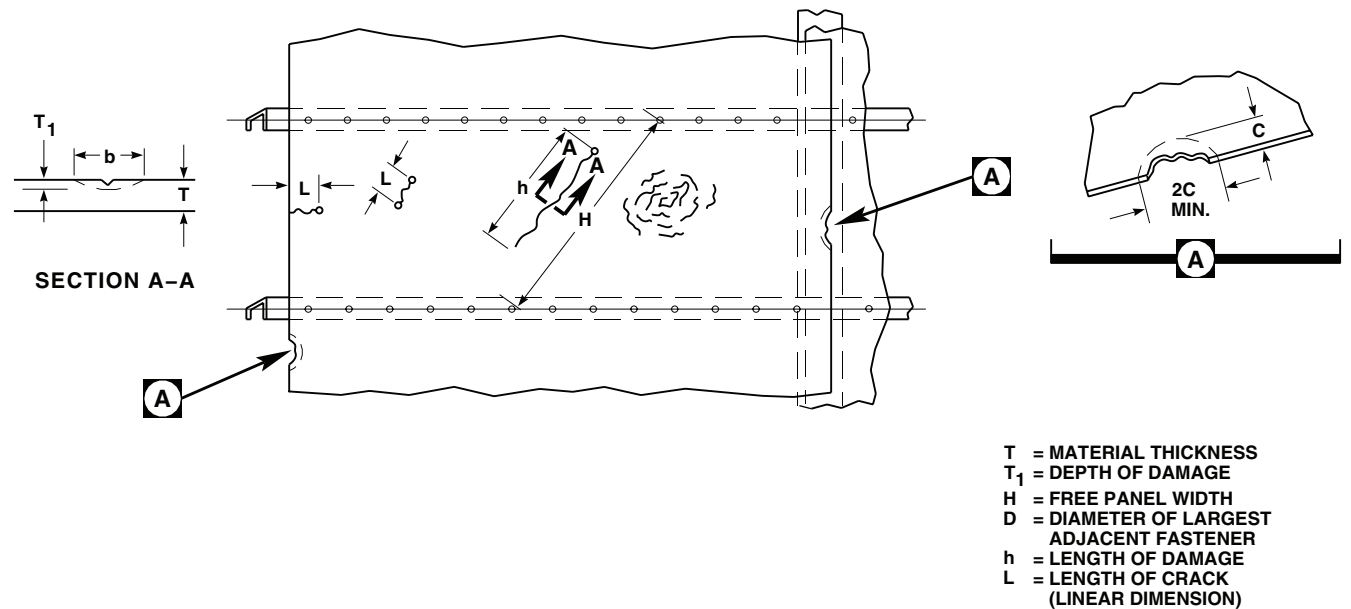
AK2099
SA

Figure 4. Skin Damage.

Negligible Damage Limits For Nonstructural Low Stress Frames

1. Nicks, Scratches, or Corrosion. Do not remove more material than necessary. Determine cross-sectional area of part. $(F + F + W) \times T = \text{area of cross-section}$. If damage reduction area (cleaned-up area, Figure 2, Section A-A) is no more than 10% of cross-section area, damage is classified as negligible. Treat nicks and scratches as cracks if reduction area is more than 10% of cross section area.
2. Cracks. No cracks allowed in radius area. Stop-drill cracks, or nicks and scratches over damage limit, using 1/8-inch diameter drill. Cleaned-up damage to be not more than "C" damage reduction (Figure 5). Consider transverse length of damage when figuring damage reduction (Figure 1). Nicks and gouges in flange edge not over 1/4-inch deep are blended out to not more than twice damage depth (Figure 2).
3. Punctures. Cleaned-up area (including radiating cracks) to be not more than "C" damage reduction. Patch holes over "C" damage limits (WP 0378 00).
4. Dents. Smooth dents, free from sharp projections or abrasions, not over 1/4-inch deep and 1 inch in diameter, are considered negligible. Treat as holes all dents over 1/4-inch deep or those with nicks or gouges. Trim out damage, sand to a smooth rounded contour, and patch (WP 0378 00).
5. Cracks in Flange Edge. When a crack ends at fastener hole, same type fastener may be added on each side of damage hole, providing "M" dimension is not less than "2D" (Figure 2). If "M" dimension is less than "2D", repair flange (WP 0378 00).

NEGLIGIBLE DAMAGE LIMITATIONS - Continued

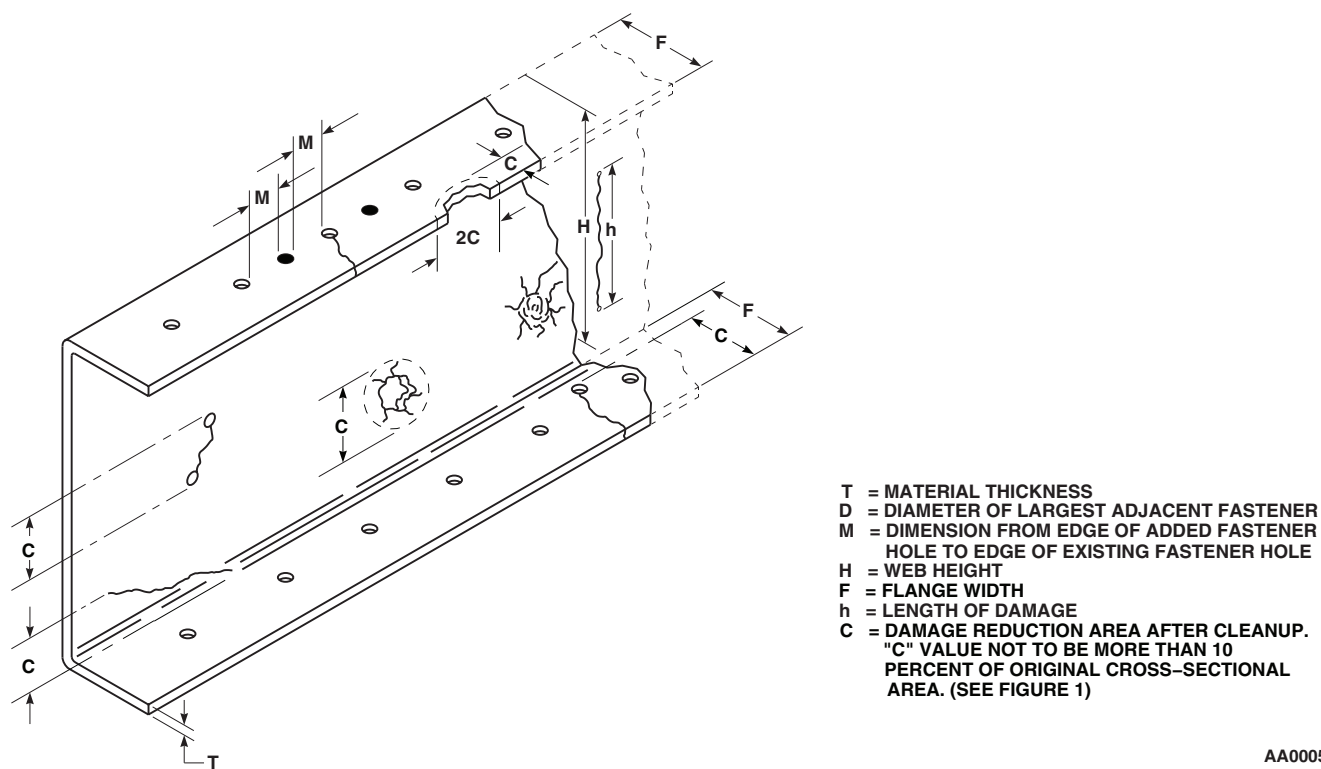
AA0005A
SA

Figure 5. Nonstructural Frame Damage.

CORROSION PROTECTION AND WATER INTEGRITY SEALING

Description

1. Without proper protection, metals used in any helicopter structure are subject to various types of corrosion damage. During manufacture and assembly, primers, sealants, and coatings are applied to the entire structure to shield against moisture, salts, chemicals, harmful vapors, and the electrolytic action caused by dissimilar metals in direct contact with each other. The sealants also prevent rain and splashed water from leaking through the skin, plating, and window joints of the helicopter. MIL-C-46168 paints used for the interior and exterior color schemes of late series helicopters resist chemical, gas, and radiation absorption.

NOTE

On earlier series helicopters, some areas will be found that were not originally sealed with sealing compound, Item 266, WP 1803 00. However, for making repairs, use the following sealing requirements for all helicopters, regardless of serial number.

2. When making repairs, the corrosion protection and water integrity of the entire structure must be carefully preserved. If old parts are being repaired or replaced, sealants and coatings which were broken or removed in cleanup must be reapplied. Likewise, when new joints, fasteners, and overlaps are added as would be done in making a skin patch, the specified corrosion protection and sealant must be carefully and thoroughly applied.
3. Clean surfaces and the correct use of materials are essential for sealants to adhere properly. For general procedures on cleaning and corrosion control, refer to TM 1-1500-344-23. Special instructions for all helicopter series are given in the following paragraphs. For paint touch up instructions, refer to TM 55-1500-345-23 and WP 0381 00.

CLEANING

CAUTION

Metal chips, filings, and grit may damage metal surfaces when wiping. Make sure to remove them before wiping surface.

1. Remove metal chips, filings, and grit from area using vacuum cleaner.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Wipe all surfaces clean with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. When cleaning large areas, change or refold cheesecloth frequently.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean one small section at a time and do not allow technical acetone, Item 5, WP 1803 00, to air dry.
4. Dry surfaces with clean cheesecloth cloth, Item 85, WP 1803 00, wiping in one direction only.

Surface Preparation and Priming

Unless otherwise noted and before sealing compound is applied, treat all bare metal surfaces with corrosion resistant coating, Item 111, WP 1803 00, and prime (WP 0381 00).

APPLICATION OF SEALING COMPOUND

1. Use sealing compound, Item 266, unless otherwise specified.
2. When using sealing compound, Item 266, WP 1803 00, select the grade that will allow repair parts to be assembled and fastened while compound is still wet.
3. Refer to Figure 6, Sheet 1, for examples of different methods of sealing compound application.
4. When hole tolerances are too close for use of sealing compound, Item 266, WP 1803 00 install bushings, bolts, inserts etc, with a wet coat of epoxy primer coating kit, Item 120, WP 1803 00 (Figure 6, Sheet 1).
5. Refer to Figure 6, Sheet 2, for structural examples of sealing compound application. Apply sealing compound according to example which matches or is most like the repair being made.
6. Before installation apply a wet coat of sealing compound, Item 266, WP 1803 00 to all rivets, bolts, and other fasteners which attach skins, fittings, plating, and accessories to the airframe exterior. Install fasteners while sealing compound is still wet.
7. When installing skins, patches, fittings and accessories on any exterior surface of the airframe, seal all overlapping and mating surfaces. Brush or roll on a coat of sealing compound, Item 266, WP 1803 00 to make a finished seal 0.002 to 0.005-inch maximum thickness.

APPLICATION OF SEALING COMPOUND - Continued

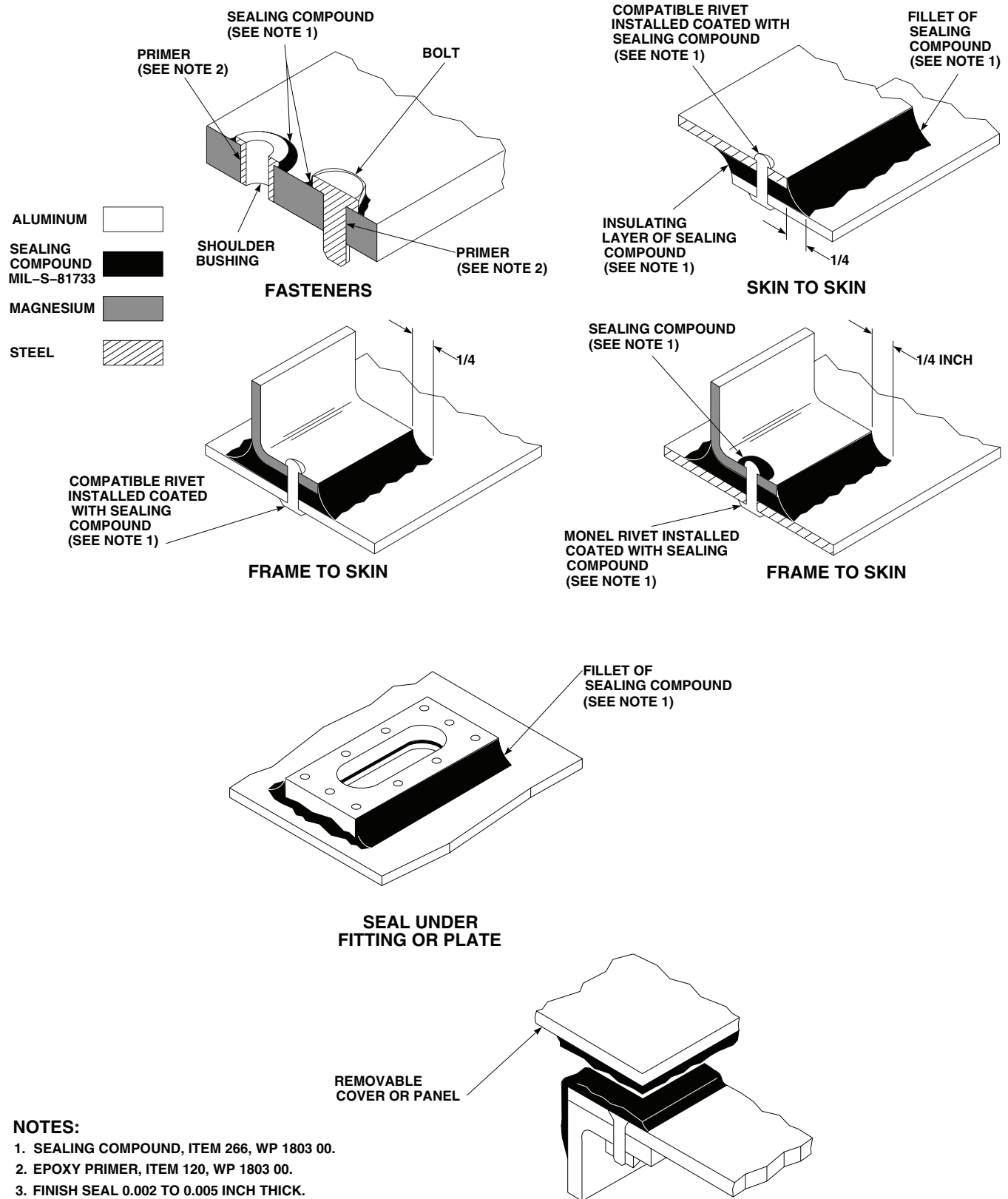
AB2439_1D
SA

Figure 6. Sealing Compound Application. (Sheet 1 of 2)

APPLICATION OF SEALING COMPOUND - Continued

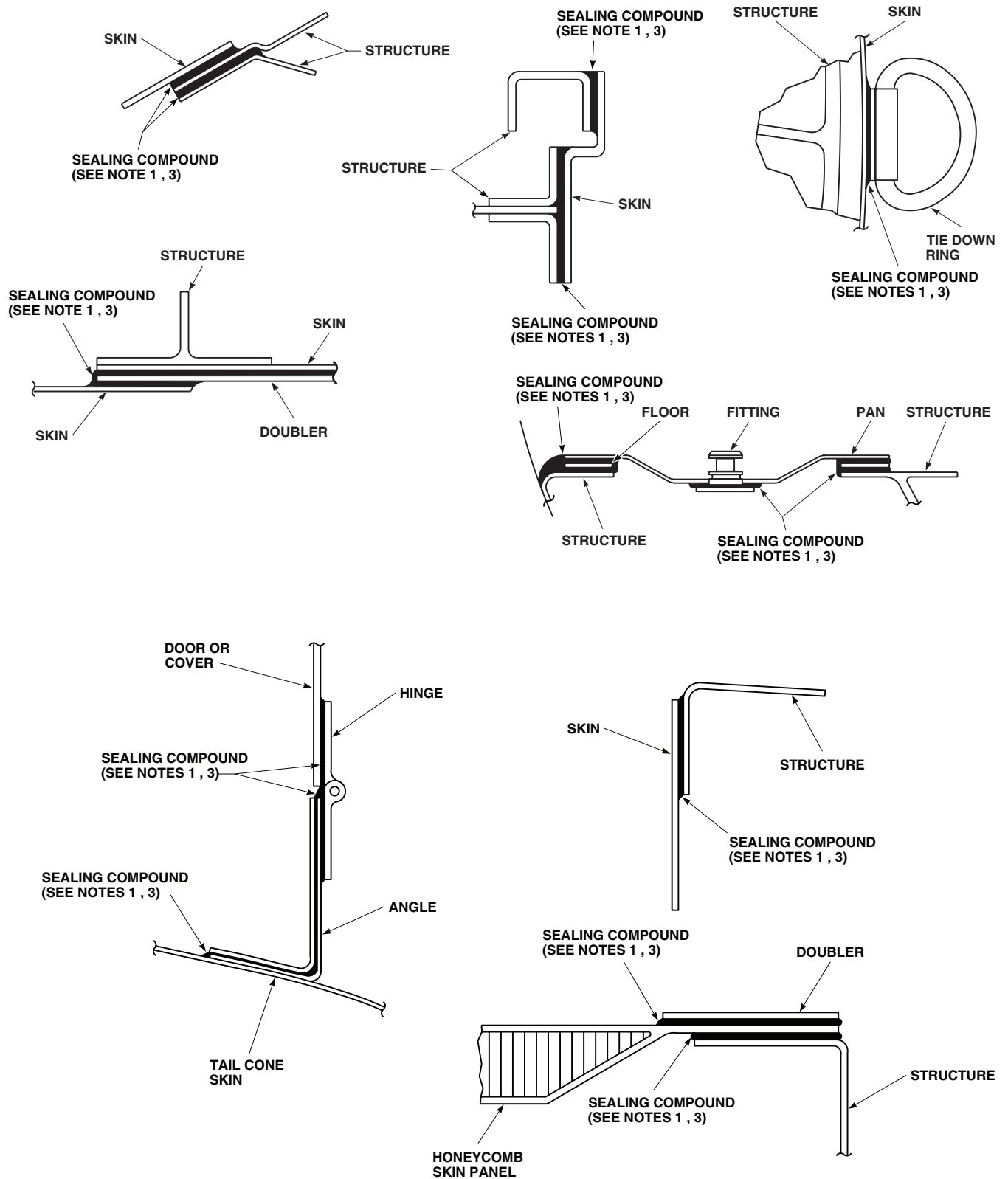
AK2111_2A
SA

Figure 6. Sealing Compound Application. (Sheet 2 of 2)

APPLICATION OF SEALING COMPOUND - Continued

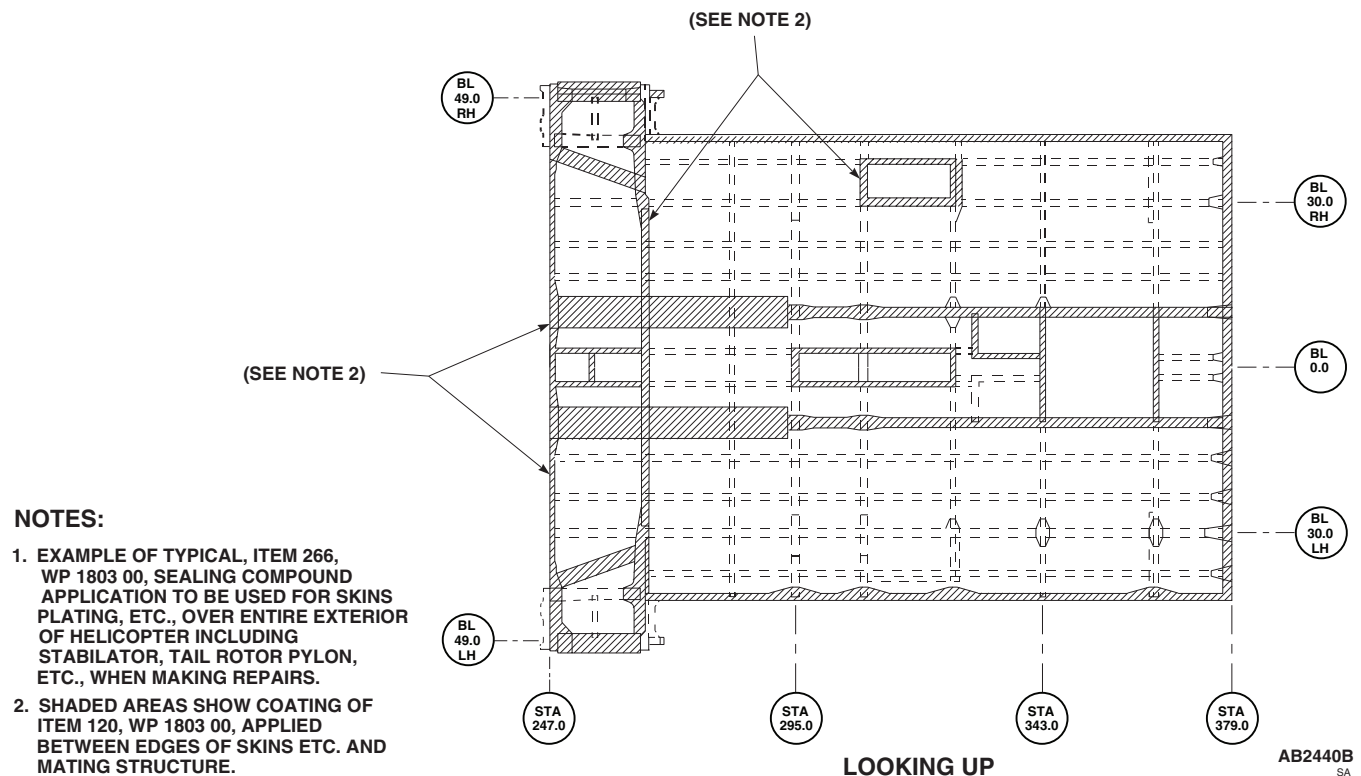


Figure 7. Exterior Plating Sealing.

8. On entire exterior of airframe, seal all edges of skins, panels, and plating to mating structure with sealing compound, Item 266, WP 1803 00. Spread compound with a brush or roller to make a finished seal 0.002 to 0.005-inch thick. Inboard locations of items that attach to structure do not need brush sealing. At all hole locations, along edges, and in inboard areas, install rivets and fasteners, and a wet coat of sealing compound, Item 266, WP 1803 00 (Figure 7).
9. Apply sealing compound, Item 266, WP 1803 00 between gussets, doublers, and mating skin surfaces. Install fasteners for these items with a wet coat of sealing compound.
10. After completing a repair, but before painting top coat, fill all gaps, joints, voids, and pockets which could trap and hold water with a fillet of sealing compound, Item 266, WP 1803 00.
11. After priming, seal all repaired surfaces on the interior of the structure from STA 162.0 to STA 485.0 and below waterlines as shown (Figure 8) with two spray coats of corrosion preventive compound, Item 102, WP 1803 00. Do not apply corrosion preventive compound to top sides of cockpit and cabin floors.
12. Refer to WP 0381 00, to find paint type and color for the helicopter.

INSPECTION OF CAMLOC FASTENERS

1. **NO MORE THAN 20% OF CAMLOC FASTENERS ON ANY FAIRING ARE ALLOWED TO BE BROKEN.**
2. Each broken fastener must have serviceable fasteners on both sides. **NO TWO ADJACENT FASTENERS CAN BE BROKEN.**
3. When fasteners are grouped in a small area the following damage limits apply:
 - a. Quantity of 1 to 3 fasteners. **NO FASTENERS SHALL BE BROKEN.**

INSPECTION OF CAMLOC FASTENERS - Continued

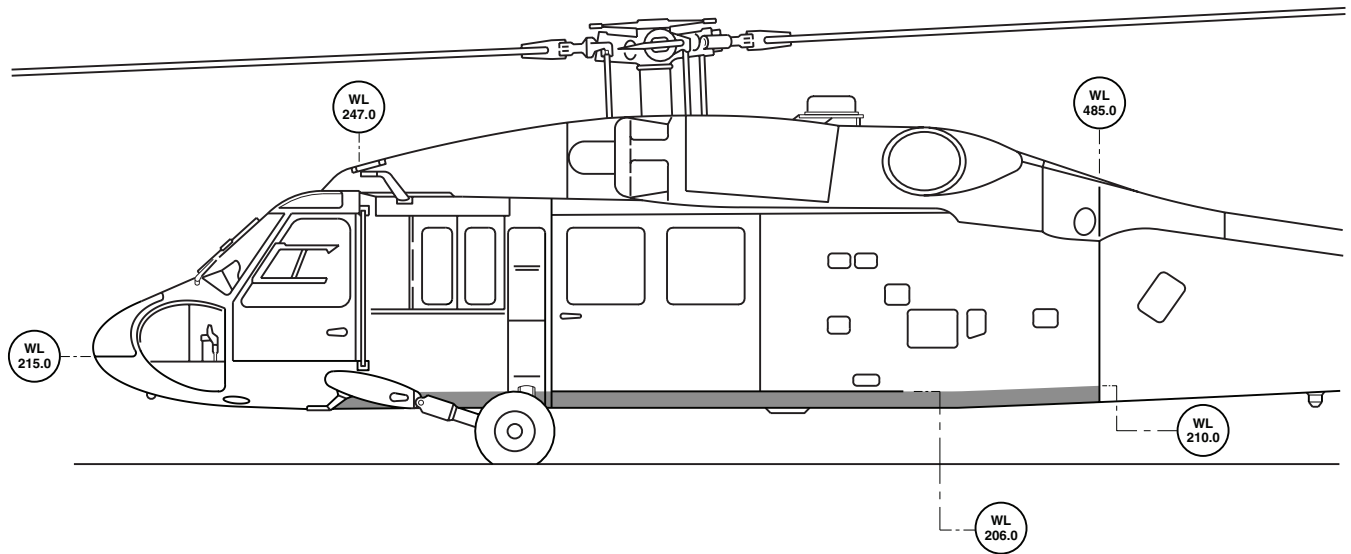
AA0009B
SA

Figure 8. Corrosion Prevention Spray.

- b. Quantity of 4 to 9 fasteners. **ONE FASTENER CAN BE BROKEN.**
- c. Quantity of 10 or more fasteners. **20% OF FASTENERS CAN BE BROKEN.**

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****METAL STRUCTURE REPAIR**

This WP supersedes WP 0378 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield

Rivet, MS20615M
Rivet, NAS1097AD
Tape, Pressure Sensitive, Adhesive, Item 332,
WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 334,
WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Hi-Shear Rivet, HS52P6
Rivet, MS20426AD
Rivet, MS20470AD
Rivet, MS20600AD

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
WP 1803 00

PRIMARY STRUCTURE SKIN PATCH - NON-FLUSH PATCH

1. Refer to Figure 1.
2. On aluminum alloy use MS20470AD rivets.
3. On stainless steel or titanium use MS20615M rivets.

PRIMARY STRUCTURE SKIN PATCH - NON-FLUSH FORMED PATCH

1. Refer to Figure 2.
2. On aluminum alloy use MS20470AD rivets.
3. On stainless steel or titanium use MS20615M rivets.

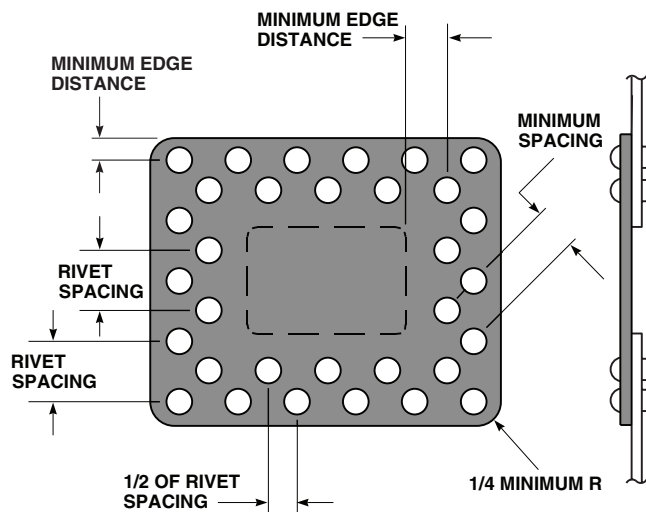
SECONDARY STRUCTURE SKIN PATCH

1. Refer to Figure 3.
2. On aluminum alloy use MS20470AD rivets.
3. On stainless steel or titanium use MS20615M rivets.

RIVETED FLUSH PATCH REPAIR

1. Refer to Figure 4.
2. Skin filler - same gage and material as original skin.
3. Skin patch - one gage thicker of same material as original skin.
4. For flush installations, dimple gages under 0.050 and machine countersink gages 0.050 and greater. Double dimple or dimple thin skin into countersunk heavy skin. Rivet head style will be flush type.

RIVETED FLUSH PATCH REPAIR - Continued

**NOTE:**

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

RIVET TABLE NON-FLUSH FORMED SHEET PATCH				
PATCH	RIVET DIAMETER	RIVET SPACING INCHES	MIN. EDGE DISTANCE INCHES	MIN. SPACING INCHES
0.012	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
0.016	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
0.020	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
0.025	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
0.032	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
0.040	$\frac{5}{32}$	$\frac{5}{8}$	$\frac{5}{16}$	$\frac{3}{4}$
0.050	$\frac{5}{32}$	$\frac{3}{4}$	$\frac{3}{8}$	$\frac{3}{4}$
0.063	$\frac{3}{16}$	$\frac{3}{4}$	$\frac{3}{8}$	$\frac{3}{4}$
0.071	$\frac{3}{16}$	$\frac{3}{4}$	$\frac{3}{8}$	$\frac{3}{4}$
0.080	$\frac{3}{16}^*$	$\frac{3}{4}$	$\frac{3}{8}$	$\frac{3}{4}$
0.090	$\frac{3}{16}^*$	$\frac{3}{4}$	$\frac{3}{8}$	$\frac{3}{4}$
0.125	$\frac{3}{16}^{**}$	$\frac{3}{4}$	$\frac{3}{8}$	$\frac{3}{4}$

* ALUMINUM LOCKBOLTS OR EQUIVALENT
 ** STEEL LOCKBOLTS OR EQUIVALENT

AB0226A
SA

Figure 1. Non-Flush Patch.

TEMPORARY PRESSURE SENSITIVE ADHESIVE TAPE PATCH REPAIR

1. Refer to Figure 5.
2. Damage one inch or less in diameter after trimming may be repaired by patching with pressure sensitive adhesive tape, Item 332, WP 1803 00.
3. Bend edge of hole slightly inward to provide smooth, flat surface for patch.
4. Mask off a rounded area on skin 1 1/8-inches from and around edge of trimmed hole, using pressure sensitive adhesive tape Item 334, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Remove paint and clean bare metal using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

TEMPORARY PRESSURE SENSITIVE ADHESIVE TAPE PATCH REPAIR - Continued

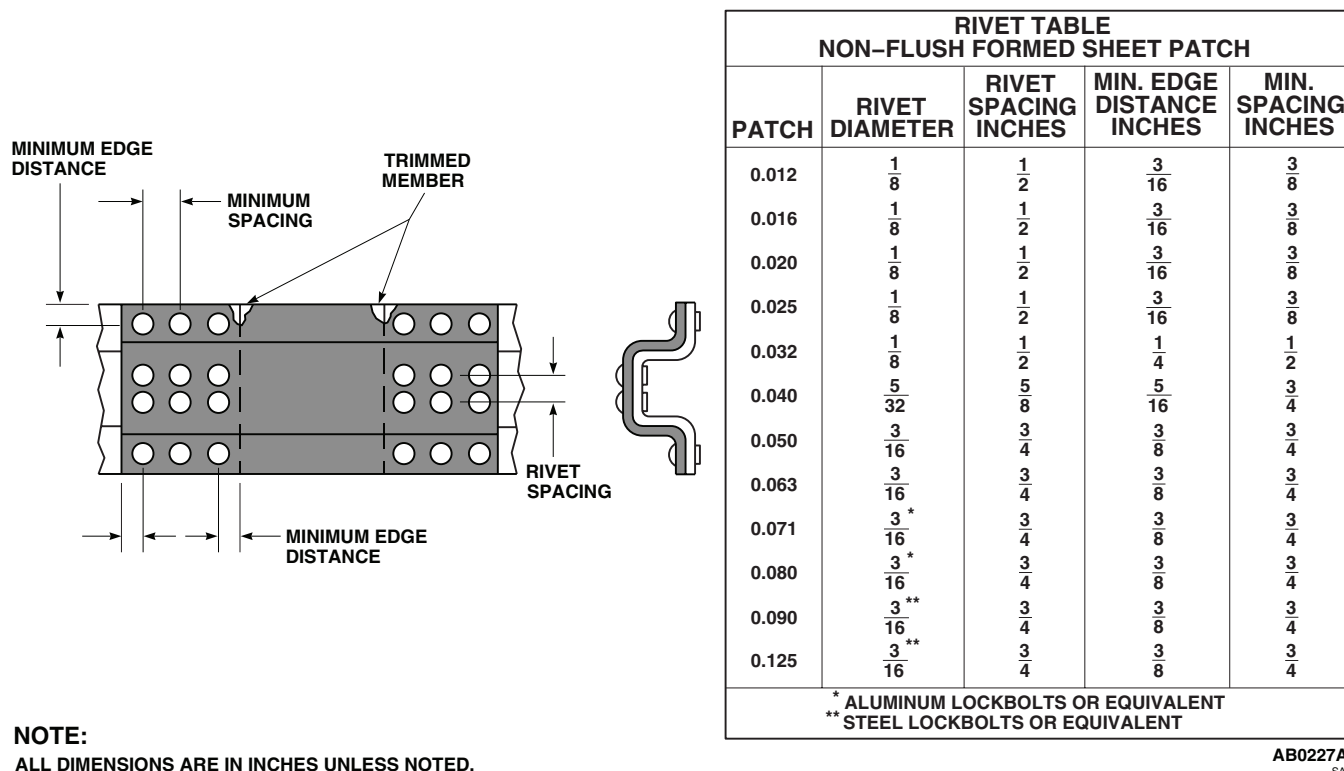


Figure 2. Non-Flush Formed Patch.

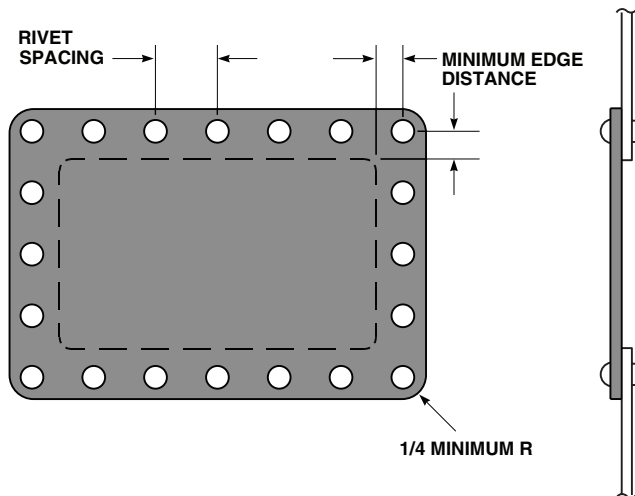
CAUTION

- Pressure sensitive tape adhesive will become contaminated if not properly handled. Do not place adhesive side of pressure sensitive tape patch against template to prevent adhesive contamination.
 - Pressure sensitive tape will not adhere to skin if surface of skin is too cold. Temperature of skin should be at least 50°F (10°C) before aluminum tape patch is applied. Warm skin area to proper temperature if temperature of skin is below minimum.
6. Make paper patch template same size as cleaned patch area inside masking tape. Trim template 1/8-inch short of masking tape edges and cut adhesive pressure sensitive tape, Item 332, WP 1803 00, to shape and size of template.
 7. To maintain patch overlap requirements on wide repairs, use two pieces of tape side-by-side with edges touching over center of hole. Add another strip of tape in same direction to cover tape joint (Figure 5, Section A-A).
 8. Carefully place pressure sensitive tape patch in place over damaged area on skin, 1/8-inch away from edges of masking tape. Rub patch with clean pad of cheesecloth cloth, Item 85, WP 1803 00. Do not rub or depress tape patch over area of trimmed hole.

REPAIR DAMAGED BEADED SECTION

1. Refer to Figure 6.
2. Patch plate bead radii should nest over or inside bulkhead beads.
3. Rivet heads should not contact bead radii.

REPAIR DAMAGED BEADED SECTION - Continued



RIVET TABLE			
PATCH GAGE	RIVET DIAMETER INCHES	RIVET SPACING INCHES	MIN. EDGE DISTANCE INCHES
0.012	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{16}$
0.016	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{16}$
0.020	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{16}$
0.025	$\frac{1}{8}$	$\frac{3}{4}$	$\frac{3}{16}$
0.032	$\frac{1}{8}$	$\frac{7}{8}$	$\frac{3}{16}$
0.040	$\frac{5}{32}$	$\frac{7}{8}$	$\frac{5}{16}$
0.050	$\frac{5}{32}$	$\frac{7}{8}$	$\frac{5}{16}$
0.063	$\frac{3}{16}$	1	$\frac{3}{8}$
0.071	$\frac{3}{16}$	1	$\frac{3}{8}$
0.080	$\frac{3}{16}$	1	$\frac{3}{8}$
0.090	$\frac{3}{16}$	1	$\frac{3}{8}$
0.125	$\frac{3}{16}$	1	$\frac{3}{8}$

NOTE:

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB0228A
SA

Figure 3. Secondary Structure Skin Patch.

REPAIR SKIN CLOSE TO A FRAME

1. Refer to Figure 7.
2. Skin patch - one gage thicker of same material as original skin.
3. Skin patch must extend 3/4-inch beyond frame.
4. Repair skin per (TM 1-1500-204-23).

REPAIR FORMED AND EXTRUDED ANGLE

1. Refer to Figure 8 and Table 1.
2. Type I Repair, Damage to One Flange- **DAMAGE MUST NOT BE MORE THAN 60% OF FLANGE WIDTH.** If damage exceeds limit, use Type II Repair.
3. Repair member to be next heavier gage of same material as original angle (clad sheet permissible for extrusion).
4. Primary Type I Repair - Use flanged repair member. Flange extension beyond radius must be equal to material thickness of repair member.
5. Alternate Type I Repair - Use unflanged repair member.
6. Lay out rivet pattern around existing original attachments. Replace original attachments with same diameter and type attachments. Use aluminum alloy fillers as necessary.
7. Repair member radii should nest or be a practicable minimum. Rivet heads should not contact bend radius.

REPAIR FORMED AND EXTRUDED ANGLE - Continued

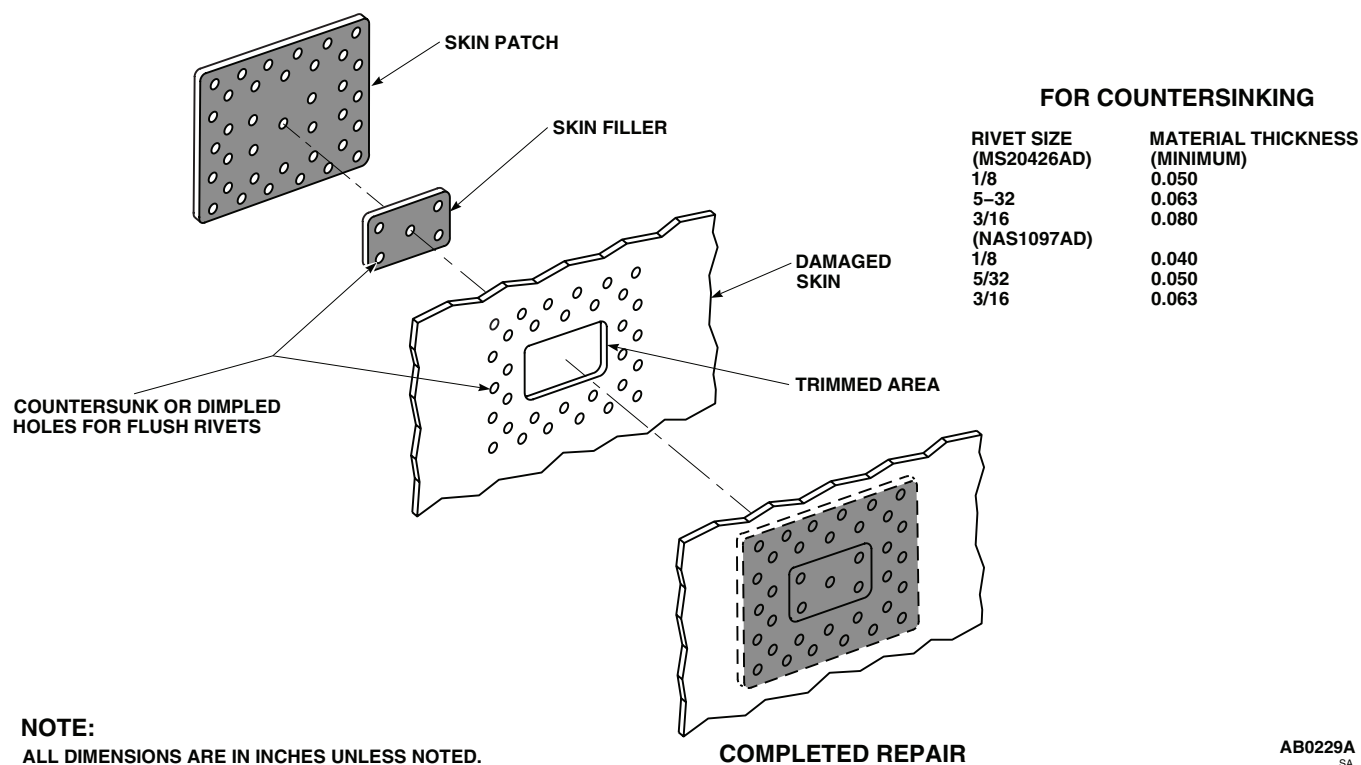
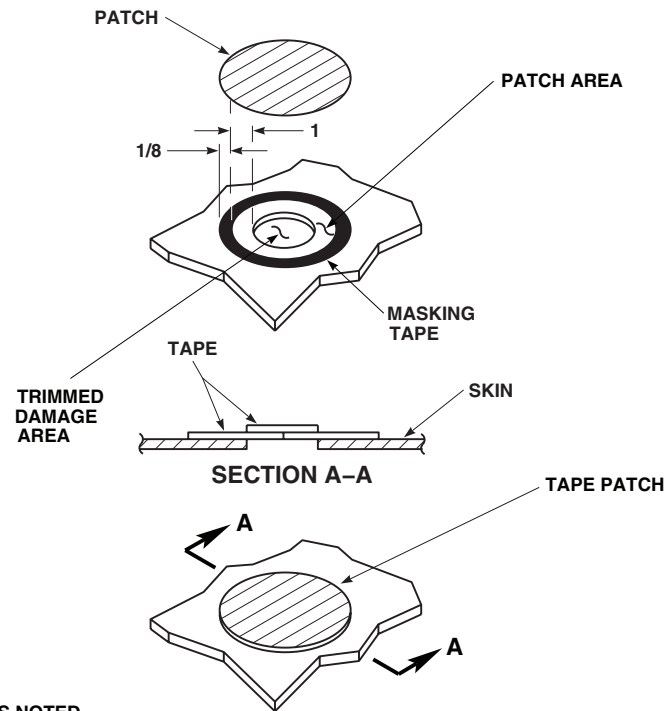


Figure 4. Flush Patch.

8. MS20600AD blind rivets or equivalent may be substituted for same diameter solid rivets, provided the number of blind rivets required is increased at a ratio of 3 to 2.
9. Rivet requirements for 7075-T6 clad material are tabulated as follows:
 - a. If original material is 2024-T4 clad, multiply tabulated rivet requirements by 0.8.
 - b. If original material is extruded, multiply tabulated rivet requirements by 1.15.

REPAIR FORMED AND EXTRUDED ANGLE - Continued

**NOTE:**

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB0230A
SA

Figure 5. Temporary Pressure Sensitive Adhesive Tape Patch.

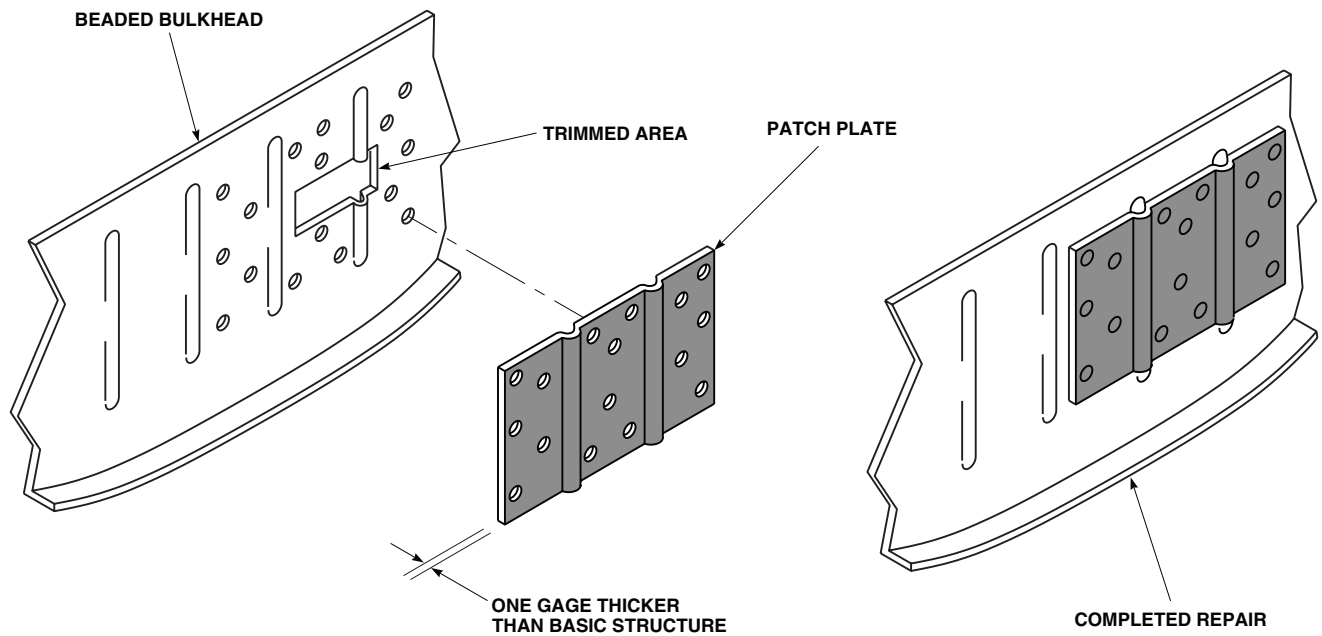
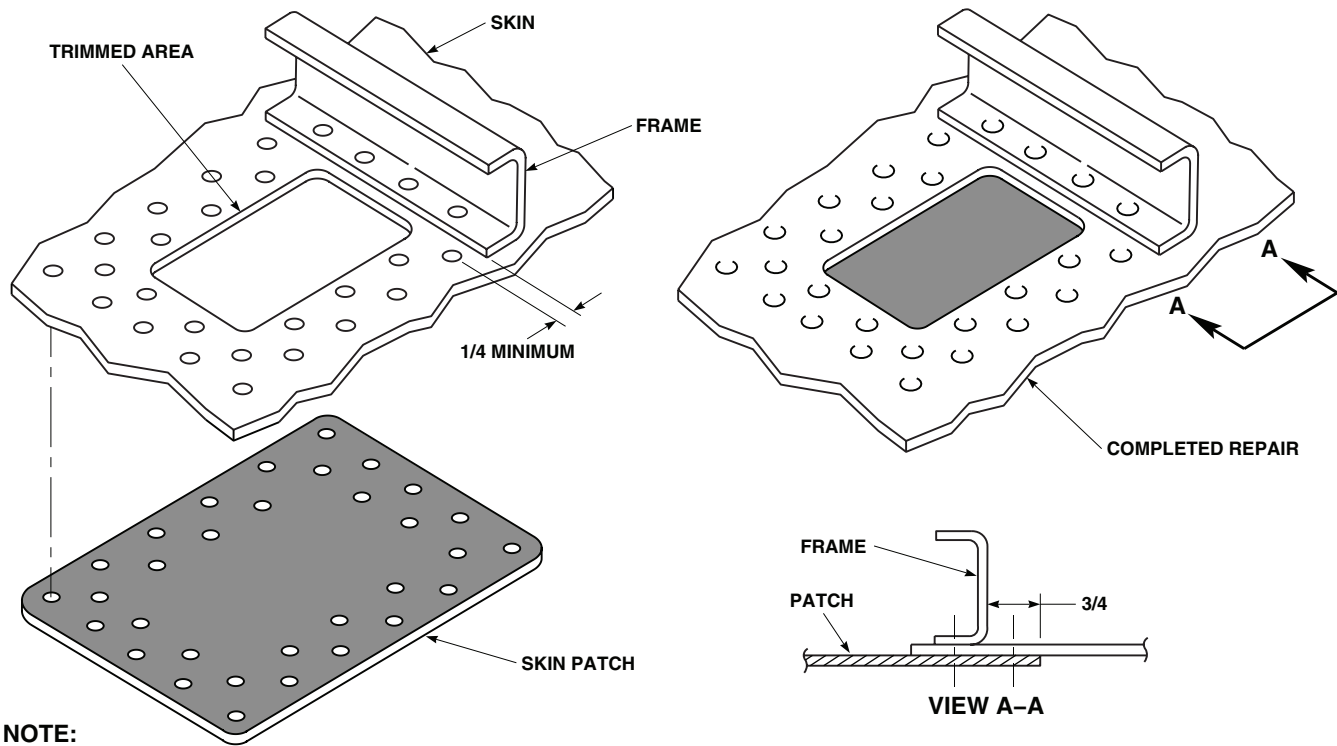
REPAIR FORMED AND EXTRUDED ANGLE - ContinuedAB0231
SA

Figure 6. Beaded Section.

REPAIR FORMED AND EXTRUDED ANGLE - Continued

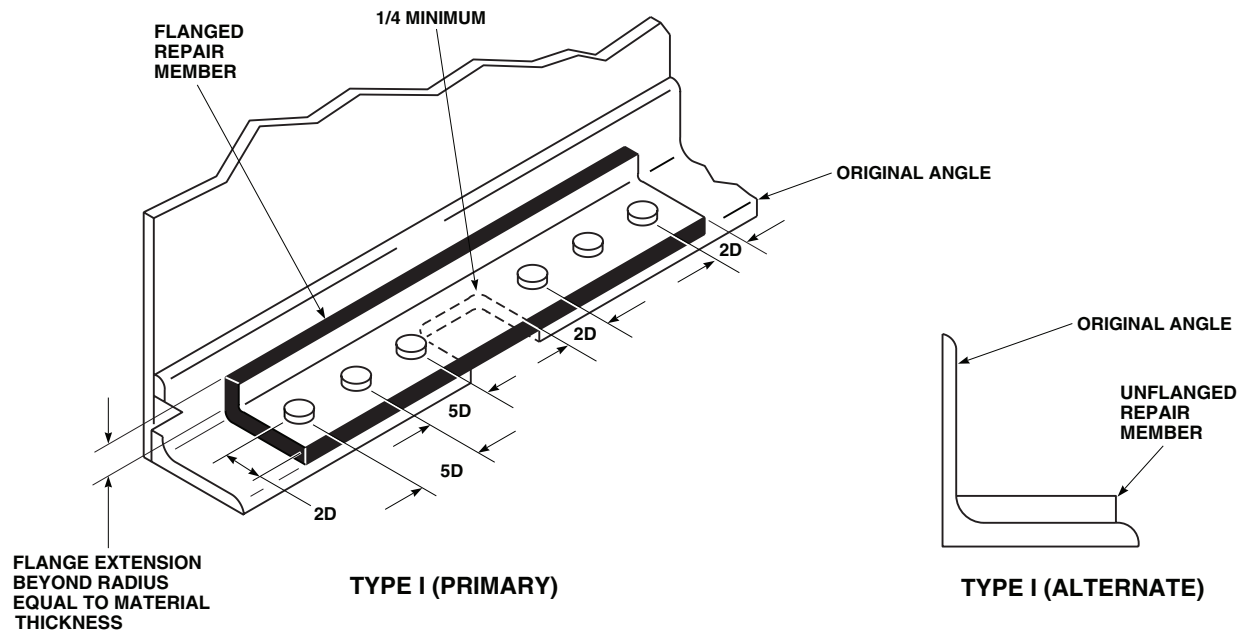


NOTE:
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

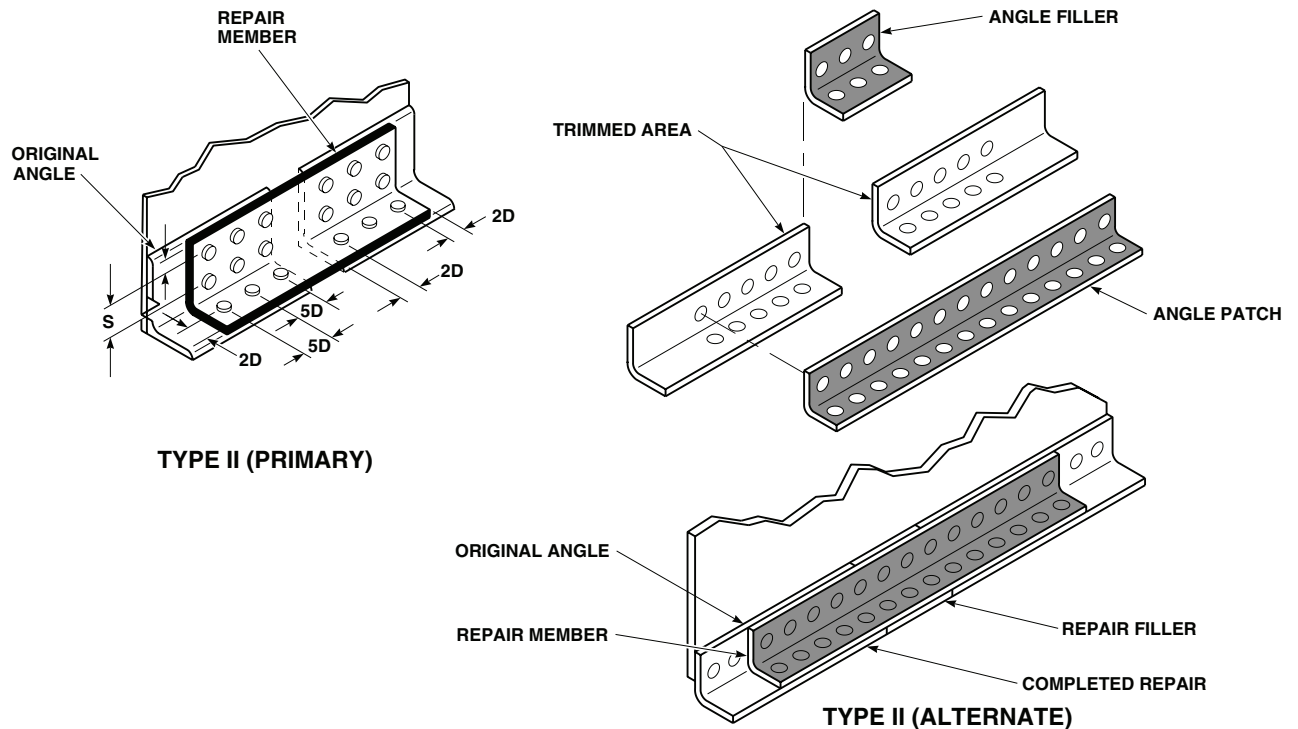
AB0232A
SA

Figure 7. Skin Close to Frame.

REPAIR FORMED AND EXTRUDED ANGLE - Continued



TYPE I REPAIR-DAMAGE TO ONE FLANGE



TYPE II REPAIR-COMPLETE FRACTURE

NOTE:

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB0233A
SA

Figure 8. Formed and Extruded Angle.

REPAIR FORMED AND EXTRUDED ANGLE - Continued**Table 1. Rivet Requirements for Formed and Extruded Angle Repair.**

ANGLE GAGE, RIVETS SIZE, LENGTHWISE SPACING (4D) MINIMUM CROSSWISE SPACING MINIMUM EDGE DISTANCE (2D)	NUMBER OF RIVETS REQUIRED WHEN EXTENT OF DAMAGE ACROSS ANGLE IS:				
	0.50-IN.	0.75-IN.	1 IN.	1.25-IN.	1.50-IN.
Gage - 0.025 Rivet - MS20470AD4 Lengthwise Spacing - 0.63 Crosswise Spacing - 0.55 Edge Distance - 0.25	3	3	4	5	6
Gage - 0.032 Rivet - MS20470AD4 Lengthwise Spacing - 0.63 Crosswise Spacing - 0.55 Edge Distance - 0.25	3	4	5	6	8
Gage - 0.040 Rivet - MS20470AD5 Lengthwise Spacing - 0.78 Crosswise Spacing - 0.67 Edge Distance - 0.31	3	3	4	5	6
Gage - 0.050 Rivet - MS20470AD5 Lengthwise Spacing - 0.78 Crosswise Spacing - 0.67 Edge Distance - 0.31	3	4	5	6	8
Gage - 0.063 Rivet - MS20470AD5 Lengthwise Spacing - 0.78 Crosswise Spacing - 0.67 Edge Distance - 0.31	3	5	6	8	9
Gage - 0.063 Rivet - MS20470AD6 Lengthwise Spacing - 0.94 Crosswise Spacing - 0.81 Edge Distance - 0.38	3	4	5	6	7

REPAIR DAMAGED TEE

1. Refer to Figure 9.
2. Repairs are applicable to the two types of tee extrusions shown in Figure 9, Section A-A and Figure 9, Section B-B.
3. Make repair angles and fillers from the same material as original section. Gage for repair angles to be one gage heavier than flanges of original section, but not less than one-half thickness of stem of original section.
4. Lay out rivet pattern around any original attachments. Replace original attachments using fasteners shown in Table 2.
5. If extent of damage exceeds 0.500-inch, insert fillers and continue rivet distribution through damage area.
6. Repair member radii should nest or be a practicable minimum. Rivet heads should not contact bend radius.

REPAIR DAMAGED TEE - Continued

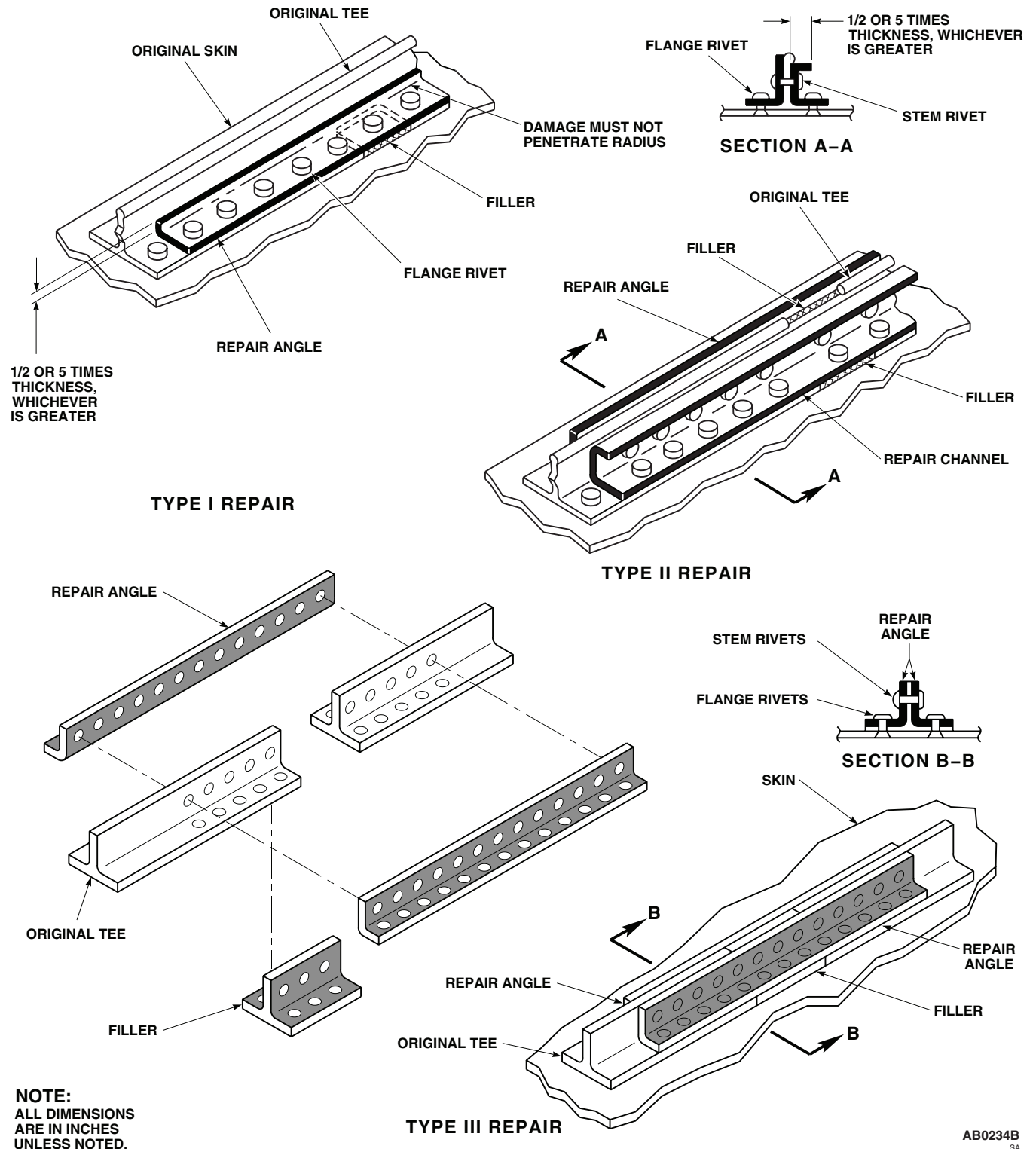


Figure 9. Damaged Tee.

REPAIR DAMAGED TEE - Continued

7. Stem Rivets, MS20470AD:
 - a. For 7075-T6, use 5 rivets on each side of damage area.
 - b. For 2024-T4, use 4 rivets on each side of damage area.
8. Flange Rivets, MS20426AD, Hi-Loks, or Lockbolts:
 - a. For 7075-T6, use 5 rivets on each side of damage area in each row.
 - b. For 2024-T4, use 4 rivets on each side of damage area in each row.

Table 2. Attachment Requirements for Damaged Tee Repair.

ORIGINAL MATERIAL THICKNESS	RIVET DIAMETER (D)	HI-LOK OR LOCKBOLT DIAMETER (D)	2D EDGE DISTANCE
0.032	4	-	0.25
0.040	5	-	0.31
0.050	5	-	0.31
0.063	6	-	0.38
0.071	6	-	0.38
0.080	-	6	0.38
0.090	-	6	0.38
0.100	-	6	0.38
0.125	-	6	0.38
0.160	-	6	0.38
0.190	-	6	0.38

REPAIR DAMAGED FORMER

1. Refer to Figure 10.
2. Former filler same material and thickness as former.
3. Former patch same material as former, one gage thicker.
4. Former patch bend radius should nest or be a practicable minimum. Rivet heads should not contact bend radius.
5. Minimum of 3 rivets required on each side of damage.

REPAIR DAMAGED RIB

1. Refer to Figure 11.

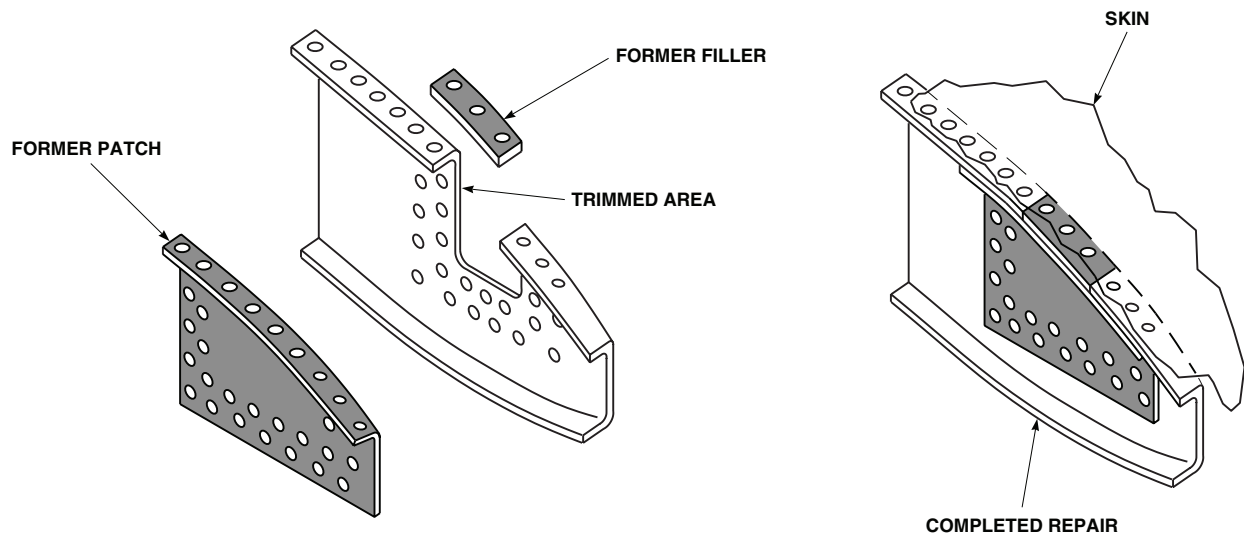
REPAIR DAMAGED RIB - ContinuedAB0235
SA

Figure 10. Damaged Former.

2. Rib Patch - Pick up existing rivet locations through bulkhead flanges. Rivets should be added between existing rivets when a minimum of 4D rivet spacing can be maintained.
3. Rivet heads should not contact bend radii.
4. Example One Repair - A minimum of 3 rivets in each row are required on each side of damage.

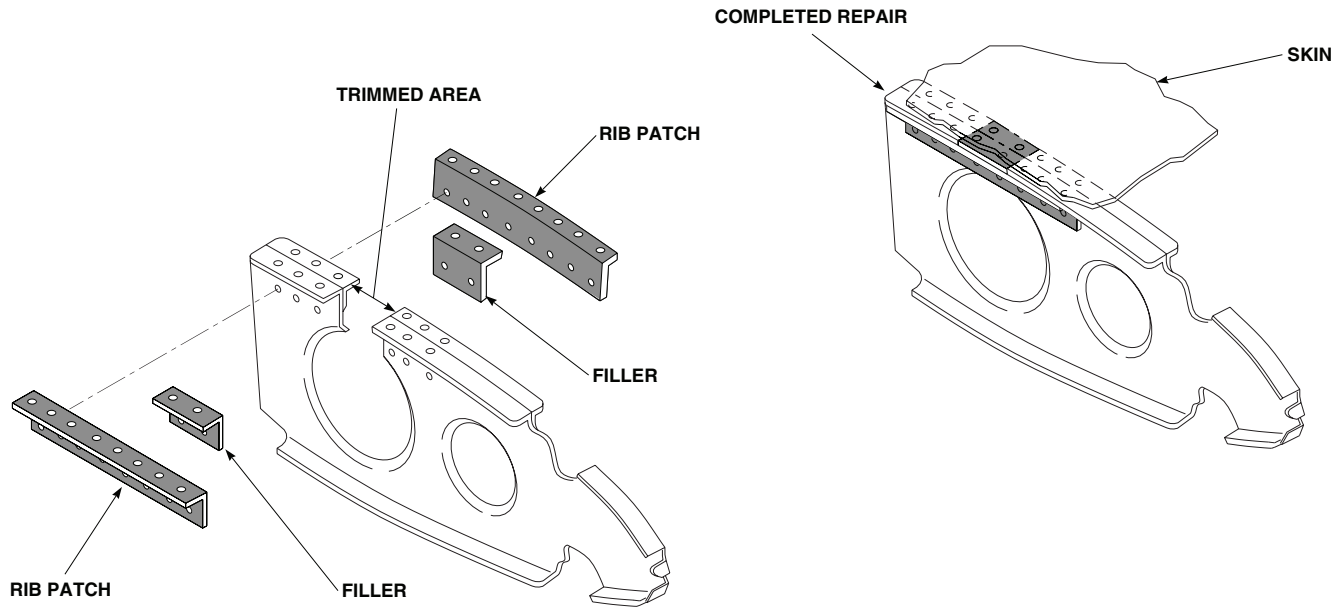
REPAIR DAMAGED CHANNEL

1. Refer to Figure 12.
2. Channel patch same material as channel, one gage thicker.
3. Channel patch bend radii should nest or be a practicable minimum.
4. Rivet heads should not contact bend radii.

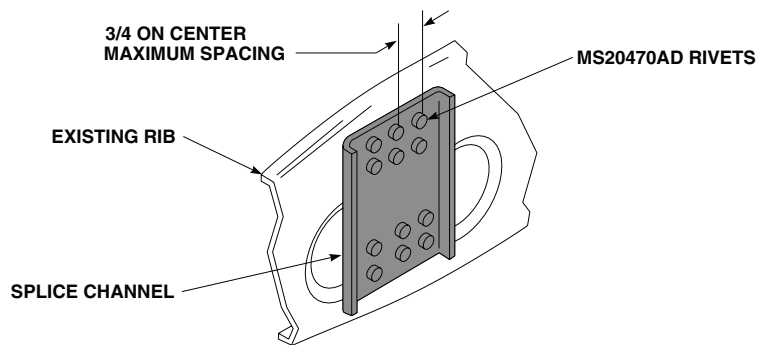
REPAIR DAMAGED ZEE

1. Refer to Figure 13, Sheet 1, Figure 13, Sheet 2, and Table 3.
2. Repair material to be same as existing material.

REPAIR DAMAGED ZEE - Continued

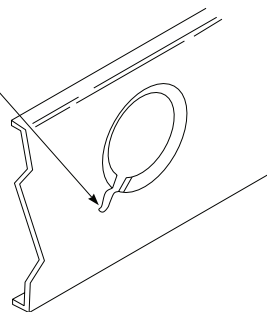


EXAMPLE ONE

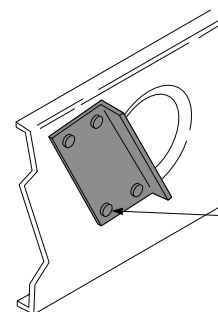


EXAMPLE TWO

DRILL 1/8 DIA
HOLE AT END
OF CRACK



EXAMPLE THREE



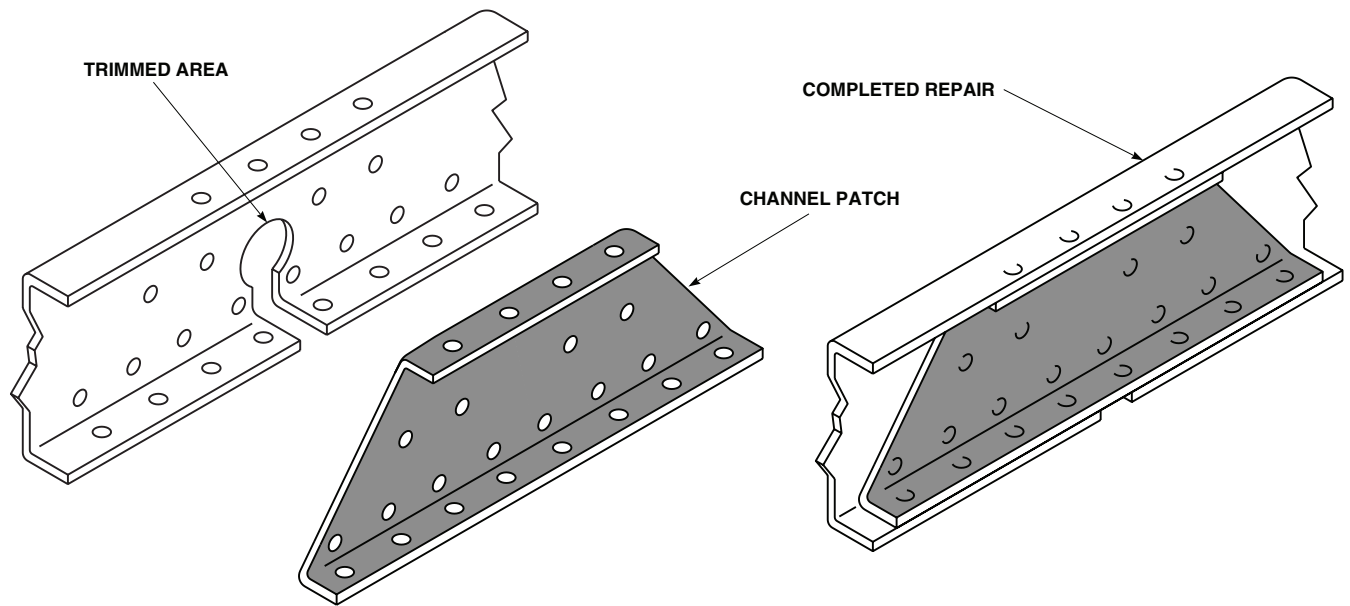
MS20470AD RIVETS
(4 REQUIRED)
APPROXIMATELY
AS SHOWN

NOTE:
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

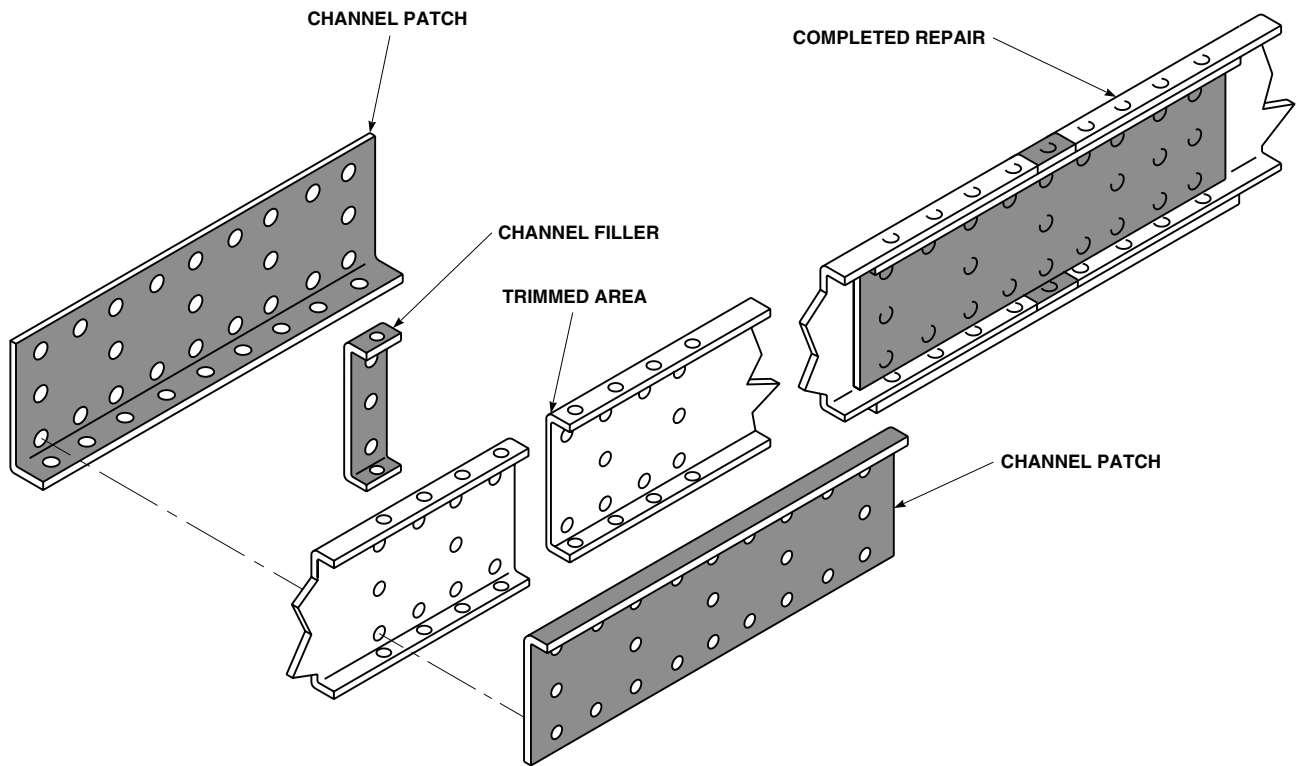
AB0236A
SA

Figure 11. Damaged Rib.

REPAIR DAMAGED ZEE - Continued



PATCHING CHANNEL CRACKS



CHANNEL INSERTION

AB0237
SA

Figure 12. Damaged Channel.

REPAIR DAMAGED ZEE - Continued**Table 3. Rivet Requirements for Damaged Zee Repair.**

ZEE GAGE, REPAIR MEMBER GAGE, RIVET SIZE	NUMBER OF RIVETS REQUIRED AFTER CALCULATING L1+L2+L3 (SEE NOTE):				
	2 IN.	2.50- IN.	3 IN.	3.5- IN.	4 IN.
Zee Gage - 0.025 Repair Member Gage - 0.032 Rivet Size - MS20470AD4	10	13	15	18	20
Zee Gage - 0.032 Repair Member Gage - 0.040 Rivet Size - MS20470AD4	20	15	18	22	25
Zee Gage - 0.040 Repair Member Gage - 0.050 Rivet Size - MS20470AD5	-	13	15	18	20
Zee Gage - 0.050 Repair Member Gage - 0.063 Rivet Size - MS20470AD5	-	15	18	21	24
Zee Gage - 0.063 Repair Member Gage - 0.071 Rivet Size - MS20470AD6	-	19	23	27	30
NOTE					
Number of rivets shown is for each side of damage. Divide equally among all rows. Add to number in table as necessary to make it divisible by number of rows used. For in-between values of L1+L2+L3 use next higher column.					

3. Rivet Requirements:

- Use same type of rivets in skin flange as originally installed. If there is no skin attached to member, use MS20470AD rivets.
- Use existing skin flange rivet spacing if flange is drilled. If member was not originally drilled, use 5 times rivet diameter for center-to-center spacing.
- Use original hole and rivet size as required. If hole is elongated, use larger diameter.

4. Filler member to be same gage and material as original zee.

5. Use same number of rows of attachments per flange as in original zee.

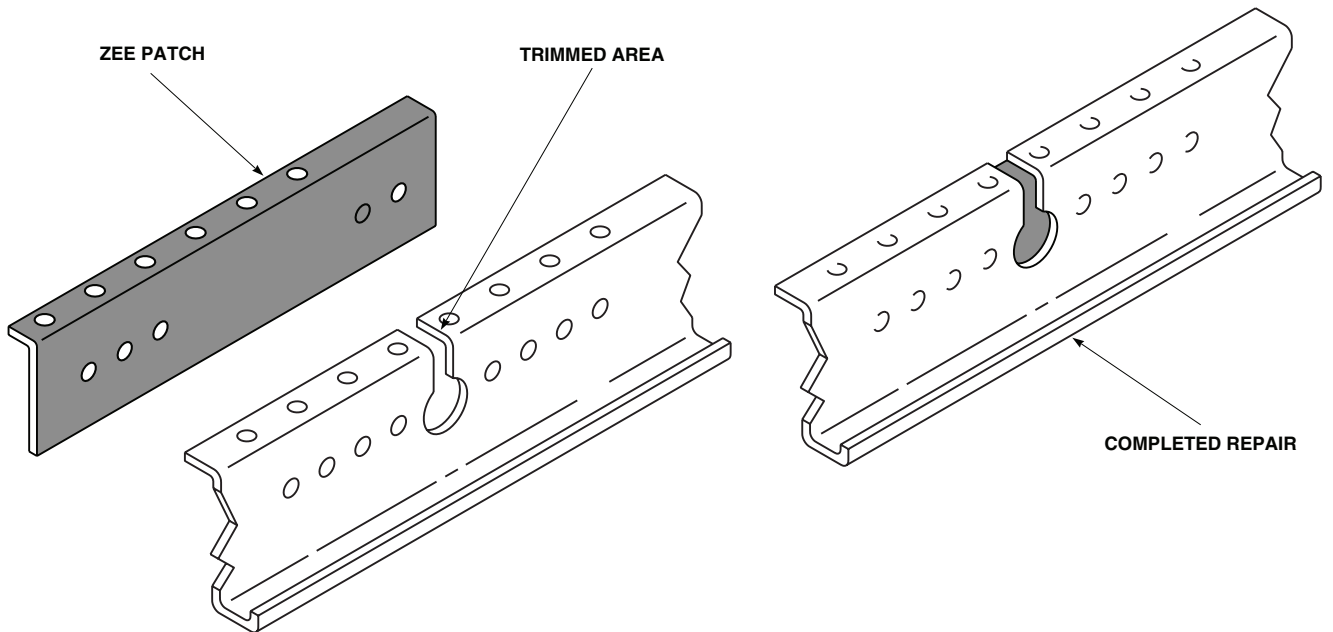
6. Flange angle with web may vary.

7. When L2 is less than 10 times diameter of required rivets, form a single zee instead of two angles and leave off web strap. Use only one row of rivets through web.

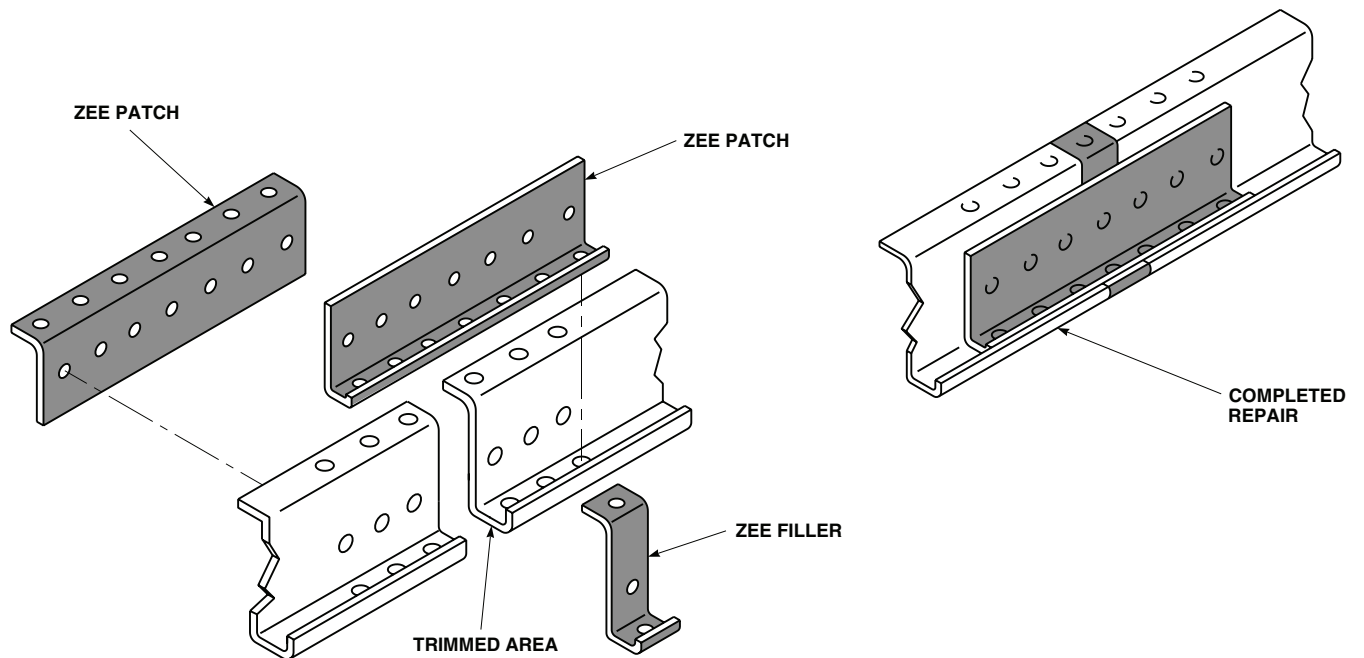
8. Repair angle and zee patch bend radii should nest or be a practicable minimum.

9. Rivet heads should not contact bend radii.

REPAIR DAMAGED ZEE - Continued



PATCHING ZEE CRACKS



ZEE INSERTION TYPE I

AB0238_1
SA

Figure 13. Damaged Zee. (Sheet 1 of 2)

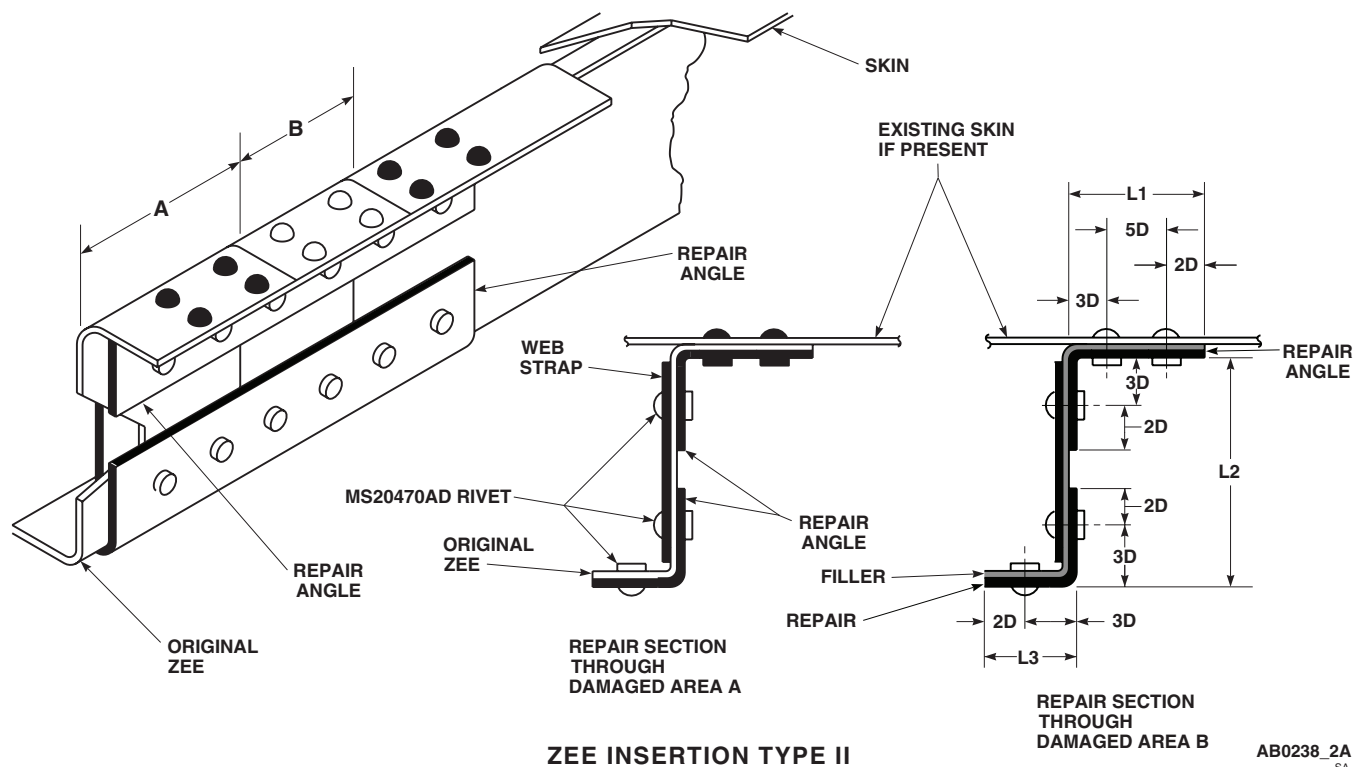
REPAIR DAMAGED ZEE - Continued

Figure 13. Damaged Zee. (Sheet 2 of 2)

REPAIR DAMAGED LONGERON

1. Refer to Figure 14.
2. Repair member bend radii should nest or be a practicable minimum.
3. Rivet heads should not contact bend radii.
4. Filler - Same gage and material as longeron.
5. Top splice zee, top splice, and bottom splice - Same material as longeron, one gage thicker.

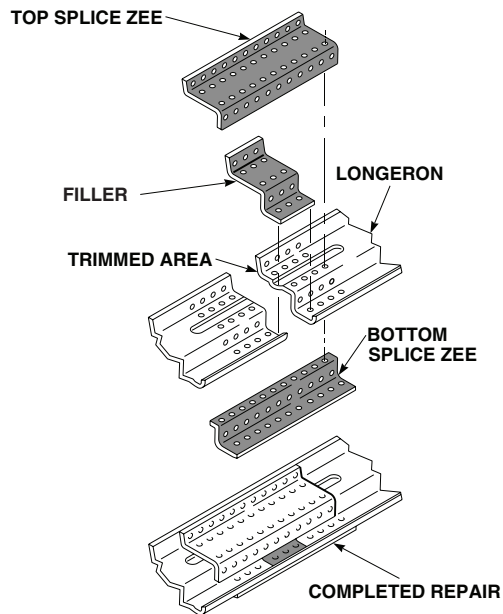
REPAIR DAMAGED FRAME

1. Refer to Figure 15.
2. Repair member bend radii should nest or be a practicable minimum.
3. Rivet heads should not contact bend radii.
4. Filler - Same gage and material as frame.
5. Patch angle and patch- Same material as frame, one gage thicker.
6. Use MS20470AD type rivet.

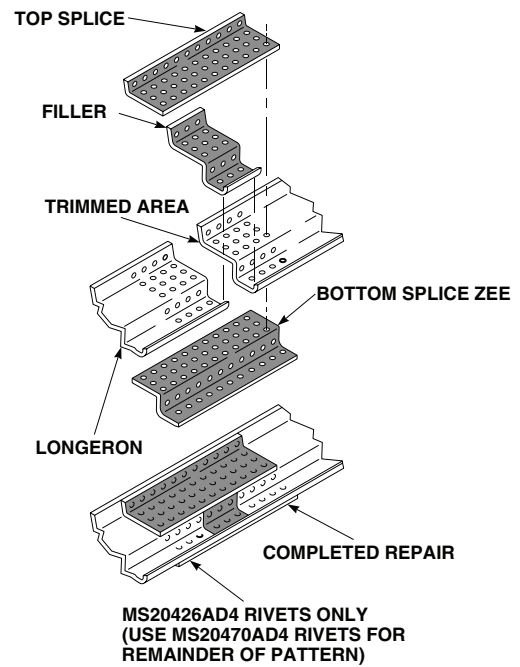
REPAIR DAMAGED LIGHTNING HOLES

1. Refer to Figure 16.
2. Repair member bend radii should nest or be a practicable minimum.

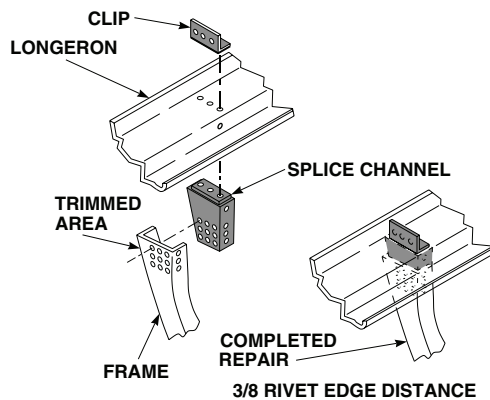
REPAIR DAMAGED LIGHTNING HOLES - Continued



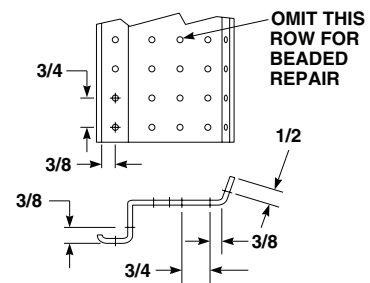
BEADED REPAIR



UNBEADED REPAIR



FRAME AND LONGERON REPAIR



RIVET PATTERN (TYPICAL)

NOTE:
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB0239A
SA

Figure 14. Damaged Longeron.

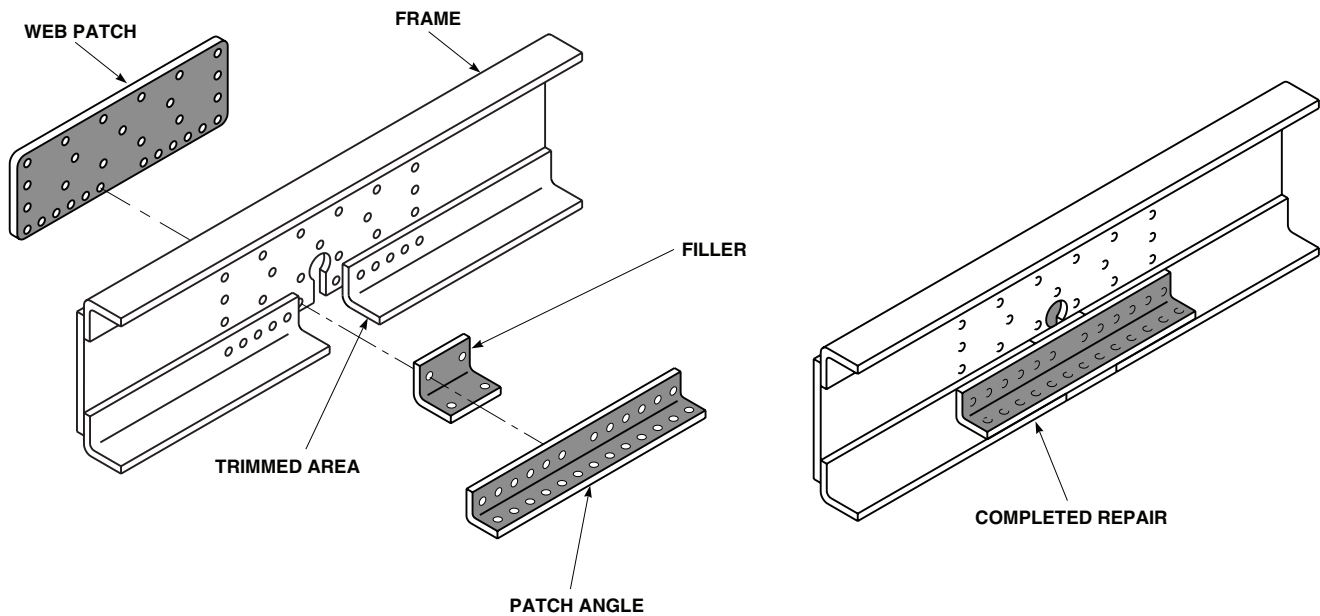
REPAIR DAMAGED LIGHTENING HOLES - ContinuedAB0240
SA

Figure 15. Damaged Frame.

3. Rivet heads should not contact bend radii.
4. Filler - Same gage and material as frame.
5. Frame patch- Same material as frame, one gage thicker.

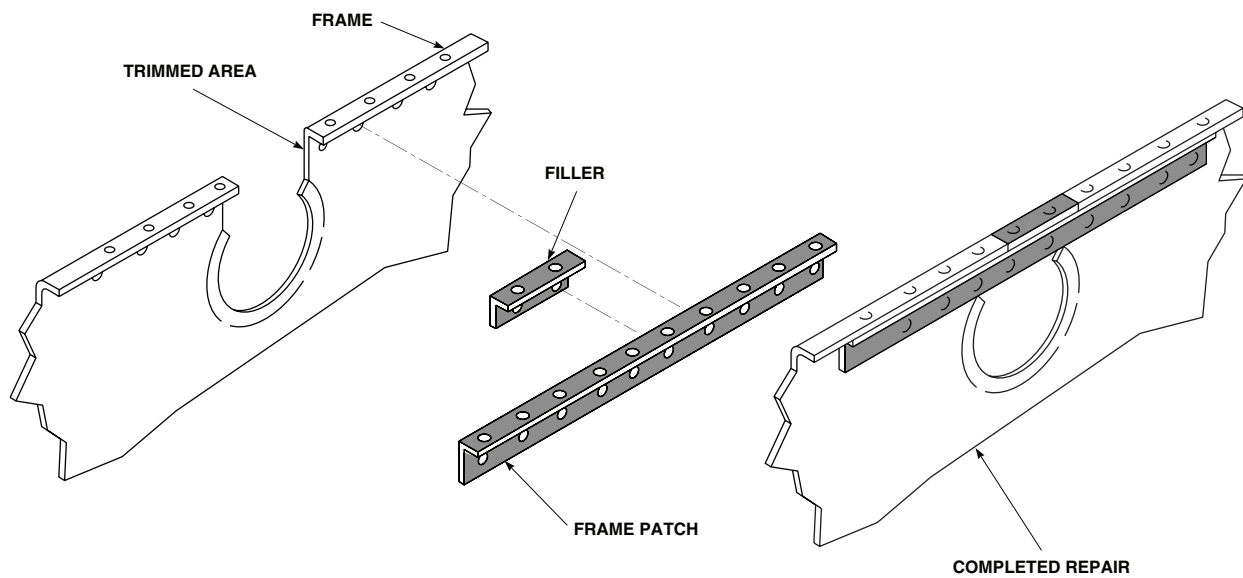
REPAIR DAMAGED HAT SECTION

1. Refer to Figure 17.
2. Hatch section patch bend radii should nest or be a practicable minimum.
3. Rivet heads should not contact bend radii.
4. Skin filler - Same gage and material as skin.
5. Hat section filler - Same gage and material as hat section.
6. Skin patch - Same material as skin, one gage thicker.
7. Hat section patch- Same material as hat section, one gage thicker.

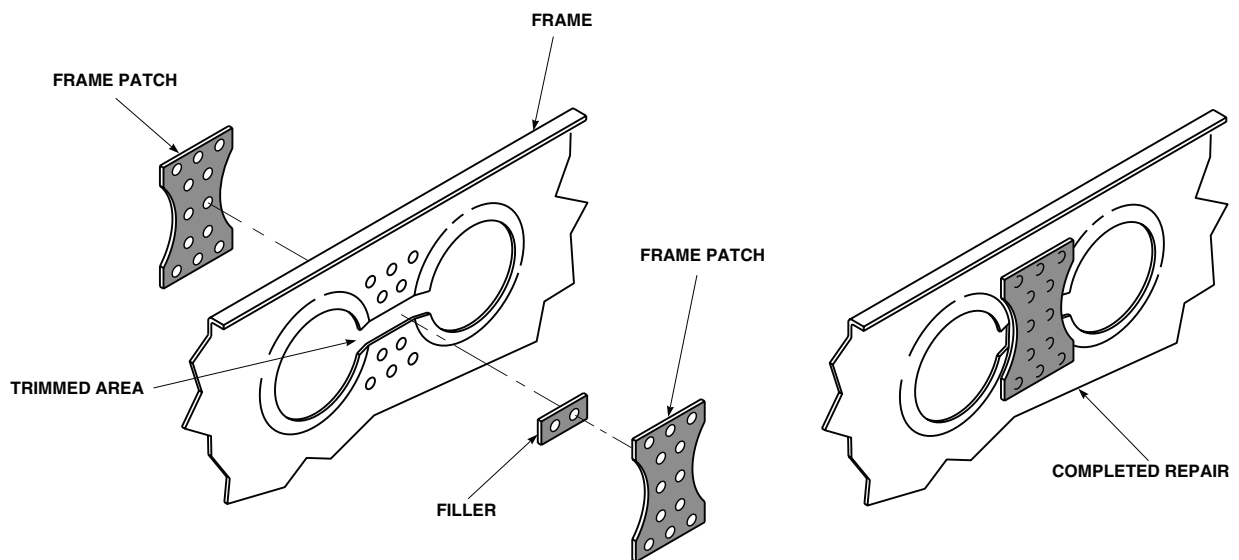
REPAIR DAMAGED CHANNEL STIFFENER

1. Refer to Figure 18.
2. Repair member bend radii should nest or be a practicable minimum.
3. Rivet heads should not contact bend radii.
4. Channel filler - Same gage and material as channel.

REPAIR DAMAGED CHANNEL STIFFENER - Continued



REPAIR OF LIGHTENING HOLE AND FLANGE



REPAIR BETWEEN LIGHTENING HOLES

AB0241
SA

Figure 16. Damaged Lightning Holes.

5. Skin filler - Same gage and material as skin.

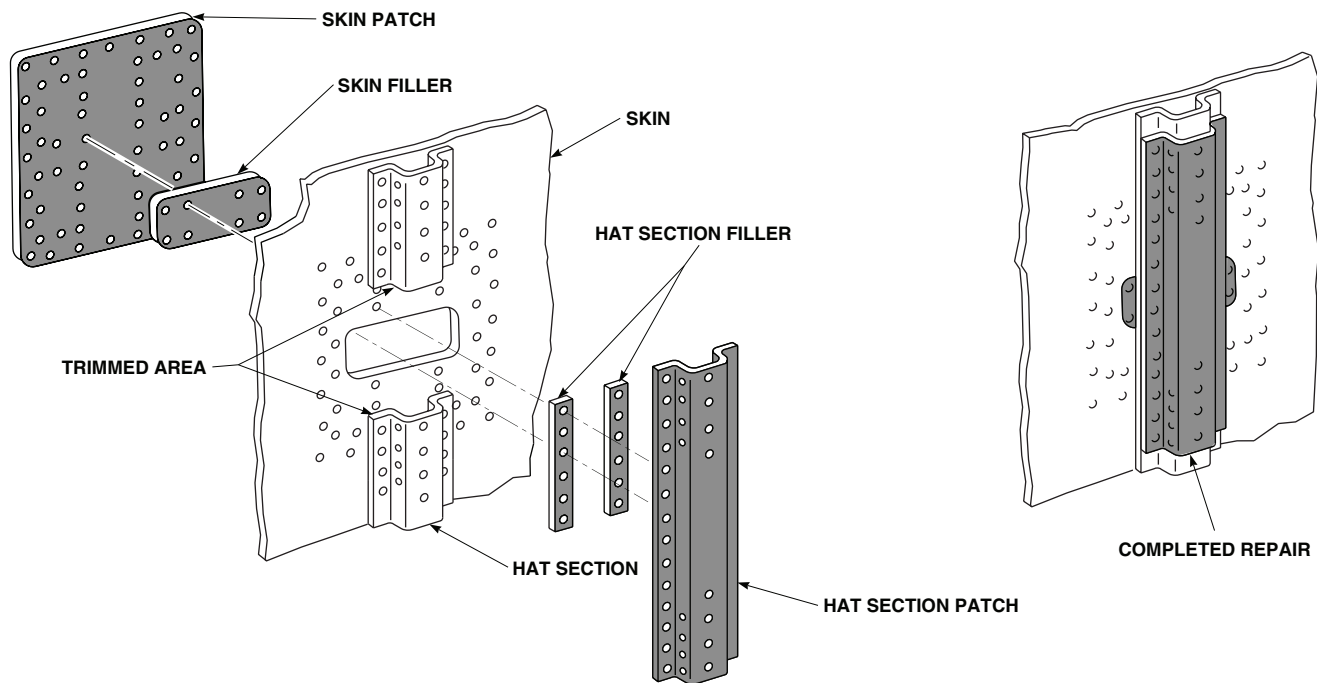
REPAIR DAMAGED CHANNEL STIFFENER - ContinuedAB0242
SA

Figure 17. Damaged Hat Section.

6. Skin patch- Same material as skin, one gage thicker.

REPAIR DAMAGED TEE AND WEB

1. Refer to Figure 19.
2. Repair member bend radii should nest or be a practicable minimum.
3. Rivet heads should not contact bend radii.
4. Web filler - Same gage and material as web.
5. Tee filler - Same gage and material as tee.
6. Patch angle - 0.125 gage, same material as tee.
7. Skin filler- Same gage and material as skin.
8. Skin Patch - Same material as skin, one gage thicker.
9. Use HS52P6 hi-shear rivets.

REPAIR DAMAGED BEAM CAPS

1. Refer to Figure 20.
2. Strap patch edge should have a radius to nest within radius of angle.
3. Rivet heads should not contact bend radii.
4. Skin filler - Same gage and material as skin.

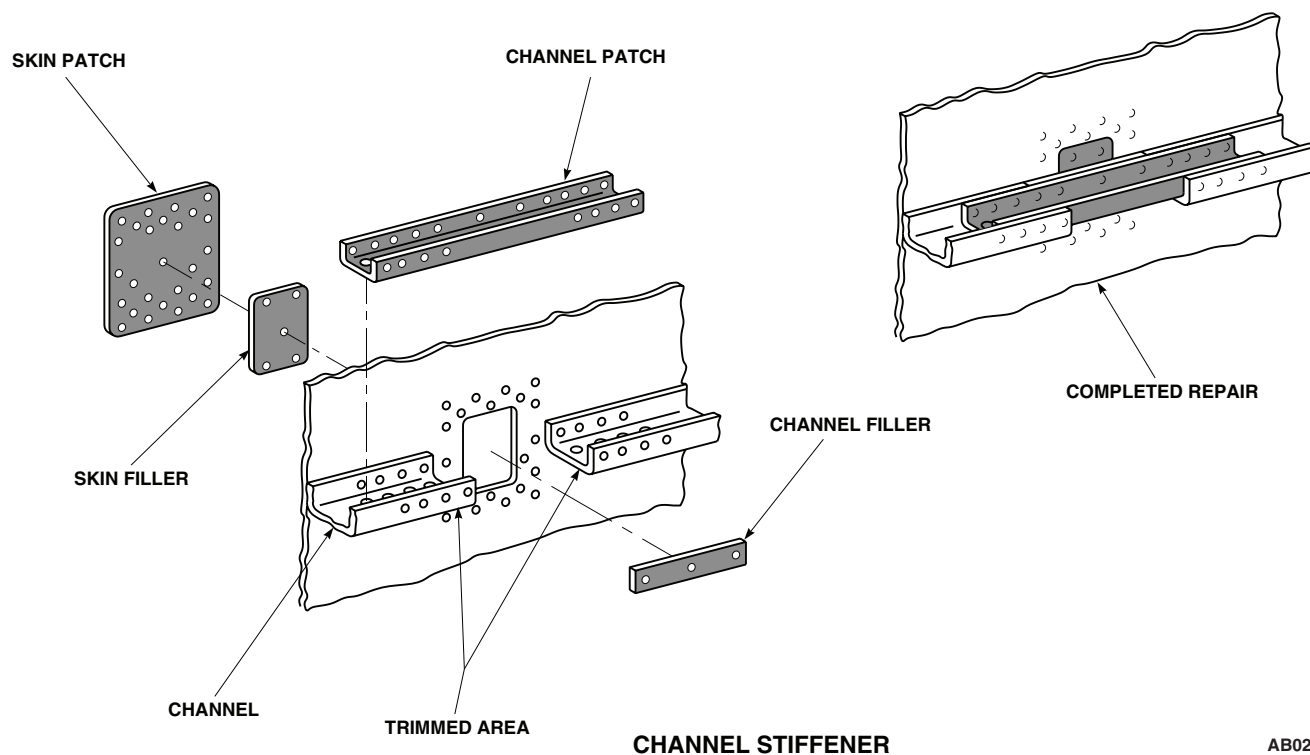
REPAIR DAMAGED BEAM CAPS - Continued

Figure 18. Damaged Channel Stiffener.

5. Angle filler - Same gage and material as angle.
6. Strap patch - Same material as angle, one gage thicker.
7. Skin patch- Same material as skin, one gage thicker.

REPAIR DAMAGED SKIN AND FRAME

1. Refer to Figure 21.
2. Repair member bend radii should nest or be a practicable minimum.
3. Rivet heads should not contact bend radii.
4. Fillers - Same gage and material as original member.
5. Patches - Same material as original member, one gage thicker.

REPAIR DAMAGED SKIN AND STRINGER

1. Refer to Figure 22.
2. Stringer patch bend radii should nest or be a practicable minimum.
3. Rivet heads should not contact bend radii.
4. Fillers - Same gage and material as original member.
5. Patches - Same material as original member, one gage thicker.

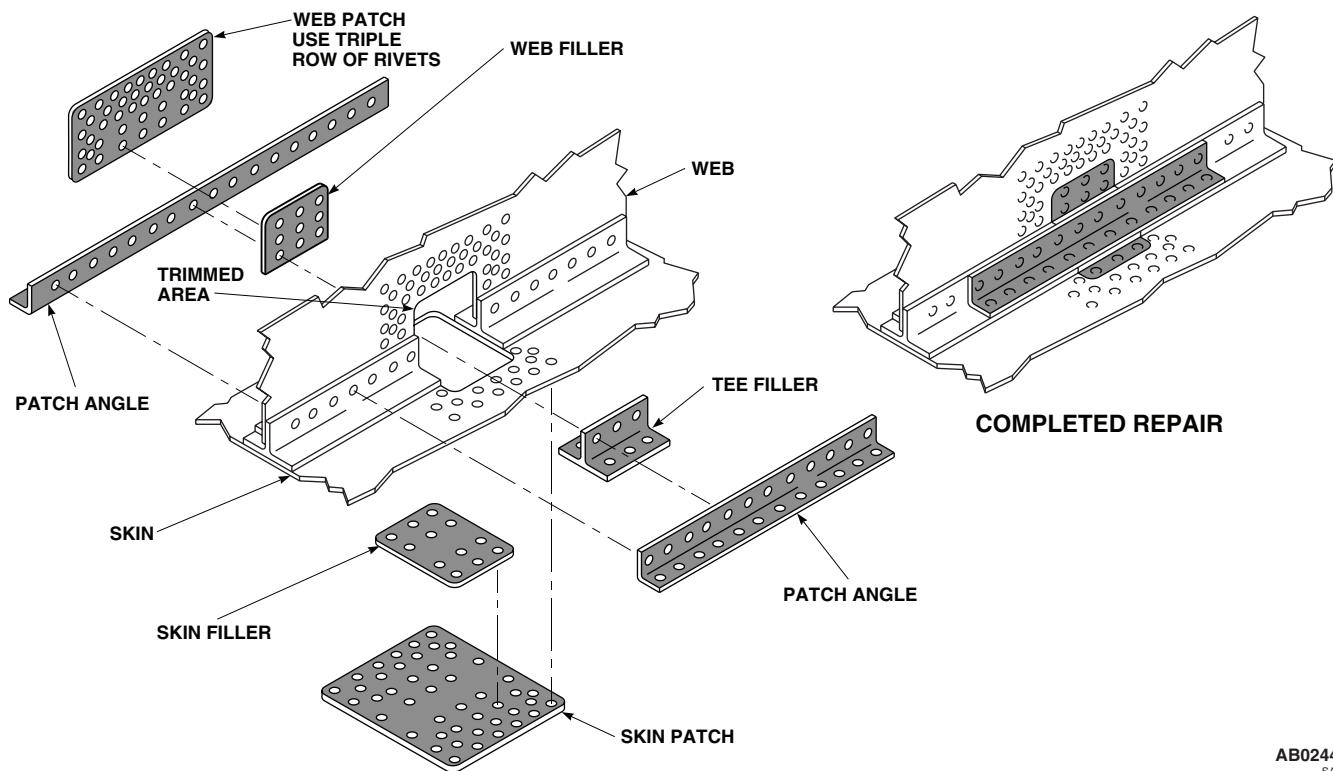
REPAIR DAMAGED SKIN AND STRINGER - ContinuedAB0244
SA

Figure 19. Damaged Tee and Web.

REPAIR DAMAGED SKIN, FRAME, AND STRINGER

1. Refer to Figure 23.
2. Repair member bend radii should nest or be a practicable minimum.
3. Rivet heads should not contact bend radii.
4. Fillers - Same gage and material as original member.
5. Splices and patches - Same material as original member, one gage thicker.

REPAIR DAMAGED BULKHEAD WEB AND SKIN

1. Refer to Figure 24.
2. Repair member bend radii should nest or be a practicable minimum.
3. Rivet heads should not contact bend radii.
4. Fillers - Same gage and material as original member.
5. Patches - Same material as original member, one gage thicker.

MISCELLANEOUS REPAIRS

1. Refer to Figure 25.
2. Corner Crack:
 - a. **LENGTH OF CRACK SHALL NOT EXCEED 25 TIMES THICKNESS OF DAMAGED MEMBER.**

MISCELLANEOUS REPAIRS - Continued

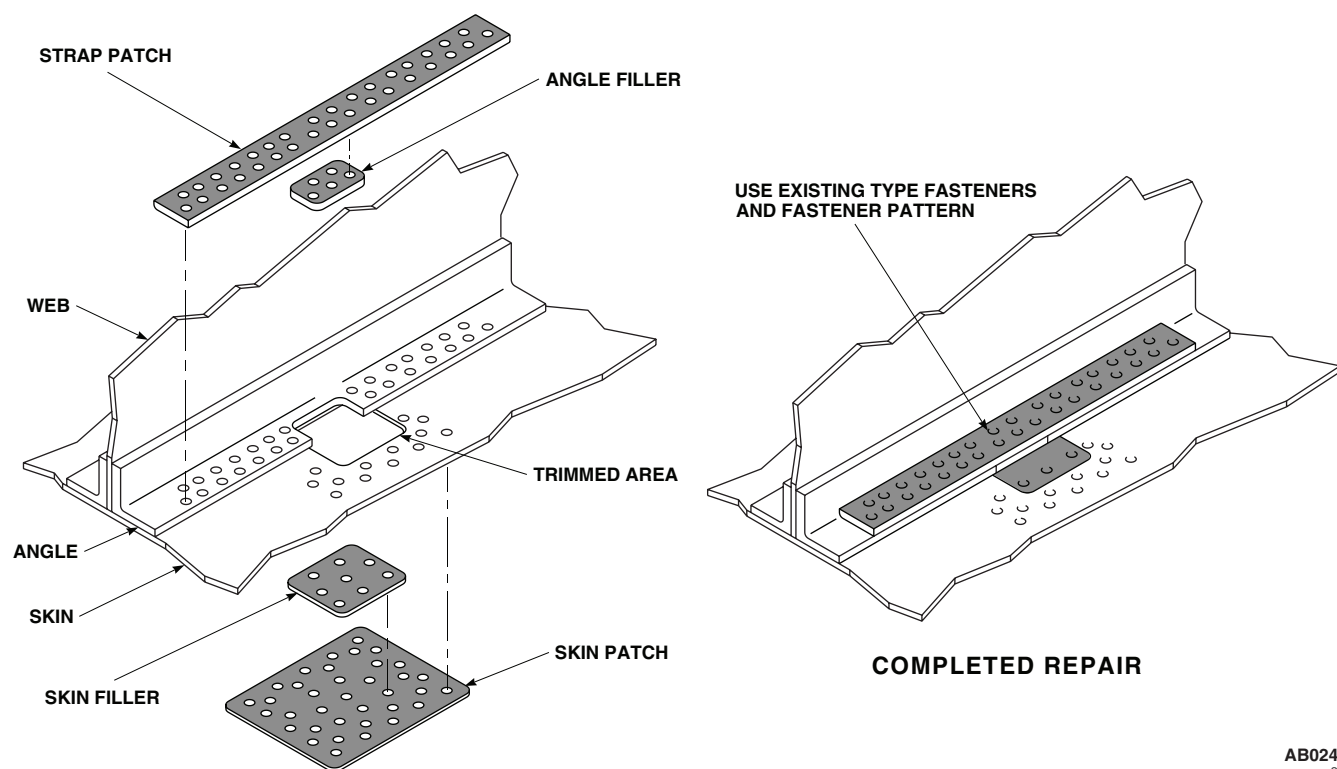
AB0245
SA

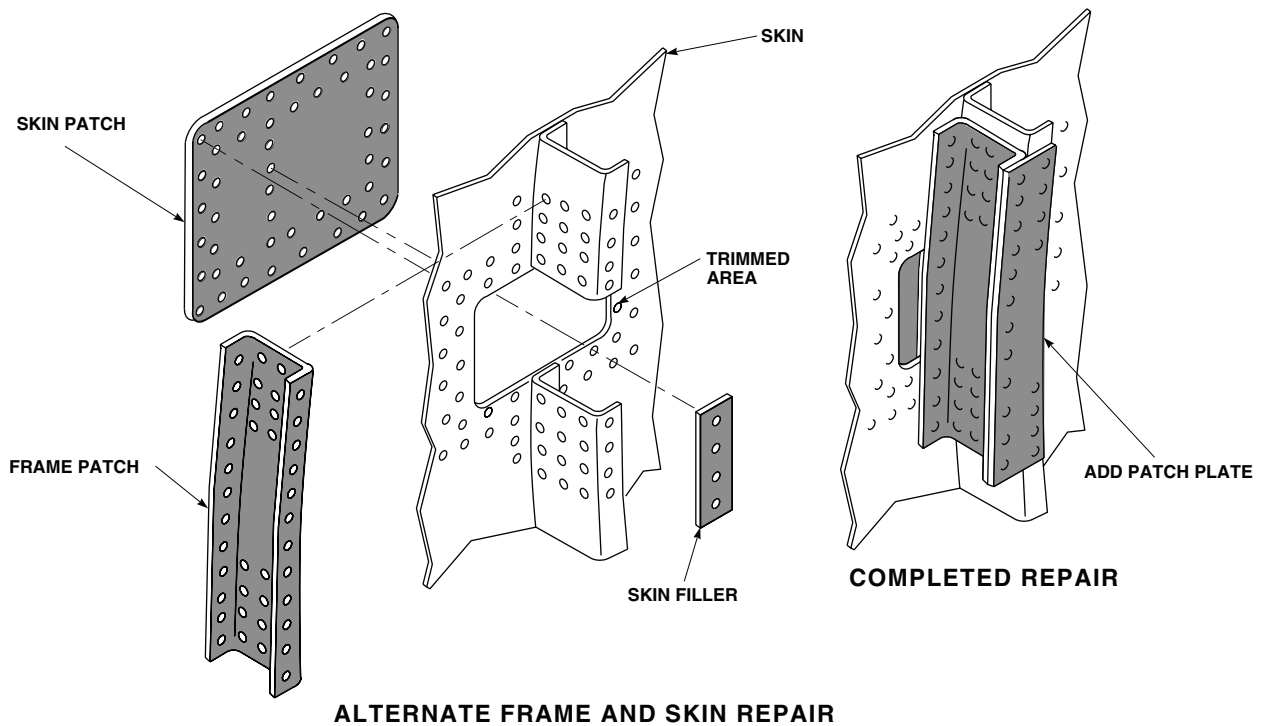
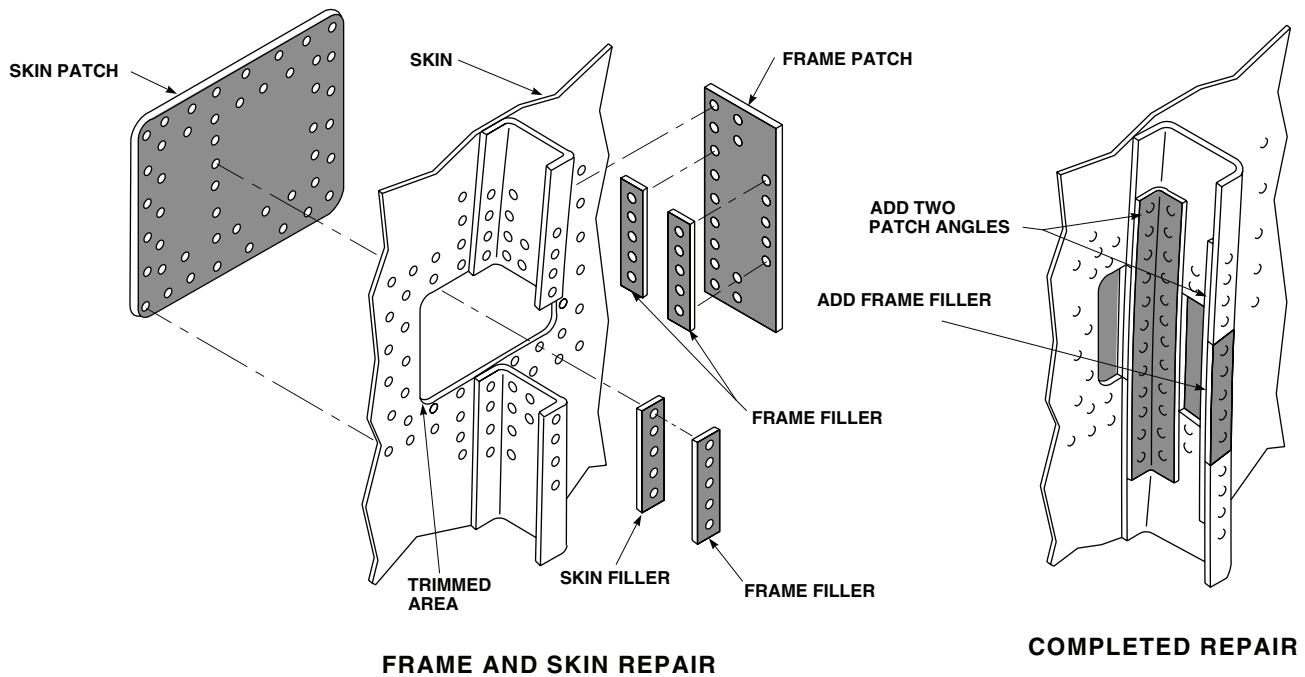
Figure 20. Damaged Beam Caps.

- b. Stop-drill hole using 1/8 (0.125)-inch diameter drill.
 - c. Repair member - Same gage and material as original member.
3. Crack at Stinger Cutout:
- a. **LENGTH OF CRACK SHALL NOT EXCEED 20 TIMES THICKNESS OF DAMAGED MEMBER.**
 - b. Stop-drill hole using 1/8 (0.125)-inch diameter drill.
 - c. Repair member - Same gage and material as original member.

Table 4. Rivet Requirements for Crack at Stringer Cutout.

ORIGINAL MATERIAL GAGE	RIVET TYPE AND SIZE
0.020	MS20470AD3
0.025	MS20470AD4
0.032, 0.040	MS20470AD5
0.050, 0.063, 0.071	MS20470AD6
0.080, 0.090	MS20470AD8

MISCELLANEOUS REPAIRS - Continued



AB0246
SA

Figure 21. Skin and Frame.

4. Crack in Bend Radius:

MISCELLANEOUS REPAIRS - Continued

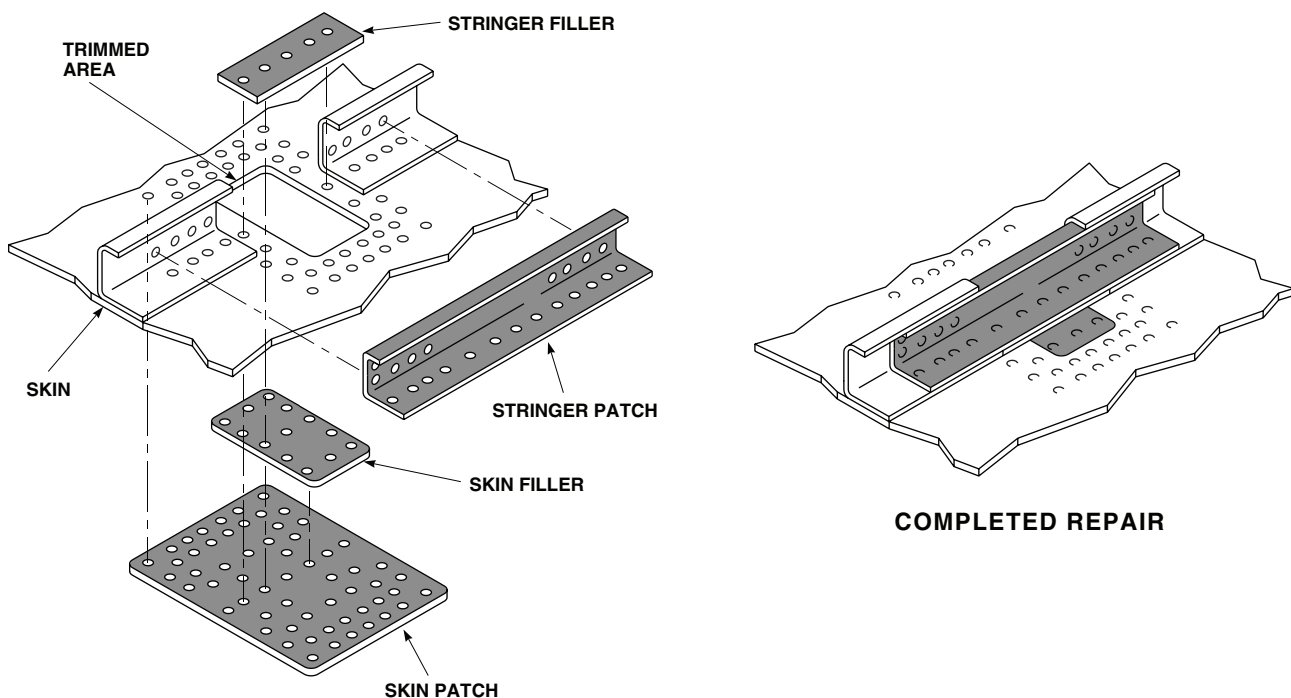
AB0247
SA

Figure 22. Skin and Stringer.

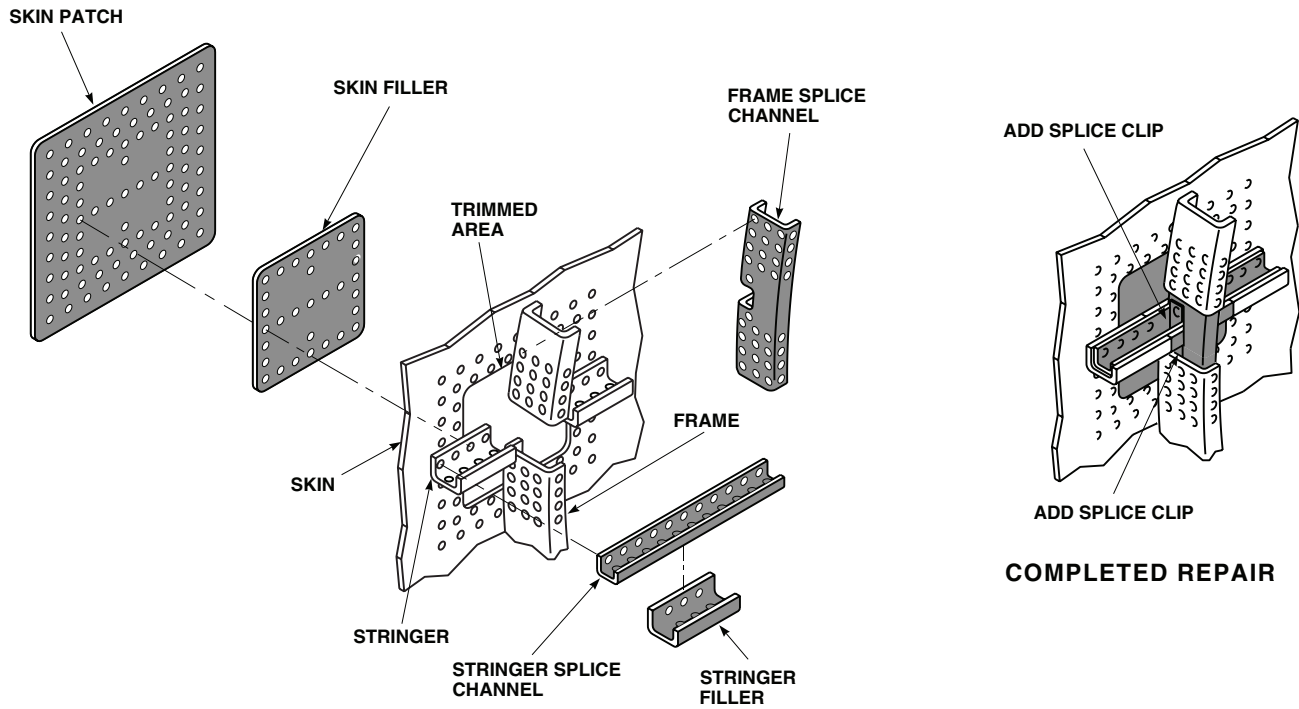
- a. **LENGTH OF CRACK SHALL NOT EXCEED 25 TIMES THICKNESS OF DAMAGED MEMBER.**
 - b. Stop-drill hole using 1/8 (0.125)-inch diameter drill.
 - c. Repair member - Same gage and material as original member.
5. Rivet Having Short Edge Distance:

NOTE

This repair is not to be used for skin splices.

- a. **REPAIR IS LIMITED TO NOT MORE THAN 5 CONSECUTIVE ATTACHMENTS OR MORE THAN 50% OF THE TOTAL RIVET PATTERN.**
- b. Install 2 rivets, one on each side of faulty rivet. Use same type, but one diameter smaller than faulty rivet.
- c. Repair member - Same gage and material as original member.

MISCELLANEOUS REPAIRS - Continued



AB0248
SA

Figure 23. Skin, Frame, and Stringer.

MISCELLANEOUS REPAIRS - Continued

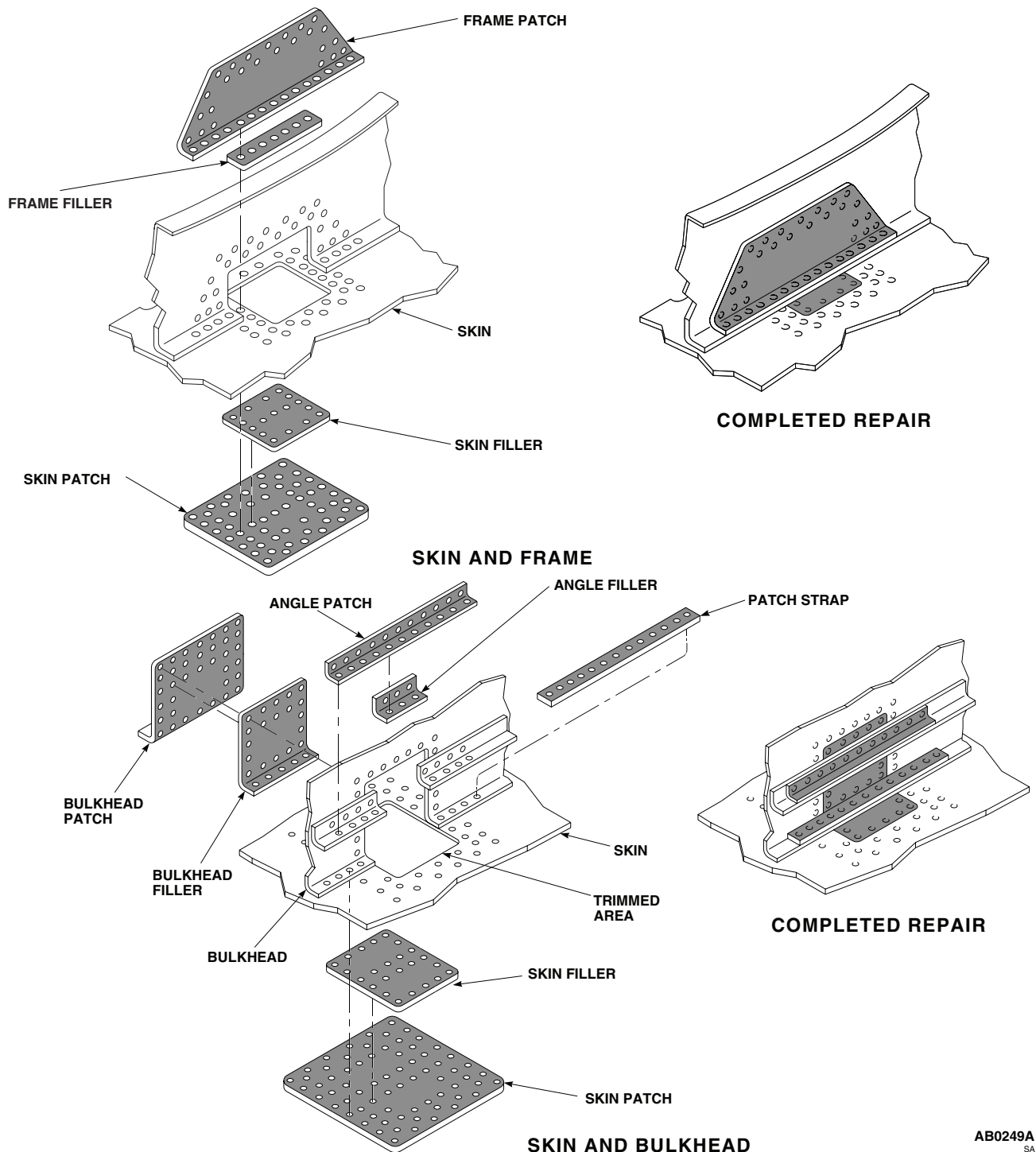
AB0249A
SA

Figure 24. Repair Damaged Bulkhead Web and Skin.

MISCELLANEOUS REPAIRS - Continued

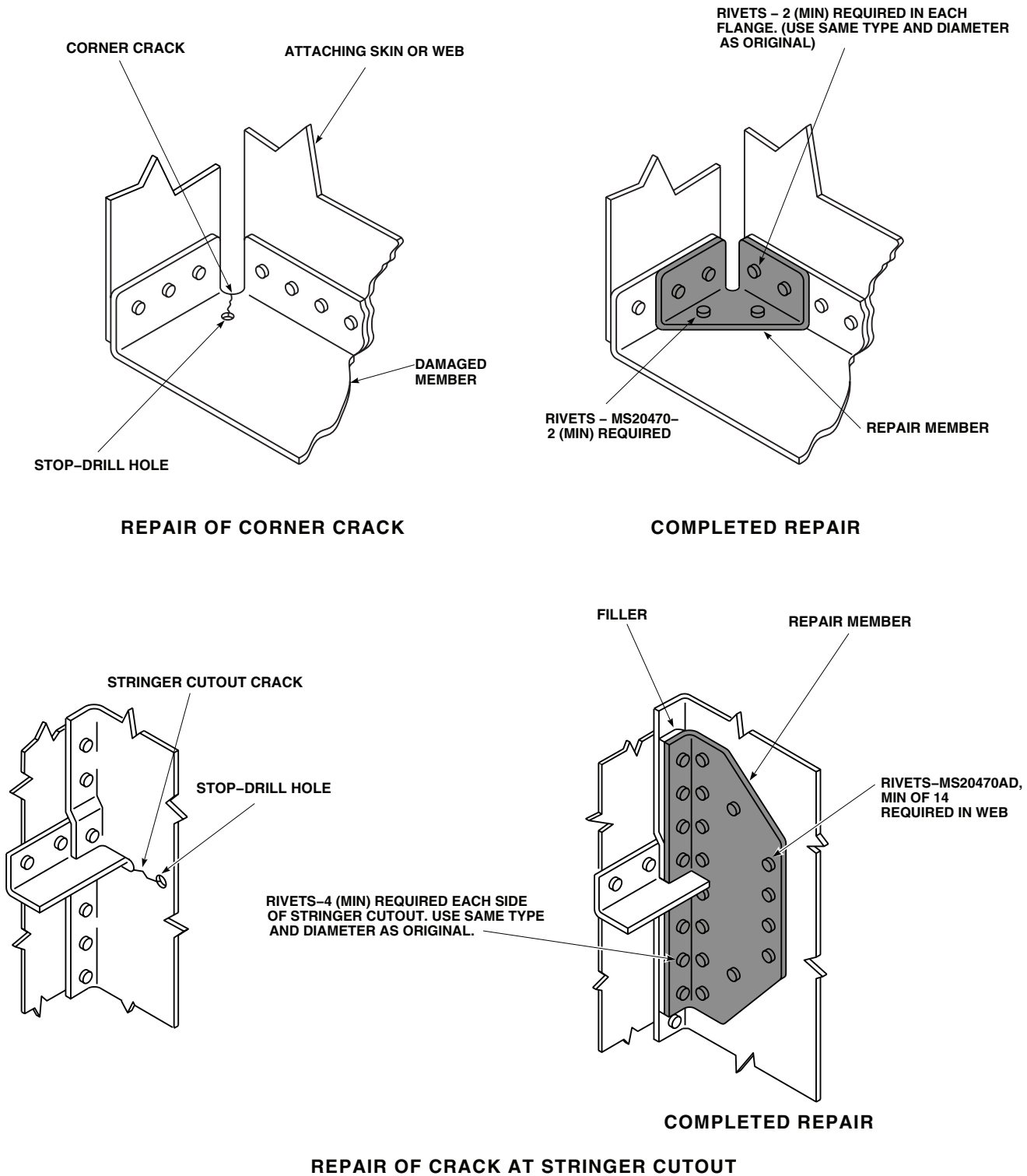
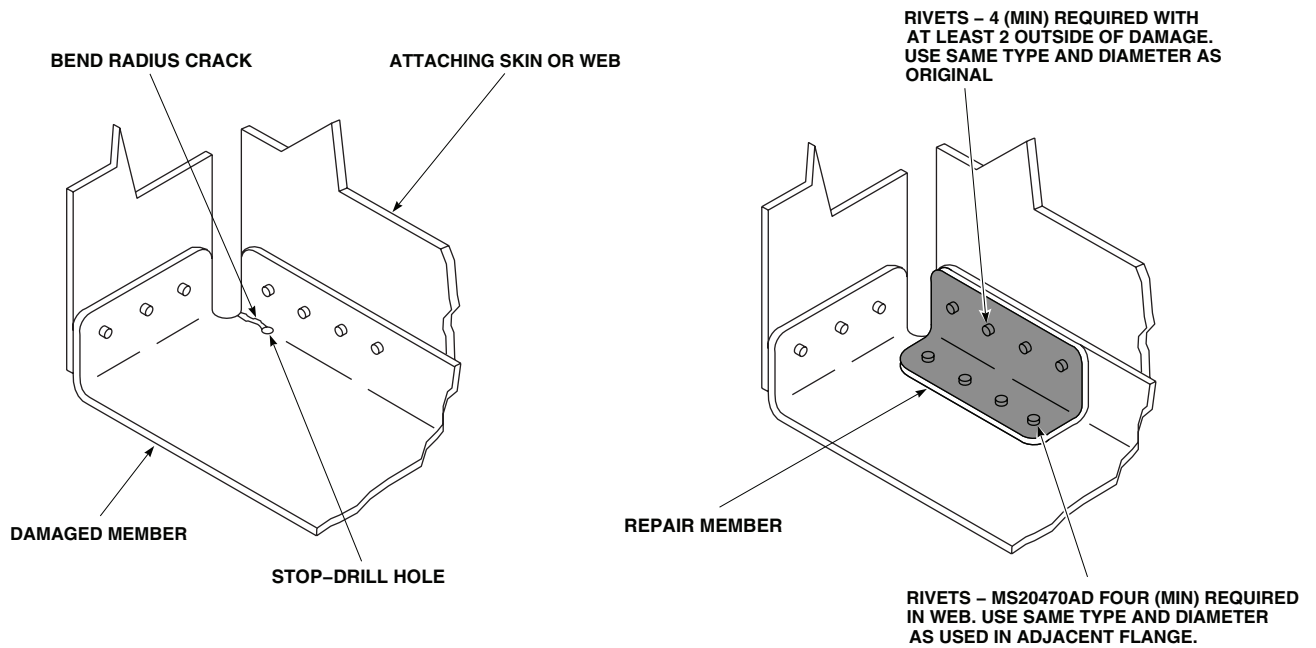
AB0250_1A
SA

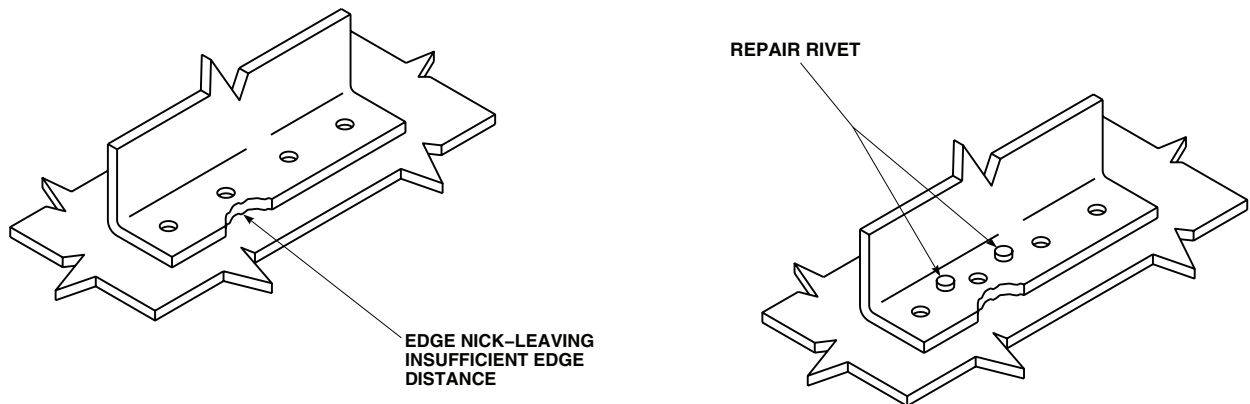
Figure 25. Miscellaneous Repairs. (Sheet 1 of 2)

MISCELLANEOUS REPAIRS - Continued



REPAIR OF CRACK IN BEND RADIUS

COMPLETED REPAIR



REPAIR OF RIVET HAVING SHORT EDGE DISTANCE

AB0250_2A
SA

Figure 25. Miscellaneous Repairs. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****COMPOSITE STRUCTURE REPAIR**

This WP supersedes WP 0379 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
C-Clamp, A-A-429
Disposable Hypodermic Syringe, Locally-Made, Figure
H-197, WP 1805 00
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Heat Gun, LFT500
Nonmetallic Scraper
Squeegee
Tack Rag
Tapping Coin

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive, Item 12, WP 1803 00
Adhesive, Item 13, WP 1803 00
Adhesive, Item 18, WP 1803 00
Adhesive, Item 19, WP 1803 00
Adhesive, Item 35, WP 1803 00
Adhesive, Item 44, WP 1803 00
Anodized Aluminum Sheet, 0.016-Inch Thick
Brush, Acid Swabbing, Item 57, WP 1803 00
Cloth, Abrasive, Item 76, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00

Cloth, Cheesecloth, Item 85, WP 1803 00
Cloth, Satin, Item 90, WP 1803 00
Corrosion Resistant Coating, Item 110, WP 1803 00
Diethylenetriamine, Technical, Item 116, WP 1803 00
Filler, Dent, Metal Surface, Item 126, WP 1803 00
Hardener, Adhesive, Item 141, WP 1803 00
Hardener, Adhesive, Item 142, WP 1803 00
Insulating Compound, Item 151, WP 1803 00
Paper, Abrasive, Item 211, WP 1803 00
Plastic, Sheet, Item 220, WP 1803 00
Sealing Compound, Item 276, WP 1803 00
Sealing Compound, Item 260, WP 1803 00
Stirring Stick, Beverage, Item 300, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 334,
WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

MIL-F-9084
MIL-R-9300
TM 1-1500-204-23
TM 55-1500-345-23
WP 0381 00
WP 0544 00
WP 1803 00
WP 1805 00

EQUIPMENT AND SUPPLIES**WARNING**

Observe all cautions and warnings on containers when using consumables. When applicable, wear necessary protective gear during handling and use. If a consumable is flammable or explosive, MAKE CERTAIN consumable and its vapors are kept away from heat, spark, or flame. MAKE CERTAIN helicopter is properly grounded and fire-fighting equipment is readily available prior to use. For additional information on toxicity, flashpoint, and flammability of chemicals, refer to the manufacturer of the product.

NOTE

- Equipment and material used in the repair of reinforced plastic should duplicate as closely as possible those used in the manufacture of the original part.
 - The following repair procedures are not to be used for rotor blade repair.
 - For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.
1. The recommended satin fabric used for reinforced plastic repairs shall conform to MIL-F-9084, Class 2, and the resin systems shall conform to Specification MIL-R-9300.
 2. Use backup forms to repair large rounded areas.
 3. Use vacuum bags or clamping plates to provide pressure during cure.
 4. Use hot bonders, heat lamps and heat guns whenever possible to cure repair areas.

CAUTION

To avoid poor bond and paint adhesion, do not use waxes or similar products as release materials. They cannot be removed completely from repair surface. Use only film release materials.

5. Use release materials such as sheet plastic, Item 219, WP 1803 00, (TM 1-1500-204-23).
6. Assemble equipment for storing resins, measuring, weighing, cutting fabric, cutting cured laminates, sanding and finishing.

WARNING

Reinforced plastic dust is a health hazard that requires protection of eyes, respiratory system, and exposed skin. Wear protective clothing, eye protection, and dust mask when working with reinforced plastic.

CAUTION

Reinforced plastic dust is highly abrasive and can damage equipment and contaminate parts with which it comes in contact. Cover or remove equipment when sanding reinforced plastic.

7. Have an exhaust system for removing reinforced plastic dust resulting from cutting and sanding operations.

EQUIPMENT AND SUPPLIES - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

8. Have technical acetone, Item 5, WP 1803 00, available for cleaning equipment and parts.
9. Control storage temperature and shelf life of resins for resin reliability. Resins should be stored at 75°F (24°C) unless otherwise indicated on containers.

KEVLAR**CAUTION**

To prevent damage to Kevlar laminate, do not allow paint removers to contact exposed fibers. Do not allow solvents to remain in contact with Kevlar laminate.

1. Kevlar fibers fray when sanded, cut, or drilled. To prevent surface fraying and provide a smooth finish, mask surface as close to repair dimensions as possible, using pressure sensitive adhesive tape, Item 334, WP 1803 00.
2. Trim edges with abrasive cloth, Item 76, WP 1803 00. Use abrasive cloth, Item 83, WP 1803 00, for removing resin gloss.
3. Do not remove masking tape until after final sanding.

TYPES OF DAMAGE

1. There are two major types of repairable damage:
 - a. Type I repair is for laminates with surface damage such as scratches or pits that do not penetrate first ply. Refer to, TYPE I REPAIRABLE DAMAGE, in this work package.
 - b. Type II repair is for laminates with damage extending into laminate fabric, or through part. Refer to, TYPE II REPAIRABLE DAMAGE, in this work package.
2. Damage to holes and attachment points may be repaired. Refer to, REPAIR HOLES AND ATTACHMENT POINTS 0.098-INCH OR LESS, in this work package.
3. Broken corners or tabs may be repaired. Refer to, REPAIR BROKEN CORNER OR TAB, in this work package.
4. Damage to foam ribs may be repaired. Refer to, REPAIR DAMAGED FOAM CORE RIB, DAMAGE LESS THAN 2 INCHES LONG, in this work package.

TYPES OF RESIN FOR REPAIR

WARNING

- The resins, curing agents, times, and temperatures listed in Table 1, are for repair of these laminates: glass fabric/polyester resin, glass fabric/epoxy resin, and Kevlar/epoxy resin. Type I resin is for low temperature curing and Type II resin is for high temperature curing.
- When using or mixing resins and curing agents, fumes from these materials cause both a fire and health hazard.
- Have good ventilation and breathing protection. Wear protective clothing and eye shield. Post NO SMOKING signs in area and keep materials away from spark or flame. Wash contacted skin with soap and water. If materials contact eyes, flush out eyes with clean water and seek immediate medical attention.
- Use properly grounded and explosion-proof electrical equipment in working area.
- Wear protective cotton and plastic gloves when handling materials.
- Wash hands before eating or smoking.

The resins and curing agents shown in Table 1 are to be used during repairs.

Table 1. Resins and Curing Agents.

MIX RATIO, RESIN TO CURING AGENT BY WEIGHT					
RESIN BASE AND CURING AGENT TYPE	RESIN BASE	CURING AGENT	POT LIFE AT 70°F (21°C) MINUTES	NOTES	CURE CYCLE MINIMUM TIME
Resin and Curing Agent - Adhesive, Item 13, WP 1803 00 (Type II)	100 ± 2	57 to 59	25 to 30	2, 5	24 hrs at 70°F (21°C) or 1 hr at 240°F (116°C) to 260°F (126°C)
Resin - Sealing Compound, Item 276, WP 1803 00 (Type II) Curing Agent - Insulating Compound, Item 151, WP 1803 00	100 ± 2	13 to 15	20	3, 5	48 hrs at 70°F (21°C)
Resin - Sealing Compound, Item 276, WP 1803 00 (Type II) Curing Agent - Sealing Compound, Item 276, WP 1803 00	100 ± 2	13 to 15	30	3, 5	336 hrs at 70°F (21°C) or 16 hrs at 70°F (21°C) plus 1 hr at 195°F (91°C) to 205°F (96°C)

TYPES OF RESIN FOR REPAIR - Continued

Table 1. Resins and Curing Agents. - Continued

MIX RATIO, RESIN TO CURING AGENT BY WEIGHT					
RESIN BASE AND CURING AGENT TYPE	RESIN BASE	CURING AGENT	POT LIFE AT 70°F (21°C) MINUTES	NOTES	CURE CYCLE MINIMUM TIME
Resin - Sealing Compound, Item 276, WP 1803 00 (Type II) Curing Agent - Epocast 1652-B	100 ± 2	11 to 13	40	3, 5	2 hrs at 70°F (21°C) plus 1 to 3 hrs at 145°F (63°C) to 155°F (68°C)
Resin - Sealing Compound, Item 260, WP 1803 00 (Type I) Curing Agent - Insulat- ing Compound, Item 151, WP 1803 00	100 ± 5	19 to 21	10 to 15	2, 3, 4	24 hrs at 70°F (21°C) or 2 hrs at 145°F (63°C) to 155°F (68°C)
Resin - Sealing Compound, Item 260, WP 1803 00 (Type I) Curing Agent - Adhesive, Item 35, WP 1803 00	100 ± 5	48 to 52	25 to 30	2, 3, 4	24 hrs at 70°F (21°C) or 2 hrs at 145°F (63°C) to 155°F (68°C)
Resin - Adhesive, Item 18, WP 1803 00 (Type I) Curing Agent - Insulating Compound, Item 151, WP 1803 00	100 ± 2	14 to 16	15 to 20	2, 4	16 hrs at 70°F (21°C) or 2 hrs at 140°F (60°C) to 160°F (71°C)
Resin - Adhesive, Item 18, WP 1803 00 (Type I) Curing Agent - Adhesive Hardener, Item 142, WP 1803 00	100 ± 2	10 to 12	40 to 70	2, 4	24 hrs at 70°F (21°C) or 2 hrs at 140°F (60°C) to 160°F (71°C)
Resin - Adhesive, Item 19, WP 1803 00 (Type I) Curing Agent - Adhesive Hardener, Item 141, WP 1803 00	100 ± 2	23 to 30	15 to 40	1, 4, 6	24 hrs at 70°F (21°C) or 1 hr at 140°F (60°C) to 160°F (71°C)

TYPES OF RESIN FOR REPAIR - Continued

Table 1. Resins and Curing Agents. - Continued

MIX RATIO, RESIN TO CURING AGENT BY WEIGHT					
RESIN BASE AND CURING AGENT TYPE	RESIN BASE	CURING AGENT	POT LIFE AT 70°F (21°C) MINUTES	NOTES	CURE CYCLE MINIMUM TIME
Resin - Adhesive, Item 44, WP 1803 00 (Type I) Curing Agent - Technical Diethylenetriamine, Item 116, WP 1803 00	100 ± 2	8 to 12	15 to 30	1, 3, 4, 6	24 hrs at 70°F (21°C) or 1 hr at 140°F (60°C) to 160°F (71°C)
Resin - Adhesive, Item 44, WP 1803 00 (Type I) Curing Agent - Adhesive Hardener, Item 141, WP 1803 00	100 ± 2	23 to 30	15 to 40	1, 3, 4, 6	24 hrs at 70°F (21°C) or 1 hr at 140°F (60°C) to 160° (71°C)
NOTES					
1.	Exterior use only.				
2.	High temperature cure follows 1 hour at room temperature.				
3.	Syntactic foam.				
4.	TYPE I (General Purpose). Kevlar/epoxy: 160°F (71°C) continuous, 250°F (121°C) short term; fiberglass/epoxy: 160°F (71°C) continuous, 300°F (149°C) short term.				
5.	TYPE II (High Temperature). Kevlar/epoxy: 425°F (218°C) maximum service; fiberglass/epoxy: 450°F (232°C) maximum service.				
6.	Do not use in fuel contact area.				

TYPE I REPARABLE DAMAGE

NOTE

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. Surface damage consisting of scratches or pits in resin coat, not affecting laminate fabric ply (Figure 1).

TYPE I REPARABLE DAMAGE - Continued

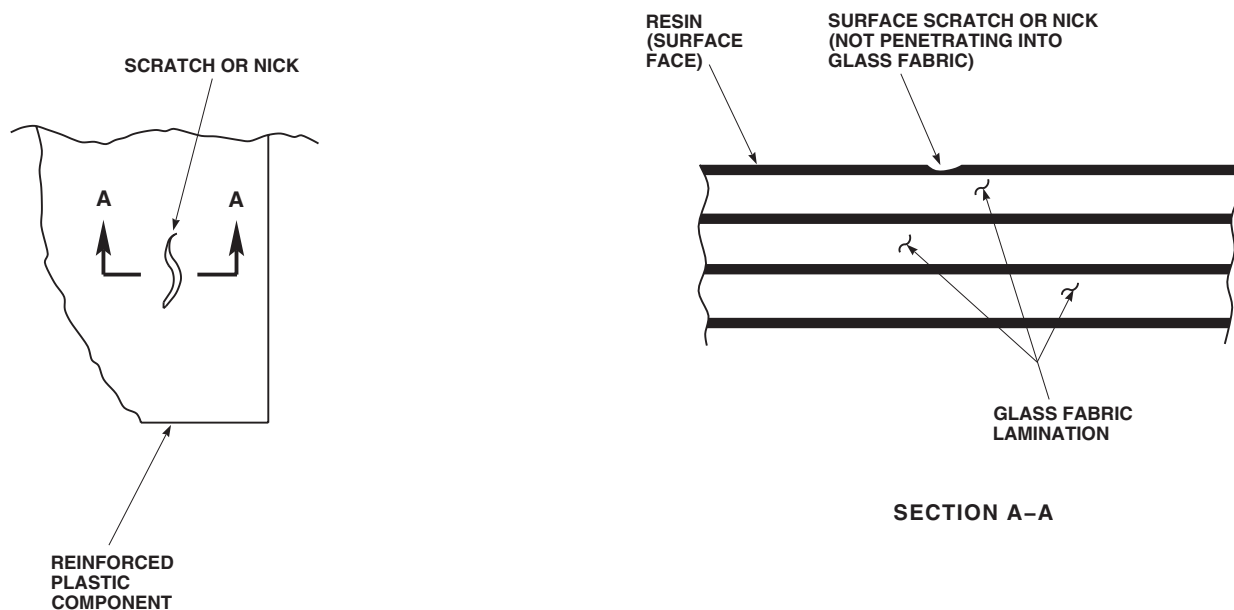
AA0036
SA

Figure 1. Type I Reparable Damage.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Remove paint finish in damaged area, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
3. Sand damaged area, plus 1/2-inch beyond, on all sides, using abrasive cloth, Item 83, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Wipe sanded area with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

TYPE I REPARABLE DAMAGE - Continued**NOTE**

Observe all warnings and cautions. Refer to, GENERAL and TYPES FOR RESIN REPAIR, in this work package.

5. Mix resin/curing agent (Table 1), adding chopped satin cloth, Item 90, WP 1803 00, to make putty-like mixture.
6. Apply thin, smooth coat of mixed resin to damaged area, using beverage stirring stick, Item 300, WP 1803 00, and squeegee to uniform thickness.
7. Cure repair (Table 1).
8. Lightly sand surface smooth with abrasive cloth, Item 83, WP 1803 00.
9. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

TYPE II REPARABLE DAMAGE**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. Remove paint from surface in repair area WP 0381 00 and TM 55-1500-345-23.

NOTE

- Observe all warnings and cautions. Refer to, GENERAL and TYPES FOR RESIN REPAIR, in this work package.
 - Do not mark on work with grease or graphite pencil, or ink pen.
2. Determine total size of repair, allowing at least 1/2-inch step (or blend) in all directions for each ply penetrated (Figure 2). The final patch must overlap at least 1 inch in all directions.
 3. Using abrasive cloth, Item 76, WP 1803 00, sand away plies as indicated.
 4. Lightly sand top surface to limits of final patch.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Thoroughly clean entire repair area with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
6. Protect surfaces from contamination while allowing to air dry.

NOTE

Cut each repair ply so that weave direction matches its laminate ply.

7. Cut plies of satin cloth, Item 90, WP 1803 00, to fit stepped plies sanded away.
8. Lay repair plies on flat surface covered with sheet plastic, Item 220, WP 1803 00.

TYPE II REPARABLE DAMAGE - Continued

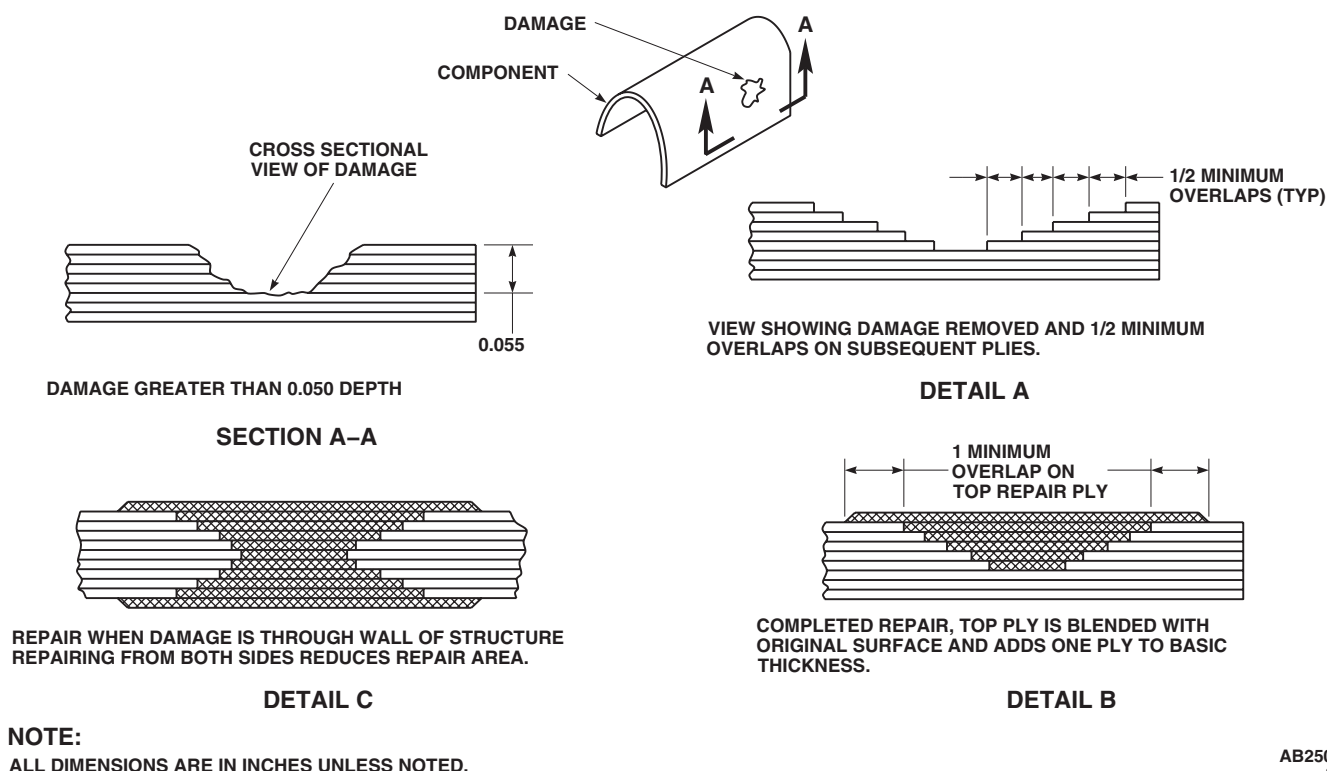


Figure 2. Type II Reparable Damage.

WARNING

Serious eye and skin irritation can be caused by resins used in laminate repair. Use protective glasses, rubber gloves, and long-sleeved clothing when making repairs. If resin contacts skin, remove promptly and wash thoroughly with soap and water.

9. Mix resin/curing agent (Table 1).
10. Apply coat of mixed resin to repair area using acid swabbing brush, Item 57, WP 1803 00.
11. Saturate repair plies of satin cloth, Item 90, WP 1803 00, with mixed resin using acid swabbing brush, Item 57, WP 1803 00, or beverage stirring stick, Item 300, WP 1803 00. Resin content of fabric, when thoroughly impregnated, must be about 50% by weight.
12. Place repair plies in damaged area, matching weave direction to laminate plies. Each repair ply overlaps preceding ply by 1/2-inch in all directions. Final ply overlaps 1 inch and adds 1 ply thickness to laminate (2 ply thickness when repairing both sides).
13. Cover repair with sheet plastic, Item 220, WP 1803 00. Work out all trapped air and excess resin with squeegee.
14. Apply pressure to repair plies by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).

TYPE II REPARABLE DAMAGE - Continued**CAUTION**

Improper control of heat source can damage laminate or weaken bond. Pay strict attention to temperature and curing time.

15. Cure repair (Table 1).
16. Remove pressure and heating equipment and polyvinyl plastic sheet.
17. Blend repair to surface of part using abrasive cloth, Item 83, WP 1803 00.
18. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HOLES AND ATTACHMENT POINTS 0.098-INCH OR LESS**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. For holes not more than 0.098-inch in diameter, with centers at least 1/8-inch from edge of part, in laminates of 12 plies or less, repair as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- a. Clean repair area thoroughly with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cheesecloth.
 - b. Mix adhesive, Item 11, WP 1803 00 (Table 2).
 - c. Fill holes with mixed adhesive using beverage stirring stick, Item 300, WP 1803 00.
 - d. Cover repair area with sheet plastic, Item 220, WP 1803 00.
 - e. Cure adhesive (Table 2).
 - f. Sand smooth to adjacent skin using abrasive cloth, Item 83, WP 1803 00.
2. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HOLES AND ATTACHMENT POINTS 0.098-INCH OR LESS - Continued**Table 2. Fillers and Accelerators.**

MIX RATIO, FILLER BASE TO ACCELERATOR BY WEIGHT				
FILLER BASE AND ACCELERATOR	FILLER BASE	ACCEL- ERATOR	POT LIFE AT 70°F (21°C) (Minutes)	CURE CYCLE MINIMUM TIME
Filler and Accelerator - Adhesive, Item 11, WP 1803 00 Accelerator -	100 ± 5	21 to 23	30	24 hrs at 70°F (21°C) or 1 hr at 140°F (60°C) to 160°F (71°C)
Filler and Accelerator - Adhesive, Item 12, WP 1803 00	100 ± 5	31 to 35	60	1 hr at 190°F (88°C) to 210°F (99°C)
Filler and Accelerator - Metal surface dent filler, Item 126, WP 1803 00	100 ± 5	2	10	2 hrs at 70°F (21°C)

REPAIR HOLES AND ATTACHMENT POINTS OVER 0.098-INCH, UNDER 3/4-INCH, THICKNESS 6 PLYS OR LESS**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package in addition to these procedures.

1. For holes larger than 0.098-inch and less than 3/4-inch in diameter, if part thickness is 6 plies or less, repair as follows (Figure 3):
 - a. On both surfaces of part, mark an outline for repair allowing 1/2-inch minimum overlap for edge of hole in all directions (allow 1 inch for 1/2 to 3/4-inch diameter hole).
 - b. Lightly sand repair areas with abrasive cloth, Item 76, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- c. Clean repair area thoroughly with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cheesecloth, Item 85, WP 1803 00.
- d. Cut ply of satin cloth, Item 90, WP 1803 00, for each side of part, following outline on part surface. Match weave direction of repair ply and surface ply.
- e. Mix resin/curing agent (Table 1) and saturate both repair plies using beverage stirring stick, Item 300, WP 1803 00.

REPAIR HOLES AND ATTACHMENT POINTS OVER 0.098-INCH, UNDER 3/4-INCH, THICKNESS 6 PLIES OR LESS - Continued

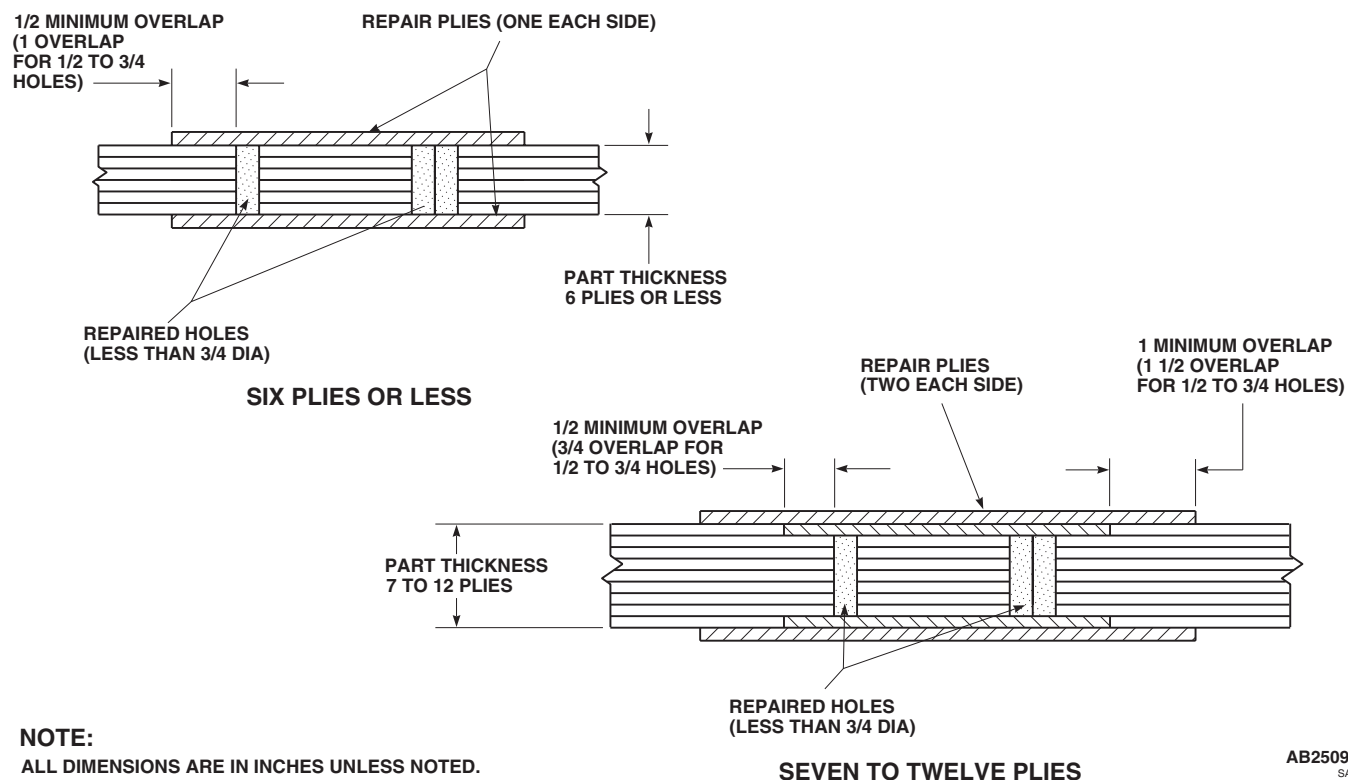


Figure 3. Hole Repair.

- f. Apply coat of mixed resin to one repair surface of part. Apply repair ply and cover with sheet plastic, Item 220, WP 1803 00.
 - g. Mix resin/curing agent (Table 1) adding chopped satin cloth, Item 90, WP 1803 00, to obtain putty-like consistency.
 - h. Fill holes with resin/cloth mixture using beverage stirring stick, Item 300, WP 1803 00.
 - i. Repeat step 1. e. and step 1. f. on other side of part.
 - j. Carefully work out all air and excess resin with squeegee.
 - k. Apply pressure to repair plies by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
 - l. Cure repair (Table 1).
 - m. Remove vacuum bags, clamping plates, or system of weights and pressure plates; remove polyvinyl plastic sheet.
 - n. Sand to remove excess resin and blend repair into adjoining surface using abrasive cloth, Item 83, WP 1803 00.
2. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HOLES AND ATTACHMENT POINTS OVER 0.098-INCH, UNDER 3/4-INCH, THICKNESS 7 TO 12 PLYS**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. For holes larger than 0.098-inch and less than 3/4-inch in diameter, if part is 7 to 12 plies thick, repair as follows:
 - a. On both surfaces of part, mark an outline for repair allowing 1 1/2-inch minimum overlap from edge of hole in all directions (allow 2 inches for 1/2 to 3/4-inch diameter hole).
 - b. Lightly sand repair areas with abrasive cloth, Item 76, WP 1803 00.
 - c. Mark outline for removal of 1 ply (from both surfaces of part) a minimum of 1/2-inch from edge of hole in all directions (allow 3/4-inch for 1/2 to 3/4-inch diameter hole).
 - d. Remove 1 ply from each surface of part within marked outline, using abrasive cloth, Item 76, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

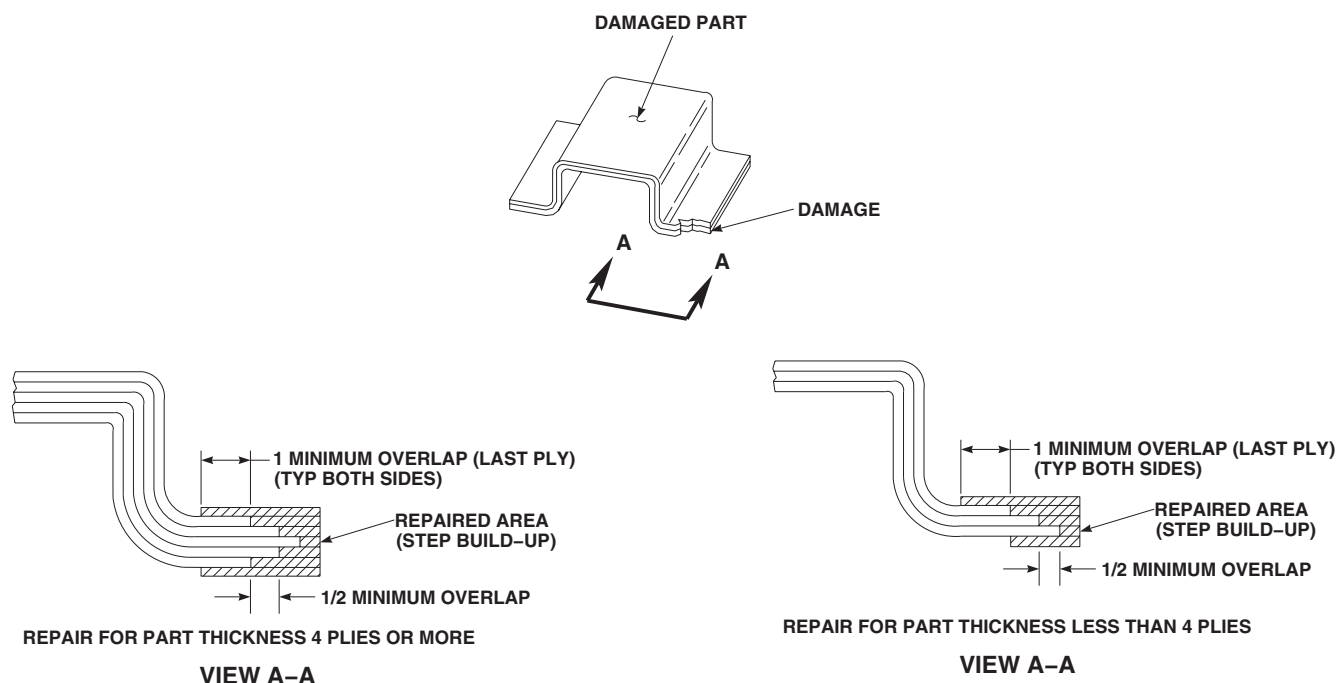
- e. Clean repair area thoroughly with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cheesecloth.

NOTE

Cut each repair ply so that weave direction matches its laminate ply.

- f. Cut plies of satin cloth, Item 90, WP 1803 00, to match outlines on part surfaces and to fit step cuts in surface layers.
 - g. Mix resin/curing agent (Table 1) and saturate all repair plies, using beverage stirring stick, Item 300, WP 1803 00.
 - h. Apply coat of mixed resin to repair area of one surface. Apply step back ply into surface ply and outer ply on top of it. Match weave direction to laminate plies.
 - i. Cover outer repair ply with sheet plastic, Item 220, WP 1803 00.
 - j. Mix resin/curing agent (Table 1), adding chopped satin cloth, Item 90, WP 1803 00, to make putty-like mixture.
 - k. Fill holes with mixed resin using beverage stirring stick, Item 300, WP 1803 00.
 - l. Repeat step 1. g. and step 1. h. on other side of part.
 - m. Carefully work out all air and excess resin with squeegee.
 - n. Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).

REPAIR HOLES AND ATTACHMENT POINTS OVER 0.098-INCH, UNDER 3/4-INCH, THICKNESS 7 TO 12 PLIES - Continued



NOTE:

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0039A
SA

Figure 4. Corner or Tab Repair.

CAUTION

To prevent damage to repaired part when using heat lamp or heat gun, pay strict attention to temperature range. Adjust distance of heat source to part, as necessary.

- o. Cure repair (Table 1).
- p. Remove vacuum bags, clamping plates, or system of weights and pressure plates; remove polyvinyl plastic sheet.
- q. Sand to remove excess resin and blend repair into adjoining surface using abrasive cloth, Item 83, WP 1803 00.
2. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR BROKEN CORNER OR TAB

NOTE

- For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.
- Damage extending 3/4-inch or less from edge of part may be repaired. More extensive damage requires part replacement.
- This repair increases edge thickness by adding one ply to each side of part. If this will cause interference, do not repair part; replace it.

1. Prepare part for repair by stepping back plies by doing the following (Figure 4):
 - a. For edge less than 4 plies thick, step back from one side only.

REPAIR BROKEN CORNER OR TAB - Continued

- b. For edge 4 plies thick or more, step back from both sides.
- c. Allow 1/2-inch overlap for each repair ply, plus 1 inch overlap for outer repair ply on each side.
- d. Mark outline of ply areas for removal.
- e. Remove laminate plies, as required, using abrasive cloth, Item 76, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- f. Clean repair area thoroughly with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Air dry.
- 2. Cut repair plies of satin cloth, Item 90, WP 1803 00, to fit removed areas of laminate, extending them beyond edge of part to allow trimming to size after curing. Match weave direction of repair ply to laminate ply.
- 3. Mix resin/curing agent (Table 1).
- 4. Saturate repair plies with mixed resin, using beverage stirring stick, Item 300, WP 1803 00, or acid swabbing brush, Item 57, WP 1803 00.
- 5. Apply coat of mixed resin to repair areas of part with acid swabbing brush, Item 57, WP 1803 00.
- 6. Put all repair plies in place, including outer plies on each side.
- 7. If necessary, use a metal strip, covered with sheet plastic, Item 220, WP 1803 00, held to one side of part by spring clamps, to keep repair buildup in alignment.
- 8. Cover repair with sheet plastic, Item 220, WP 1803 00. Work out trapped air and excess resin with squeegee.
- 9. Apply pressure by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
- 10. Cure repair (Table 1).
- 11. Remove vacuum bags, clamping plates, or system of weights and pressure plates; remove polyvinyl plastic sheet.
- 12. Trim edge of repair to match part.
- 13. Blend outer plies into adjoining surfaces using abrasive cloth, Item 83, WP 1803 00.
- 14. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR DAMAGED FOAM CORE RIB, DAMAGE LESS THAN 2 INCHES LONG**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

- 1. Stiffening ribs are formed by laminated plies of glass cloth or Kevlar over foam cores. Ribs with covering plies cut, cracked, delaminated, punctured, or torn, with damage to foam core, may be repaired. Damaged area must be less than 2 inches long; only 2 repairs, not less than 1 foot apart, may be made in the same rib.

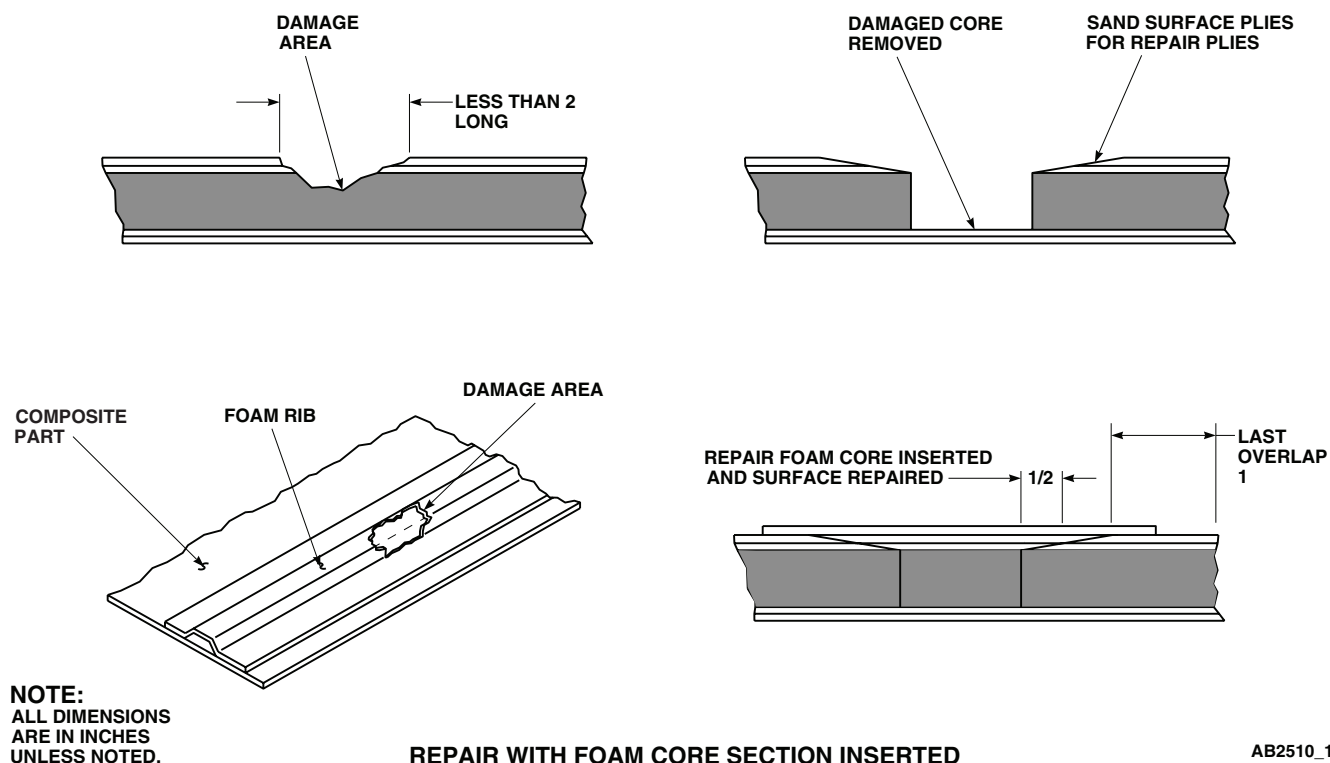
REPAIR DAMAGED FOAM CORE RIB, DAMAGE LESS THAN 2 INCHES LONG - Continued

Figure 5. Repair Damaged Rib. (Sheet 1 of 2)

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

The foam core material can be damaged by contact with technical acetone, Item 5, WP 1803 00. Do not allow technical acetone to contact foam core.

2. Clean damage area and adjacent surfaces with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
3. Mark points where rib can be cut to remove damaged material (Figure 5, Sheet 1).
4. Mark limits of required laminate blending, allowing for patch overlaps of 1/2-inch per ply.
5. Cut away damage and blend laminate using abrasive cloth, Item 76, WP 1803 00.
6. Cut repair core from foam strip that matches rib core.

REPAIR DAMAGED FOAM CORE RIB, DAMAGE LESS THAN 2 INCHES LONG - Continued

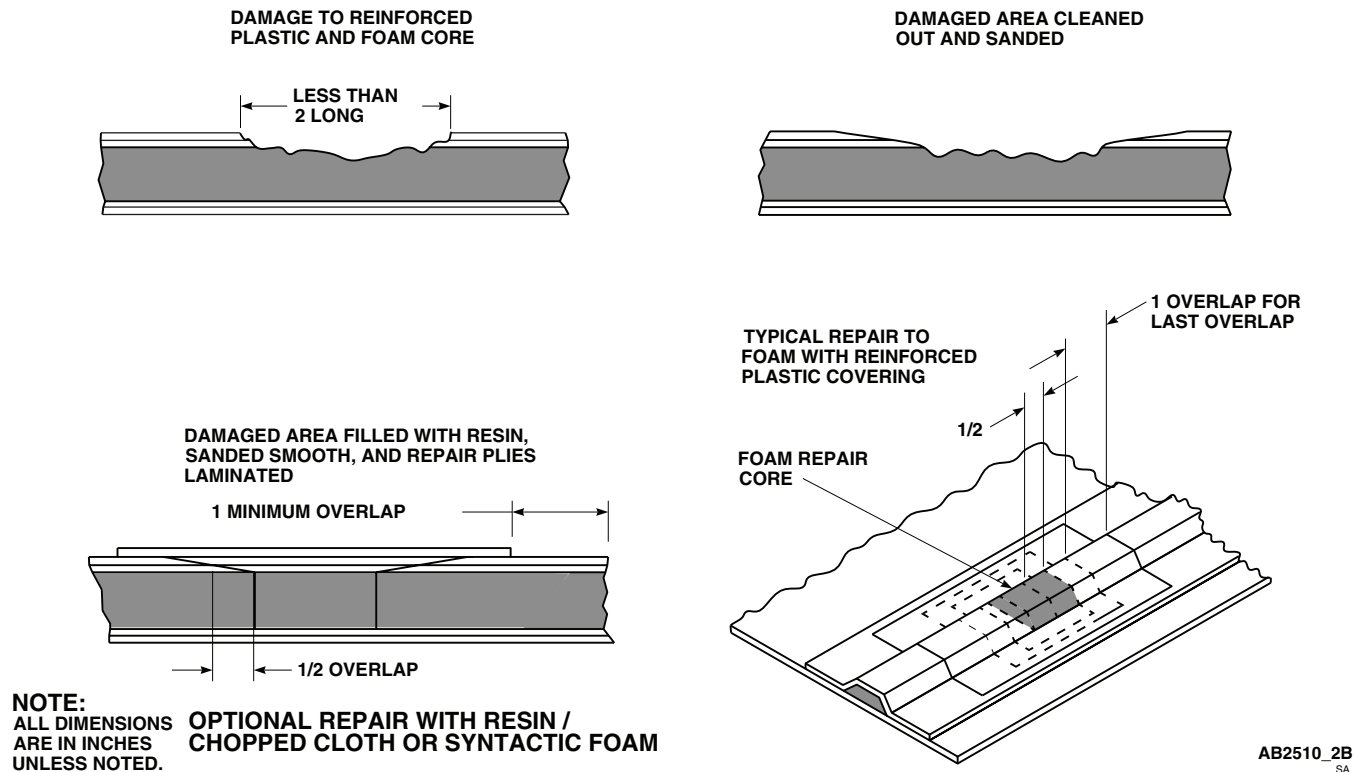
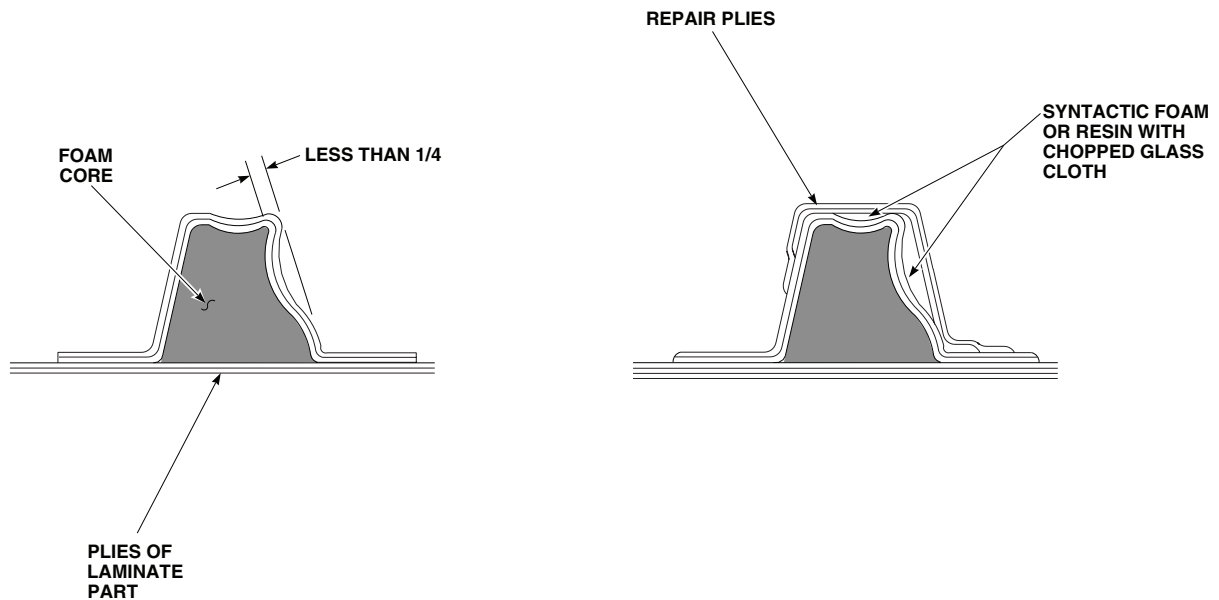


Figure 5. Repair Damaged Rib. (Sheet 2 of 2)

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Clean repair area, except for foam surfaces, with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Let air dry.
8. Mix resin/curing agent (Table 1).
9. Apply coat of mixed resin to repair core and cutaway section of rib with acid swabbing brush, Item 57, WP 1803 00.
10. Insert repair core into rib cutout.
11. For optional method, add chopped satin cloth, Item 90, WP 1803 00, to mixed resin, or use syntactic foam, to repair core (Table 1) (Figure 5, Sheet 2).
12. Cure resin bond of inserted core, or built-up repair of resin/chopped cloth or syntactic foam (Table 1).
13. Sand repair to match undamaged core, using abrasive cloth, Item 83, WP 1803 00.
14. Cut repair plies of satin cloth, Item 90, WP 1803 00, to size, allowing 1/2-inch minimum overlap on corresponding ply of rib.
15. Cut an extra top ply to overlap top patch ply by a minimum of 1 inch.

REPAIR DAMAGED FOAM CORE RIB, DAMAGE LESS THAN 2 INCHES LONG - Continued**NOTE:**

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2511
SA

Figure 6. Repair Crushed Rib.

16. Saturate all patch plies with mixed resin, using acid swabbing brush, Item 57, WP 1803 00, or beverage stirring stick, Item 300, WP 1803 00.
17. Apply coat of mixed resin to repair area of rib.
18. Place all repair plies in position and cover with sheet plastic, Item 220, WP 1803 00.
19. Work out trapped air and excess resin, using squeegee.
20. Apply pressure to patches by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
21. Cure repair (Table 1).
22. When cure is complete, remove vacuum bag, clamping plates, or weights; remove polyvinyl plastic sheet.
23. Remove excess resin and blend repair into adjacent surface using abrasive cloth, Item 83, WP 1803 00.
24. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR DAMAGED FOAM CORE, CRUSHED RIB**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. Crushed ribs may be repaired when one side is undistorted and other side and/or top, has been pushed in less than 1/4-inch. No part of the rib may project beyond normal dimension (Figure 6).

REPAIR DAMAGED FOAM CORE, CRUSHED RIB - Continued

2. Determine outside limits of repair area:
 - a. Total number of repair plies must be the same as number of plies in rib.
 - b. First ply must overlap beyond crushed area by at least 1/2-inch; each succeeding ply overlaps by the same amount.
 - c. Final ply must overlap preceding ply by at least 1 inch.
 - d. Outline limits of final ply.
3. Scuff entire repair surface lightly with abrasive cloth, Item 76, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

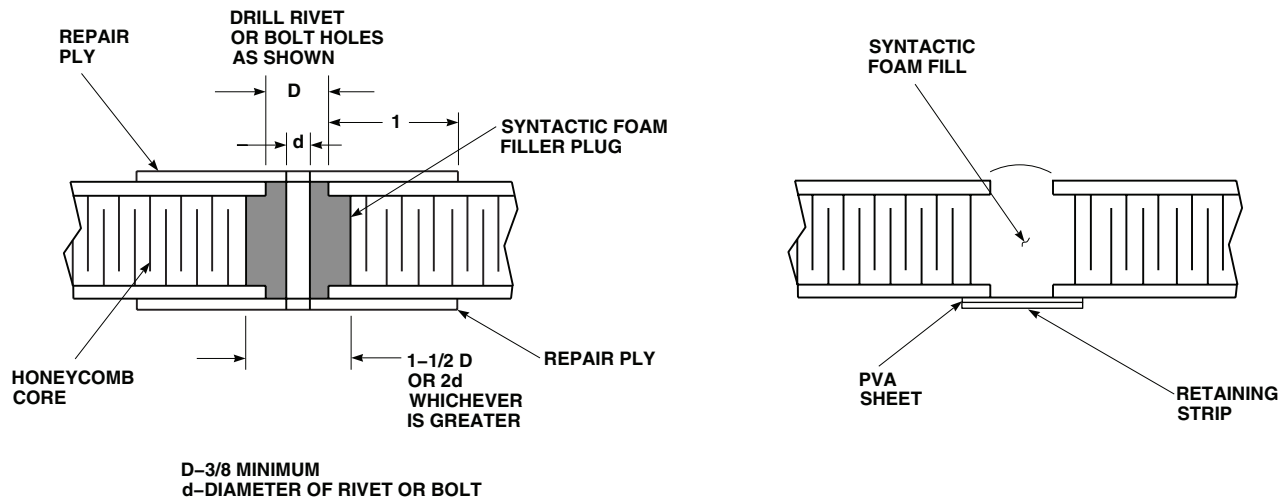
4. Clean repair area thoroughly with cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Air dry.
5. To make a mold for repair, attach wood or metal strips to rib sides with pressure sensitive adhesive tape, Item 334, WP 1803 00, or clamps. Cover inside of mold with sheet plastic, Item 220, WP 1803 00.
6. Mix syntactic foam (Table 1).
7. Fill cavity between deformed rib and mold with syntactic foam by pouring or with beverage stirring stick, Item 300, WP 1803 00. Release trapped air by working beverage stirring stick up and down. Completely fill cavity.
8. Cure repair (Table 1).
9. Remove mold and shape repair to undamaged part of rib with abrasive cloth, Item 76, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

10. Clean repair area with cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Let air dry.
11. Cut repair plies from satin cloth, Item 90, WP 1803 00 in sizes to meet requirement for 1/2-inch overlaps (1 inch for final patch).
12. Mix resin/curing agent (Table 1).
13. Saturate all repair plies with mixed resin using acid swabbing brush, Item 57, WP 1803 00, or beverage stirring stick, Item 300, WP 1803 00.
14. Coat repair area of rib with mixed resin.
15. Match all repair plies in position and cover with sheet plastic, Item 220, WP 1803 00.
16. Work out trapped air and excess resin with squeegee.

REPAIR DAMAGED FOAM CORE, CRUSHED RIB - Continued

**NOTE:**

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2512
SA

Figure 7. Honeycomb Panel Mounting Hole.

17. Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
18. Cure repair (Table 1).
19. Remove vacuum bag, clamping plates, or system of weights and pressure plates, and polyvinyl plastic sheet.
20. Use abrasive cloth, Item 83, WP 1803 00, to remove excess resin and blend repair into adjoining surfaces.
21. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR/REPLACE MOUNTING HOLE IN HONEYCOMB PANEL**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. Mark hole location on top skin of honeycomb panel. Extend lines or reference points out at least 4 inches from center.
2. Mark outline of repair plies at least 1 inch in all directions beyond drilled hole and mask around repair area with pressure sensitive adhesive tape, Item 334, WP 1803 00.
3. Drill hole through panel skin using 3/8-inch diameter drill.
4. Using a ground-down Allen wrench, or L-shaped tool attached to drill motor, rout out honeycomb core material through top skin hole. Make cavity 1 1/2 times larger than drilled hole (Figure 7).

REPAIR/REPLACE MOUNTING HOLE IN HONEYCOMB PANEL - Continued

5. Fill cavity with syntactic foam, as follows:
 - a. Cover skin opening on one side of panel with sheet plastic, Item 220, WP 1803 00, backed by wood or metal retaining strip held in place by clamps or pressure sensitive adhesive tape, Item 334, WP 1803 00.
 - b. Mix syntactic foam (Table 1).
 - c. Fill cavity of panel with syntactic foam by pouring or with beverage stirring stick, Item 300, WP 1803 00.
 - d. Release trapped air by working a beverage stirring stick, Item 300, WP 1803 00, up and down. Completely fill cavity, slightly beyond outer surface of skin to allow for shrinkage.
 - e. Cure syntactic foam (Table 1).
 - f. Remove retaining strip and polyvinyl plastic sheet.
6. Sand exposed syntactic foam flush with skin surface, using abrasive cloth, Item 76, WP 1803 00.
7. Cut satin cloth, Item 90, WP 1803 00, for repair plies to extend at least 1 inch in all directions beyond edge of drilled hole. Use same number of plies as in skin. Match ply weave direction to skin.
8. Apply and cure repair plies as follows:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Delamination of honeycomb panel will result if too much solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

- a. Clean repair area using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
 - b. Mix resin/curing agent (Table 1).
 - c. Spread repair plies on flat surface covered with sheet plastic, Item 220, WP 1803 00.
 - d. Saturate repair plies with mixed resin using acid swabbing brush, Item 57, WP 1803 00, or beverage stirring stick, Item 300, WP 1803 00.
 - e. Apply coat of mixed resin to repair area on both sides of panel, using acid swabbing brush, Item 57, WP 1803 00.
 - f. Put repair plies in place, cover with polyvinyl plastic sheet, Item 220, WP 1803 00, and force out trapped air and excess resin with squeegee.
 - g. Apply pressure to repair plies by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
 - h. Cure repair (Table 1).
9. After curing, remark hole location lines on patch to pick up center of plug and drill appropriate size hole for intended fastener through repaired area.
 10. Sand repair to blend with surrounding skin using abrasive cloth, Item 83, WP 1803 00.
 11. Remove masking tape.

REPAIR/REPLACE MOUNTING HOLE IN HONEYCOMB PANEL - Continued

12. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, DAMAGE TO SKIN NOT PENETRATING TO HONEYCOMB CORE**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. Repair minor damage to skin covering per TYPE I REPAIRABLE DAMAGE, in this work package.
2. Repair damage to skin covering affecting one or more plies per TYPE II REPAIRABLE DAMAGE, in this work package.

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, IMPACT DAMAGE NOT PENETRATING TO HONEYCOMB CORE**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. Damage less than 1/8-inch deep, not puncturing skin, with skin not disbonded from core, may be repaired.
2. Coin tap damage area to verify absence of disbonding (WP 0544 00).
3. Mark outline of damage area with at least 1 inch added in all directions.
4. Mask surface beyond marked area with pressure sensitive adhesive tape, Item 334, WP 1803 00.
5. Lightly sand repair area with abrasive cloth, Item 83, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Thoroughly clean repair area with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Air dry.
7. Mix metal surface dent filler, Item 126, WP 1803 00 (Table 2).
8. Apply metal surface dent filler, Item 126, WP 1803 00, with beverage stirring stick, Item 300, WP 1803 00.
9. Smooth filler and remove excess with squeegee.
10. Let filler dry for at least 2 hours.
11. Sand repair to blend into adjoining surface using abrasive cloth, Item 83, WP 1803 00.
12. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, DAMAGE TO ONE SKIN PENETRATING INTO HONEYCOMB CORE**NOTE**

- Indentations deeper than 1/8-inch can be repaired by this procedure.
 - For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.
1. Punctures from one surface, less than 4 inches in length or diameter, located at least 1 inch from edge of honeycomb core, may be repaired (core not more than 1 inch thick) (Figure 8).

CAUTION

Damage to inner skin will result if cutting tools are used improperly. Use care when using cutting tools.

2. Open up puncture (or dent) with drill or router, removing ragged edges and broken fibers.
3. Clean out crushed core, undercutting skin by at least 1/8-inch. Remove debris with vacuum cleaner.
4. Allowing at least 1 inch in all directions beyond trimmed edge of damage, mark perimeter of repair patch.
5. Mask surface just outside marked area with pressure sensitive adhesive tape, Item 334, WP 1803 00.

WARNING

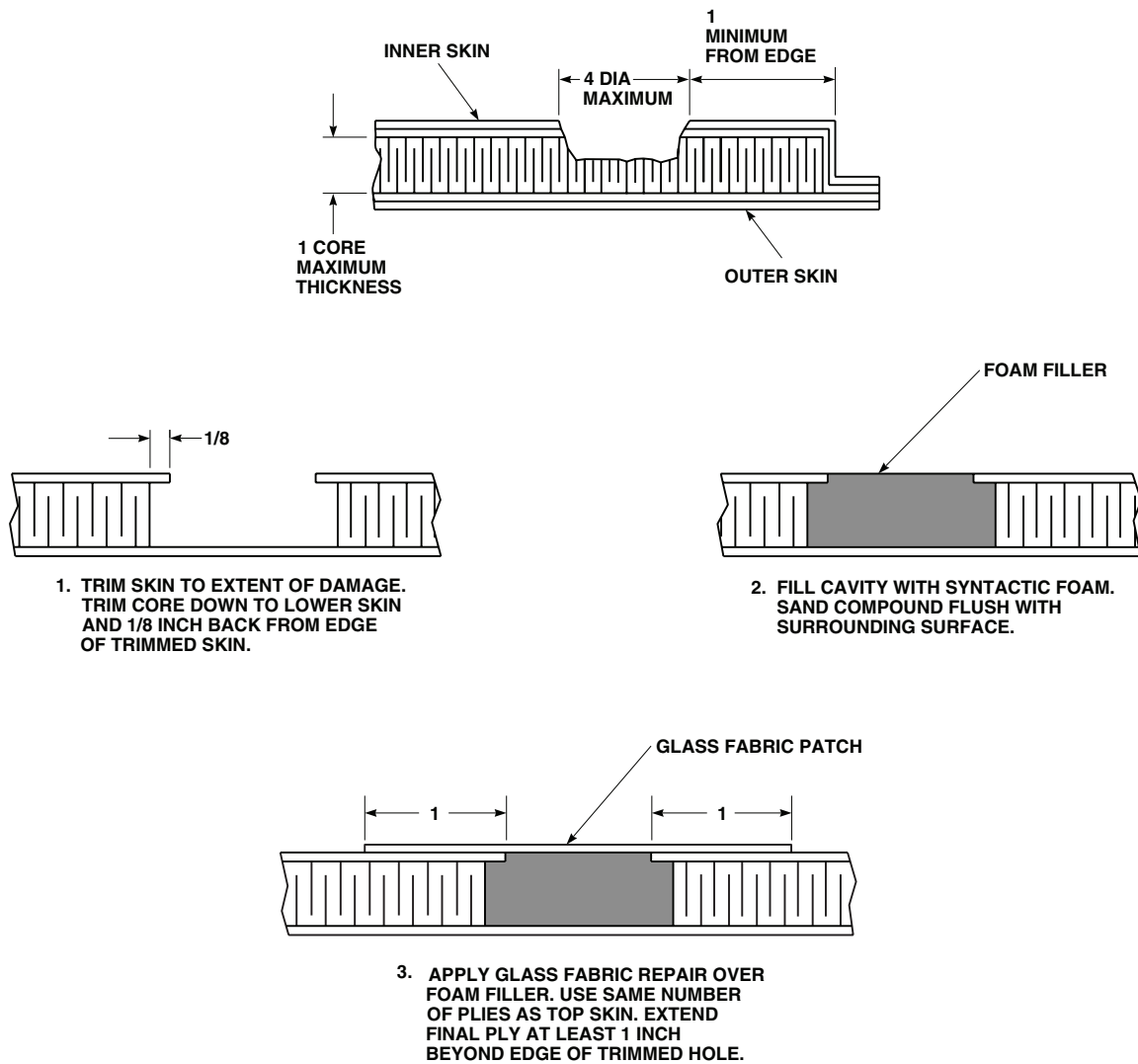
Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Delamination of honeycomb panel will result if too much solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

6. Remove paint finish inside masked area and clean surface using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
7. Mix syntactic foam (Table 1).
8. Fill cavity of panel with syntactic foam by pouring or with a beverage stirring stick, Item 300, WP 1803 00. Release trapped air by working a beverage stirring stick, Item 300, WP 1803 00, up and down. Completely fill cavity slightly beyond skin surface to allow for shrinkage.
9. Cure syntactic foam (Table 1).
10. Sand cured foam flush with skin, using abrasive cloth, Item 76, WP 1803 00.

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, DAMAGE TO ONE SKIN PENETRATING INTO HONEYCOMB CORE - Continued



NOTE:
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2513
SA

Figure 8. Damage to One Skin and Core.

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, DAMAGE TO ONE SKIN PENETRATING INTO HONEYCOMB CORE - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

11. Clean repair area with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
12. Cut repair plies from satin cloth, Item 90, WP 1803 00, to shape of marked area, number of plies to equal number of plies in skin.
13. Mix resin/curing agent (Table 1).
14. Spread repair plies on flat surface covered with sheet plastic, Item 220, WP 1803 00.
15. Saturate repair plies with mixed resin using acid swabbing brush, Item 57, WP 1803 00, or beverage stirring stick, Item 300, WP 1803 00.
16. Apply coat of mixed resin to unmasked skin and foam areas of part, using acid swabbing brush, Item 57, WP 1803 00.
17. Put repair plies in place over cavity, overlapping undamaged skin at least 1 inch in all directions.
18. Cover repair with sheet plastic, Item 220, WP 1803 00.
19. Squeegee trapped air and excess resin out of repair plies.
20. Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
21. Cure repair (Table 1).
22. Remove vacuum bag, clamping plates, or system of weights and pressure plates and polyvinyl plastic sheet.
23. Sand to remove excess resin and blend repair into adjoining surface, using abrasive cloth, Item 83, WP 1803 00.
24. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, DAMAGE PENETRATING BOTH SKINS AND HONEYCOMB CORE**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

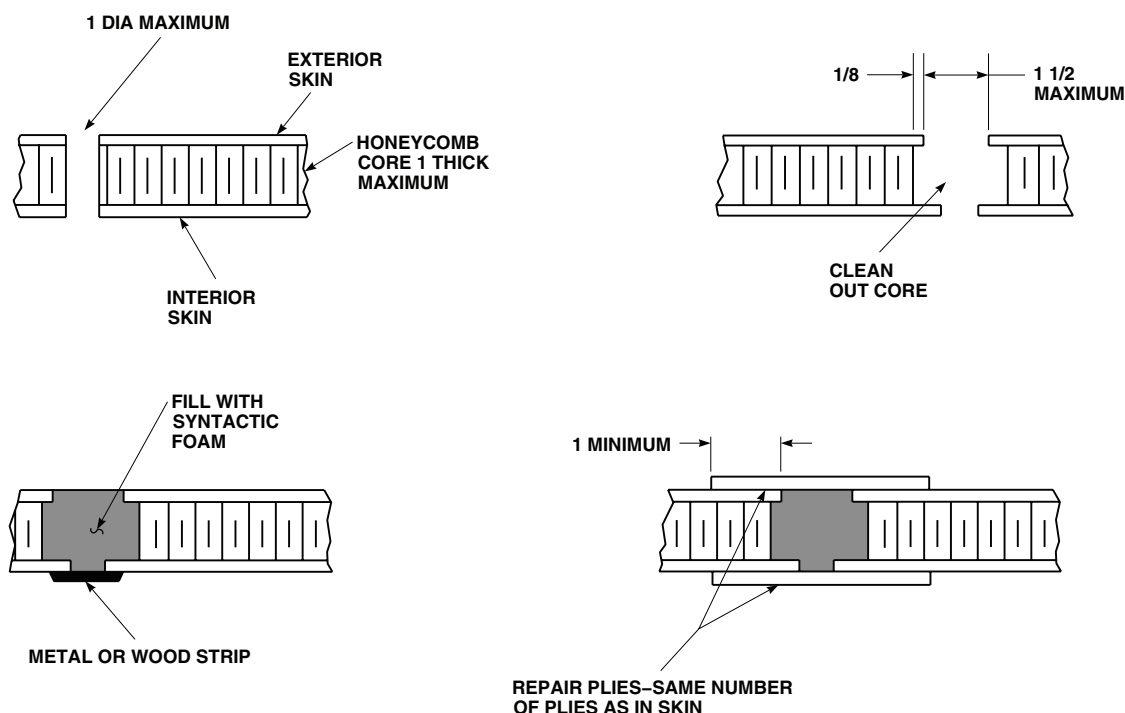
1. Penetrations completely through panel (including misdrilled holes), less than 1 inch in length or diameter and at least 2 1/2-inches from honeycomb core edge, may be repaired.

CAUTION

Damage to inner skin will result if cutting tools are used improperly. Use care when using cutting tools.

2. Open up puncture, with drill, router, or hole saw to remove ragged edges and broken fibers, to a maximum of 1 inch diameter.
3. Counterbore hole in exterior skin to 1 1/2 times original hole diameter. Do not exceed 1 1/2-inches (Figure 9).

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, DAMAGE PENETRATING BOTH SKINS AND HONEYCOMB CORE - Continued



NOTE:

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2514
SA

Figure 9. Damage to Both Skins and Core.

4. Clean out crushed core, undercutting at least 1/8-inch below exterior skin. Remove debris with vacuum cleaner.
5. Mark outline of repair at least 1 inch beyond hole edge, on both exterior and interior skins.
6. Mask around outlines with pressure sensitive adhesive tape, Item 334, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Delamination of honeycomb panel will result if too much solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

7. Remove paint finish inside masked areas and clean skin with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
8. Cover opening in interior skin with sheet plastic, Item 220, WP 1803 00, backed up by wood or metal strip held in place by clamps or pressure sensitive adhesive tape, Item 334, WP 1803 00.
9. Mix syntactic foam with curing agent (Table 1).
10. Fill cavity with syntactic foam by pouring or with beverage stirring stick, Item 300, WP 1803 00.

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, DAMAGE PENETRATING BOTH SKINS AND HONEYCOMB CORE - Continued

11. Release trapped air by working beverage stirring stick, Item 300, WP 1803 00, up and down. Completely fill cavity, slightly beyond skin surface to allow for shrinkage.
12. Cure syntactic foam (Table 1).
13. Sand foam repair even with exterior skin using abrasive cloth, Item 83, WP 1803 00.
14. Remove wood or metal strip and polyvinyl plastic sheet, from interior skin surface.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

15. Clean foam, repair and both skins inside masked areas with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
16. Prepare resin mix (Table 1).
17. Cut repair plies of satin cloth, Item 90, WP 1803 00, to fit areas on both skins within masked area. Number of plies in each repair must equal number of plies in skin.
18. Place repair plies on flat surface protected by plastic sheet, Item 220, WP 1803 00.
19. Saturate repair plies with resin mix, using acid swabbing brush, Item 57, WP 1803 00, or beverage stirring stick, Item 300, WP 1803 00.
20. Apply coat of resin mix to repair area of both skins, using acid swabbing brush, Item 57, WP 1803 00.
21. Put repair plies in place, cover with plastic sheet, Item 220, WP 1803 00, and force out trapped air and excess resin with squeegee.
22. Apply pressure to repair plies by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
23. Cure repair (Table 1).
24. Remove pressure and heat equipment, plastic sheet, and masking tape.
25. Blend repair into surrounding skin surfaces with abrasive cloth, Item 83, WP 1803 00.
26. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, SKIN-TO-CORE DISBOND**NOTE**

For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. Determine extent of disbond using coin tapping method described in (WP 0544 00).
2. Disbonded areas not more than 2 inches in diameter are repairable by method A or method B.

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, SKIN-TO-CORE DISBOND - Continued

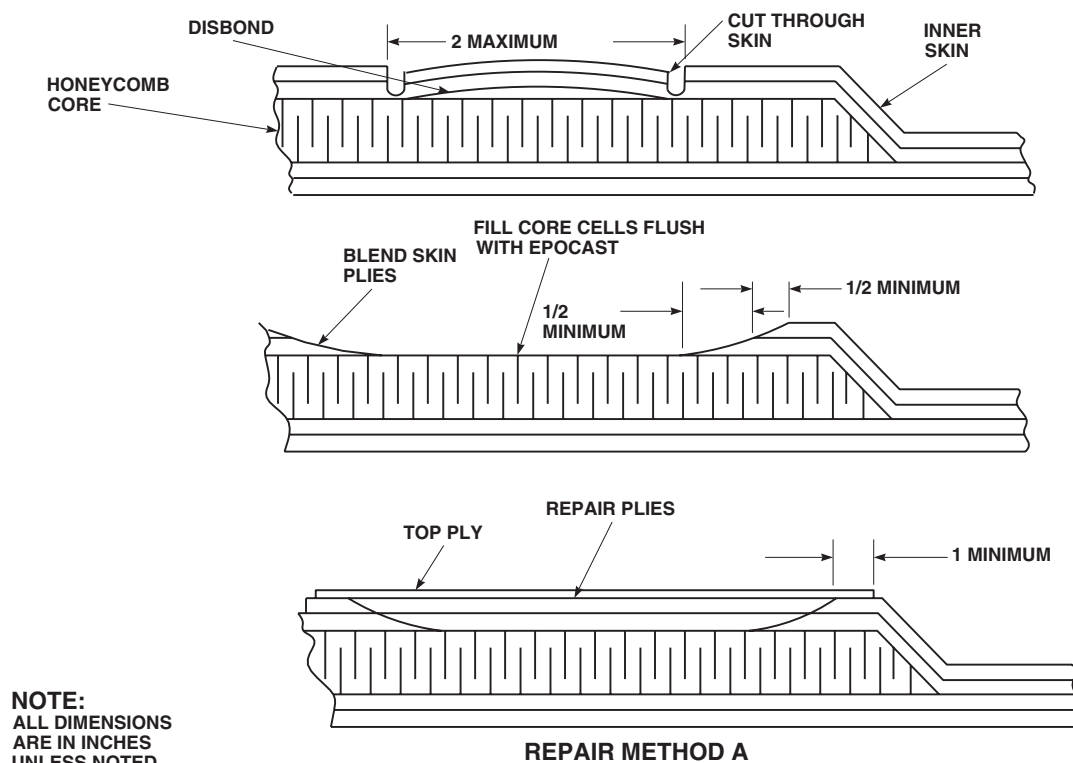


Figure 10. Skin-to-Core Disbond. (Sheet 1 of 2)

REPAIR METHOD A

CAUTION

Honeycomb core can be easily damaged while cutting through skin. Use extreme care to limit damage.

1. Cut through skin along marked outline of disbonded area, using router bit. Remove and discard cutout skin.
2. Allowing 1/2-inch blend per ply, mark the outside limit of repair.
3. Blend skin plies as shown in Figure 10, Sheet 1, using abrasive cloth, Item 76, WP 1803 00.
4. Mark outline of overlap ply extending 1 inch beyond blended area.
5. Sand overlap area lightly with abrasive cloth, Item 83, WP 1803 00.
6. Remove loose particles from cavity with vacuum cleaner and wipe up sanding dust with clean tack rag.
7. Mask surface just outside marked area with pressure sensitive adhesive tape, Item 334, WP 1803 00.

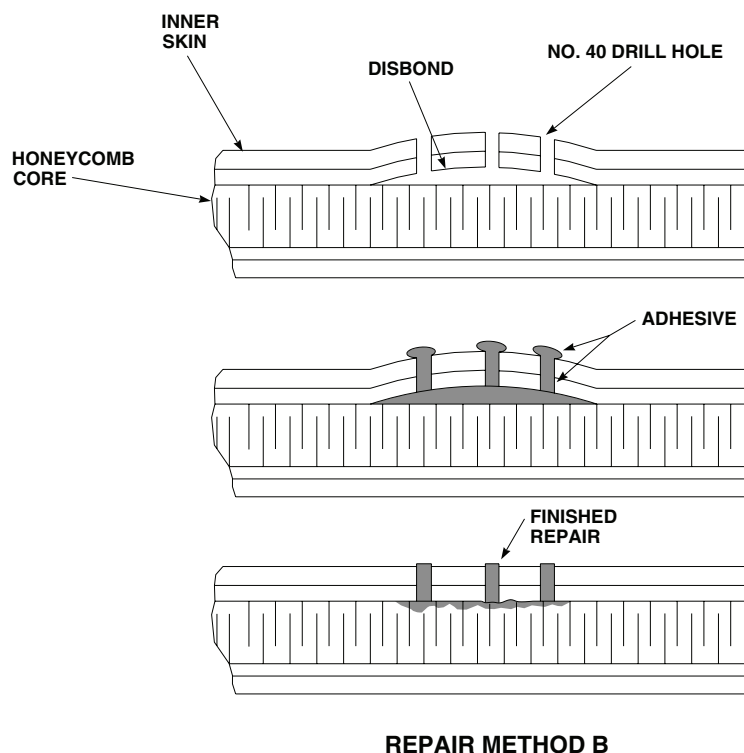
REPAIR METHOD A - Continued**REPAIR METHOD B**AB2541_2
SA

Figure 10. Skin-to-Core Disbond. (Sheet 2 of 2)

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

Delamination of honeycomb panel will result if too much solvent is allowed to seep into panel. Use solvent sparingly to prevent delamination.

8. Clean skin inside masked area using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
9. Mix sealing compound, Item 276, WP 1803 00, or sealing compound, Item 260, WP 1803 00 (Table 1).
10. Fill exposed core with sealing compound, Item 276, WP 1803 00, or sealing compound, Item 260, WP 1803 00, using beverage stirring stick, Item 300, WP 1803 00. Force Epocast into honeycomb to eliminate air pockets, filling core slightly higher than original surface to allow for shrinkage.

REPAIR METHOD A - Continued**CAUTION**

Damage to part will result from expansion of wet Epocast when heated. Let Epocast harden to touch before heat-curing.

11. Cure sealing compound, Item 276, WP 1803 00, or sealing compound, Item 260, WP 1803 00 (Table 1).

NOTE

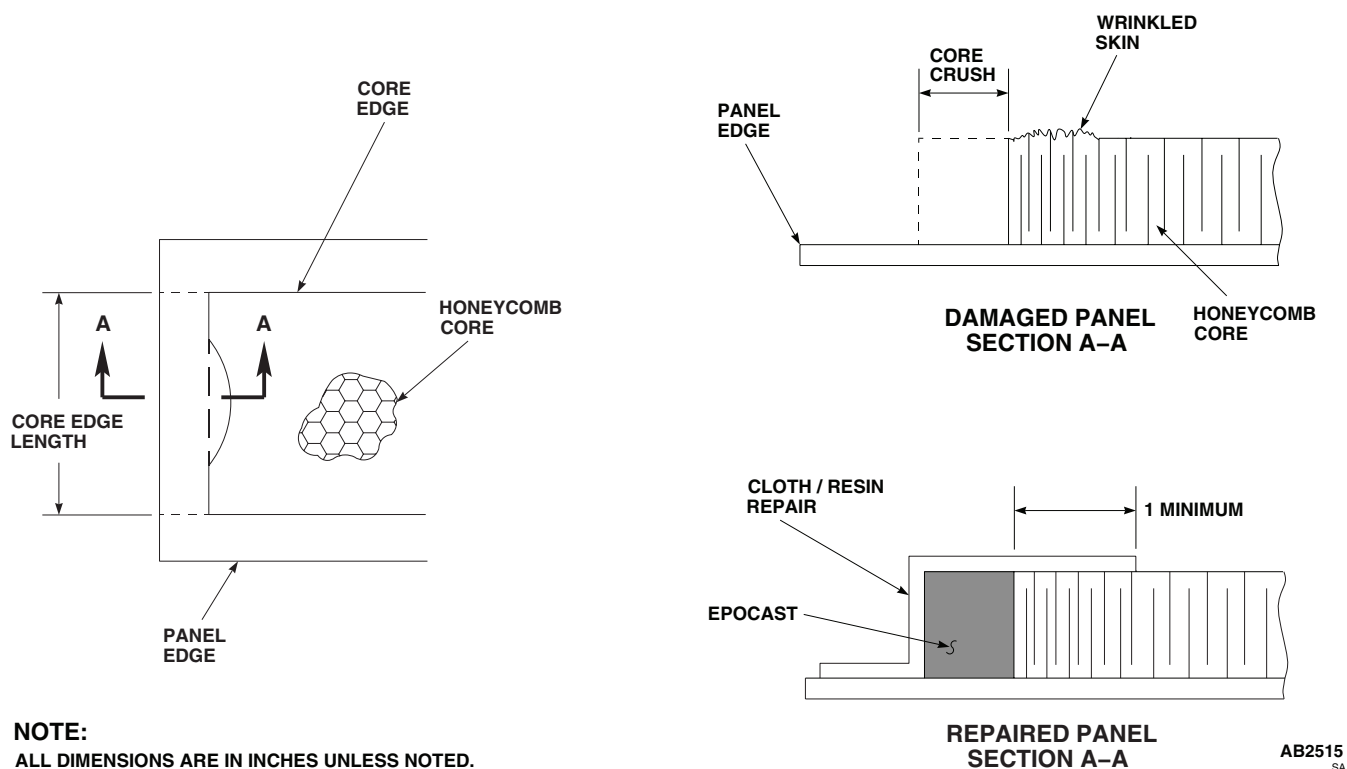
Cut weave of each repair ply to match its laminate ply.

12. Cut the same number of repair plies as in original skin from satin cloth, Item 90, WP 1803 00.
13. Cut top ply from satin cloth, Item 90, WP 1803 00, to overlap repair by 1 inch, with fabric direction the same as top skin ply.
14. Mix resin/curing agent (Table 1).
15. Spread repair plies on flat surface covered with sheet plastic, Item 220, WP 1803 00.
16. Saturate repair plies with mixed resin using acid swabbing brush, Item 57, WP 1803 00, or beverage stirring stick, Item 300, WP 1803 00.
17. Apply coat of mixed resin to unmasked skin and foam areas of part, using acid swabbing brush, Item 57, WP 1803 00.
18. Put repair plies in place with top ply, overlapping undamaged skin at least 1 inch in all directions.
19. Cover repair with sheet plastic, Item 220, WP 1803 00.
20. Squeegee trapped air and excess resin out of repair patch.
21. Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
22. Cure repair (Table 1).
23. Remove vacuum bag, clamping plates, or system of weights and pressure plates, and sheet.
24. Sand to remove excess resin and blend repair into adjoining surface, using abrasive cloth, Item 83, WP 1803 00.
25. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR METHOD B

1. Drill 2 or more holes, through disbonded skin only, with No. 40 drill, at least 1/2-inch apart (Figure 10, Sheet 2).
2. Remove loose particles in and around drill holes with vacuum cleaner.
3. Select adhesive, Item 44, WP 1803 00, or adhesive, Item 13, WP 1803 00, or adhesive, Item 12, WP 1803 00. Mix Epon (Table 1) or adhesive (Table 2).
4. Locally make hypodermic syringe (WP 1805 00).
5. Inject adhesive/resin into each drilled hole until it overflows using locally made hypodermic syringe (WP 1805 00).
6. Remove excess adhesive/Epon from skin surface and place sheet plastic, Item 220, WP 1803 00, over repair.
7. Apply pressure to disbonded area by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
8. Cure repair (Table 1 or Table 2).
9. Remove vacuum bag, clamping plates, or system of weights and pressure plates. Remove polyvinyl plastic sheet.

REPAIR METHOD B - Continued

**NOTE:**

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

Figure 11. Crushed Core Edge.

10. Sand to remove excess resin and blend repair into adjoining surface, using abrasive cloth, Item 83, WP 1803 00.
11. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, REPAIR CORE EDGE**NOTE**

- Core edges crushed inward more than 1/2-inch but less than 1 1/2-inches, not exceeding 50 percent of total core edge length, are reparable by this method.
- For Kevlar repair, refer to, KEVLAR, in this work package, in addition to these procedures.

1. Mark outline of repair area, indicating original contour of core, using scribe (Figure 11).
2. Adding at least 1 inch in all directions for feathering repair, mask outside of area with pressure sensitive adhesive tape, Item 334, WP 1803 00.
3. Sand smooth any wrinkles of inner skin caused by crushing of core, using abrasive cloth, Item 76, WP 1803 00, and abrasive cloth, Item 83, WP 1803 00.
4. Lightly sand entire repair area with abrasive cloth, Item 83, WP 1803 00.

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, REPAIR CORE EDGE - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean repair area thoroughly with cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Air dry.
6. To make a mold for repair, attach wood or metal strip across crushed area with pressure sensitive adhesive tape, Item 334, WP 1803 00. Cover inside of strip with sheet plastic, Item 220, WP 1803 00.
7. Mix sealing compound, Item 276, WP 1803 00, or sealing compound, Item 260, WP 1803 00 (Table 1).
8. Apply Epocast with beverage stirring stick, Item 300, WP 1803 00, to fill any exposed core and build core up to original contour.

CAUTION

Damage to part will result from expansion of wet Epocast when heated. Let Epocast harden to touch before heat-curing.

9. Cure sealing compound, Item 276, WP 1803 00, or sealing compound, Item 260, WP 1803 00 (Table 1).
10. Remove mold strip and polyvinyl plastic sheet.
11. Sand Epocast flush with adjacent core edges with abrasive cloth, Item 76, WP 1803 00, and abrasive cloth, Item 83, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

12. Clean repair and overlap area with cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Let air dry.

NOTE

Match weave direction of each repair ply to its laminate ply.

13. Cut repair plies from satin cloth, Item 90, WP 1803 00, to extend at least 1 inch beyond actual repair in all directions. Use same number of plies as in original skin.
14. Mix resin/curing agent (Table 1).
15. Spread repair plies on flat surface covered with sheet plastic, Item 220, WP 1803 00.
16. Saturate repair plies with mixed resin using acid swabbing brush, Item 57, WP 1803 00, or beverage stirring stick, Item 300, WP 1803 00.
17. Coat repair area and 1 inch overlap with mixed resin, using acid swabbing brush, Item 57, or beverage stirring stick, Item 300, WP 1803 00.
18. Place all repair plies in position and cover with sheet plastic, Item 220, WP 1803 00.

REPAIR HONEYCOMB CORE - COMPOSITE SKIN PANEL, REPAIR CORE EDGE - Continued

19. Work out trapped air and excess resin with squeegee.
20. Apply pressure to repair by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
21. Cure repair (Table 1).
22. Remove vacuum bag, clamping plates, or system of weights and pressure plates, and polyvinyl plastic sheet.
23. Use abrasive cloth, Item 83, WP 1803 00, to remove excess resin and blend repair into adjoining surfaces.
24. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR INNER SKIN-TO-CORE DISBOND**NOTE**

A disbond between inner skin and honeycomb core, not more than 2 square inches in area, is repairable. Multiple disbonds on one panel, at least 4 inches between areas, are repairable, but total area must not exceed 10% of total panel surface.

1. Locate and outline disbond areas by coin-tapping method described in (WP 0544 00).

CAUTION

Helicopter fuel contacting repair can adversely affect adhesive. Do not make this repair on bonded panels exposed to helicopter fuel.

2. Drill 2 or more holes through inner skin in disbonded area, using No. 40 drill bit. Space holes at least 1/2-inch apart (Figure 12).
3. Mix adhesive, Item 11, WP 1803 00 (Table 2).
4. Locally make hypodermic syringe (WP 1805 00).
5. Inject adhesive/resin into each drilled hole until it overflows using locally made hypodermic syringe (WP 1805 00).
6. Apply pressure sensitive adhesive tape, Item 334, WP 1803 00, over all drilled holes.
7. Turn panel upside down to let adhesive flow throughout disbonded area for 10 to 15 minutes.
8. Apply pressure to disbonded area by vacuum bag, clamping plates, or system of weights and pressure plates (TM 1-1500-204-23).
9. Cure repair (Table 2).
10. Remove vacuum bag, clamping plates, or weights and pressure plates.
11. Remove masking tape.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

12. Remove excess adhesive using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cloth.

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR INNER SKIN-TO-CORE DISBOND - Continued

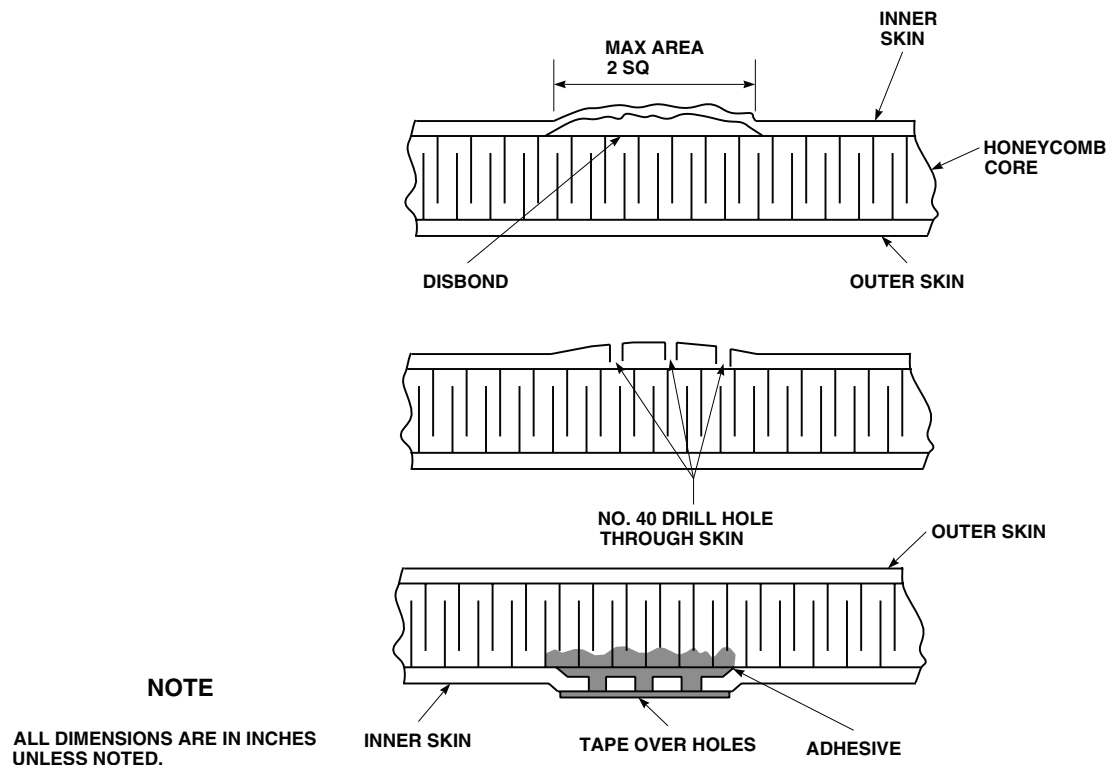


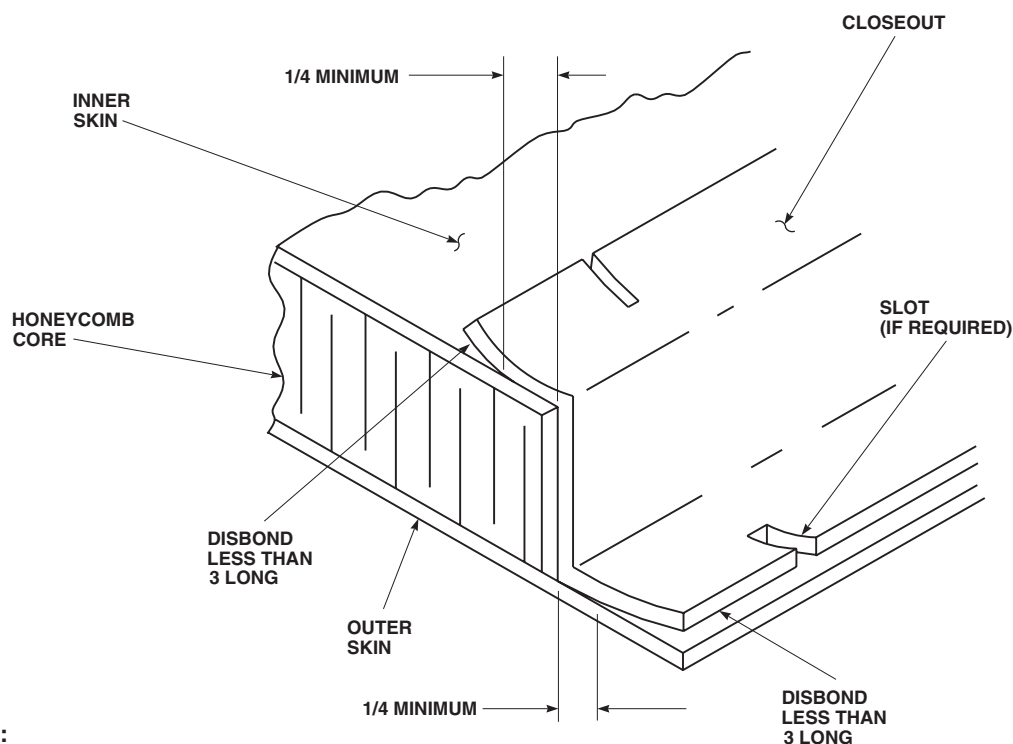
Figure 12. Inner Skin-to-Core Disbond.

WARNING

- Serious burns to skin and eyes will result if contact made with corrosion resistant coating. Avoid breathing in its mist, dust, or vapor.
- Wear rubber gloves, rubber apron, and protective goggles, when using corrosion resistant coating.
- In case of chemical contact, flush exposed areas with water for 15 minutes and get immediate medical attention.

13. If skin surface has been abraded during repair, apply corrosion resistant coating, Item 110, WP 1803 00, to skin surface and dry thoroughly (TM 1-1500-204-23).
14. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR INNER SKIN-TO-CORE DISBOND - Continued



NOTE:

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0048A
SA

Figure 13. Closeout Disbond.

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR CLOSEOUT DISBOND

CAUTION

- Helicopter fuel contacting repair can adversely affect adhesive. Do not make this repair on bonded panels exposed to helicopter fuel.
- Use care in slotting closeout to avoid damaging skin or cutting into bonded area.

NOTE

Disbonds with at least 1/4-inch contact (measured from vertical core edge toward outer edge of closeout) and less than 3 inches long are repairable.

1. If disbonded area of closeout bulges away from panel, cut a slot near center of disbond to permit flattening closeout (Figure 13).
2. Mix adhesive, Item 11, WP 1803 00 (Table 2).
3. Force adhesive into crack of disbond, using beverage stirring stick, Item 300, WP 1803 00 (sharpened, if required) or thin plastic tool, until disbond is filled.
4. Press closeout tightly against panel, using C-clamps to maintain pressure.

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR CLOSEOUT DISBOND - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Remove excess adhesive using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
6. Cure repair (Table 2).
7. Remove clamps. Inspect visually and by coin tapping (WP 0544 00) to verify bond.
8. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR INNER SKIN PUNCTURE, UNDER 1/2-INCH**CAUTION**

- Helicopter fuel contacting repair can adversely affect adhesive. Do not make this repair on bonded panels exposed to helicopter fuel.
- If outer skin is punctured during work, do not complete repair. Use extreme care to avoid this.

NOTE

Punctures to inner skin less than 1/2-inch in length or diameter, located at least 1 inch from nearest panel closeout, are repairable by this method.

1. Clean up puncture with drill or hole saw to remove ragged edges and eliminate cracks. Deburr hole edge (Figure 14, Sheet 1).
2. If finished hole is more than 1/4-inch in diameter, clean out crushed core with router and undercut skin at least 1/8-inch.
3. Remove all loose particles from cavity with vacuum cleaner.

WARNING

- Serious burns to skin and eyes will result if contact made with corrosion resistant coating. Avoid breathing in its mist, dust, or vapor.
 - Wear rubber gloves, rubber apron, and protective goggles, when using Alodine.
 - In case of chemical contact, flush exposed areas with water for 15 minutes and get immediate medical attention.
4. Apply corrosion resistant coating, Item 110, WP 1803 00, to cut skin and dry thoroughly.
 5. Mix adhesive, Item 11, WP 1803 00 (Table 2).
 6. Fill cavity completely with adhesive, slightly above skin surface to allow for shrinkage.
 7. Cure repair (Table 2).
 8. Lightly sand repair flush with skin surface, using abrasive paper, Item 211, WP 1803 00.

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR INNER SKIN PUNCTURE, UNDER 1/2-INCH

- Continued

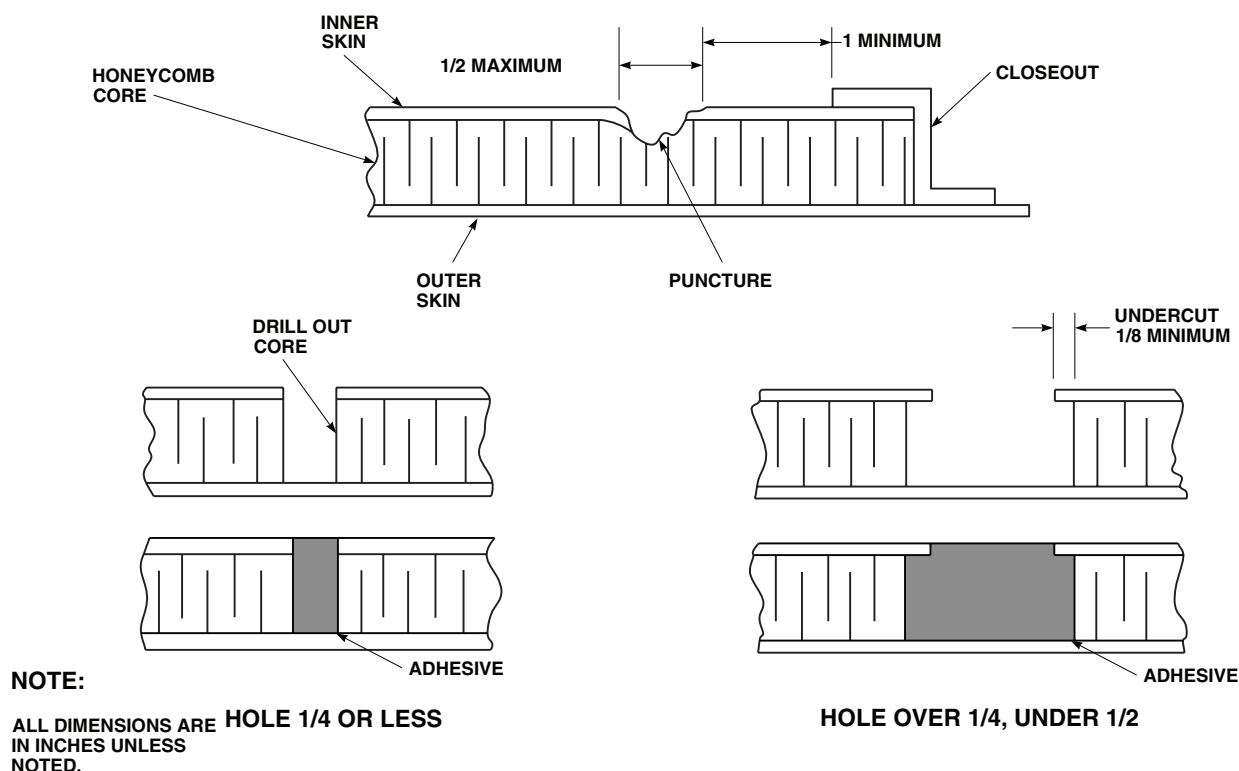


Figure 14. Inner Skin Puncture. (Sheet 1 of 2)

9. If surrounding skin is scratched or abraded, apply corrosion resistant coating, Item 110, WP 1803 00, and dry thoroughly.
10. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR INNER SKIN PUNCTURE, 1/2 TO 1 1/2-INCHES

CAUTION

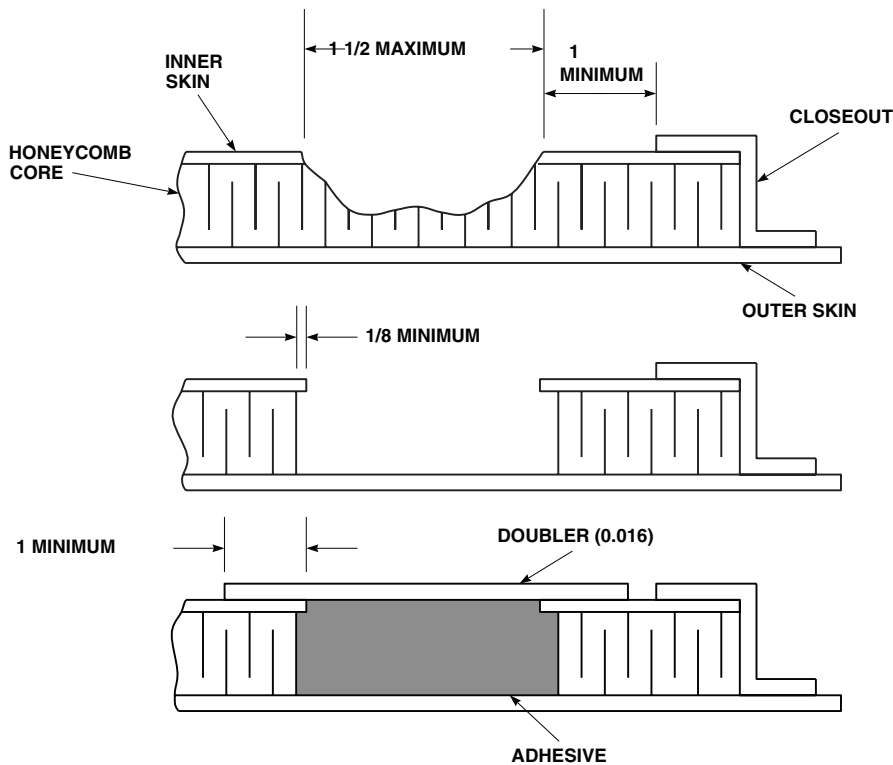
- Helicopter fuel contacting repair can adversely affect adhesive. Do not make this repair on bonded panels exposed to helicopter fuel.
- If outer skin is punctured during work, do not complete repair. Use extreme care to avoid this condition.

NOTE

Punctures to inner skin from 1/2-inch to 1 1/2-inches in length or diameter, located at least 1 inch from nearest panel closeout, are repairable by this method.

1. Clean up puncture with drill or hole saw to remove ragged edges and eliminate cracks. Deburr hole edge (Figure 14, Sheet 2).
2. Using router, remove damaged core and undercut inner skin by at least 1/8-inch.
3. Remove all loose particles from cavity with vacuum cleaner.

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR INNER SKIN PUNCTURE, 1/2 TO 1 1/2-INCHES - Continued



NOTE:
ALL DIMENSIONS
ARE IN INCHES
UNLESS NOTED.

AB2542_2B
SA

Figure 14. Inner Skin Puncture. (Sheet 2 of 2)

WARNING

- Serious burns to skin and eyes will result if contact made with corrosion resistant coating. Avoid breathing in its mist, dust, or vapor.
- Wear rubber gloves, rubber apron, and protective goggles, when using Alodine.
- In case of chemical contact, flush exposed areas with water for 15 minutes and get immediate medical attention.

4. Apply corrosion resistant coating, Item 110, WP 1803 00, to cut skin and dry thoroughly.
5. Make doubler as follows:
 - a. Cut from anodized 2024-T3 aluminum sheet, 0.016-inch thick, extending at least 1 inch beyond skin opening in all directions.
 - b. If doubler is rectangular, round corners to 0.120-inch radius, minimum.
 - c. Deburr all edges.
 - d. Apply corrosion resistant coating, Item 110, WP 1803 00, to cut edges and dry thoroughly.

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR INNER SKIN PUNCTURE, 1/2 TO 1 1/2-INCHES - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Thoroughly clean skin area to be covered by doubler with cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Air dry.
7. Mix adhesive, Item 11, WP 1803 00, per Table 2.
8. Fill cavity with adhesive until level with skin surface.
9. Spread adhesive over skin surface to be covered by doubler, using beverage stirring stick, Item 300, WP 1803 00, or acid swabbing brush, Item 57, WP 1803 00.
10. Press doubler firmly in place.
11. Remove adhesive squeeze out, using nonmetallic scraper, and cover with plastic sheet, Item 220, WP 1803 00.
12. Apply pressure to doubler by vacuum bag, clamping plates, or weight and pressure plate (TM 1-1500-204-23).
13. Cure repair (Table 2).
14. Remove pressure equipment and plastic sheet.
15. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR OUTER SKIN PUNCTURE**NOTE**

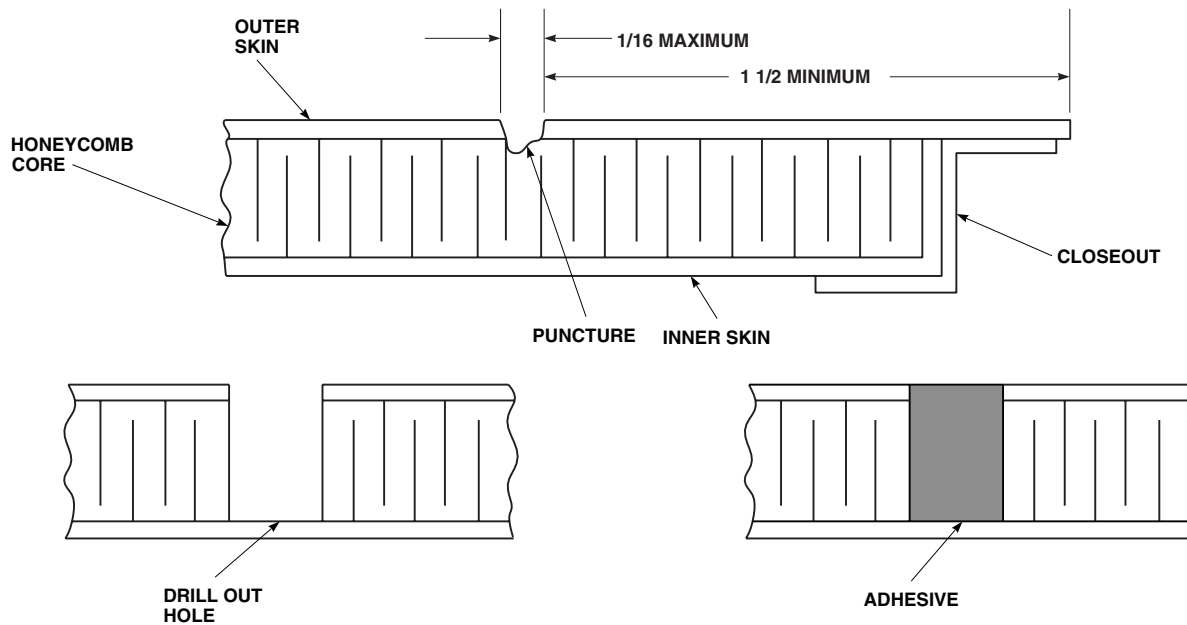
Punctures to outer skin 1/16-inch or less in length or diameter, located at least 1 1/2-inches from skin edge, are repairable by this method.

1. Using drill, enlarge puncture hole enough to remove ragged edges and eliminate any cracks (Figure 15). Deburr hole edge.
2. Remove all loose particles from cavity with vacuum cleaner.

WARNING

- Serious burns to skin and eyes will result if contact made with corrosion resistant coating. Avoid breathing in its mist, dust, or vapor.
 - Wear rubber gloves, rubber apron, and protective goggles, when using Alodine.
 - In case of chemical contact, flush exposed areas with water for 15 minutes and get immediate medical attention.
3. Apply corrosion resistant coating, Item 110, WP 1803 00, to cut skin and dry thoroughly.
 4. Mix adhesive, Item 11, WP 1803 00, per Table 2.
 5. Fill cavity completely with adhesive, Item 11, WP 1803 00, flush with outer skin surface.
 6. Cure repair (Table 2).
 7. Lightly sand repair flush with outer skin surface, using abrasive paper, Item 211, WP 1803 00.

REPAIR HONEYCOMB CORE - METAL SKIN PANEL, REPAIR OUTER SKIN PUNCTURE - Continued

**NOTE:**

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0051A
SA

Figure 15. Outer Skin Puncture.

8. If surrounding skin surface is scratched or abraded, apply corrosion resistant coating, Item 110, WP 1803 00, and dry thoroughly.
9. Touch up paint, as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

EXTRUSION CHARTS

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

References

TM 1-1500-204-23

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

TYPICAL RIVET SPACING - SHEET METAL SUBSTITUTE

1. Use required rivet spacing and edge distance, when making sheet stock substitution for extrusion (Figure 1 and TM 1-1500-204-23).
2. Use No. 30 drill for MS20470AD4 rivets and No. 21 drill for MS20470AD5 rivets.
3. Use flush rivets, as required, to reduce interference with adjacent structural members.

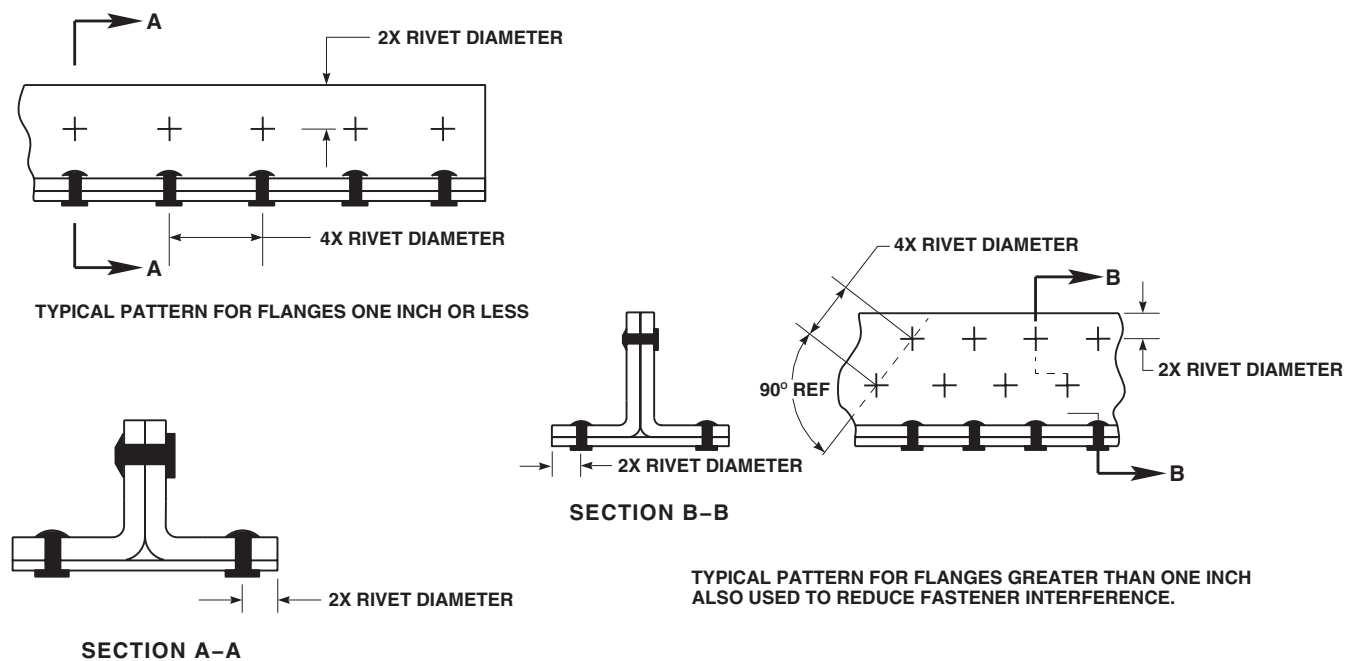
AA0052A
SA

Figure 1. Rivet Spacing and Edge Distance.

TYPICAL RIVET SPACING - SHEET METAL SUBSTITUTE - Continued**EQUAL LEG ANGLE**

Use Figure 2, Sheet 1 and Figure 2, Sheet 2 for equal leg angle.

UNEQUAL LEG ANGLE

Use Figure 3, Sheet 1, Figure 3, Sheet 2, and Figure 3, Sheet 3 for unequal leg angle.

TEE SECTION

Use Figure 4, Sheet 1, Figure 4, Sheet 2, and Figure 4, Sheet 3 for tee section.

UNEQUAL THICKNESS ANGLE

Use Figure 5 for unequal thickness angle.

TAPERED LEG ANGLE

Use Figure 6 for tapered leg angle.

CLOSED ANGLE TEE

Use Figure 7 for closed angle tee.

ZEE SECTION

Use Figure 8 for zee section.

CHANNEL

Use Figure 9 for channel.

CLOSED LEG ANGLE

Use Figure 10 for closed leg angle.

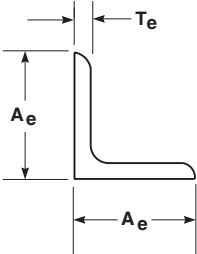
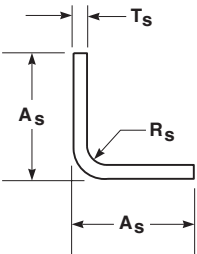
OPEN LEG ANGLE

Use Figure 11 for open leg angle.

MISCELLANEOUS SHAPES

Use Figure 12, Sheet 1, Figure 12, Sheet 2, Figure 12, Sheet 3, Figure 12, Sheet 4, and Figure 12, Sheet 5 for miscellaneous shapes.

MISCELLANEOUS SHAPES - Continued

EXTRUSION	SUBSTITUTION
	

EXTRUSION				SUBSTITUTION		
DIE NUMBER	MATERIAL	A_e	T_e	A_s	T_s	R_s
AND10133-0601	7075-T6511	0.75	0.063	0.750	0.071	7/32
AND10133-0701	7075-T6511	0.875	0.063	0.875	0.071	7/32
AND10133-0702	7075-T6	0.875	0.094	0.875	0.125	3/8
AND10133-1001	7075-T6	1.000	0.063	1.000	0.071	7/32
AND10133-1002	7075-T6511	1.000	0.094	1.000	0.125	3/8
AND10133-1003*	7075-T6	1.000	0.125	1.000	0.125	0.188**
AND10133-1201	7075-T6	1.250	0.063	1.250	0.071	7/32
AND10133-1202	7075-T6	1.250	0.094	1.250	0.125	3/8
AND10133-1401	7075-T6511	1.500	0.063	1.500	0.071	7/32
AND10133-1404	7075-T6	1.500	0.188	MACHINE FROM STOCK		
ALCOA 77U*	7075-T6	1.750	0.188	1.750	0.188	0.188**
ALCOA 78F	7075-T6	1.000	0.094	1.000	0.125	3/8

AA0053.1
SA

Figure 2. Equal Leg Angle. (Sheet 1 of 2)

MISCELLANEOUS SHAPES - Continued

(CONT)

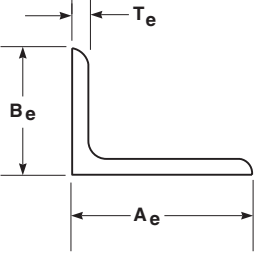
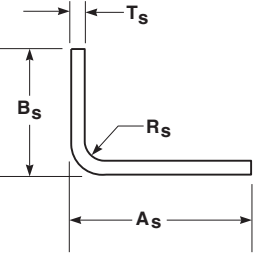
EXTRUSION				SUBSTITUTION		
DIE NUMBER	MATERIAL	A _e	T _e	A _s	T _s	R _s
ALCOA 78J	7075-T6511	1.000	0.062	1.000	0.071	7/32
ALCOA 78L	7075-T6511	1.750	0.094	1.750	0.125	3/8
ALCOA 79CC	7075-T6	1.000	0.094	1.000	0.125	3/8
ALCOA 5401	7075-T6511	0.750	0.062	0.750	0.071	7/32
ALCOA 9802	7075-T6511	0.750	0.050	0.750	0.063	3/16
ALCOA 10854	7075-T6511	0.625	0.040	0.625	0.050	5/32
ALCOA 22477	7075-T6511	0.750	0.040	0.750	0.050	5/32
ALCOA 23299	7075-T6511	0.875	0.063	0.875	0.071	7/32
ALCOA 26290*	7075-T6511	2.000	0.156	2.000	0.156	0.156**
ALCOA 30845	7075-T6	1.000	0.076	1.000	0.090	9/32
ALCOA 32059	7075-T6	0.875	0.081	0.875	0.090	9/32
ALCOA 44076	7075-T6511	0.750	0.050	0.750	0.063	3/16
ALCOA 59409	7075-T6	1.000	0.050	1.000	0.063	3/16
ALCOA 67580	7075-T6511	0.750	0.050	0.750	0.063	3/16
ALCOA 71369	7075-T6	1.000	0.072	1.000	0.090	9/32
ALCOA 82300	7075-T6511	1.000	0.062	1.000	0.071	7/32
HARVEY 22264	7075-T6511	1.000	0.080	1.000	0.090	9/32
HARVEY 50276	7075-T6	1.000	0.040	1.000	0.050	5/32
REYNOLDS 19237	7075-T6511	0.875	0.050	0.875	0.063	3/16
REYNOLDS 22647	7075-T6511	0.875	0.040	0.875	0.050	5/32
REYNOLDS 23888	7075-T6511	1.000	0.090	1.000	0.100	5/16
TIERNAY 60-4645	7075-T6511	1.000	0.100	1.000	0.125	3/8

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

Figure 2. Equal Leg Angle. (Sheet 2 of 2)

MISCELLANEOUS SHAPES - Continued

EXTRUSION	SUBSTITUTION
	

EXTRUSION					SUBSTITUTION			
DIE NUMBER	MATERIAL	A _e	B _e	T _e	A _s	B _s	T _s	R _s
AND10134-0501	7075-T6511	0.625	0.500	0.063	0.625	0.500	0.071	7/32
AND10134-0702	7075-T6511	0.875	0.750	0.063	0.875	0.750	0.071	7/32
AND10134-1001	7075-T6	1.000	0.625	0.063	1.000	0.625	0.071	7/32
AND10134-1002*	7075-T6	1.000	0.625	0.125	1.000	0.625	0.063	0.125**
AND10134-1003	7075-T6511	1.000	0.750	0.063	1.000	0.750	0.071	7/32
AND10134-1004*	7075-T6	1.000	0.750	0.125	1.000	0.750	0.125	0.125**
AND10134-1005	7075-T6	1.000	0.875	0.063	1.000	0.875	0.071	7/32
AND10134-1006	7075-T6511	1.000	0.875	0.094	1.000	0.875	0.125	3/8
AND10134-1204	7075-T6	1.250	1.000	0.063	1.250	1.000	0.071	7/32
AND10134-1205	7075-T6511	1.250	1.000	0.094	1.250	1.000	0.125	3/8
AND10134-1206*	7075-T6	1.250	1.000	0.125	1.250	1.000	0.125	0.125**
AND10134-1401	7075-T6	1.500	0.750	0.094	1.500	0.750	0.125	3/8
AND10134-1407	7075-T6511	1.500	1.250	0.125	NO SUBSTITUTION			
AND10134-1604	7075-T6511	1.750	1.500	0.156	NO SUBSTITUTION			
AND10134-2402	7075-T6511	2.500	1.250	0.188	NO SUBSTITUTION			

AA0054 1
SA

Figure 3. Unequal Leg Angle. (Sheet 1 of 3)

MISCELLANEOUS SHAPES - Continued

EXTRUSION					SUBSTITUTION			
DIE NUMBER	MATERIAL	A _e	B _e	T _e	A _s	B _s	T _s	R _s
ALCOA 1288	7075-T6	1.000	0.750	0.070	1.000	0.750	0.080	1/4
ALCOA 6494	7075-T6511	1.750	1.125	0.064	1.750	1.125	0.080	1/4
ALCOA 9413	7075-T6	0.750	2.000	0.125	0.750	2.000	0.125	3/8
ALCOA 10188	7075-T6	2.500	1.250	0.062	1.500	1.250	0.071	7/32
ALCOA 11472	7075-T6	1.500	1.000	0.062	1.500	1.000	0.071	7/32
ALCOA 12888	7075-T6511	0.750	0.625	0.051	0.750	0.625	0.063	3/16
ALCOA 13633	7075-T6511	4.000	2.500	0.125	MACHINE FROM STOCK			
ALCOA 14263	7075-T6511	1.750	1.000	0.094	1.750	1.000	0.125	3/8
ALCOA 22385	7075-T6	1.000	0.750	0.062	1.000	0.750	0.071	7/32
ALCOA 22509	7075-T6511	1.000	0.750	0.040	1.000	0.750	0.050	5/32
ALCOA 22526	7075-T6	1.000	0.750	0.050	1.000	0.750	0.063	3/32
ALCOA 22580	7075-T6	1.250	1.125	0.062	1.250	1.125	0.071	7/32
ALCOA 22878	7075-T6511	0.875	0.750	0.062	0.875	0.750	0.071	7/32
ALCOA 22885	7075-T6511	1.000	0.750	0.062	1.000	0.750	0.071	7/32
ALCOA 22990	7075-T6	1.500	0.750	0.062	1.500	0.750	0.071	7/32
ALCOA 24262	7075-T6	0.875	0.625	0.051	0.875	0.625	0.063	3/32
ALCOA 24920	7075-T6	1.250	1.000	0.051	1.250	1.000	0.063	3/32
ALCOA 27076	7075-T6511	1.250	0.750	0.062	1.250	0.750	0.071	7/32
ALCOA 44562	7075-T6	0.750	0.500	0.050	0.750	0.500	0.063	3/16
ALCOA 46675	7075-T6511	1.250	1.000	0.080	1.250	1.000	0.090	9/32
ALCOA 52106	7075-T6	0.750	0.680	0.040	0.750	0.680	0.050	5/32
ALCOA 52147	7075-T6511	1.250	1.000	0.109	1.250	1.000	0.125	3/8
ALCOA 52421		1.000	1.250	0.120	NO SUBSTITUTION			
ALCOA 52735	7075-T6	1.250	0.750	0.051	1.250	0.750	0.063	3/16
ALCOA 57238	7075-T6511	3.000	1.250	0.125	3.000	1.250	0.125	3/8
ALCOA 59128	7075-T6511	1.120	1.000	0.094	1.120	1.000	0.125	3/8
ALCOA 59194	7075-T6	1.000	0.800	0.050	1.000	0.800	0.063	3/16
ALCOA 59195	7075-T6511	1.750	1.420	0.064	1.750	1.420	0.071	7/32
ALCOA 59342	7075-T6511	1.000	0.830	0.060	1.000	0.830	0.071	7/32
ALCOA 60242	7075-T6	1.700	1.000	0.070	1.700	1.000	0.080	1/4
ALCOA 61992	7075-T6	1.800	3.000	0.062	1.800	3.000	0.071	7/32

AA0054_2
SA

Figure 3. Unequal Leg Angle. (Sheet 2 of 3)

MISCELLANEOUS SHAPES - Continued

EXTRUSION					SUBSTITUTION			
DIE NUMBER	MATERIAL	A _e	B _e	T _e	A _s	B _s	T _s	R _s
ALCOA 66084	7075-T6511	1.000	0.830	0.050	1.000	0.830	0.063	3/16
ALCOA 78849	7075-T6	0.870	0.750	0.040	0.870	0.750	0.050	5/32
ALCOA 80415	7075-T6	1.812	0.938	0.062	1.812	0.938	0.071	7/32
ALCOA 91655	7075-T6	2.500	1.000	0.078	2.500	1.000	0.090	9/32
ALCOA 91655	2024-T3511	2.500	1.000	0.078	2.500	1.000	0.090	9/32
ALCOA 141532	7075-T6	2.500	1.062	0.125	NO SUBSTITUTION			
HARVEY 14518	7075-T6511	1.000	0.750	0.078	1.000	0.750	0.090	9/32
HARVEY 15078	7075-T6	0.875	0.750	0.051	0.875	0.750	0.063	3/16
HARVEY 27407	7075-T6511	0.800	0.750	0.062	0.800	0.750	0.071	7/32
PIONEER 4191	7075-T6	1.000	1.180	0.050	1.000	1.180	0.063	3/16
REYNOLDS 9204	7075-T6	1.125	1.000	0.062	1.125	1.000	0.071	7/32
REYNOLDS 16381	7075-T6511	0.875	0.750	0.051	0.875	0.750	0.063	3/16
REYNOLDS 18275	7075-T6511	2.625	0.875	0.125	NO SUBSTITUTION			
REYNOLDS 19767	7075-T6511	2.500	1.250	0.094	2.500	1.250	0.125	3/8
S6105-2012-2	7075-T6	1.562	0.750	0.062	1.562	0.750	0.071	7/32
TIERNAY 60-4645	7075-T6511	1.000	1.000	0.100	1.000	1.000	0.125	3/8

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

Figure 3. Unequal Leg Angle. (Sheet 3 of 3)

MISCELLANEOUS SHAPES - Continued

EXTRUSION	SUBSTITUTION

EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A_e	B_e	T_1	T_2	A_s	B_s	T_1	T_2	R_s
AND10136-1302	7075-T6	1.375	1.000	0.063	0.063	1.375	1.000			
AND10136-1303	7075-T6	1.375	1.000	0.078	0.078	1.375	1.000			
AND10136-1304	7075-T6	1.375	1.000	0.094	0.094	1.375	1.000			
AND10136-1401	7075-T6	1.500	0.750	0.063	0.063	1.500	0.750			
AND10136-1405	7075-T6511	1.500	1.250	0.078	0.078	1.500	1.250			
AND10136-1407	7075-T6511	1.500	1.250	0.125	0.125		MACHINE FROM STOCK			
AND10136-1504	7075-T6	1.625	1.125	0.125	0.125	1.625	1.125	0.125	0.125	3/8
AND10136-1603	7075-T6511	1.750	1.125	0.094	0.094	1.750	1.125	0.090	0.090	9/32
AND10136-2001	7075-T6511	2.000	1.000	0.078	0.078	2.000	1.000	0.080	0.080	1/4
AND10136-2002	7075-T6	2.000	1.000	0.094	0.094	2.000	1.000	0.090	0.090	9/32
AND10136-2003	7075-T6	2.000	1.000	0.125	0.125	2.000	1.000	0.125	0.125	3/8
AND10136-2004	7075-T6511	2.000	1.250	0.078	0.078	2.000	1.250	0.080	0.080	1/4
AND10136-2005	7075-T6511	2.000	1.250	0.094	0.094	2.000	1.250	0.090	0.090	9/32
AND10136-2006	7075-T6	2.000	1.250	0.125	0.125	2.000	1.250	0.125	0.125	3/8
AND10136-2008	7075-T6	2.000	1.750	0.125	0.125	2.000	1.750	0.125	0.125	3/8
AND10136-2402	7075-T6	2.500	1.250	0.125	0.125	2.500	1.250	0.125	0.125	3/8

AA0055_1
SA

Figure 4. Tee Section. (Sheet 1 of 3)

MISCELLANEOUS SHAPES - Continued

EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	T ₁	T ₂	A _s	B _s	T ₁	T ₂	R _s
AND10136-2404	7075-T6	2.500	1.625	0.094	0.094	2.500	1.625	0.090	0.090	9/32
AND10136-2407	7075-T6511	2.000	1.756	0.094	0.094	MACHINE FROM STOCK				
AND10136-2408	7075-T6511	2.500	2.000	0.156	0.156	MACHINE FROM STOCK				
AND10136-2409	7075-T6	2.500	2.000	0.188	0.188	MACHINE FROM STOCK				
ALCOA 3094	7075-T6	1.250	1.550	0.050	0.050	1.250	1.550	0.050	0.050	5/32
ALCOA 11485	7075-T6511	2.000	3.000	0.093	0.093	2.000	3.000	0.090	0.090	9/32
ALCOA 13630	7075-T6511	1.937	3.250	0.125	0.125	1.937	3.250	0.125	0.125	3/8
ALCOA 16136	7075-T6	1.500	0.750	0.062	0.062	1.500	0.750	0.063	0.063	5/32
ALCOA 22675	7075-T6	2.375	2.625	0.080	0.080	2.375	2.625	0.080	0.080	1/4
ALCOA 22899	7075-T6511	1.875	1.000	0.050	0.050	1.875	1.000	0.050	0.050	5/32
ALCOA 22982	7075-T6	1.500	0.812	0.050	0.050	1.500	0.812	0.050	0.050	5/32
ALCOA 27427	7075-T6511	1.750	3.500	0.156	0.125	NO SUBSTITUTION				
ALCOA 28957	7075-T6	1.500	0.750	0.050	0.050	1.500	0.750	0.050	0.050	1/8
ALCOA 29119	7075-T6	2.000	2.000	0.094	0.094	2.000	2.000	0.090	0.090	9/32
ALCOA 31510	7075-T6	2.000	2.562	0.125	0.125	2.000	2.562	0.125	0.125	3/8
ALCOA 33481	7075-T6	2.250	1.750	0.062	0.062	2.250	1.750	0.063	0.063	5/32
ALCOA 39689	7075-T6511	1.750	2.500	0.091	0.091	1.750	2.500	0.090	0.090	9/32
ALCOA 43025*	7075-T6511	2.600	3.850	0.380	0.190	2.600	3.850	0.380	0.190	0.120**
ALCOA 43874	7075-T6511	2.500	3.500	0.250	0.125	NO SUBSTITUTION				
ALCOA 44103*	7075-T6511	2.625	4.562	0.500	0.125	2.625	4.562	0.500	0.125	0.250**
ALCOA 44282	7075-T6	2.140	3.700	0.100	0.100	2.140	3.700	0.100	0.100	5/16
ALCOA 44532	7075-T6511	4.000	2.000	0.250	0.250	MACHINE FROM STOCK				
ALCOA 45180	7075-T6	3.000	1.750	0.094	0.094	3.000	1.750	0.090	0.090	9/32
ALCOA 45277	2024-0	2.680	1.780	0.064	0.064	2.680	1.780	0.063	0.063	3/16
ALCOA 51777	7075-T6	1.562	1.000	0.062	0.062	1.562	1.000	0.063	0.063	3/16
ALCOA 52414	7075-T6	2.000	1.200	0.062	0.062	2.000	1.200	0.063	0.063	3/16
ALCOA 52986	7075-T6	3.700	2.850	0.125	0.125	3.700	2.850	0.125	0.125	3/8
ALCOA 55780	7075-T6	1.760	2.000	0.062	0.062	1.760	2.000	0.063	0.063	3/16
ALCOA 55919	7075-T6511	3.240	1.460	0.078	0.078	3.240	1.460	0.080	0.080	1/4
ALCOA 55956	7075-T6511	2.800	2.200	0.078	0.078	2.800	2.200	0.080	0.080	1/4
ALCOA 59051	7075-T6511	1.500	1.125	0.050	0.050	1.500	1.125	0.050	0.050	5/32
ALCOA 59223	7075-T6	1.800	3.250	0.080	0.080	1.800	3.250	0.080	0.080	1/4

AA0055_2
SA

Figure 4. Tee Section. (Sheet 2 of 3)

MISCELLANEOUS SHAPES - Continued

EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	T ₁	T ₂	A _s	B _s	T ₁	T ₂	R _s
ALCOA 59862	7075-T6	1.500	1.500	0.062	0.062	1.500	1.500	0.063	0.063	3/16
ALCOA 60038	7075-T6	1.750	1.750	0.125	0.125	1.750	1.750	0.125	0.125	3/8
ALCOA 61633	7075-T6	1.260	2.250	0.060	0.060	1.260	2.250	0.063	0.063	3/16
ALCOA 66336	7075-T6511	2.120	3.000	0.125	0.125	2.120	3.000	0.125	0.125	3/8
ALCOA 69162	7075-T6511	2.600	2.000	0.100	0.100	2.600	2.000	0.100	0.100	5/16
ALCOA 71132	7075-T6	2.600	2.030	0.080	0.080	2.600	2.030	0.080	0.080	1/4
ALCOA 72221	7075-T6511	1.780	2.875	0.078	0.078	1.780	2.875	0.080	0.080	1/4
ALCOA 76759	2024-T3511	1.500	0.800	0.040	0.040	1.500	0.800	0.040	0.040	1/8
ALCOA 80760	7075-T6511	2.250	1.000	0.050	0.050	2.250	1.000	0.050	0.050	5/32
ALCOA 85740	7075-T6511	3.000	3.250	0.090	0.090	3.000	3.250	0.090	0.090	9/32
ALCOA 156895	7075-T6	2.000	0.750	0.040	0.040	2.000	0.750	0.040	0.040	1/8
HARVEY 14712	7075-T6511	1.750	1.750	0.050	0.050	1.750	1.750	0.050	0.050	5/32
HARVEY 14678	7075-T6511	2.000	3.000	0.125	0.125	2.000	3.000	0.125	0.125	3/8
HARVEY 18648	7075-T6511	2.000	2.600	0.188	0.100	MACHINE FROM STOCK				
HARVEY 21423	7075-T6511	1.780	3.125	0.094	0.094	1.780	3.125	0.090	0.090	9/32
HARVEY 21701	7075-T6	2.000	1.100	0.050	0.050	2.000	1.100	0.050	0.050	5/32
HARVEY 22398	7075-T6511	4.000	1.500	0.062	0.062	4.000	1.500	0.063	0.063	3/16
PIONEER 6568	7075-T6511	1.500	1.000	0.062	0.062	1.500	1.000	0.063	0.063	3/16
REYNOLDS 11416	7075-T6511	2.000	1.000	0.062	0.062	2.000	1.000	0.063	0.063	3/16
REYNOLDS 16917	7075-0	9.943	1.250	0.063	0.063	9.943	1.250	0.063	0.063	3/16
REYNOLDS 11731	7075-T6511	2.000	3.000	0.125	0.100	2.000	3.000	0.125	0.100	3/8
REYNOLDS 19274	7075-T6	1.500	2.500	0.062	0.062	1.500	2.500	0.063	0.063	3/16
REYNOLDS 19446	7075-T6	1.700	2.250	0.062	0.062	1.700	2.250	0.063	0.063	3/16
REYNOLDS 23203	7075-T6	2.000	1.000	0.062	0.062	2.000	1.000	0.063	0.063	3/16
REYNOLDS 23243	7075-T6511	1.500	2.500	0.125	0.102	1.500	2.500	0.125	0.125	9/32
REYNOLDS 39689	7075-T6511	1.750	2.500	0.091	0.091	1.750	2.500	0.090	0.090	9/32
TIERNAY 60-2617	7075-T6511	1.500	1.000	0.050	0.050	1.500	1.000	0.050	0.050	5/32
TIERNAY 60-4394	7075-T6511	2.450	1.450	0.050	0.050	2.450	1.450	0.050	0.050	5/32
TIERNAY 60-5243	7075-T6511	1.500	2.125	0.125	0.125	1.500	2.125	0.125	0.125	3/8
TIERNAY 60-6081	7075-T6511	1.500	3.375	0.125	0.062	1.500	3.375	0.125	0.125	3/8

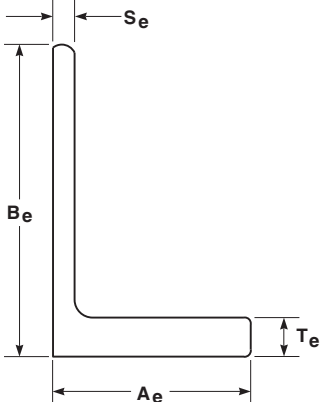
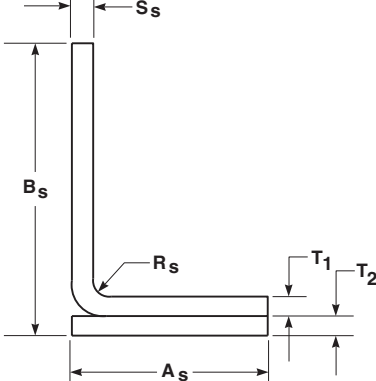
* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

AK5004_3
SA

Figure 4. Tee Section. (Sheet 3 of 3)

MISCELLANEOUS SHAPES - Continued

EXTRUSION	SUBSTITUTION
	

EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	S _e	T _e	A _s	B _s	T ₁	T ₂	R _s
ALCOA 14800	7075-T6511	1.625	3.250	0.094	0.188	1.000	MACHINE FROM STOCK			3/8
ALCOA 27159	7075-T6511	1.000	2.500	0.094	0.156		2.500	0.125	0.063	
ALCOA 30737	7075-T6511	2.312	4.094	0.125	0.188		MACHINE FROM STOCK			
ALCOA 44350	7075-T6	1.625	3.250	0.094	0.188		MACHINE FROM STOCK			
ALCOA 46202	7075-T6	2.000	1.500	0.375	0.312	1.000	A MACHINE FROM STOCK			3/8
ALCOA 52301	7075-T6511	1.000	2.500	0.094	0.125		2.500	0.125	0.050	
ALCOA 76145	7075-T6	1.250	1.400	0.080	0.150		1.400	0.100	0.100	
REYNOLDS 7825	7075-T6511	1.250	1.125	0.188	0.125		A MACHINE FROM STOCK			
REYNOLDS 8334*	7075-T6511	1.500	1.750	0.125	0.188	1.500	1.750	0.188	0.125	0.125**
REYNOLDS 8334*	7075-T2024-T4	1.500	1.750	0.125	0.188	1.500	1.750	0.188	0.125	0.125**
REYNOLDS 17676*	7075-T6	2.000	1.500	0.375	0.312	MACHINE FROM STOCK				

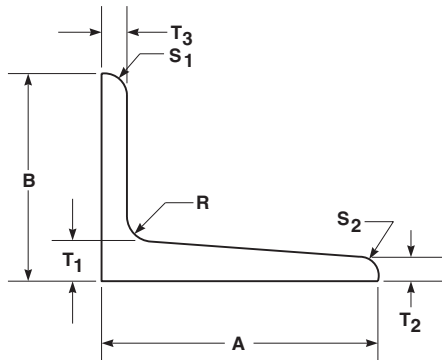
* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

AA0056
SA

Figure 5. Unequal Thickness Angle.

MISCELLANEOUS SHAPES - Continued

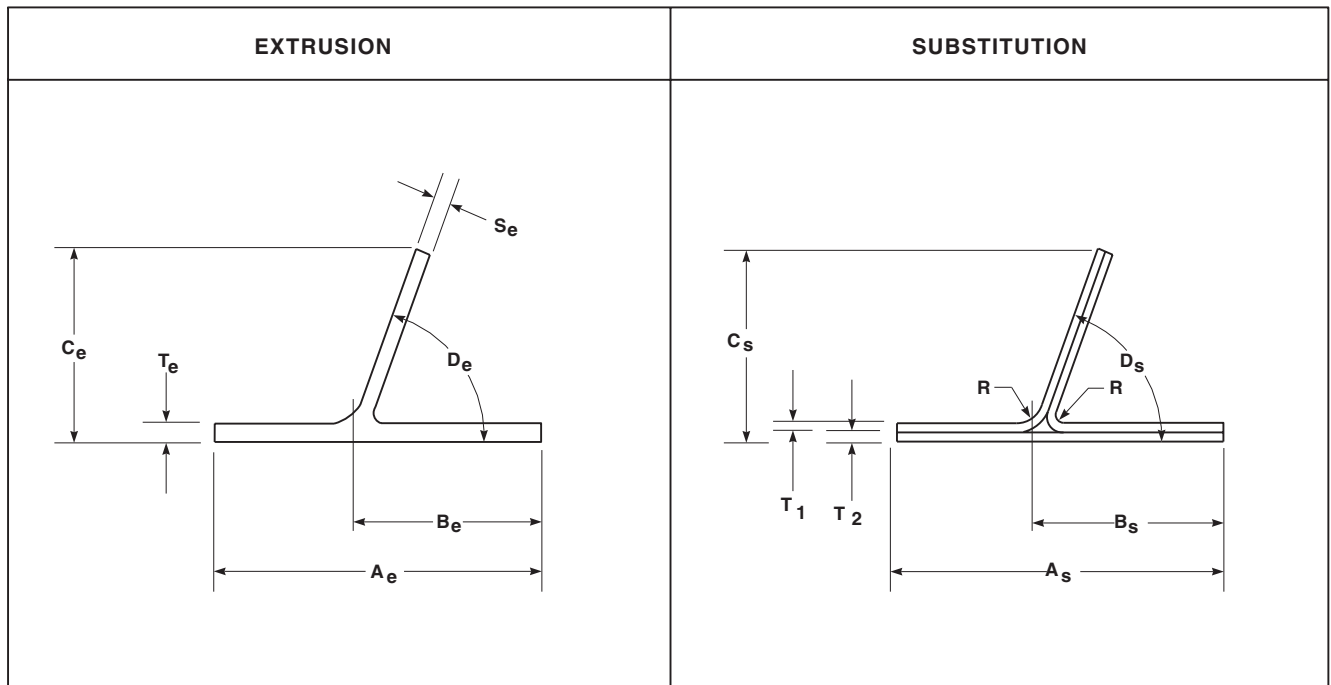
EXTRUSION	SUBSTITUTION
	MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL

EXTRUSION						SUBSTITUTION			
DIE NUMBER	MATERIAL	A	B	T ₁	T ₂	T ₃	R	S ₁	S ₂
ALCOA 14030	7075-T6	2.000	1.500	0.078	0.047	0.100	0.125	0.100	0.047
ALCOA 14030	2024-T4	2.000	1.500	0.078	0.047	0.100	0.125	0.100	0.047
ALCOA 22384	7075-T6	3.125	1.125	0.188	0.078	0.188	0.188		
ALCOA 24698	7075-T6	3.125	1.062	0.288	0.094	0.312	0.250	0.031	0.031
ALCOA 31928	7075-T6	6.500	1.625	0.375	0.094	0.500	0.500	0.062	0.062
ALCOA 60169	7075-T6511	2.000	1.125	0.125	0.062	0.125	0.125	0.031	0.031
ALCOA 76114	7075-T6	2.800	1.250	0.180	0.080	0.300	0.180	0.031	0.031

AA0057
SA

Figure 6. Tapered Leg Angle.

MISCELLANEOUS SHAPES - Continued

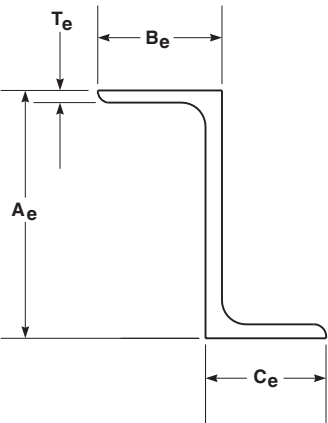
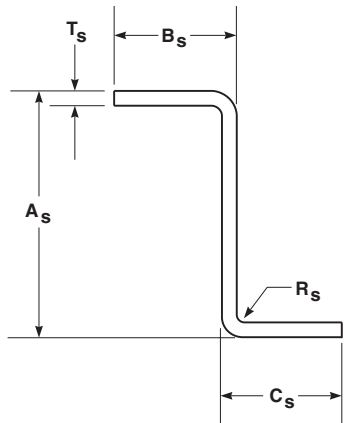


EXTRUSION								SUBSTITUTION						
DIE NUMBER	MATERIAL	A _e	B _e	C _e	D _e	T _e	S _e	A _s	B _s	C _s	D _s	T ₁	T ₂	R _s
ALCOA 44271	7075-T6511	2.000	1.000	1.121	85°	0.125	0.125	2.000	1.000	1.121	85°			
ALCOA 76941	7075-T6511	2.290	1.274	2.498	88°33'	0.064	0.064	2.290	1.274	2.498	88°33'	0.071	0.063	7/32
ALCOA 93560	2024-T4	2.250	1.125	1.997	94°30'	0.062	0.062	2.250	1.125	1.997	94°30'	0.071	0.071	7/32
PIONEER 11032	7075-T6511	1.500	0.750	1.994	95°	0.094	0.094	1.500	0.750	1.994	95°	0.125	0.125	3/8
SIKORSKY 38010-70026	7075-T6511	2.000	1.025	0.920	48°42'	0.050	0.050	NO SUBSTITUTION						

AA0058
SA

Figure 7. Closed Angle Tee.

MISCELLANEOUS SHAPES - Continued

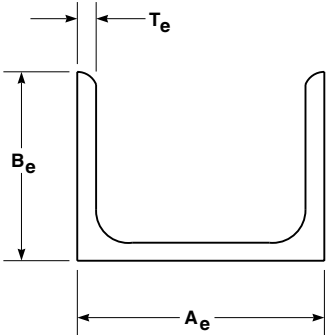
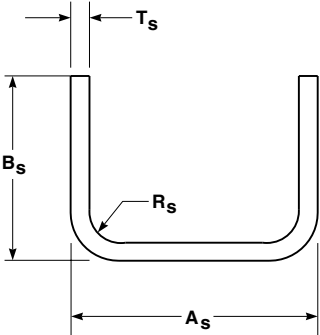
EXTRUSION	SUBSTITUTION
	

EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	C _e	T _e	A _s	B _s	C _s	T _s	R _s
AND 10138-1004	7075-T6511	1.000	0.750	0.750	0.063	1.000	0.750	0.750	0.071	7/32
AND 10138-1402	7075-T6511	1.500	1.000	1.000	0.078	1.500	1.000	1.000	0.090	9/32
AND 10139-1205	7075-T6511	1.250	0.750	1.250	0.078	1.250	0.750	1.250	0.090	9/32
ALCOA 3172	7075-T6511	0.563	0.875	1.500	0.125	MACHINE FROM STOCK				
REYNOLDS 18226	7075-T6511	0.354	0.880	0.500	0.064	0.354	0.880	0.500	0.071	7/32
REYNOLDS 38973	2024-T3511	0.922	0.625	0.625	0.040	0.922	0.625	0.625	0.050	5/32

AA0059
SA

Figure 8. Zee Section.

MISCELLANEOUS SHAPES - Continued

EXTRUSION	SUBSTITUTION
	

EXTRUSION					SUBSTITUTION			
DIE NUMBER	MATERIAL	A _e	B _e	T _e	A _s	B _s	T _s	R _s
AND10137-1012*	7075-T6	1.000	1.250	0.125	1.000	1.250	0.125	0.125**
SIKORSKY 38010-70025	7075-T6511	0.587	0.937	0.050	0.587	0.937	0.063	3/16

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINED SUBSTITUTE RADIUS.

AK5005
SA

Figure 9. Channel.

MISCELLANEOUS SHAPES - Continued

EXTRUSION	SUBSTITUTION

EXTRUSION							SUBSTITUTION					
DIE NUMBER	MATERIAL	A _e	B _e	C _e	T ₁	T ₂	A _s	B _s	C _s	T ₁	T ₂	R _s
38010-70029-001*	7075-T6511	1.300	1.300	80°30'	0.125	1.125	1.30	1.300	88°30'	0.125	120.125	0.156**
HARVEY 16161*	7075-T6511	1.880	1.000	88°30'	0.090	0.188	1.880	1.000	88°30'	0.090	0.188	0.156**
REYNOLDS 19234	7075-T6511	1.000	1.000	84°	0.076	0.076	1.000	1.000	84°	0.090	0.090	9/32

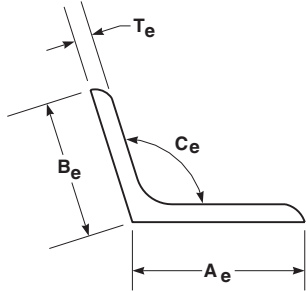
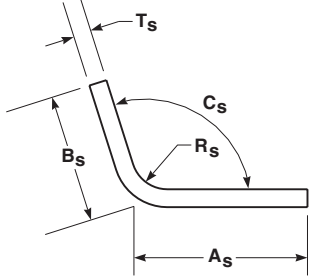
* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

** MACHINE SUBSTITUTE RADIUS.

AA0061
SA

Figure 10. Closed Leg Angle.

MISCELLANEOUS SHAPES - Continued

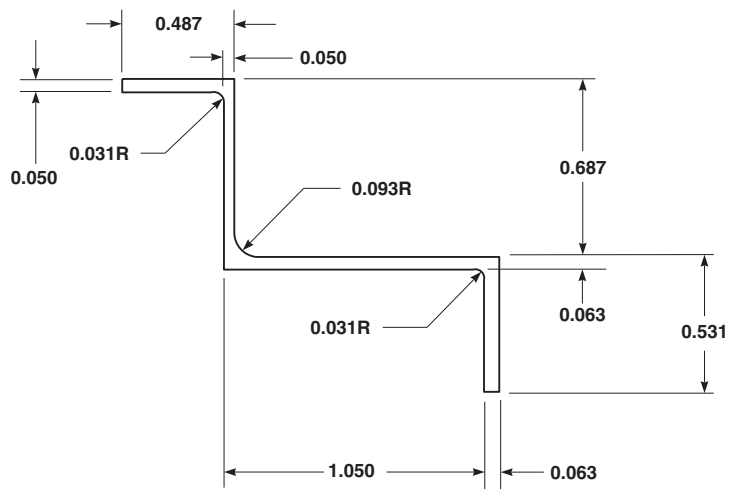
EXTRUSION	SUBSTITUTION
	

EXTRUSION						SUBSTITUTION				
DIE NUMBER	MATERIAL	A _e	B _e	C _e	T _e	A _s	B _s	C _s	T _s	R
REYNOLDS 11465	7075-T6	0.950	0.840	95°30'	0.050	0.950	0.840	95°30'	0.063	3/16
REYNOLDS 19272	7075-T6511	1.750	1.250	115°	0.125	2.000	MACHINE FROM STOCK			3/8
SIKORSKY 38010-70014	7075-T6511	2.500	1.750	109°	0.125		A MACHINE FROM STOCK			
SIKORSKY 38010-70020	7075-T6511	2.000	1.250	100°45'	0.094		1.250	100°45'	0.125	
SIKORSKY 38010-70021	7075-T6511	1.094	1.688	132°	0.153	A MACHINE FROM STOCK				

AA0062
SA

Figure 11. Open Leg Angle.

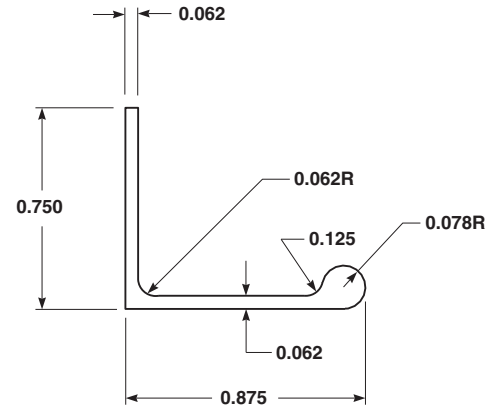
MISCELLANEOUS SHAPES - Continued



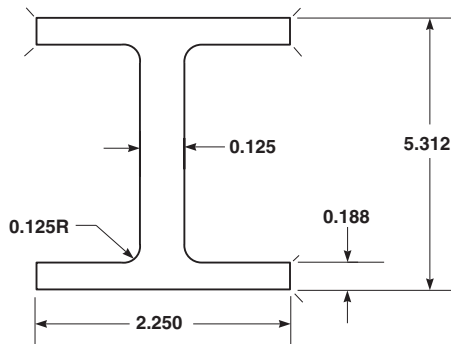
EFFECTIVITY

UH60A 82-23671 - SUBQ

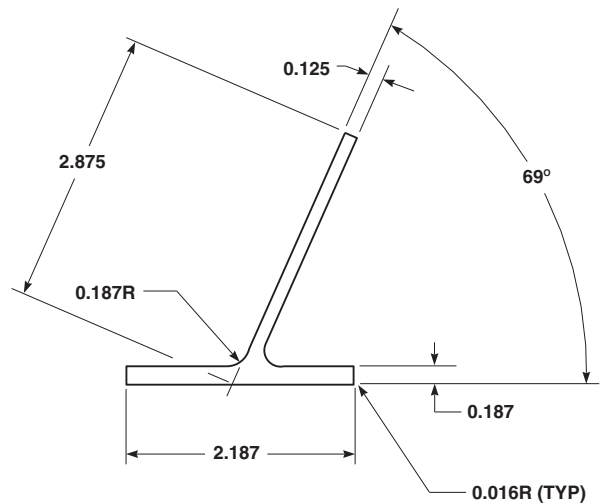
38010-70045
UPPER DOOR TRACK
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



ALCOA 158245
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



REYNOLDS 6494
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL

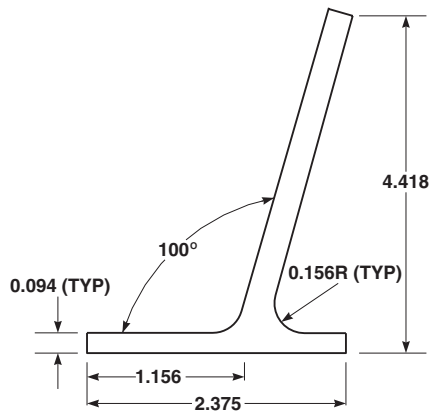


38010-70010
SPECIFICATIONS QQ-A-200-31
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL

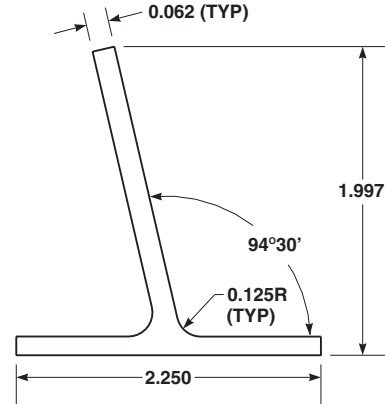
AA0063_2B
SA

Figure 12. Miscellaneous Shapes. (Sheet 2 of 5)

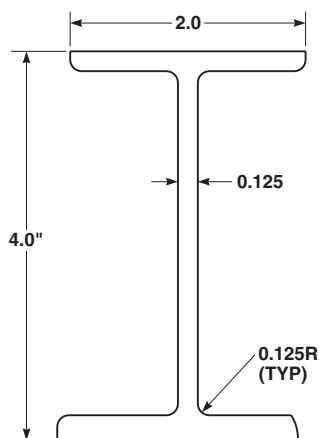
MISCELLANEOUS SHAPES - Continued



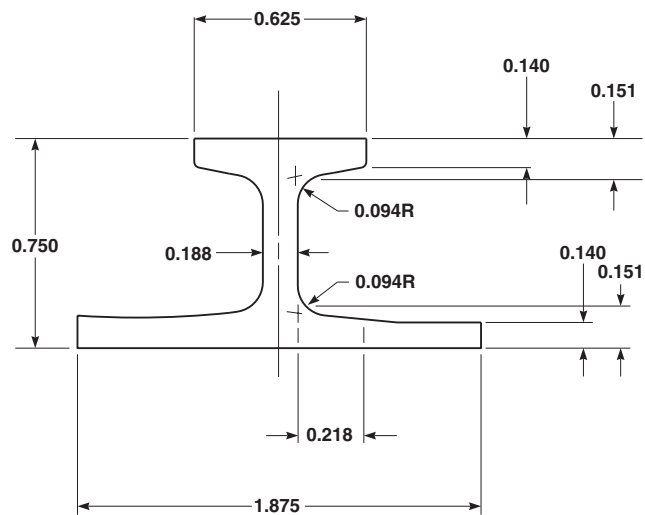
ALCOA 41129
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



ALCOA 93560
SPECIFICATION QQ-A-200-3
MATERIAL 2024-T4
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



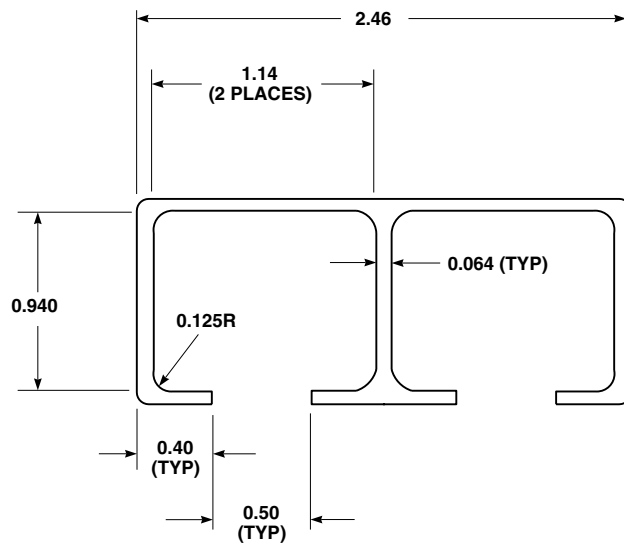
AND10140-4003
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



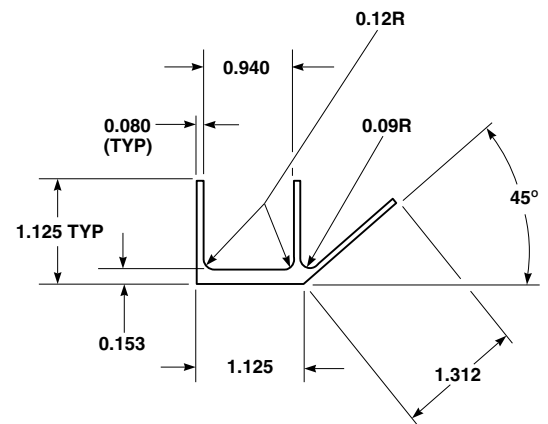
REYNOLDS 34557
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL

Figure 12. Miscellaneous Shapes. (Sheet 3 of 5)

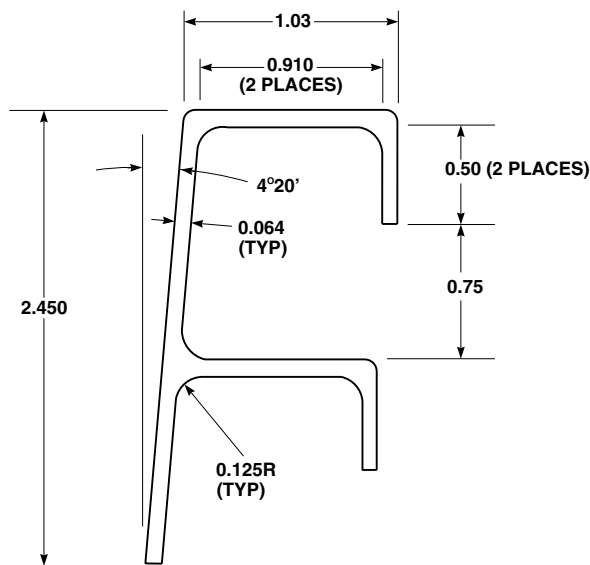
MISCELLANEOUS SHAPES - Continued



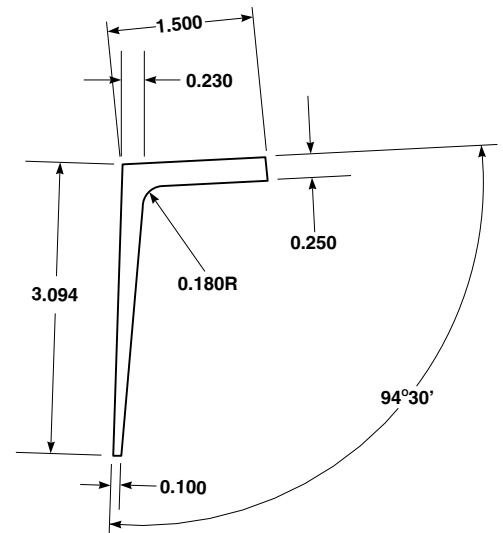
30810-70016
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



30810-70022
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL



38010-70017
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL

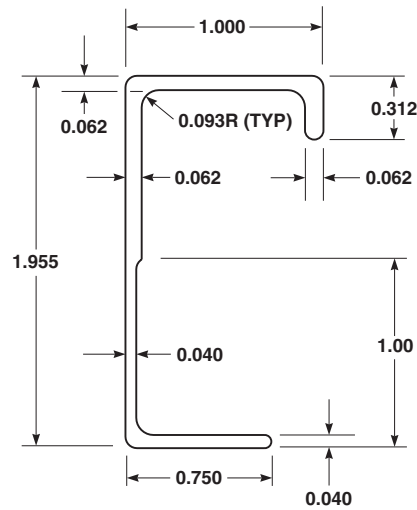


38010-70019
SPECIFICATION QQ-A-200-11
MATERIAL 7075-T6511
MACHINE SUBSTITUTE FROM
BAR STOCK OF SAME MATERIAL

AA0063_4
SA

Figure 12. Miscellaneous Shapes. (Sheet 4 of 5)

MISCELLANEOUS SHAPES - Continued



38010-70001
 SPECIFICATION QQ-A-200-11
 MATERIAL 7075-T6511
 MACHINE SUBSTITUTE FROM
 BAR STOCK OF SAME MATERIAL

EXTRUSION					SUBSTITUTION	
DIE NUMBER	MATERIAL	A	B	C	T ₁	T ₂
70207-02725-001*	7075-T6511	1.218	0.531	0.437	0.050	0.063
70207-02725-002*	7075-T6511	1.294	0.607	0.513	0.16	0.139

* MACHINE SUBSTITUTE FROM BAR STOCK OF SAME MATERIAL.

AA0063 5
 SA

Figure 12. Miscellaneous Shapes. (Sheet 5 of 5)

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****HELICOPTER PAINTED SURFACES TOUCHUP REPAIR**

This WP supersedes WP 0381 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
 Gloves
 Goggles/Face Shield

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cleaning Compound, Aluminum, Item 68,
 WP 1803 00
 Cloth, Abrasive, Item 80, WP 1803 00
 Cloth, Abrasive, Item 81, WP 1803 00
 Cloth, Cheesecloth, Item 85, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Sealing Compound, Item 261, WP 1803 00
 Sealing Compound, Item 287, WP 1803 00

References

MIL-C-22750
 MIL-C-46168
 MIL-C-7439
 MIL-C-83286
 MIL-C-85054
 MIL-L-19538
 MIL-L-46159
 TM 1-1500-344-23
 TM 1-1500-345-23
 TT-L-20
 WP 0377 00
 WP 0576 00
 WP 0585 00
 WP 1803 00

TOUCH UP HELICOPTER PAINTED SURFACES**NOTE**

This information does not apply to touch up repair of the main and tail rotor blades.
 Finishes and materials for rotor blades are covered in WP 0576 00 and WP 0585 00.

1. Touch up protective finishes (metal treatment and paint) whenever finish has been damaged or removed by repair, corrosion, or wear.

CAUTION

Protect areas not to be refinished with covers to prevent damage. Do not allow paint, corrosion removers, solvents, or corrosion resistant coating to contact transparent plastic panels, bonded joints, reinforced plastic surfaces, or rubber seals.

2. Treat bare aluminum, cut or trimmed edges, and drilled holes with corrosion resistant coating, Item 111, WP 1803 00, before applying primer.
3. Seal and prime sheet metal parts before assembly.
4. Refer to TM 1-1500-344-23 and TM 1-1500-345-23, for general surface cleaning, paint and corrosion removal, surface preparation, metal treatment, sealing compounds, and paint application.
5. When touching up composite plastic structures, prepare surfaces as follows:

TOUCH UP HELICOPTER PAINTED SURFACES - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

CAUTION

Technical acetone will damage bonded joints and bond lines of reinforced plastic components. Do not allow technical acetone to run into bonded joints and bond lines or bond may be weakened or separated. Use technical acetone sparingly.

- a. Clean surface with a clean cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
- b. Dry surface with another clean cheesecloth cloth, Item 85, WP 1803 00, by wiping in single strokes then turning and refolding cheesecloth to a fresh side for the next stroke.
- c. Sand surface with abrasive cloth, Item 80, WP 1803 00, and abrasive cloth, Item 81, WP 1803 00.
- d. Make a solution of aluminum cleaning compound, Item 68, WP 1803 00, mixed to a ratio of two parts water to one part cleaner. Use solution at room temperature.
- e. Using a bristle brush or abrasive pad, scrub surface thoroughly with solution.
- f. Thoroughly rinse solution from surface with clear cold tap water.
- g. Dry with filtered compressed air or clean cheesecloth cloth, Item 85, WP 1803 00.
- h. Do a water-break test on cleaned surface (TM 1-1500-344-23).

Table 1. Paint Requirements.

AREA OR DETAILS	HELICOPTER EFFECTIVITY	PAINT COLOR, TYPE, AND SPECIFICATION	NUMBER OF COATS
Interior and Exterior - Overall and all Details	All Serial Nos	Epoxy Primer Coating Kit, Item 120, WP 1803 00	One
Interior Airframe of Tub (STA 162.0 to STA 485.0)	All Serial Nos	Corrosion Preventive Compound, MIL-C-85054 or Clear Polyurethane, MIL-C-83286	One
Exterior Top Coat-Overall	Serial Nos Prior to 82-23755	Olive Drab Acrylic Lacquer, No. 34087, MIL-L-46159	Two
	Serial Nos 82-23756 and Subsequent	Aircraft Green, Aliphatic Polyurethane, MIL-DTL-53039B.	Two
Cabin Interior Top Coat	Serial Nos Prior to 84-23933	Dark Gull Gray, Acrylic Lacquer, No. 36231, TT-L-20	Two

TOUCH UP HELICOPTER PAINTED SURFACES - Continued

Table 1. Paint Requirements. - Continued

AREA OR DETAILS	HELICOPTER EFFECTIVITY	PAINT COLOR, TYPE, AND SPECIFICATION	NUMBER OF COATS
Cockpit Interior Top Coat	Serial Nos 84-23933 and Subsequent	Gray, Aliphatic Polyurethane, MIL-C-46168	Two
	Serial Nos Prior to 84-23933	Instrument Black, Acrylic Lacquer, No. 5157A, MIL-L-46159	Two
Cabin Emergency Handles and Letter Markings	Serial Nos 84-23933 and Subsequent	Black, Aliphatic Polyurethane, No. 37038, MIL-C-46168	Two
	Serial Nos Prior to 84-23933	Orange Yellow Nitrocellous Lacquer, No. 13538, Fed. Spec TT-L-20	One
Oil Level Gage Handle	Serial Nos 84-23933 and Subsequent	Orange Yellow Aliphatic Polyurethane, No. 33538, MIL-C-46168	One
	All Serial Nos	Yellow Epoxy Paint, No. 13655, MIL-C-22750	Two
Cockpit Emergency Handles and Letter Markings, Emergency Exit Markings	Serial Nos Prior to 84-23933	Orange Yellow Nitrocellous Lacquer, No. 33538, Fed. Spec TT-L-20	One
	Serial Nos 84-23933 and Subsequent	Orange Yellow Aliphatic Polyurethane, No. 33538, MIL-C-46168	One
Cockpit Glare Shield Stencil	All Serial Nos	White Epoxy Paint, No. 37875, MIL-C-22750	One
Stencils and Step Markings	Serial Nos Prior to 84-23933	Lusterless Black, Acrylic Lacquer, No. 37038, MIL-L-46159	Two
	Serial Nos 84-23933 and Subsequent	Lusterless Black Aliphatic Polyurethane, No. 37038, MIL-C-46168	Two
Stabilator Tips Danger Markings	Serial Nos Prior to 84-23933	Camouflage Red Acrylic Lacquer, No. 31136, MIL-L-19538	Two
	Serial Nos 84-23933 and Subsequent	Camouflage Red Aliphatic Polyurethane, No. 31136, MIL-C-46168	Two

TOUCH UP HELICOPTER PAINTED SURFACES - Continued

Table 1. Paint Requirements. - Continued

AREA OR DETAILS	HELICOPTER EFFECTIVITY	PAINT COLOR, TYPE, AND SPECIFICATION	NUMBER OF COATS
Engine Inlet Duct and Exhaust Area (NOTE: Black paint is not required on exhaust area with HIRSS module installed)	Serial Nos Prior to 84-23933	Lusterless Black, Epoxy Polyamide, No. 37038, MIL-L-22750	Two
	Serial Nos 84-23933 and Subsequent	Lusterless Black Aliphatic Polyurethane, No. 37038, MIL-C-46168	Two
Oil Cooler Support Installation	All Serial Nos	Lusterless Black Aliphatic Polyurethane, No. 37038, MIL-C-46168	Two
Battery Area	All Serial Nos	Protective Neoprene Coating, No. N700-9, MIL-C-7439	Two
Walkway Areas: Cabin Floor Strips, Steps, and Gunners Window Sill	All Serial Nos	Dark Gull Gray Non-Skid Coating, No. 36231,	Two
Exterior Skin Under Pylon Sliding Cover	Serial Nos Prior to 84-23933	Yellow Acrylic Lacquer, No. 33531, MIL-L-19538	Two
	Serial Nos 84-23933 and Subsequent	Water Dispersible Aliphatic Polyurethane, No. 33446, CARC, MIL-DTL-64159 or Aliphatic Polyurethane, No. 33446, Single Component CARC, MIL-DTL-53039	Two
Internal Mechanical Parts, Control System Linkages and Supports, Bell Cranks, Levers, Servos, Etc.	All Serial Nos	Yellow, MIL-L-46159 or Lusterless Gray, MIL-C-46168, No. 36231, (SEE NOTE)	Two
Transmission and Gear Housings	All Serial Nos	Silver Gloss Epoxy Polyamide Coating, MIL-C-22750, or Gull Gray Aliphatic Polyurethane, No 36231, MIL-C-46168	Two
Main Transmission Oil Level Gage Handle	All Serial Nos	Yellow Epoxy Polyamide, No. 13655, MIL-C-22750	Two

TOUCH UP HELICOPTER PAINTED SURFACES - Continued**Table 1. Paint Requirements. - Continued**

AREA OR DETAILS	HELICOPTER EFFECTIVITY	PAINT COLOR, TYPE, AND SPECIFICATION	NUMBER OF COATS
NOTE Late model helicopter parts and replacement parts are painted gray, MIL-C-46168. These parts and older parts painted yellow, MIL-L-46159, are interchangeable and need not be repainted in the field. Yellow painted parts may be touched up with MIL-C-46168 gray if first primed with epoxy primer coating kit, Item 120, WP 1803 00			

- i. Prime per Table 1.

CAUTION

Interruption of proper airflow can affect engine operation. Do not block any slots or holes with paint in the engine inlet duct or exhaust areas.

NOTE

- Black anodized surfaces in engine inlet duct areas are not to be painted.
- Paint must be applied within eight (8) hours of primer application. If paint is not applied within this time component must be reprimed per step 5. i.
- To prevent a sheen on part surface reduce component B of paint by 10%.

6. Paint per Table 1, as follows:

- a. Mix paint according to manufacturers instructions.
- b. Apply paint to surface.

7. After primer touch up, treat repairs on interior surfaces of the structure below floor levels with corrosion preventive compound, Item 106, WP 1803 00 (WP 0377 00).

TOUCH UP HELICOPTER CONDUCTIVE PAINT

1. Lightly sand repaired area using abrasive cloth, Item 81, WP 1803 00.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

2. Wipe area with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
3. Apply one coat epoxy primer coating kit, Item 120, WP 1803 00. Allow primer to dry at least 1 hour.
4. Apply three coats of sealing compound, Item 287, WP 1803 00, to make 0.003 to 0.004-inch thick coat.
5. Install any grounding attachments.
6. Touch up paint area. Refer to, TOUCHUP HELICOPTER PAINTED SURFACES, in this work package.

CORROSION RESISTANT COATING, MIL-C-5541, CLASS 3**Apply****CAUTION**

Improper surface preparation and application of corrosion resistant coating, Item 111, WP 1803 00, will result in excessive electrical resistance. Excessive electrical resistance will degrade electromagnetic environment (EME) protection. Make sure the following procedures are performed with extreme care.

1. Remove paint and corrosion resistant coating from an area slightly larger than surface to be treated (TM 1-1500-345-23).
2. Clean surface to be treated (TM 1-1500-344-23).

WARNING

- Chemical conversion coating solution is corrosive and will cause serious burns to skin and eyes on contact. Avoid breathing in its mist, dust, or vapor.
 - Wear rubber gloves, rubber apron, and protective goggles, when using corrosion resistant coating.
 - In case of chemical contact, flush exposed areas with water for 15 minutes and get immediate medical attention.
3. Apply corrosion resistant coating, Item 111, WP 1803 00, as follows:
 - a. Mix corrosion resistant coating per manufacturer's instructions.
 - b. Apply corrosion resistant coating solution per manufacturer's instructions.

Inspect

Inspect coated surface for breaks or scratches. **NONE ALLOWED.** If coated surface has breaks or scratches. Refer to, APPLY, in this work package.

TORQUE STRIPE APPLICATION**WARNING**

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
- Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible. Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.

NOTE

- Application of torque stripe is required for all fasteners with critical characteristics of torque or installation, regardless of requirements for safety wire, cotter pins, and other safety devices.
 - Cracking or chipping of torque stripe caused by accidental contact is not reason for reapplication of torque value; however, the torque stripe shall be reapplied.
1. Remove old torque stripe using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
 2. After final required torque value has been obtained, clean areas to be torque striped, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

NOTE

- Torque stripe shall be applied on head end unless inaccessible, then nut end shall be torque striped.
 - If part is subsequently sealed or painted, torque stripe shall be reapplied per step 6.
3. Using sealing compound, Item 261, WP 1803 00, apply torque stripe width about half of fastener diameter (not more than 3/16-inch), extending at least 1/2-inch on base part, or to edge of part which ever is smaller.
 4. Touch up paint as required. Refer to, TOUCH UP HELICOPTER PAINTED SURFACES, in this work package.
 5. Seal parts as required (WP 0377 00).
 6. If required reapply torque stripe as follows:
 - a. Clean areas to be torque striped, using cheesecloth cloth, Item 85, WP 1803 00.
 - b. Repeat step 3.

END OF WORK PACKAGE

UNIT LEVEL
AIRFRAME

NOSE DOOR REPAIR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-R-6855

QQ-A-200

QQ-A-225

QQ-A-250

TM 1-1500-204-23

TM 55-1500-345-23

Material/Parts

Cloth, Abrasive, Item 78, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

Assistant - (1)

WP 0377 00

WP 0378 00

WP 0379 00

WP 0380 00

WP 0381 00

WP 1803 00

References

MIL-I-631

MIL-P-15035

MIL-R-5847

REPAIR NOSE DOOR

1. The nose door is a bonded sandwich-constructed assembly made from aluminum honeycomb core and Kevlar (reinforced plastic) skins. Outboard edges of nose door are graphite reinforced with Kevlar covering. Cockpit cooling ducts are a part of the bonded assembly. Metal components include hinges, angle brackets, and latches (Figure 1, Sheet 1 and Figure 1, Sheet 2).
2. Refer to WP 0379 00 in combination with WP 0378 00 honeycomb repair for evaluating damage and repairing the nose door. Refer also to TM 1-1500-204-23, Sandwich Construction Repair.
3. Repair of graphite is beyond the scope of this manual.
4. Evaluate damage to metal components (WP 0377 00). Negligible damage to hinge fittings is limited to nicks, scratches, and corrosion not over 5% of material thickness at point of damage within 1 inch of hinge line, and 10% of material thickness at point of damage beyond 1 inch of hinge line. Blend and polish damage with fine grit abrasive cloth, Item 78, WP 1803 00, to same finish as surrounding area.
5. Replace aluminum alloy extrusions (WP 0380 00).
6. Seal repaired area for watertightness (WP 0377 00).
7. Touch up paint in repaired area (WP 0381 00 and TM 55-1500-345-23).

REPAIR NOSE DOOR - Continued

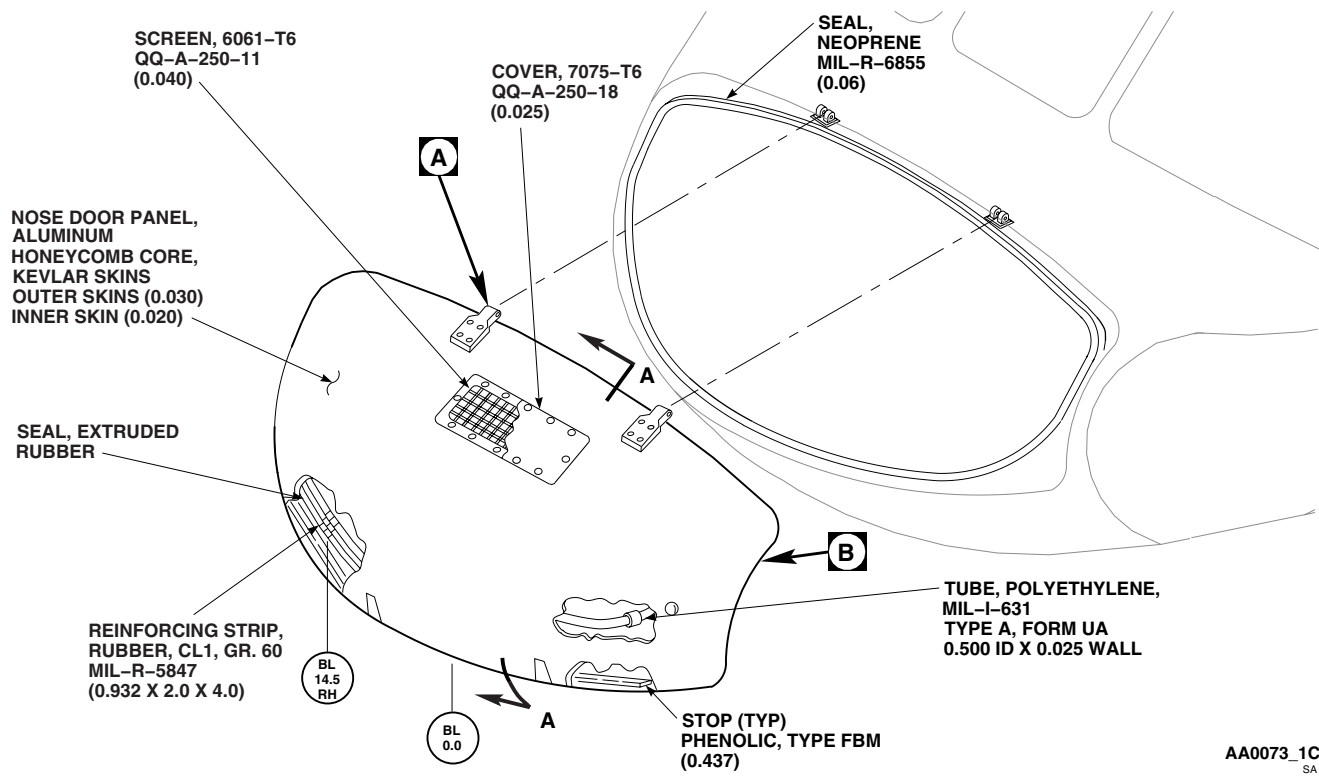
AA0073_1C
SA

Figure 1. Repair Nose Door. (Sheet 1 of 2)

REPAIR NOSE DOOR - Continued

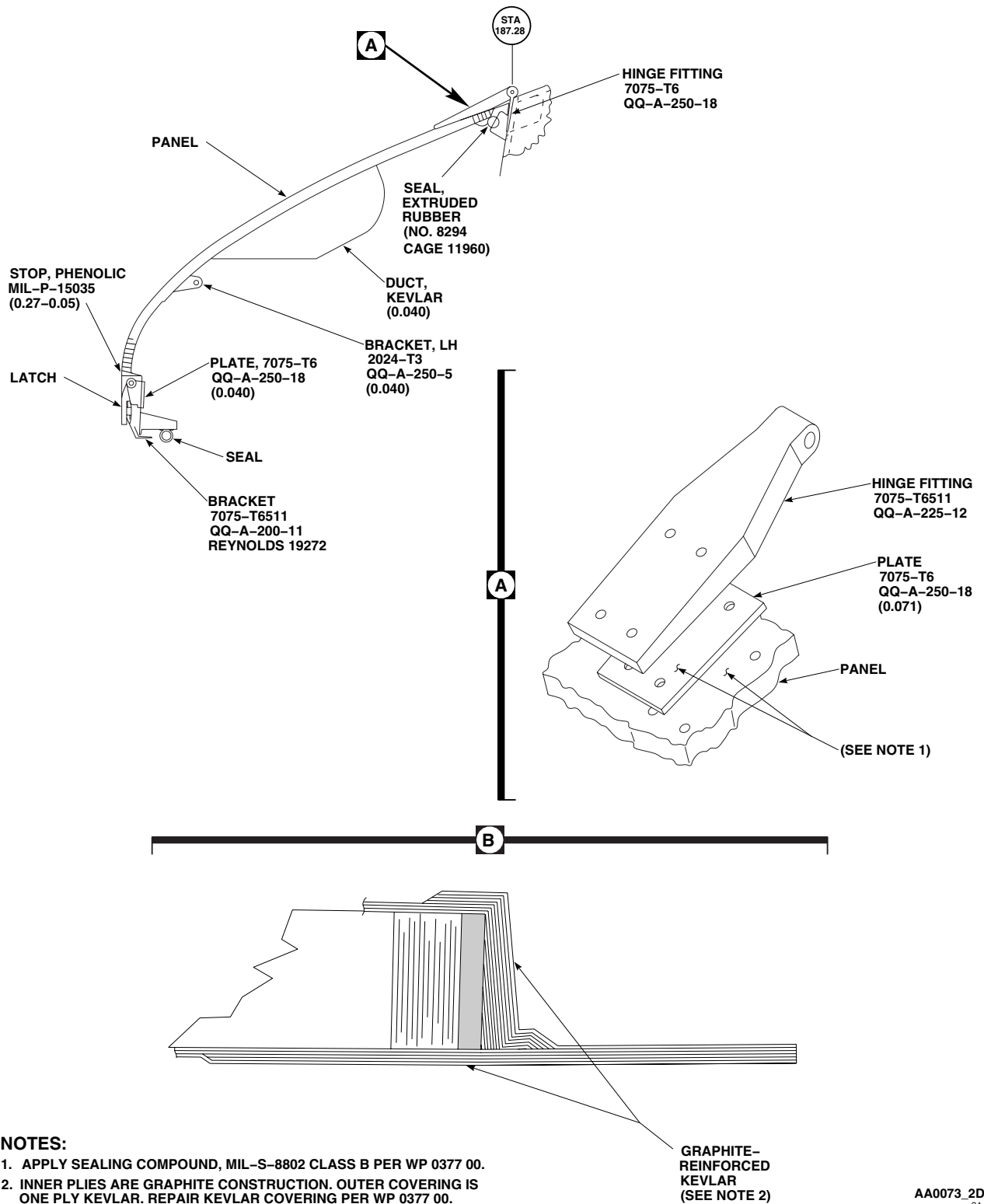


Figure 1. Repair Nose Door. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

COCKPIT DOOR REPAIR

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-P-25690

MIL-S-5059

MIL-S-6758

MIL-STD-670

QQ-A-200

QQ-A-250

QQ-A-367

TM 1-1500-204-23

TM 55-1500-345-23

WP 0377 00

WP 0378 00

WP 0379 00

WP 0380 00

WP 0381 00

WP 1803 00

Material/Parts

Adhesive, Item 14, WP 1803 00

Adhesive, Item 40, WP 1803 00

Bushing, NAS 77-4-19

Primer, Adhesive, Item 237, WP 1803 00

Sealing Compound, Item 273, WP 1803 00

Sealing Compound, Item 287, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

AMS 4260

MIL-P-15035

REPAIR COCKPIT DOOR

1. The bonded door assembly is made of Kevlar (reinforced plastic) framing and skins, with aluminum angles, plates, and channels. Hinge and latch mechanism attachment points are stiffened with Kevlar channels and honeycomb. Door window is made from acrylic sheet plastic. The door seal is extruded rubber bonded to the door with adhesive, Item 14, WP 1803 00. Hook and pile fastener tape sewn to the data case cover keeps it closed. The door structure is coated with sealing compound, Item 287, WP 1803 00. The door is electrically bonded to airframe with conductive seals (Figure 1, Sheet 1).
2. Evaluate damage to aluminum parts (WP 0377 00).
3. Use WP 0378 00 and WP 0380 00 for typical repairs to aluminum parts.
4. Evaluate damage and make repairs to Kevlar and composite plastic components (WP 0379 00 and TM 1-1500-204-23).
5. Clean, polish, wax, and maintain acrylic plastic windows (TM 1-1500-204-23).
6. Touch up paint in repaired area (WP 0381 00 and TM 55-1500-345-23).

REPAIR COCKPIT DOOR - Continued

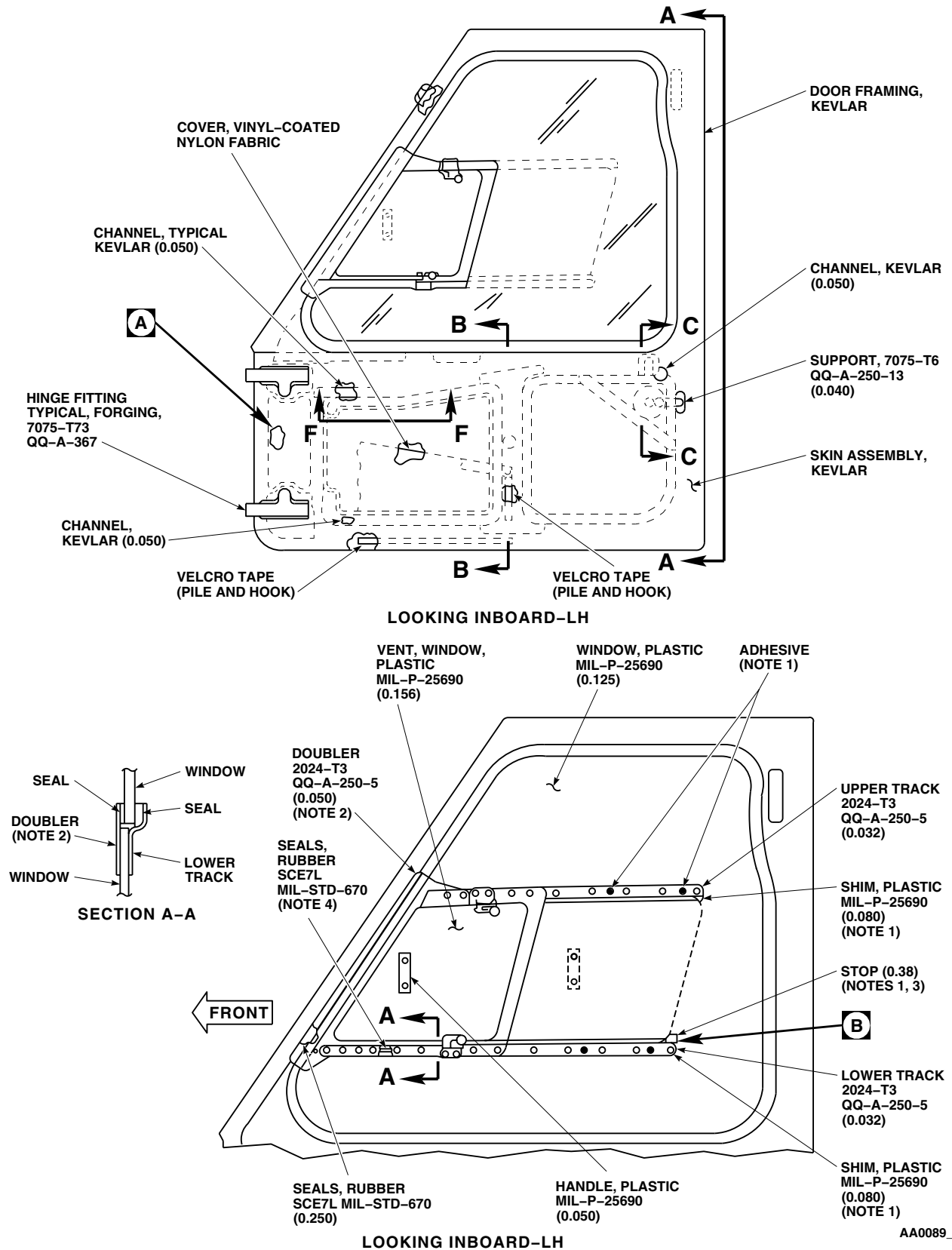


Figure 1. Repair Cockpit Door. (Sheet 1 of 4)

REPAIR COCKPIT DOOR - Continued

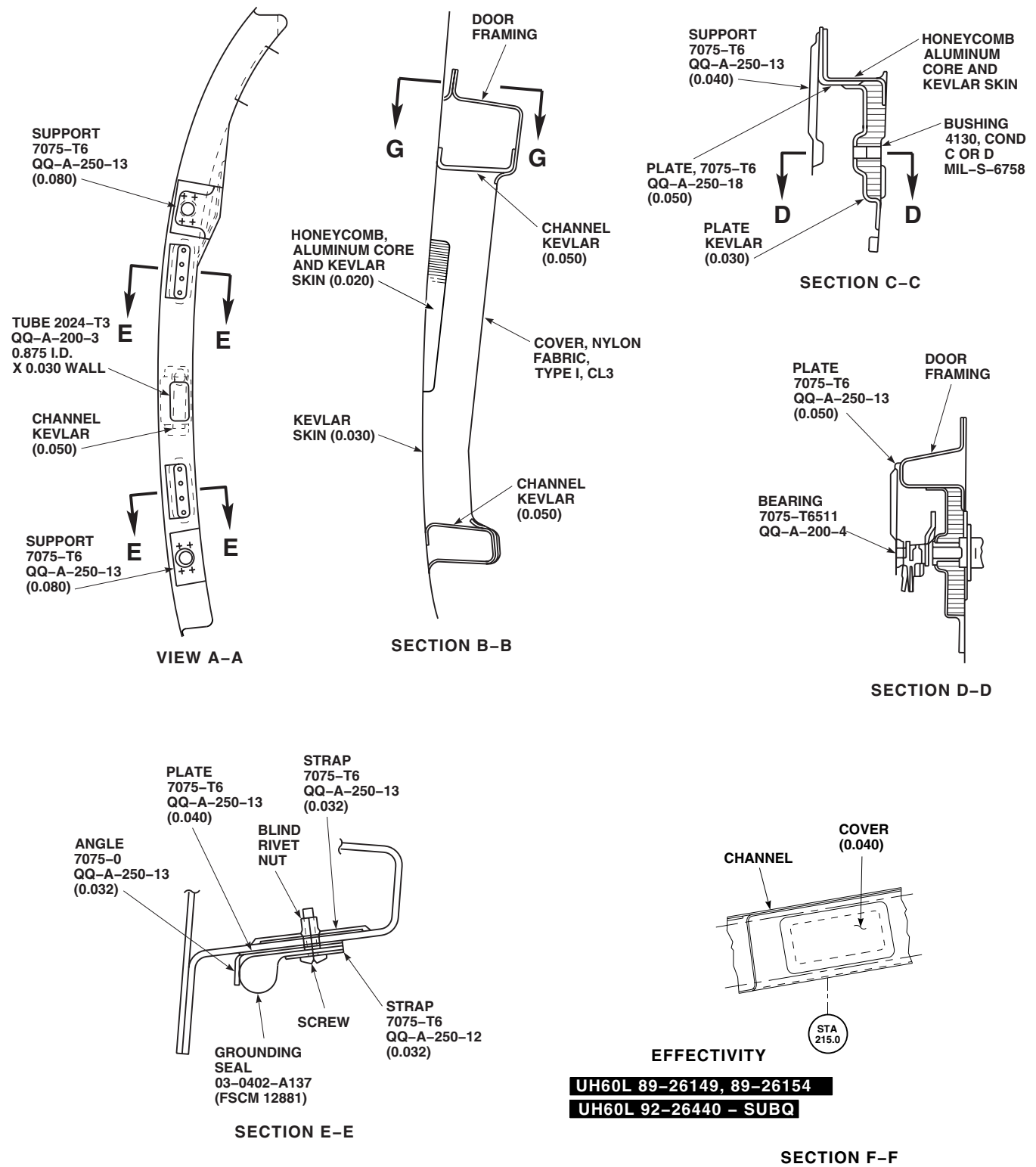
AA0089_2C
SA

Figure 1. Repair Cockpit Door. (Sheet 2 of 4)

REPAIR COCKPIT DOOR - Continued

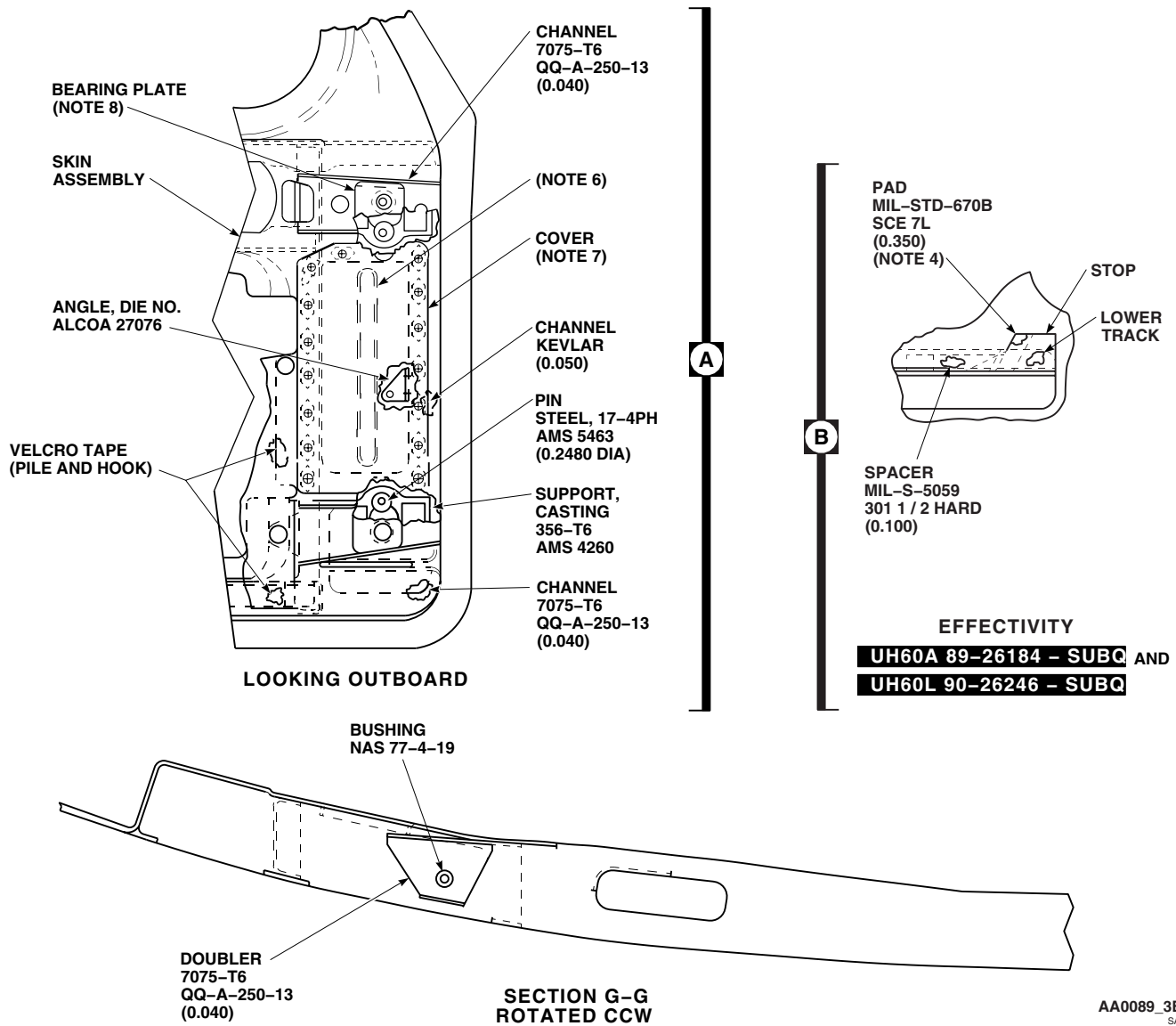


Figure 1. Repair Cockpit Door. (Sheet 3 of 4)

REPAIR COCKPIT DOOR - Continued

NOTES:

1. BONDED TO WINDOW WITH ADHESIVE, ITEM 40, WP 1803 00 IN 1/4-INCH DIAMETER SPOTS WHERE SHOWN.
2. SEAL MATING SURFACES WITH SEALING COMPOUND, ITEM 274, WP 1803 00.
3. **W/O MWO 50-42** → PLASTIC MIL-P-25690. ◀
MWO 50-52 | **UH60L** | **EH60A** PHENOLIC MIL-P-15035, TYPE FMB. ◀
4. **UH60A 77-22714 - 86-24551** → **EH60A 84-24017 - 85-24482** (0.062). ◀
UH60A 86-24552 - SUBQ → **EH60A 86-24561 - SUBQ** | **UH60L** (0.125). ◀
5. BOND WITH ADHESIVE PRIMER, ITEM 237, WP 1803 00.
6. SKIN MAY OR MAY NOT BE BEADED.
7. EITHER KEVLAR (0.060) OR 7075-T6, QQ-A-250-13, (0.125).
8. **UH60A 77-22714 - 80-23246** → 7075-T6 QQ-A-250-13 (0.125). ◀
UH60A 80-23247 - SUBQ → **UH60L** | **EH60A** 4130 COND A OR N (0.062). ◀
9. BOND TO STOP AND TRACK WITH ADHESIVE, ITEM 14, WP 1803 00.
10. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0089_4D
SA

Figure 1. Repair Cockpit Door. (Sheet 4 of 4)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

NOSE SECTION FRAMING REPAIR

This WP supersedes WP 0384 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-P-15035

MIL-S-18729

MIL-S-22499

Material/Parts

Collar, NAS 1080C06

QQ-A-200

Grommet, MS 21266-1N

QQ-A-225

Hinge, MS 20257P3-1462

QQ-A-250

Pin, NAS 1446C03

QQ-A-255

Sealing Compound, Item 273, WP 1803 00

QQ-A-367

Sill, 70207-01009-101, -102

TM 1-1500-204-23

Spacer, NAS 42DD3-6

TM 1-1500-344-23

Washer, MS 25440-3

TM 55-1500-345-23

WP 0377 00

Personnel Required

WP 0378 00

Aircraft Structural Repairer MOS 15G (1)

WP 0379 00

WP 0380 00

References

MIL-A-21180

WP 0381 00

MIL-C-7438

WP 1803 00

REPAIR NOSE SECTION FRAMING

1. All nose section framing, except reinforced plastic members, is primary structure. Beams at LBL 10.0 and RBL 10.0, are reinforced with Boron straps bonded to cap angles. Damage to angles bonded to Boron straps is limited to negligible damage repairs using WP 0377 00. Repair of Boron is beyond the scope of this manual. Fittings are considered critical primary structure. Negligible damage to fittings is limited to nicks, scratches, and corrosion no deeper than 5% of material thickness at point of damage. Blend damage smoothly into surrounding area using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage to fittings greater than negligible limits must be reviewed by engineering authority for determining repair or replacement. All repairs to nose section framing must maintain design strength and alignment (Figure 1, Sheet 1).
2. Evaluate damage in noncritical areas (WP 0377 00).
3. Refer to WP 0378 00 and WP 0380 00 for typical framing repairs. Coordinate repairs with TM 1-1500-204-23 requirements.
4. Refer to WP 0379 00 for evaluating damage and repairing reinforced plastic. Coordinate repairs with TM 1-1500-204-23 requirements.
5. For corrosion protection and watertight sealing, seal all repairs between framing and plating (WP 0377 00 and TM 1-1500-344-23).
6. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR NOSE SECTION FRAMING - Continued

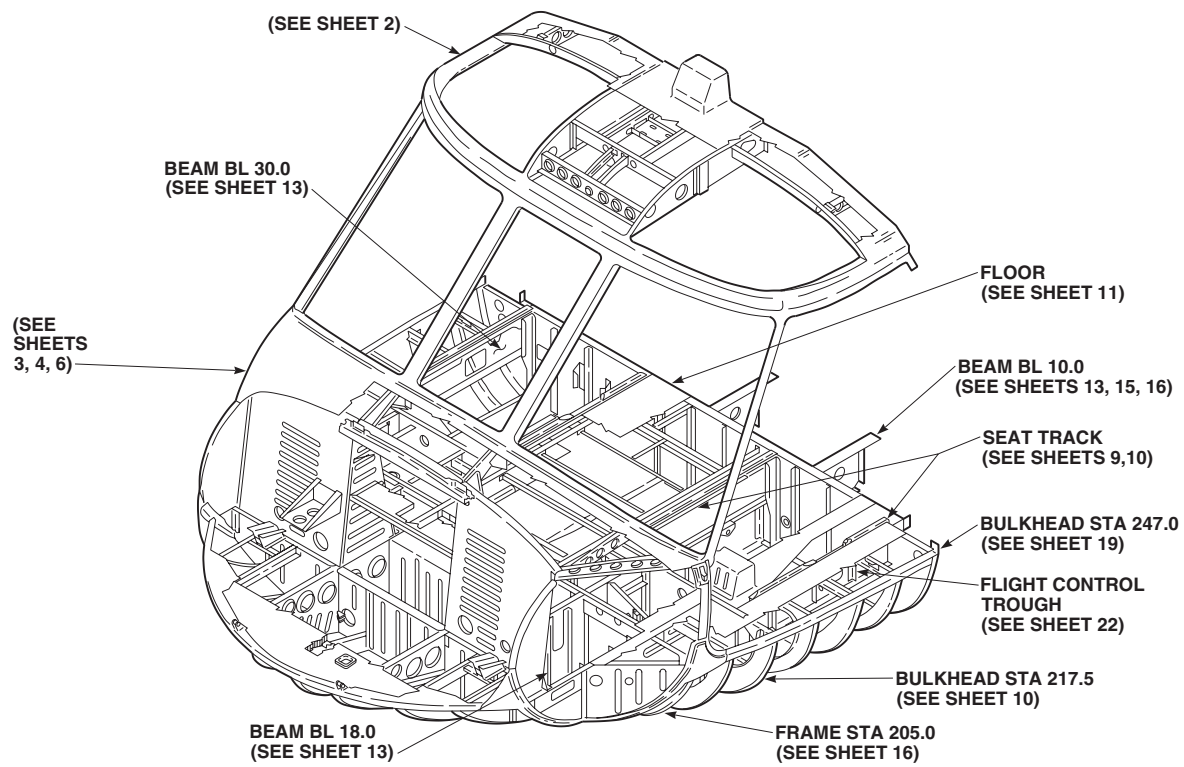
AB2650_1
SA

Figure 1. Nose Section Framing. (Sheet 1 of 22)

REPAIR NOSE SECTION FRAMING - Continued

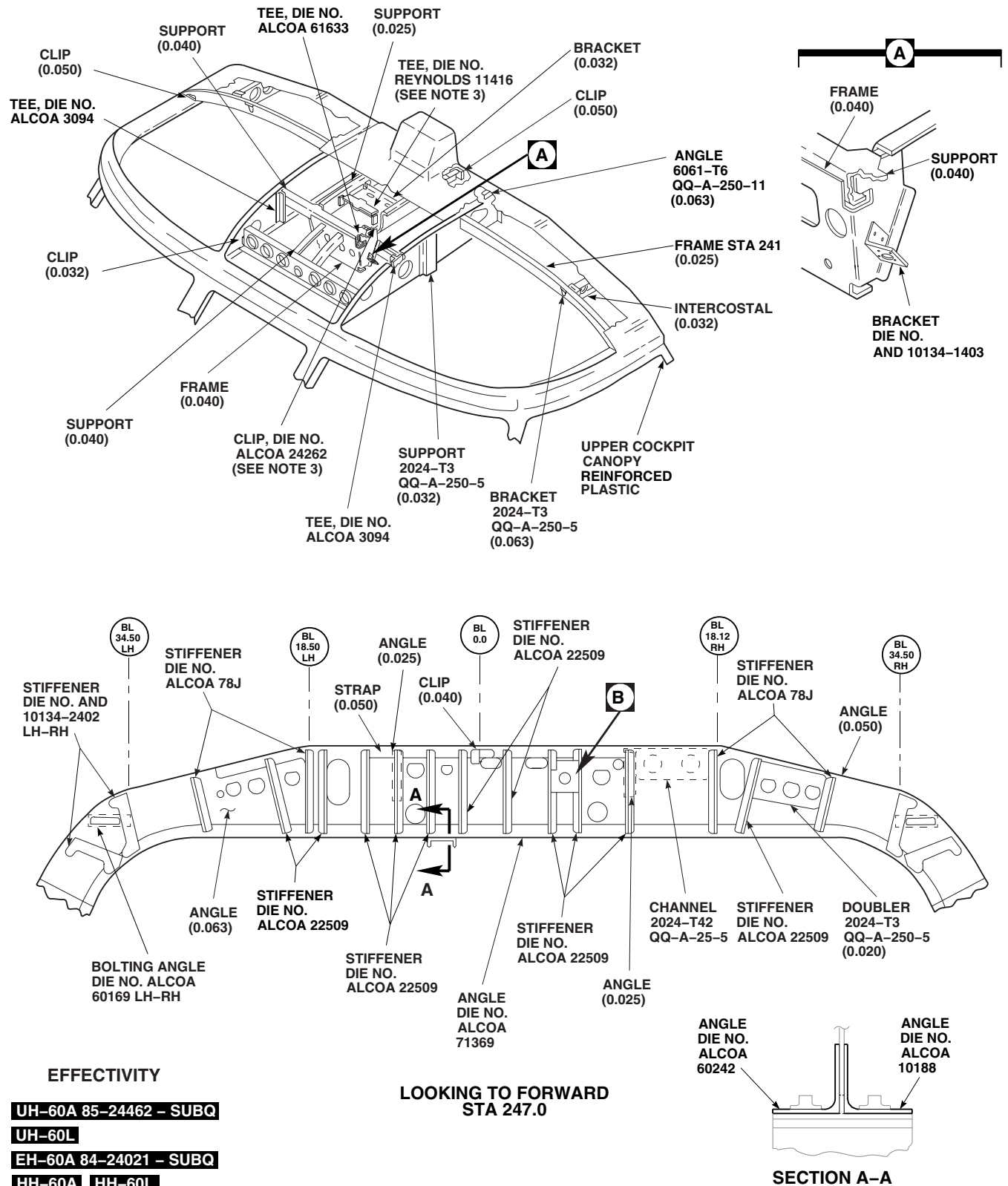
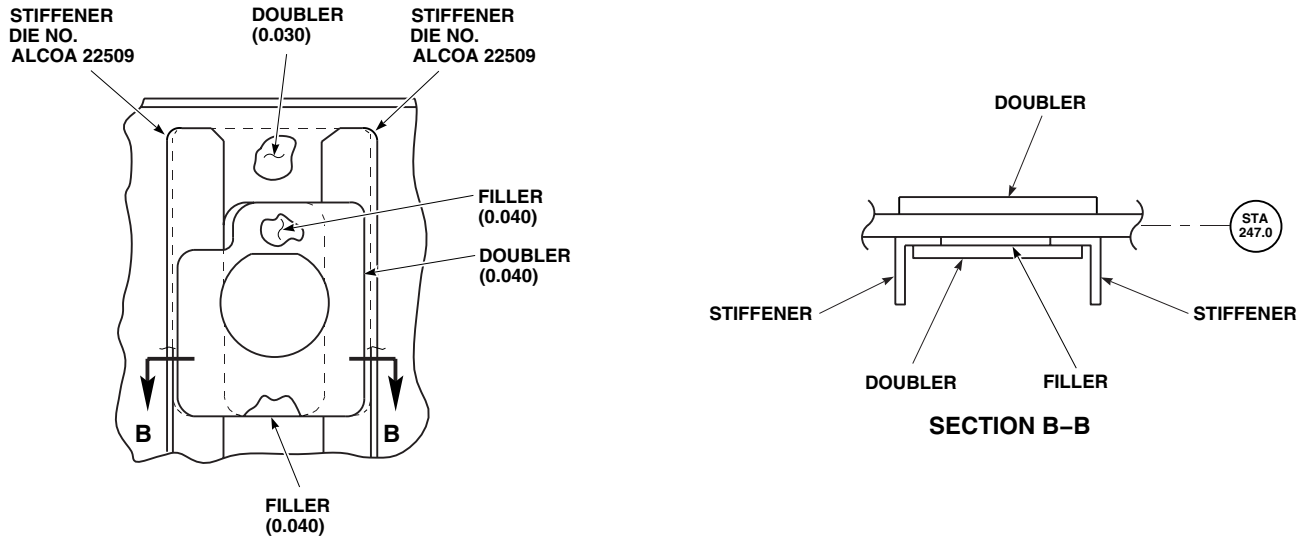


Figure 1. Nose Section Framing. (Sheet 2 of 22)

REPAIR NOSE SECTION FRAMING - Continued



B

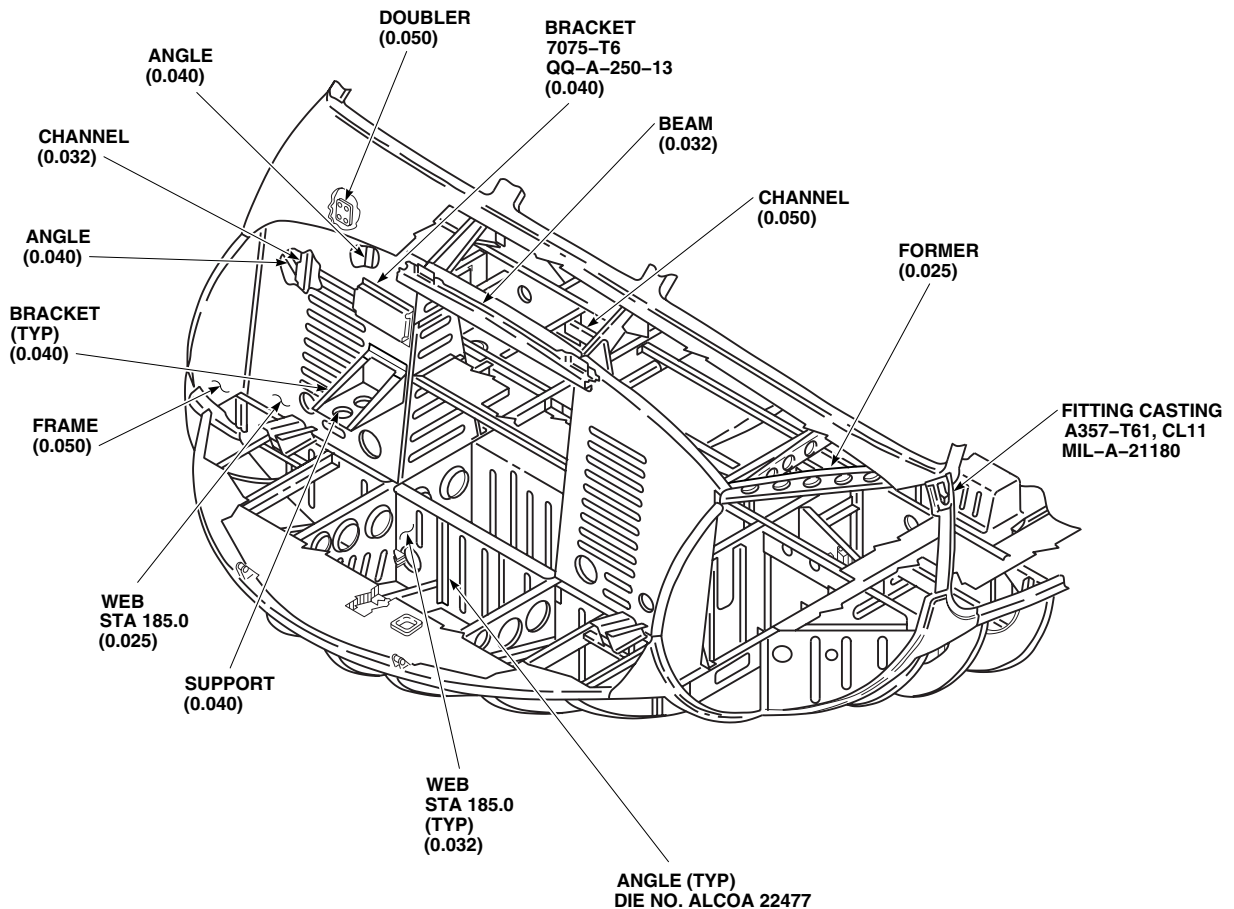
AB2650_3
SA

Figure 1. Nose Section Framing. (Sheet 3 of 22)

REPAIR NOSE SECTION FRAMING - Continued

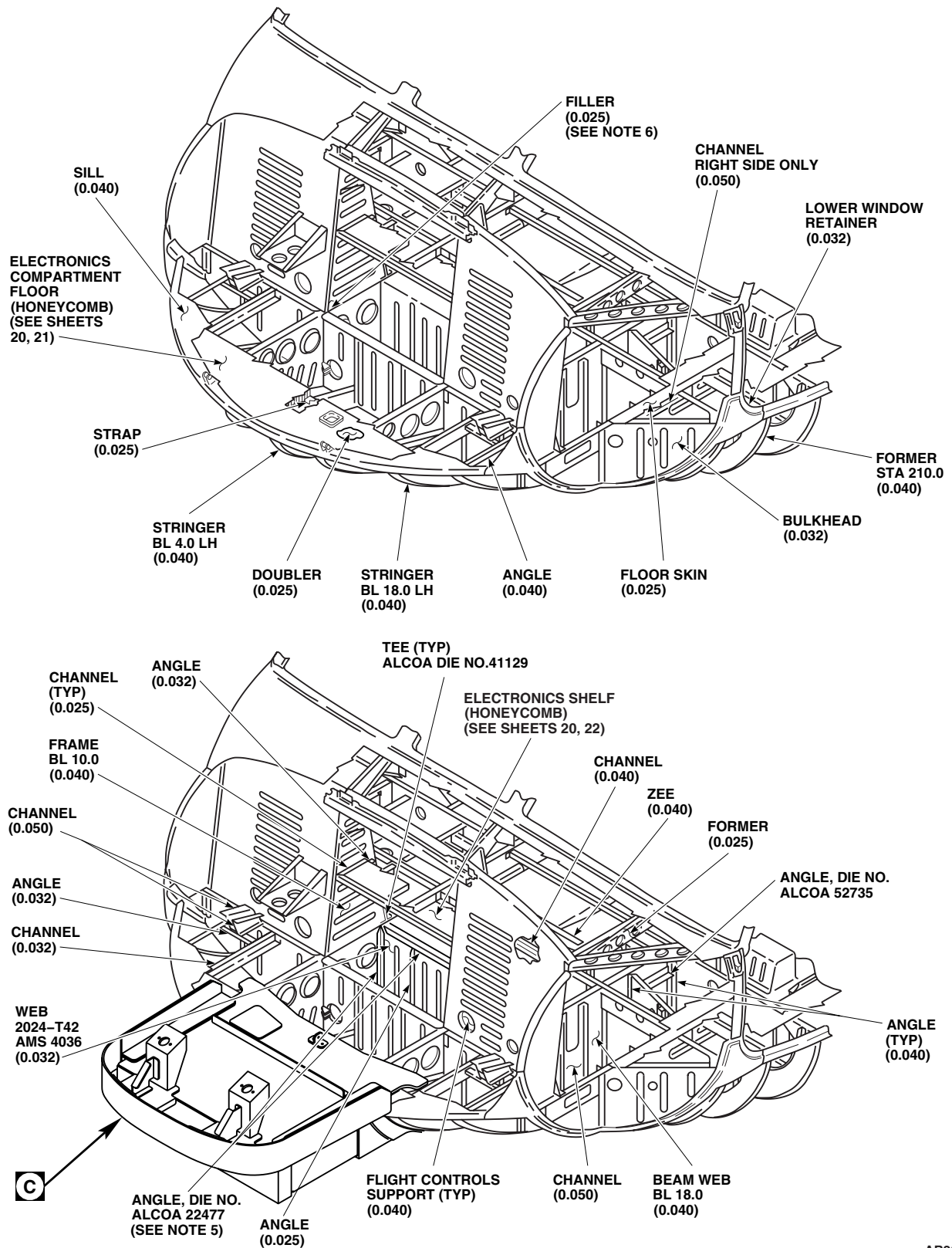
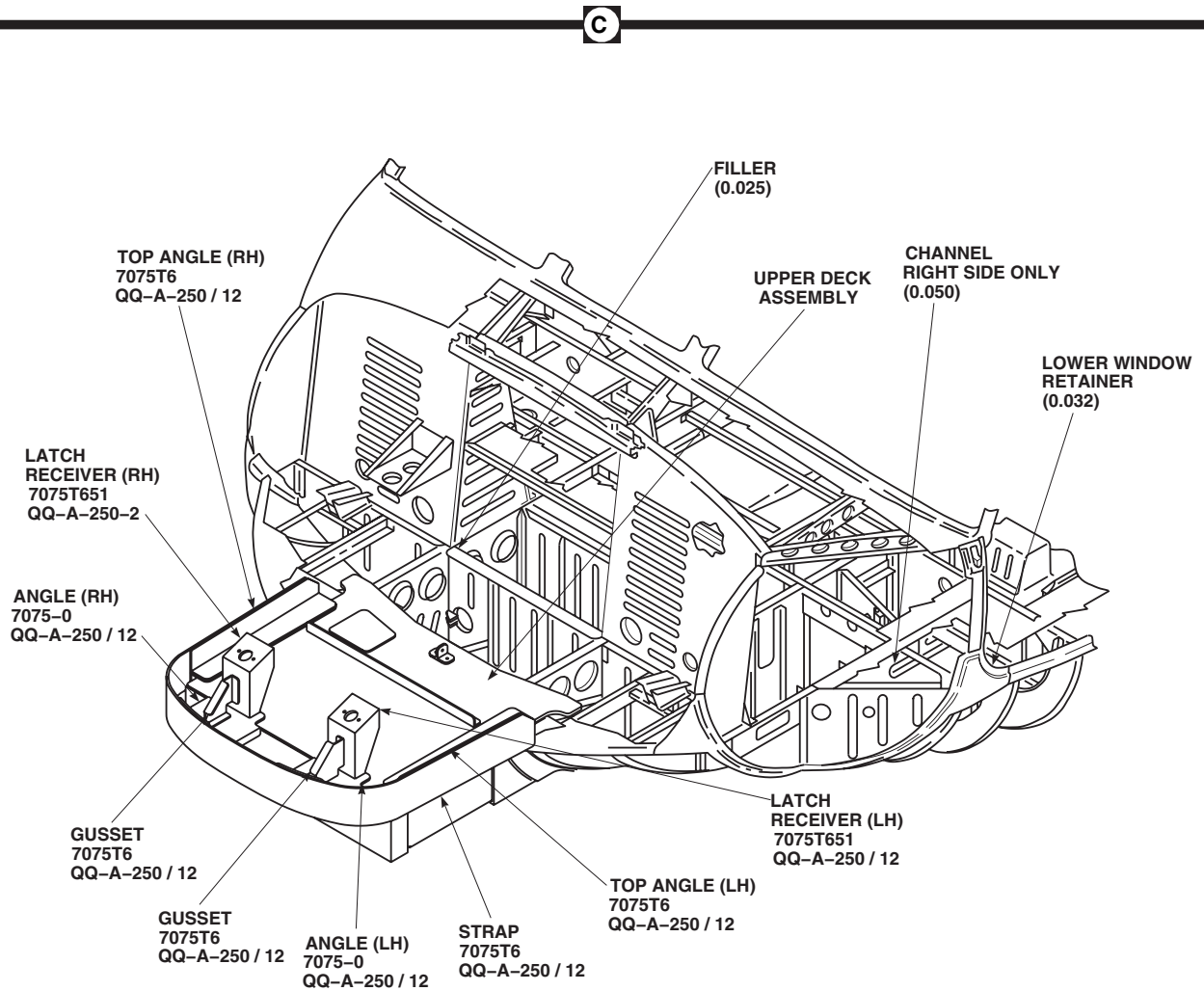
AB2650_4
SA

Figure 1. Nose Section Framing. (Sheet 4 of 22)

REPAIR NOSE SECTION FRAMING - Continued



EFFECTIVITY

HH-60A HH-60L

AB2650_5A
SA

Figure 1. Nose Section Framing. (Sheet 5 of 22)

REPAIR NOSE SECTION FRAMING - Continued

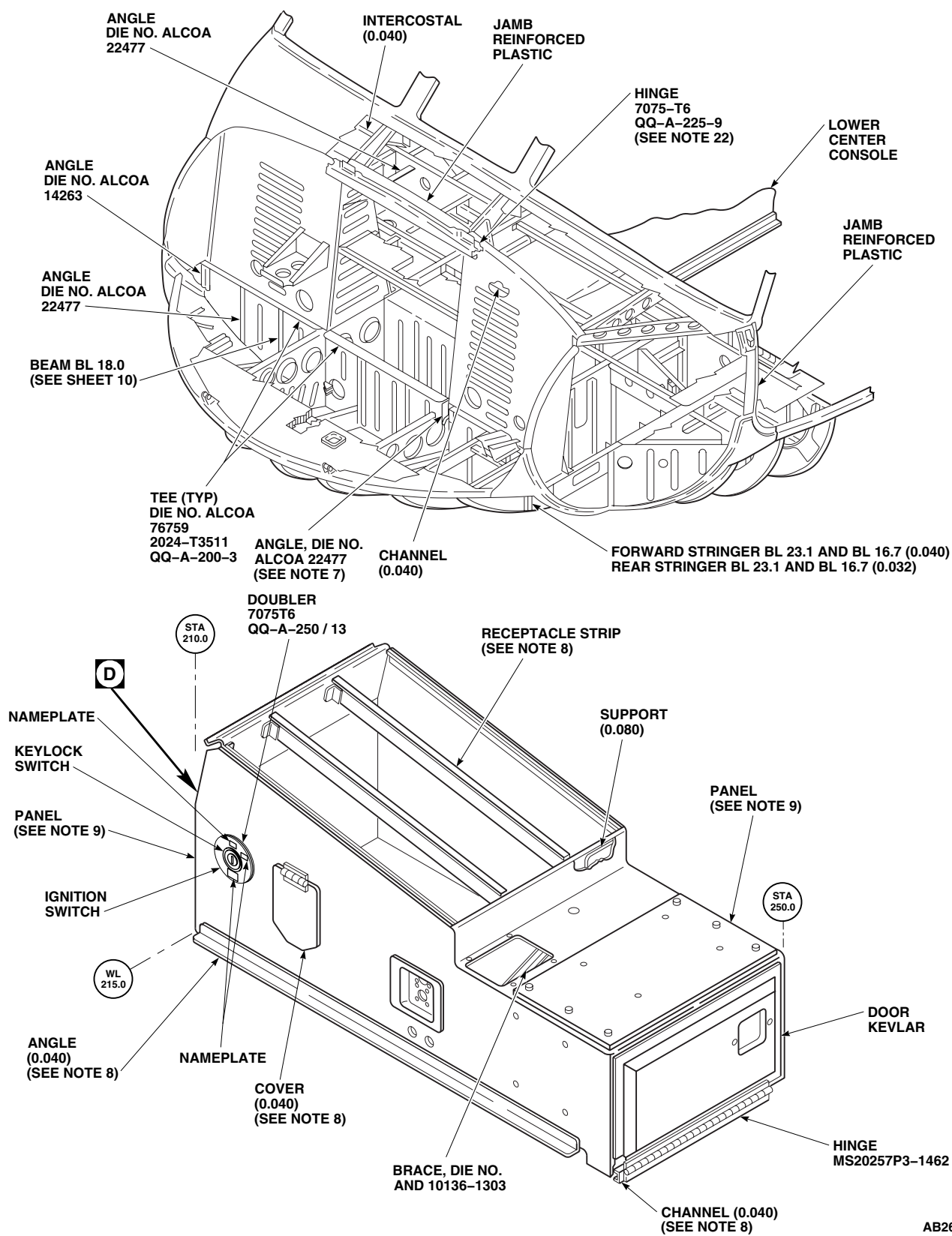
AB2650_6
SA

Figure 1. Nose Section Framing. (Sheet 6 of 22)

REPAIR NOSE SECTION FRAMING - Continued

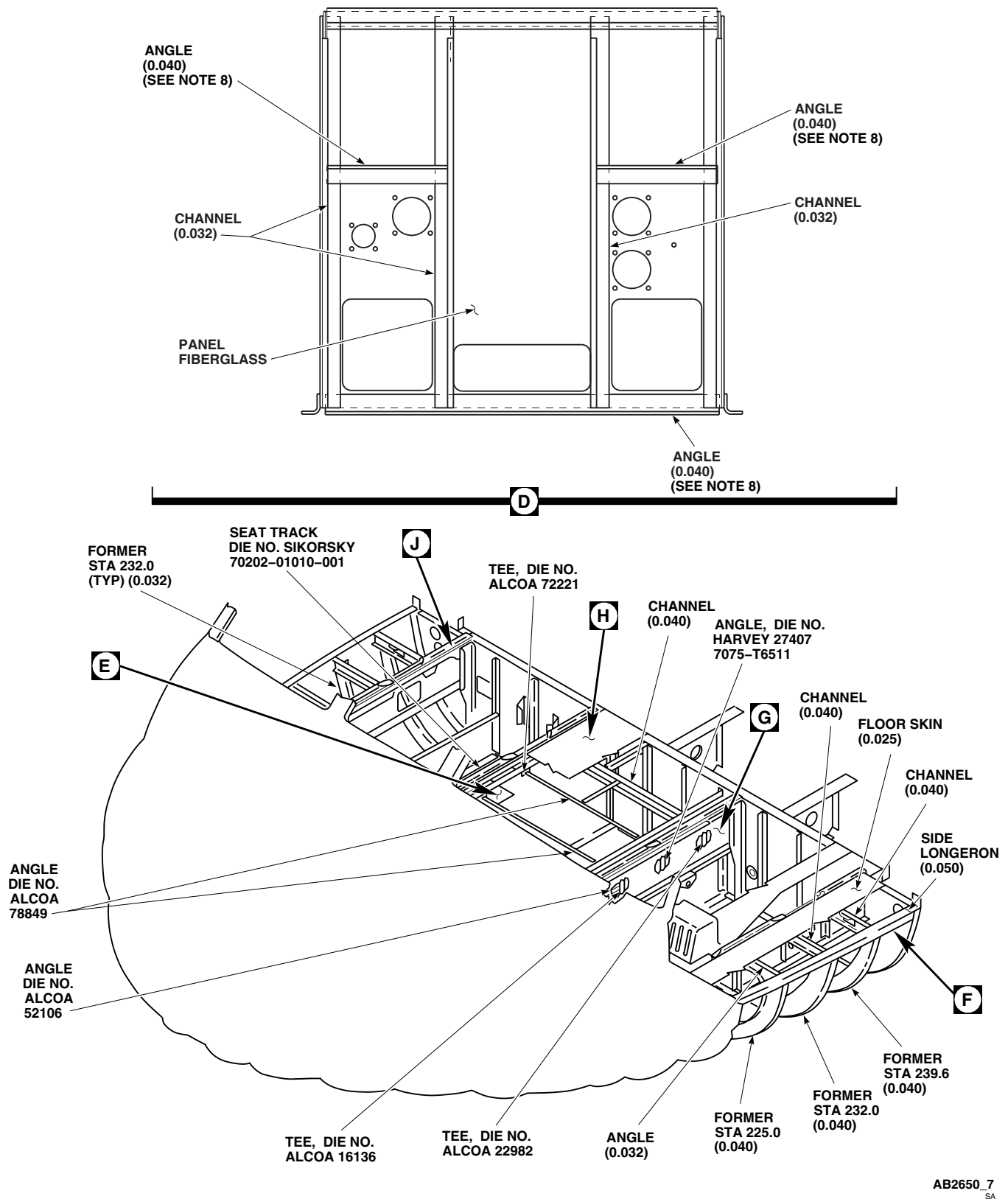
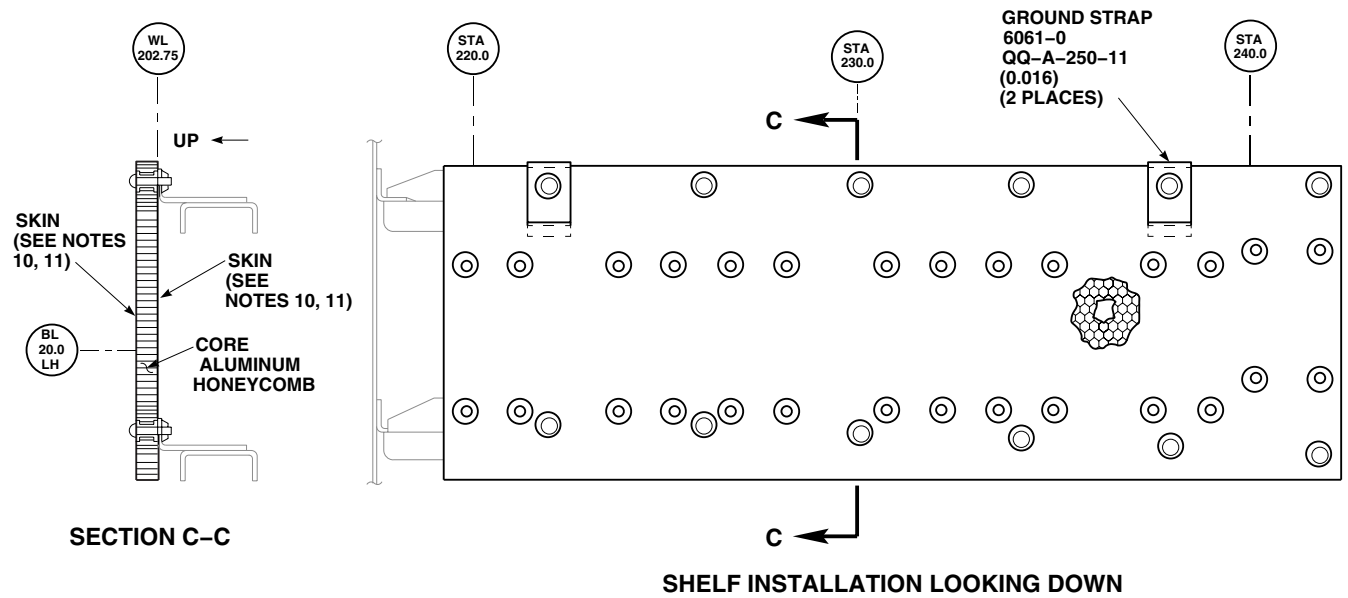


Figure 1. Nose Section Framing. (Sheet 7 of 22)

REPAIR NOSE SECTION FRAMING - Continued

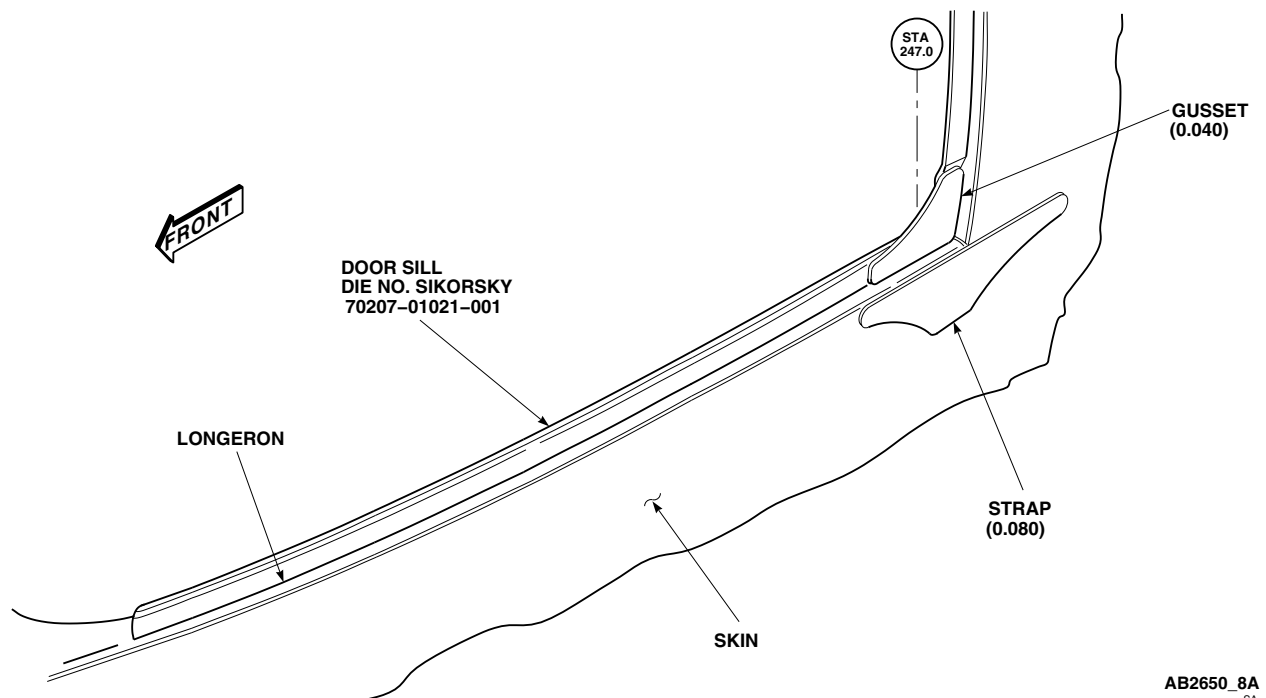
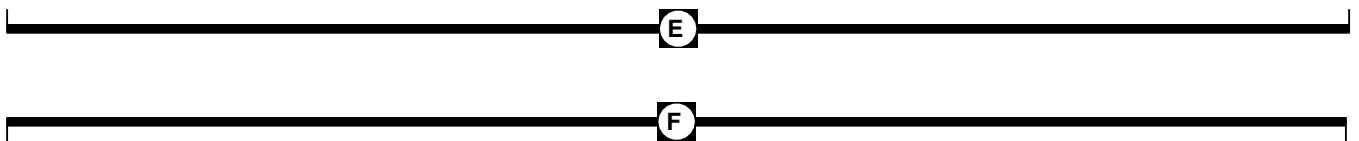


EFFECTIVITY

UH-60A 84-23953, 84-23962 - SUBQ

UH-60L EH-60A

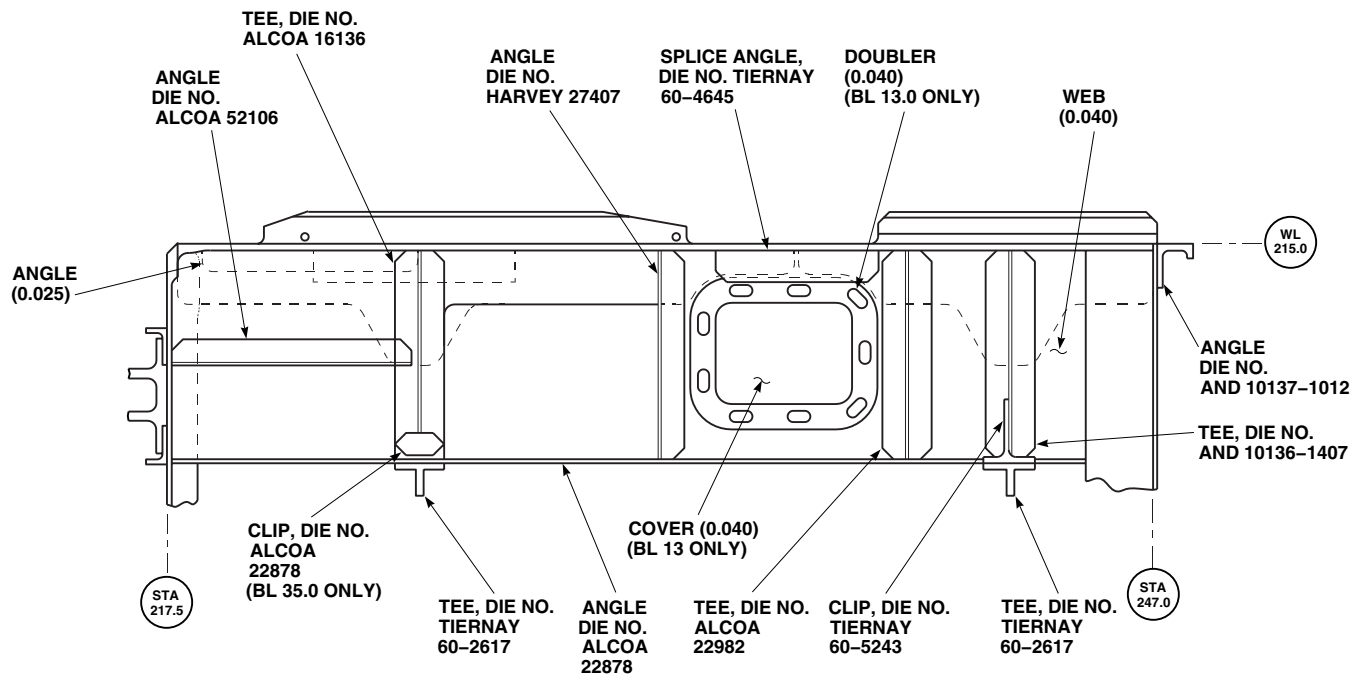
HH-60A HH-60L



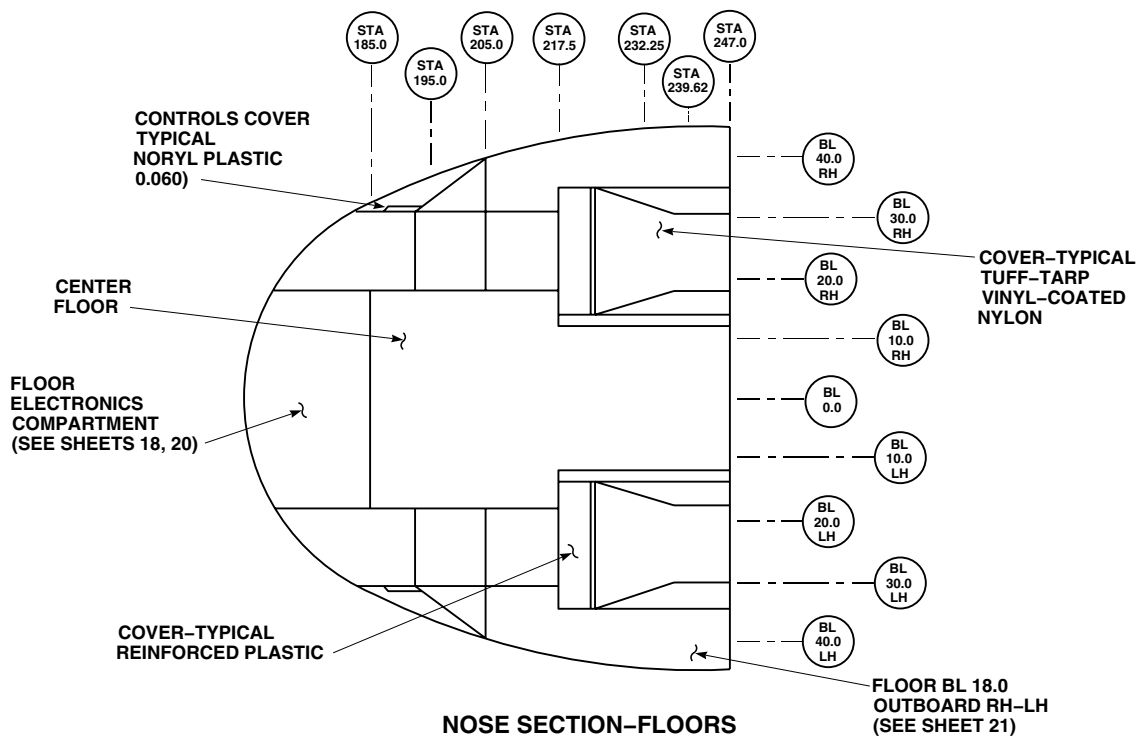
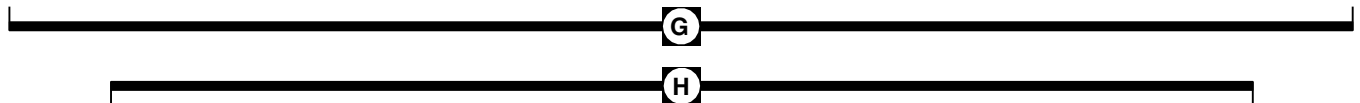
AB2650_8A
SA

Figure 1. Nose Section Framing. (Sheet 8 of 22)

REPAIR NOSE SECTION FRAMING - Continued



BEAM-SEAT TRACK - BL 13.0 LH AND BL 35.0 LH



NOSE SECTION-FLOORS

AB2650_9
SA

Figure 1. Nose Section Framing. (Sheet 9 of 22)

ANGLE (0.063)

ANGLE DIE NO. ALCOA 22878

BL 30.0 LH

VIEW A

ANGLE (0.032)

ANGLE (0.040)

ANGLE, DIE NO. ALCOA 80415

ANGLE, DIE NO. REYNOLDS 19237

TEE, DIE NO. ALCOA 55780

WEB (0.050)

CLIP 2024-T3 QQ-A-250-5 (0.032)

WEB (0.050)

RADIUS BLOCK 7075-T6511 QQ-A-200-11 (0.125)

ANGLE, DIE NO. PIONEER 4191

ANGLE (0.050)

ANGLE (0.040)

ANGLE, DIE NO. REYNOLDS 16381

WEB (0.050)

BRACKET (0.032)

WEB (0.050)

BL 30.0 RH

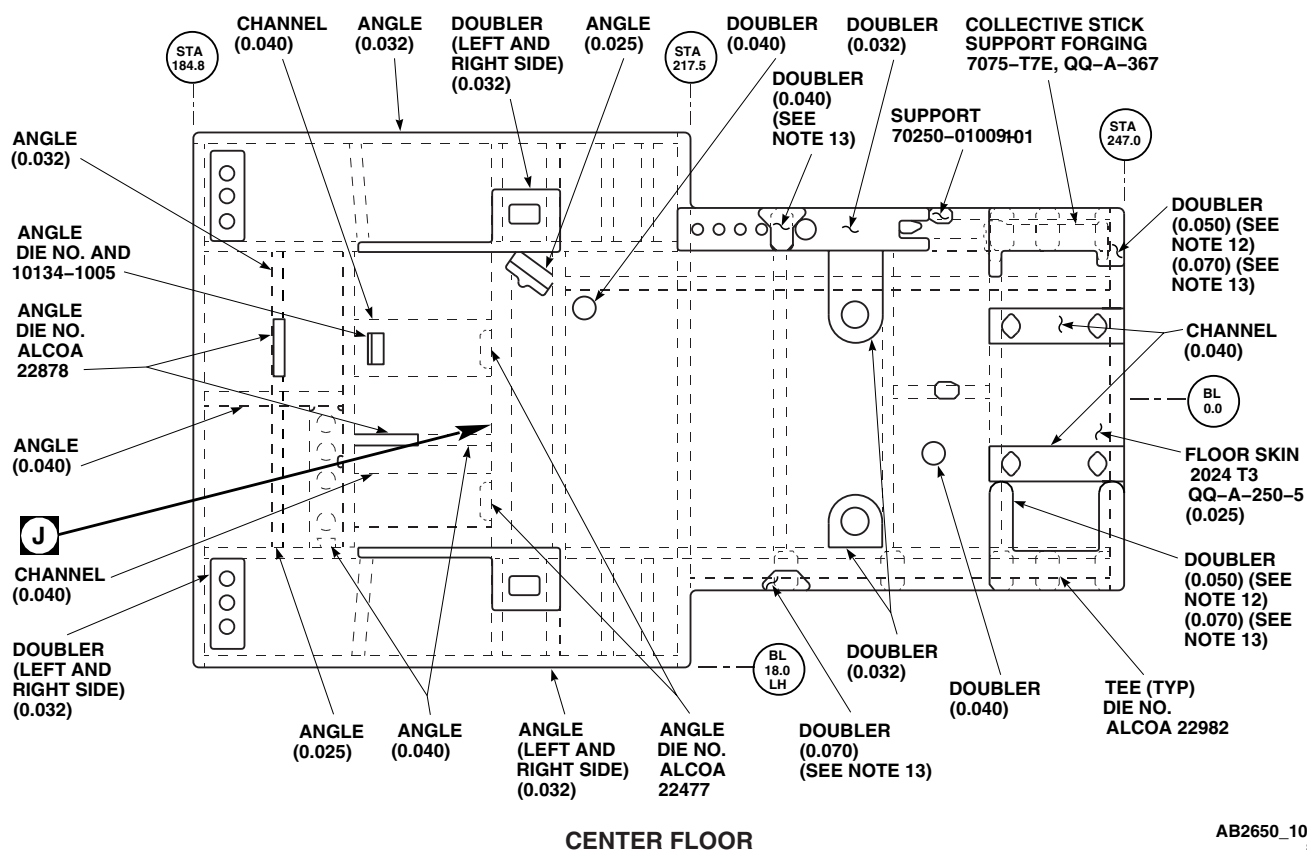
RIGHT SIDE ONLY

EFFECTIVITY

UH-60A 88-26043 - SUBQ
UH-60L
EH-60A 87-24690 - SUBQ
HH-60A HH-60L

BULKHEAD STA 217.5
LEFT SIDE SHOWN, RIGHT SIDE OPPOSITE

VIEW A



AB2650_10A
SA

Figure 1. Nose Section Framing. (Sheet 10 of 22)

REPAIR NOSE SECTION FRAMING - Continued

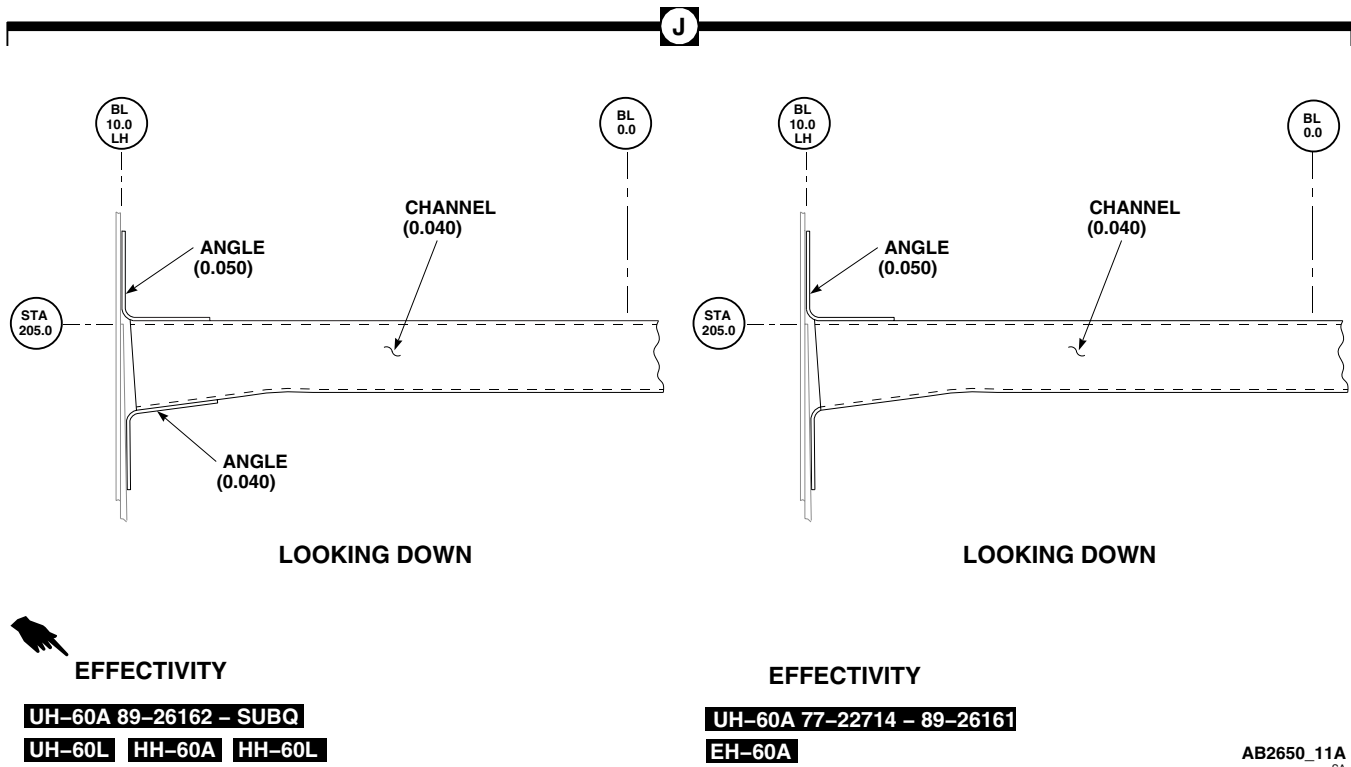
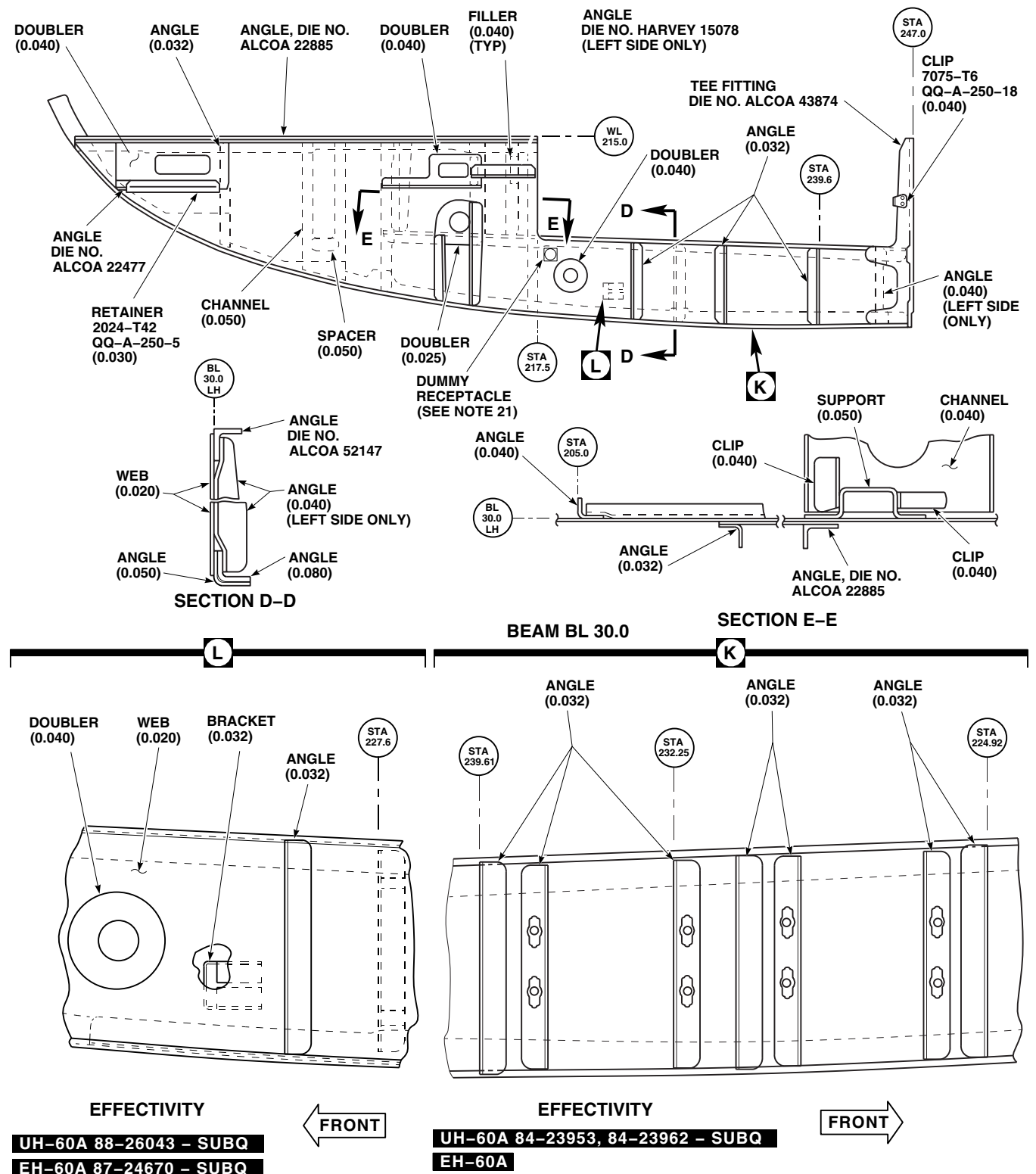


Figure 1. Nose Section Framing. (Sheet 11 of 22)

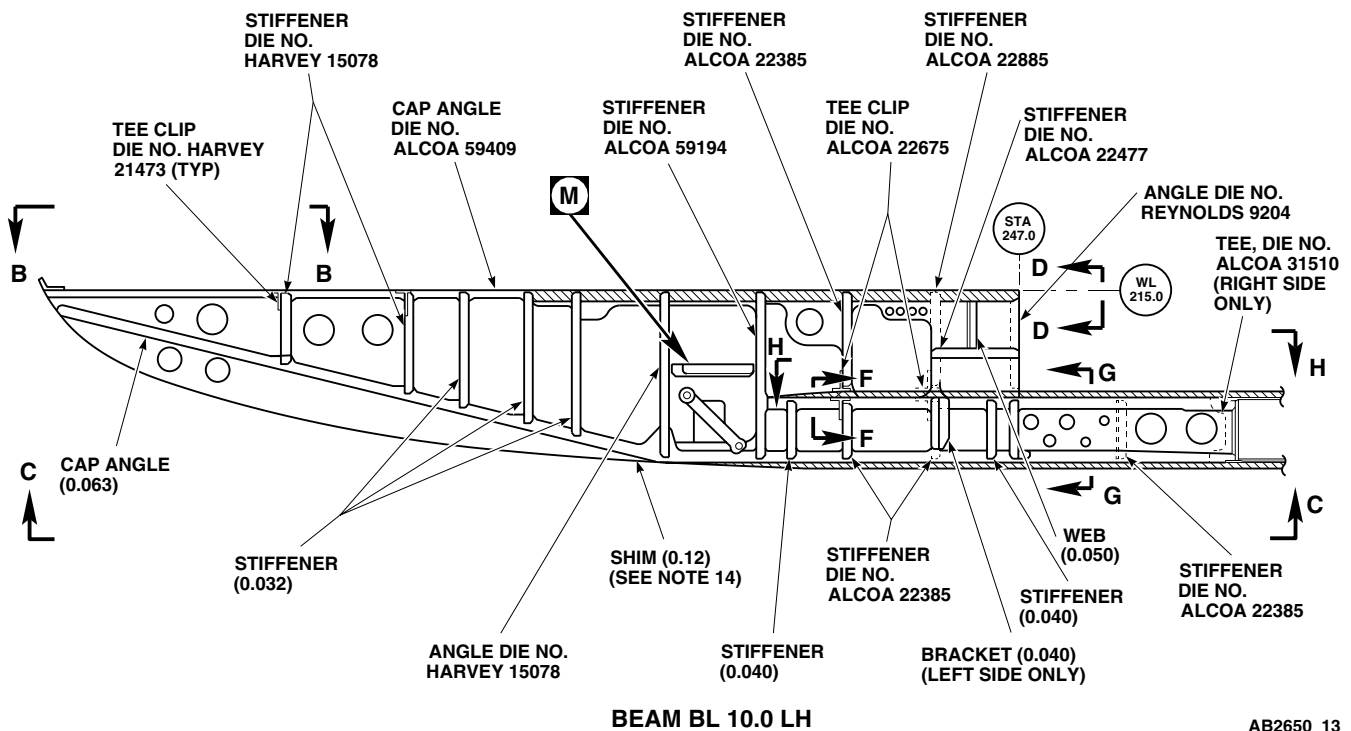
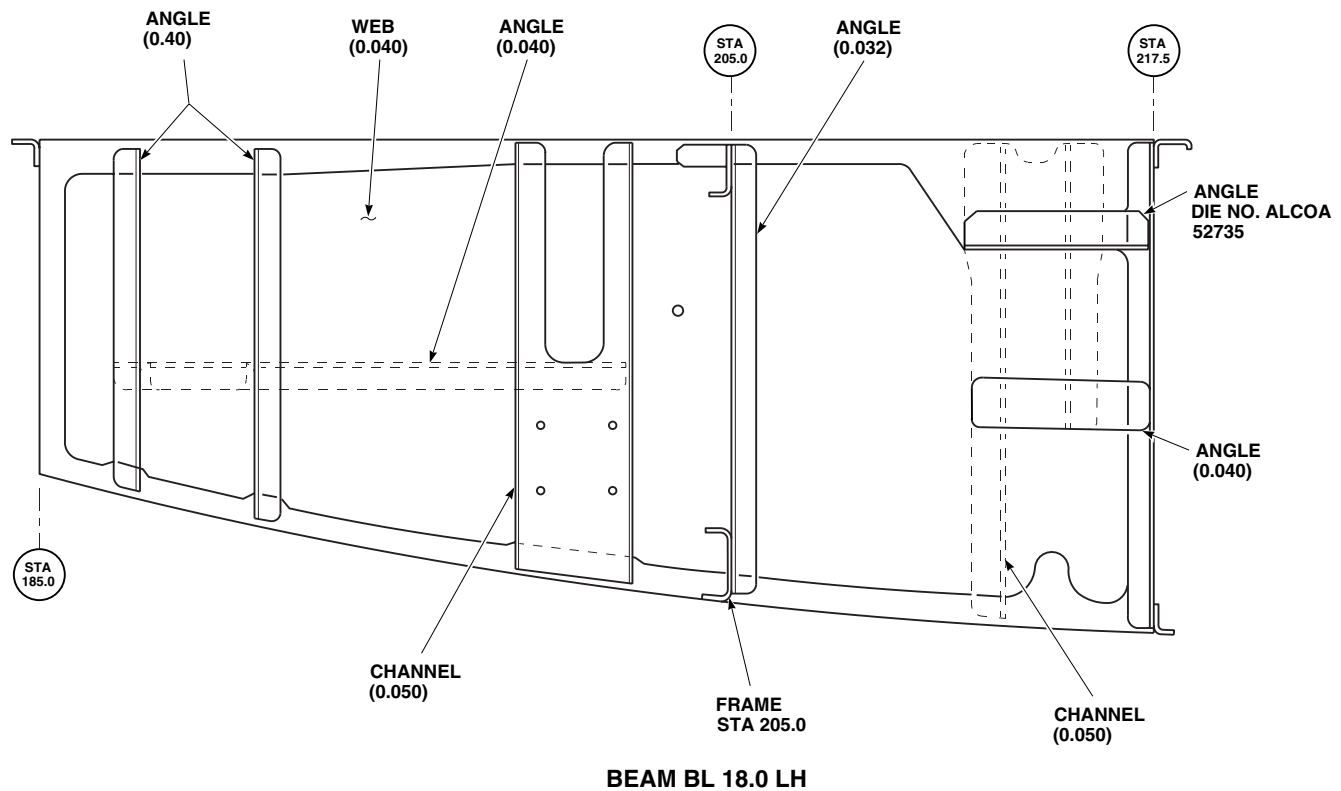
REPAIR NOSE SECTION FRAMING - Continued



AB2650_12
SA

Figure 1. Nose Section Framing. (Sheet 12 of 22)

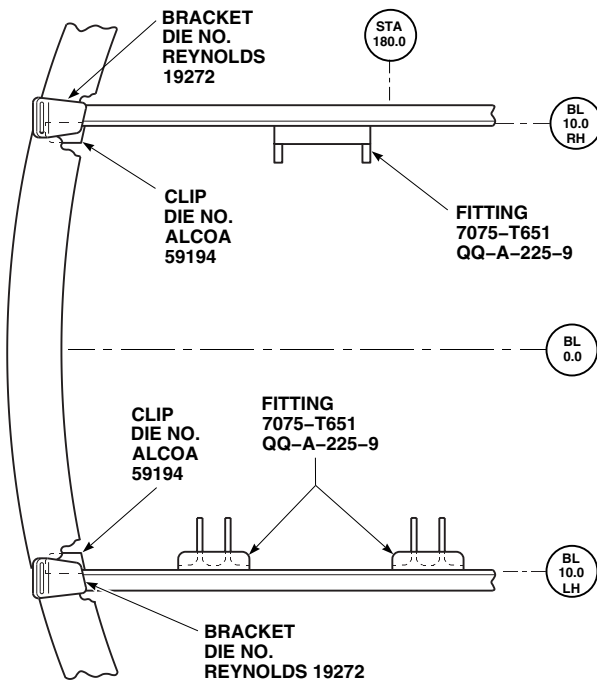
REPAIR NOSE SECTION FRAMING - Continued



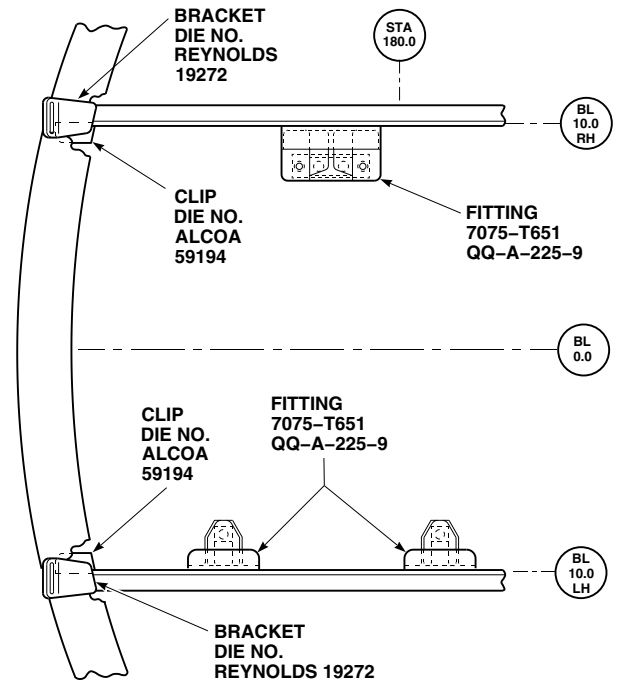
AB2650_13
SA

Figure 1. Nose Section Framing. (Sheet 13 of 22)

REPAIR NOSE SECTION FRAMING - Continued



VIEW B-B



EFFECTIVITY

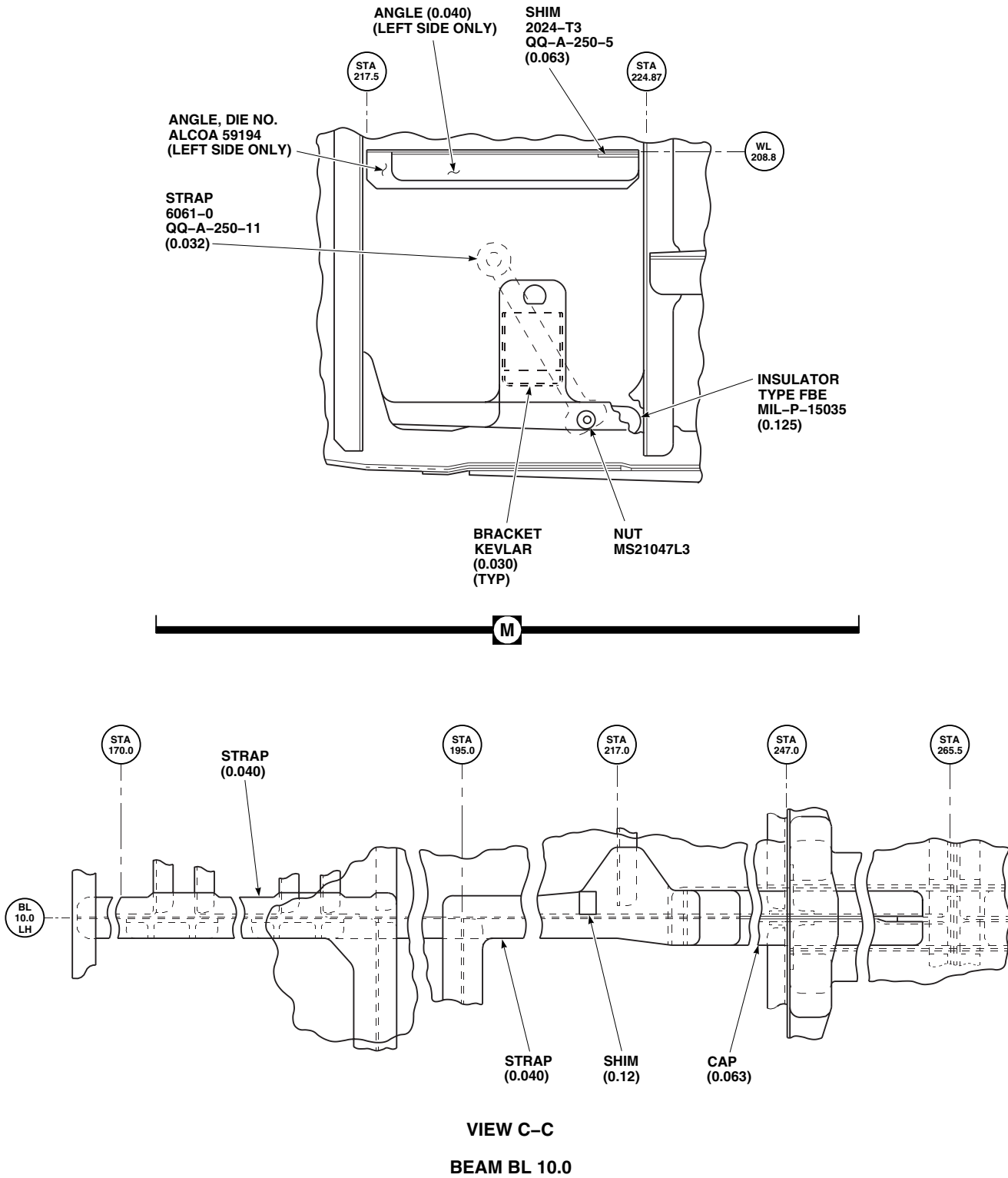
94-26617

VIEW B-B

AB2650_14
SA

Figure 1. Nose Section Framing. (Sheet 14 of 22)

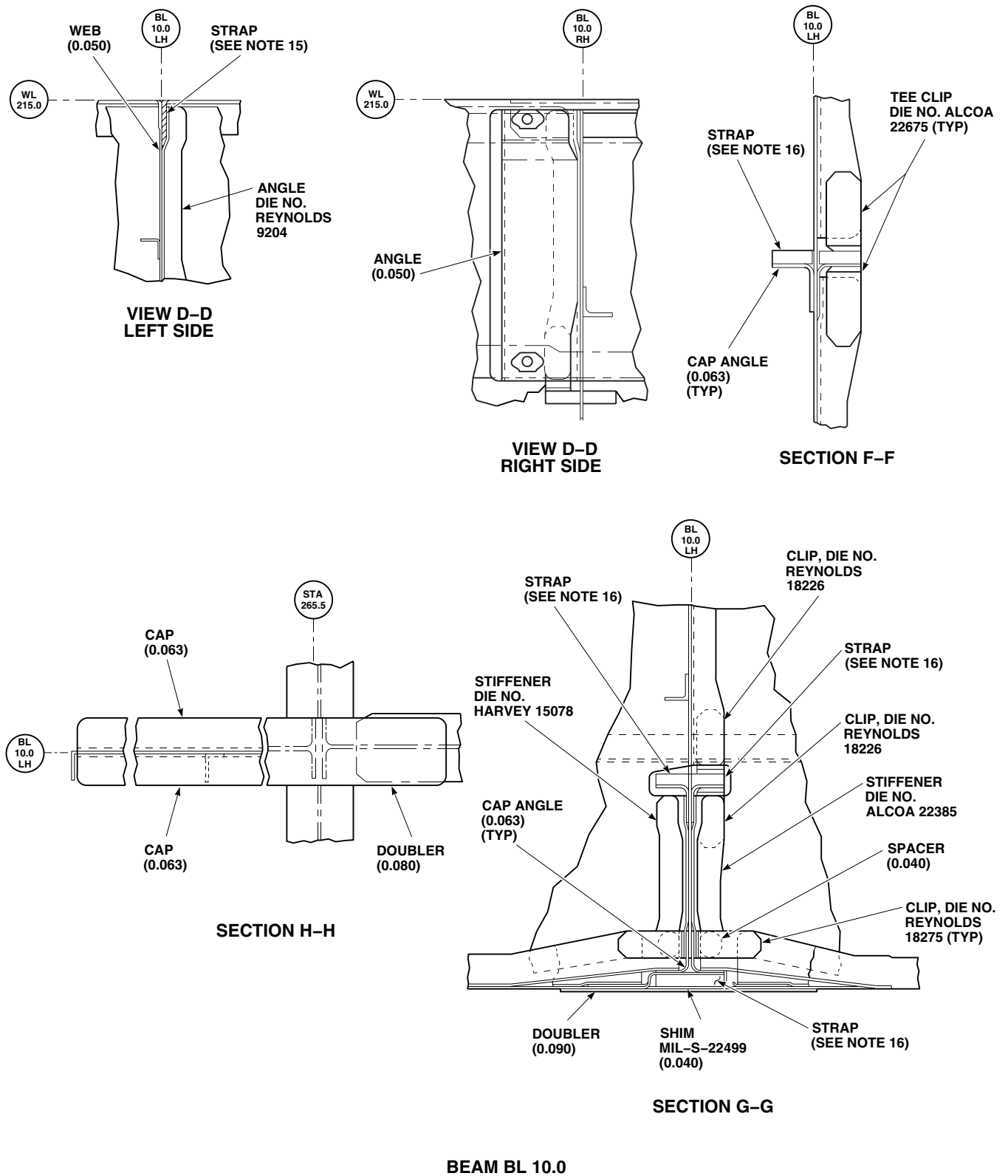
REPAIR NOSE SECTION FRAMING - Continued



AB2650_15
SA

Figure 1. Nose Section Framing. (Sheet 15 of 22)

REPAIR NOSE SECTION FRAMING - Continued



AB2650_16
SA

Figure 1. Nose Section Framing. (Sheet 16 of 22)

REPAIR NOSE SECTION FRAMING - Continued

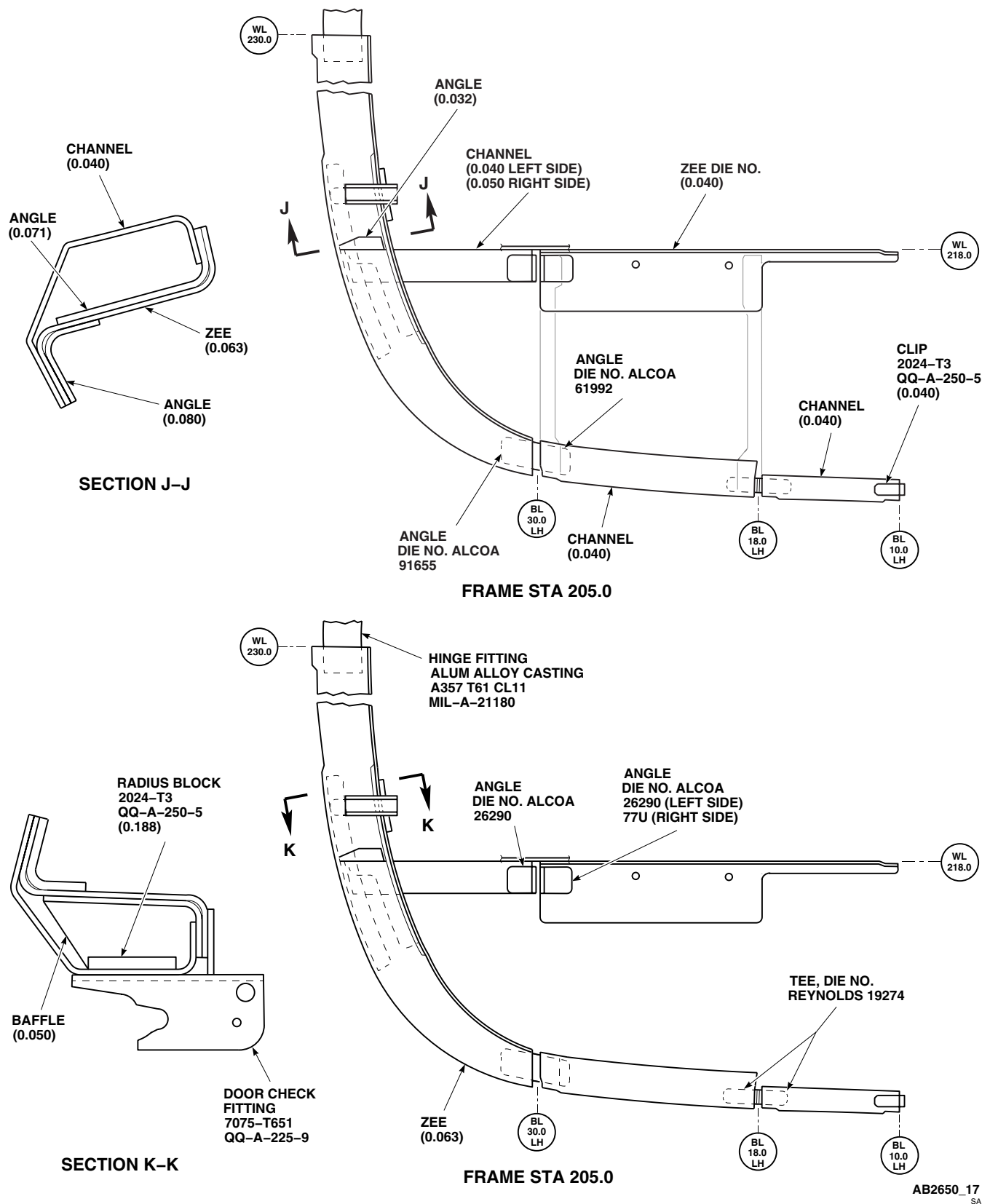
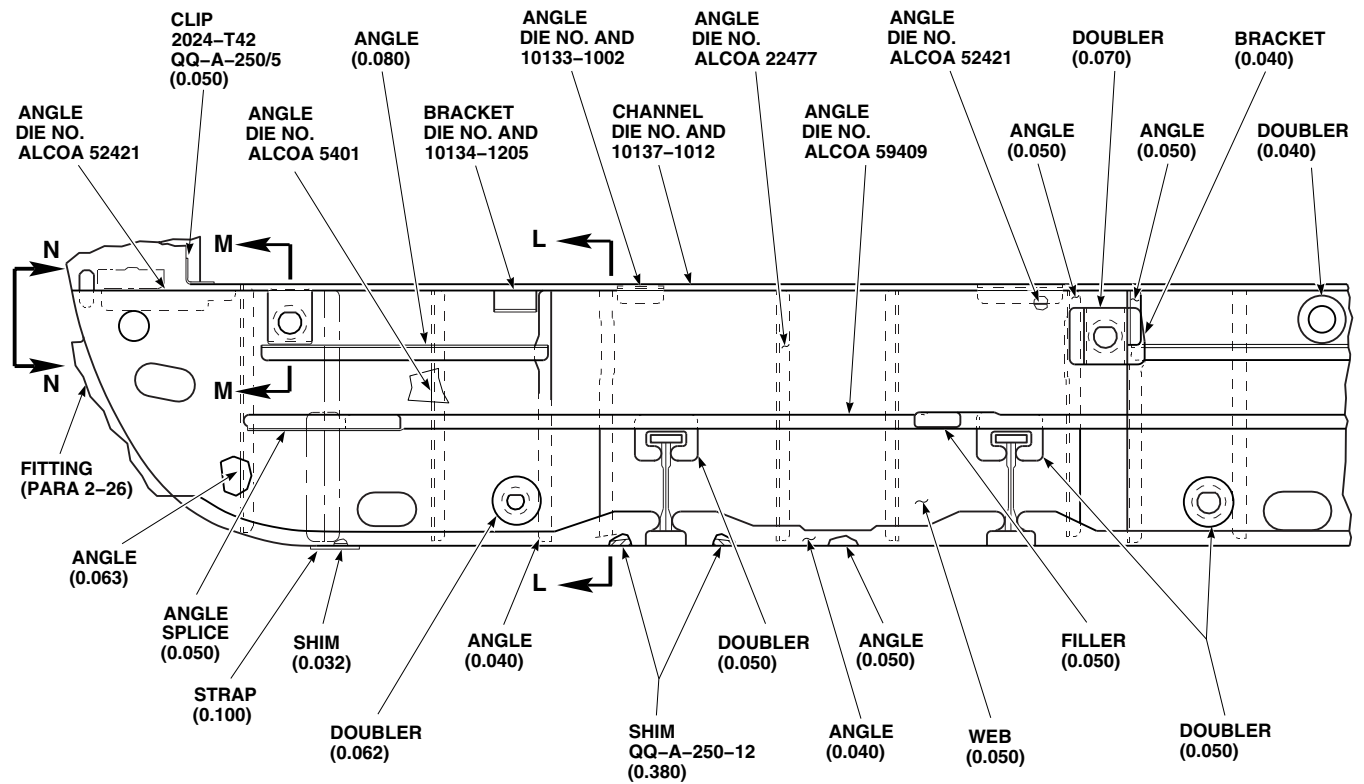
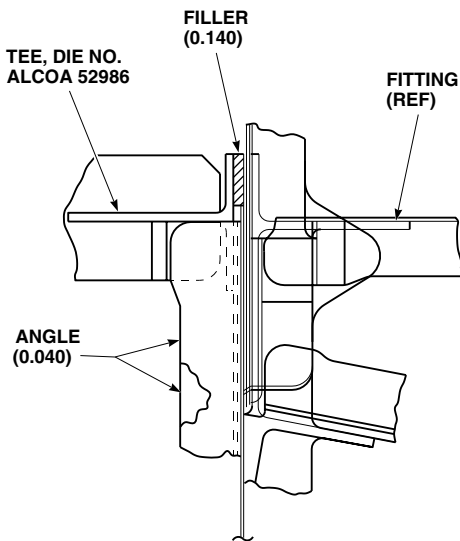


Figure 1. Nose Section Framing. (Sheet 17 of 22)

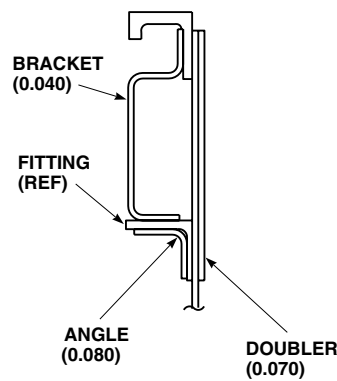
REPAIR NOSE SECTION FRAMING - Continued



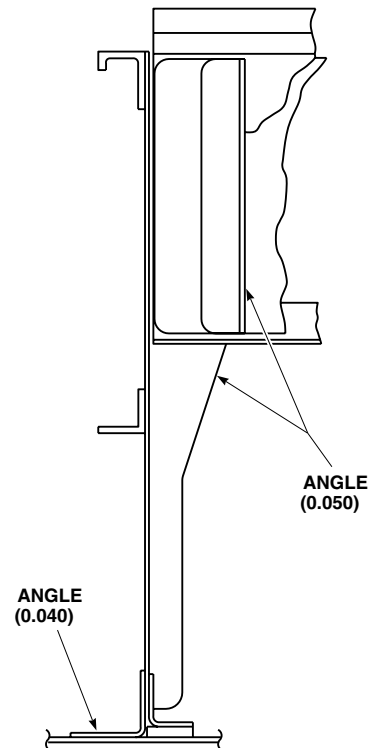
BULKHEAD STA 247.0



VIEW N-N



SECTION M-M



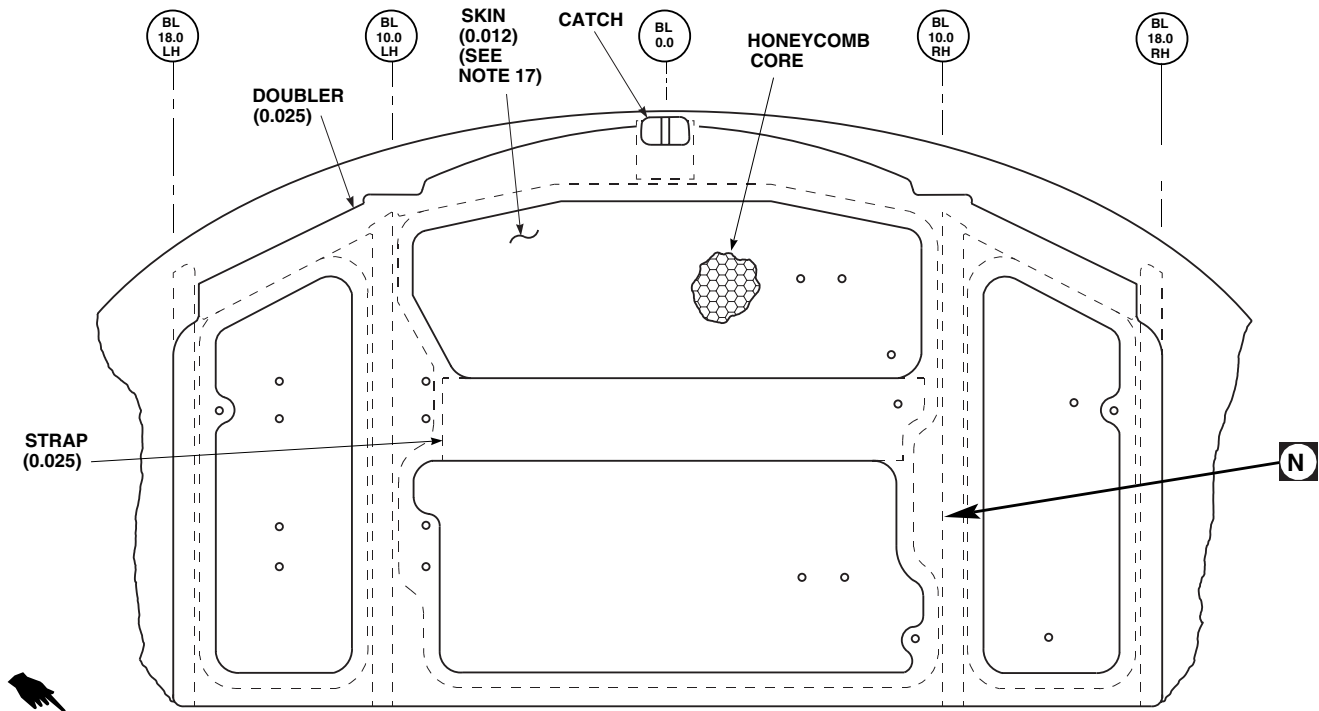
SECTION L-L

BULKHEAD STA 247.0

AB2650_18
SA

Figure 1. Nose Section Framing. (Sheet 18 of 22)

REPAIR NOSE SECTION FRAMING - Continued

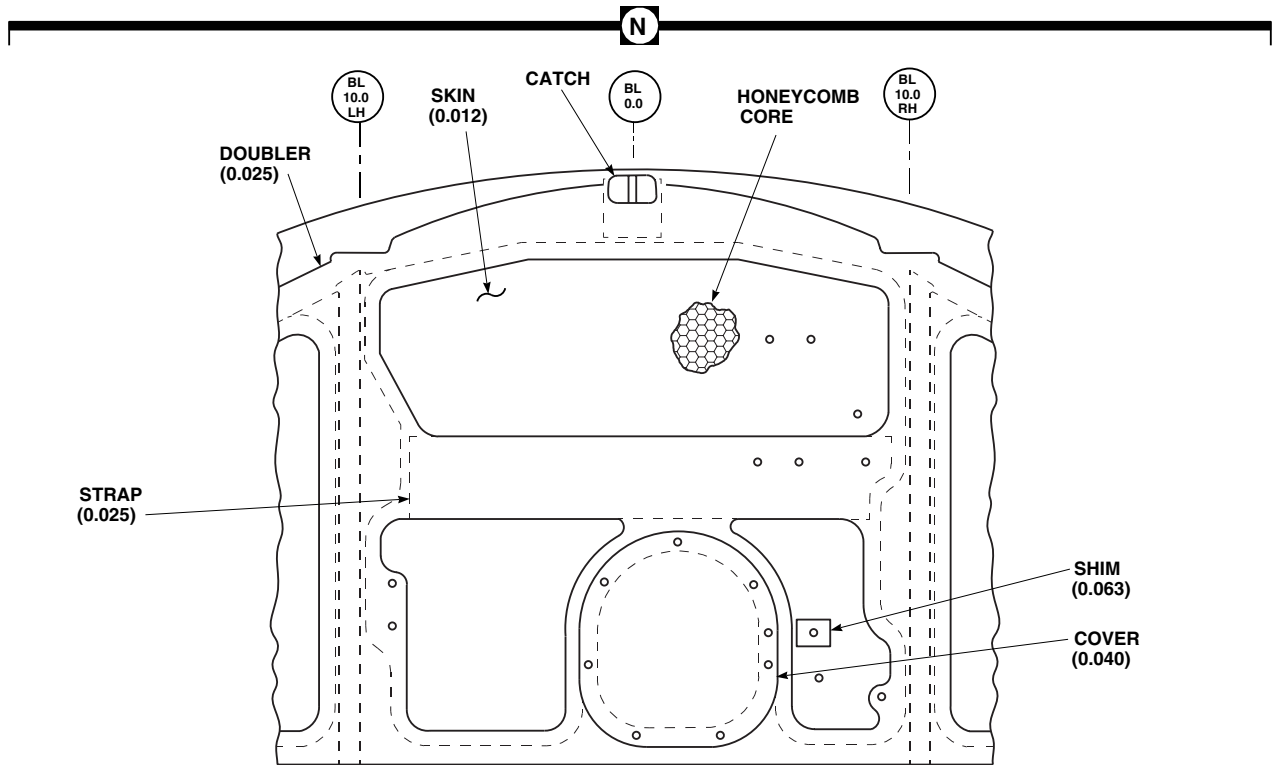


EFFECTIVITY

UH-60A 79-23340 - SUBQ

UH-60L HH-60A HH-60L

LOOKING DOWN
ELECTRONICS COMPARTMENT FLOOR



EFFECTIVITY

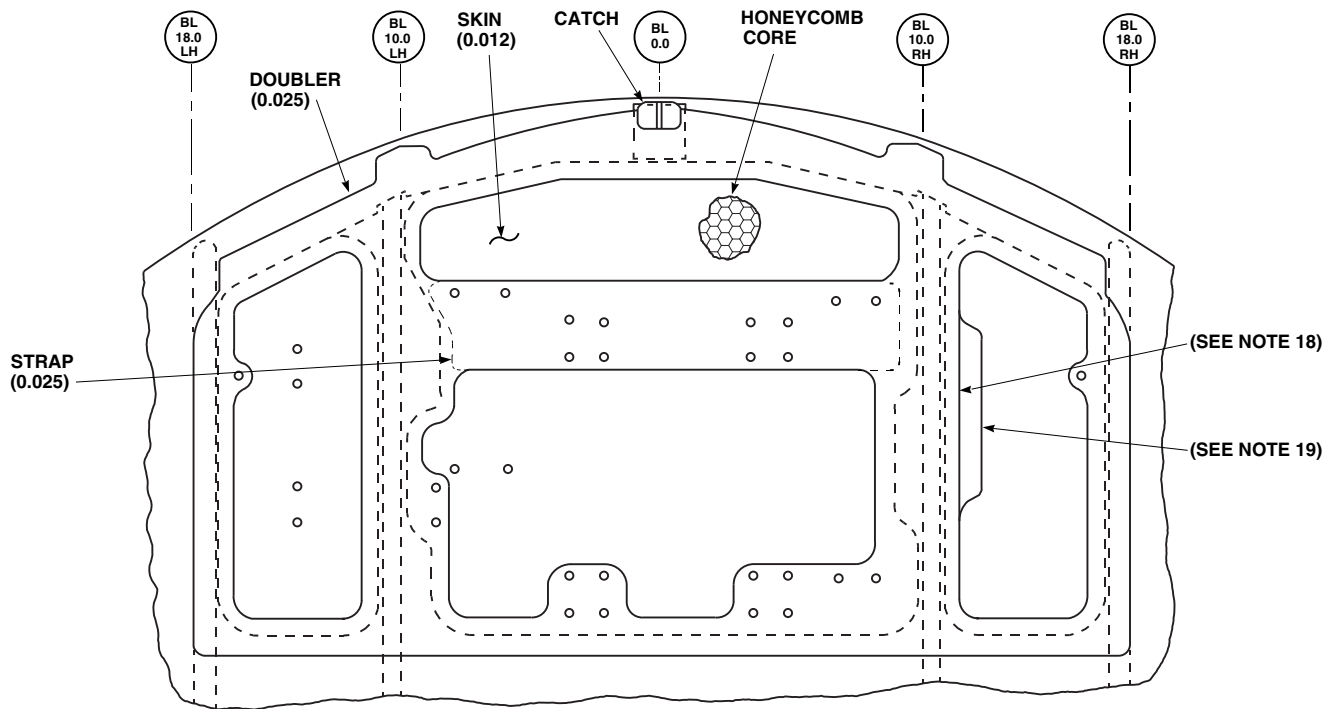
UH-60A 79-23282 - 79-23339

LOOKING DOWN
ELECTRONICS COMPARTMENT FLOOR

AB2650_19A
SA

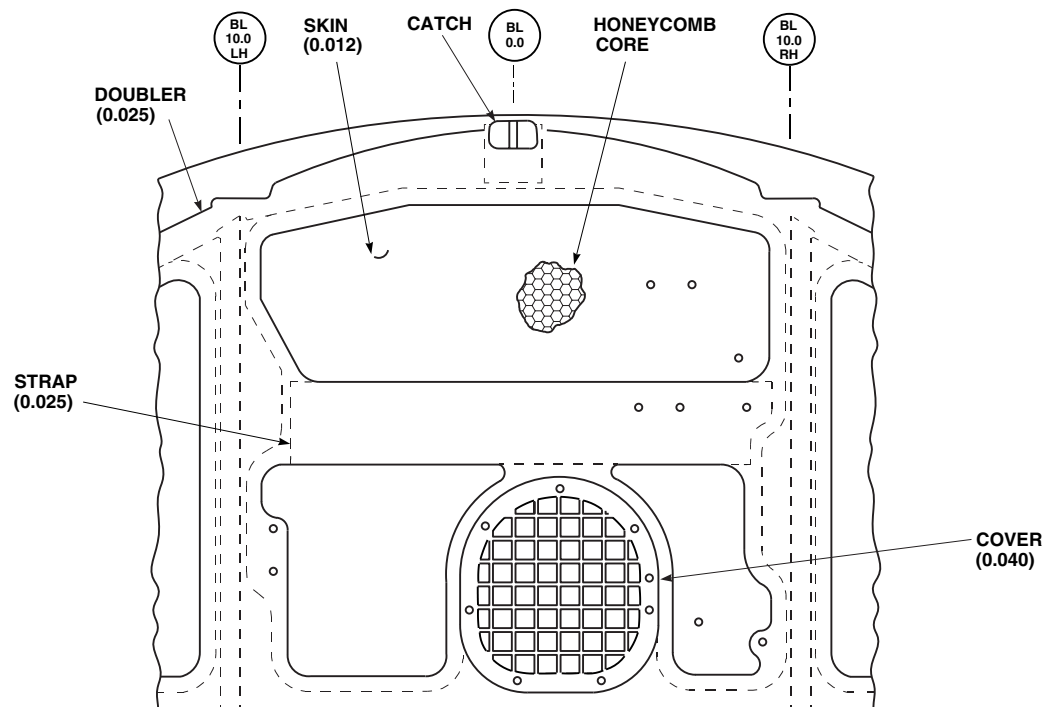
Figure 1. Nose Section Framing. (Sheet 19 of 22)

REPAIR NOSE SECTION FRAMING - Continued



EFFECTIVITY
EH-60A

LOOKING DOWN
ELECTRONICS COMPARTMENT FLOOR



EFFECTIVITY
UH-60A 77-22714 - 79-23281

LOOKING DOWN
ELECTRONICS COMPARTMENT FLOOR

AB2650_20
SA

Figure 1. Nose Section Framing. (Sheet 20 of 22)

REPAIR NOSE SECTION FRAMING - Continued

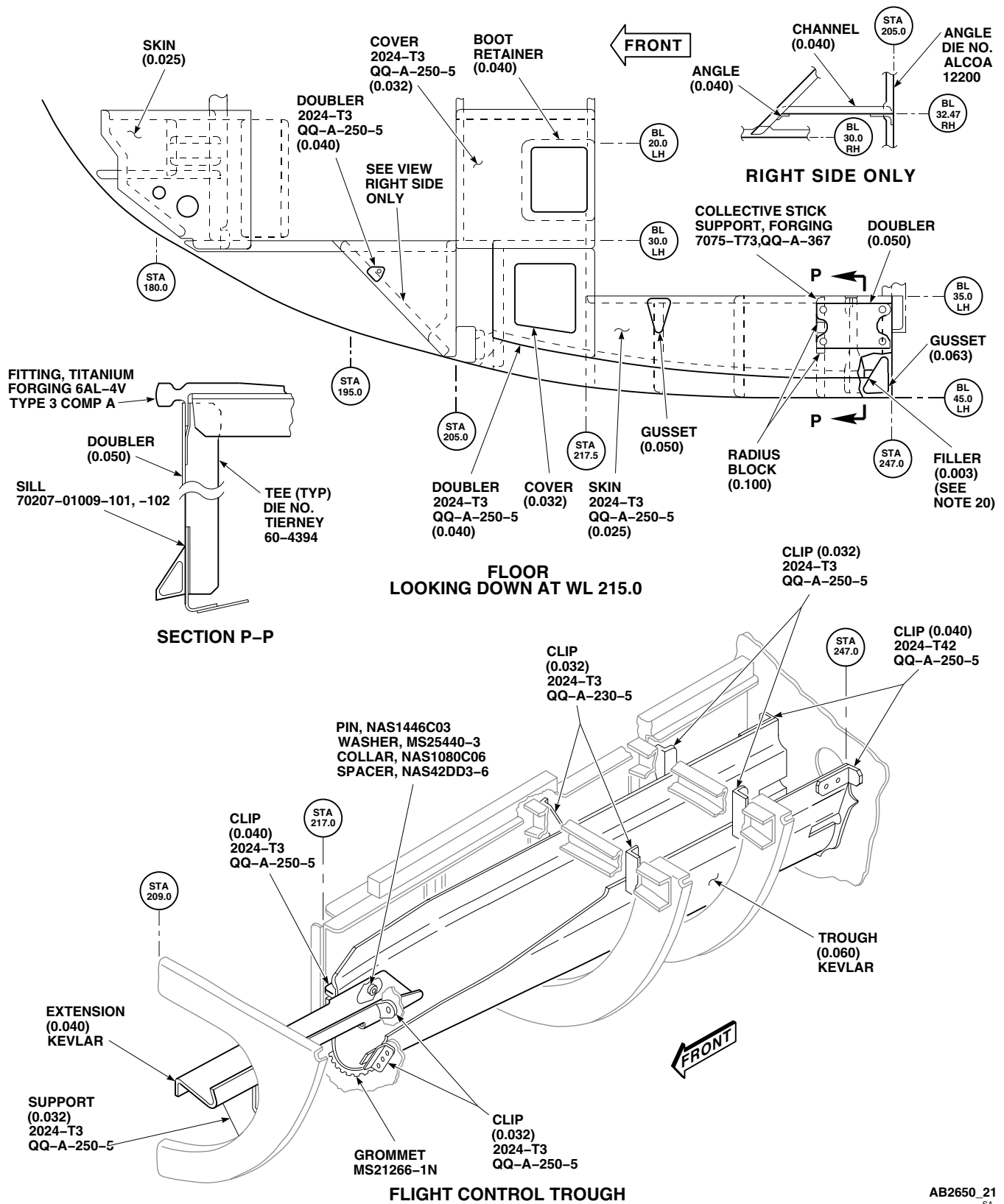


Figure 1. Nose Section Framing. (Sheet 21 of 22)

REPAIR NOSE SECTION FRAMING - Continued

NOTES:

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY, 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. UH-60A 86-24516 - SUBQ UH-60L EH-60A 85-24469 - SUBQ
4. UH-60A 87-26005 - SUBQ UH-60L UH-60A 87-24662 - SUBQ MWO 50-42
5. UH-60A 88-26085 - SUBQ UH-60L
6. UH-60A 88-26048 - 89-26173 UH-60L 89-26144 - 89-26149 (0.063). ◀
7. UH-60A 89-26134 - SUBQ UH-60L ▶
8. ALCLAD 2024-T3, QQ-A-250-5.
9. SANDWICH CONSTRUCTION: ALUMINUM HONEYCOMB MIL-C-7438 AND ALUMINUM SKINS.
10. UH-60A 84-23953 - 84-23962 - SUBQ ▶ EH-60A 84-24017 - 85-24478 TOP SKIN IS 7075-T6, QQ-A-250-13 (0.020); BOTTOM SKIN IS 7075-T6, QQ-A-250-13 (0.016). ◀
11. UH-60A 86-24648 - SUBQ ▶ UH-60L EH-60A 85-24479 - SUBQ TOP AND BOTTOM SKINS ARE 2024-T3, QQ-A-250-5 (0.020). ◀
12. UH-60A 77-22714 - 78-22979
13. UH-60A 78-22979 - SUBQ UH-60L
14. UH-60A 86-24554 - SUBQ UH-60L EH-60A 86-24561 - SUBQ
15. UH-60A 77-22714 - 78-22965 BORON EPOXY. ◀ UH-60A 78-22966 - SUBQ UH-60L EH-60A STEEL, 4130 COND N, MIL-S-18729 (0.090). ◀
16. UH-60A 77-22714 - 78-22960 BORON EPOXY. ◀ UH-60A 78-22961 - SUBQ UH-60L EH-60A STEEL (0.016) AND ALUMINUM (0.190). ◀
17. BOTTOM SKIN: 6061-T6, QQ-A-250-11 (0.020).
18. EH-60A 84-24017 - 86-24561
19. EH-60A 86-24562 - SUBQ
20. UH-60L 92-26419 - SUBQ
21. MWO 50-64
22. APPLY SEALING COMPOUND, ITEM 273, WP 1803 00, AROUND ENTIRE PERIMETER OF HINGE, WHERE ACCESSIBLE.
23. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2650_22B
SA

Figure 1. Nose Section Framing. (Sheet 22 of 22)

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****NOSE SECTION PLATING REPAIR**

This WP supersedes WP 0385 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

TM 1-1500-344-23

TM 55-1500-345-23

Material/Parts

Insulating Compound Item 151, WP 1803 00

Sealing Compound, Item 273, WP 1803 00

Sealing Compound, Item 276, WP 1803 00

WP 0377 00

WP 0378 00

WP 0379 00

WP 0381 00

WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

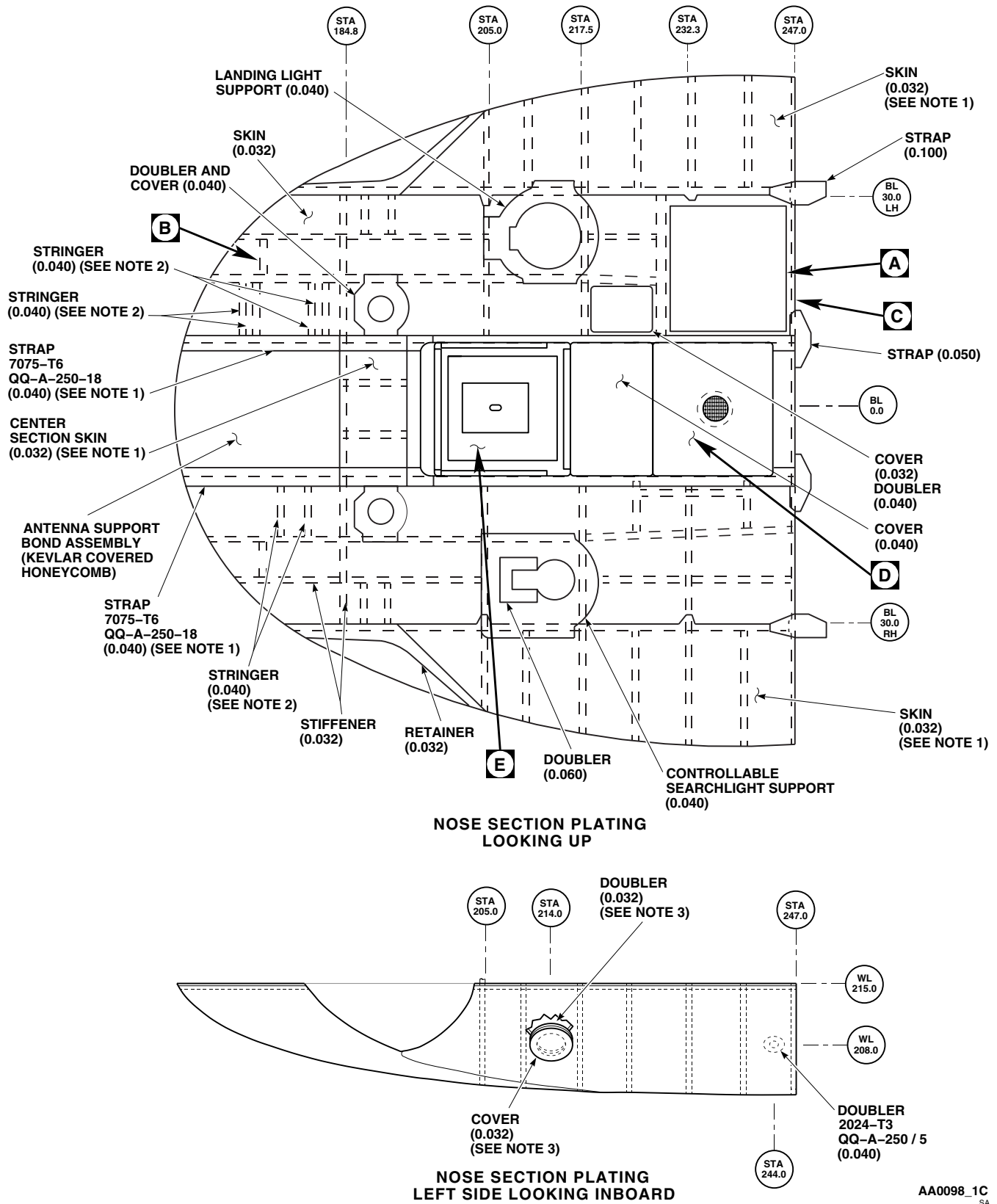
QQ-A-250

TM 1-1500-204-23

REPAIR

1. All plating with the exception of reinforced plastic covers is primary structure (Figure 1). Evaluate damage (WP 0377 00).
2. Refer to WP 0378 00 for typical skin repairs. Coordinate repairs with TM 1-1500-204-23 requirements.
3. Refer to WP 0379 00 for typical reinforced plastic repairs. Coordinate repairs with TM 1-1500-204-23 requirements.
4. For corrosion protection and watertightness, seal all plating repairs (WP 0377 00 and TM 1-1500-344-23).
5. Touch up paint (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued



AA0098_1C
SA

Figure 1. Repair Nose Section Plating. (Sheet 1 of 5)

REPAIR - Continued

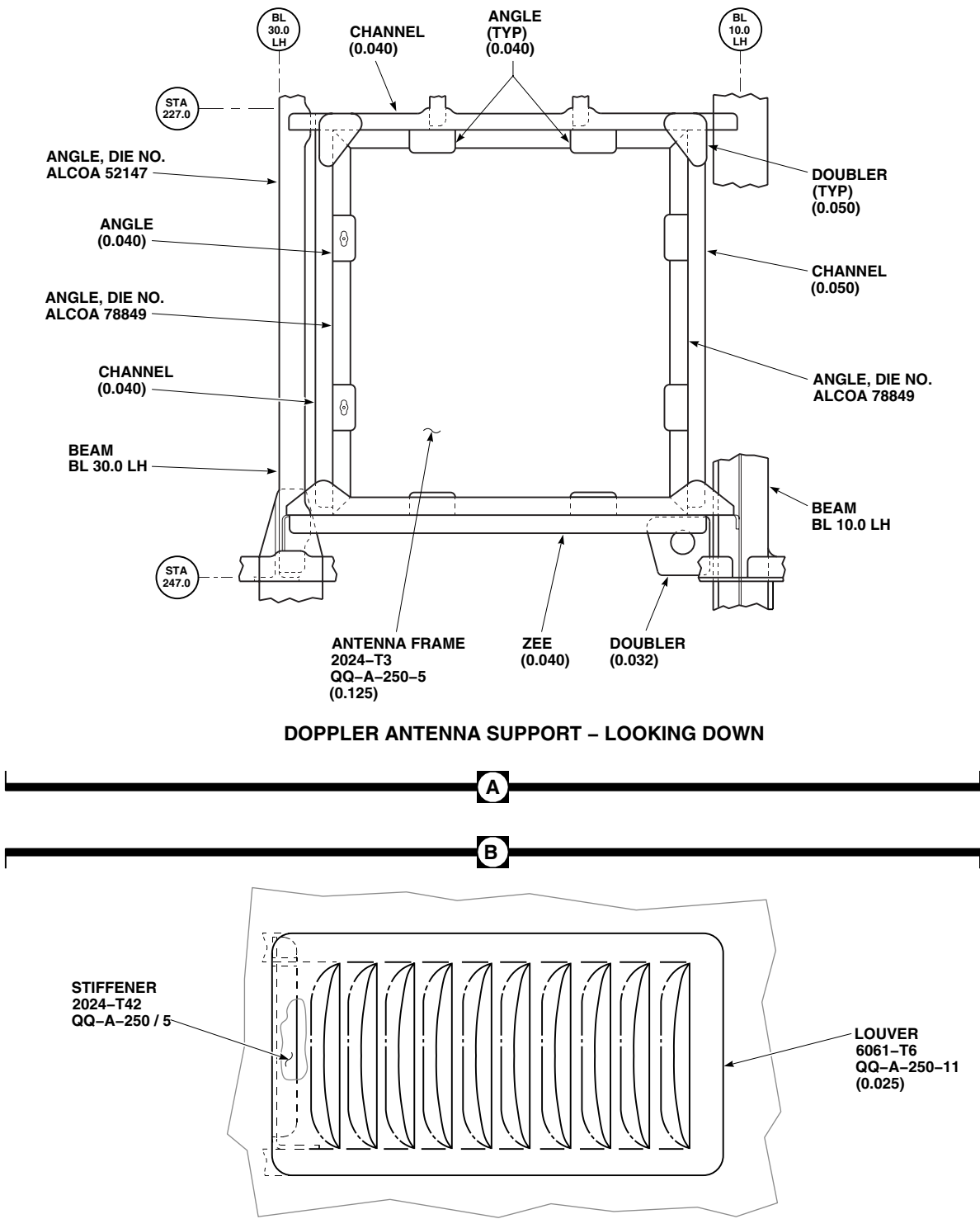
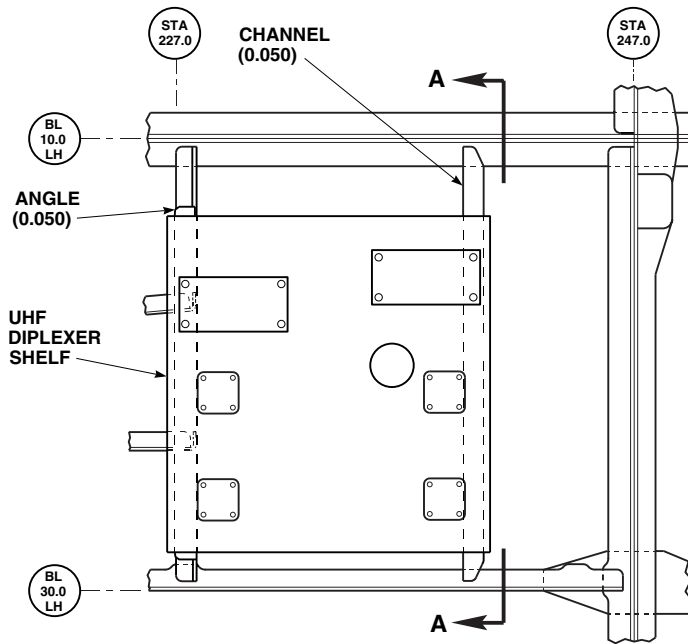


Figure 1. Repair Nose Section Plating. (Sheet 2 of 5)

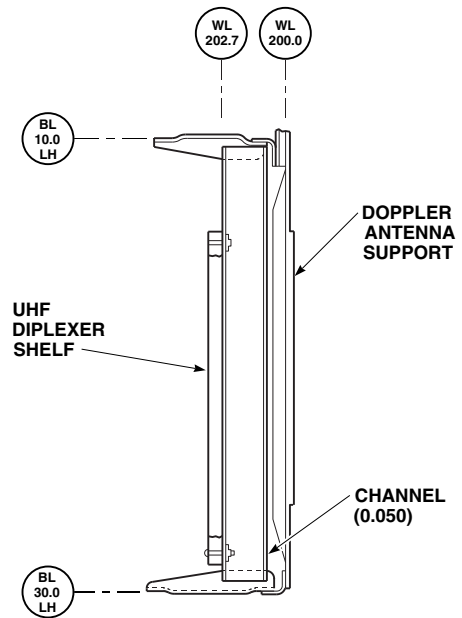
REPAIR - Continued



EFFECTIVITY

EH60A 84-24017 - 87-24664

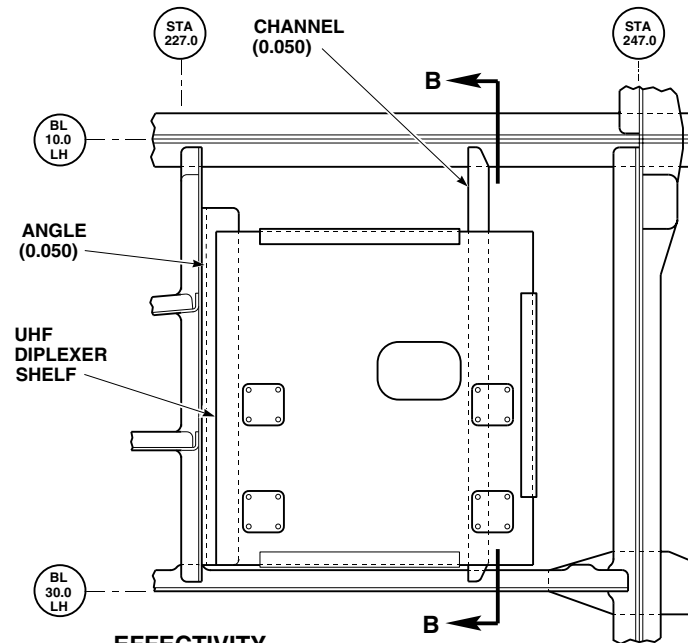
UHF DIPLEXER SHELF - LOOKING DOWN



SECTION A-A

C

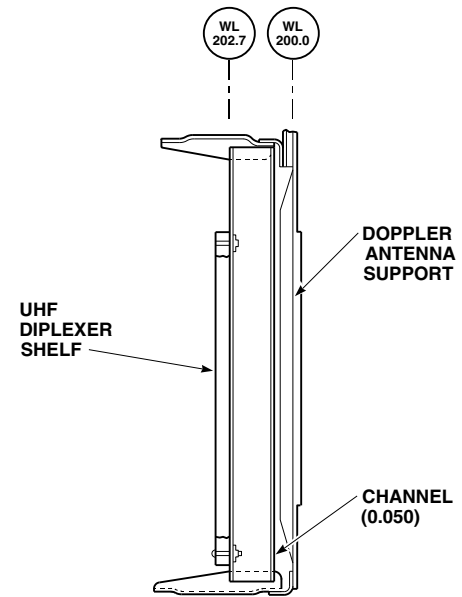
C



EFFECTIVITY

EH60A 87-24665 - SUBQ

UHF DIPLEXER SHELF - LOOKING DOWN



SECTION B-B

AA0098_3B
SA

Figure 1. Repair Nose Section Plating. (Sheet 3 of 5)

REPAIR - Continued

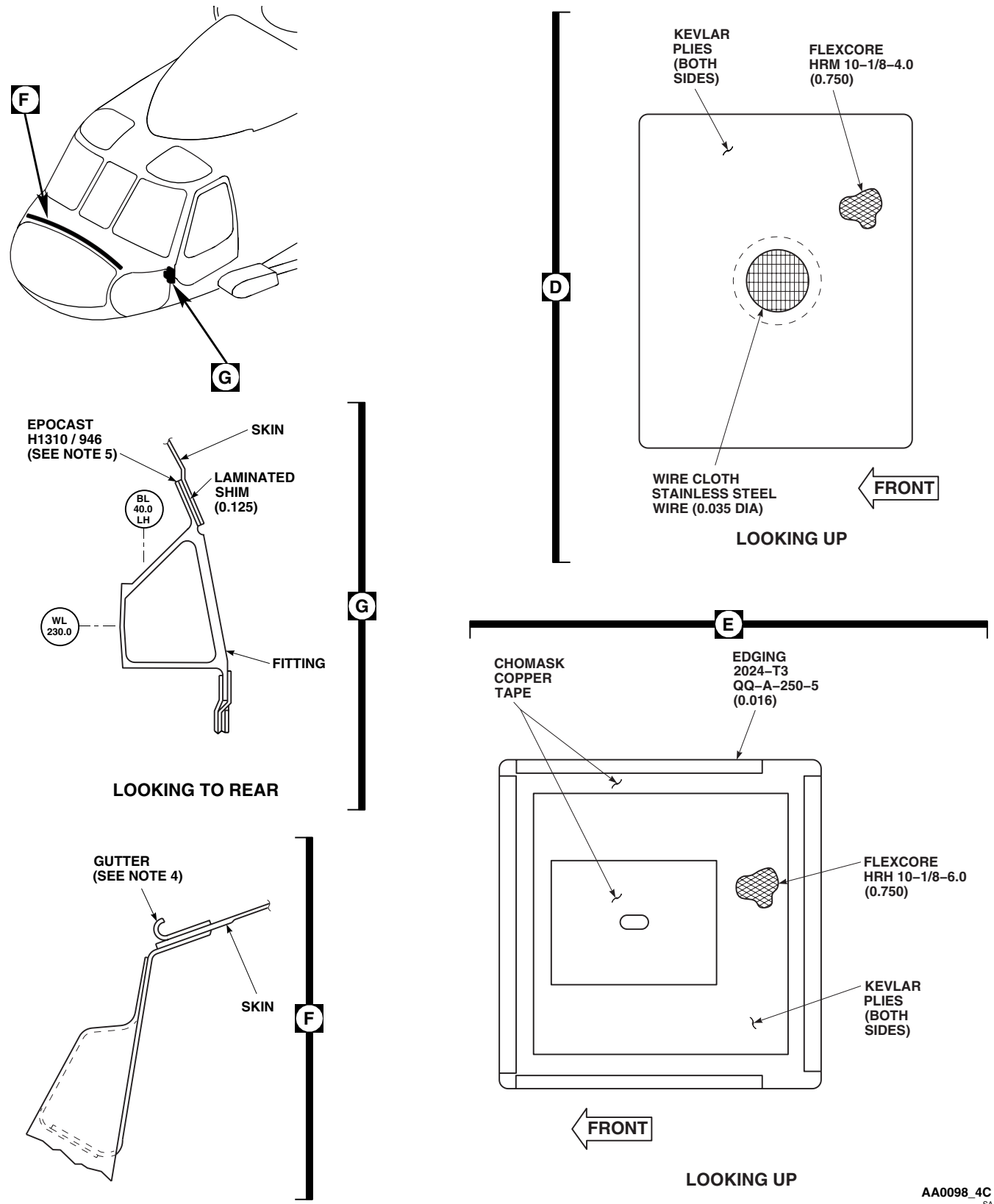


Figure 1. Repair Nose Section Plating. (Sheet 4 of 5)

REPAIR - Continued**NOTES:**

1. SEAL WITH SEALING COMPOUND, ITEM 273, WP 1803 00. REFER TO WP 0377 00.
2. **UH60A 84-23974 - SUBQ** **UH60L** **EH60A**
3. **UH60A 81-23571 - SUBQ** **UH60L** **EH60A**
4. **UH60A 87-24162 - SUBQ** **UH60L** **EH60A 86-24573 - SUBQ**
5. USE INSULATING COMPOUND, ITEM 151, WP 1803 00 AND SEALING COMPOUND, ITEM 276, WP 1803 00 FOR SHIMMING AS REQUIRED.
6. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0098 ^{5E}
SA

Figure 1. Repair Nose Section Plating. (Sheet 5 of 5)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

CABIN SECTION FORWARD FRAMING REPAIR

This WP supersedes WP 0386 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-STD-18729

MIL-STD-670

Material/Parts

Adhesive, Item 11, WP 1803 00

Pin/Collar Assembly, HL20PB86TW12-11

Sealing Compound, Item 273, WP 1803 00

QQ-A-200

QQ-A-225

QQ-A-250

QQ-A-367

TM 1-1500-204-23

TM 1-1500-344-23

TM 55-1500-345-23

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

WP 0377 00

References

AMS 5643

AMS 6359

MIL-S-22499

WP 0378 00

WP 0380 00

WP 0381 00

WP 1803 00

REPAIR CABIN SECTION FORWARD FRAMING

1. Forward cabin framing includes formed and extruded members made from 7075-T6, QQ-A-250-13, Alclad sheet stock, and 7075-T6511, QQ-A-200-11, aluminum alloy extrusion. Landing gear attachment fittings, bellcrank support fittings, and frame fittings at STA 295.0 and STA 308.0 are made from forged aluminum alloy and heat-treated to 7075-T73, QQ-A-367 condition. Tracks for the sliding cover (main rotor pylon) and gunner's window are made from 7075-T6511 extruded sections. Fittings under the gunner's window are made from 4340 steel and 4130 steel (Figure 1).
2. All forward cabin framing is primary structure. Fittings, however, are considered critical primary structure. Negligible damage to fittings is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area, using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage greater than negligible limits in critical areas must be reviewed by engineering authority for damage evaluation, and either repair or replacement. All repairs to the forward cabin structure must maintain the design strength and alignment.
3. Evaluate damage in noncritical areas (WP 0377 00).
4. Use WP 0378 00 and WP 0380 00 for typical framing repairs. Refer to TM 1-1500-204-23 for repair materials and fastener information.
5. Coordinate repairs with TM 1-1500-204-23 requirements.
6. For corrosion protection and watertightness, seal all framing repairs (WP 0377 00 and TM 1-1500-344-23).
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR CABIN SECTION FORWARD FRAMING - Continued

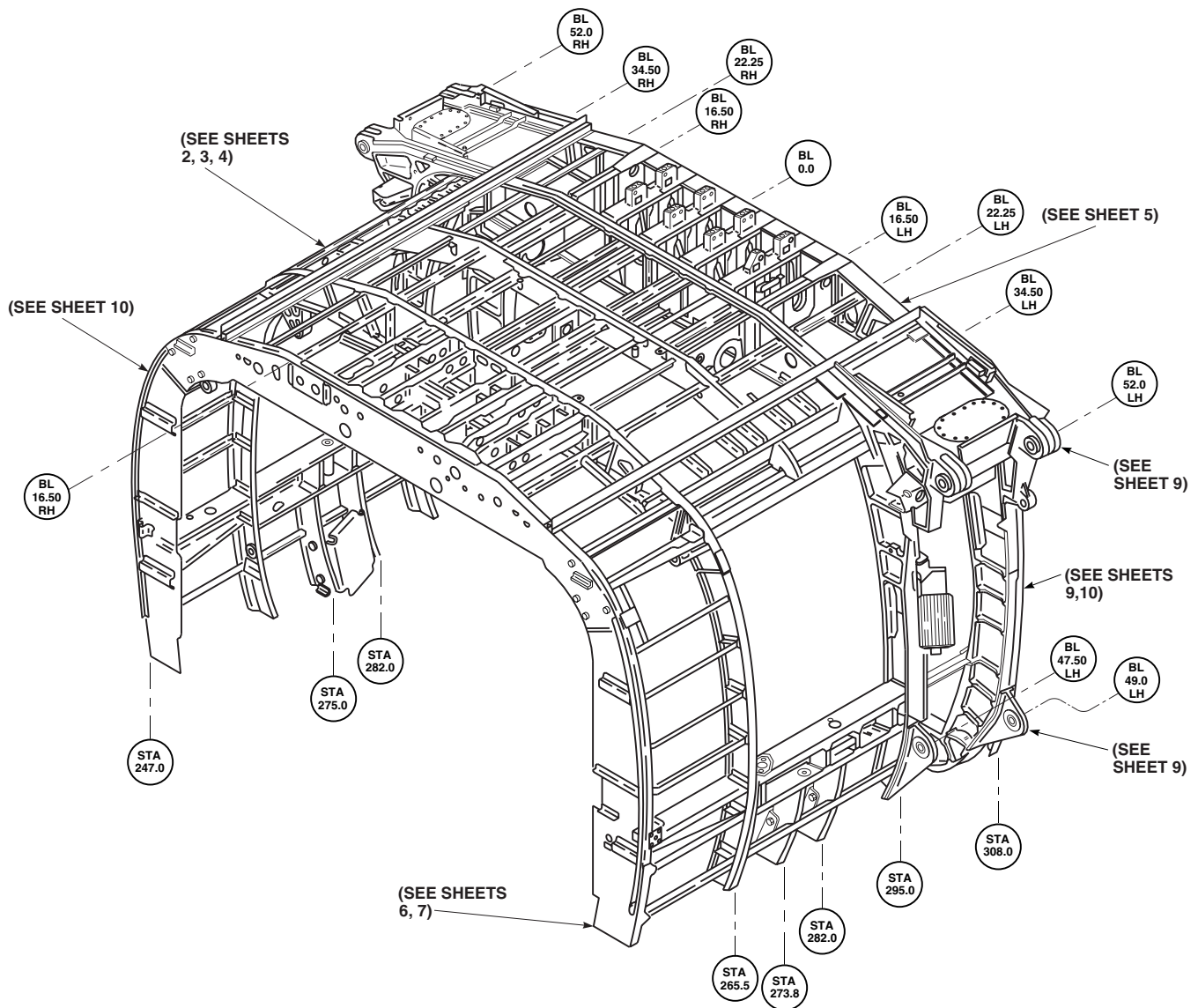
AA0100_1A
SA

Figure 1. Repair Cabin Section Forward Framing. (Sheet 1 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

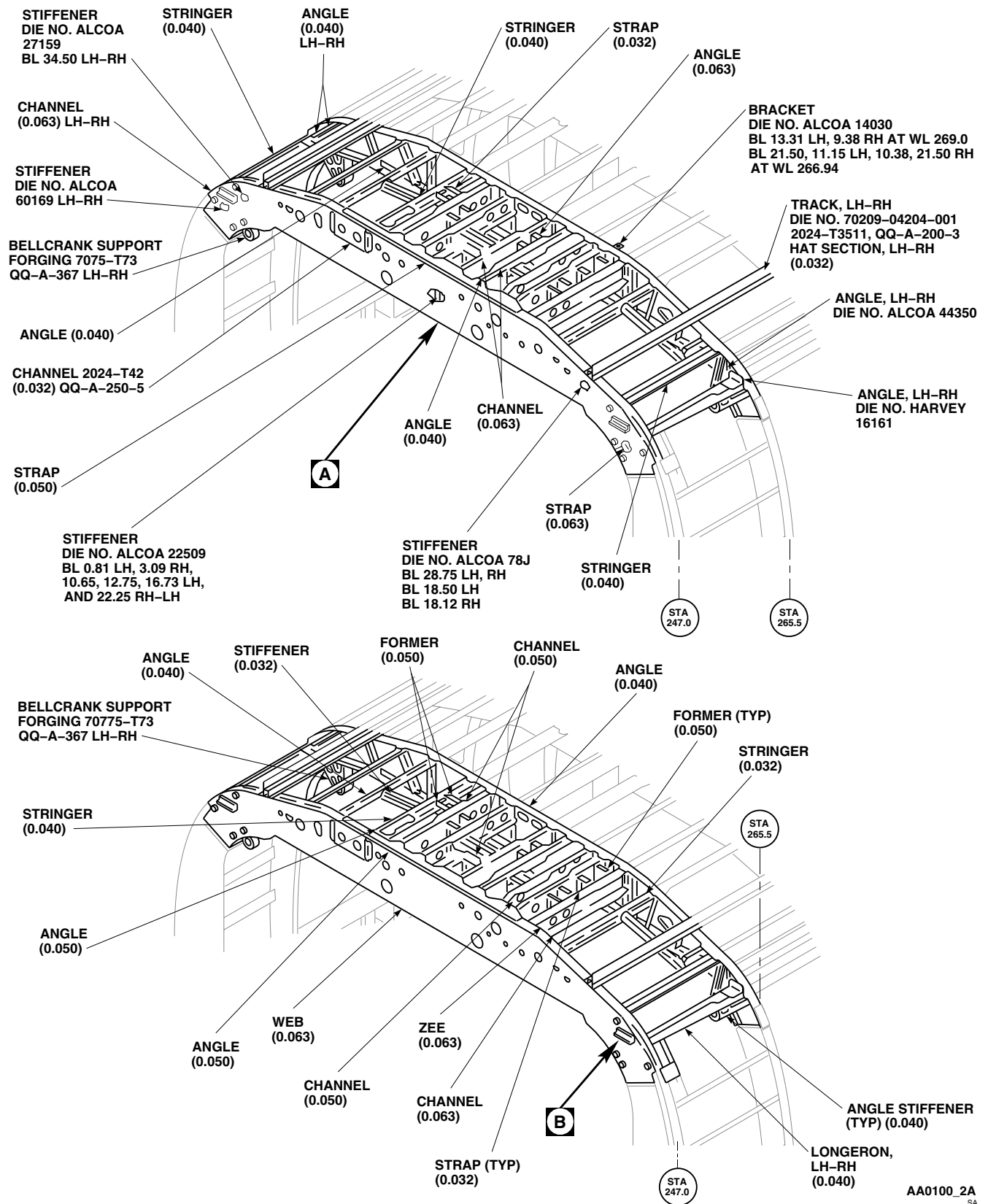


Figure 1. Repair Cabin Section Forward Framing. (Sheet 2 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

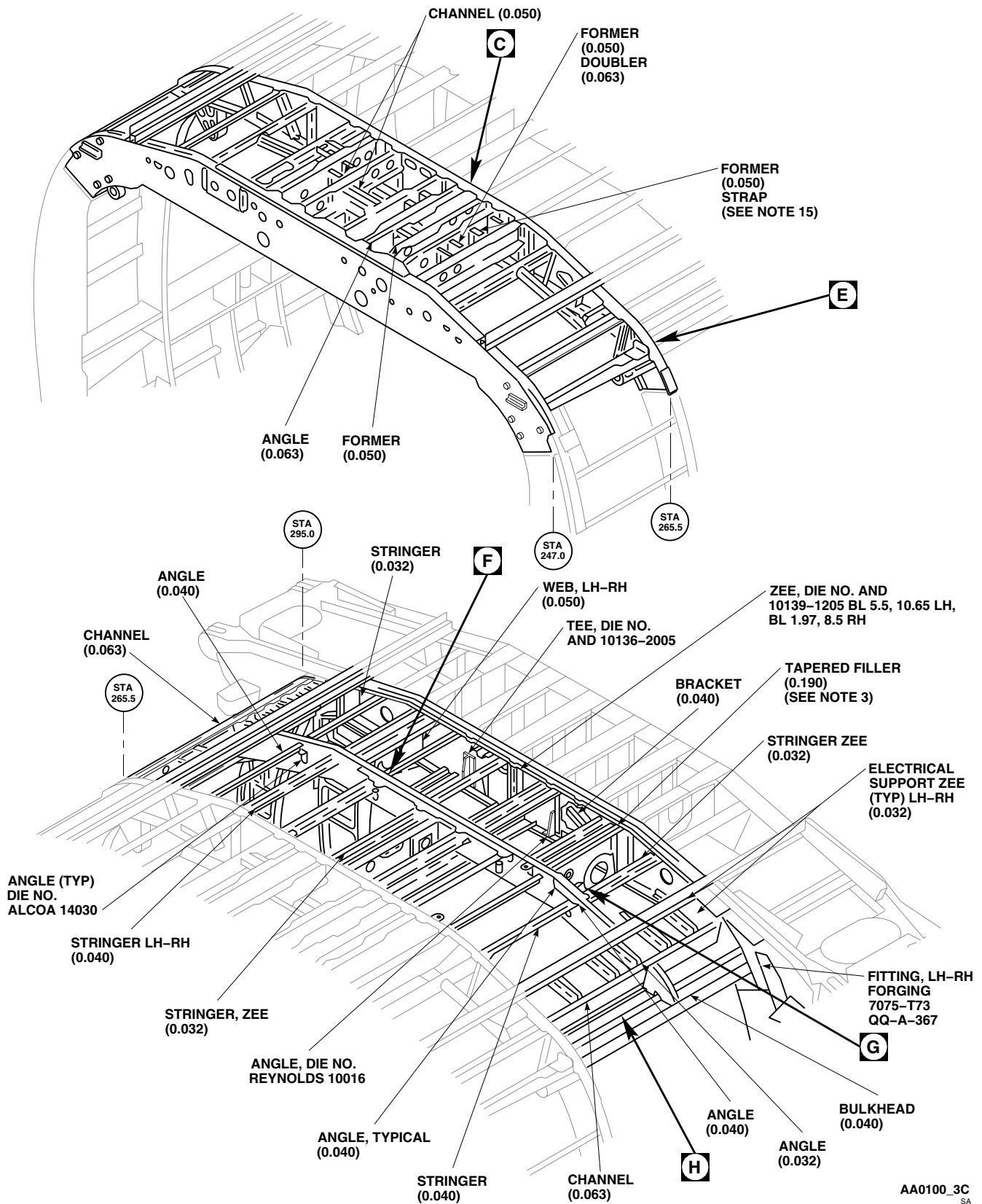


Figure 1. Repair Cabin Section Forward Framing. (Sheet 3 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

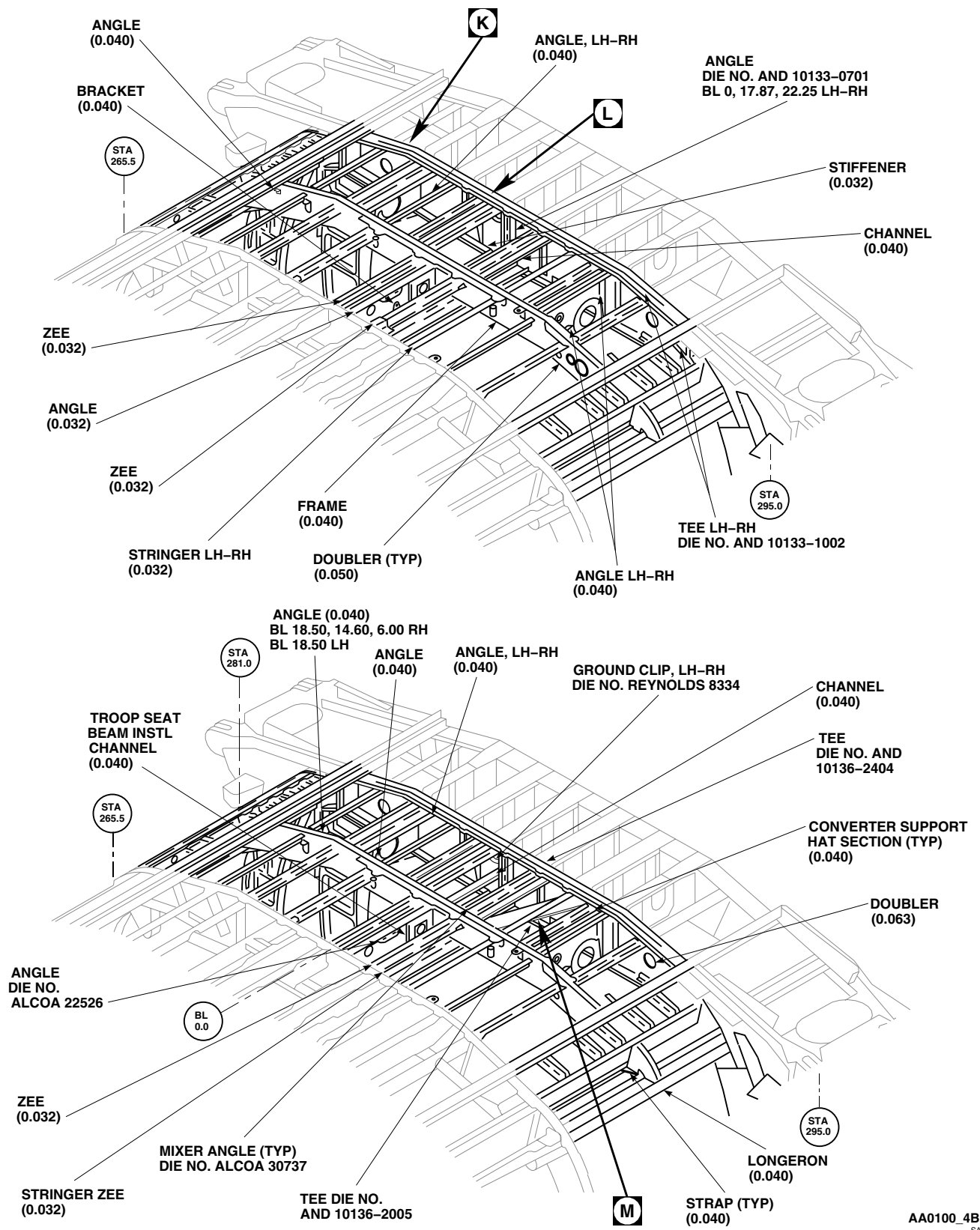


Figure 1. Repair Cabin Section Forward Framing. (Sheet 4 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

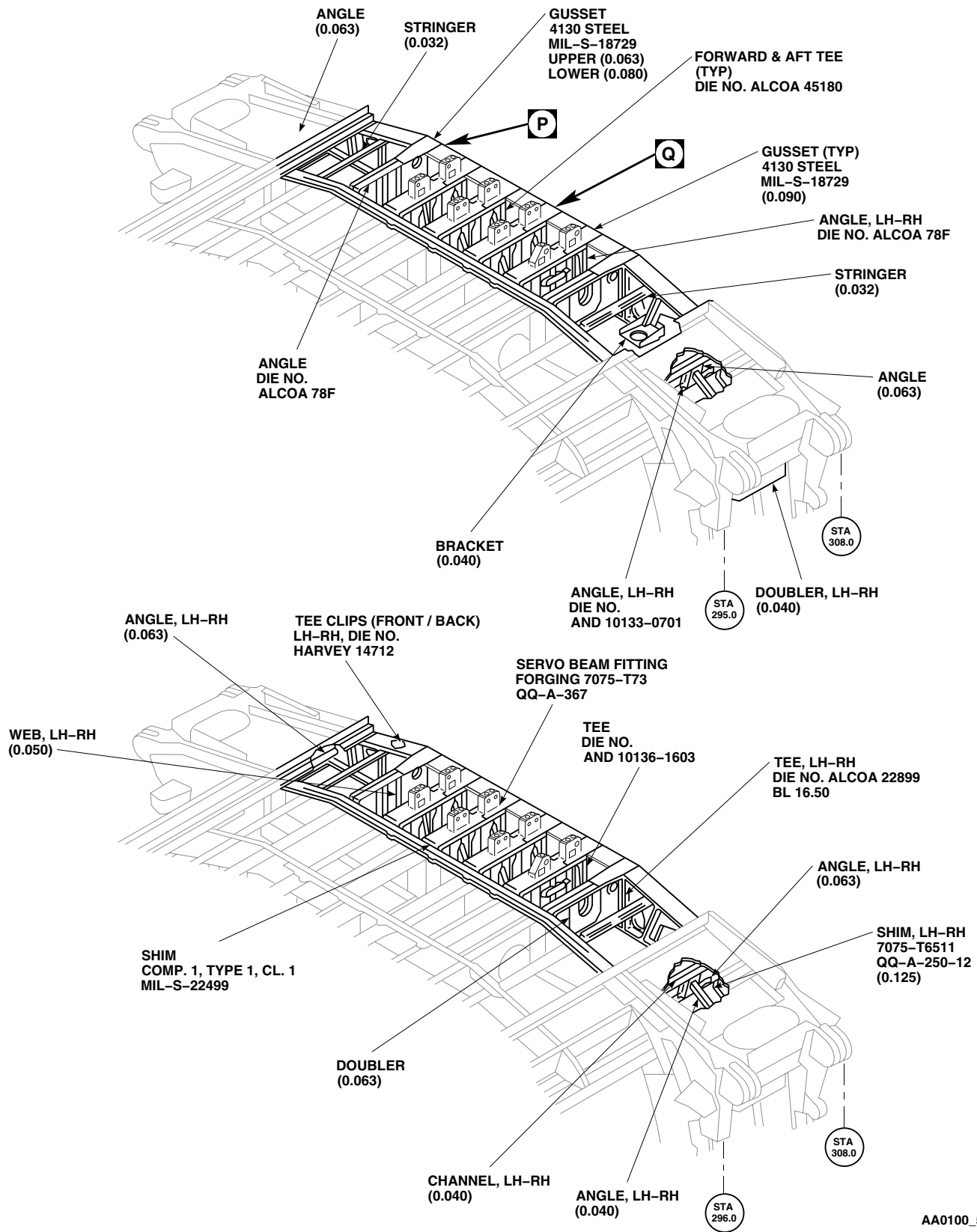


Figure 1. Repair Cabin Section Forward Framing. (Sheet 5 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

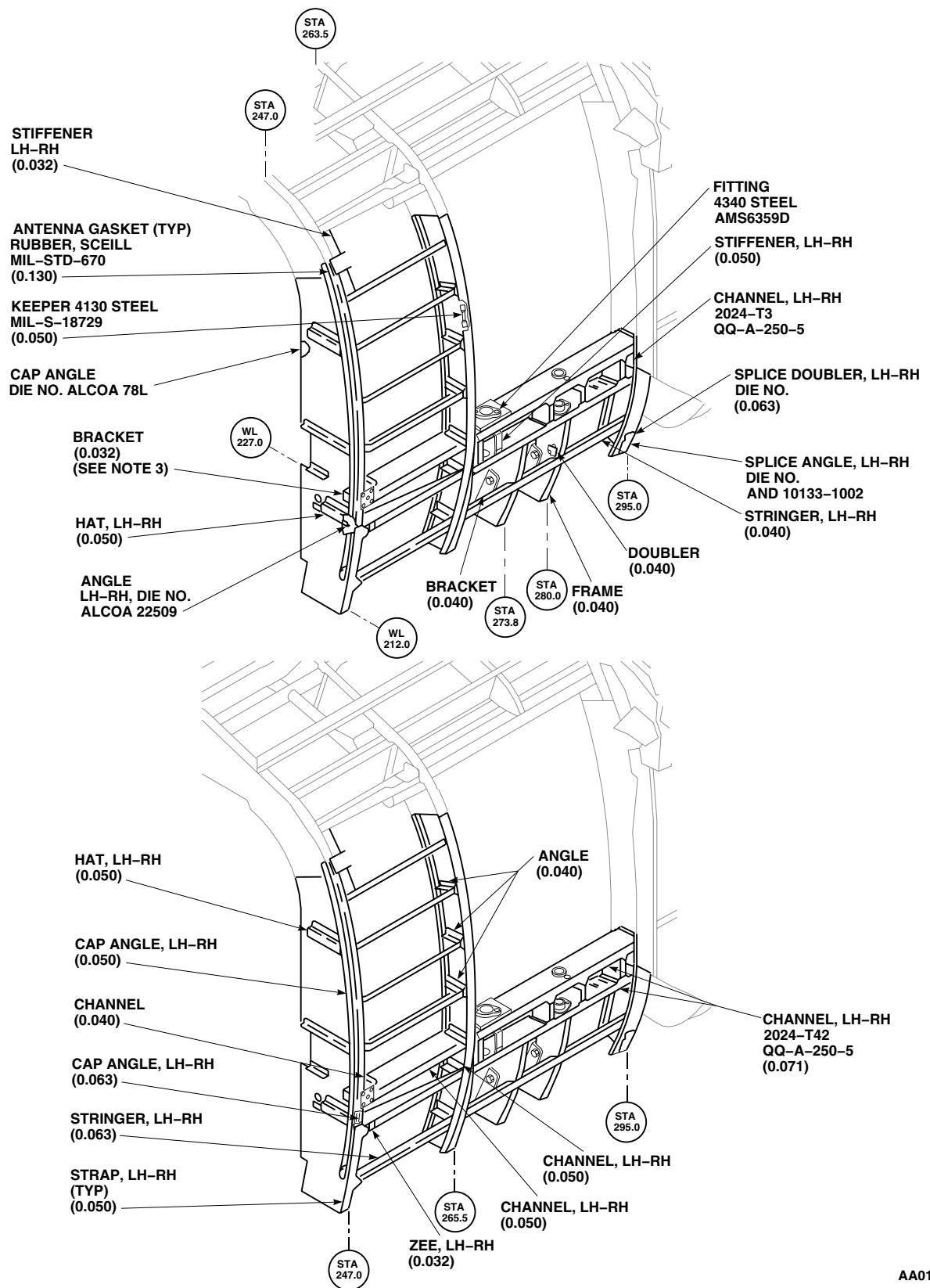


Figure 1. Repair Cabin Section Forward Framing. (Sheet 6 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

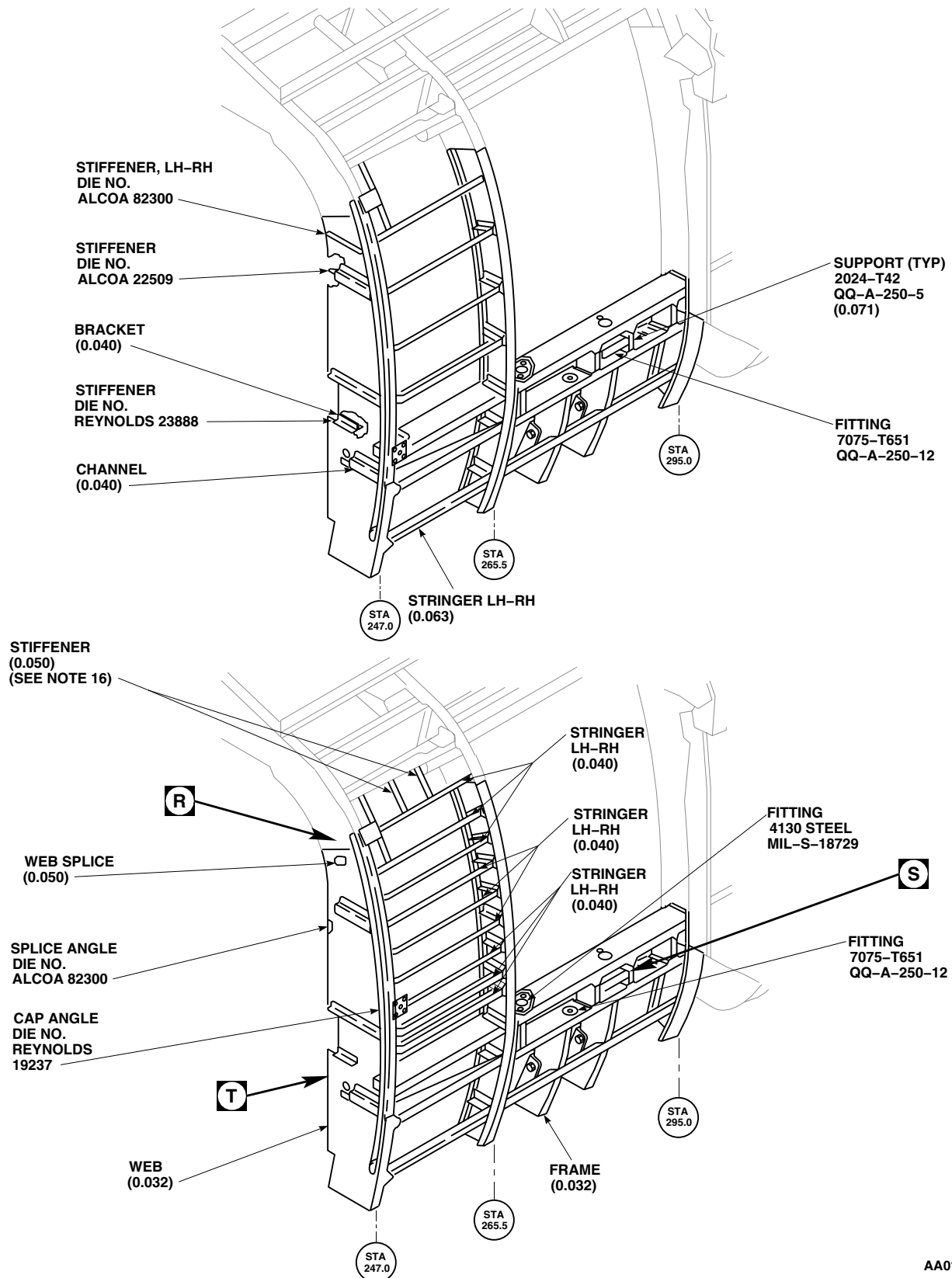
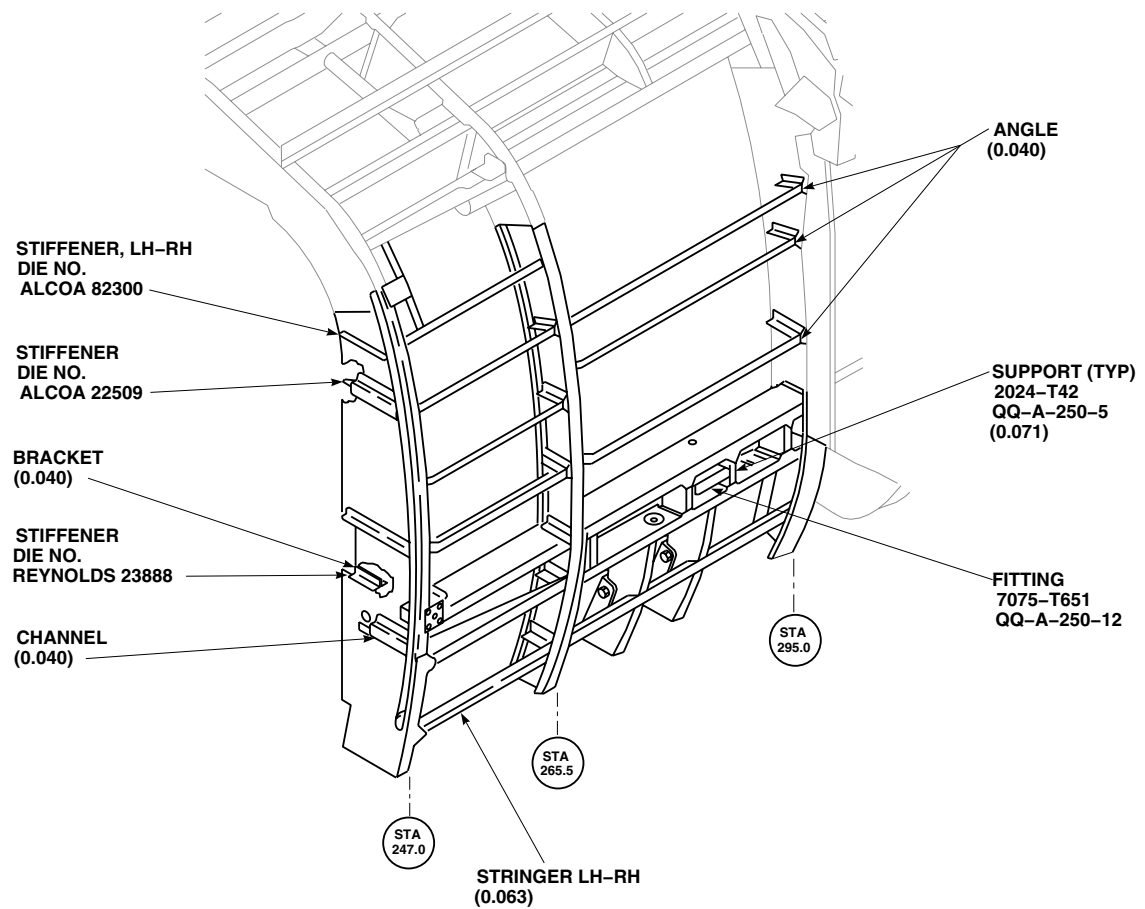


Figure 1. Repair Cabin Section Forward Framing. (Sheet 7 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued



EFFECTIVITY



HH-60A HH-60L

AA0100_8F
SA

Figure 1. Repair Cabin Section Forward Framing. (Sheet 8 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

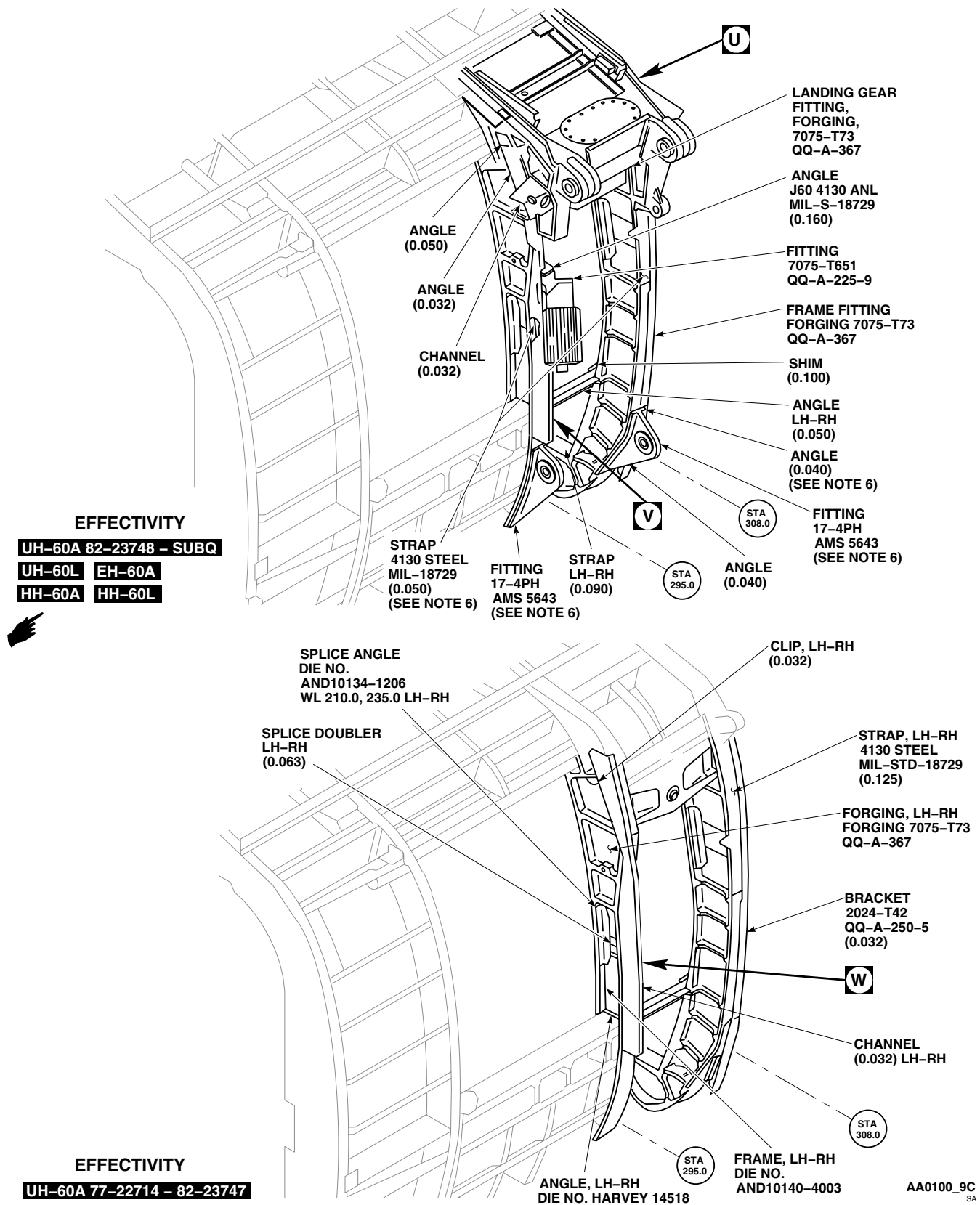


Figure 1. Repair Cabin Section Forward Framing. (Sheet 9 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

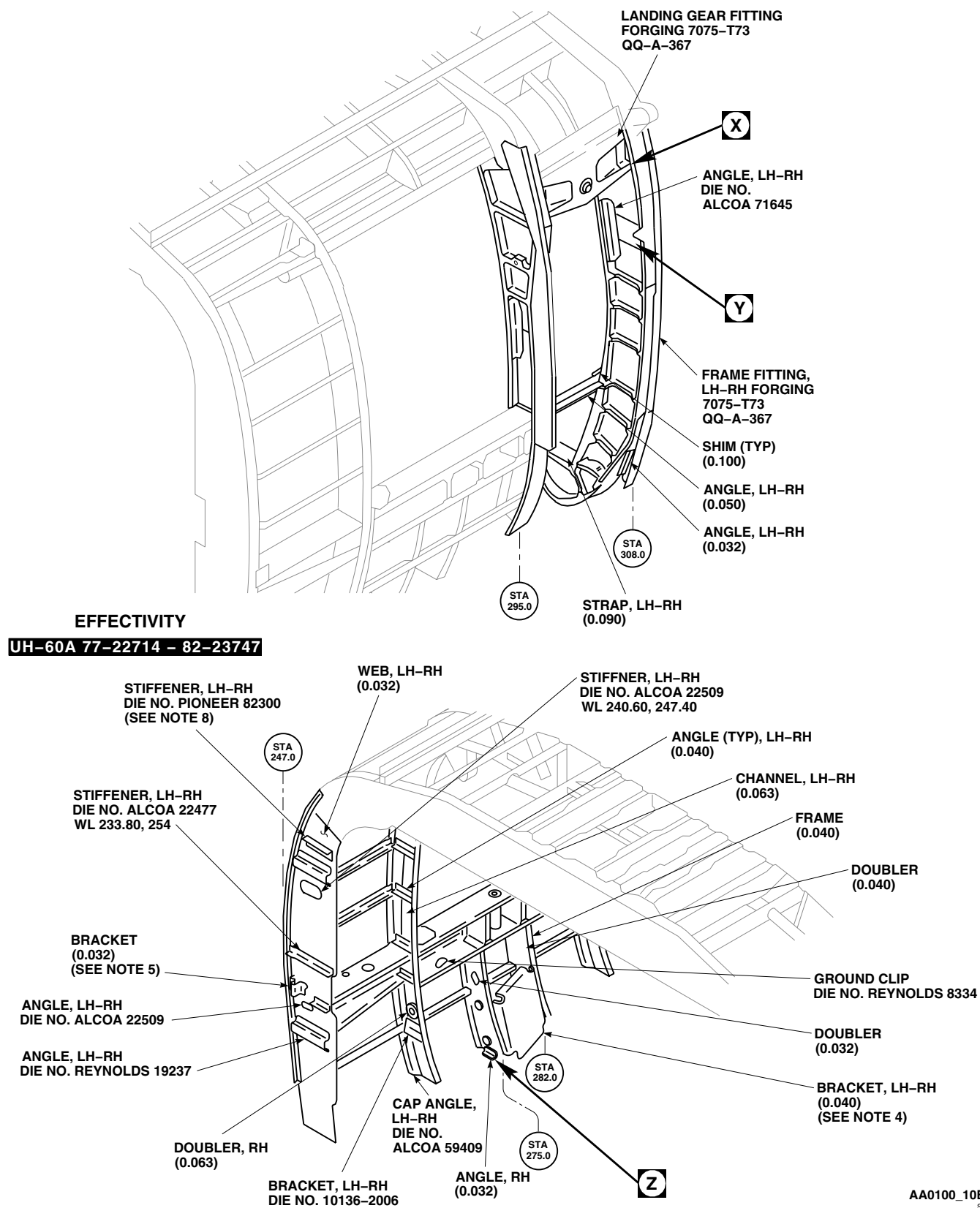


Figure 1. Repair Cabin Section Forward Framing. (Sheet 10 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

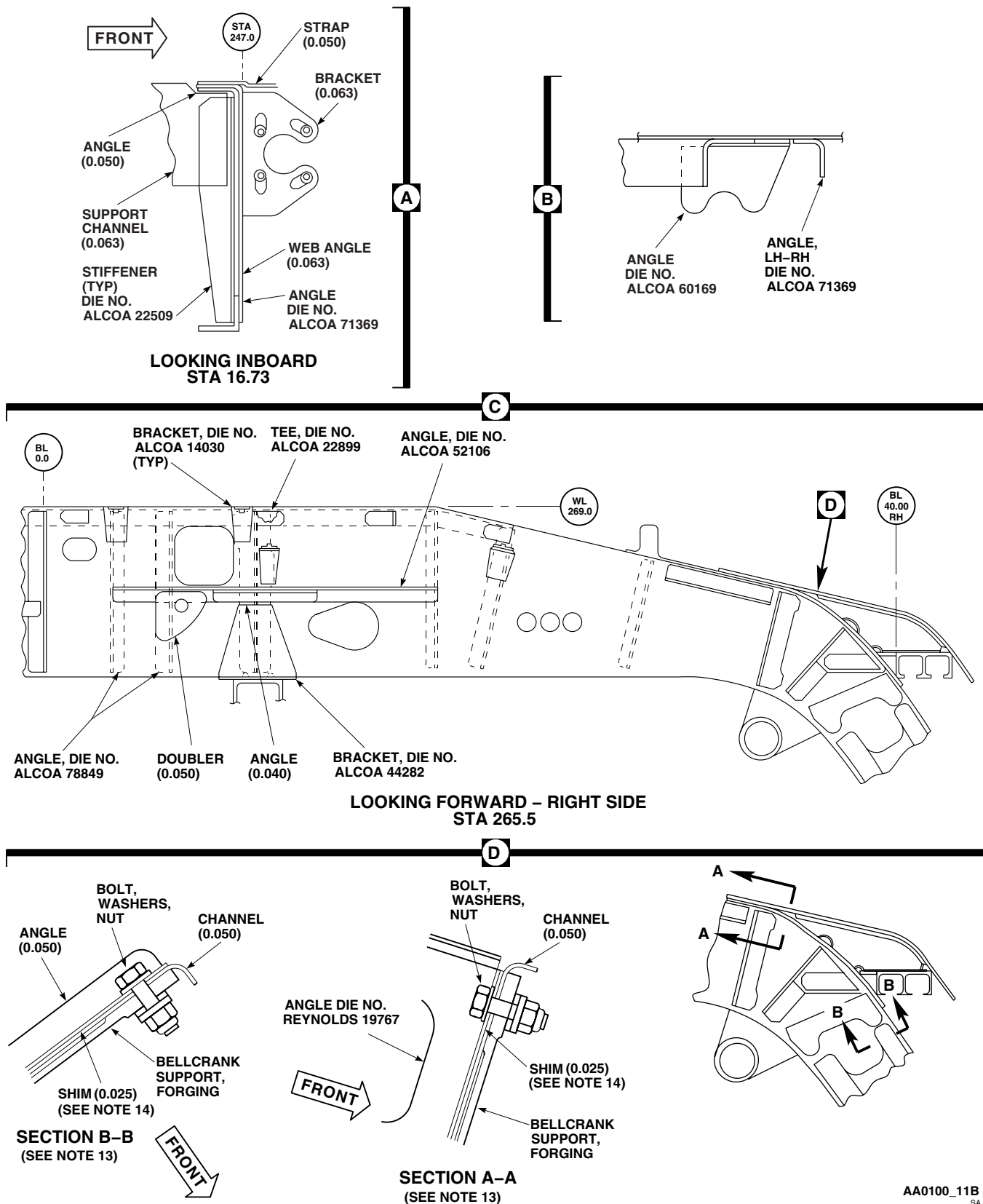
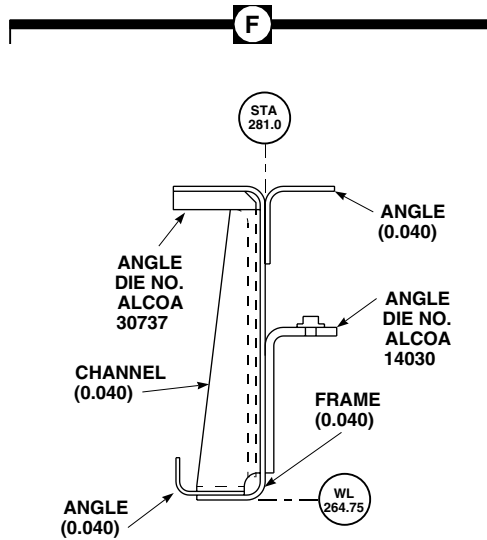
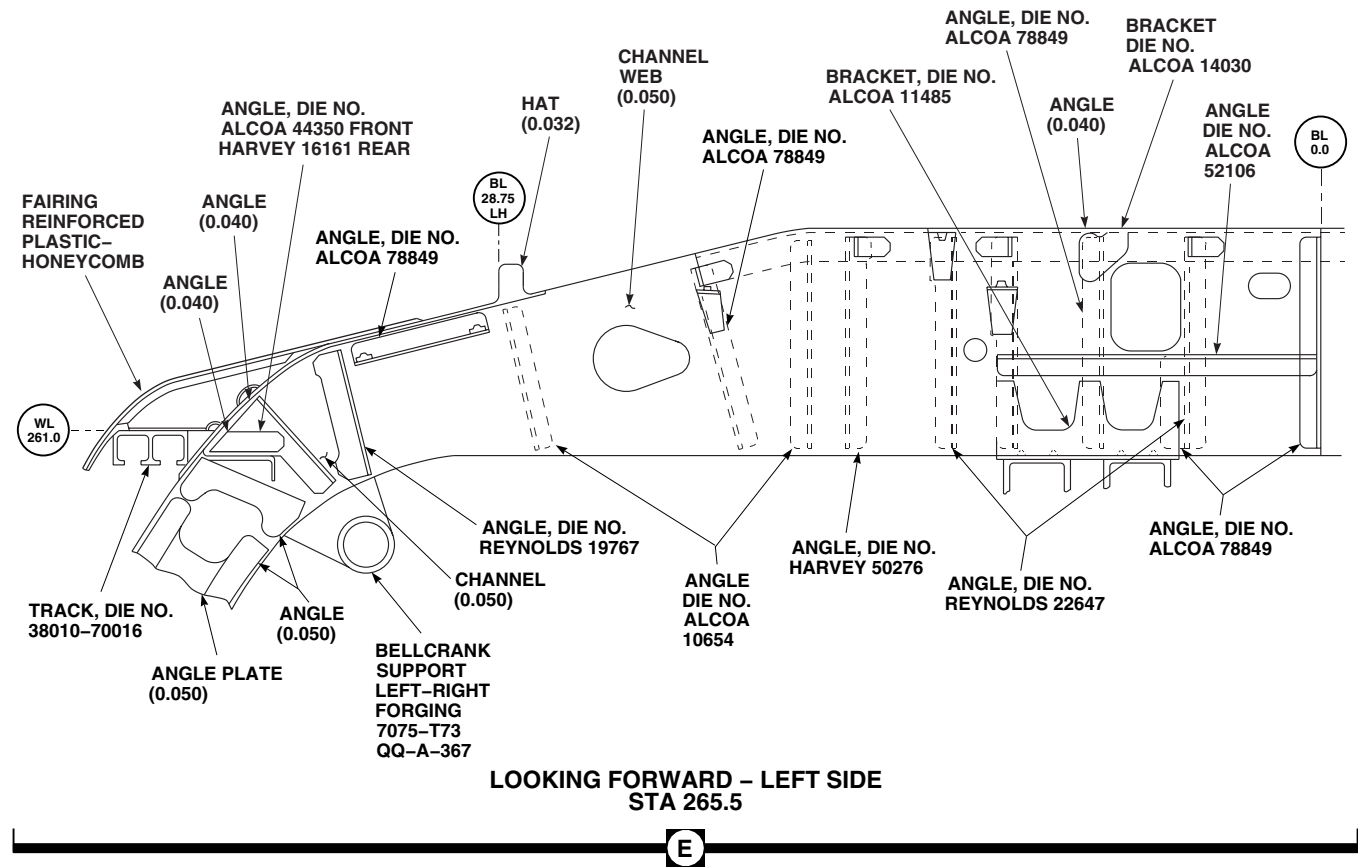


Figure 1. Repair Cabin Section Forward Framing. (Sheet 11 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued



AA0100_12B
SA

Figure 1. Repair Cabin Section Forward Framing. (Sheet 12 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

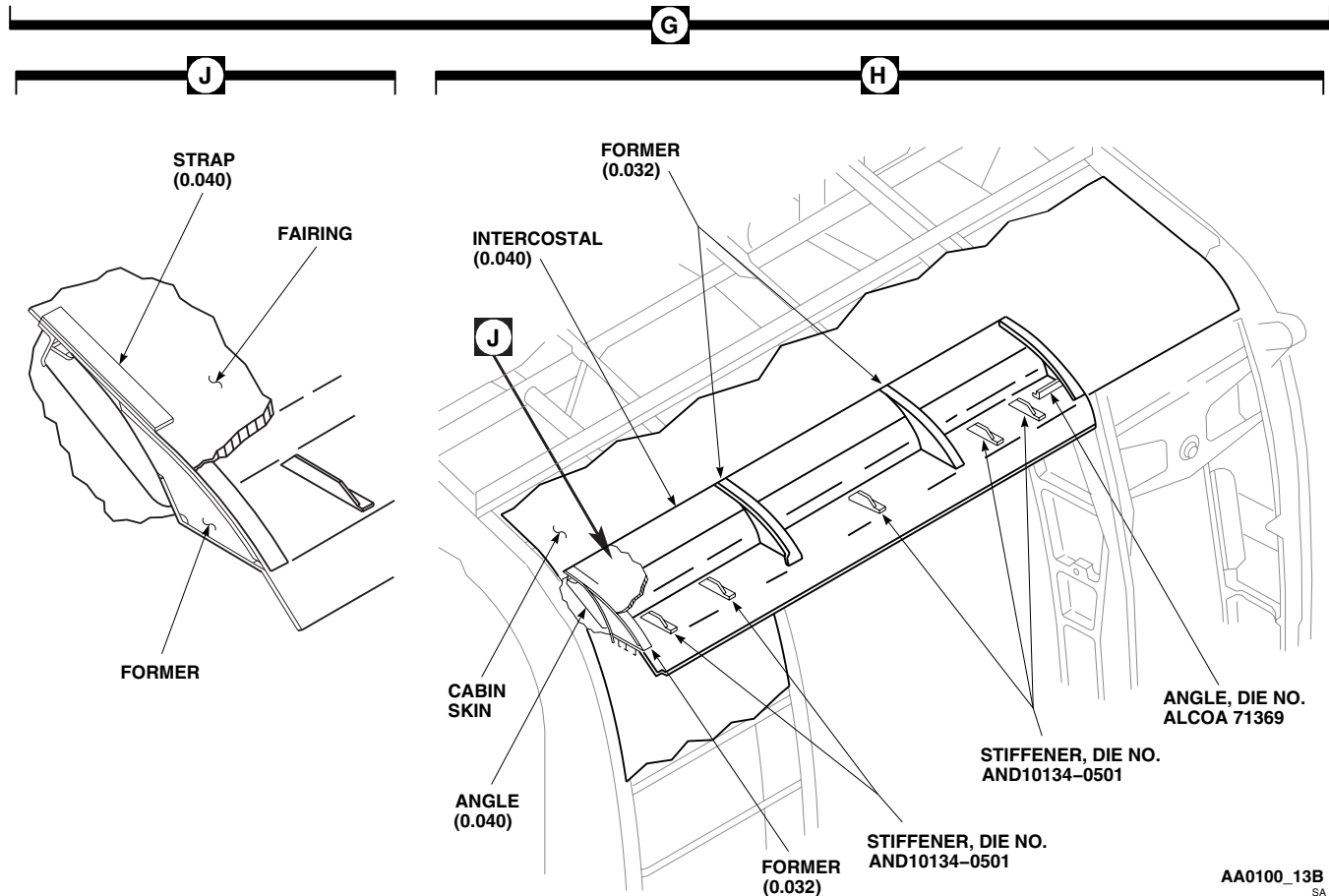
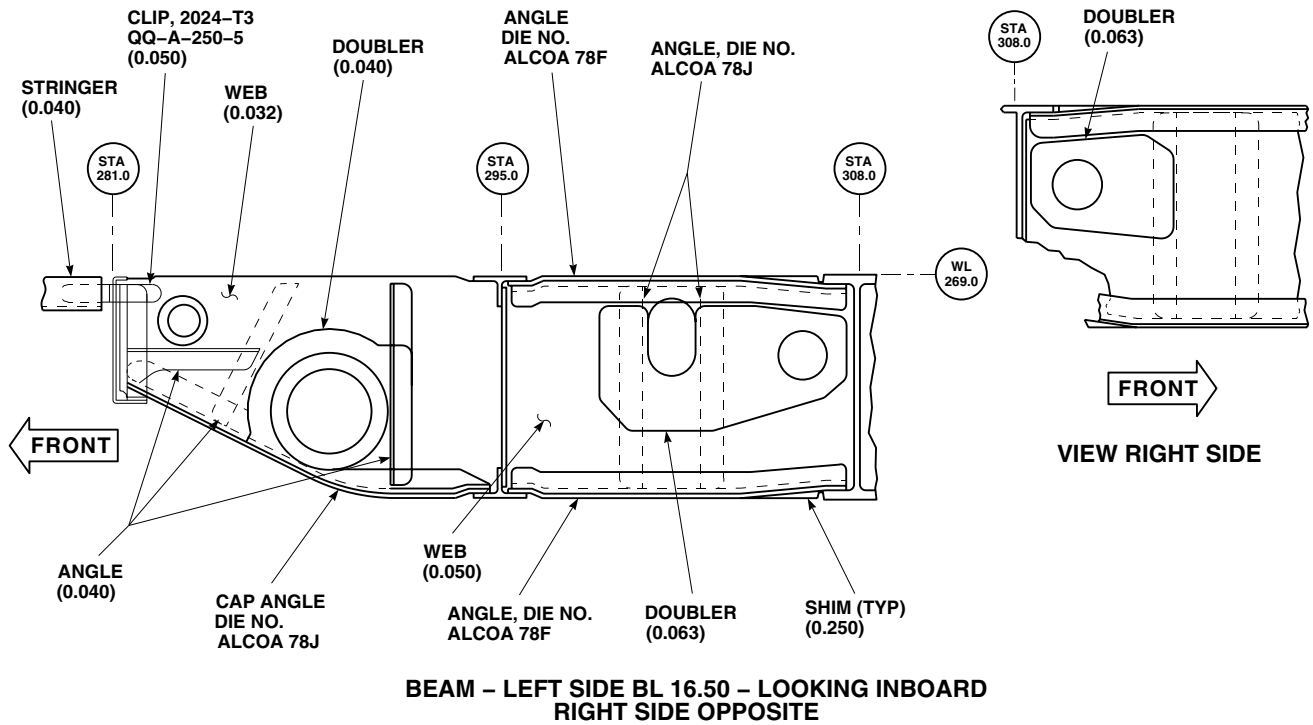


Figure 1. Repair Cabin Section Forward Framing. (Sheet 13 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

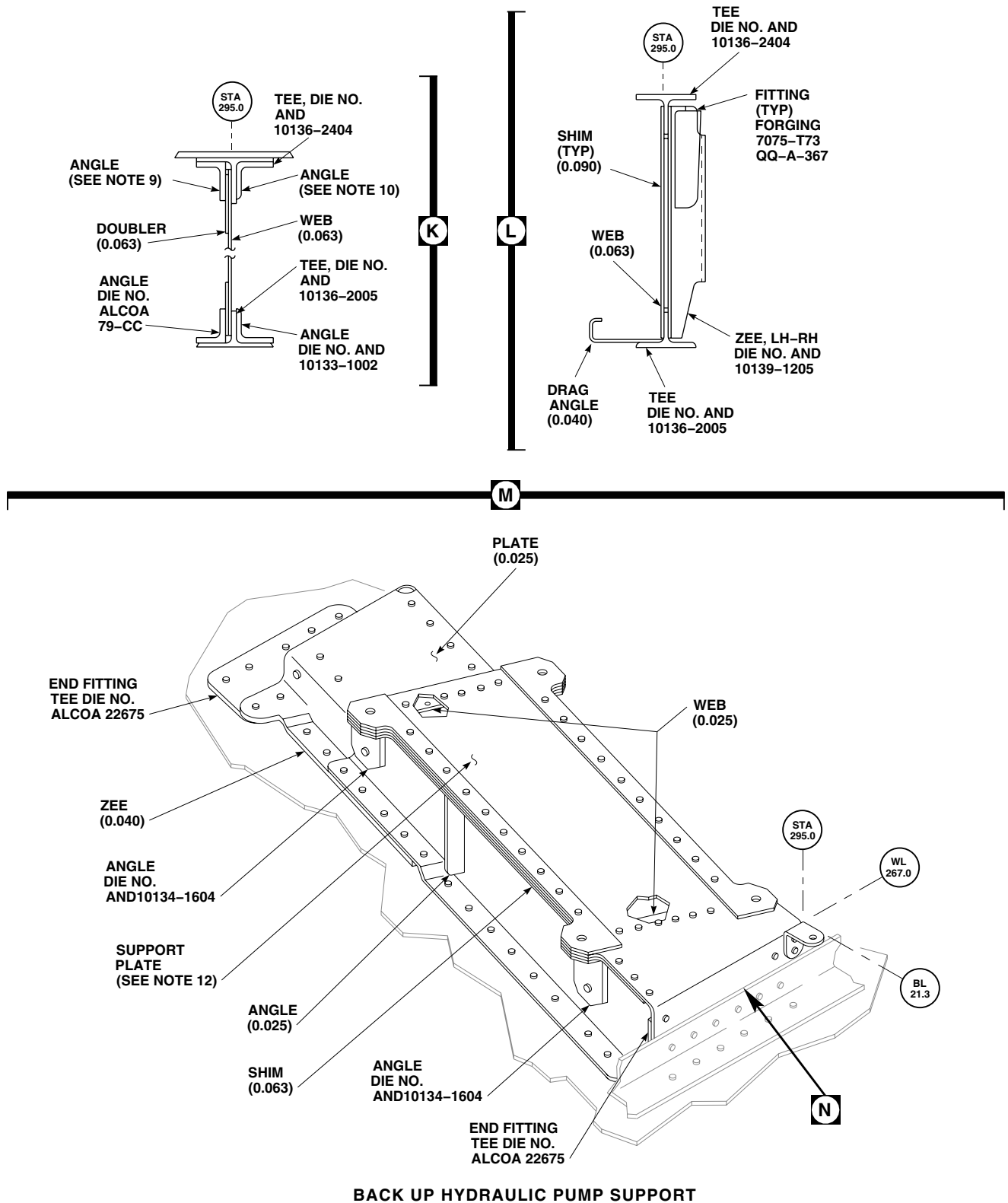
AA0100_14E
SA

Figure 1. Repair Cabin Section Forward Framing. (Sheet 14 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

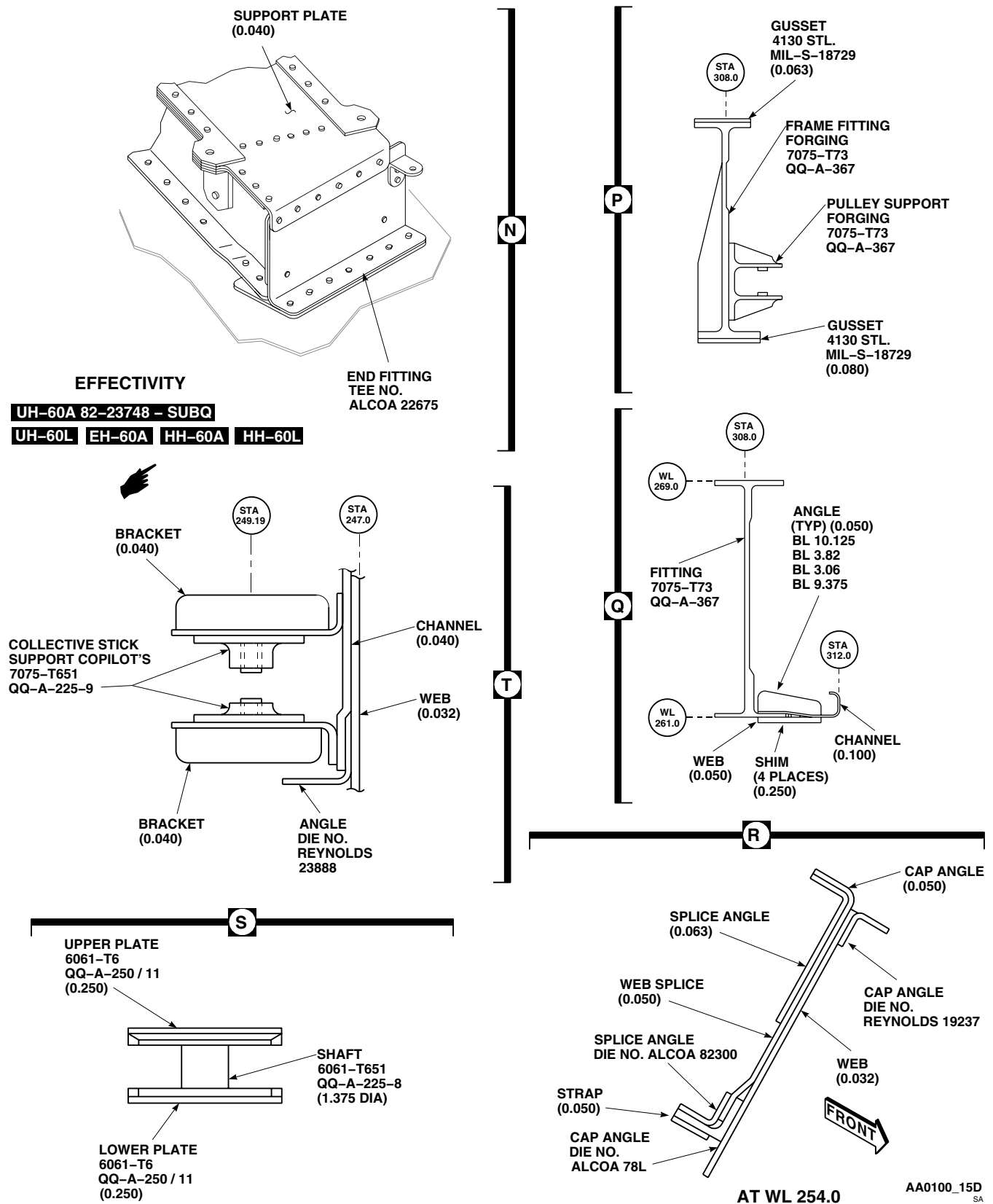


Figure 1. Repair Cabin Section Forward Framing. (Sheet 15 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

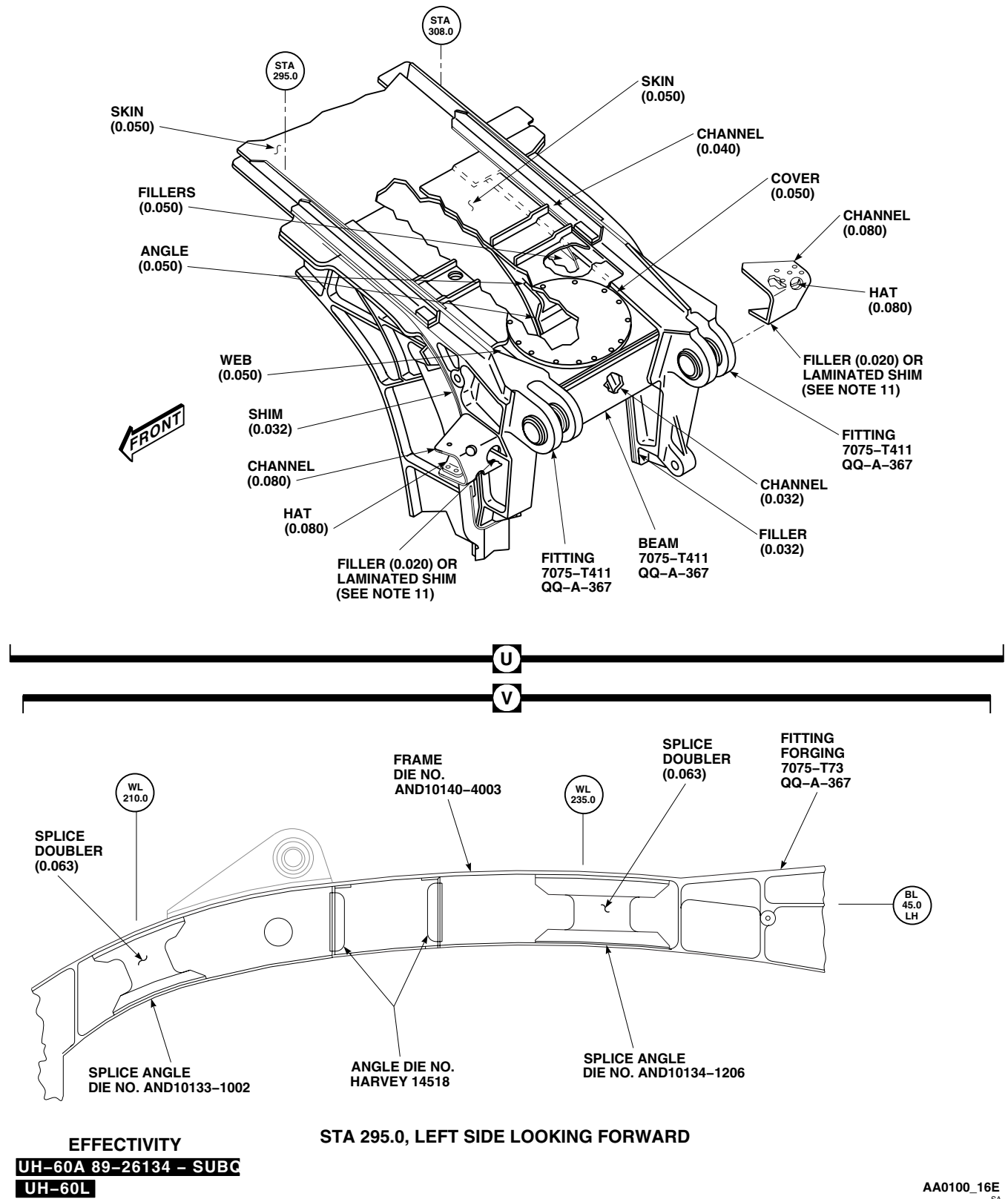
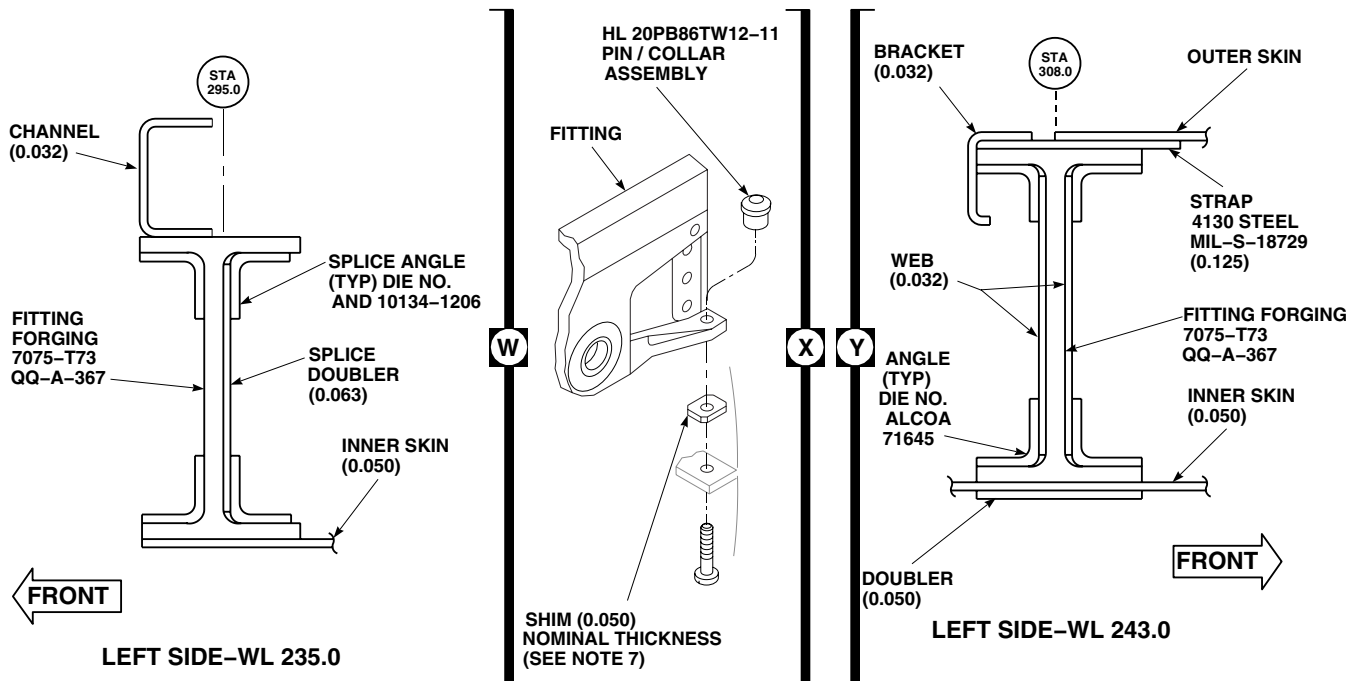
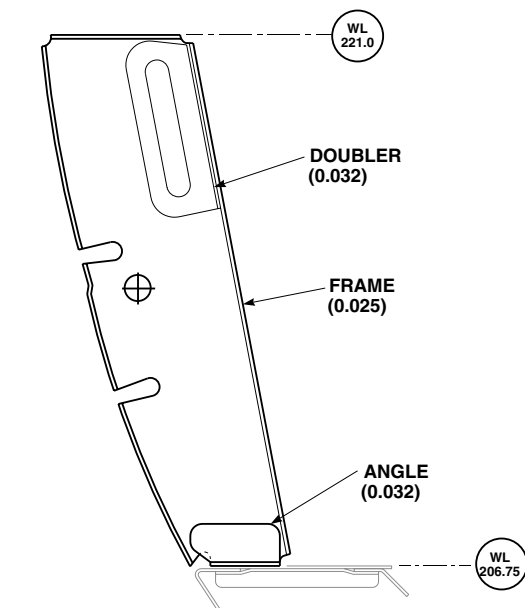


Figure 1. Repair Cabin Section Forward Framing. (Sheet 16 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

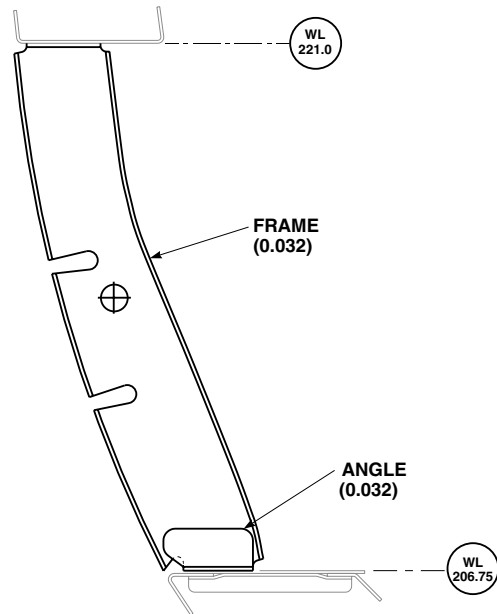


Z



EFFECTIVITY

UH-60A 77-22714 - 79-23294



EFFECTIVITY

UH-60A 79-23295 - SUBQ
UH-60L EH-60A
HH-60L HH-60A

LOOKING REAR - RIGHT SIDE - STA 275.0

AA0100_17E
SA

Figure 1. Repair Cabin Section Forward Framing. (Sheet 17 of 18)

REPAIR CABIN SECTION FORWARD FRAMING - Continued

NOTES:

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. **ESSS**
4. **UH-60A 82-23664 - SUBQ** **UH-60L** **EH-60A**
5. REFER TO WP 0381 00 FOR CONDUCTIVE PAINT AND GROUNDING REQUIREMENTS.
6. SEAL BETWEEN SURFACES OF STRAPS, FITTINGS, AND OTHER PARTS WITH SEALING COMPOUND, ITEM 273, WP 1803 00. USE ENOUGH SEALING COMPOUND TO FILL GAPS BETWEEN SURFACES.
7. APPLY ENOUGH SEALING COMPOUND, ITEM 273, WP 1803 00 TO SQUEEZE OUT INTO GAP BETWEEN FITTING AND STRUCTURE WP 0377 00.
8. **UH-60A 82-23664 - SUBQ** **UH-60L** **EH-60A**
9. **UH-60A 79-23309 - SUBQ** **UH-60L** **EH-60A**
10. **W/O ESSS** DIE NO. ALCOA 79-CC. **ESSS** **UH-60L** **EH-60A** 4130 STEEL (0.080).
11. **W/O ESSS** DIE NO. AND1033-1002. **ESSS** **UH-60L** **EH-60A** 4130 STEEL.
12. USE 0.020 SOLID STOCK OR LAMINATED SHIM, MIL-S-22499, COMP 1, TYPE 1, CLASS 2, AS REQUIRED TO FILL GAP.
13. **W/O ESSS** 0.025 **ESSS** **UH-60L** **EH-60A** 0.040
14. LEFT SIDE OPPOSITE.
15. BOND TO FRAME WITH ADHESIVE ITEM 11, WP 1803 00.
16. **UH-60L 96-26723 - SUBQ**
17. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0100_18B
SA

Figure 1. Repair Cabin Section Forward Framing. (Sheet 18 of 18)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

CABIN SECTION UPPER FRAMING REPAIR

This WP supersedes WP 0387 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-S-18729

MIL-S-22499

Material/Parts

Acetone, Technical Item 5, WP 1803 00

Adhesive, Item 7, WP 1803 00

Brush, Acid Swabbing, Item 57, WP 1803 00

Cloth, Abrasive Item 83, WP 1803 00

Corrosion Resistant Coating Item 111, WP 1803 00

Epoxy Primer Coating Kit Item 120, WP 1803 00

Remover, Paint Item 250, WP 1803 00

Sealing Compound Item 273, WP 1803 00

Sealing Compound Item 263, WP 1803 00

QQ-A-200

QQ-A-225

QQ-A-250

TM 1-1500-204-23

TM 1-1500-344-23

TM 55-1500-345-23

WP 0377 00

WP 0378 00

WP 0380 00

WP 0381 00

WP 1328 00

WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

AMS 4149

ASTM-E1417

REPAIR CABIN SECTION UPPER FRAMING

1. Upper cabin framing (STA 308.0 to STA 398.0) includes formed and extruded members made from 7075-T6, QQ-A-250-13, Alclad sheet stock and 7075-T6511, QQ-A-200-11, aluminum alloy extrusion. Gussets at STA 308.0 are made from 4130 steel, MIL-S-18729. Main gear box support structure and the STA 379.0 frame fitting are forged aluminum alloy, 7175-T736, AMS 4149 (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, Figure 1, Sheet 7, Figure 1, Sheet 8, Figure 1, Sheet 9, Figure 1, Sheet 10, Figure 1, Sheet 11 and Figure 1, Sheet 12).
2. All upper cabin framing is primary structure, however, the main gear box support structure and STA 379.0 frame fitting are considered critical primary structure. Negligible damage to the forged fittings is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness, at point of damage. Blend damage smoothly to surrounding area, using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage greater than negligible limits in critical areas must be reviewed by engineering authority for damage evaluation and either repair or replacement. All repairs to the upper cabin framing must maintain the design strength and alignment.
3. Evaluate damage in noncritical areas (WP 0377 00).
4. Use WP 0378 00 and WP 0380 00 for typical framing repairs. Refer to TM 1-1500-204-23 for repair materials and fastener information.
5. Coordinate repairs with TM 1-1500-204-23 requirements.

REPAIR CABIN SECTION UPPER FRAMING - Continued

6. For corrosion protection and watertightness, seal all repairs between framing and plating (WP 0377 00 and TM 1-1500-344-23).
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR RESCUE HOIST AIRFRAME FITTING**Remove**

1. Turn off electrical power.
2. Remove rescue hoist (WP 1328 00).
3. Remove paint, sealing compound, and primer from inner bore and outside surface area of socket using paint remover, Item 250, WP 1803 00 (Figure 2, Sheet 1).

Inspect/Repair

1. Inspect airframe fitting socket Area E for cracks, using eddy current paying particular attention to bolt hole areas per TM 55-1500-335-23 (Figure 2, Sheet 1). If no cracks are found proceed to step 2. If cracks are found repair as follows:

NOTE

Damage greater than limits must be reviewed by engineering authority for damage evaluation.

- a. Damage to Area E may be blended out up to 0.060-inch deep over an area not to exceed 9% of circumferential cross section (total arc length of 0.85-inch). Material may be removed from remaining surface area provided maximum bore diameter does not exceed 2.992-inch (0.002-inch into wall thickness) (Figure 2, Sheet 1).
- b. Blend reparable damage using abrasive cloth, Item 83, WP 1803 00.
- c. After blending repeat eddy current inspection to verify crack removal (ASTM-E1417).
2. Inspect Area A (upper flange) as follows:
 - a. Check Area A (upper flange) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.050-inch deep (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.
 - b. Check Area A (upper flange) for corrosion. Damage may be blended out up to 0.020-inch deep over 100% of cross sectional width (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.
3. Inspect Area B (lower flange) as follows:
 - a. Check Area B (lower flange) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.050-inch deep (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.
 - b. Check Area B (lower flange) for corrosion. Damage may be blended out up to 0.020-inch deep over 100% of cross sectional width (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.
4. Inspect Area C (stiffeners) as follows:
 - a. Check Area C (stiffeners) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.050-inch deep (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.
 - b. Check Area C (stiffeners) for corrosion. Damage may be blended out up to 0.020-inch deep over 100% of cross sectional width (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.
5. Inspect Area D (webs) as follows:
 - a. Check Area D (webs) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.040-inch deep (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued

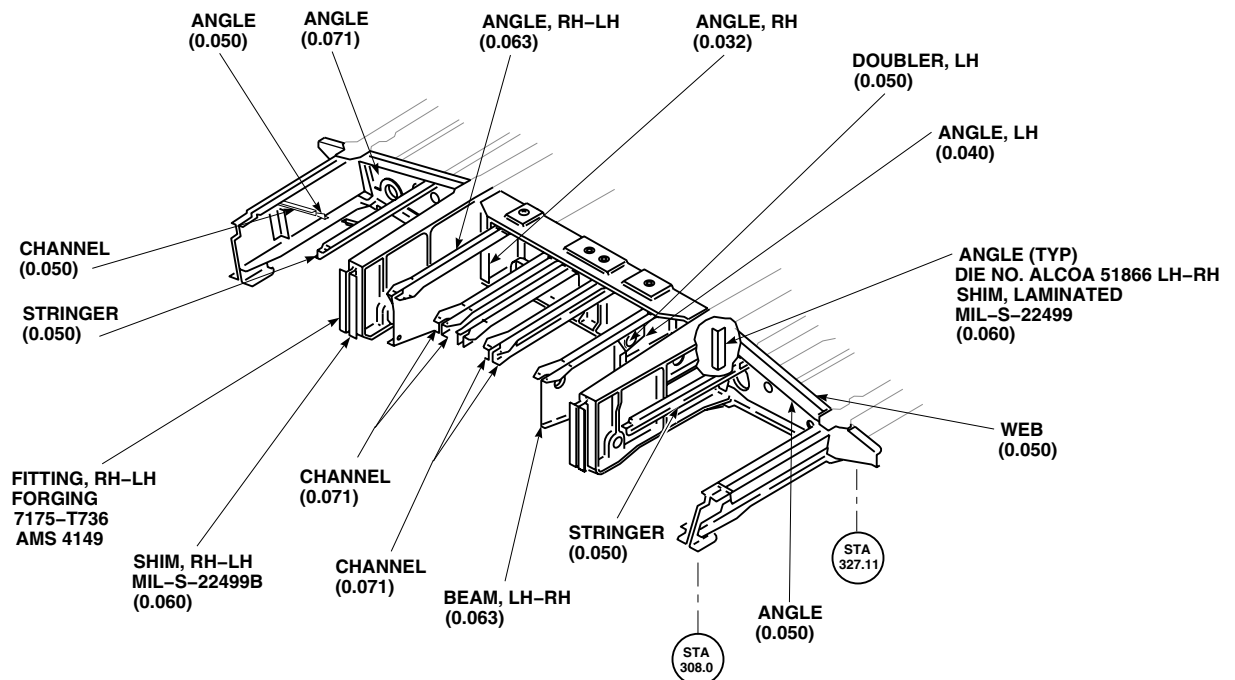
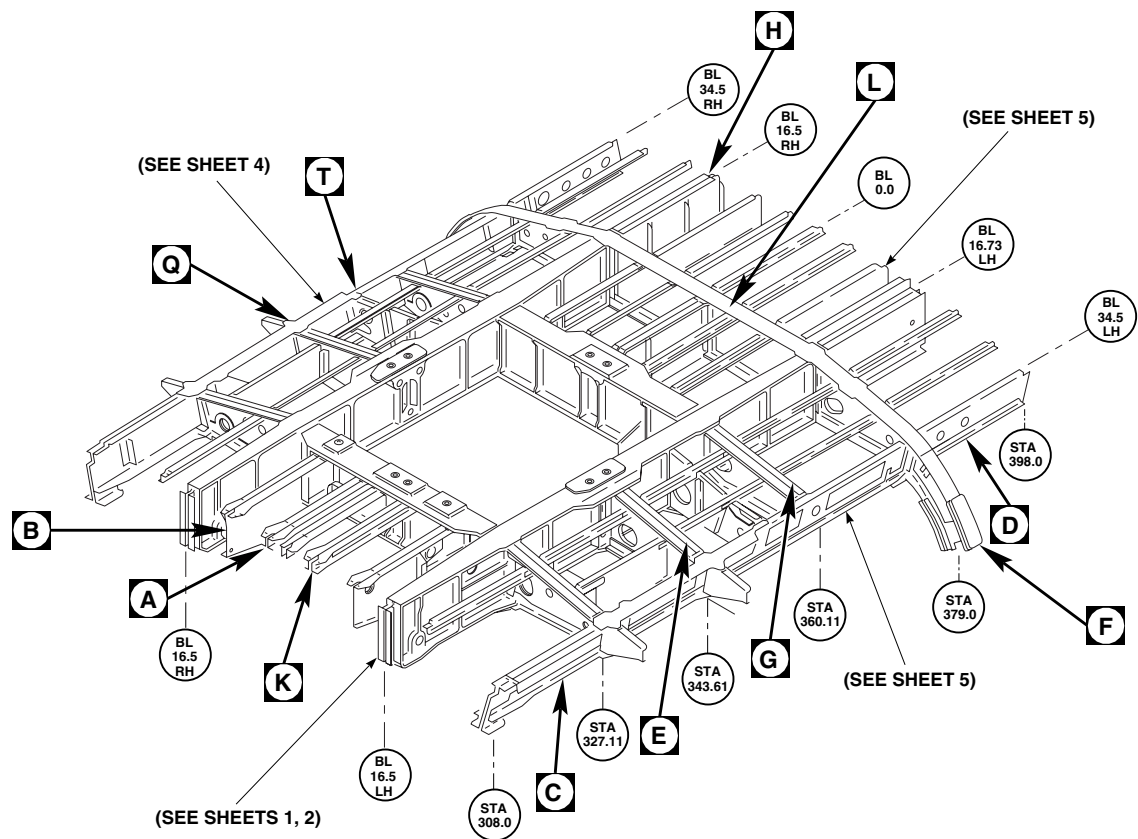
AA0101_1B
SA

Figure 1. Repair Cabin Section Upper Framing. (Sheet 1 of 13)

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued

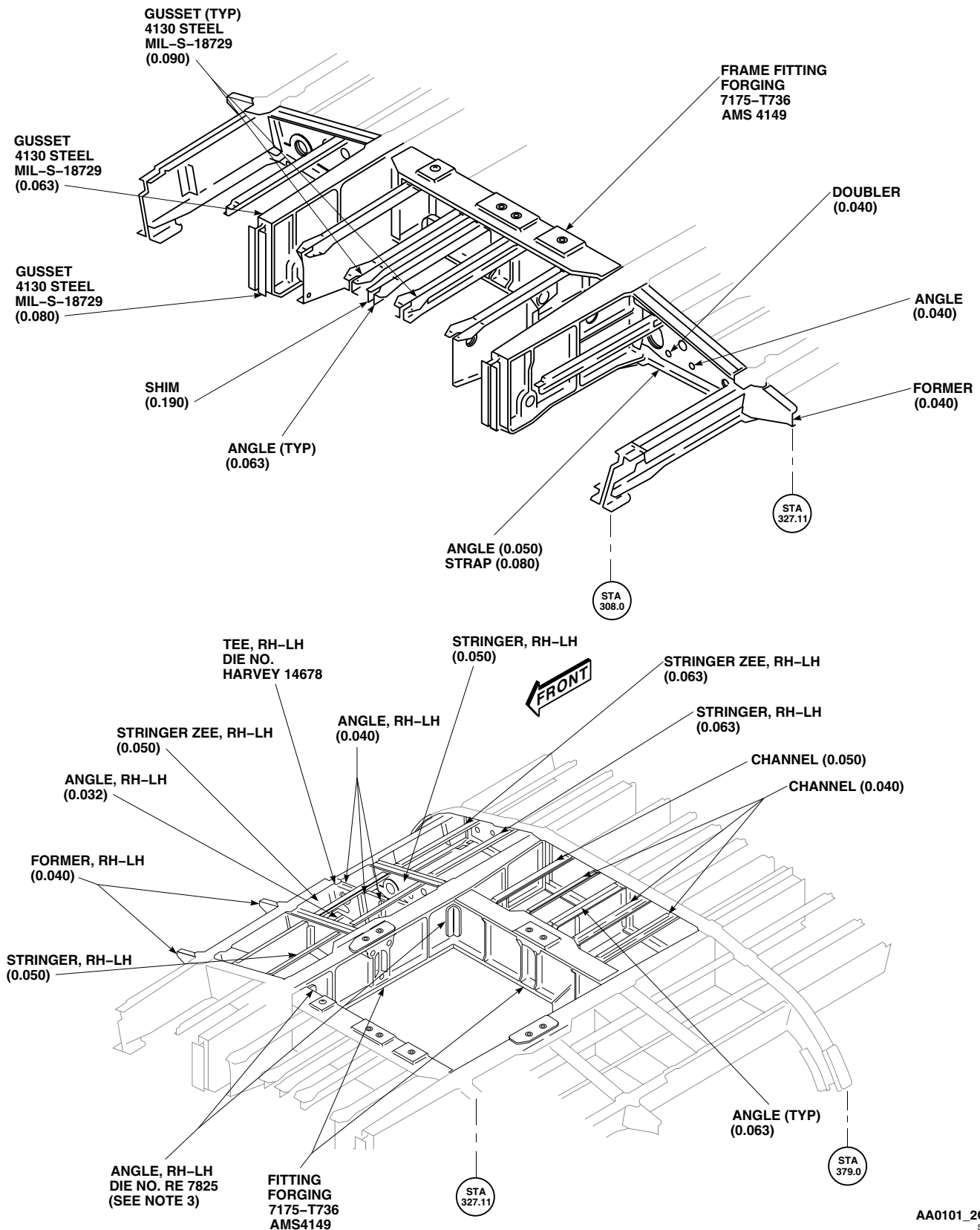
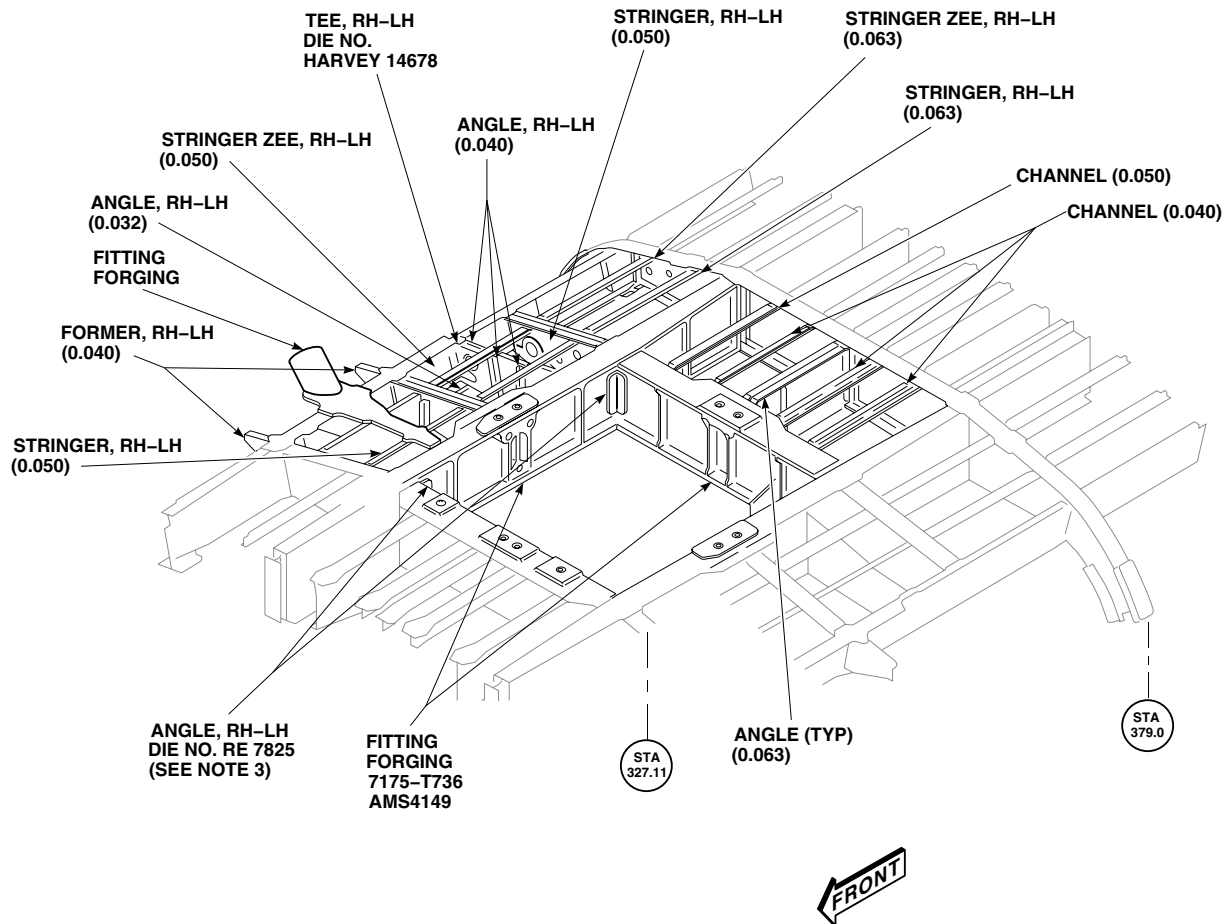


Figure 1. Repair Cabin Section Upper Framing. (Sheet 2 of 13)

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued



EFFECTIVITY

HH-60A HH-60L

AA0101_3D
SA

Figure 1. Repair Cabin Section Upper Framing. (Sheet 3 of 13)

Technical drawing of a bridge girder assembly, showing various components and their dimensions. The drawing includes a perspective view of the girder structure, with labels for various parts and their dimensions.

Components and Dimensions:

- GUSSET (TYP) 4130 STEEL MIL-S-18729 (0.090)
- GUSSET 4130 STEEL MIL-S-18729 (0.063)
- GUSSET 4130 STEEL MIL-S-18729 (0.080)
- SHIM (0.190)
- ANGLE (TYP) (0.063)
- ANGLE (0.050) STRAP (0.080)
- FRAME FITTING FORGING 7175-T736 AMS 4149
- DOUBLER (0.040)
- ANGLE (0.040)
- FORMER (0.040)
- STA 327.11
- STA 308.0
- TEE, RH-LH DIE NO. HARVEY 14678
- STRINGER, RH-LH (0.050)
- STRINGER ZEE, RH-LH (0.063)
- ANGLE, RH-LH (0.040)
- STRINGER ZEE, RH-LH (0.050)
- ANGLE, RH-LH (0.032)
- FORMER, RH-LH (0.040)
- STRINGER, RH-LH (0.050)
- CHANNEL (0.050)
- CHANNEL (0.040)
- CHANNEL (0.040)
- ANGLE, RH-LH (0.040)
- FITTING FORGING 7175-T736 AMS4149
- ANGLE, TYPICAL (0.063)
- STA 327.11
- STA 379.0

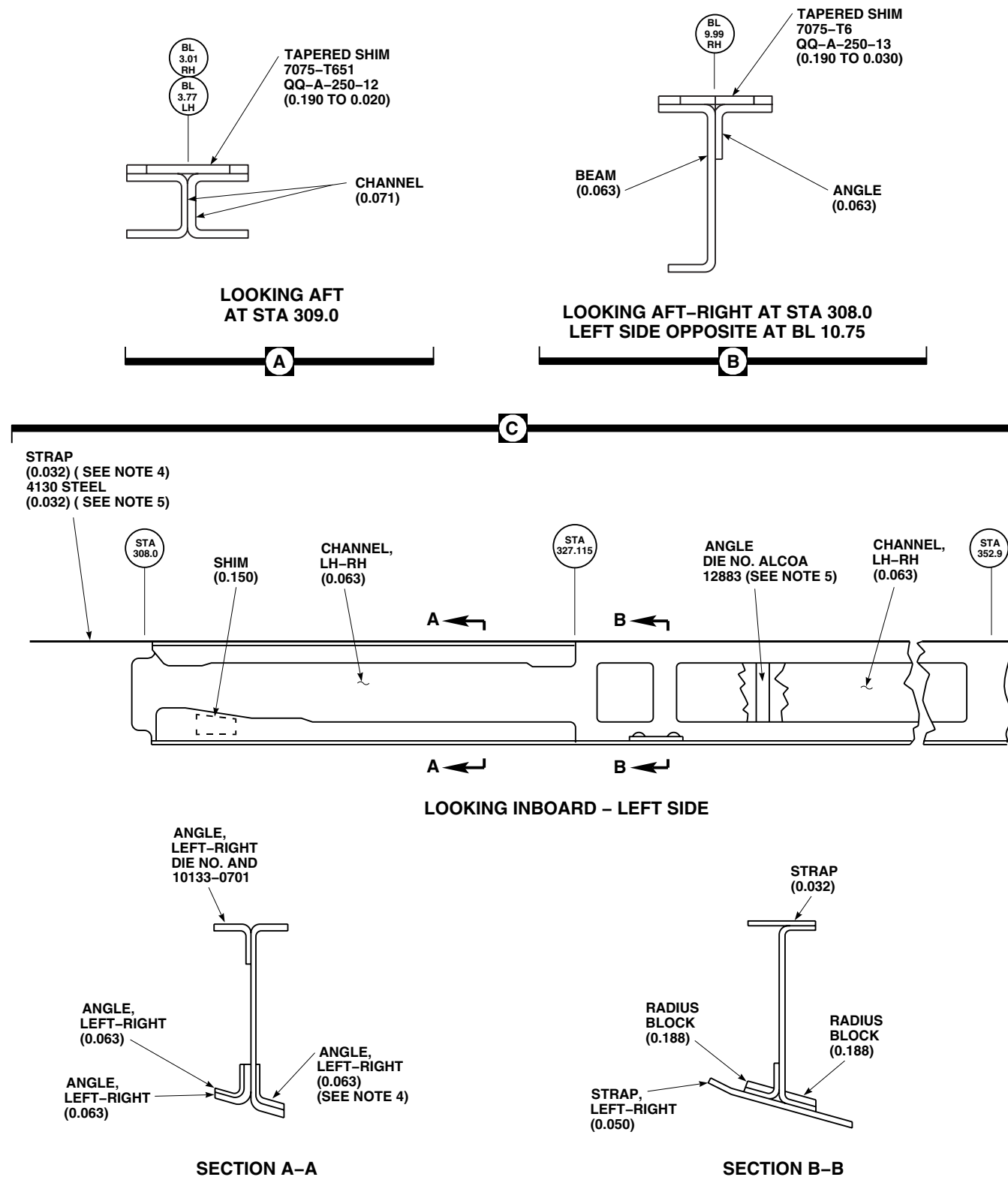
Front View: A perspective view of the girder structure, showing the internal components and their dimensions. The front view is indicated by a "FRONT" arrow.

Dimensions: The dimensions are given in inches, with some components having dimensions in parentheses.

Notes: The drawing includes several notes, including "SEE NOTE 3" and "HARVEY 14678".

Change 1

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued



AA0101_5A
SA

Figure 1. Repair Cabin Section Upper Framing. (Sheet 5 of 13)

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued

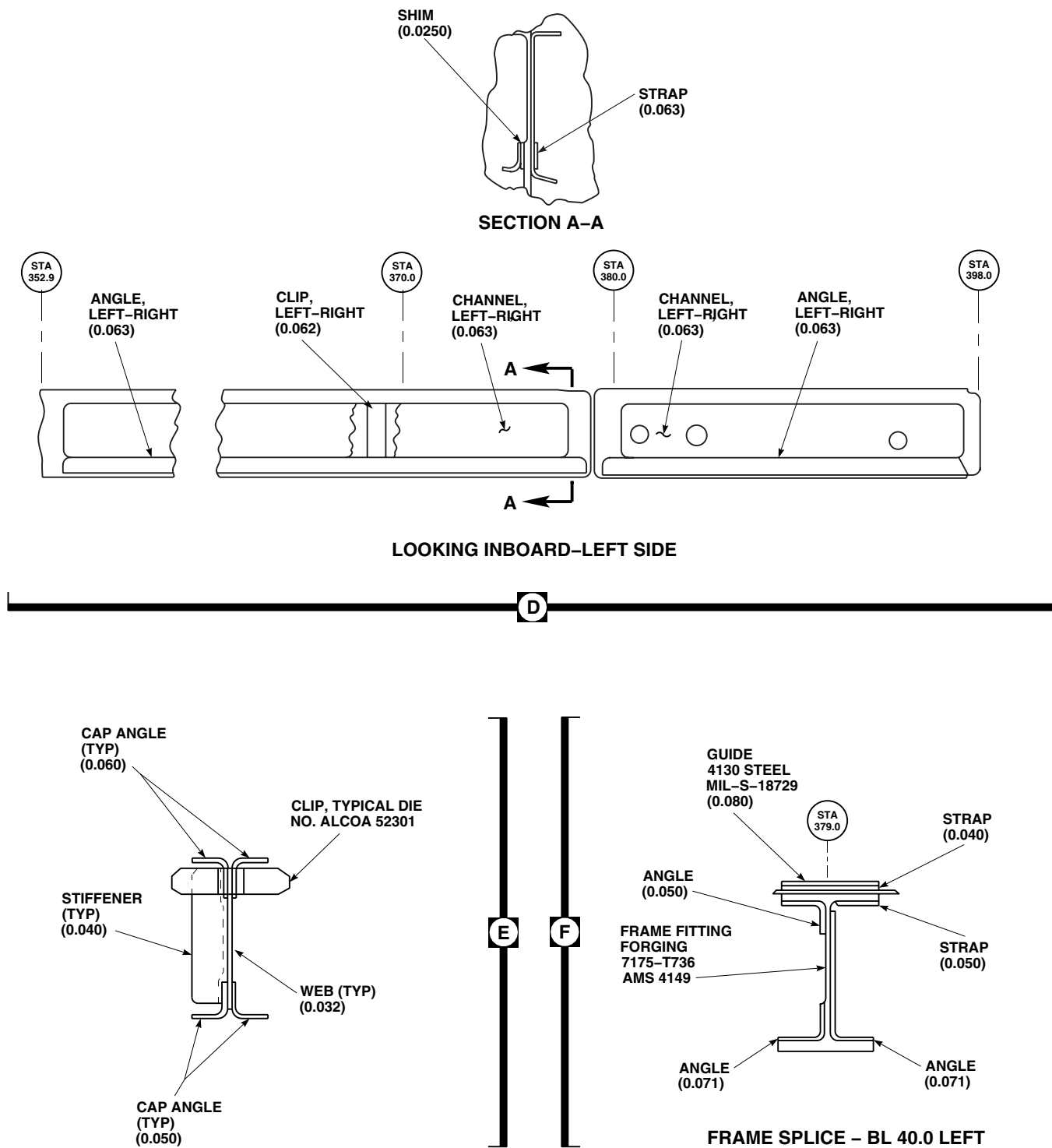
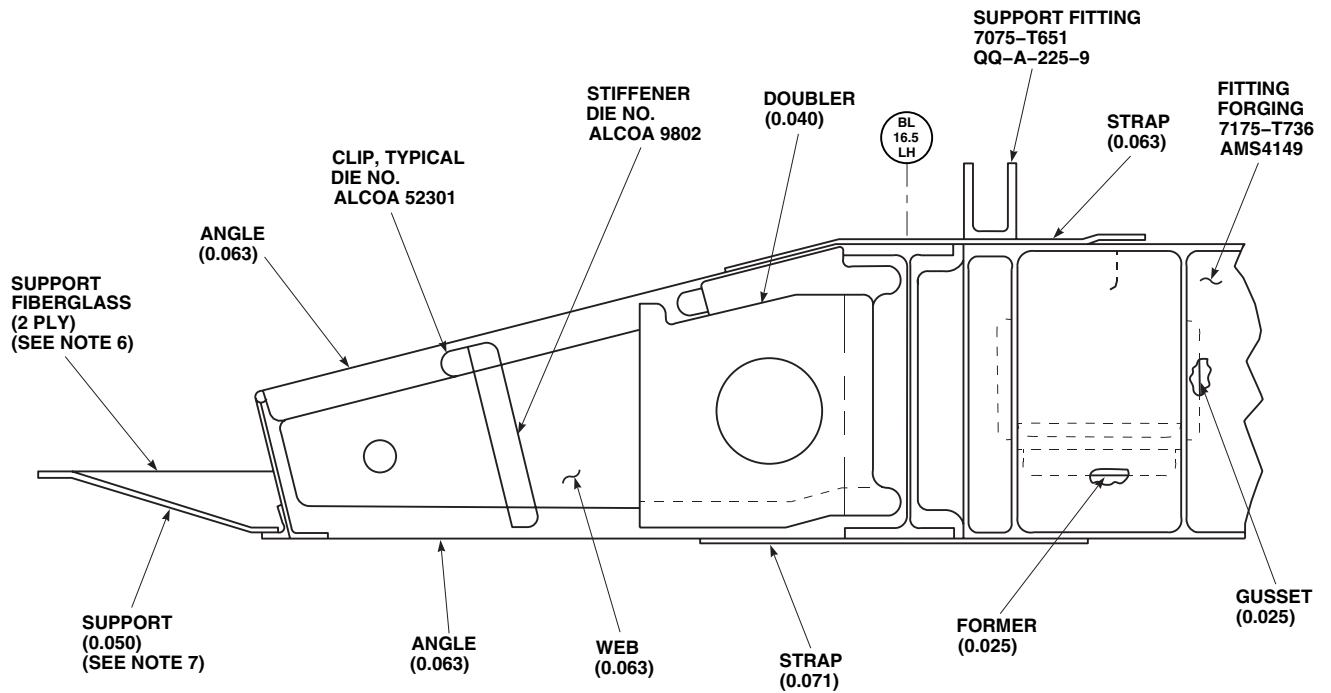
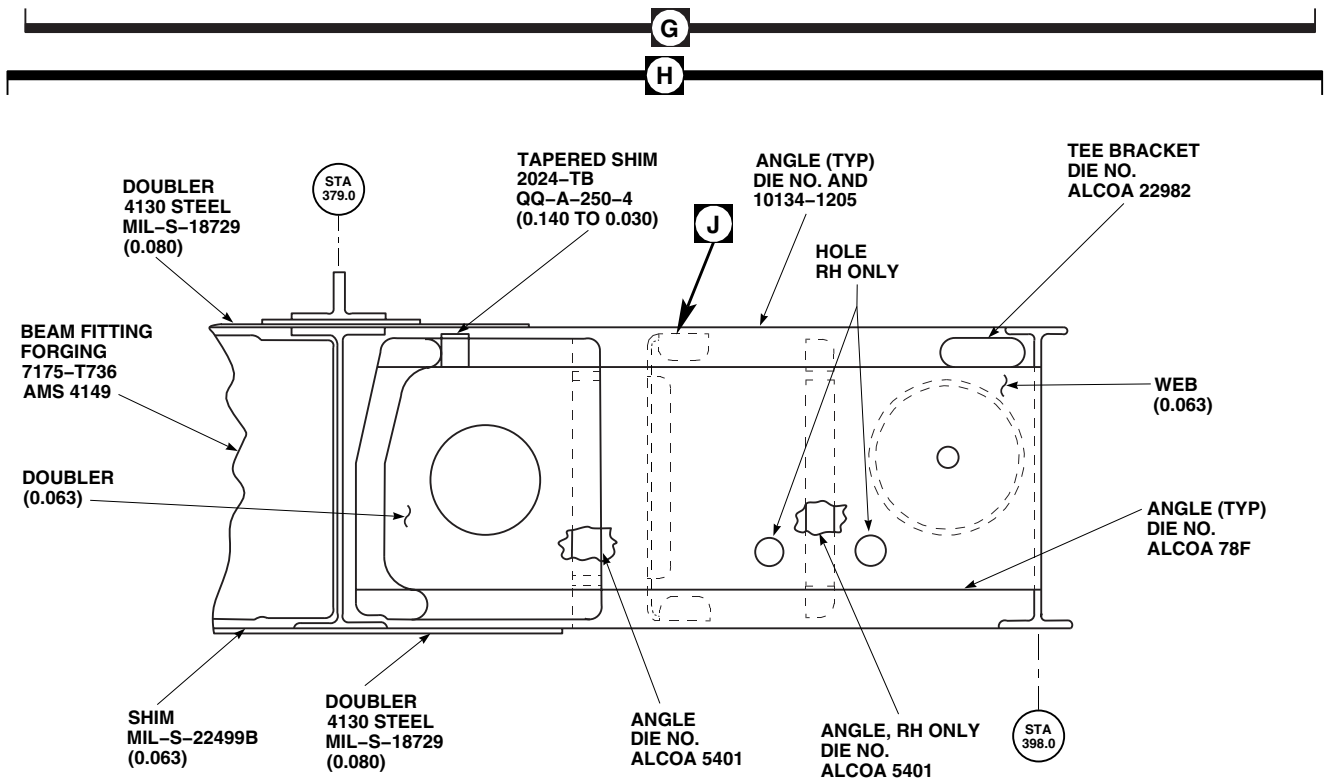
AA0101_6B
SA

Figure 1. Repair Cabin Section Upper Framing. (Sheet 6 of 13)

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued



LOOKING FORWARD - STA 360.115 LEFT SIDE



LOOKING INBOARD - BL 16.73 LEFT SIDE - RIGHT SIDE OPPOSITE

AA0101_7C
SA

Figure 1. Repair Cabin Section Upper Framing. (Sheet 7 of 13)

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued

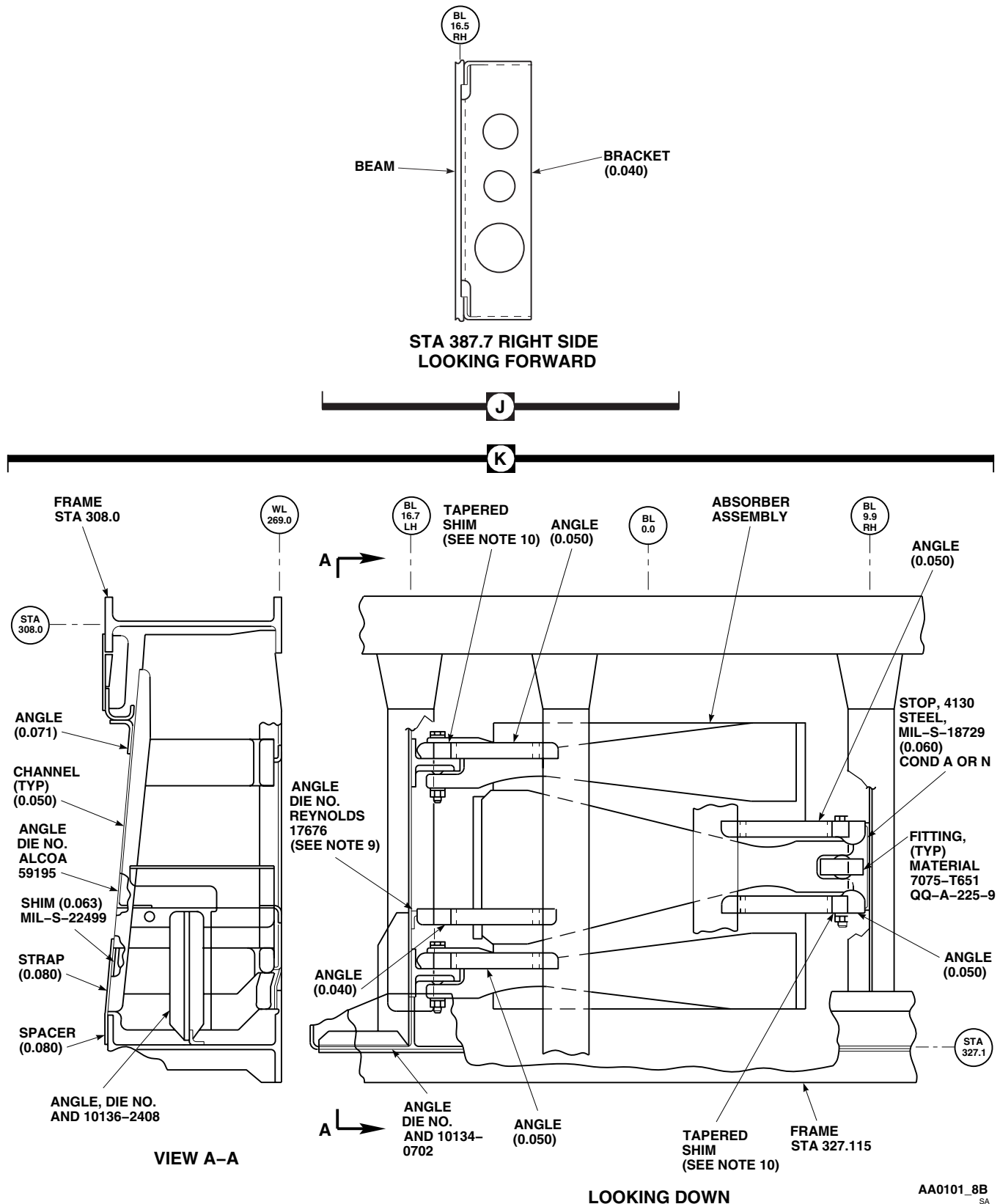


Figure 1. Repair Cabin Section Upper Framing. (Sheet 8 of 13)

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued

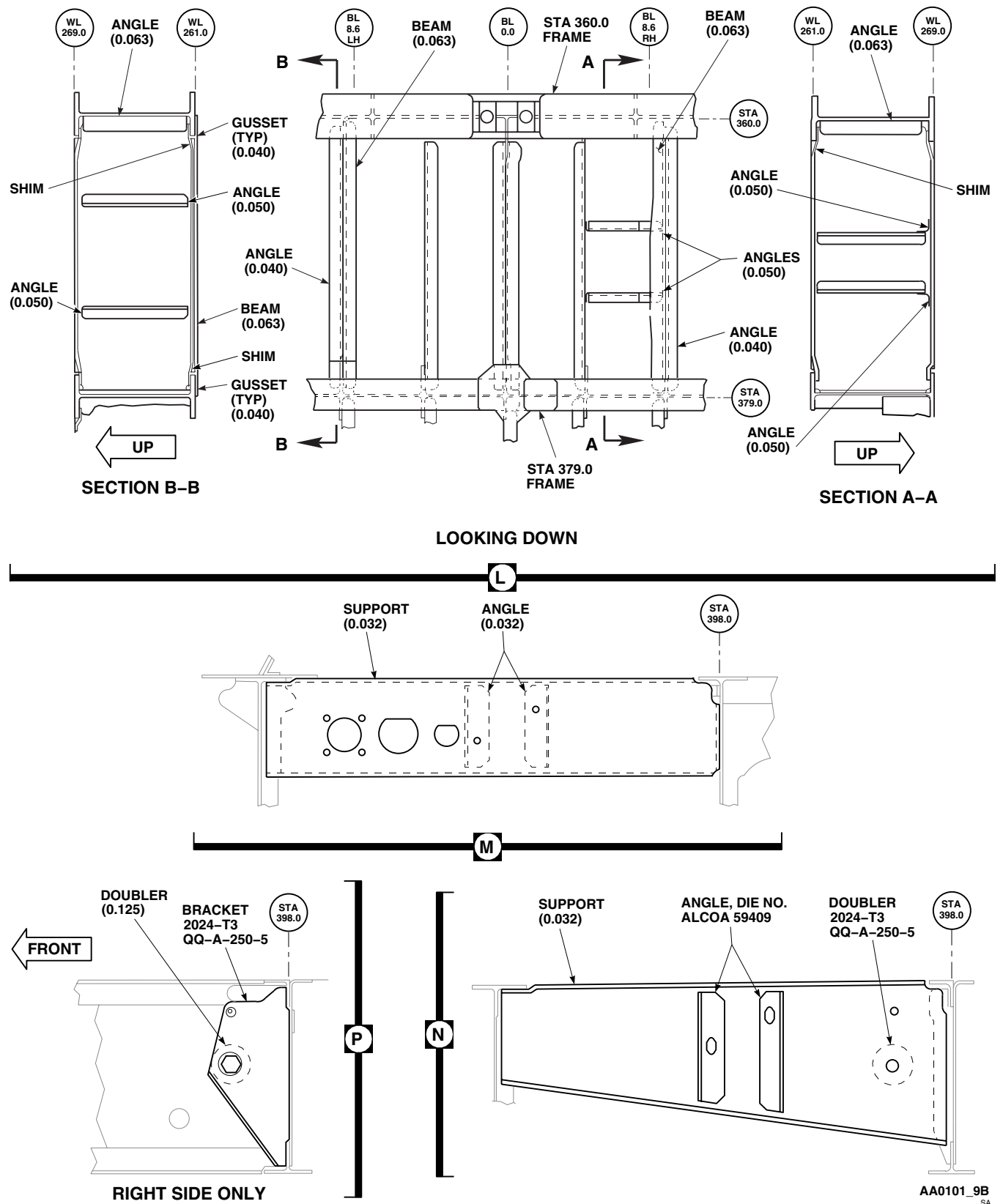
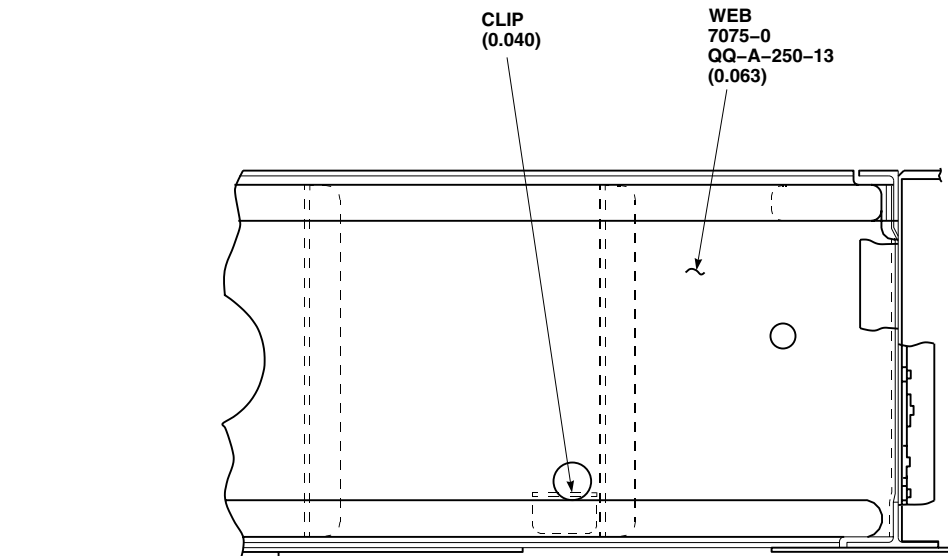
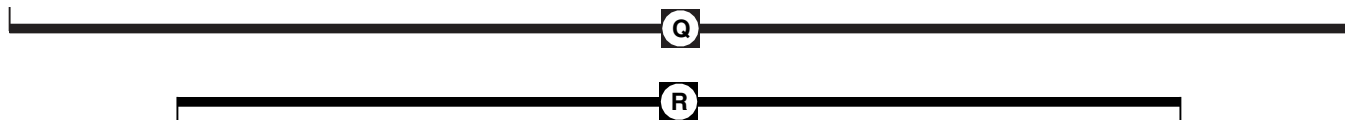
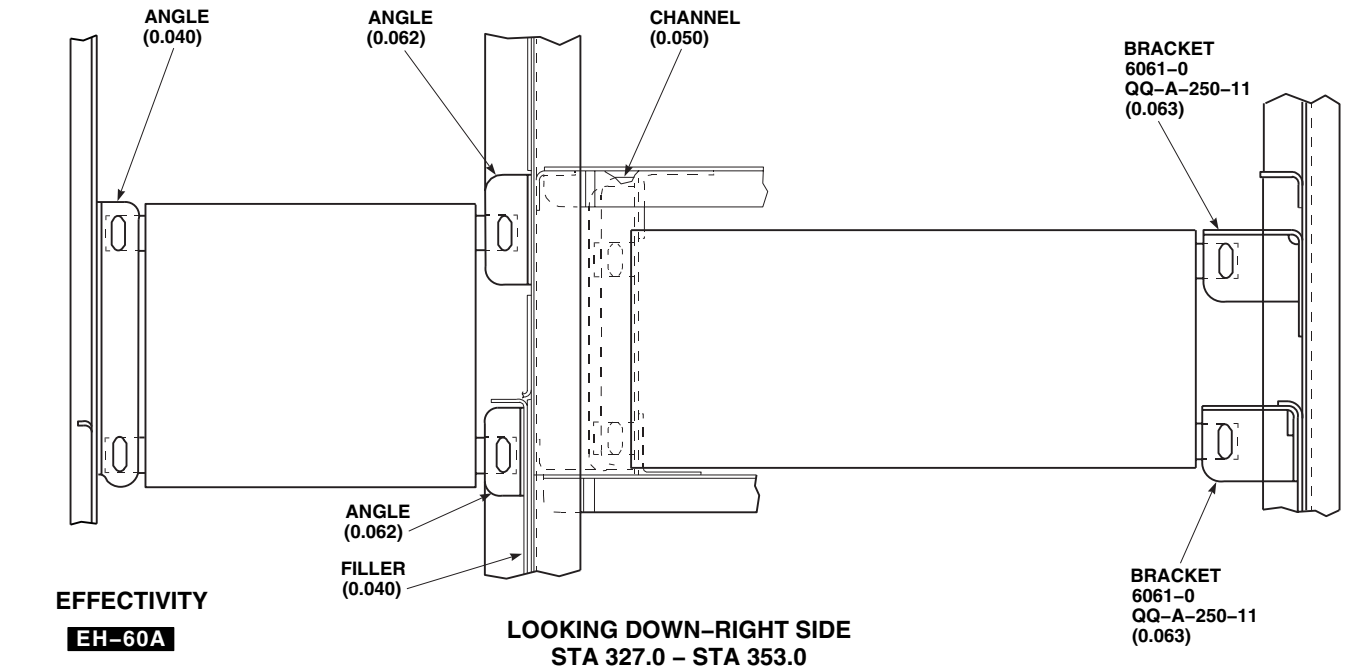


Figure 1. Repair Cabin Section Upper Framing. (Sheet 9 of 13)

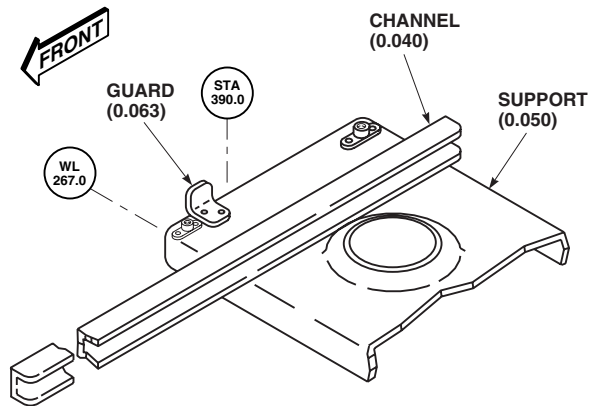
REPAIR RESCUE HOIST AIRFRAME FITTING - Continued



AA0101_10B
SA

Figure 1. Repair Cabin Section Upper Framing. (Sheet 10 of 13)

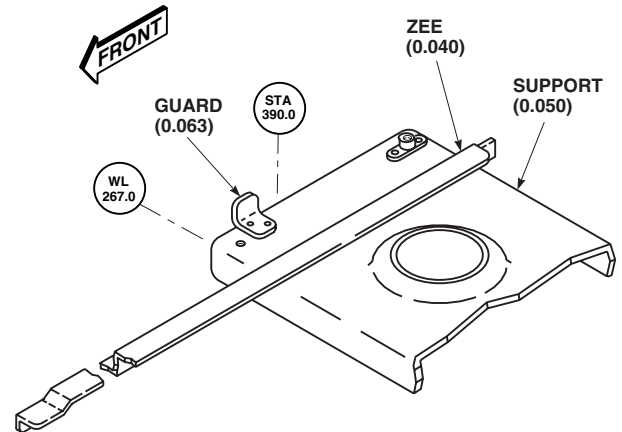
REPAIR RESCUE HOIST AIRFRAME FITTING - Continued



EFFECTIVITY

UH-60A 77-22714 - 89-26133

EH-60A

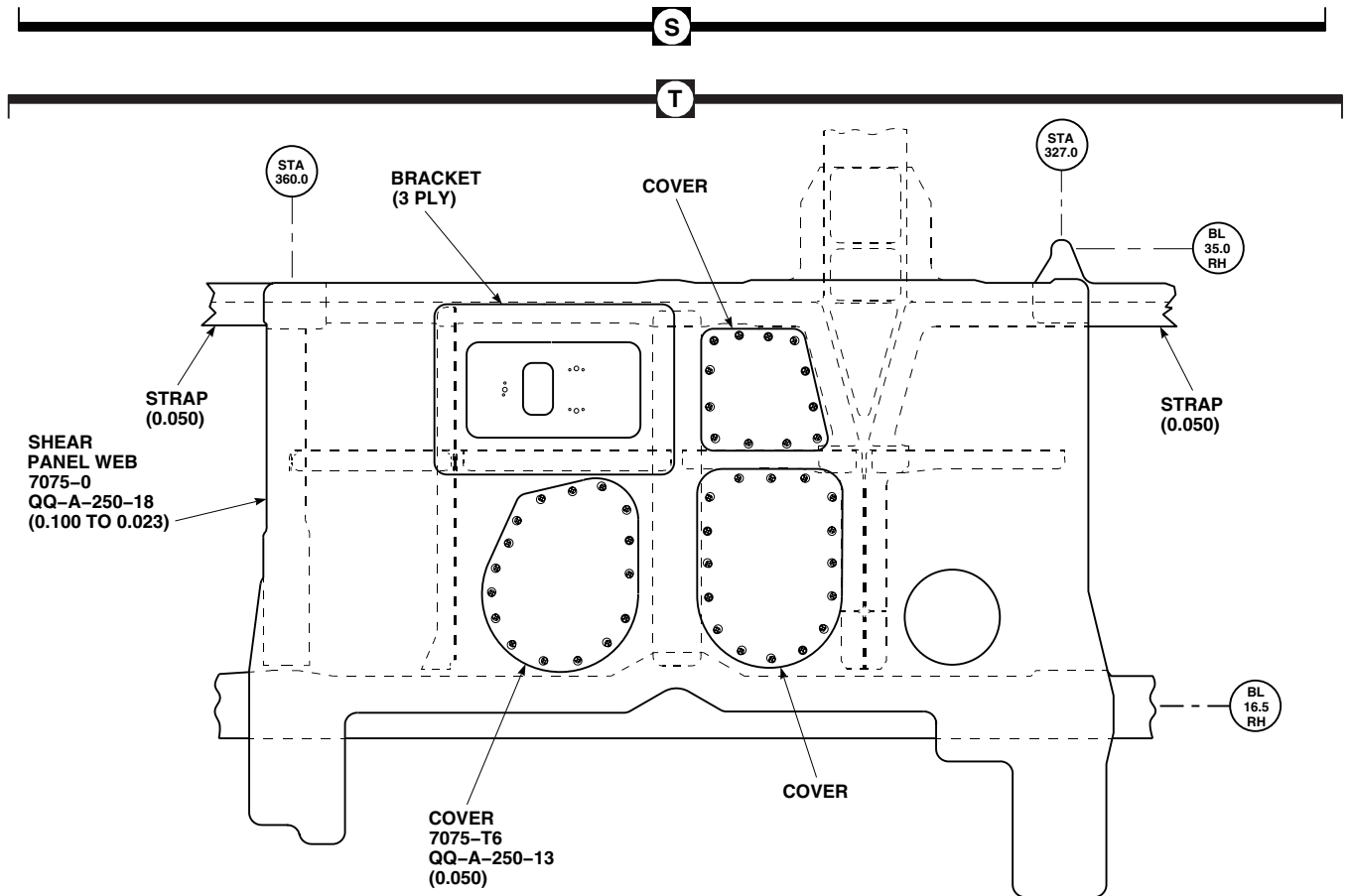


EFFECTIVITY

UH-60A 89-26134 - SUBQ

UH-60L

RIGHT SIDE ONLY

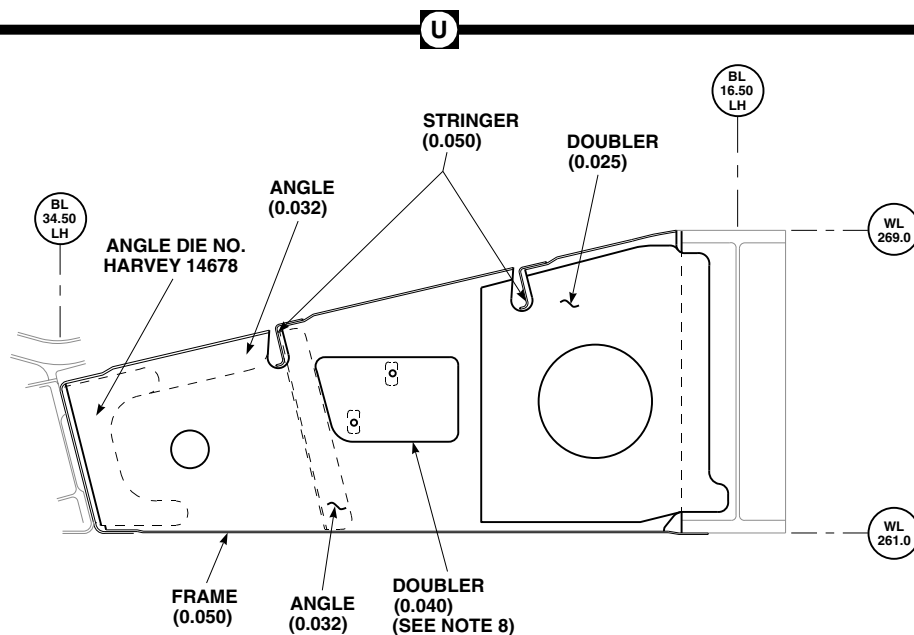


LOOKING UP-RIGHT SIDE
WL 261.0

AA0101_11B
SA

Figure 1. Repair Cabin Section Upper Framing. (Sheet 11 of 13)

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued



LOOKING FORWARD – LEFT SIDE STA 352.90

AA0101_12B
SA

Figure 1. Repair Cabin Section Upper Framing. (Sheet 12 of 13)

- b. Check Area D (webs) for corrosion. Damage may be blended out up to 0.020-inch deep over 1 inch diameter circle (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.
6. Inspect Area F (socket/web interface) as follows:
 - a. Check Area F (socket/web interface) for nicks, dents, scratches, and gouges. Damage may be blended out up to 0.060-inch over a 1 inch length and 0.1875-inch width (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.
 - b. Check Area F (socket/web interface) for corrosion. Damage may be blended out up to 0.020-inch deep over 100% of cross sectional width (Figure 2, Sheet 1). Blend damage using abrasive cloth, Item 83, WP 1803 00.
7. Treat reworked areas on fitting as follows:
 - a. Apply corrosion resistant coating, Item 111, WP 1803 00 to reworked areas with acid swabbing brush, Item 57, WP 1803 00.
 - b. Maintain surface wetness for 3 to 5 minutes.
 - c. Thoroughly rinse treated areas with cold water.
 - d. Allow reworked areas to dry completely.
 - e. Apply one coat of epoxy primer coating kit, Item 120, WP 1803 00 to all reworked areas. **DO NOT PAINT BOLT HOLES.**
 - f. Touch up finish on all reworked areas (WP 0381 00 and TM 55-1500-345-23).

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued**NOTES:**

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. APPLY SEALING COMPOUND, ITEM 273, WP 1803 00, TO ALL MATING SURFACES. AFTER INSTALLING FASTENERS, RECLEAN AREA, PLUG GAPS WITH SEALING COMPOUND, ITEM 263, WP 1803 00, AND APPLY SEALING COMPOUND, ITEM 273, WP 1803 00, OVER PLUGS AND AROUND EDGES OF MATING PARTS.
4. **W/O ESSS**
5. **ESSS UH-60L EH-60A**
6. **UH-60A 77-22714 - 82-23879**
7. **UH-60A 82-23880 - SUBQ UH-60L EH-60A**
8. **UH-60L**
9. LIQUID SHIM BETWEEN ANGLE AND MATING SURFACE, UP TO 0.060-INCH THICKNESS, USING, TECHNICAL ACETONE, ITEM 5, WP 1803 00.
10. SS81-139-1 (5 PLACES): BOND IN PLACE WITH ADHESIVE, ITEM 7, WP 1803 00, BEFORE DRILLING FASTENER HOLE.
11. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0101_13B
SA

Figure 1. Repair Cabin Section Upper Framing. (Sheet 13 of 13)

Install

1. Install rescue hoist (WP 1328 00).
2. Turn on electrical power.

Diagram of the right side of the engine block looking forward. The diagram shows the cylinder heads, valves, and the crankshaft area. Labels include: A (top edge), B (bottom edge), C (left side), D (right side), E (top right corner), F (valve clearance), and SOCKET (crankshaft area). Dimensions WL 281.0 and BL 16.550 RH are indicated.

AREA	INSPECTION	MAXIMUM REPARABLE LIMITS
A – UPPER FLANGE	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.050. (SEE NOTES 1, 2, AND 3)
	CORROSION	NOT DEEPER THAN 0.020 OVER 100% OF CROSS SECTIONAL WIDTH. (SEE NOTES 1 AND 5)
B – LOWER FLANGE	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.050. (SEE NOTES 1, 2, AND 3)
	CORROSION	NOT DEEPER THAN 0.020 OVER 100% OF CROSS SECTIONAL WIDTH. (SEE NOTES 1 AND 5)
C – STIFFENERS	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.050. (SEE NOTES 1, 2, AND 3)
	CORROSION	NOT DEEPER THAN 0.020 OVER 100% OF CROSS SECTIONAL WIDTH. (SEE NOTES 1 AND 5)
D – WEBS	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.040. (SEE NOTES 1, 2, AND 4)
	CORROSION	NOT DEEPER THAN 0.020 OVER A 1.00 DIAMETER CIRCLE (SEE NOTES 1 AND 5)
E – SOCKET ID	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.050 OVER 9% OF CIRCUMFERENTIAL CROSS SECTION (TOTAL ARC LENGTH OF 0.85). (SEE NOTES 1 AND 2)
	CORROSION	NOT DEEPER THAN 0.002 INTO WALL OVER 100% OF ID. MAXIMUM BORE DIAMETER NOT TO EXCEED 2.992. (SEE NOTES 1 AND 5)
F – SOCKET / WEB INTERFACE	NICKS, DENTS, SCRATCHES, OR GOUGES	NOT DEEPER THAN 0.060 OVER A 1.00 LENGTH AND 0.1875 WIDTH. (SEE NOTES 1 AND 2)
	CORROSION	NOT DEEPER THAN 0.020 OVER 100% OF CROSS SECTIONAL WIDTH. (SEE NOTES 1 AND 5)

REPAIR RESCUE HOIST AIRFRAME FITTING - Continued**NOTES:**

1. THE MAXIMUM REPARABLE LIMITS FOR NICKS, DENTS, SCRATCHES, GOUGES, AND CORROSION ARE NOT CUMULATIVE AT ANY ONE CROSS SECTION.
2. SCRATCHES WHICH FALL IN THE CORROSION REPARABLE LIMITS ARE ACCEPTABLE.
3. AFTER REWORK FOR NICKS, DENTS, AND GOUGES, BLENDED AREAS FOR FLANGES AND STIFFENERS SHALL NOT EXCEED 0.375 INCH DIAMETER AND BE NO CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLE.
4. AFTER REWORK FOR NICKS, DENTS, AND GOUGES, BLENDED AREAS FOR WEBS SHALL NOT EXCEED 0.500 INCH DIAMETER AND BE NO CLOSER THAN ONE HOLE DIAMETER FROM ANY EXISTING FASTENER HOLES.
5. ISOLATED CORROSION PITS MAY BE REPAIRED USING THE LIMITS SPECIFIED IN NICKS, DENTS, AND GOUGES FOR THE PARTICULAR ITEM.
6. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2505_2
SA

Figure 2. Repair Rescue Hoist Airframe Fitting. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

CABIN SECTION LOWER FRAMING REPAIR

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-A-21180

MIL-C-81773

MIL-R-6855

MIL-S-5059

MIL-S-6758

MIL-T-9046

Material/Parts

Adhesive, Item 14, WP 1803 00

Adhesive, Item 15, WP 1803 00

Bushing, NAS 75-8-032

Cloth, Abrasive, Item 83, WP 1803 00

Sealing Compound, Item 273, WP 1803 00

Sealing Compound, Item 292, WP 1803 00

QQ-A-200

QQ-A-250

QQ-A-367

TM 1-1500-204-23

TM 1-1500-344-23

TM 55-1500-345-23

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

WP 0377 00

References

AMS 4149

ASTM A231

L-P-410

WP 0378 00

WP 0380 00

WP 0381 00

WP 1803 00

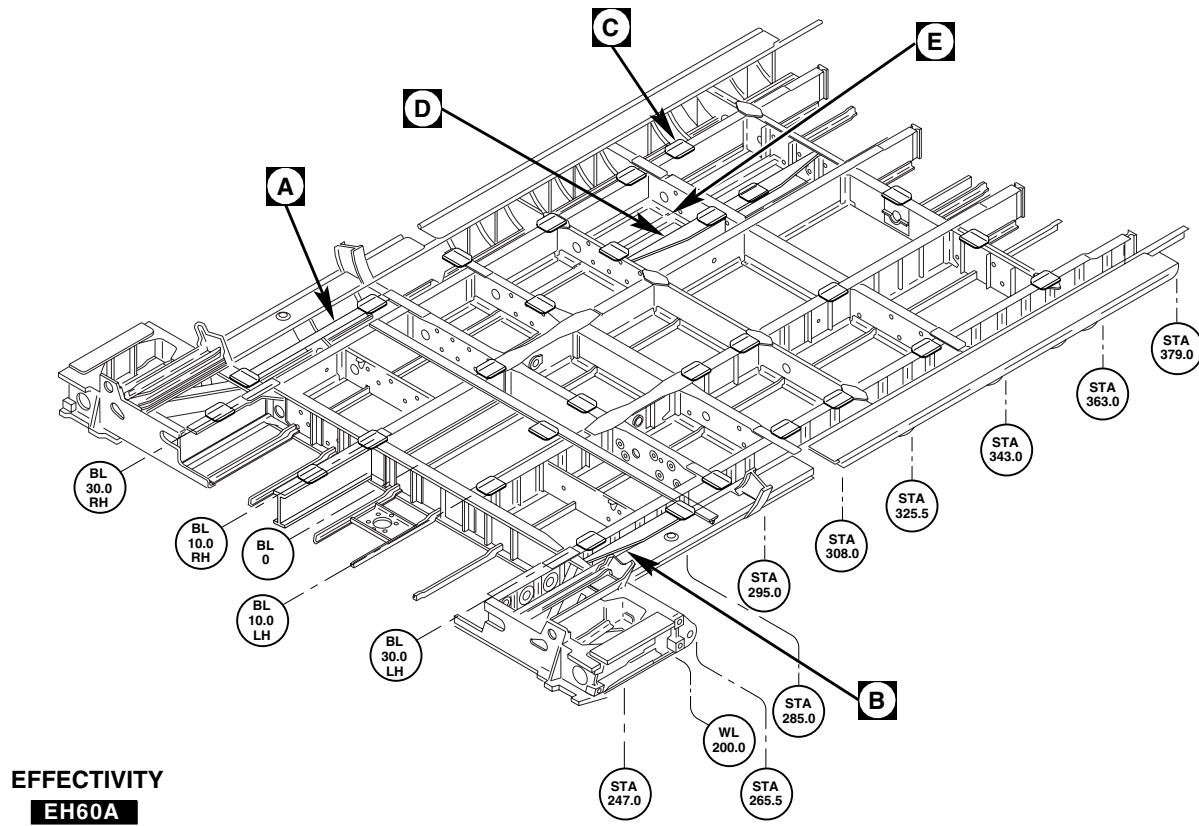
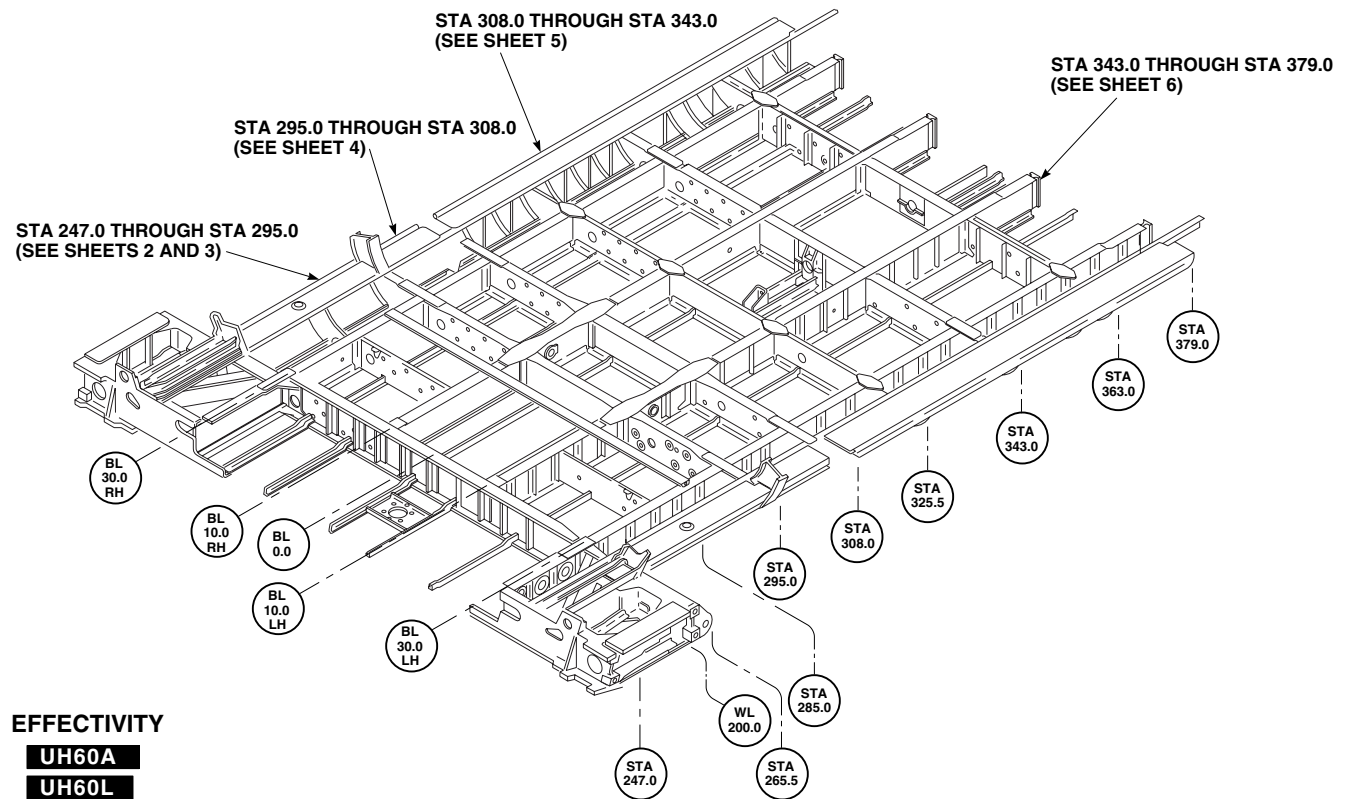
REPAIR LOWER CABIN FRAMING

1. Lower cabin framing (STA 247.0 to STA 379.0) includes formed and extruded members made from 7075-T6, QQ-A-250-13, Alclad sheet stock and 7075-T6511, QQ-A-200-11, aluminum alloy extrusion. Landing gear support fittings and frame fittings between STA 247.0 and STA 295.0 are forged aluminum alloy, 7075-T73, QQ-A-367. Support fittings for the cargo hook are forged aluminum alloy, 7175-T736, AMS 4149 (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, Figure 1, Sheet 7, Figure 1, Sheet 8, Figure 1, Sheet 9, Figure 1, Sheet 10, Figure 1, Sheet 11, and Figure 1, Sheet 12).
2. All lower cabin framing is primary structure; however, the support and frame fittings, and the fittings in the cargo hook area, are considered critical primary structure. Negligible damage to the forged fittings is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area, using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage greater than negligible limits in critical areas must be reviewed by engineering authority for damage evaluation and either repair or replacement. All repairs to the lower cabin framing must maintain the design strength and alignment.
3. Inspect lower cabin framing for corrosion. Remove surface corrosion using abrasive cloth, Item 83, WP 1803 00. **CORROSION REMOVAL SHALL NOT EXCEED 5% OF MATERIAL THICKNESS.**
4. Inspect for corroded fasteners. Remove corroded fasteners. Remove corrosion beneath fasteners using abrasive cloth, Item 83, WP 1803 00. **CORROSION REMOVAL SHALL NOT EXCEED 5% OF MATERIAL THICKNESS.** Replace fasteners (TM 1-1500-204-23).
5. Evaluate damage in noncritical areas (WP 0377 00).
6. Use WP 0378 00 and WP 0380 00 for typical framing repairs. Refer to TM 1-1500-204-23 for repair materials and fastener information.

REPAIR LOWER CABIN FRAMING - Continued

7. For corrosion protection and watertightness, seal all repairs between framing and plating (WP 0377 00 and TM 1-1500-344-23).
8. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR LOWER CABIN FRAMING - Continued



AA0102_1B
SA

Figure 1. Repair Lower Cabin Framing. (Sheet 1 of 13)

REPAIR LOWER CABIN FRAMING - Continued

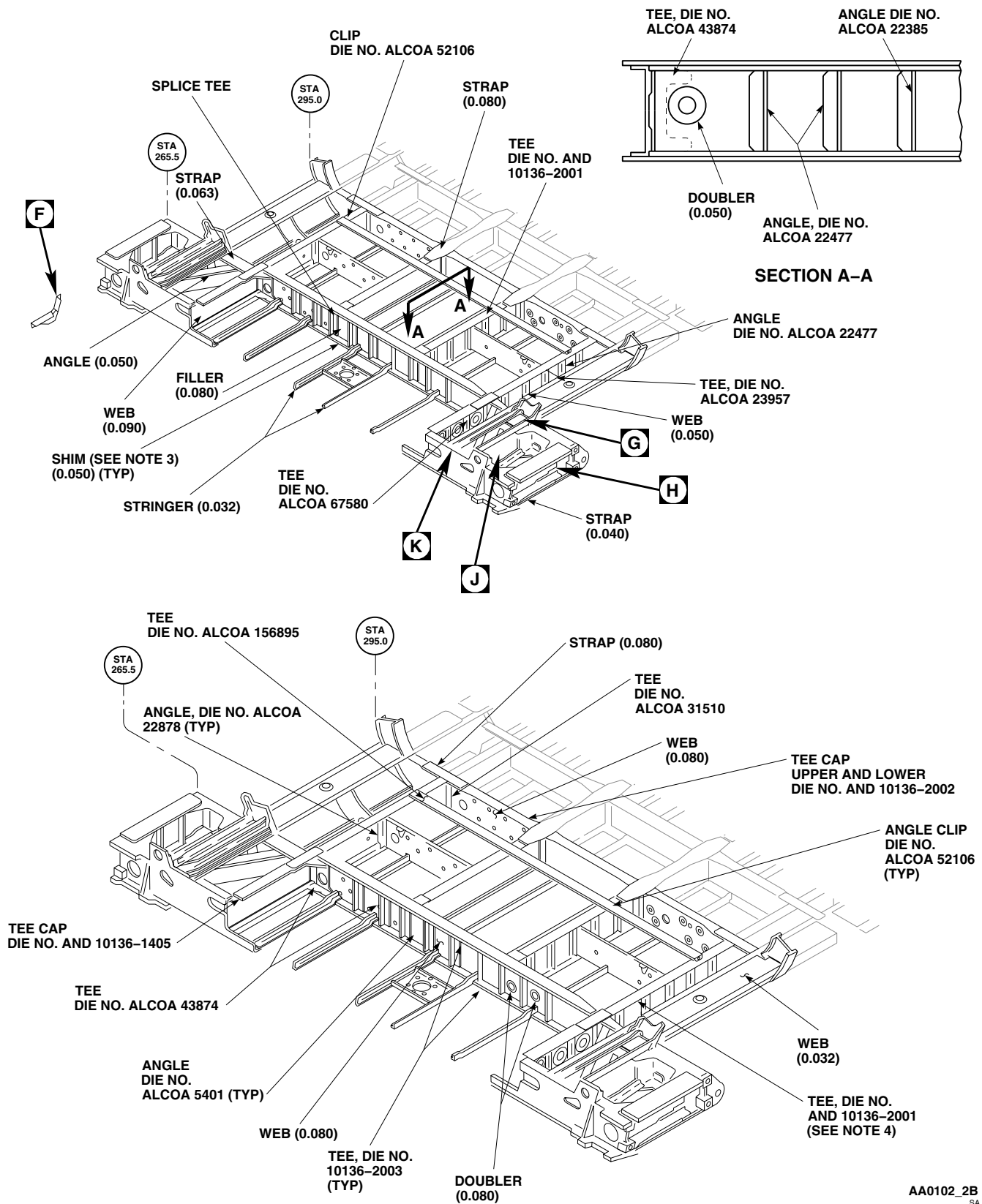


Figure 1. Repair Lower Cabin Framing. (Sheet 2 of 13)

REPAIR LOWER CABIN FRAMING - Continued

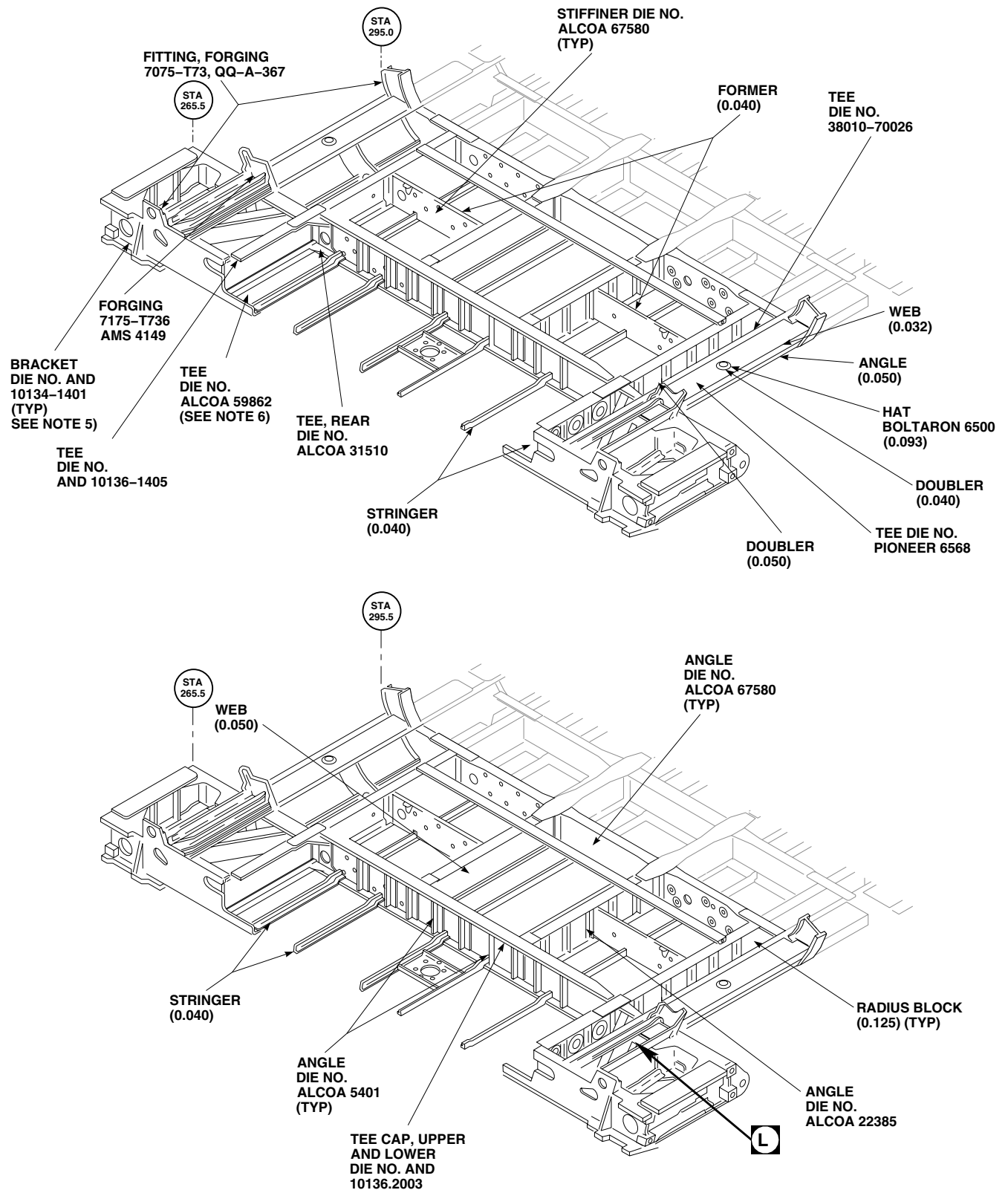
AA0102_3B
SA

Figure 1. Repair Lower Cabin Framing. (Sheet 3 of 13)

REPAIR LOWER CABIN FRAMING - Continued

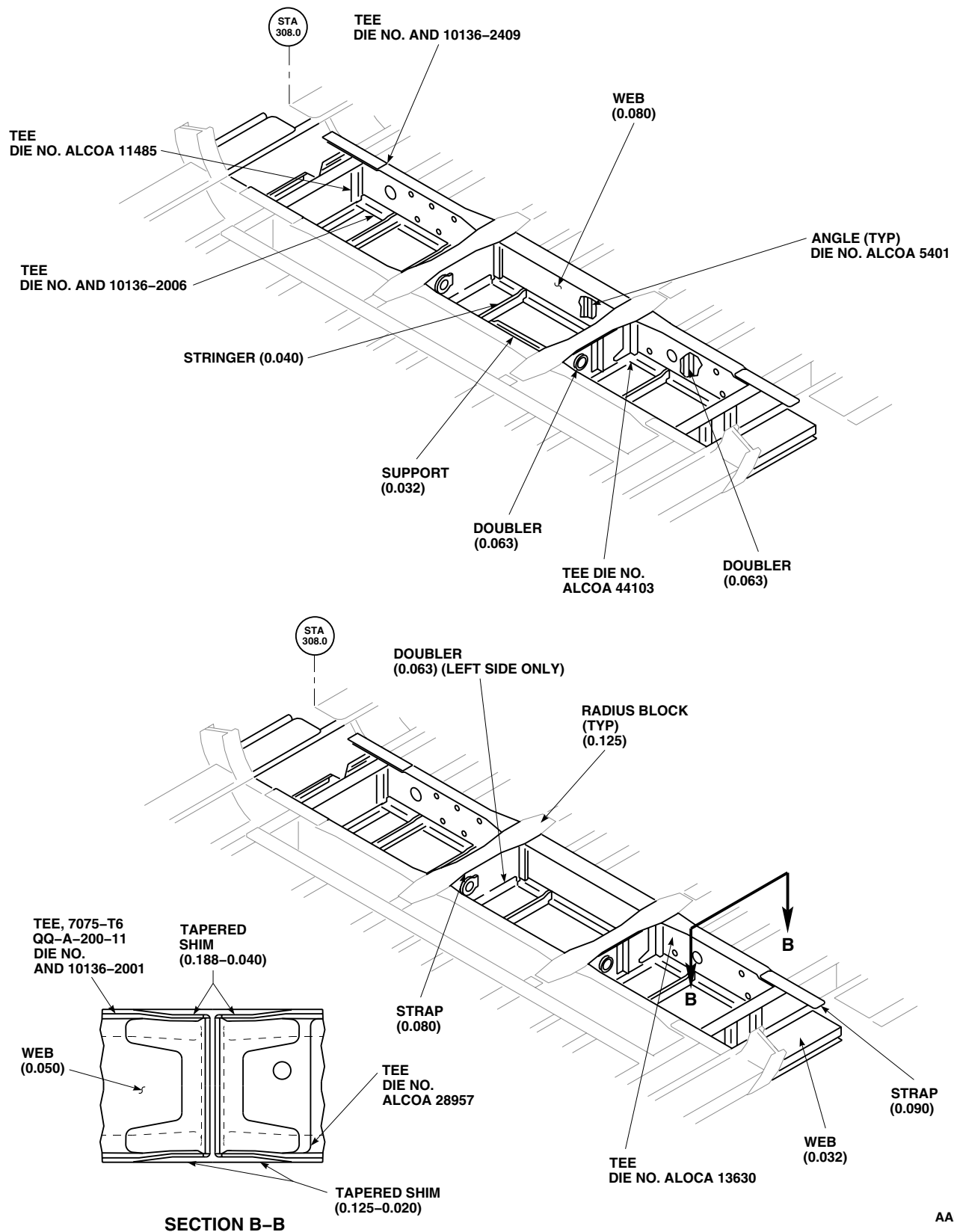
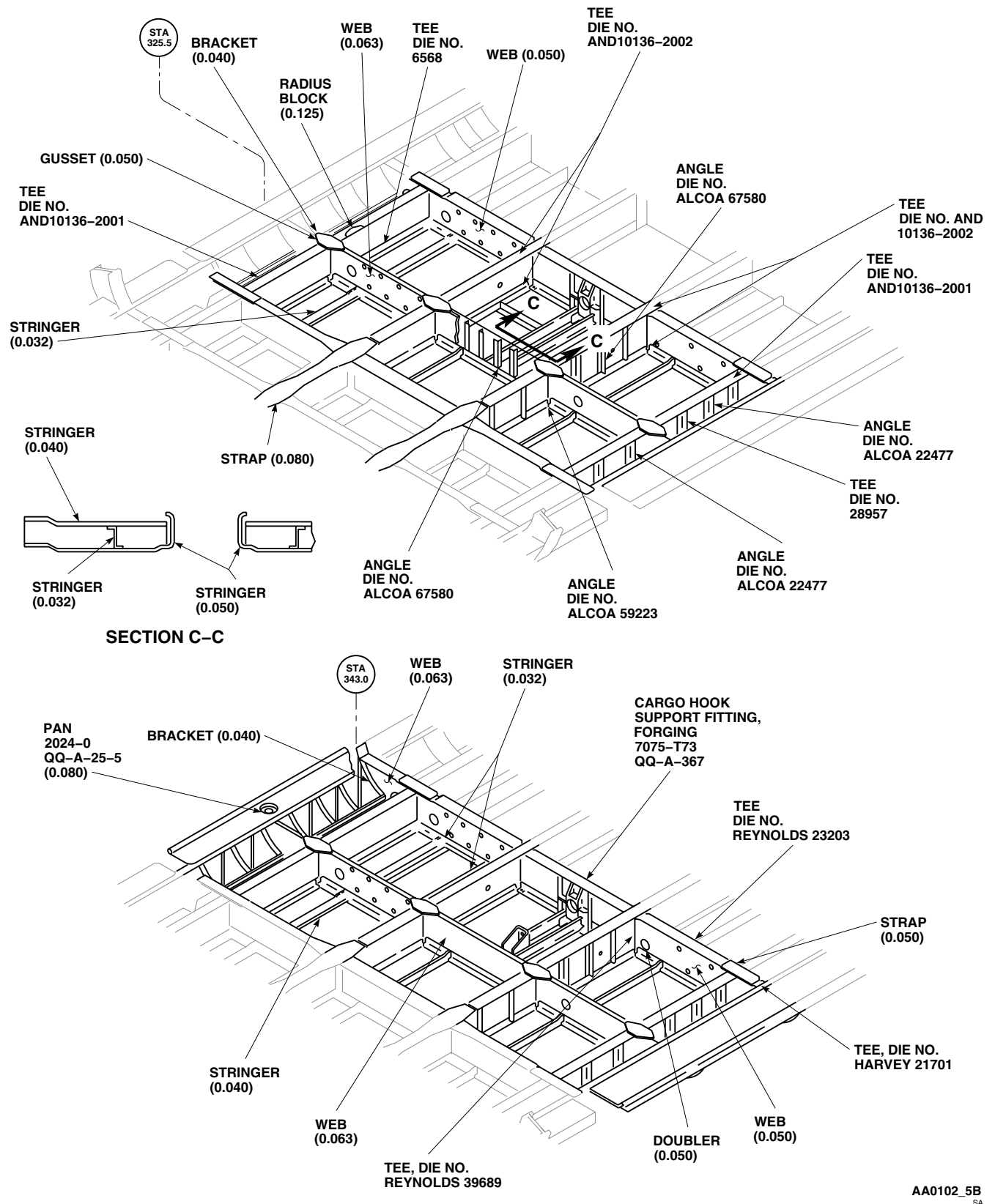


Figure 1. Repair Lower Cabin Framing. (Sheet 4 of 13)

REPAIR LOWER CABIN FRAMING - Continued



AA0102_5B
SA

Figure 1. Repair Lower Cabin Framing. (Sheet 5 of 13)

REPAIR LOWER CABIN FRAMING - Continued

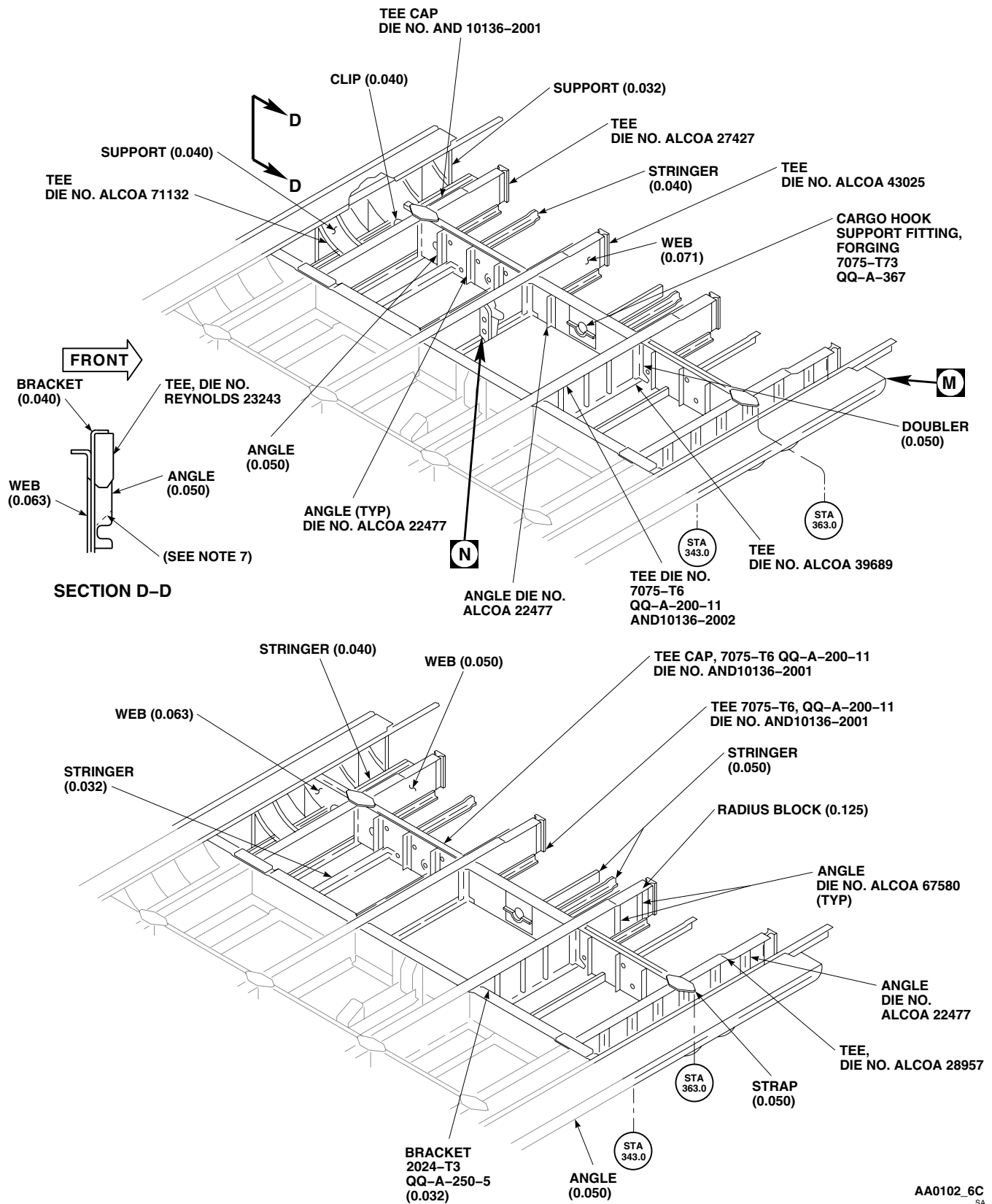
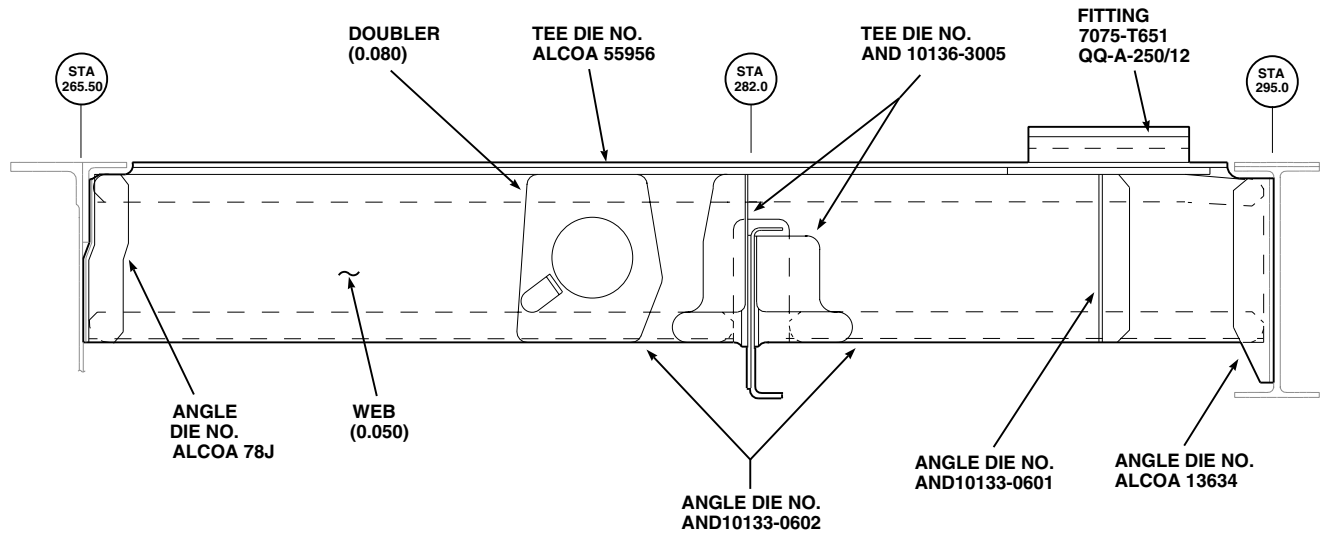


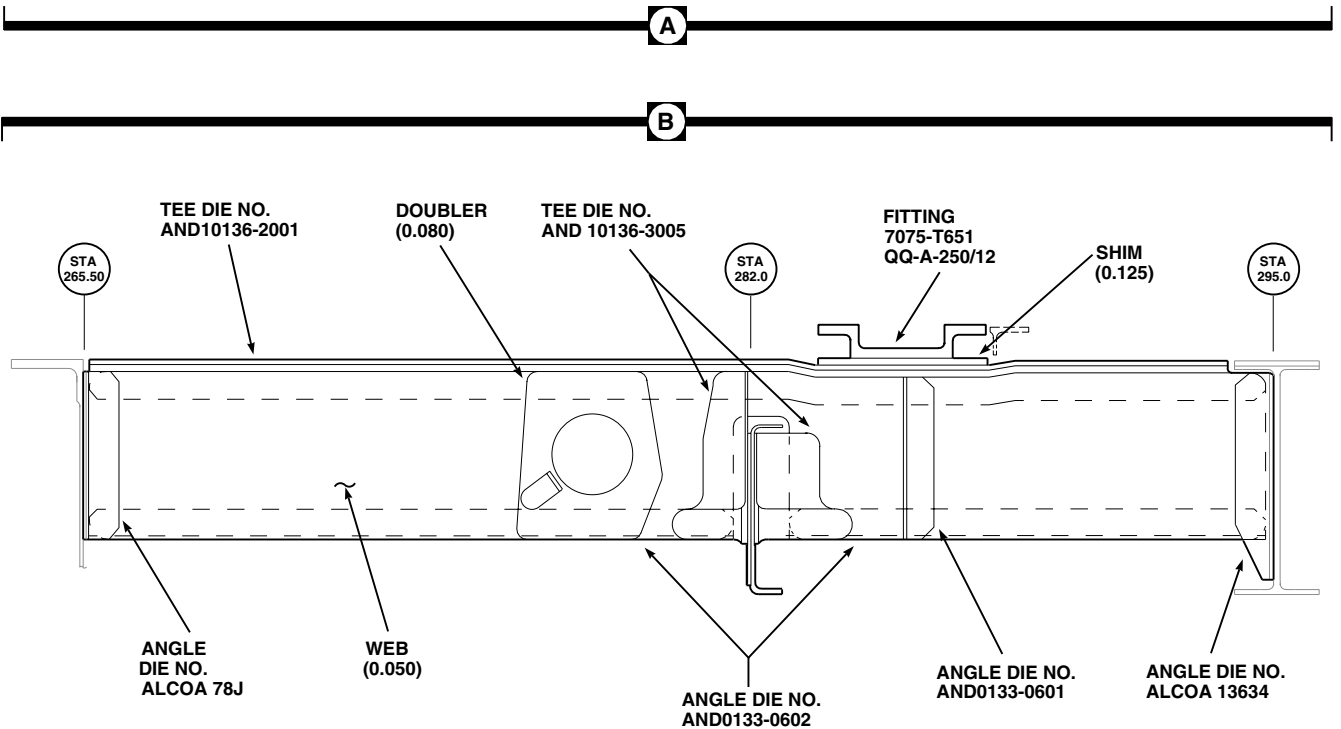
Figure 1. Repair Lower Cabin Framing. (Sheet 6 of 13)

REPAIR LOWER CABIN FRAMING - Continued



EFFECTIVITY

EH60A



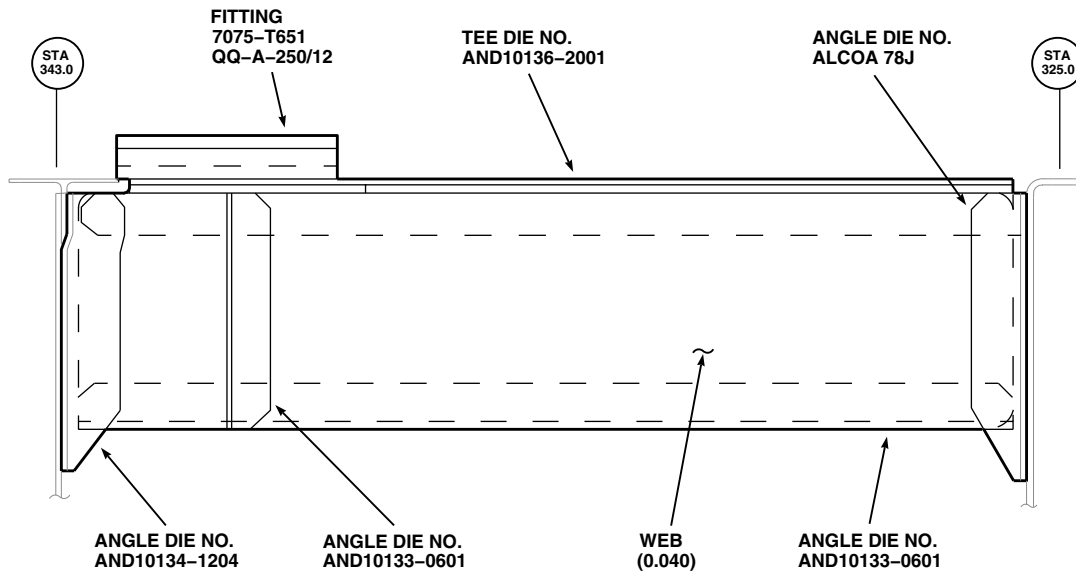
EFFECTIVITY

EH60A

AA0102_7C
SA

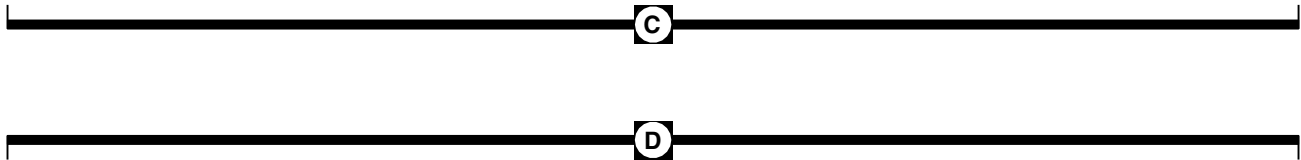
Figure 1. Repair Lower Cabin Framing. (Sheet 7 of 13)

REPAIR LOWER CABIN FRAMING - Continued



EFFECTIVITY

EH60A



EFFECTIVITY

EH60A

AA0102_8B
SA

Figure 1. Repair Lower Cabin Framing. (Sheet 8 of 13)

REPAIR LOWER CABIN FRAMING - Continued

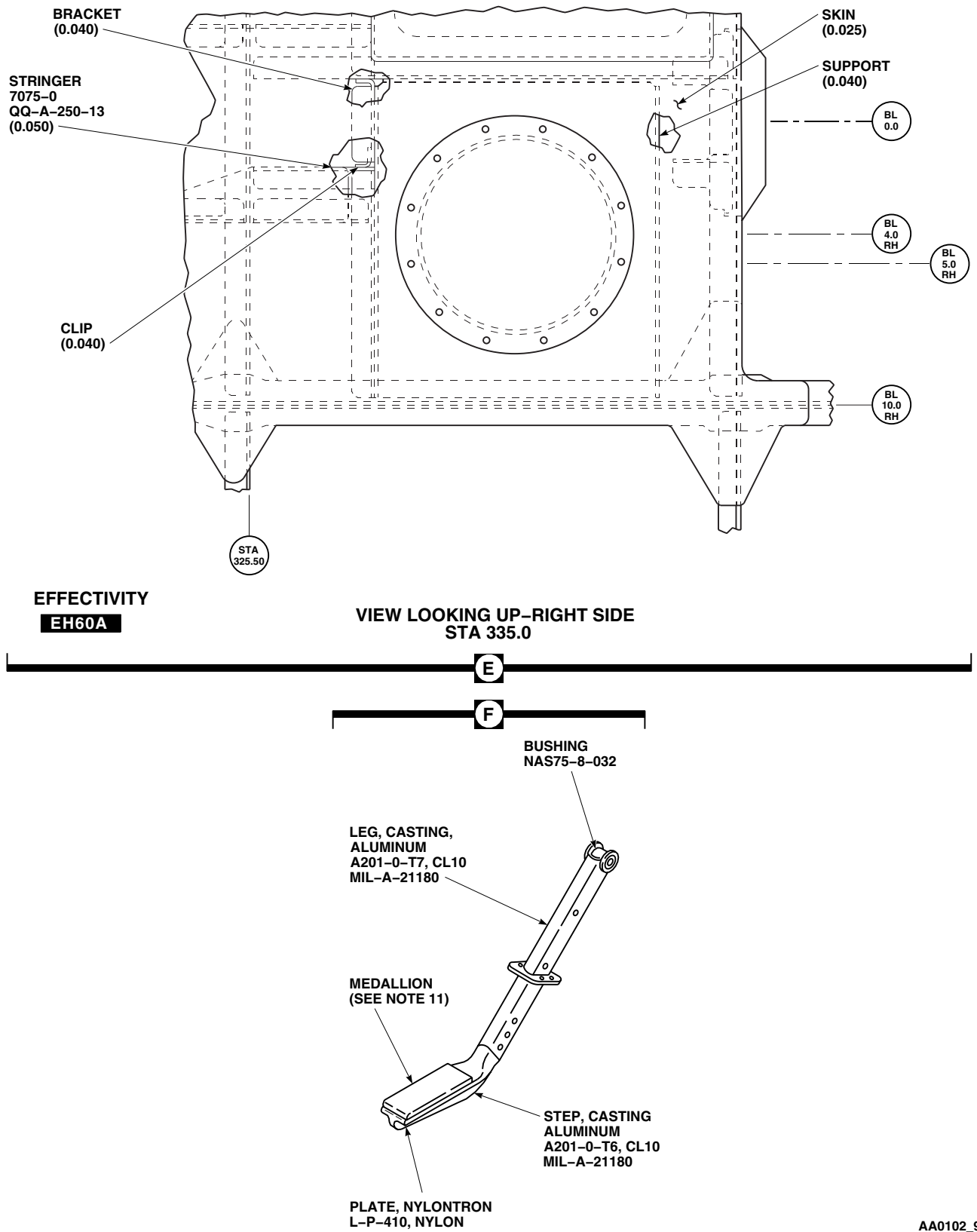
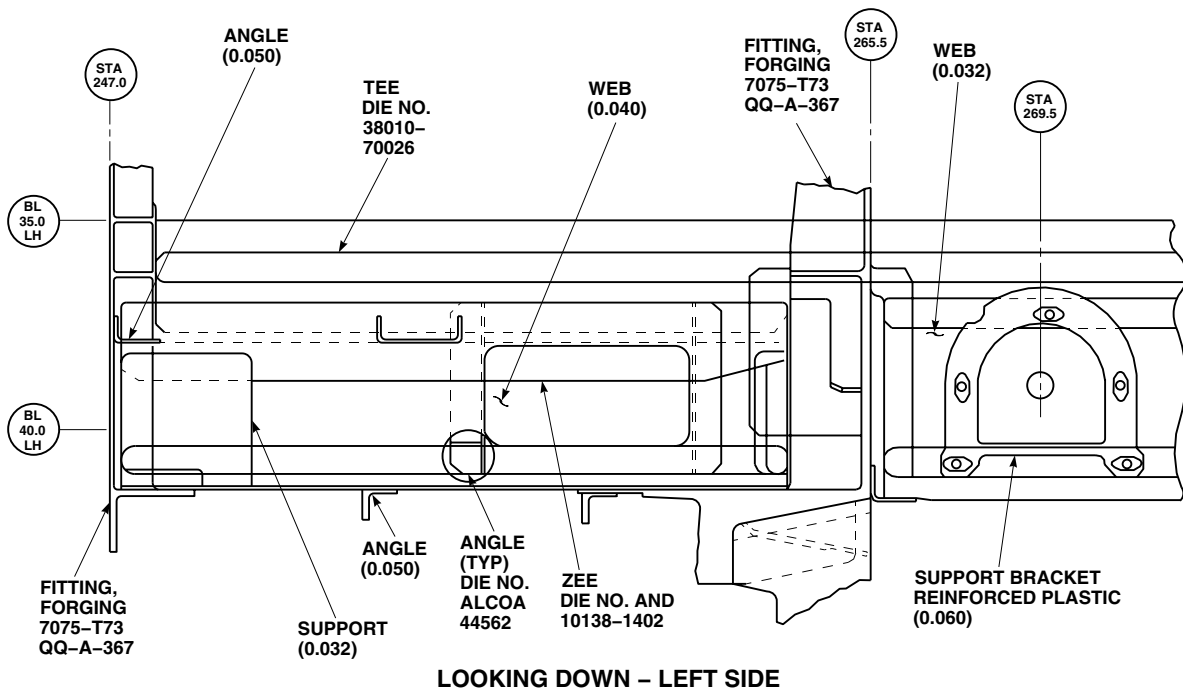


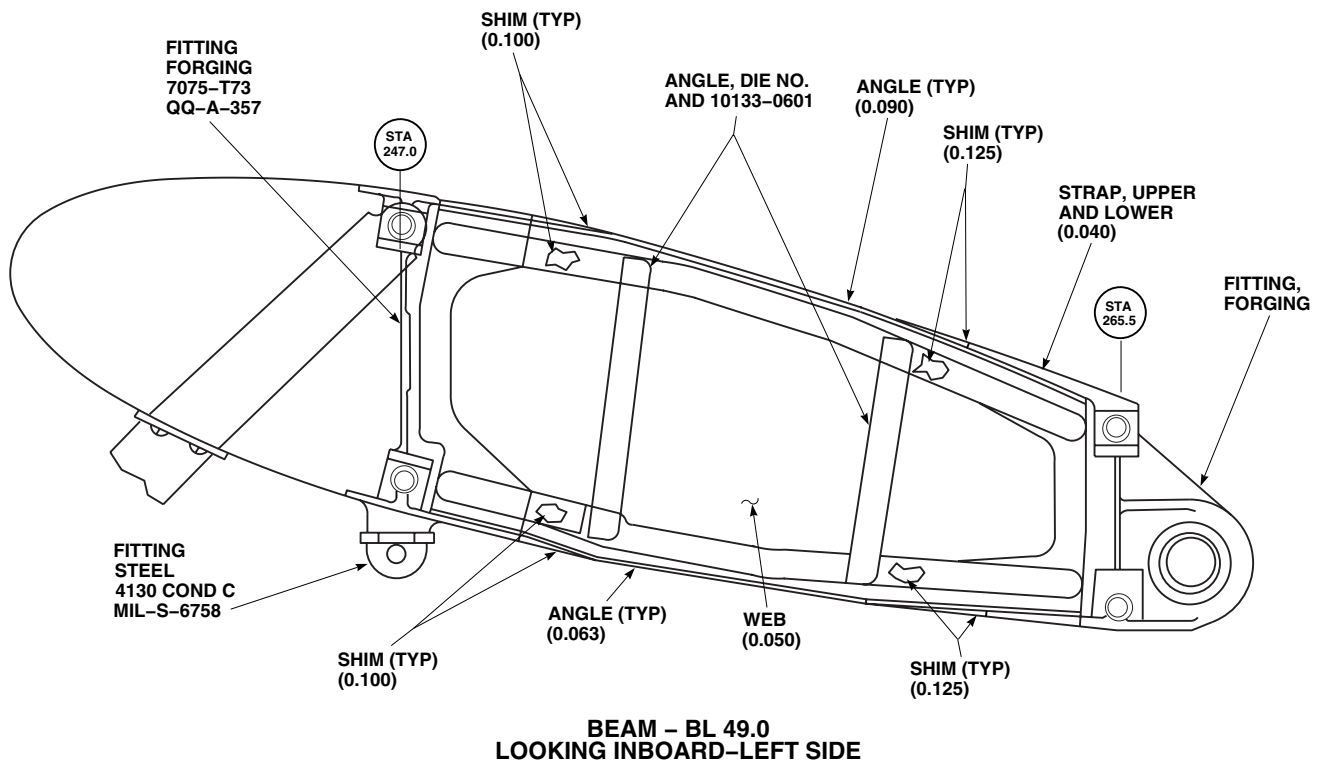
Figure 1. Repair Lower Cabin Framing. (Sheet 9 of 13)

REPAIR LOWER CABIN FRAMING - Continued



G

H



AA102_10B
SA

Figure 1. Repair Lower Cabin Framing. (Sheet 10 of 13)

REPAIR LOWER CABIN FRAMING - Continued

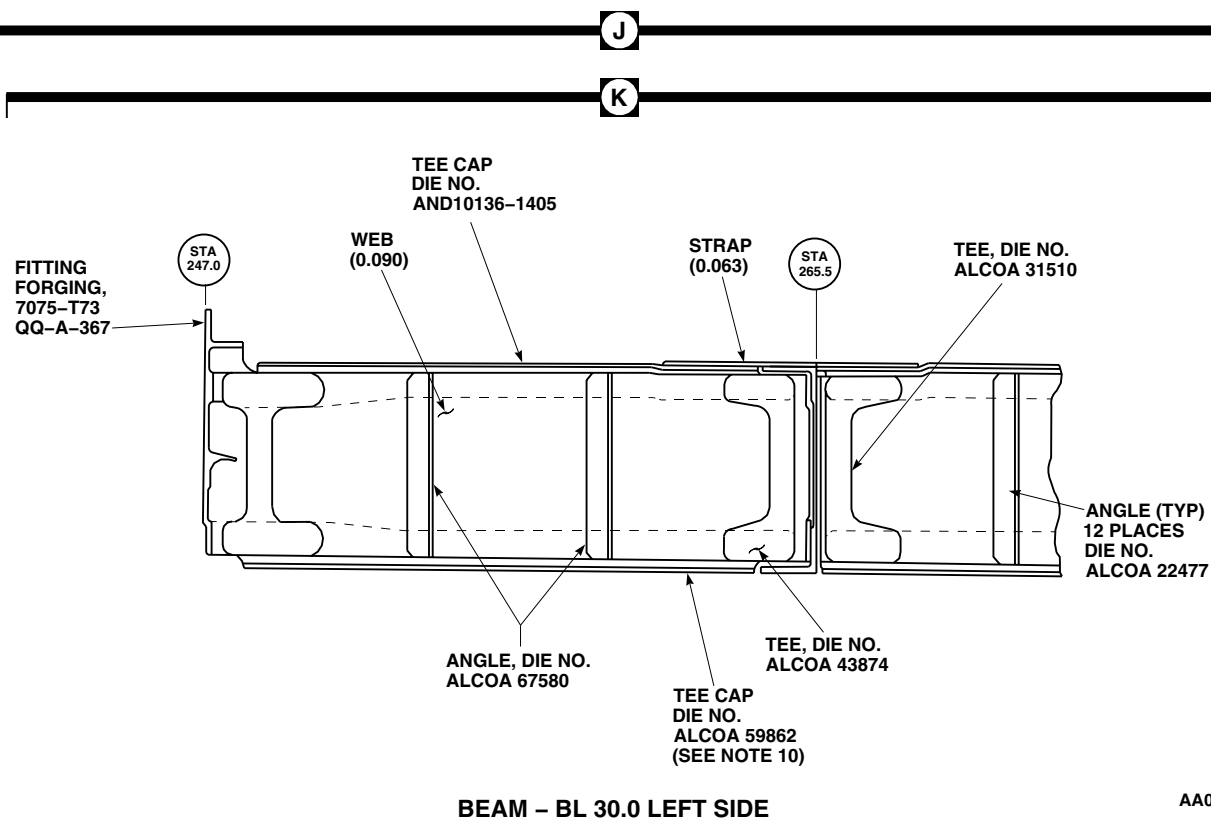
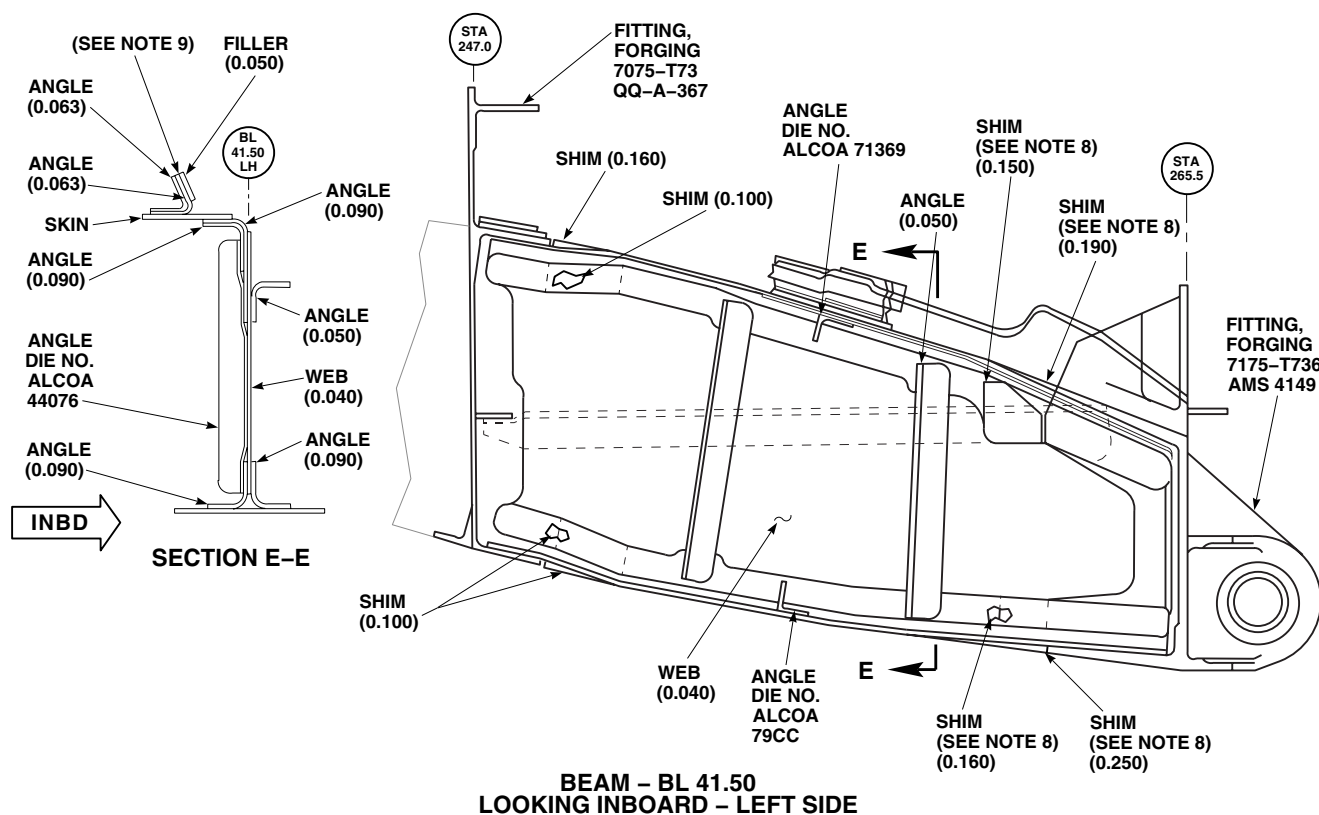
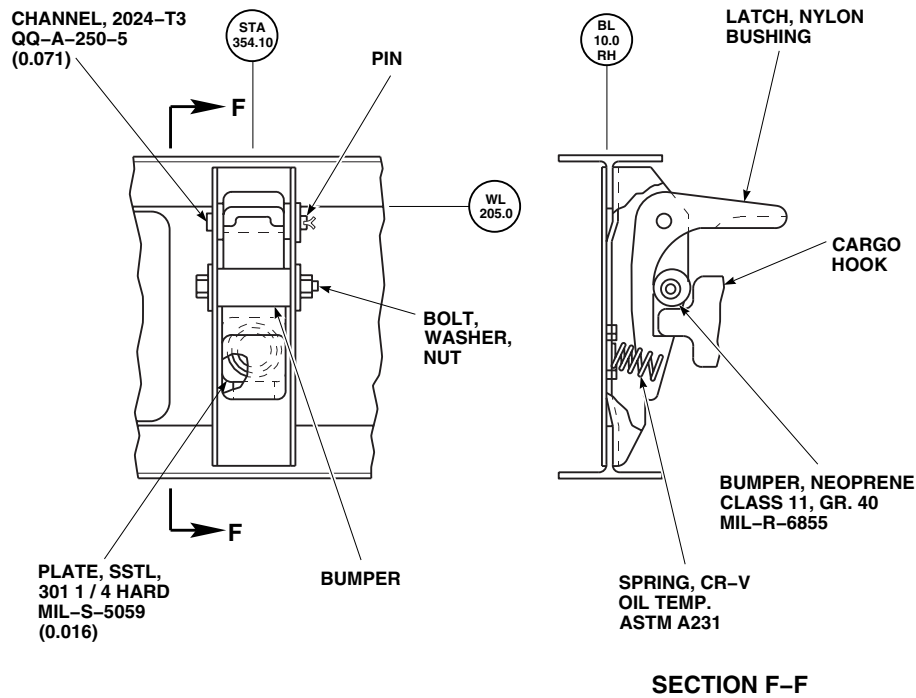
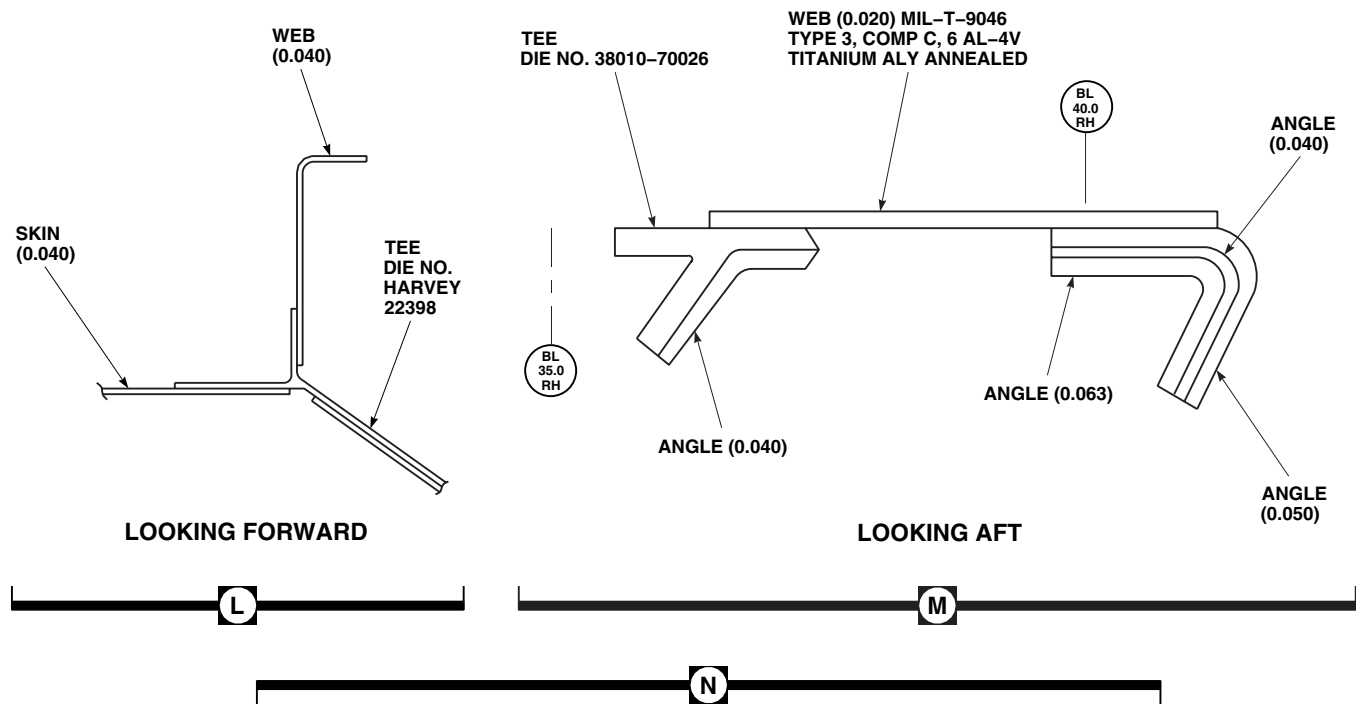


Figure 1. Repair Lower Cabin Framing. (Sheet 11 of 13)

REPAIR LOWER CABIN FRAMING - Continued



VIEW LOOKING OUTBOARD - RIGHT SIDE

AA0102_12B
SA

Figure 1. Repair Lower Cabin Framing. (Sheet 12 of 13)

REPAIR LOWER CABIN FRAMING - Continued**NOTES:**

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. BOND WITH SEALING COMPOUND, ITEM 273, WP 1803 00.
4. **UH60L** DIE NO. AND10136-2004. ◀
5. **W/O ESSS**
6. **UH60A 88-26084 - SUBQ** ▶ **UH60L** DIE NO. ALCOA NO. 33481. ◀
7. **UH60A 80-23432 - SUBQ** ▶ **UH60L** **EH60A** EDGE OF ANGLE. ◀
8. BOND SHIM TO STRUCTURE WITH ADHESIVE, ITEM 15, WP 1803 00.
9. SEAL AREA WITH SEALING COMPOUND, ITEM 273, WP 1803 00.
10. **UH60L 90-26309 - SUBQ** ▶ DIE NO. ALCOA 33481. ◀
11. BOND TO PLATE WITH ADHESIVE, ITEM 15, WP 1803 00. COAT WITH CLEAR URETHANE, MIL-C-81773, (0.001 THICK).
12. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0102_13D
SA

Figure 1. Repair Lower Cabin Framing. (Sheet 13 of 13)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

CABIN SECTION PLATING REPAIR

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Sump, 70302-02159-045

MIL-S-18729

MIL-S-5000

MIL-S-5059

MIL-STD-670

MIL-T-9046

Material/Parts

Adhesive, Item 14, WP 1803 00

Cloth, Abrasive, Item 75, WP 1803 00

Grommet, MS35489-5

Hinge, MS2057P3

Sealing Compound, Item 273, WP 1803 00

QQ-A-200

QQ-A-250

TM 1-1500-204-23

TM 1-1500-344-23

TM 55-1500-345-23

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

WP 0377 00

WP 0378 00

WP 0379 00

References

MIL-R-6855

MIL-S-1030

WP 0381 00

WP 1803 00

REPAIR CABIN SECTION PLATING

1. Cabin section plating is Alclad aluminum 7075-T6, QQ-A-250-13, except in high-temperature areas where titanium alloy, MIL-T-9046, is used. Certain fairings and panels are made from reinforced plastic (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, Figure 1, Sheet 7, Figure 1, Sheet 8, and Figure 1, Sheet 9).
2. All plating, except for reinforced plastic members, is primary structure. Evaluate damage (WP 0377 00).
3. Use WP 0378 00 for typical plating repairs. When repairing titanium plating, use same gage titanium stock, or the equivalent gage 301/302 stainless steel, MIL-S-5059, for patching. Use Monel or stainless steel fasteners for attaching titanium or stainless steel patches. Refer to TM 1-1500-204-23 for cutting, drilling, forming, and fastening titanium or stainless steel sheet stock.
4. Use WP 0379 00 for evaluating damage and repairing reinforced plastic.
5. Make sure drain holes in lower cabin plating are clear of debris.
6. Inspect lower cabin plating for surface corrosion. Remove surface corrosion using abrasive cloth, Item 75, WP 1803 00. **CORROSION REMOVAL SHALL NOT EXCEED 5% OF MATERIAL THICKNESS.**
7. Inspect for corroded fasteners. Remove corroded fasteners. Remove corrosion beneath fasteners using abrasive cloth, Item 75, WP 1803 00. **CORROSION REMOVAL SHALL NOT EXCEED 5% OF MATERIAL THICKNESS.** Replace fasteners and coordinate repairs with TM 1-1500-204-23 requirements.
8. For corrosion protection and watertightness, seal all repairs on the cabin section plating (WP 0377 00 and TM 1-1500-344-23).
9. Seal all areas where cabin ceiling is penetrated with sealing compound, Item 273, WP 1803 00.

REPAIR CABIN SECTION PLATING - Continued

10. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

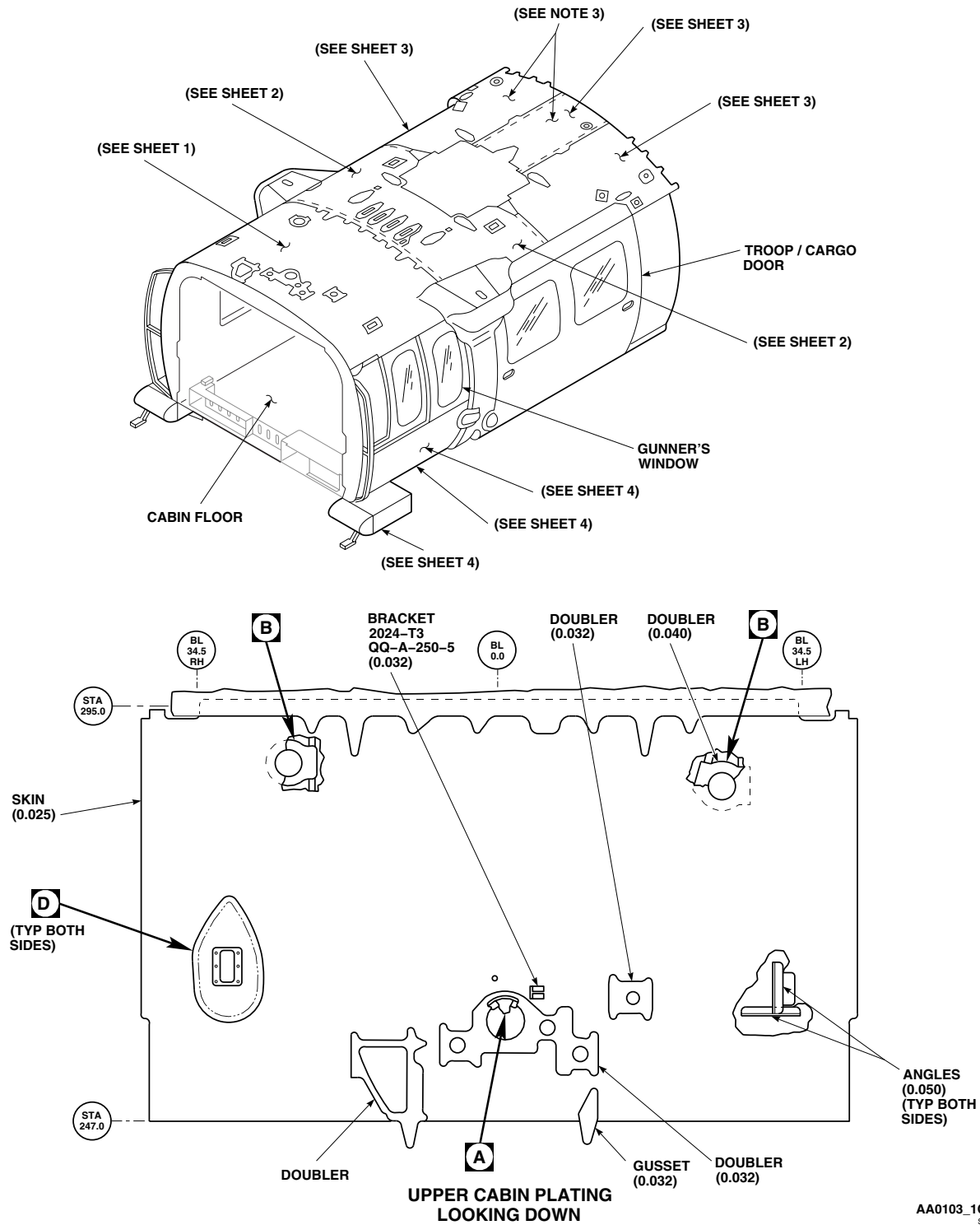


Figure 1. Repair Cabin Section Plating. (Sheet 1 of 10)

REPAIR CABIN SECTION PLATING - Continued

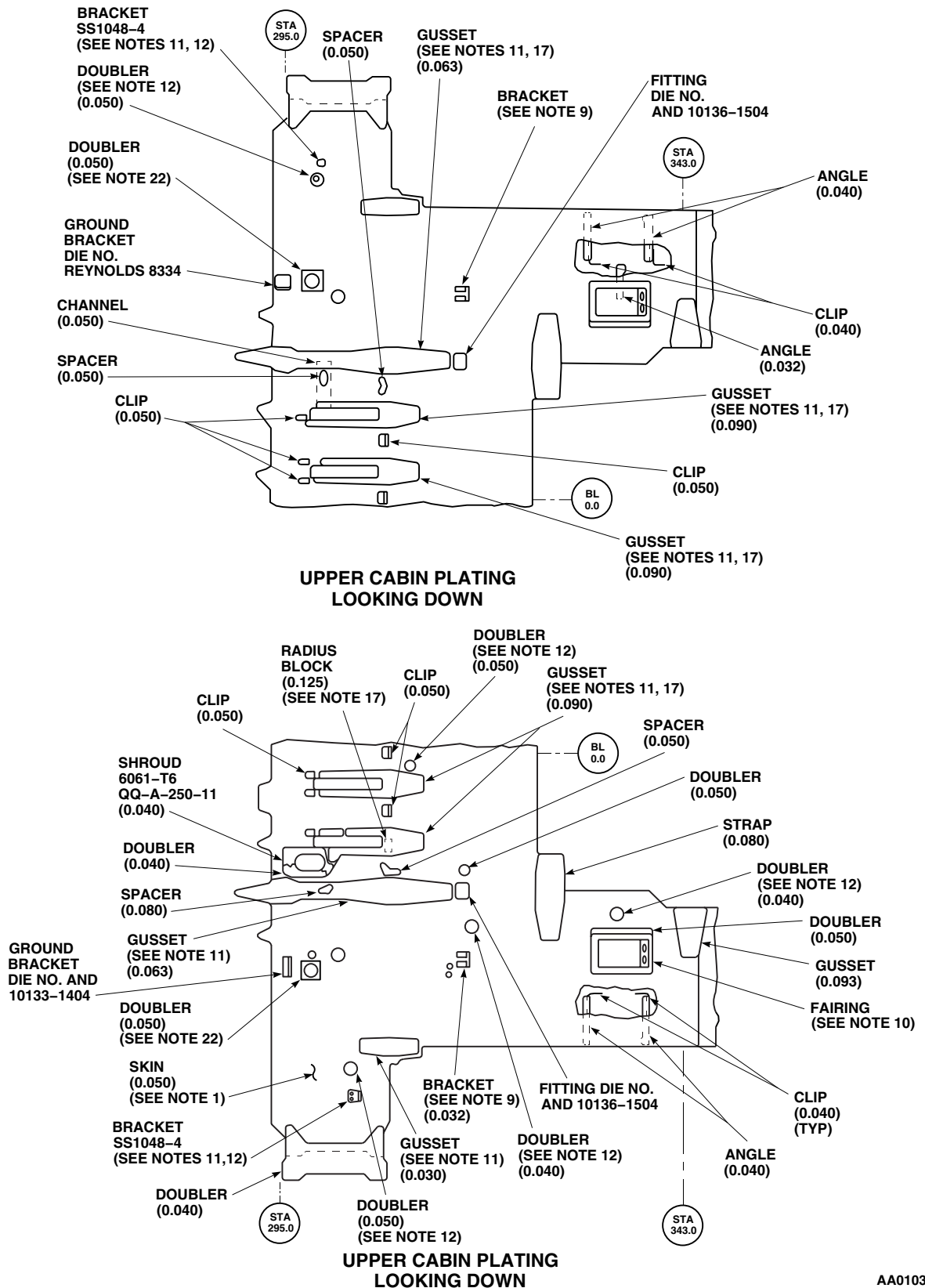
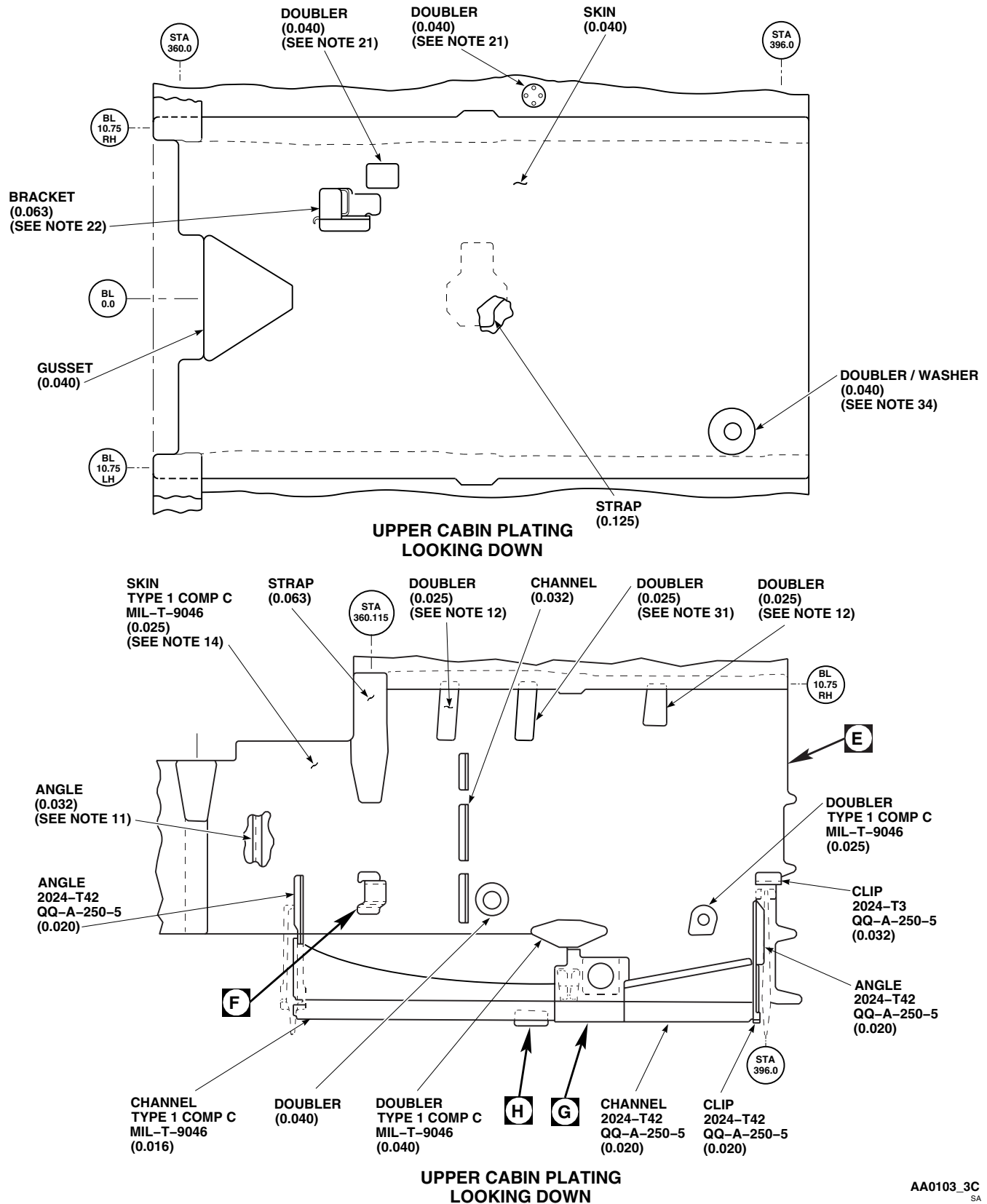


Figure 1. Repair Cabin Section Plating. (Sheet 2 of 10)

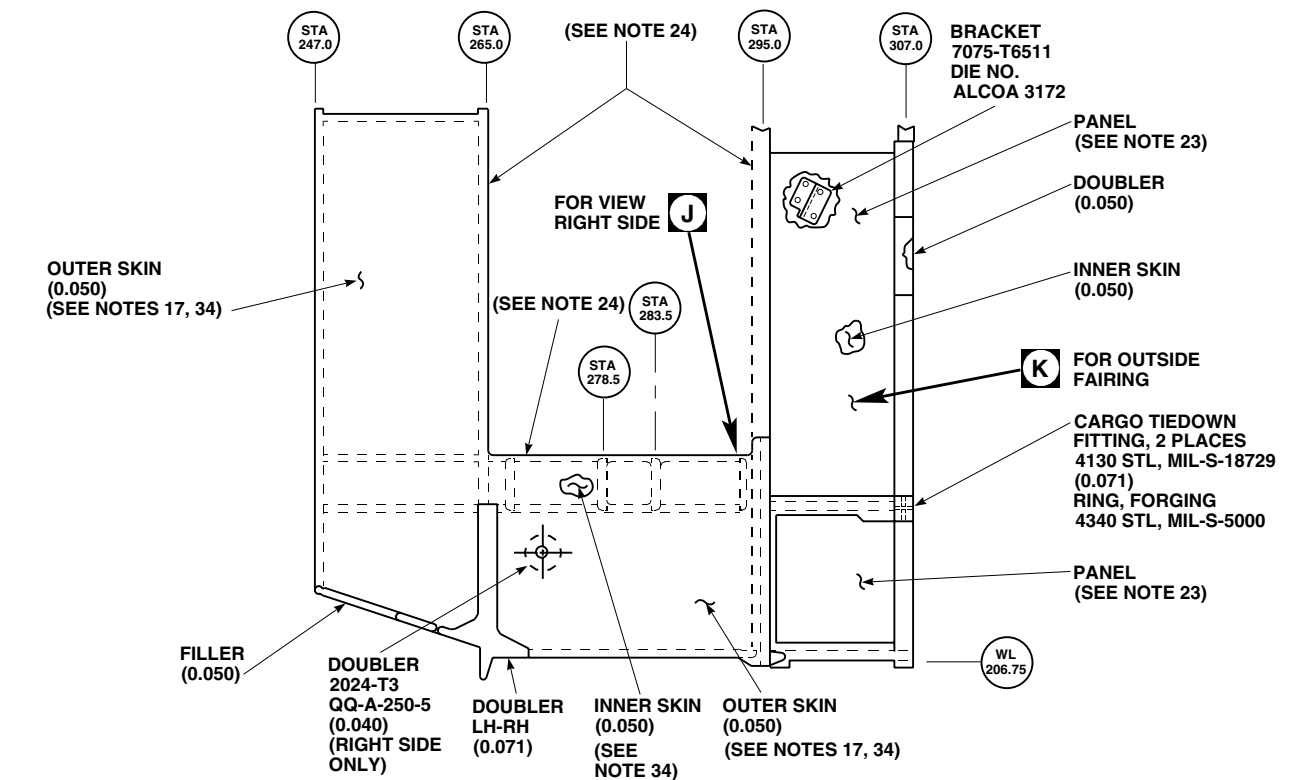
REPAIR CABIN SECTION PLATING - Continued



AA0103_3C
SA

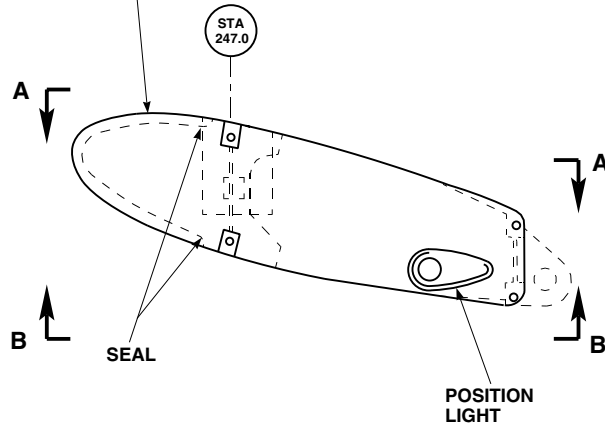
Figure 1. Repair Cabin Section Plating. (Sheet 3 of 10)

REPAIR CABIN SECTION PLATING - Continued



LOOKING INBOARD - LEFT SIDE
RIGHT SIDE SAME EXCEPT AS NOTED

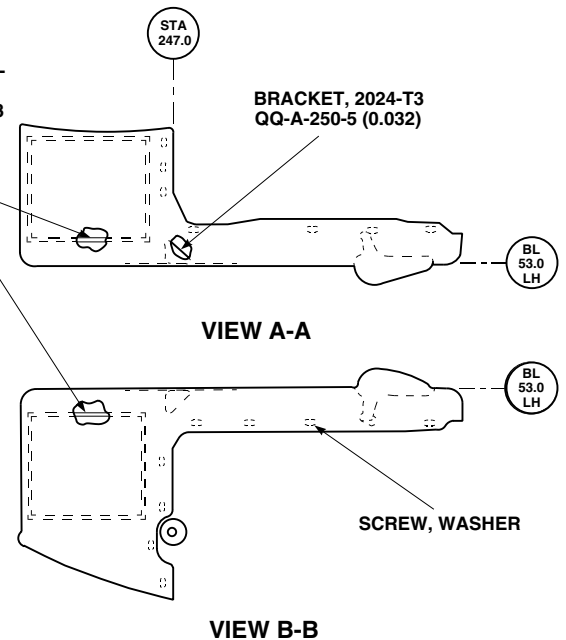
FAIRING, FLEXCORE (0.375)
KEVLAR SKIN (0.050 to 0.070)



LOOKING INBOARD-LEFT SIDE

SEAL, STYRENE-BUTADIENE RUBBER, CH-383 (CAGE 83473) (SEE NOTE 30)

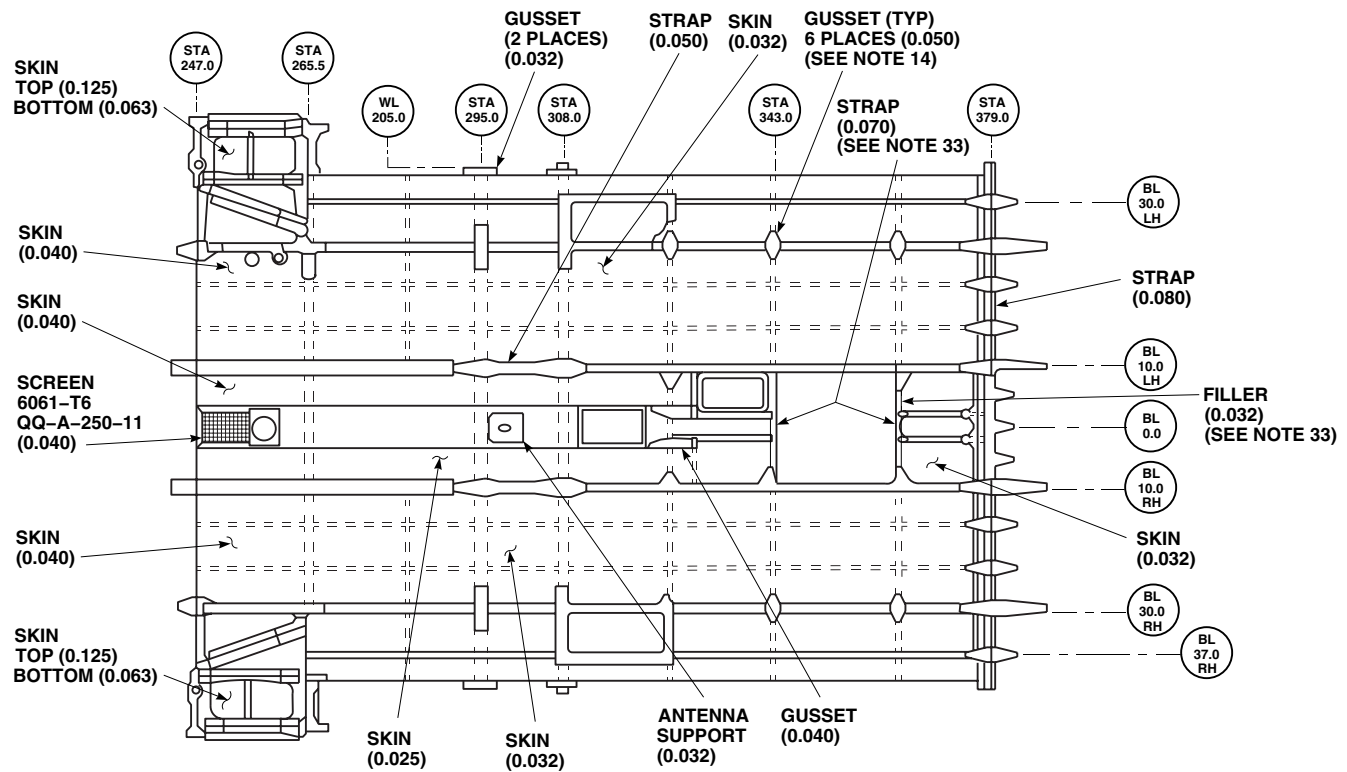
BRACKET, 2024-T3 QQ-A-250-5 (0.032)



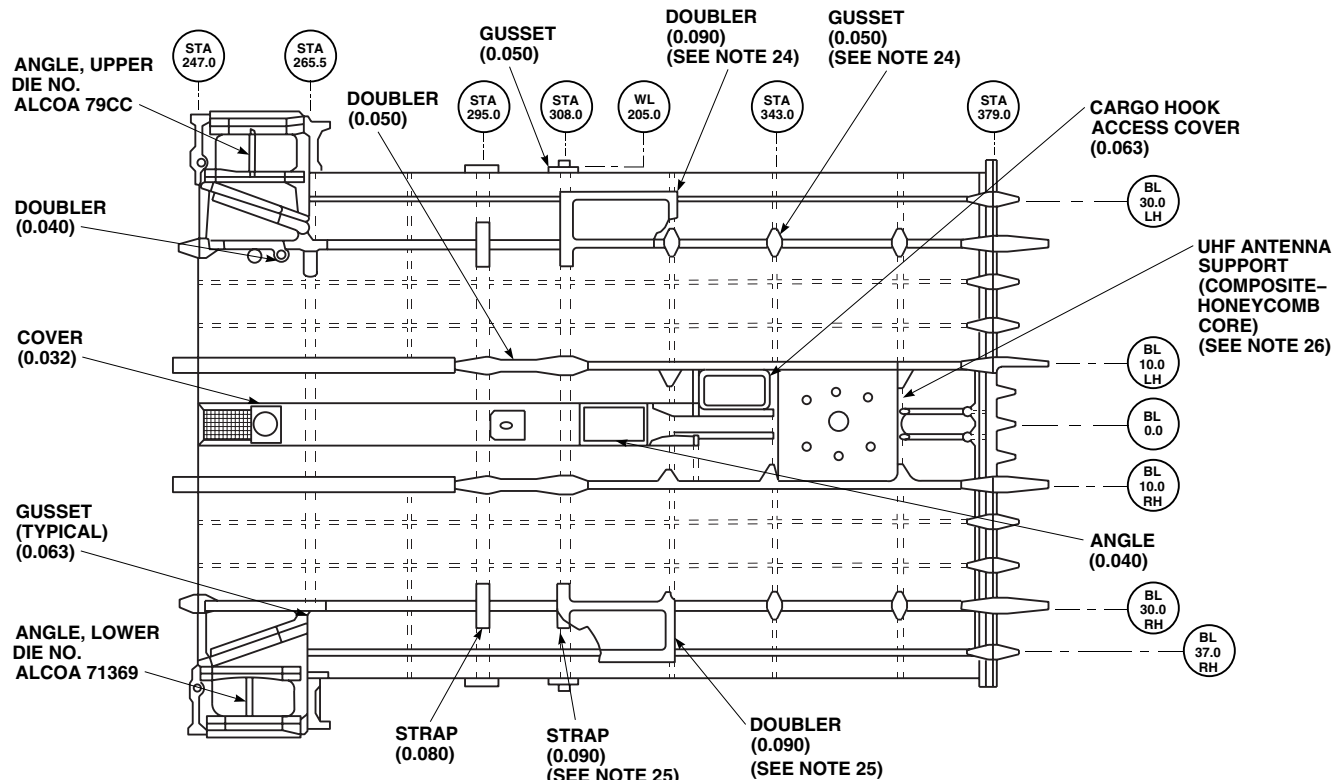
AA0103_4F
SA

Figure 1. Repair Cabin Section Plating. (Sheet 4 of 10)

REPAIR CABIN SECTION PLATING - Continued



LOWER CABIN PLATING-LOOKING UP



LOWER CABIN PLATING-LOOKING UP

AA0103_5C
SA

Figure 1. Repair Cabin Section Plating. (Sheet 5 of 10)

REPAIR CABIN SECTION PLATING - Continued

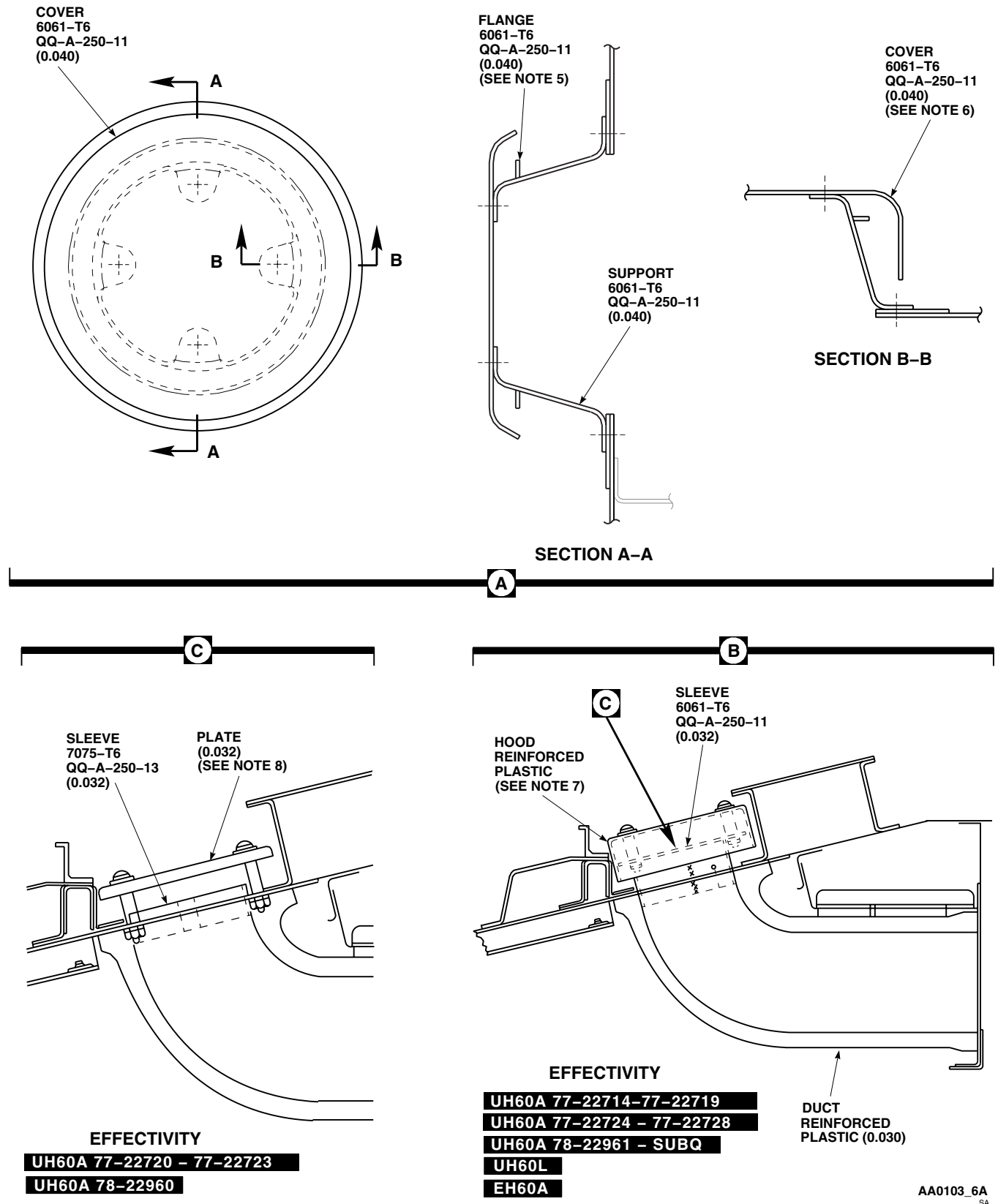


Figure 1. Repair Cabin Section Plating. (Sheet 6 of 10)

REPAIR CABIN SECTION PLATING - Continued

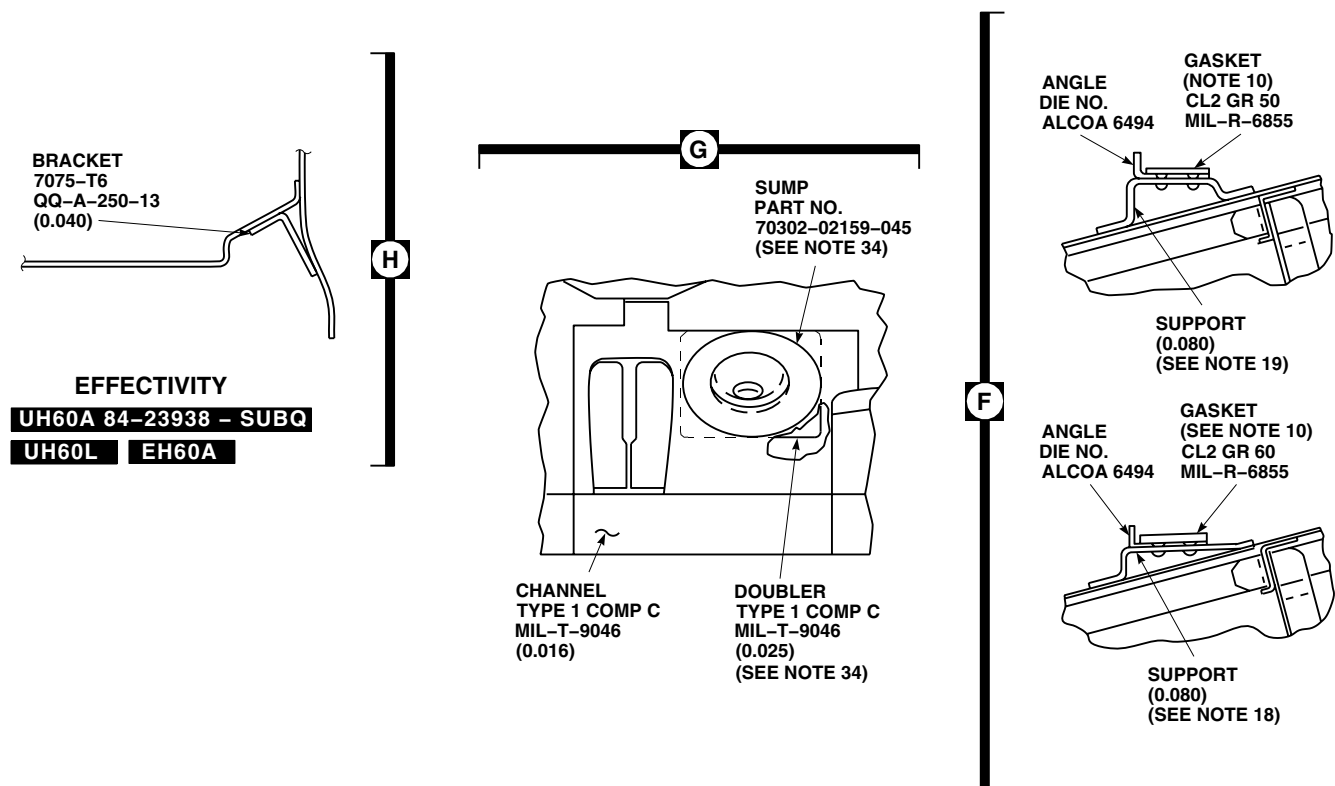
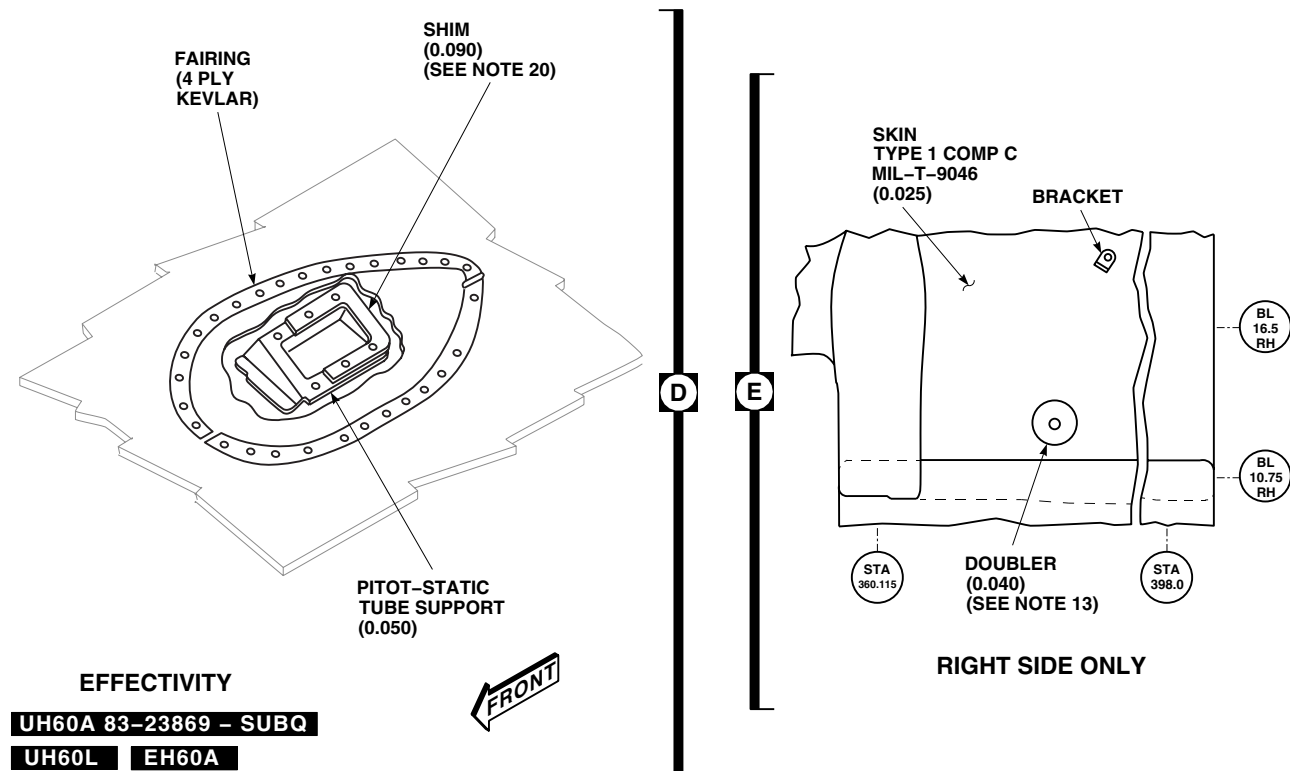
AA0103_7C
SA

Figure 1. Repair Cabin Section Plating. (Sheet 7 of 10)

ENCLOSURE (SEE NOTE 28)

LATCH (SEE NOTE 27)

DOUBLER (0.040)

CLIP, 301 ST. STL, 1 / 4H MIL-S-5059 (0.050)

SHIM (0.090)

HINGE

OUTER SKIN (0.050)

COVER (0.040)

STA 281.0

STA 276.0

FRONT

LOOKING INBOARD - RIGHT SIDE

J

SEAL, NEOPRENE FOAM, DCE 3L MIL-STD-670 (0.250)

VELCRO LOOP

VELCRO HOOK

RETAINER (0.025)

EFFECTIVITY

UH60A 77-22714 - 78-22976

L

CAP FAIRING 6061-T6 QQ-A-250-11 (0.040)

ANGLE (0.063)

FILLER (0.040)

STOP MIL-R-6855 CL1, GR40 (0.250)

FILLER (0.040)

STOP MIL-R-6855 CLI, GR40 (0.500)

FILLER (0.040)

HINGE

ANGLE (0.040)

FAIRING KEVLAR

GUSSET (0.040)

TAPERED SHIM (0.063)

ANGLE (0.040)

DOOR 6061-T6 (0.040)

PANEL (0.040)

COVER (0.040)

GROMMET

STA 295.0

STA 308.0

FRONT

LOOKING INBOARD - LEFT SIDE

K

BUMPER, NEOPRENE FOAM, SCE 3L MIL-STD-670 (0.380)

ACCESS DOOR REINFORCED PLASTIC (0.050 TO 0.080)

FILLER (0.032)

LAND (0.040)

GROMMET MS35489-5

COVER (0.040)

FAIRING REINFORCED PLASTIC (0.050 TO 0.070)

ANGLE (0.040)

SEAL, RUBBER AMS 3195

FORMER (0.040)

PANEL (0.040)

HINGE MS2057P3

WL 260.0

STA 295.0

STA 308.0

LOOKING INBOARD - LEFT SIDE

EFFECTIVITY

UH60A 77-22714 - 82-23747

L

AA0103_8B

0389 00-9

REPAIR CABIN SECTION PLATING - Continued

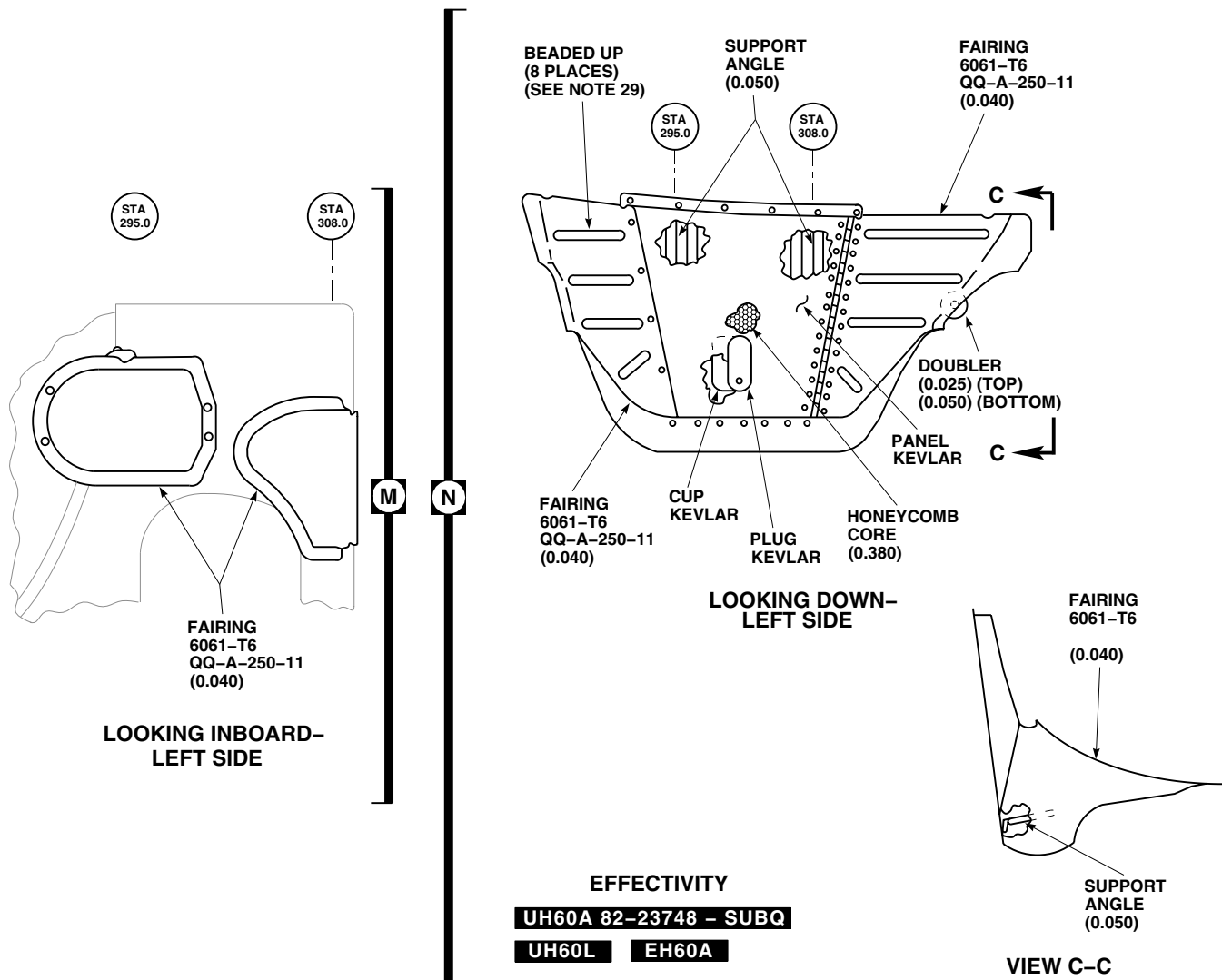
AA0103_9B
SA

Figure 1. Repair Cabin Section Plating. (Sheet 9 of 10)

REPAIR CABIN SECTION PLATING - Continued

NOTES:

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHERE NOTED.
3. SEAL ALL FAYING SURFACES WITH SEALING COMPOUND, ITEM 273, WP 1803 00 (WP 0377 00).
4. **UH60A 77-22714 - 78-22964** 0.032 **UH60A 78-22965 - SUBQ** **UH60L** **EH60A** 0.058.
5. **UH60A 78-22988 - SUBQ** **UH60L** **EH60A**
6. **UH60A 81-23547 - SUBQ** **UH60L** **EH60A**
7. **UH60A 83-23868 - SUBQ** **UH60L** **EH60A**
8. **UH60A 77-22714 - 83-23868**
9. 2024-T3, QQ-A-250-15.
10. REINFORCED PLASTIC 0.030; ALTERNATE MATERIAL: 6061-T6, QQ-A-250-11 0.032.
11. STEEL, 4130, COND N, MIL-S-18729.
12. **UH60A 77-22727 - 77-22728** **UH60A 78-22963 - SUBQ** **UH60L** **EH60A**
13. **ESSS** **UH60L** **EH60A**
14. **UH60L 90-26309 - SUBQ** 0.020, MIL-T-9046, TYPE III, COMP C
15. BOND WITH ADHESIVE, ITEM 36, WP 1803 00.
16. CLEAN AND SEAL WITH SEALING COMPOUND, ITEM 273, WP 1803 00 (WP 0377 00).
17. CLEAN AND SEAL WITH SEALING COMPOUND, ITEM 273, WP 1803 00 (WP 0377 00).
18. **UH60A 77-22714 - 79-23290**
19. **UH60A 79-23291 - SUBQ** **UH60L** **EH60A**
20. **UH60A 86-24514 - SUBQ** **UH60L** **EH60A 85-24469 - SUBQ**
21. **UH60L**
22. **UH60A 88-26092 - 88-26048** **UH60A 88-26119 - SUBQ** **MOD** **UH60L**
23. DAMPING FOIL SJ2017X (CAGE 76381).
24. PLUG OPENINGS WITH SEALING TAPE, MIL-S-1030. SEAL WITH SEALING COMPOUND, ITEM 273, WP 1803 00 FILLET SEALING COMPOUND ALONG EDGE OF WINDOW FRAME AND SKIN (WP 0377 00).
25. **W/O ESSS**
26. **EH60A**
27. H-5010K1941-040-125 (CAGE 83014).
28. **UH60A 77-22714 - 82-23572** FIBERGLASS, 0.040 TO 0.080 **UH60A 82-23573 - SUBQ** **UH60L** **EH60A** FIBERGLASS AND 6061-T6 ENCLOSURES ARE INTERCHANGEABLE.
29. **UH60A 82-23748 - 87-24581** **EH60A 84-24017 - 86-24562**
30. BOND TO FAIRING WITH ADHESIVE, ITEM 14, WP 1803 00.
31. **UH60A 89-26171 - SUBQ**
32. **UH60L 92-26421 - SUBQ** **UH60L**
33. APPLY SEALING COMPOUND, ITEM 273, WP 1803 00, TO MATING SURFACES AND FASTENERS. INSTALL WHILE WET.
34. **UH60L 96-26723 - SUBQ** SKINS ARE NOT CHEMICALLY MILLED.
35. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0103_10G
SA

Figure 1. Repair Cabin Section Plating. (Sheet 10 of 10)

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

CABIN SECTION BEADED INNER FUSELAGE PLATING STA 295.0 THROUGH STA 308.0 REPAIR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

WP 0481 00

WP 0630 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-344-23

WP 0377 00

REPAIR**NOTE**

Only a total of 5 beads are permitted to be repaired per panel, after which the panel must be replaced at next scheduled preventive maintenance periodic inspection.

1. Evaluate repairs using WP 0377 00 and TM 1-1500-344-23.
2. Prior to repair, remove engines, (WP 0481 00) and transmission, (WP 0630 00). If shoring is preferred, permission must be received from the responsible maintenance engineer.
3. Cut away damaged bead to a point 1/16-inch beyond bead bend.
4. Round corners to 0.25-inch radius.
5. Attach narrow doubler to large doubler plate with a single row of rivets (Figure 1).
6. With narrow doubler fitted into bead cutout, butt large doubler against existing strap fastened to STA 308.0 frame.
7. Attach repair large doubler to existing skin using a double row of rivets on STA 295.0 side of cutout (Figure 1).
8. Space rivets 0.38-inch from edge of large doubler and 1 inch apart.
9. Position shim over strap where front doubler will attach.
10. Attach front doubler plate over shim and large doubler, picking up existing rivets at STA 308.0.

REPAIR - Continued

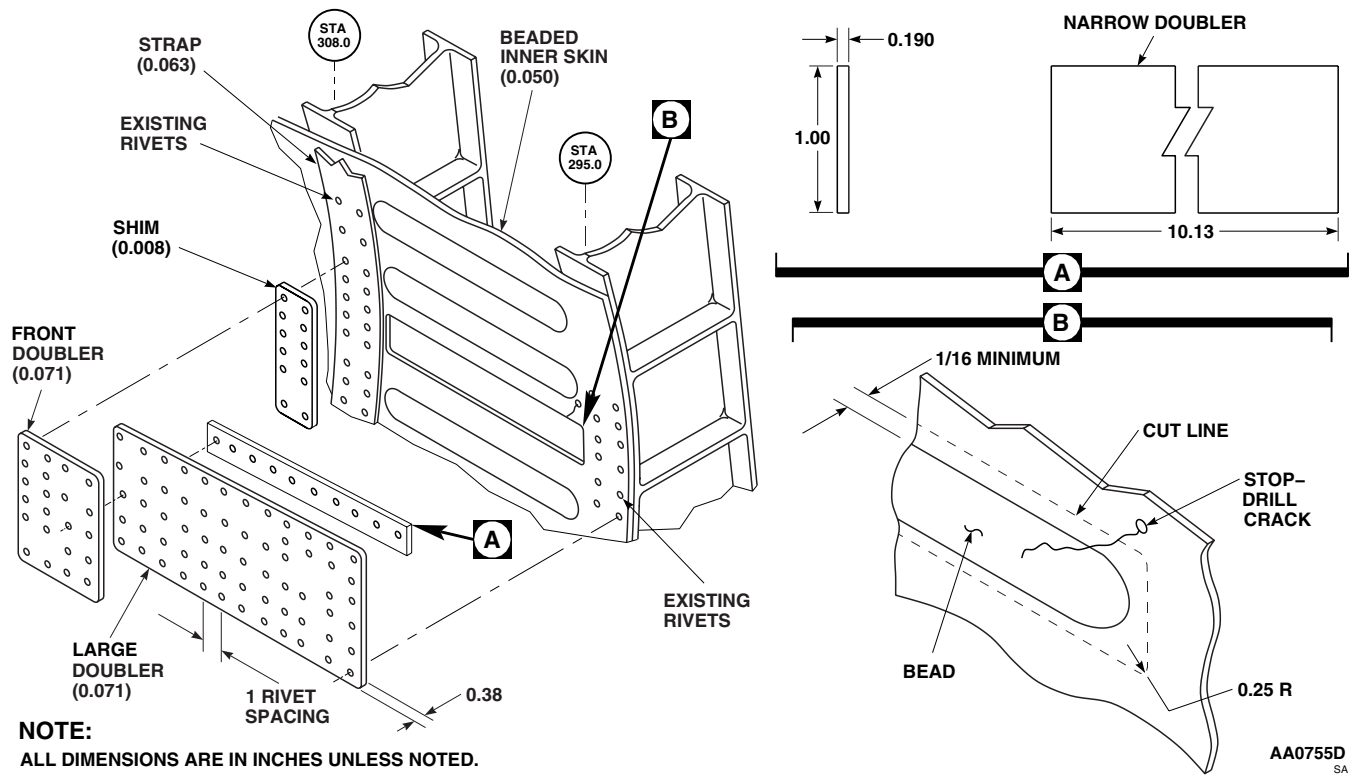


Figure 1. Repair Cabin Section Beaded Inner Fuselage Plating, STA 295.0 Through 308.0.

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

CARGO DOOR UPPER TRACK REPAIR

This WP supersedes WP 0391 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
 Drill Set, GGG-D-751
 Gloves
 Goggles/Face Shield
 Torque Wrench, 30 - 200 in. lbs.

References

AMS 4013
 MIL-S-18729
 MIL-S-22499
 TM 1-1500-204-23
 TM 1-1500-344-23
 TM 55-1500-345-23
 WP 0244 00
 WP 0254 00
 WP 0313 00
 WP 0381 00
 WP 0489 00
 WP 1803 00
 WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 11, WP 1803 00
 Corrosion Preventive Compound, Item 107,
 WP 1803 00
 Plastic, Sheet, Item 220, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wear Strip - Cabin Door Upper Track, Locally-Made,
 Figure 142 , WP 1805 00

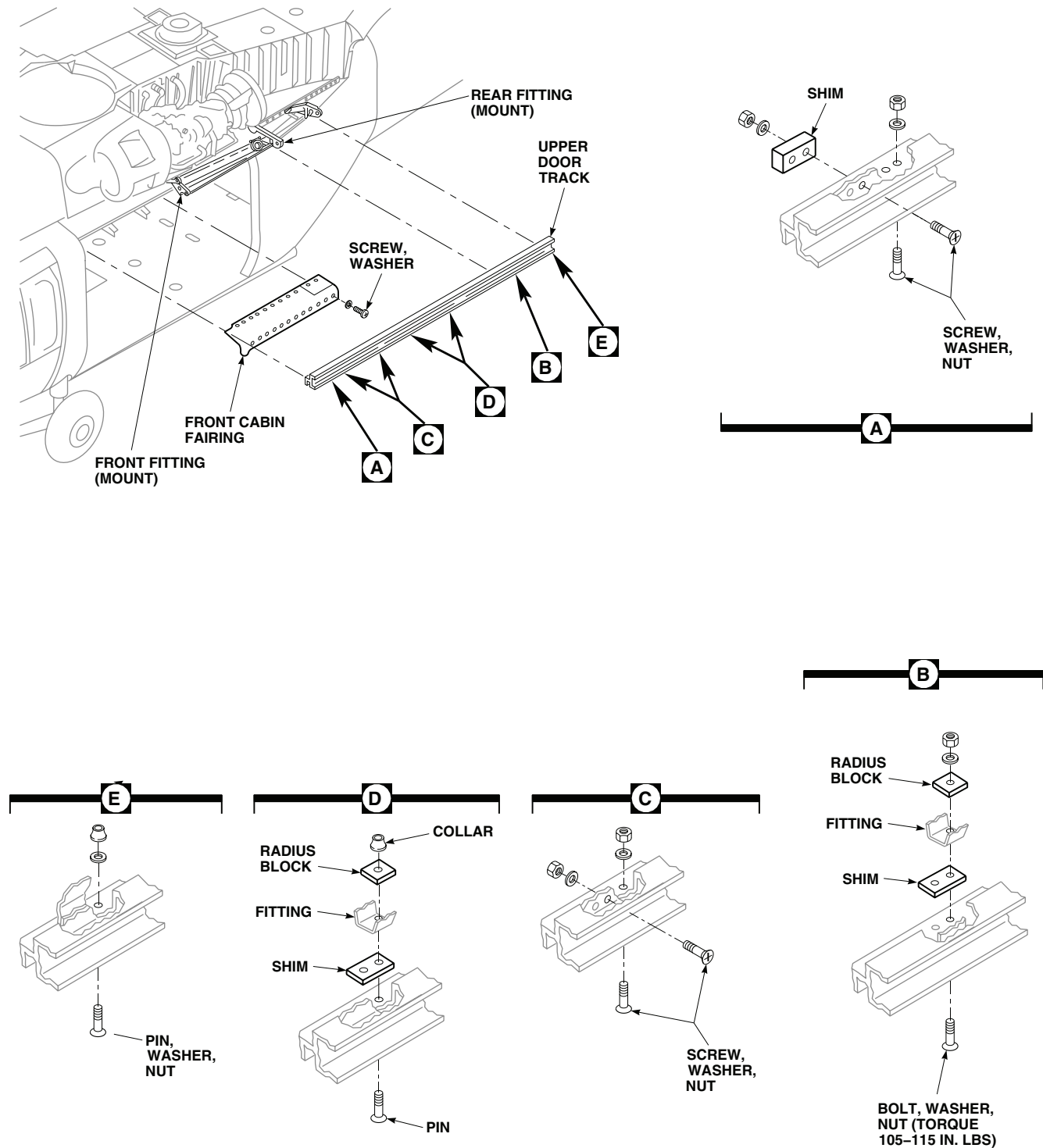
Personnel Required

Aircraft Structural Repairer MOS 15G (1)
 Assistant - (1)

REMOVE

1. Remove troop/cargo door (WP 0244 00).
2. Remove upper door track fairings (WP 0254 00).
3. Remove engine cowling (WP 0313 00).
4. Remove exhaust module (WP 0489 00).
5. Remove screws, washers, and front cabin fairing. Remove sealing compound from fairing and fuselage.
6. Remove fasteners holding soundproofing to straps riveted on track.
7. **ESSS** Drill out and remove rivets from bracket and track at STA 314.0 (Figure 1, Sheet 2, Detail G).
8. Working from front to rear, remove hardware holding track to helicopter structure (Figure 1, Sheet 1, Figure 1, Sheet 2, and Figure 1, Sheet 3). Have assistant hold track at front.
9. Remove fasteners holding upper track to helicopter structure.

REMOVE - Continued



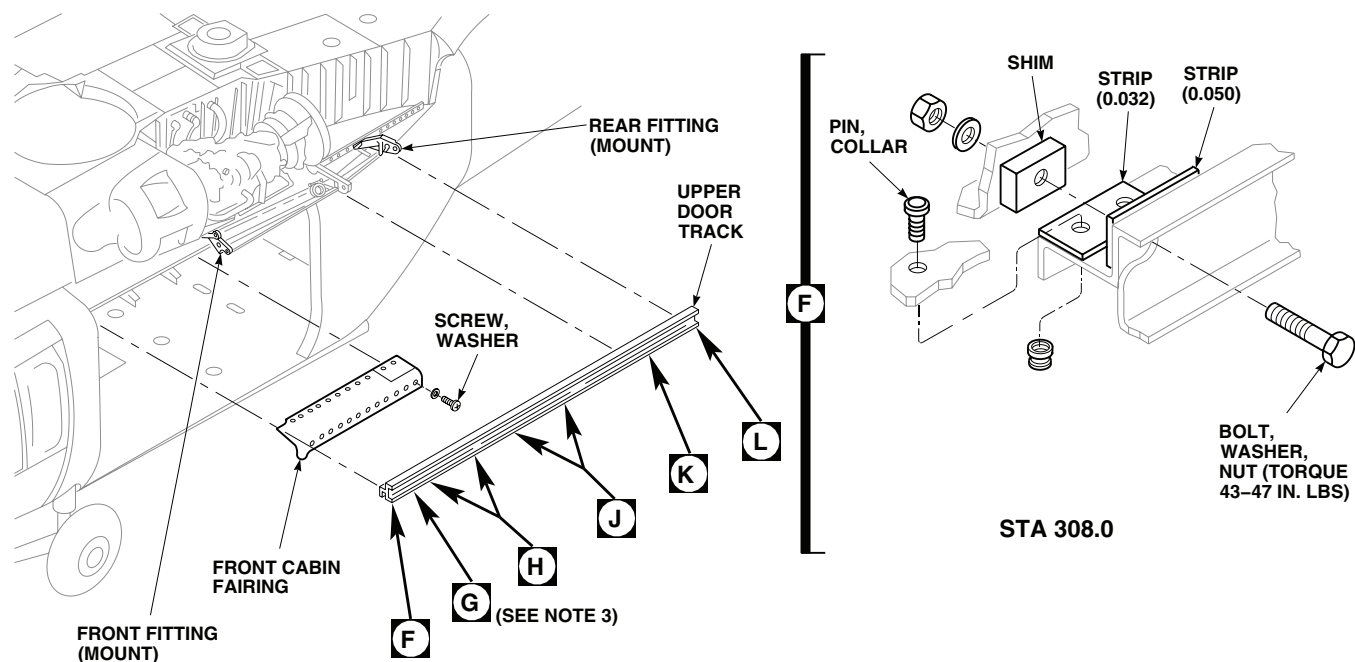
EFFECTIVITY

UH60A 77-22714 - 81-23560

AB1597_1
SA

Figure 1. Repair Cargo Door Upper Track. (Sheet 1 of 4)

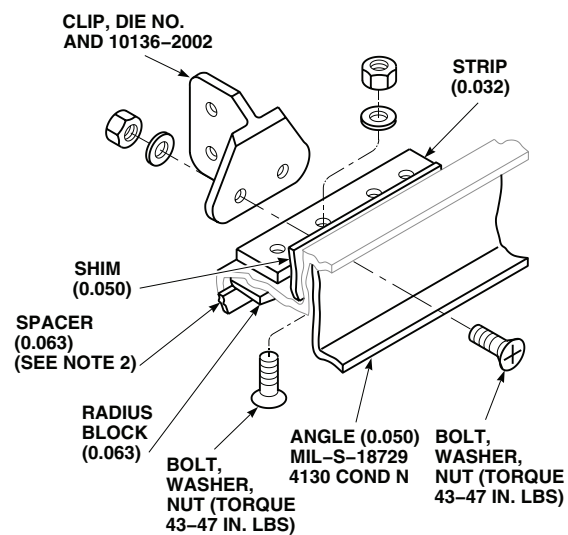
REMOVE - Continued



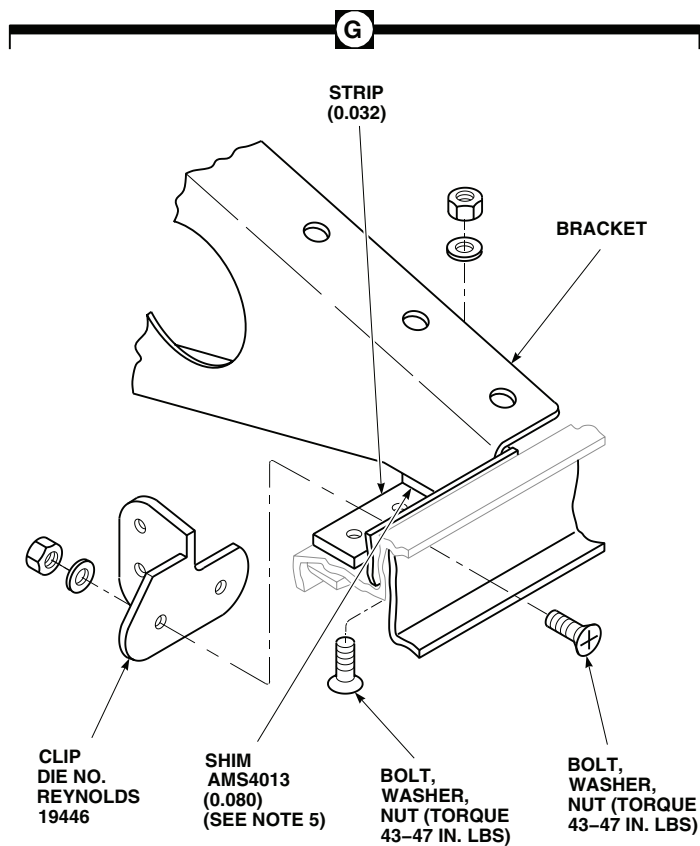
EFFECTIVITY

UH60A 81-23561 - SUBQ

MWO 50-12



STA 327.0 AND STA 343.0

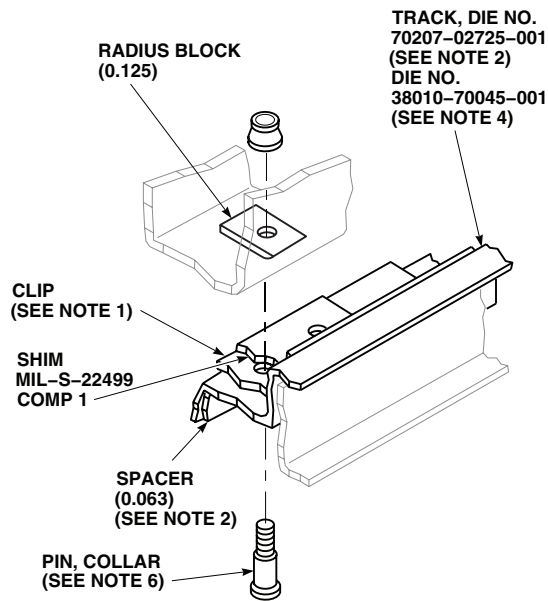


STA 314.0

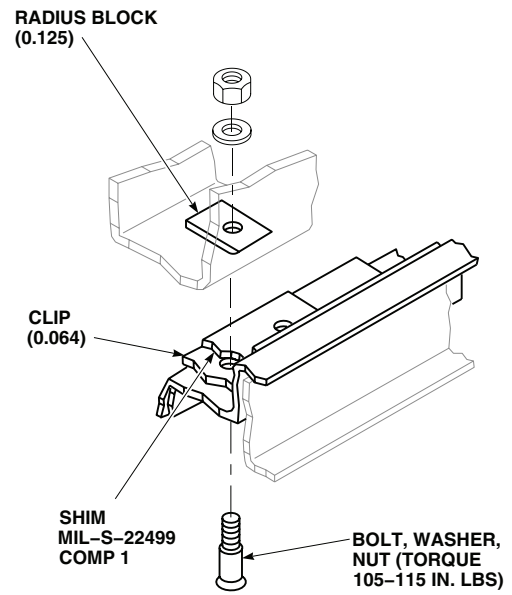
AB1597_2A
SA

Figure 1. Repair Cargo Door Upper Track. (Sheet 2 of 4)

REMOVE - Continued



STA 352.0 AND STA 379.0

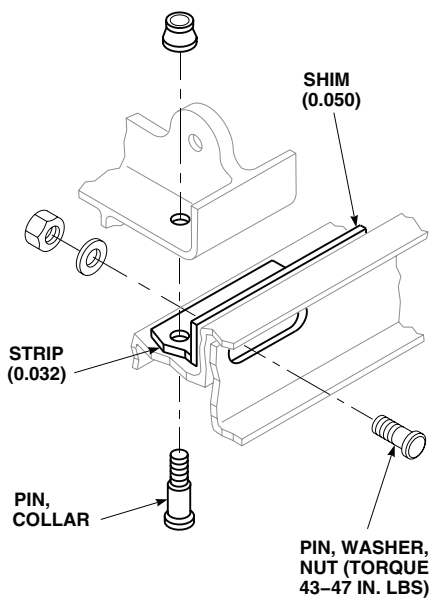


EFFECTIVITY

UH60A 83-23920 - SUBQ

STA 379.0

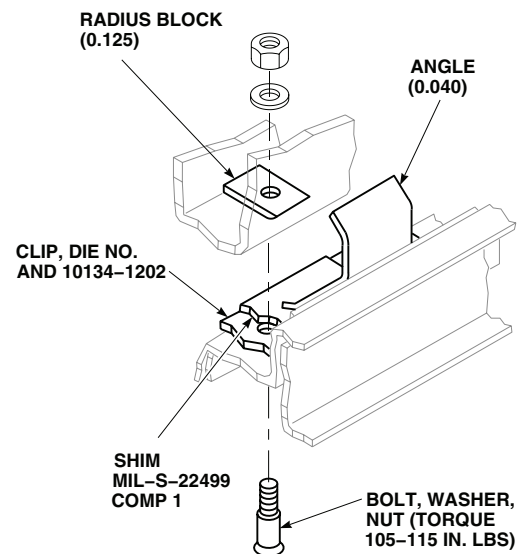
J



STA 421.0

L

K



STA 398.0

AB1597_3A
SA

Figure 1. Repair Cargo Door Upper Track. (Sheet 3 of 4)

REMOVE - Continued**NOTES:**

1. DIE NO. AND 10134-1202 AT STA 352.0, MIL-S-18729, 4130 COND A/N AT STA 379.0.
2. **UH60A 81-23561 - 82-23670**
3. **ESSS**
4. **UH60A 82-23670 - SUBQ**
5. MAY OR MAY NOT BE PRESENT.
6. **UH60A 77-22714 - 83-23919**
7. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB1597_4B
SA

Figure 1. Repair Cargo Door Upper Track. (Sheet 4 of 4)

NOTE

Note number and location of shims and grip length of bolts and pins for track reinstallation.

10. Remove upper track from helicopter.

INSPECT

1. Visually check track for cracks. **NO CRACKS ALLOWED.** Replace cracked tracks.
2. Check for worn or missing wear strips. **REPLACE WEAR STRIPS WORN TO LESS THAN 0.010-INCH THICK.** Replace missing wear strips.
3. Check for wear in track. **REPLACE TRACK IF WORN TO LESS THAN 0.040-INCH THICK.**
4. Inspect areas around fasteners for surface corrosion. Refer to TM 1-1500-344-23 for corrosion removal and treatment. Touch up area (WP 0381 00 and TM 55-1500-345-23).

REPAIR**NOTE**

- Locally make replacement wear strip only when replacing wear strips that vertical rollers rest on. For other wear strips, bond normal replacement wear strip, obtained through supply channels, as follows, but do not rivet to track.
 - Use these procedures to bond wear strips to both upper and lower door tracks.
1. Remove worn wear strip.
 2. Fill in, flush with original surface, any worn spots in track that do not go over repair limits. Use adhesive, Item 11, WP 1803 00.
 3. If required, locally make replacement wear strip (WP 1805 00).
 4. Place wear strip over area where old wear strip was, centering it on area.

REPAIR - Continued**NOTE**

When door is on wear strips (either fully open, or fully closed) rollers will rest on center of strip.

5. If required, mark and drill rivet holes in track.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

6. Clean bonding area on track using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
7. Mix adhesive, Item 11, WP 1803 00, and apply to bonding area on track, and on wear strip.
8. Apply wear strip to track.
9. Cut sheet plastic, Item 220, WP 1803 00, to size and place on area where wear strip is to be bonded.
10. Immediately clamp wear strip in place. Allow adhesive to cure for 1 hour at 160°F(71°C) or 24 hours at 75°F(24°C).

NOTE

Make sure rivets do not protrude above surface of wear strip.

11. Install rivets in strip.

INSTALL

1. Apply sealing compound, Item 273, WP 1803 00, to mating surfaces of upper track and fuselage.
2. Have assistant hold upper track in place on helicopter while sealing compound is still wet.
3. At STA 398.0, apply sealing compound, Item 273, WP 1803 00, to bolts, washers, nuts, shims, and radius blocks. Install wet. **TORQUE NUTS TO 105 - 115 INCH-POUNDS (Figure 1, Sheet 3, Detail K).**
4. **UH-60A 77-22714 - 83-23919** At STA 379.0 apply sealing compound, Item 273, WP 1803 00, to pins, collars, and radius blocks (Figure 1, Sheet 3, Detail J). Install wet. ■
5. **UH-60A 83-23920 - SUBQ** At STA 379.0 apply sealing compound, Item 273, WP 1803 00, to bolts, washers, nuts, shims, and radius blocks (Figure 1, Sheet 3, Detail J). Install wet. **TORQUE NUTS TO 105 - 115 INCH-POUNDS.** ■
6. At STA 308.0, apply sealing compound, Item 273, WP 1803 00, to pins, collars, bolt, washer, nut, and shim (Figure 1, Sheet 2, Detail F). Install wet. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
7. At STA 314.0, STA 327.0, and STA 343.0, apply sealing compound, Item 273, WP 1803 00, to bolts, washers, nuts, shims, and radius blocks (Figure 1, Sheet 2, Detail G and Detail H). Install wet. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
8. At STA 352.0, apply sealing compound, Item 273, WP 1803 00, to pins, collars, spacer, shims, and radius blocks (Figure 1, Sheet 3, Detail J). Install wet.
9. At STA 421.0, apply sealing compound, Item 273, WP 1803 00, to pins, collars, washers, nuts, and shim (Figure 1, Sheet 3, Detail L). Install wet. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

INSTALL - Continued

10. If installing new track, do this:
 - a. Drill out rivets, and remove straps for soundproofing from old track.
 - b. Install straps on soundproofing with fasteners.
 - c. Drill and rivet straps to track (TM 1-1500-204-23).
 - d. Install shims as required.
11. Apply corrosion preventive compound, Item 107, WP 1803 00 to fairing attachment hardware.
12. Install front cabin fairing with screws and washers. Seal edges of fairing with sealing compound, Item 273, WP 1803 00.
13. Install exhaust module (WP 0489 00).
14. Install engine cowling (WP 0313 00).
15. Install upper door track fairings (WP 0254 00).
16. Install troop/cargo door (WP 0244 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

CARGO DOOR LOWER TRACK FAIRING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751

Material/Parts

Corrosion Preventive Compound, Item 107,
WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

QQ-A-250
TM 1-1500-204-23
TM 55-1500-345-23
WP 0244 00
WP 0262 00
WP 0377 00
WP 0381 00
WP 1803 00

REMOVE**NOTE**

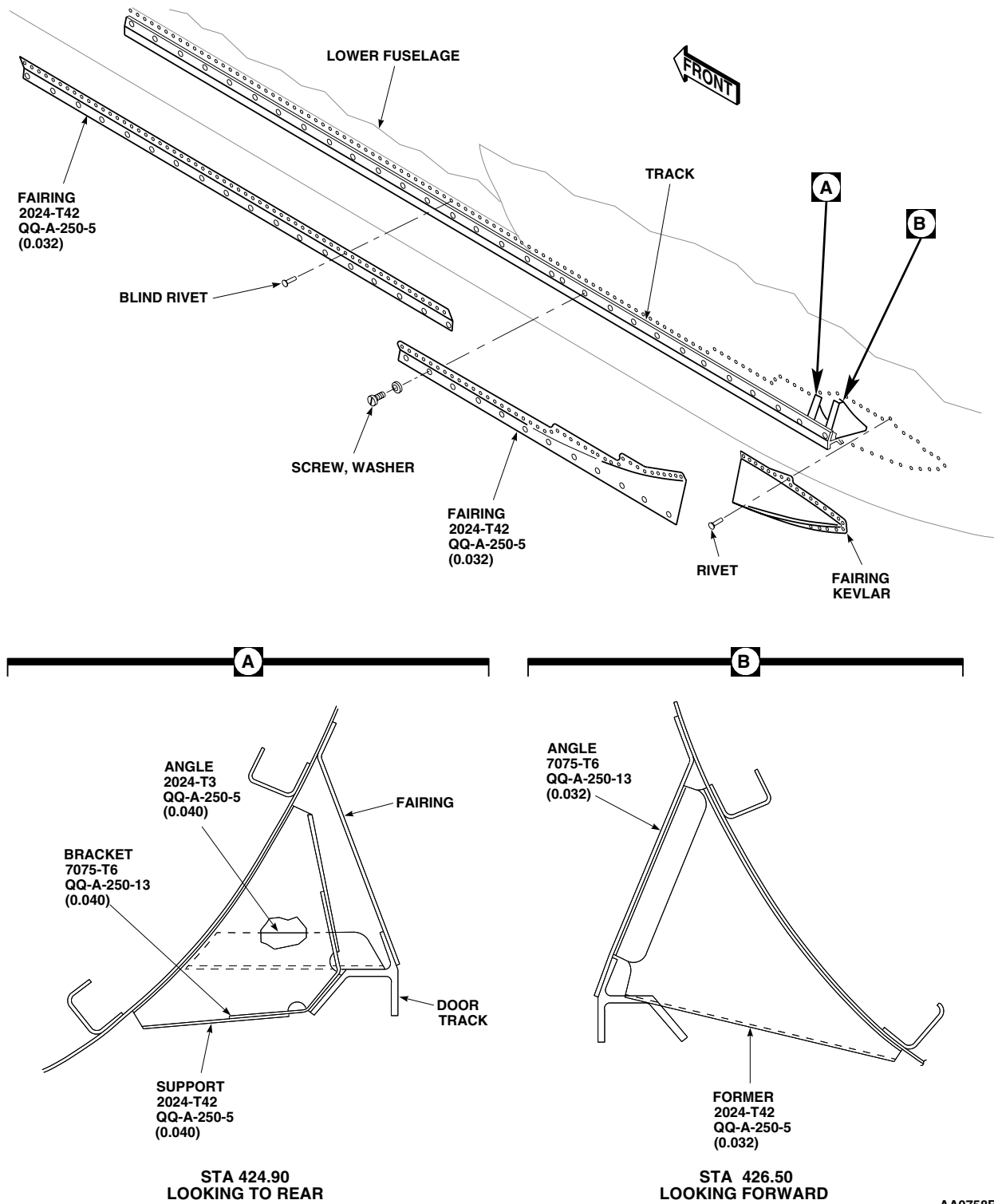
The following instructions apply to either right or left door lower track fairings.

1. Remove troop/cargo door (WP 0244 00).
2. Remove floor panels (WP 0262 00).
3. Remove screws attaching fairing to door track (Figure 1).
4. Carefully drill out and remove rivets in fairing securing it to fuselage and supports.

INSTALL

1. Position fairing on fuselage and locate holes (TM 1-1500-204-23).
2. Clean and seal under fairing (WP 0377 00).
3. Install rivets attaching fairing to fuselage.
4. Apply corrosion preventive compound, Item 112, WP 1803 00 to fairing attachment hardware.
5. Install screws attaching fairing to lower track.
6. Make sure area is clean and free of foreign material.
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
8. Install floor panels (WP 0262 00).
9. Install troop/cargo door (WP 0244 00).

INSTALL - Continued



AA0758B
SA

Figure 1. Repair Cargo Door Lower Track Fairing.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****CARGO DOOR LOWER TRACK WEAR STRIPS REPAIR****This WP supersedes WP 0393 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield

Wear Strip - Cabin Door Lower Track, Locally-Made,
Figure 143, WP 1805 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 32, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

References

TM 1-1500-204-23
WP 1803 00
WP 1805 00

REPAIR

1. Replace nonriveted wear strips as follows:

- a. Mark location of wear strip(s). Peel damaged wear strip from track (Figure 1).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- b. Using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, clean remaining adhesive from track. Wipe dry using clean machinery towel.
 - c. Locally make replacement wear strips (WP 1805 00).
 - d. Position new wear strip on outboard side of track 0.07-inch from bottom edge (Figure 1, Detail A) within location marks made before removing old strip. Using wear strip as a template, drill two 0.125-inch diameter holes through track. Remove wear strip and remove burrs from drilled holes.
 - e. Put a thin coat of adhesive, Item 32, WP 1803 00, on mating surfaces of track and wear strips (TM 1-1500-204-23).
 - f. Put two wear strips on track, centered on 0.125-inch diameter holes, one on inside and one on outside of track. Rivet wear strips to track with flush rivets. Heads must be flush with wear strip surface (Figure 1, Detail B and TM 1-1500-204-23).

REPAIR - Continued

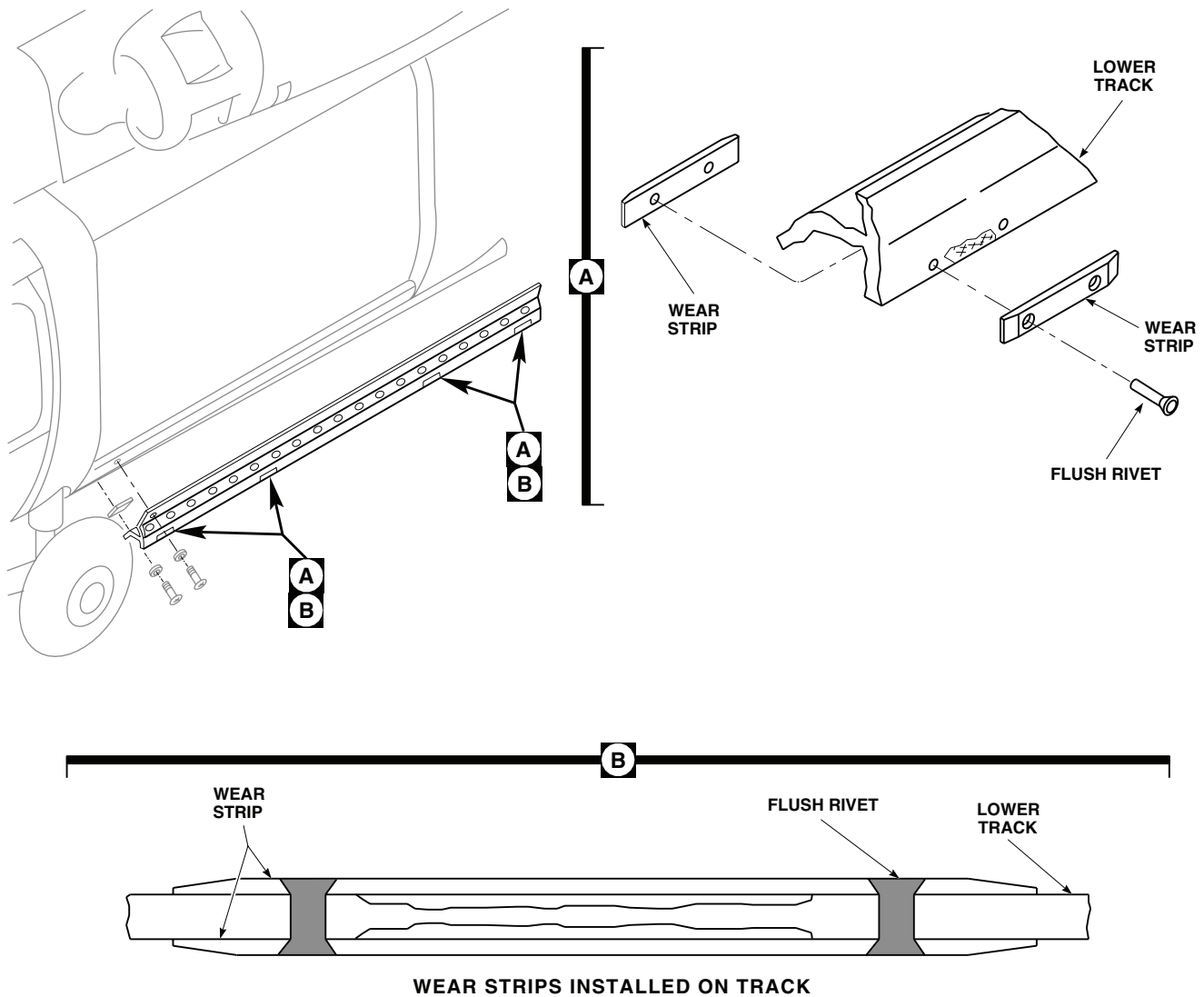
AA0759A
SA

Figure 1. Repair Cargo Door Lower Track Wear Strips.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- g. Using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, remove excess adhesive from track and wear strips. Wipe dry using clean machinery towel.
2. Replace riveted wear strips as follows:
 - a. Drill out rivets and peel damaged wear strips from track (TM 1-1500-204-23).

REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- b. Using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, clean remaining adhesive from track. Wipe dry.
- c. Make replacement wear strips (WP 1805 00). Do not drill countersunk holes now.
- d. Position two undrilled wear strips on inside of track, 0.07-inch from bottom edge (Figure 1, Detail A) and equally spaced each side of rivet holes in track. Clamp wear strips to track.
- e. Using track rivet holes as pilot holes, drill two holes through wear strips using a 0.125-inch diameter drill.
- f. Remove wear strips from track and countersink drilled holes (WP 1805 00).
- g. Put a thin coat of adhesive, Item 32, WP 1803 00, on mating surfaces of track and wear strips (TM 1-1500-204-23).
- h. Put two wear strips on track centered on 0.125-inch diameter holes, one on inside and one on outside of track. Rivet wear strips to track with flush rivets. Heads must be flush with wear strip surface (Figure 1, Detail B and TM 1-1500-204-23).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- i. Using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, remove excess adhesive from track and wear strips. Wipe dry using clean machinery towel.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

EXTERNAL POWER RECEPTACLE COVER

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
No. 24 (0.152-Inch Diameter) Drill

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

REPLACE LATCH/HINGE/COVER - EXTERNAL POWER RECEPTACLE**NOTE**

Use each removal and installation procedure as required.

1. Replace latch as follows:
 - a. Carefully drill out and remove four rivets securing latch to cover assembly using No. 24 drill and 1/8-inch diameter drive pin punch (Figure 1, Detail A).
 - b. Position latch on cover assembly and hold in place with 5/32-inch diameter clecos in two diagonal holes.
 - c. Install rivets in between clecos. Remove clecos and install remaining rivets.
2. Replace cover assembly as follows:
 - a. Carefully drill out and remove securing cover assembly to hinge using No. 24 drill and 1/8-inch diameter drive pin punch.
 - b. Position cover assembly on hinge and install two washers in two end holes between hinge and cover assembly and secure with rivets.
3. Replace hinge as follows:
 - a. If hinge spring is broken, but hinges are serviceable, straighten one crimped end of hinge pin and drive out hinge pin and separate hinges.
 - b. Replace spring, mate hinges and install hinge pin. Crimp ends of hinge pin.
 - c. To replace hinge, drill out rivets securing hinge to inside of receptacle enclosure.
 - d. Drill out rivets securing hinge to cover assembly.
 - e. Install replacement hinge to inside of receptacle enclosure and secure using rivets with washers under the heads of the rivets.
 - f. Install cover assembly on hinge.

REPLACE LATCH/HINGE/COVER - EXTERNAL POWER RECEPTACLE - Continued

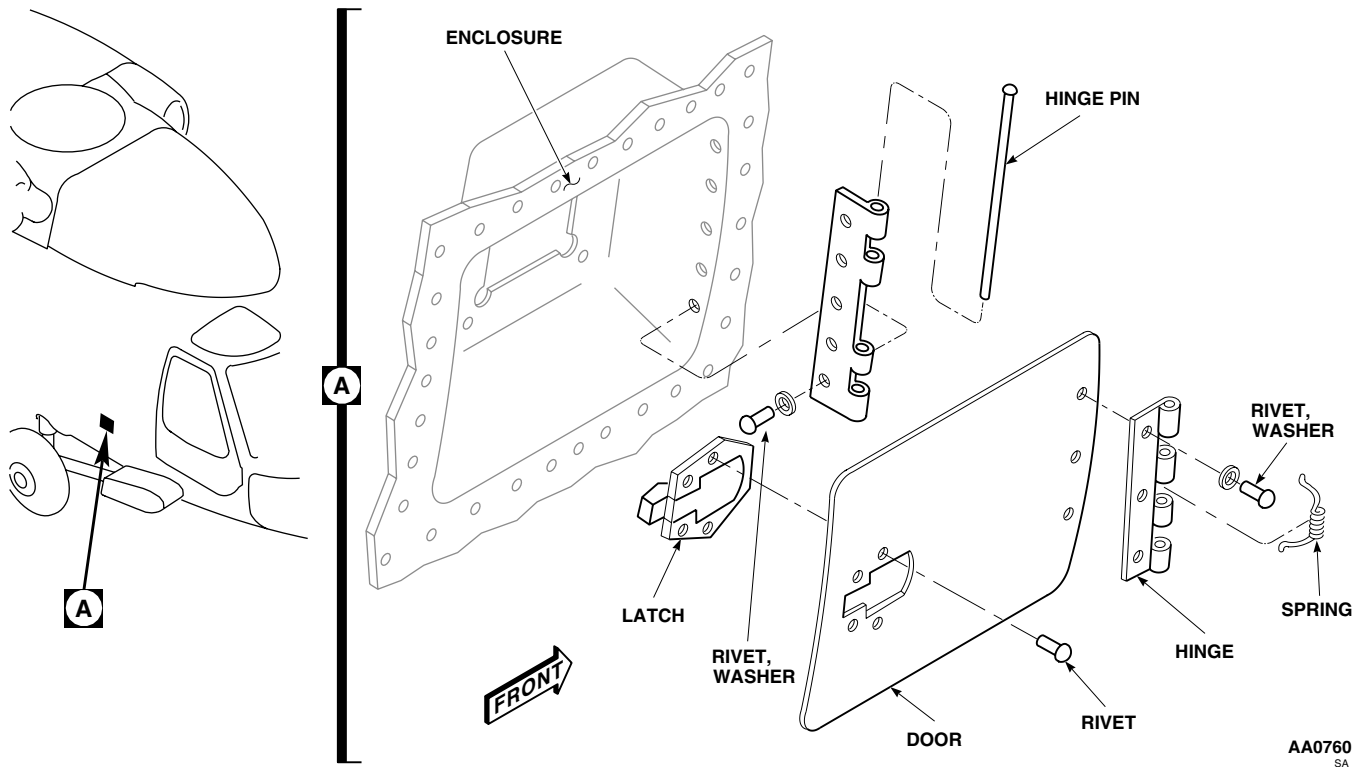


Figure 1. Replace Latch/Hinge/Cover - External Power Receptacle.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****EXTERNAL STORES SUPPORT SYSTEM (ESSS) FUSELAGE FITTING BEARING**

This WP supersedes WP 0395 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Bearing Installation/Removal Tool, 70700-77533-041
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Epoxy Primer Coating Kit, Item 120, WP 1803 00
Grease, Molybdenum Disulfide, Item 138,
WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Preventive Compound, Item 106,
WP 1803 00

References

WP 1803 00

NOTE

Strut fittings at both STA 295.0 and STA 308.0 have similar construction and identical bearings (Figure 1, Sheet 1, Detail A).

REMOVE**CAUTION**

Damage to fitting chamfer will occur if staked lip of bearing is improperly removed.
Be careful not to damage fitting chamfer when machining off staked lip of bearing.

1. Apply molybdenum disulfide grease, Item 138, WP 1803 00 to mandrel of bearing installation/removal tool, 70700-77533-041.
2. Machine roll-staked edge of bearing using bearing installation/removal tool, 70700-77533-041 (Figure 1, Sheet 2, Step 1).
3. Press out bearing using bearing installation/removal tool, 70700-77533-041 (Figure 1, Sheet 2, Step 2).

REMOVE - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean primer from bore and chamfer, using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.

INSPECT

1. Measure fitting ID. **MAXIMUM ALLOWABLE ID IS 1.7495-INCHES.** Replace fitting if limit is exceeded.
2. Inspect fitting bore for corrosion. **PITS MUST NOT EXCEED 0.005-INCH DEEP OR COVER MORE THAN 25% OF FITTING ID.** Replace fitting if limits are exceeded.
3. If corrosion is within limits remove using scouring pad, Item 205, WP 1803 00.
4. Inspect fitting bore for damage. **FITTING BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.75-INCH LONG.** Replace fitting if limits are exceeded.
5. If damage is within limits blend out using scouring pad, Item 205, WP 1803 00. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** Replace fitting if limit is exceeded.

INSTALL

1. Make sure that fitting bore, chamfers, and adjacent surfaces are clean and dry.
2. Apply a light coat of primer coating, Item 243, WP 1803 00, to inside of fitting bore and to outside surface of outer race of bearing.
3. Apply molybdenum disulfide grease, Item 138, WP 1803 00 to mandrel of bearing installation/removal tool, 70700-77533-041.
4. Press bearing into fitting using bearing installation/removal tool, 70700-77533-041 (Figure 1, Sheet 2, Step 3).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

5. Clean primer, if any, from bearing v-grooves, using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe dry.
6. Swage v-groove lip over chamfer in fitting bore using bearing installation/removal tool, 70700-77533-041 (Figure 1, Sheet 2, Step 4).
7. Repeat swaging process on other side of bearing using installation/removal tool, 70700-77533-041 (Figure 1, Sheet 2, Step 5).
8. Inspect staking impressions as follows:
 - a. Check for gap between staked lip of bearing and bearing bore chamfer (Figure 1, Sheet 1, Detail D). **GAP SHALL NOT EXCEED 0.005-INCH MAXIMUM.** Replace bearing if gap exceeds maximum limit.

INSTALL - Continued

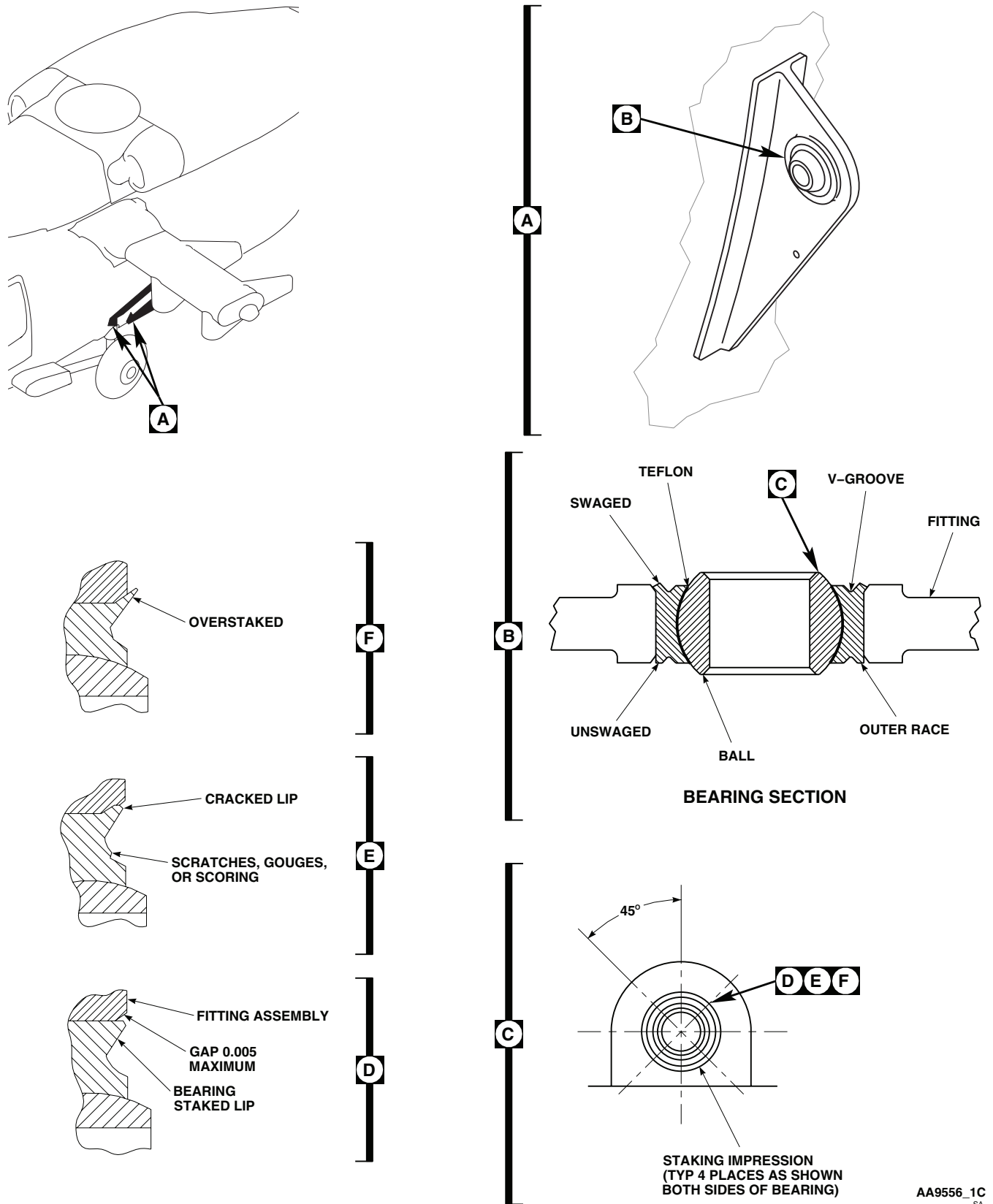
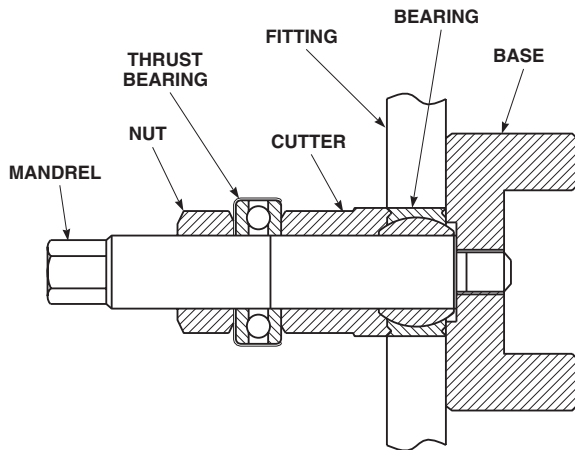
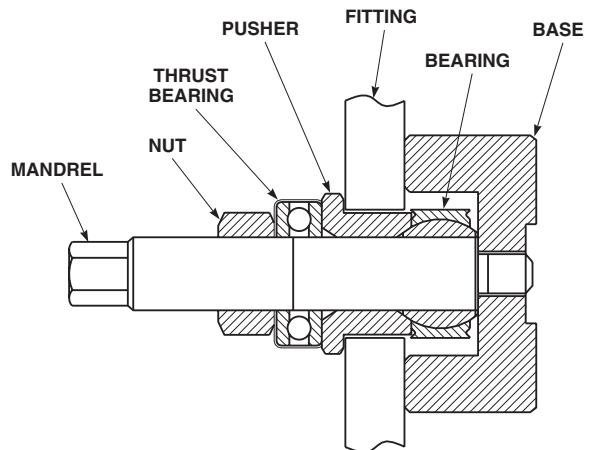
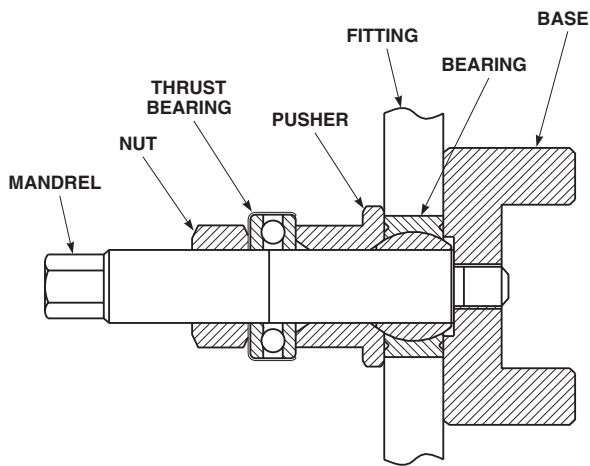
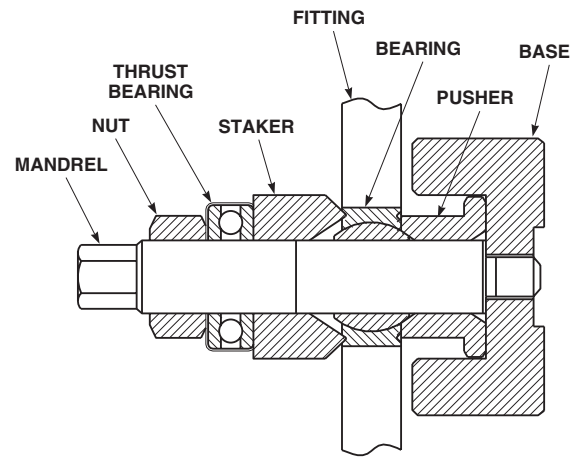
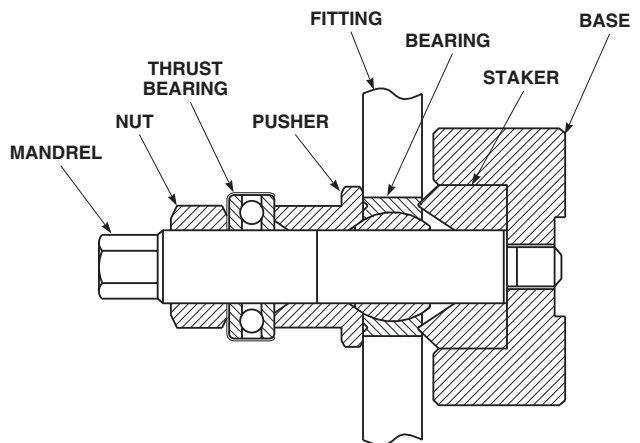


Figure 1. Replace External Stores Support System (ESSS) Fuselage Fitting Bearing. (Sheet 1 of 2)

INSTALL - Continued**STEP 1: REMOVE STAKING****STEP 2: PRESS OUT BEARING****STEP 3: PRESS IN NEW BEARING****STEP 4: STAKE FORWARD SIDE OF FITTING****STEP 5: STAKE REAR SIDE OF FITTING****NOTES:**

1. APPLY MOLYBDENUM DISULFIDE GREASE, ITEM 138, WP 1803 00, TO MANDREL PRIOR TO EACH USE.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA9556_2F
SA

Figure 1. Replace External Stores Support System (ESSS) Fuselage Fitting Bearing. (Sheet 2 of 2)

INSTALL - Continued

- b. Check for cracks in staked lip (Figure 1, Sheet 1, Detail E). **NO CRACKS ALLOWED.** Replace bearing if cracks are found.
 - c. Check for scratches, gouges, or scouring caused by staking groove (Figure 1, Sheet 1, Detail E). **NONE ALLOWED.** Replace bearing if damage is found.
 - d. Check staked lip for over staking, forming a feathered or knife edge beyond face of fitting (Figure 1, Sheet 1, Detail F). **NO OVER STAKING ALLOWED.** Replace bearing if damage is found.
- 9. Touch up bearing surfaces with corrosion preventive compound, Item 106, WP 1803 00.
 - 10. Touch up fitting surfaces with epoxy primer coating kit, Item 120, WP 1803 00.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****GUNNER'S WINDOW LOWER TRACK**

This WP supersedes WP 0396 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0237 00
WP 0240 00
WP 0377 00
WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

REMOVE

1. Remove gunner's windows (WP 0237 00).
2. Remove seals from track (WP 0240 00).
3. Remove swage locking pins fastening lower track to fuselage (TM 1-1500-204-23). Note location, diameter, and length of pins for proper replacement (Figure 1).
4. Drill out rivets fastening track to fuselage.
5. Remove lower track from fuselage.

INSTALL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean mating surfaces of replacement track and fuselage, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
2. Apply even coat of sealing compound, Item 273, WP 1803 00, to mating surface of lower track.
3. Place lower track against fuselage and align fastener holes, with sealing compound still wet (Figure 1).
4. Apply sealing compound, Item 273, WP 1803 00, to fastening hardware and install wet.
5. Install swage locking pins and collars, and rivets (WP 0377 00 and TM 1-1500-204-23).
6. Install new seals (WP 0240 00).

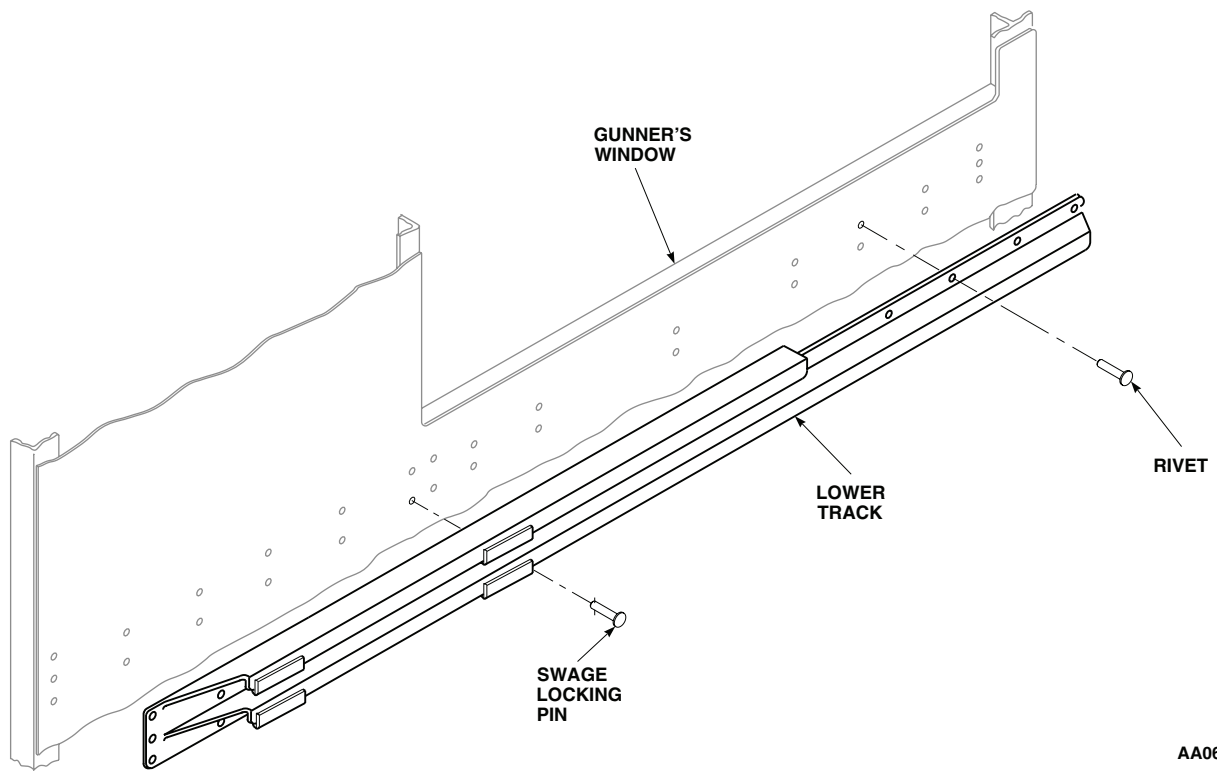
INSTALL - ContinuedAA0619
SA

Figure 1. Replace Gunner's Window Lower Track.

7. Install gunner's windows (WP 0237 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****GUNNER'S WINDOW TRACK REPAIRS**

This WP supersedes WP 0397 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Airframe Repairer's Toolkit, SC 5180-99-B02
 Gloves
 Goggles/Face Shield
 Heat Gun, LFT500

Stirring Stick, Beverage, Item 300, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wear Strip, Locally-Made, Figure 269, WP 1805 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 11, WP 1803 00
 Cloth, Abrasive, Item 81, WP 1803 00
 Primer, Adhesive, Item 240, WP 1803 00

References

WP 0237 00
 WP 1803 00
 WP 1805 00

GUNNER'S WINDOW LOWER TRACK**Repair**

1. On helicopters without wear strips, if track wear in area where rear rollers rest is less than limit, do this:
 - a. Locally make wear strips, 70207-02727-125 (WP 1805 00). Locate wear strip so that rear rollers bear on center of wear strips when windows are fully open. Mark location on track and remove wear strip.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- b. Clean mating surfaces of track and wear strips using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe area with a clean machinery towel.
- c. Fill damaged track surface with adhesive, Item 11, WP 1803 00, if necessary. Smooth adhesive even with undamaged track surface, using beverage stirring stick, Item 300, WP 1803 00.
- d. Using one wear strip as a template against outer track surface, drill two 0.125-inch diameter holes through track. Remove burrs from drilled holes.
- e. Roughen flat surface of wear strips using abrasive cloth, Item 81, WP 1803 00. Wipe clean with machinery wiping towel, Item 344, WP 1803 00.
- f. Apply light coat of adhesive, Item 11, WP 1803 00, to mating surfaces of track and wear strips.
- g. Place inside and outside wear strips against track and align rivet holes.
- h. Install wear strips to lower track with rivets. Make sure heads are flush with wear strip (Figure 1, Detail D).

GUNNER'S WINDOW LOWER TRACK - Continued

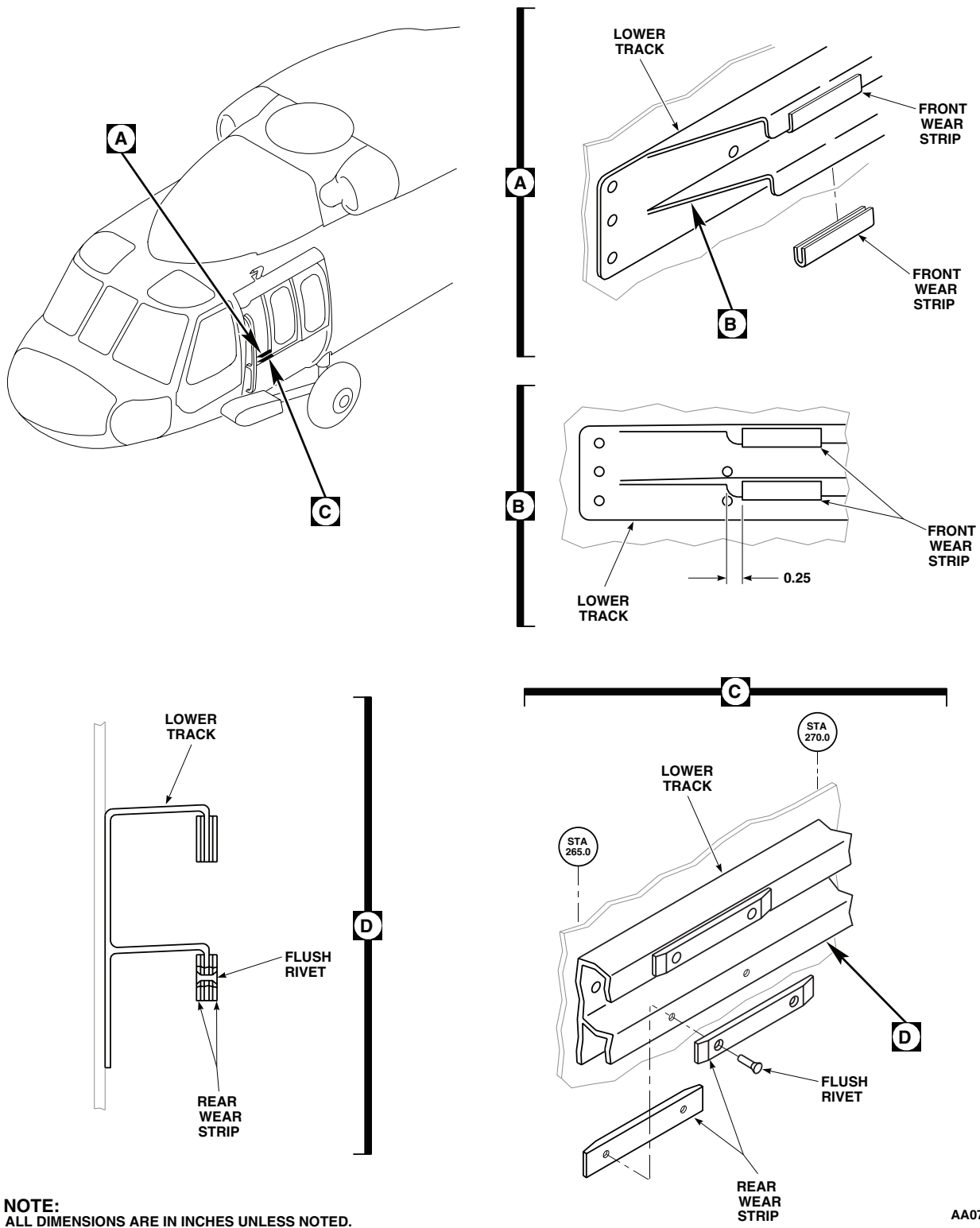


Figure 1. Repair Gunner's Window Lower Track.

GUNNER'S WINDOW LOWER TRACK - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- i. Clean adhesive from track and wear strips using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
2. If wear strips are loose, worn beyond limits, or missing do this:

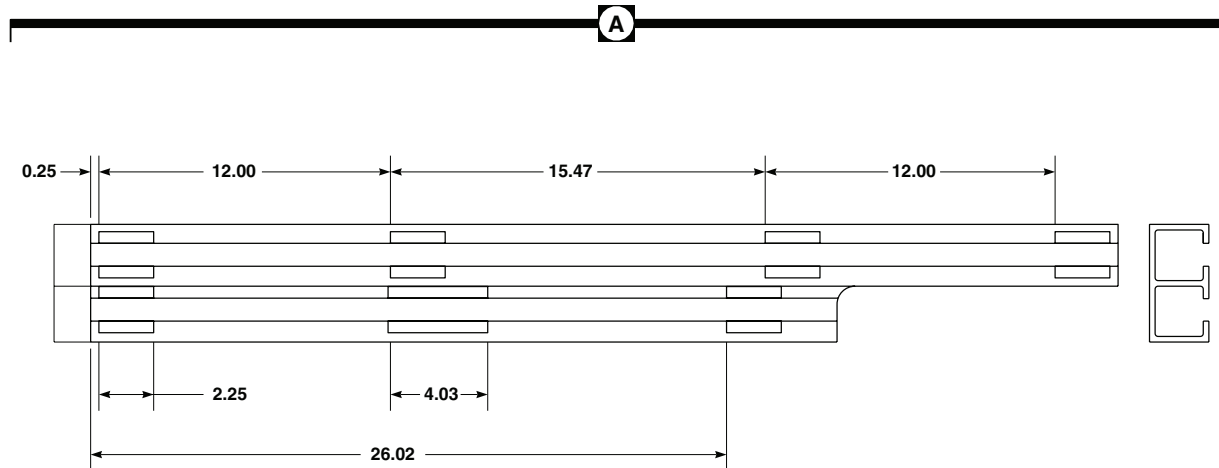
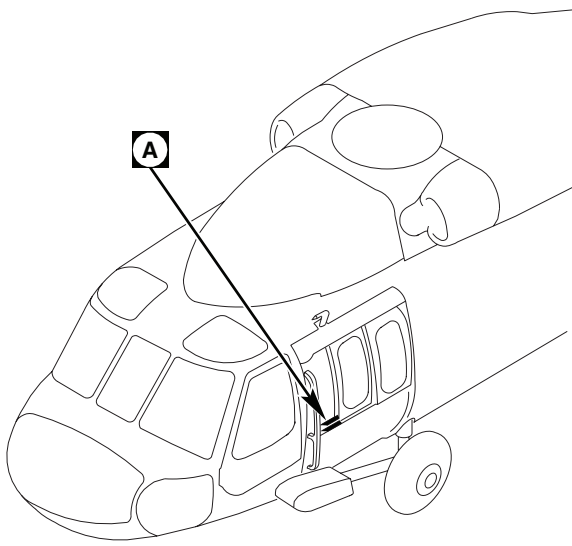
WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- a. Clean adhesive from lower track using a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe area with a clean machinery towel.
- b. Roughen inside edges of wear strip using abrasive cloth, Item 81, WP 1803 00. Wipe inside edges with a clean machinery wiping towel, Item 344, WP 1803 00.
- c. Apply a light coat of adhesive primer, Item 240, WP 1803 00, on lower door track and inside edges of wear strip. Allow primer to cure at 70°F(21°C) to 80°F(27°C) for 1 hour, followed by 1 hour at 240°F(116°C) to 275°F(135°C) using heat gun.
- d. Apply a light coat of adhesive, Item 11, WP 1803 00, on lower door track and inside edges of wear strip.
- e. Install front wear strip on lower track with front end of wear strip 0.25-inch from front end of lower track (Figure 1, Detail D).
- f. Install rear wear strip on lower track with front end of wear strip 11.28-inches from front end of lower track.
- g. Allow adhesive, Item 11, WP 1803 00, to cure for at least 1 hour at 160°F(71°C), or 24 hours at 75°F(24°C).
3. Install gunner's window (WP 0237 00).

GUNNER'S WINDOW UPPER TRACK**Repair**

1. On helicopters without wear strips installed which show signs of wear, or on helicopters with wear strips installed which are loose or worn perform the following procedure:
 - a. Locally make six (6) each outer web wear strips with a width of 0.243 +/- 0.050 inch, six (6) inner web wear strips with a width of 0.205 +/- 0.050 inch, one (1) each inner wear web strip with a width of 0.205 +/- 0.050 inch, and one (1) each outer web wear strip with a width of 0.243 +/- 0.50inch (Figure 269, WP 1805 00).
 - b. Locate wear strips so that the center of roller mounting brackets are aligned with the center of wear strips in the open or closed positions for each section of window (Figure 2, Detail A).
 - c. Visually check that the wear strips sit flat and do not interfere with the track interior radius. Trim the wear strip, as required, to prevent interference with the track interior radius.

GUNNER'S WINDOW UPPER TRACK - Continued**NOTE**

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

AB3680
SA

Figure 2. Repair Gunner's Window Upper Track.

GUNNER'S WINDOW UPPER TRACK - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- d. Clean mating surfaces of track and wear strips using machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe area with a clean machinery towel.
- e. Fill damaged track surface with adhesive, Item 11, WP 1803 00, if necessary. Smooth adhesive even with undamaged track surface, using beverage stirring stick, Item 300, WP 1803 00.
- f. Allow adhesive Item 11, WP 1803 00 to cure for 24 hours at room temperature 70°F(21°C) to 80°F(27°C), or for 1 hour at 160°F(71°C).
- g. Roughen mating surfaces of wear strips using abrasive cloth, No. 280 grit, Item 81, WP 1803 00. Wipe clean with machinery wiping towel, Item 344, WP 1803 00.
- h. Apply a light coat of adhesive primer, Item 23, WP 1803 00, to the inside surface of the wear strips and to mating surfaces of upper door tracks.
- i. Allow the adhesive primer, Item 23, WP 1803 00, to cure for a minimum of 1 hour at room temperature 75°F(24°C).
- j. Bond wear strips to the upper door track using Item 11, WP 1803 00. Squeeze-out any excess adhesive, as required, around the edges of wear strips, and smooth adhesive even, using beverage stirring stick, Item 300, WP 1803 00, to provide fillet around wear strips
- k. Allow adhesive Item 11, WP 1803 00 to cure for 24 hours at room temperature 70°F(21°C) to 80°F(27°C), or for 1 hour at 160°F(71°C).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean adhesive from track and wear strips using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
2. Install gunner's window (WP 0237 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****GUNNER'S WINDOW UPPER TRACK**

This WP supersedes WP 0398 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0237 00
WP 0241 00
WP 0399 00
WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

REMOVE

1. Remove gunner's windows (WP 0237 00).
2. Remove upper track fairing (WP 0399 00).
3. Remove seals from track (WP 0241 00).
4. Drill out rivets fastening upper track to intercostal support (Figure 1).
5. Remove upper track.

INSTALL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean mating surfaces of replacement track and intercostal support, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
2. Apply even coat of sealing compound, Item 273, WP 1803 00, to mating surface of upper track.
3. Place track, with sealing compound still wet, against intercostal support and align fastener holes (Figure 1).
4. Coat rivets with sealing compound, Item 273, WP 1803 00, and install rivets wet.
5. Remove excess sealing compound from working surfaces of track.
6. Install new seals (WP 0241 00).
7. Install upper track fairing (WP 0399 00).

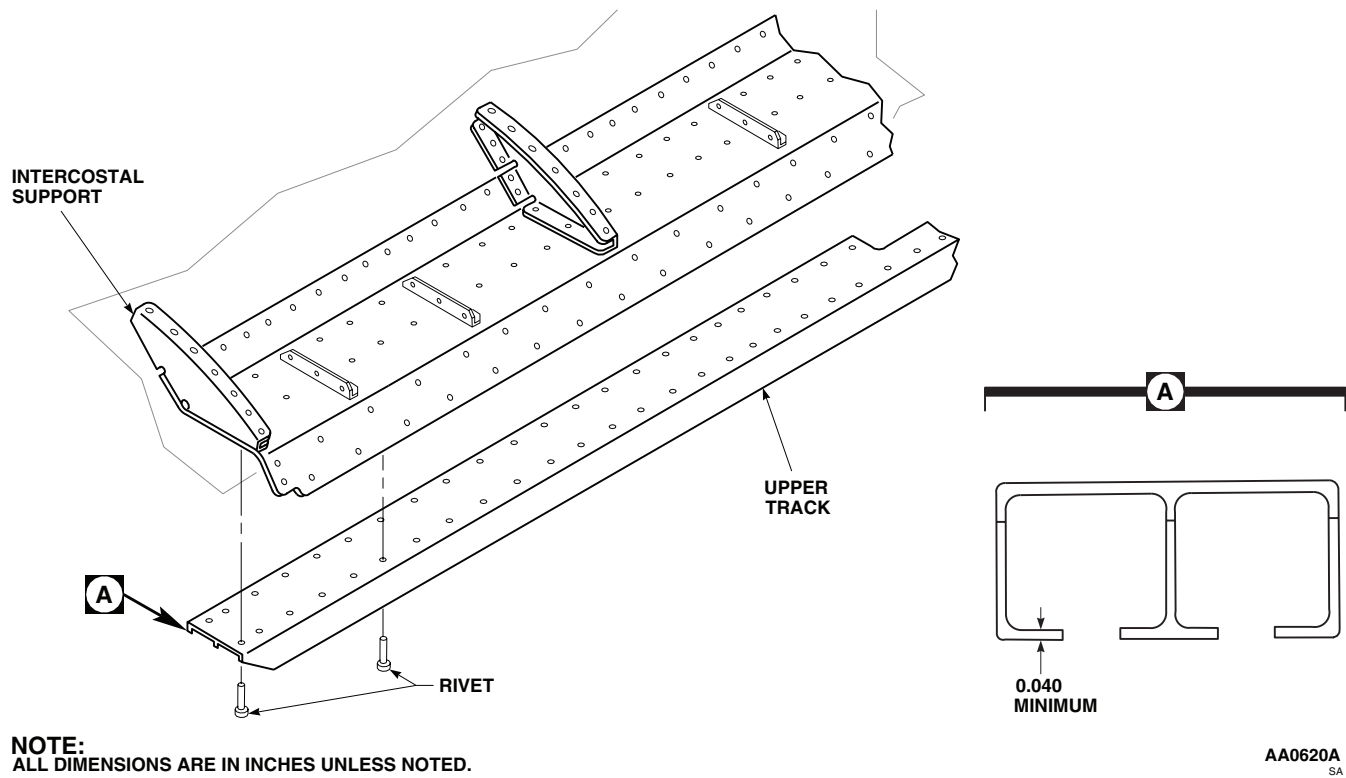
INSTALL - Continued

Figure 1. Replace Gunner's Window Upper Track.

8. Install gunner's windows (WP 0237 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****GUNNER'S WINDOW UPPER TRACK FAIRING**

This WP supersedes WP 0399 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Set, GGG-D-751
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0377 00
WP 0381 00
WP 0389 00
WP 0767 00
WP 1803 00

REMOVE

1. Remove pitot-static head (WP 0767 00).
2. Remove pitot-static head support fairing and head support (WP 0389 00 and WP 0767 00).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Remove sealing compound from fairing cap sealed edges, screw heads, bolthead, and rivets, using a nonmetallic scraper and machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
4. Remove screws and bolt fastening front cap to upper fairing and fuselage (Figure 1).
5. Drill out rivet heads holding cap to upper fairing, strap, and fuselage (TM 1-1500-204-23). Punch out rivets.
6. Remove front cap.

REMOVE - Continued

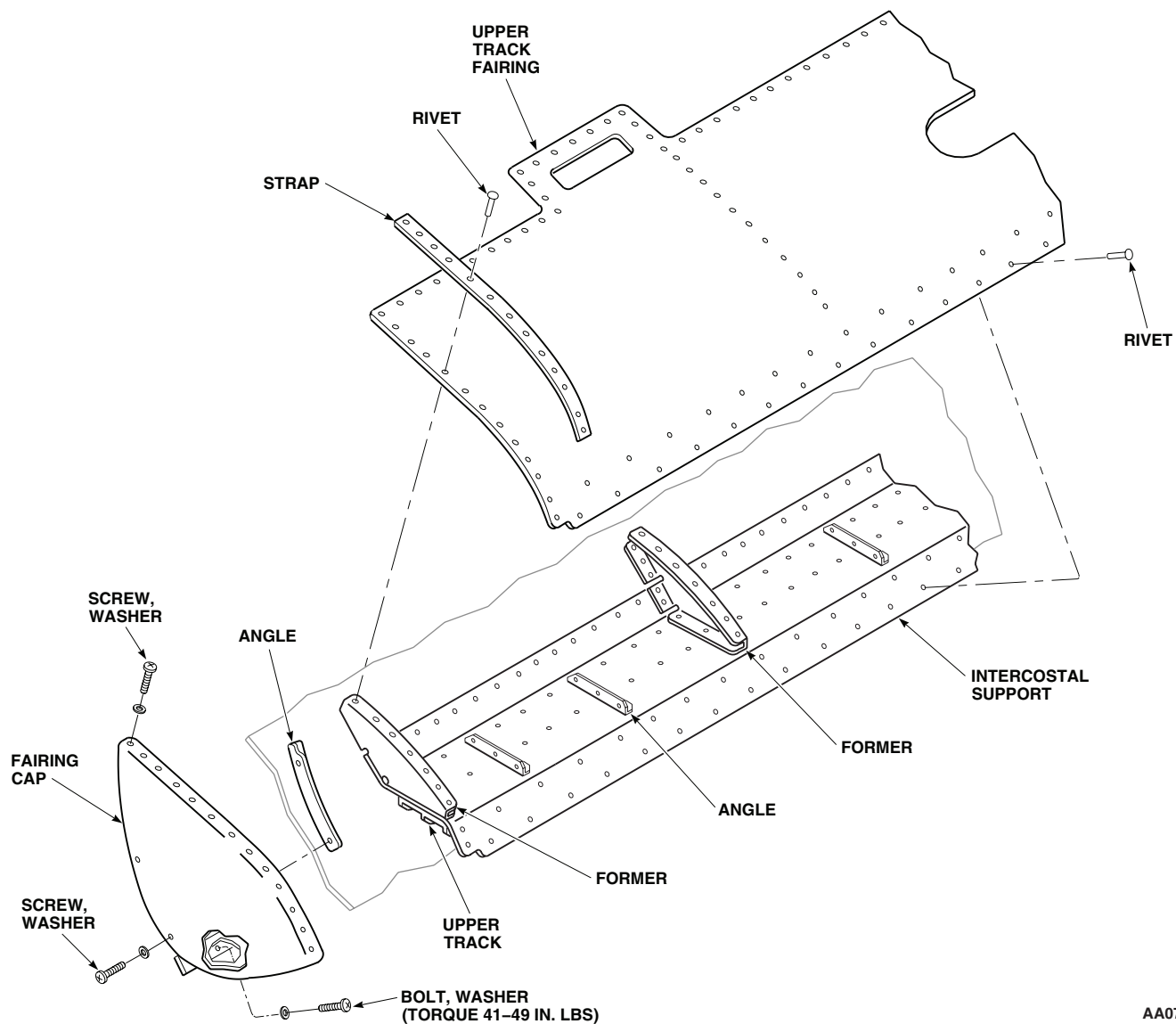
AA0769
SA

Figure 1. Replace Gunner's Window Upper Track Fairing.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

7. Remove sealing compound from upper fairing sealed edges and rivets, using a nonmetallic scraper and machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
8. Drill out rivet heads holding fairing to formers, intercostal support, and fuselage (TM 1-1500-204-23). Punch out rivets.

REMOVE - Continued

9. Remove upper fairing.

INSTALL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Thoroughly clean all mating surfaces of upper fairing, front cap, fuselage, formers, and intercostal support, using nonmetallic scraper and machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
2. Coat all mating surfaces of upper fairing, fuselage, formers, and intercostal support with liberal amount of sealing compound, Item 273, WP 1803 00.
3. Put upper fairing in place while sealing compound is wet and align fastener holes (Figure 1).
4. Install rivets wet with sealing compound, Item 273, WP 1803 00 (WP 0377 00).
5. Coat mating surfaces of front cap, fuselage, strap, and upper fairing liberally with sealing compound, Item 273, WP 1803 00.
6. While sealing compound is wet, install strap and front cap with screws, bolt, and rivets wet with sealing compound, Item 273, WP 1803 00 (WP 0377 00). **TORQUE BOLTS 41 - 49 INCH-POUNDS.**
7. Seal all around exposed edges of front cap and fairing with sealing compound, Item 273, WP 1803 00 (WP 0377 00).
8. Install pitot-static head support and fairing (WP 0389 00 and WP 0767 00).
9. Install pitot-static head (WP 0767 00).
10. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

GUNNER'S WINDOW REPAIR

This WP supersedes WP 0400 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
 Drill Set, GGG-D-751
 Gloves
 Goggles/Face Shield
 Nonmetallic Scraper

MIL-P-25690
 MIL-R-6855
 MIL-S-18729
 QQ-A-250
 TM 1-1500-204-23
 TM 55-1500-345-23

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 31, WP 1803 00
 Fastener Tape, Pile, Item 125, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

WP 0237 00
 WP 0240 00
 WP 0377 00
 WP 0378 00
 WP 0379 00
 WP 0380 00
 WP 0381 00
 WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

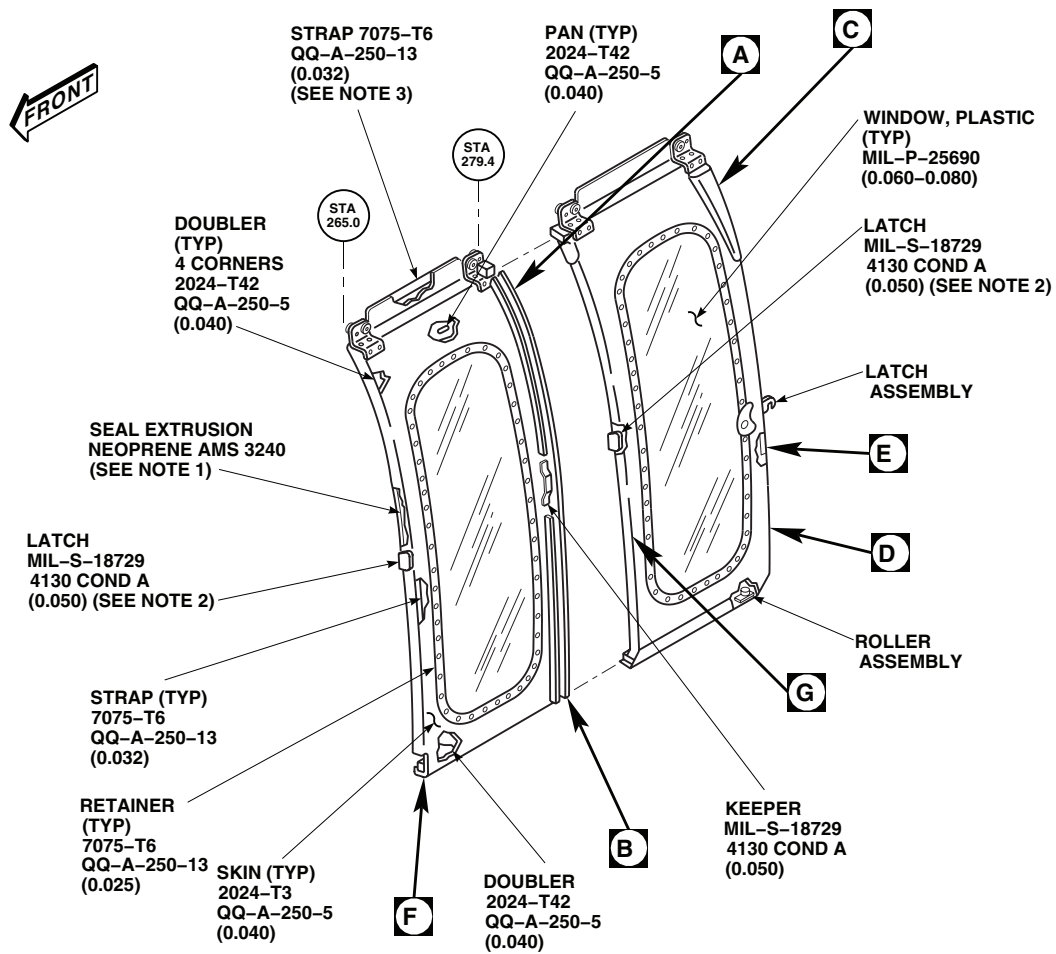
References

AMS 3195
 AMS 3240

REPAIR GUNNER'S WINDOW

1. All window structure, except for plastic panels and seals, is aluminum alloy, 2024-T42, QQ-A-250-5 or 7075-T6, QQ-A-250-13. Windows are stretched Plexiglass and seals are either silicone sponge rubber or neoprene sheet or extrusion charts. Roller supports are made from 4130 steel and the rear window fairing is reinforced plastic. The window structure is secondary structure (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, and Figure 1, Sheet 5).
2. Remove window if required (WP 0237 00).
3. Evaluate damage to skin and framing (WP 0377 00).
4. Use WP 0378 00 and WP 0380 00 for typical structural repairs.
5. Use WP 0379 00 for evaluating and repairing reinforced plastic.
6. Rebond seals if required (WP 0240 00).
7. Seal roller assembly attachment surfaces for dissimilar metals insulation and watertightness using sealing compound, Item 273, WP 1803 00.
8. Seal upper window track mating surfaces for watertightness using sealing compound, Item 273, WP 1803 00.
9. Touch up paint in repaired area, as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR GUNNER'S WINDOW - Continued

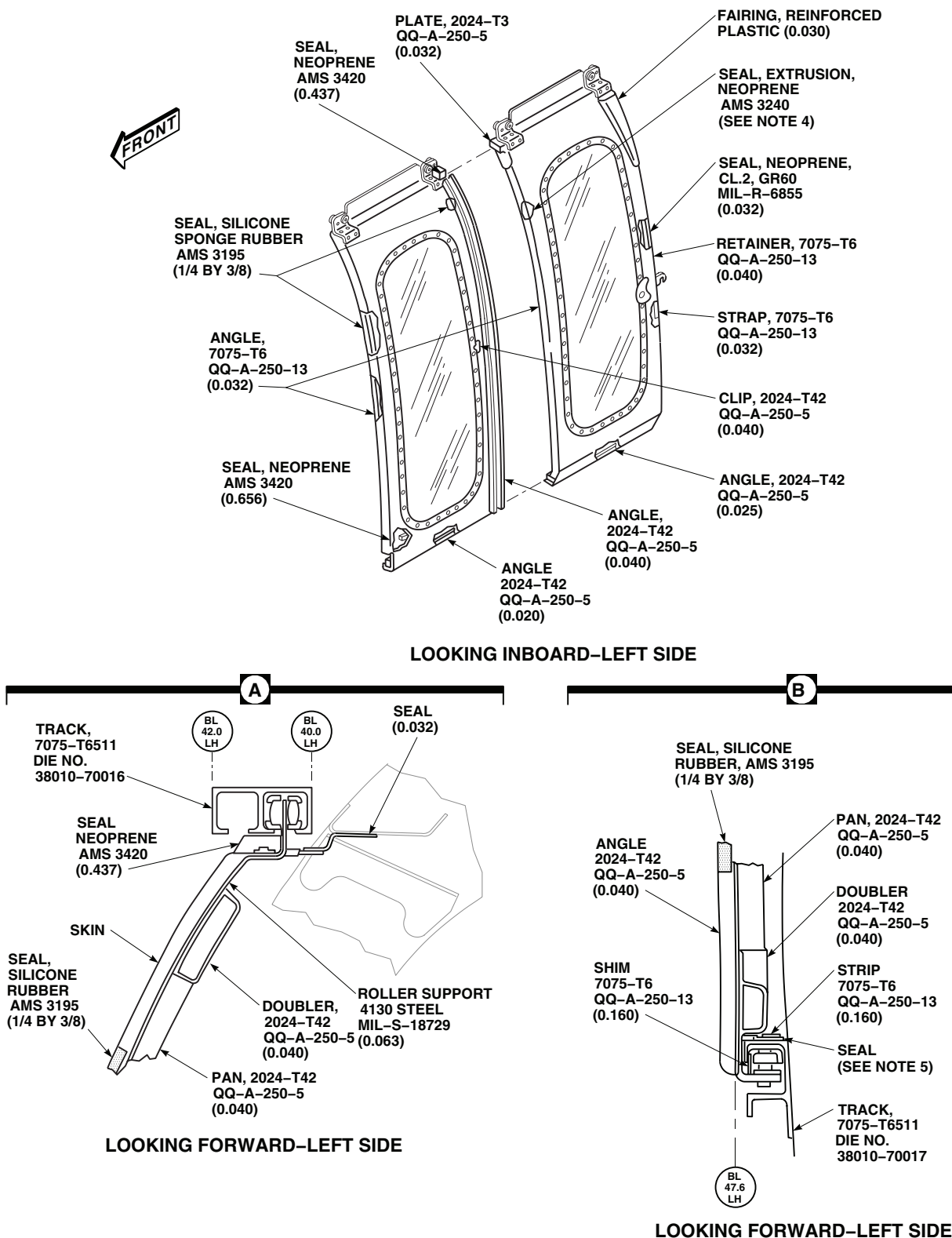


LOOKING INBOARD-LEFT SIDE

AA0770_1C
SA

Figure 1. Repair Gunner's Window. (Sheet 1 of 5)

REPAIR GUNNER'S WINDOW - Continued



AA0770_2C
SA

Figure 1. Repair Gunner's Window. (Sheet 2 of 5)

REPAIR GUNNER'S WINDOW - Continued

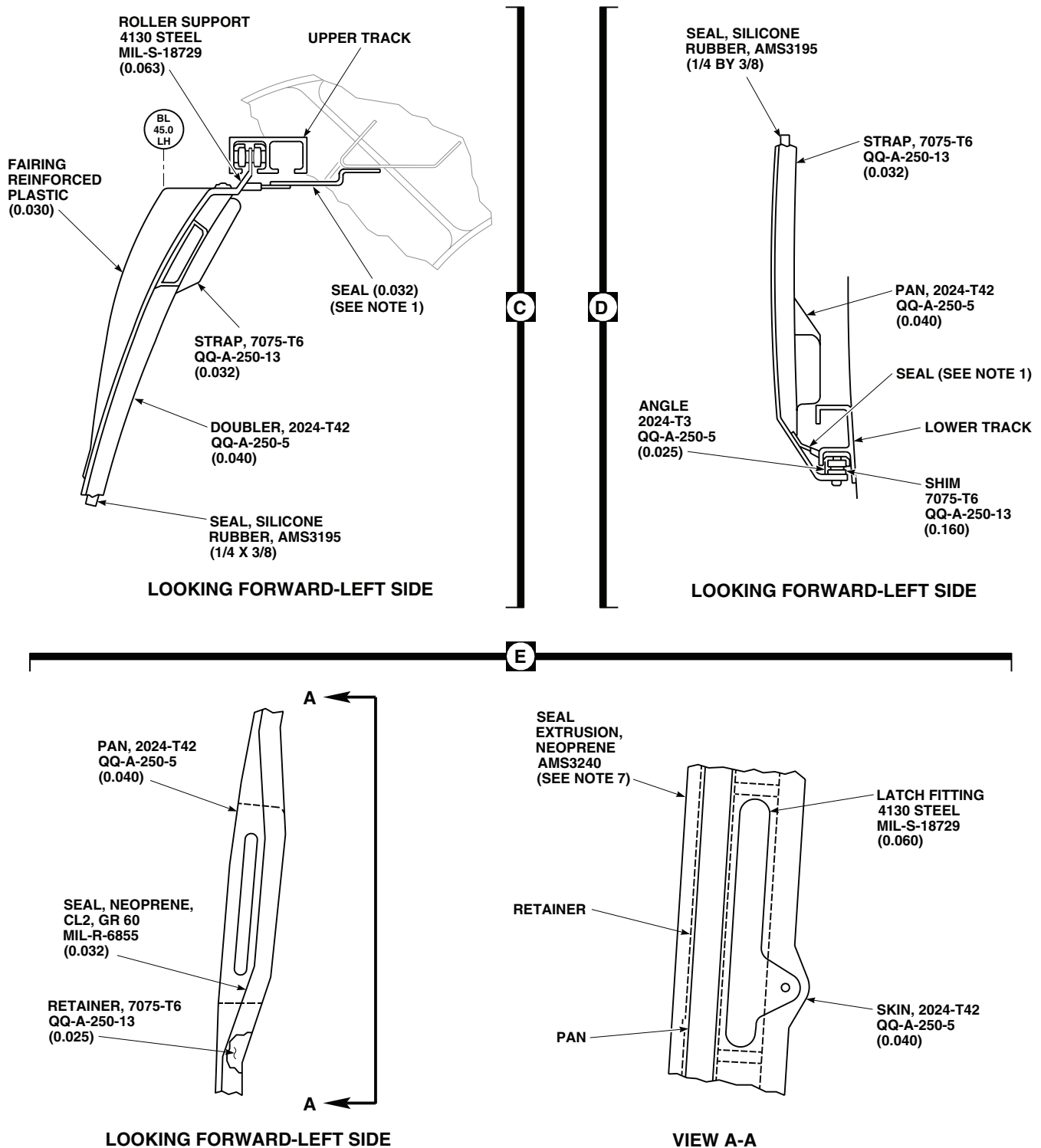
AA0770_3C
SA

Figure 1. Repair Gunner's Window. (Sheet 3 of 5)

REPAIR GUNNER'S WINDOW - Continued

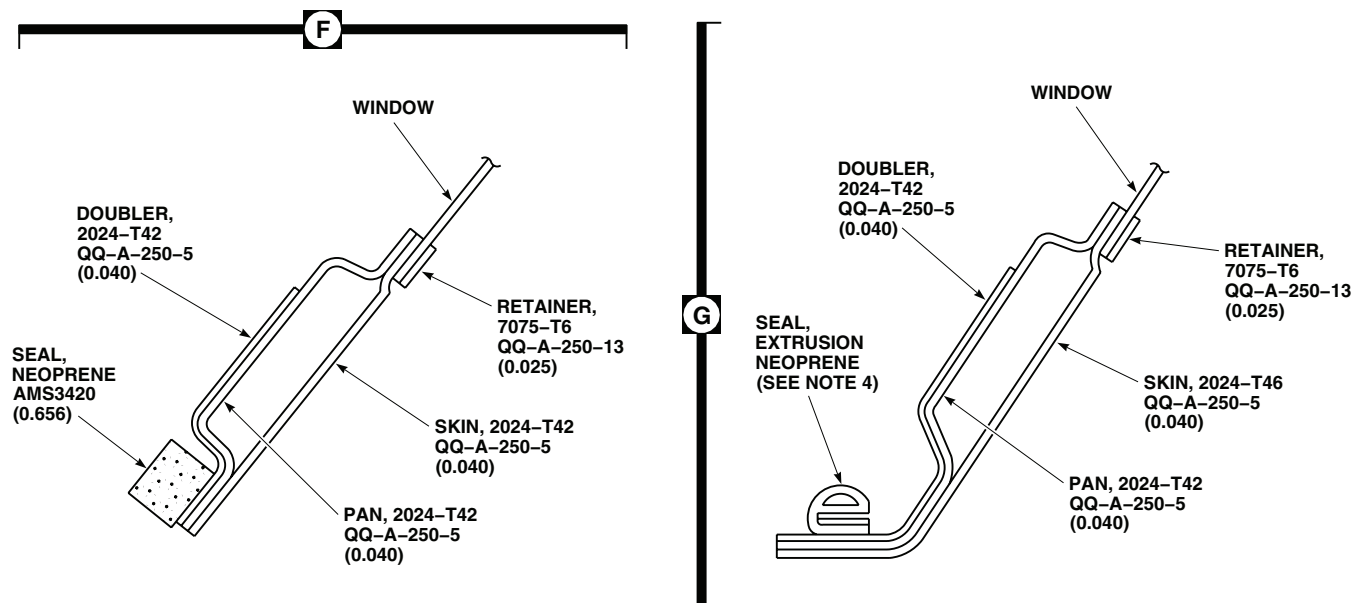
AA0770_4
SA

Figure 1. Repair Gunner's Window. (Sheet 4 of 5)

10. If window has minor scratches, hazing, crazing, damage by airborne abrasives or surface imperfections, polish window per TM 1-1500-204-23.
11. Install window if required (WP 0237 00).

REPLACE PLASTIC WINDOW PANELS

Remove

1. Remove pile fastener tape around inside of window opening, using nonmetallic scraper.
2. Carefully drill out and remove rivets securing window panel to skin, using 1/8-inch diameter drill and 3/32-inch diameter drive pin punch (TM 1-1500-204-23).
3. Remove retainer and carefully work window panel loose from skin using nonmetallic scraper (Figure 2).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean adhesive from skin and inner surface with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00, and a nonmetallic scraper. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.

REPLACE PLASTIC WINDOW PANELS - Continued

NOTES:

1. SIKORSKY P/N S6122-82279-2
2. UH-60A 84-23935 - 84-24016
3. UH-60A 84-24017 - 84-24028 UH-60A 85-24428 - SUBQ
4. SIKORSKY P/N S6122-82279-2
5. UH-60A 81-23608 - SUBQ
6. SIKORSKY P/N S6122-82279-1
7. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0770_5A
SA

Figure 1. Repair Gunner's Window. (Sheet 5 of 5)

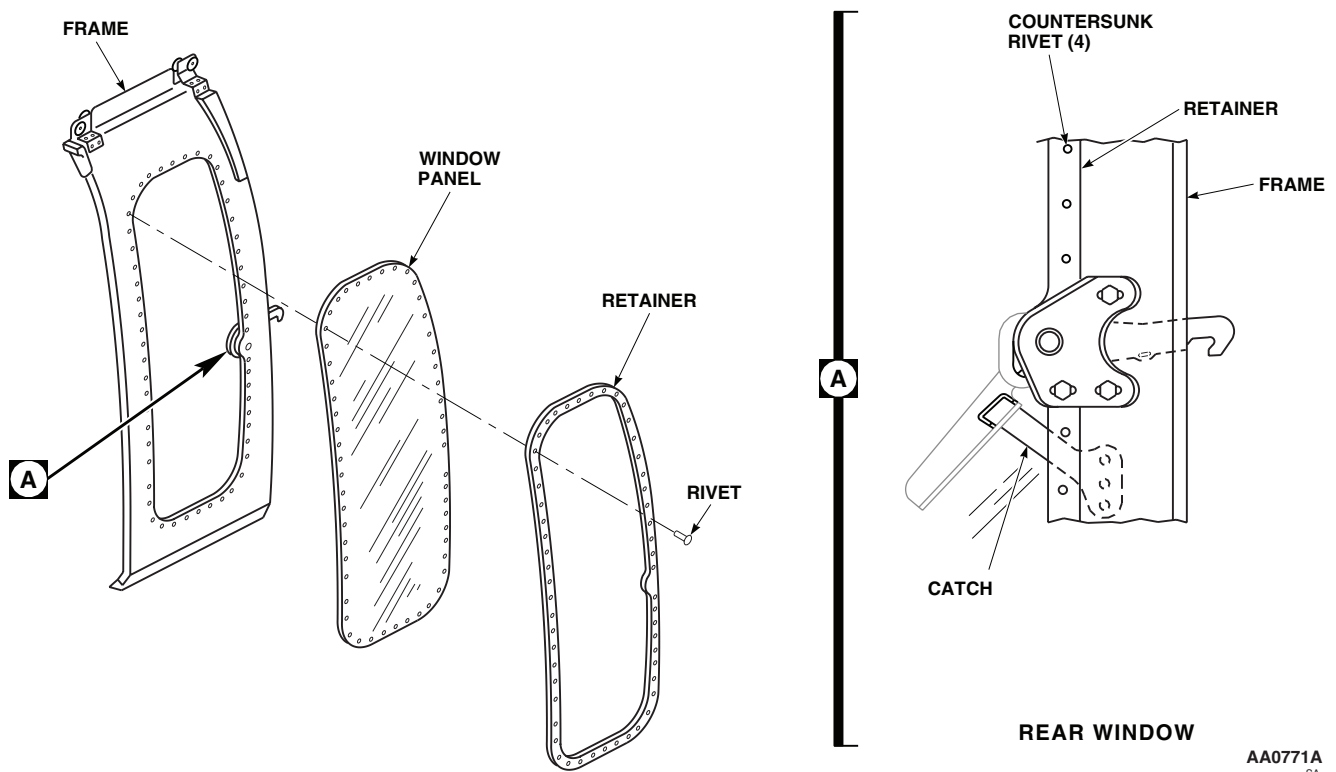
AA0771A
SA

Figure 2. Replace Plastic Window Panels.

REPLACE PLASTIC WINDOW PANELS - Continued

5. Inspect skin and retainer for damage using WP 0377 00 negligible damage limits.
6. Replace retainer and repair skin if damage is greater than negligible damage (WP 0237 00).

Install**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean sealing surface of skin with machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry with a clean machinery wiping towel, Item 344, WP 1803 00.
2. Temporarily place window panel over opening. Mark and drill 0.1405-inch holes where rivets will go (Figure 2).
3. Wash sealing side of window with mild soap and water. Wipe dry with clean, soft cloth.
4. Apply smooth coating of sealing compound, Item 273, WP 1803 00, on flange of window opening. Apply enough sealing compound for good squeezeout.
5. Position window panel in opening and press firmly in place.
6. Place retainer on outside of window panel, line up holes, and temporarily fasten entire unit with 1/8-inch diameter clecos in every third hole.
7. Check sealing compound squeezeout on outside of window for continuous bead with no voids. If there are voids, remove window panel and add more sealing compound, Item 273, WP 1803 00.

NOTE

Four countersunk rivets are installed above latch installation on rear window with head against retainer.

8. Install universal head rivets in between clecos, advancing around window with each rivet.
9. Remove clecos and install remaining rivets.
10. Wipe squeezed-out sealing compound to form a smooth unbroken fillet 1/8 to 1/4-inch wide around entire outside edge of window panel.
11. Allow sealant to cure for 72 hours at room temperature 72°F(22°C) or for 2 hours at 160°F(71°C) to 180°F(82°C).
12. Apply even coat of adhesive, Item 31, WP 1803 00, to bonding surfaces of window frame and pile fastener tape, Item 125, WP 1803 00.
13. Let adhesive air dry for 10 to 15 minutes.
14. Install pile fastener tape, Item 125, WP 1803 00, flush with edge of window opening, pressing it firmly into place.
15. Let adhesive cure at room temperature 72°F(22°C) for 12 to 16 hours.
16. Touch up paint in repaired area, as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****TROOP/CARGO DOOR REPAIR**

This WP supersedes WP 0401 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-S-22499

MIL-STD-670

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

QQ-A-200

QQ-A-225

QQ-A-250

References

AMS 3240

AMS 4229

LP-410

MIL-P-15035

MIL-P-25690

MIL-R-6855

MIL-S-18729

TM 1-1500-204-23

TM 55-1500-345-23

WP 0247 00

WP 0377 00

WP 0378 00

WP 0379 00

WP 0381 00

REPAIR

1. Troop/cargo door structure is a bonded sandwich construction of Nomex honeycomb and aluminum skins, reinforced with aluminum angles, straps, and channels. Transparent window panels are riveted to window frames which are secured to door by jettison mechanism. Window and door seals are formed rubber. Rear seal on fuselage is riveted to plating. Jettison/latching mechanism is aluminum alloy and 4130 steel. Pans under windows, housing jettison components, are reinforced plastic. Track extrusion and roller assembly support are aluminum alloy. Forward and rear fairings are Kevlar reinforced plastic. Troop/cargo door is secondary structure (Figure 1, Sheet 1, Figure 1, Sheet 2, and Figure 1, Sheet 3).
2. Evaluate and repair damage to honeycomb structure (WP 0379 00 and TM 1-1500-204-23).
3. Evaluate and repair damage to aluminum reinforcement parts and track (WP 0377 00 and WP 0378 00). Refer to TM 1-1500-204-23 for repair materials and fastener information.
4. Evaluate and repair damage to reinforced plastic pans and Kevlar fairings (WP 0379 00).
5. Refer to WP 0247 00 for door seal installation.
6. Clean transparent windows per TM 1-1500-204-23. If plastic window has minor scratches, hazing, crazing, damage by airborne abrasives or surface imperfections polish window per TM 1-1500-204-23.
7. Touch up paint in repaired area as required (WP 0381 00 and TM 55-1520-345-23).

REPAIR - Continued

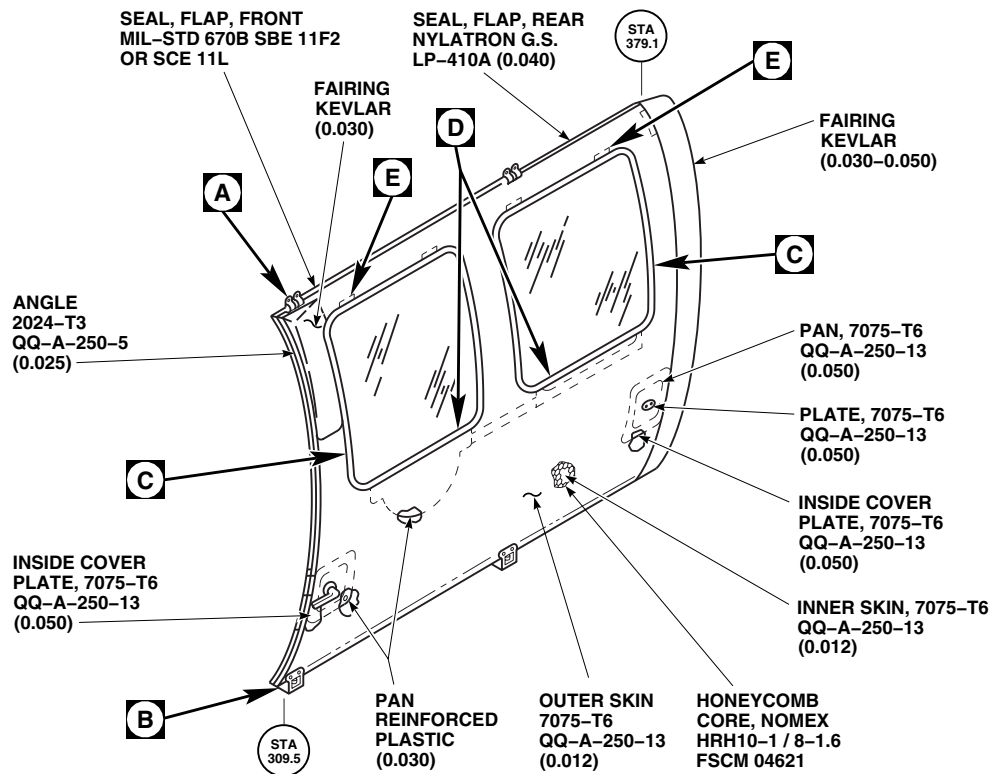
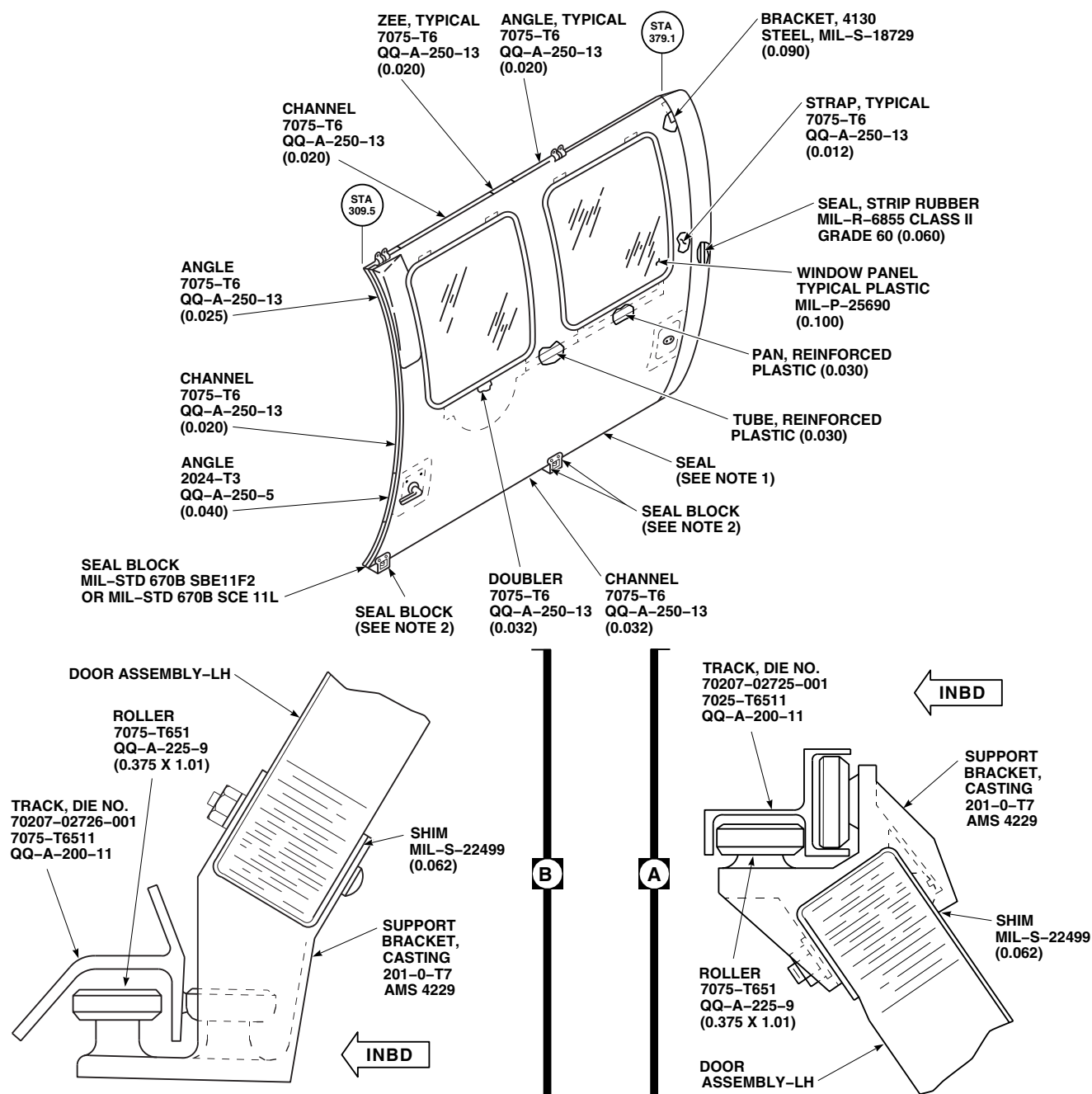
AA0785_1
SA

Figure 1. Repair Troop/Cargo Door. (Sheet 1 of 3)

REPAIR - Continued

**NOTES:**

- ON HELICOPTERS WITH DOORS, 70217-02701-041 THROUGH -044 SHEATH 7075-T6 QQ-A-250-13 (0.020).
ON HELICOPTERS WITH DOORS, 70217-02071-045, -046 SEAL STRIP, NYLATRON (0.040).
ON UH-60L HELICOPTERS SERIAL NOS 91-26401 AND SUBSEQUENT, SEAL STRIP URETHANE (0.060).
- HELICOPTERS WITH DOOR PART NOS 70217-02701-041, -042 -043 AND -044 SEAL BLOCKS MIL-STD 670B SBE 7LCD1Q80.

AA0785_2
SA

Figure 1. Repair Troop/Cargo Door. (Sheet 2 of 3)

REPAIR - Continued

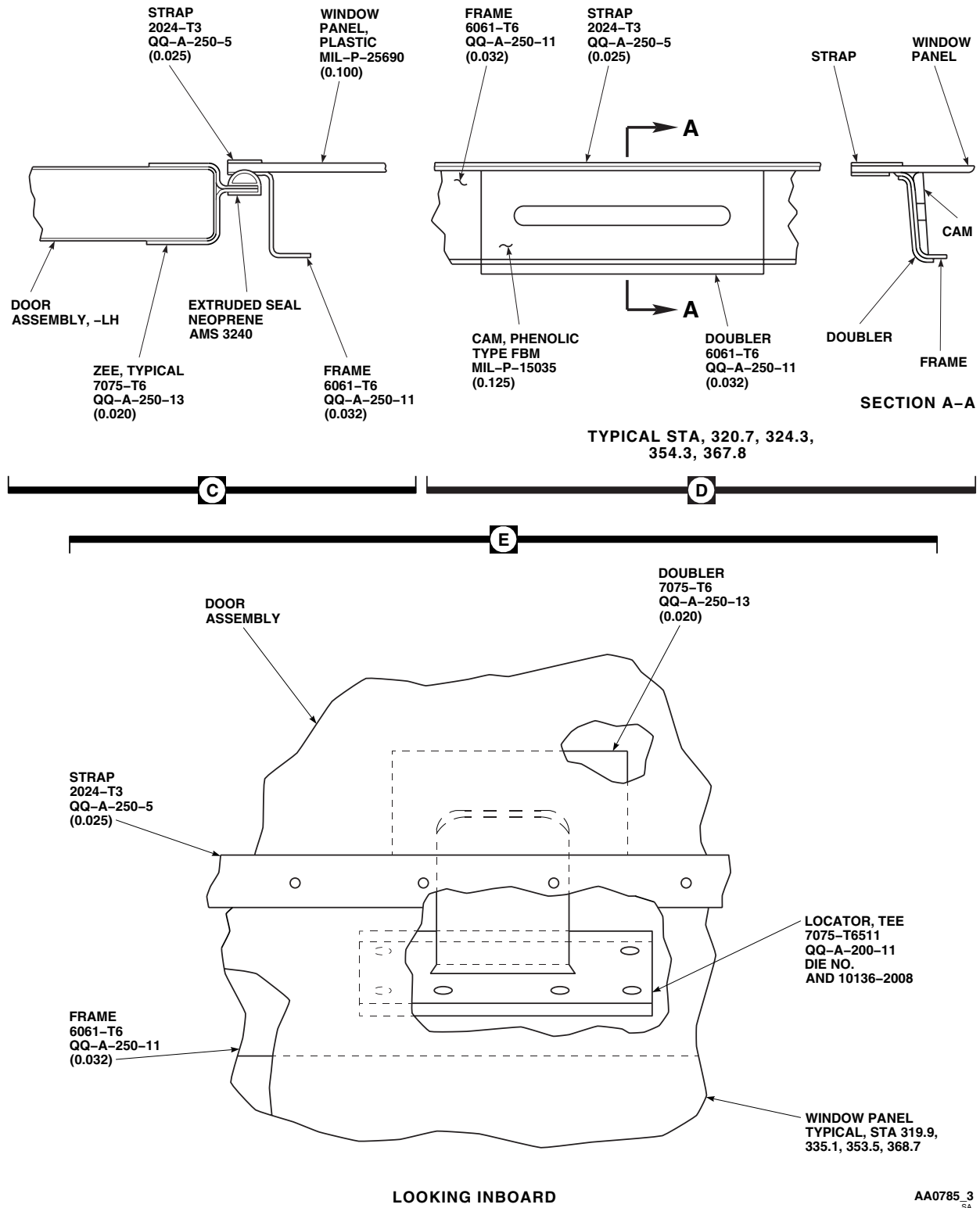


Figure 1. Repair Troop/Cargo Door. (Sheet 3 of 3)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

CABIN FLOOR REPAIR

This WP supersedes WP 0402 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Airframe Repairer's Toolkit, SC 5180-99-B02
 Hacksaw, 00181
 Torque Wrench, 0 - 30 in. lbs.

Material/Parts

Adhesive, Item 39, WP 1803 00
 Corrosion Preventive Compound, Item 106,
 WP 1803 00
 Corrosion Removing Compound, Item 108,
 WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Pad, Scouring, Item 205, WP 1803 00
 Primer, Adhesive, Item 241, WP 1803 00
 Primer Coating, Item 243, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

AMS 5528
 MIL-R-6855
 MIL-S-18799
 MIL-S-22499
 MIL-STD-670
 MIL-T-9046
 QQ-A-200
 QQ-A-225
 QQ-A-250
 TM 1-1500-204-23
 TM 55-1500-345-23
 WP 0377 00
 WP 0378 00
 WP 0379 00
 WP 0380 00
 WP 0381 00
 WP 1803 00

REPAIR CABIN FLOOR

1. Cabin floor panels and cargo hook access door are of bonded sandwich construction made from Nomex/Hexcel honeycomb core and fiberglass (reinforced plastic) skins. Provisions for cargo tiedown, troop seat, and litter fitting installations are incorporated in each panel (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, and Figure 1, Sheet 7, and Figure 1, Sheet 8).
2. Use WP 0378 00 and WP 0379 00 for evaluating damage and repairing honeycomb floor panels. Refer also to TM 1-1500-204-23.
3. Evaluate damage to metal components of floor panels (WP 0377 00). Replace extrusions (WP 0380 00). Refer to TM 1-1500-204-23 for repair materials and fastener information.
4. Pan replacement requires bonding pan to floor panel using adhesive, Item 39, WP 1803 00. Apply adhesive primer, Item 241, WP 1803 00, to metal pans (aluminum or steel) before applying adhesive.
5. Floor attachment screws that are countersunk in dense honeycomb areas are to be seated. **WRENCH TORQUE IS NOT TO BE MORE THAN 10 INCH-POUNDS ABOVE SEATING TORQUE.**
6. For corrosion protection and watertightness, seal all repairs on the floor panels (WP 0377 00).
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
8. After priming, apply two spray coats of corrosion preventive compound, Item 106, WP 1803 00, to seal repairs on bottom side of floor panels (WP 0377 00).

REPAIR CABIN FLOOR - Continued

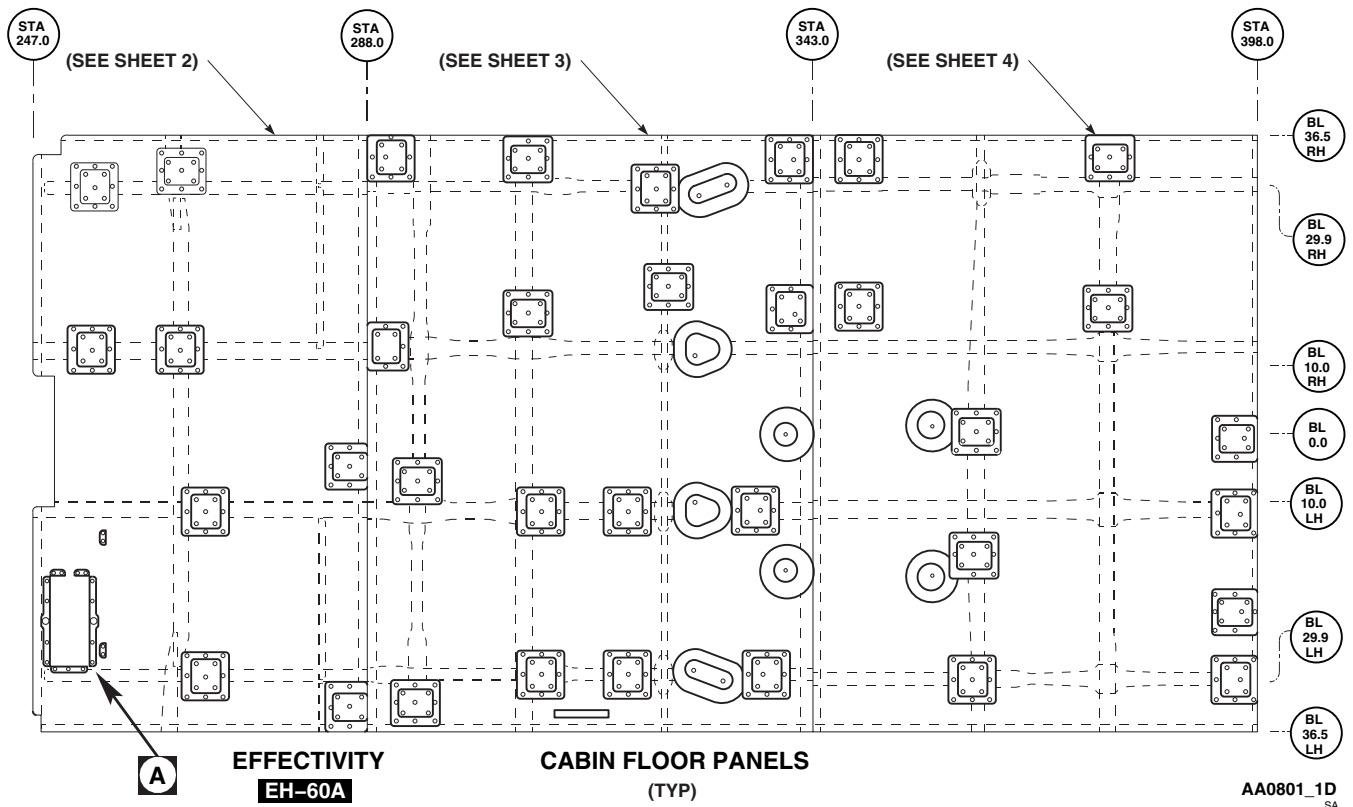
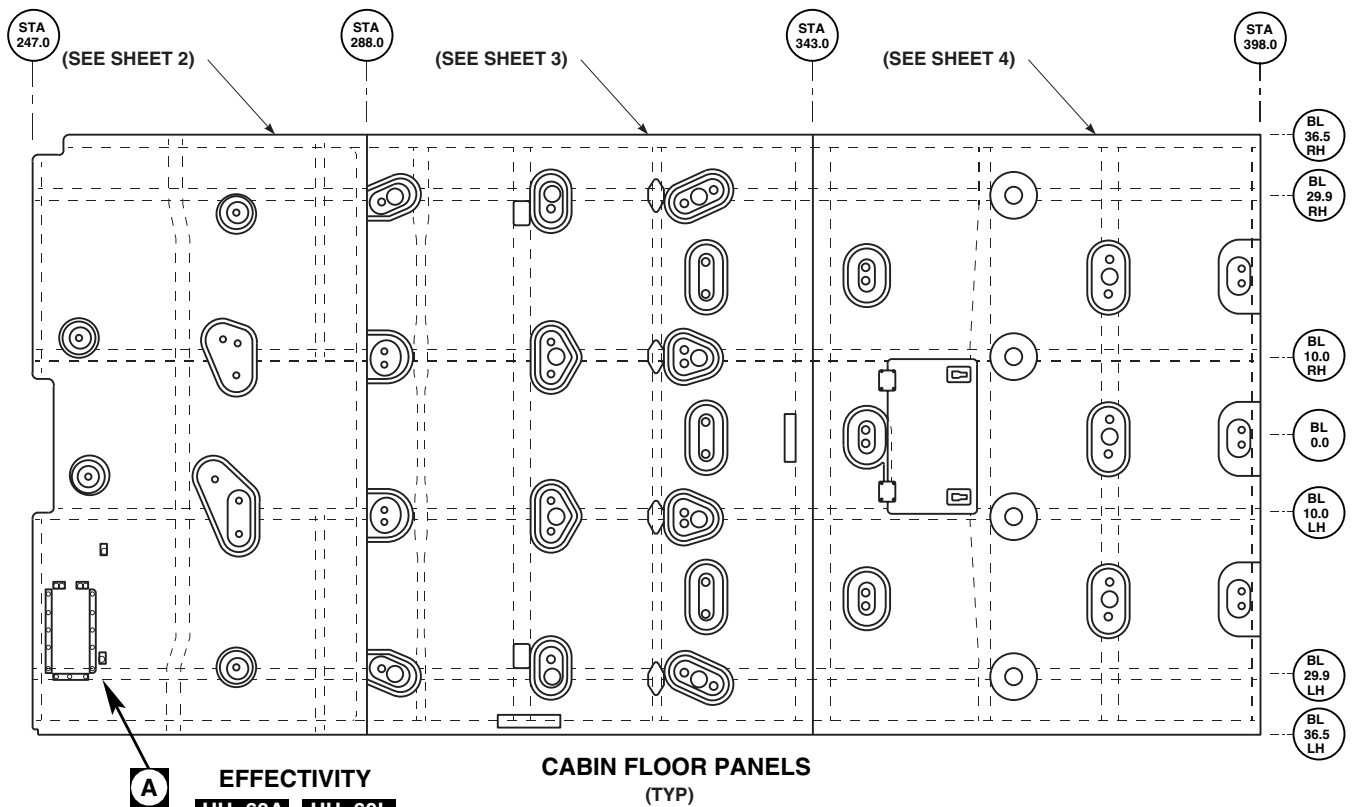
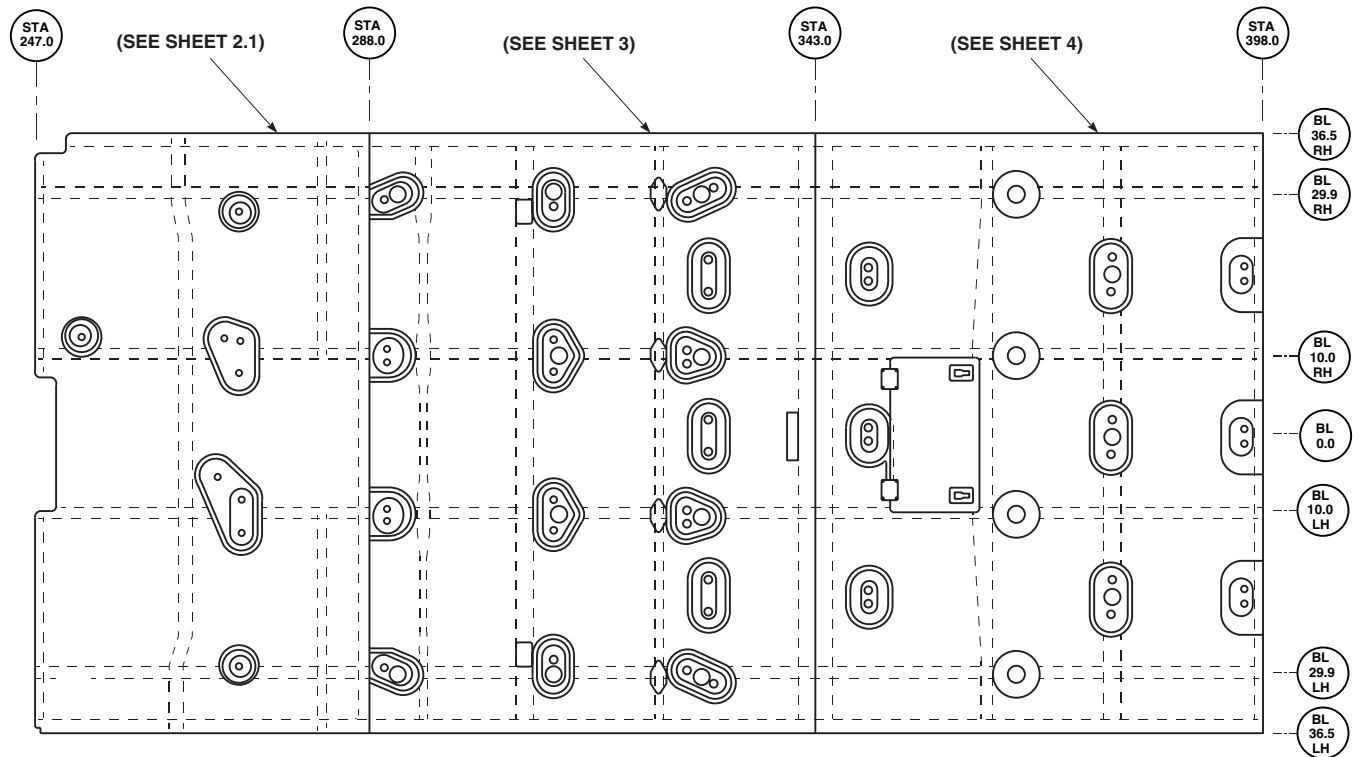


Figure 1. Repair Cabin Floors. (Sheet 1 of 8)

REPAIR CABIN FLOOR - Continued



CABIN FLOOR PANELS
(TYP)

EFFECTIVITY

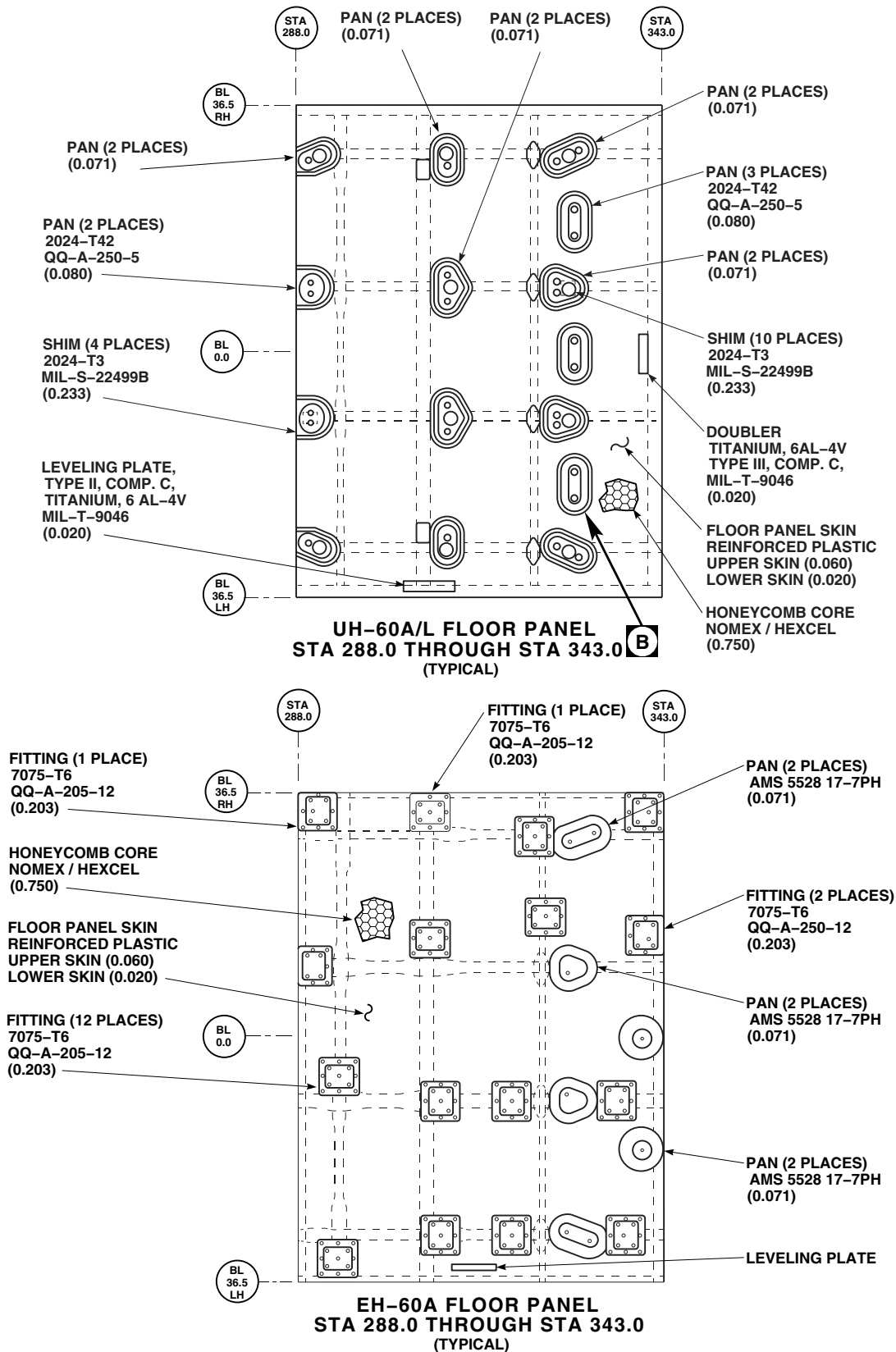
HH-60A HH-60L



AA0801_2E
SA

Figure 1. Repair Cabin Floors. (Sheet 2 of 8)

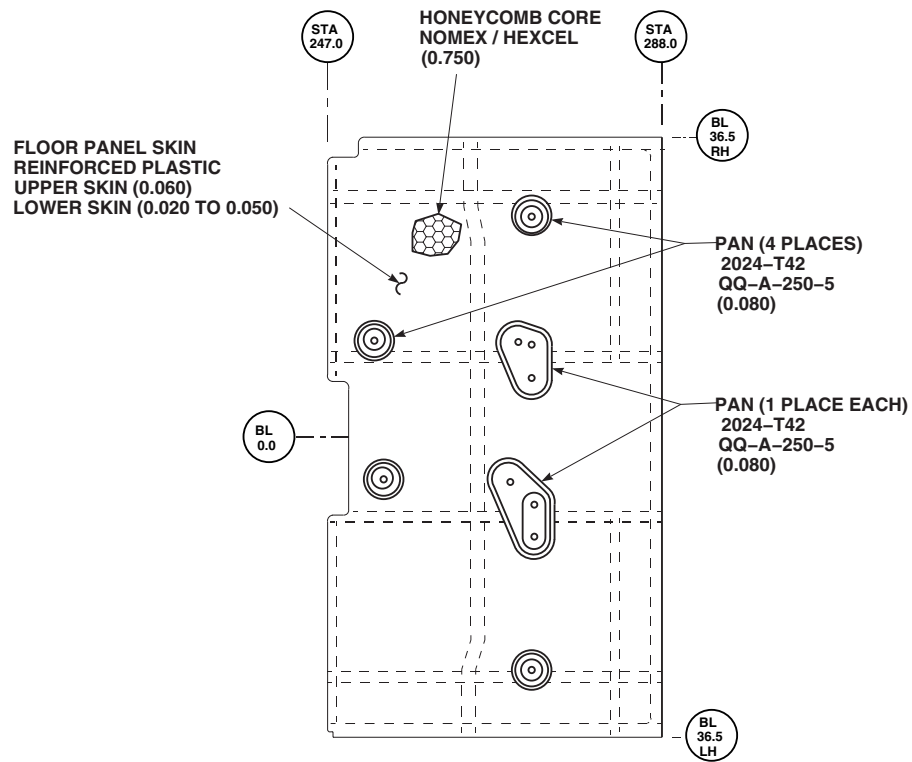
REPAIR CABIN FLOOR - Continued



AA0801_3D
SA

Figure 1. Repair Cabin Floors. (Sheet 3 of 8)

REPAIR CABIN FLOOR - Continued



FLOOR PANEL
STA 247.0 THROUGH STA 288.0

EFFECTIVITY

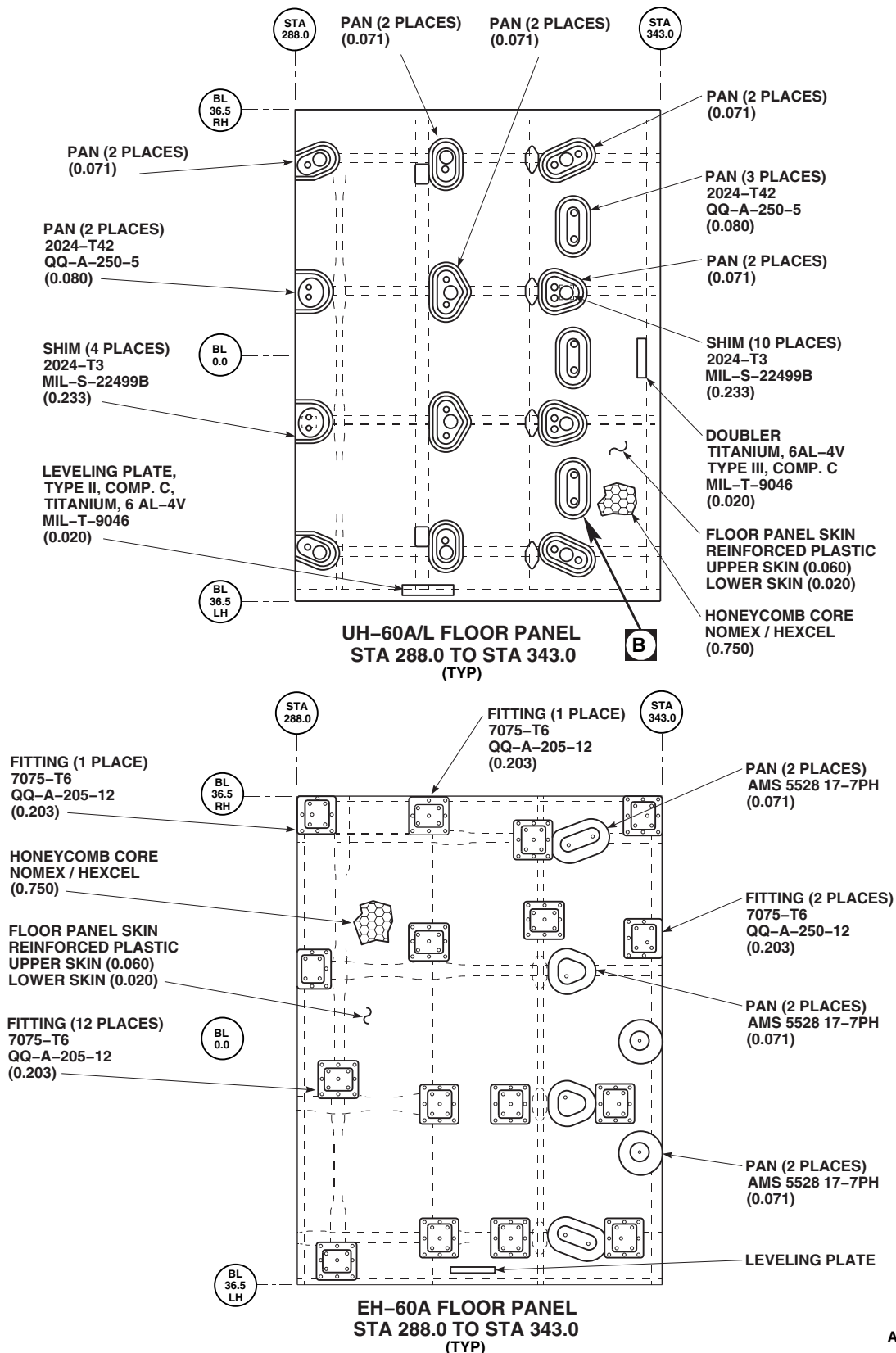
HH-60A HH-60L



AA0801_4F
SA

Figure 1. Repair Cabin Floors. (Sheet 4 of 8)

REPAIR CABIN FLOOR - Continued



AA0801_5A
SA

Figure 1. Repair Cabin Floors. (Sheet 5 of 8)

REPAIR CABIN FLOOR - Continued

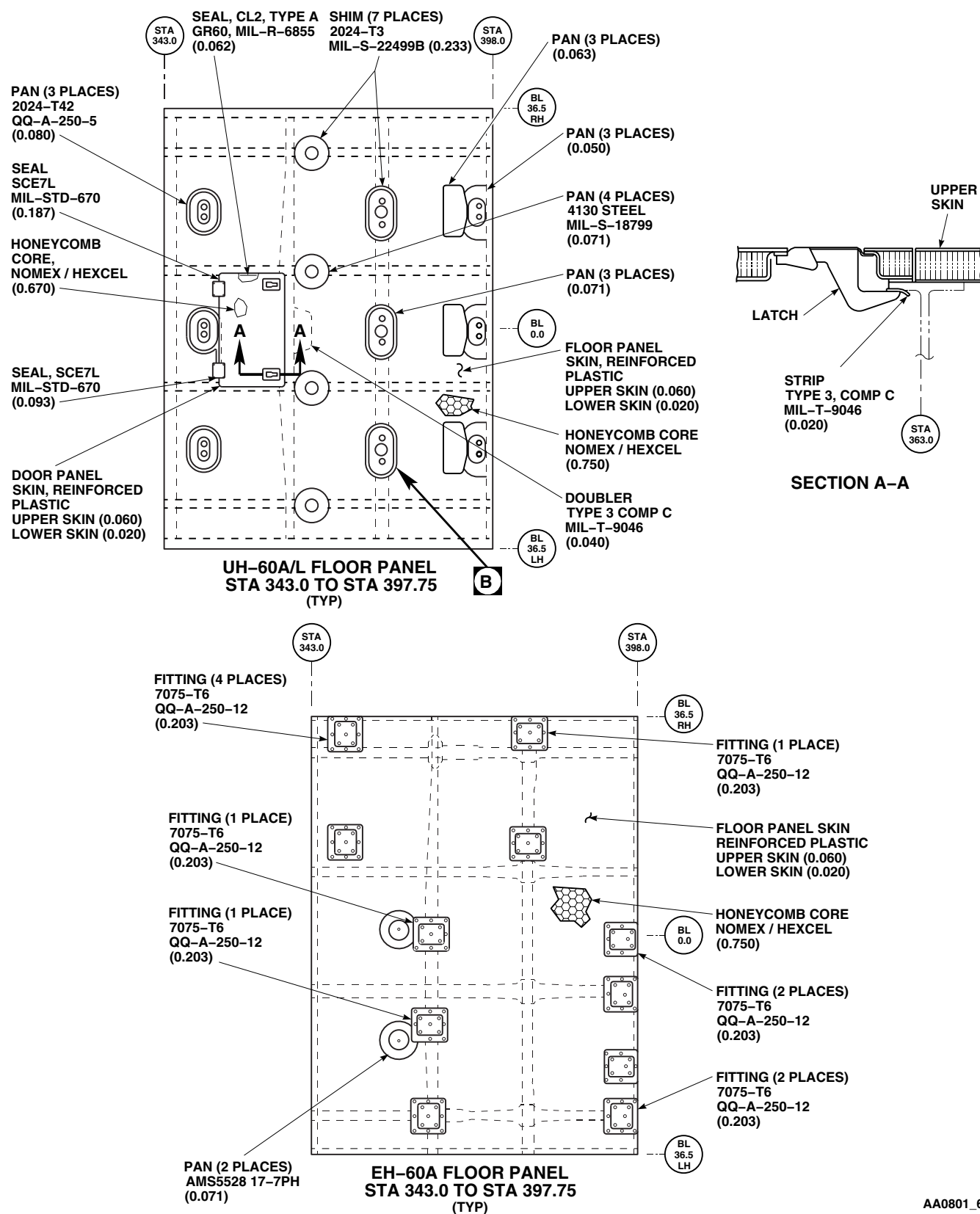


Figure 1. Repair Cabin Floors. (Sheet 6 of 8)

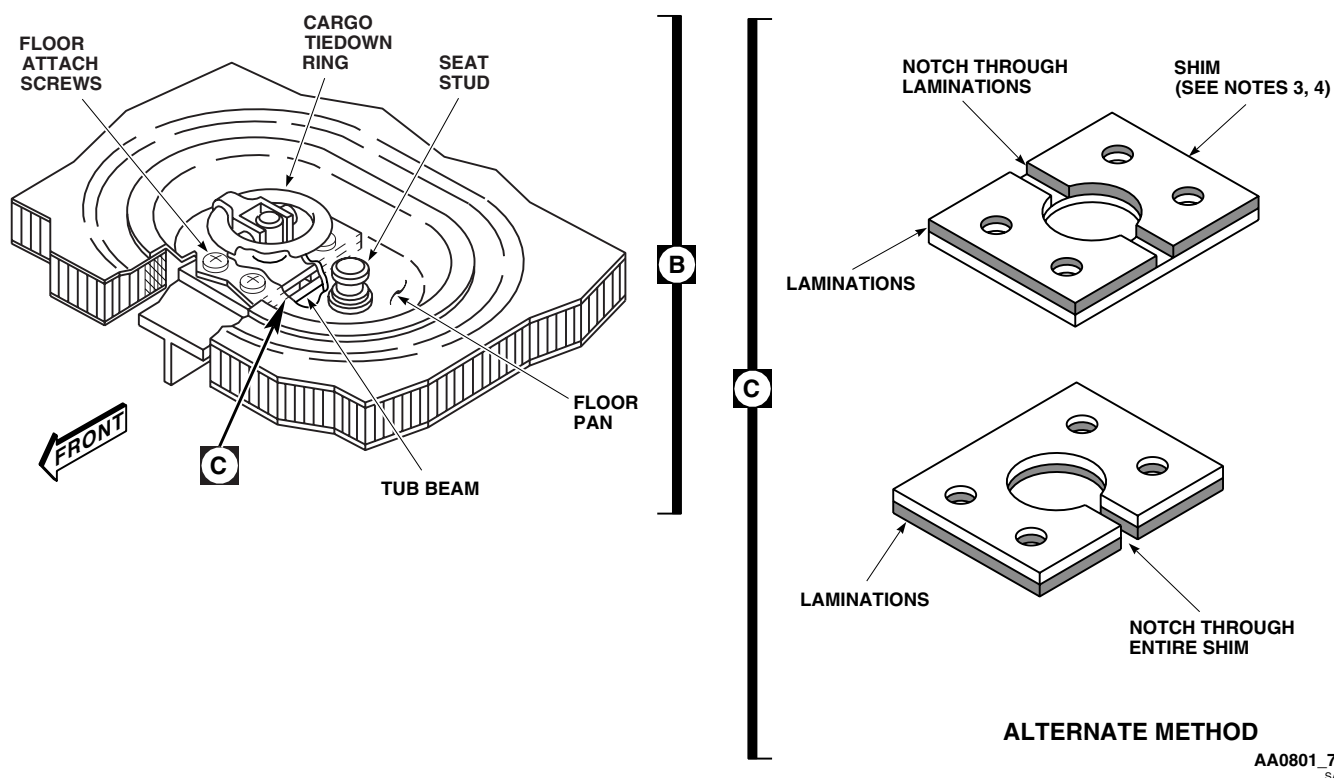
REPAIR CABIN FLOOR - Continued

Figure 1. Repair Cabin Floors. (Sheet 7 of 8)

REPAIR CARGO TIEDOWN RINGS AND FLOOR PANS

1. Remove corrosion manually using scouring pad, Item 205, WP 1803 00. **PITS IN WELD AREA SHALL NOT EXCEED 0.010-INCH DIAMETER AFTER CLEAN UP.**
2. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, to 1 part water.
3. Apply compound solution to corroded areas of tiedown rings and floor pans.
4. Rinse thoroughly with clean water.
5. Touch up paint tiedown rings and floor pans with epoxy primer coating kit, Item 120, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).

REPAIR FLOOR ATTACH SCREWS**NOTE**

If possible, replace corroded screws. Cadmium plating will be removed during corrosion removal.

1. Remove corrosion manually from floor attach screws using scouring pad, Item 205, WP 1803 00. **PITS SHALL NOT EXCEED 0.050-INCH DEEP AFTER CLEAN UP.**
2. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, to 1 part water.
3. Apply compound solution to corroded areas of floor attach screws.
4. Rinse thoroughly with water.

REPAIR FLOOR ATTACH SCREWS - Continued**NOTES:**

1. **UH-60A 77-22714 - 84-23939**
2. **UH-60L 93-26513 - SUBQ**
3. SHIM REMOVED FROM FLOOR PAN FOR CLARITY.
4. SHIM SHOWN UPSIDE DOWN. NONLAMINATED PORTION IS BONDED TO FLOOR PLAN.
5. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0801_8
SA

Figure 1. Repair Cabin Floors. (Sheet 8 of 8)

REPAIR FLOOR SHIMS

1. Remove floor pans, if required.
2. Turn floor pan over. Cut a slot in laminated shim 1/8-inch wide and 1/8-inch deep using a hacksaw. Extend slot across entire surface of shim (Figure 1, Sheet 5, Detail C).
3. Peel laminations from shim to bottom of cut.
4. Alternate method is to cut a slot in laminated shim 1/8-inch wide through shim. Extend slot from edge of shim to center hole in shim using a hacksaw. Do not cut into floor pan.
5. Touch up paint floor shims with primer coating, Item 243, WP 1803 00, (WP 0381 00 and TM 55-1500-345-23).

REPAIR CARGO HOOK DOOR **HH-60A HH-60L**

1. Refer to WP 0379 00 to repair any skin or structural damage to door.

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

MAIN ROTOR PYLON FRAMING REPAIR

This WP supersedes WP 0403 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
 Drill Set, GGG-D-751
 Gloves
 Goggles/Face Shield

MIL-P-46060

MIL-R-6855

MIL-S-18729

MIL-S-5000

MIL-S-5059

MIL-S-6721

MIL-S-6758

MIL-S-7947

MIL-S-8844

MIL-S-9047

MIL-STD-670

MIL-T-9046

QQ-A-200

QQ-A-225

QQ-A-250

QQ-A-367

TM 1-1500-204-23

TM 1-1500-344-23

TM 55-1500-345-23

WP 0296 00

WP 0377 00

WP 0378 00

WP 0380 00

WP 0381 00

WP 1803 00

WP 1805 00

WW-T-700

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 10, WP 1803 00
 Adhesive, Item 11, WP 1803 00
 Adhesive, Item 29, WP 1803 00
 Adhesive, Item 40, WP 1803 00
 Adhesive, Item 47, WP 1803 00
 Adhesive Primer, Item 23, WP 1803 00
 Cloth, Abrasive, Item 80, WP 1803 00
 Cloth, Cheesecloth, Item 85, WP 1803 00
 Hinge, MS20001P5-1000
 Rivet, NAS1097AD-4
 Sealing Compound, Item 263, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Track Repair Angle, Locally-Made, Figure 186,
 WP 1805 00
 Track Repair Plate, Locally-Made, Figure 185,
 WP 1805 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

AMS 5355
 L-P-410

REPAIR

1. Main rotor pylon framing includes formed and extruded members made from 7075-T6, QQ- A-250-13, Alclad sheet stock and 7075-T6511, QQ-A-200-11, aluminum alloy extrusion. Firewalls and heat-resistant components are made from 6AL-4V titanium alloy, MIL-T-9046; 321 corrosion-resistant stainless steel, MIL-S-6721; and 301 corrosion-resistant steel MIL-S-5059 (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, Figure 1, Sheet 7, Figure 1, Sheet 8, Figure 1, Sheet 9, Figure 1, Sheet 10, Figure 1, Sheet 11, and Figure 1, Sheet 12).

REPAIR - Continued

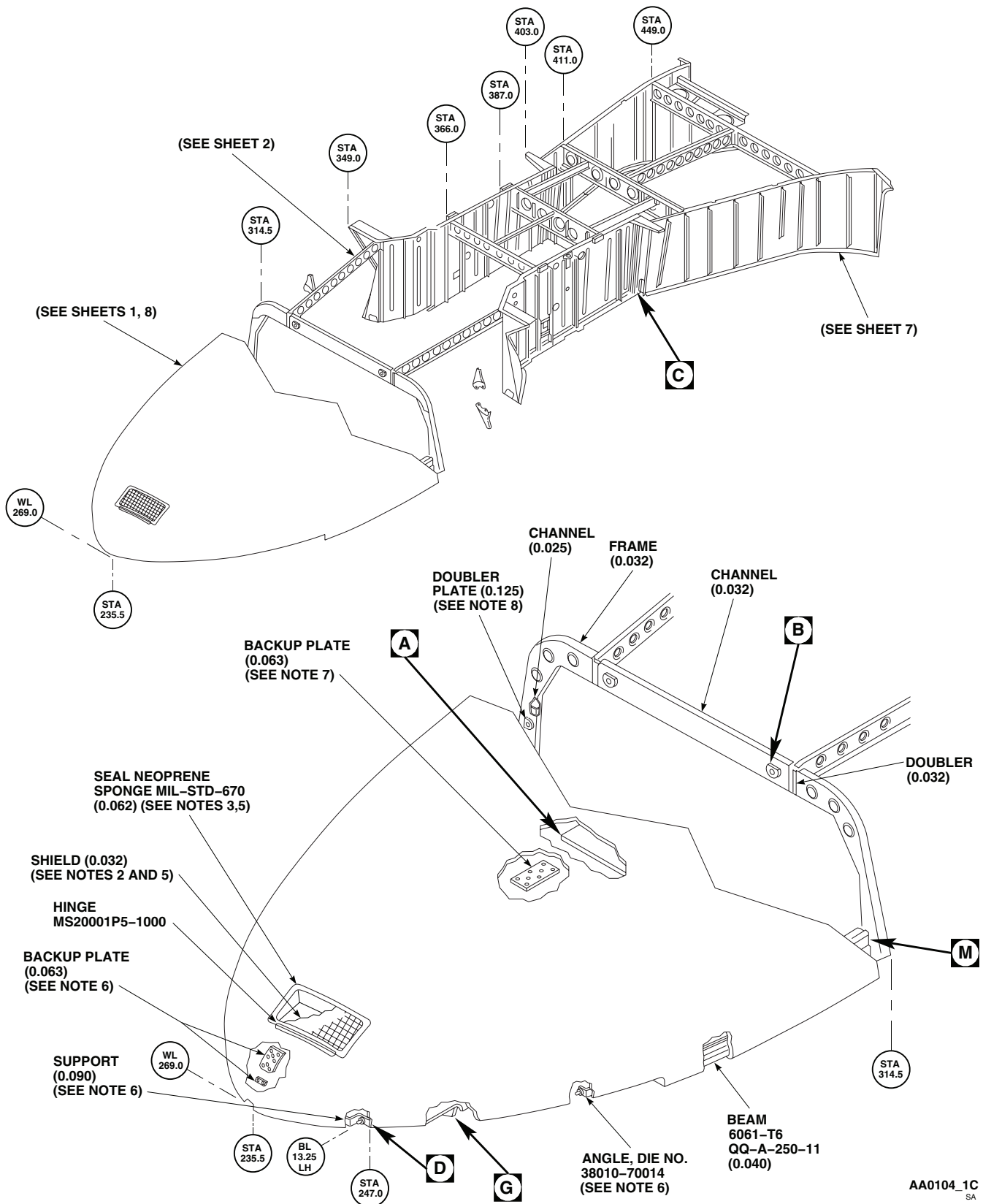


Figure 1. Main Rotor Pylon Framing Repair. (Sheet 1 of 12)

[illegible]

Change 3

REPAIR - Continued

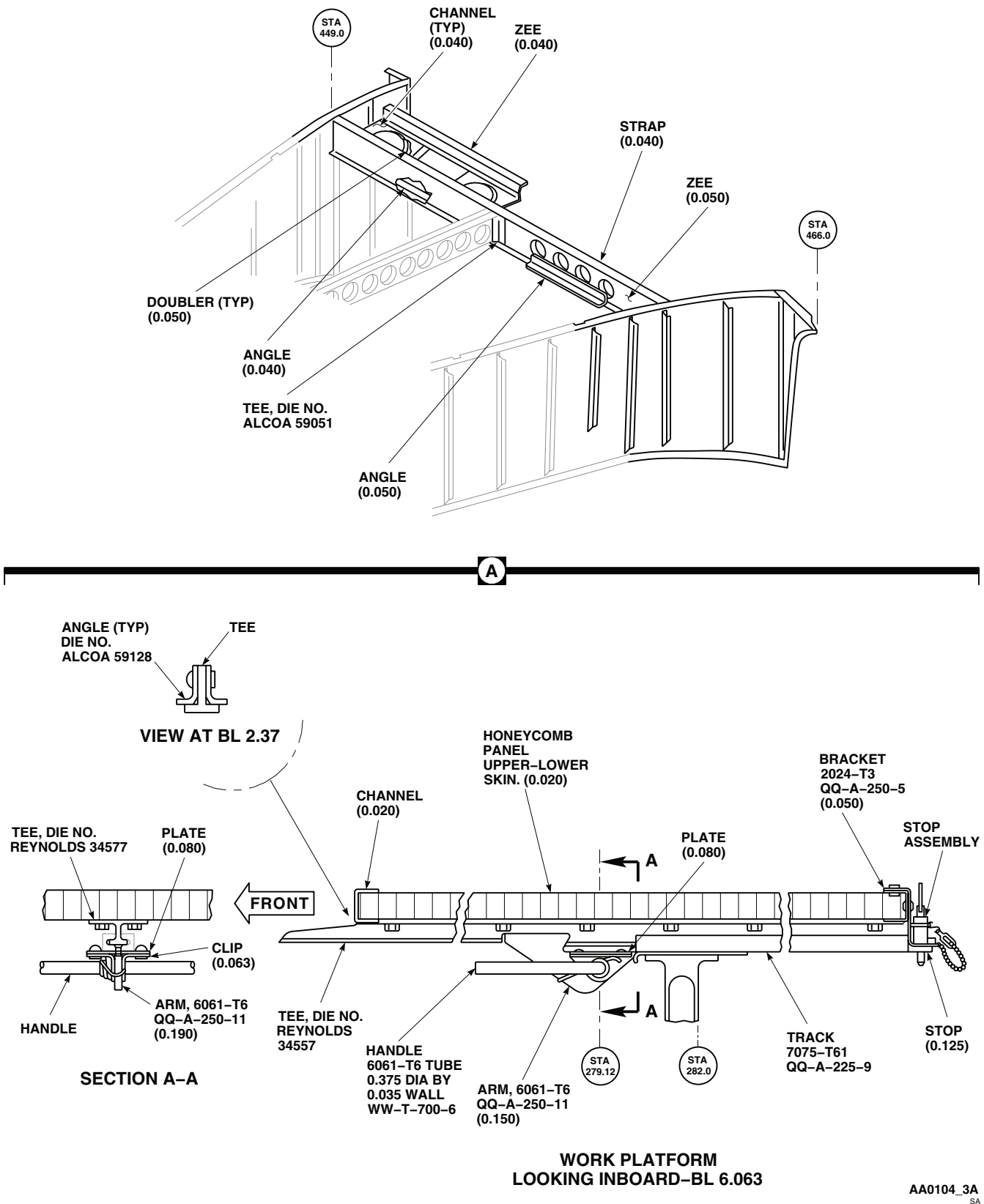


Figure 1. Main Rotor Pylon Framing Repair. (Sheet 3 of 12)

REPAIR - Continued

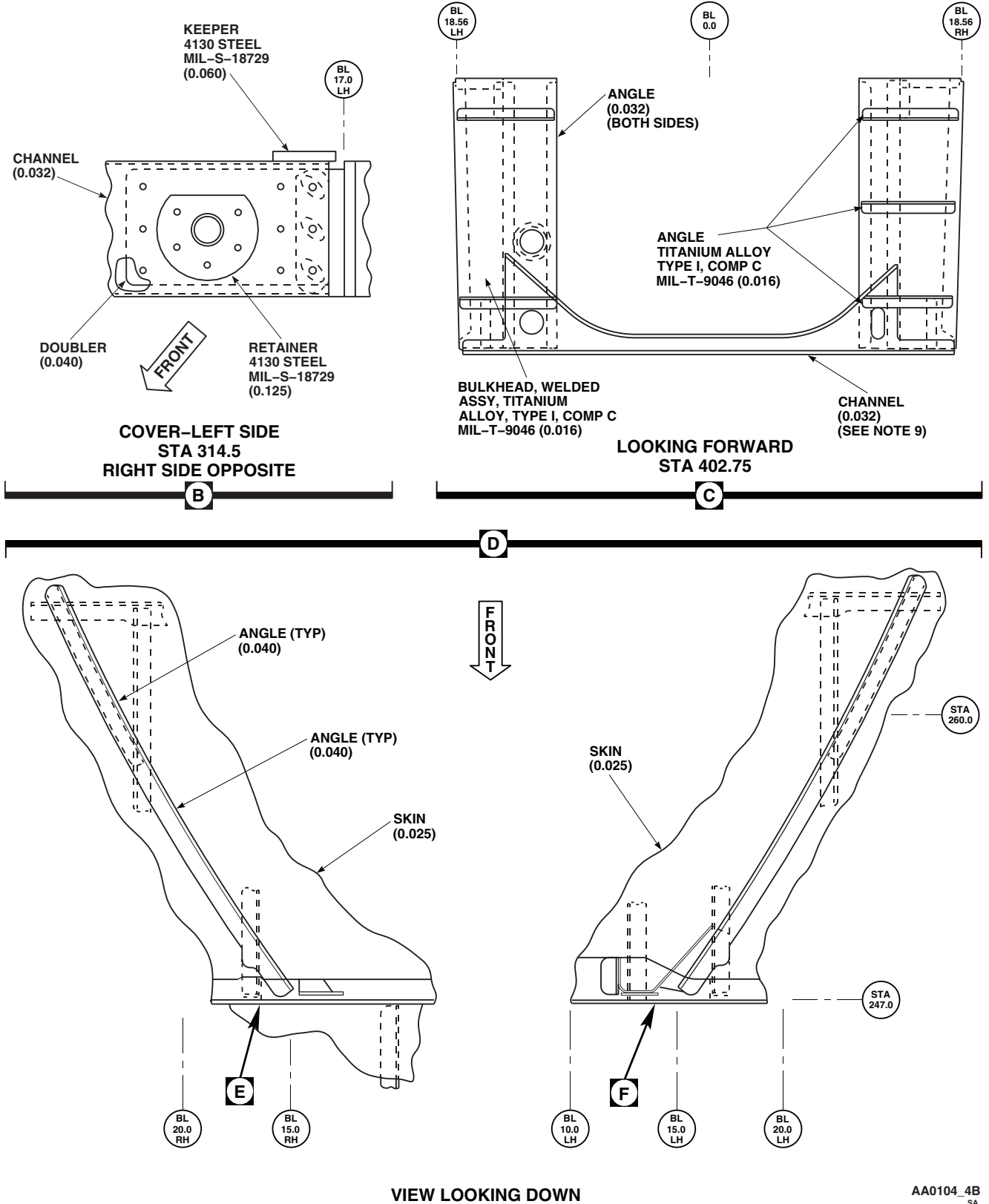


Figure 1. Main Rotor Pylon Framing Repair. (Sheet 4 of 12)

REPAIR - Continued

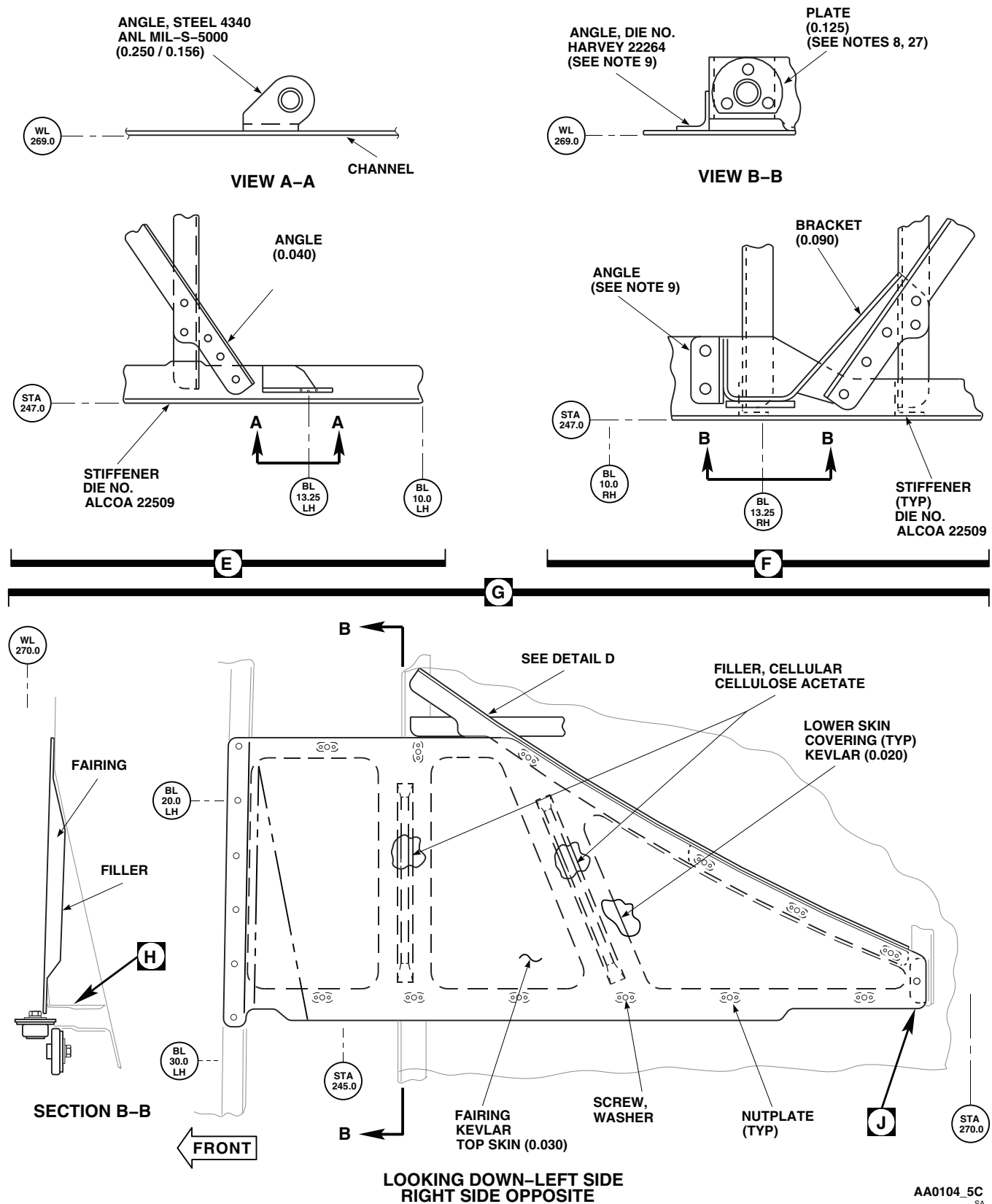


Figure 1. Main Rotor Pylon Framing Repair. (Sheet 5 of 12)

REPAIR - Continued

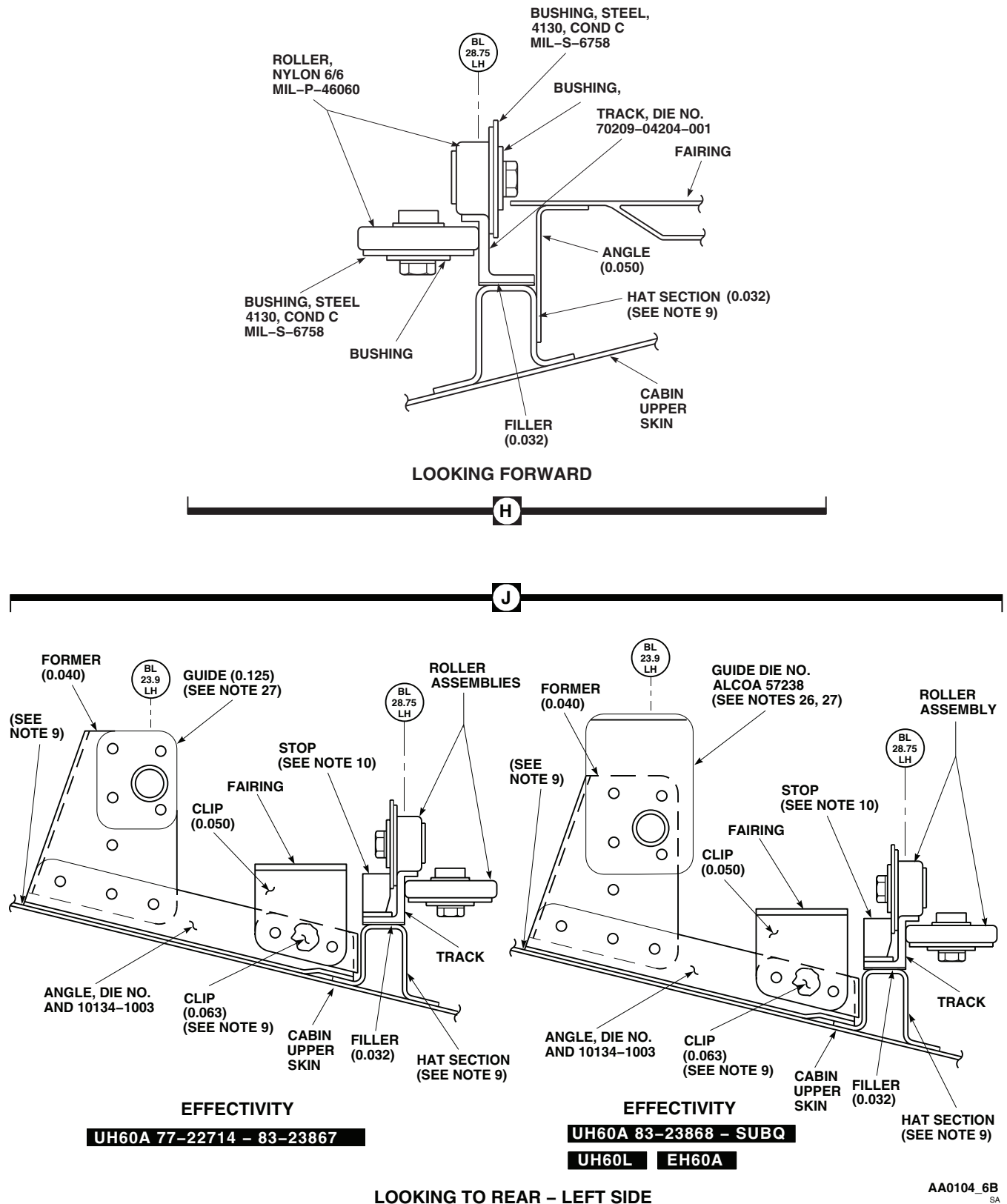


Figure 1. Main Rotor Pylon Framing Repair. (Sheet 6 of 12)

REPAIR - Continued

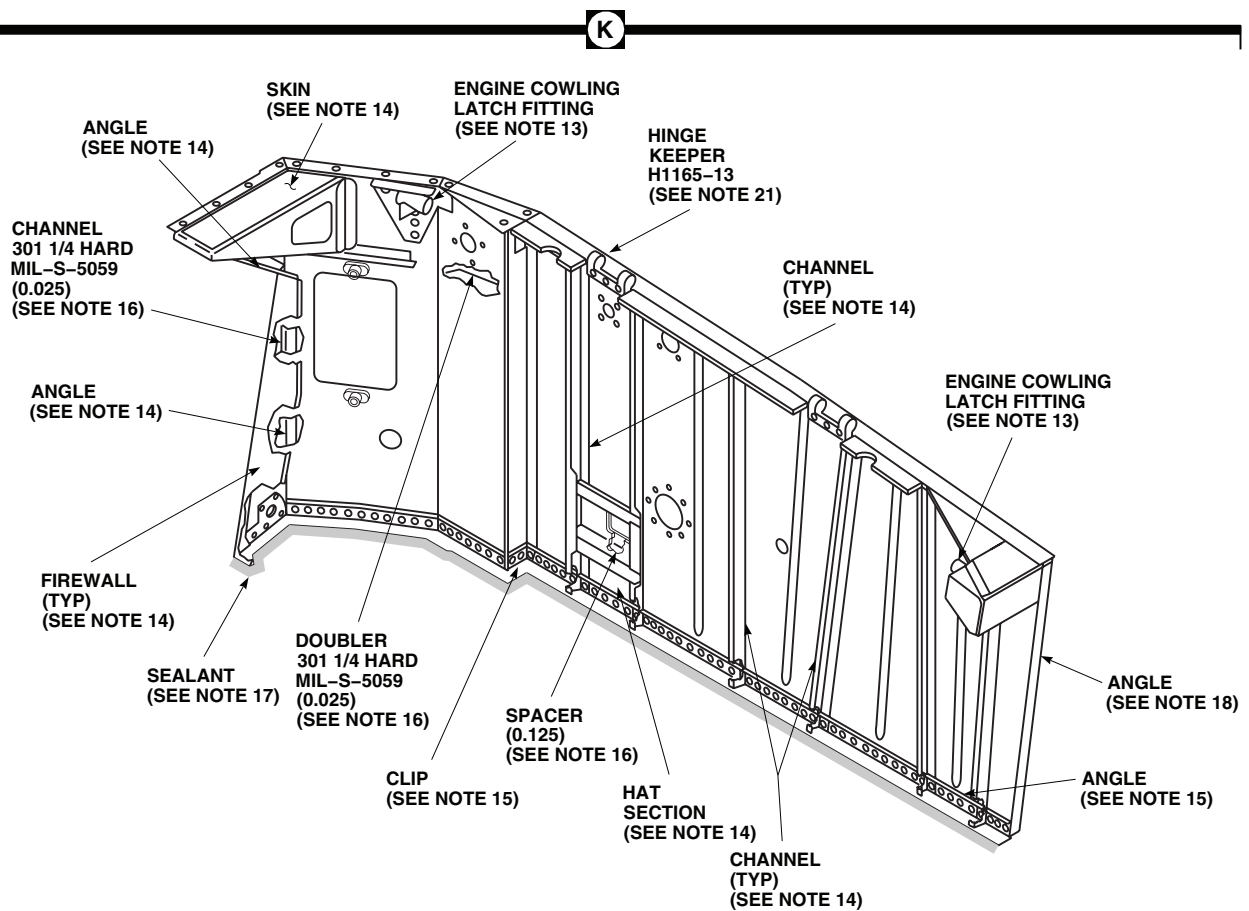
AA0104_7C
SA

Figure 1. Main Rotor Pylon Framing Repair. (Sheet 7 of 12)

REPAIR - Continued

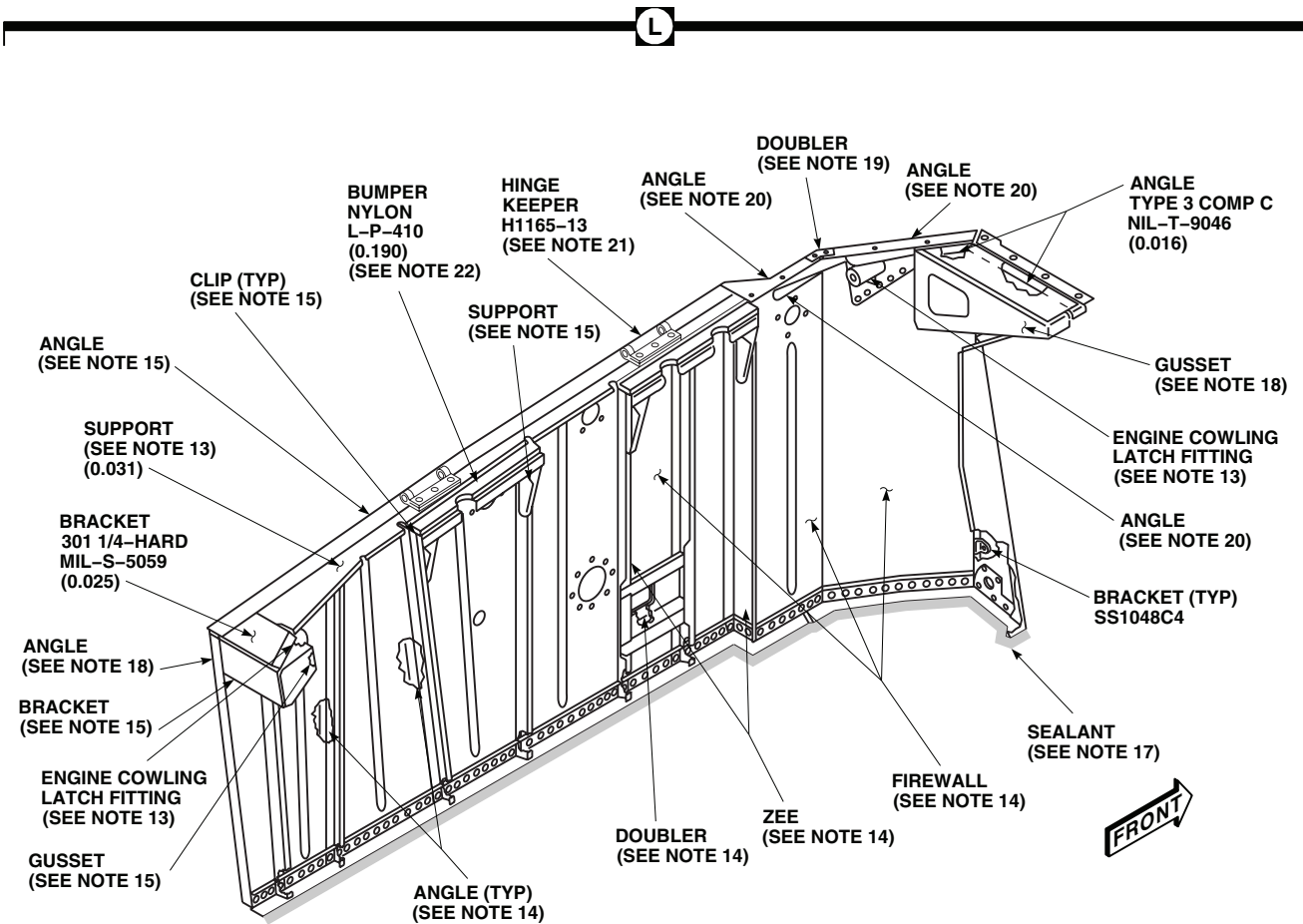
AA0104_8C
SA

Figure 1. Main Rotor Pylon Framing Repair. (Sheet 8 of 12)

REPAIR - Continued

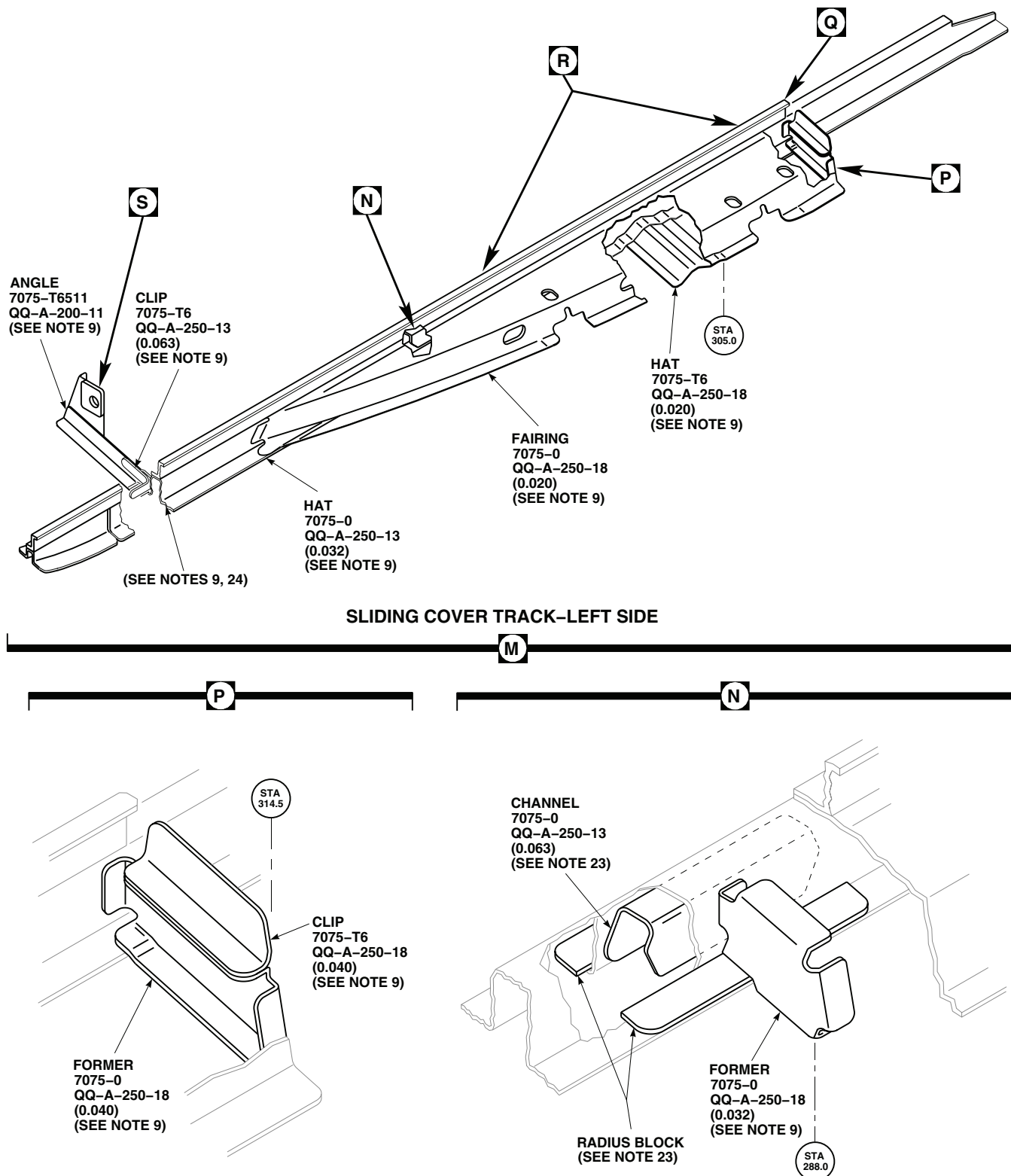
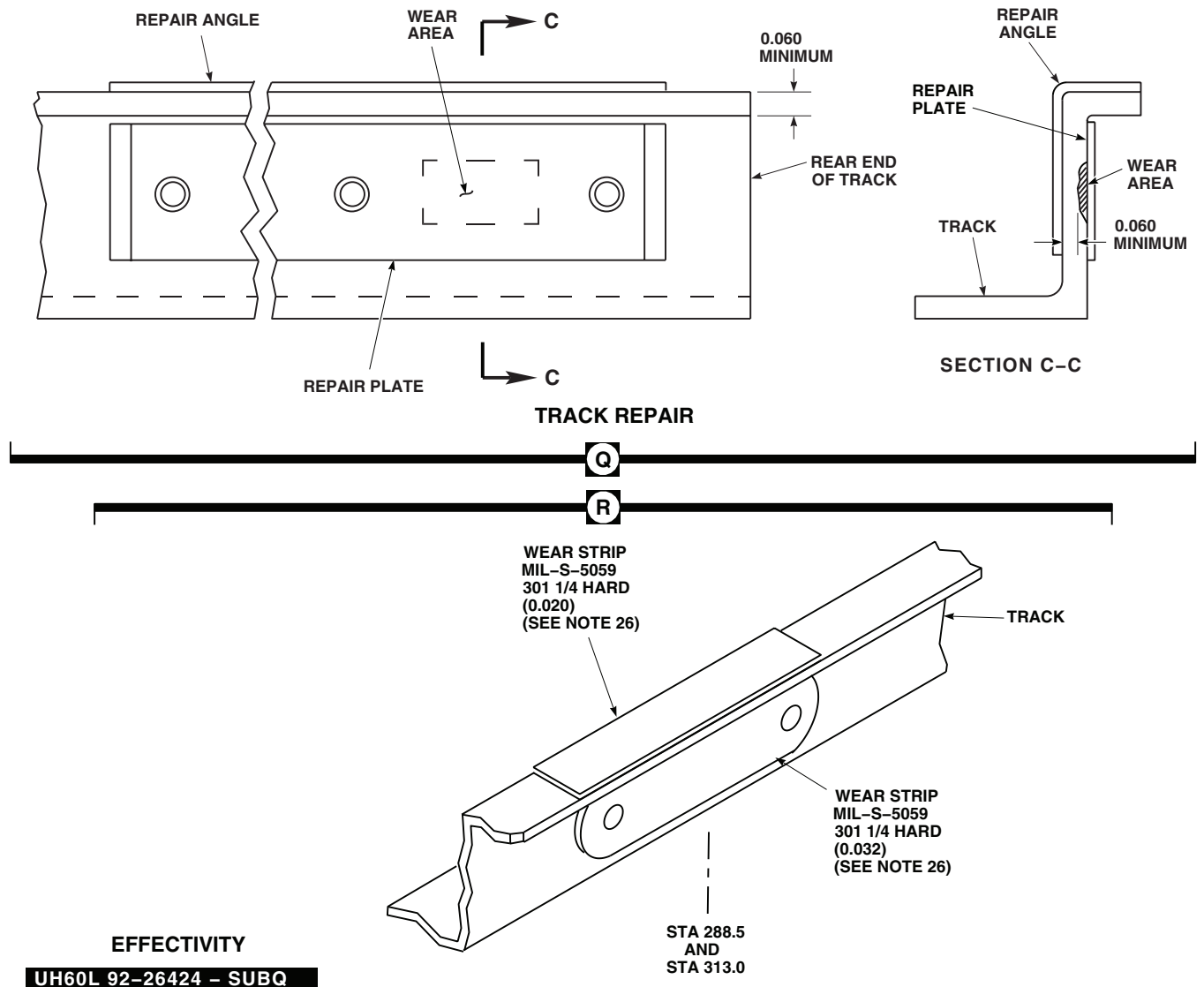
AA0104_9C
SA

Figure 1. Main Rotor Pylon Framing Repair. (Sheet 9 of 12)

REPAIR - Continued

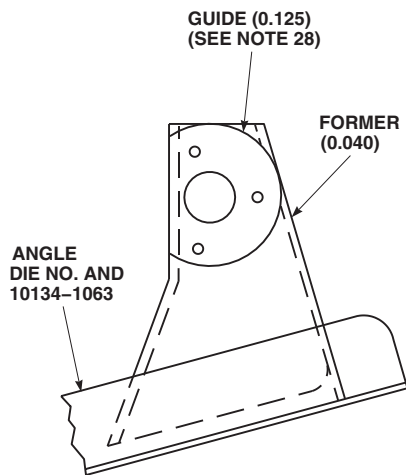


AA0104_10B
SA

Figure 1. Main Rotor Pylon Framing Repair. (Sheet 10 of 12)

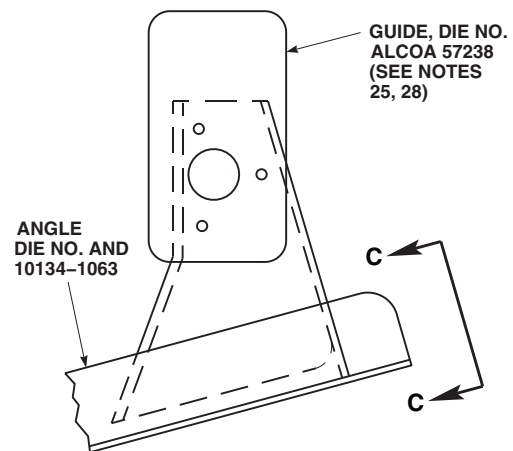
REPAIR - Continued

S



EFFECTIVITY

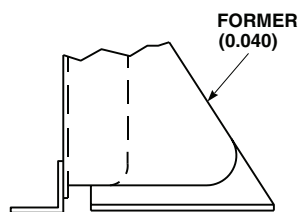
UH60A 77-22714 - 83-23867



EFFECTIVITY

UH60A 83-23868 - SUBQ

UH60L EH60A



EFFECTIVITY

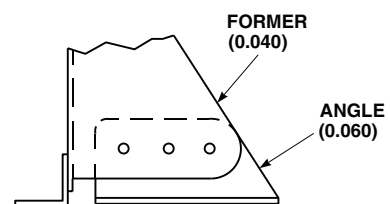
UH60A

UH60L 89-26149, 89-26154

UH60L 89-26179 - 92-26430

EH60A

VIEW C-C



EFFECTIVITY

UH60L 92-26431 - SUBQ

VIEW C-C

LOOKING TO REAR - RIGHT SIDE

AA0104_11A
SA

Figure 1. Main Rotor Pylon Framing Repair. (Sheet 11 of 12)

REPAIR - Continued

NOTES:

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. BOND TO LIP OF CUTOUT WITH ADHESIVE, ITEM 40, WP 1803 00.
4. BOND SHIELD TO EDGE OF SCREEN WITH ADHESIVE, ITEM 40, WP 1803 00.
5. **UH60A 83-23867 - SUBQ** **UH60L** **EH60A**
6. FILL GAP ON MOUNTING SURFACE WITH ADHESIVE, ITEM 11, WP 1803 00.
7. **UH60A 86-24516 - SUBQ** **UH60L** **EH60A 85-24469 - SUBQ**
8. **UH60A 89-26134 - SUBQ** **UH60L** 4130 STEEL, MIL-S-18729. ◀
9. CLEAN AND SEAL WITH SEALING COMPOUND, ITEM 273, WP 1803 00.
10. MADE EITHER FROM STEEL, MIL-S-7947, OR FROM CASTING, 356-T6.
11. 6AL-4V TITANIUM ALLOY ANNEALED, TYPE III, COMP C.
12. **UH60A 77-22714 - 86-24490** **EH60A 84-24017 - 84-24026** 0.016. ◀
13. **UH60A 77-22714 - 83-23834** WELDMENT OF TYPE 1, COMP, MIL-T-9046 AND TYPE 1, COMP A, MIL-T-9047. ◀
UH60A 83-23840 - SUBQ **UH60L** **EH60A** CASTING AMS5355 17-4 PH H-1000. ◀
14. TYPE III, COMP C, MIL-T-9046 (0.016).
15. TYPE III, COMP C, MIL-T-9046 (0.025).
16. **UH60A 81-23643 - SUBQ** **UH60L** **EH60A**
17. CLEAN AND BOND WITH ADHESIVE, ITEM 29, WP 1803 00.
18. **UH60A 77-22714 - 84-23468** TYPE III, COMP C, MIL-T-9046 (0.016). ◀
UH60A 84-23469 - SUBQ **UH60L** **EH60A** (0.025). ◀
19. **UH60A 77-22714 - 89-26168** **UH60L 89-26149 - 89-26154** **EH60A**
TYPE III, COMP C, MIL-T-9046 (0.016) ◀ **UH60A 89-26169 - SUBQ** **UH60L 89-26179 - SUBQ**
TYPE III, COMP C, MIL-T-9046 (0.032). ◀
20. **UH60A 77-22714 - 89-26168** **EH60A** TYPE III, COMP C, MIL-T-9046 (0.025). ◀
UH60A 89-26169 - SUBQ **UH60L** TYPE III, COMP C, MIL-T-9046 (0.040). ◀
21. **UH60A 79-23288 - SUBQ** **UH60L** **EH60A** PART NO. 70304-02117-101. ◀
22. ALTERNATE MATERIAL: RUBBER, MIL-R-6855, CLASS I, GRADE 40.
23. CLEAN AND SEAL WITH SEALING COMPOUND, ITEM 273, WP 1803 00.
24. FILL GAPS WITH SEALING COMPOUND, ITEM 263, WP 1803 00, AND SEAL WITH SEALING COMPOUND, ITEM 274, WP 1803 00 (WP 0377 00).
25. **UH60L 92-26424 - SUBQ** GUIDE PLATE IS STEEL, MIL-S-5000, 4340 BAR ANL. ◀
26. WEAR PLATES INSTALLED WITH ADHESIVE, ITEM 11, WP 1803 00.
27. **UH60L 92-26424 - SUBQ** PLATE OF STEEL, MIL-S-8844. ◀
28. ALUMINUM PART MAY BE REPLACED BY STEEL PLATE FOR REPAIR. WEAR OF LOCATOR PIN CLEARANCE HOLE IN STRUCTURE, UP TO 0.38 INCH IN RADIUS FROM ORIGINAL HOLE CENTER, IS ALLOWED.
29. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

REPAIR - Continued

2. Hinge and support fittings for the engine cowling/work platform are made from 6AL-4V titanium alloy and forged aluminum alloy, 7075-T73, QQ-A-367. Negligible damage limits to the fittings are limited to nicks, scratches, corrosion, and wear, no deeper than 10% of material thickness at point of damage. Blend damage smoothly to surrounding area using long, even strokes. Polish out all surface imperfections and restore protective finish.
3. Main rotor pylon framing members are considered secondary structure.

NOTE

All structure and hardware passing through cabin skin must be sealed to deck using sealing compound, Item 273, WP 1803 00.

4. Evaluate damage to the framing members and firewalls (WP 0377 00).
5. Refer to WP 0378 00 and WP 0380 00 for typical framing repairs. When repairing titanium firewalls use same gage titanium stock or the equivalent gage 301/302 stainless steel, MIL-S-5059, for patching. Use Monel or stainless steel rivets, or spot or fusion welding, for attaching titanium and stainless steel patches. Refer to TM 1-1500-204-23 for information on cutting, drilling, forming, and fastening titanium and stainless steel sheet stock.
6. Repair sliding cover track, as follows:
 - a. Locally make repair plate and repair angle (WP 1805 00).
 - b. Inspect track web for wear from horizontal roller contact. If damage is deeper than 0.020-inch, prepare for installation of repair plate, as follows:
 - (1) Remove nicks and gouges by blending with abrasive cloth, Item 80, WP 1803 00.
 - (2) Measure remaining thickness of web in damage area. **REPLACE TRACK IF WEB THICKNESS IS LESS THAN 0.060-INCH.**
 - c. Inspect track flange for wear from vertical roller contact. If damage is deeper than 0.020-inch, prepare for installation of repair angle, as follows:
 - (1) Remove track flange material in wear area, using file, so that repair angle seats flush against flange.
 - (2) Measure remaining thickness of flange in repair area. **REPLACE TRACK IF WEB THICKNESS IS LESS THAN 0.060-INCH.**
 - d. Install repair plate and repair angle, as follows:
 - (1) Chamfer track upper corner the minimum amount needed to let repair angle fit snugly.
 - (2) Place repair plate against track web, with wear area between rear fastener holes (Figure 1, Sheet 10, Detail Q). Hold in place with edge-grip clecos.

CAUTION

To avoid weakening track, new fastener holes must be at least 0.375-inch (3 times rivet diameter) from any existing holes in track web.

- (3) Place repair angle against track, aligned with repair plate. Hold in place with edge-grip clecos.
- (4) Drill 1/8-inch holes, centered on repair plate holes, through track web and repair angle.
- (5) Remove clecos and repair plate and angle.

REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- (6) Clean all bonding surfaces using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
- (7) Apply thin coat of adhesive primer, Item 23, WP 1803 00, to all bonding surfaces. Let dry for 1 hour.
- (8) Apply adhesive, Item 11, WP 1803 00, as liquid shim to web repaired area to bring it level with web surface.
- (9) Apply even coat of adhesive, Item 11, WP 1803 00, to all bonding surfaces.
- (10) Immediately assemble repair plate and angle to track and install 3 rivets, NAS1097AD-4.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- (11) Clean up excess adhesive, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
7. **UH-60L 92-26424 - SUBQ** Remove and replace wear strips as follows (Figure 1, Sheet 10, Detail Q): ■
- a. Drill out rivets, where required.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- b. Remove wear strip and old adhesive, using nonmetallic scraper and cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, or equivalent.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- c. Thoroughly clean bonding surfaces of track and wear strip with cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cheesecloth.
- d. Apply light coat of adhesive primer, Item 23, WP 1803 00, and let cure for 1 hour at room temperature.

REPAIR - Continued

- e. Coat bonding surfaces of track and wear strip with adhesive, Item 11, WP 1803 00. Install in position immediately.
 - f. Cure at room temperature for 24 hours.
 - g. Install rivets where required.
8. Replace sliding cover track by doing this:
- a. Remove sliding cover (WP 0296 00).
 - b. Drill out rivets holding track, filler, and stop to hat (TM 1-1500-204-23).
 - c. Rivet replacement track, filler, and stop to hat (TM 1-1500-204-23).
9. Bond rubber bumpers to firewalls with adhesive, Item 10, WP 1803 00, or adhesive Item 47, WP 1803 00.
10. Seal for corrosion protection and watertightness (WP 0377 00 and TM 1-1500-344-23).
11. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

MAIN ROTOR PYLON FAIRING REPAIR

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-C-7438

MIL-R-6855

MIL-S-18729

MIL-S-22499

MIL-S-6721

MIL-S-7047

MIL-S-9047

MIL-STD-670

MIL-T-9046

QQ-A-250

QQ-A-367

TM 1-1500-204-23

TM 1-1500-344-23

TM 55-1500-345-23

WP 0377 00

WP 0378 00

WP 0379 00

WP 0381 00

WP 1803 00

WW-T-700

Material/Parts

Adhesive, Item 14, WP 1803 00

Adhesive, Item 21, WP 1803 00

Bolt, AN23-16

Bolt, AN24-38

Bushing, NAS77A4-24-2

Bushing, NAS77A4-36-2

Nut, MS17825-3

Nut, MS17825-4

Nut, MS21044N3

Screw, MS27039C1-08

Screw, MS27039C1-09

Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

AMS 4260

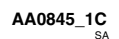
AMS 5643

REPAIR

1. Main rotor pylon fairing and covers are of bonded sandwich construction made from aluminum honeycomb core and Kevlar laminate skins, except for the latch fairings, IR jammer fairing, and rear fairing, which are Kevlar laminates. Screens and the welded bracket assembly for the IR jammer fairing are aluminum alloy, 6061-T6, QQ-A-250-11. Skin cover and screens for the oil coolers are aluminum alloy, 2024-T3, QQ-A-250-5. Doublers for the hinges and latches on doors are titanium, MIL-T-9046, Type III, comp C, or graphite laminates (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, and Figure 1, Sheet 5).
2. Repair of graphite is beyond the scope of this manual.
3. Evaluate damage and make repairs to titanium parts (WP 0378 00).
4. Evaluate damage and make repairs to Kevlar laminates and honeycomb panels (WP 0379 00 and TM 1-1500-204-23).
5. Evaluate damage to aluminum alloy parts (WP 0377 00) and make repairs using typical repair examples (WP 0378 00).
6. Evaluate damage to engine cowling/work platform hinge and support fitting using limits (WP 0377 00 and WP 0378 00).

Diagram illustrating the components and assembly of the fairing and hinged cover for the nose of a vehicle. The components are labeled as follows:

- CHAFING STRIP (TYP) RUBBER CLII, GR60 MIL-R-6855 (0.015)
- FAIRING AND HINGED COVER REINFORCED PLASTIC (0.040-0.060)
- BRACKET WELDED ASSEMBLY 6061-T6 QQ-A-250-11 (0.071)
- COVER, 7075-T6 QQ-A-250-13 (0.040)
- FAIRING REINFORCED PLASTIC (0.050)
- STRAP, 7075-T6 QQ-A-250-13 (0.040)
- COVER (SEE NOTE 1)
- SCREEN 6061-T6 QQ-A-250-11 (0.063)
- SKIN AND SCREEN 2024-T3 QQ-A-250-5 (0.063)
- ZEE, LEFT AND RIGHT 7075-T6 QQ-A-250-13 (0.040)
- PANEL (SEE NOTE 1)
- FAIRING, LEFT AND RIGHT (SEE NOTE 1)
- LATCH FAIRING REINFORCED PLASTIC (0.030) (SEE NOTE 2) 6061-T6, QQ-A-250-11 (0.032) (SEE NOTE 3)
- SCREEN 6061-T6 QQ-A-250-11 (0.040)
- SLIDING FAIRING (SEE NOTE 1)
- COVER, LEFT AND RIGHT 7075-T6 QQ-A-250-13 (0.032)
- COVER (SEE NOTE 1)



0404 00-2

REPAIR - Continued

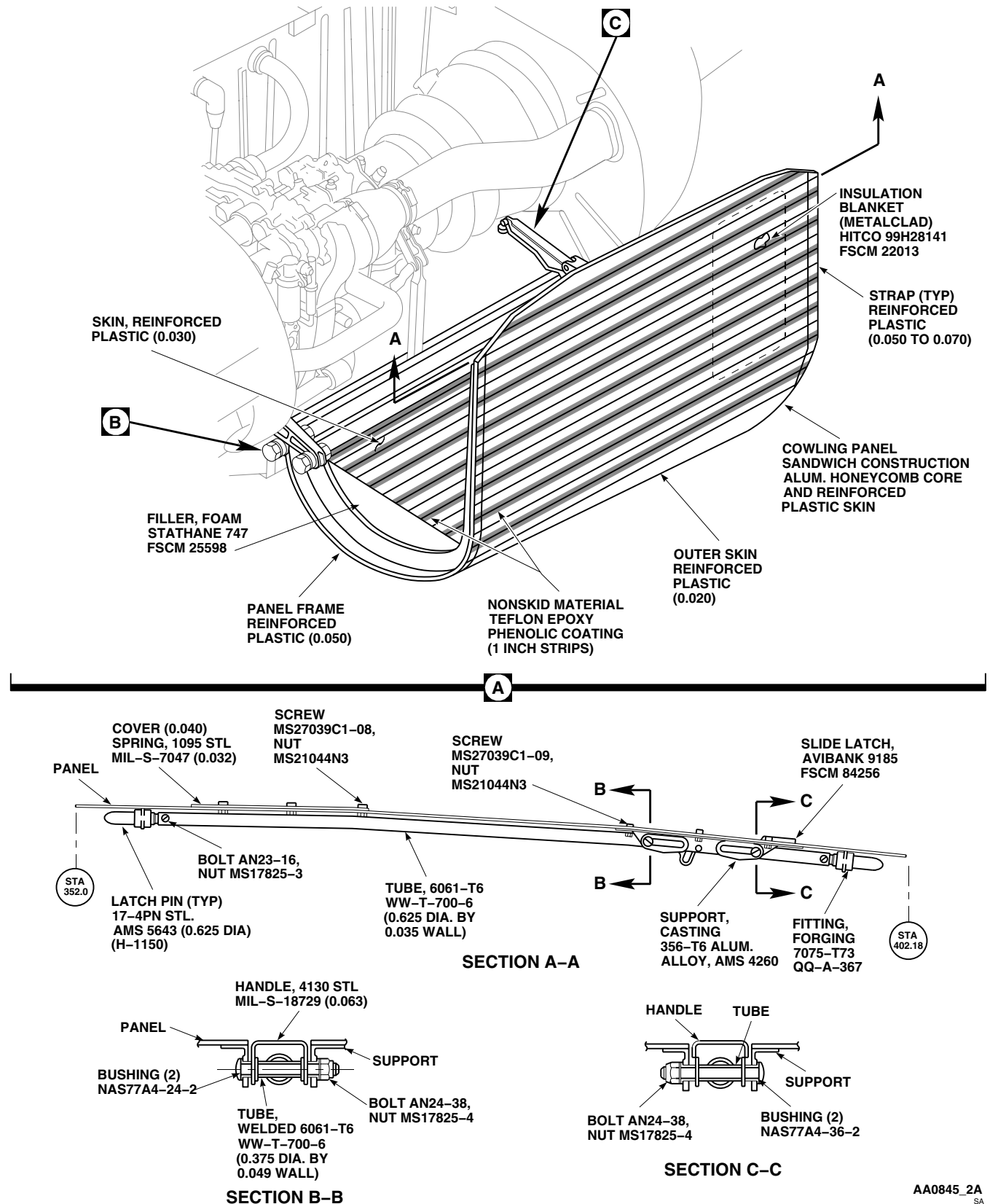


Figure 1. Main Rotor Pylon Fairing Repair. (Sheet 2 of 5)

REPAIR - Continued

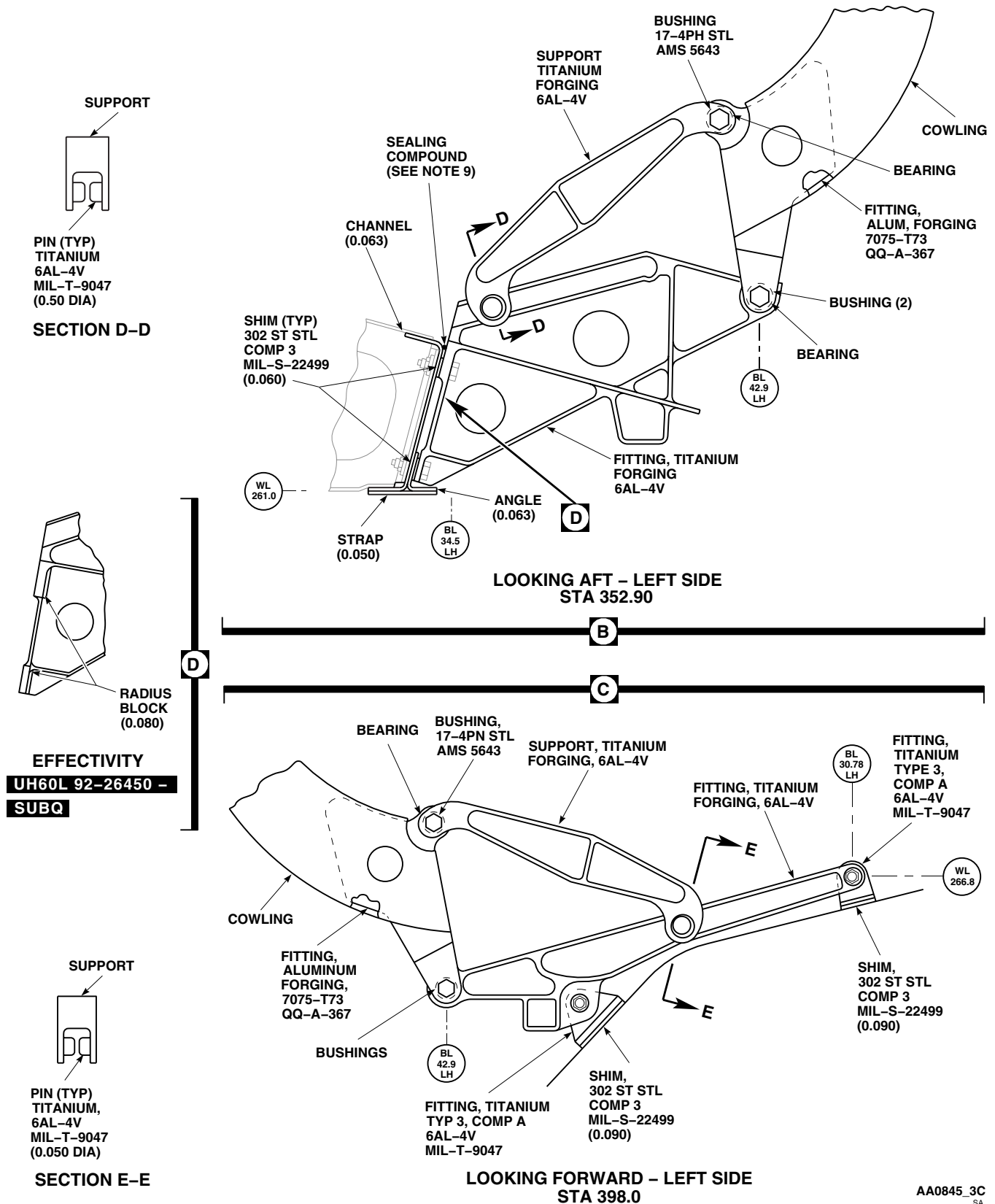


Figure 1. Main Rotor Pylon Fairing Repair. (Sheet 3 of 5)

REPAIR - Continued

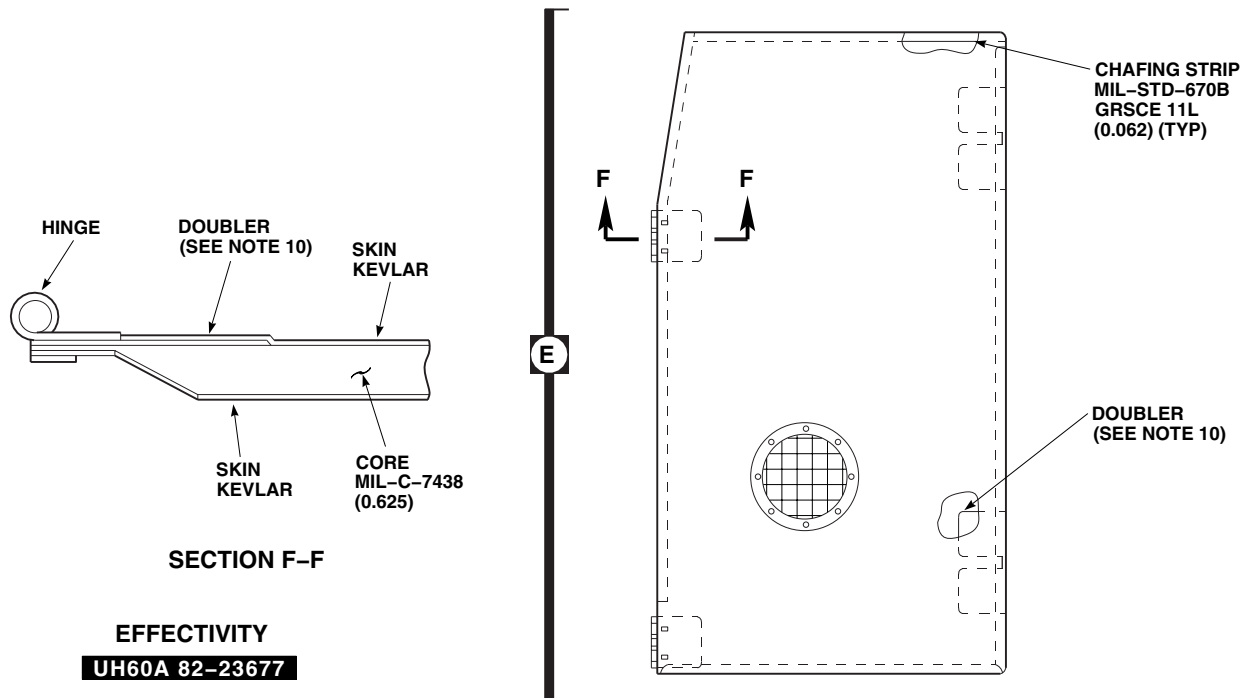
AA0845_4B
SA

Figure 1. Main Rotor Pylon Fairing Repair. (Sheet 4 of 5)

7. Rebond rubber chafing strips on hinged panels over oil coolers, using adhesive, Item 14, WP 1803 00, or adhesive, Item 21, WP 1803 00.
8. Seal for corrosion protection and watertightness (WP 0377 00 and TM 1-1500-344-23).
9. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued**NOTES:**

1. SANDWICH CONSTRUCTION ALUMINUM HONEYCOMB (MIL-C-7438) AND KEVLAR SKINS.
2. UH60A 77-22714 - 82-23671
3. UH60A 82-23671 - SUBQ
4. UH60A 77-22714 - 81-23546
5. UH60A 81-23547 - SUBQ
6. UH60A 86-24525 - SUBQ
7. EITHER TITANIUM, MIL-T-9046 TYPE 111 COMP C, OR GRAPHITE 5 PLIES COVERED BY 2 PLIES KEVLAR.
8. UH60A 82-23677
9. APPLY BEAD OF SEALING COMPOUND, ITEM 245, APPENDIX D.
10. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0845_5C
SA

Figure 1. Main Rotor Pylon Fairing Repair. (Sheet 5 of 5)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TRANSITION SECTION FRAMING REPAIR

This WP supersedes WP 0405 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
 Drill Set, GGG-D-751
 Gloves
 Goggles/Face Shield
 Nonmetallic Scraper

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Brush Adhesive Promoter Item 387, WP 1803 00
 Cleaning Compound, Solvent, Item 74, WP 1803 00
 Corrosion Preventive Compound, Item 103,
 WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 332,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

References

AMS 5355
 MIL-C-7438
 MIL-R-6855
 MIL-S-18729
 QQ-A-200
 QQ-A-250
 QQ-A-367
 TM 1-1500-204-23
 TM 55-1500-345-23
 WP 0378 00
 WP 0379 00
 WP 0380 00
 WP 0381 00
 WP 0963 00
 WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

TRANSITION SECTION FRAMING REPAIR

1. Transition section framing includes formed and extruded members made from 7075-T6, QQ-A-250-13 Alclad sheet stock and 7075-T6511, QQ-A-200-11 aluminum alloy extrusion. Frame fitting at STA 379.0 and attachment fittings at STA 485.0 are made from forged aluminum alloy, 7075-T73, QQ-A-367.
2. All transition section framing, except the reinforced plastic/honeycomb core panel, is primary structure. The forged fittings, however, are considered critical primary structure. Negligible damage to the fittings is limited to nicks, scratches, and corrosion no deeper than 5% of material thickness at point of damage. Blend damage smoothly into surrounding area using long, even strokes. Polish out all surface imperfections and restore protective finish. Damage to the fittings greater than negligible limits must be reviewed by engineering authority for damage analysis and, either repair or replacement. All repairs to transition section framing must maintain the design strength and alignment (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, Figure 1, Sheet 7, Figure 1, Sheet 8, Figure 1, Sheet 9, Figure 1, Sheet 10, Figure 1, Sheet 11, Figure 1, Sheet 12, Figure 1, Sheet 13, Figure 1, Sheet 14, Figure 1, Sheet 15, Figure 1, Sheet 16, Figure 1, Sheet 17, Figure 1, Sheet 18, Figure 1, Sheet 19, Figure 1, Sheet 20, Figure 1, Sheet 21, Figure 1, Sheet 22, Figure 1, Sheet 23, Figure 1, Sheet 24, Figure 1, Sheet 25, Figure 1, Sheet 26, Figure 1, Sheet 27, Figure 1, Sheet 28, Figure 1, Sheet 29, and Figure 1, Sheet 30).
 - a. Use WP 0378 00 and WP 0380 00 for typical framing repairs.
 - b. Use WP 0379 00 and TM 1-1500-204-23 for evaluating damage and repairing reinforced plastic/honeycomb core panels.

TRANSITION SECTION FRAMING REPAIR - Continued

- c. For corrosion protection and watertightness, seal all repairs between frame members and plating (WP 0377 00).
- d. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

FUEL CELL LINER

- 1. Visually inspect fuel cell liner. Any signs of cracking, corrosion or degradation is cause for replacement of fuel cell liner.
- 2. Replace fuel cell liner as follows (Figure 1, Sheet 29, Detail AV):
 - a. Turn off all electrical power.
 - b. Remove main fuel cells (WP 0963 00).
 - c. Remove fuel cell liners from frame, using nonmetallic scraper.
 - d. Remove foam blocks from framing structure using nonmetallic scraper. Inspect foam blocks for corrosion or deterioration. Replace all foam blocks if corrosion or deterioration is found on any block.

WARNING

- Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.
 - Provide proper ventilation and protective clothing, including eye shield, when using solvents. Avoid breathing vapors and skin contact as much as possible. Wash contacted skin with soap and water. If solvent contacts eyes, flush them with clean water and get immediate medical help.
- e. Thoroughly clean tub structure and fuel cell liner bonding surfaces, using machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 74, WP 1803 00.
 - f. Coat cleaned area of tub with corrosion preventive compound, Item 103, WP 1803 00. **DO NOT ALLOW CORROSIVE PREVENTIVE COMPOUND TO CONTACT ANY AREA USED FOR BONDING FOAM BLOCKS AND FUEL CELL LINERS TO FRAME.** Allow sufficient time for corrosion preventive compound to dry completely.
 - g. Insert foam blocks into framing structure. Apply sealing compound, Item 273, WP 1803 00, bonding foam blocks to frame.
 - h. Apply sealing compound, Item 273, WP 1803 00, to bonding areas of frame. Sealing compound must be in direct contact with liner and adjoining structure at all points. **NO GAPS ALLOWED.** Place liner in position against frame.
 - i. Where required to hold liner firmly against frame, drill 3/32 - inch holes and insert No. 40 clecos.
 - j. Prior to installation of pressure sensitive adhesive tape, Item 332, WP 1803 00, brush adhesive promoter, Item 387, WP 1803 00, to all surfaces requiring tape and let dry.
 - k. Tape all inside edges of liner with pressure sensitive adhesive tape, Item 332, WP 1803 00.
 - l. Cure sealing compound for 24 hours at room temperature, or 2 hours at 150°F(55°C) to 180°F(82°C).
 - m. Remove clecos, if required. Fastener holes may be left unfilled.

FUEL CELL LINER - Continued**WARNING**

Acetone is extremely flammable and toxic to eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- n. Remove excess sealing compound using nonmetallic scraper and machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
- o. Acceptable bond joints shall meet these criteria:
 - (1) **VOIDS OF SEALING COMPOUND IN JOINT SHALL NOT EXCEED A TOTAL OF 3 INCHES IN ANY 6 INCH LENGTH.**
 - (2) **VOIDS OF SEALING COMPOUND IN JOINTS SHALL NOT EXCEED, IN LENGTH, 10% OF UPPER EDGE BONDLINE, 25% OF SIDE AND BOTTOM BONDLINE, OR 75% OF CORNER BONDLINE.**
- 3. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

FUEL CELL LINER - Continued

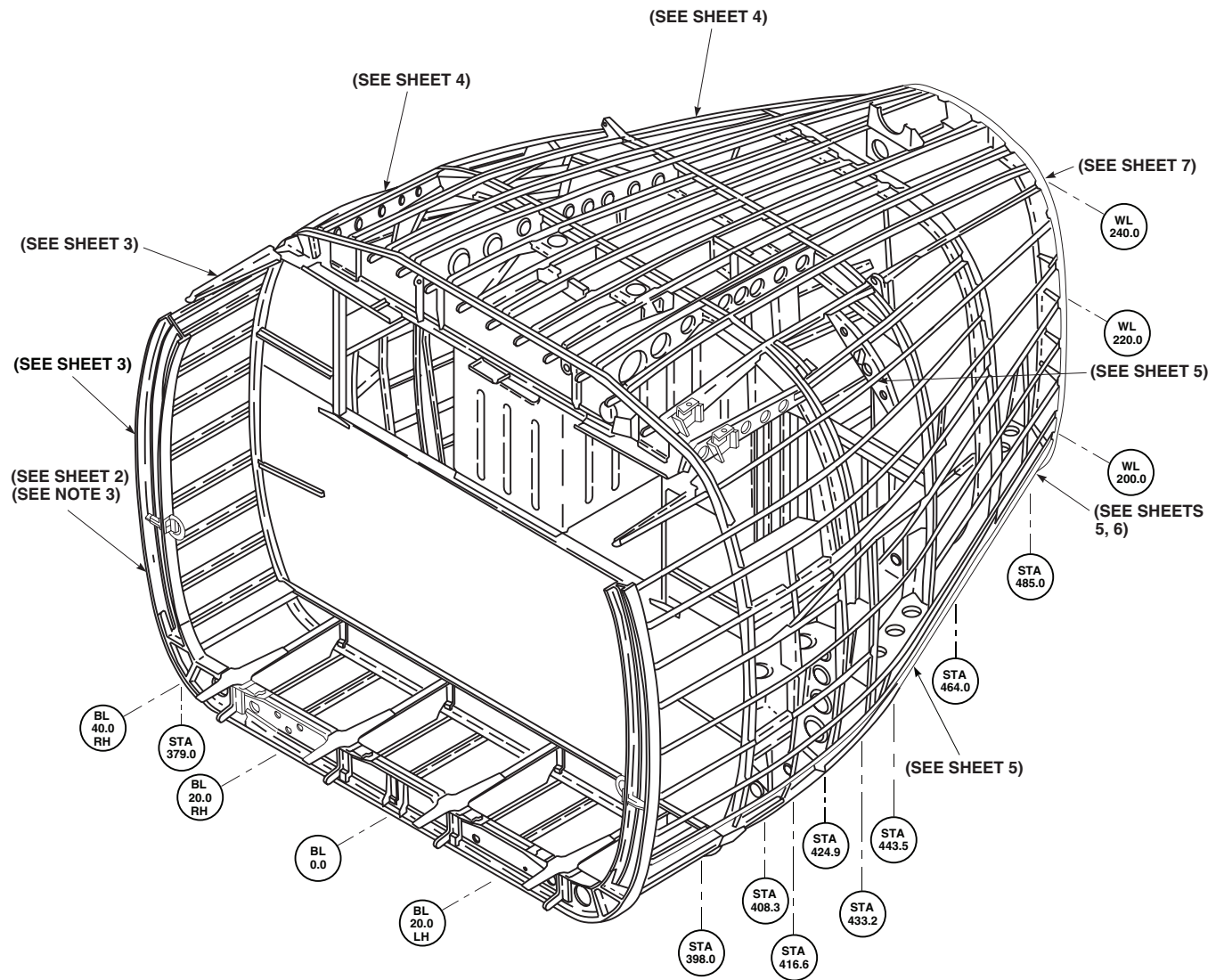
AB2674_1
SA

Figure 1. Transition Section Framing Repair. (Sheet 1 of 30)

FUEL CELL LINER - Continued

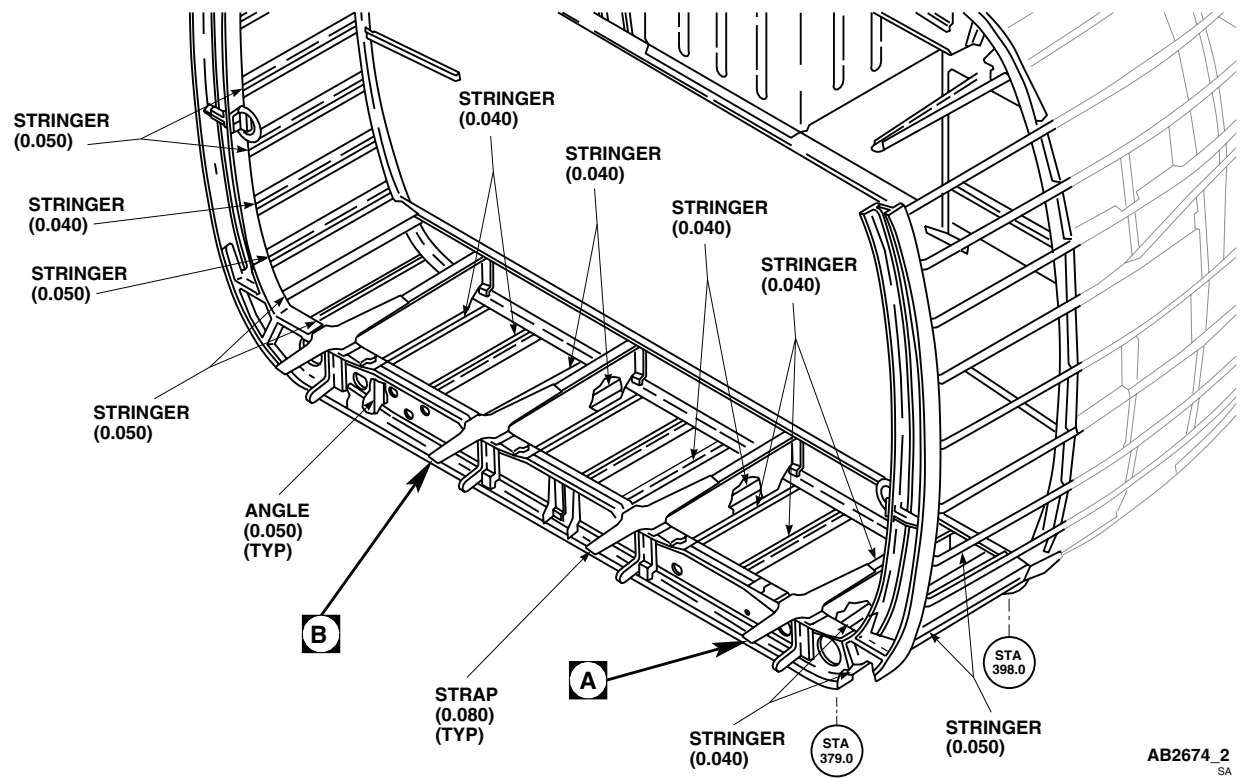


Figure 1. Transition Section Framing Repair. (Sheet 2 of 30)

FUEL CELL LINER - Continued

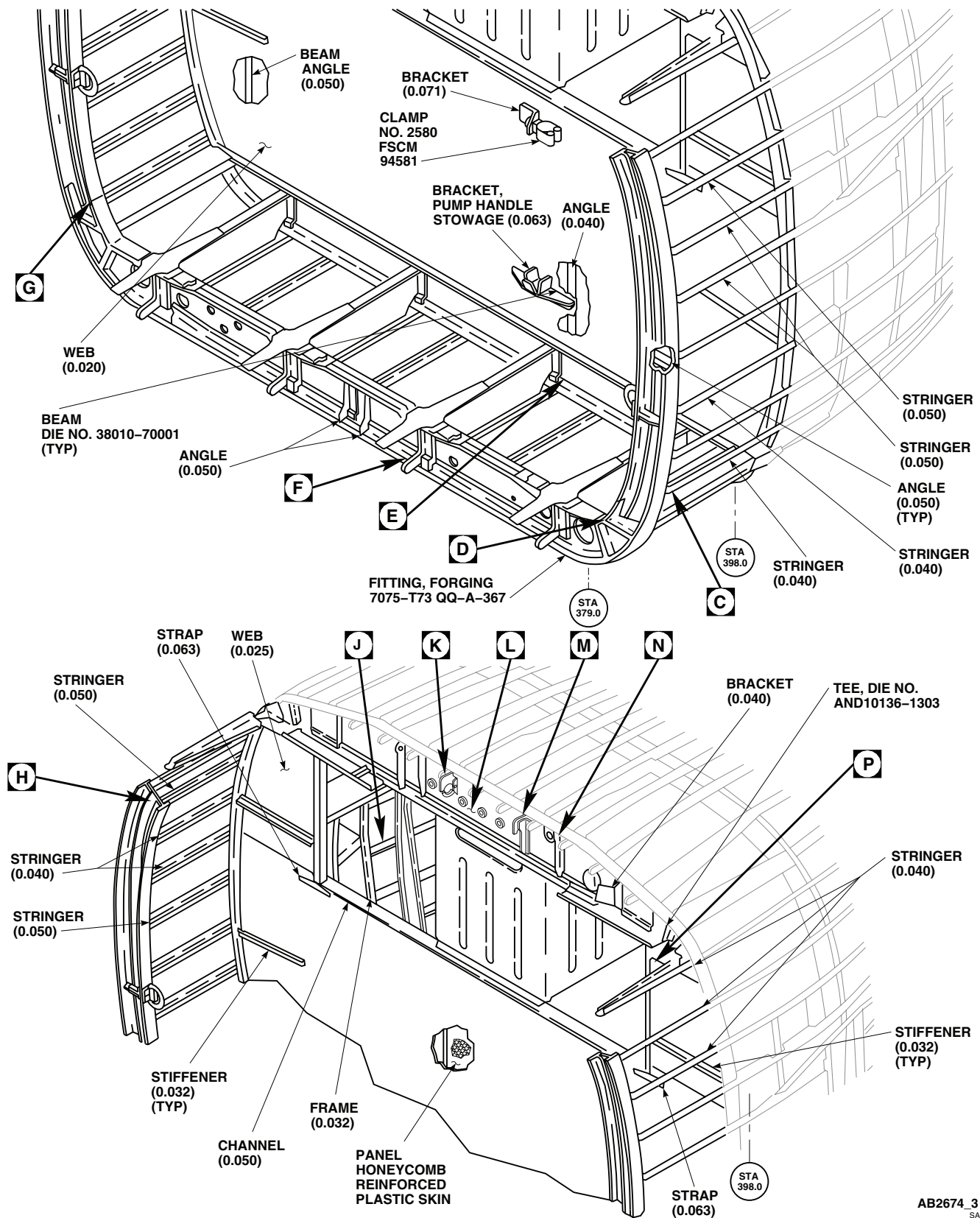
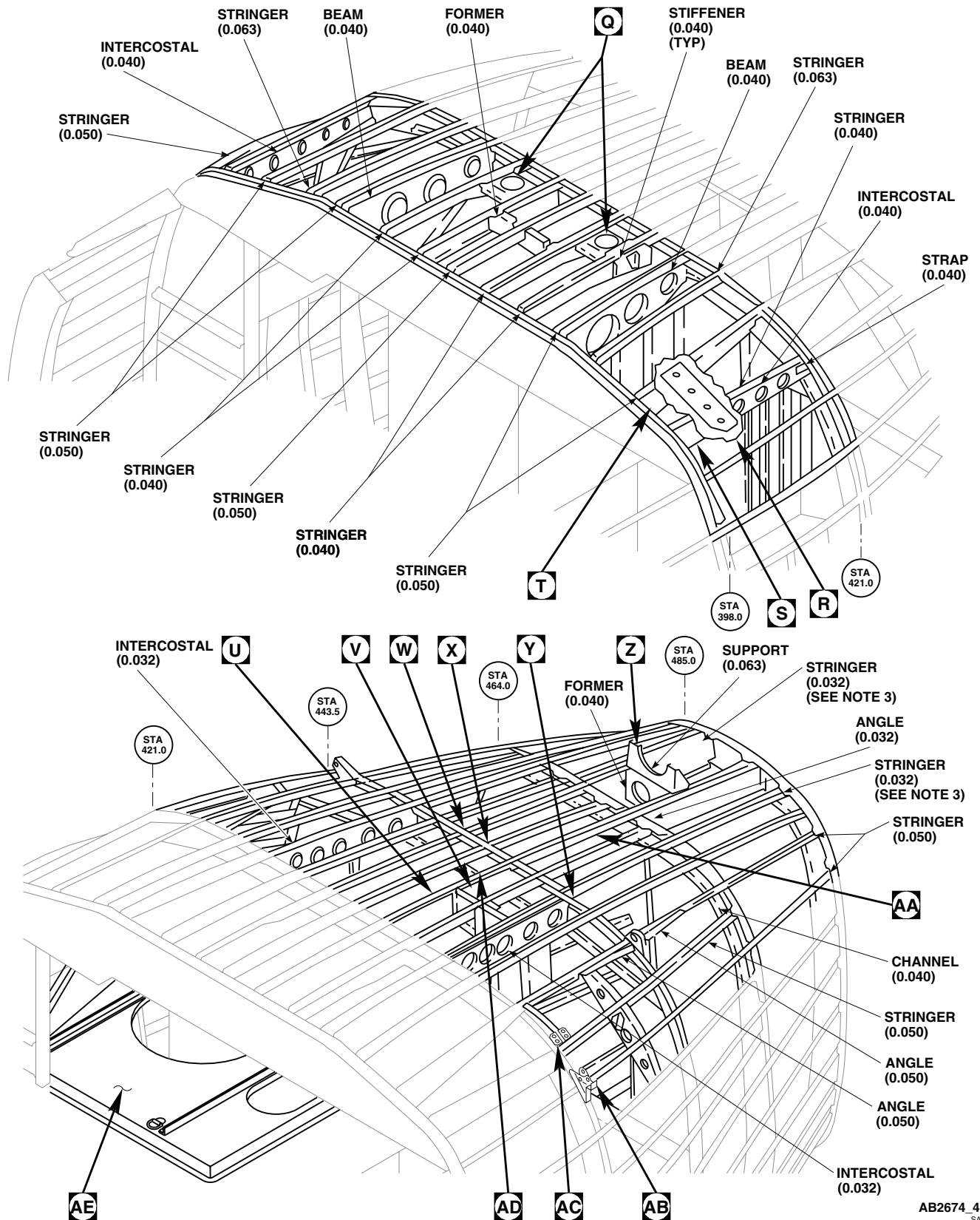


Figure 1. Transition Section Framing Repair. (Sheet 3 of 30)

FUEL CELL LINER - Continued



AB2674_4
SA

Figure 1. Transition Section Framing Repair. (Sheet 4 of 30)

FUEL CELL LINER - Continued

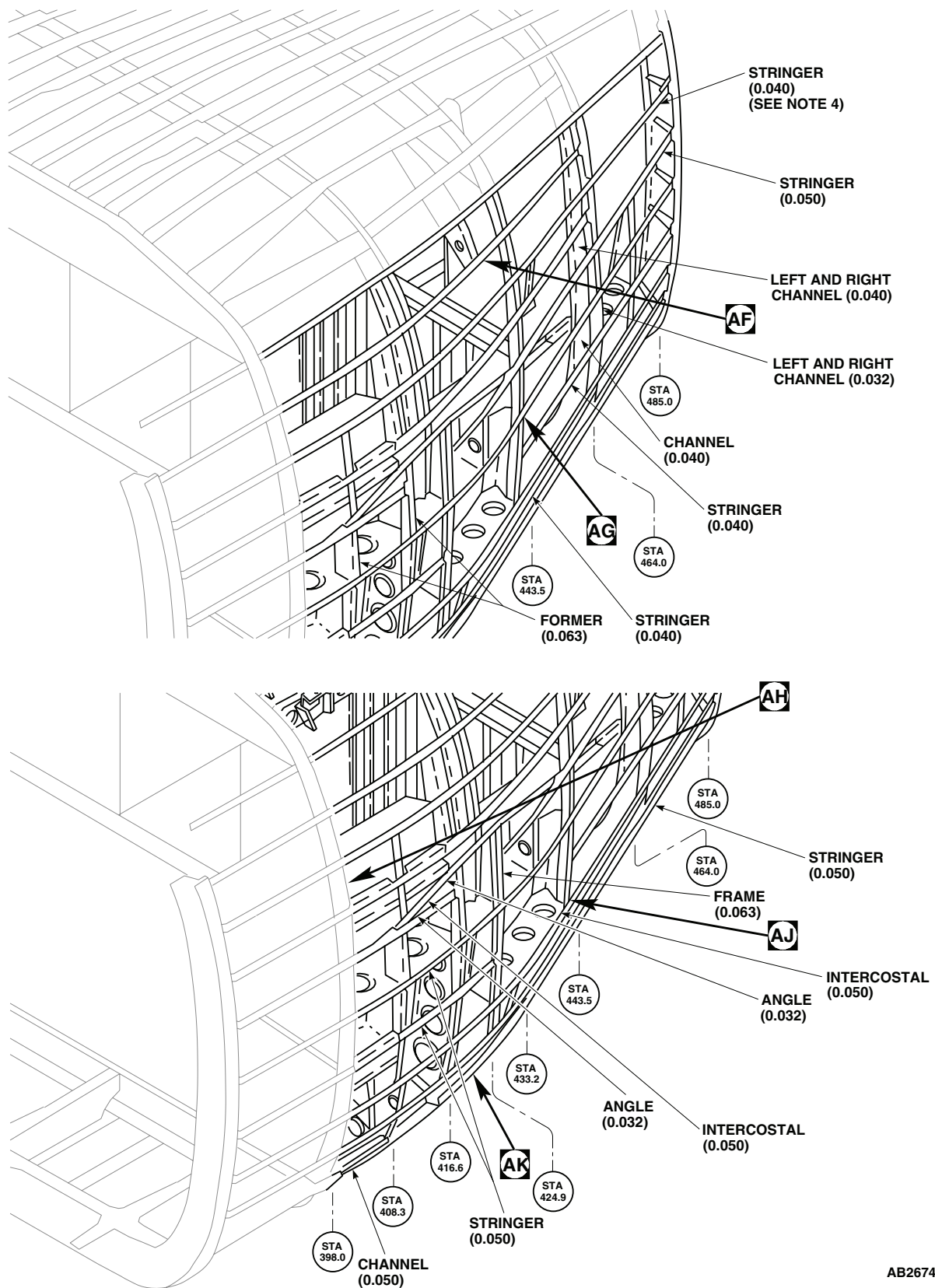
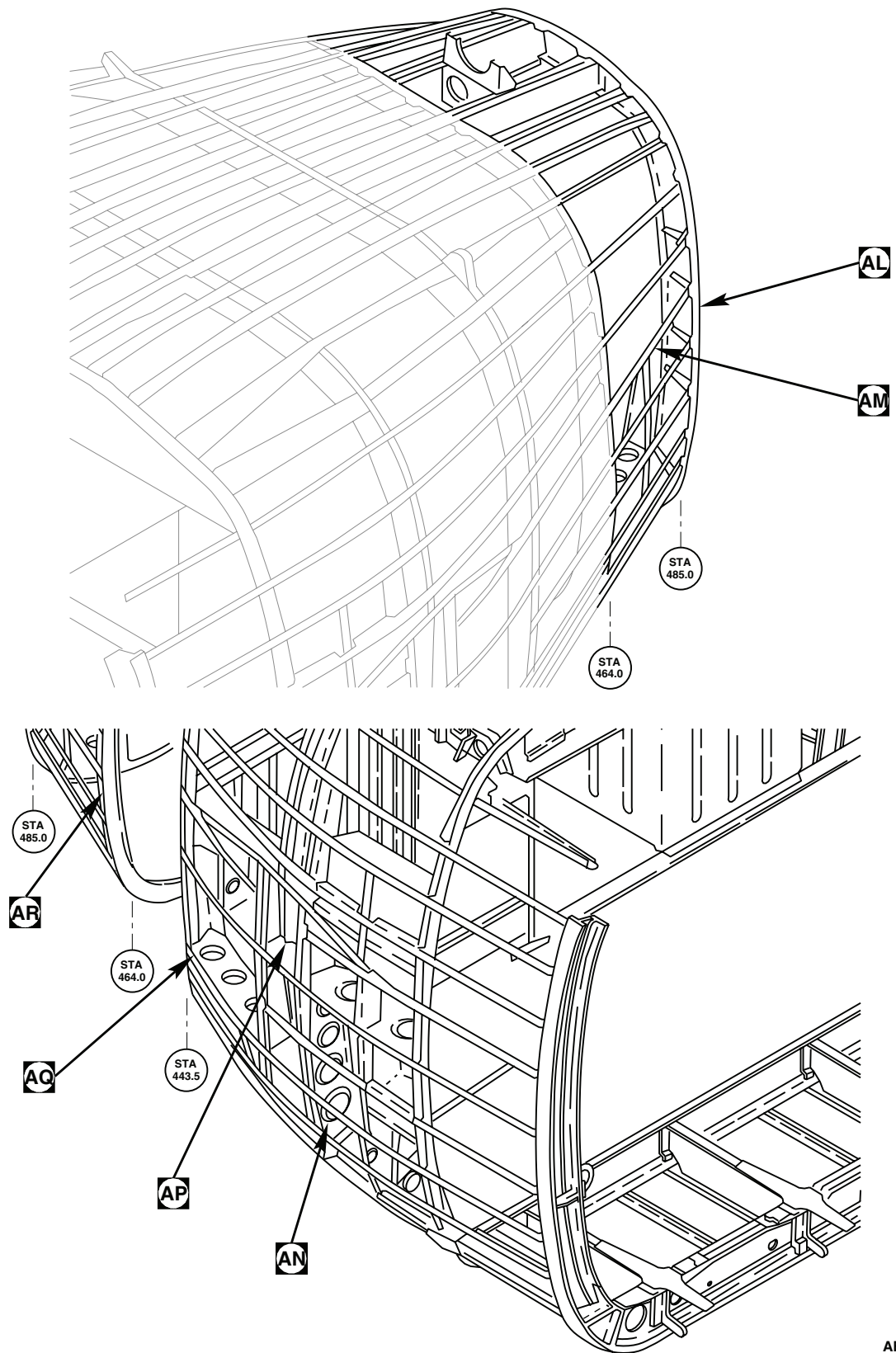
AB2674_5
SA

Figure 1. Transition Section Framing Repair. (Sheet 5 of 30)

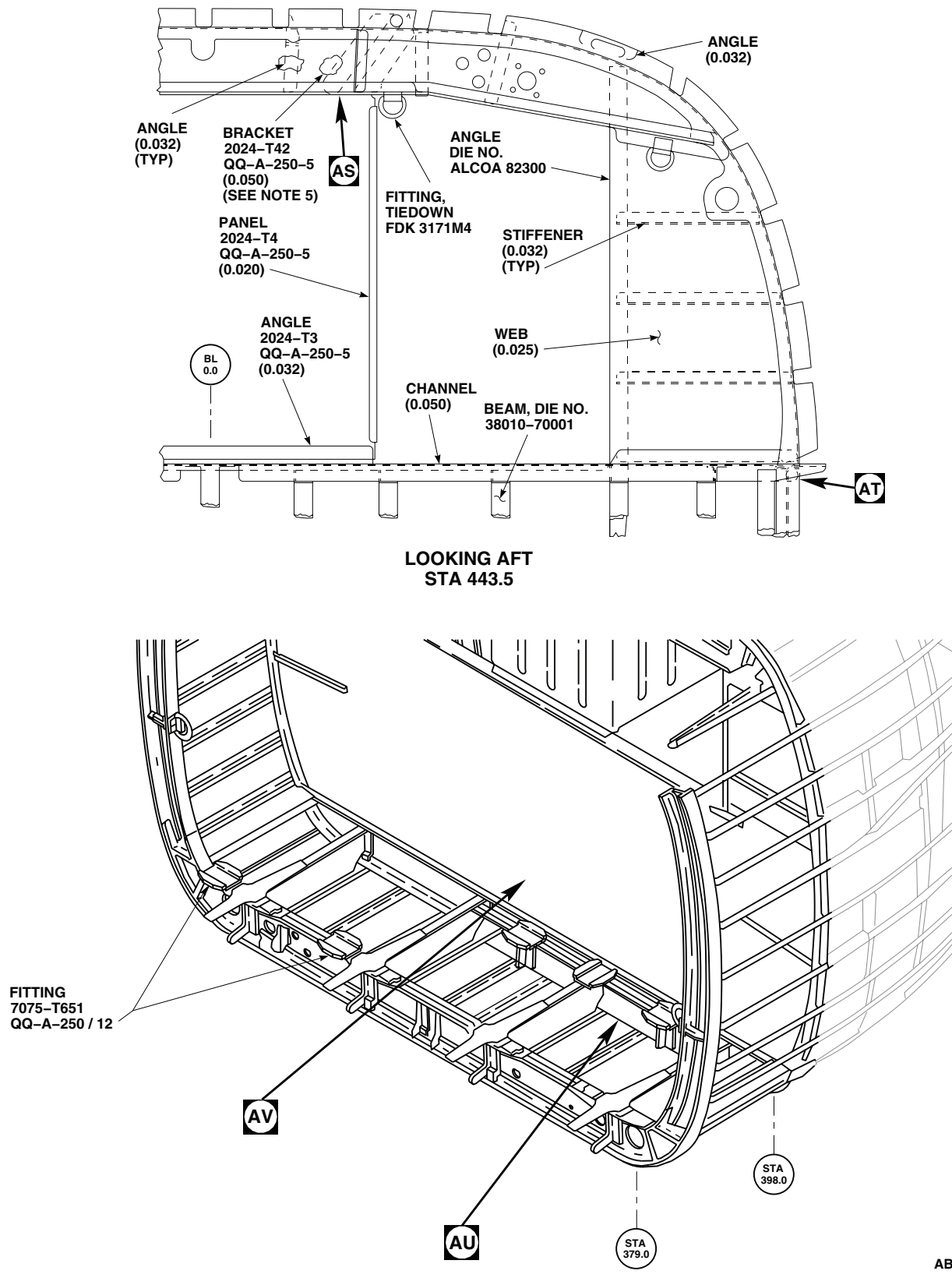
FUEL CELL LINER - Continued



AB2674_6
SA

Figure 1. Transition Section Framing Repair. (Sheet 6 of 30)

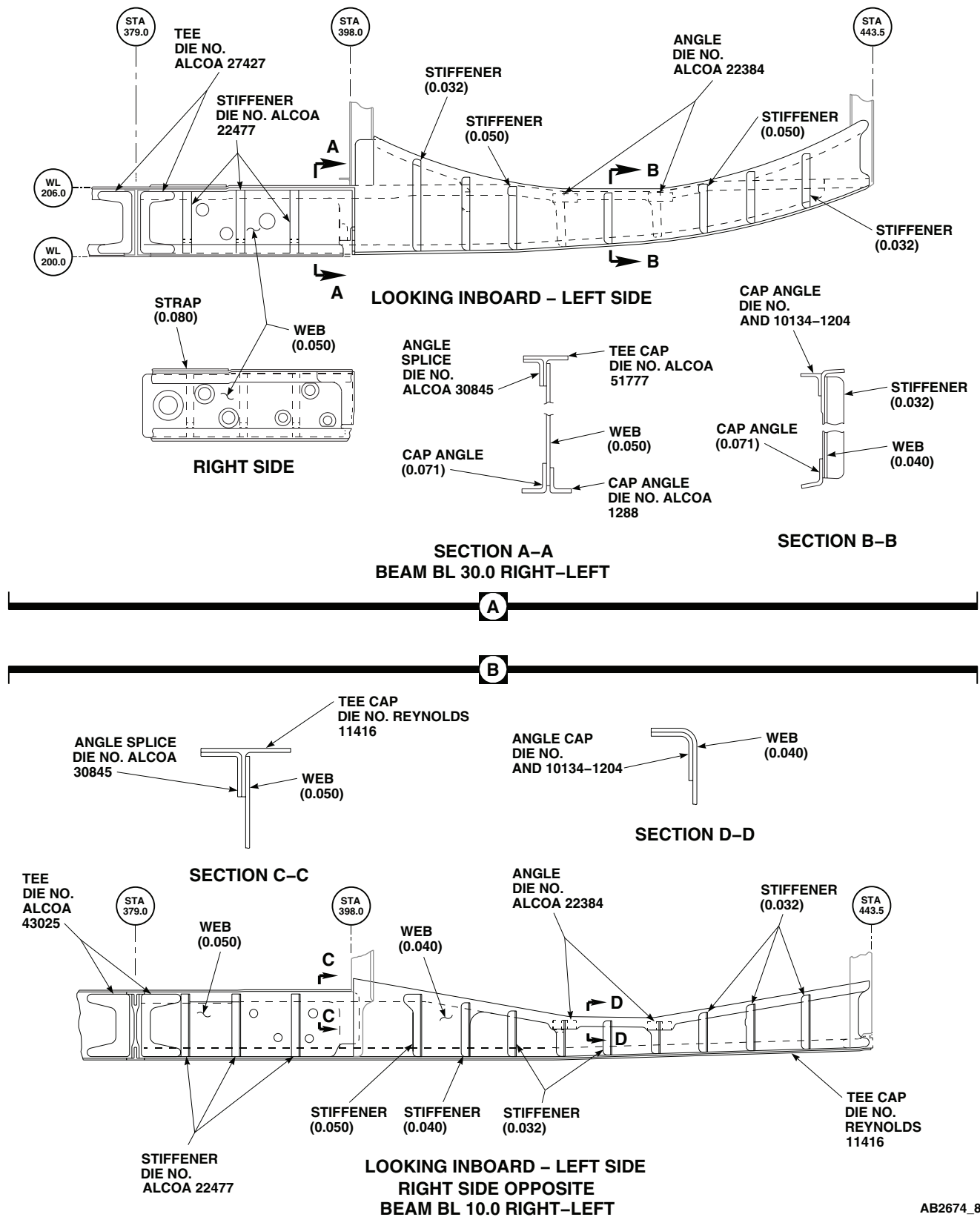
FUEL CELL LINER - Continued



AB2674_7
SA

Figure 1. Transition Section Framing Repair. (Sheet 7 of 30)

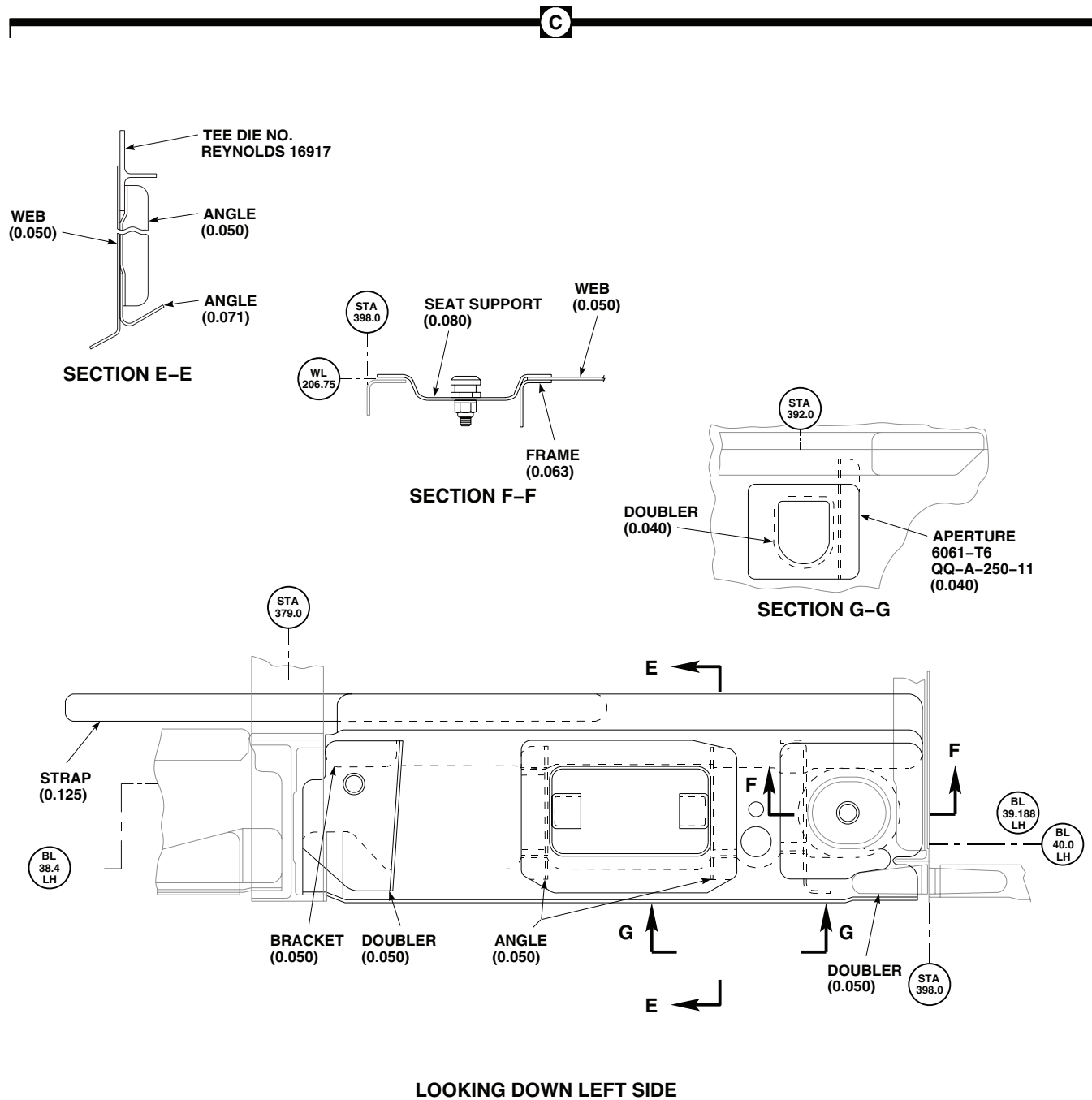
FUEL CELL LINER - Continued



AB2674_8
SA

Figure 1. Transition Section Framing Repair. (Sheet 8 of 30)

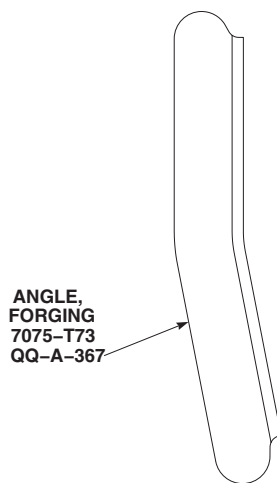
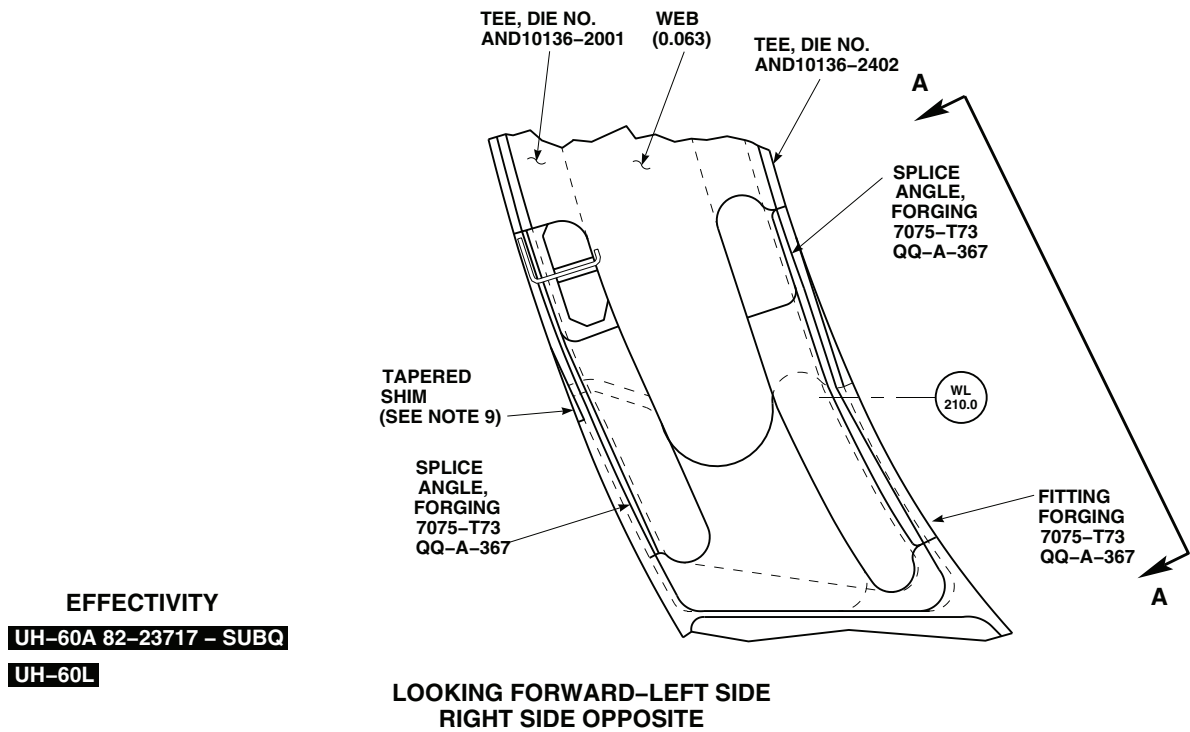
FUEL CELL LINER - Continued



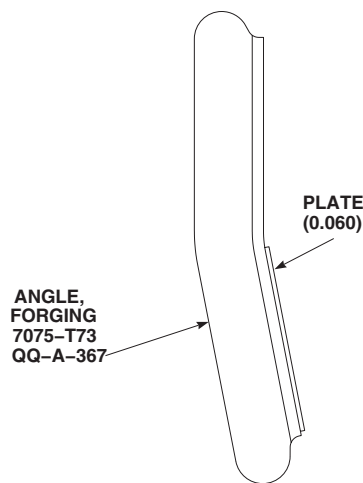
AB2674_9
SA

Figure 1. Transition Section Framing Repair. (Sheet 9 of 30)

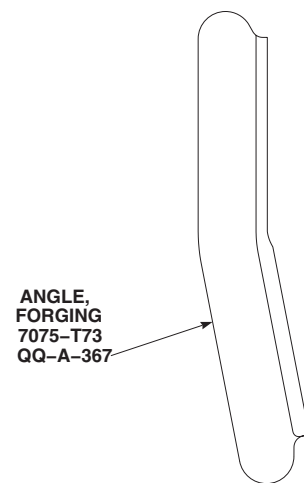
FUEL CELL LINER - Continued



EFFECTIVITY
 UH-60A 77-22714 - 79-23285



EFFECTIVITY
 UH-60A 79-23286 - 82-23716



EFFECTIVITY
 UH-60A 82-23717 - SUBQ
 UH-60L

VIEW A-A

AB2674_10
 SA

Figure 1. Transition Section Framing Repair. (Sheet 10 of 30)

FUEL CELL LINER - Continued

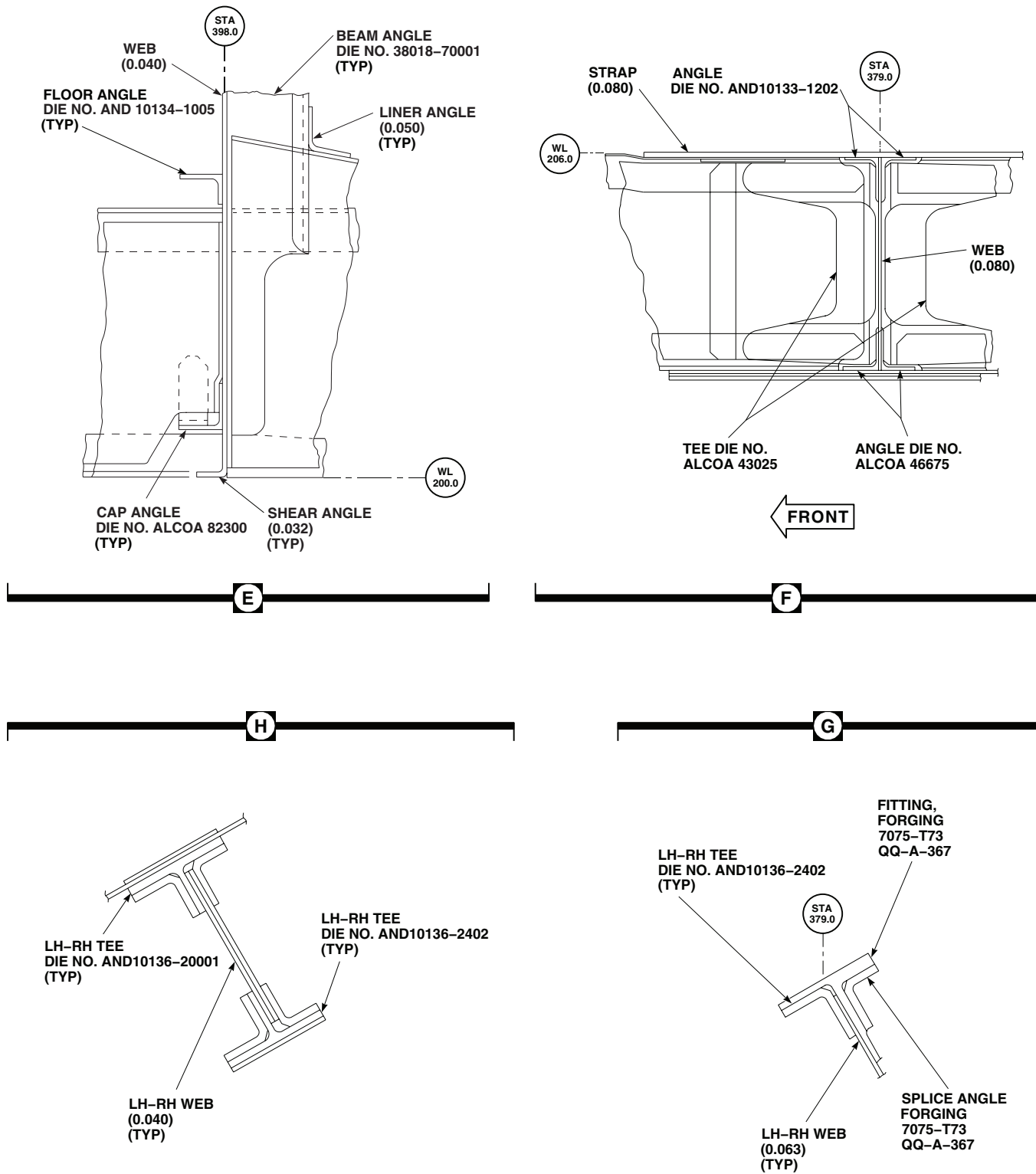
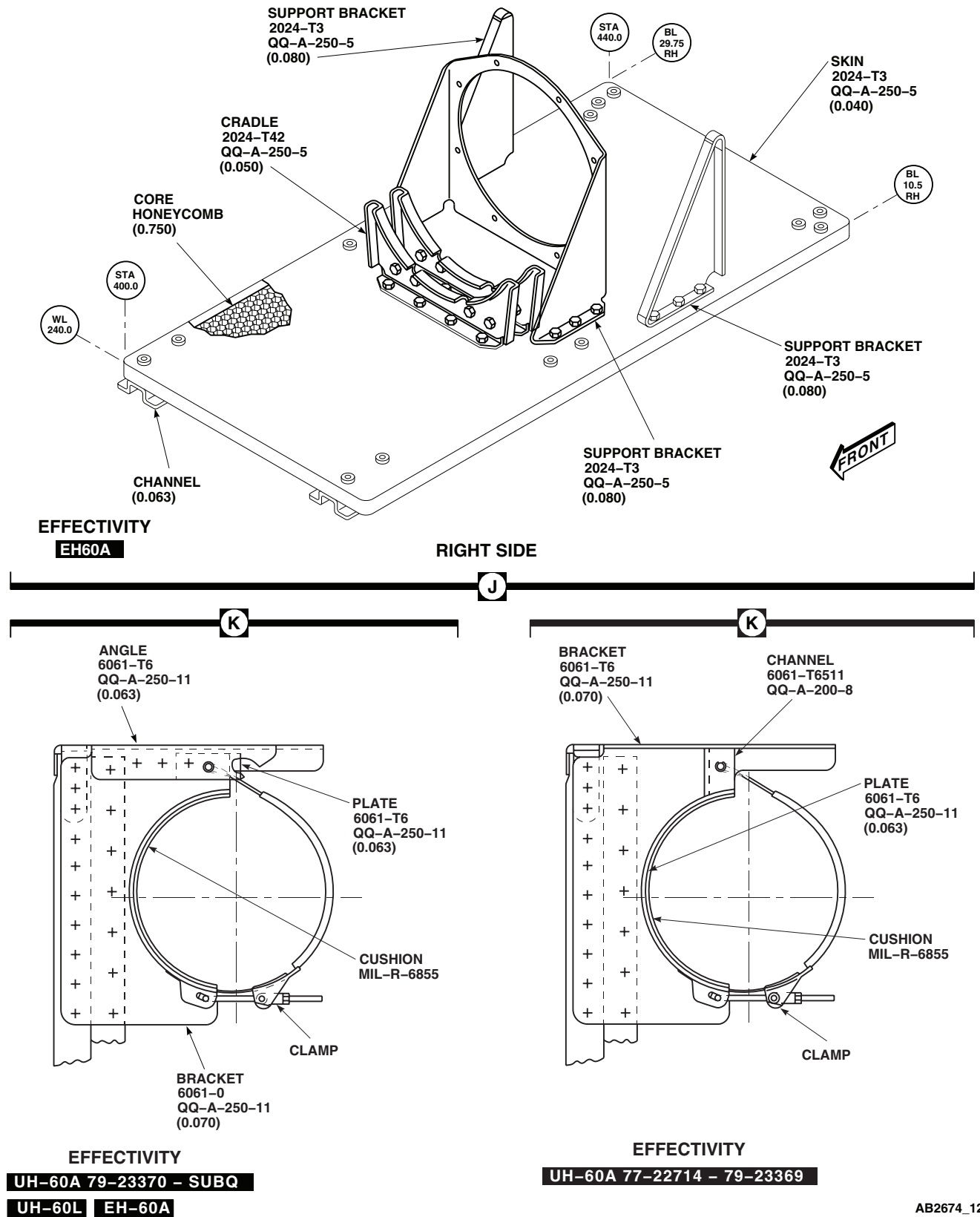
AB2674_11
SA

Figure 1. Transition Section Framing Repair. (Sheet 11 of 30)

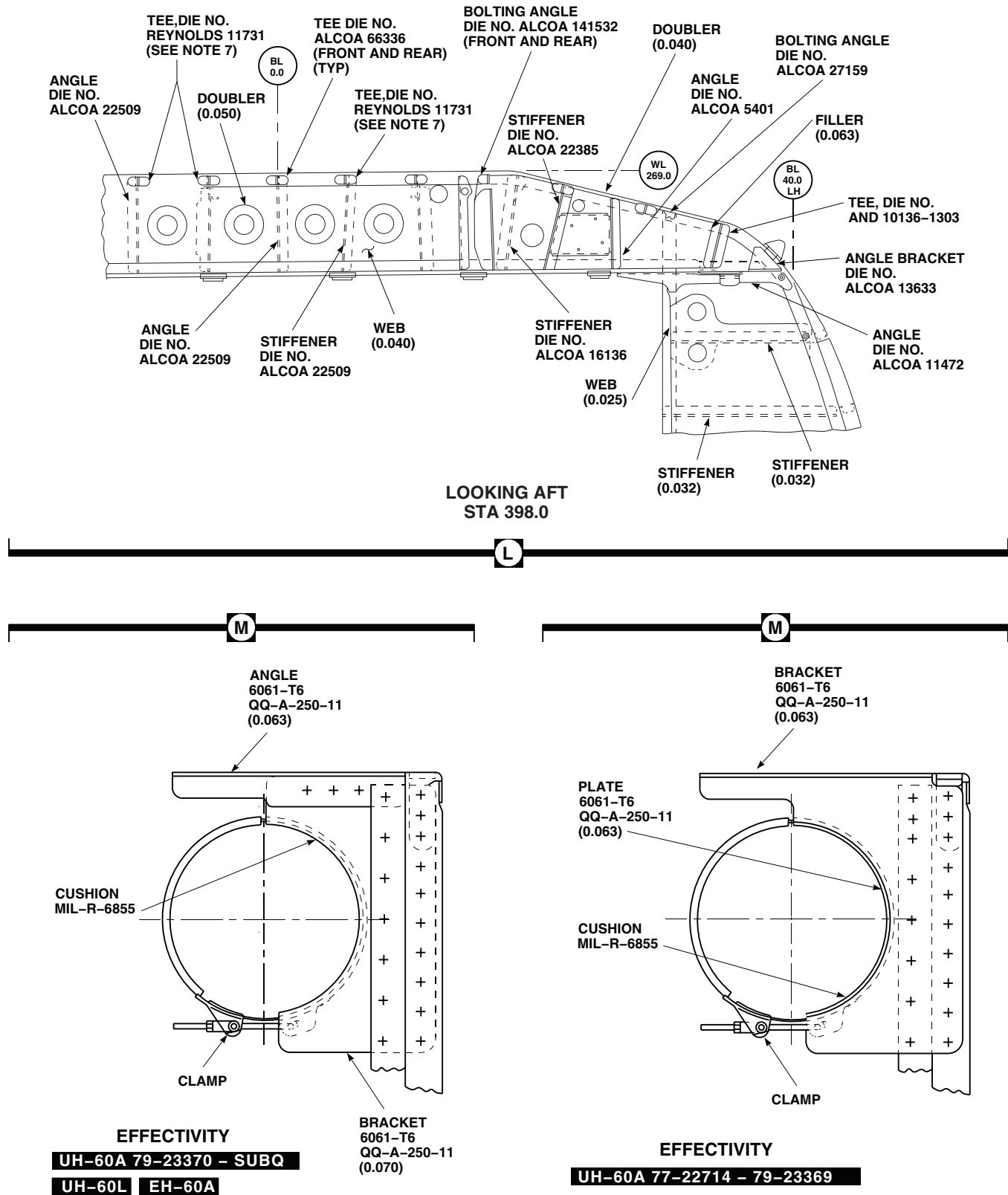
FUEL CELL LINER - Continued



AB2674_12
SA

Figure 1. Transition Section Framing Repair. (Sheet 12 of 30)

FUEL CELL LINER - Continued

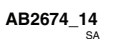


AB2674_13
SA

Figure 1. Transition Section Framing Repair. (Sheet 13 of 30)

Diagram illustrating the assembly components and their dimensions:

- ANGLE DIE NO. ALCOA 71369
- STA 398.0
- STIFFENER (0.040)
- WEB (0.040)
- TEE DIE NO. ALCOA 52414
- BEAM (0.050)
- RADIUS BLOCK (0.125)
- SUPPORT 4130 STEEL, MIL-S-18729 (0.071)
- FRONT



Change 3

FUEL CELL LINER - Continued

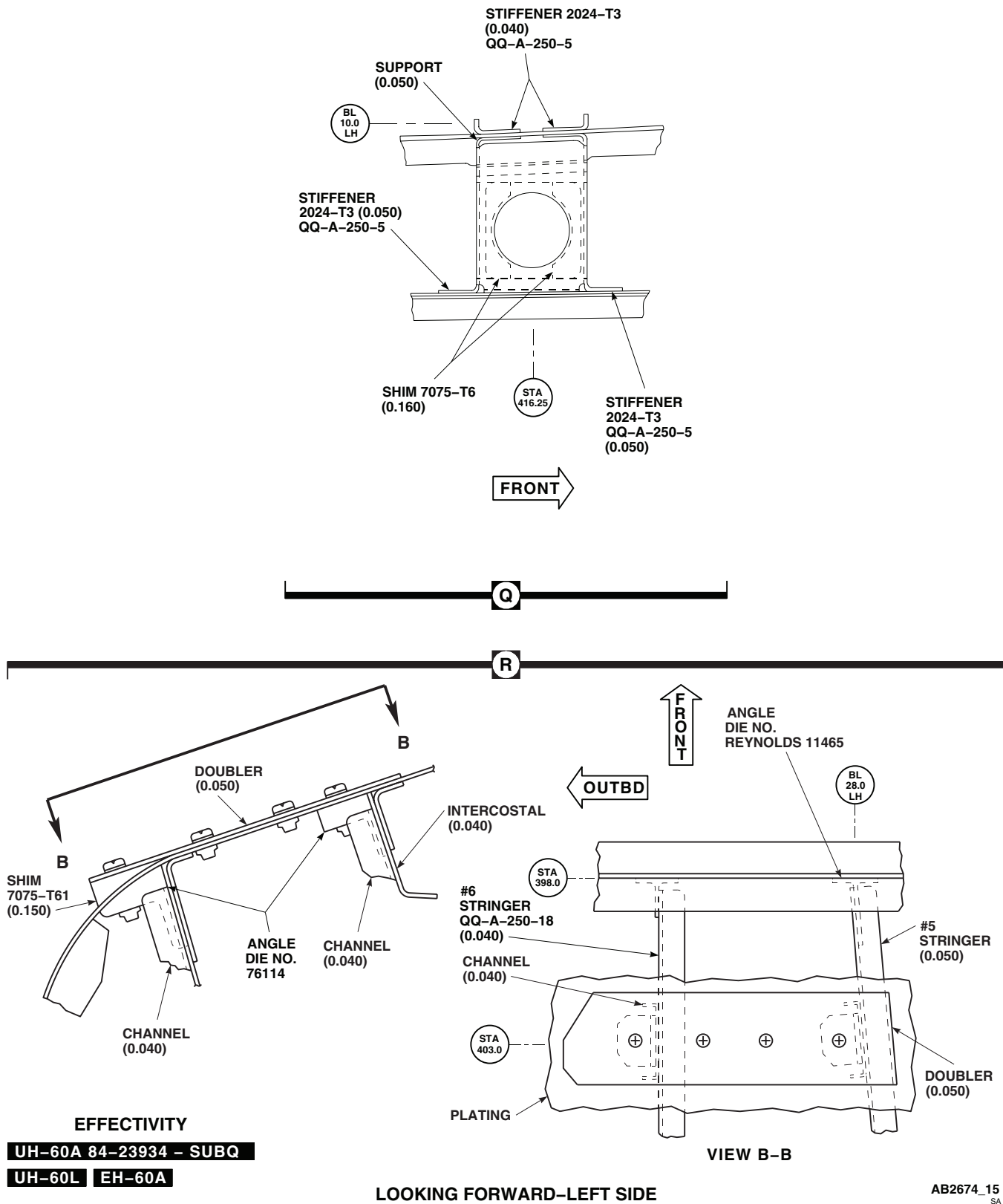


Figure 1. Transition Section Framing Repair. (Sheet 15 of 30)

FUEL CELL LINER - Continued

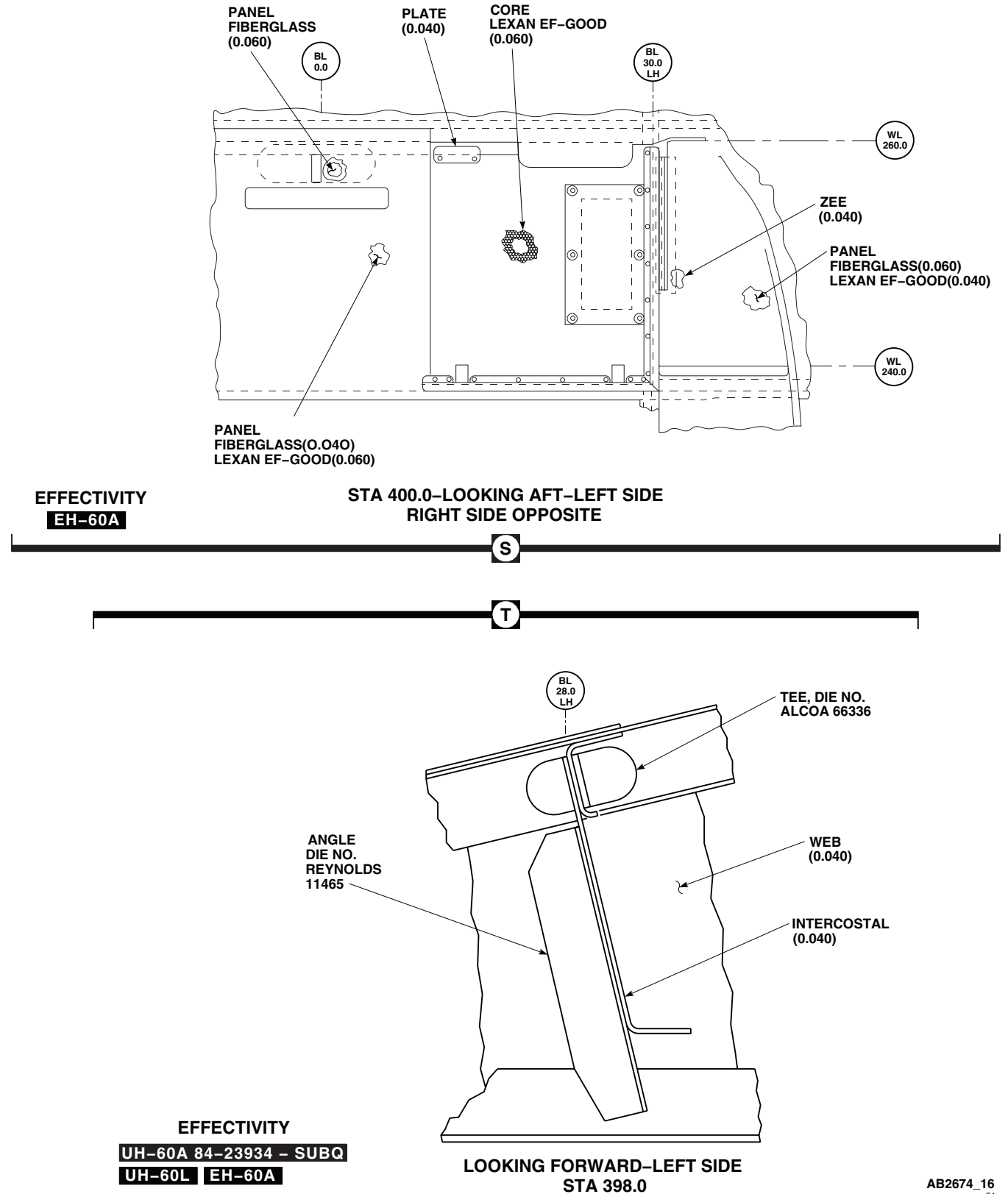


Figure 1. Transition Section Framing Repair. (Sheet 16 of 30)

FUEL CELL LINER - Continued

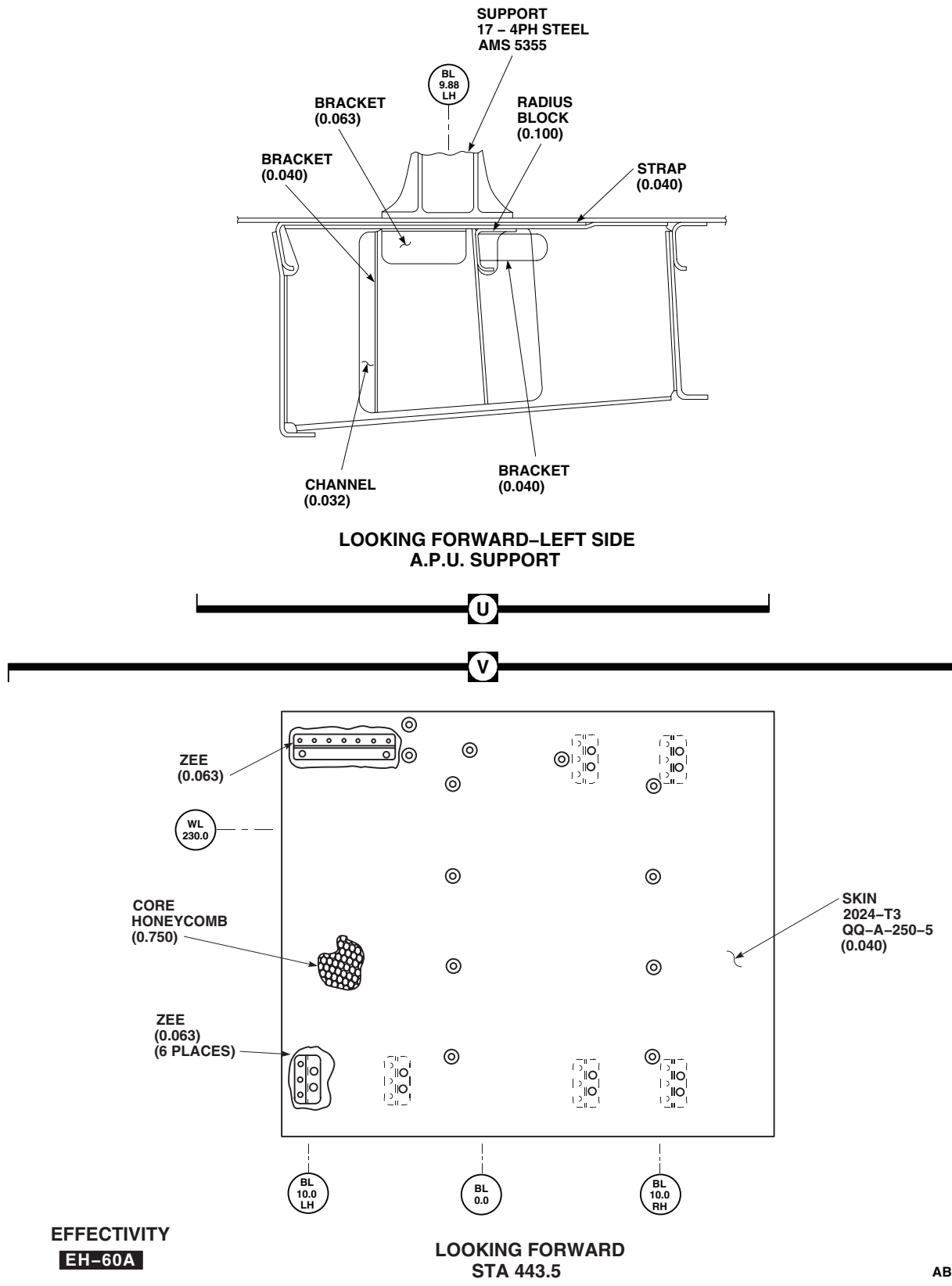


Figure 1. Transition Section Framing Repair. (Sheet 17 of 30)

FUEL CELL LINER - Continued

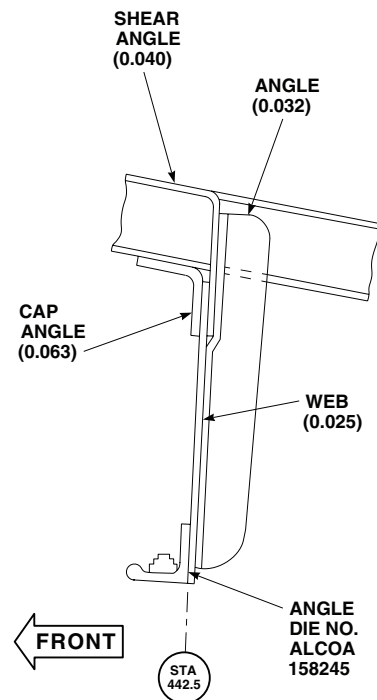
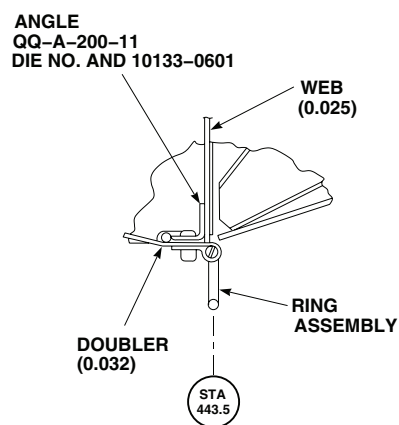
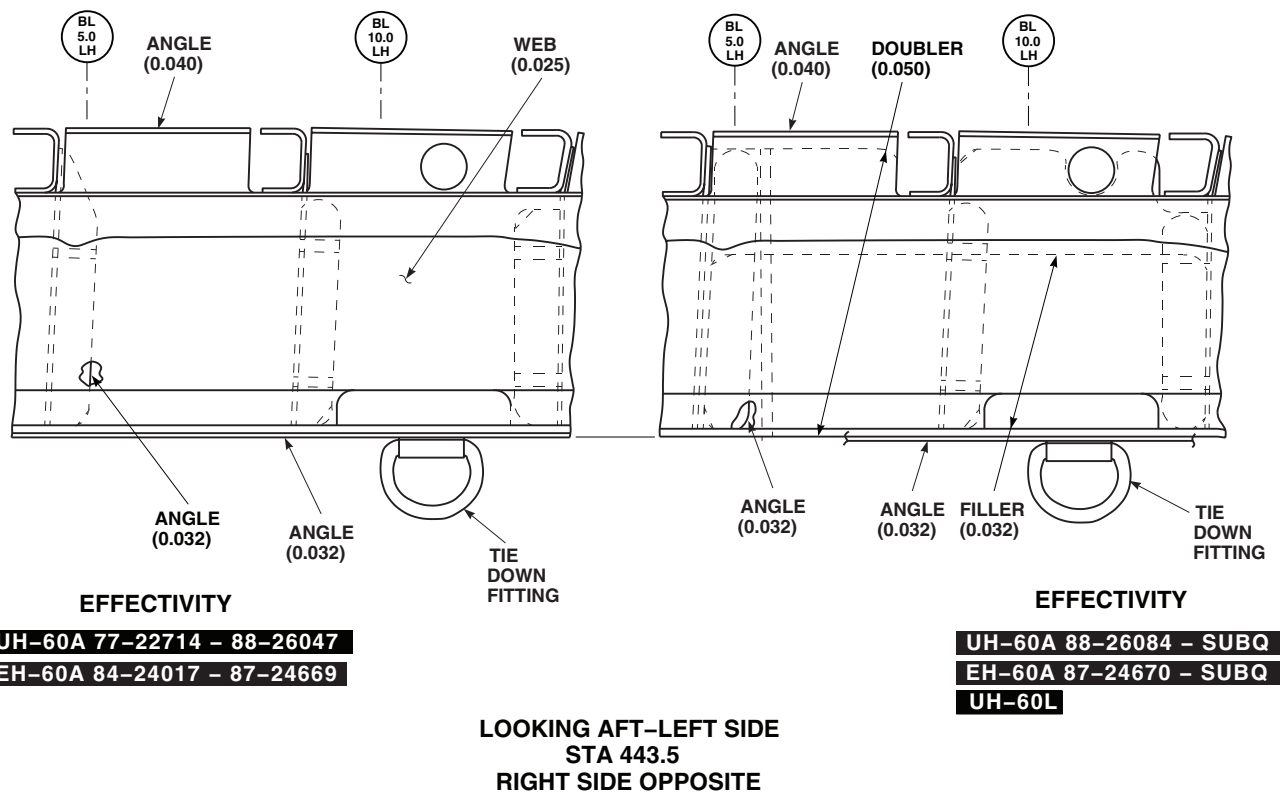
AB2674_18
SA

Figure 1. Transition Section Framing Repair. (Sheet 18 of 30)

FUEL CELL LINER - Continued

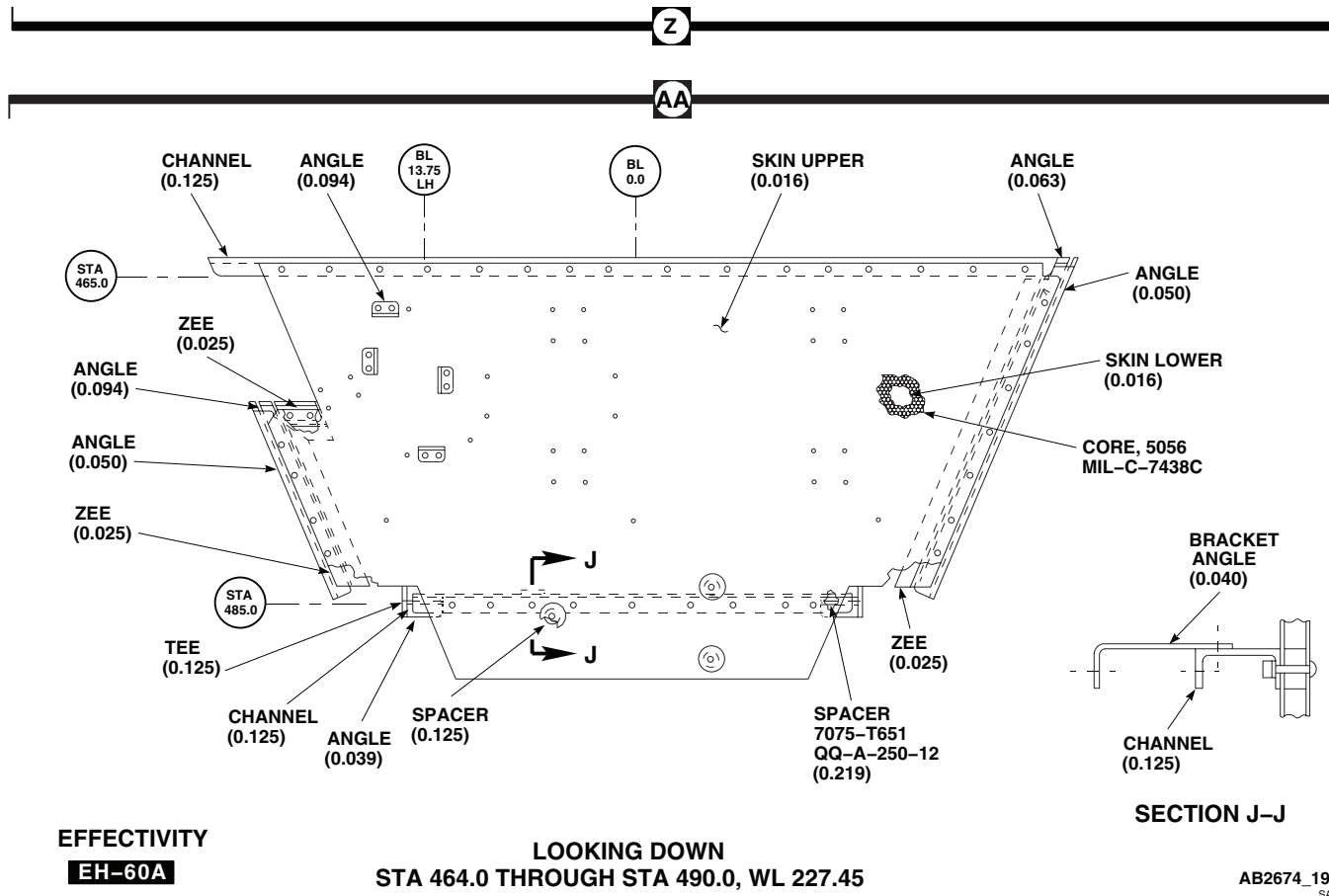
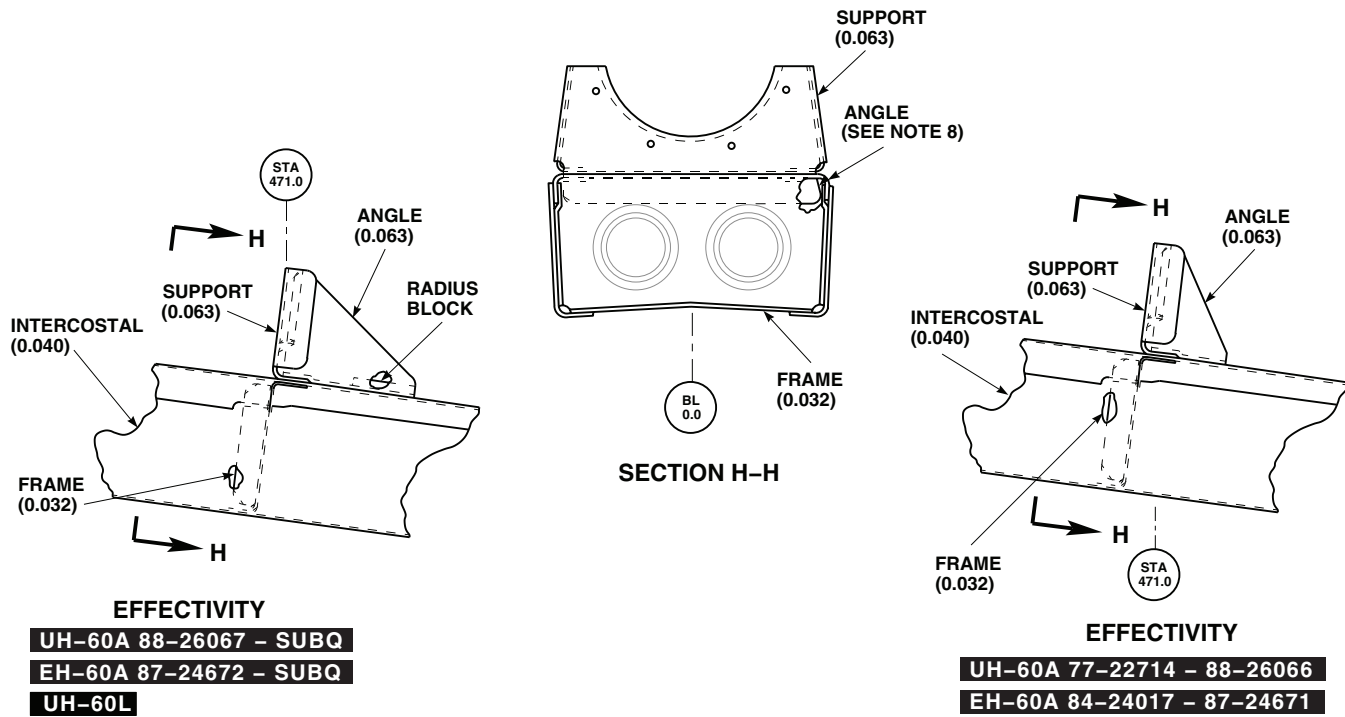


Figure 1. Transition Section Framing Repair. (Sheet 19 of 30)

FUEL CELL LINER - Continued

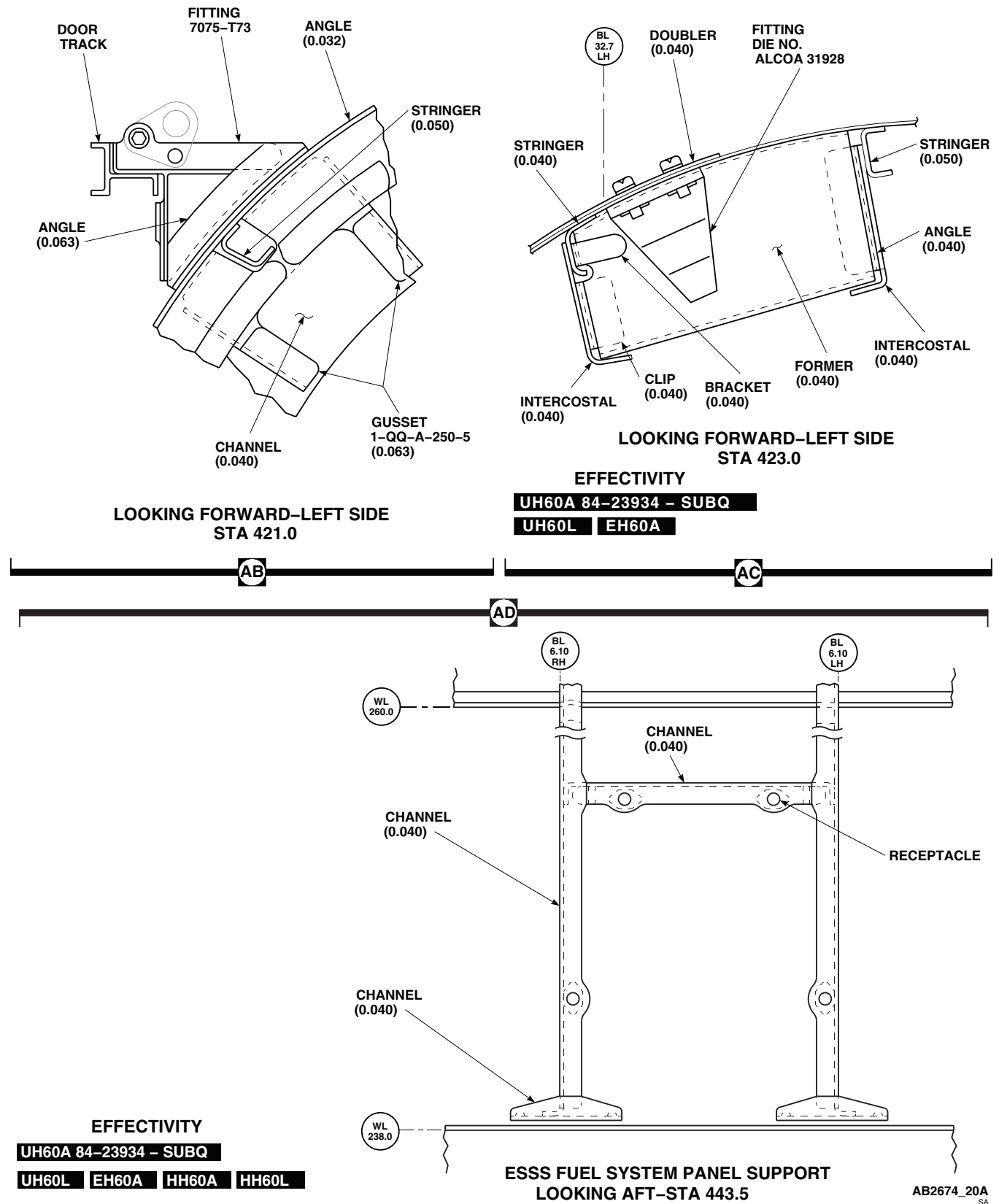


Figure 1. Transition Section Framing Repair. (Sheet 20 of 30)

FUEL CELL LINER - Continued

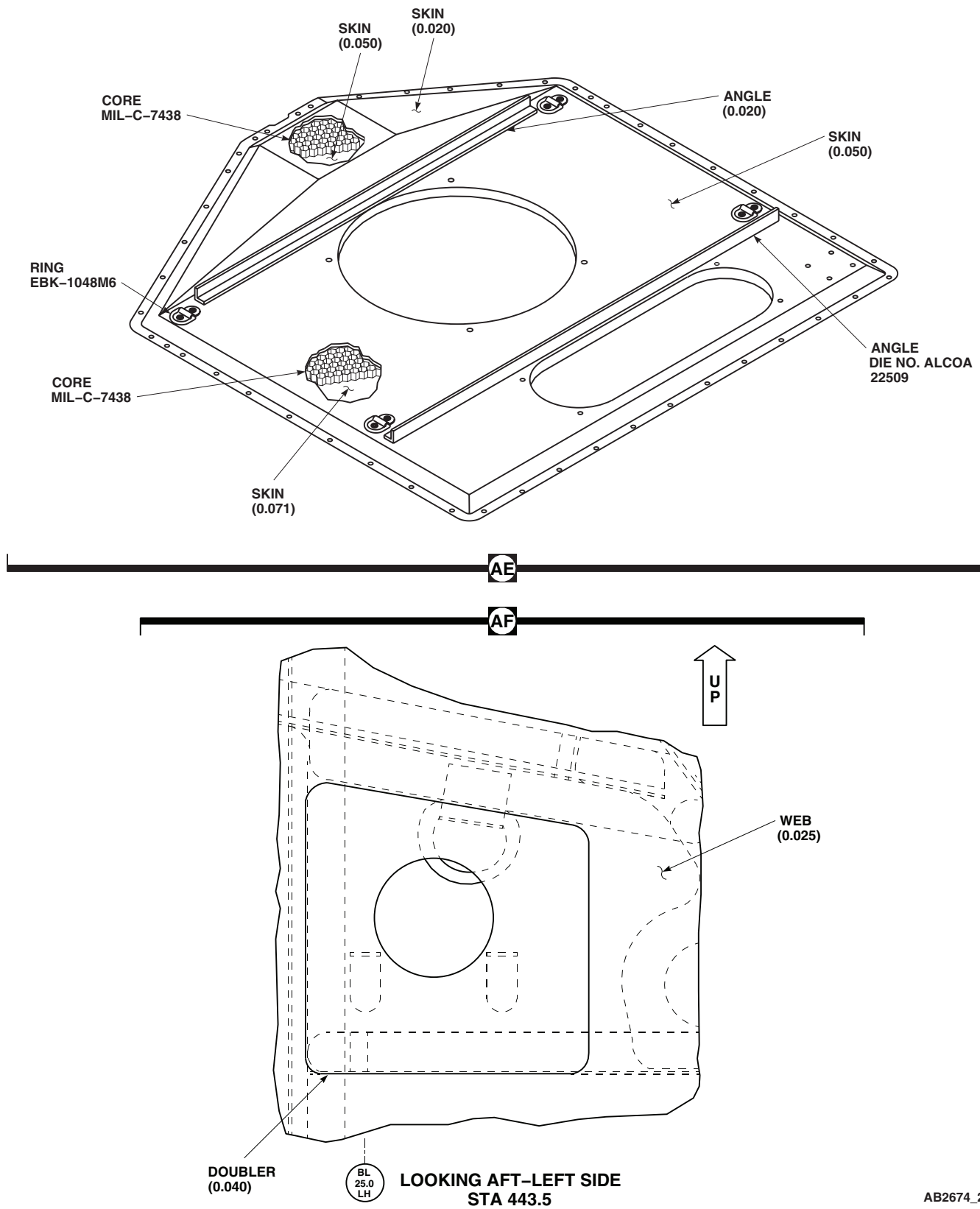


Figure 1. Transition Section Framing Repair. (Sheet 21 of 30)

FUEL CELL LINER - Continued

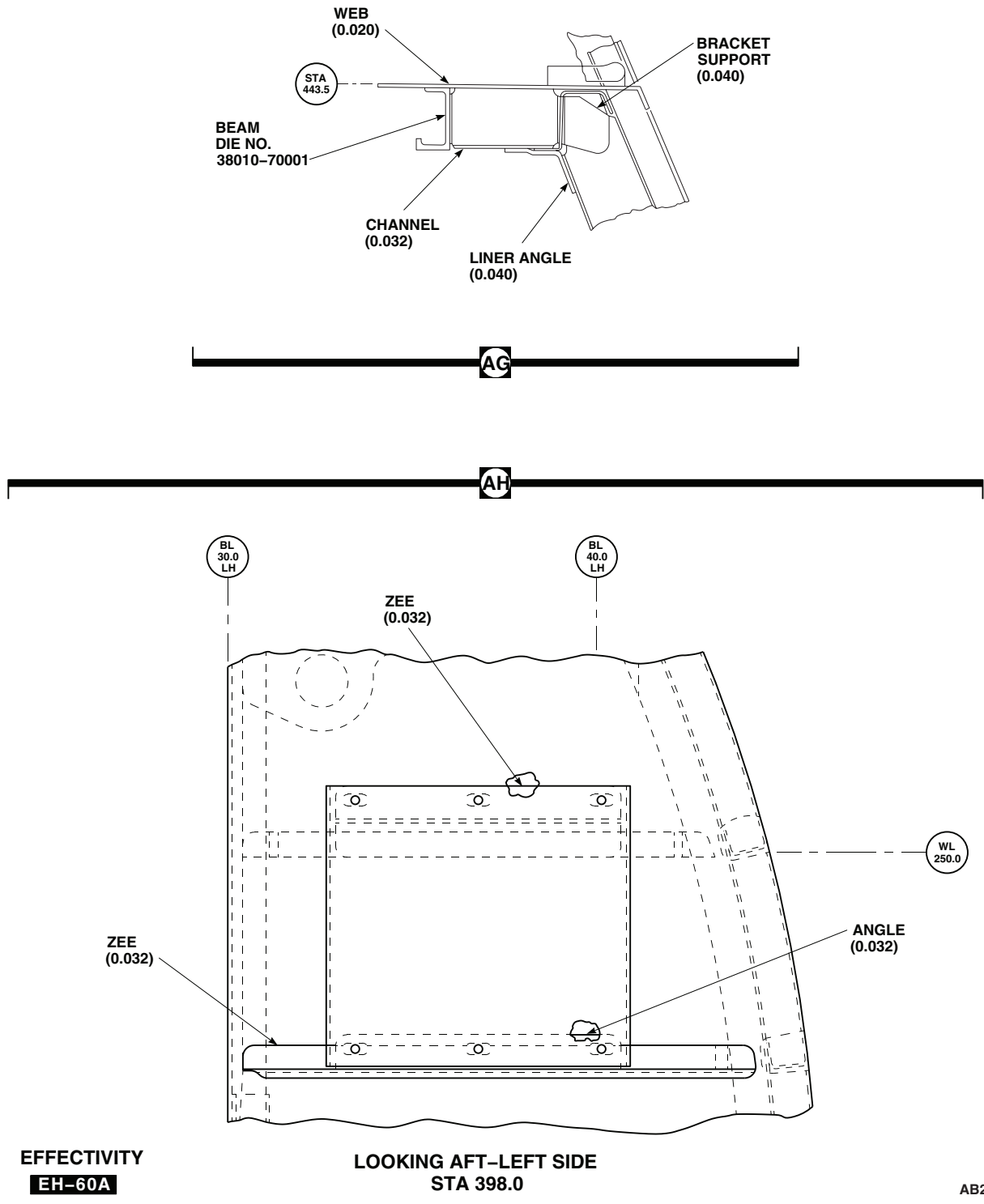


Figure 1. Transition Section Framing Repair. (Sheet 22 of 30)

FUEL CELL LINER - Continued

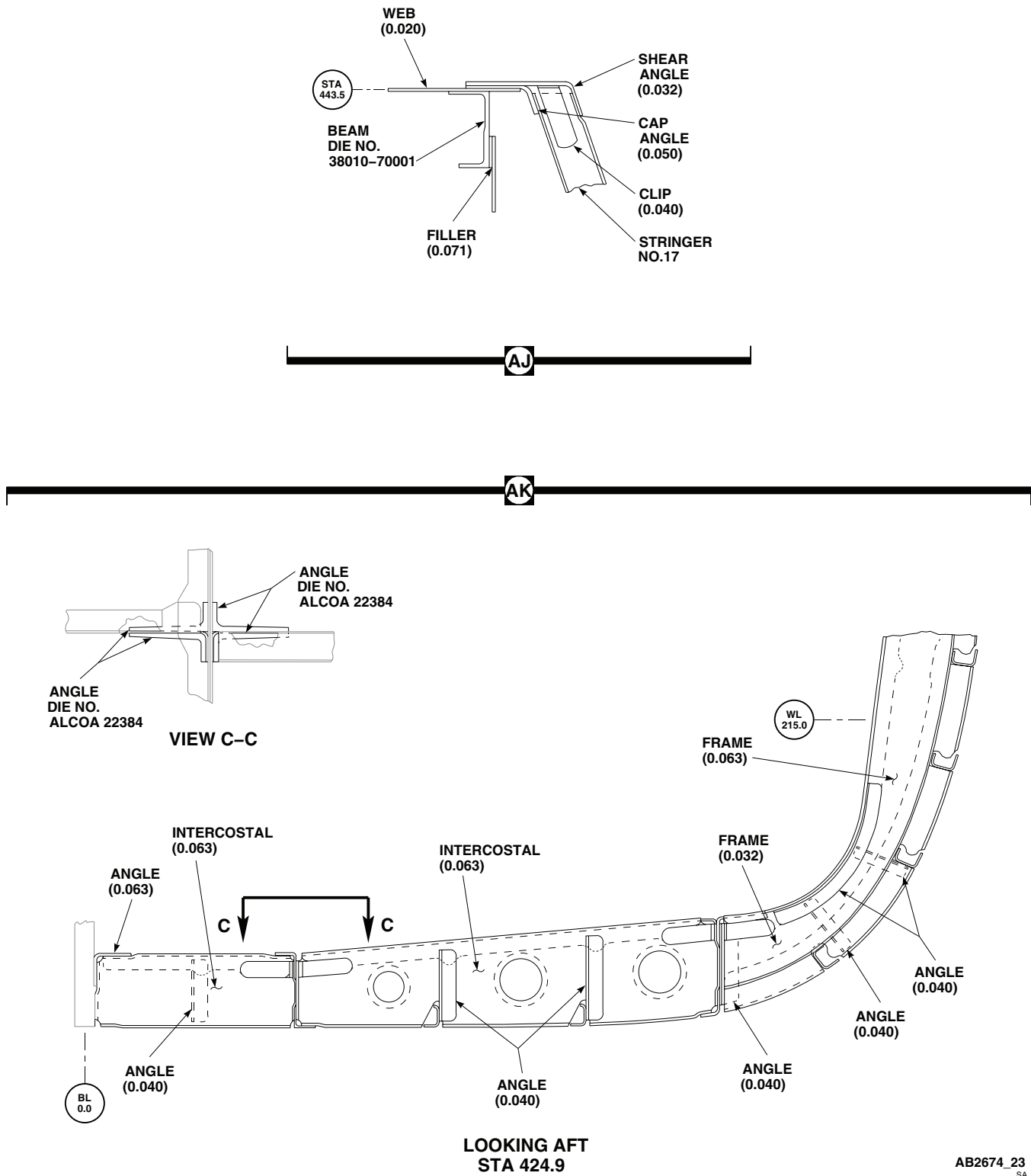


Figure 1. Transition Section Framing Repair. (Sheet 23 of 30)

FUEL CELL LINER - Continued

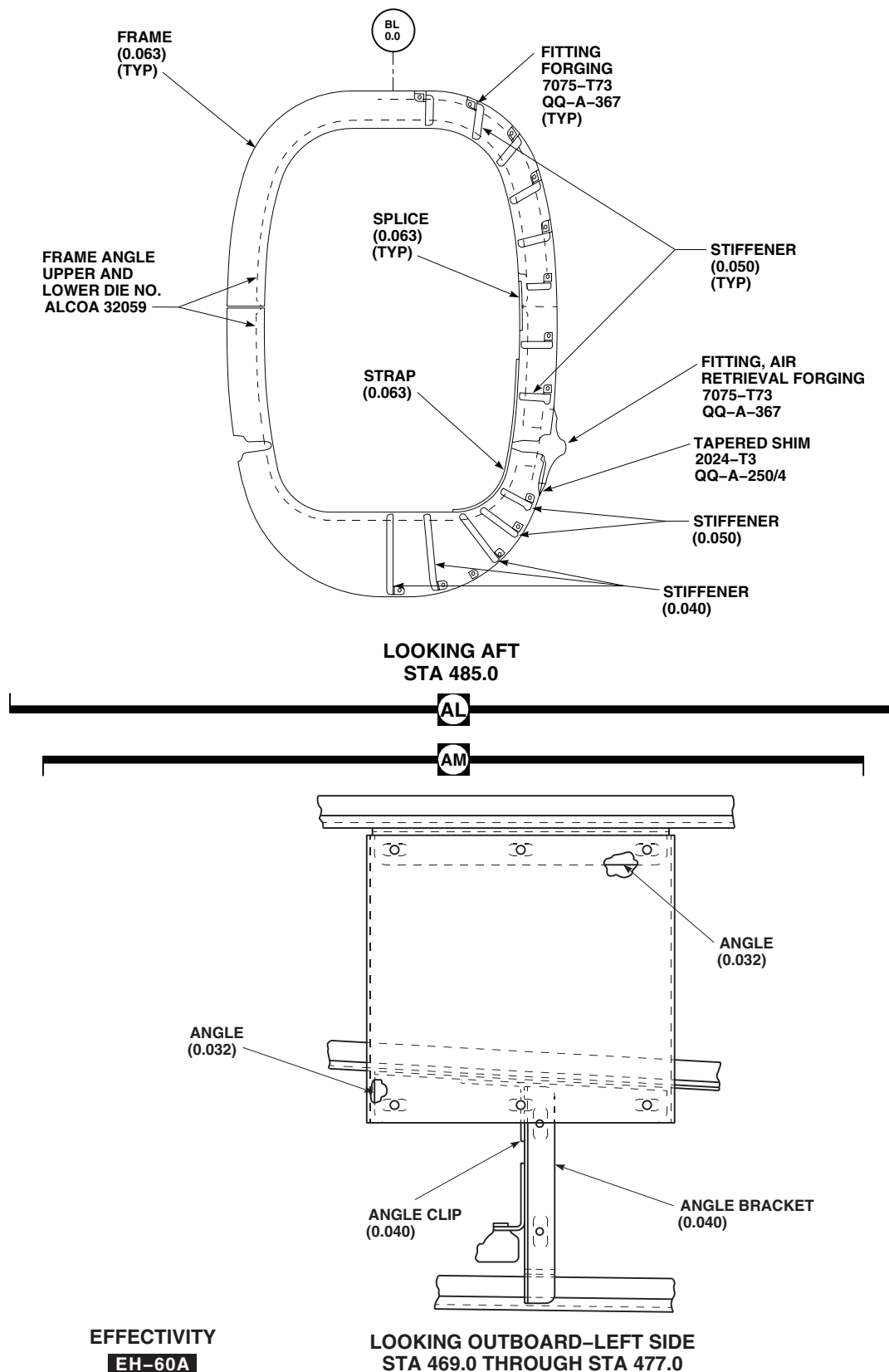


Figure 1. Transition Section Framing Repair. (Sheet 24 of 30)

FUEL CELL LINER - Continued

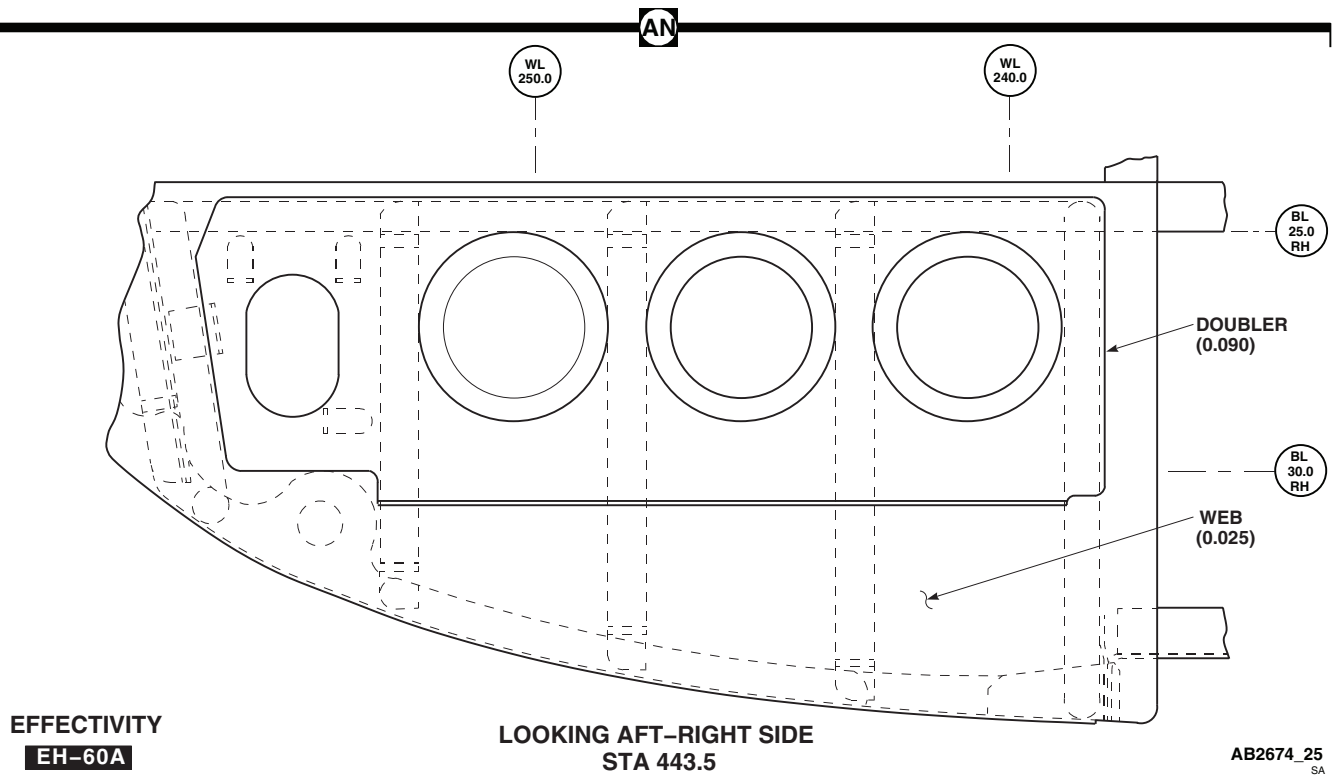
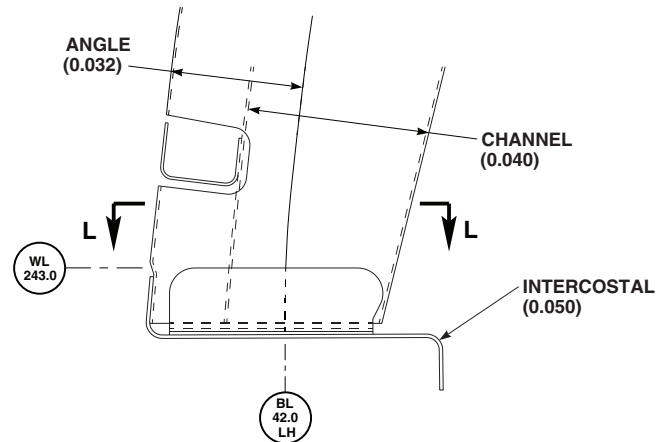


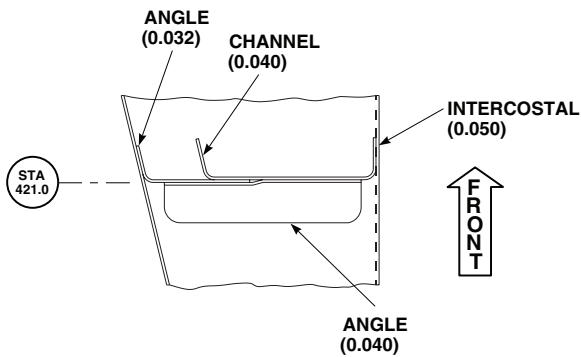
Figure 1. Transition Section Framing Repair. (Sheet 25 of 30)

FUEL CELL LINER - Continued

AP



LOOKING FORWARD-LEFT SIDE
STA 421.0

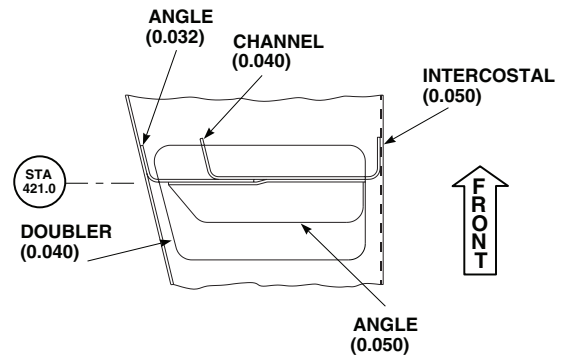


SECTION L-L

EFFECTIVITY

UH-60A 77-22714 - 89-26206

UH-60L 89-26149, 89-26154



SECTION L-L

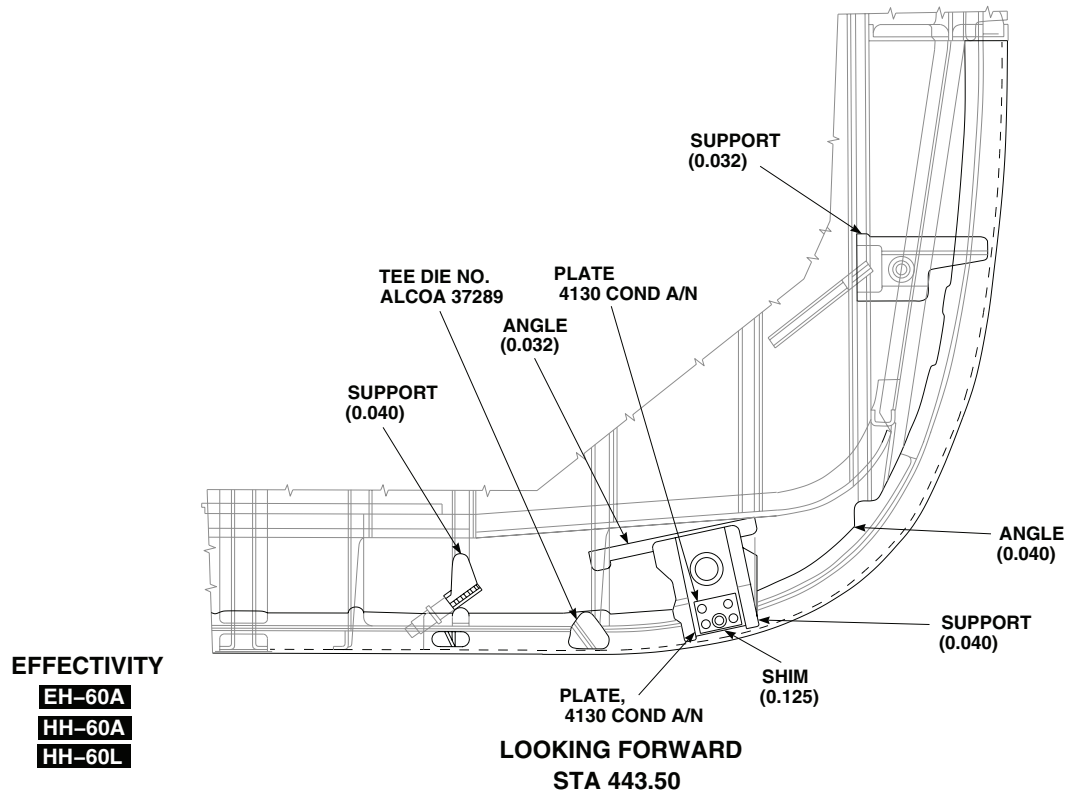
EFFECTIVITY

UH-60A 89-26207 - SUBQ

UH-60L 89-26179 - SUBQ

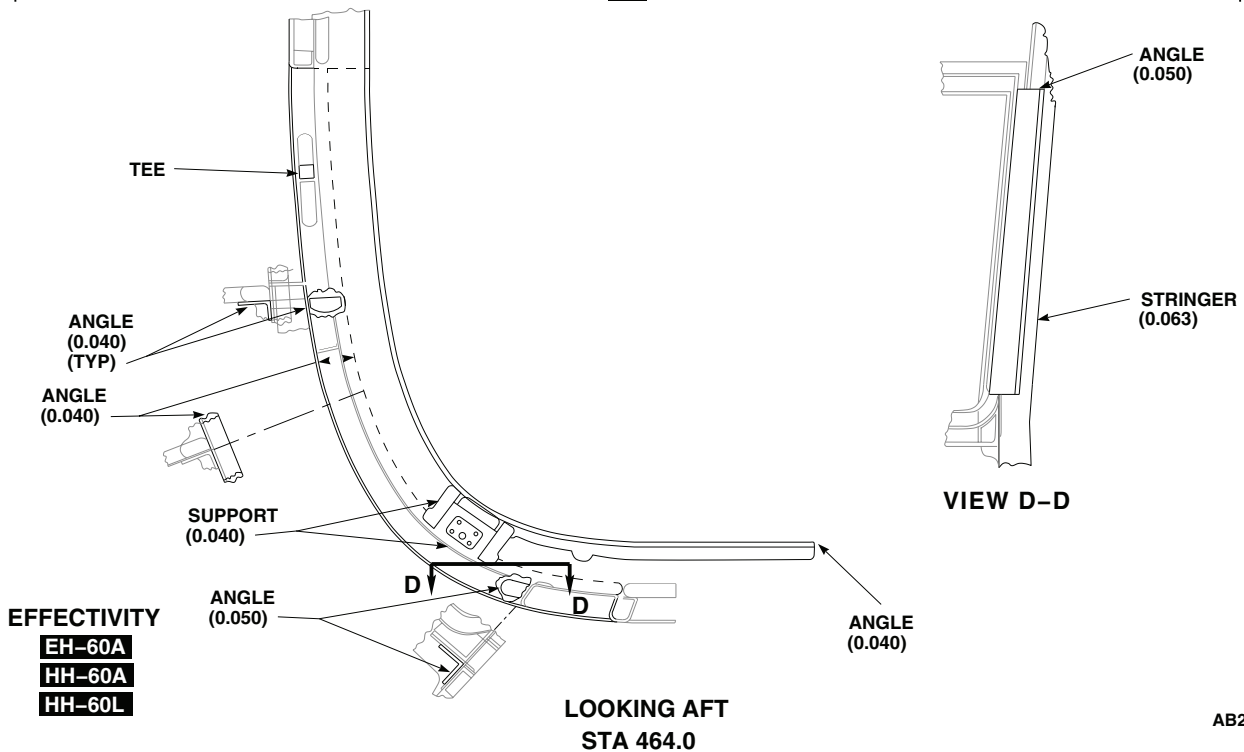
Figure 1. Transition Section Framing Repair. (Sheet 26 of 30)

FUEL CELL LINER - Continued



AQ

AR



AB2674_2A
 SA

Figure 1. Transition Section Framing Repair. (Sheet 27 of 30)

FUEL CELL LINER - Continued

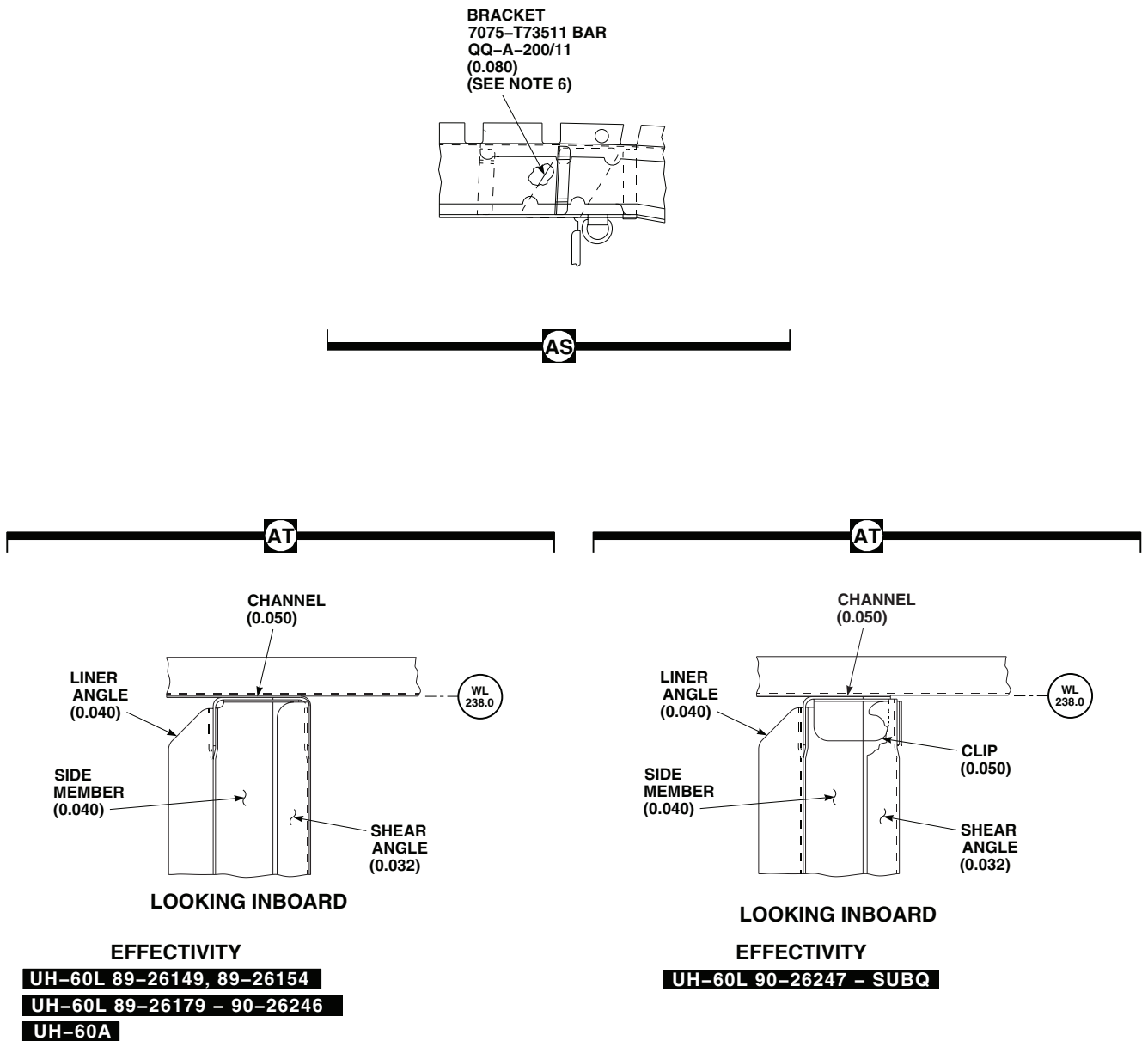
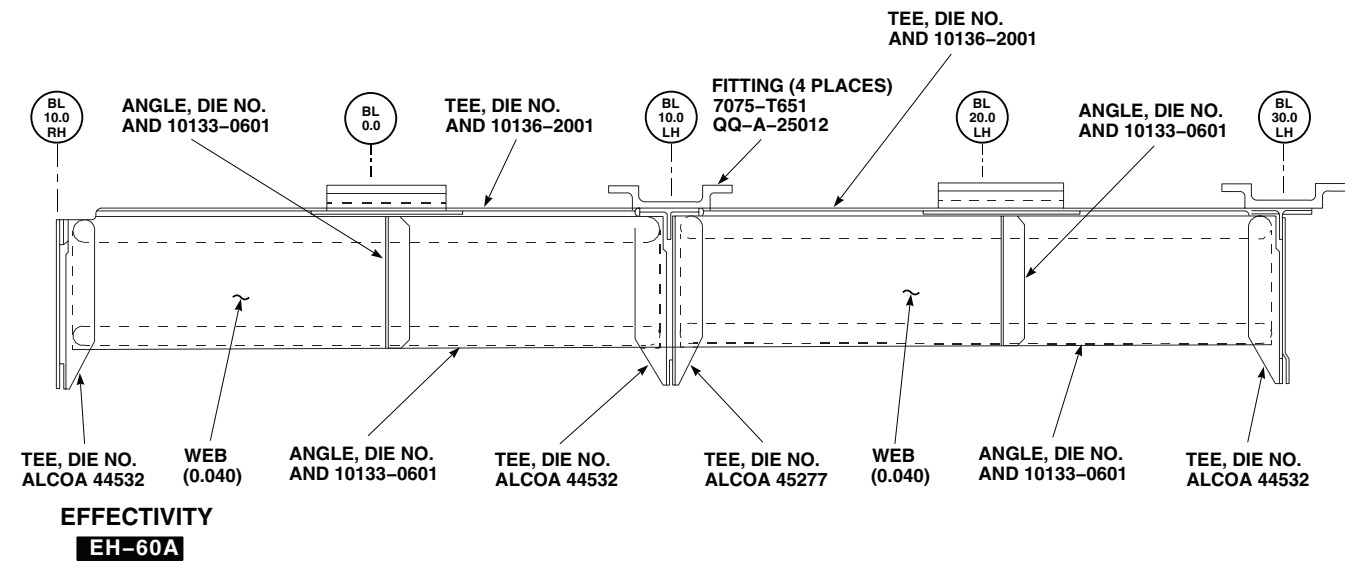


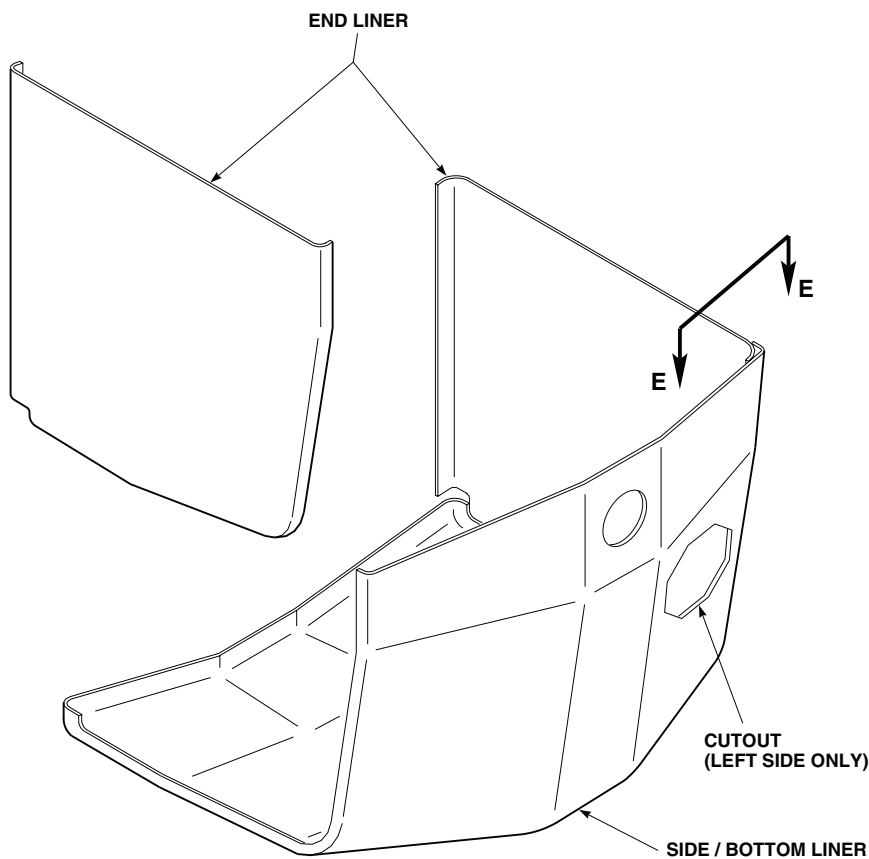
Figure 1. Transition Section Framing Repair. (Sheet 28 of 30)

FUEL CELL LINER - Continued



AU

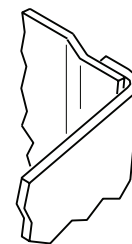
AV



VIEW E-E

EFFECTIVITY

UH-60A 77-22714 - 78-23014



VIEW E-E

EFFECTIVITY

UH-60A 78-23015 - SUBQ

UH-60L UH-60L

RIGHT SIDE OPPOSITE, EXCEPT AS SHOWN

AB2674_29
SA

Figure 1. Transition Section Framing Repair. (Sheet 29 of 30)

FUEL CELL LINER - Continued**NOTES:**

1. UNLESS OTHERWISE NOTED, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. **EH-60A 85-24478 - SUBQ**
4. **UH-60A 80-23429 - SUBQ** **UH-60L** **EH-60A**
5. **UH-60A 77-22714 - 88-26048** **EH-60A 84-24017 - 87-24670**
6. **UH-60A 88-26084 - SUBQ** **UH-60L** **EH-60A 87-24670 - SUBQ**
7. **UH-60A 89-26162 - SUBQ** **UH-60L**
8. 0.050 **UH-60A 77-22714 - 86-24521** **EH-60A 84-24017 - 85-24471** **UH-60L**
EH-60A 85-24471 - SUBQ **UH-60A 86-24521 - SUBQ**
9. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2674_30
SA

Figure 1. Transition Section Framing Repair. (Sheet 30 of 30)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TRANSITION SECTION PLATING REPAIR

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-S-18729

MIL-S-5095

MIL-STD-670

Material/Parts

Cloth, Abrasive, Item 78, WP 1803 00

Hinge, MS20001P4-375

Primer, Adhesive, Item 240, WP 1803 00

Sealing Compound, Item 273, WP 1803 00

Sealing Compound, Item 263, WP 1803 00

MIL-T-9046

MIL-W-6858

MS20001

QQ-A-200

QQ-A-230

QQ-A-250

QQ-A-362

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

TM 1-1500-204-23

TM 55-1500-345-23

References

AMS 4036

AMS 4260

AMS 5334

MIL-C-7637

MIL-P-15035

WP 0377 00

WP 0378 00

WP 0379 00

WP 0381 00

WP 1803 00

REPAIR

1. Transition section plating is Alclad aluminum 7075-T6, QQ-A-250-13, except in high temperature areas where titanium alloy, MIL-T-9046, is used (Figure 1, Sheet 1).
2. All plating, with the exception of reinforced plastic members, is primary structure. Evaluate damage (WP 0377 00).
3. Use WP 0378 00 for typical plating repairs. When repairing titanium plating, use same gage titanium stock, or the equivalent gage 301/302 stainless steel, MIL-S-5059. Use Monel or stainless steel fasteners for attaching titanium or stainless steel patches. Refer to TM 1-1500-204-23 for cutting, drilling, forming, and fastening titanium or stainless steel sheet stock.
4. The avionics compartment door is a bonded sandwich-construction assembly made from aluminum alloy honeycomb core and Kevlar (reinforced plastic) skins. Metal components include hinges straps, latches, and strut fitting.
5. Evaluate damage in noncritical areas per WP 0377 00.
6. Evaluate damage to latch assemblies and strut fitting per WP 0377 00.
7. Negligible damage to hinge straps is limited to nicks, scratches and corrosion not over 0.005-inch if point of damage is within 1 inch of hinge line and 0.010-inch if point of damage is beyond 1 inch of hinge line (Figure 1, Sheet 6, Detail F). Blend and polish damage with fine grit abrasive cloth, Item 78, WP 1803 00, to same finish as surrounding area.
8. Use WP 0379 00 for typical reinforced plastic repairs.
9. Refer to Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, Figure 1, Sheet 7 and Figure 1, Sheet 8 for access door panel repair and TM 1-1500-204-23 requirements.

REPAIR - Continued

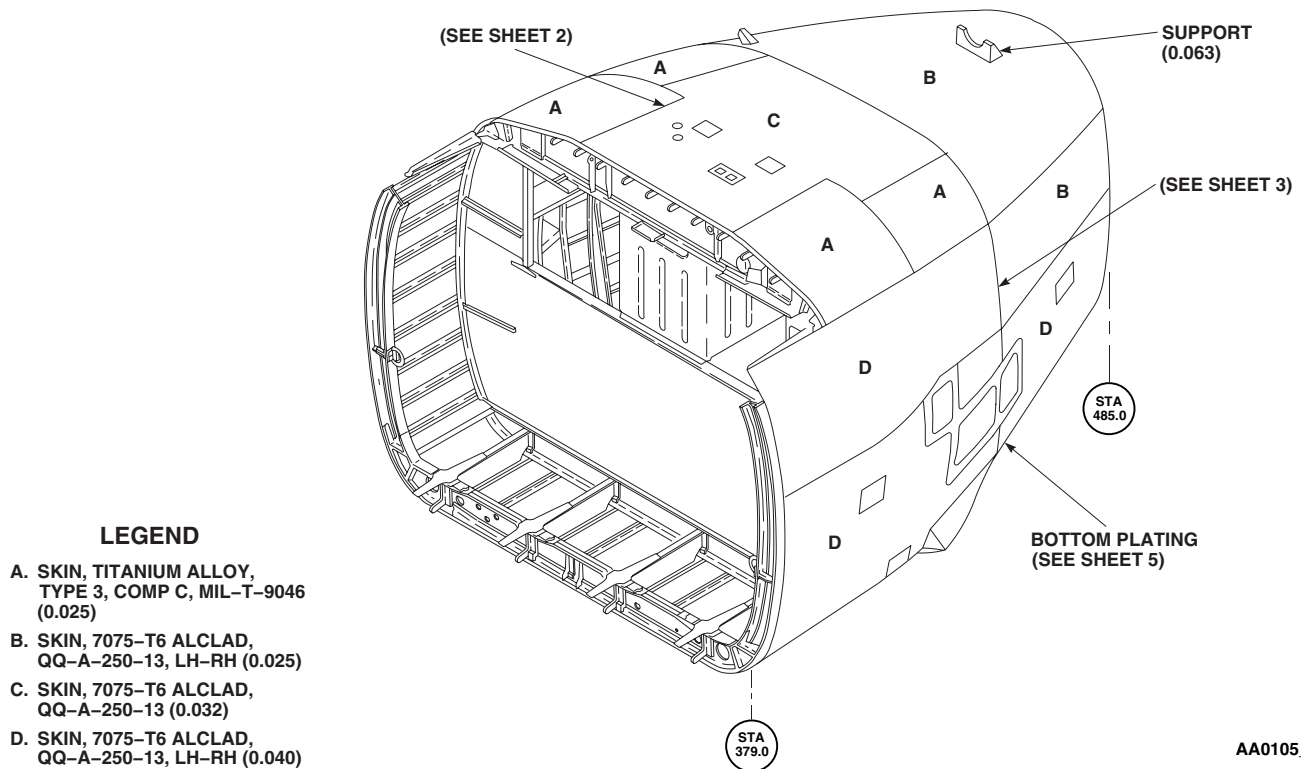


Figure 1. Transition Section Plating Repair. (Sheet 1 of 9)

10. For corrosion protection and watertightness, seal all repairs of the transition section plating (WP 0377 00).
11. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued

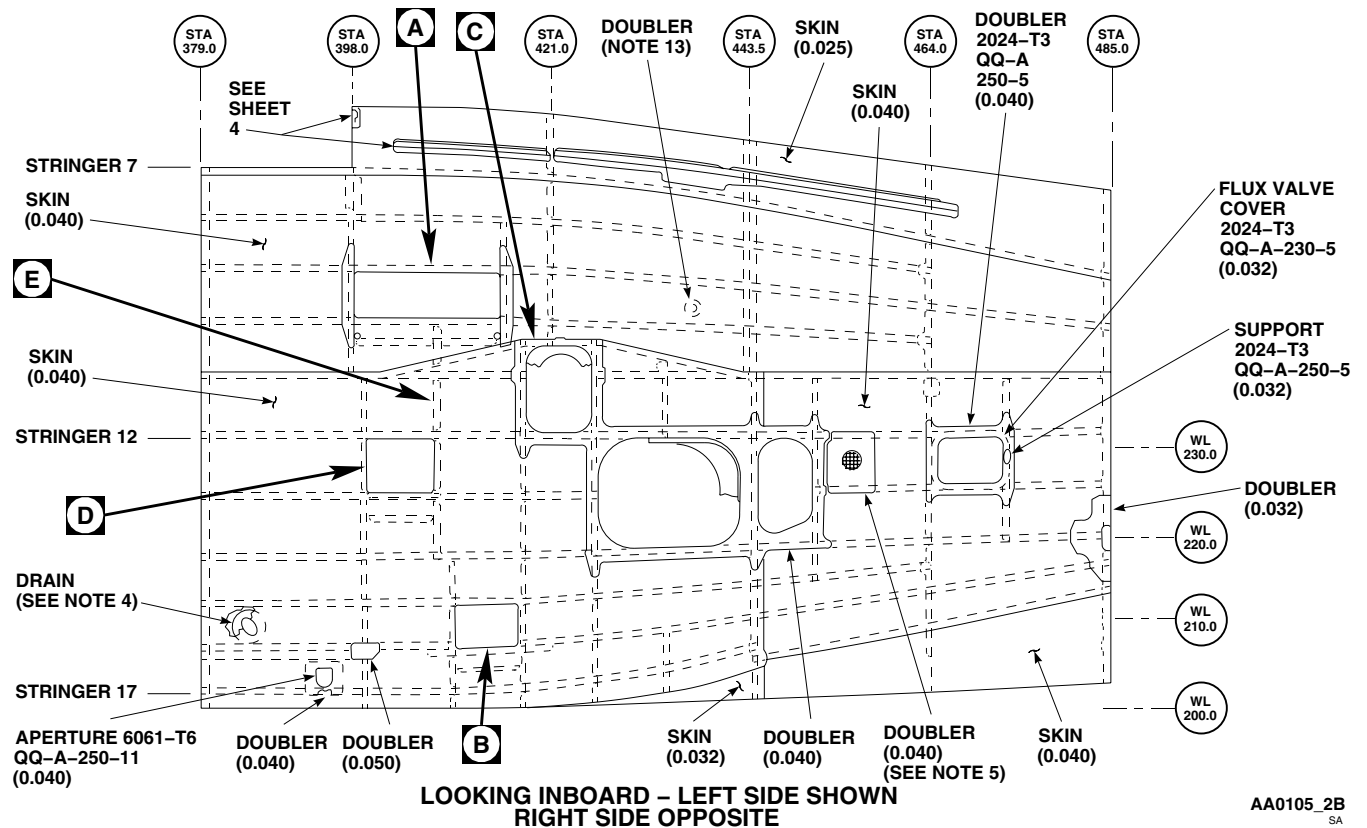
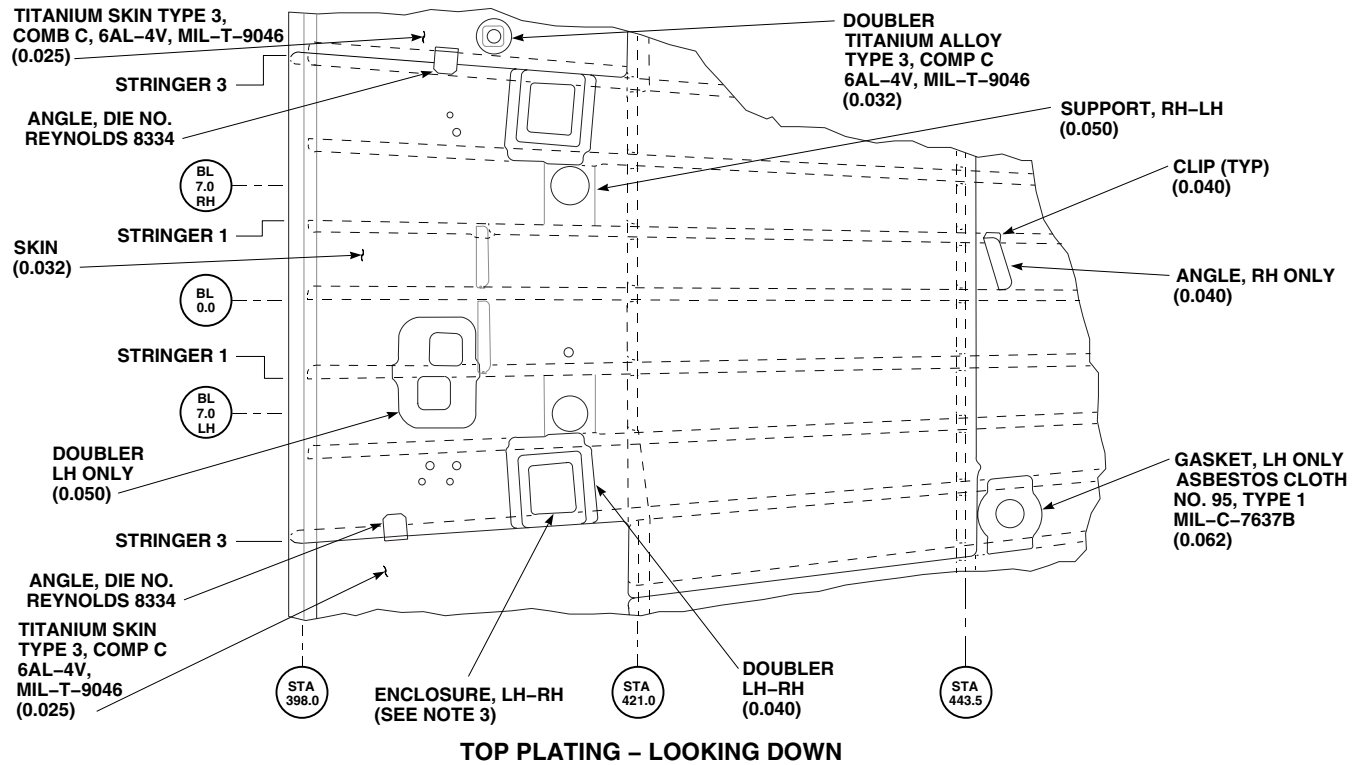


Figure 1. Transition Section Plating Repair. (Sheet 2 of 9)

REPAIR - Continued

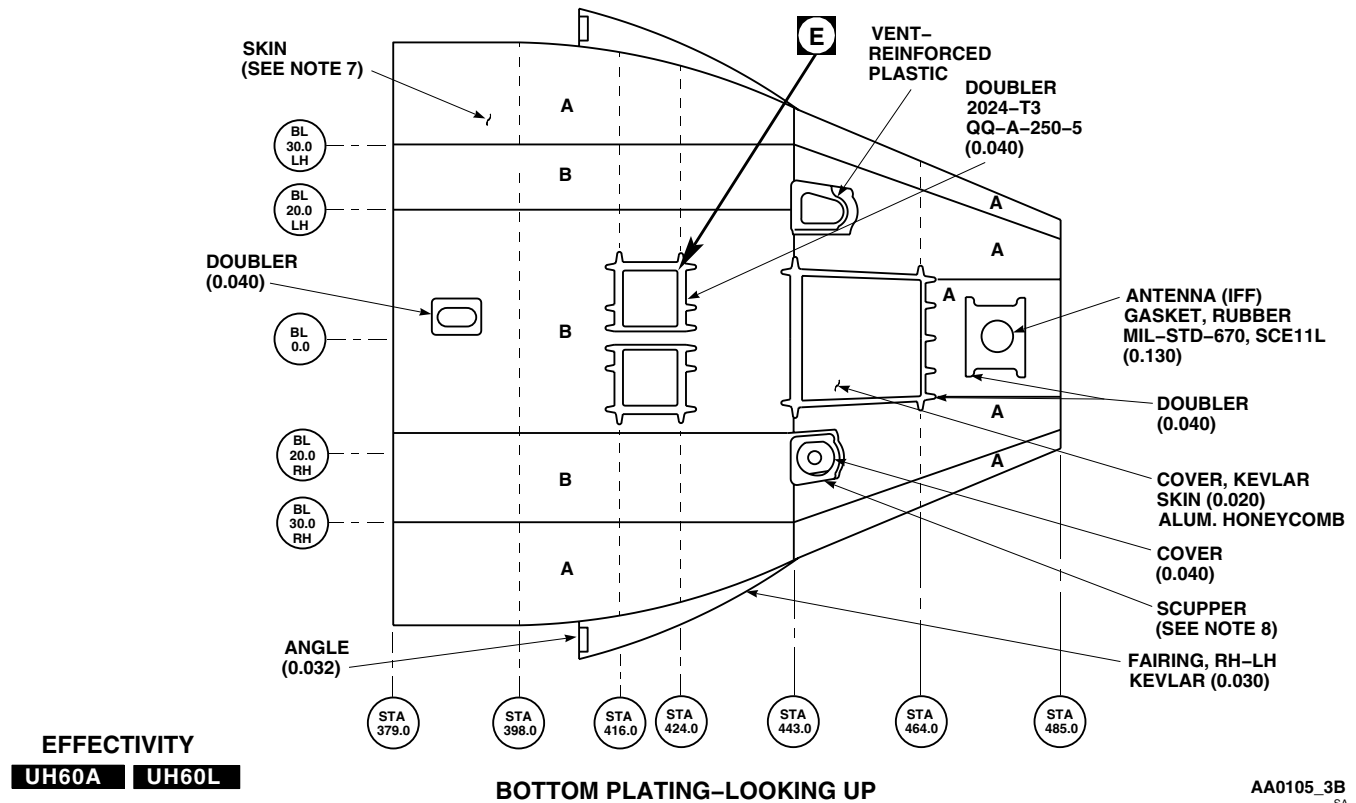
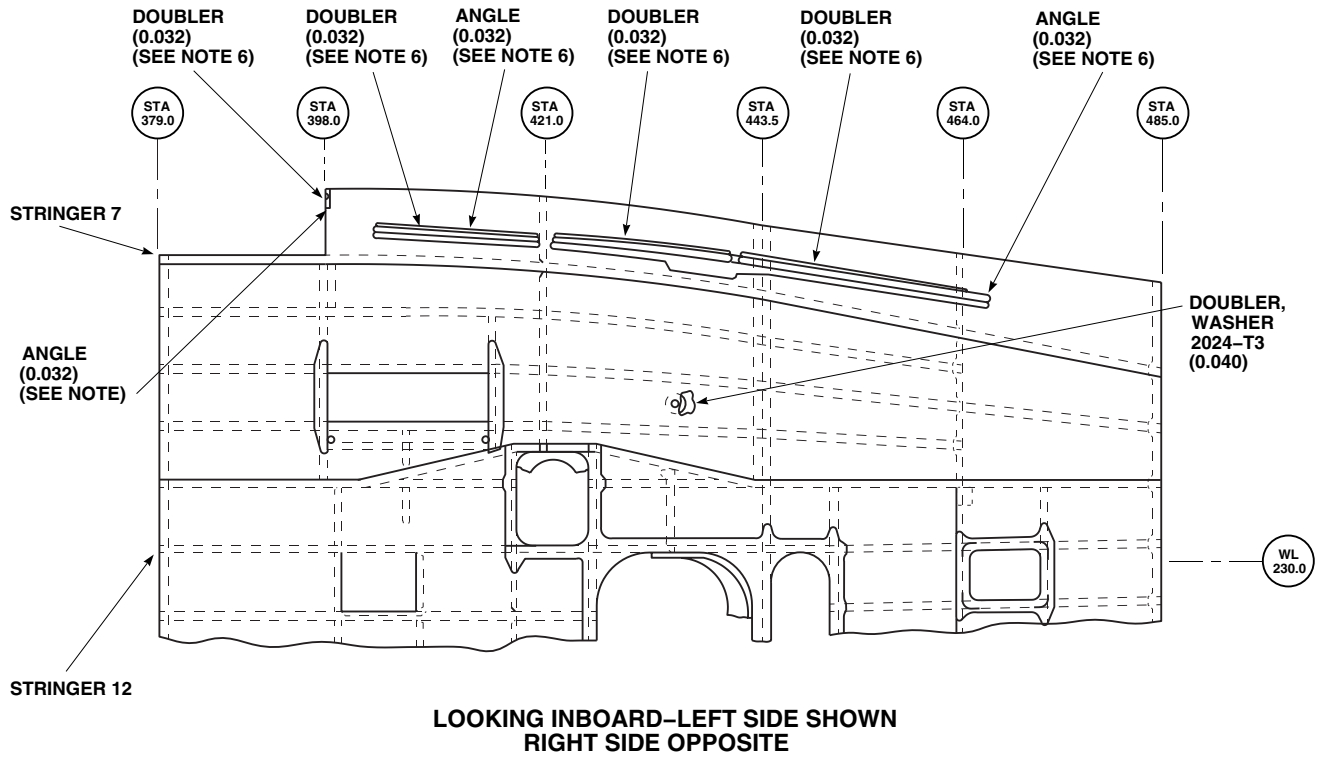


Figure 1. Transition Section Plating Repair. (Sheet 3 of 9)

REPAIR - Continued

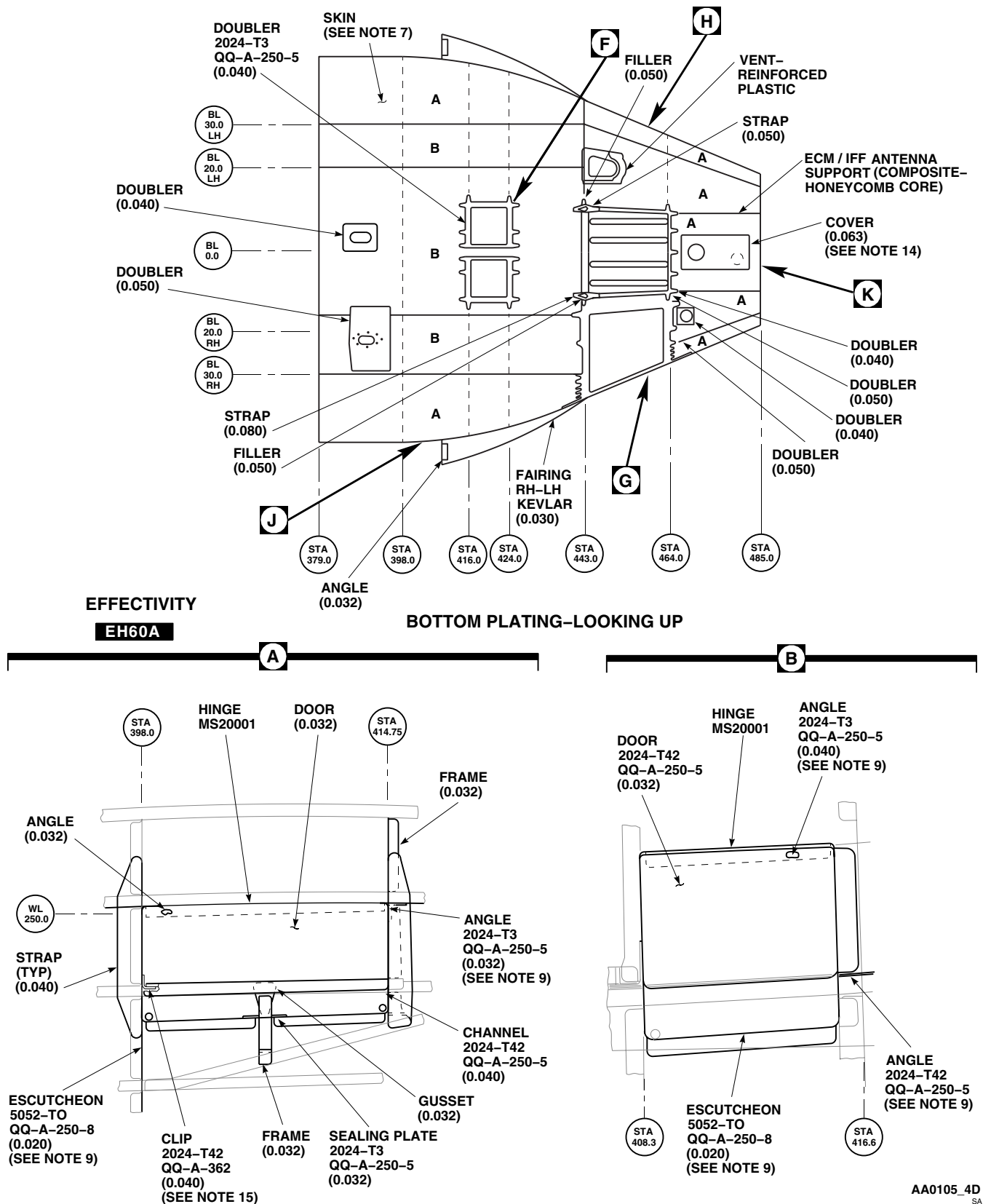
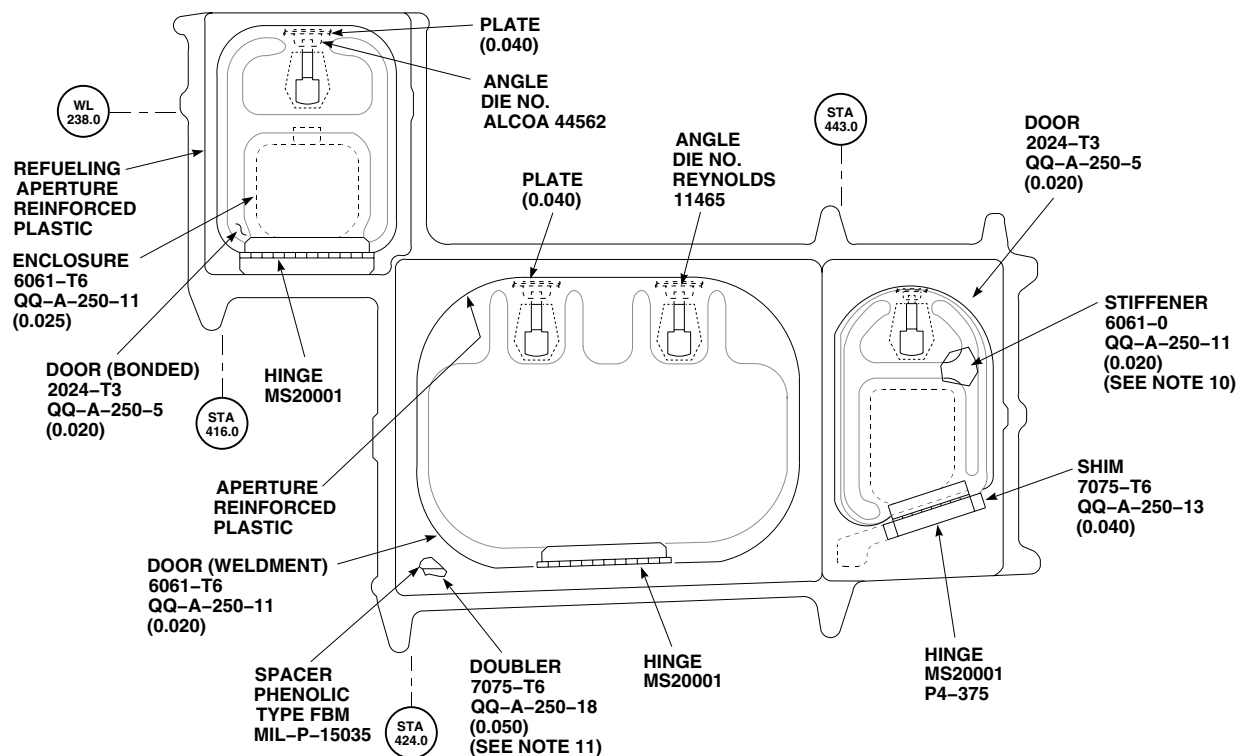


Figure 1. Transition Section Plating Repair. (Sheet 4 of 9)

REPAIR - Continued



C

D

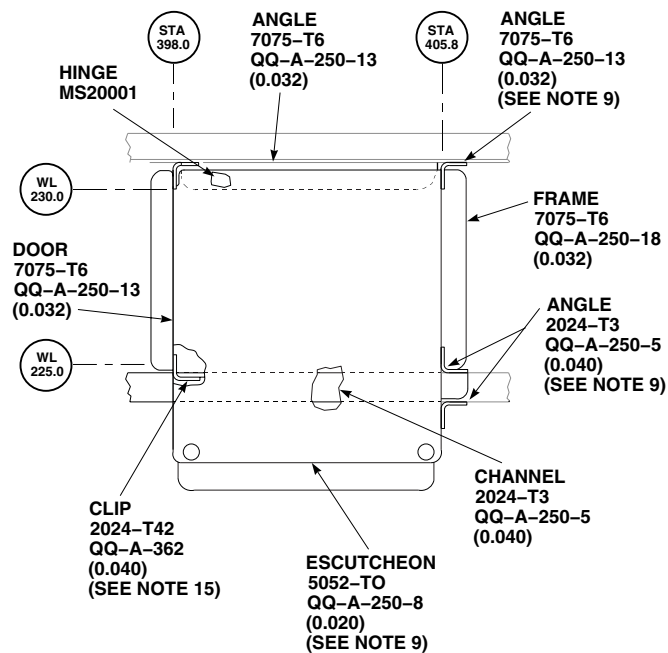
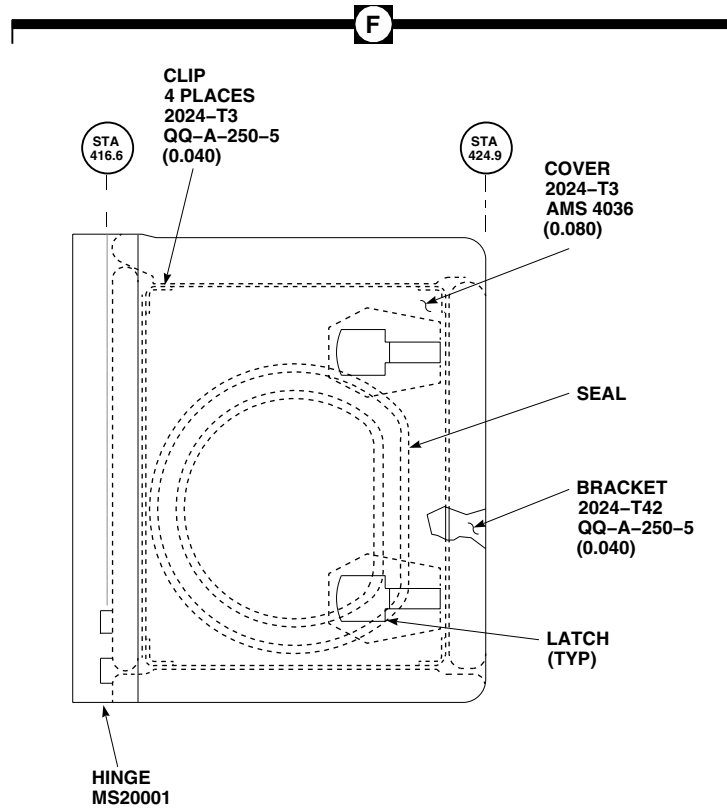
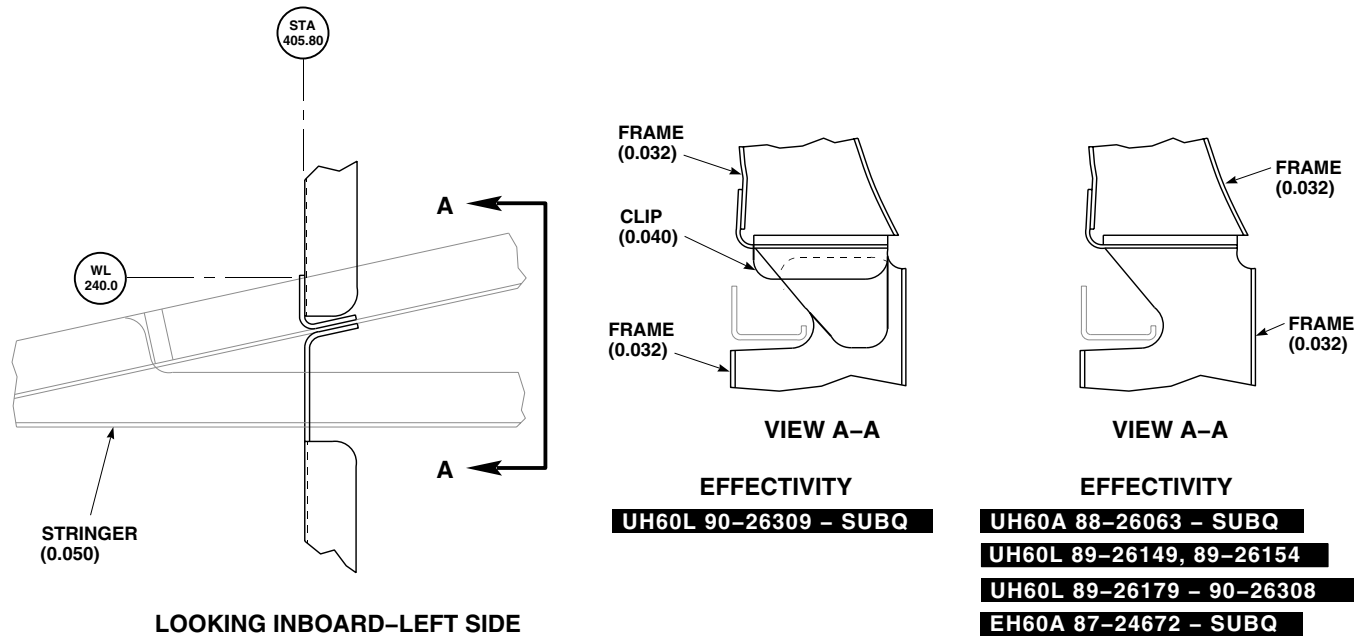
AA0105_5B
SA

Figure 1. Transition Section Plating Repair. (Sheet 5 of 9)

REPAIR - Continued



AA0105_6B
SA

Figure 1. Transition Section Plating Repair. (Sheet 6 of 9)

REPAIR - Continued

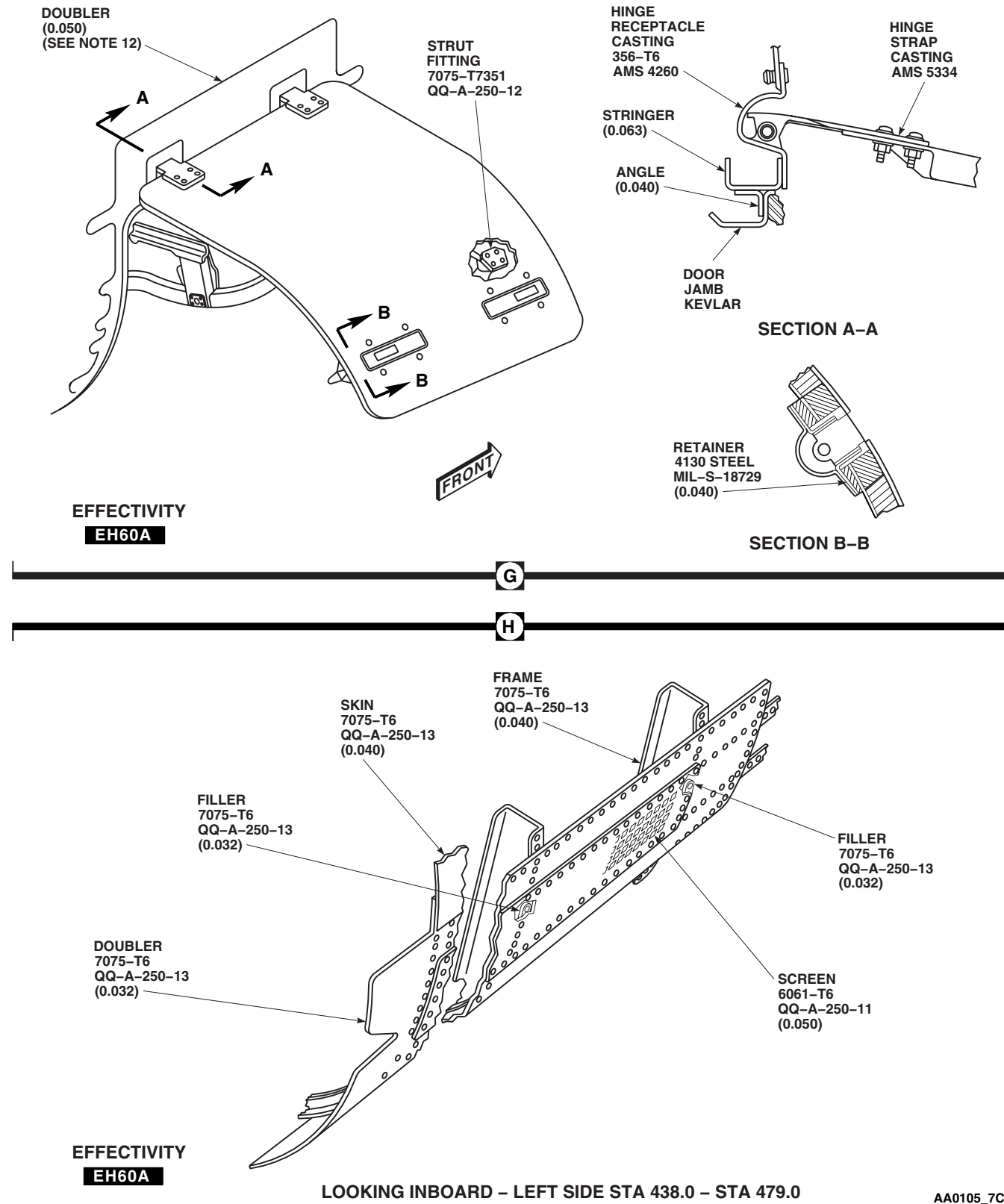


Figure 1. Transition Section Plating Repair. (Sheet 7 of 9)

REPAIR - Continued

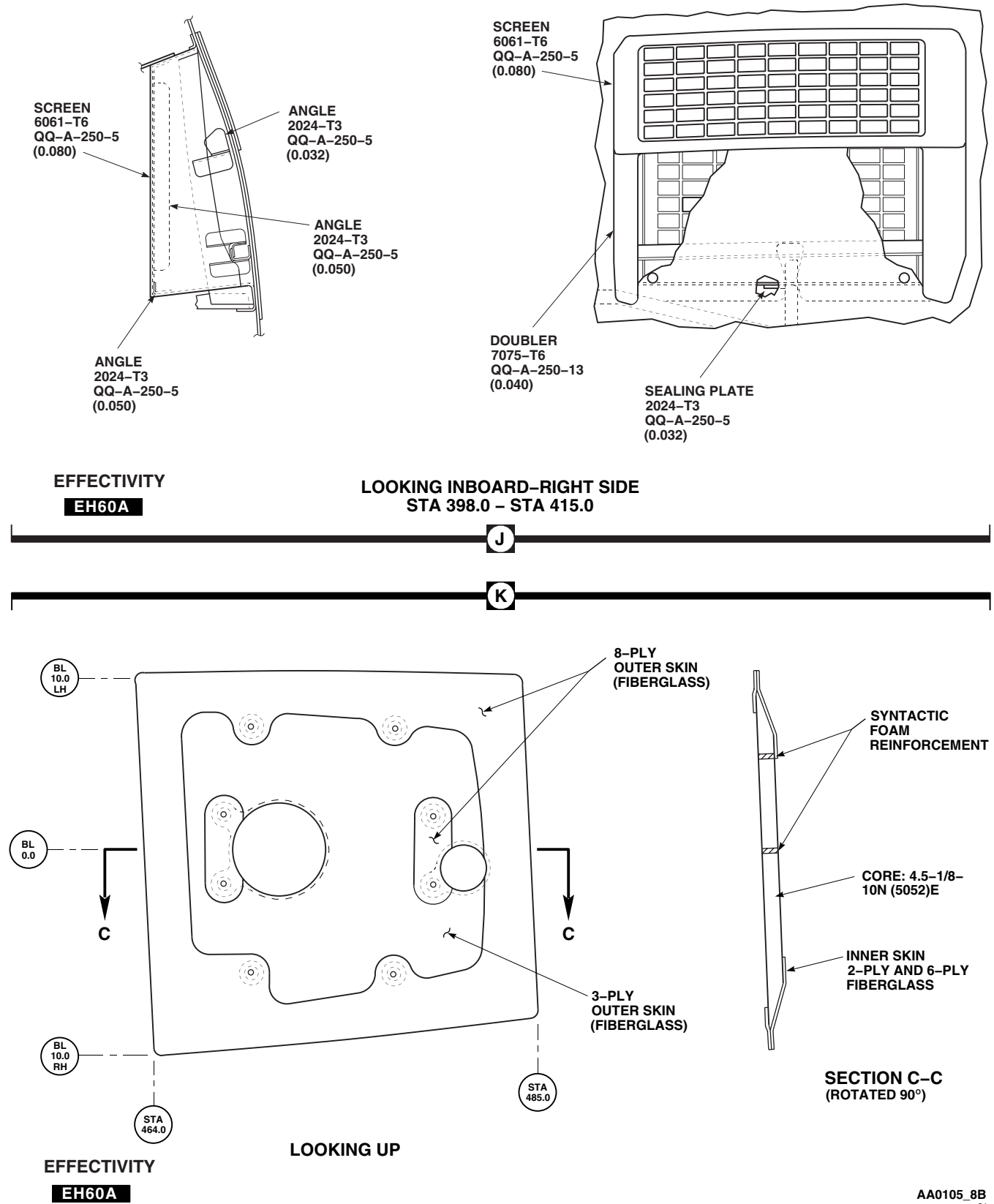


Figure 1. Transition Section Plating Repair. (Sheet 8 of 9)

REPAIR - Continued

NOTES:

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY, 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. **UH60A 77-22714 - 80-23504** → **UH60A 80-23505 - SUBQ** KEVLAR, 0.040 TO 0.060. ◀
UH60L **EH60A** 6061-T6, QQ-A-250-11, 0.032. ◀
4. **UH60A 77-22714 - 85-24454** → **EH60A 84-24017 - 84-24018** WELD ASSEMBLY, 6061-T6, QQ-A-250-11. ◀
UH60A 85-24455 - SUBQ → **UH60L** **EH60A 84-24019 - SUBQ** CASTING, AM54260, 356-T6.
 EITHER MAY BE USED. ◀
5. **EH60A** LEFT SIDE ONLY.
6. TITANIUM, 6AL-4YT1, MIL-T-9046.
7. A SKIN, 0.040; B SKIN, 0.032.
8. **UH60A 77-22714 - 81-23562** → KEVLAR. ◀ **UH60A 81-23563 - SUBQ** → **UH60L** 6061-T6, QQ-A-250-11, 0.032. ◀
 6061-T6, QQ-A-250-11, 0.032.
9. CLEAN AND SEAL WITH SEALING COMPOUND, ITEM 273, WP 1803 00. FILL GAPS WITH SEALING COMPOUND, ITEM 275, WP 1803 00 AND SEAL WITH SEALING COMPOUND, ITEM 273, WP 1803 00 (WP 0377 00).
10. PRIMARY CONSTRUCTION: SPOT WELDING PER MIL-W-6858, CLADD C; ALTERNATE CONSTRUCTION: BONDED.
11. **UH60A 78-23015 - SUBQ** **UH60L** **EH60A**
12. GAP OF 0.040 - 0.120 ALLOWED ALL AROUND BETWEEN DOOR AND DOUBLER.
13. **UH60L** → **UH60L 84-26149 - 89-26154** **UH60L 89-26179 - 90-26277** SS430X20A010825. ◀
UH60L 90-26278 - SUBQ → QQ-A-250-13, 7075-T6, 0.040. ◀
14. **EH60A 86-24577 - SUBQ**
15. CLEAN BONDING SURFACES, APPLY ADHESIVE PRIMER, ITEM 48, WP 1803 00, AND BOND WITH ADHESIVE, ITEM 46 WP 1803 00, BEFORE FASTENING WITH RIVETS.
16. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0105_9D
SA

Figure 1. Transition Section Plating Repair. (Sheet 9 of 9)

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****TRANSITION SECTION STEP DOOR HINGE WL 210.0 AND WL 230.0 REPAIR**

This WP supersedes WP 0407 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Rivet Kit, G-740-ELM

Screw, MS27039-0806

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Aircraft Transport Helicopter Repairer MOS 15T (1)

Material/Parts

Body Plug, BP-16232
Hinge Half, MS20001PX4-0450
Hinge Half, MS20001PY4-0450
Nutplate, MS21075-08
Rivet, NAS1097AD3
Rivet, NAS1739B4

References

QQ-W-470
TM 55-1500-345-23
WP 0377 00
WP 0381 00

REMOVE

1. Remove four screws securing door to angle and remove door (Figure 1, Detail A).
2. Open and secure hinge assembly to keep hinges open.
3. Carefully drill out and remove five rivets securing upper hinge to step housing and stringer, using 1/8-inch diameter drill and 3/32-inch diameter drive pin punch.
4. Remove hinge assembly and angle.
5. If spring is broken but hinges are serviceable, do not remove angle from lower hinge. If hinges are to be replaced, drill out rivets in lower hinge same as upper hinge.
6. Straighten hinge pin and remove. Remove spring.
7. Clean up rivet debris.

INSTALL

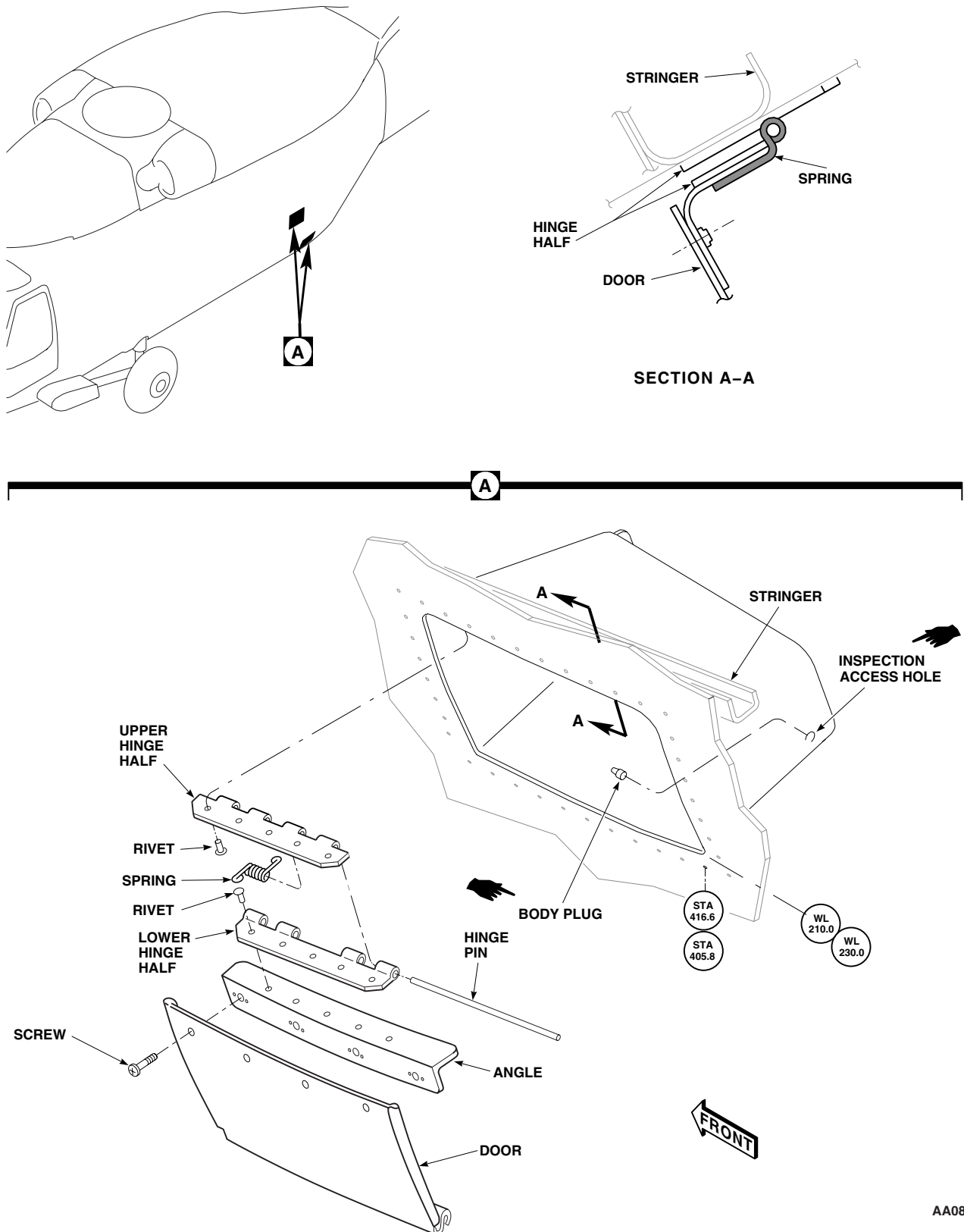
1. Inspect nutplates on angle for damage. If damaged, replace with nutplates, MS21075-08, and rivet to angle using rivets, NAS1097AD3.

NOTE

Hinge halves are made from MS20001PY4-0450 and MS20001PX4-0450. Remove center tang on lower hinge for spring.

2. Layout holes in replacement hinges and drill five holes, using No. 30 drill.
3. If lower hinge was removed from angle, position replacement hinge on angle and hold in place using 1/8-inch diameter clecos in two holes (Figure 1, Detail A).
4. Install blind rivets, NAS1739B4, in between clecos. Remove clecos and install remaining rivets.

INSTALL - Continued



AA0865B
SA

Figure 1. Replace Transition Section Step Door Hinge WL 210.0 and WL 230.0.

INSTALL - Continued**NOTE**

Spring, 70209-03023-101, is made from 0.049-inch diameter 1095 steel, QQ-W-470.

5. Position replacement spring in hinges (Figure 1, Detail A, Section A-A) and insert new hinge pin into hinge assembly. (Cut pin 1 inch longer than hinge).
6. Allow pin to extend 1/2-inch beyond hinge on each end for bending.
7. Bend pin ends enough to hold pin in place.
8. Clamp hinge assembly to remain open. Position upper hinge against step housing and stringer, and hold in place using 1/8-inch diameter clecos.
9. Install blind rivets, NAS1739B4, in between clecos. Remove clecos and install remaining rivets.
10. Release hinge assembly.
11. For corrosion protection and watertightness, seal hinges and other parts which fasten to structure (WP 0377 00).
12. Position door on angle and secure to angle with four screws, MS27039-0806.
13. Check open/close function of door.
14. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPLACE BODY PLUG**REMOVE**

1. Locate body plug, as shown in Figure 1, Detail A.
2. Grasp body plug by hand and pull body plug out while twisting plug.

INSTALL

1. Locate inspection access hole as shown in Figure 1, Detail A
2. Grasp body plug by hand and insert into inspection access hole.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

REPLACE TRANSITION SECTION STEP DOOR HINGE WL 250.0

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Hinge Half, MS20001PX4-0450

Hinge Half, MS20001PY4-0450

Nutplate, MS21075-08

Rivet, NAS1097AD3

Rivet, NAS1097AD4

Screw, MS27039-0806

Sealing Compound, Item 273, WP 1803 00

References

QQ-W-470

TM 55-1500-345-23

WP 0233 00

WP 0258 00

WP 0377 00

WP 0381 00

WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

Assistant - (1)

REMOVE

1. Remove six screws securing door to angle and remove door (Figure 1, Detail A).
2. Open and secure hinge assembly to keep hinges open.
3. Carefully drill out and remove ten rivets (five each assembly) securing upper hinges to step housing and stringer, using 1/8-inch diameter drill and 3/32-inch diameter drive pin punch.
4. Remove hinge assemblies and angle.
5. If spring is broken but hinges are serviceable, do not remove angle from lower hinges. If hinges are to be replaced, drill out rivets in lower hinges, same as upper hinges.
6. Straighten hinge pin and remove. Remove spring.
7. Clean up rivet debris.

INSTALL

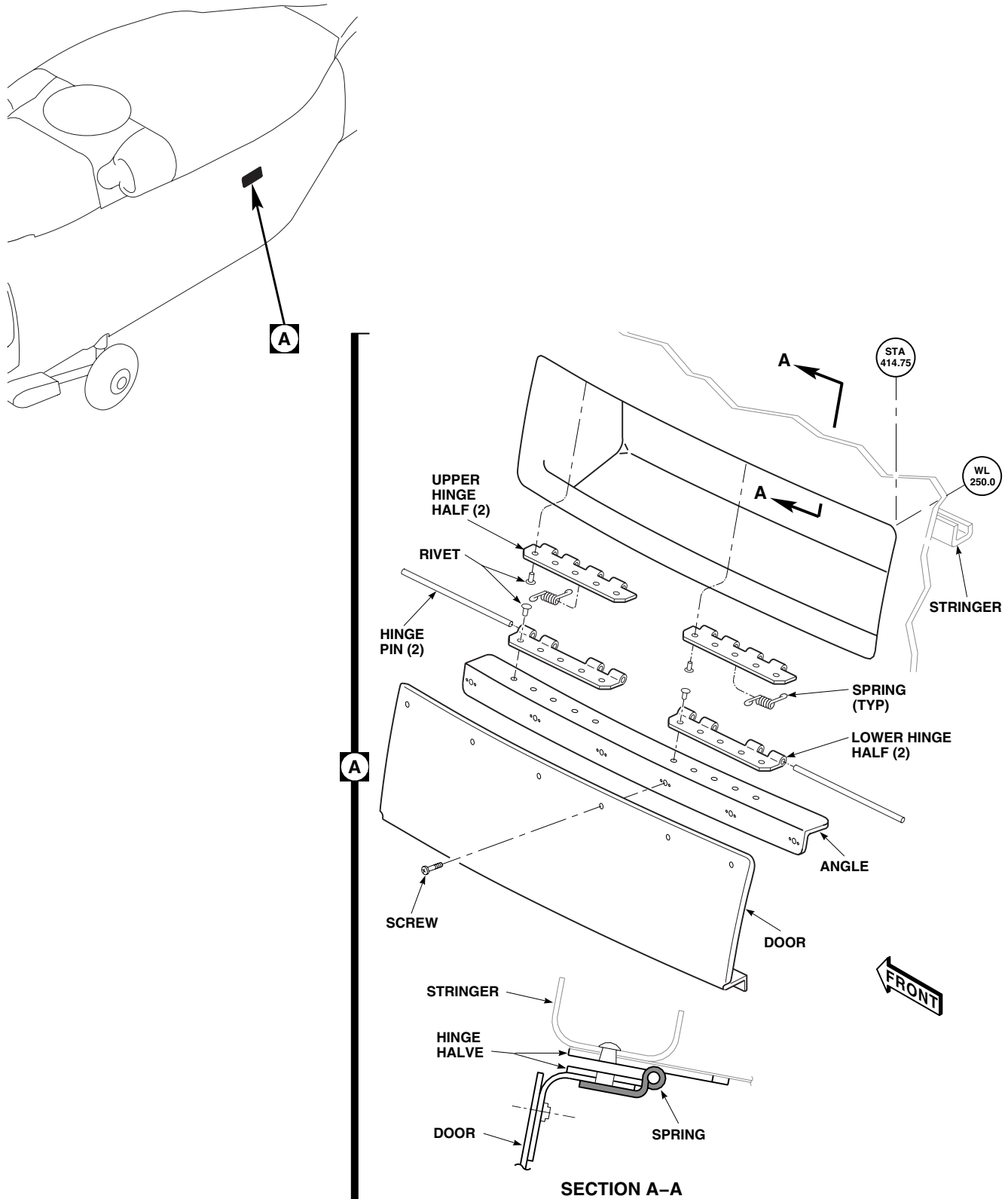
1. Inspect nutplates on angle for damage. If damaged, replace with nutplates, MS21075-08, and rivet to angle, using rivets, NAS1097AD3.

NOTE

Hinge halves are made from MS200001PY4-450 and MS20001PX4-450. Remove center tang on lower hinges for springs.

2. Lay out holes in replacement hinges and drill five holes in each hinge using No. 30 drill.
3. If lower hinges were removed from angle, position replacement hinges on angle and hold in place using 1/8-inch diameter clecos in end holes of each hinge (Figure 1, Detail A).

INSTALL - Continued



AA0866A
SA

Figure 1. Replace Transition Section Step Door Hinge WL 250.0.

INSTALL - Continued

4. Install rivets, NAS1097AD4, in between clecos. Remove clecos and install remaining rivets.

NOTE

Spring, 70209-03023-101, is made from 0.049-inch diameter 1095 steel, QQ-W-470.

5. Position replacement springs in hinges (Figure 1, Detail A, Section A-A) and insert new hinge pins into hinge assemblies.
6. Allow pin to extend at least 1/2-inch beyond hinge on each end for flattening or bending.
7. Flatten or bend pin ends enough to hold pins in place.

NOTE

To get to rear compartment, remove rear cabin troop seats remove soundproofing panel from left (or right) stowage compartment and unhook cargo netting (WP 0233 00 and WP 0258 00).

8. Have assistant inside rear compartment to buck rivets and clean up rivet debris.
9. Clamp hinge assemblies in open position. Position upper hinges against step housing and stringer, and hold in place using 1/8-inch diameter clecos.
10. Install rivets, NAS1097AD4 (five each hinge), in between clecos. Remove clecos and install remaining rivets.
11. Release hinge assemblies.
12. Remove tools, clean up working area, and install cargo hook netting, stowage compartment soundproofing panels, and rear cabin troop seats (WP 0233 00 and WP 0258 00).
13. Seal edges of hinge installation for corrosion protection and watertightness using sealing compound, Item 273, WP 1803 00 (WP 0377 00).
14. Position door on angle and secure to angle with six screws, MS27039-0806.
15. Check open/close function of door.
16. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****TRANSITION SECTION FUEL SUMP DRAIN DOOR**

This WP supersedes WP 0409 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
No. 24 (0.152-Inch Diameter) Drill
Nonmetallic Scraper

Rivet, MS20470AD5

Rivet, NAS1738B5

Sealing Compound, Item 273, WP 1803 00

Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

Material/Parts

Accelerator, Adhesive Curing, Item 3, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Adhesive, Item 33, WP 1803 00
Hinge, MS20001P4-1012
Latch, H-5010-081-196

References

TM 55-1500-345-23
WP 0377 00
WP 0381 00
WP 1803 00

NOTE

Use each removal and installation procedure, as required.

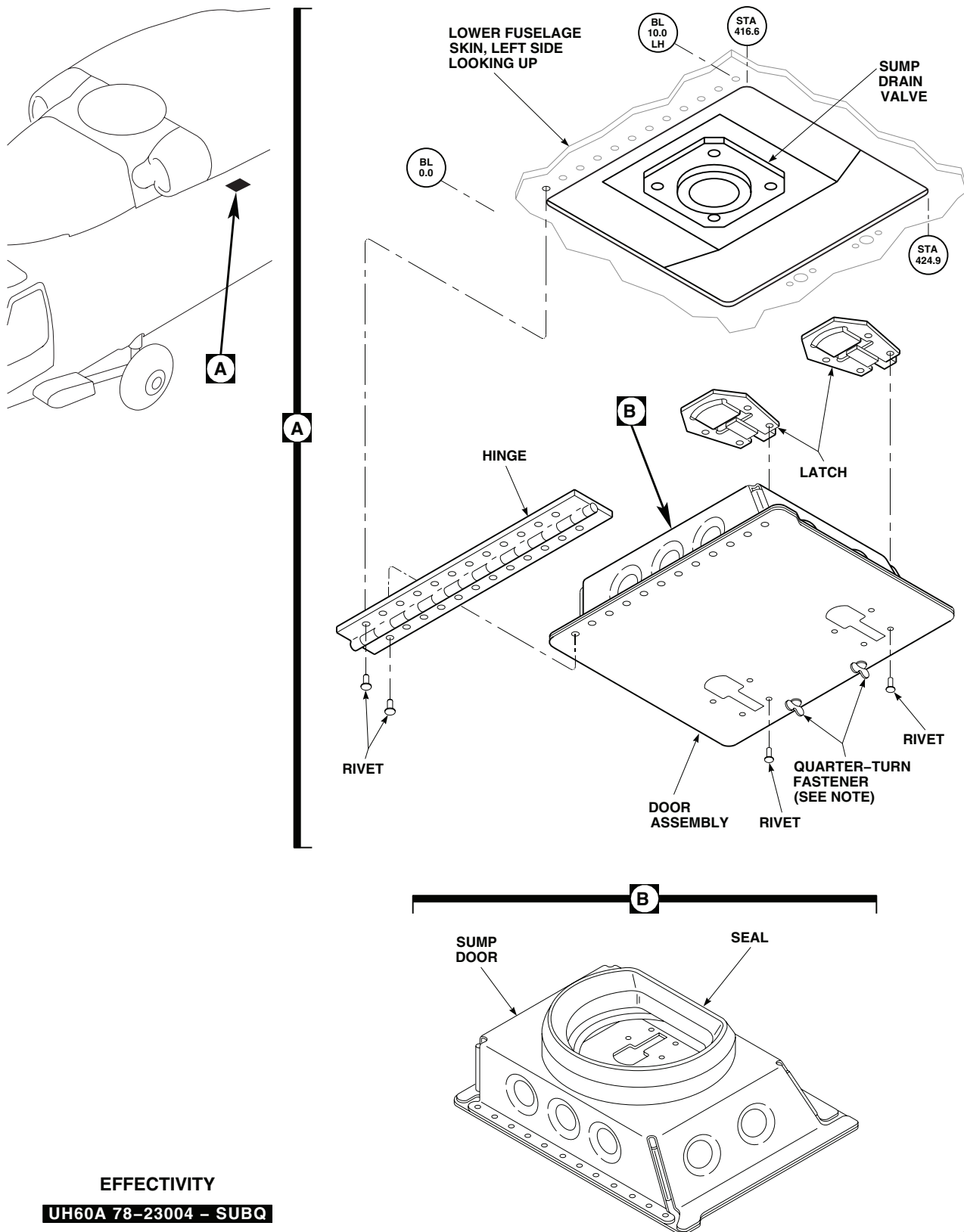
REMOVE

1. Carefully drill out and remove 11 rivets securing hinge to helicopter structure, using No. 24 drill and 1/8-inch diameter drive pin punch (Figure 1, Detail A).
2. Release latches and quarter-turn fasteners and remove door assembly.
3. Carefully drill out and remove 11 rivets securing hinge to door assembly, using No. 24 drill and 1/8-inch diameter drive pin punch.
4. Remove hinge from door assembly.
5. Carefully drill out and remove four rivets securing each latch to door assembly, using No. 24 drill and 1/8-inch diameter drive pin punch.
6. Remove latch or latches, as applicable.
7. Clean up rivet debris.

REPAIR

1. Remove seal from door with nonmetallic scraper (Figure 1, Detail B).

REPAIR - Continued



AA0867C
SA

Figure 1. Replace Latch/Hinge/Door Fuel Sump Drain.

REPAIR - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean bonding surface of door with machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.
3. Mix adhesive, Item 11, WP 1803 00, per manufacturer's instructions.
4. Apply adhesive to door and new seal. Place seal on door.
5. Let adhesive cure for 2 hours at 110°F(43°C) to 130°F(54°C), or for 24 hours at 68°F(20°C).

NOTE

Seal may also be bonded with adhesive, Item 33, WP 1803 00.

6. To bond seal to door with adhesive, Item 33, WP 1803 00, do this:

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition. Volatile and toxic fumes occur when using solvents, causing both a fire and a health hazard.

- a. Clean area to be bonded with machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
 - b. Apply adhesive curing accelerator, Item 3, WP 1803 00, to seal and door. Allow primer to dry at least 3 minutes at 68°F(20°C).
 - c. Apply ribbon of adhesive, Item 33, WP 1803 00, to seal bonding area, and firmly press seal in place on door. Allow to cure for 15 minutes at 68°F(20°C).
 - d. Seal bond joint with sealing compound, Item 273, WP 1803 00 (WP 0377 00).
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL

1. Replace latch, H-5010-081-196, as required.
2. Position latch on door assembly, and hold in place with 5/32-inch diameter clecos in two diagonal corners (Figure 1, Detail A).
3. Install rivets, MS20470AD5, in between clecos. Remove clecos and install remaining rivets.

NOTE

Replace hinge, MS20001P4-1012, as required.

4. Position hinge on door assembly, and hold in place with 5/32-inch diameter clecos in every third hole.
5. Install rivets, MS20470AD5, in between hole holders. Remove clecos and install remaining rivets.

INSTALL - Continued

6. Position door assembly on bottom skin structure, secure latches, and hold door in place with 5/32-inch diameter clecos in every third hole.
7. Check open/close function of door. Adjust door, as required.
8. Install blind rivets, NAS1738B5, in between clecos. Remove clecos and install remaining rivets.
9. Recheck open/close function of door and latches.
10. Secure quarter-turn fasteners.
11. Seal for corrosion protection and watertightness (WP 0377 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

TRANSITION SECTION GRAVITY REFUEL DOOR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Rivet Kit, G-740-ELM

Washer, AN960PD8
Washer, AN960PD8L

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

Material/Parts

Adhesive, Item 14, WP 1803 00
Hinge, MS20001P4-600
Latch, H-5010-064-179
Rivet, MS20470AD5
Rivet, NAS1097AD5
Rivet, NAS1739B5

References

TM 55-1500-345-23
WP 0377 00
WP 0381 00
WP 1803 00

NOTE

Use each removal and installation procedure, as required.

REMOVE

1. Carefully drill out and remove seven rivets securing upper half of hinge to door, using No. 24 drill and 1/8-inch diameter drive pin punch (Figure 1, Detail A).
2. Release latch and remove door assembly.
3. Carefully drill out and remove eight rivets securing lower half of hinge to fuselage structure, using No. 24 drill and 1/8-inch diameter drive pin punch.
4. Remove hinge, plate, and eight washers. Save washers and plate for reinstallation.
5. Carefully drill out and remove four rivets securing latch to door, using No. 24 drill and 1/8-inch diameter drive pin punch.
6. Inspect parts for damage per WP 0377 00. Replace hinge and latch damaged beyond negligible limits.
7. Clean up rivet debris.

INSTALL

1. For corrosion protection and watertightness, seal parts during assembly refer to WP 0377 00.
2. Replace latch, H-5010-064-179, as required.
3. Position latch on door assembly and hold in place with 5/32-inch diameter clecos in two diagonal corners (Figure 1, Detail A).
4. Install rivets, MS20470AD5, between clecos. Remove clecos and install remaining rivets.

INSTALL - Continued

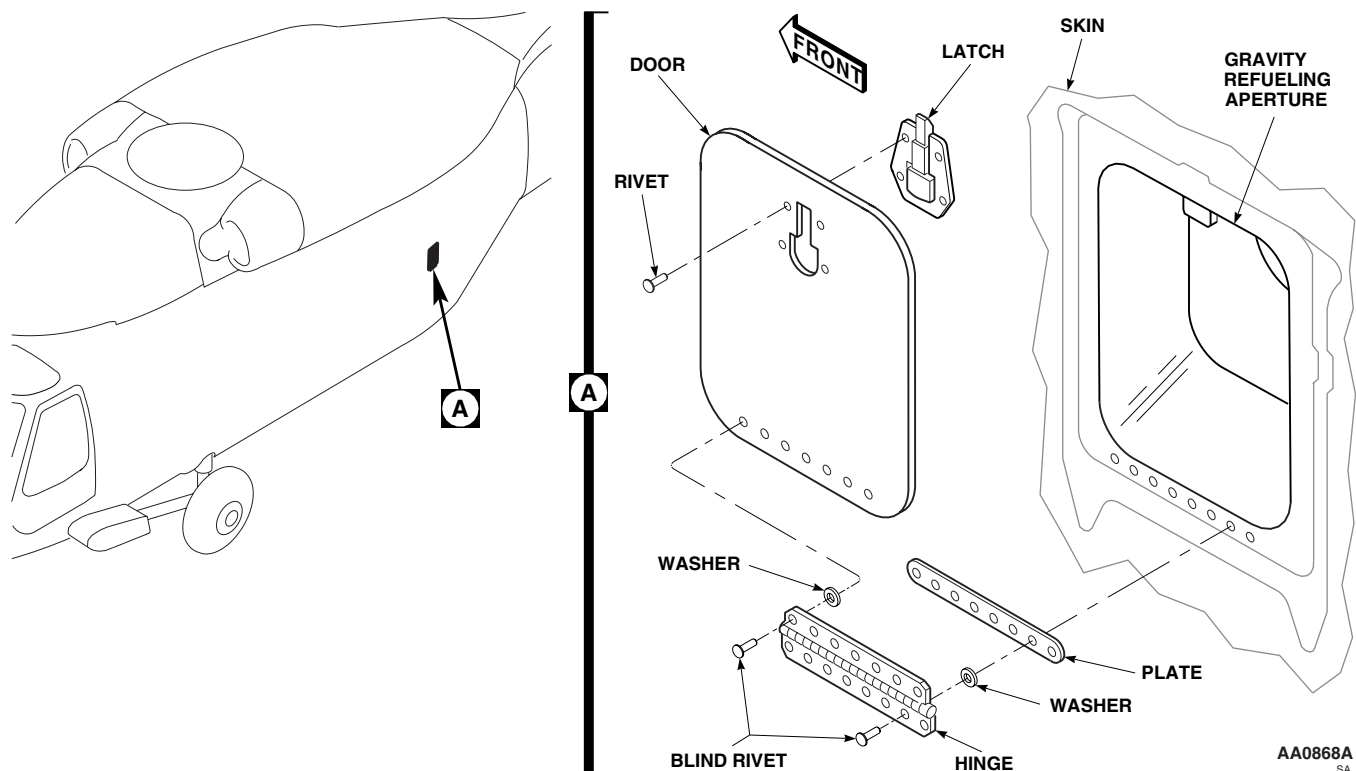


Figure 1. Replace Latch/Hinge/Door - Gravity Refueling.

NOTE

Replace hinge, MS20001PH-600, as required.

5. Position upper half of hinge on door. Place, under hinge, two washers, AN960PD8, on end holes and two washers, AN960PD8L, on holes second from the end. Hold in place with 5/32-inch diameter clecos.
6. Install countersunk rivets, NAS1097AD5, in between clecos. Remove clecos and install remaining rivets.
7. Bond plate to housing with adhesive, Item 14, WP 1803 00, per manufacturer's instructions.
8. Position door assembly and four washers over plate on aperture housing. Latch door to housing. Place between hinge and plate two washers, AN960PD8, on end holes and two washers, AN960PD8L, on holes second from end. Hold in place with 5/32-inch diameter clecos.
9. Check open/close function of door. Adjust, as required.
10. Install countersunk blind rivets, NAS1739B5, in between clecos. Remove clecos and install remaining rivets.
11. Recheck open/close function of door.
12. For corrosion protection and watertightness, seal edges around assembled parts (WP 0377 00).
13. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
14. Close and latch door.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****TRANSITION SECTION PRESSURE REFUEL DOOR**

This WP supersedes WP 0411 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Blind Rivet Kit, G-740-ELM

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

Material/Parts

Hinge, MS20001P4-600
Latch, H-5010-064-179
Rivet, MS20470AD5
Rivet, NAS1738B5

References

TM 55-1500-345-23
WP 0377 00
WP 0378 00
WP 0381 00

NOTE

Use each removal and installation procedure, as required.

REMOVE

1. Carefully drill out and remove six rivets securing door assembly to upper half of hinge, using No. 24 drill and 1/8-inch diameter drive pin punch (Figure 1, Detail A).
2. Release latches and remove door assembly.
3. Carefully drill out and remove four rivets securing each latch to door assembly, using No. 24 drill and 1/8-inch diameter drive pin punch.
4. Remove latches from door.
5. Carefully drill out and remove six rivets securing lower half of hinge to aperture housing, using No. 24 drill and 1/8-inch diameter drive pin punch. Remove hinge.
6. Inspect parts for damage (WP 0377 00). Replace latches and hinge damaged beyond negligible limits. Refer to, (WP 0378 00) for repair of door.
7. Clean up rivet debris.
8. Touch up protective finishes as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL

1. For corrosion protection and watertightness seal parts during assembly (WP 0377 00).
2. Replace latch, H-5010-064-179, as required.
3. Position latch (2) on door assembly and hold in place with 5/32-inch diameter clecos in two diagonal holes (Figure 1, Detail A).
4. Install rivets, MS20470AD5, in between clecos. Remove clecos and install remaining rivets.

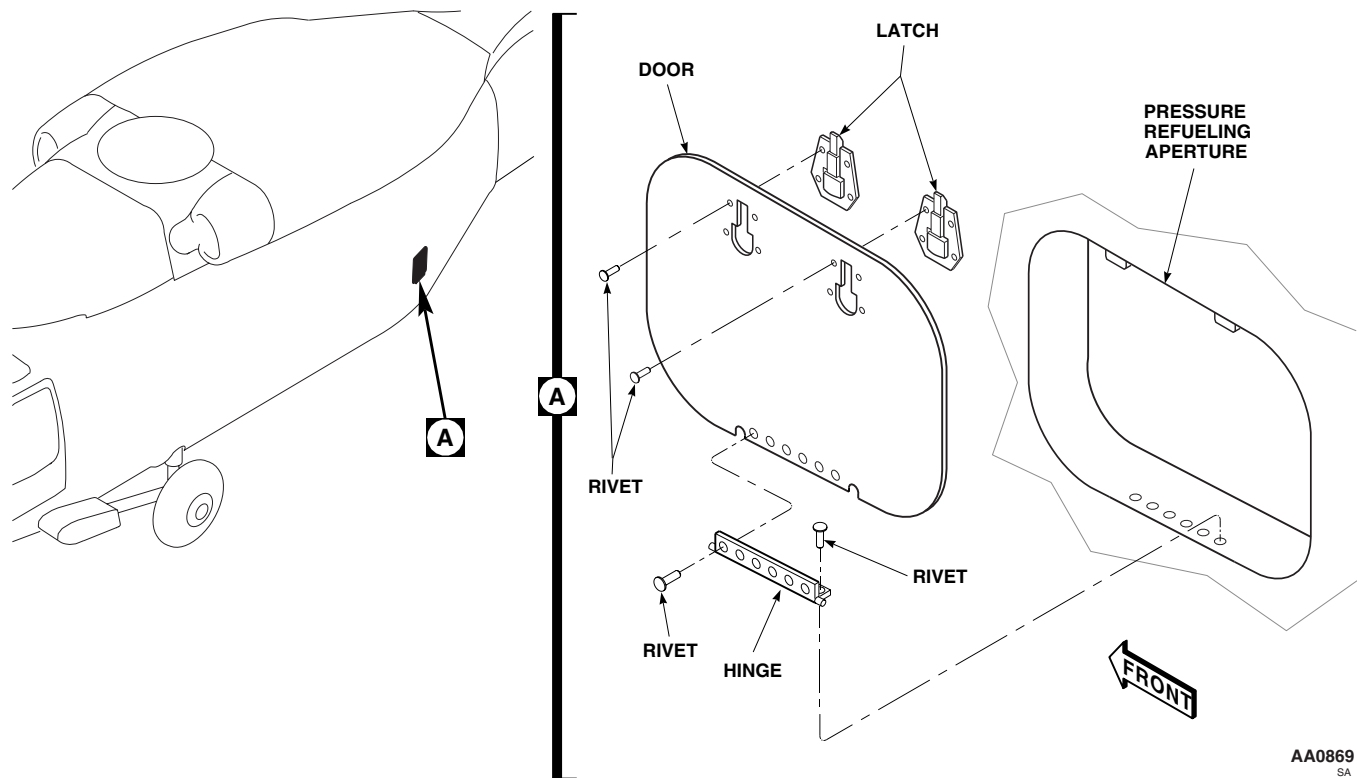
INSTALL - Continued

Figure 1. Replace Latch/Hinge/Door Pressure Refueling.

NOTE

Replace hinge, MS20001P4-600, as required.

5. Position lower half of hinge on aperture housing and hold in place with 5/32-inch diameter clecos in end holes.
6. Install countersunk blind rivets, NAS1738B5, in between clecos. Remove clecos and install remaining rivets.
7. Position door assembly on aperture housing and engage latch. Line up upper half of hinge with fastener holes in door and hold in place using 5/32-inch diameter clecos in end holes.
8. Check open/close function of door. Adjust hinge, as required.
9. Install rivets, MS20470AD5, between clecos. Remove clecos and install remaining rivets.
10. Recheck open/close function of door and latches.
11. For corrosion protection and watertightness, seal edges of assembly (WP 0377 00).
12. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
13. Close and latch door.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

TRANSITION SECTION PNEUMATIC GROUND START DOOR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Latch, H-5010-064-179

Rivet, MS20470AD4

Rivet, MS20470AD5

Sealing Compound, Item 273, WP 1803 00

References

TM 1-1500-204-23

TM 55-1500-345-23

WP 0377 00

WP 0381 00

WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

NOTE

Use each removal and installation procedure, as required.

REMOVE

1. Carefully drill out and remove six rivets securing door assembly to upper half of hinge (TM 1-1500-204-23) (Figure 1, Detail A).
2. Release latch and remove door.
3. Carefully drill out and remove six rivets securing lower half of hinge to housing, using No. 32 drill and 3/32-inch diameter drive pin punch. Remove hinge.
4. Carefully drill out and remove four rivets securing latch to door assembly (TM 1-1500-204-23). Remove latch.
5. Inspect parts for damage (WP 0377 00). Replace latch and hinge damaged beyond negligible limits. Refer to, INSTALL, in this work package for repair of door.
6. Clean up rivet debris.
7. Touch up protective finishes as required (WP 0381 00 and TM 55-1500-345-23).

INSTALL

1. For corrosion protection and watertightness, seal hinge parts and rivets during assembly (WP 0377 00).
2. Replace latch, H-5010-064-179, as required.
3. Position latch on door assembly and hold in place with 5/32-inch diameter clecos in two diagonal holes.
4. Install rivets, MS20470AD5, in between clecos (Figure 1, Detail A). Remove clecos and install remaining rivets.

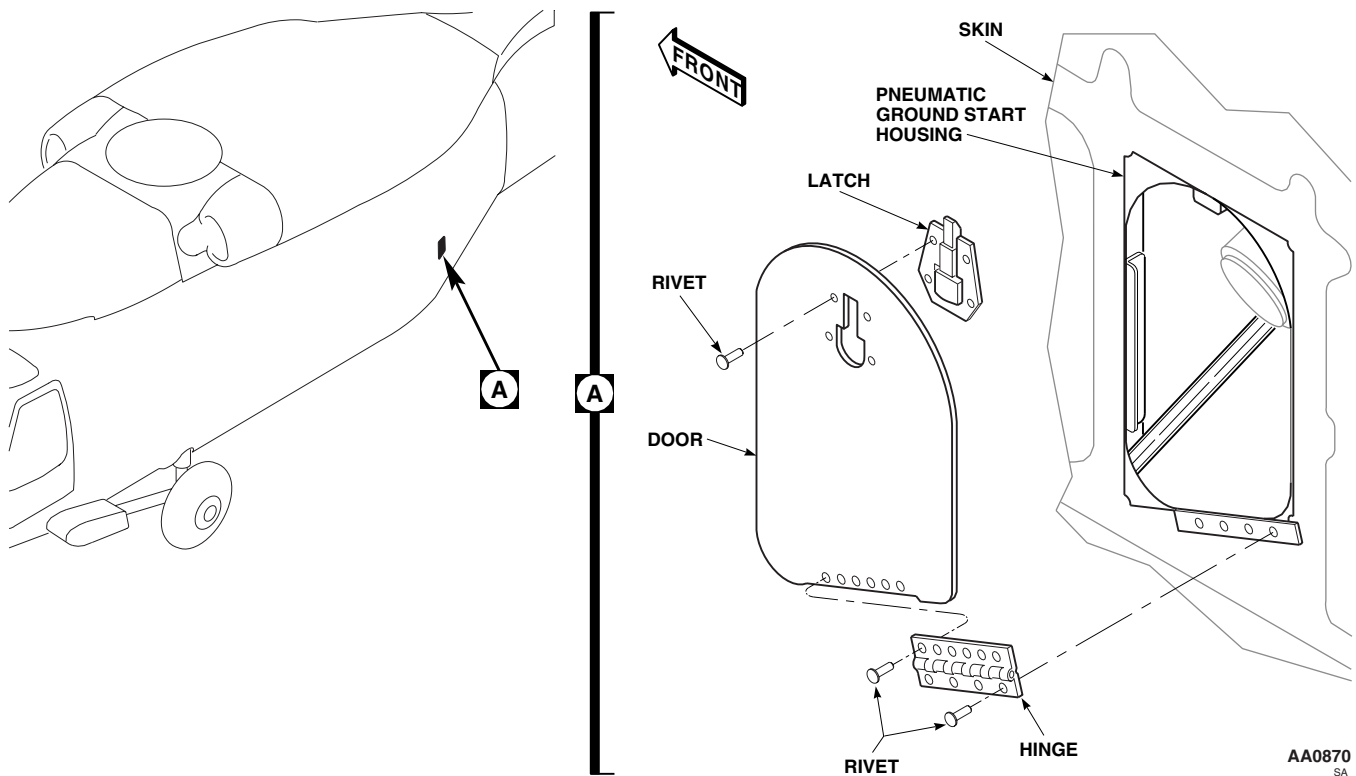
INSTALL - Continued

Figure 1. Replace Latch/Hinge/Door Pneumatic Ground Start.

NOTE

Replace hinge, MS20001P4-375, as required.

5. Position lower half of hinge on housing and hold in place with 5/32-inch diameter clecos in end holes of hinge.
6. Install rivets, MS20470AD5, in between clecos. Remove clecos and install remaining rivets.
7. Position door assembly on housing and engage latch. Line up upper half of hinge with fastener holes in door and hold in place with 1/8-inch diameter clecos in end holes.
8. Check open/close function of door. Adjust hinge, as required.
9. Install rivets, MS20470AD4, in between clecos. Remove clecos and install remaining rivets, check open/close function of door and latch.
10. Seal for watertightness using sealing compound, Item 273, WP 1803 00 (WP 0377 00).
11. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
12. Close and latch door.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****TRANSITION SECTION APU ACCESS DOOR SCREWS REPAIR.**

This WP supersedes WP 0413 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield

Sealing Compound, Item 286, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Sealing Compound, Item 284, WP 1803 00

References

WP 1803 00

REPAIR

1. Remove APU access door hinge fitting screws and center beam screws.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Clean screw and nutplate threads with technical acetone, Item 5, WP 1803 00.
3. Treat screw and nutplate threads with sealing compound, Item 286, WP 1803 00. Allow to dry for at least 5 minutes.
4. Apply sealing compound, Item 284, WP 1803 00.
5. Install screws and washers.

END OF WORK PACKAGE

UNIT LEVEL
AIRFRAME
GROUNDING RECEPTACLE

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Digital Low-Resistance Ohmmeter, DLRO247000

References

TM 55-1500-345-23
WP 0381 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

NOTE

Replacement procedures are typical for all grounding receptacles.

INSPECT

1. Measure resistance between grounding receptacle and an adjacent point on helicopter skin using digital low-resistance multimeter. **RESISTANCE SHALL NOT BE MORE THAN 1 OHM.**
2. If resistance is more than 1 ohm, remove grounding receptacle. Refer to, REMOVE, in this work package. Replace chemical conversion coated surfaces (WP 0381 00).

REMOVE

1. Remove nut, lockwasher, and washer from grounding receptacle (Figure 1, Detail A).
2. Remove grounding receptacle from helicopter structure.
3. Remove washer from grounding receptacle.

INSTALL

1. Install washer on grounding receptacle (Figure 1, Detail A).
2. Position grounding receptacle through helicopter structure.
3. Install nut, lockwasher, and washer on grounding receptacle. Tighten nut on grounding receptacle.
4. Inspect grounding receptacle. Refer to, INSPECT, in this work package.
5. Touch up paint finish, as required (WP 0381 00 and TM 55-1500-345-23).

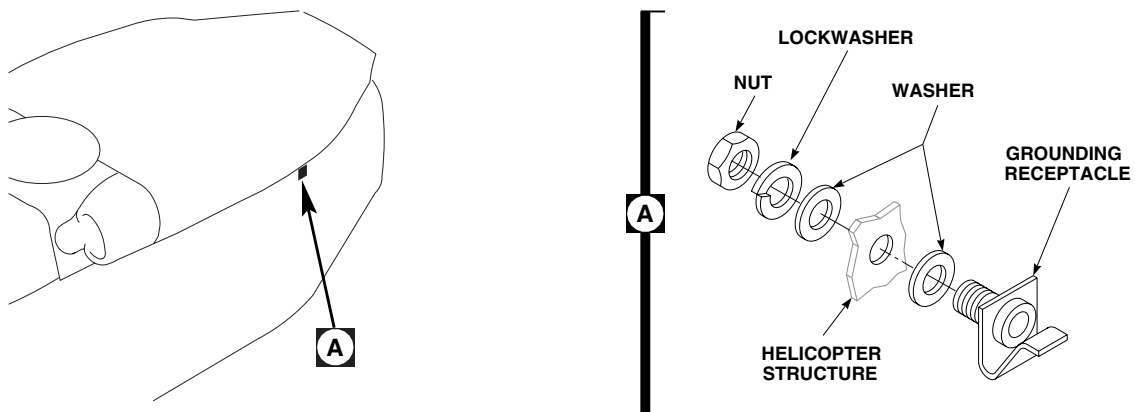
INSTALL - ContinuedAB2701
SA

Figure 1. Replace Grounding Receptacle.

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TAILCONE FRAMING REPAIR

This WP supersedes WP 0415 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
 Gloves
 Goggles/Face Shield
 Nonmetallic Scraper
 Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cloth, Abrasive, Item 83, WP 1803 00
 Corrosion Removing Compound, Item 108,
 WP 1803 00
 Hinge, MS20001-4
 Latch, H80-1 (83014)
 Pad, Scouring, Item 205, WP 1803 00
 Primer, Adhesive, Item 240, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 330,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

MIL-S-18729
 MIL-S-5059
 QQ-A-200
 QQ-A-225
 QQ-A-250
 QQ-A-367
 TM 1-1500-344-23
 TM 55-1500-345-23
 WP 0377 00
 WP 0378 00
 WP 0380 00
 WP 0381 00
 WP 0418 00
 WP 1803 00

REPAIR

1. Tailcone framing includes formed and extruded members made from 7075-T6, QQ-A-250-13, Alclad sheet stock and 7075-T6511, QQ-A-200-11 aluminum alloy extrusion. Fittings at STA 605.0 (landing gear support) and STA 648.0 (canted hinge bulkhead fitting) are made from forged aluminum alloy, 7075-T73, QQ-A-367. Blade stowage fittings (STA 585.0 and STA 624.0) are made from 7075-T6, QQ-A-200-11 (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, Figure 1, Sheet 7, Figure 1, Sheet 8, Figure 1, Sheet 9, Figure 1, Sheet 10, Figure 1, Sheet 11, and Figure 1, Sheet 12).
2. All tailcone framing is primary structure, except the tail drive shaft cover supports and formers which are secondary structure. The forged fittings (STA 605.0 and STA 648.0) are considered critical primary structure. Negligible damage to these fittings is limited to 5% of material thickness at point of damage. Blend damage smoothly to surrounding area using long even strokes. Polish with abrasive cloth, Item 83, WP 1803 00, and restore protective finish. Damage greater than negligible limits to the fittings must be reviewed by engineering authority for damage analysis, and either repair or replacement. All repairs to tailcone framing must maintain the designed strength and alignment.
3. Evaluate damage in noncritical areas (WP 0377 00).
4. Use WP 0378 00 and WP 0380 00 for typical framing repairs.
5. Refer to WP 0418 00 for drive shaft cover repairs.
6. For corrosion protection and watertightness, seal all repairs of mating tailcone framing with plating (WP 0377 00).

REPAIR - Continued

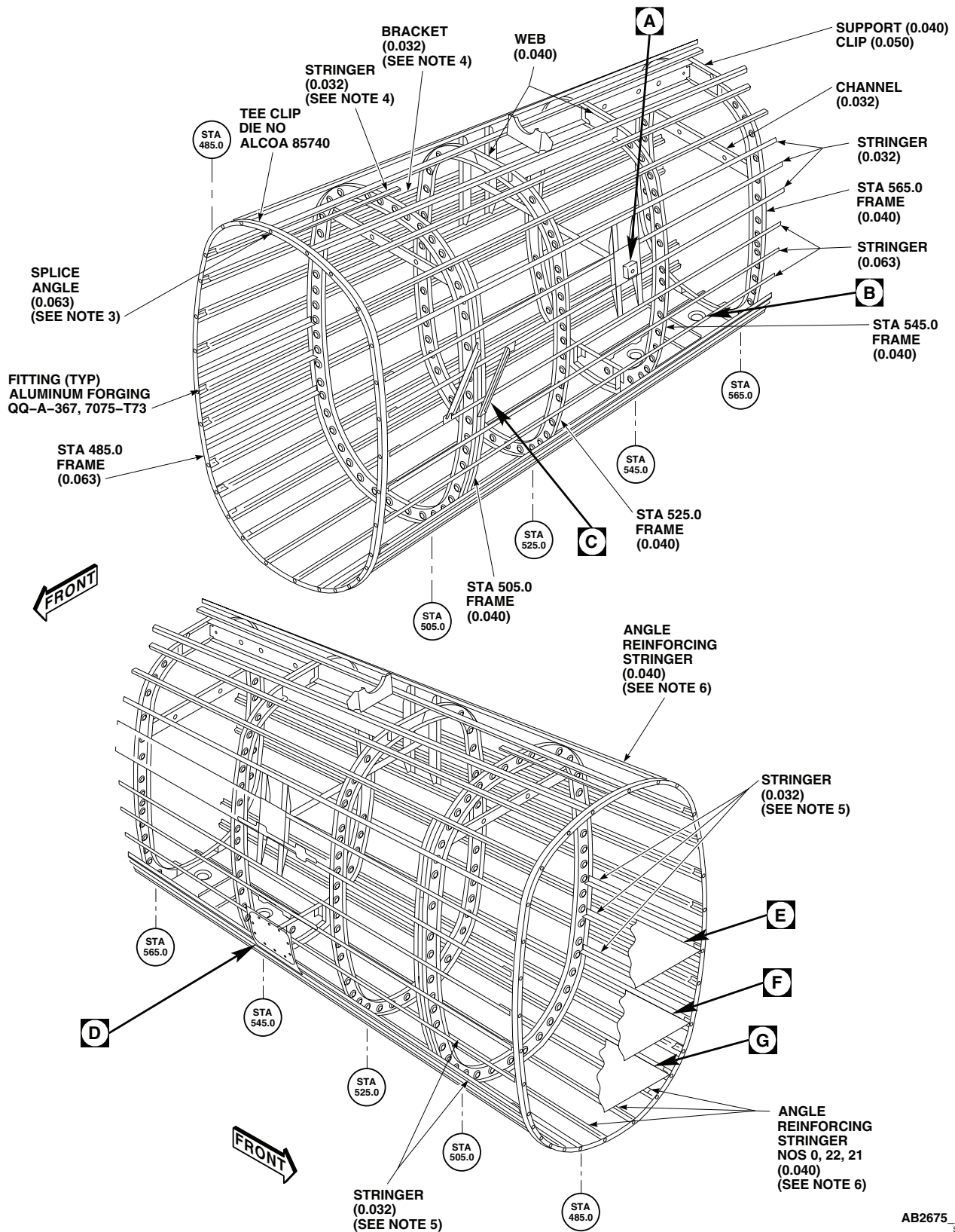
AB2675_1
SA

Figure 1. Repair Tailcone Framing. (Sheet 1 of 12)

REPAIR - Continued

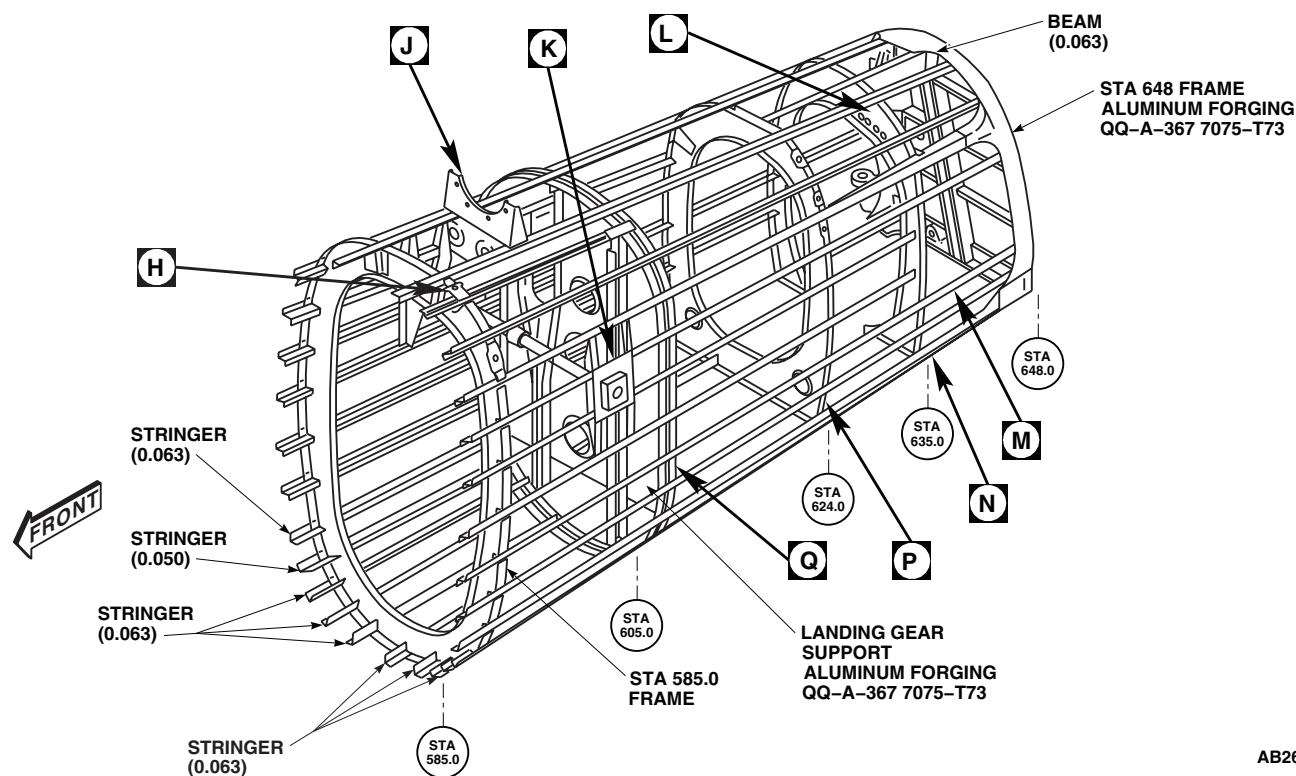
AB2675_2
SA

Figure 1. Repair Tailcone Framing. (Sheet 2 of 12)

7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
8. To repair polyethylene abrasion strip on anticollision light support, do this:
 - a. Stow anticollision light.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- b. Remove old abrasion strip with nonmetallic scraper. Clean area with technical acetone, Item 5, WP 1803 00, and wipe area dry with machinery wiping towel, Item 344, WP 1803 00.
 - c. Mask area where tape is to be applied.
 - d. Spray area with one coat of adhesive primer, Item 240, WP 1803 00.
 - e. Apply pressure sensitive adhesive tape, Item 330, WP 1803 00, to primed area.
9. Visually inspect exposed area of blade stowage fittings for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film.
10. To remove corrosion from blade stowage fittings do this:
 - a. Remove corrosion manually, using scouring pad, Item 205, WP 1803 00.

Exploded view diagram of a rectangular assembly. The central component is a rectangular block with a circular feature in the center, labeled **FITTING QQ-A-250-9 7075-T651**. Surrounding this central block are eight **BRACKET (0.040)** components, four on each long side. These brackets are shown in an exploded state, with dashed lines indicating their alignment and position relative to the central block. The entire assembly is mounted on a larger, irregularly shaped base, which is labeled **FORMER (0.040)** at two locations. The diagram uses solid lines for the components and dashed lines for alignment guides.

AB2675_3
SA

Change 3

REPAIR - Continued

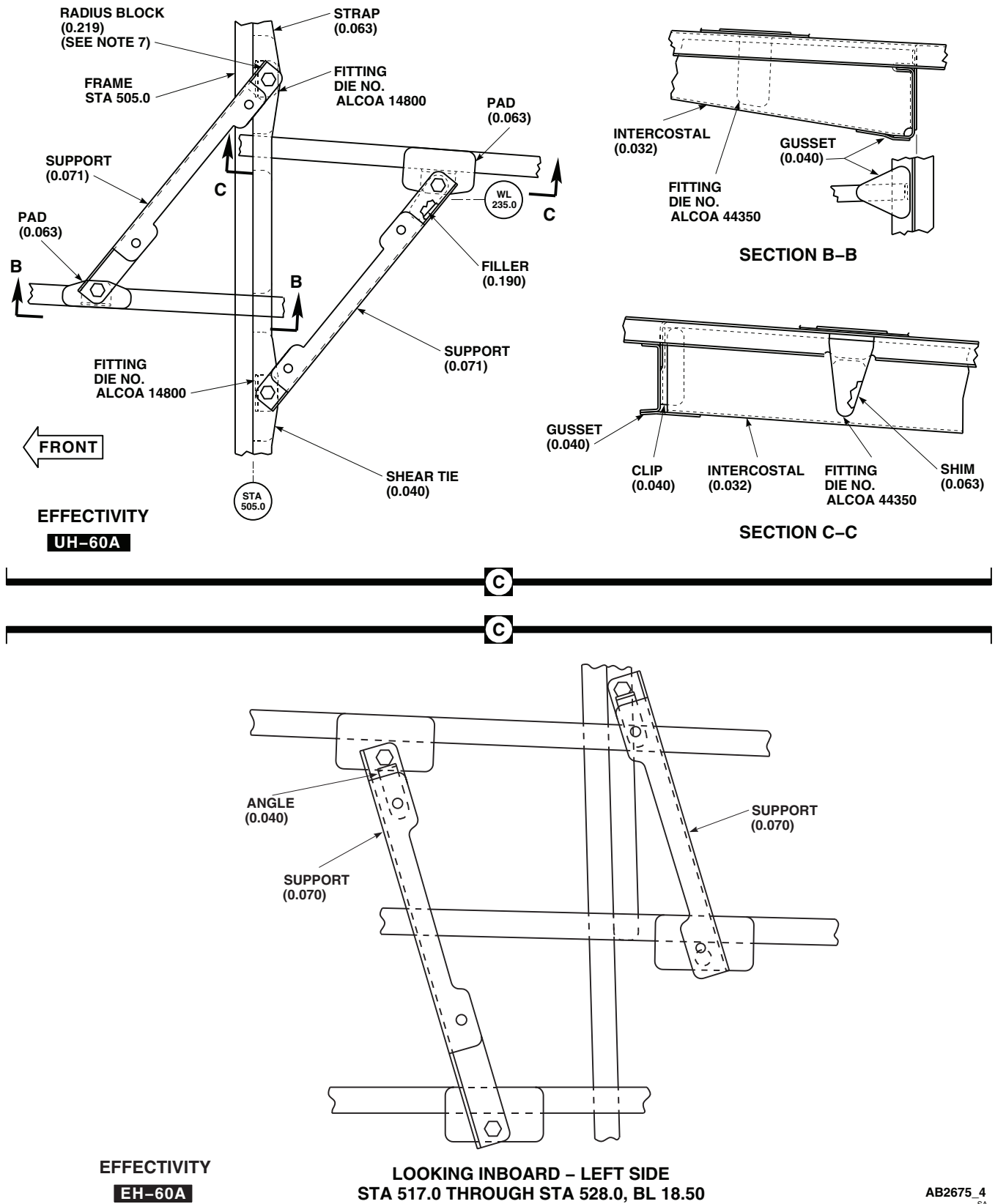


Figure 1. Repair Tailcone Framing. (Sheet 4 of 12)

REPAIR - Continued

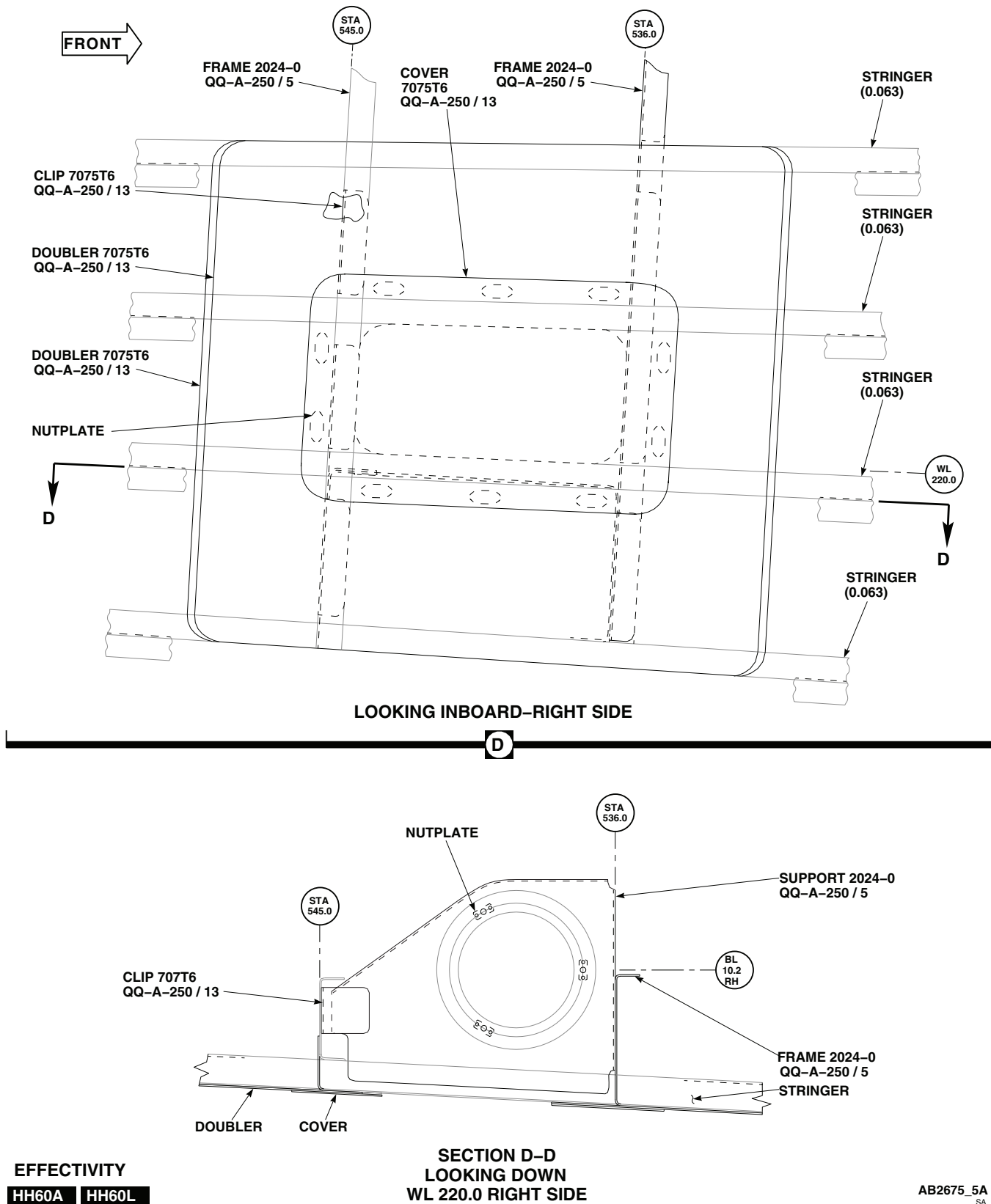
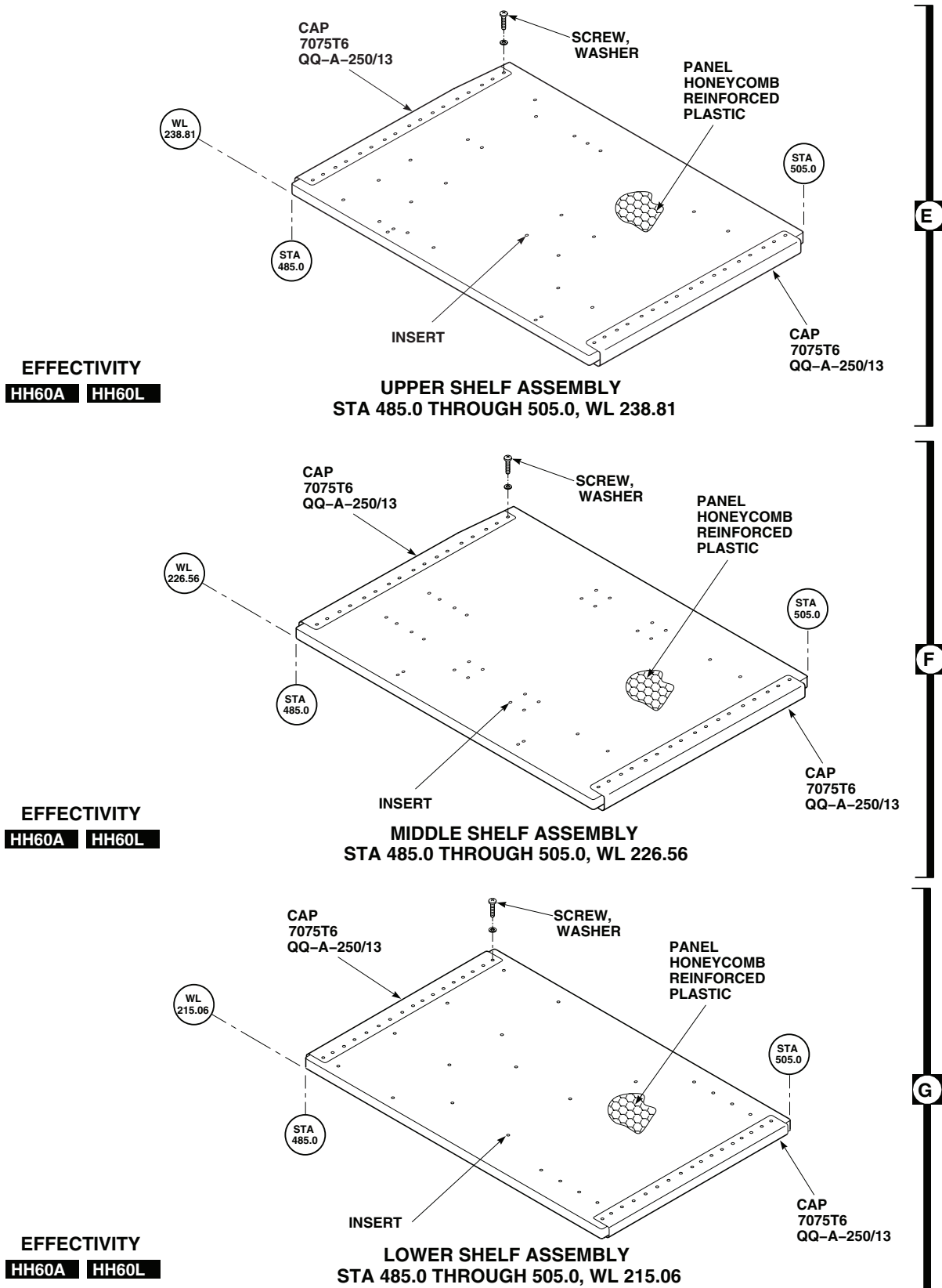


Figure 1. Repair Tailcone Framing. (Sheet 5 of 12)

REPAIR - Continued



AB2675_6A
SA

Figure 1. Repair Tailcone Framing. (Sheet 6 of 12)

REPAIR - Continued

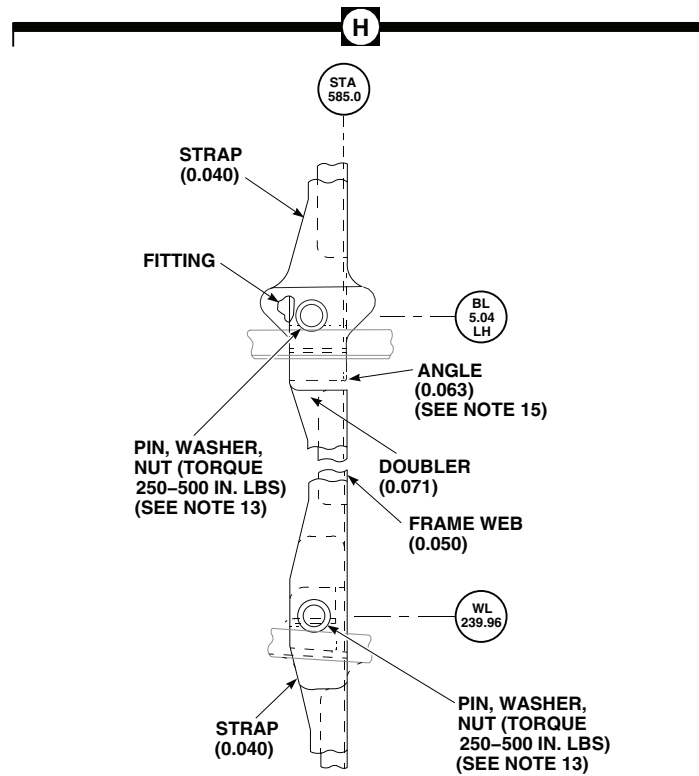
AB2675_7
SA

Figure 1. Repair Tailcone Framing. (Sheet 7 of 12)

- b. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, to 1 part water.
 - c. Apply compound solution to corroded areas.
 - d. Rinse thoroughly with clean water.
11. **REPLACE BLADE STOWAGE FITTING WHEN BLADE SUPPORT RETAINER NO LONGER FITS SECURELY IN FITTING.**

REPAIR - Continued

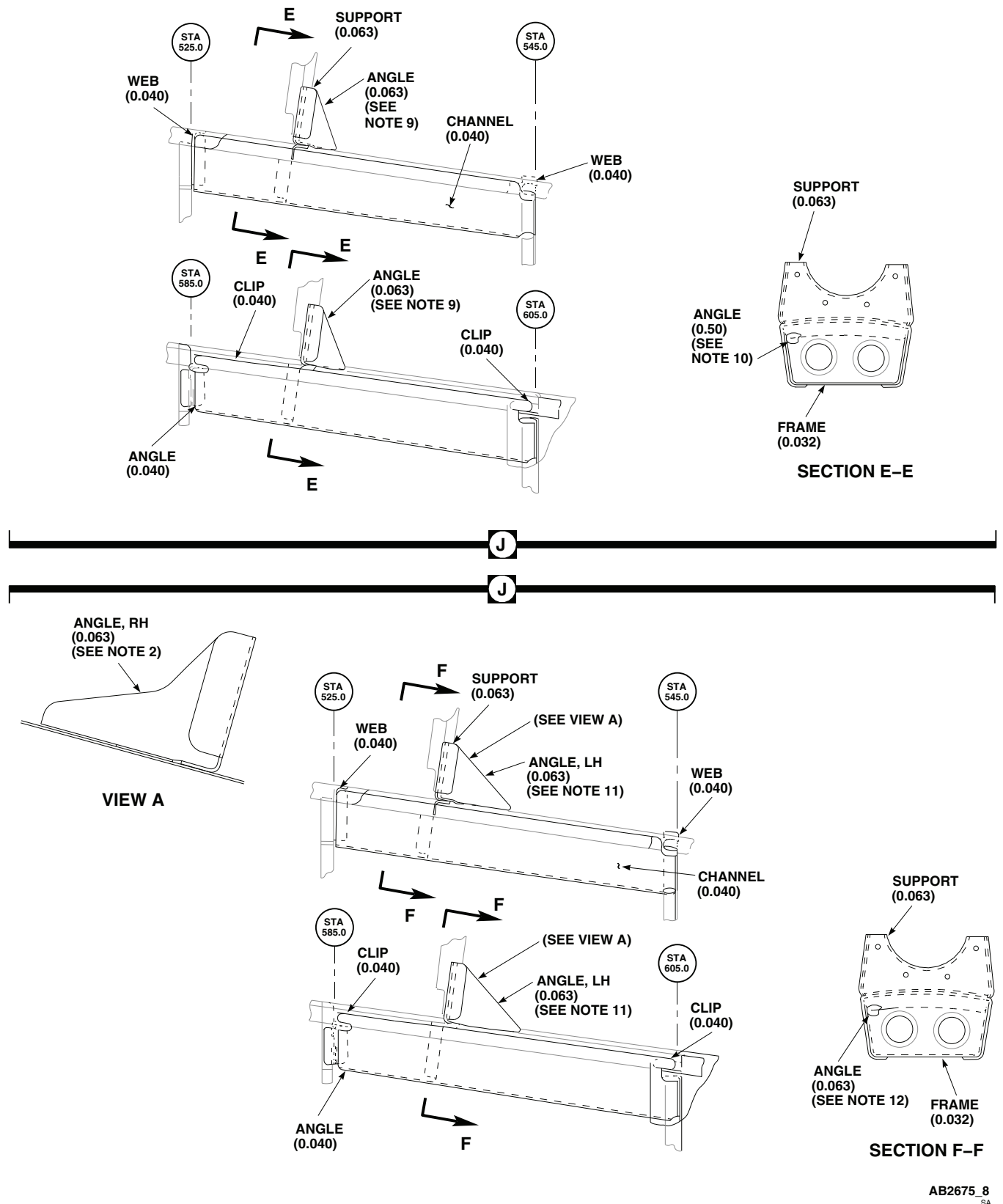
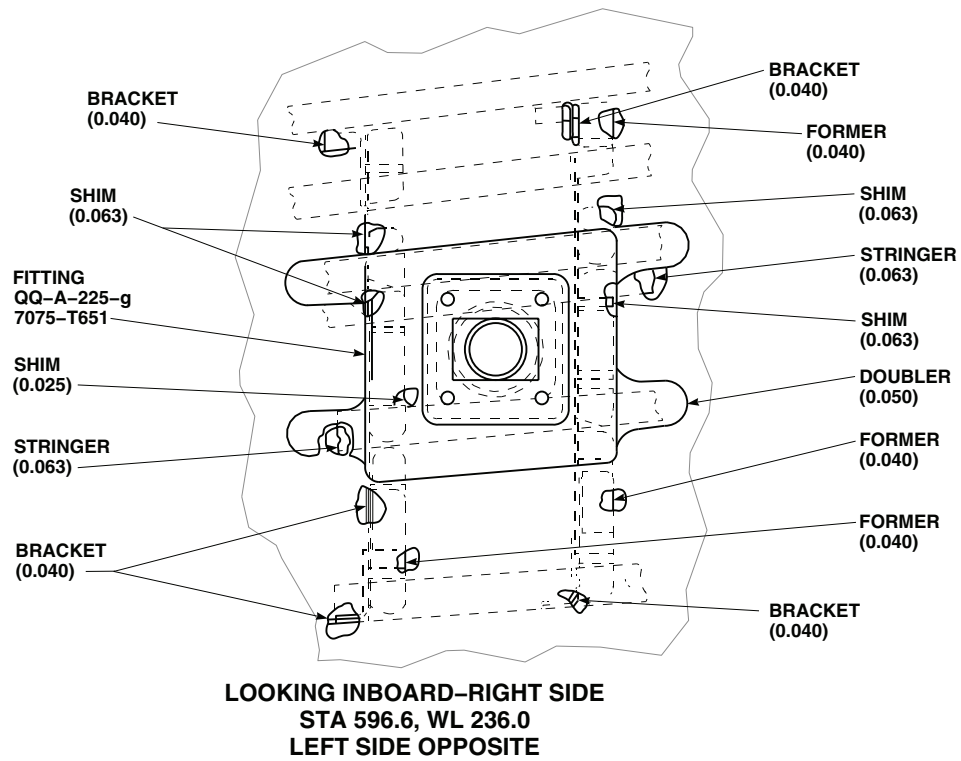
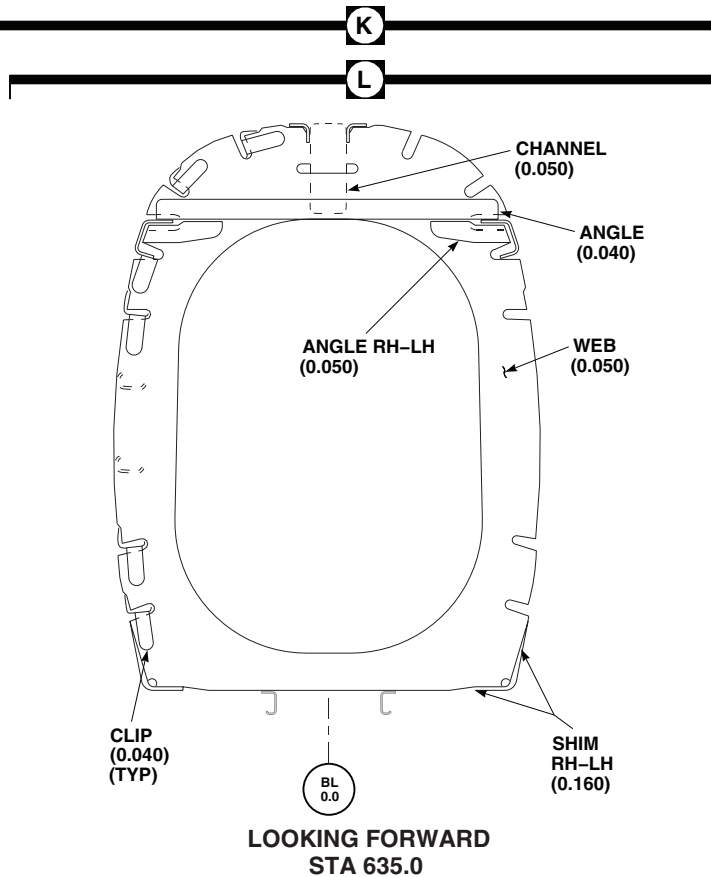


Figure 1. Repair Tailcone Framing. (Sheet 8 of 12)

REPAIR - Continued



EFFECTIVITY
EH-60A



AB2675_9
SA

Figure 1. Repair Tailcone Framing. (Sheet 9 of 12)

REPAIR - Continued

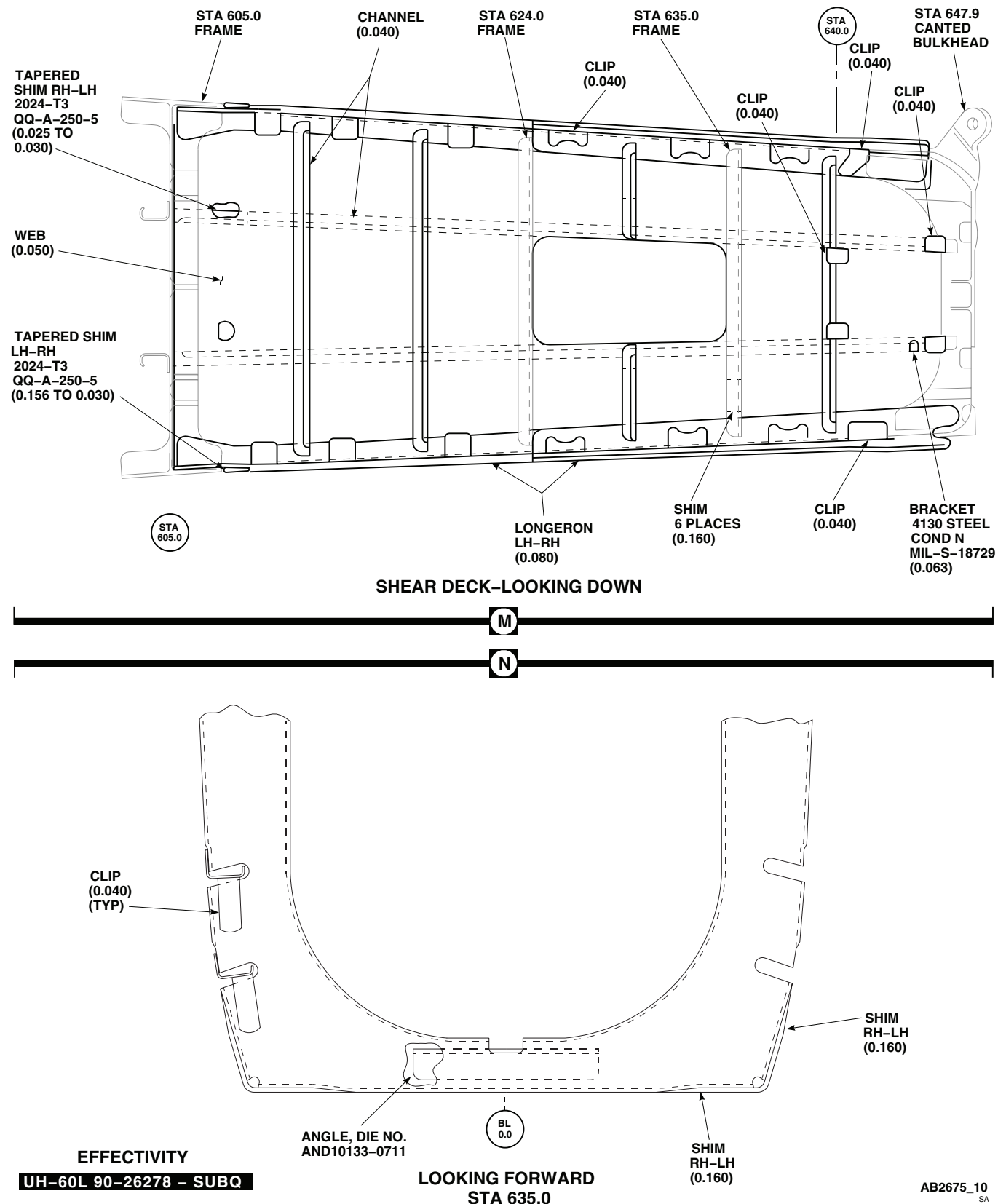


Figure 1. Repair Tailcone Framing. (Sheet 10 of 12)

REPAIR - Continued

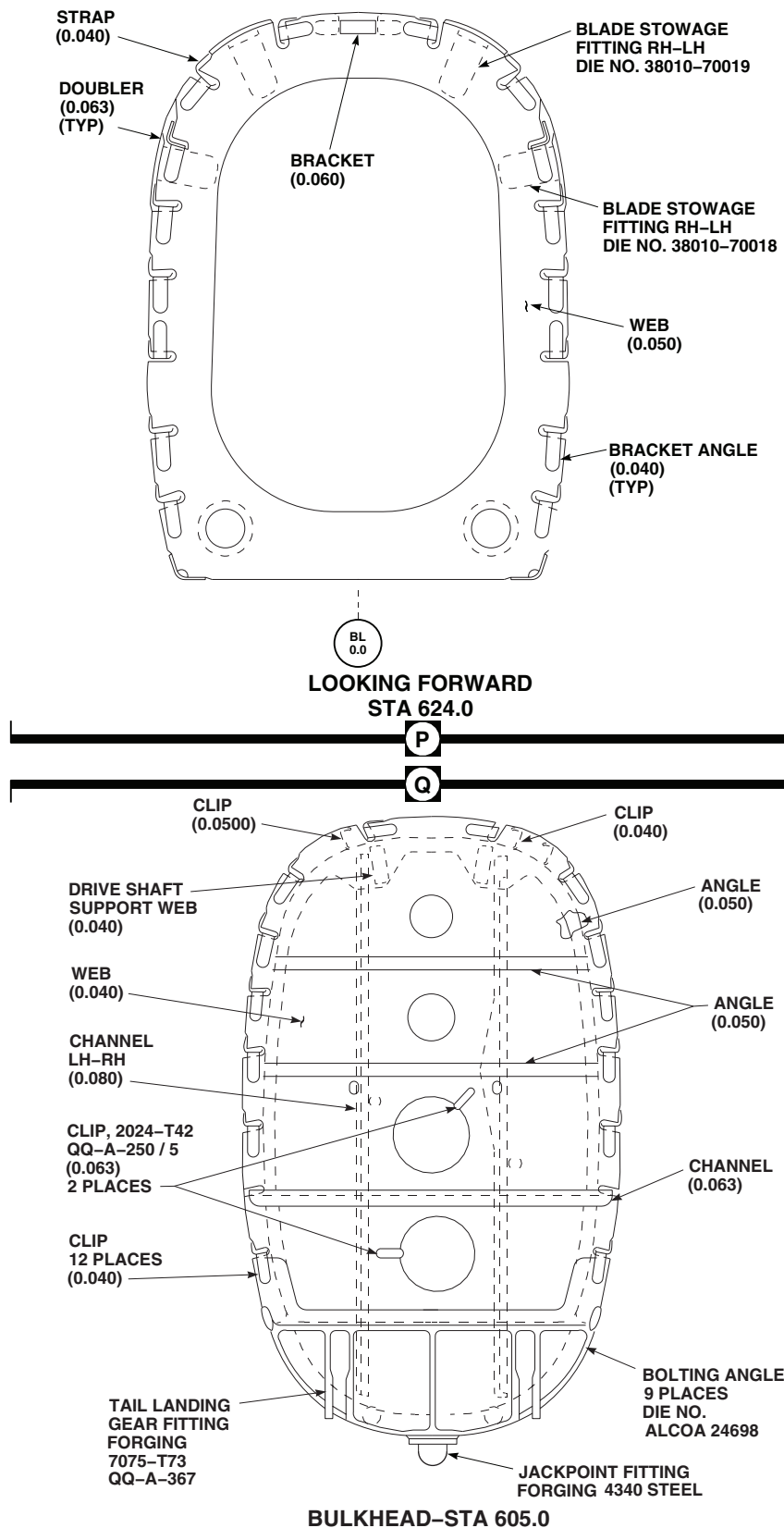
AB2675_11
SA

Figure 1. Repair Tailcone Framing. (Sheet 11 of 12)

REPAIR - Continued**NOTES:**

1. UNLESS OTHERWISE NOTED ON SHEETS OF THIS FIGURE, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSIONS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT AS NOTED.
3. **UH-60A 77-22714 - 79-23274** TOP ONLY. **UH-60A 79-23275 - SUBQ** **UH-60L** **EH-60A**
TOP AND BOTTOM.
4. **UH-60A 86-24545 - SUBQ** **UH-60L** **EH-60A**
5. **UH-60A 86-24547 - SUBQ** **UH-60L** **EH-60A**
6. **EH-60A**
7. **UH-60A 81-23556 - SUBQ** **EH-60A**
8. **UH-60A 79-23343 - SUBQ** **UH-60L** **EH-60A**
9. **UH-60A 77-22714 - 88-26062** **EH-60A 84-24017 - 87-24671**
10. **UH-60A 77-22714 - 86-24521** **EH-60A 84-24017 - 85-24471**
11. **UH-60A 88-26063 - SUBQ** **UH-60L** **EH-60A 87-24672 - SUBQ**
12. **UH-60A 86-24521 - SUBQ** **UH-60L** **EH-60A 85-24471 - SUBQ**
13. IF HARDWARE IS STAINLESS STEEL, ASSEMBLE WET WITH SEALING COMPOUND, ITEM 273, WP 1803 00.
14. **UH-60A 89-26156 - SUBQ** **UH-60L**
15. **UH-60A 89-26156 - SUBQ** **UH-60L 89-26179 - SUBQ**

AB2675_12B
SA

Figure 1. Repair Tailcone Framing. (Sheet 12 of 12)

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

TAILCONE PLATING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

WP 0377 00

WP 0378 00

WP 0379 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

WP 0381 00

WP 0418 00

References

QQ-A-250

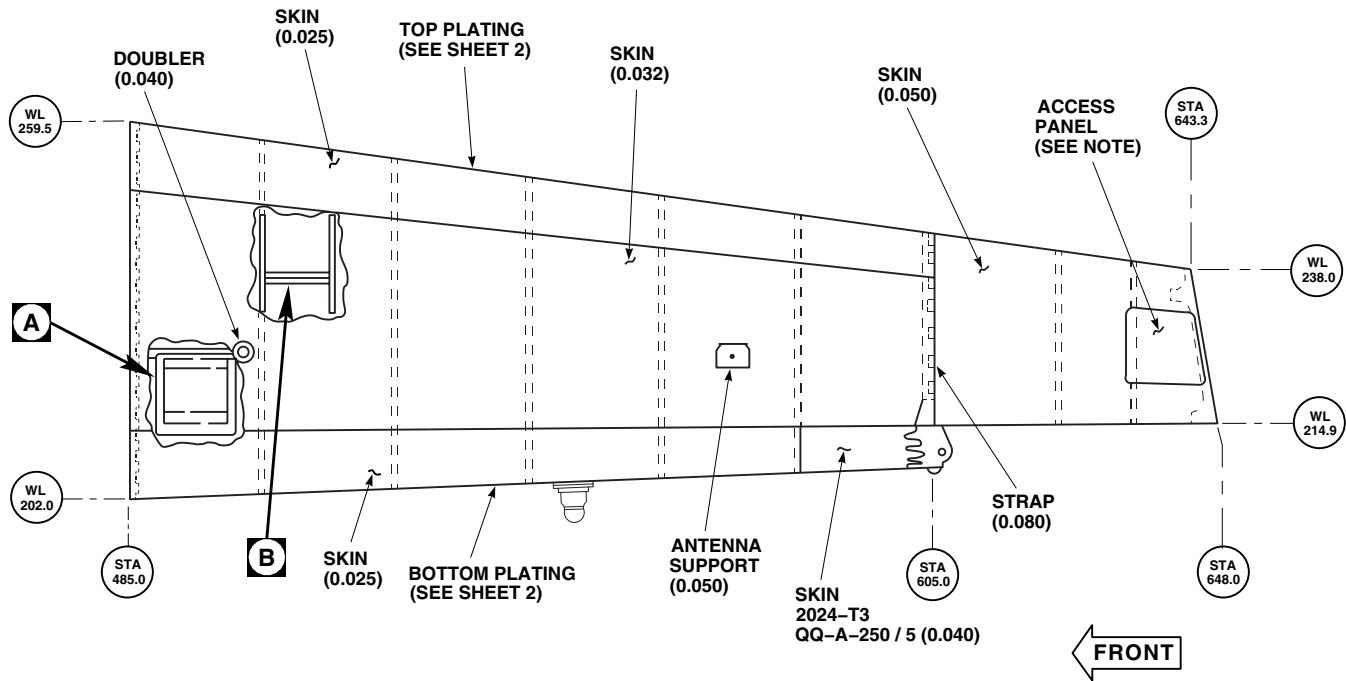
WP 1541 00

TM 55-1500-345-23

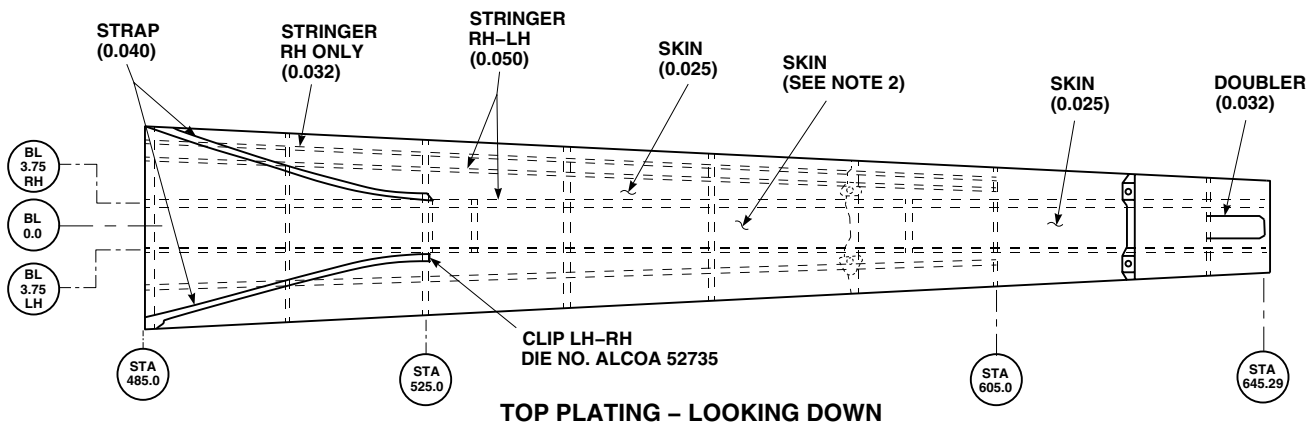
REPAIR

1. Tailcone plating is Alclad aluminum, 7075-T6. QQ-A-250-13, and except for access panels and reinforced plastic components, is primary structure. Evaluate damage to plating (WP 0377 00) (Figure 1, Sheet 1, Figure 1, Sheet 2, and Figure 1, Sheet 3).
2. If required, remove the chaff/flare supports (WP 1541 00).
3. Use WP 0378 00 for typical primary skin repairs.
4. Use WP 0379 00 for typical reinforced plastic repairs.
5. Refer to WP 0418 00 for tail drive shaft fairing repair information.
6. For corrosion protection and watertightness, seal all repairs of tailcone plating (WP 0377 00).
7. If previously removed, reinstall the chaff/flare supports (WP 1541 00).
8. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

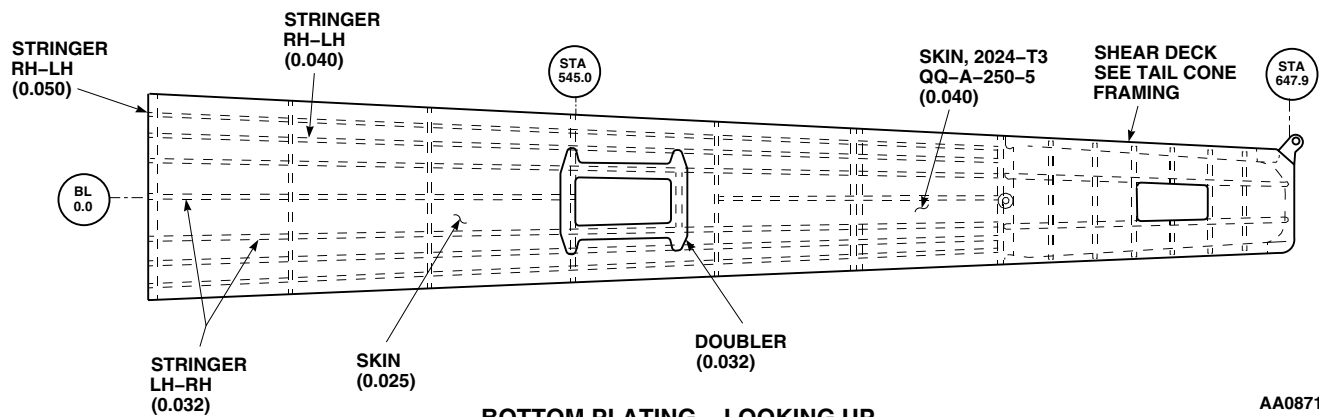
REPAIR - Continued



SIDE PLATING
LEFT SIDE, RIGHT SIDE OPPOSITE



TOP PLATING - LOOKING DOWN

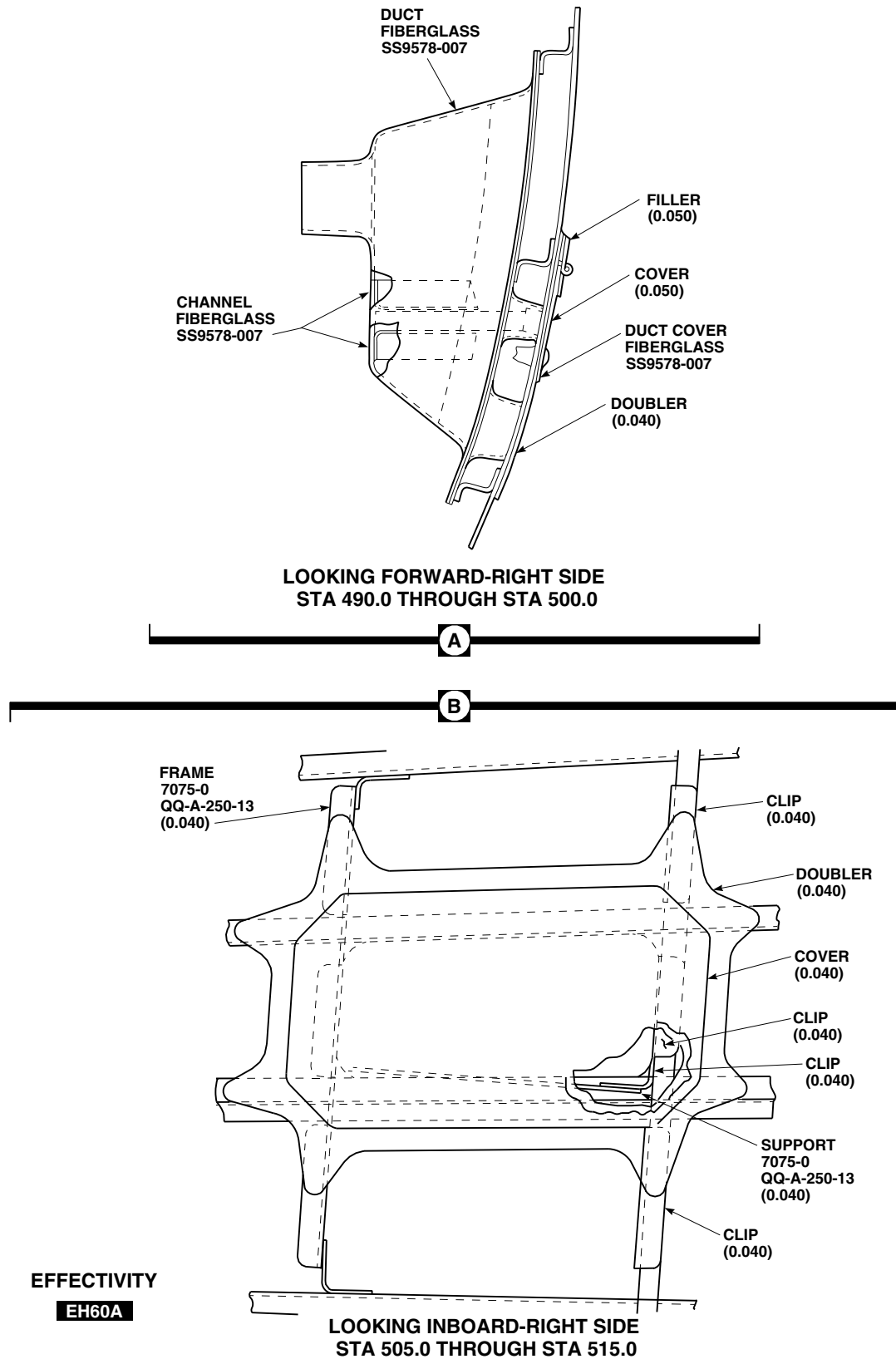


BOTTOM PLATING - LOOKING UP

AA0871_1B
SA

Figure 1. Repair Tailcone Plating. (Sheet 1 of 3)

REPAIR - Continued



AA0871_2C
SA

Figure 1. Repair Tailcone Plating. (Sheet 2 of 3)

REPAIR - Continued**NOTES:**

1. **UH60A 77-22714 - 85-24387** → STEEL WELDMENT 0.064. ◀
UH60A 85-24388 - SUBQ → ALUMINUM 7075-T6 0.063. ◀
2. **UH60A 77-22714 - 87-24620** → 0.025 ◀ **UH60A 87-24621 - SUBQ** 0.040. ◀
3. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0871_3A
SA

Figure 1. Repair Tailcone Plating. (Sheet 3 of 3)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TAILCONE CANTED HINGE BULKHEAD

This WP supersedes WP 0417 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield

Nut, AN310-9

Sealing Compound, Item 273, WP 1803 00
Washer, AN960-916

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 11, WP 1803 00
Alcohol, Denatured, Item 48, WP 1803 00
Bolt, AN8-13
Bushing, NAS75-6-008
Cloth, Abrasive, Item 76, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00
Corrosion Removing Compound, Item 108,
WP 1803 00
Insert, NAS1395-5L
Lubricating Oil, Helicopter Transmission, Item 187,
WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

QQ-A-250
QQ-A-367
TM 1-1500-344-23
TM 55-1500-345-23
WP 0377 00
WP 0381 00
WP 0544 00
WP 1681 00
WP 1803 00

REPAIR

WARNING

CSI

The canted hinge bulkhead fitting has critical areas with critical dimensions. These critical dimensions are indicated in Figure 1, Sheet 2. The limits specified must not be exceeded.

1. The canted hinge bulkhead fitting for the tailcone is machined from forged aluminum alloy, 7075-T73, QQ-A-367. This fitting is considered critical primary structure. Negligible damage is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly into surrounding area with abrasive cloth, Item 76, WP 1803 00, using long, even strokes. Polish out all surface imperfections with abrasive cloth, Item 83, WP 1803 00, and restore protective finish. Polish with abrasive cloth, Item 83, WP 1803 00, and restore protective finish. Refer to WP 0381 00 and TM 55-1500-345-23 for primer and paint requirements. Damage to fitting greater than negligible limits must be reviewed by engineering authority for damage analysis and recommended repair or replacement. NAS75-6-008 bushings (Figure 1, Sheet 1), are press-fit into rear side of fitting and NAS1395-5L inserts are threaded into front side of fitting. Bushings for upper and lower hinge lugs are cadmium-plated steel and are press-fit into lugs (Figure 1, Sheet 1, Figure 1, Sheet 2, and Figure 1, Sheet 3). Line ream bushing to 0.530 - 0.536 inch diameter after installation. Brush cadmium plating onto machined areas of bushing after line reaming.
2. Open and close tail rotor pylon (WP 1681 00).

REPAIR - Continued

A - BUSHING
INSERT
8 PLACES

T1 - OUTER WEB THICKNESS - 0.312

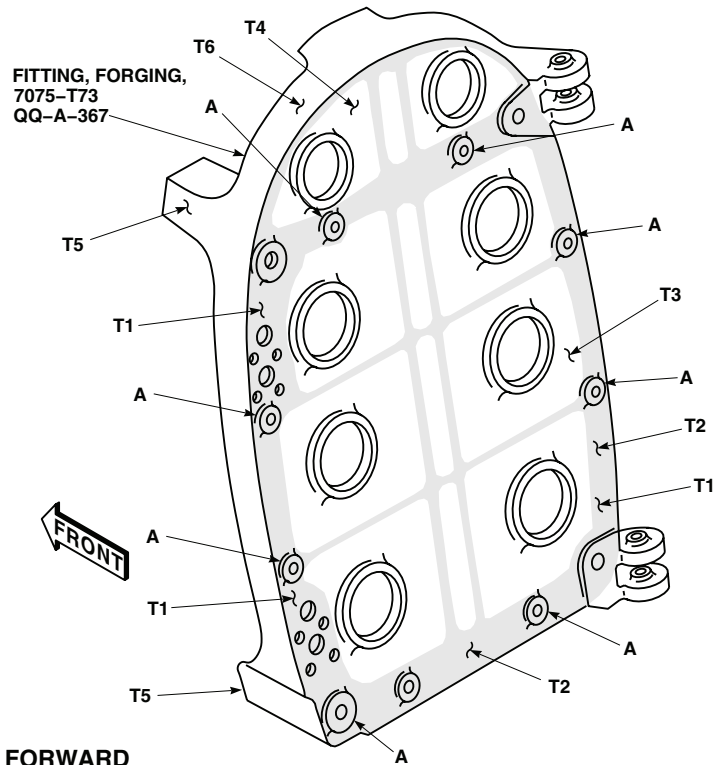
T2 - OUTER WEB THICKNESS, TYPICAL
EXCEPT AS NOTED - 0.120

T3 - INSIDE WEB THICKNESS, TYPICAL
AS NOTED - 0.090 / 0.060

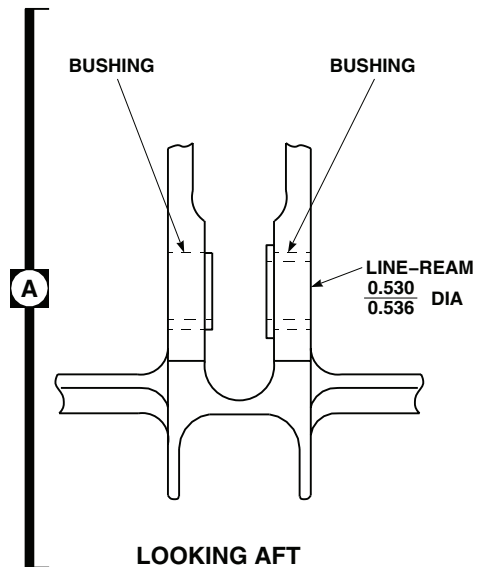
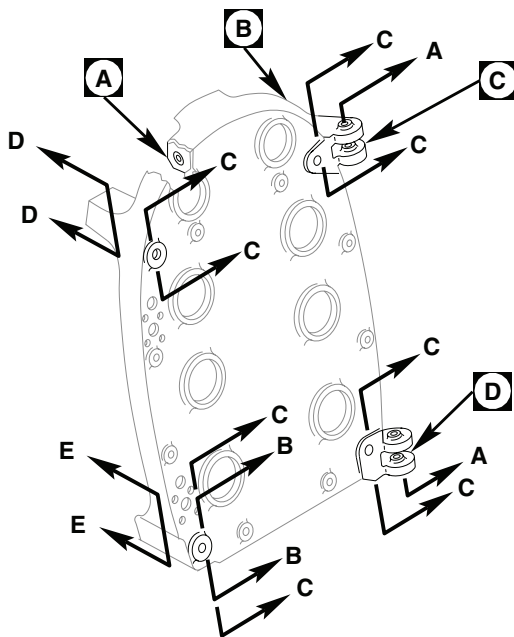
T4 - UPPER WEB THICKNESS - 0.100

T5 - FLANGE THICKNESS - 0.280 (SEE NOTE 2)
0.160 (SEE NOTE 1)

T6 - FLANGE THICKNESS, TYPICAL EXCEPT
AS NOTED - 0.100



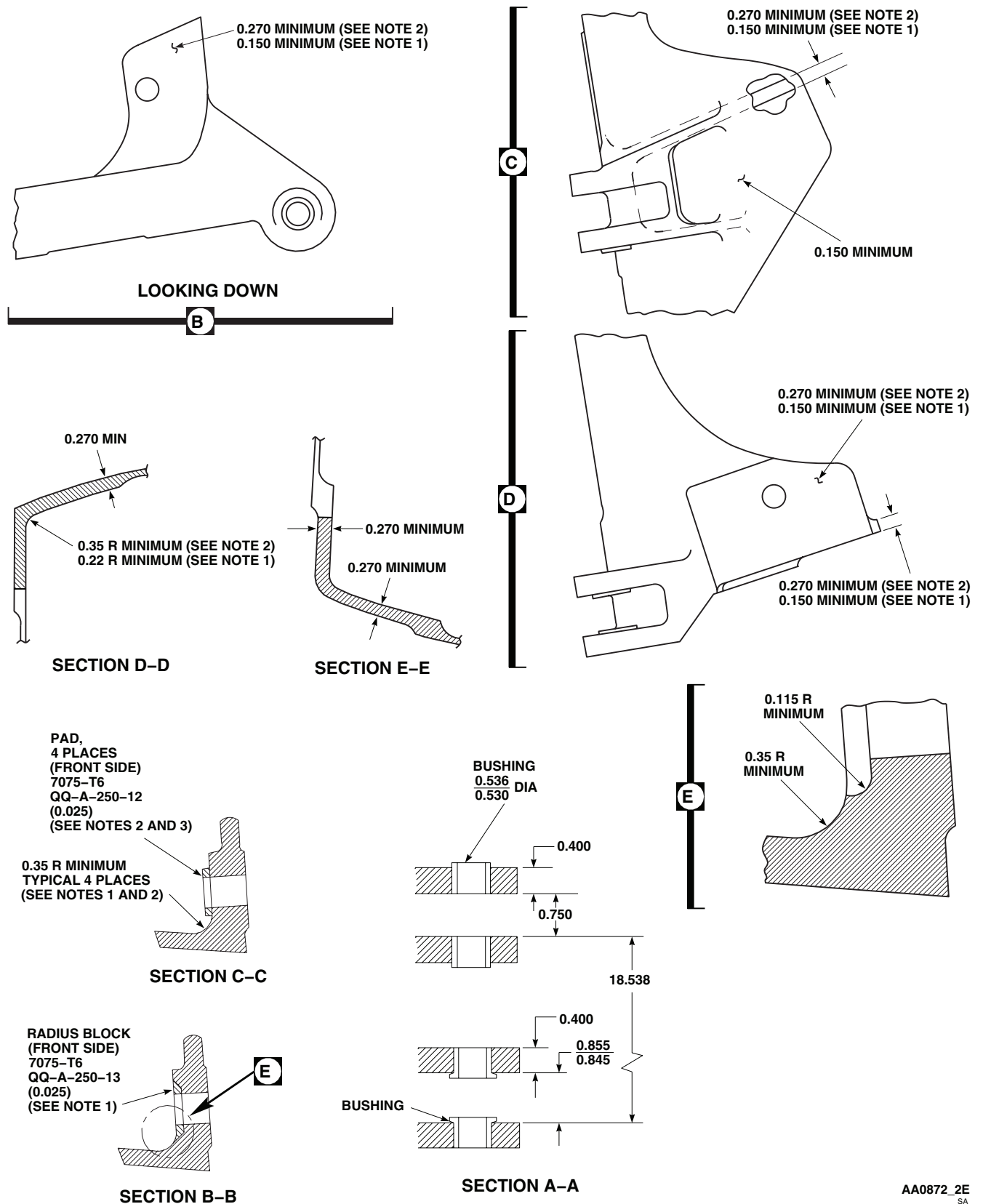
LOOKING FORWARD



AA0872_1C
SA

Figure 1. Repair Canted Hinge Bulkhead. (Sheet 1 of 3)

REPAIR - Continued



AA0872_2E
SA

Figure 1. Repair Canted Hinge Bulkhead. (Sheet 2 of 3)

REPAIR - Continued**NOTES:**

1. UH60A 77-22714 - 77-22726
2. UH60A 77-22727 - SUBQ
3. EACH PAD HAS UNIQUE SHAPE TO FIT ITS SEAT IN BULKHEAD.
4. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0872_3B
SA

Figure 1. Repair Canted Hinge Bulkhead. (Sheet 3 of 3)

3. Seal fitting for corrosion protection and watertightness using sealing compound, Item 273, WP 1803 00 (WP 0377 00).
4. Touch up paint and primer as required (WP 0381 00 and TM 55-1500-345-23).
5. Prior to bonding pads on bulkhead do this:
 - a. Open hole in pads to 0.593 to 0.599-inch diameter.
 - b. Break all sharp edges.
 - c. Touch up paint and primer as required (WP 0381 00 and TM 55-1500-345-23).

WARNING**CSI**

Bonding of pads and radius block is a critical characteristic. The bonding procedure given here must be followed exactly to assure proper bonding.

6. To bond pads or radius block on bulkhead, do this:
 - a. Treat bond surfaces on bulkhead and radius side of pad with corrosion removing compound, Item 108, WP 1803 00, followed by a thorough rinsing with clean water.
 - b. While still wet, check that surfaces are water-break free (TM 1-1500-344-23).
 - c. Repeat step 6. a. if necessary.
 - d. Rinse surfaces with a 1 part solution of denatured alcohol, Item 48, WP 1803 00, and 1 part water. Allow to dry.
 - e. Coat bolt, AN8-13, with a light film of helicopter transmission lubricating oil, Item 187, WP 1803 00.
 - f. Carefully place pad or radius block on bolt, making sure treated bond surface is not contaminated by oil or touched with fingers.

REPAIR - Continued**NOTE**

Pot life of adhesive, Item 11, WP 1803 00, is 25 minutes at 75°F(24°C).

- g. Prepare adhesive, Item 11, WP 1803 00, according to instructions on container.
- h. Apply a coat of adhesive on pad or radius block.
- i. Put bolt with pad or radius block through mounting hole in bulkhead.
- j. Turn pad so that radius edges make maximum contact with radius in bulkhead.
- k. Install a washer, AN960-916, and nut, AN310-9, on bolt and lightly tighten nut just enough to pull pad or radius block solidly against bulkhead. Do not overtighten.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

- l. Clean up adhesive squeezeout with cloth moistened in technical acetone, Item 5, WP 1803 00.
- m. After curing, remove bolt, nut, and washer.
- n. Check bond for voids by coin-tapping pad or radius block as for rotor blade inspection (WP 0544 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

TAIL ROTOR DRIVE SHAFT COVERS REPAIR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

Material/Parts

Rivets, MS20470AD4

Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23

TM 55-1500-345-23

WP 0377 00

WP 0378 00

WP 0379 00

WP 0381 00

WP 1803 00

REPAIR TAIL ROTOR DRIVE SHAFT FAIRING

1. Hinged fairing covers for the tail rotor drive shaft are made of Kevlar reinforced plastic with aluminum alloy clips, angles, hinges, and turnlock fasteners (Figure 1).
2. Evaluate damage and make repairs to Kevlar covers (WP 0379 00).
3. Evaluate damage and make repairs to metal components (WP 0378 00).
4. Replace fasteners (TM 1-1500-204-23).
5. Make sure area is clean and free of foreign material.
6. Touch up paint in repaired area as required (WP 0381 00 and TM 55-1500-345-23).

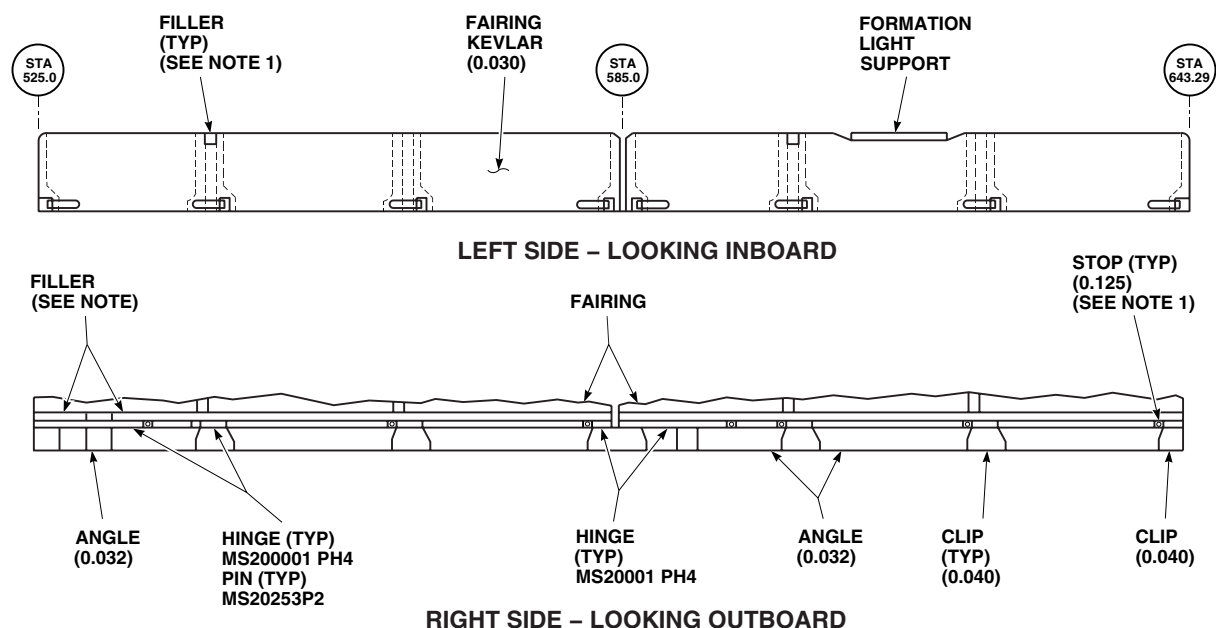
REPLACE TAIL ROTOR DRIVE SHAFT FAIRING UPPER HINGE**Remove**

1. Carefully drill out and remove rivets securing upper hinge half to fairing using 1/8-inch diameter drill (Figure 2).
2. Open tail rotor drive shaft cover and remove nut, washer, and screw securing stop to fairing. Remove stop.
3. Remove hinge pin and upper hinge half.
4. Make sure area is clean and free of foreign material.

Install**NOTE**

Use old hinge half for hole jig, or fairing indexing tool (TM 1-1500-204-23).

1. Lay out and drill holes in replacement hinge half, using No. 30 drill.
2. For corrosion protection, apply sealing compound, Item 273, WP 1803 00, under hinge and clip before assembling (WP 0377 00).
3. Position hinge half on fairing and temporarily hold in place, using 1/8-inch diameter clecos.
4. Line up hinge with lower hinge half and install hinge pin with swaged end on pin stop side (Figure 2).

REPLACE TAIL ROTOR DRIVE SHAFT FAIRING UPPER HINGE - Continued**NOTES:**

1. FILLERS MADE FROM CELLULAR CELLULOSE ACETATE BONDED INTO FAIRING.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0874B
SA

Figure 1. Repair Tail Rotor Drive Shaft Fairing.

5. Coat all fasteners with sealing compound, Item 273, WP 1803 00, before installing (WP 0377 00).
6. Attach stop with screw, washer, and nut.
7. Install required rivets, MS20470AD4, between clecos. Remove clecos and install remainder of rivets.
8. Check open/close function of cover.
9. Make sure area is clean and free of foreign material.
10. Touch up paint in repaired area (WP 0381 00 and TM 55-1500-345-23).

REPLACE TAIL ROTOR DRIVE SHAFT FAIRING LOWER HINGE**Remove****NOTE**

Rivets securing clip to angle should not be removed unless clip is to be replaced.

1. Carefully drill out and remove rivets securing lower half hinge to angle and clip, using 1/8-inch diameter drill (Figure 2).
2. Open tail rotor drive shaft cover and remove nut, washer, and screw securing stop to cover. Remove stop.
3. Remove hinge pin and lower hinge half.
4. Make sure area is clean and free of foreign material.

REPLACE TAIL ROTOR DRIVE SHAFT FAIRING LOWER HINGE - Continued

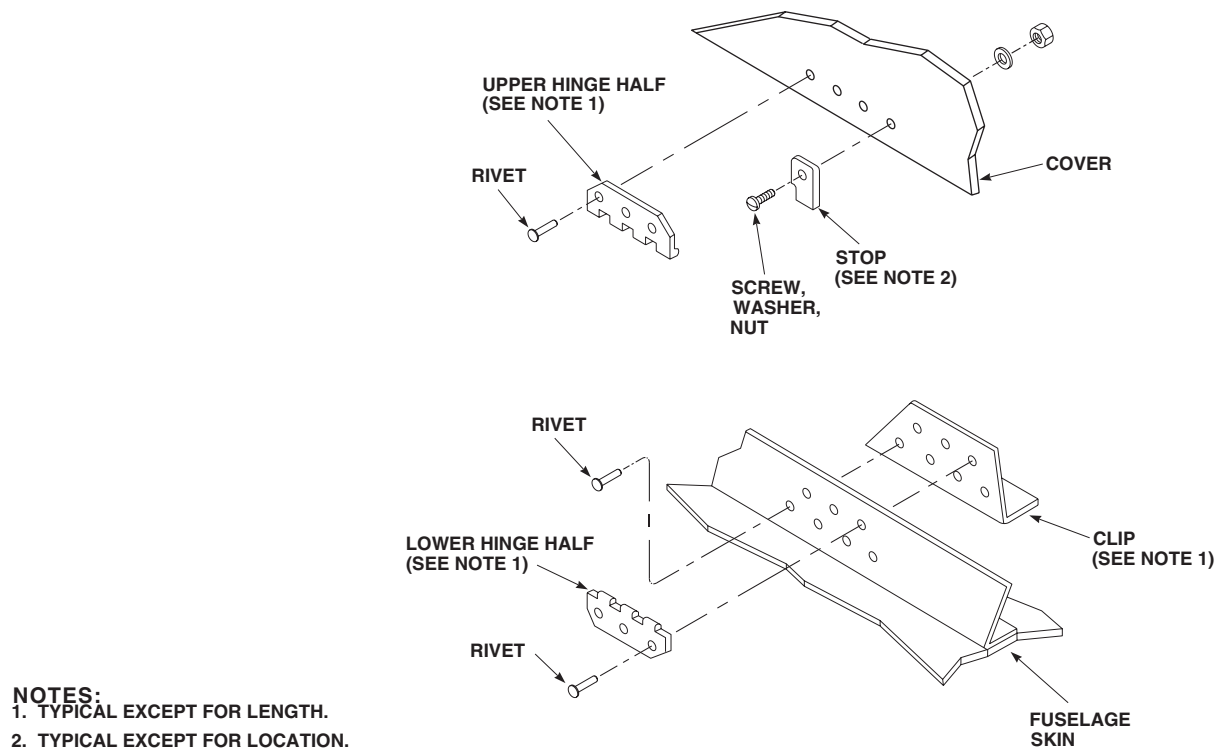
AA0875
SA

Figure 2. Replace Tail Rotor Drive Shaft Fairing Hinge.

Install

NOTE

Use old hinge half for hole jig, or cover indexing tool (TM 1-1500-204-23).

1. Lay out and drill holes in replacement hinge half, using No. 30 drill.
2. For corrosion protection, apply sealing compound, Item 273, WP 1803 00, under hinge and clip before assembling (WP 0377 00).
3. Position hinge half on angle and temporarily hold in place, using 1/8-inch diameter clecos.
4. Line up hinge with upper hinge half and install hinge pin with swaged end on pin stop side (Figure 2).
5. Coat shank of all fasteners with sealing compound, Item 273, WP 1803 00, before installing (WP 0377 00).
6. Attach stop to cover with screw, washer, and nut.
7. Install required rivets, MS20470AD4, between clecos. Remove clecos and install remainder of rivets.
8. Check open/close function of cover.
9. Make sure area is clean and free of foreign material.
10. Touch up paint in repaired area (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TAIL ROTOR PYLON FRAMING REPAIR

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02

MIL-S-5000

QQ-A-200

QQ-A-225

Material/Parts

Cloth, Abrasive, Item 83, WP 1803 00

QQ-A-250

Clip, AN743Z13

QQ-A-367

TM 55-1500-345-23

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

WP 0377 00

WP 0378 00

WP 0380 00

References

AMS 4260

WP 0381 00

AMS 5334

WP 0426 00

MIL-S-18729

WP 1803 00

WW-T-700

REPAIR

1. Tail rotor pylon framing includes formed and extruded members made from 7075-T6, QQ-A-250-13 Alclad aluminum sheet stock and 7075-T6511, QQ-A-200-11 aluminum alloy extrusion. Support and attachment fittings made from 7075-T73, QQ-A-367 aluminum alloy forgings, are used for the canted hinge frame, intermediate and tail gear box supports, spar and stabilator attachment, pylon lower step, and stabilator actuator support (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, Figure 1, Sheet 7, Figure 1, Sheet 8, Figure 1, Sheet 9, Figure 1, Sheet 10, Figure 1, Sheet 11, and Figure 1, Sheet 12).
2. All tail rotor pylon framing is primary structure. Fittings, however, are considered critical primary structure. Negligible damage to the fittings is limited to nicks, scratches, and corrosion no deeper than 5% material thickness at point of damage. Blend damage smoothly to surrounding area using long, even strokes. Polish with abrasive cloth, Item 83, WP 1803 00, and restore protective finish. Damage greater than negligible limits must be reviewed by engineering authority for damage evaluation and either repair or replacement. All tail rotor pylon framing repairs must maintain design strength and alignment.
3. Evaluate damage in noncritical areas (WP 0377 00).
4. Use WP 0378 00 and WP 0380 00 for typical framing repairs.
5. Refer to WP 0426 00 for canted hinge frame fitting repair information.
6. For corrosion protection and watertightness, seal all repairs between frame members and pylon plating (WP 0377 00).
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued

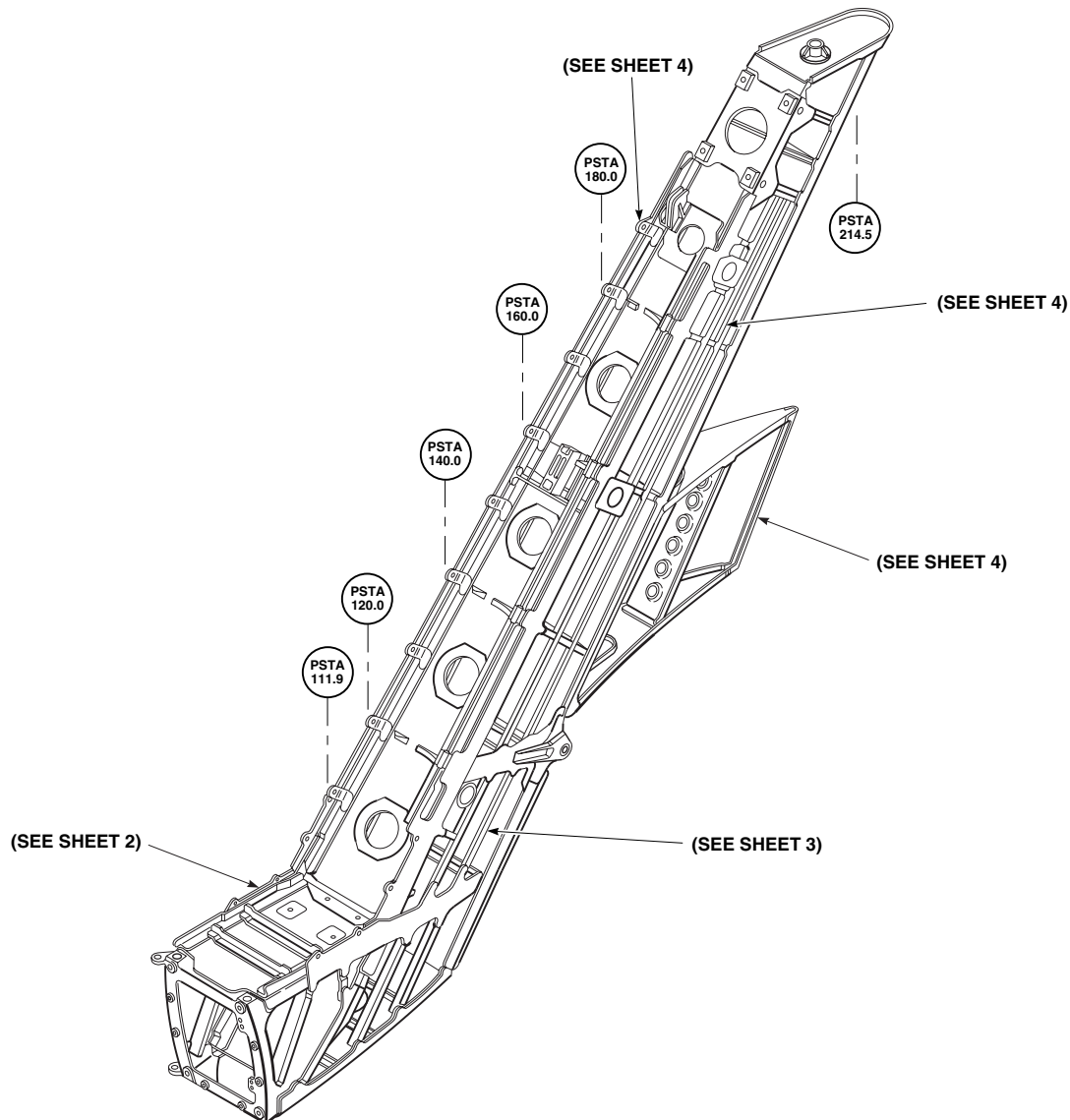


Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 1 of 12)

REPAIR - Continued

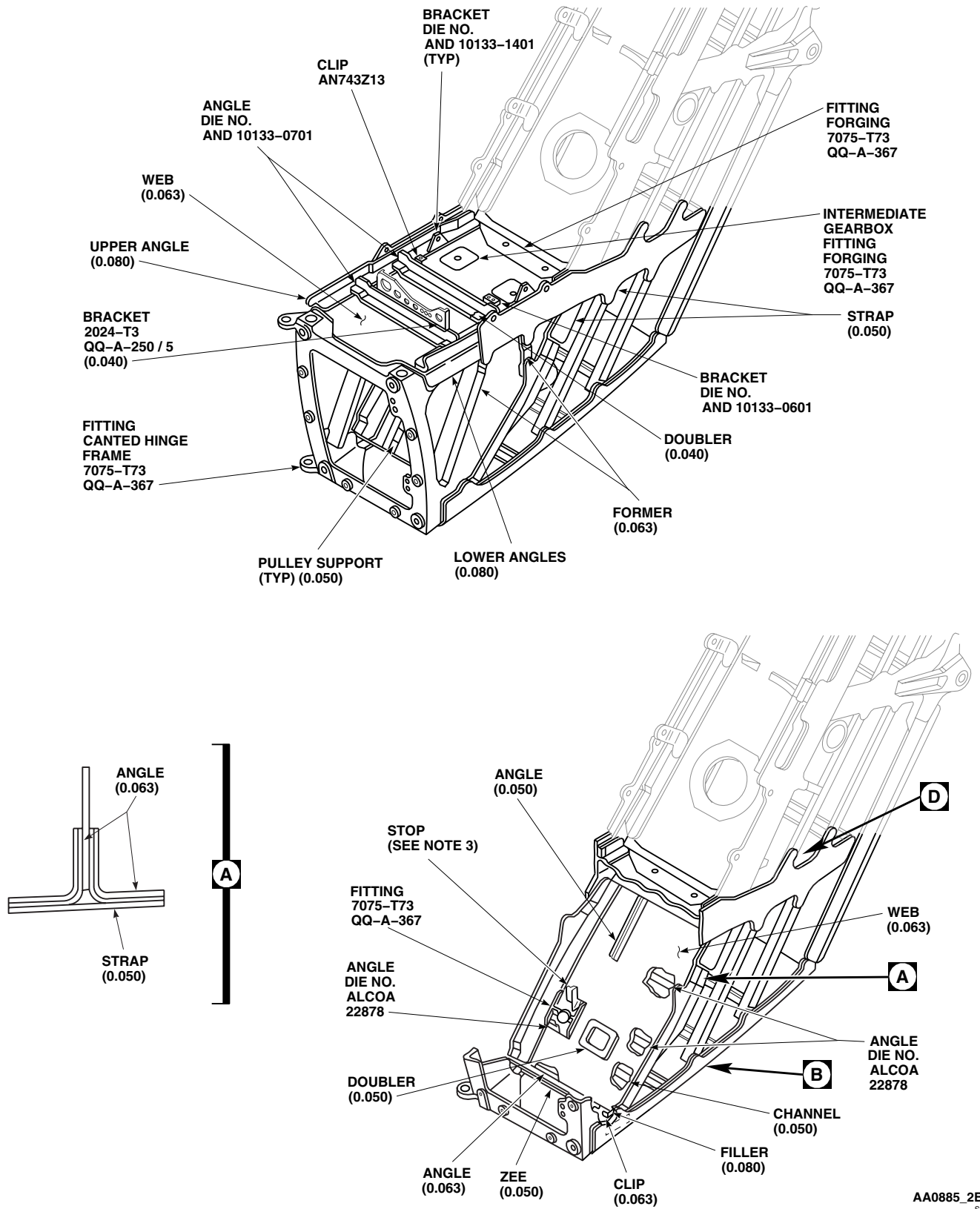


Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 2 of 12)

REPAIR - Continued

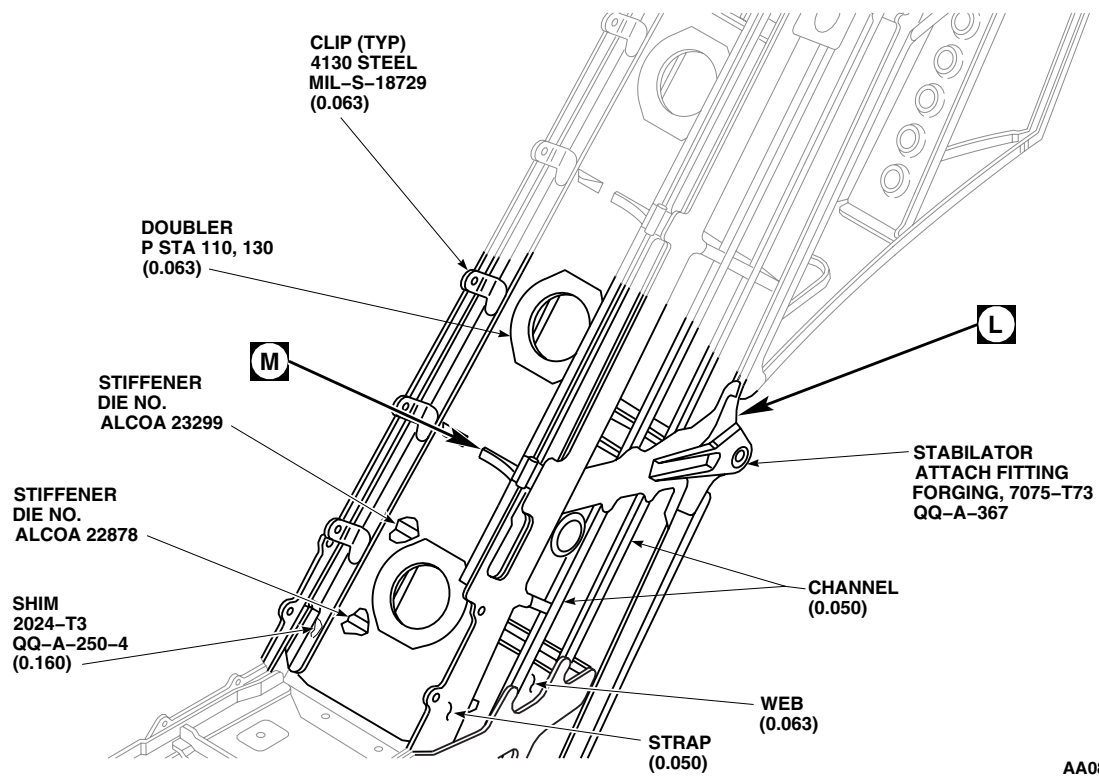
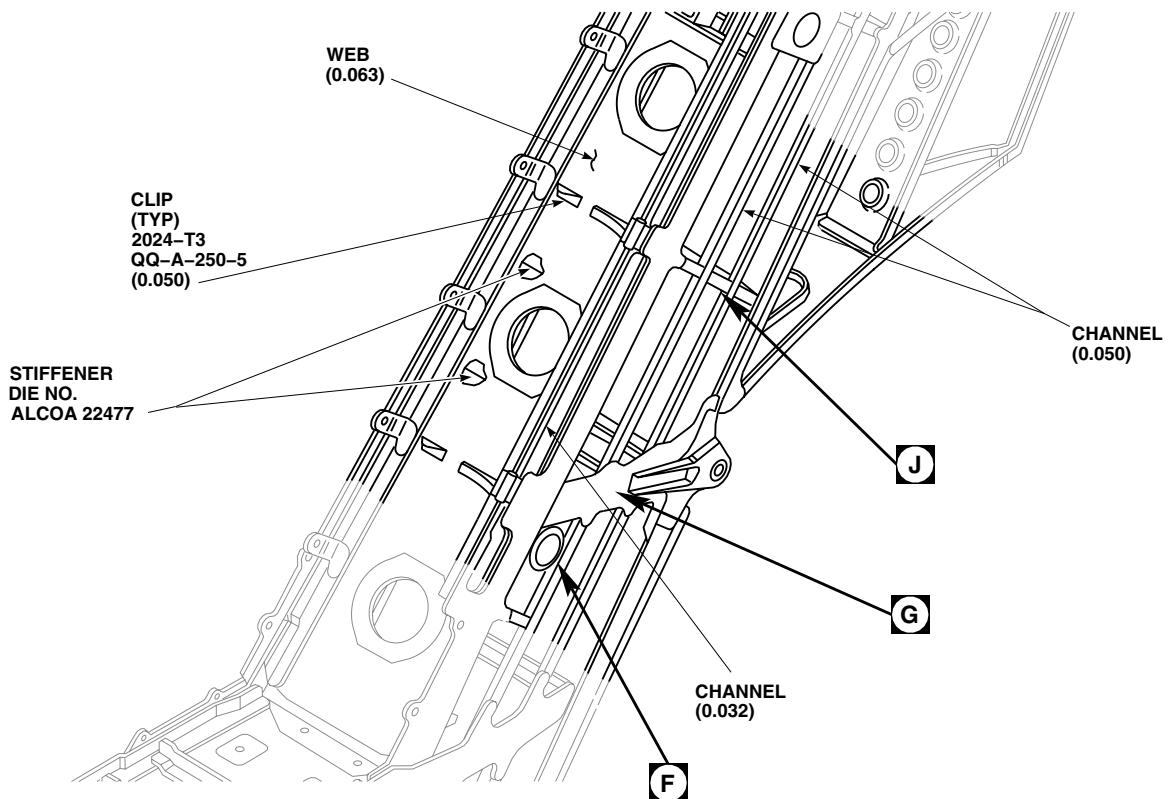
AA0885_3B
SA

Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 3 of 12)

REPAIR - Continued

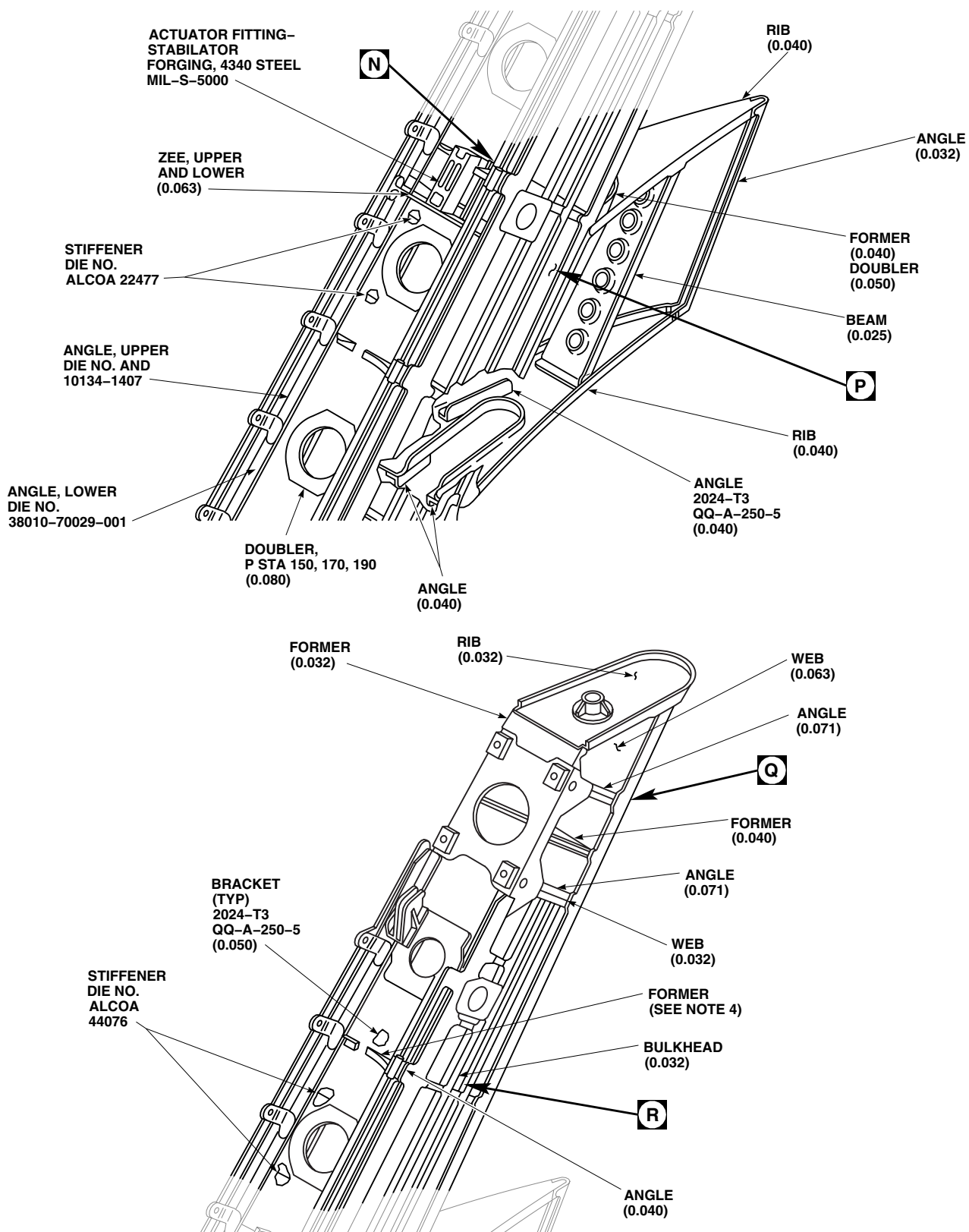
AA0885_4B
SA

Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 4 of 12)

REPAIR - Continued

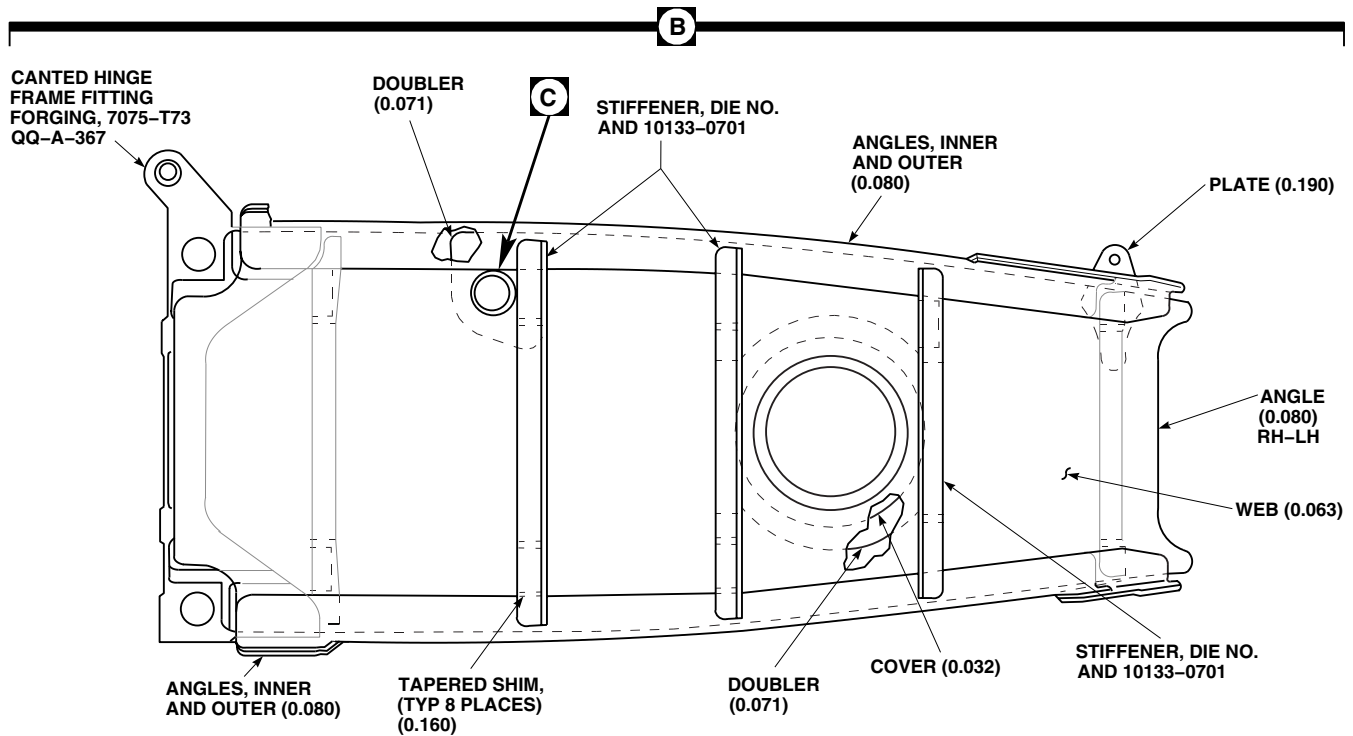
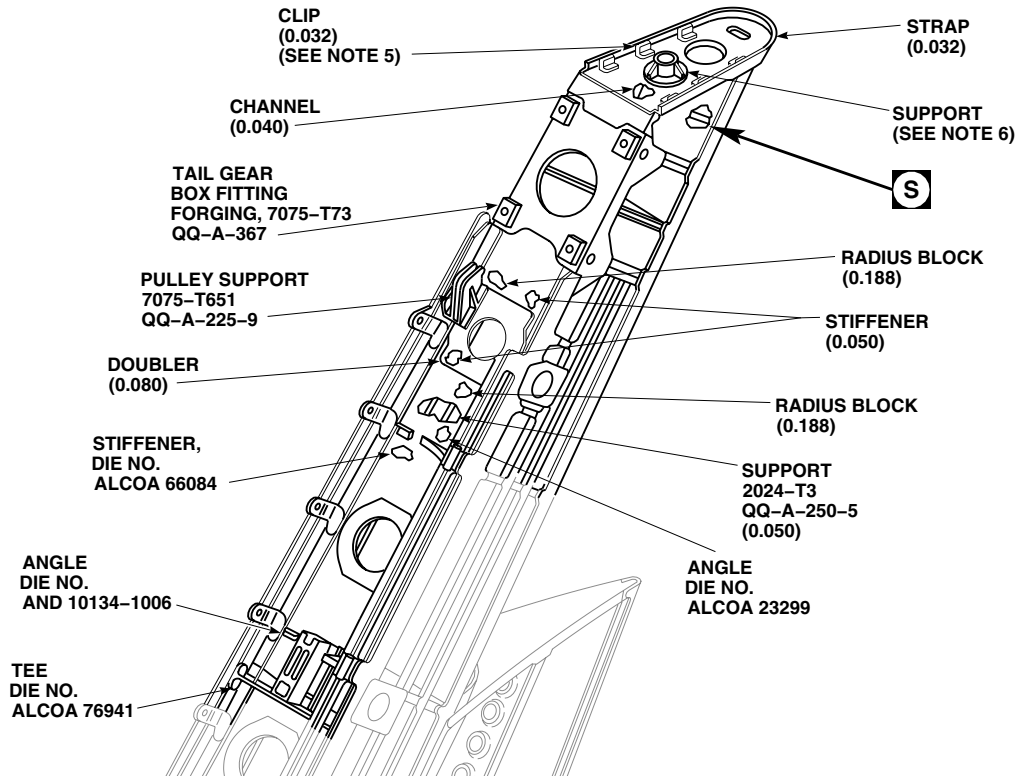
AA0885_5B
SA

Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 5 of 12)

REPAIR - Continued

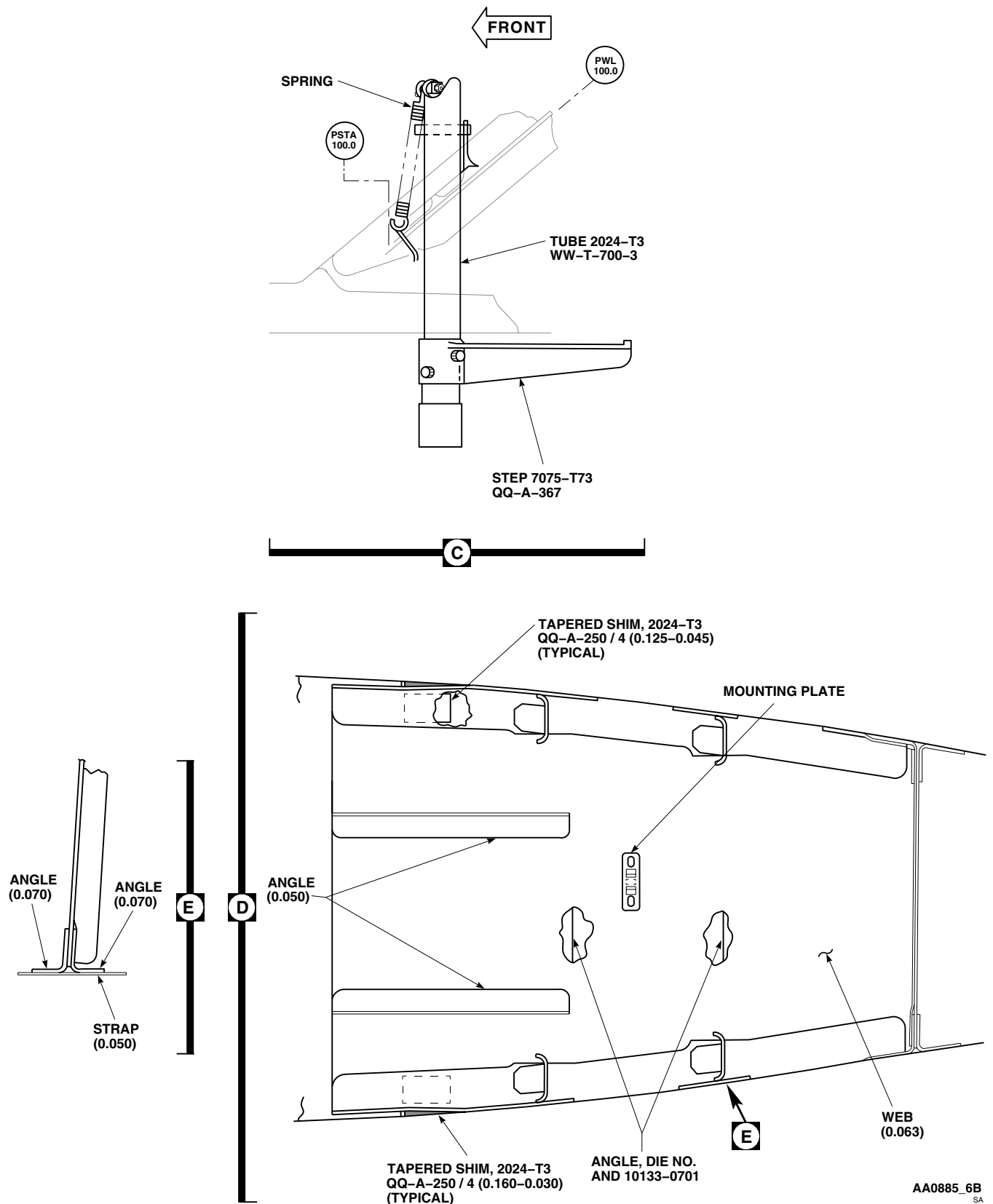
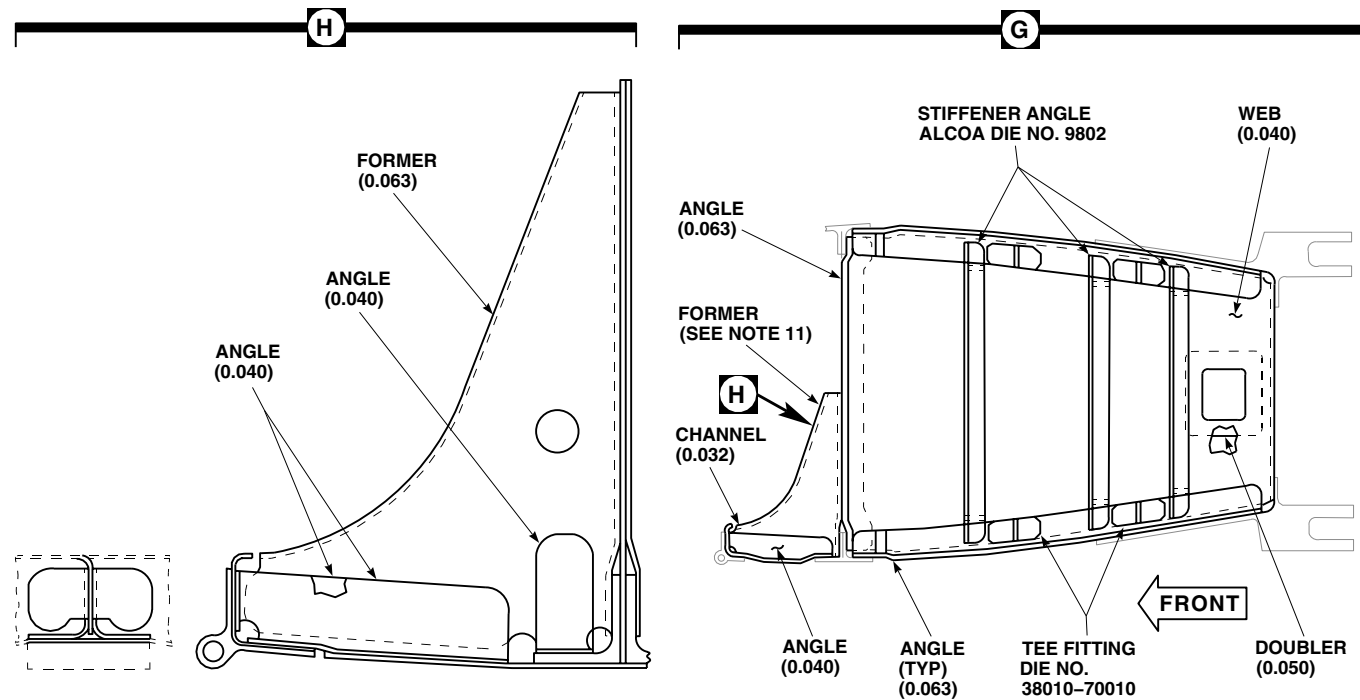
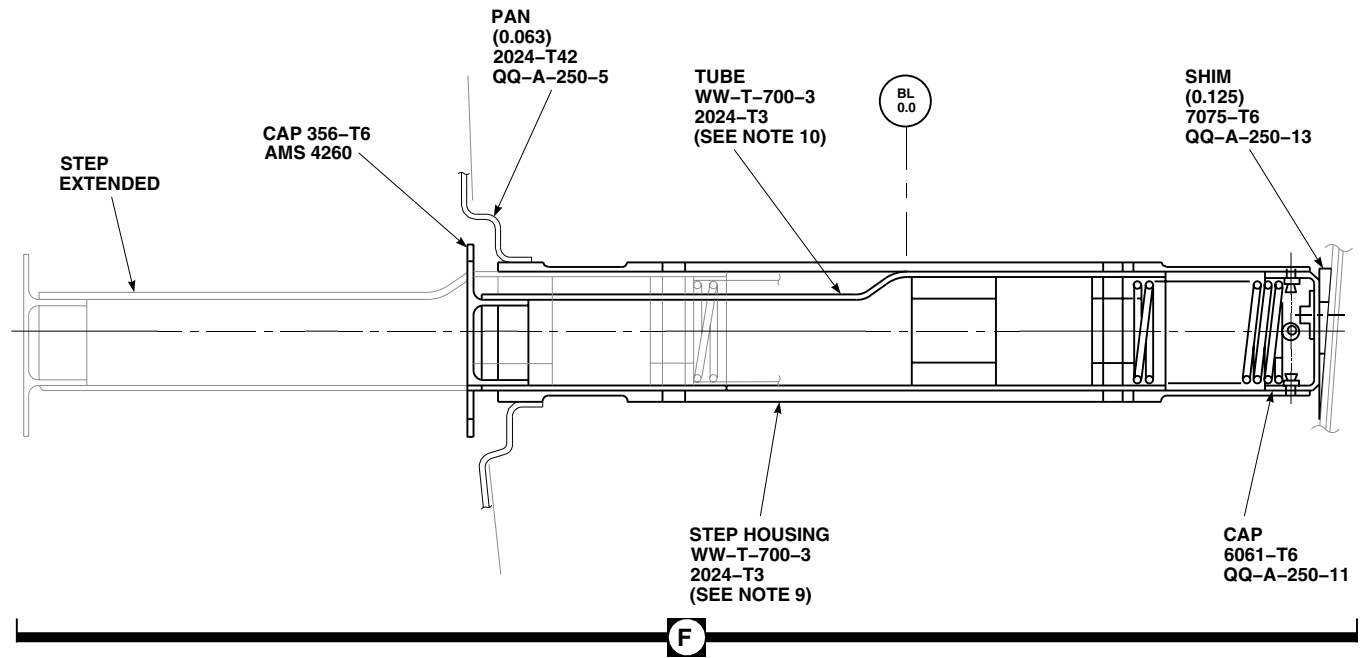


Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 6 of 12)

REPAIR - Continued



EFFECTIVITY

UH60A 89-26134 - SUBQ

AA0885_7B
SA

Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 7 of 12)

REPAIR - Continued

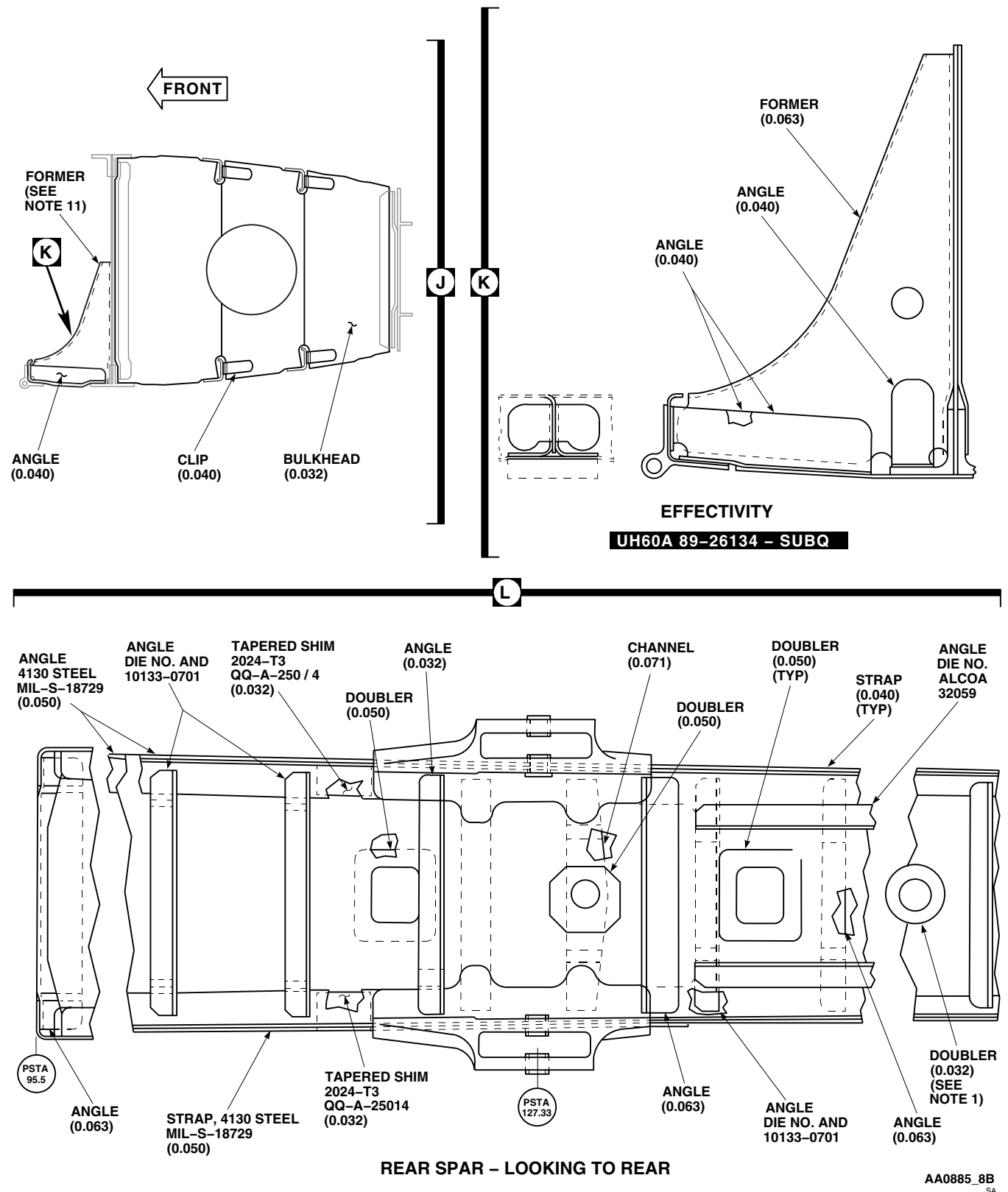
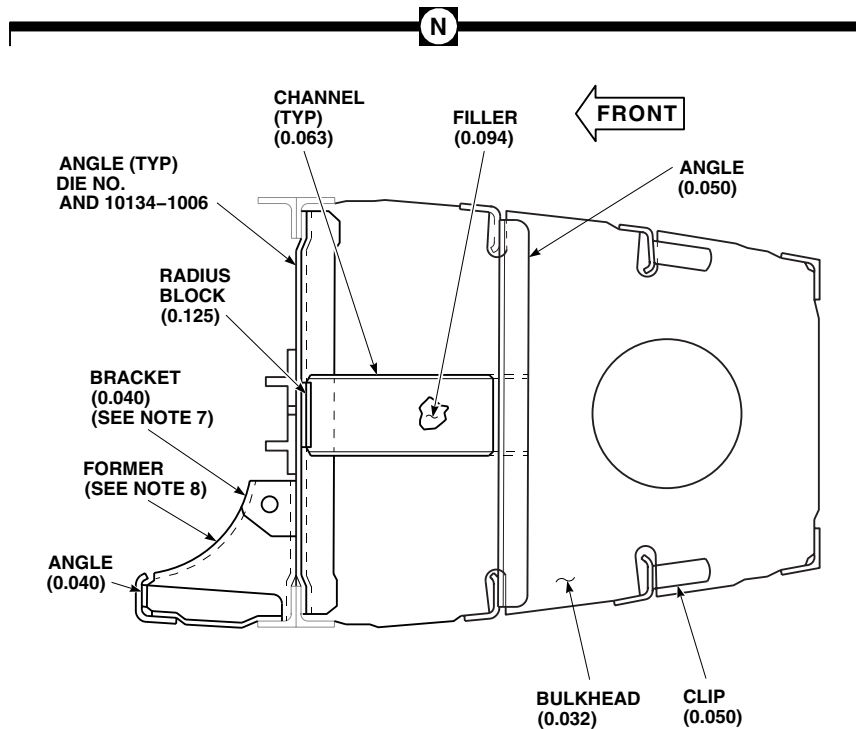
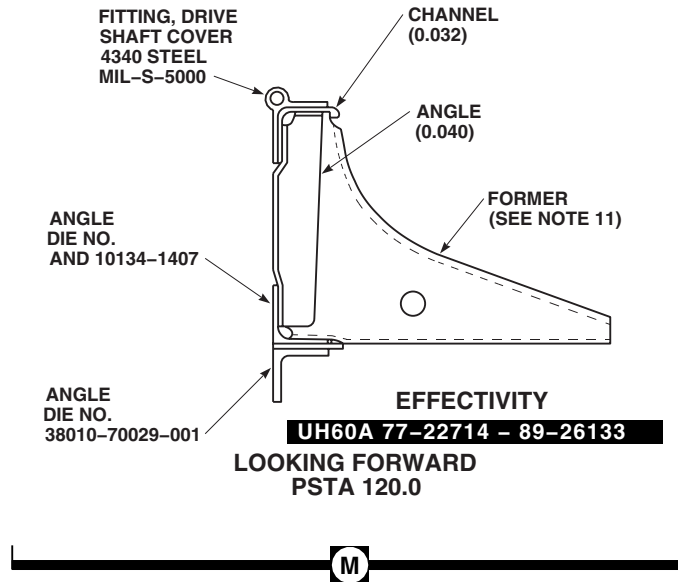


Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 8 of 12)

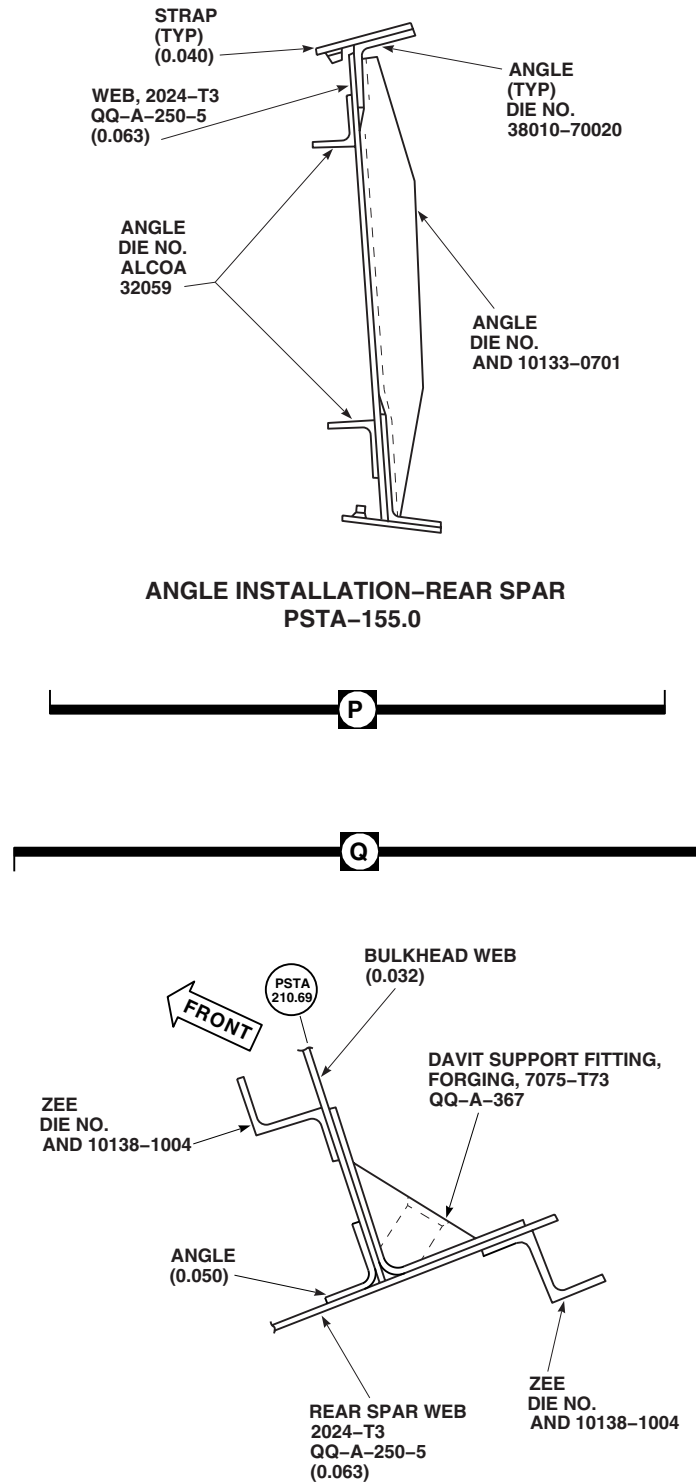
REPAIR - Continued



AA0885_9B
SA

Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 9 of 12)

REPAIR - Continued



AA0885_10A
SA

Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 10 of 12)

REPAIR - Continued

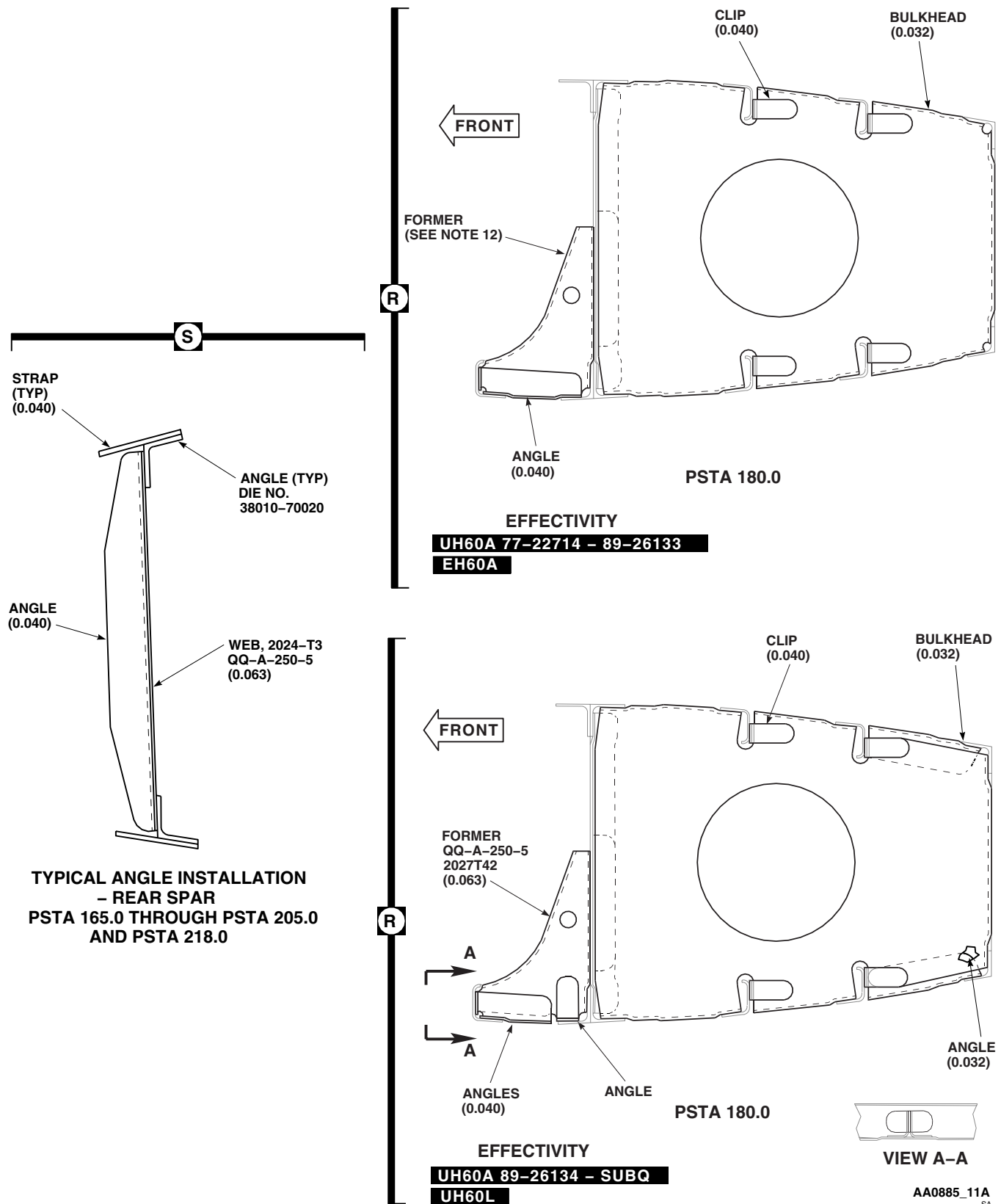


Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 11 of 12)

REPAIR - Continued**NOTES:**

1. UNLESS OTHERWISE NOTED, ALL SHEET METAL PARTS SHOWN ARE ALCLAD 7075-T6, QQ-A-250-13, AND ALL EXTRUSION PARTS ARE ALUMINUM ALLOY 7075-T6511, QQ-A-200-11.
2. LEFT AND RIGHT SIDES ARE THE SAME EXCEPT WHEN NOTED.
3. EFFECTIVITY: PYLONS PRIOR TO SERIAL NO. A0330-0816 USE 2024-T3, QQ-A-250-5 (0.050). PYLONS SERIAL NOS A0330-0816 AND SUBSEQUENT USE MIL-S-18729 4130, COND A (0.050).
4. **UH60A 82-23664 - SUBQ** **UH60L** **EH60A**
5. **UH60A 81-23610 - SUBQ**
6. **UH60A 77-22714 - 78-22959** **WELDMENT, 4130 STEEL MIL-S-18729.** **UH60A 78-22960 - SUBQ** **CASTING, AMS 5334.**
7. **UH60A 82-23664 - SUBQ** **UH60L** **EH60A**
8. **UH60A 77-22714 - 85-24464** **EH60A 84-24017 - 85-24482** 7075-T6, QQ-A-250-13 (0.040). **UH60A 85-24483 - SUBQ** **UH60L** **EH60A 86-24561 - SUBQ** 2024-T42, QQ-A-250-5 (0.063).
9. MADE FROM 38-10-70012-001 SEAMLESS TUBING.
10. MADE FROM TUBING 1.375 IN. OD 0.065 IN WALL.
11. **UH60A 77-22714 - 85-24464** **EH60A 84-24017 - 85-24482** 7075-T6, QQ-A-250-13 (0.040). **UH60A 86-24483 - 89-26133** **EH60A 86-24561 - SUBQ** 2074-T42, QQ-A-250-3-5 (0.063).
12. **UH60A 77-22714 - 85-24464** **EH60A 84-24017 - 85-24482** 7075-T6, QQ-A-250-13 (0.040). **UH60A 86-24483 - 89-26133** **EH60A 84-24017 - 85-24482** 2074-T42, QQ-A-250-3-5 (0.063).
13. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0885_12B
SA

Figure 1. Repair Tail Rotor Pylon Framing. (Sheet 12 of 12)

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

TAIL ROTOR PYLON PLATING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

TM 55-1500-345-23

WP 0377 00

WP 0378 00

Material/Parts

Hinge, MS20257HP3-500

WP 0379 00

Sealing Compound, Item 273, WP 1803 00

WP 0381 00

WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

MIL-S-18729

QQ-A-250

REPAIR

1. Tail rotor pylon upper plating, between the forward and rear spars, is Alclad aluminum, 2024-T3, QQ-A-250-5. Lower plating, between spars, is Alclad aluminum 7075-T6, QQ-A-250-13. Leading and trailing edge fairings, and tail rotor and intermediate gear box fairings, are reinforced plastic (Figure 1). Reinforcement straps in the lower pylon are 4130 steel, MIL-S-18729. Upper and lower plating between spars is primary structure. All reinforced plastic or aluminum skin fairings and the cambered fairing are secondary structure. Evaluate damage to the plating, aluminum fairing, and steel straps (WP 0377 00).
2. Use WP 0378 00 for typical plating and aluminum fairing repairs. Replace steel straps damaged beyond negligible damage limits (Figure 1, Detail A).
3. On cambered fairing, wet install any replaced skin fasteners, using sealing compound, Item 273, WP 1803 00.
4. Use WP 0379 00 for evaluating damage and repairing reinforced plastic.
5. For corrosion protection and watertightness, seal all repairs of pylon plating (WP 0377 00).
6. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued

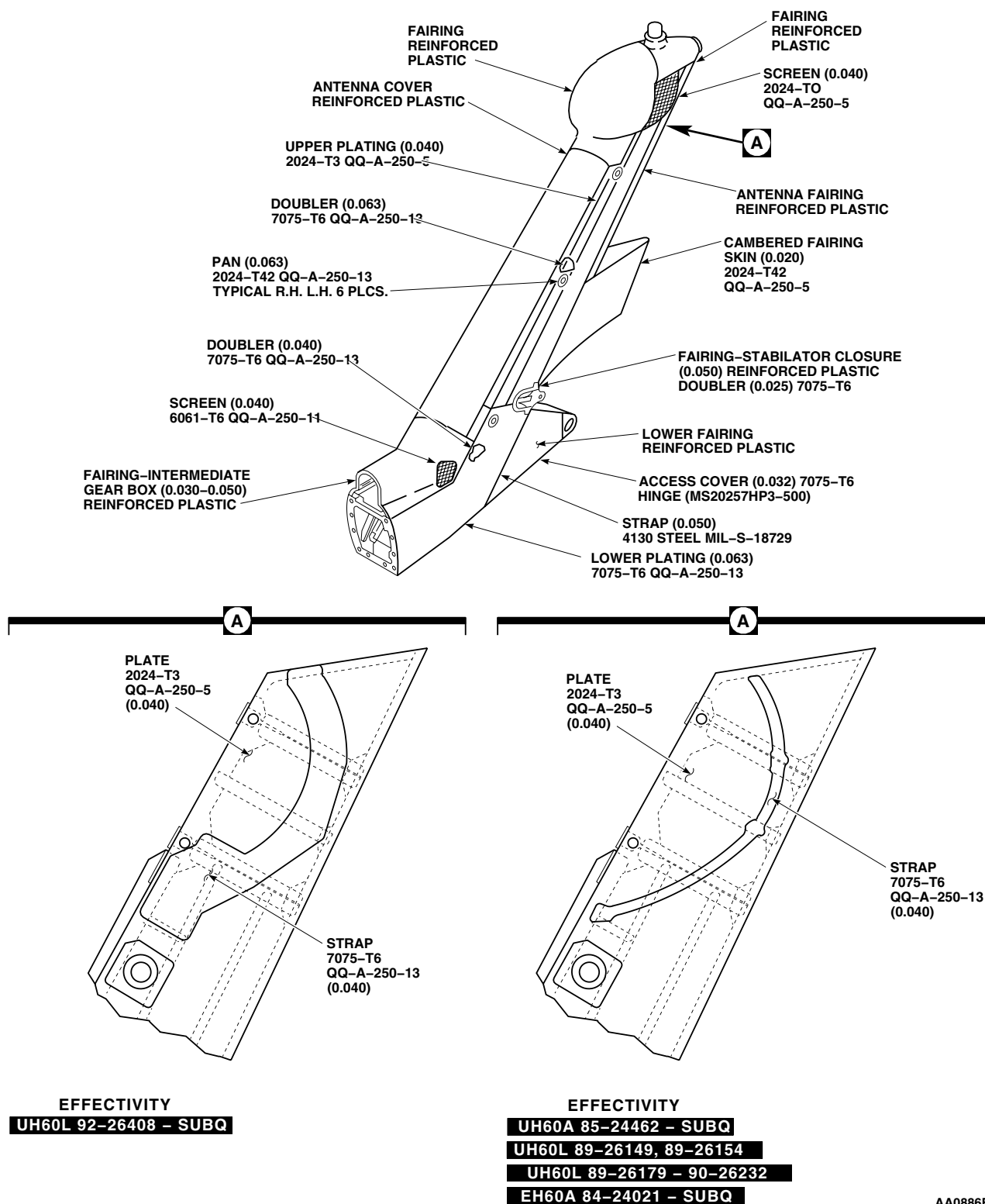
AA0886B
SA

Figure 1. Repair Tail Rotor Pylon Plating.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****TAIL ROTOR PYLON LEFT STABILATOR ATTACH FITTING**

This WP supersedes WP 0421 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Gloves
Goggles/Face Shield
Stabilator/Pylon Hinge Tool, 38076-00017-002

Lubricant, Solid Film, Item 180, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Sealing Compound, Item 293, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Removing Compound, Item 108,
WP 1803 00

References

WP 0328 00
WP 0338 00
WP 1803 00

REMOVE

1. Remove stabilator (WP 0338 00).
2. Remove damaged bushings using stabilator/pylon hinge tool (Figure 1, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

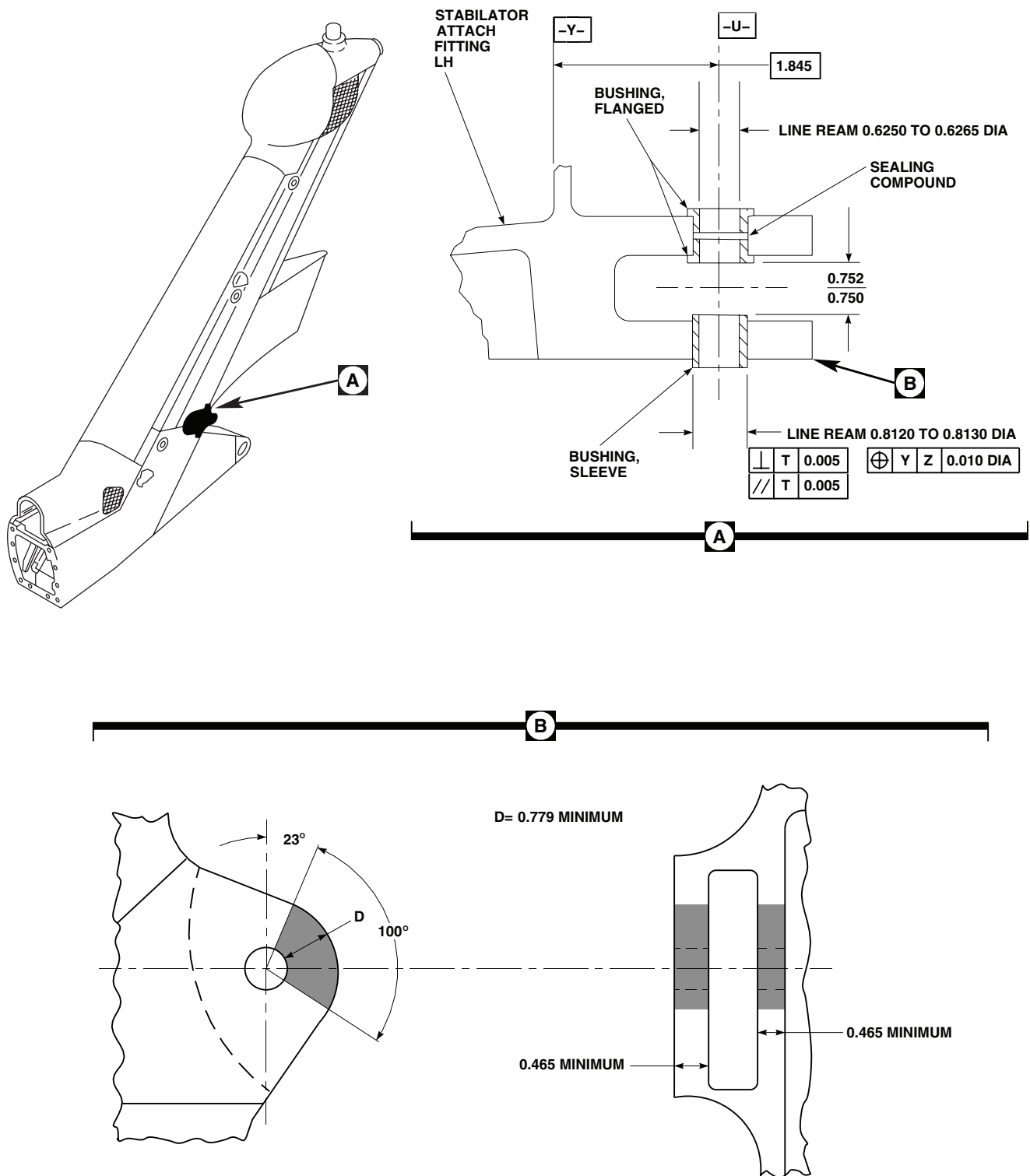
3. Clean lug bore surfaces using cheesecloth cloth, Item 85, WP 1803 00 and technical acetone, Item 5, WP 1803 00.

INSPECT**WARNING****CSI**

- The left stabilator attachment fittings have critical areas that must not be damaged during inspection or repair. Refer to Figure 1 for critical characteristic areas.
- Bonding of liners in fitting is a critical characteristic. Proper bonding must be verified.
- Damage to fitting will occur if feeler gage is improperly used. Liner may become dis-bonded. Use care when using feeler gage to check for liner edge separation.

1. Inspect lug bore surfaces for damage and corrosion as follows:

INSPECT - Continued



NOTES:

1. CRITICAL CHARACTERISTIC AREAS SHOWN IN SHADED AREA, INCLUDING HIDDEN SURFACES.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB0935A
SA

Figure 1. Repair Left Stabilator Attach Fitting Bushings.

INSPECT - Continued

- a. Inspect bonding of liners in fitting. Probe for edge separation using 0.0015-inch feeler gage. **VOIDS MUST NOT EXCEED 5% OF JOINT LINE LENGTH.** Replace pylon if limits are exceeded (WP 0338 00).
- b. Inspect liner bore for corrosion. **PITS MUST NOT EXCEED 0.003-INCH DEEP OR COVER MORE THAN 25% OF LINER BORE.** Replace pylon if limits are exceeded (WP 0338 00).
- c. If corrosion is within limits remove using scouring pad, Item 205, WP 1803 00.
- d. Inspect liner bore for damage. **LINER BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.40-INCH WIDE, AND 0.75-INCH LONG.** Replace pylon if limits are exceeded (WP 0338 00).
- e. If damage is within limits blend out using scouring pad, Item 205, WP 1803 00. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** Replace pylon if limits are exceeded (WP 0338 00).
- f. Blend out nicks, dents, scratches, or other damage, using abrasive cloth, Item 82, WP 1803 00. **MINIMUM THICKNESS BETWEEN BLENDED SURFACE AND OPPOSITE LUG FACE MUST BE 0.465-INCH (Figure 1, Detail B).** If limits are exceeded, replace pylon (WP 0328 00). If damage is within limits blend out using scouring pad, Item 205, WP 1803 00. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** Replace pylon if limits are exceeded (WP 0338 00).

INSTALL

1. Install bushings using sealing compound, Item 293, WP 1803 00, and stabilator/pylon hinge tool to a depth of 0.752 - 0.750 inches (Figure 1, Detail A).
2. Apply sealing compound, Item 293, WP 1803 00, around flange and end of bushing. Pay particular attention to make sure gap between bushings of inboard lug is filled.
3. Allow sealing compound to cure.
4. Line ream bushings 0.6250 to 0.6265-inch diameter.
5. Deburr bushings.
6. Apply a coating of corrosion removing compound, Item 108, WP 1803 00.
7. Apply a coating of solid film lubricant, Item 180, WP 1803 00.
8. Install stabilator (WP 0338 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

TAIL ROTOR PYLON RIGHT STABILATOR ATTACH FITTING

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Bushing /Bearing Removal/Installation Set, 70700-20129-041
Feeler Gage, GGG-G-17
Fluorescent Penetrant Inspection Kit, XMA101

Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

TM 1-1520-265-23
TM 55-1500-345-23
WP 0328 00
WP 0338 00
WP 0381 00
WP 1803 00

Material/Parts

Cloth, Abrasive, Item 82, WP 1803 00
Corrosion Preventive Compound, Item 100, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00

REMOVE**WARNING****CSI**

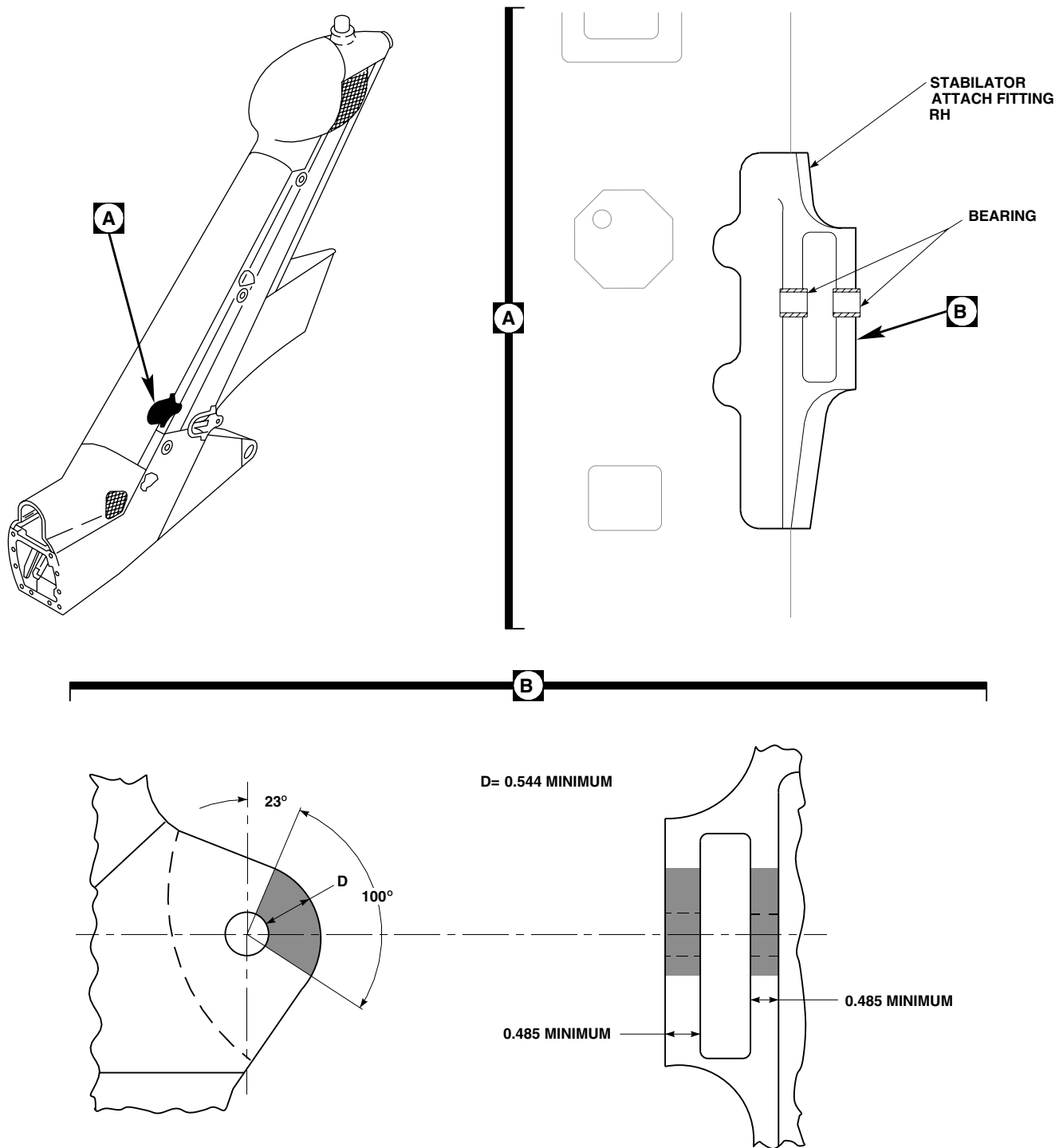
Right stabilator attach fittings on pylon and attach bolts, and actuator attachment fitting (on stabilator) and attach bolt, have critical characteristics. Extreme care must be used to prevent damage during stabilator removal.

1. Remove stabilator (WP 0338 00).
2. Remove spherical bearings from right fitting, using bearing removal/installation set (Figure 1, Detail A), as follows:

NOTE

Nuts installed on either end of tools should only be snug.

- a. If bearing is roller staked assemble tool (toolkit instructions step 1), and remove staking flanges on outside of outer race, by rotating arbor and cutter while applying light pressure. Continue until cutter has penetrated staking 0.020-inch.
- b. Using installation/removal tool, remove staking flanges from outside of other bearing using procedure in step 2.
 - a.
- c. Using installation/removal tool, remove bearings from fitting (toolkit instructions step 2 and step 3).

REMOVE - Continued**NOTES:**

1. CRITICAL CHARACTERISTIC AREAS SHOWN IN SHADED AREA, INCLUDING HIDDEN SURFACES.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB0936A
SA

Figure 1. Replace Rotor Pylon Pylon Right Stabilator Attach Fitting Bearings.

INSPECT**WARNING****CSI**

The left and right stabilator attachment fittings have critical areas that must not be damaged during inspection or repair. Refer to Figure 1, Detail B for critical characteristic areas.

1. Visually inspect critical characteristic areas of left and right fittings (Figure 1).
2. Blend out nicks, dents, scratches, or other damage, using abrasive cloth, Item 82, WP 1803 00. **MINIMUM THICKNESS BETWEEN BLENDED SURFACE AND OPPOSITE LUG FACE MUST BE 0.485-INCH.** If limits are exceeded, replace pylon (WP 0328 00).
3. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
4. Check bushing bores on left fitting for galling and corrosion. Blend out any galling or corrosion with abrasive cloth, Item 82, WP 1803 00. **DAMAGE AFTER BLENDING SHALL NOT EXCEED 0.010-INCH OR 25% OF BORE AREA.**
5. Inspect lug area of left and right fitting for cracks, using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace pylon if cracks are found (WP 0328 00).

WARNING**CSI**

- Bonding of liners in fitting is a critical characteristic. Proper bonding must be verified.
 - Damage to fitting will occur if feeler gage is improperly used. Liner may become disbonded. Use care when using feeler gage to check for liner edge separation.
6. Inspect bonding of liners in fitting. Probe for edge separation using 0.0015-inch feeler gage. **VOIDS MUST NOT EXCEED 5% OF JOINT LINE LENGTH.** Replace pylon if limits are exceeded (WP 0328 00).
 7. Inspect liner bore for corrosion. **PITS MUST NOT EXCEED 0.003-INCH DEEP OR COVER MORE THAN 25% OF LINER BORE.** Replace pylon if limits are exceeded (WP 0328 00).
 8. If corrosion is within limits remove using scouring pad, Item 205, WP 1803 00.
 9. Inspect liner bore for damage. **LINER BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.75-INCH LONG.** Replace pylon if limits are exceeded (WP 0328 00).
 10. If damage is within limits blend out using scouring pad, Item 205, WP 1803 00. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** Replace pylon if limit is exceeded (WP 0328 00).

INSTALL

1. Install spherical bearings (Figure 1, Detail A), as follows:

INSTALL - Continued**NOTE**

- For each installation operation, assemble tool from bushing/bearing removal/installation set according to information plates in tool case.
 - Stabilator attach fitting liner may need to be chamfered if ball bearings were previously installed. If not previously chamfered, chamfer per step 4 and step 5 of toolkit instructions.
 - Prior to installing bearings, apply corrosion preventive compound Item 100, WP 1803 00.
 - a. Assemble tool and install bearing.
 - b. Using tool, stake both bearings on inner and outer faces (toolkit instructions step 8 and step 9). Stagger stake points 45 degrees from those on first side. Staking flare is complete when 0.005-inch feeler gage cannot be inserted between flare lip and fitting.
2. On left attach fitting, seal flanged bushings around flanges, using sealing compound, Item 273, WP 1803 00.
 3. Install stabilator (WP 0338 00).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

TAIL ROTOR PYLON STABILATOR ACTUATOR ATTACH FITTING

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
WP 0336 00
WP 0342 00
WP 0669 00

REMOVE

1. Remove tail rotor drive shaft (WP 0669 00).
2. Remove stabilator actuator (WP 0342 00).
3. Remove pylon step (WP 0336 00).
4. Remove swage lock pins from fitting and zee (TM 1-1500-204-23).
5. Remove threaded pins and bolts from fitting and spar (Figure 1, Detail A).

WARNING**CSI**

The stabilator actuator attach fittings have critical areas that must not be damaged during installation. **NICKS, DENTS, SCRATCHES AND TOOL OR SCUFF MARKS WITHIN CRITICAL CHARACTERISTIC AREA SHALL NOT EXCEED ACCEPTANCE LIMITS.**

6. Remove fitting from pylon.

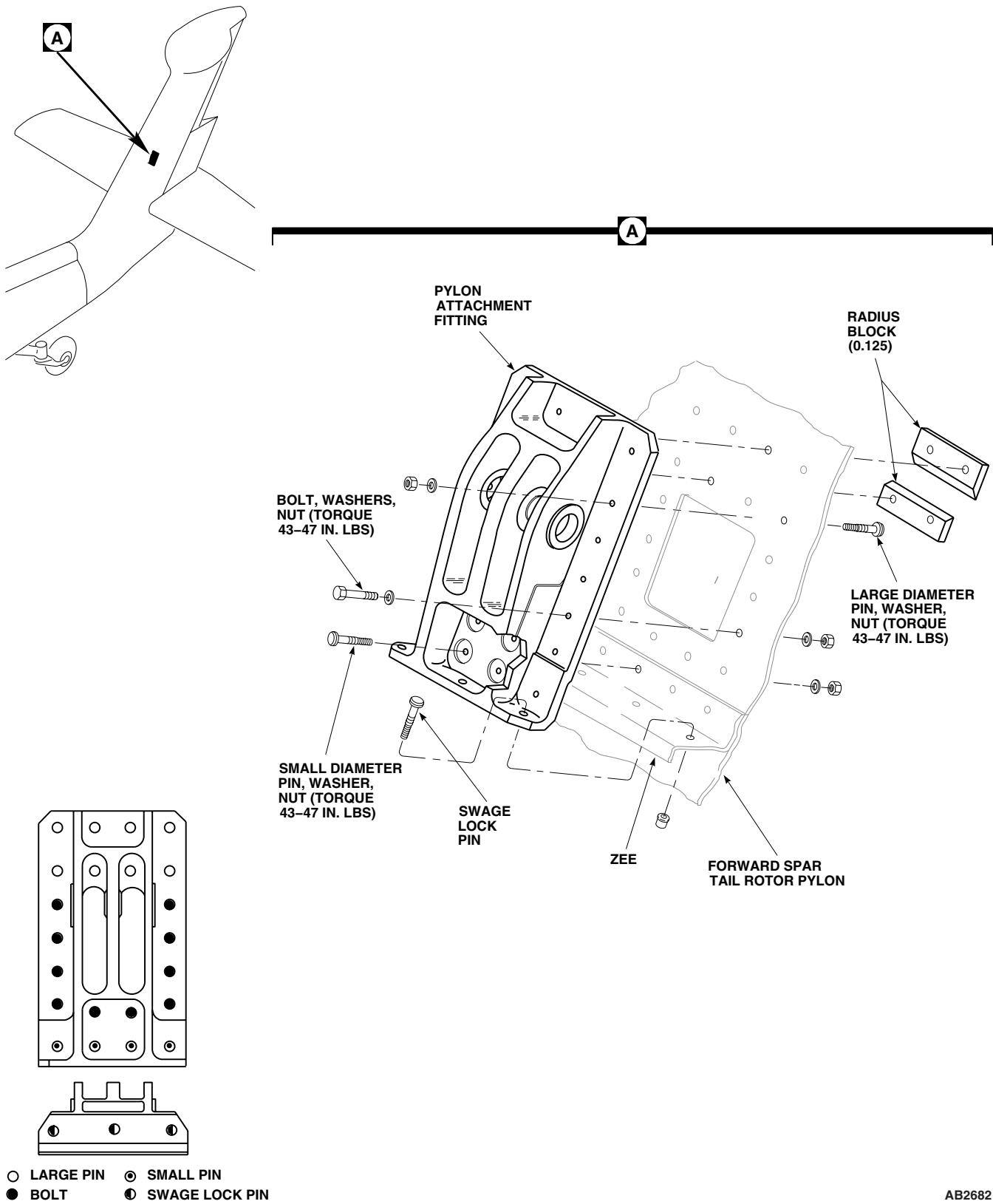
INSTALL

WARNING**CSI**

The stabilator actuator attach fittings have critical areas that must not be damaged during installation. **NICKS, DENTS, SCRATCHES AND TOOL OR SCUFF MARKS WITHIN CRITICAL CHARACTERISTIC AREA SHALL NOT EXCEED ACCEPTANCE LIMITS.**

1. Place fitting on pylon spar, with bushings toward top of pylon and flanged and against zee (Figure 1, Detail A). Align fastener holes.
2. Install radius blocks under heads of 4 large pins.
3. Install threaded pins, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
4. Install bolts, washers, and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**

INSTALL - Continued



AB2682
SA

Figure 1. Replace Stabilator Actuator Attach Fitting.

INSTALL - Continued

5. Install swage lock pins (TM 1-1500-204-23).
6. Install pylon step (WP 0336 00).
7. Install stabilator actuator (WP 0342 00).
8. Install tail rotor drive shaft (WP 0669 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****TAIL ROTOR PYLON STABILATOR ACTUATOR ATTACH FITTING BEARINGS**

This WP supersedes WP 0424 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Bushings/Bearing Removal/Installation Set, 70700-20129-041
Gloves
Goggles/Face Shield
Feeler Gage, GGG-G-17
Open End Wrench, A-A-1356

Cloth, Cheesecloth, Item 85, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Primer, Sealing Compound, Item 248, WP 1803 00
Sealing Compound, Item 293, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

References

WP 0342 00
WP 0423 00
WP 0669 00
WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Brush, Acid Swabbing, Item 57, WP 1803 00
Cloth, Abrasive, Item 81, WP 1803 00

REMOVE

1. Remove tail rotor drive shaft (WP 0669 00).
2. Remove stabilator actuator (WP 0342 00).

NOTE

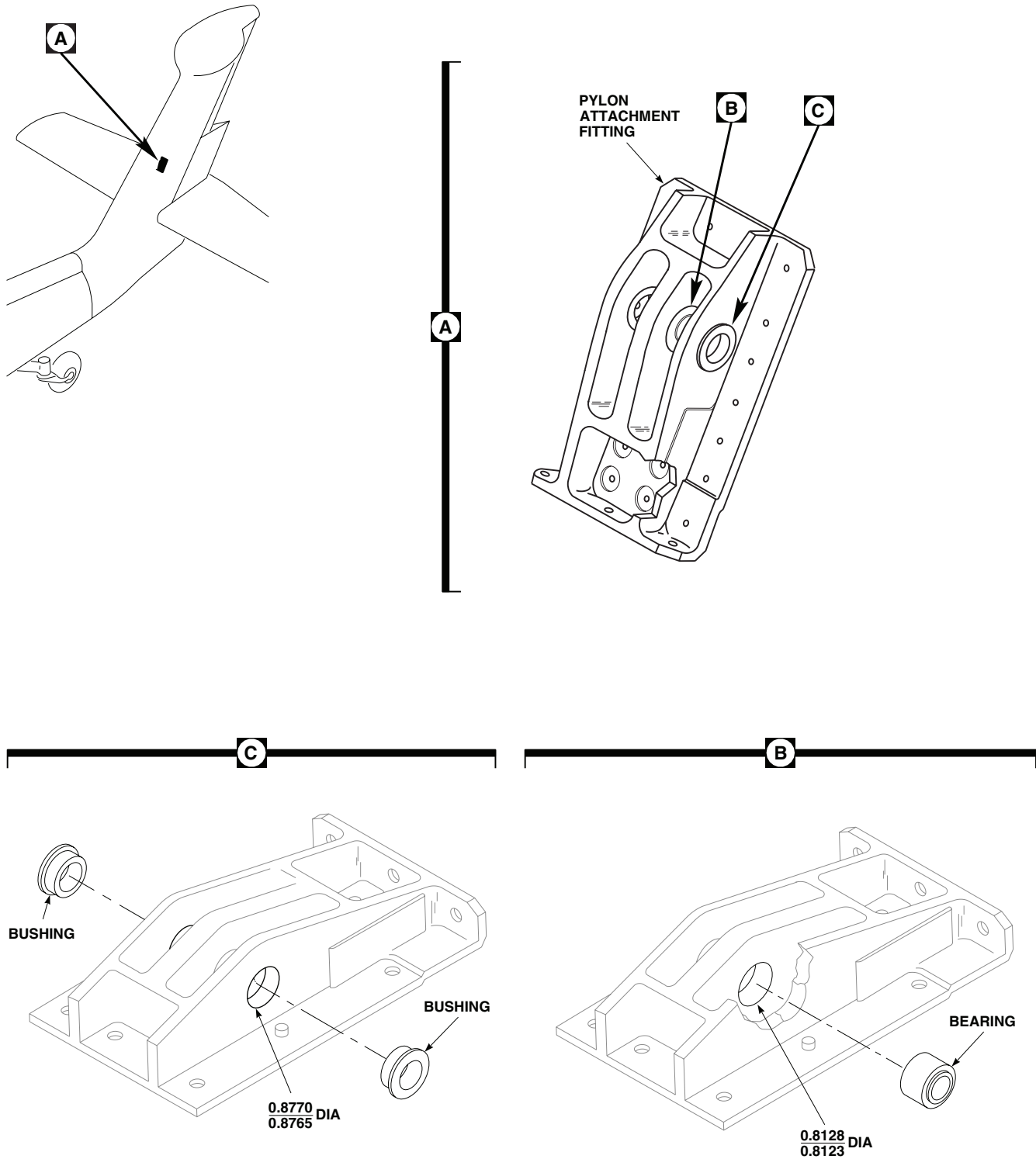
- If either bushings or bearing need replacement, replace both.
 - For the following steps refer to bushing/bearing removal/installation set instructions for removal of bushings and bearings.
3. Assemble tool and remove flanged bushings from fitting (toolkit instructions step 1 and step 2) (Figure 1, Detail B and Detail C).
 4. If bearing is roller staked assemble tool and remove staking (toolkit instructions step 3 and step 4), from one side of bearing outer race by rotating arbor and cutter, applying light pressure. Continue until cutter has penetrated staking 0.020-inch.

WARNING**CSI**

The stabilator actuator attach fittings have critical areas that must not be damaged during removal. **NICKS, DENTS, SCRATCHES AND TOOL OR SCUFF MARKS WITHIN CRITICAL CHARACTERISTIC AREA SHALL NOT EXCEED ACCEPTANCE LIMITS.**

5. Assemble tool and remove bearing from fitting (toolkit instructions step 5 and step 6).

REMOVE - Continued



NOTE:
ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB2517
SA

Figure 1. Replace Tail Rotor Pylon Stabilator Actuator Attach Fitting Bearings.

INSPECT**WARNING****CSI**

- Bonding of liners in center lobe of stabilator actuator attach fitting is a critical characteristic. Proper bonding must be verified.
 - Damage to fitting will occur if feeler gage is improperly used. Liner may become dis-bonded. Use care when using feeler gage to check for liner edge separation.
1. Inspect bonding of liners in center lobe of fitting. Probe for edge separation using 0.0015-inch feeler gage. **VOIDS MUST NOT EXCEED 5% OF JOINT LINE LENGTH.** Replace fitting if limits are exceeded (WP 0423 00).
 2. Inspect liner bore for corrosion. **PITS MUST NOT EXCEED 0.005-INCH DEEP OR COVER MORE THAN 25% OF LINER BORE.** Replace fitting if limits are exceeded (WP 0423 00).
 3. If corrosion is within limits remove using scouring pad, Item 205, WP 1803 00.

WARNING**CSI**

The stabilator actuator attach fittings have critical areas that must not be damaged.
NICKS, DENTS, SCRATCHES AND TOOL OR SCUFF MARKS WITHIN CRITICAL CHARACTERISTIC AREA SHALL NOT EXCEED ACCEPTANCE LIMITS.

4. Inspect liner bore for damage. **LINER BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.75-INCH LONG.** Replace fitting if limits are exceeded (WP 0423 00).
5. If damage is within limits blend out using scouring pad, Item 205, WP 1803 00. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** Replace fitting if limit is exceeded (WP 0423 00).
6. Measure liner/bearing bore in fitting. **LINER /BEARING BORE IN FITTING MUST BE 0.8128 TO 0.8123-INCH DIAMETER** (Figure 1, Detail B). Replace fitting if limits are exceeded (WP 0423 00).
7. Measure bushing bore in fitting. **BUSHING BORE IN FITTING MUST BE 0.8770 TO 0.8765 -INCH DIAMETER** (Figure 1, Detail C). Replace fitting if limits are exceeded (WP 0423 00).

INSTALL

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

CAUTION

To avoid damage to adhesive retaining liners in fitting bore and in bearing, do not allow technical acetone to contact these parts.

NOTE

For the following steps, refer to bushing/bearing removal/installation set instructions for installation of bushings and bearings.

1. Clean mating surfaces of bearing outer race and fitting bore, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
2. Remove residual sealing compound from fitting surfaces and bores, using abrasive cloth, Item 81, WP 1803 00. Sand very lightly.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

3. Clean sanded surfaces with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Let air dry.
4. Apply thin coat of sealing compound primer, Item 248, WP 1803 00, to bore in fitting center lobe, using clean acid swabbing brush, Item 57, WP 1803 00.

WARNING

CSI

The stabilator actuator attach fittings have critical areas that must not be damaged during installation. **NICKS, DENTS, SCRATCHES AND TOOL OR SCUFF MARKS WITHIN CRITICAL CHARACTERISTIC AREA SHALL NOT EXCEED ACCEPTANCE LIMITS.**

5. Let primer dry for 15 to 30 minutes until visibly dry. Protect drying surface from any contamination or hand contact.

INSTALL - Continued**CAUTION**

Installation of bearing must be completed promptly to avoid premature curing of sealing compound by the frictional heat of insertion. If bearing is not fully installed within 4 minutes after applying sealing compound, repeat entire cleaning, priming, and compound application procedure.

6. Apply even coat of sealing compound, Item 293, WP 1803 00, using clean acid swabbing brush, Item 57, WP 1803 00, to mating surfaces of bearing and fitting bore.

NOTE

For each installation operation, assemble tool from bushing/bearing removal/installation set according to information plates in tool case.

7. Assemble tool and install bearing (toolkit instructions step 8 and step 9), holding arbor stationary while turning nut with open end wrench until sharp increase in torque is felt (Figure 1, Detail B and Detail C).
8. Inspect to verify that bearing is centered in fitting.
9. Allow 15 minutes for curing of adhesive holding bearing in fitting.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

10. Clean up excess sealing compound, Item 293, WP 1803 00, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
11. Assemble tool and stake bearing (toolkit instructions step 9 and step 10). Staking flare is complete when 0.005-inch feeler gage cannot be inserted between flare lip and fitting. Stagger stake points 45 degrees from those on first side.
12. Assemble tool and install flanged bushings in fittings (toolkit instructions step 11 and step 12).
13. Install stabilator actuator (WP 0342 00).
14. Install tail rotor drive shaft (WP 0669 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****TAIL ROTOR PYLON CABLE PULLEY SUPPORTS REPAIR**

This WP supersedes WP 0425 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Bushing Installation Kit, GGG-P-643
Gloves
Goggles/Face Shield
Hand Reamer Set
Puller Kit, GGG-P-781
Sizing Block
Slide Hammer, C646-2

Cloth, Cheesecloth, Item 85, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 1146 00
WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Carbon Dioxide, Item 64, WP 1803 00

REMOVE

1. Remove pulley wheel from support (WP 1146 00).
2. Remove bushings from support, using puller set and slide hammer (Figure 1, Detail A and Detail B and TM 1-1500-204-23).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/Face Shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

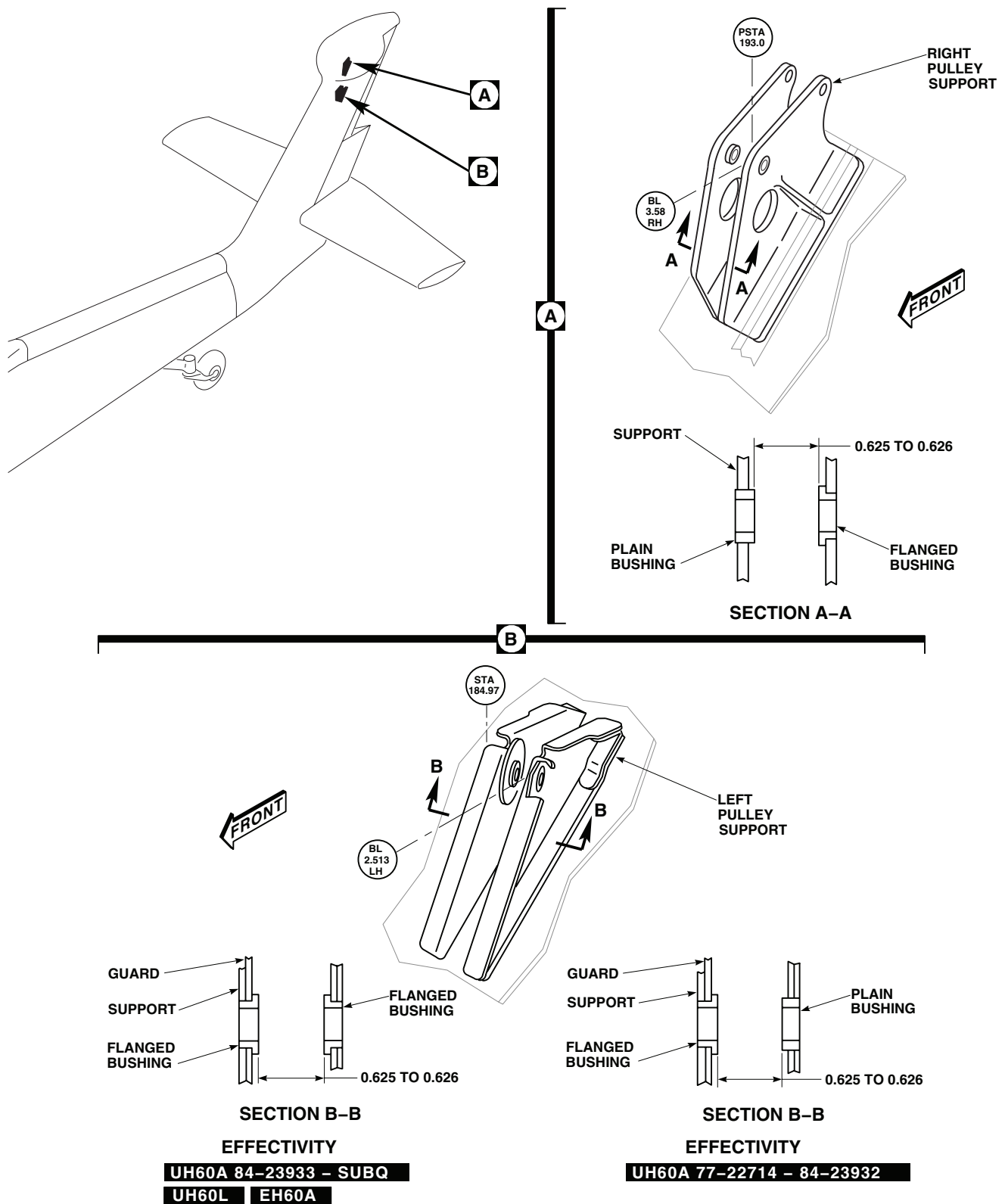
3. Thoroughly clean bores in support, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.

INSTALL**WARNING**

Personal injury will result if carbon dioxide or chilled parts touch bare skin. Wear protective gloves when working with carbon dioxide.

1. Put replacement bushings in container of carbon dioxide, Item 64, WP 1803 00, for 30 minutes.
2. Immediately before installing, coat outside of each bushing with primer coating, Item 243, WP 1803 00.
3. Install bushings in support while primer is still wet, using bushing installation kit and slide hammer (Figure 1, Detail A and Detail B and TM 1-1500-204-23).

INSTALL - Continued



NOTE:

ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0890C
SA

Figure 1. Repair Tail Rotor Pylon Cable Pulley Supports.

INSTALL - Continued**NOTE**

Obtain correct inside spacing between bushings by using properly sized sizing block.

4. Line ream bushings to 0.3120 to 0.3125-inch inside diameter, using hand reamer set.
5. Install pulley wheel (WP 1146 00).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****TAIL ROTOR PYLON CANTED HINGE FRAME REPAIR****This WP supersedes WP 0426 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

References

QQ-A-367

TM 55-1500-345-23

WP 0377 00

WP 0381 00

WP 1681 00

WP 1803 00

Material/Parts

Cloth, Abrasive, Item 76, WP 1803 00

Cloth, Abrasive, Item 83, WP 1803 00

Corrosion Preventive Compound, Item 100,
WP 1803 00

Sealing Compound, Item 273, WP 1803 00

Personnel RequiredAircraft Structural Repairer MOS 15G (1)

REPAIR

1. The canted hinge frame for the tail rotor pylon is machined from forged aluminum alloy, 7075-T73, QQ-A-367 (Figure 1). This fitting is considered critical primary structure. Negligible damage to the fitting is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area with abrasive cloth, Item 76, WP 1803 00, using long, even strokes. Polish with abrasive cloth, Item 83, WP 1803 00, and restore protective finish. Fitting is prime-coated with corrosion preventive compound, Item 100, WP 1803 00. Damage to the fitting greater than negligible limits must be reviewed by engineering authority for damage analysis and recommended repair or replacement. Bushings, eight places, are made from 17-4PH corrosion-resistant steel, heat-treated to H-925 condition and press fit into fittings. Bushings for upper and lower hinge lugs are cadmium-plated steel and are press-fit into lugs (Figure 1, Section A-A). Line ream bushings to 0.530 - 0.536 inch ID after installation. Brush cadmium plating onto machined areas of bushings after final reaming. Seal hinge bushings around flanges, using sealing compound, Item 273, WP 1803 00.
2. Open and close tail rotor pylon (WP 1681 00).
3. Seal fitting for corrosion protection and watertightness using sealing compound, Item 273, WP 1803 00 (WP 0377 00).
4. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

REPAIR - Continued

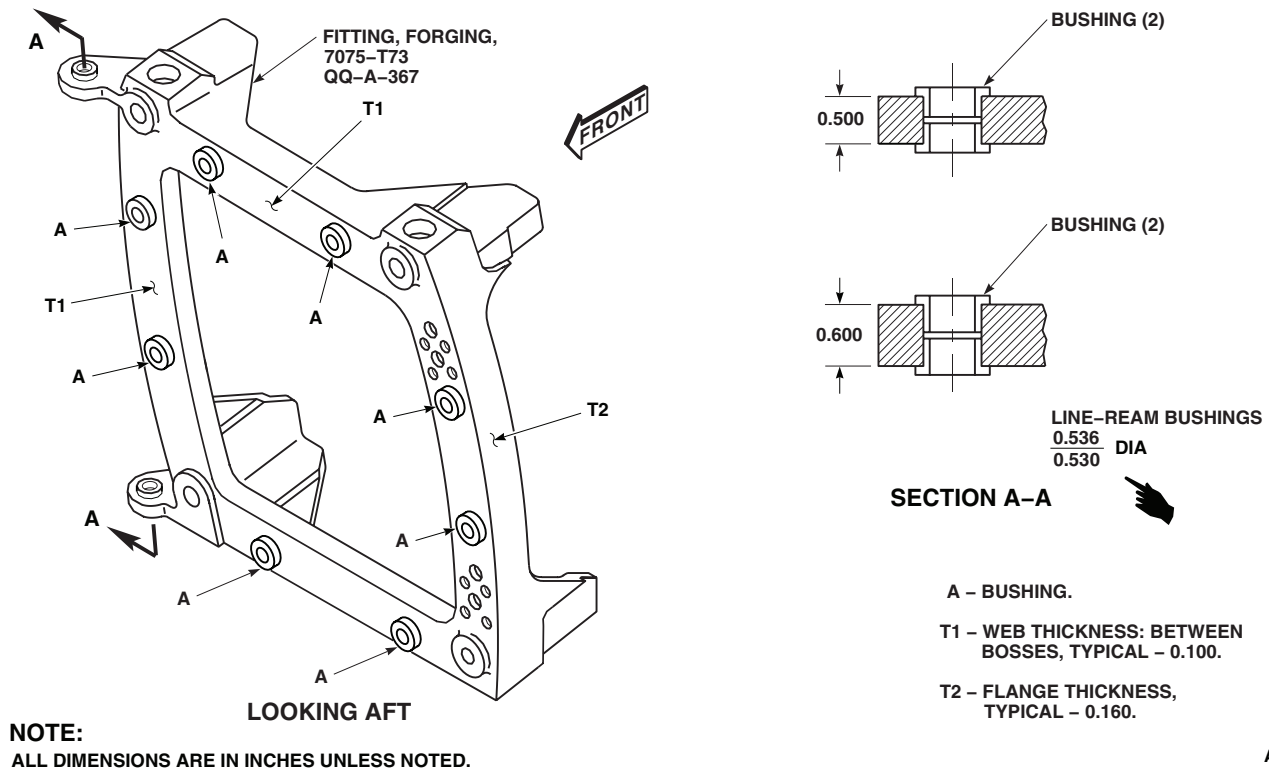


Figure 1. Repair Canted Hinge Frame Fitting.

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME**

STABILATOR FRAMING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B01
 Powertrain Repairer's Toolkit, SC 5180-99-B13

MIL-S-22499

MIL-S-5000

QQ-A-200

QQ-A-250

Material/Parts

Cloth, Abrasive, Item 83, WP 1803 00

TM 1-1500-344-23

TM 1-1520-237-PMI

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)

Aircraft Structural Repairer MOS 15G (1)

TM 55-1500-345-23

WP 0377 00

WP 0378 00

WP 0380 00

References

MIL-P-25515

MIL-S-18729

WP 0381 00

WP 1803 00

REPAIR**WARNING**

Boron strap material presents a health hazard to personnel when cut or drilled. Do not drill, repair, or rework Boron strap.

1. All stabilator framing between the front and rear spars is primary structure. The spars, between the Boron reinforcement straps bonded to the upper and lower cap angles, are limited to negligible damage repairs (WP 0377 00). No repair of the Boron straps is authorized. The front and rear spars are considered critical primary structure. All ribs and beams, between front and rear spars and LBL 82.72 and RBL 82.72 ribs, are considered primary structure. Negligible damage in these areas is limited to nicks, scratches, and corrosion, no deeper than 5% of material thickness at point of damage. Blend damage smoothly to surrounding area using long, even strokes. Polish with abrasive cloth, Item 83, WP 1803 00, and restore protective finish. Damage greater than negligible limits in the critical areas must be reviewed by engineering authority for determining repair or replacement. All repairs to stabilator framing must maintain designed strength and alignment criteria (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, Figure 1, Sheet 4, Figure 1, Sheet 5, Figure 1, Sheet 6, and Figure 1, Sheet 7).

WARNING**CSI**

Stabilator actuator attach fittings have critical characteristic areas. No damage is allowed. Dimension limits must not be exceeded.

2. Critical surfaces of actuator attach fitting are shown in Figure 1, Sheet 3, Detail D.
3. Evaluate damage in noncritical areas (WP 0377 00).
4. Repair nicks, scratches, and gouges in stabilator hinge fitting and actuator attach fittings up to 0.005-inch deep, 0.040-inch wide, and 0.500-inch long. Blend damage with minimum blend radius of 3/8-inch. Blended area shall be less than 0.008-inch below unrepaired surface after repair.

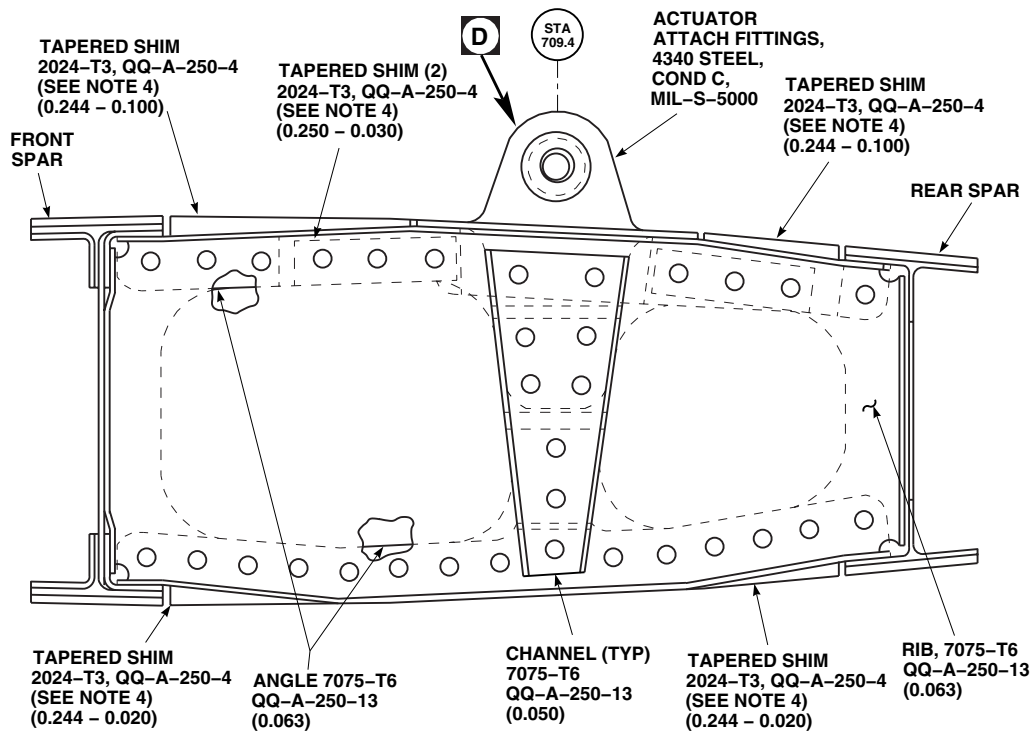
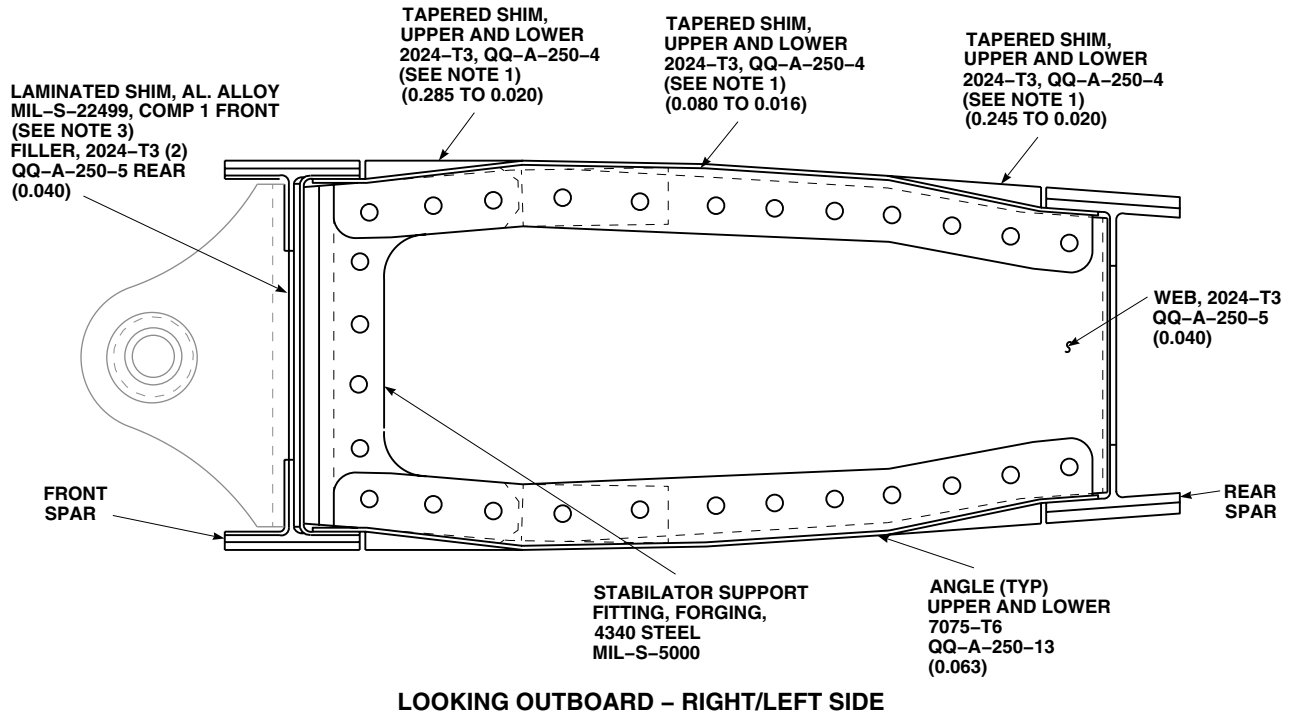
Exploded view diagram of a composite beam assembly. The diagram shows the following components and dimensions:

- BEAM, (TYP) (0.032)**
- WEB (0.063)**
- BL 9.0 RH**
- BL 0.0**
- BL 9.0 LH**
- RIB (0.032)**
- RIB (0.032) TAPERED SHIM 4 PLACES, 2024-T3 (0.206 TO 0.030)**
- BL 28.0 LH**
- ANGLE (TYP) BETWEEN BL 9.0 AND BL 75.0 (0.032)**
- BL 47.0 LH**
- BL 66.0 LH**
- BL 82.72 LH**
- ANGLE (0.040)**
- ANGLE (0.040)**
- ANGLE (0.040)**
- ANGLE (TYP) BETWEEN BL 9.0 AND BL 7.0 (0.040)**
- ANGLE (0.032)**
- ANGLE (0.032)**
- DOUBLER (0.032)**
- ANGLE (0.032)**
- RIB (0.032)**
- TIP CAP REINFORCED PLASTIC**
- WEB, 4130 STEEL MIL-S-18729 (0.100)**
- TAPERED SHIM 2 PLACES, RH-LH 2024-T3 QQ-A-250-4 (0.284 TO 0.030)**
- TAPERED SHIM 6 PLACES, RH-LH 2024-T3 QQ-A-250-4 (0.244 TO 0.030)**

AA0895_1B
SA

0427 00-2

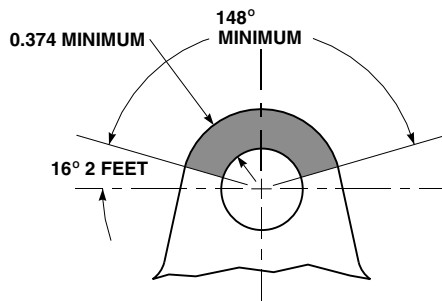
REPAIR - Continued



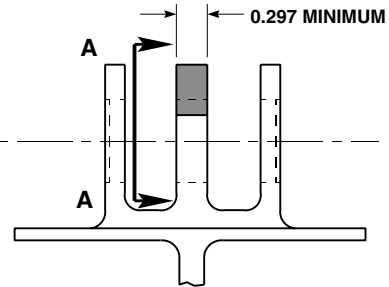
AA0895_2B
SA

Figure 1. Repair Stabilator Framing. (Sheet 2 of 7)

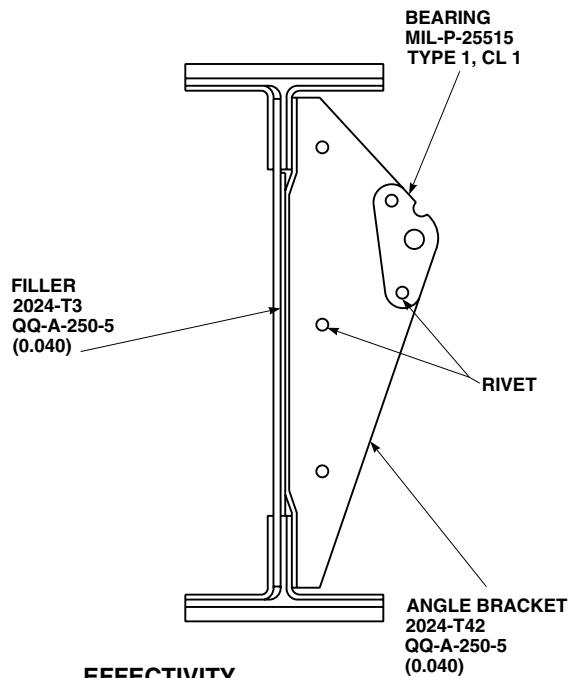
REPAIR - Continued



VIEW A-A

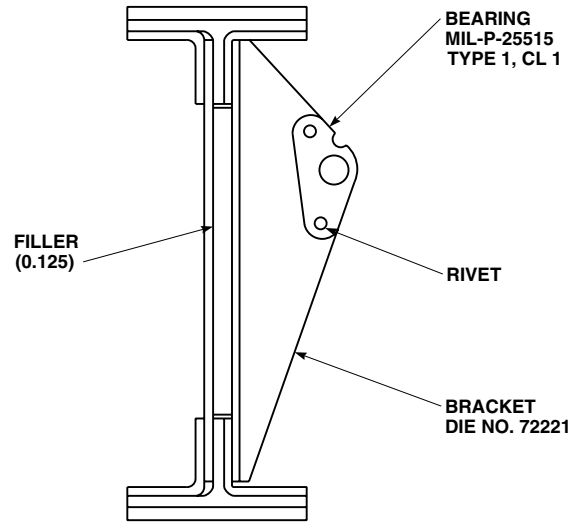


SHADED AREA CONTAINS
CRITICAL SURFACES



EFFECTIVITY

UH60A 77-22714 - 80-23494



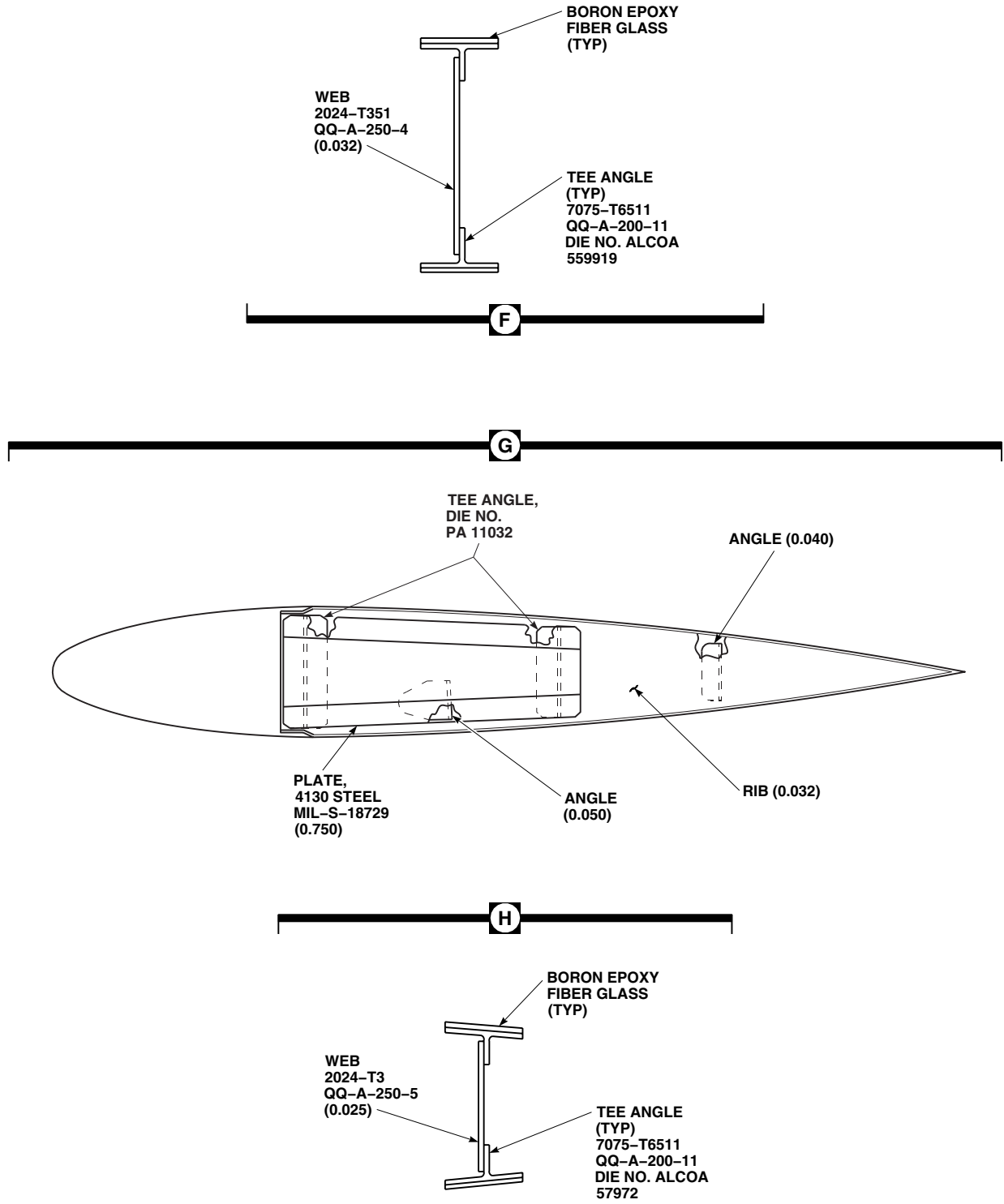
EFFECTIVITY

UH60A 80-23495 - SUBQ

UH60L EH60A

Figure 1. Repair Stabilator Framing. (Sheet 3 of 7)

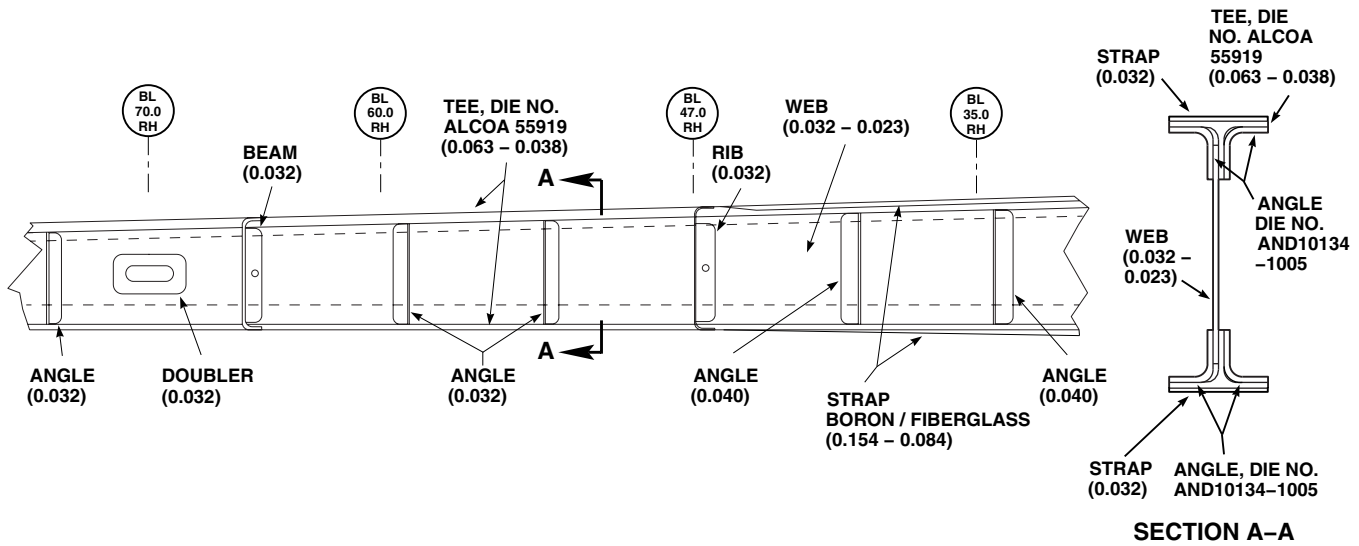
REPAIR - Continued



AA0895_4B
SA

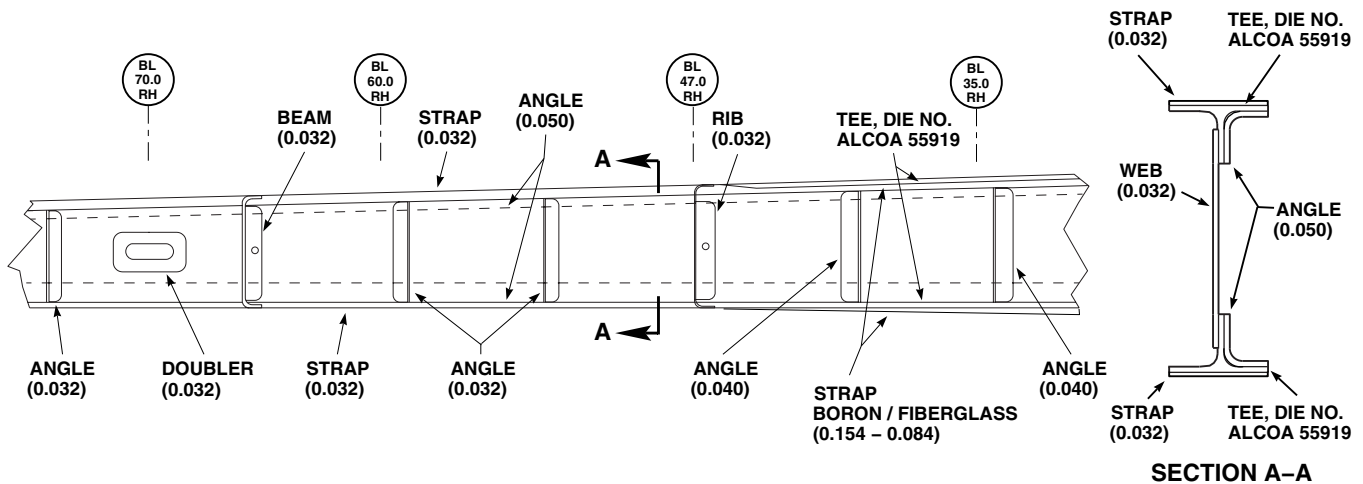
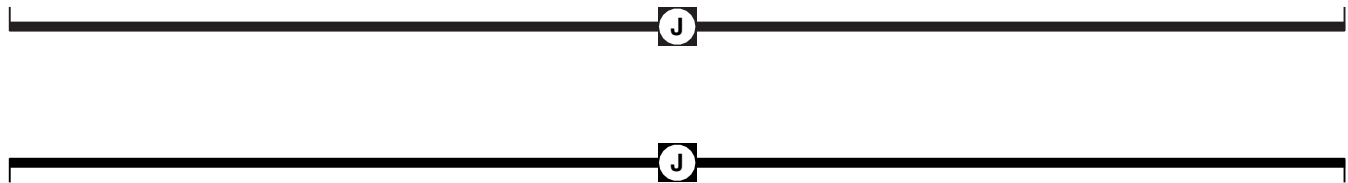
Figure 1. Repair Stabilator Framing. (Sheet 4 of 7)

REPAIR - Continued



EFFECTIVITY

UH60A 77-22714 - 89-26151



EFFECTIVITY

UH60A 89-26152 - SUBQ

Figure 1. Repair Stabilator Framing. (Sheet 5 of 7)

REPAIR - Continued

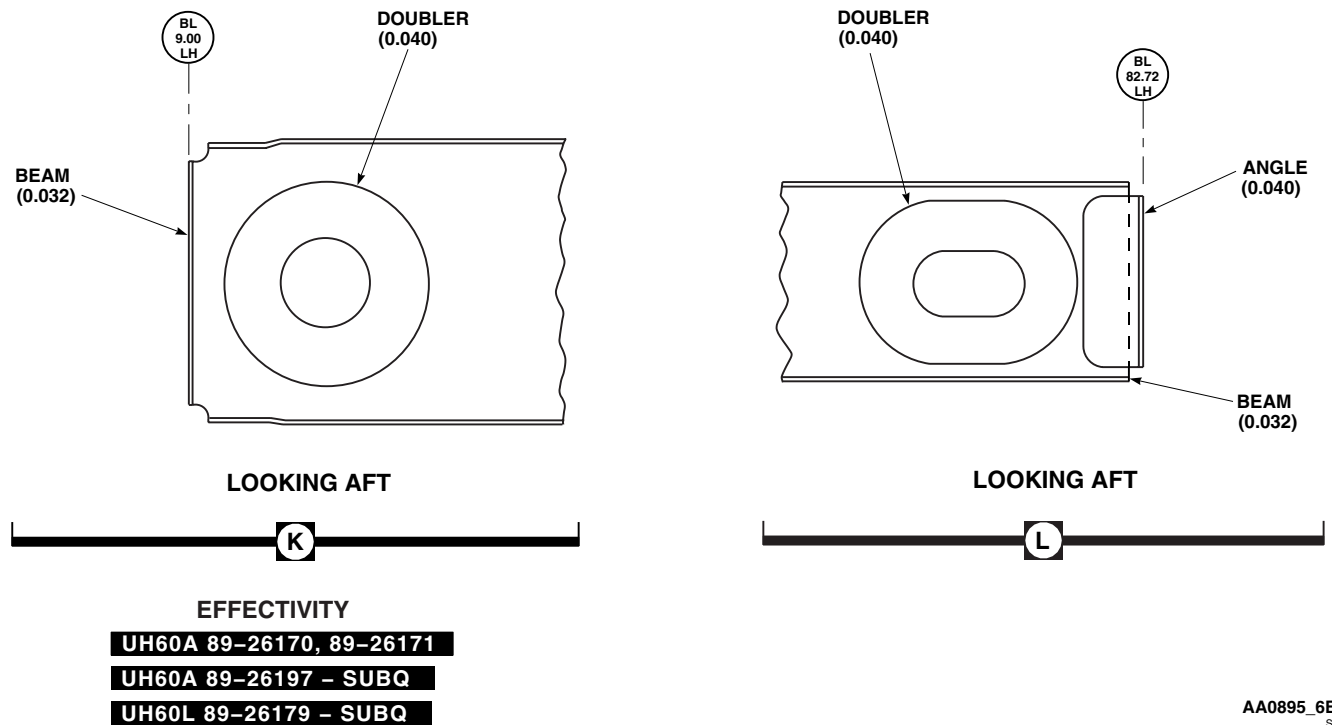


Figure 1. Repair Stabilator Framing. (Sheet 6 of 7)

5. Use WP 0378 00 and WP 0380 00 for typical framing repairs.
6. For corrosion protection and watertightness, seal all repairs between frame and plating (WP 0377 00 and TM 1-1500-344-23).
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
8. Stop-drill repairs are a temporary repair, and a follow up repair is required at the next periodic maintenance inspection (TM 1-1520-237-PMI).

REPAIR - Continued**NOTES:**

1. ALL SHEET METAL PARTS SHOWN ARE 7075-T6 ALCAD. QQ-A-250-13, EXCEPT AS NOTED.
2. TAPERED SHIMS ARE 2024-T3, QQ-A-250-4.
3. PEEL SHIM TO ELIMINATE ANY GAP BETWEEN FITTING AND SPAR.
4. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0895_7A
SA

Figure 1. Repair Stabilator Framing. (Sheet 7 of 7)

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

STABILATOR PLATING REPAIR

This WP supersedes WP 0428 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
 Drill Set, GGG-D-751
 Powertrain Repairer's Toolkit, SC 5180-99-B13

References

MIL-S-18729
 MIL-S-22499
 QQ-A-250
 TM-1500-345-23
 WP 0377 00
 WP 0378 00
 WP 0379 00
 WP 0381 00
 WP 0429 00
 WP 1803 00

Material/Parts

Adhesive, Item 11, WP 1803 00
 Corrosion Resistant Coating, Item 111, WP 1803 00
 Rivet, MS20470AD
 Rivet, M7885
 Rivet, NAS1738B
 Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 Aircraft Structural Repairer MOS 15G (1)

REPAIR STABILATOR PLATING

1. Stabilator plating is aluminum alloy 7075-T6, QQ-A-250-18, and 6061-T6, QQ-A-250-11, except Kevlar reinforced plastic leading edge and tip cap. Doublers attached to skins are 7075-T6, QQ-A-250-13 (Figure 1, Sheet 1, Figure 1, Sheet 2, Figure 1, Sheet 3, and Figure 1, Sheet 4).
2. All plating, except reinforced plastic members, trailing edge skin, and stabilator mount bolt access covers is primary structure. All repairs must maintain designed strength and alignment criteria. Evaluate damage (WP 0377 00).
3. To repair damaged stabilator trailing edge plating inboard of LBL 9.0 and RBL 9.0 within limits described in Table 1, follow the instructions in the repair trailing edge beaded panel section of (WP 0428 00), otherwise do this:
 - a. Remove damaged panel (Figure 1, Sheet 2, Detail D).
 - b. Make plate support angles of 0.040-inch 7075T6 stock.
 - c. Make four clips from 0.040-inch stock.
 - d. Break all edges and radius corners.
 - e. Coat parts with primer (WP 0381 00 and TM 55-1500-345-23).
 - f. Attach angles to ribs using clips and four rivets, NAS1738B, per clip.
 - g. Make replacement plating from 0.025-inch 7075T6 stock.
 - h. Attach plating to angles using rivets, NAS1738B, spaced 1 inch apart.
 - i. Attach plating to structure with M7885 rivets.
4. Use WP 0378 00 for typical plating repairs.

REPAIR STABILATOR PLATING - Continued

5. Use WP 0379 00 for typical reinforced plastic repairs.
6. Refer to WP 0429 00 for stabilator tip cap repair.
7. For corrosion protection and watertightness, seal all repairs on stabilator plating (WP 0377 00).
8. To repair damaged stabilator beaded plating, do this:
 - a. Cut away damaged plating to at least 1/16-inch beyond any bead bend radius (Figure 1, Sheet 2, Detail C).
 - b. Make a doubler of 0.187-inch-thick stock 1 1/4-inch wide, and the same length as bead being replaced.
 - c. Size and cut a patch from 0.025-inch-thick 7075T6 stock.
 - d. Break all edges and radius corners on all parts.
 - e. Coat parts with primer (WP 0381 00 and TM 55-1500-345-23).
 - f. Rivet doubler to patch using a single row of rivets, MS20470AD, spaced 1 inch apart.
 - g. Rivet patch to undamaged plating with rivets, NAS1738B. Space rivets 1 inch apart with a 1/4-inch edge distance.
9. Touch up paint in repaired area as required (WP 0381 00 and TM 55-1500-345-23).
10. See Table 1 for damage limits and repair actions.

REPAIR STABILATOR PLATING - Continued

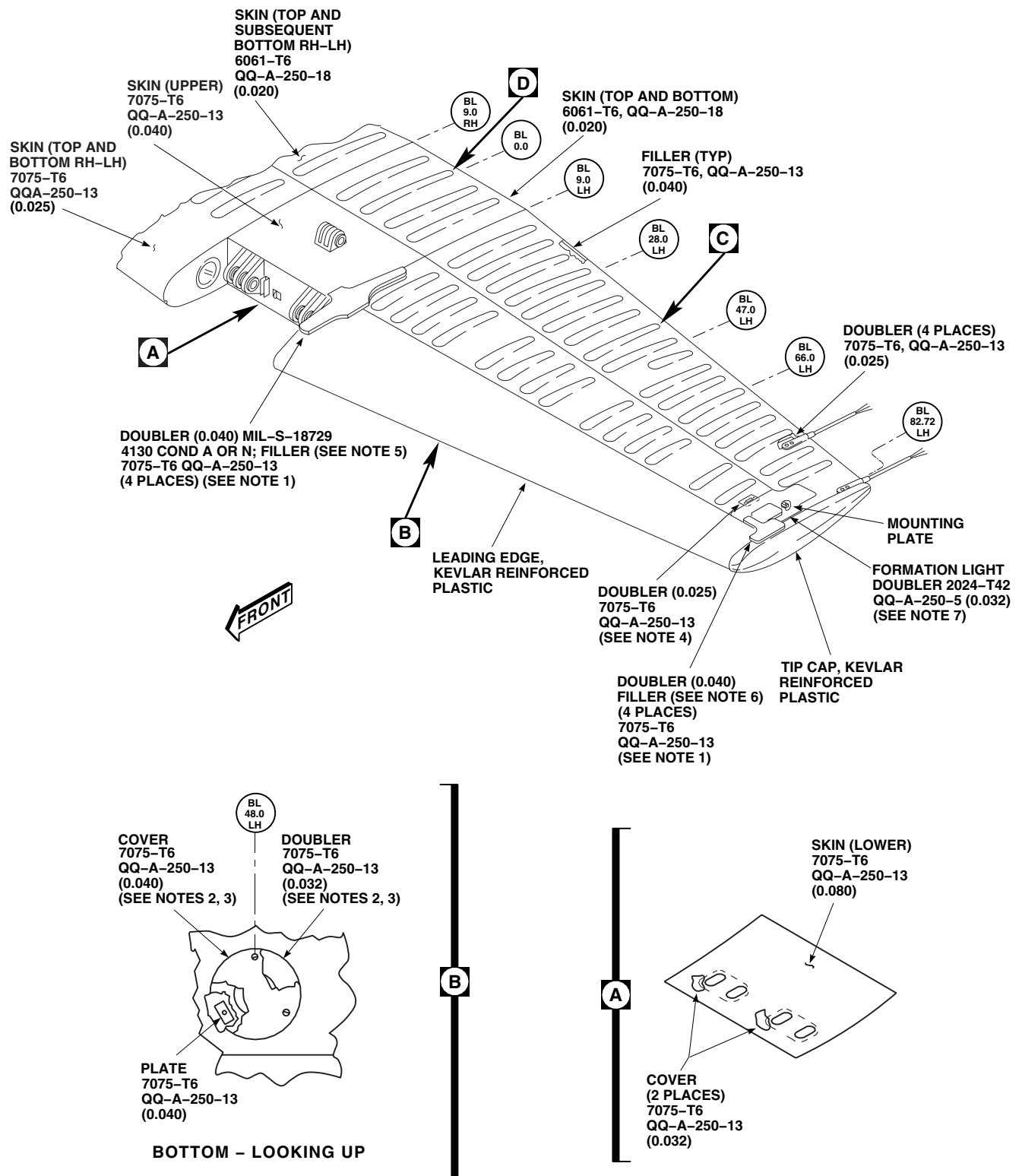
AA0896_1B
SA

Figure 1. Repair Stabilator Plating. (Sheet 1 of 4)

REPAIR STABILATOR PLATING - Continued

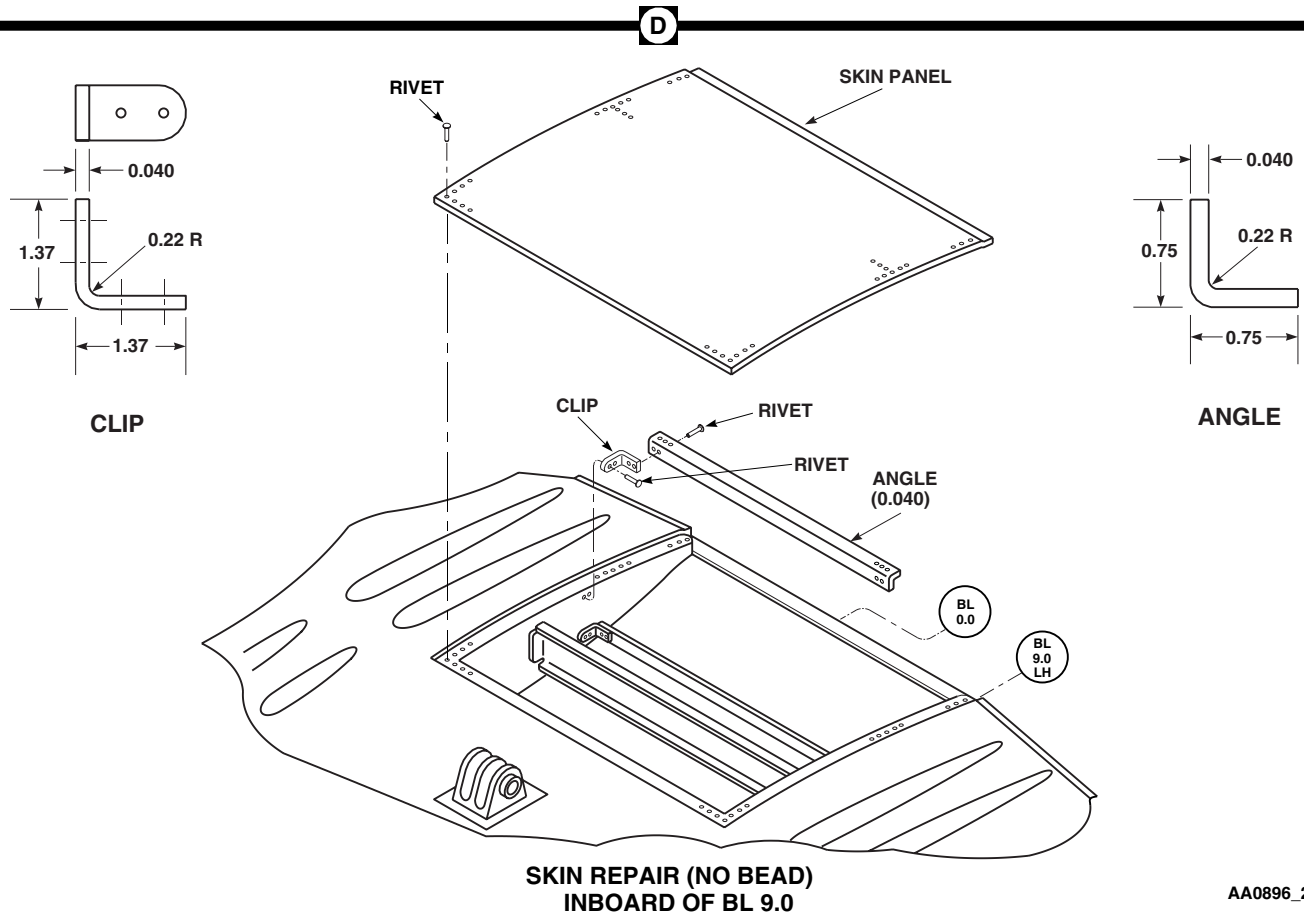
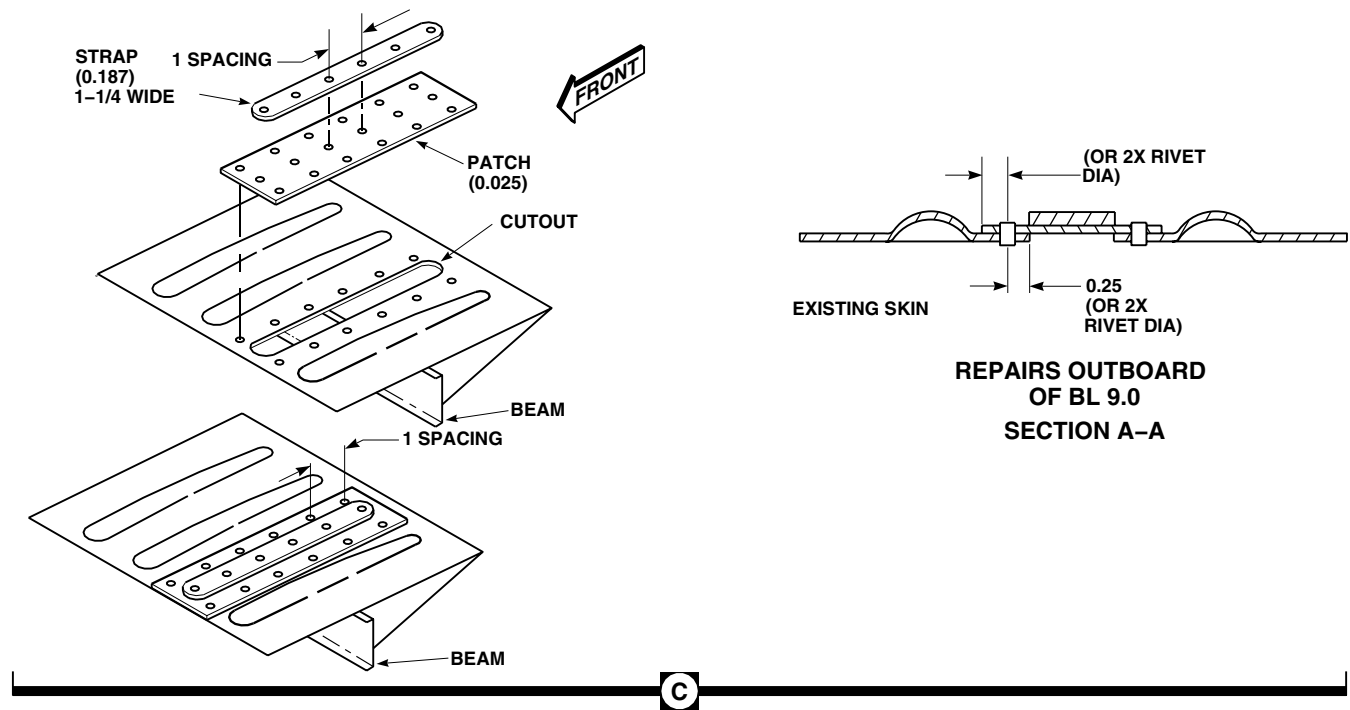
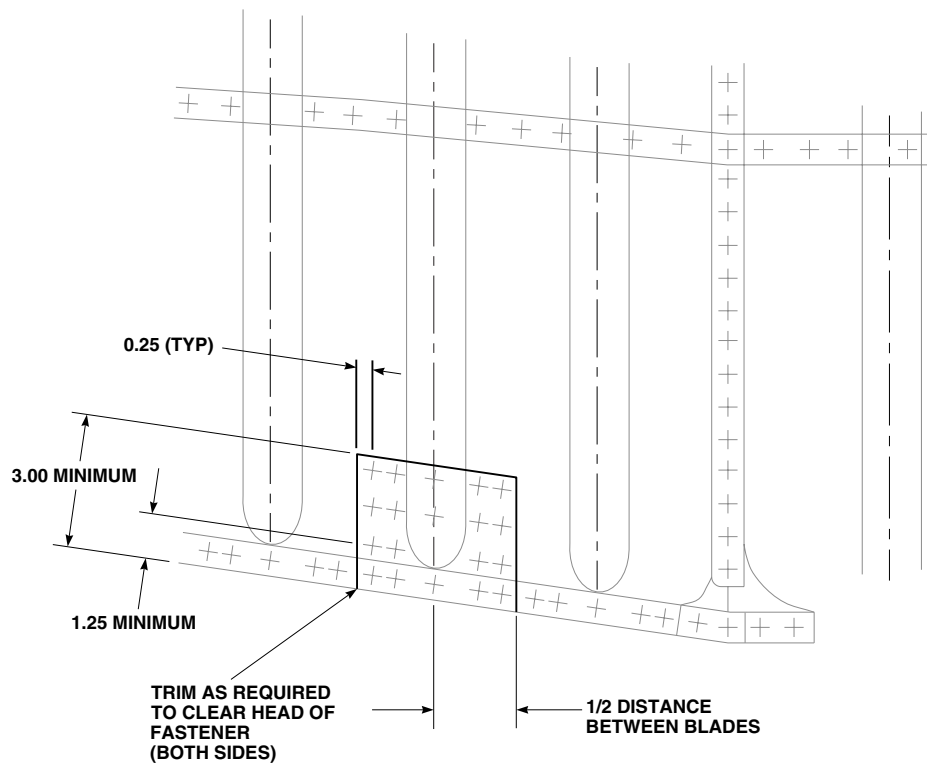


Figure 1. Repair Stabilator Plating. (Sheet 2 of 4)

REPAIR STABILATOR PLATING - Continued



LOOKING DOWN
UPPER SKIN LEFT SIDE

AA0896_3C
SA

Figure 1. Repair Stabilator Plating. (Sheet 3 of 4)

REPAIR STABILATOR PLATING - Continued

NOTES:

1. UH60A 80-23492 - SUBQ UH60L EH60A
2. UH60A 84-23935 - SUBQ UH60L EH60A MWO 50-72 RFRB
3. BOND WITH ADHESIVE, ITEM 11, WP 1803 00 AT ASSEMBLY.
4. UH60A 85-24462 - SUBQ UH60L EH60A 85-24465 - SUBQ
5. UH60A EH60A UH60L 89-26149 - 89-26154 UH60L 89-26179 - 91-26366 (0.025). ◀
UH60L 91-26367 - SUBQ (0.050). ◀
6. UH60A EH60A UH60L 89-26149 - 89-26154 UH60L 89-26179 - 91-26366 (0.032). ◀
UH60L 91-26367 - SUBQ (0.050). ◀
7. UH60A 80-23492 - SUBQ UH60L EH60A MOD
8. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AA0896_4B
SA

Figure 1. Repair Stabilator Plating. (Sheet 4 of 4)

Table 1. Negligible Damage Limits.

COMPONENT	TYPE AND SIZE	REPAIR ACTION	REPAIR LOCATION	
			ON ACFT	OFF ACFT
Fiberglass/Epoxy Leading Edge	Scratches - - Surface scratches not penetrating into fabric	No repair required	x	
	- - Scratches penetrating not more than 1 ply, less than 2 inches long	No repair required	x	
	- - Scratches penetrating not more than 2 plies, less than 0.50-inch long, with at least 1 inch between scratches	No repair required	x	
	- - Scratches penetrating all 3 plies but not into honeycomb core	Repair (WP 0379 00)	x	
	Damage penetrating into honeycomb core, maximum diameter 6 inches	Repair using fiberglass instead of Kevlar (WP 0379 00)	x	

REPAIR STABILATOR PLATING - Continued

Table 1. Negligible Damage Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	REPAIR LOCATION	
			ON ACFT	OFF ACFT
Primary Skin Between Beads	Dents - - Up to 0.50-inch diameter and 0.060-inch deep, with at least 2 inches of undamaged surface between dents. No dents allowed within 1 inch of leading edge nose (Figure 2, Detail A)	No repair required. If damage exceeds limits, treat as damage penetrating into core	x	
	Cracks not ending at fastener holes, maximum length (L) 0.50-inch	Stop-drill using 0.125-inch drill bit	x	
	Edge damage, cleaned-up area (C) not more than 0.50-inch deep	Blend damage to smooth, rounded contour not more than twice the depth long	x	
	Nicks, scratches, and corrosion cleaned-up area (Figure 2, Section A-A) not to exceed: Depth (T1) 0.004-inch (0.15 x T) Width (b) 0.062-inch (2.5 x T) Length (h) 40% of H	Remove no more material than necessary during cleanup. Treat nicks and scratches as cracks if exceeding T1, b, or h limits (Figure 2)	x	
	Dents. Smooth dents, without sharp projections or abrasions, not more than 0.250-inch deep	No repair required. Treat all dents deeper than 0.250-inch, or those with nicks and gouges, as punctures	x	
	Punctures, maximum dimension 0.50-inch after cleanup	Trim damage to smooth, rounded contour. Repair using WP 0378 00	x	
Primary Skin Beads	Damage exceeding limits (Figure 2)	Repair using WP 0378 00 or WP 0428 00 as applicable		x
	Cracks, cleaned-up damage not to exceed: Width (W) 0.1875-inch Length (L) 0.50-inch	Stop-drill using 0.125-inch drill bit	x	

REPAIR STABILATOR PLATING - Continued

Table 1. Negligible Damage Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	REPAIR LOCATION	
			ON ACFT	OFF ACFT
Secondary Skin Between Beads	Nicks, scratches, and corrosion cleaned-up area (Figure 2), not to exceed: Depth (T1) 0.004-inch (0.15 x T) Width (b) 0.062-inch (2.5 x T) Width (W) 0.50-inch Length (h) 0.50-inch	Remove no more material than necessary during cleanup. Treat nicks and scratches as cracks if exceeding T1, b, or h limits (Figure 2)	x	
	Dents - - Smooth dents without sharp projections or abrasions not deeper than 0.050-inch	Treat all dents deeper than 0.050-inch, or those with nicks and gouges, as punctures	x	
	Punctures - - Maximum diameter 0.1875-inch after cleanup	Drill out damage using maximum drill diameter of 0.1875-inch	x	
	- - Damage exceeding limits (Figure 2)	Repair using WP 0378 00 or WP 0428 00 as applicable		x
	Cracks not ending at fastener holes, maximum length (L) 1 inch	Stop-drill using 0.125-inch drill bit	x	
	Edge damage, cleaned-up area (C) not to exceed 1 inch	Blend damage to smooth, rounded contour not longer than twice the depth	x	
	Nicks, scratches, and corrosion cleaned-up area (Figure 2, Section A-A) not to exceed: Depth (T1) 0.007-inch (0.35 x T) Width (b) 0.050-inch (2.5 x T) Length (h) 40% of H	Remove no more material than necessary during cleanup. Treat nicks and scratches as cracks if exceeding T1, b, or h limits (Figure 2)	x	
	Dents - - Smooth dents without sharp projections or abrasions not deeper than 0.250-inch	Treat all dents deeper than 0.250-inch, or those with nicks and gouges, as punctures	x	
	Punctures - - Maximum dimension 1 inch after cleanup	Trim damage to smooth, rounded contour and repair using WP 0378 00 or WP 0428 00 as applicable	x	

REPAIR STABILATOR PLATING - Continued

Table 1. Negligible Damage Limits. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION	REPAIR LOCATION	
			ON ACFT	OFF ACFT
Secondary Skin Beads	- - Damage exceeding limits (Figure 2)	Repair using WP 0378 00 or WP 0428 00 as applicable		x
	Cracks -- Cleaned-up damage maximum dimensions: Width (W) 0.50-inch Length (L) 1 inch	Stop-drill using 0.125-inch drill bit	x	
	Nicks, scratches, and corrosion cleaned-up area (Figure 2, Section A-A), not to exceed: Depth (T1) 0.007-inch (0.35 x T) Width (b) 0.050-inch (2.5 x T)	Remove no more material than necessary during cleanup. Treat nicks and scratches as cracks if exceeding T1, or b limits (Figure 2)	x	
	Dents -- Smooth dents without sharp projections or abrasions not deeper than 0.125-inch deep	Treat all dents deeper than 0.125-inch, or those with nicks and gouges, as punctures	x	
	Punctures -- Maximum dimension 0.50-inch after cleanup	Trim damage to smooth, rounded contour and repair using WP 0378 00 or WP 0428 00 as applicable	x	
	- - Damage exceeding limits (Figure 2)	Repair using WP 0378 00 or WP 0428 00 as applicable		x

REPAIR TRAILING EDGE BEADED PANEL

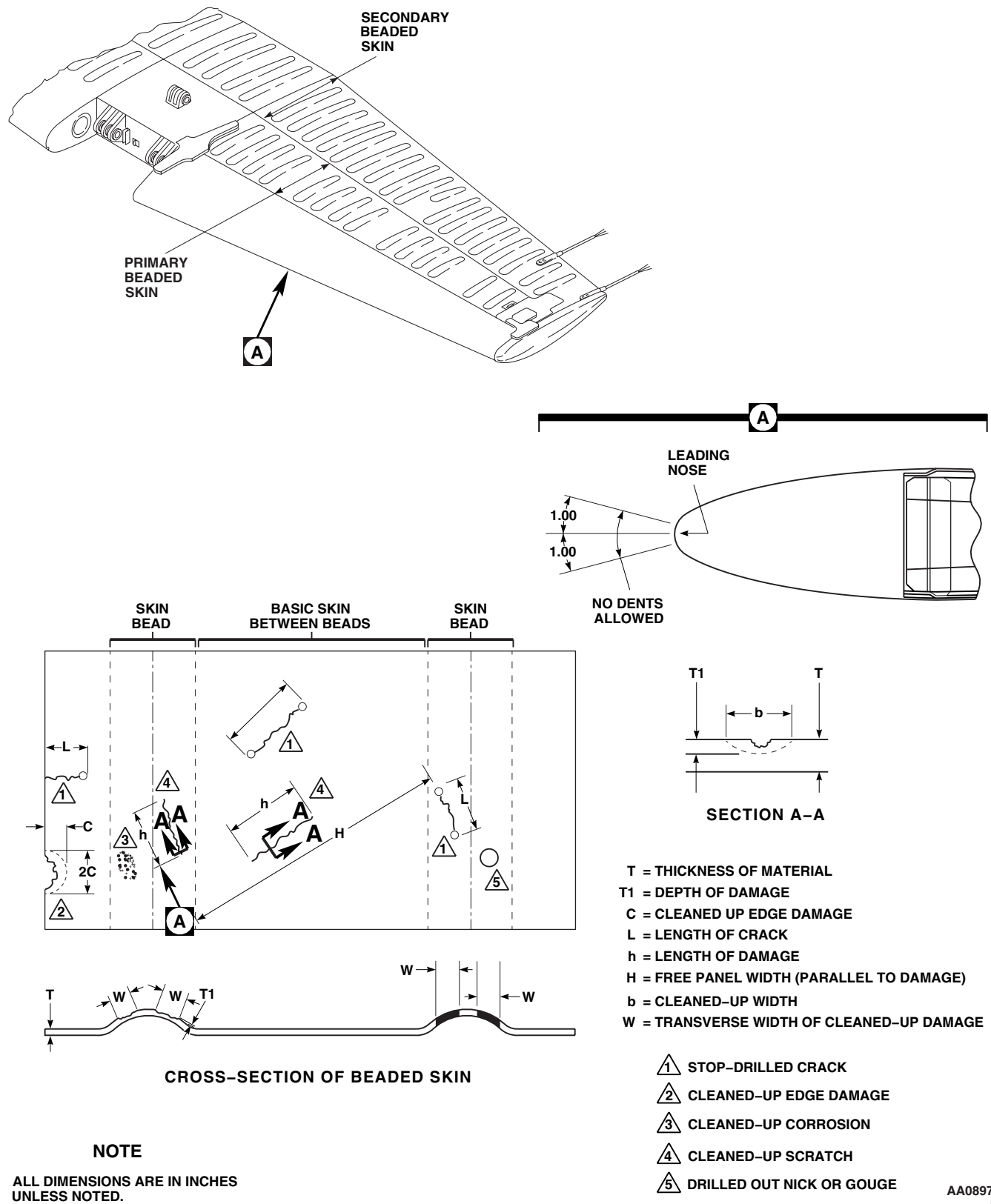
NOTE

Use a 70203-07051-142 (RH) or a -143 (LH) skin panel to fabricate a patch. To improve fit, cut the patch from approximately the same area of the skin that the patch will be used on the stabilator.

1. Prepare patch:

- a. Determine the area and length of the bead crack. Stop drill the end of the crack(s) use a No. 40 drill.

REPAIR TRAILING EDGE BEADED PANEL - Continued



AA0897A
SA

Figure 2. Repair Limits.

REPAIR TRAILING EDGE BEADED PANEL - Continued

- b. Trim the -142 or -143 to cover the entire crack plus half way between the inboard and outboard bead. The patch should be a minimum of 3 inches long as measured from the trailing edge and extended a minimum of 1 inch beyond crack.
 - c. Use the existing fastener pattern of the trailing edge.
 - d. Cut and size patch for installation maintaining a 0.25-inch edge distance of either side of patch not to exceed 0.25-inch distance from the fillet of the radius of bead. No fasteners should be located closer than 1.25-inch of the trailing edge.
 - e. Break all edges and radius corners of patch.
 - f. Touch up patch with corrosion resistant coating, Item 111, WP 1803 00.
2. Apply sealing compound, Item 273, WP 1803 00, between patch and existing skin.

CAUTION

Damage to the opposite side of skin can be caused by tail of fastener. Use care when installing fasteners.

3. Fasteners are to be wet installed.
- a. Use MS20470AD rivets as required at trailing edge.
 - b. Use NAS1738B-4-1 rivets as required to fasten patch to existing skin.
4. Touch up paint in repair area as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

UNIT LEVEL

AIRFRAME

STABILATOR TIP CAP REPAIR

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B01
 Powertrain Repairer's Toolkit, SC 5180-99-B13

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
 Aircraft Structural Repairer MOS 15G (1)

References

TM 55-1500-345-23
 WP 0341 00
 WP 0377 00
 WP 0379 00
 WP 0381 00

REPAIR

1. The stabilator tip cap is made from Kevlar reinforced plastic and is riveted to the outboard ends of the stabilator. Material thickness is 0.030-inch (Figure 1).
2. Evaluate and repair Kevlar tip cap (WP 0379 00). Damage beyond negligible limits will require removal of tip cap for repair or replacement. Repair cracks in attachment area or in contour of cap (WP 0379 00).
3. Remove and replace tip cap (WP 0341 00).

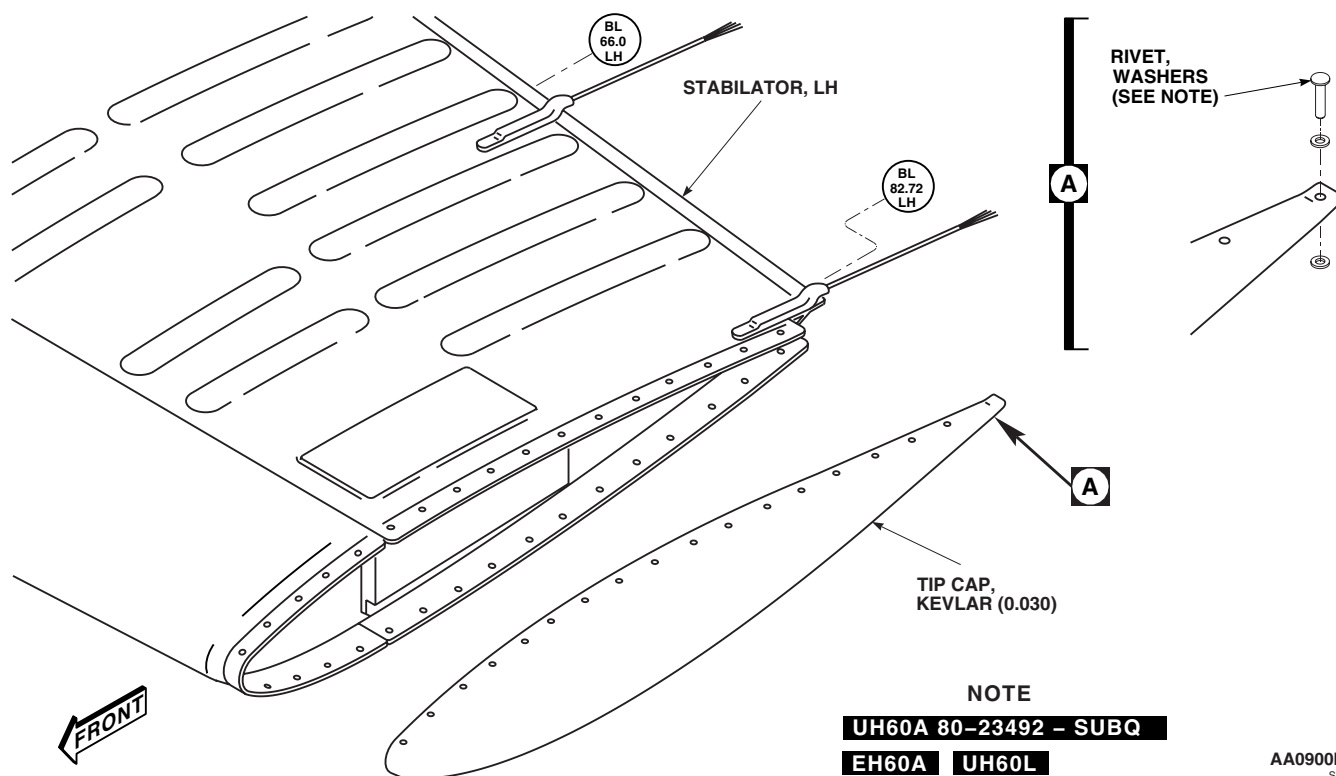


Figure 1. Repair Tip Cap.

REPAIR - Continued

4. For corrosion protection and watertightness, seal tip cap to stabilator (WP 0377 00).
5. Touch up paint finish as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

UNIT LEVEL**AIRFRAME****STABILATOR ACTUATOR ATTACH FITTING REPAIR**

This WP supersedes WP 0430 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Bushing/Bearing Removal/Installation Set, 70700-20129-041
Feeler Gage, GGG-G-17
Gloves
Goggles/Face Shield
Powertrain Repairer's Toolkit, SC 5180-99-B13

Pad, Scouring, Item 205, WP 1803 00
Primer, Sealing Compound, Item 247, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Sealing Compound, Item 292, WP 1803 00

Personnel Required

Aircraft Powertrain Repairer MOS 15D (1)
Aircraft Structural Repairer MOS 15G (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Brush, Acid Swabbing, Item 57, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00

References

WP 0338 00
WP 0342 00
WP 0423 00
WP 1803 00

REMOVE

1. Turn on all electrical power.
2. Slew down stabilator.
3. Support stabilator with padded support.
4. Turn off all electrical power.
5. Remove nut and bolt attaching actuator clevis to stabilator (WP 0342 00).
6. Lower stabilator slightly and disengage actuator clevis from stabilator fitting.
7. Secure actuator assembly away from fitting.

NOTE

- If either bushings or bearing need replacement, replace both.
 - For the following steps refer to toolkit instructions for removal/installation from 70209-06054/70209-07053 fitting.
8. Assemble tool and remove flanged bushings from fittings (toolkit instructions step 1 and step 2).
 9. If bearing is roller staked, assemble tool and remove staking (toolkit instructions step 3 and step 4) from one side of bearing outer race by rotating arbor and cutter, applying light pressure. Continue until cutter has penetrated staking 0.020-inch.
 10. Assemble tool and remove bearing from fitting (toolkit instructions step 5 and step 6).

INSPECT**WARNING****CSI**

- Bonding of liners in center lobe of stabilator actuator attach fitting is a critical characteristic. Proper bonding must be verified.
 - Damage to fitting will occur if feeler gage is improperly used. Liner may become dis-bonded. Use care when using feeler gage to check for liner edge separation.
1. Visually inspect bonding of liners in center lobe of fitting. Probe for edge separation using 0.0015-inch feeler gage. **VOIDS MUST NOT EXCEED 5% OF JOINT LINE LENGTH.** Replace fitting if limits are exceeded (WP 0423 00).
 2. Inspect liner bore for corrosion. **PITS MUST NOT EXCEED 0.005-INCH DEEP OR COVER MORE THAN 25% OF LINER BORE.** Replace stabilator fitting if limit is exceeded (WP 0338 00).
 3. If corrosion is within limits remove using scouring pad, Item 205, WP 1803 00. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** Replace stabilator fitting if limit is exceeded (WP 0338 00).
 4. Inspect liner bore for damage. **LINER BORE DAMAGE MUST NOT EXCEED 0.005-INCH DEEP, 0.040-INCH WIDE, AND 0.62-INCH LONG.** Replace stabilator fitting if limit is exceeded (WP 0338 00).
 5. If damage is within limits blend out using scouring pad, Item 205, WP 1803 00. **MAXIMUM BLENDED DEPTH IS 0.005-INCH.** Replace stabilator fitting if limit is exceeded (WP 0338 00).

INSTALL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

CAUTION

To avoid damage to adhesive retaining liners in fitting bore and in bearing, do not allow technical acetone to contact these parts.

NOTE

For the following steps refer to toolkit instructions for removal/installation from fitting, 70209-06054/70205-07053.

1. Clean mating surfaces of bearing outer race and fitting bore, using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
2. Remove residual sealing compound from fitting surfaces and bores, using abrasive cloth, Item 82, WP 1803 00. Sand very lightly.

INSTALL - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

3. Clean sanded surfaces with cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Let air dry.
4. Apply thin coat of sealing compound primer, Item 247, WP 1803 00, to bore in fitting center lobe, using clean acid swabbing brush, Item 57, WP 1803 00.
5. Let primer dry for 15 to 30 minutes until visibly dry. Protect drying surface from any contamination or hand contact.

CAUTION

Installation of bearing must be completed promptly to avoid premature curing of sealing compound by the frictional heat of insertion. If bearing is not fully installed within 4 minutes after applying sealing compound, repeat entire cleaning, priming, and compound application procedure.

6. Apply even coat of sealing compound, Item 292, WP 1803 00, using clean acid swabbing brush, Item 57, WP 1803 00, to mating surfaces of bearing and fitting bore.

NOTE

For each installation operation, assemble tool from bushing/bearing removal/installation set according to information plates in tool case.

7. Assemble tool and install bearings (toolkit instructions step 7 and step 8), holding arbor stationary while turning nut with wrench until sharp increase in torque is felt.
8. Inspect to verify bearing is centered in fitting.
9. Allow 15 minutes for curing of adhesive holding bearing in fitting.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

10. Clean up excess compound using cheesecloth cloth, Item 85, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
11. Assemble tool and stake bearing (toolkit instructions step 9 and step 10). Staking flare is complete when 0.005-inch feeler gage cannot be inserted between flare lip and fitting. Stagger stake points 45 degrees from those on first side.
12. Assemble tool and install flanged bushings in fitting (toolkit instruction step 11 and step 12).
13. Apply 1/4-inch minimum fillet of sealing compound, Item 273, WP 1803 00, around flanged edges of both bushings.
14. Engage actuator clevis with stabilator fitting and install bolt, bushing, washers, nut, and cotter pin (WP 0342 00, Install).

INSTALL - Continued

15. Remove stabilator support.

END OF WORK PACKAGE

UNIT LEVEL

LANDING GEAR

MAIN LANDING GEAR DRAG BEAM AND AXLE INSPECTIONS

This WP supersedes WP 0431 00, dated 15 December 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Micrometer, 0 to 1 Inch
 Micrometer, 1 to 2 Inch
 Torque Wrench, 150 - 1000 in. lbs.

Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 334,
 WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0381 00
 WP 0438 00
 WP 0441 00
 WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Brush, Acid Swabbing, Item 57, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 81, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Abrasive, Item 78, WP 1803 00
 Corrosion Removing Compound, Item 108,
 WP 1803 00

INSPECT MAIN LANDING GEAR DRAG BEAM

WARNING

CSI

Shot peening of exterior surfaces of drag beam is a critical characteristic. On drag beams 70250-12062-102, and 70250-12062-104, cadmium plating and baked resin coating of axle bore are critical characteristics. On drag beam 70250-22011-041, epoxy primer in axle bore is a critical characteristic. Use extreme care to avoid damage to these surfaces during repair of axle.

1. Inspect drag beam pivot pin holes, attach bolt hole and jackpad hole for scratches, nicks, or pits. **NONE ALLOWED.** Replace drag beam if damage is found.
2. On helicopters with drag beam 70250-22011-041, inspect drag beam pivot pin bushings for scratches, nicks, or pits. Replace bushings if damaged (WP 0441 01).
3. On helicopters with drag beam 70250-22011-041, inspect drag beam shock strut bushings for scratches, nicks, or pits. Replace bushings if damaged (WP 0441 01).
4. Inspect drag beams 70250-12062-102 and 70250-12062-104, or 70250-22011-041, around jackpad and tiedown ring mounting holes for cracks, using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace drag beam if cracks are found.

INSPECT MAIN LANDING GEAR DRAG BEAM - Continued**WARNING****CSI**

On drag beam 70250-22011-101, epoxy primer in jackpad mounting hole is a critical characteristic. Use extreme care to avoid damage to these surfaces during repair of jackpad.

5. On helicopters with drag beam 70250-22011-101, inspect drag beam jackpad mounting hole for complete coverage of epoxy primer. Apply two coats of epoxy primer coating kit, Item 120, WP 1803 00, to areas missing epoxy primer.
6. Inspect drag beam for dents. **ROUND SMOOTH DENTS SHALL NOT EXCEED 1 INCH IN DIAMETER AND 0.010-INCH DEEP.** Replace drag beam if dents are more than limits.

CAUTION

Damage to drag beam will result if a machine grinder is used for repairs. Machine grinder will go beyond repair limits. Hand polish all repairs to drag beam.

7. Inspect drag beam for scratches, nicks, or pits and repair using abrasive cloth, Item 82, WP 1803 00, as follows:
 - a. Repair drag beam area which includes fuselage attachment lugs as follows:
 - (1) **BLEND SCRATCHES, NICKS, OR PITS UP TO 0.010-INCH DEEP. NO MORE THAN 1 REPAIR ON EACH LUG ALLOWED.**
 - (2) Replace drag beam if repairs are more than limits.
 - b. Repair drag beam main body, except diameter that includes jackpad as follows:
 - (1) **BLEND LOCAL SCRATCHES, NICKS, OR PITS UP TO 0.010-INCH DEEP.**
 - (2) **BLEND SCRATCHES WHICH RUN AROUND DRAG BEAM MAIN BODY NO MORE THAN 20% OF DRAG BEAM CIRCUMFERENCE UP TO 0.003-INCH DEEP.**
 - (3) **BLEND SCRATCHES PARALLEL TO CENTERLINE NO MORE THAN 3 INCHES LONG UP TO 0.003-INCH DEEP.**
 - (4) Replace drag beam if repairs are more than limits.
 - c. Repair drag beam in diameter that includes jackpad as follows:
 - (1) **BLEND SCRATCHES, NICKS, OR PITS UP TO 0.015-INCH DEEP.**
 - (2) Replace drag beam if repairs are more than limits.
 - d. Repair drag beam flat surfaces adjacent to jackpad holes as follows:
 - (1) **BLEND SCRATCHES, NICKS, OR PITS UP TO 0.005-INCH DEEP.**
 - (2) Inspect reworked area for cracks using fluorescent penetrant inspection kit, (XMA101) (TM 1-1520-265-23). **NO CRACKS ALLOWED.**
 - (3) Inspect drag beam flat surfaces for worn plating. Apply two coats of epoxy primer coating kit, Item 120, WP 1803 00, to worn plating.
 - (4) Replace drag beam if repairs are more than limits.
 - e. Repair drag beam area which includes shock strut attachment lugs as follows:
 - (1) **BLEND SCRATCHES, NICKS, OR PITS UP TO 0.010-INCH DEEP. NO MORE THAN 1 REPAIR ON EACH LUG ALLOWED.**

INSPECT MAIN LANDING GEAR DRAG BEAM - Continued

- (2) Replace drag beam if repairs are more than limits.
- f. Remove corrosion on drag beam that is reddish brown in appearance and without evidence of pitting using abrasive cloth, Item 82, WP 1803 00. Blend edges of repair. **EVIDENCE OF PITTING IS CAUSE FOR REMOVAL OF THE DRAG BEAM.**
- 8. **TOTAL SUM OF REPAIRED AREAS ALLOWED ON DRAG BEAM MUST NOT EXCEED 3 SQUARE INCHES.** Replace drag beam if repairs are more than limits.
- 9. Mask pivot pin holes, attach bolt hole, and jackpad hole using pressure sensitive adhesive tape, Item 334, WP 1803 00.
- 10. Clean repaired areas with machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
- 11. Apply corrosion removing compound, Item 108, WP 1803 00, to repaired areas using acid swabbing brush, Item 57, WP 1803 00. Allow to dry for 30 seconds. Wipe dry with machinery wiping towel, Item 344, WP 1803 00.
- 12. Touch up repaired areas (WP 0381 00 and TM 55-1500-345-23).

INSPECT MAIN LANDING GEAR AXLE**WARNING****CSI**

Verification of baked resin coverage and thickness is a critical characteristic.

- 1. On helicopters with drag beam 70250-12062-102, and 70250-12062-104, inspect interior surface of drag beam axle bore for breaks in baked resin coating (Figure 1). **NONE ALLOWED.** Replace drag beam if any break in baked resin coating is visible (WP 0438 00).
- 2. On helicopters with drag beams 70250-22010-041, and 70250-22010-042, inspect interior surface of drag beam axle bore for breaks in epoxy primer. Apply two coats of epoxy primer coating kit, Item 120, WP 1803 00, to areas missing epoxy primer allowing two hours between application. Minimum bore diameter to be 2.000-inch after primer application.
- 3. Visually inspect axle threads for stripped, crossed, or flattened threads. **IF DAMAGE IS NOT MORE THAN 0.005-INCH DEEP, REPAIR BY BLENDING OUT DAMAGE.**
- 4. Visually inspect bearing journals for missing, scored, or damaged nickel plating. **IF DAMAGE IS NOT MORE THAN 0.001-INCH DEEP AND NOT MORE THAN 10% OF THE TOTAL AREA REPAIRS BY BLENDING OUT DAMAGE NOT TO EXCEED THE DEPTH OF THE NICKEL PLATING. IF BASE METAL IS INADVERTENTLY EXPOSED DURING BLENDING CLEAN AREA WITH TECHNICAL ACETONE, ITEM 5, WP 1803 TOUCH UP EXPOSED AREA WITH EPOXY PRIMER COATING KIT, ITEM 120, WP 1803 00. PRIMER MAY NOT EXTEND ABOVE THE SURFACE OF THE JOURNAL.**
- 5. Using a 10x magnifying glass, inspect axle for cracks. **NO CRACKS ALLOWED.**
- 6. Visually inspect axle surface for nicks and scratches perpendicular to centerline. **SCRATCHES PERPENDICULAR TO CENTERLINE MUST NOT EXCEED 20% OF THE CIRCUMFERENCE AND BE NO DEEPER THAN 0.005-INCH AFTER BLENDING.**
- 7. Visually inspect axle for isolated nicks, scratches, gouges, or corrosion. **DEPTH AFTER BLENDING NOT TO EXCEED 0.010-INCH.**

INSPECT MAIN LANDING GEAR AXLE - Continued

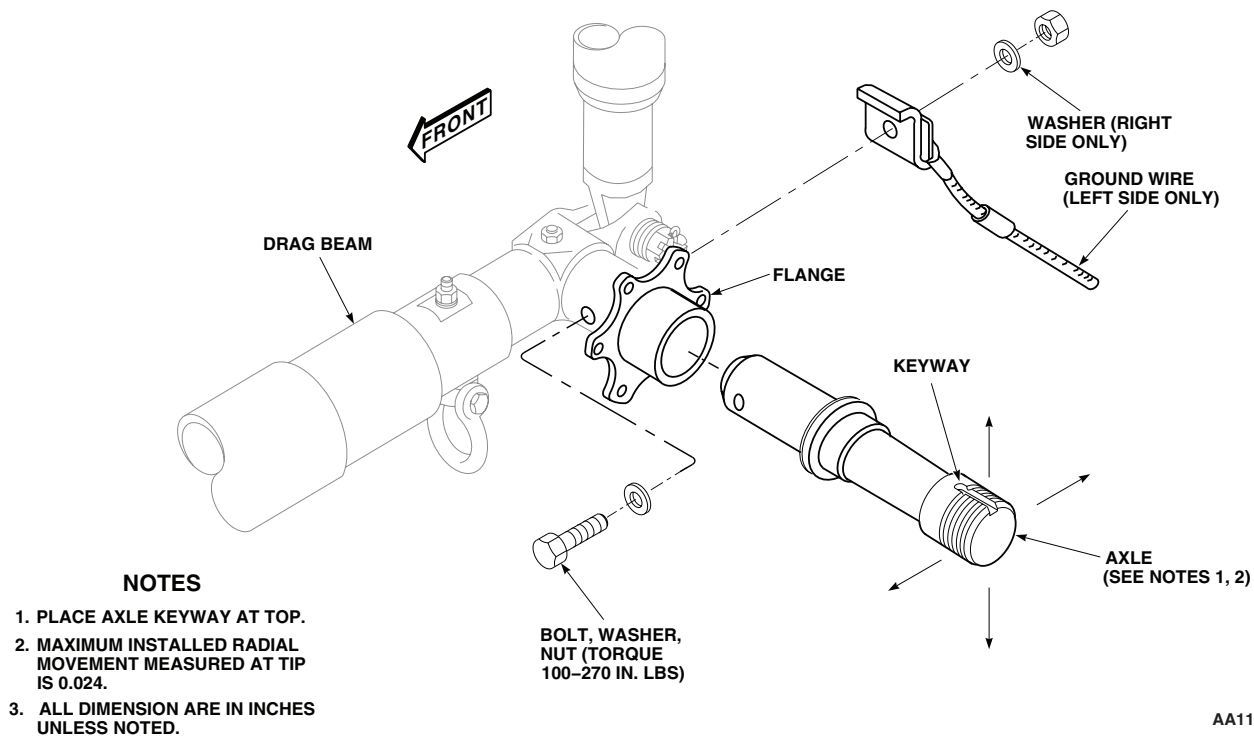


Figure 1. Inspect Main Landing Gear Axle.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

8. Blend out damaged area using abrasive cloth, Item 78, WP 1803 00 or abrasive cloth, Item 81, WP 1803 00. **DO NOT GRIND.** After blending, clean area with technical acetone, Item 5, WP 1803 00. Apply two coats of epoxy primer coating kit, Item 120, WP 1803 00, allowing two hours between applications.

END OF WORK PACKAGE

UNIT LEVEL**LANDING GEAR**

MAIN LANDING GEAR INSPECTION SHOCK STRUT QUICK SERVICE CHECK

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

WP 1597 00

Personnel Required

Assistant - (1)

UH-60 Helicopter Repairer MOS 15T (1)

CHECK**NOTE**

This procedure is a quick way to check if shock strut needs servicing.

1. Measure shock strut for dimension X (Figure 1).
2. Determine weight of helicopter DA Form 365-4 (Weight and Balance Clearance Form F-Transport/Tactical).
3. Using Figure 1, check if dimension X is within acceptable range.

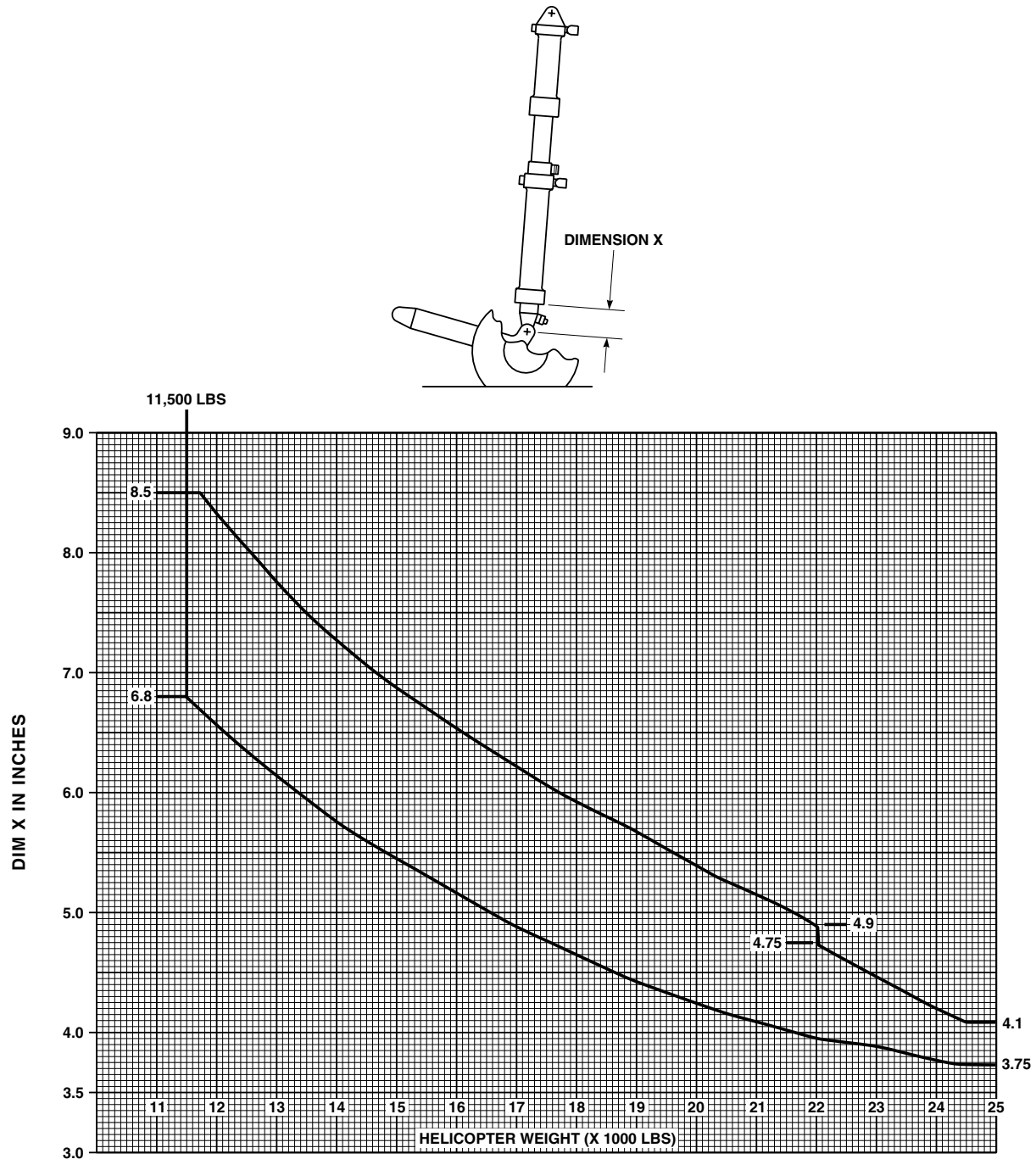
EXAMPLE**RESULT**

Helicopter weight is 16,500 lbs. Dimension X is 5 3/4-inches. Acceptable range is 5 inches to 6 3/8-inches.

Dimension X is within acceptable range. No need to service shock strut.

4. If dimension X is above or below acceptable range, service shock strut (WP 1597 00).

CHECK - Continued



SHOCK STRUT DIMENSION X VERSUS HELICOPTER WEIGHT

AA8101A
SA

Figure 1. Shock Strut Quick Service Check.

END OF WORK PACKAGE

UNIT LEVEL

LANDING GEAR

MAIN LANDING GEAR TIRE INSPECTION

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

TM 55-2620-200-24

WP 0450 00

WP 1598 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT

WARNING

Injury to personnel will result if foreign objects embedded in tire are removed while tire is inflated. Deflate tire before removing embedded foreign objects.

NOTE

Replace any tire that is damaged beyond limits per WP 0450 00.

1. Inspect tire for damage (TM 55-2620-200-24).
2. Inspect tire sidewall for blisters. **NO BLISTERS ALLOWED.**
3. Inspect tire for wear. **REPLACE TIRE IF WEAR HAS REACHED FIRST FABRIC PLY OF TREAD GROOVE OR IF WORN FOR A CONTINUOUS LENGTH OF 12 INCHES AT BOTTOM OF TREAD GROOVE.**
4. Inspect tire tread and shoulder for cuts or damage. **CUTS OR DAMAGE MAY BE ANY LENGTH PROVIDED DAMAGE DOES NOT CUT FIRST LAYER OF CORD BODY. IF CUT PENETRATES FIRST LAYER OF CORD BODY AND IS LESS THAN 1 INCH LONG, DEPTH OF CUT SHALL NOT EXCEED CUT LIMIT SHOWN ON SIDEWALL OF TIRE. IF TIRE GROOVE NEAREST CUT IS WORN SMOOTH AND CUT PENETRATES CORD BODY REPLACE TIRE.** Replace tire that exceeds limits.
5. Inspect tire sidewalls for cuts, scratches, abrasions or scuffs. **REPLACE TIRE IF DAMAGE PENETRATES FIRST LAYER OF CORD BODY.** Replace tire that exceeds limits.
6. Inspect tire surface for weather checking. **NO SEPARATION OF RUBBER BETWEEN CORDS ALLOWED. MAXIMUM DEPTH OF CRACK BENEATH TREAD RIB IS 1/8-INCH DEEP AND 1/4-INCH LONG.** Replace tire that exceeds limits.
7. Inspect tire for proper pressure. **PRESSURE SHOULD BE 130 TO 140 PSI.** Service tire if beyond limits (WP 1598 00).
8. Repair tire (TM 55-2620-200-24).

END OF WORK PACKAGE

UNIT LEVEL
LANDING GEAR
MAIN LANDING GEAR WHEEL BRAKE INSPECTION

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT

1. Check brake for wear as follows:
 - a. Apply parking brake.
 - b. Press on wear pin. **WEAR PIN MUST PROTRUDE ABOVE HOUSING NUT.** If wear pin is flush with housing nut, replace brake (Figure 1).

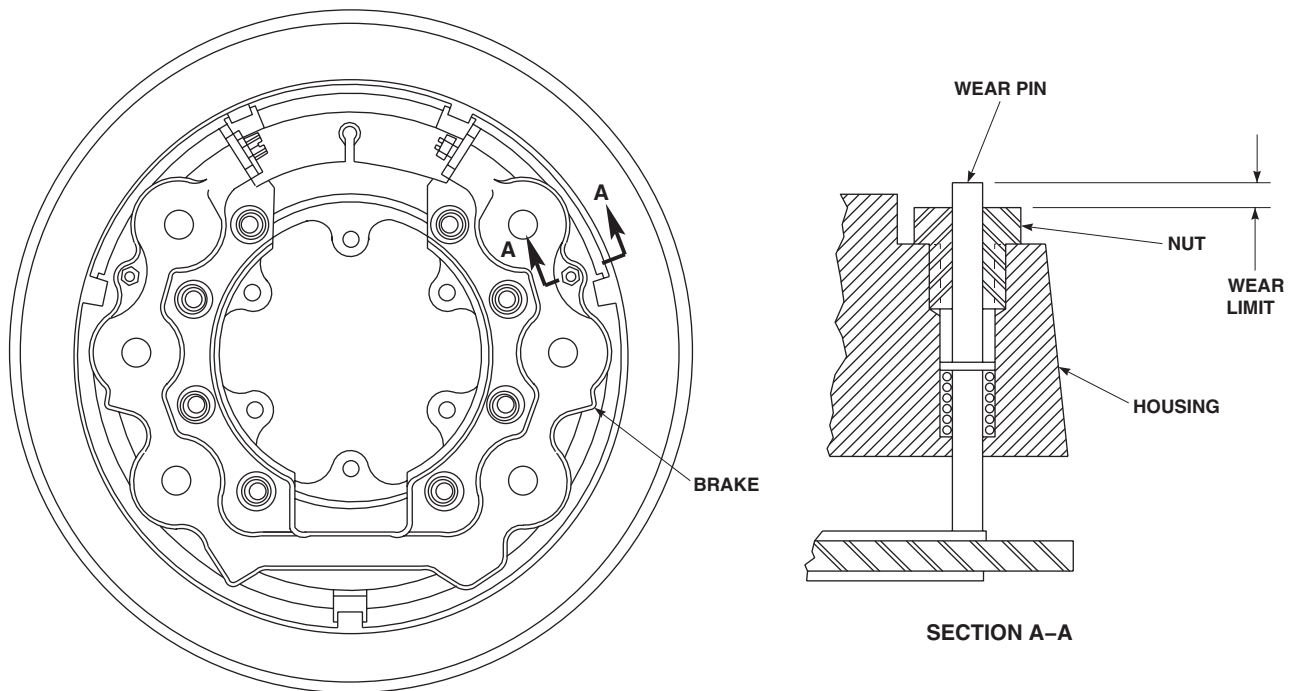
AA3382_1
SA

Figure 1. Inspect Main Landing Gear Wheel Brake.

END OF WORK PACKAGE

UNIT LEVEL**LANDING GEAR****PARKING BRAKE ROD INSPECTION**

This WP supersedes WP 0435 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Pad, Scouring, Item 205, WP 1803 00

Material/Parts

Corrosion Preventive Compound, Item 106,
WP 1803 00
Corrosion Removing Compound, Item 108,
WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
WP 1803 00

INSPECT

1. Visually inspect rod for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. **PITS SHALL NOT EXCEED 0.050-INCH DEEP.**
2. Remove corrosion manually from rod, using scouring pad, Item 205, WP 1803 00.
3. Mix solution of 1 part corrosion removing compound, Item 108, WP 1803 00, to 1 part water.
4. Apply corrosion compound solution to corroded areas of rod.
5. Rinse thoroughly with clean water.
6. Coat rod with corrosion preventive compound, Item 106, WP 1803 00.

END OF WORK PACKAGE

INSPECT - Continued

3. Check lateral movement of yoke assembly. If lateral movement exceeds 0.047-inch, remove yoke (AVIM) (WP 0462 00) and replace bushings (WP 0474 00). If after installation of new yoke bushings lateral movement exceeds 0.039-inch, replace fuselage/yoke fitting bushing (AVIM) (WP 0474 00).

END OF WORK PACKAGE

UNIT LEVEL**LANDING GEAR**

TAIL LANDING GEAR TIRE INSPECTION

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

TM 55-2620-200-24

WP 0465 00

WP 1600 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT**NOTE**

Replace any tire that is worn or damaged beyond limits per WP 0465 00.

1. Inspect tire for damage (TM 55-2620-200-24).
2. Check tire for proper pressure. **PRESSURE SHOULD BE 115 TO 125 PSI.** Service tire if beyond limits (WP 1600 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR**MAIN LANDING GEAR DRAG BEAM**

This WP supersedes WP 0438 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Micrometer, 0 to 1 Inch
 Micrometer, 1 to 2 Inch
 Nonmetallic Scraper
 Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Corrosion Preventive Compound, Item 105,
 WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Hydraulic Fluid, Fire Resistant, Item 144,
 WP 1803 00
 Hydraulic Fluid, Petroleum Base, Item 145,
 WP 1803 00
 Protective Plugs and Caps
 Sealing Compound, Item 273, WP 1803 00
 Sealing Compound, Item 271, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wire, Nonelectrical, Item 353, WP 1803 00
 Wire, Nonelectrical, Item 355, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
 TM 55-1500-345-23
 WP 0274 00
 WP 0275 00
 WP 0381 00
 WP 0431 00
 WP 0439 00
 WP 0440 00
 WP 0441 00
 WP 0442 00
 WP 0443 00
 WP 0444 00
 WP 0447 00
 WP 0451 00
 WP 0457 00
 WP 1675 00
 WP 1803 00

REMOVE**WARNING****CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during removal of drag beam.

1. **WSPS** Remove main landing gear cable cutter and clamps (WP 0274 00 and WP 0275 00). ■
2. Jack helicopter until shock strut is fully extended (WP 1675 00).
3. Remove wheel and tire (WP 0447 00).
4. Remove wheel brake (WP 0451 00).
5. Disconnect brake hose at quick-disconnect coupling on drag beam (Figure 1, Sheet 1).
6. Remove screw, washer, and bonding jumper from dust cover on drag beam.
7. Remove shock strut from drag beam (WP 0444 00).

REMOVE - Continued

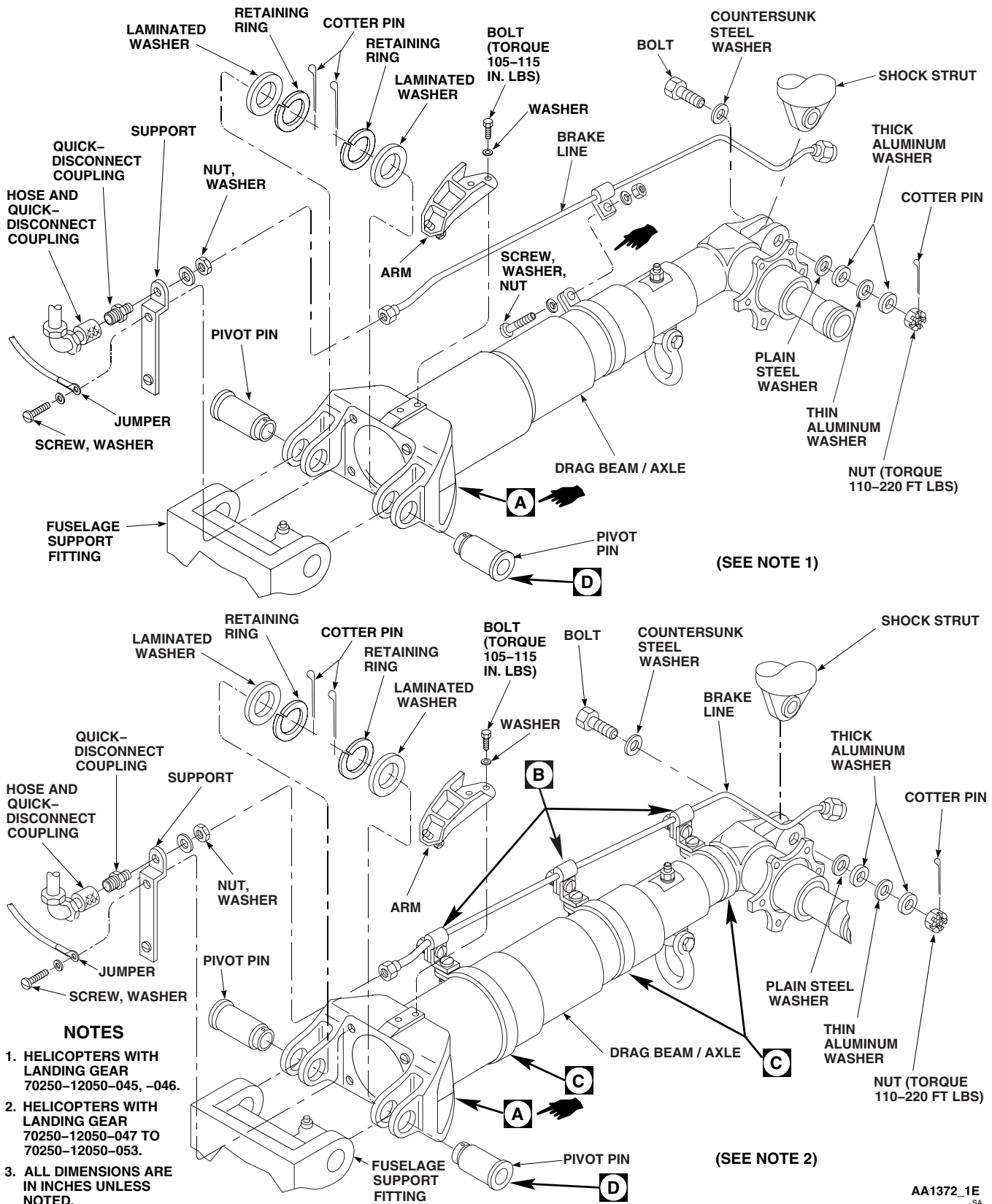


Figure 1. Replace Main Landing Gear Drag Beam 70250-12057-045/-046/-047/-053. (Sheet 1 of 2)

REMOVE - Continued

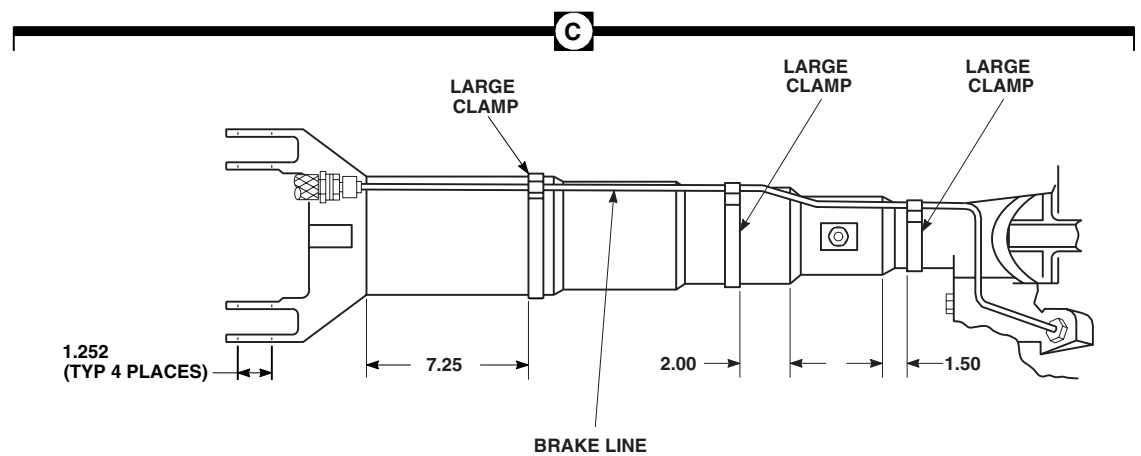
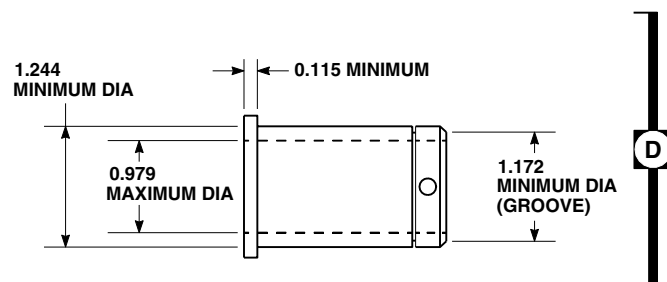
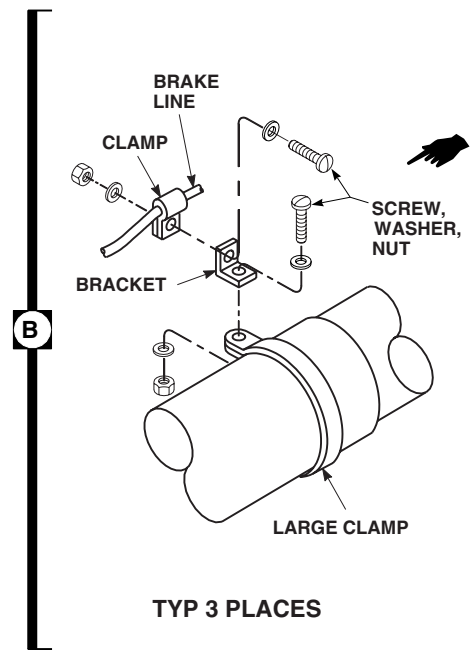
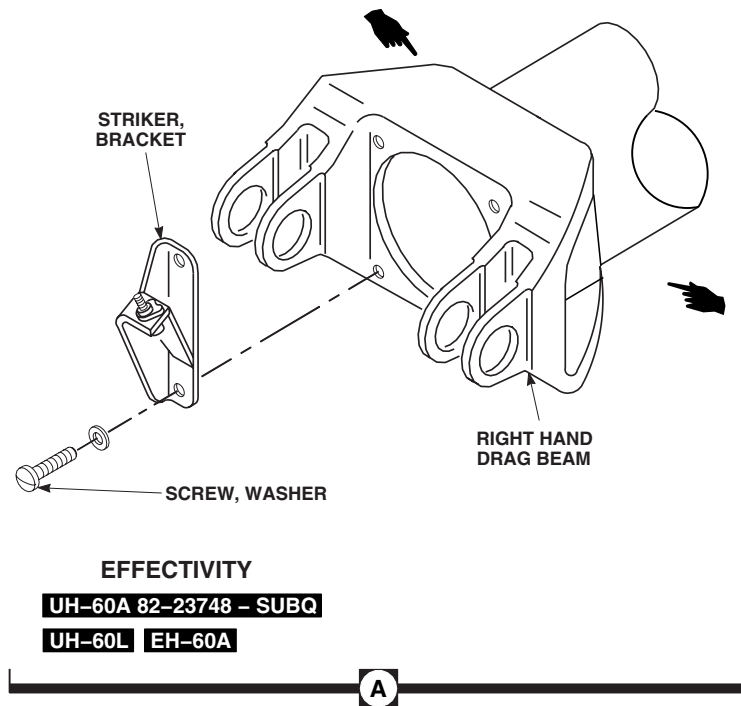
AA1372_2E
SA

Figure 1. Replace Main Landing Gear Drag Beam 70250-12057-045/-046/-047/-053. (Sheet 2 of 2)

REMOVE - Continued

8. Remove retaining rings and laminated washers from pivot pins.
9. Remove pivot pins and remove drag beam and axle from fuselage support fitting.

DISASSEMBLE

1. On helicopters with landing gear, 70250-12050-045, and 70250-12050-046, remove brake line as follows:
 - a. Disconnect brake line from coupling (Figure 1, Sheet 1).
 - b. Remove screw, washer, and nut from clamp. Remove clamp from brake line on drag beam. Cap brake line.
 - c. Remove large clamp from drag beam.
 - d. Remove quick-disconnect coupling half from support by removing nut and washer. Cap coupling.

WARNING**CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during removal of drag beam.

2. On helicopters with landing gear, 70250-12050-047 through 70250-12050-049, remove brake line as follows:
 - a. Disconnect brake line from quick-disconnect coupling.
 - b. Remove screw, washer, and nut from clamp. Remove clamp from brake line on drag beam. Cap brake line.
 - c. Remove three large clamps and brackets from drag beam (Figure 1, Sheet 2, Detail B).
 - d. Remove quick-disconnect coupling from support by removing nut and washer. Cap coupling (Figure 1, Sheet 1).

WARNING**CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during removal of drag beam.

3. On helicopters with landing gear, 70250-22009-041, and 70250-22009-042, remove brake line as follows:
 - a. Disconnect brake line from quick-disconnect coupling (Figure 1, Sheet 1). Cap coupling and plug brake line.
 - b. Disconnect brake line from elbow in inlet port of brake. Cap elbow and plug brake line.
 - c. Remove screws, washers, and nuts from clamps and brackets.
 - d. Remove brake line from drag beam.
 - e. Remove screws, washers, and nuts and remove standoffs and clamps from drag beam.
 - f. Remove nut and washer from quick-disconnect and remove quick-disconnect from supportst.
4. On landing gear, 70250-22009-042, drag beam, remove screws, washers, and ground wire and remove striker bolt support (Figure 1, Sheet 2, Detail A).
5. On landing gear, 70250-22009-041 or 70250-22009-042, drag beam, remove screws, washers, and ground wire and remove quick-disconnect support (Figure 1, Sheet 1).
6. On left drag beam, remove bolts and washers and remove arm.
7. On right drag beam, remove striker bolt support.

DISASSEMBLE - Continued

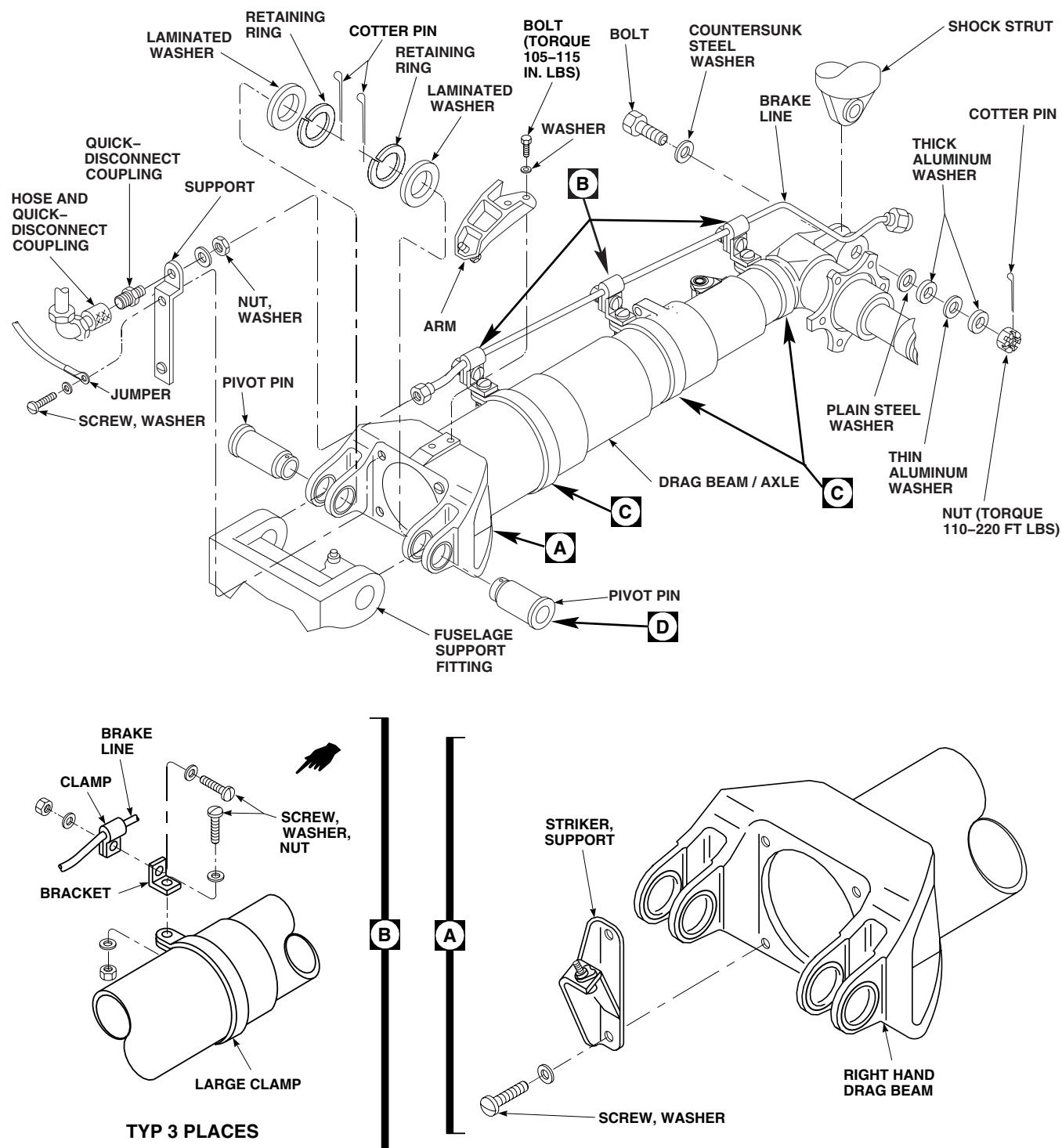
8. **UH-60A 82-23748 - SUBQ EH-60A UH-60L** On right drag beam, remove screws and washers and remove bracket with striker (Figure 1, Sheet 2, Detail A).
9. Remove screws and washers and remove brake line support.
10. Remove axle (WP 0439 00).
11. Remove jackpad and tiedown ring (WP 0442 00).
12. Remove brake flange spacers, as required (WP 0440 00).

INSPECT**WARNING****CSI**

Verification of baked resin coverage and thickness is a critical characteristic.

1. On helicopters with drag beam, 70250-12062-102, remove drag beam access cover and inspect interior surfaces for breaks in baked resin coating. **NONE ALLOWED.** Replace drag beam if any breaks in baked resin coating is visible.
2. Inspect drag beam for nicks, scratches, and corrosion (WP 0431 00). On drag beams, 70250-12057-045, 70250-12057-046, 70250-12057-047, 70250-12057-048, 70250-12057-049, use a nonmetallic scraper, remove sealing compound from around jack pad and clean area around upper and lower holes. With jack pad in place, using magnifying glass, inspect 3 inch radius around jack pad holes for evidence of cracking. **NO CRACKS ALLOWED.** Apply sealing compound, Item 271, WP 1803 00 (WP 0442 00).
3. Inspect fuselage support fitting assembly, 70219-02526-041 and 70219-02526-042, for cracks, nicks, scratches, and corrosion (WP 0441 00).
4. Check bearing wear in fuselage support fitting. **WEAR SHALL NOT EXCEED 0.010-INCH OVERSIZE AND/OR OUT-OF-ROUND. BEARING BORE DIAMETER SHALL NOT EXCEED 1.2615-INCHES AT ANY POINT OF BORE.** Replace bearing if limits are exceeded (WP 0441 00).
5. Inspect Teflon coating on bearings. **TEFLON SHALL NOT BE MISSING OR SHOW SIGNS OF METAL CONTACT.** Replace bearings if Teflon is missing or shows signs of metal contact (WP 0441 00).
6. Visually inspect pivot pins for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting. Remove corrosion from pivot pins as follows:
 - a. Blend out minor corrosion or pits using abrasive cloth, Item 82, WP 1803 00.
 - b. **DIMENSIONAL LIMITS SHOWN IN FIGURE 1, SHEET 2, DETAIL D, SHALL NOT BE EXCEEDED.** Replace pin if dimensions are exceeded.
7. On drag beams, 70250-12062-102 and 70250-12062-104, measure drag beam mounting pin bore. **DIMENSION LIMIT SHOWN IS 1.252-INCH, FIGURE 1, SHEET 2, DETAIL C, SHALL NOT BE EXCEEDED. IF MEASUREMENT IS BEYOND LIMITATION, REPLACE DRAG BEAM.**
8. On drag beams, 70250-22011-101, measure drag beam mounting pin bore bushing. **DIMENSION LIMIT SHOWN IS 1.252-INCH, FIGURE 2, SHEET 2, DETAIL C, SHALL NOT BE EXCEEDED.** If measurement is beyond limits, replace bushing (WP 0441 01 00).
9. Touch up non-mating surfaces of pivot pin with epoxy primer coating kit, Item 120, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23). After epoxy primer coating kit touch up, apply corrosion preventive compound, Item 105, WP 1803 00.

INSPECT - Continued



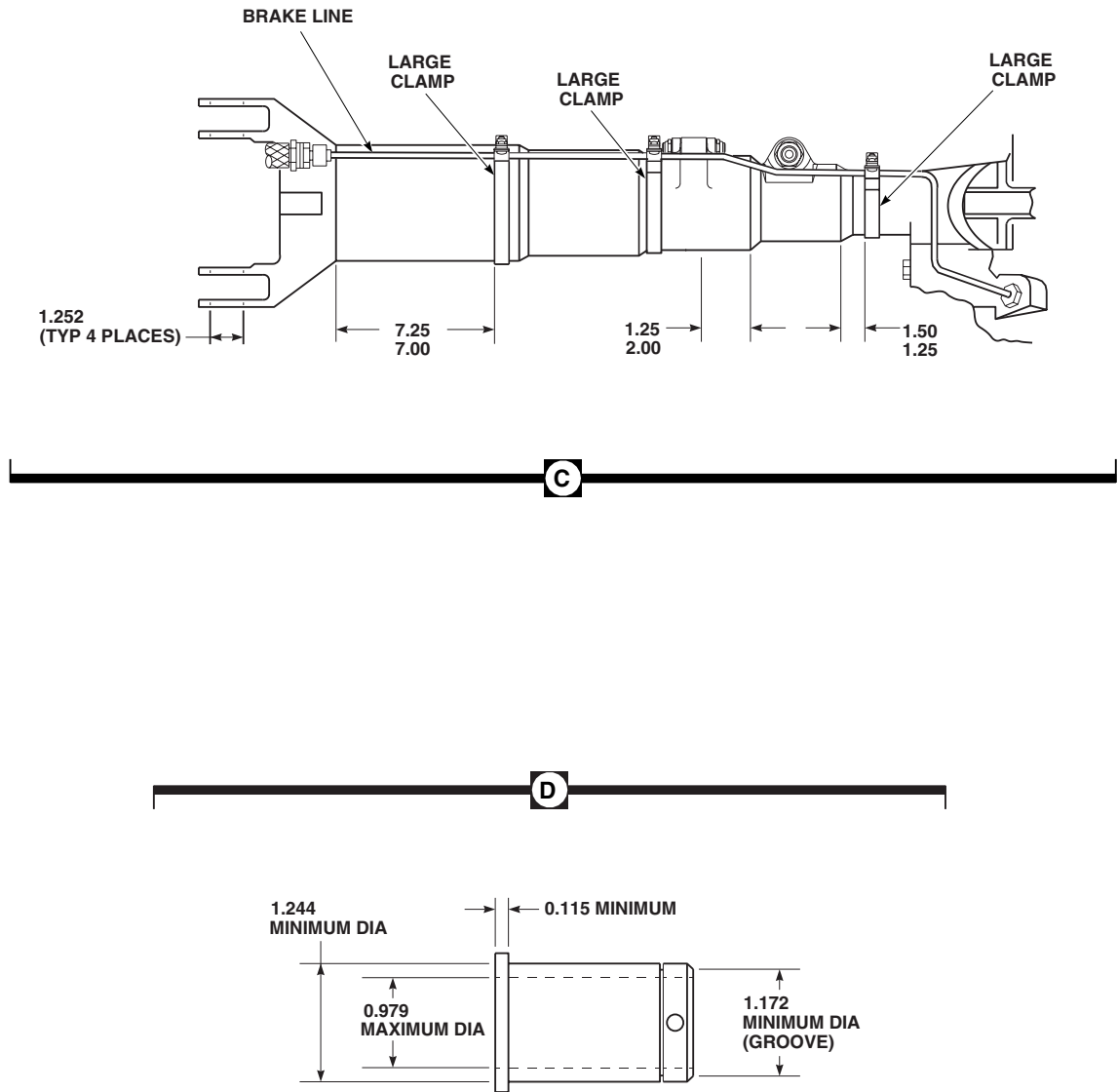
NOTES

1. HELICOPTERS WITH LANDING GEAR 70250-22009-041, -042.
2. ALL DIMENSIONS ARE IN INCHES UNLESS NOTED.

AB3381_1A
SA

Figure 2. Replace Main Landing Gear Drag Beam, 70250-22009-041/-042. (Sheet 1 of 2)

INSPECT - Continued



AB3381_2
SA

Figure 2. Replace Main Landing Gear Drag Beam, 70250-22009-041/-042. (Sheet 2 of 2)

ASSEMBLE

1. **QA** On left drag beam, install arm with bolts and washers. **TORQUE TO 105 - 115 INCH-POUNDS.** Using nonelectrical wire, Item 355, WP 1803 00, lockwire bolts. (Figure 1, Sheet 1 or Figure 2, Sheet 1).
2. **QA UH-60A 82-23748 - SUBQ EH-60A UH-60L** On right drag beam, install striker bracket with screws and washers (Figure 1, Sheet 2, Detail A).
3. On landing gear, 70250-22009-042, drag beam, install striker bolt support and ground wire with screws and washers (Figure 2).
4. On landing gear, 70250-22009-041, or 70250-22009-042, drag beam, install ground wire and quick-disconnect support with screws and washers. Using nonelectrical wire, Item 355, WP 1803 00, lockwire screws from striker bolt support to quick-disconnect support (Figure 2, Sheet 1, Detail A).
5. Install axle (WP 0439 00).
6. Install jackpad and tiedown ring (WP 0442 00).
7. Install brake flange spacers, as required (WP 0440 00).
8. Install brake line support using two inboard dust cover retaining screws. Do not disturb cover. Use thin washers on inboard screws and thick washers on outboard screws (Figure 1, Sheet 1).

WARNING**CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of brake line.

9. On helicopters with landing gear, 70250-12050-047 through 70250-12050-049, install brake line as follows:
 - a. Remove cap and install quick-disconnect coupling in support, and install washer and nut (Figure 1, Sheet 1).
 - b. Install three large clamps and brackets on drag beam. Fit each clamp to the correct diameter on drag beam and correct distance from shoulder (Figure 1, Sheet 2, Detail B and Figure 1, Sheet 2, Detail C).

WARNING

Injury to personnel will result if proper eye protection is not worn when working with compressed air. Eye protection must be worn when using or working in the area while compressed air is being used.

- c. Remove caps and blow brake line clear with dry compressed air. Replace cap at wheel brake end.
- d. Coat threads of quick-disconnect coupling with fire resistant hydraulic fluid, Item 147, or petroleum base hydraulic fluid, Item 145, WP 1803 00, and connect brake line to quick-disconnect coupling.
- e. Clamp brake line to large clamp brackets with screw, washer, and nut.

WARNING**CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of brake line.

10. On helicopters with landing gear, 70250-12050-045 and 70250-12050-046, install brake line as follows:
 - a. Remove cap and install quick-disconnect coupling in support, and install washer and nut (Figure 1, Sheet 1).

ASSEMBLE - Continued

- b. Install large clamp on drag beam.

WARNING

Injury to personnel will result if proper eye protection is not worn when working with compressed air. Eye protection must be worn when using or working in the area while compressed air is being used.

- c. Remove caps and blow brake line clear with dry compressed air. Replace cap at wheel brake end.
 - d. Coat threads of quick-disconnect coupling with fire resistant hydraulic fluid, Item 147 or petroleum base hydraulic fluid, Item 145, WP 1803 00, and connect brake line to quick-disconnect coupling.
 - e. Clamp brake line to large clamp with screw, washer and nut.
11. Install three large clamps and brackets on drag beam. Fit each clamp to the correct diameter on drag beam and correct distance from shoulder.

WARNING

Injury to personnel will result if proper eye protection is not worn when working with compressed air. Eye protection must be worn when using or working in the area while compressed air is being used.

12. Remove caps and blow brake line clear with dry compressed air. Replace cap at wheel brake end.
13. Coat threads of quick-disconnect coupling with fire resistant hydraulic fluid, Item 147, or petroleum base hydraulic fluid, Item 145, WP 1803 00 and connect brake line to quick-disconnect coupling.
14. Clamp brake line to large clamp brackets with screw, washer, and nut.

WARNING**CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of brake line.

15. On helicopters with landing gear, 70250-22009-041, and 70250-22009-042, install brake line as follows:
- a. Remove cap and install quick-disconnect coupling in support, and install washer and nut (Figure 2, Sheet 1).

WARNING

Injury to personnel will result if proper eye protection is not worn when working with compressed air. Eye protection must be worn when using or working in the area while compressed air is being used.

- b. Remove plugs and blow brake lines clear with dry compressed air.
- c. Coat threads of quick-disconnect coupling with fire resistant hydraulic fluid, Item 147, or petroleum base hydraulic fluid, Item 145, WP 1803 00, and connect brake line to quick-disconnect coupling.
- d. Remove cap from elbow in inlet port of brake.
- e. Coat threads of elbow with fire resistant hydraulic fluid, Item 147, or petroleum base hydraulic fluid, Item 145, WP 1803 00, and connect brake line to quick-disconnect coupling.
- f. Connect brake line to elbow in inlet port of brake.
- g. Install clamps around brake line.

ASSEMBLE - Continued

- h. Install three large clamps and brackets on drag beam. Fit each clamp to the correct diameter on drag beam and correct distance from shoulder (Figure 2, Sheet 2, Detail C).
- i. Connect three large clamps and brackets with screws, washers and nuts (Figure 2, Sheet 2, Detail C).

INSTALL**WARNING****CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of drag beam.

- 1. **QA** Place replacement drag beam and axle on fuselage support fitting. Install pivot pins and retaining rings (Figure 1, Sheet 1).
- 2. Measure endplay between retaining rings and drag beam lugs. **PEEL LAMINATED WASHERS AS REQUIRED FOR 0.001 TO 0.005-INCH TOTAL ENDPLAY.**
- 3. **QA** Install laminated washers between lugs and retaining rings, with laminations facing lugs.
- 4. **QA** Install cotter pins in pivot pins.
- 5. Install shock strut to drag beam (WP 0444 00).
- 6. Using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00, clean area where jumper is to be connected.
- 7. **QA** Connect jumper with screw and washer. Using nonelectrical wire, Item 353, WP 1803 00, lockwire top screws together.
- 8. Coat head of screw, washer, and jumper terminal with sealing compound, Item 273, WP 1803 00.
- 9. **QA** Connect hose at quick-disconnect coupling on drag beam.
- 10. Check adjustment of drag beam switch (WP 0443 00).
- 11. Install wheel brake (WP 0451 00).
- 12. Bleed brake system (WP 0457 00).

WARNING

The 7 lug wheel (5003889) may experience lug failure when operating at or above gross weights of 20,500 pounds. Lug failure may result in flying debris during ground handling and/or unexpected tire failure. The 14 lug wheel (040-25300) should be utilized when operating at or above gross weights of 20,500 pounds.

- 13. Install wheel and tire (WP 0447 00).
- 14. Lower helicopter and remove jacks (WP 1675 00).

INSTALL - Continued**WARNING****CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of main landing gear cutter and clamps.

15. **WSPS** Install main landing gear cable cutter and clamps (WP 0274 00 and WP 0275 00). ■

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR**MAIN LANDING GEAR AXLE**

This WP supersedes WP 0439 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Dial Indicator, MIL-I-18422
 Nonmetallic Scraper
 Puller Kit, Mechanical, GGG-P-781
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Cloth, Abrasive, Item 82, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Sealing Compound, Item 267, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References**Material/Parts**

Acetone, Technical, Item 5, WP 1803 00
 Brush, Acid Swabbing,, Item 57, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00

WP 0438 00
 WP 0447 00
 WP 1675 00
 WP 1803 00

REMOVE**WARNING****CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. On drag beams 70250-12062-102, and 70250-12062-104, cadmium plating and baked resin coating of axle bore are critical characteristics. Use extreme care to avoid damage to these surfaces during removal of axle.

1. Jack main landing gear assembly (WP 1675 00).
2. Remove wheel and tire (WP 0447 00).
3. Break sealing compound around axle bolt and nut. Use a nonmetallic scraper to remove sealing compound.
4. Remove axle bolt, nut, and washer(s). On left side, also remove clip and ground wire (Figure 1).
5. Remove sealing compound around drag beam flange using a nonmetallic scraper.
6. Attach puller to axle and remove axle from drag beam flange.

INSPECT**WARNING****CSI**

Verification of baked resin coverage and thickness is a critical characteristic.

1. On helicopters with drag beams 70250-12062-102, and 70250-12062-104, inspect interior surface of drag beam axle bore for breaks in baked resin coating(Figure 1). **NONE ALLOWED.** Replace drag beam if any break in baked

INSPECT - Continued

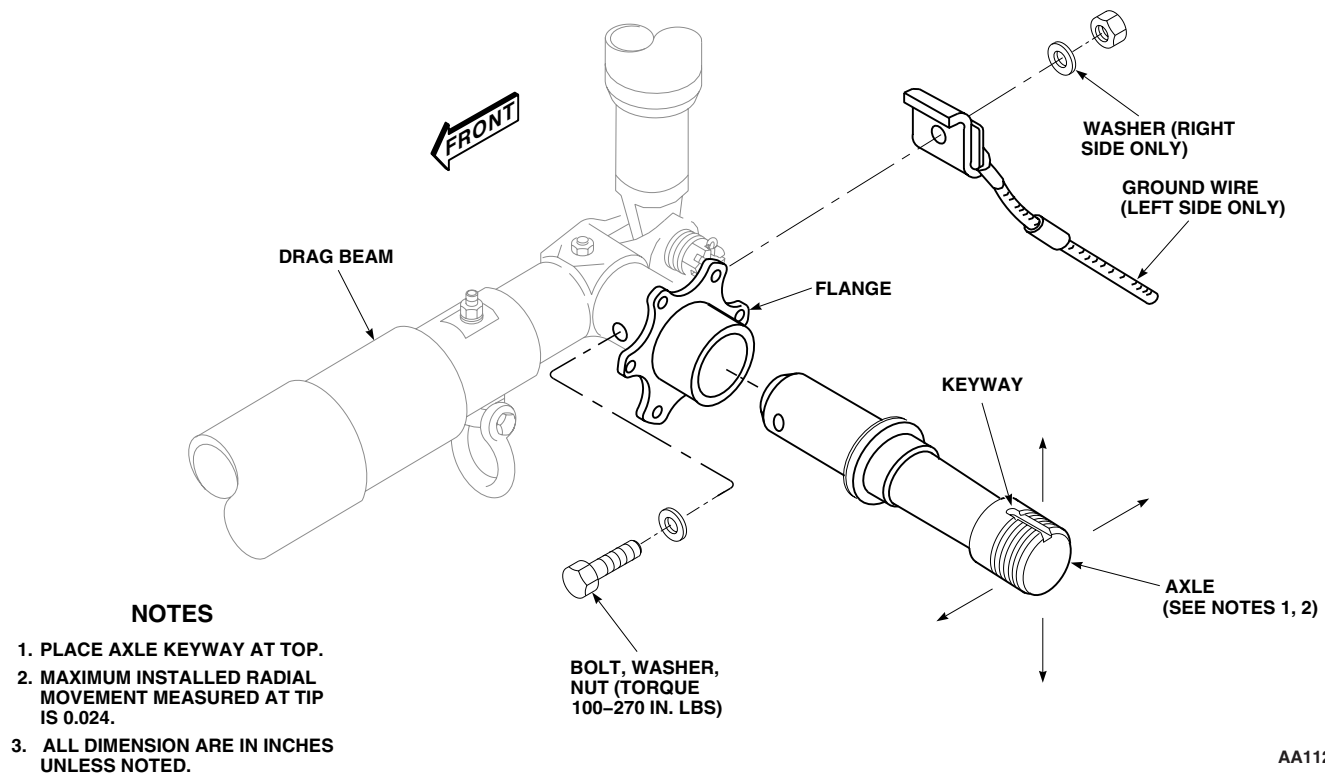


Figure 1. Replace Main Landing Gear Axle.

resin coating is visible (WP 0438 00). On helicopters with drag beam 70250-22010-041, and 70250-22010-042, inspect interior surface of drag beam axle bore for brakes in epoxy primer. Apply two coats of epoxy primer coating kit, Item 120, WP 1803 00, to areas missing epoxy primer allowing two hours between application. Minimum bore diameter to be 2.000 after primer application.

2. Visually inspect axle threads for stripped, crossed, or flattened threads. **IF DAMAGE IS NOT MORE THAN 0.005-INCH DEEP, REPAIR BY BLENDING OUT DAMAGE.**
3. Visually inspect bearing journals for missing, scored, or damaged nickel plating. **IF DAMAGE IS NOT MORE THAN 0.001-INCH DEEP AND NOT MORE THAN 10% OF THE TOTAL AREA, REPAIR BY BLENDING OUT DAMAGE NOT TO EXCEED THE DEPTH OF THE NICKLE PLATING. IF BASE METAL IS INADVERTENTLY EXPOSED DURING BLENDING, CLEAN AREA WITH TECHNICAL ACETONE, ITEM 5, WP 1803. TOUCH UP EXPOSED AREA WITH EPOXY PRIMER COATING KIT, ITEM 120, WP 1803. PRIMER MAY NOT EXTEND ABOVE THE SURFACE OF THE JOURNAL.**
4. Inspect inside diameter of axle for damaged or deteriorated foam. Repair foam by bonding and filling as necessary with sealing compound, Item 267, WP 1803 00. Allow to cure completely prior to further assembly.
5. Inspect axle for fractures and cracks using magnifying glass. **NONE ALLOWED.** Replace axle.
6. Inspect axle for nicks and scratches perpendicular to centerline, not to exceed 20% of circumference. Blend nicks and scratches using abrasive cloth, Item 82, WP 1803 00. Clean blended area using cleaning compound solvent, Item 71, WP 1803 00. Apply two coats of epoxy primer coating kit, Item 120, WP 1803 00. Blending not to exceed 0.005-inch after repair.

INSTALL**WARNING****CSI**

Shot peening of exterior surfaces of drag beam is a critical characteristic. On drag beams 70250-12062-102, and 70250-12062-104, cadmium plating and baked resin coating of axle bore are critical characteristics. On drag beam 70250-22011-041, epoxy primer in axle bore is a critical characteristic. Use extreme care to avoid damage to these surfaces during installation of axle.

1. Remove old sealing compound from around drag beam flange, and from bolthole. Loosen up sealing compound with cleaning compound solvent, Item 71, WP 1803 00, and scrape off with nonmetallic scraper.
2. **QA** Insert replacement axle into drag beam axle bore with keyway slot in axle threads at top and boltholes lined up (Figure 1).
3. Using dial indicator, inspect axle for radial movement at tip of axle. **MAXIMUM RADIAL MOVEMENT IS 0.024-INCH.**
4. **QA** Install bolt, clip and ground wire (left side only), washer(s), and nut. **TORQUE NUT TO 100 - 270 INCH-POUNDS.** Make sure axle tip is angled slightly downward.
5. Apply sealing compound, Item 273, WP 1803 00, around bolthead, nut, and drag beam flange.

WARNING

The 7 lug wheel (5003889) may experience lug failure when operating at or above gross weights of 20,500 pounds. Lug failure may result in flying debris during ground handling and/or unexpected tire failure. The 14 lug wheel (040-25300) should be utilized when operating at or above gross weights of 20,500 pounds.

6. Install wheel and tire (WP 0447 00).
7. Lower helicopter and remove jacks (WP 1675 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****BRAKE FLANGE SPACERS**

This WP supersedes WP 0440 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Adhesive, Item 32, WP 1803 00
Cloth, Abrasive, Item 81, WP 1803 00
Corrosion Resistant Coating, Item 110, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Material/Parts

Abrasive Mat, Item 2, WP 1803 00
Accelerator, Adhesive Curing, Item 3, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1803 00

REMOVE

1. Remove spacers from brake flange with a sharp nonmetallic scraper. Carefully tap around bonded edges of spacer to prevent damage to spacer or flange. Spacers can be reused if not damaged (Figure 1).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

2. Remove old adhesive from spacers and brake flange with technical acetone, Item 5, WP 1803 00, and abrasive mat, Item 2, WP 1803 00. Wipe clean and dry with machinery wiping towel, Item 344, WP 1803 00.

INSPECT/REPAIR

1. Brake flange spacers may be repaired if inspection shows damage to be within these limits:
 - a. **CORROSION UP TO 0.005-INCH DEEP AND 20% OF TOTAL AREA.**
 - b. **SCRATCHES UP TO 0.010-INCH DEEP, 0.040-INCH WIDE, AND 0.050-INCH LONG.**
 - c. **NICKS AND GOUGES UP TO 0.020-INCH DEEP AND 1/4-INCH IN DIAMETER.**
 - d. Replace brake flange spacer, if limits are exceeded.
2. Use abrasive cloth, Item 81, WP 1803 00, to blend damage to a smooth contour, removing as little material as possible.
3. Touch up repaired surfaces with corrosion resistant coating, Item 110, WP 1803 00.

INSPECT/REPAIR - Continued

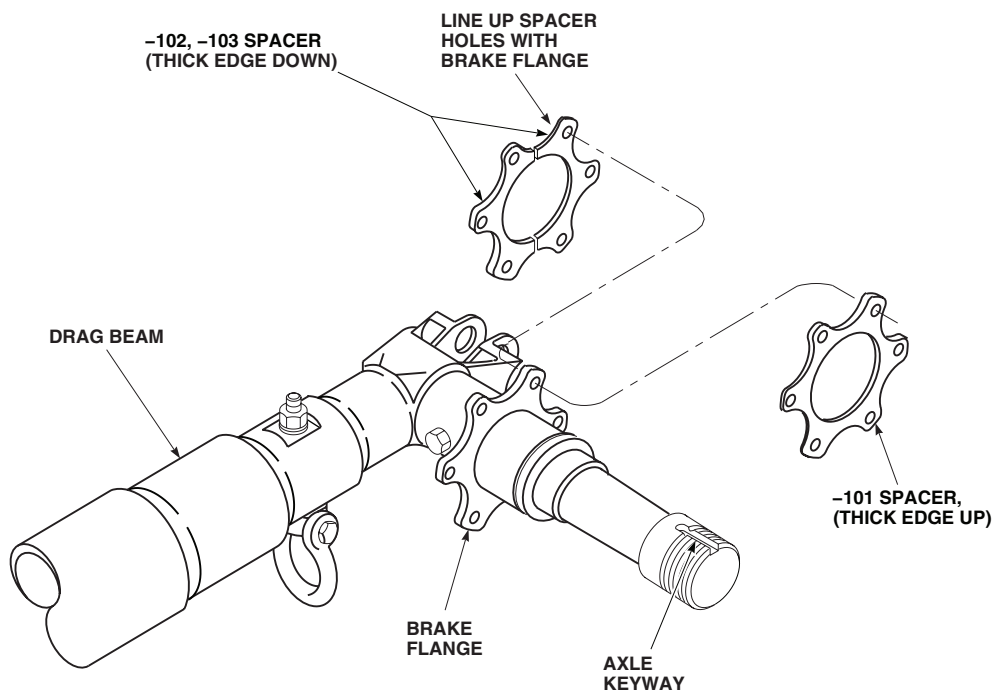
AA3384
SA

Figure 1. Replace Brake Flange Spacers.

INSTALL

NOTE

- When rebonding spacers while drag beam is mounted on the helicopter, use same procedure as listed except as follows: Directions "UP" and "DOWN" used in positioning spacers are determined in relationship with drag beam centerline. When the helicopter is supported on jacks, the axle end of drag beam rotates downward 35 degrees. The axle keyway may be used as a reference to position spacers, regardless of the drag beam angle.
- When changing drag beam from left/right to right/left, the bonded spacers, 70250-12098-101 through 70250-12098-103, on the brake flange must be repositioned (Figure 1).

1. Position drag beam as follows:

- On helicopters with landing gear, 70250-12045-045 and 70250-12045-047, place drag beam horizontally on bench with axle bore to the right and facing you.
- On helicopters with landing gear, 70250-12045-046, 70250-12045-048, and 70250-12045-049, place drag beam horizontally on bench with axle bore to the left and facing you.

2. Bond spacers to brake flange by doing this:

- Apply an even coat of adhesive curing accelerator, Item 3, WP 1803 00, to each surface to be bonded. Allow to dry thoroughly for at least 3 minutes at room temperature.

INSTALL - Continued

- b. Apply a bead of adhesive, Item 32, WP 1803 00, on one bonding surface only. Position spacer and bond in place immediately. Use slight contact pressure to hold spacer in place and allow bond to cure for 15 minutes.
3. With drag beam properly positioned, bond spacers, 70250-12098-102 and 70250-12098-103, to inboard surface of brake flange (Figure 1). Locate spacers with thick edge down and part number exposed. Gap between spacers, 70250-12098-102 and 70250-12098-103, should line up with axle keyway.
4. Bond spacer, 70250-12098-101, to outboard side of brake flange with thick edge up and part number exposed. If spacers are correctly positioned, the exposed edges should be parallel to each other.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

LANDING GEAR

DRAG BEAM SUPPORT FITTING BEARINGS

This WP supersedes WP 0441 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Brass Drift
Drag Beam Fuselage Support Fitting Bearing Drift,
Locally-Made, Figure 178, WP 1805 00
Fluorescent Penetrant Inspection Kit, XMA101
Gloves
Goggles/Face Shield
Micrometer, 1 to 2 Inch
Puller Kit, Manual, GGG-P-781

Remover, Paint, Item 250, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 327,
WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1520-265-23
WP 0438 00
WP 1803 00
WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Primer Coating, Item 243, WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Remove drag beam (WP 0438 00).
3. Locally made drag beam fuselage support fitting bearing drift (WP 1805 00).

NOTE

Puller jaws may require some alteration to allow jaws to fit gap (about 1/16-inch)
between inner and outer bearings.

4. Remove outer bearing from outboard lug of fuselage support fitting with slide hammer and internal pulling jaws from puller kit (Figure 1, Detail A) (TM 1-1500-204-23).
5. Drive inner bearing from outboard lug using brass drift and nonmetallic hammer.
6. Remove inner and outer bearings from inboard lug of fitting using above procedures.

INSPECT

1. Strip paint from lug area of fitting with paint remover, Item 250, WP 1803 00.
2. Check bearing bores in fuselage support fitting for scoring and corrosion. Blend out any damage with abrasive cloth, Item 82, WP 1803 00. **REPLACE FITTING IF DAMAGE IS DEEPER THAN 0.010-INCH OR BLENDED AREA IS MORE THAN 25% OF BORE AREA.**
3. Fluorescent penetrant inspect lug area of fuselage support fitting for cracks, using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.**

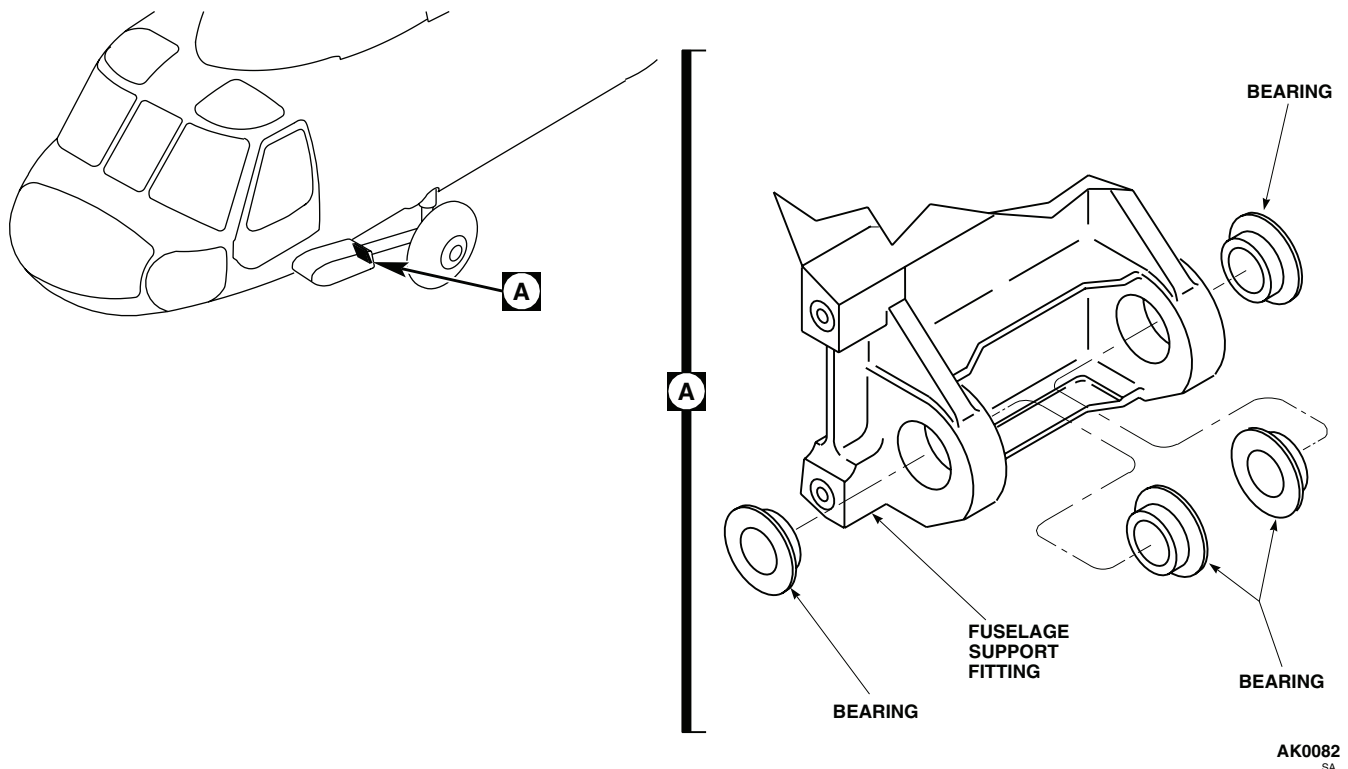
INSPECT - Continued

Figure 1. Replace Drag Beam Support Fitting Bearings.

4. Using micrometer, check diameter of bearing bores in fuselage support fitting. **BORE DIAMETER SHALL NOT EXCEED 1.4385-INCHES.**

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

5. If fuselage support fitting damage is within acceptable limits, clean repaired area with machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe with clean machinery wiping towel until dry.
6. Mask fitting bearing bores with pressure sensitive adhesive tape, Item 327, WP 1803 00. Touch up repaired area with three coats of primer coating, Item 243, WP 1803 00. Allow primer to dry and remove masking tape.

INSTALL

1. Turn off all electrical power.

INSTALL - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

2. Clean bearing bores in fitting and outside surface of new bearings with machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe dry.
3. Apply thin even coat of sealing compound, Item 273, WP 1803 00, to outside surface of new bearings. Keep sealing compound off Teflon coating of bearing.
4. While sealing compound, Item 273, WP 1803 00, is wet, line up bearing with bearing bore in lug and drive it in with soft-face hammer (Figure 1, Detail A). Make sure bearing is fully seated.

CAUTION

Damage to bearing will result if sealing compound is allowed to remain on Teflon coating. Remove all sealing compound that come into contact with Teflon coating.

5. Seal edge between bearing and fitting lug, and gap between inner ends of bearings, with sealing compound, Item 273, WP 1803 00.
6. Install drag beam (WP 0438 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR**MAIN LANDING GEAR DRAG BEAM BUSHINGS**

This WP supersedes WP 0441 01, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Gloves
 Goggles/Face Shield
 Fluorescent Penetrant Inspection Kit, XMA101
 Hand Reamer Set, Adjustable, GG-R-180
 Micrometer, 1 to 2 Inch

Carbon Dioxide, Item 64, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Brush Plating Solution, Cadmium, Item 62,
 WP 1803 00

References

TM 1-1520-265-23
 WP 0438 00
 WP 1803 00

REPLACE MAIN LANDING GEAR DRAG BEAM PIVOT PIN LUG BUSHINGS**Remove**

1. Turn off all electrical power.
2. Remove drag beam (WP 0438 00).
3. Place drag beam in arbor press and carefully remove pivot pin bushings (Figure 1).

Inspect

1. Check bearing bores in drag beam pivot pin lugs for scoring and corrosion. Blend out any damage with abrasive cloth, Item 82, WP 1803 00. **REPLACE FITTING IF DAMAGE IS DEEPER THAN 0.010-INCH OR BLENDED AREA IS MORE THAN 25% OF BORE AREA.**
2. Fluorescent penetrant inspect lug area of drag beam pivot pin fitting for cracks, using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.**
3. Using micrometer, check diameter of bearing bores in drag beam pivot pin lug. **BORE DIAMETER SHALL NOT EXCEED 1.361-INCHES.**

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

4. If drag beam pivot pin lug damage is within acceptable limits, clean repaired area with machinery wiping towel, Item 344, WP 1803 00, wet with technical Acetone, Item 5, WP 1803 00. Wipe with clean machinery wiping towel until dry.

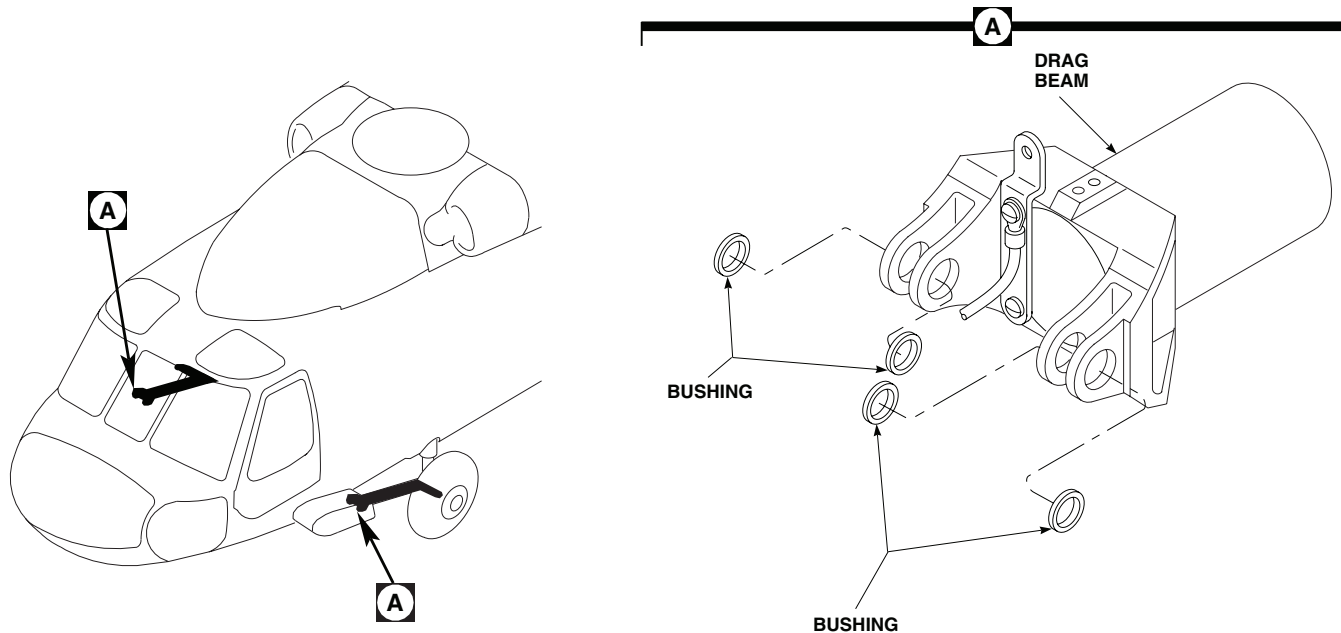
REPLACE MAIN LANDING GEAR DRAG BEAM PIVOT PIN LUG BUSHINGS - ContinuedAB3452
SA

Figure 1. Replace Drag Beam Pivot Pin Lug Bushings.

Install

1. Turn off all electrical power.
2. Clean bushing bores in lug and outside surface of new bushings with machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe dry.

WARNING

Injury to personnel will occur if protective gloves are not worn when handling carbon dioxide or chilled parts. Wear gloves when handling carbon dioxide or chilled parts.

3. Chill bushings in a container of carbon dioxide, Item 64, WP 1803 00, for 45 minutes.
4. Remove bushing from carbon dioxide and apply thin coat of sealing compound, Item 273, WP 1803 00.
5. Install bushing in drag beam pivot pin lug while sealing compound is still wet, making sure bushing is flush with inside face of lug.
6. Allow bushing to return to room temperature. Line-ream bushings to 1.2511 to 1.2521-inches using hand reamer set.
7. Apply cadmium brush plating solution, Item 62, WP 1803 00, to ID of bushing.
8. Apply fillet of sealing compound, Item 273, WP 1803 00 to edge of bushing and face of lug.
9. Install drag beam (WP 0438 00).

REPLACE MAIN LANDING GEAR DRAG BEAM PIVOT PIN LUG BUSHINGS - Continued

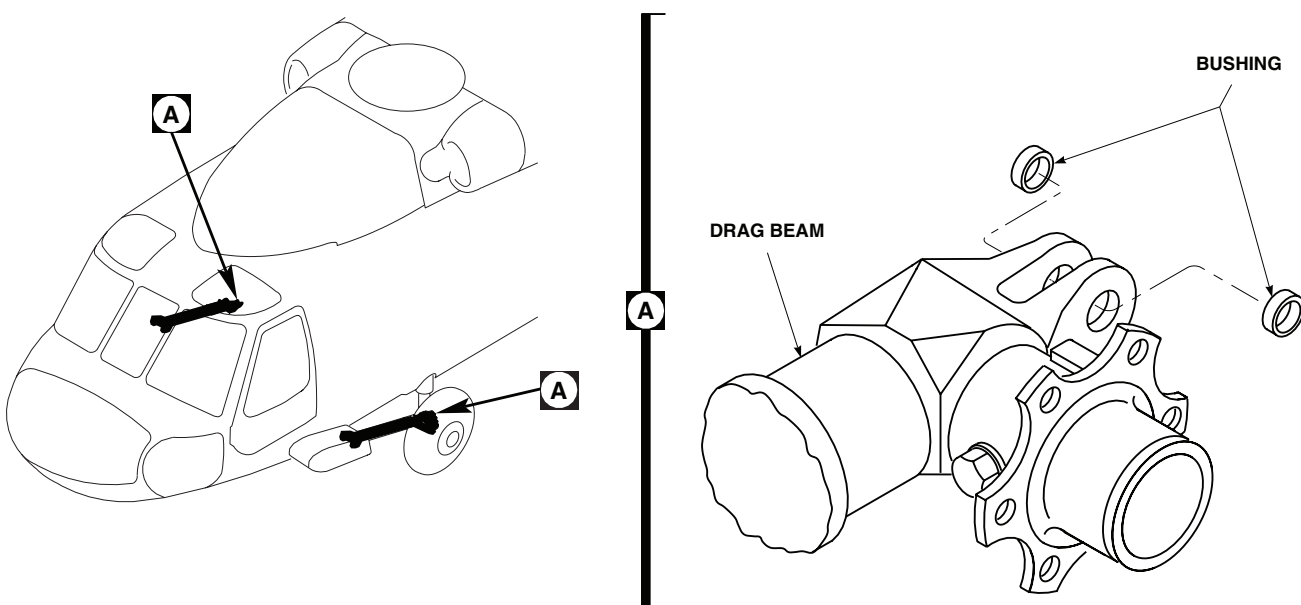
AB3455
SA

Figure 2. Replace Drag Beam Shock Strut Lug Bushings.

REPLACE MAIN LANDING GEAR DRAG BEAM SHOCK STRUT LUG BUSHINGS

Remove

1. Turn off all electrical power.
2. Remove drag beam (WP 0438 00).
3. Place drag beam in arbor press and carefully remove shock strut bushings (Figure 2).

Inspect

1. Check bearing bores in drag beam shock strut lug for scoring and corrosion. Blend out any damage with abrasive cloth, Item 82, WP 1803 00. **REPLACE FITTING IF DAMAGE IS DEEPER THAN 0.010-INCH OR BLENDED AREA IS MORE THAN 25% OF BORE AREA.**
2. Fluorescent penetrant inspect lug area of drag beam shock strut fitting for cracks, using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.**
3. Using micrometer, check diameter of bearing bores in drag beam shock strut lugs. **BORE DIAMETER SHALL NOT EXCEED 1.1108-INCHES.**

REPLACE MAIN LANDING GEAR DRAG BEAM SHOCK STRUT LUG BUSHINGS - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

4. If drag beam pivot pin lug damage is within acceptable limits, clean repaired area with machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe with clean machinery wiping towel until dry.

Install

1. Turn off all electrical power.
2. Clean bushing bores in lug and outside surface of new bushings with machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe dry.

WARNING

Injury to personnel will occur if protective gloves are not worn when handling carbon dioxide or chilled parts. Wear gloves when handling carbon dioxide or chilled parts.

3. Chill bushings in a container of carbon dioxide, Item 64, WP 1803 00, for 45 minutes.
4. Remove bushing from carbon dioxide and apply thin coat of sealing compound, Item 273, WP 1803 00.
5. Install bushing in drag beam shock strut lug while sealing compound is still wet, making sure bushing is flush with inside face of lug.
6. Allow bushing to return to room temperature. Line-ream bushings to 0.9990 to 1.0016-inches using hand reamer set.
7. Apply cadmium brush plating solution, Item 62, WP 1803 00, to ID of bushing.
8. Install drag beam (WP 0438 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING**

This WP supersedes WP 0442 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Micrometer, 0 to 1 Inch
Micrometer, 1 to 2 Inch
Nonmetallic Drift
Nonmetallic Scraper
Torque Wrench, 150 - 1000 in. lbs.

Corrosion Preventive Compound, Item 104,
WP 1803 00
Sealing Compound, Item 271, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 55-1500-345-23
WP 0381 00
WP 0431 00
WP 1803 00

Material/Parts

Abrasive Mat, Item 2, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00

REPLACE MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING (DRAG BEAM/AXLE, 70250-12105-043)**Remove**

1. Turn off all electrical power.
2. Clean sealing compound from around nut and jackpad with nonmetallic scraper.

NOTE

Jackpad removal may require driving the pad shank out of drag beam with nonmetallic drift, due to foam adhering to jackpad.

3. Remove dust plug, nut, and sealing washer from jackpad. Remove jackpad from drag beam (Figure 1).
4. Remove nut, washers, and bolt, and remove tiedown ring from jackpad. Discard nut.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

5. Clean up remaining sealing compound on drag beam with nonmetallic scraper, technical acetone, Item 5, WP 1803 00, and abrasive mat, Item 2, WP 1803 00.

Inspect

1. Inspect drag beam jackpad hole and adjacent flat surfaces (WP 0431 00).

REPLACE MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING (DRAG BEAM/AXLE, 70250-12105-043) - Continued

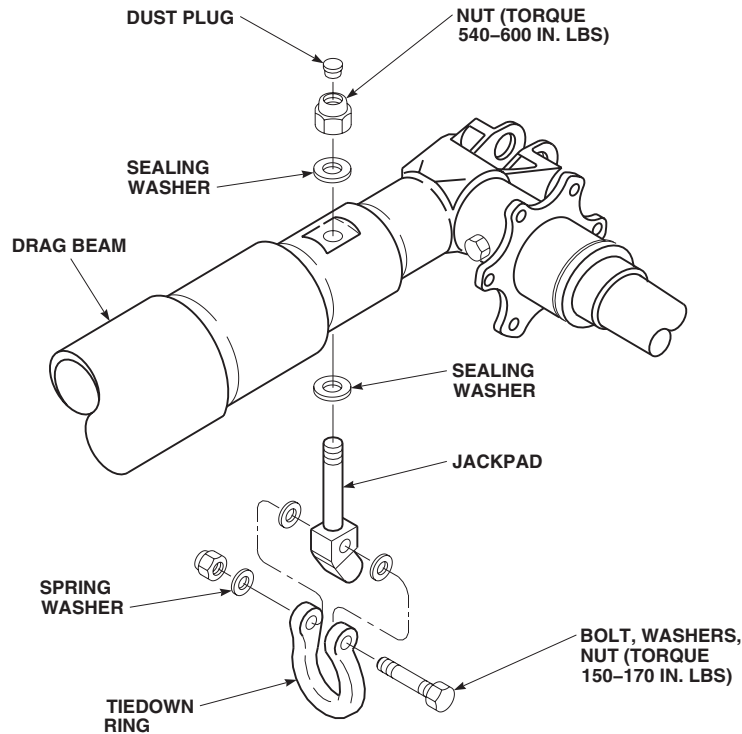
AK0075B
SA

Figure 1. Replace Main Landing Gear Drag Beam, 70250-12105-043, Jackpad and Tiedown Ring.

2. Inspect cadmium plated circular shank portion of jackpad for scratches, nicks, or pits. If scratches penetrate through cadmium plating, replace jackpad.
3. Inspect remaining portion of jackpad for scratches, nicks, or pits and repair using abrasive cloth, Item 82, WP 1803 00, as follows:
 - a. Repair flat surfaces of jackpad as follows:
 - (1) **BLEND SCRATCHES, NICKS, OR PITS UP TO 0.020-INCH DEEP.**
 - (2) Replace jackpad if repairs are more than limits.
 - b. Repair spherical ball portion of jackpad as follows:
 - (1) **BLEND SCRATCHES, NICKS, OR PITS UP TO 0.025-INCH DEEP. BLENDED AREAS SHALL NOT EXCEED 25% OF THE TOTAL SPHERICAL SURFACE.**
 - (2) Replace jackpad if repairs are more than limits.
4. Touch up finish on all repaired surfaces (WP 0381 00 and TM 55-1500-345-23).

Install

1. Turn off all electrical power.
2. **QA** Fit washers between tiedown ring and jackpad (Figure 1). Install bolt wet with corrosion preventive compound, Item 104, WP 1803 00, spring washer, and nut. **TORQUE NUT TO 150 - 170 INCH-POUNDS.**
3. Fill hole in drag beam with sealing compound, Item 271, WP 1803 00. While sealing compound is wet, install jackpad as follows:

REPLACE MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING (DRAG BEAM/AXLE, 70250-12105-043) - Continued

- a. Apply bead of sealing compound, Item 271, WP 1803 00, to both sides of sealing washer. Position sealing washer on bolt end of jackpad.
- b. **QA** Insert bolt end of jackpad through drag beam with rotating motion, pushing out extra sealing compound. Position pad angled back toward axle end of beam.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

- c. Clean off sealing compound forced out of drag beam by bolt with machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe area clean, including threaded portion of bolt.
- d. Apply bead of sealing compound, Item 271, WP 1803 00, to one side of sealing washer. Install sealing washer, sealant toward drag beam, over bolt end of jackpad.
4. **QA** Install new nut on jackpad. **TORQUE NUT TO 540 - 600 INCH-POUNDS.**
5. Apply sealing compound, Item 271, WP 1803 00, over washer on bottom side of jack pad and over threads, nut, and washers on top side. **SEALING COMPOUND IS NOT TO EXTEND MORE THAN 0.380-INCH BEYOND EDGE OF SEALING WASHERS.**
6. **QA** Install dust plug on threaded end of jackpad.
7. Cover dust plug with sealing compound, Item 271, WP 1803 00.
8. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).

REPLACE MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING (DRAG BEAM AXLE, 70250-22010-041 AND 70250-22010-042)**Remove**

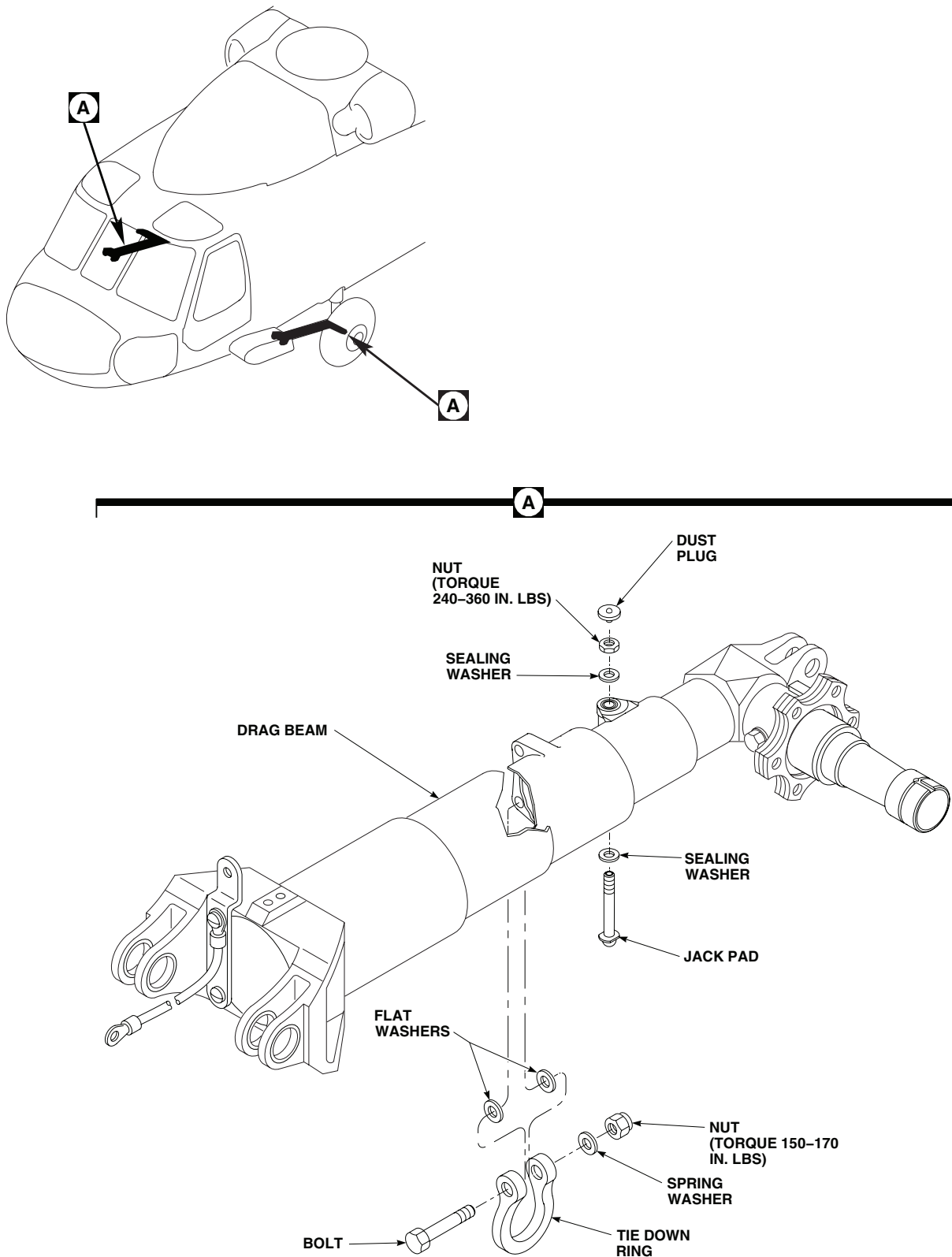
1. Turn off all electrical power.
2. Clean sealing compound from around nut and jackpad with nonmetallic scraper.

NOTE

Jackpad removal may require driving the pad shank out of drag beam with nonmetallic drift, due to foam adhering to jackpad.

3. Remove dust plug, nut, and sealing washer from jackpad. Remove jackpad from drag beam (Figure 2, Detail A).
4. Remove nut, washers, and bolt, and remove tiedown ring from jackpad. Discard nut.

REPLACE MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING (DRAG BEAM AXLE, 70250-22010-041 AND 70250-22010-042) - Continued



AB3380
SA

Figure 2. Replace Main Landing Gear Drag Beam, 70250-22010-041/-042, Jackpad and Tiedown Ring.

REPLACE MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING (DRAG BEAM AXLE, 70250-22010-041 AND 70250-22010-042) - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

5. Clean up remaining sealing compound on drag beam with nonmetallic scraper, technical acetone, Item 5, WP 1803 00, and abrasive mat, Item 2, WP 1803 00.

Inspect

1. Inspect drag beam jackpad hole and adjacent flat surfaces (WP 0431 00).
2. Inspect cadmium plated circular shank portion of jackpad for scratches, nicks, pits and corrosion. If scratches penetrate through cadmium plating, replace jackpad.
3. Inspect remaining portion of jackpad for scratches, nicks, pits and corrosion, and repair using abrasive cloth, Item 82, WP 1803 00, as follows:
 - a. Repair flat surfaces of jackpad as follows:
 - (1) **BLEND SCRATCHES, NICKS, PITS OR CORROSION UP TO 0.020-INCH DEEP.**
 - (2) Replace jackpad if repairs are more than limits.
 - b. Repair spherical ball portion of jackpad as follows:
 - (1) **BLEND SCRATCHES, NICKS, PITS OR CORROSION, UP TO 0.025-INCH DEEP. BLENDED AREAS SHALL NOT EXCEED 25% OF THE TOTAL SPHERICAL SURFACE.**
 - (2) Replace jackpad if repairs are more than limits.
4. Touch up finish on all repaired surfaces (WP 0381 00 and TM 55-1500-345-23).

Install

1. Turn off all electrical power.
2. **QA** Fit washers between tiedown ring and jackpad (Figure 2, Detail A). Install bolt wet with corrosion preventive compound, Item 104, WP 1803 00, spring washer, and nut. **TORQUE NUT TO 150 - 170 INCH-POUNDS.**
3. Fill hole in drag beam with sealing compound, Item 271, WP 1803 00. While sealing compound is wet, install jackpad as follows:
 - a. Apply bead of sealing compound, Item 271, WP 1803 00, to both sides of sealing washer. Position sealing washer on bolt end of jackpad.
 - b. **QA** Insert bolt end of jackpad through drag beam with rotating motion, pushing out extra sealing compound. Position pad angled back toward axle end of beam.

REPLACE MAIN LANDING GEAR DRAG BEAM JACKPAD AND TIEDOWN RING (DRAG BEAM AXLE, 70250-22010-041 AND 70250-22010-042) - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

- c. Clean off sealing compound forced out of drag beam by bolt with machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe area clean, including threaded portion of bolt.
- d. Apply bead of sealing compound, Item 271, WP 1803 00, to one side of sealing washer. Install sealing washer, sealant toward drag beam, over bolt end of jackpad.
4. **QA** Install new nut on jackpad. **TORQUE NUT TO 240- 360 INCH-POUNDS.**
5. Apply sealing compound, Item 271, WP 1803 00, over washer on bottom side of jack pad and over threads, nut, and washers on top side. **SEALING COMPOUND IS NOT TO EXTEND MORE THAN 0.380-INCH BEYOND EDGE OF SEALING WASHERS.**
6. **QA** Install dust plug on threaded end of jackpad.
7. Cover dust plug with sealing compound, Item 271, WP 1803 00.
8. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR**

MAIN LANDING GEAR DRAG BEAM SWITCH

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Multimeter, 60741-310
Torque Wrench, 30 - 200 in. lbs.

References

WP 0286 00
WP 1675 00
WP 1703 00
WP 1803 00

Material/Parts

Wire, Nonelectrical, Item 353, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

NOTE

Replacement of left and right switches is the same except as noted.

REMOVE

1. Turn off all electrical power.
2. Remove landing gear support fairing (WP 0286 00).
3. Remove bolts, radius block (if installed on right switch), washers, and nuts holding switch attachment bracket to fuselage support. Remove switch attachment bracket to get at drag beam switch (Figure 1, Detail B or Figure 2, Detail A).
4. Remove upper jamnut, lockwasher, and key-washer from drag beam switch. Remove switch from switch attachment bracket.

NOTE

Drag beam switch is part of a molded cable assembly. If switch malfunctions, replace entire assembly.

5. Disconnect electrical connector, P163 (left) or P164 (right), from receptacle on bracket at front of fuselage support (Figure 1, Detail A).
6. Remove cable from fuselage support.

INSTALL

1. Turn off all electrical power.
2. Feed replacement cable assembly through fuselage support structure.
3. Inspect and treat electrical connector (WP 1703 00).

INSTALL - Continued

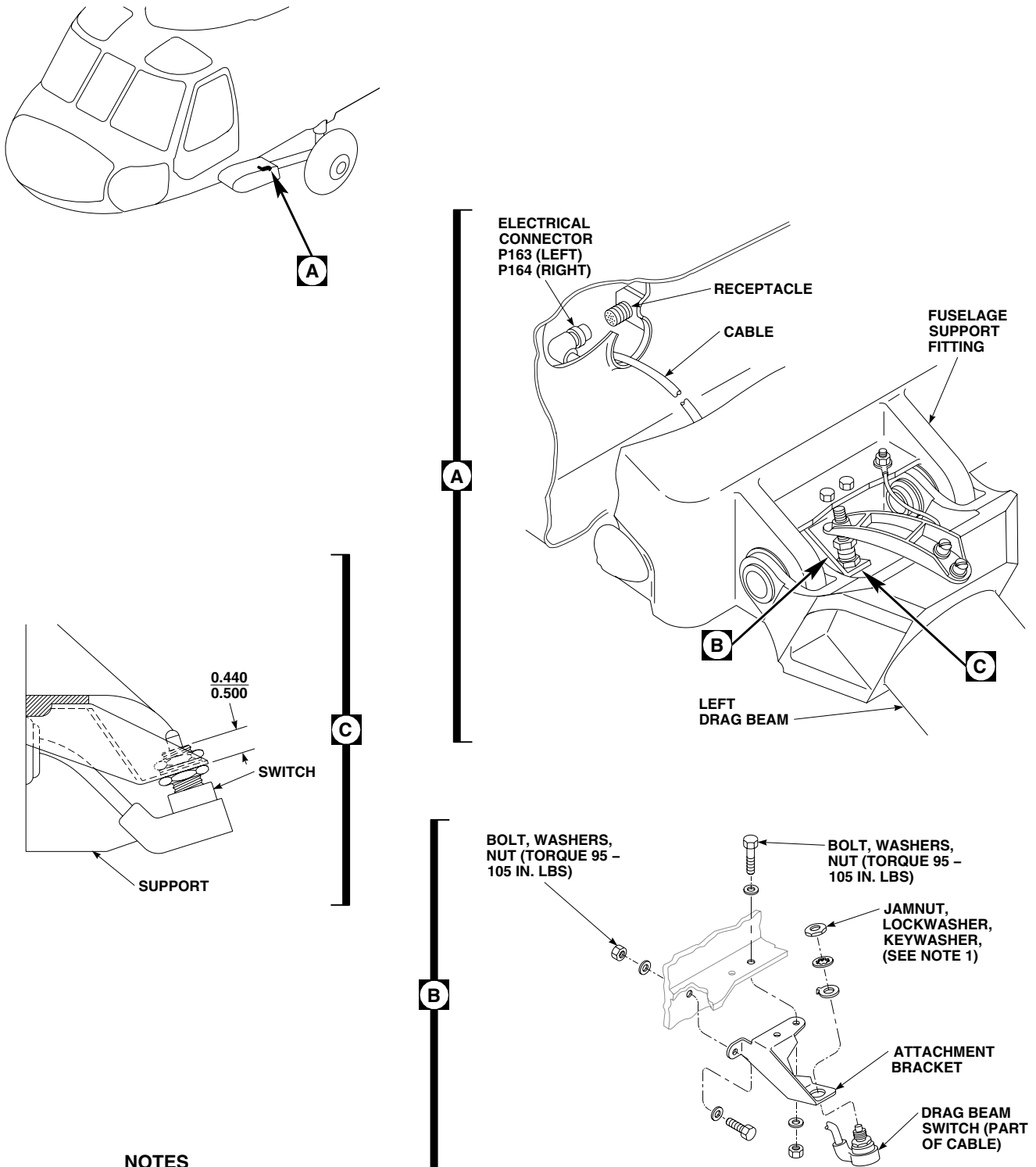


Figure 1. Replace Left Main Landing Gear Drag Beam Switch.

AB2524
SA

INSTALL - Continued

4. **QA** Connect electrical connector, P163 (left) or P164 (right).
5. Remove upper jamnut and lockwasher from switch.
6. **QA** Position drag beam switch in attachment bracket and reinstall keywasher, lockwasher and jamnut (Figure 1, Detail B or Figure 2, Detail A).
7. **ADJUST LEFT DRAG BEAM SWITCH POSITION, WITHIN ATTACHMENT BRACKET, TO GET 0.440 TO 0.500-INCH MEASUREMENT BETWEEN BASE OF BRACKET AND END OF THREADED PART OF SWITCH (Figure 1, Detail C).**
8. **ADJUST RIGHT DRAG BEAM SWITCH POSITION, WITHIN ATTACHMENT BRACKET, TO GET 0.515 TO 0.575-INCH MEASUREMENT BETWEEN BASE OF BRACKET AND END OF THREADED PART OF SWITCH (Figure 2, Detail B).**
9. **TIGHTEN JAMNUT 1/6 TO 1/3 TURN BEYOND POINT WHERE A SHARP RISE IN TORQUE IS FELT.**
10. **QA** Install left attachment bracket and drag beam switch on fuselage support with bolts, washers, and nuts (Figure 1, Detail B). **TORQUE NUTS ON LEFT DRAG BEAM SWITCH BRACKET TO 95 - 105 INCH-POUNDS.**
11. **QA** Install right attachment bracket on fuselage support with bolts, radius block (if required), washers, and nuts (Figure 2, Detail A). **TORQUE NUTS ON RIGHT DRAG BEAM SWITCH BRACKET TO 43 - 47 INCH-POUNDS.**
12. Using nonelectrical wire, Item 353, WP 1803 00, lockwire jamnut to attachment bracket.

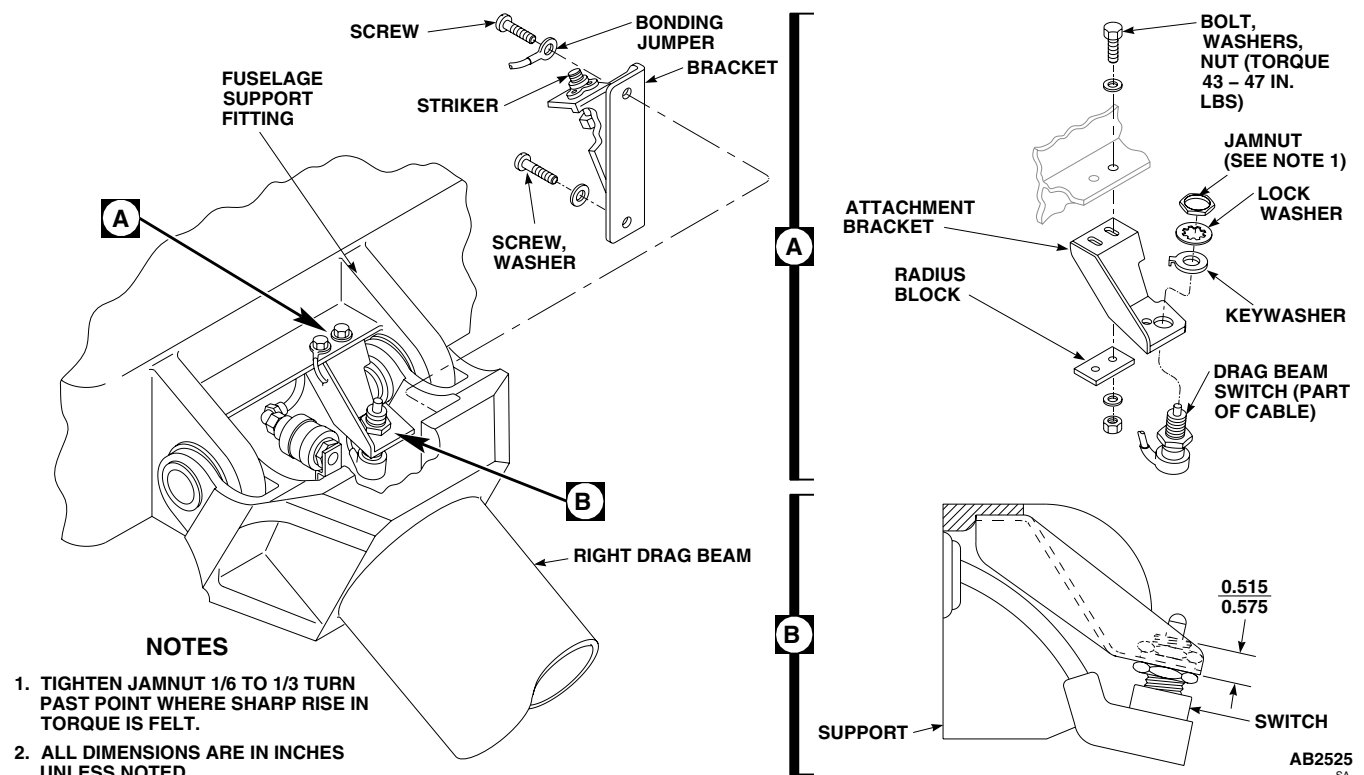


Figure 2. Replace Right Main Landing Gear Drag Beam Switch.

TEST

NOTE

Check of left or right drag beam switch is the same, except as noted.

1. To check drag beam switch, do this:
 - a. Simulate switch ground position by extending switch plunger.
 - b. Using multimeter, check for continuity between pin A and pin B and pin E and pin D of electrical connector, P163, for left switch or connector, P164, for right switch.
 - c. Using multimeter, check for integrity of the ground of both the right and left drag beam switch by checking continuity from pin A of J164 and pin D of J163 to helicopter ground.
2. If continuity is not shown, replace drag beam switch assembly.
3. Install drag beam switch attachment bracket with bolts, washers, and nuts (Figure 1, Detail B or Figure 2, Detail A).
 - a. Simulate switch flight position by compressing switch plunger.
 - b. Check for continuity between pin A and pin C and pin F and pin D of electrical connector, P163, for left switch or P164, for right switch.
 - c. Using multimeter, check for integrity of the ground of both the right and left drag beam switch by checking continuity from pin A of J164 and pin D of J163 to helicopter ground.
4. If continuity is not shown, replace drag beam switch assembly.
5. Connect electrical connector, P163 (left), or P164 (right), to bracket at front of landing gear support.
6. Adjust drag beam switch. Refer to, ADJUST, in this work package.

ADJUST

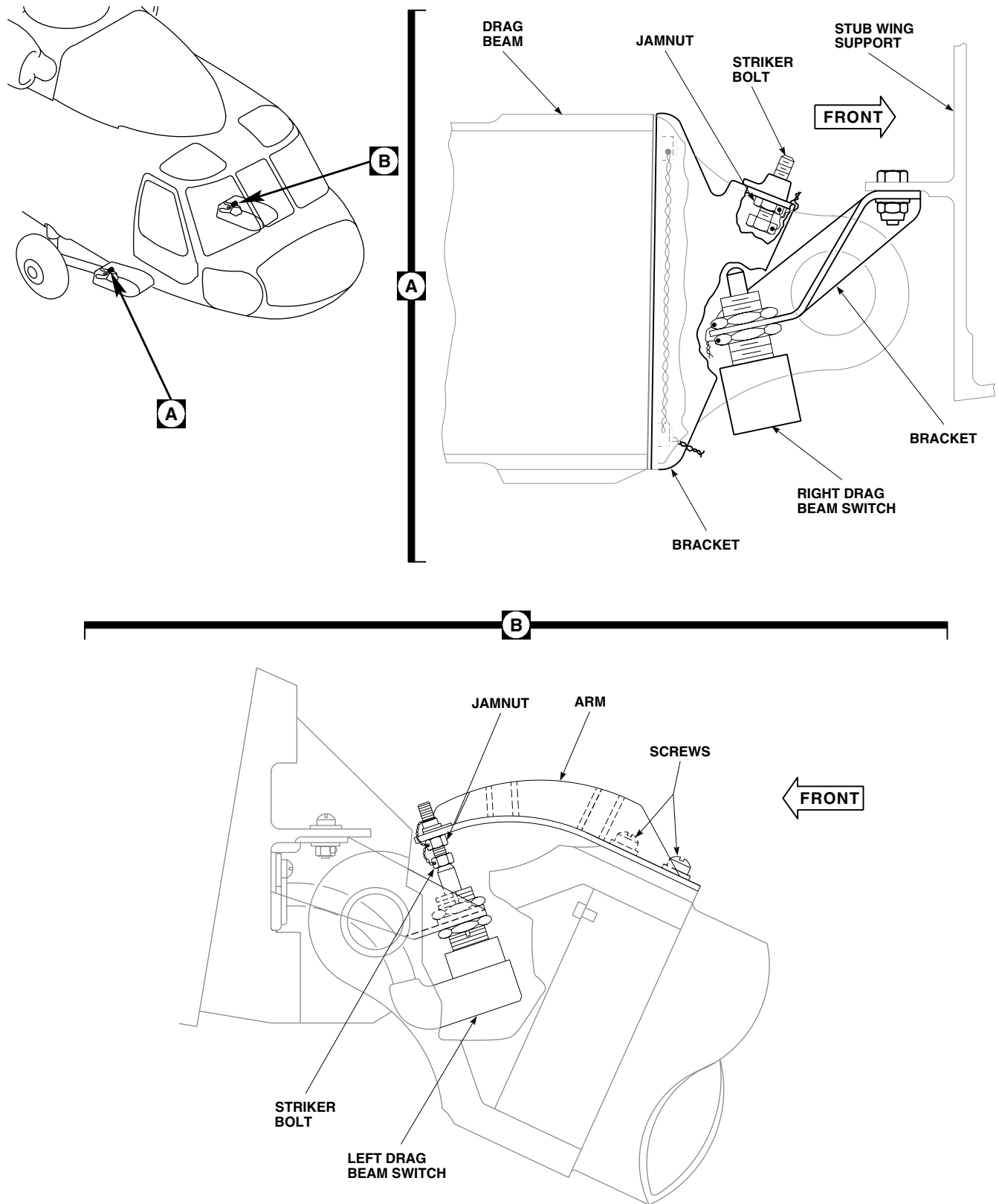
1. Turn off all electrical power.
2. Place a jack under stub wing jackpad on side where drag beam switch is being adjusted.
3. Jack helicopter until main gear wheel is clear of ground with shock strut fully extended (WP 1675 00).

WARNING

Injury to personnel and damage to equipment will result if shock strut is stuck or not fully extended. Kick or shake main landing gear wheel to make sure shock strut is fully extended.

4. On left main landing gear, loosen two screws attaching arm to drag beam. Position arm so that striker bolt (on arm) lines up with plunger of drag beam switch (Figure 3, Detail B).
5. **QA** Tighten two screws attaching arm to drag beam.
6. On right main landing gear a fixed position striker arm is attached to forward face of drag beam. Switch and striker bolt are adjustable up and down (Figure 3, Detail A).
7. Loosen jamnut of striker bolt and adjust bolt so it does not touch drag beam switch.
8. **ADJUST STRIKER BOLT UNTIL HEAD OF BOLT TOUCHES DRAG BEAM SWITCH. TURN BOLT TOWARD DRAG BEAM SWITCH ANOTHER FOUR FULL TURNS.**

ADJUST - Continued



AK0630A
SA

Figure 3. Adjust Main Landing Gear Drag Beam Switch.

ADJUST - Continued

9. **QA** Tighten jamnut on striker bolt. **TIGHTEN 1/6 TO 1/3 TURN BEYOND POINT WHERE A SHARP RISE IN TORQUE IS FELT.**
10. **QA** Using nonelectrical wire, Item 353, WP 1803 00, lockwire jamnut, starting at head of striker bolt, to jamnut, to threaded shank of bolt.
11. **QA** Using nonelectrical wire, Item 353, WP 1803 00, lockwire two screws attaching arm to drag beam on left main landing gear.
12. Using nonelectrical wire, Item 353, WP 1803 00, lockwire two screws attaching striker bracket to drag beam on right main landing gear.
13. Lower helicopter and remove jack (WP 1675 00).
14. Make sure area is clean and free of foreign material before closing up.
15. Install main landing gear support fairings (WP 0286 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR**

MAIN LANDING GEAR SHOCK STRUT

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 2-Gallon
Torque Wrench, 0 - 600 ft lbs.

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cloth, Abrasive, Item 84, WP 1803 00
Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Nitrogen, Item 199, WP 1803 00
Petrolatum, Item 216, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Tubing, Nonmetallic Tygothane, Item 345,
WP 1803 00
Union, AN815-4

References

TM 1-1520-237-PMI
TM 55-1500-345-23
WP 0286 00
WP 0359 00
WP 0381 00
WP 0445 00
WP 1597 00
WP 1675 00
WP 1803 00

PREPARE TO REMOVE

1. Turn off all electrical power.
2. Remove main landing gear support fairings from fuselage (WP 0286 00).
3. Jack helicopter with jack beneath pad on drag beam support (WP 1675 00).

INSPECT

1. Place containers to collect fluid from upper and lower stages, identify as upper or lower.
2. Remove upper/lower stage valve caps (Figure 1).
3. Slip nonmetallic tygothane tubing, Item 345, WP 1803 00, over valve stem. Place free end in correct container to capture hydraulic fluid as nitrogen is released from shock strut chambers.
4. Turn outer hexnut counterclockwise to release nitrogen from upper/lower shock strut chamber.
5. After all nitrogen pressure is released, turn inner hexnut counterclockwise to remove.
6. Allow any remaining hydraulic fluid in upper/ lower shock strut chambers to drain into containers.
7. **IF DISCHARGE FROM UPPER OR LOWER STAGE IS GREATER THAN 200 MILLILITERS (CUBIC CENTIMETERS), REPLACE SHOCK STRUT.**
8. **IF DISCHARGE IS BETWEEN 100 TO 200 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 200 FLIGHT HOURS.**
9. **IF DISCHARGE IS LESS THAN 100 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT NEXT PERIODIC MAINTENANCE INSPECTION (TM 1-1520-237-PMI)..**

REMOVE**WARNING**

Injury to personnel and damage to equipment will result if shock strut is not supported during removal. Shock strut weighs 60 pounds. Make sure shock strut is supported during removal.

1. Remove bolts, washers, and nuts from upper and lower end of shock strut (Figure 1).
2. Remove shock strut.

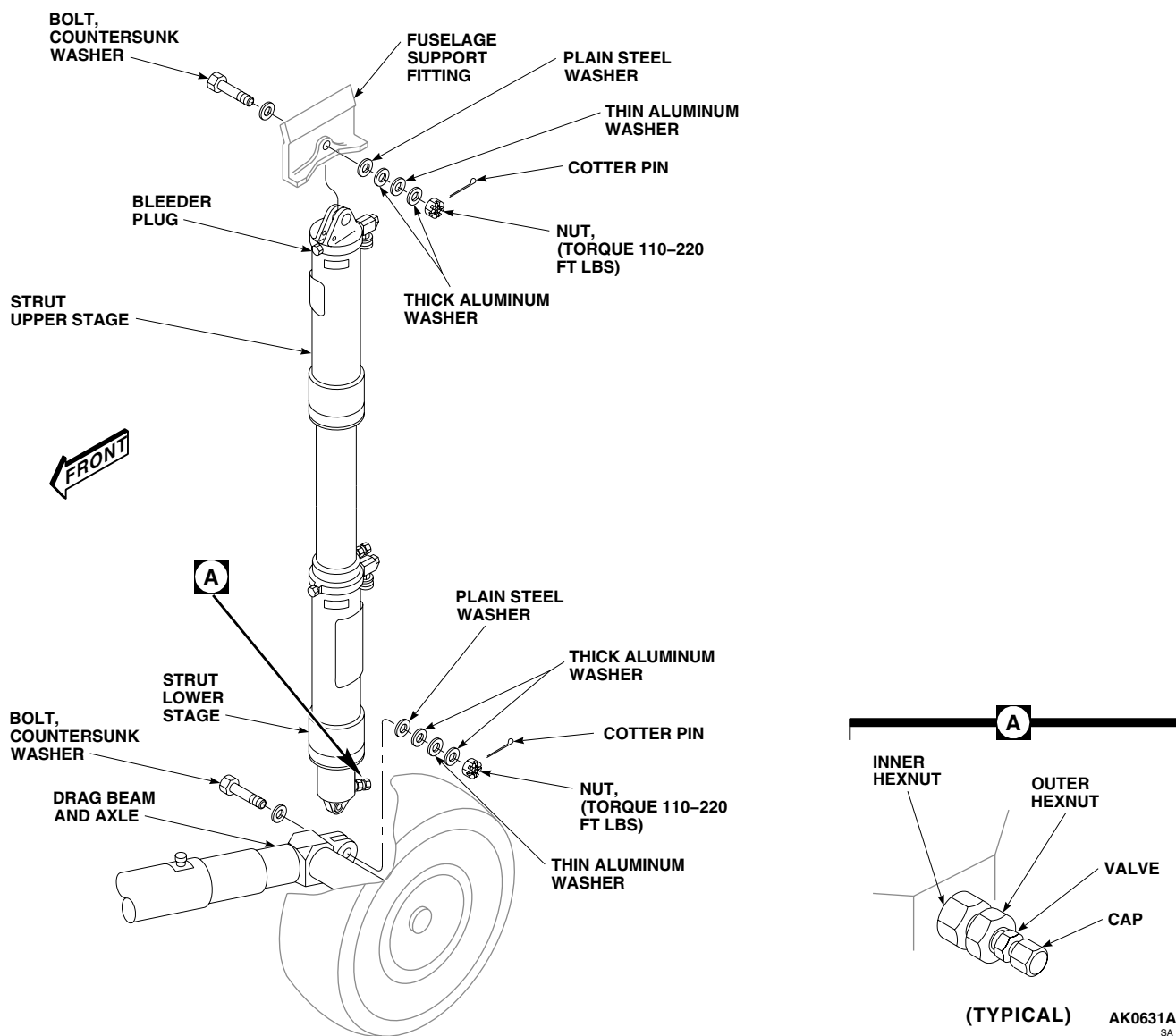


Figure 1. Replace Main Landing Gear Shock Strut.

REMOVE - Continued

3. Check the upper fuselage fitting assembly spherical bearing for maximum radial and axial play. **FUSELAGE FITTING ASSEMBLY SPHERICAL BEARING MAY HAVE UP TO 0.010-INCH MAXIMUM RADIAL PLAY AND 0.030-INCH MAXIMUM AXIAL PLAY.** Replace fitting assembly spherical bearing if out of tolerance (WP 0359 00).

INSTALL

1. Turn off all electrical power.
2. If replacement shock strut assembly is filled with hydraulic preservative fluid, MIL-H-46170, drain as follows:
 - a. Remove cap from upper valve (Figure 1, Detail A).

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing valve. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valve.

- b. Release nitrogen pressure in shock strut by turning outer hexnut counterclockwise. A hissing sound can be heard when releasing nitrogen. Remove valve from strut.
- c. Remove upper bleeder plug from shock strut.
- d. Install union, AN815-4, with packing off bleeder plug, in bleeder port. Connect nonmetallic tygothane tubing, Item 345, WP 1803 00, to union. Place overflow line from bleeder port into 2 gallon container.
- e. Install valve in shock strut upper stage (WP 0445 00).
- f. Attach servicing chuck from bottled nitrogen, Item 199, WP 1803 00, to upper valve.
- g. Lay shock strut on side with bleeder port facing down.

WARNING

Injury to personnel and damage to equipment will result if an unregulated nitrogen source is used. Apply pressure gradually from a regulated source of nitrogen.

NOTE

Do not pressurize strut with more than 100 psi pressure.

- h. Increase nitrogen pressure until strut upper stage is fully extended.

NOTE

There are 72.94-inches between center of upper and lower attachment boltholes when both stages of shock strut are fully extended.

- i. Disconnect nitrogen bottle from valve and continue draining shock strut until hydraulic fluid cannot be seen coming from bleeder port.
- j. Remove union from bleeder port.
- k. **QA** Install packing, lubricated with petrolatum, Item 216, WP 1803 00, on bleeder plug.
- l. **QA** Screw plug into cylinder until contact is made with cylinder surface.
- m. Drain preservative fluid from lower stage of strut using the same method as with upper stage.

INSTALL - Continued**WARNING**

Injury to personnel and damage to equipment may occur when using shock struts, 70250-12051-042, make sure both left and right sides have same part number. Shock struts, 70250-12051-043, through 70250-12051-045 are fully interchangeable, and may be mixed with each other on left and right sides.

3. Position upper and lower shock strut lugs on fuselage and drag beam support fittings (Figure 1).
4. **QA** Bolt upper end of shock strut to fuselage support fitting and lower end of shock strut to drag beam fitting as follows:
 - a. Install countersunk washer on bolt with countersunk side toward head.
 - b. Install bolt with head facing inboard.
 - c. Install plain steel washer on bolt.
 - d. Install thick aluminum washer on bolt.
 - e. Install thin aluminum washer on bolt.
 - f. Install thick aluminum washer on bolt.
 - g. Install nut.
5. **QA TORQUE NUT TO 110 - 220 FOOT-POUNDS.** Install cotter pin.

NOTE

Minor seepage in service is not considered cause for replacement.

6. Service main landing gear shock strut upper and lower stages (WP 1597 00).
7. After replacing shock strut with new strut, do a static leak check as follows:
 - a. Wipe polished piston of shock strut dry using machinery wiping towel, Item 344, WP 1803 00.
 - b. Wait 1 hour and check polished piston for hydraulic fluid.
 - c. If there is any evidence of petroleum base hydraulic fluid replace shock strut.
 - d. If no fluid is present, soak a clean machinery wiping towel, Item 344, WP 1803 00, with petroleum base hydraulic fluid, Item 145, WP 1803 00, and wipe all exposed polished portions of piston.
8. Make sure area is clean and free of foreign material.
9. Lower helicopter and remove jack (WP 1675 00).
10. Install main landing gear support fairings (WP 0286 00).

REPAIR

1. Blend out any minor scratches on housing with abrasive cloth, Item 84, WP 1803 00. **SCRATCHES AT RIGHT ANGLES TO CENTERLINE MUST NOT BE OVER 20% OF HOUSING CIRCUMFERENCE. SCRATCHES PARALLEL TO CENTERLINE MUST NOT BE MORE THAN 1.5-INCHES LONG. DEEPEST PART OF SCRATCH MUST NOT BE MORE THAN 0.005-INCH AFTER BLENDING.**
2. Blend out any small scratches or nicks on upper stage housing and lug with abrasive cloth, Item 84, WP 1803 00. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.020-INCH AFTER BLENDING.**
3. Blend out any small scratches or nicks on lower stage housing with abrasive cloth, Item 84, WP 1803 00. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.010-INCH AFTER BLENDING.**

REPAIR - Continued

4. No repair is allowed to upper or lower piston surface. **ALLOWABLE DEPTH OF DAMAGE IS 0.010-INCH, LENGTH NOT MORE THAN 1 INCH, NOT OVER 20% OF PISTON CIRCUMFERENCE.**
5. **ALLOWABLE WEAR IS LIMITED TO 0.0015-INCH OR SMALLEST OD OF 3.4935-INCHES FOR UPPER PISTON, AND 2.4935-INCHES FOR LOWER PISTON.**
6. Lower piston lug scratches or nicks may be blended out using abrasive cloth, Item 84, WP 1803 00. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.010-INCH AFTER BLENDING.**
7. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR
MAIN LANDING GEAR SHOCK STRUT KNEELING VALVE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Hydraulic Repairer's Toolkit, SC 5180-99-A03
Torque Wrench, 30 - 200 in. lbs.

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0286 00
WP 1597 00
WP 1675 00
WP 1803 00

Material/Parts

Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Pneudraulics Repairer MOS 15H (1)
Assistant - (1)

REPLACE MAIN LANDING GEAR SHOCK STRUT KNEELING VALVE**Remove****CAUTION**

Damage to equipment will result (the rear, lower ESSS hard point will contact the main landing gear tire) if both lower and upper stages of main landing gear shock strut are collapsed. Collapse only the upper or the lower stages of main landing gear shock strut at one time.

NOTE

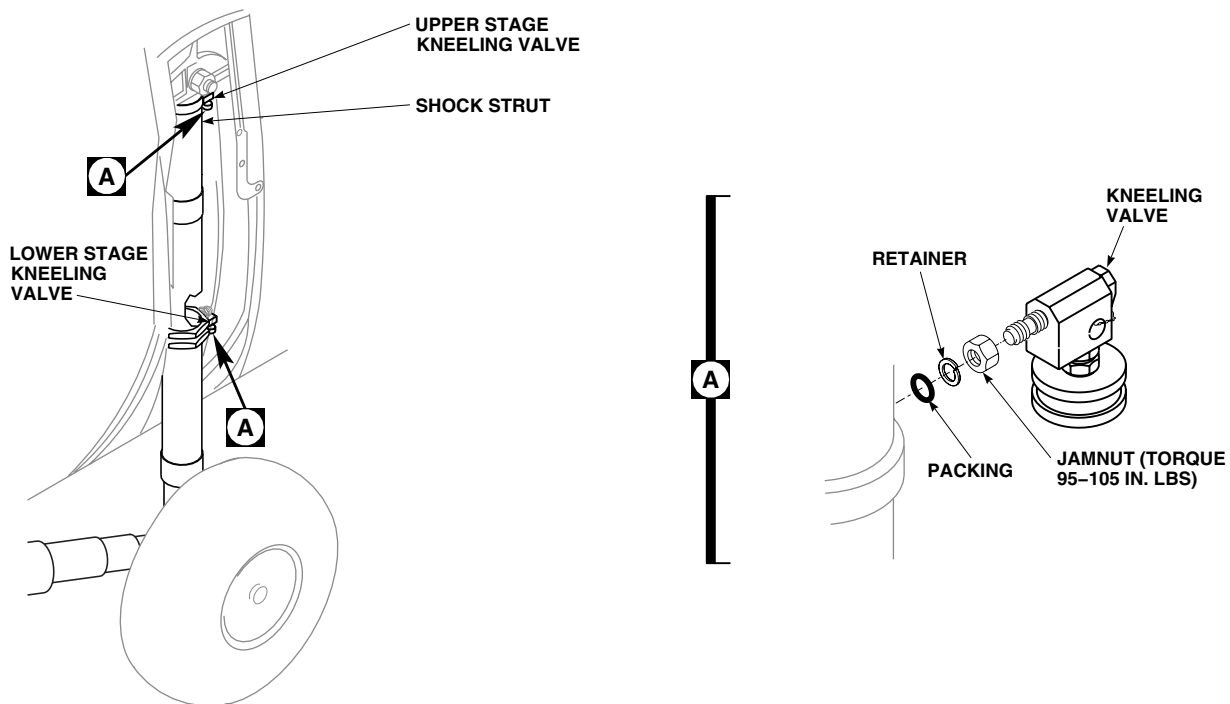
Replacement of either kneeling valve (upper or lower) is the same.

1. Remove main landing gear support fairings from fuselage WP 0286 00
2. Jack helicopter with jack beneath pad on drag beam support (WP 1675 00).
3. Deservice shock strut upper stage (WP 1597 00).
4. Loosen jamnut and remove coupling half from upper stage kneeling valve.
5. Loosen jamnut and remove kneeling valve from strut. Remove packing, retainer, and jamnut from valve (Figure 1, Detail A). Discard packing. Keep jamnut for installation of replacement valve.

Install

1. Lubricate packing, retainer, and male threads of replacement kneeling valve with petroleum base hydraulic fluid, Item 145, WP 1803 00.
2. **QA** Install jamnut, retainer, and packing onto kneeling valve. Turn jamnut and retainer down until packing is pushed firmly against the lower threaded section of kneeling valve (Figure 1, Detail A).

REPLACE MAIN LANDING GEAR SHOCK STRUT KNEELING VALVE - Continued



AB0154
SA

Figure 1. Replace Main Landing Gear Shock Strut Kneeling Valve.

3. **QA** Install kneeling valve in strut until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN KNEELING VALVE AND JAMNUT 1 1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN KNEELING VALVE NO MORE THAN 1 ADDITIONAL TURN TO POSITION COUPLING HALF PORT FACING DOWN.**
4. **PREVENT KNEELING VALVE FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
5. **QA** Install coupling half in kneeling valve.
6. Service shock strut upper stage (WP 1597 00).
7. Make sure area is free of foreign material before closing up.
8. Lower helicopter and remove jack (WP 1675 00).
9. Install main landing gear support fairings (WP 0286 00).

REPAIR MAIN LANDING GEAR SHOCK STRUT KNEELING VALVE

Disassemble

1. Remove knob stop pin from body (Figure 2).
2. Remove shaft by turning knob counterclockwise.
3. Remove packing and retainers from shaft. Discard packing.
4. Loosen jamnut and remove coupling half from valve body.
5. Remove packing, retainer, and jamnut from coupling half. Discard packing.

REPAIR MAIN LANDING GEAR SHOCK STRUT KNEELING VALVE - Continued

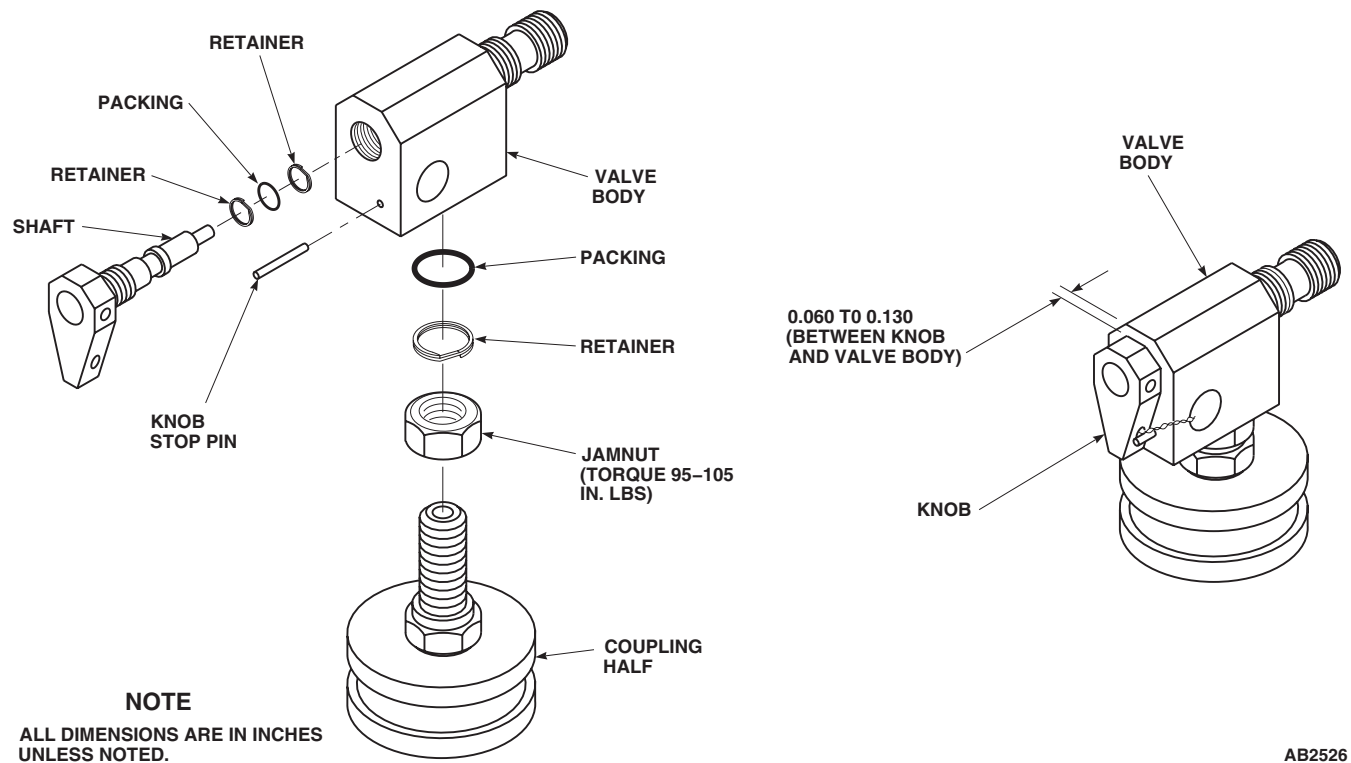


Figure 2. Repair Main Landing Gear Shock Strut Kneeling Valve.

Inspect

1. Check shaft for corrosion or deformation. Replace if corroded or deformed.
2. Check shaft bore in valve body. Replace body if corroded or deformed.

Assemble

1. Lubricate packing, retainer, and male threads of coupling half with petroleum base hydraulic fluid, Item 145, WP 1803 00.
2. **QA** Install jamnut, retainer, and packing onto coupling half. Turn nut and retainer down until packing is pushed firmly against the lower threaded section of coupling half (Figure 2).
3. **QA** Install coupling half into valve body until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN COUPLING HALF AND JAMNUT 1 1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
4. **PREVENT COUPLING HALF FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
5. Lubricate packing and retainers with petroleum base hydraulic fluid, Item 145, WP 1803 00.
6. **QA** Install retainer, packing, and second retainer on shaft.
7. **QA** Insert shaft carefully into valve body.
8. **QA** Turn knob clockwise to set specified clearance between knob and valve body. **CLEARANCE MUST RANGE FROM 0.060 TO 0.130-INCH.**
9. **QA** Install knob stop pin, in valve body.

REPAIR MAIN LANDING GEAR SHOCK STRUT KNEELING VALVE - Continued

10. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire knob to hole in valve body.

Test

Test for operation on helicopter, no bench test required.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR
MAIN LANDING GEAR SHOCK STRUT AIR VALVE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Torque Wrench, 30 - 200 in. lbs.

References

WP 0286 00
 WP 1597 00

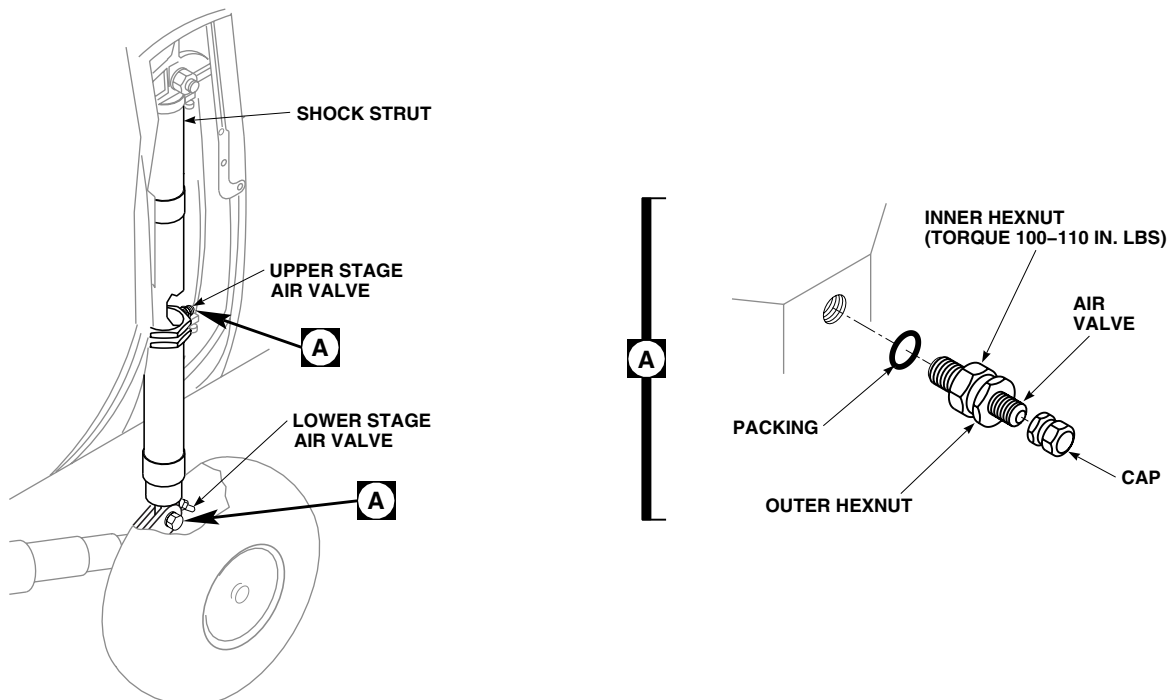
Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

Replacement of either air valve (upper or lower) is the same.

1. Remove lower fairing and cover from fuselage (WP 0286 00).
2. Remove cap from air valve being replaced (Figure 1, Detail A).



AB0153
SA

Figure 1. Replace Main Landing Gear Shock Strut Air Valve.

REMOVE - Continued

3. Turn outer hexnut counterclockwise until you hear air hissing from valve.

WARNING

Injury to personnel and damage to equipment will result if all nitrogen pressure is not released prior to removing valves. Shock strut is under high pressure. Make sure all nitrogen pressure is released prior to removing valves.

4. When air valve stops hissing, turn inner hexnut counterclockwise and remove air valve.
5. Check condition of packing. If packing is determined to be source of air leak, replace packing.
6. If valve is determined to be malfunctioning, replace valve.

INSTALL**NOTE**

Replacement air valve includes new packing.

1. **QA** Install replacement air valve in strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH-POUNDS (FIGURE 1, DETAIL A).**
2. Service stage of shock strut from which valve was replaced (WP 1597 00).
3. Make sure area is free of foreign material before closing up.
4. Install main landing gear fairings (WP 0286 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****MAIN LANDING GEAR WHEEL AND TIRE**

This WP supersedes WP 0447 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Tire Cage, NSN 4920-01-545-0763 SK2000TC-6
High Pressure Nitrogen Servicing Cart, PD-EIG-90-2
Tire Inflator Assembly, M85352/1 or M008348-1-1
Torque Wrench, 0 - 300 ft lbs.
Wrench, Crowfoot, 2 1/4-in., GGG-C-001507

Sealing Compound, Item 273, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Grease, Aircraft, Item 131, WP 1803 00
Nitrogen, Item 199, WP 1803 00

References

WP 0450 00
WP 1598 00
WP 1675 00
WP 1803 00

REMOVE

1. Jack main landing gear assembly (WP 1675 00).
2. Set parking brake. Leave brake on until wheel is replaced.

WARNING

Injury to personnel will result if valve core is removed from valve stem before the tire is completely deflated. Release pressure from tire before removing valve core from valve stem.

3. Remove valve cap from valve stem. Completely deflate tire, then remove valve core from valve stem. Reinstall valve core. Attach deflation tag (Figure 1).

CAUTION

Damage to bearings will result if bearings are dropped while removing wheel. Make sure bearings do not drop while removing wheel.

4. Remove cotter pin, nut, and key washer using 2 1/4-inch crowfoot, 6 inch extension, and 1/2-inch drive handle. Remove seal and outer bearing from wheel (Figure 1).

NOTE

In the following step, inner bearing and seal may remain on axle when removing wheel. Make sure bearing does not fall on ground.

5. Carefully slide wheel and tire assembly from axle.
6. Remove inner bearing and seal from axle or wheel.

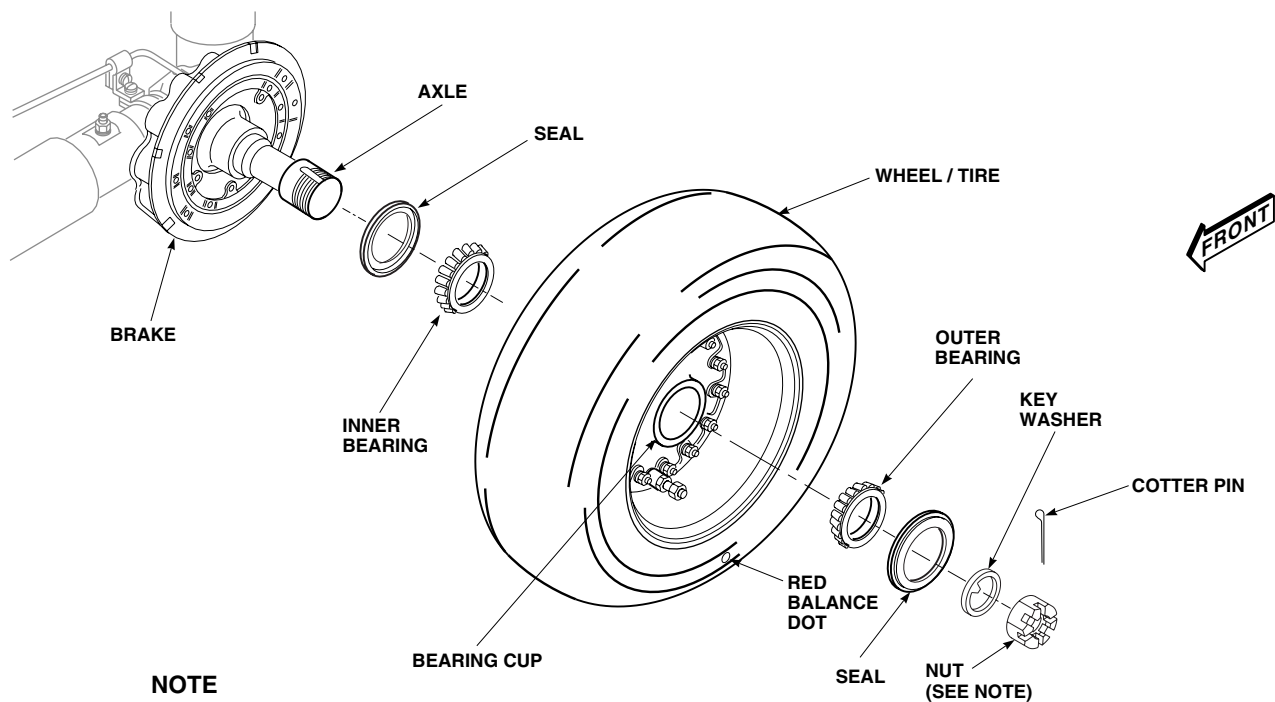
REMOVE - ContinuedFB4375
SA

Figure 1. Replace Main Landing Gear Wheel and Tire.

INSPECT

1. Check bearing seals for damaged reinforcing ring, visible damage, or deformation. **NO DAMAGE OR DEFORMATION ALLOWED.** Replace any seal found unserviceable.

CAUTION

Damage to bearings will result if they are allowed to spin while drying. Secure inner and outer race of bearings while drying.

2. Clean bearings with cleaning compound solvent, Item 71, WP 1803 00. Dry bearings with compressed air, if available, or let air-dry.
3. Inspect bearing for evidence of wear, corrosion, galling, scratches, pits, water stains, heat discoloration, or cracked and distorted bearing cage. **NO DAMAGE ALLOWED.** Replace damaged bearings.
4. If bearing is determined serviceable, repack it with aircraft grease, Item 131, WP 1803 00.
5. Review inspection requirements for wheel (WP 0450 00).

INSTALL**WARNING**

Injury to personnel and damage to equipment will result if tire pressure is more than 20 psi prior to installation. Make sure tire pressure is 20 psi or less prior to installation.

NOTE

10 ply and 12 ply tires are fully interchangeable and may be mixed with each other on left and right sides.

1. Inspect wheel for obvious damage before installing.
2. Clean axle and bearing cups using machinery wiping towel, Item 344, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00.
3. Lubricate axle (but not threads), with aircraft grease, Item 131, WP 1803 00.

CAUTION

The part number for the seal and inner bearing is different depending on if a 7 or 14 bolt wheel configuration is being installed. Make sure correct part number for the seal and inner bearing is installed.

4. **QA** Install seal and inner bearing on axle. Lip of seal is toward bearing (Figure 1).
5. Lubricate bearing cups in wheel and lips of seals with aircraft grease, Item 131, WP 1803 00.
6. Support wheel and tire assembly and line up brake discs.

WARNING

The 7 lug wheel (5003889) may experience lug failure when operating at or above gross weights of 20,500 pounds. Lug failure may result in flying debris during ground handling and/or unexpected tire failure. The 14 lug wheel (040-25300) should be utilized when operating at or above gross weights of 20,500 pounds.

7. **QA** Carefully slide wheel and tire assembly on axle. Make sure wheel disc lugs are through brake discs and inner bearing is properly seated.

CAUTION

The part number for the seal and outer bearing is different depending on if a 7 or 14 bolt wheel configuration is being installed. Make sure correct part number for the seal and outer bearing is installed.

8. **QA** Place outer bearing and seal on axle. Install bearing in bearing cup in wheel. Install seal in bearing cup. Lip of seal is toward bearing.
9. **QA** Install keywasher and nut on axle.
10. Release parking brake.
11. **QA USING 2 1/4-INCH CROWFOOT, 6 INCH EXTENSION, AND 1/2-INCH DRIVE HANDLE, TURN WHEEL WHILE TIGHTENING AXLE NUT WITH TORQUE WRENCH UNTIL AN INCREASE IN ROTATIONAL WHEEL DRAG IS FELT.**

INSTALL - Continued

12. **QA** While continuing to turn wheel, **TORQUE NUT TO 70 - 90 FOOT-POUNDS**, to make sure full seating of wheel bearings.
13. **QA** Back off axle nut until no contact with wheel.
14. **QA** While continuously turning wheel, slowly **TORQUE NUT TO 35 - 45 FOOT-POUNDS**.
15. **QA** Back off to next castellation.
16. Push wheel inboard and outboard to insure there is no free play in bearings. If play exist, repeat step 12. through step 15.
17. **QA** If play still exists, advance nut to next castellation. Wheel must rotate freely with no binding. Install cotter pin.

WARNING

- Injury to personnel or damage to equipment will result if nitrogen source is not vented at 160 psi maximum. Make sure to check tire inflator assembly nitrogen source is vented at 160 psi maximum.
 - Injury to personnel or damage to equipment will result if tire is inflated beyond recommended pressure. Do not inflate tire beyond its rated inflation pressure.
 - Injury to personnel or damage to equipment will result if an unregulated pressure source is used. Use an regulated pressure source to service tire.
 - Injury to personnel will result if the tire sidewall is facing personnel. Make sure personnel face tire tread and not tire sidewall when inflating tire.
 - Injury to personnel, or damage to equipment will result, if a tire inflation safety cage is not used while servicing tires. Use a tire inflation safety cage when inflating tire.
 - Failure to use proper procedures during servicing of main or tail landing gear tires may result in serious injury or death.
18. **INFLATE TIRE TO 130 TO 140 PSI** with nitrogen, Item 199, WP 1803 00. Use a regulated nitrogen pressure source connected to tire inflator assembly. Use primary or alternate methods for inflation (WP 1598 00).
 19. After a 5 to 10 minute waiting period, check wheel for leakage as follows:
 - a. Moisten air valve and around tire beads.
 - b. If bubbles appear, deflate, disassemble, and correct at point of leakage (WP 0450 00).
 20. Seal around nut and axle with sealing compound, Item 273, WP 1803 00.
 21. Install valve cap on valve stem.
 22. Lower helicopter and remove jack (WP 1675 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****MAIN LANDING GEAR WHEEL VALVE**

This WP supersedes WP 0448 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Tire Cage, NSN 4920-01-545-0763 SK2000TC-6
High Pressure Nitrogen Servicing Cart, PD-EIG-90-2
Tire Inflator Assembly, M85352/1 or M008348-1-1
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 1598 00
WP 1675 00
WP 1803 00

Material/Parts

Grease, Aircraft, Item 131, WP 1803 00
Nitrogen, Item 199, WP 1803 00

REMOVE

1. Jack main landing gear assembly (WP 1675 00).
2. Apply parking brake.

WARNING

Injury to personnel will result if valve core from valve stem is removed, before the tire is completely deflated. Release pressure from tire before removing valve core from valve stem.

3. Remove valve cap from valve stem. Completely deflate tire, then remove valve core from valve stem (Figure 1, Detail A).
4. Unscrew and remove valve stem from wheel.
5. Remove and discard packing from valve stem.

INSTALL

1. **QA** On wheel, 5003889, do this:
 - a. Lightly lubricate new packing with aircraft grease, Item 131, WP 1803 00.
 - b. **QA** Install packing on valve stem (Figure 1, Detail A).
 - c. **QA** Install valve stem into outboard wheel half. **TORQUE VALVE STEM TO 60 - 75 INCH-POUNDS.**
2. **QA** On wheel, 040-25300, do this:
 - a. Lightly lubricate new packing with aircraft grease, Item 131, WP 1803 00.
 - b. **QA** Install packing on valve stem (Figure 1, Detail A).
 - c. **QA** Install valve stem into outboard wheel half. **TORQUE VALVE STEM TO 40 - 50 INCH-POUNDS.**
3. **QA** Install new valve core in valve stem.

INSTALL - Continued

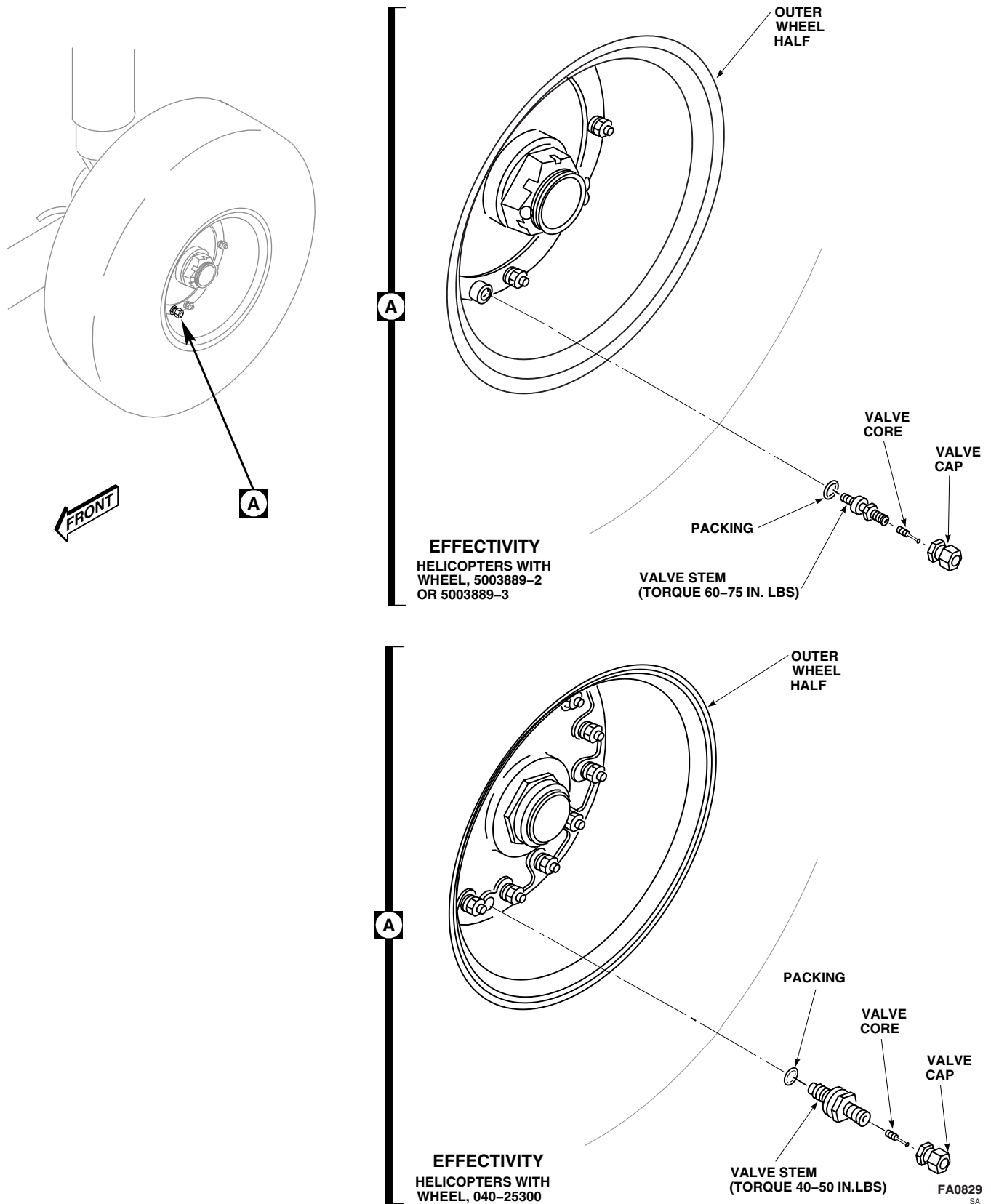


Figure 1. Replace Main Landing Gear Wheel Valve.

INSTALL - Continued**WARNING**

- Injury to personnel or damage to equipment will result if nitrogen source is not vented at 160 psi maximum. Make sure to check tire inflator assembly nitrogen source is vented at 160 psi maximum.
 - Injury to personnel or damage to equipment will result if tire is inflated beyond recommended pressure. Do not inflate tire beyond its rated inflation pressure.
 - Injury to personnel or damage to equipment will result if an unregulated pressure source is used. Use an regulated pressure source to service tire.
 - Injury to personnel will result if the tire sidewall is facing personnel. Make sure personnel face tire tread and not tire sidewall when inflating tire.
 - Injury to personnel, or damage to equipment will result, if a tire inflation safety cage is not used while servicing tires. Use a tire inflation safety cage when inflating tire.
 - Failure to use proper procedures during servicing of main or tail landing gear tires may result in serious injury or death.
4. **INFLATE TIRE TO 130 TO 140 PSI** with nitrogen, Item 199, WP 1803 00. Use a regulated nitrogen pressure source connected to tire inflator assembly. Use primary or alternate methods for inflation (WP 1598 00).
 5. After a 5 to 10 minute waiting period, check valve stem for leakage as follows:
 - a. Moisten valve stem area.
 - b. If bubbles appear, deflate and correct at point of leakage.
 6. Install valve cap.
 7. Release parking brake.
 8. Lower helicopter and remove jack (WP 1675 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR
MAIN LANDING GEAR WHEEL DISK DRIVE KEYS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Micrometer, 0 to 1 Inch
Torque Wrench, 30 - 200 in. lbs.

References

WP 0468 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. On inner wheel half of 7 bolt configured wheel, remove screw staking with 0.125-inch drill.
2. On 7 bolt and 14 bolt configuration remove screws and disc drive keys (Figure 1).

INSPECT

1. Inspect wheel half (WP 0468 00).
2. Check disc drive keys for wear. If one or more keys measure less than 0.696-inch between disc contacting surfaces, replace all keys on wheel half.

INSTALL

1. **QA** Place disc drive keys on wheel key bosses (Figure 1).
2. **QA** On inner wheel half of 7 bolt configured wheel, install and stake screws.
3. **QA** On inner wheel half of 14 bolt configured wheel install screws. **TORQUE SCREW TO 30 - 35 INCH-POUNDS.**

INSTALL - Continued

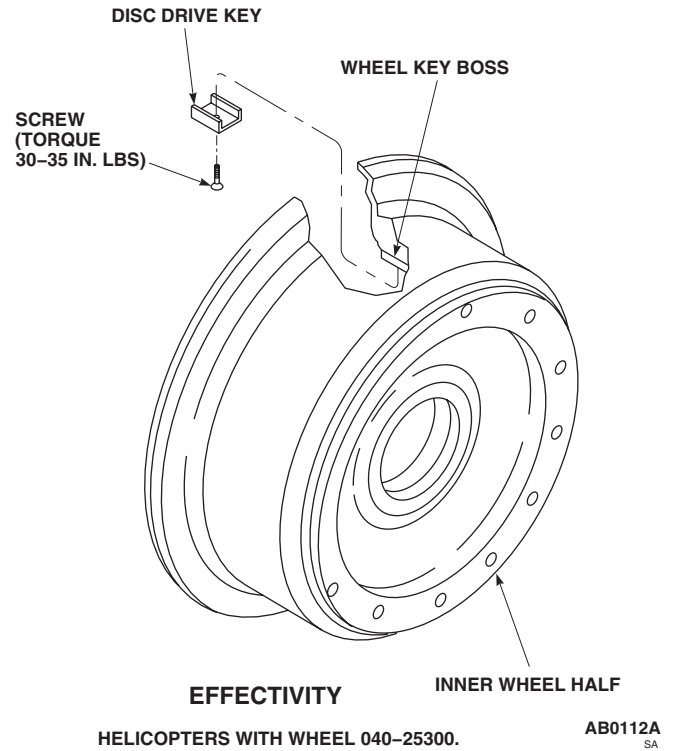
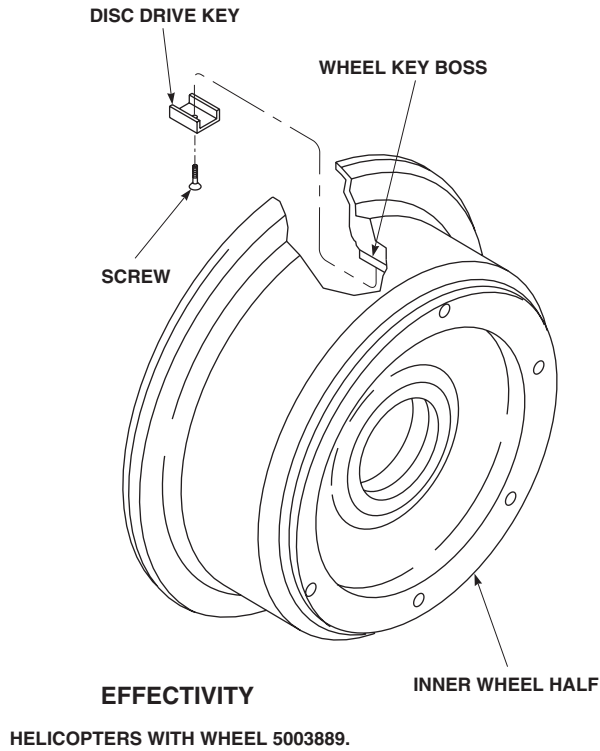


Figure 1. Replace Main Landing Gear Wheel Disc Drive Keys.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR**MAIN LANDING GEAR TIRE**

This WP supersedes WP 0450 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Tire Cage, NSN 4920-01-545-0763 SK2000TC-6
 Fluorescent Penetrant Inspection Kit, XMA101
 High Pressure Nitrogen Servicing Cart, PD-EIG-90-2
 Tire Bead Breaker, 5372
 Tire Inflator Assembly, M85352/1 or M008348-1-1
 Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Alcohol, Denatured, Item 48, WP 1803 00
 Antiseize Compound, Item 51, WP 1803 00
 Bolt, NAS625-96
 Bolt, NAS626-96
 Grease, Aircraft, Item 131, WP 1803 00
 Nitrogen, Item 199, WP 1803 00

Tags

Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (2)

References

TM 1-1520-265-23
 TM 55-1500-345-23
 TM 55-2620-200-24
 WP 0381 00
 WP 0448 00
 WP 0468 00
 WP 1598 00
 WP 1803 00

REMOVE**WARNING**

- Injury to personnel will result if valve core from valve stem is removed, before the tire is completely deflated. Release pressure from tire before removing valve core from valve stem.
- The 7 lug wheel (5003889) may experience lug failure when operating at or above gross weights of 20,500 pounds. Lug failure may result in flying debris during ground handling and/or unexpected tire failure. The 14 lug wheel (040-25300) should be utilized when operating at or above gross weights of 20,500 pounds.

1. Completely deflate tire and remove valve core (Figure 1).

CAUTION

- Damage to component's protective finish will result if the bead breaker is not used on a clean flat surface. Disassemble and assemble wheel on a clean flat surface to protect finish.
 - Damage to components will occur if hand tools are used to pry between tire bead and wheel. Do not pry between tire bead and wheel with hand tools.
2. Break tire beads away from wheel flange by applying even pressure to sides. Use tire bead breaker.

REMOVE - Continued

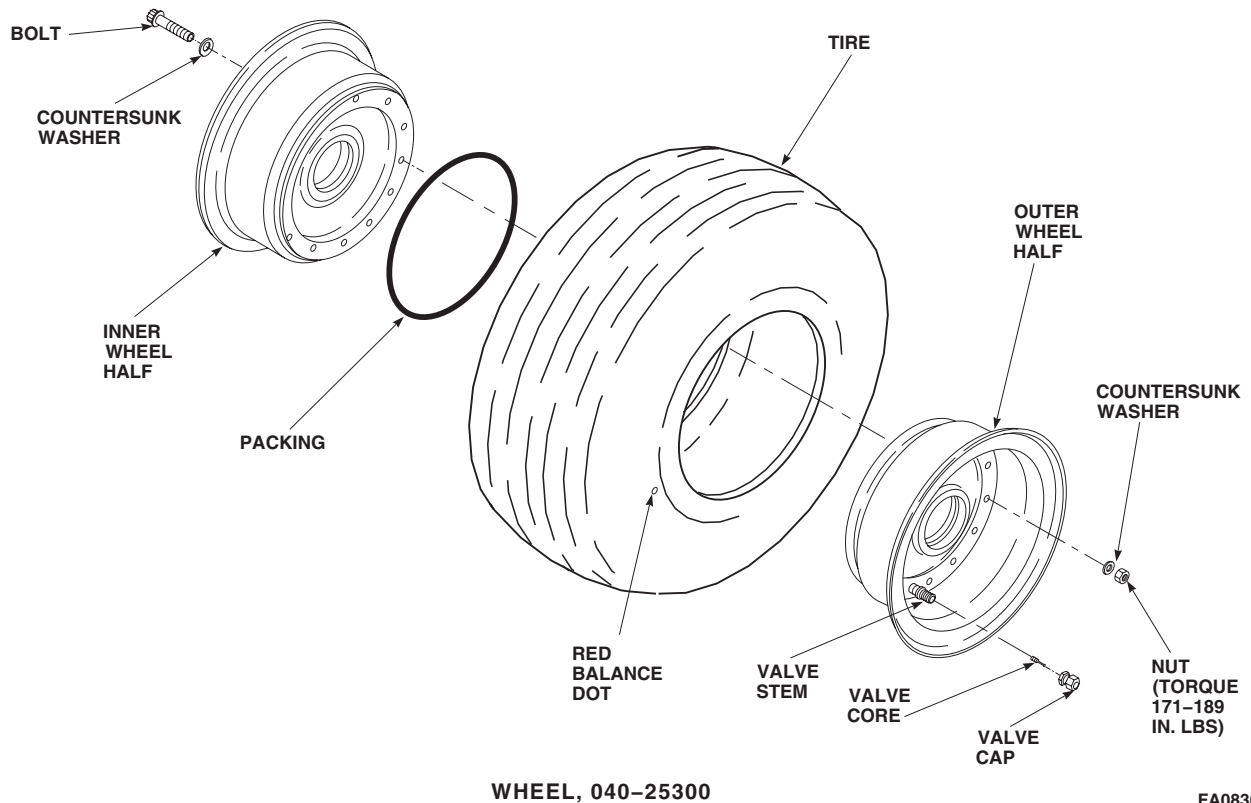
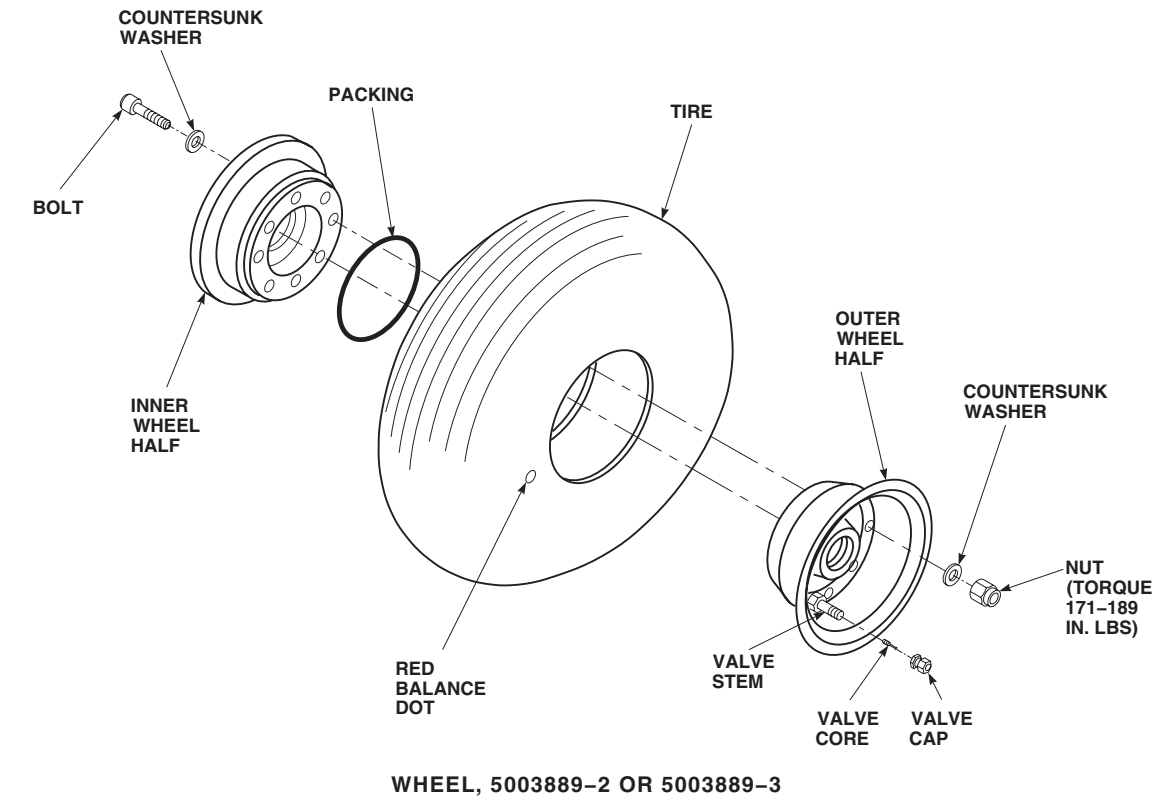
FA0830
SA

Figure 1. Replace Main Landing Gear Tire.

REMOVE - Continued**CAUTION**

Damage to equipment will result if impact or power wrenches are used to remove bolts and nuts. Wheel assembly will be damaged. Do not use impact or power wrenches to remove bolts and nuts.

NOTE

Hardware on wheel, 040-25300, may be reused if no cracks are found.

3. Remove bolts, countersunk washers, and nuts. Separate wheel halves and remove tire. Discard hardware on wheel, 5003889.
4. Remove packing from inner wheel half.

INSPECT

1. Inspect tire for damage (TM 55-2620-200-24).
2. Inspect tire for wear. **REPLACE TIRE IF WEAR HAS REACHED FIRST FABRIC PLY ON TREAD GROOVE OR IF WORN FOR A CONTINUOUS LENGTH OF 12 INCHES AT BOTTOM OF TREAD GROOVE.**
3. Inspect tire tread and shoulder for cuts or damage. **CUTS OR DAMAGE MAY BE ANY LENGTH PROVIDED DAMAGE DOES NOT CUT FIRST LAYER OF CORD BODY. IF CUT PENETRATES FIRST LAYER OF CORD BODY AND IS LESS THAN 1 INCH LONG, DEPTH OF CUT SHALL NOT EXCEED CUT LIMIT SHOWN ON SIDEWALL OF TIRE. IF TIRE GROOVE NEAREST CUT IS WORN SMOOTH AND CUT PENETRATES CORD BODY REPLACE TIRE.** Replace tire that exceeds limits.
4. Inspect tire sidewalls for cuts, scratches, abrasions or scuffs. **CUTS OR DAMAGE MAY BE ANY LENGTH PROVIDED DAMAGE DOES NOT PENETRATE FIRST LAYER OF CORD BODY.** Replace tire that exceeds limits.
5. Inspect tire surface for weather checking. **NO SEPARATION OF RUBBER BETWEEN CORDS ALLOWED. MAXIMUM DEPTH OF CRACK BENEATH TREAD RIB IS 1/8-INCH DEEP AND 1/4-INCH LONG.** Replace tire that exceeds limits.
6. Check condition of valve. If damaged, install a new valve (WP 0448 00).
7. Inspect new tire beads for damage. Clean tire beads with machinery wiping towel, Item 344, WP 1803 00, dampened with denatured alcohol Item 48, WP 1803 00.
8. Repair tire (TM 55-2620-200-24).

WARNING

Injury to personnel or damage to equipment will result if hardware is reused on wheel, 5003889. Replace all bolts and nuts on wheel, 5003889.

NOTE

Hardware on wheel, 040-25300, may be reused if no cracks are found.

9. On wheel, 040-25300, fluorescent penetrant inspect bolts for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NONE ALLOWED.** Replace cracked bolts.

INSPECT - Continued**CAUTION**

Damage to equipment will result if wheel halves are not inspected whenever wheel is disassembled. Inspect wheel halves whenever wheel is disassembled.

NOTE

Fluorescent penetrant inspection is not required on wheel halves, 040-25300.

10. Inspect and repair wheel (WP 0468 00).

INSTALL

1. Clean wheel bead seat with machinery wiping towel, Item 344, WP 1803 00, dampened with denatured alcohol Item 48, WP 1803 00.
2. Lubricate packing with a light coat of aircraft grease, Item 131, WP 1803 00.
3. **QA** Install packing in groove on inner wheel half (Figure 1).
4. **QA** Place tire on inner wheel half.
5. **QA** Place outer wheel half in tire and line up boltholes in both wheel halves. Line up red balance dot on sidewall with valve stem.
6. Put antiseize compound, Item 49, WP 1803 00, on new bolt and nut threads, face of nut, under boltheads, and both sides of washers.

WARNING

Injury to personnel or damage to equipment may occur if hardware is reused on wheel, 5003889. Replace all bolts and nuts on wheel, 5003889.

CAUTION

Damage to equipment will result if impact or power wrenches are used to tighten or torque bolts or nuts. Wheel assembly and hardware will be damaged. Do not use impact or power wrenches when tightening or torquing bolts or nuts.

NOTE

- Hardware on wheel, 040-25300, may be reused if no cracks are found.
- Short bolts and stiff tires sometimes make it hard to draw wheel halves together enough to start nuts.

7. If nuts cannot be started on bolts of wheel, 5003889, do this:
 - a. Temporarily install two NAS626-96 bolts, 180 degrees apart, through both wheel halves. Install washers and 0.375-24 nuts.
 - b. Tighten nuts evenly, a little at a time, to draw wheel halves together.
 - c. **QA** Once wheel is together, install regularly used bolts, washers, and nuts. Remove temporary hardware.
8. If nuts cannot be started on bolts of wheel, 040-25300, do this:
 - a. Temporarily install two NAS625-96 bolts, 180 degrees apart, through both wheel halves. Install washers and 0.375-24 nuts.
 - b. Tighten nuts evenly, a little at a time, to draw wheel halves together.

INSTALL - Continued

- c. **QA** Once wheel is together, install regularly used bolts, washers, and nuts. Remove temporary hardware.
9. On wheel, 040-25300, bolt wheel halves together, with boltheads on inner side of wheel. Install countersunk washers so countersunk sides are toward boltheads and nut faces respectively. **TIGHTEN NUTS IN A CRISSCROSS PATTERN, IN EVEN UNITS OF 25 INCH-POUNDS, UNTIL A FINAL TORQUE OF 171 - 190 INCH-POUNDS IS OBTAINED.**
10. Apply torque stripes (WP 0381 00 and TM 55-1500-345-23) to wheel nuts.
11. Install valve core.

WARNING

- Injury to personnel or damage to equipment will result if nitrogen source is not vented at 160 psi maximum. Make sure to check tire inflator assembly nitrogen source is vented at 160 psi maximum.
 - Injury to personnel or damage to equipment will result if tire is inflated beyond recommended pressure. Do not inflate tire beyond its rated inflation pressure.
 - Injury to personnel or damage to equipment will result if an unregulated pressure source is used. Use an regulated pressure source to service tire.
 - Injury to personnel will result if the tire sidewall is facing personnel. Make sure personnel face tire tread and not tire sidewall when inflating tire.
 - Injury to personnel, or damage to equipment will result, if a tire inflation safety cage is not used while servicing tires. Use a tire inflation safety cage when inflating tire.
12. Place wheel in a tire inflation safety cage (TM 55-2620-200-24).
13. Connect tire inflation assembly, to tire valve.
14. Using a regulated source of nitrogen, Item 199, WP 1803 00, inflate tire with enough pressure to seat tire beads.

WARNING

Failure to use proper procedures during servicing of main or tail landing gear tires may result in serious injury or death.

15. **INFLATE TIRE TO 130 TO 140 PSI PRESSURE** with nitrogen, Item 199, WP 1803 00. Use a regulated nitrogen pressure source connected to tire inflator assembly. Use primary or alternate methods of inflation (WP 1598 00).
16. After a 5 to 10 minute waiting period, check wheel for leakage as follows:
- a. Moisten air valve, around tire beads, and bolts and nuts.
 - b. If bubbles appear, deflate, disassemble, and correct at point of leakage.

NOTE

Tire will be inflated to its rated inflation pressure after wheel is installed on helicopter.

17. Reduce tire pressure to 20 psi. Tag wheel, stating psi in tire.
18. Install valve cap on valve stem.
19. Remove wheel from safety cage.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****MAIN LANDING GEAR WHEEL BRAKE****This WP supersedes WP 0451 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 1-Quart
Fluorescent Penetrant Inspection Kit, XMA101
Micrometer, 0 - 1 inch
Plastic or Wooden Pry Bar
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Paint, Heat Resisting, Item 207, WP 1803 00
Protective Caps and Plugs
Tape, Pressure Sensitive, Adhesive, Item 324,
WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Antiseize Compound, Item 51, WP 1803 00
Brush, Paint, Item 61, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 81, WP 1803 00
Corrosion Removing Compound, Item 108,
WP 1803 00
Hydraulic Fluid, Petroleum Base, Item 143,
WP 1803 00

References

TM 1-1520-265-23
WP 0070 00
WP 0440 00
WP 0447 00
WP 0457 00
WP 1803 00

REMOVE

1. Remove wheel and tire (WP 0447 00).
2. Release parking brake handle.
3. Disconnect hose at quick-disconnect coupling on drag beam (Figure 1).
4. Disconnect tube from inlet port of brake and drag beam. Cap and plug ports. Use 1 quart container to catch fluid.
5. Remove bolts, washers, and nuts attaching brake to drag beam. Carefully slide brake off axle.
6. If brake is being replaced, remove elbow, nut, retainer and packing from inlet port (Figure 1, Detail A). Discard packing.
7. Remove shield from axle and keep shield for reuse.

DISASSEMBLE

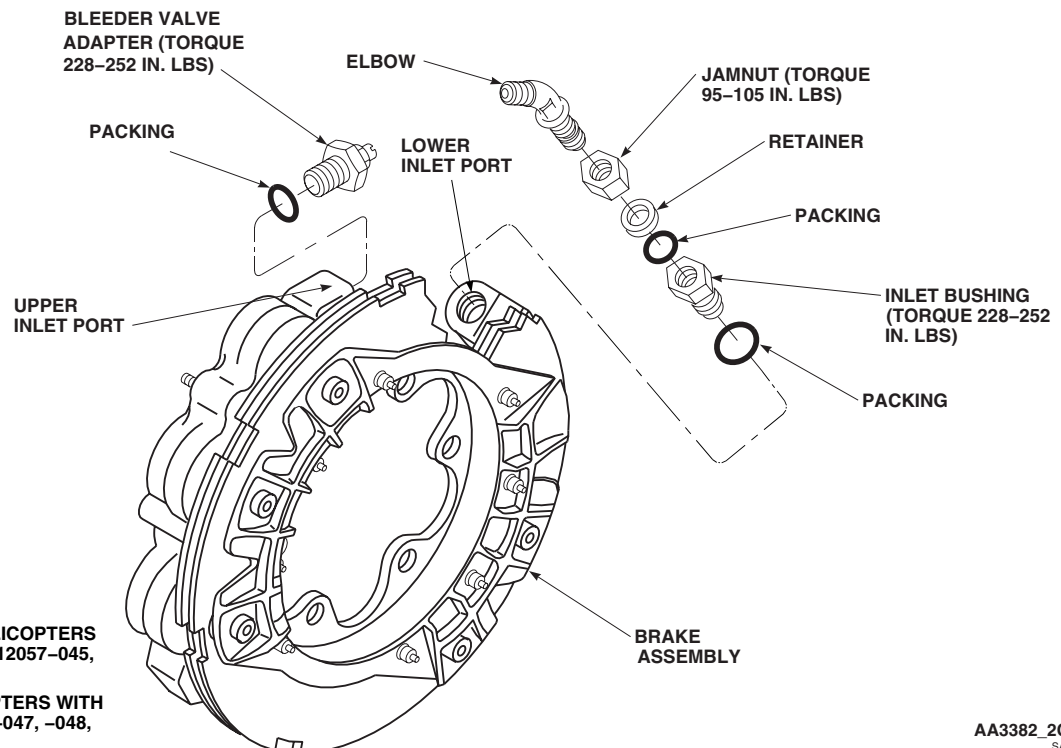
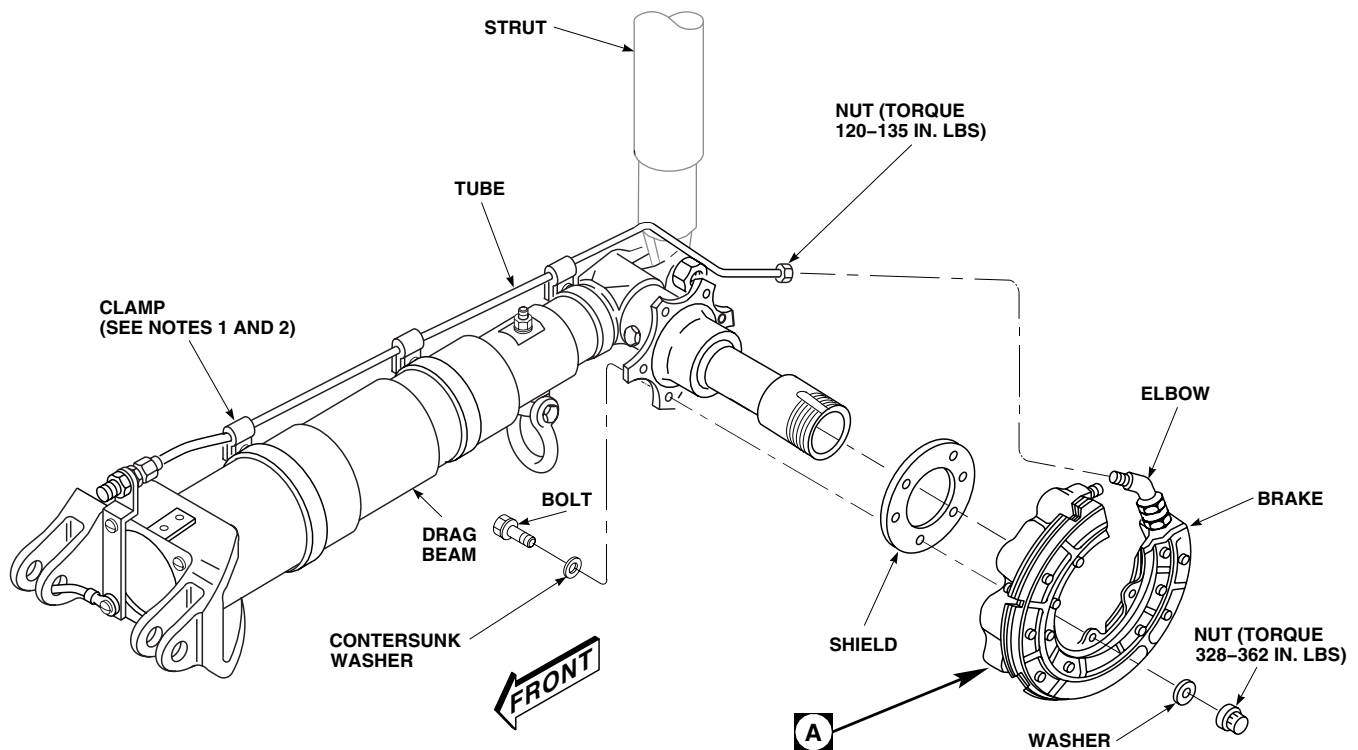
1. Place brake on a clean, flat surface with housing down.

CAUTION

Damage to brake will result if impact or power wrenches are used to remove brake bolts or nuts. Do not use impact or power wrenches to remove brake bolts or nuts.

2. Remove eight nuts holding brake assembly together. Do not remove bolts now (Figure 2).

DISASSEMBLE - Continued



NOTES

1. ONE CENTER CLAMP: HELICOPTERS WITH DRAG BEAM 70250-12057-045, -046.
2. THREE CLAMPS: HELICOPTERS WITH DRAG BEAM 70250-12057-047, -048, -049.

AA3382_2C
SA

Figure 1. Replace Main Landing Gear Wheel Brake.

DISASSEMBLE - Continued

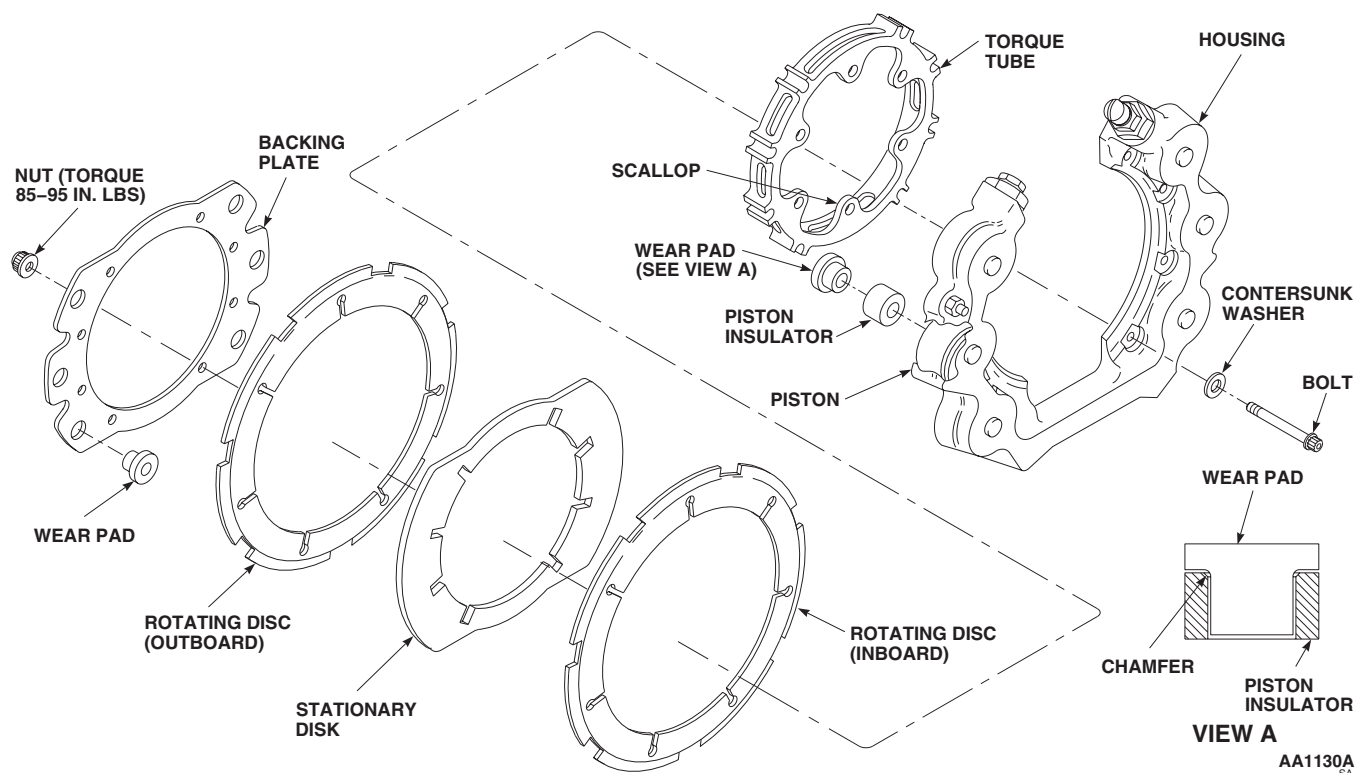


Figure 2. Main Landing Gear Wheel Brake Parts Replacement.

3. Remove backing plate, with six wear pads attached.
4. Remove outboard rotating disc.
5. Remove stationary disc.
6. Remove inboard rotating disc.
7. Remove torque tube from housing.
8. Remove wear pads and piston insulators.
9. Remove bolts.

CLEAN

CAUTION

Damage to discs will result if discs come into contact with cleaning solvent. Do not clean discs with solvent.

1. Clean rotating and stationary discs with compressed air (if available) and paint brush, Item 61, WP 1803 00.
2. Clean backing plate, torque tube, and housing with cleaning compound solvent, Item 71, WP 1803 00, and paint brush, Item 61, WP 1803 00. Dry parts with compressed air or by wiping off with machinery wiping towel, Item 344, WP 1803 00.

INSPECT

1. Check backing plate for obvious damage and warpage. **IF WARPAGE EXCEEDS 0.015-INCH OR BACKING PLATE IS DAMAGED, REPLACE PLATE.** Install new wear pads.
2. Check stationary disc for cracks, especially at key slot corners. **NO CRACKS ALLOWED.** Check disc for thickness. **REPLACE DISC IF WORN TO 0.140-INCH OR LESS ON OUTER EDGE.**
3. Check stationary disc key slots for wear and battering. **REPLACE DISC IF WIDTH OR ANY KEY SLOT IS LESS THAN 0.515-INCH OR LARGER THAN 0.560-INCH.**
4. Check stationary disc for shrinkage and distortion. **REPLACE DISC IF DISHED 0.015-INCHES OR MORE; IF 1/8-INCH OR MORE OUT-OF-ROUND; OR IF SHRUNK TO AN ID OF 6.32-INCHES OR LESS, MEASURED BETWEEN ENDS OF KEY SLOT TANGS.**
5. Check rotating discs for wear and chipping. **REPLACE DISC IF WORN TO 1/8-INCH OR LESS. REPLACE DISC IF UNCHIPPED MIX IS WORN TO 0.010-INCH THICK AT ANY POINT ON EITHER FACE OF DISC. AS MUCH AS 20% OF MIX MAY BE LOST THROUGH CHIPPING IF MIX LOSS IS SCATTERED. NO MORE THAN 25% LOSS OF MIX IS ALLOWED IN ANY SEGMENT.**
6. Check rotating discs for shrinkage and distortion. **REPLACE DISCS THAT ARE DISHED MORE THAN 0.015-INCH, ARE 1/8-INCH OUT-OF-ROUND, OR HAVE SHRUNK TO AN ID OF 6.940-INCHES OR LESS AT ANY POINT.**
7. Check rotating discs for corrosion at relief slot terminal holes. Cover face of discs with pressure sensitive adhesive tape, Item 324, WP 1803 00, and remove corrosion using abrasive cloth, Item 81, WP 1803 00 or abrasive wheel, Item 350, WP 1803 00. **CORROSION REMOVAL SHALL NOT EXCEED 0.015-INCH FOR RELIEF SLOTS AND 0.281-INCH FOR TERMINAL HOLES.** Clean area with corrosion removing compound, Item 108, WP 1803 00. Water rinse discs and dry. Paint repair area with heat resisting paint, Item 207, WP 1803 00.
8. Check rotating disc drive slots for wear and battering. **REPLACE DISC IF ONE OR MORE TANGS ARE BENT, OR IF ANY SLOT IS 0.780-INCH OR MORE IN WIDTH.**
9. Check rotating discs for cracks, especially at relief slot terminal holes. **NO CRACKS ALLOWED.** Replace any disc found cracked. **NO CRACKS ALLOWED.**
10. Make a prying tool out of phenolic, plastic, wood, or other similar material.
11. Mark location of each wear pad as it is removed, so it can be put back in same spot.
12. Remove six pads and piston insulators on pistons.
13. Using prying tool remove wear pads on piston insulators.
14. Inspect piston insulators for deterioration and wear. **REPLACE PISTON INSULATORS WORN TO 0.635-INCH OR LESS IN LENGTH, OR WITH BLISTERS AND RAISED AREAS, OR WITH END FACES NOT PARALLEL. BLISTERS LESS THAN 0.010-INCH CAN BE FILED DOWN PROVIDING 0.635-INCH LENGTH IS MAINTAINED.** Replace any cracked or gouged piston insulator.

CAUTION

Damage to brake will result if new wear pads are mixed with used pads. If any wear pad on piston or on backing plate is worn beyond limits, replace that set of six wear pads.

15. Check thickness of backplate wear pads. **REPLACE ALL SIX IF ONE OR MORE ARE WORN TO 0.380-INCH THICK OR LESS.**
16. Check thickness of piston wear pads. **REPLACE ALL SIX IF ANY ONE PAD IS WORN TO 0.380-INCH THICK OR LESS.**

INSPECT - Continued

17. Check brake bolts specifically for cracks at radius under bolthead and within threads next to bolt shank using fluorescent penetrant inspection kit (TM 1-1520-265-23). Replace cracked brake bolts.
18. Check torque tube for cracks. Replace cracked tube.
19. Check width of torque tube key. **REPLACE TUBE IF KEY WIDTH IS 0.470-INCH OR LESS AT ANY POINT ALONG LENGTH OF ANY KEY.**

ASSEMBLE

1. Put a countersunk washer under each bolthead (countersunk side faces bolthead) and insert bolt through housing (Figure 2).
2. Coat threads of brake bolts, nuts, and bearing surfaces of boltheads, washers and self locking nuts with antiseize compound, Item 49, WP 1803 00.
3. Place brake on a clean, flat surface with housing down.
4. Insert wear pad into chamfered end of piston insulation (Figure 2, View A).
5. Install six piston insulators and six wear pads in housing (Figure 2, View A).
6. **QA** Place torque tube over bolts. Make sure torque tube scallops face housing.
7. **QA** Install inboard rotating disc over torque tube. Add stationary disc and outboard rotating disc on top of tube.
8. **QA** Install backing plate, flat face down, against inboard rotating disc, with bolts sticking through backing plate.
9. Coat threads of nuts with antiseize compound, Item 49, WP 1803 00, and install bolts. Tighten each nut a little at a time until snug. **TORQUE NUTS TO 85 - 95 INCH-POUNDS.**

INSTALL**NOTE**

- Brakes are received in left configuration with bleeder valve adapter, bleeder valve, and bleeder screw in lower inlet port. If replacing right brake assembly, bleeder valve adapter, bleeder valve, and bleeder screw must be moved to upper inlet port.
- Refer to WP 0440 00 for installing brake flange spacers.

1. **QA** If replacing right brake with new brake, do this:
 - a. Remove bleeder valve as an assembly from lower inlet port of brake (Figure 1). Remove packing from bleeder valve adapter. Discard packing.
 - b. Remove inlet bushing from upper inlet port of brake. Remove packing from bushing. Discard packing.
 - c. **QA** Lubricate new packing, with petroleum base hydraulic fluid, Item 143, WP 1803 00. Install packing on bleeder valve adapter (Figure 1).
 - d. **QA** Install bleeder valve as an assembly in upper inlet port of brake. **TORQUE BLEEDER VALVE ADAPTER TO 228 - 252 INCH-POUNDS.**
 - e. **QA** Lubricate new packing, with petroleum base hydraulic fluid, Item 143, WP 1803 00. Install packing on inlet bushing.
 - f. **QA** Install inlet bushing in lower inlet port of brake. **TORQUE INLET BUSHING TO 228 - 252 INCH-POUNDS.**
 - g. Lubricate packing, retainer, and male threads of elbow with petroleum base hydraulic fluid, Item 143, WP 1803 00.

INSTALL - Continued

- h. **QA** Install jamnut, retainer, and packing, onto elbow. Turn nut and retainer down until packing is pushed firmly against the lower threaded section of elbow.
- i. **QA** Install elbow in inlet bushing until a sharp rise in torque is felt while keeping jamnut from turning.
TIGHTEN ELBOW AND JAMNUT 1-1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN ELBOW NO MORE THAN 1 ADDITIONAL TURN TO POSITION ELBOW FACING BRAKE LINE.
- j. **PREVENT ELBOW FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
- 2. **QA** Position shield over axle and against drag beam flange (Figure 1).
- 3. **QA** Place brake on drag beam flange. Line up brake so tube can be connected and shield is properly positioned.
- 4. **QA** Make sure countersunk side of washers face bolt head. Install bolts, washers, and nuts, making sure boltheads are on inner side of brake.
- 5. **TORQUE NUTS IN A CRISSCROSS MANNER, IN 100 INCH-POUND STEPS, TO 328 - 362 INCH-POUNDS.**
- 6. **QA** Install tube on elbow. **TORQUE NUT TO 120 - 135 INCH-POUNDS.**
- 7. **QA** Connect hose at quick-disconnect coupling in drag beam.

WARNING

The 7 lug wheel (5003889) may experience lug failure when operating at or above gross weights of 20,500 pounds. Lug failure may result in flying debris during ground handling and/or unexpected tire failure. The 14 lug wheel (040-25300) should be utilized when operating at or above gross weights of 20,500 pounds.

- 8. Install wheel and tire (WP 0447 00).
- 9. Bleed appropriate pilot's brake (WP 0457 00).
- 10. Do an operational check of brakes (WP 0070 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****MASTER BRAKE CYLINDER**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Protective Caps and Plugs
Towel, Machinery Wiping, Item 344, WP 1803 00

References

TM 55-1500-345-23
WP 0070 00
WP 0381 00
WP 0457 00
WP 1043 00
WP 1803 00

REMOVE

1. Chock helicopter wheels. Release parking brake handle.
2. Using release lever, recline pilot's/copilot's seat.
3. Remove pilot's/copilot's yaw control pedal boots (WP 1043 00).
4. Remove screws and washers from pilot's/copilot's floor panels around pedals. Remove panels.
5. Disconnect hose from elbow on master cylinder. Use machinery wiping towel, Item 344, WP 1803 00, to catch any fluid that might leak or drip. Plug hose with protective plug.
6. Remove bolt, washer, and nut. Remove master cylinder from pedal assembly and support assembly (Figure 1).
7. If master cylinder is being replaced, remove swivel fitting and packing from master cylinder.

INSTALL

1. **QA** If removed, lubricate packing, with petroleum base hydraulic fluid, Item 145, WP 1803 00. Install packing on swivel fitting (Figure 1).
2. **QA** If removed, install swivel fitting in lower part of master cylinder. **TORQUE NUT TO 135 - 145 INCH-POUNDS.**
3. Fully extend master cylinder piston. **MEASURE MASTER CYLINDER ATTACHMENT HOLES, CENTER-TO-CENTER, FOR 15.36-INCHES.** Adjust rod end on piston as required to obtain proper dimension.

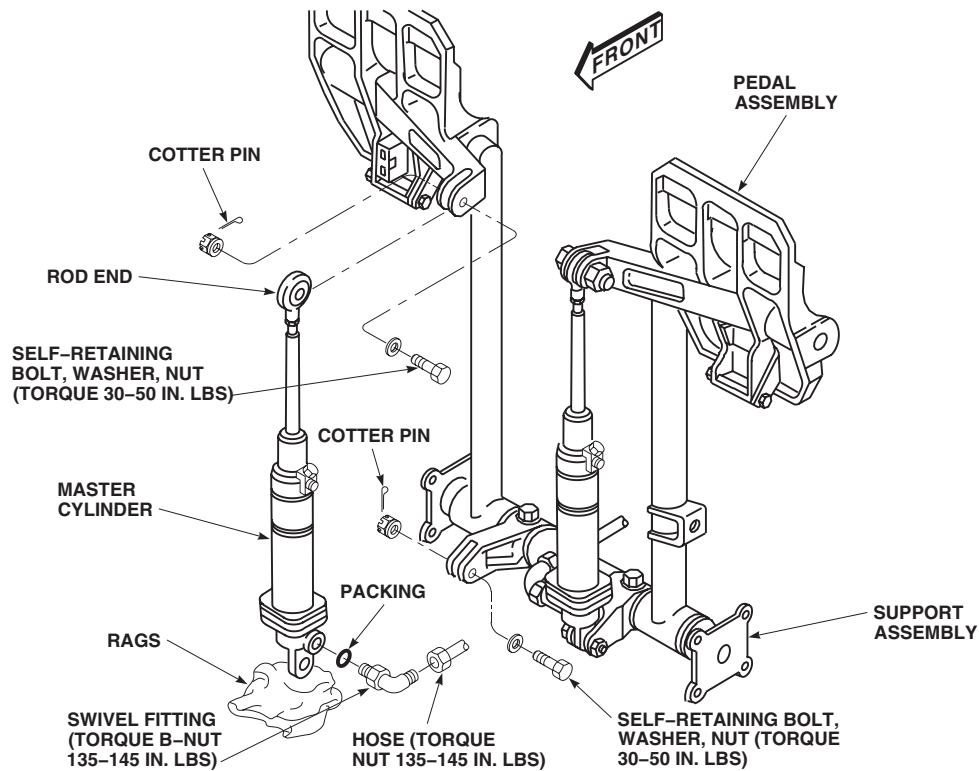
INSTALL - ContinuedAK0649
SA

Figure 1. Replace Master Brake Cylinder.

WARNING

Injury to personnel and damage to equipment may occur when using improper hardware and torque values. Verification of proper hardware installation, torque, safety installation, and proof torque of self-retaining bolts are critical characteristics.

NOTE

Installation of self-retaining bolts securing upper and lower ends of master cylinder is the same except as noted.

4. Position master cylinder on pedal assembly and support assembly and install self-retaining bolts as follows:
 - a. Select washer combination per Table 1 to provide correct grip length and cotter pin position. Install washer combination under bolthead.
 - b. **QA** Install self-retaining bolt and washer combination.
 - c. Check bolt for free axial motion and that impedance feature is not restrained in part. If bolt binds or impedance feature is depressed, change washer stackup using Table 1.

Table 1. Washer Stackup.

COMBINATION	QUANTITY	WASHER THICKNESS (Inch)
A	2	0.016

INSTALL - Continued**Table 1. Washer Stackup. - Continued**

COMBINATION	QUANTITY	WASHER THICKNESS (Inch)
B	3	0.016
C	1	0.063
D	1	0.016
	1	0.063
E	2	0.016
	1	0.063

- d. **QA** Install nut on bolt. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.

NOTE

If unable to proof torque bolthead, use a crowfoot attachment to torque nut, without disturbing cotter pin and without restraining bolthead.

- e. **PROOF TORQUE BOLTHEAD TO 15 - 17 INCH-POUNDS.** If bolt rotates, adjust using washer combinations in Table 1 or use next longer or shorter bolt. Apply torque stripe (WP 0381 00 and TM 55-1500-345-23).
5. Unplug and connect hose to swivel fitting. **TORQUE NUT TO 135 - 145 INCH-POUNDS.**
 6. Bleed the appropriate master cylinder (WP 0457 00).
 7. Make sure area is clean and free of foreign material before closing up.
 8. Install pilot's/copilot's yaw control pedal boots (WP 1043 00).
 9. **QA** Install pilot/copilot floor panel around pedals, with screws and washers.
 10. **QA** Return pilot's/copilot's seat to upright position.
 11. Do an operational check of brake system (WP 0070 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR**

MAIN LANDING GEAR WHEEL BRAKE LINES AND HOSES

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Container, 1-Quart
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0070 00
WP 0457 00
WP 1803 00

Material/Parts

Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Protective Caps and Plugs

NOTE

- The following procedures apply to all brake lines and hoses, except as noted.
- **UH60A 77-22714 - 81-23546** At STA 267.0, on left side only, a swivel fitting connects brake tube and brake hose. An elbow is installed on right side. ■
- **UH60A 81-23547 - SUBQ EH-60A UH-60L** An elbow is installed on both sides. ■

REMOVE

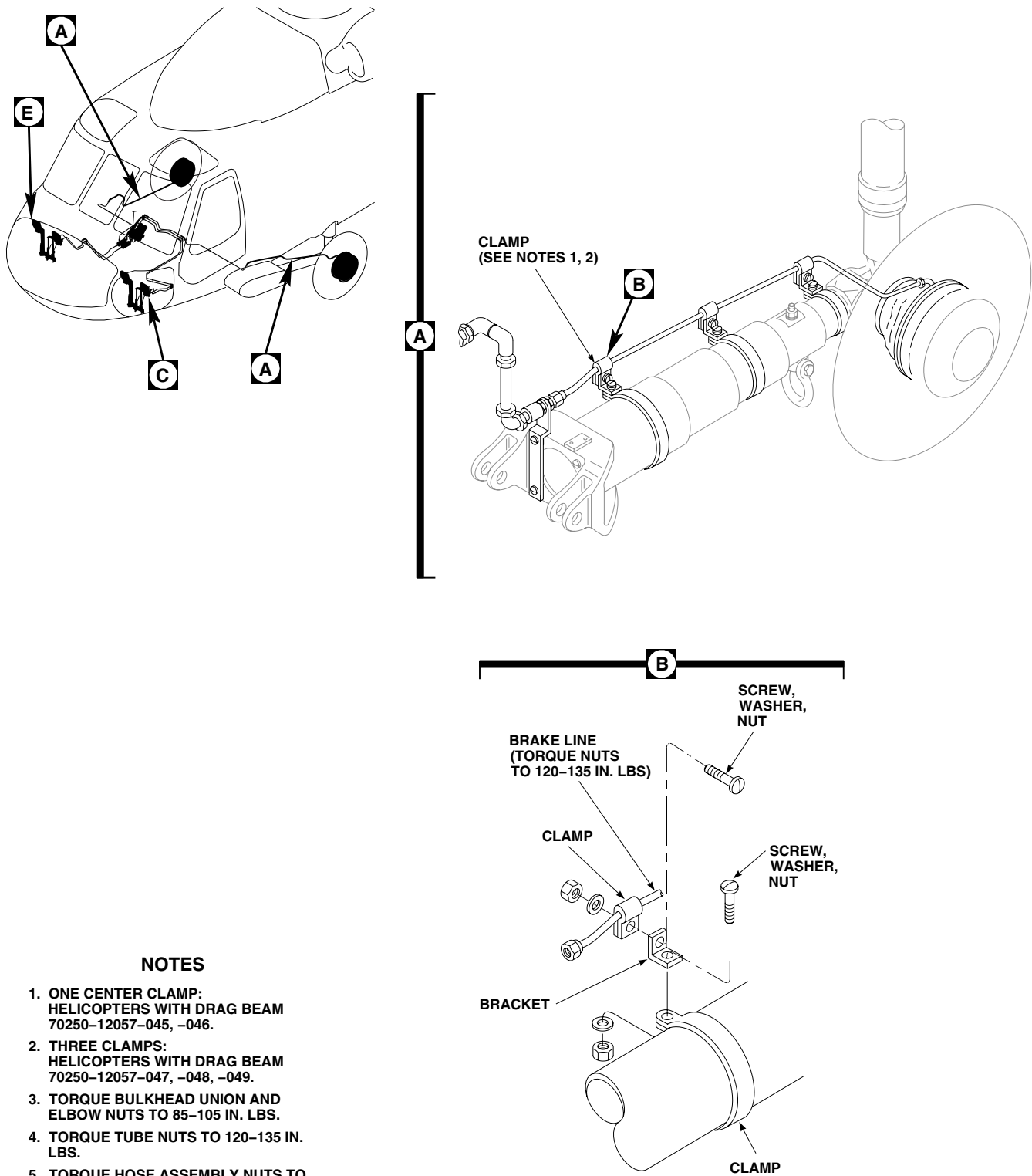
1. Chock helicopter wheels. Release parking brake handle.
2. Place a 1 quart container under nut at one end of line or hose being replaced and disconnect line or hose. Drain lines or hoses.
3. Disconnect hose or line from union or elbow and remove from helicopter. Cap union or elbow (Figure 1, Sheet 1, Detail B, Figure 1, Sheet 2, Detail C, and Figure 1, Sheet 3, Detail E).
4. Remove nut, washer, and unions or elbows.

INSTALL**NOTE**

Make sure line is clear and clean.

1. **QA** Install bulkhead unions or elbows using washer and nut. **TORQUE NUT TO 85 - 105 INCH-POUNDS** (Figure 1, Sheet 2, Detail C, and Figure 1, Sheet 3, Detail E).
2. **QA** Install packings lubricated with petroleum base hydraulic fluid, Item 145, WP 1803 00, on parking brake valve and slave mixer valve unions. Install unions to valves. **TORQUE UNIONS TO 95 - 105 INCH-POUNDS.**
3. Uncap and lubricate union or elbow threads with petroleum base hydraulic fluid, Item 145, WP 1803 00.
4. **QA** Install tubes on fittings. **TORQUE NUTS TO 120 - 135 INCH-POUNDS.**

INSTALL - Continued



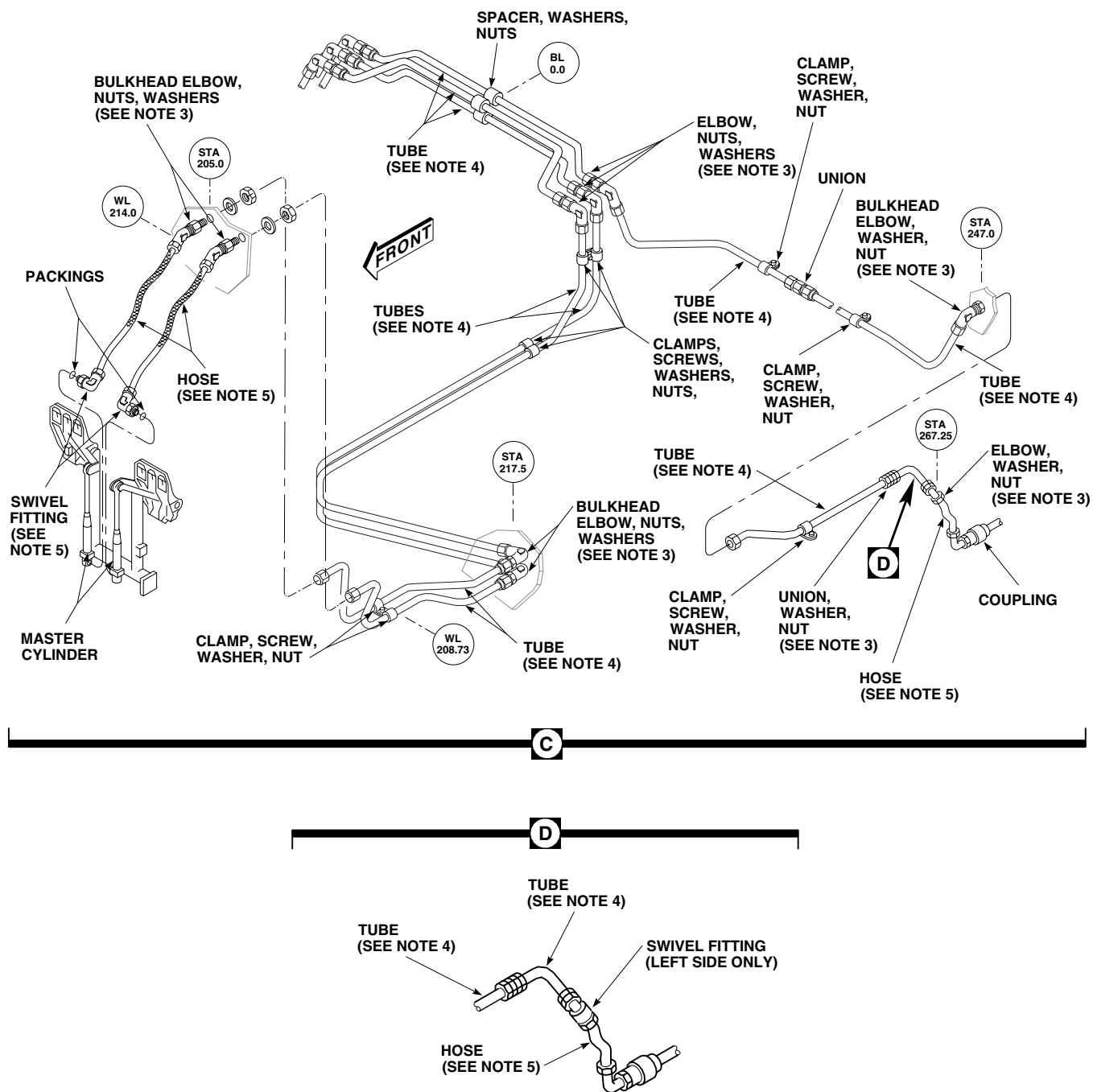
NOTES

1. ONE CENTER CLAMP:
HELICOPTERS WITH DRAG BEAM
70250-12057-045, -046.
2. THREE CLAMPS:
HELICOPTERS WITH DRAG BEAM
70250-12057-047, -048, -049.
3. TORQUE BULKHEAD UNION AND
ELBOW NUTS TO 85-105 IN. LBS.
4. TORQUE TUBE NUTS TO 120-135 IN.
LBS.
5. TORQUE HOSE ASSEMBLY NUTS TO
135-145 IN. LBS.
6. TORQUE UNIONS TO 95-105 IN. LBS.
7. ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

AA2218_1B
SA

Figure 1. Replace Main Landing Gear Wheel Brake Lines and Hoses. (Sheet 1 of 3)

INSTALL - Continued

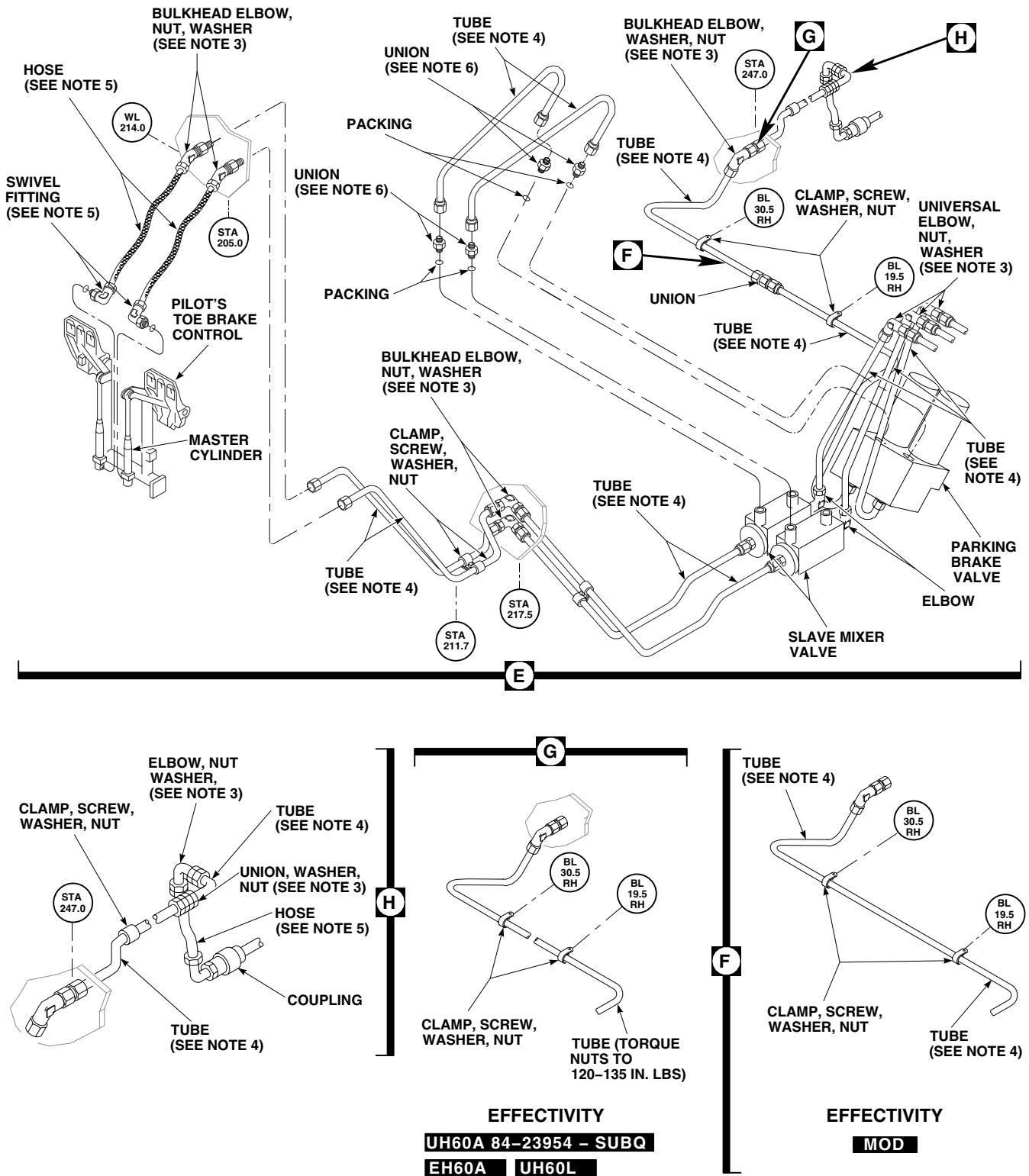


EFFECTIVITY

UH60A 77-22714 - 81-23547

Figure 1. Replace Main Landing Gear Wheel Brake Lines and Hoses. (Sheet 2 of 3)

INSTALL - Continued



AA2218_3A
SA

Figure 1. Replace Main Landing Gear Wheel Brake Lines and Hoses. (Sheet 3 of 3)

INSTALL - Continued

5. **QA** Install hose assemblies. **TORQUE NUTS TO 135 - 145 INCH-POUNDS.**
6. Bleed brakes (WP 0457 00).
7. Do an operational check of brakes (WP 0070 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****PARKING BRAKE VALVE**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Electrician MOS 15F (One)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Protective Plugs and Caps

References

TM 11-1520-237-23
WP 0070 00
WP 0457 00
WP 1803 00

REMOVE

1. Chock helicopter wheels. Release parking brake handle.
2. Turn off all electrical power.
3. Using release handle, recline pilot's seat.
4. Remove fabric cover from under pilots seat.
5. Disconnect and remove radios and radio shelf in seat well (TM 11-1520-237-23).
6. Unpin rod from parking valve lever (Figure 1).
7. Place machinery wiping towel, Item 344, WP 1803 00, beneath parking valve.
8. Disconnect tubes from inlet and outlet ports. Install protective plugs in tubes, cap ports.
9. Remove bolt, washer, and nut. Remove parking valve from support brackets.
10. Remove switch from parking valve. Do not disconnect wiring from switch.
11. Remove unions, install protective plugs in inlet ports of parking valve.

INSTALL

1. Turn off all electrical power.
2. Remove protective plugs from inlet ports.
3. **QA** Install unions and packings lubricated with petroleum base hydraulic fluid, Item 145, WP 1803 00, in inlet ports (Figure 1). **TORQUE UNIONS TO 95 - 105 INCH-POUNDS.**
4. **QA** Install switch on parking valve with screws.
5. **QA** Install valve to support brackets using bolts, washers, and nuts. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
6. Uncap outlet port fittings.

INSTALL - Continued

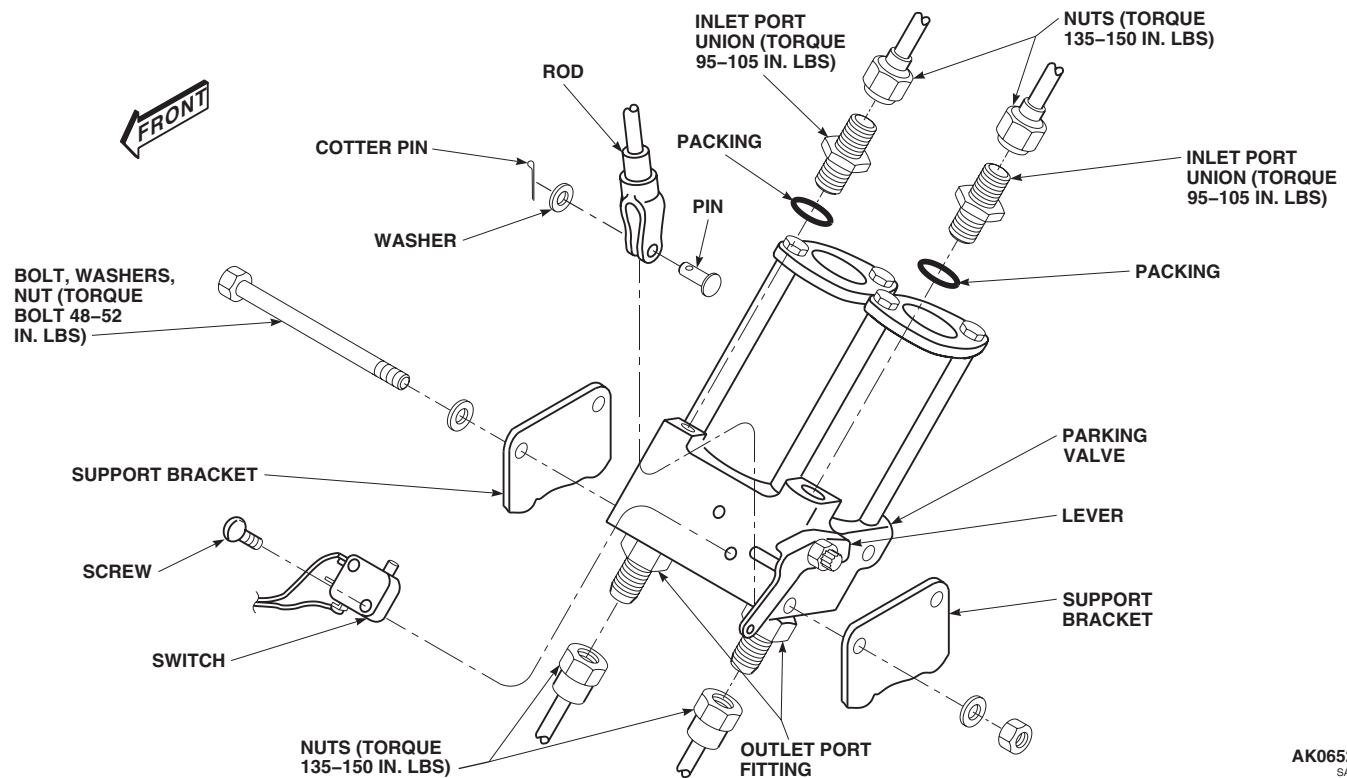
AK0652
SA

Figure 1. Replace Parking Brake Valve.

7. Unplug and connect tubes to inlet unions and outlet fittings. **WHILE HOLDING UNIONS, TORQUE NUTS TO 135 - 150 INCH-POUNDS.**
8. **QA** Pin rod to parking valve lever. Install washer and cotter pin.
9. Bleed pilot's brakes (WP 0457 00).
10. Do operational check of parking brake system (WP 0070 00).
11. Make sure area is clean and free of all foreign matter.
12. Install radio shelf and radios in seat well (TM 11-1520-237-23).
13. Install fabric cover on cockpit floor compartment.
14. **QA** Move pilot's seat upright.
15. Set parking brake.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****PARKING BRAKE VALVE SWITCH**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Tags

References

WP 0070 00

REMOVE

1. Chock helicopter wheels. Release parking brake.
2. Turn off all electrical power.
3. Using release handle, recline pilot's seat.
4. Remove fabric cover from cockpit floor compartment.
5. Remove screws and remove switch (Figure 1).
6. Disconnect and tag wire from common terminal.
7. Disconnect wire from normally closed terminal.

INSTALL

1. Turn off all electrical power.
2. **QA** Connect untagged wire to normally closed terminal (Figure 1).
3. **QA** Connect tagged wire to common terminal.
4. **QA** Install switch on parking valve with screws.
5. Make sure area is clean and free of all foreign matter.
6. Install fabric cover on cockpit floor compartment.
7. **QA** Move pilot's seat upright.
8. Do operational check of parking brake system (WP 0070 00).
9. Put on parking brake.

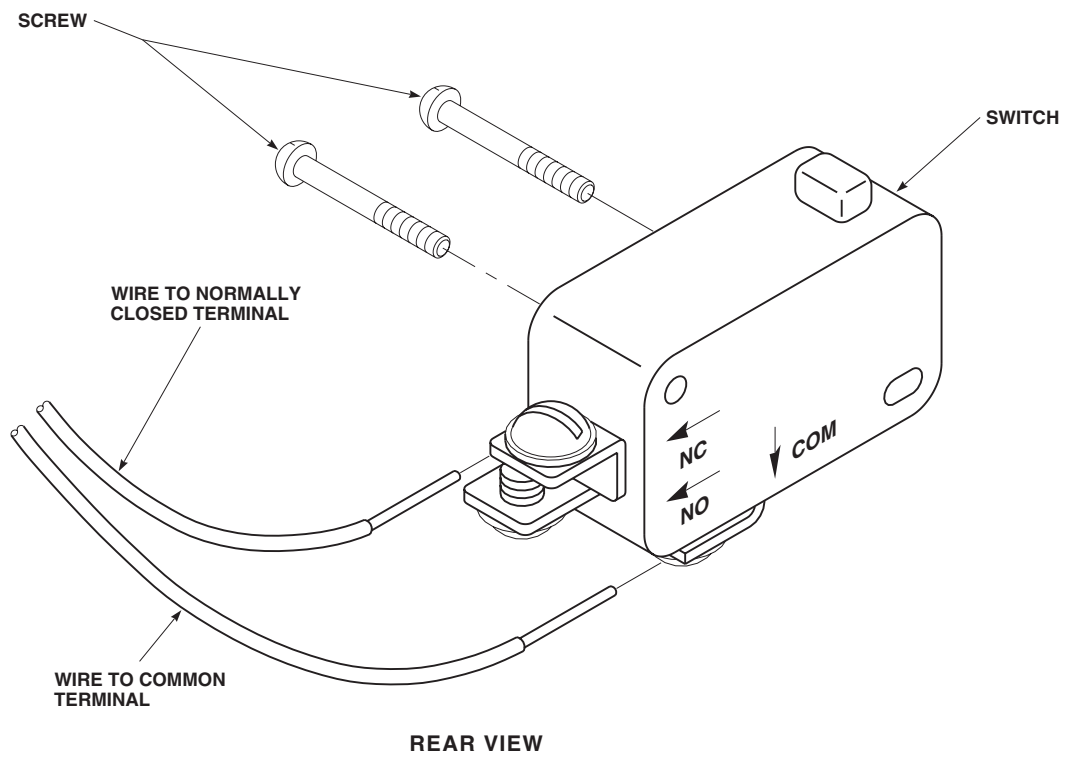
INSTALL - ContinuedAK0653
SA

Figure 1. Replace Parking Brake Valve Switch.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR**

SLAVE MIXER VALVE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Protective Plugs and Caps

References

WP 0070 00
WP 0457 00
WP 1803 00

REMOVE

1. Chock helicopter wheels. Release parking brake handle.
2. Turn off all electrical power.
3. Using release handle, recline pilot's seat.
4. Remove fabric cover from cockpit floor compartment.
5. Place machinery wiping towel, Item 344, WP 1803 00, beneath slave mixer valves.
6. Disconnect tubes from unions and elbow (Figure 1). Plug tubes with protective plugs and cap ports.
7. Disconnect slave mixer valve from support bracket.
8. If valve is being replaced, remove unions, elbow, jamnut, packings, and retainer from slave mixer valve.
9. Plug slave valve ports.

INSTALL

1. Turn off all electrical power.
2. Remove protective plugs or caps.
3. Lubricate packings with petroleum base hydraulic fluid, Item 145, WP 1803 00. Install packings on unions.
4. **QA** Install unions in inlet port 1 and outlet port 3 of slave mixer valve (Figure 1). **TORQUE UNIONS TO 95 - 105 INCH-POUNDS.**
5. Lubricate packing, retainer and male threads of elbow with petroleum base hydraulic fluid, Item 145, WP 1803 00.
6. **QA** Install jamnut, retainer, and packing onto elbow. Turn jamnut and retainer down until packing is pushed firmly against the lower threaded section of elbow.
7. **QA** Install elbow into slave mixer valve until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN ELBOW AND JAMNUT 1-1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN ELBOW NO MORE THAN 1 ADDITIONAL TURN TO POSITION ELBOW FACING UP.**

INSTALL - Continued

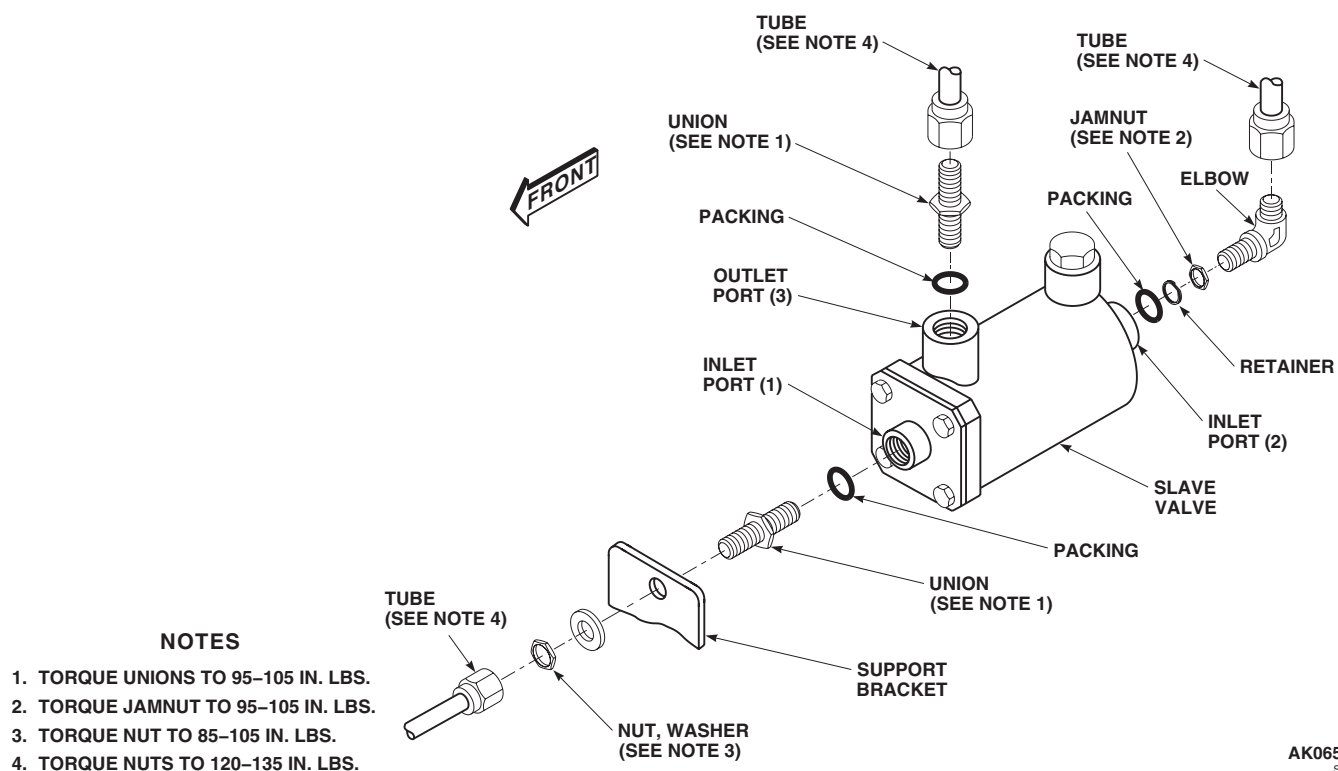


Figure 1. Replace Slave Mixer Valve.

AK0654
SA

8. **PREVENT ELBOW FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
9. **QA** Install slave mixer valve into support bracket with washer and nut. **TORQUE NUT TO 85 - 105 INCH-POUNDS.**
10. **QA** Unplug and connect tubes to unions and elbow. **TORQUE NUTS TO 120 - 135 INCH-POUNDS.**
11. Bleed pilot's and copilot's brakes (WP 0457 00).
12. Make sure area is clean and free of all foreign matter.
13. Install fabric cover on cockpit floor compartment.
14. **QA** Move pilot's seat upright.
15. Do operational check of parking brake system (WP 0070 00).
16. Set parking brake.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR

BLEED MAIN LANDING GEAR WHEEL BRAKE SYSTEM

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Container, 1-Gallon
 Hydraulic Fluid Dispenser, AF5
 Hydraulic Servicing Unit, 630AS100-11
 Hypodermic Syringe, MIL-S-36157
 Squirt Can, GGG-O-591

Tubing, Nonmetallic Tygothane, Item 345,
 WP 1803 00
 Union, AN815-3D
 Wire, Nonelectrical, Item 353, WP 1803 00

Personnel Required

Assistant - (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Hydraulic Fluid, Fire Resistant, Item 144, WP 1803 00
 Hydraulic Fluid, Petroleum Base, Item 145,
 WP 1803 00
 Reducer, AN919-2

References

WP 0070 00
 WP 1043 00
 WP 1803 00

PREPARE TO BLEED**NOTE**

- Alternate procedures for bleeding main landing gear wheel brakes are by using either method A (hydraulic fluid dispenser, BLEED COPILOT'S BRAKES, METHOD A and BLEED PILOT'S BRAKES METHOD A, in this work package) or method B (manual method, BLEED COPILOT'S BRAKES METHOD B and BLEED PILOT'S BRAKES METHOD B, in this work package) with a syringe or squirt can to add hydraulic fluid.
- Under emergency conditions, fire resistant hydraulic fluid, Item 144, WP 1803 00, may be used to service main landing gear wheel brakes. However, reservice main landing gear wheel brakes with petroleum base hydraulic fluid, Item 145, WP 1803 00, as soon as convenient.

1. Chock helicopter wheels. Release parking brake handle.
2. Using release lever, recline pilot's/copilot's seat.
3. Remove pilot's/copilot's yaw control pedal boots (WP 1043 00).
4. Remove screws and washers, remove pilot's/copilot's floor panels around pedals.

BLEED COPILOT'S BRAKES, METHOD A

1. Remove bleeder plug and packing from outer slave mixer valve (Figure 1, Sheet 1).

BLEED COPILOT'S BRAKES, METHOD A - Continued**NOTE**

- Make sure fluid dispenser is full (minimum 4 quarts) to avoid interruption of bleeding process. Interruption may cause air to enter system, and require additional bleeding.
 - Reducer is normally supplied with fluid dispenser. Select a reducer with a 7/16-20-inch thread to fit mixer valve.
2. Install packing and reducer. Connect hydraulic fluid dispenser filled with petroleum base hydraulic fluid, Item 145, WP 1803 00, to reducer.
 3. Remove filler cap from copilot's right master cylinder. Remove packing from cap.
 4. Install packing removed from filler cap on union, AN815-3D. Install union and nonmetallic tygothane tubing, Item 345, WP 1803 00, in the master cylinder filler cap port.
 5. Place open end of tubing in a clean 1 quart container and make sure drain end of bleed tubing is completely submerged in clean hydraulic fluid.

NOTE

During bleeding process, cycle parking brake ON-OFF several times to purge air from parking brake valve accumulators.

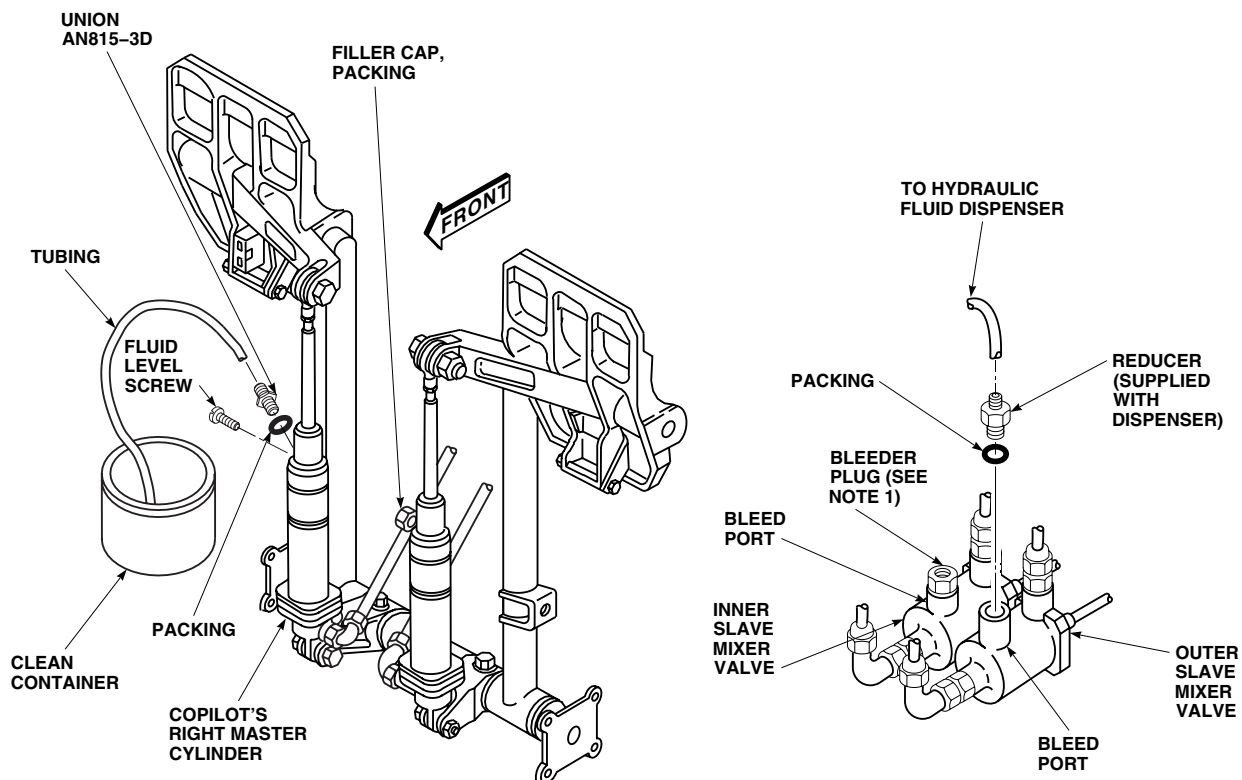
6. While pumping copilot's right master cylinder, have an assistant operate hydraulic fluid dispenser to obtain about 0.2 gpm flow at 10 psi but not more than 15 psi. Stop pressure flow when air-free fluid flows in tubing, and cylinder is in brake-off position.
7. Pinch off overflow line by kinking. Have assistant disconnect hydraulic fluid dispenser from outer slave mixer valve. Remove reducer. Install bleeder plug and packing. **TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using nonelectrical wire, Item 353, WP 1803 00, lock-wire bleeder plug.
8. Remove tubing from master cylinder.
9. Remove fluid level screw from master cylinder. Allow fluid to drain until level is even with bottom edge of screw-hole.
10. **QA** Install fluid level screw. Install filler cap and packing. Using nonelectrical wire, Item 353, WP 1803 00, lock-wire cap and screw together.

NOTE

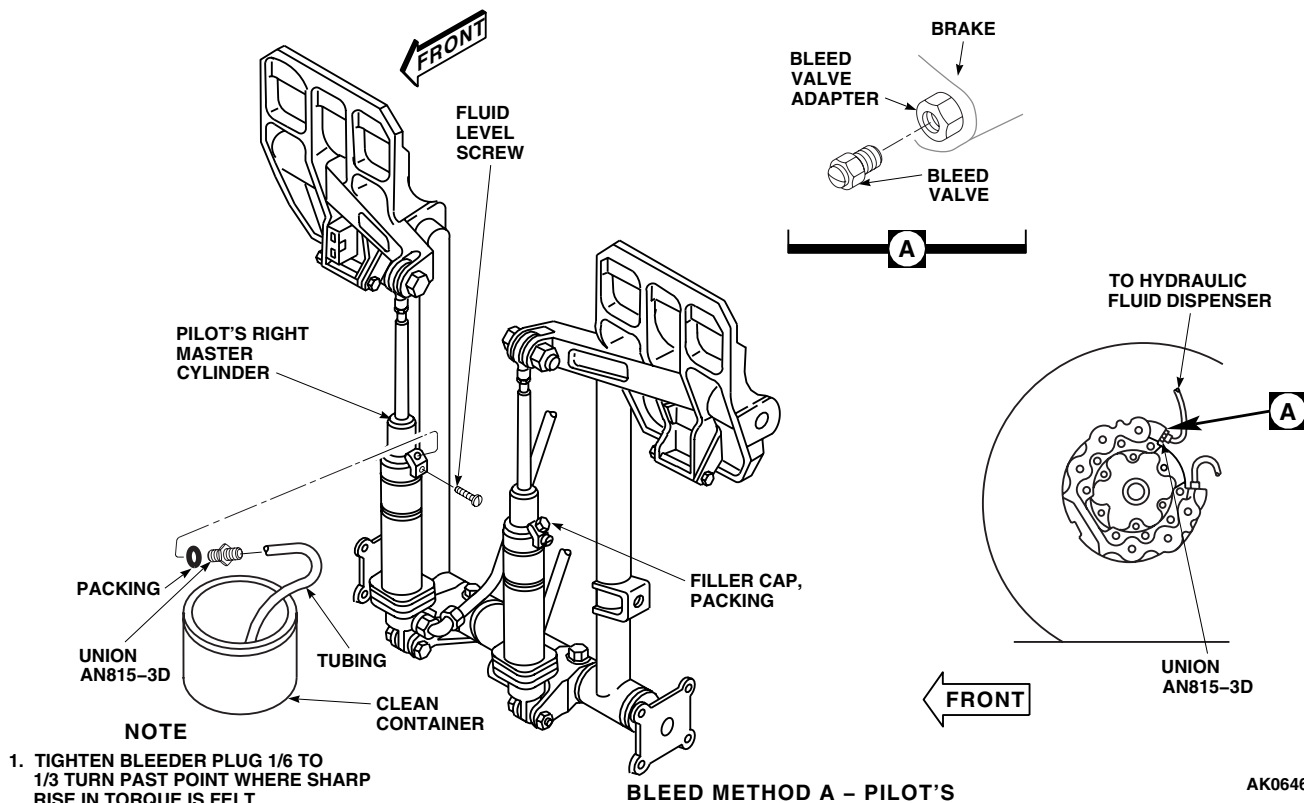
Make sure fluid dispenser is full (minimum 4 quarts) to avoid interruption of bleeding process. Interruption may cause air to enter system, and require additional bleeding.

11. Remove bleeder plug and packing from inner slave mixer valve. Install packing, and reducer. Connect hydraulic fluid dispenser to reducer.
12. Bleed copilot's left master cylinder using same method as with right master cylinder.
13. Make sure area is clean and free of foreign material before closing up.
14. Install pilot's/copilot's yaw control pedal boots (WP 1043 00).
15. Return pilot's/copilot's seat to upright position.
16. Do an operational check of brakes (WP 0070 00).

BLEED COPILOT'S BRAKES, METHOD A - Continued



BLEED METHOD A - COPILOT'S

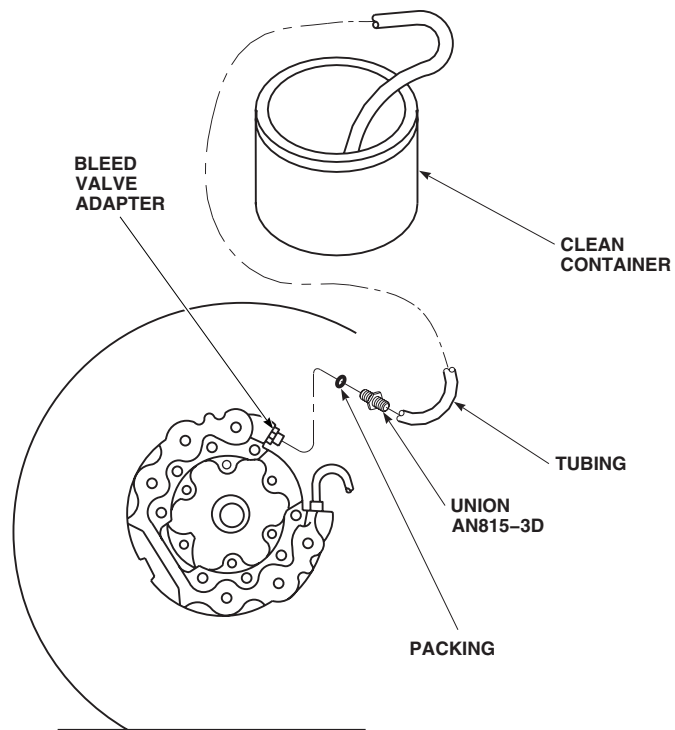


BLEED METHOD A - PILOT'S

AK0646_1
SA

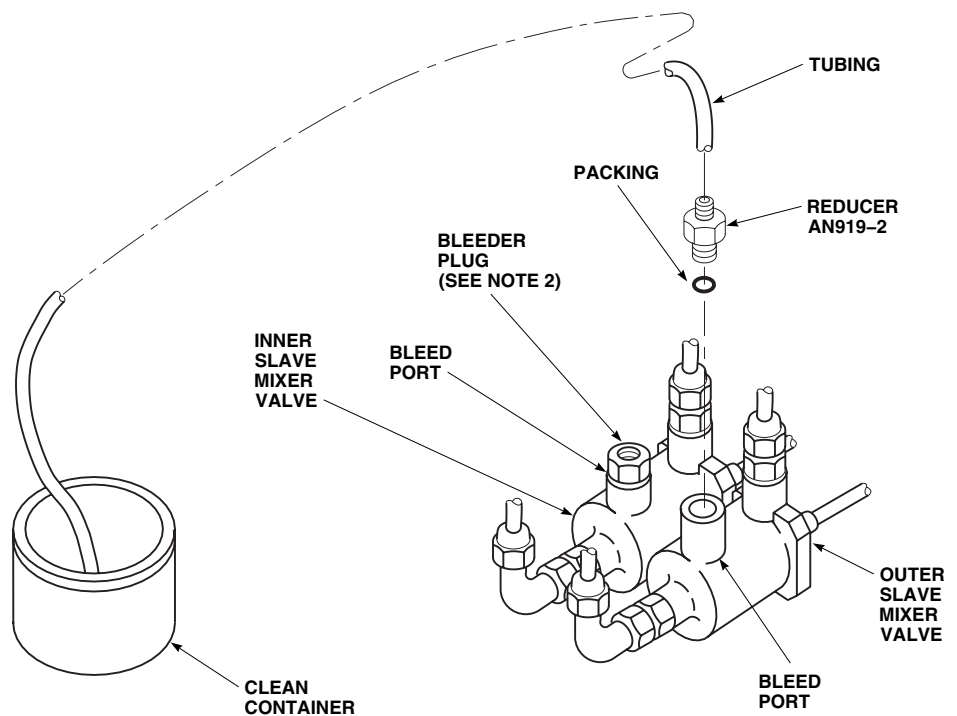
Figure 1. Bleed Main Landing Gear Wheel Brake System. (Sheet 1 of 2)

BLEED COPILOT'S BRAKES, METHOD A - Continued



BLEED METHOD B - PILOT'S

- NOTE**
2. TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.



BLEED METHOD B - COPILOT'S

AK0646_2
SA

Figure 1. Bleed Main Landing Gear Wheel Brake System. (Sheet 2 of 2)

BLEED PILOT'S BRAKES, METHOD A

1. Remove bleed valve from bleed valve adapter on right wheel brake. Install union, AN815-3D, in valve adapter (Figure 1, Sheet 1, Detail A).

NOTE

Make sure fluid dispenser is full (minimum 4 quarts) to avoid interruption of bleeding process. Interruption may cause air to enter system, and require additional bleeding.

2. Connect hydraulic servicing unit to union.
3. Disconnect hose at drag beam quick-disconnect coupling.
4. To remove air, bleed the brake and line using the hydraulic servicing unit.
5. Connect hose at drag beam quick disconnect coupling and proceed to bleed system.
6. Remove filler cap on pilot's right master cylinder. Remove packing from cap.
7. Install packing removed from filler cap and install on union, AN815-3D. Install union and nonmetallic tygothane tubing, Item 345, WP 1803 00, in the open port.
8. Place open end of tubing in a clean 1 quart container and make sure drain end of bleed tubing is completely submerged in hydraulic fluid.

NOTE

During bleeding process, cycle parking brake ON-OFF several times to purge air from parking brake valve accumulators.

9. Pump hydraulic servicing unit, keeping a constant pressure from dispenser of 0.2 gpm at 15 to 25 psi, but not over 30 psi, until air-free fluid flows in tubing, with cylinder in brake-off position. While pumping fluid dispenser, have an assistant push pilot's right brake pedal to allow pressure to build up, then slowly release pedal while holding fluid dispenser pressure. Continue this until fluid flows air-free in tubing.
10. Disconnect hydraulic fluid dispenser from union installed in bleed valve adapter on brake.
11. **QA** Install bleed valve in bleed valve adapter. Using nonelectrical wire, Item 353, WP 1803 00, lockwire screw in bleed valve to brake.
12. Remove tubing from master cylinder.
13. Remove fluid level screw from master cylinder. Allow fluid to drain until level is even with bottom edge of screw-hole.
14. **QA** Install fluid level screw. Install filler cap and packing. Using nonelectrical wire, Item 353, WP 1803 00, lockwire cap and screw together.
15. Bleed pilot's left master cylinder using same method as with right master cylinder.

NOTE

Hydraulic servicing unit is to be connected to left wheel brake.

16. Disconnect hose at drag beam quick-disconnect coupling.
17. To remove air, bleed the brake and line using the hydraulic servicing unit.
18. **QA** Connect hose at drag beam quick-disconnect coupling and proceed to bleed system.
19. Make sure area is clean and free of foreign materials before closing up.

BLEED PILOT'S BRAKES, METHOD A - Continued

20. Install pilot's/copilot's yaw control pedal boots (WP 1043 00).
21. **QA** Return pilot's/copilot's seat to upright position.
22. Do an operational check of brakes (WP 0070 00).

BLEED COPILOT'S BRAKES, METHOD B

1. Remove bleeder plug and packing from outer slave mixer valve (Figure 1, Sheet 2). Discard packing.
2. Install packing, and reducer, AN919-2. Install nonmetallic tygothane tubing, Item 345, WP 1803 00, and place open end in clean 1 gallon container.
3. Remove filler cap from copilot's right master cylinder.
4. Fill master cylinder to top with petroleum base hydraulic fluid, Item 145, WP 1803 00, using syringe or squirt can.
5. Install filler cap and packing.
6. Press right brake pedal until master cylinder piston bottoms out. Maintain pedal pressure.
7. Remove filler cap and fill master cylinder to top with petroleum base hydraulic fluid, Item 145, WP 1803 00.
8. Install filler cap and packing.
9. Release right brake pedal pressure.
10. Repeat step 3. through step 9. until no air bubbles appear in tubing.
11. After pedal pressure is released, remove filler cap and fill right master cylinder with petroleum base hydraulic fluid, Item 145, WP 1803 00, if required.
12. Remove fluid level screw from right master cylinder and let fluid drain until level with bottom of screw hole.
13. **QA** Install fluid level screw.
14. **QA** Install filler cap and packing.
15. **QA** Using nonelectrical wire, Item 353, WP 1803 00, lockwire filler cap to fluid level screw.
16. Remove tubing and reducer from outboard slave mixing valve.
17. **QA** Install bleeder plug and packing. **TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
18. **QA** Using nonelectrical wire, Item 353, WP 1803 00, lockwire bleeder plug.
19. Repeat step 1. through step 18. for bleeding inner slave mixer valve and copilot's left master cylinder.
20. Make sure area is clean and free of foreign material before closing up.
21. Install pilot's/copilot's yaw control pedal boots (WP 1043 00).
22. **QA** Return pilot's/copilot's seat to upright position.
23. Do an operational check of brakes (WP 0070 00).

BLEED PILOT'S BRAKES, METHOD B

1. Remove bleed valve from bleed valve adapter on right main wheel brake (Figure 1, Sheet 2).
2. Install union, AN815-3D, and packing, in bleed valve adapter.
3. Install nonmetallic tygothane tubing, Item 345, WP 1803 00, on union and place open end in clean 1 gallon container.

BLEED PILOT'S BRAKES, METHOD B - Continued

4. Remove filler cap from pilot's right master cylinder.
5. Fill master cylinder to top with petroleum base hydraulic fluid, Item 145, WP 1803 00, using syringe or squirt can.
6. Install filler cap and packing.
7. Press right brake pedal until master cylinder piston bottoms out. Maintain pedal pressure.
8. Remove filler cap and fill master cylinder to top with petroleum base hydraulic fluid, Item 145, WP 1803 00.
9. Install filler cap and packing.
10. Release right brake pedal pressure.
11. Repeat step 4. through step 10. until no air bubbles appear in tubing.
12. After pedal pressure is released, remove filler cap and fill master cylinder with petroleum base hydraulic fluid, Item 145, WP 1803 00, if required.
13. Remove fluid level screw from master cylinder and let fluid drain until level with bottom of screw hole.
14. Install fluid level screw.
15. **QA** Install filler cap and packing.
16. **QA** Using nonelectrical wire, Item 353, WP 1803 00, lockwire filler cap to fluid level screw.
17. Remove tubing and union from right wheel brake.
18. **QA** Install bleed valve and packing, in bleed valve adapter.
19. **QA** Using nonelectrical wire, Item 353, WP 1803 00, lockwire bleed valve screw to brake housing.
20. Repeat procedures in step 1. through step 19. for bleeding pilot's left master cylinder and left wheel brake.
21. Make sure area is clean and free of foreign material before closing up.
22. Install pilot's/copilot's yaw control pedal boots (WP 1043 00).
23. **QA** Return pilot's/copilot's seat to upright position.
24. Do an operational check of brakes (WP 0070 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****PARKING BRAKE HANDLE AND MECHANISM**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0230 00

WP 0231 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 11-1520-237-23

WP 0070 00

REPLACE PARKING BRAKE HANDLE AND MECHANISM 77-27714-96-26722**Remove**

1. Chock helicopter wheels.
2. Turn off all electrical power.
3. Open rear access door on lower console.
4. Remove SAS/FPS computer (TM 11-1520-237-23).
5. Unpin brake handle rod end from bracket (Figure 1, Detail A).
6. Remove screws and washers fastening brake handle housing to lower console. Remove brake handle from lower console.
7. Unpin and remove rod from bracket and parking valve lever.

NOTE

To aid in removing upper pin, access cover beneath pilot's seat may be removed if necessary.

8. Remove screws and inner support from cockpit floor.
9. Unpin and remove bracket and washer from outer support on cockpit floor.

Install

1. Turn off all electrical power.
2. Place bracket in outer support on cockpit floor. Install inner support on bracket and fasten to cockpit floor with screws. Install washer and cotter pin in bracket at outer support (Figure 1, Detail A).
3. **QA** Pin rod to bracket and parking valve lever with pins, washers, and cotter pins.
4. **QA** Install brake handle on lower console with screws and washers.
5. **QA** Pin brake handle rod end to bracket with pin, washers, and cotter pin.
6. Install SAS/FPS computer (TM 11-1520-237-23).

REPLACE PARKING BRAKE HANDLE AND MECHANISM 77-27714-96-26722 - Continued

7. Make sure area is clean and free of all foreign matter.
8. **QA** If removed, install access cover on side of center console, beneath pilot's seat, with screws.
9. **QA** Close rear access door on lower console.
10. Do operational check of parking brake system (WP 0070 00).
11. Set parking brake.

REPLACE HANLDE PARKING BRAKE MECHANISM UH60L 96-26723-SUBQ**Remove**

1. Chock helicopter wheels.
2. Turn off all electrical power.
3. Remove pilot's seat if replacing lower rod (WP 0231 00).
4. Remove pilot's floor cover (WP 0230 00).
5. Unpin brake handle rod end from bracket (Figure 2).
6. Remove screws and washers fastening brake handle housing to lower console. Remove brake handle from lower console.
7. Unpin and remove rod from bracket and parking valve lever.
8. Remove screws, washers, and outer support from cockpit floor.

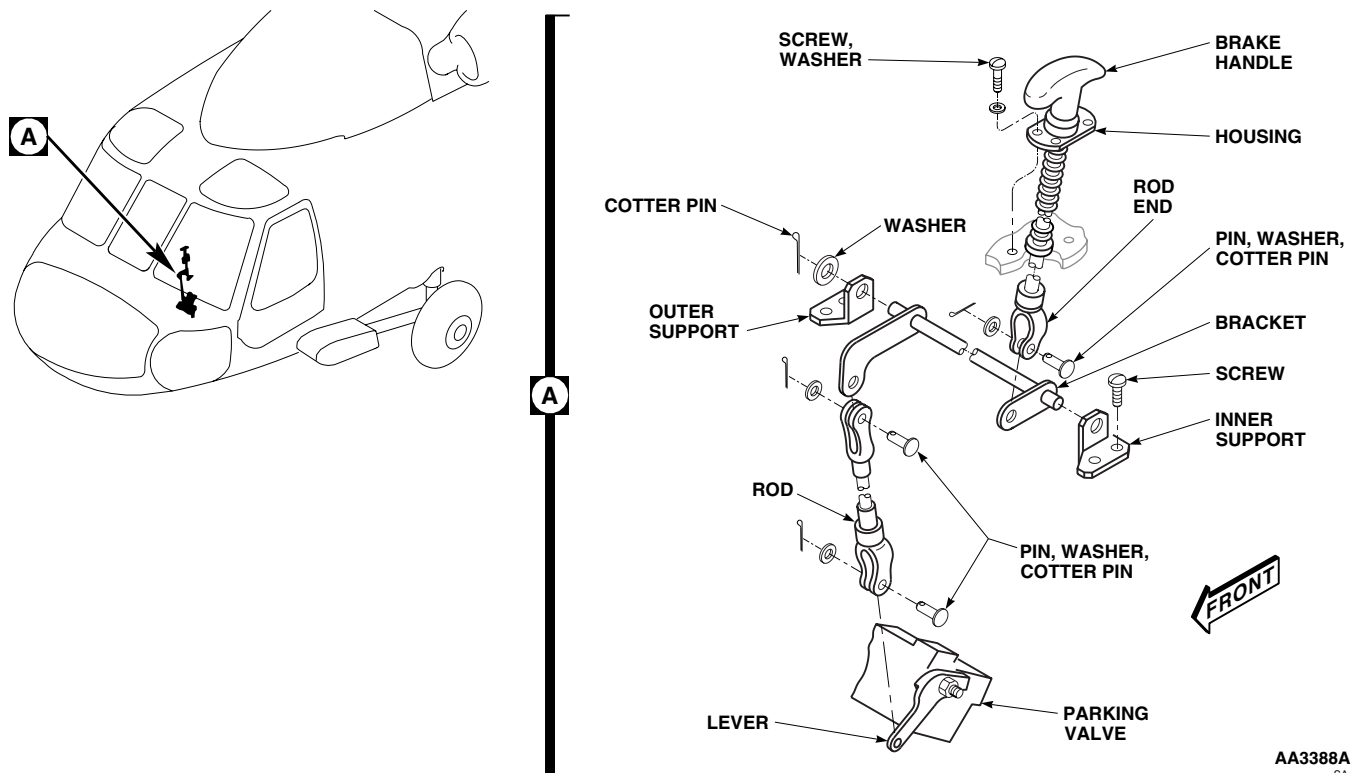
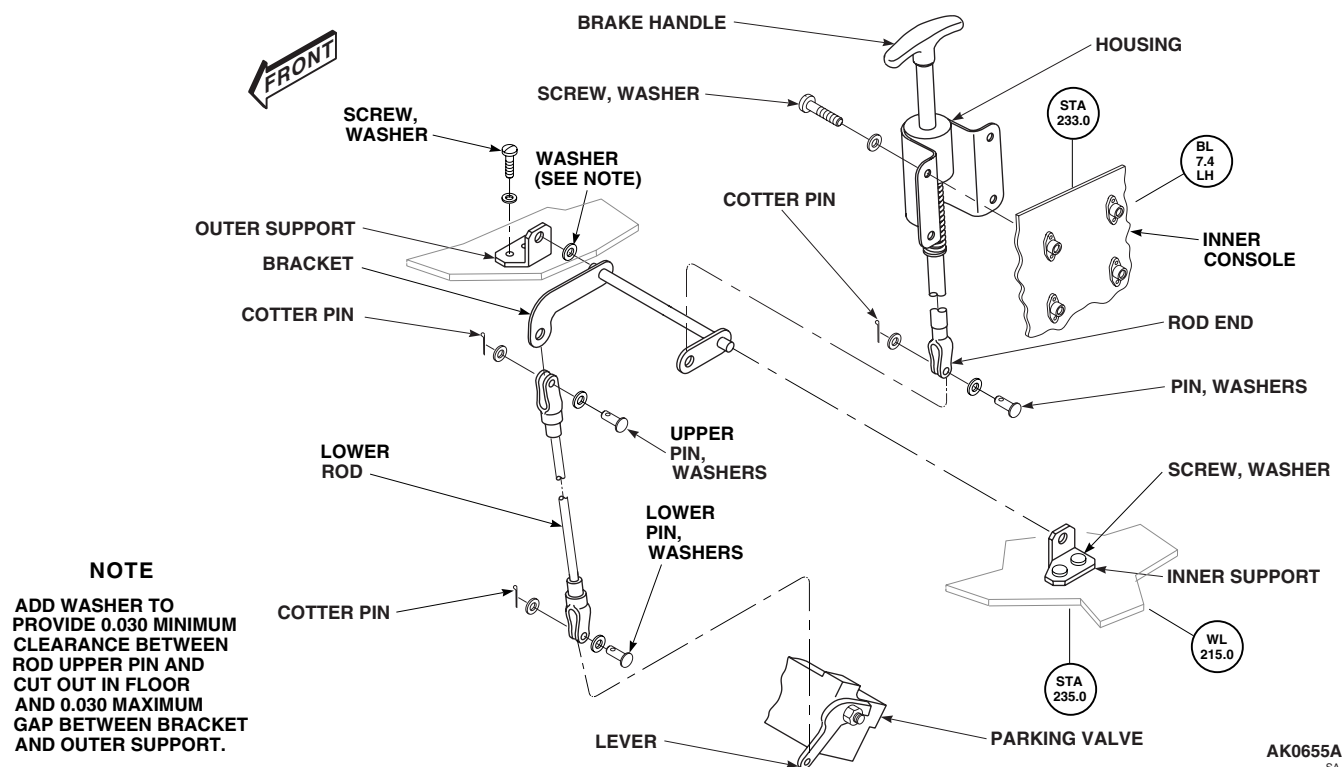
AA3388A
SA

Figure 1. Replace Parking Brake Handle Mechanism 77-27714-96-26722 .

REPLACE HANLDE PARKING BRAKE MECHANISM UH60L 96-26723-SUBQ - Continued

 Figure 2. Replace Parking Brake Handle Mechanism **UH60L 96-26723-SUBQ** .

9. Remove bracket and washers, if installed, from inner support on cockpit floor.

Install

1. Turn off all electrical power.
2. Place bracket in inner support on cockpit floor. Install outer support on bracket and fasten to cockpit floor with screws and washers. Do not tighten screws now (Figure 2).
3. **QA** Pin rod to bracket and parking valve lever with pins, washers, and cotter pins.
4. Check for clearance between rod upper pin and cutout in floor. **MINIMUM CLEARANCE IS 0.030-INCH.** Check for gap between outer support and bracket arm. **MAXIMUM GAP IS 0.030-INCH.** Add washers between outer support and bracket as required to maintain minimum clearance between rod upper pin and cutout in floor and maximum gap between outer support and bracket arm. Tighten screws securing outer support to floor.
5. **QA** Install brake handle on lower console with screws and washers.
6. **QA** Pin brake handle rod end to bracket with pin, washers, and cotter pin.
7. Make sure area is clean and free of all foreign matter.
8. Install pilot's floor cover (WP 0230 00).
9. Install pilot's seat if removed (WP 0231 00).
10. Do operational check of parking brake system (WP 0070 00).

REPLACE HANLDE PARKING BRAKE MECHANISM UH60L 96-26723-SUBQ - Continued

11. Set parking brake.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR**

TAIL LANDING GEAR SHOCK STRUT

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Containers (2), 2-Gallon
Tail Landing Gear Strut Limiter, Locally-Made, Figure 176, WP 1805 00
Torque Wrench, 150 - 1000 in. lbs.

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-237-PMI
TM 55-1500-345-23
WP 0291 00
WP 0381 00
WP 0473 00
WP 1599 00
WP 1675 00
WP 1803 00
WP 1805 00

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 83, WP 1803 00
Cloth, Abrasive, Item 84, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Hydraulic Fluid, Petroleum Base, Item 145, WP 1803 00
Polyurethane Coating, Item 232, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Tubing, Nonmetallic Tygothane, Item 345, WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Chock main landing gear wheels.
3. Put on parking brake.
4. Jack tail landing gear (WP 1675 00).
5. Remove access panel on left and right side of tailcone.
6. Remove tail landing gear fairing (WP 0291 00).
7. Place containers to collect fluid from upper and lower stages, identify as upper or lower.
8. Remove cap from upper and lower stage air valve (Figure 1, Detail A).
9. Slip nonmetallic tygothane tubing, Item 345, WP 1803 00, over valve stem. Place free end in correct container to capture hydraulic fluid as nitrogen is released from shock strut chambers.
10. Turn outer hexnut counterclockwise until you hear nitrogen hissing from valve.
11. Turn outer hexnut counterclockwise to release nitrogen from upper/lower shock strut chamber.
12. After all nitrogen pressure is released, turn inner hexnut counterclockwise to remove.
13. Allow any remaining hydraulic fluid in upper/lower shock strut chambers to drain into containers.
14. **IF DISCHARGE OF OIL FROM UPPER OR LOWER STAGE IS GREATER THAN 80 MILLILITERS (CUBIC CENTIMETERS), REPLACE SHOCK STRUT.**

REMOVE - Continued

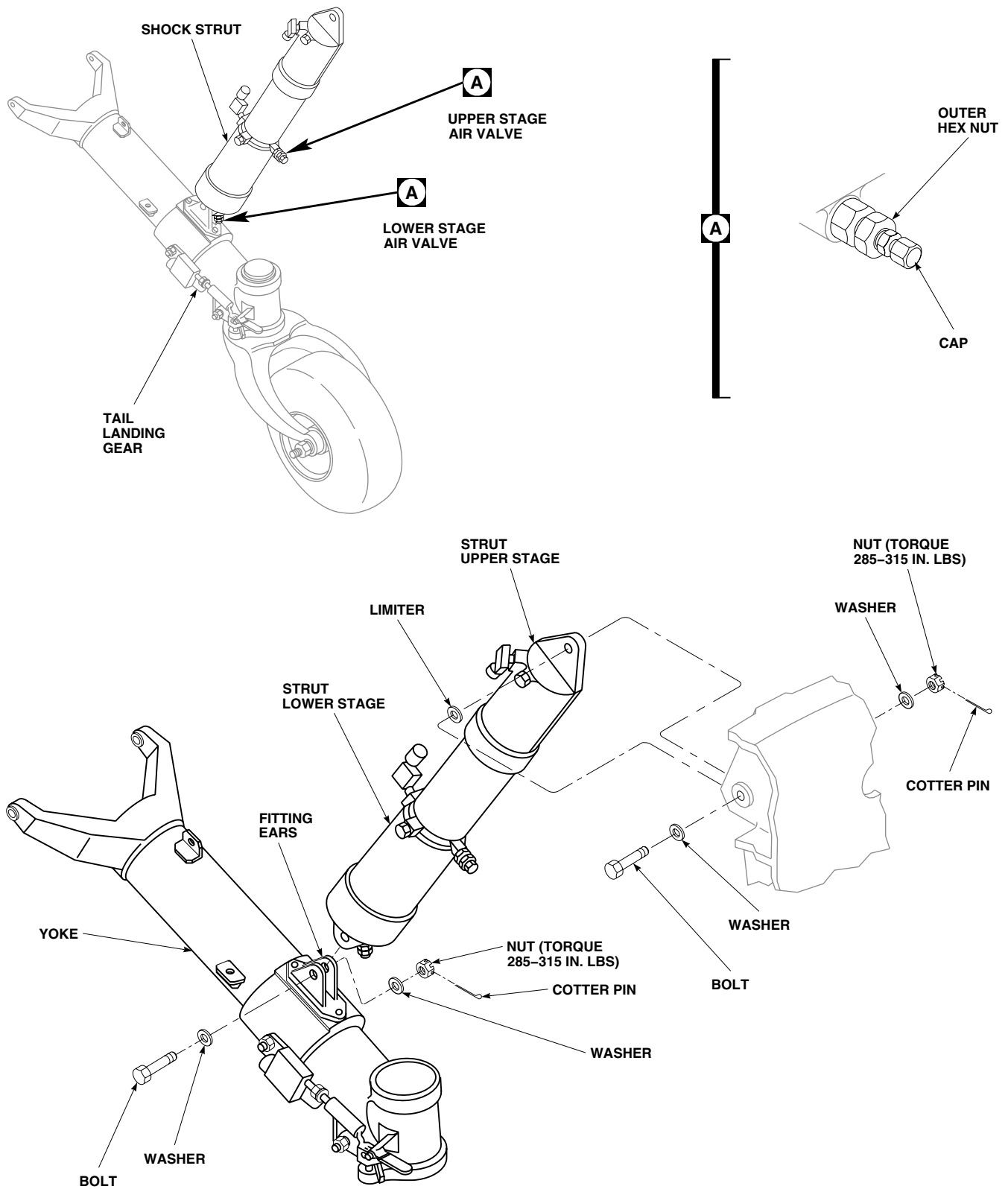


Figure 1. Replace Tail Landing Gear Shock Strut.

REMOVE - Continued

15. **IF DISCHARGE OF OIL IS BETWEEN 40 AND 80 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT 200 FLIGHT-HOURS.**
16. **IF DISCHARGE OF OIL IS LESS THAN 40 MILLILITERS (CUBIC CENTIMETERS), RESERVICE STRUT AND REINSPECT SHOCK STRUT AT NEXT PERIODIC MAINTENANCE INSPECTION (TM 1-1520-237-PMI).**
17. Install cap on both air valves after hissing has stopped.
18. Remove nut and washer from bolt attaching strut upper stage to fuselage support fitting. Do not remove bolt (Figure 1).
19. Remove nut, washers, and bolt attaching lower stage to fitting on yoke.

NOTE

Strut weighs about 20 pounds. Support it during removal.

20. With assistant lifting up on shock strut, remove bolt attaching top of strut to fuselage fitting and remove strut from helicopter. Remove limiter.

INSPECT

1. Check the upper and lower stage spherical bearing for maximum radial and axial play. **SHOCK STRUT FITTING ASSEMBLY SPHERICAL BEARING MAY HAVE UP TO 0.010-INCH MAXIMUM RADIAL PLAY AND 0.030-INCH MAXIMUM AXIAL PLAY.** Repair shock strut assembly if out of tolerance (WP 0473 00).
2. Check the upper and lower stage spherical bearing Teflon liner for disbonding. **NO DISBONDING ALLOWED.** Replace shock strut assembly if disbonding is found.
3. Inspect fuselage support fitting for chafing. Blend out chafing damage using abrasive cloth, Item 83, WP 1803 00. **DAMAGE SHALL NOT EXCEED 0.045-INCH AFTER BLENDING.**
4. Touch up protective coating (WP 0381 00 and TM 55-1500-345-23).
5. Inspect tail landing gear strut limiter for wear. If worn through, locally make tail landing gear strut limiter (WP 1805 00).

INSTALL

1. Place shock strut lower stage between fitting ears on yoke (Figure 1).

NOTE

Lower stage is identified by attachment point containing an air valve.

2. Place limiter in fuselage support fitting clevis.
3. Place lug of shock strut upper stage in clevis.
4. **QA** Install bolt, washers, and nut attaching shock strut to fuselage support fitting. **TORQUE NUT TO 285 - 315 INCH-POUNDS.** Install cotter pin.
5. **QA** With assistant raising up yoke, bolt shock strut lower stage to fitting on yoke with bolt, washers, and nut. **TORQUE NUT TO 285 - 315 INCH-POUNDS.** Install cotter pin.
6. Service tail landing gear shock strut (WP 1599 00).
7. After replacing shock strut with new strut, do a static leak check as follows:
 - a. Wipe polished piston of shock strut dry using a clean machinery wiping towel, Item 344, WP 1803 00.

INSTALL - Continued

- b. Wait 1 hour and check polished piston for hydraulic fluid.
- c. If fluid leaks appear before 1 hour after servicing, replace strut.
- d. If no fluid is present, soak a clean machinery wiping towel, Item 344, WP 1803 00, with petroleum base hydraulic fluid, Item 145, WP 1803 00, and wipe all exposed polished portions of strut.

NOTE

Minor seepage in service is not considered cause for replacement.

- 8. Lower helicopter and remove jack (WP 1675 00).
- 9. Install tail landing gear fairing (WP 0291 00).
- 10. Install access panels on tailcone.

REPAIR

- 1. Blend out any minor scratches on housing with abrasive cloth, Item 84, WP 1803 00. **SCRATCHES AT RIGHT ANGLES TO CENTERLINE MUST NOT BE OVER 20% OF CIRCUMFERENCE OF HOUSING. DEEPEST PART OF SCRATCH MUST NOT BE MORE THAN 0.005-INCH AFTER BLENDING.**
- 2. Blend out any small scratches or nicks on upper stage housing and lug with abrasive cloth, Item 84, WP 1803 00. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.010-INCH AFTER BLENDING.**
- 3. Blend out any small scratches or nicks on lower stage housing with abrasive cloth, Item 84, WP 1803 00. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.010-INCH AFTER BLENDING.**
- 4. No repair is allowed to upper or lower piston surface. **ALLOWABLE WEAR IS LIMITED TO 0.0015-INCH, OR SMALLEST DIAMETER OF 2.9935-INCHES.**
- 5. Lower piston lug scratches or nicks may be blended out using abrasive cloth, Item 84, WP 1803 00. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.010-INCH AFTER BLENDING.**
- 6. Clean any repaired area with cleaning compound solvent, Item 71, WP 1803 00.
- 7. Touch up with epoxy primer coating kit, Item 120, WP 1803 00, and polyurethane coating, Item 232, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR
TAIL LANDING GEAR SHOCK STRUT KNEELING VALVE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Hydraulic Repairer's Toolkit, SC 5180-99-A03
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Pneudraulics Repairer MOS 15H (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0291 00
WP 1599 00
WP 1803 00

REPLACE TAIL LANDING GEAR SHOCK STRUT KNEELING VALVE**Remove****NOTE**

Replacement of either kneeling valve (upper or lower) is the same.

1. Unscrew and remove left and right access panels from tailcone.
2. Remove tail landing gear fairing (WP 0291 00).
3. Deservice shock strut upper or lower stage as required (WP 1599 00).
4. Loosen jamnut and remove kneeling valve from strut (Figure 1, Detail A).
5. Remove packing, retainer, and jamnut from valve. Discard packing. Keep jamnut for installation of replacement valve.

Install

1. Lubricate packing, retainer, and male threads of replacement kneeling valve with petroleum base hydraulic fluid, Item 145, WP 1803 00.
2. **QA** Install jamnut, retainer and packing onto kneeling valve. Turn jamnut and retainer down until packing is pushed firmly against the lower threaded section of kneeling valve (Figure 1, Detail A).
3. Install kneeling valve in strut until a sharp rise in torque is felt while keeping jamnut from turning. **TIGHTEN KNEELING VALVE AND JAMNUT 1 1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT. TIGHTEN KNEELING VALVE NO MORE THAN 1 ADDITIONAL TURN TO POSITION COUPLING HALF PORT FACING DOWN FOR UPPER STAGE OR UP FOR LOWER STAGE.**
4. **TORQUE JAMNUT TO 95 - 105 INCH-POUNDS WHILE KEEPING KNEELING VALVE FROM TURNING.**
5. Service stage of shock strut in which valve was replaced (WP 1599 00).
6. Make sure area is clean and free of foreign material.
7. Install tail landing gear fairing (WP 0291 00).
8. **QA** Install left and right access panels on tailcone.

REPLACE TAIL LANDING GEAR SHOCK STRUT KNEELING VALVE - Continued

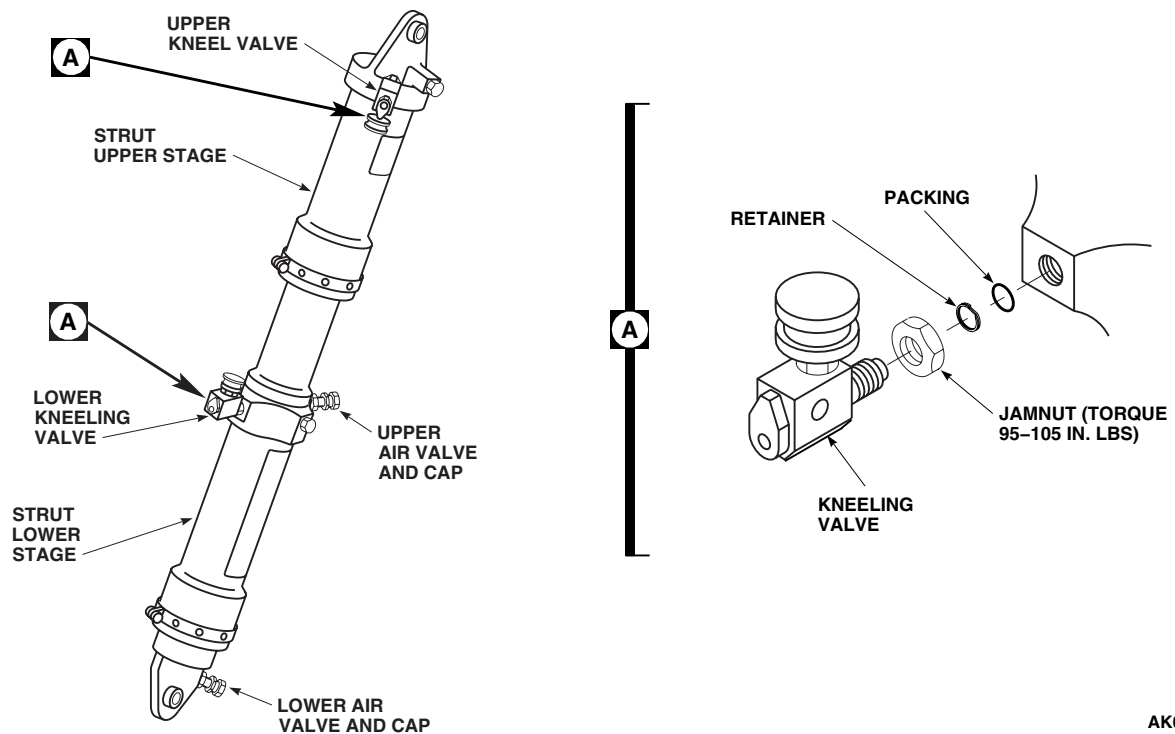
AK0665
SA

Figure 1. Replace Tail Landing Gear Shock Strut Kneeling Valve.

REPAIR**Disassemble**

1. Remove knob stop pin from body (Figure 2).
2. Remove shaft by turning knob counterclockwise.
3. Remove packing and retainers from shaft.
4. Loosen jamnut and remove coupling half from valve body.
5. Remove packing, retainer, and jamnut from coupling half.

Inspect

1. Check shaft for corrosion or deformation. Replace if corroded or deformed.
2. Check shaft bore in valve body. Replace body if corroded or deformed.

Assemble

1. Lubricate packing, retainer, and male threads of coupling half with petroleum base hydraulic fluid, Item 145, WP 1803 00.
2. **QA** Install jamnut, retainer, and packing onto coupling half. Turn nut and retainer down until packing is pushed firmly against the lower threaded section of coupling half (Figure 2).
3. **QA** Install coupling half into valve body until a sharp rise in torque is felt while keeping jamnut from turning.
TIGHTEN COUPLING HALF AND JAMNUT 1 1/2 TURNS PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.

REPAIR - Continued

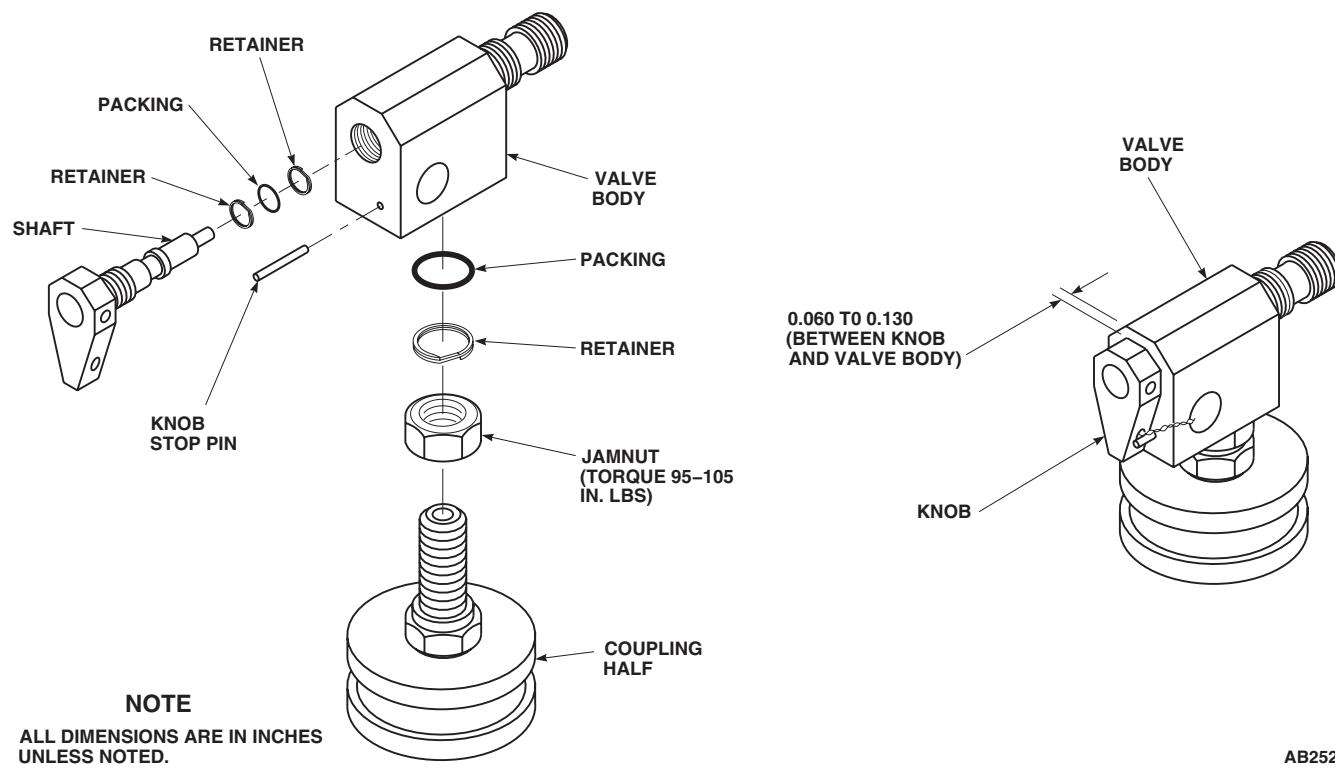


Figure 2. Repair Tail Landing Gear Shock Strut Kneeling Valve.

4. **PREVENT COUPLING HALF FROM TURNING. TORQUE JAMNUT TO 95 - 105 INCH-POUNDS.**
5. Lubricate packing and retainers with petroleum base hydraulic fluid, Item 145, WP 1803 00.
6. **QA** Install retainer, packing, and second retainer on shaft.
7. **QA** Insert shaft carefully into valve body.
8. Turn knob clockwise to set specified clearance between knob and valve body. **CLEARANCE MUST RANGE FROM 0.060 TO 0.130-INCH.**
9. **QA** Install knob stop pin, in valve body.
10. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire knob to hole in valve body.

Test

Test for operation on helicopter, no bench test required.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR
TAIL LANDING GEAR SHOCK STRUT AIR VALVES

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

References

WP 0291 00
WP 1599 00
WP 1803 00

Material/Parts

Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

Replacement of either valve (upper or lower) is the same.

1. Remove screw and remove left and right access panels from tailcone.
2. Remove tail landing gear fairing (WP 0291 00).
3. Remove cap from air valve of stage with malfunctioning air valve (Figure 1, Detail A).

WARNING

Injury to personnel and damage to equipment may occur when working with shock strut containing high pressure. To prevent injury, make sure stage in which valve is being removed is completely deflated before removing valve.

4. Turn outer hexnut counterclockwise until you hear nitrogen hissing from valve.
5. When hissing stops, turn inner hexnut counterclockwise and slowly remove air valve.
6. Check condition of packing. If packing is determined to be source of leak, replace packing.
7. If valve is determined to be malfunctioning, replace valve.

INSTALL

1. **QA** Install replacement air valve in strut. **TORQUE INNER HEXNUT TO 100 - 110 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire inner hexnut to shock strut (Figure 1, Detail A).
2. Service stage of shock strut from which valve was replaced (WP 1599 00).
3. Make sure area is clean and free of foreign material before closing up.
4. Install tail landing gear fairing (WP 0291 00).
5. **QA** Install left and right access panels on tailcone.

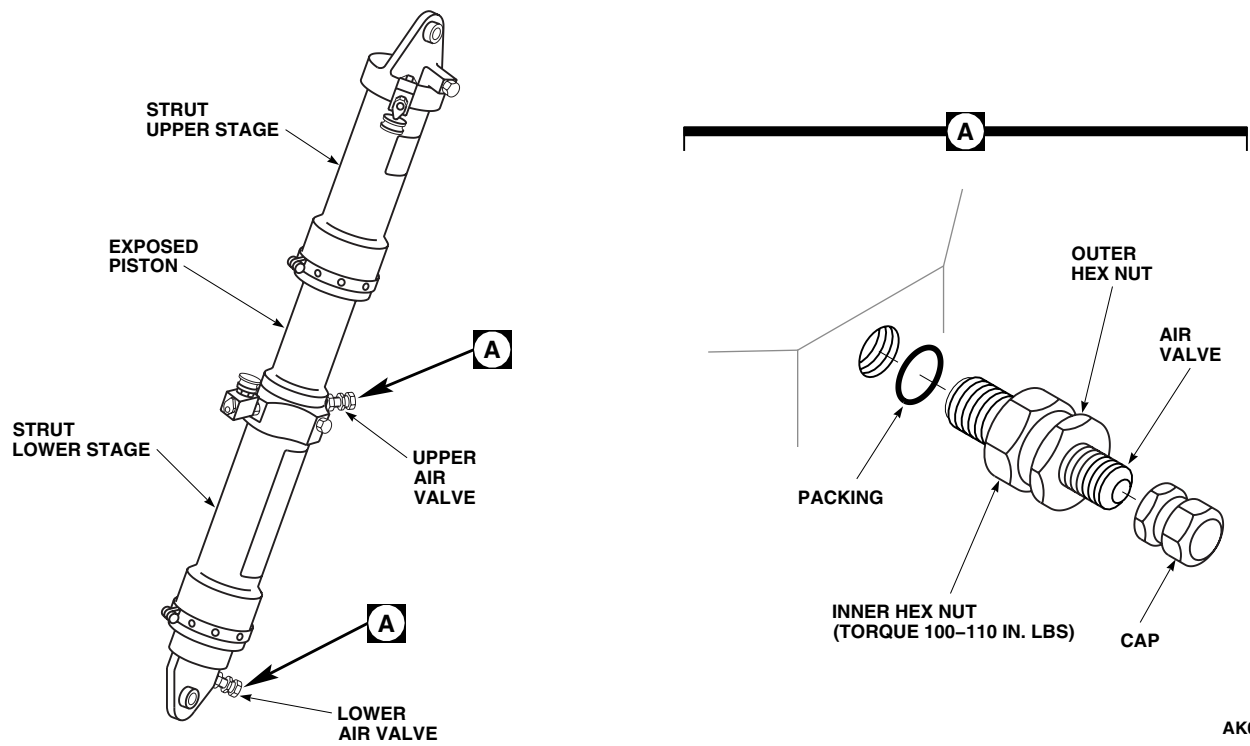
INSTALL - Continued

Figure 1. Replace Tail Landing Gear Shock Strut Air Valves.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR**TAIL LANDING GEAR YOKE**

This WP supersedes WP 0462 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 C-Clamp, GGG-C-406
 Electrical Repairer's Toolkit, SC 5180-99-B06
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Micrometer, 0 to 1 inch
 Multimeter, 60741-310
 Nonmetallic Drift
 Nonmetallic Scraper
 Torque Wrench, 0 - 30 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Abrasive Mat, Item 2, WP 1803 00
 Acetone, Technical, Item 5, WP 1803 00
 Adhesive Item 11, WP 1803 00
 Brush, Paint, Item 61, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 81, WP 1803 00
 Cloth, Abrasive, Item 77, WP 1803 00
 Cloth, Cheesecloth, Item 85, WP 1803 00
 Corrosion Resistant Coating, Item 110, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Insulation Sleeving, Electrical, Item 168, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00

Towel, Machinery Wiping, Item 344, WP 1803 00
 Wire, Nonelectrical, Item 354, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
 TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0071 00
 WP 0278 00
 WP 0279 00
 WP 0291 00
 WP 0381 00
 WP 0415 00
 WP 0463 00
 WP 0464 00
 WP 0465 00
 WP 0474 00
 WP 1599 00
 WP 1675 00
 WP 1703 00
 WP 1803 00

REPLACE TAIL LANDING GEAR YOKE ASSEMBLY**Remove**

1. Turn off all electrical power.
2. Set parking brake.
3. Engage tailwheel lock.
4. **WSPS** Remove tail landing gear deflector and clamp (WP 0278 00 and WP 0279 00). ■
5. Remove tail landing gear fairing (WP 0291 00).
6. Jack tail landing gear (WP 1675 00).
7. Remove caps from lower and upper stage air valves on shock strut.

REPLACE TAIL LANDING GEAR YOKE ASSEMBLY - Continued

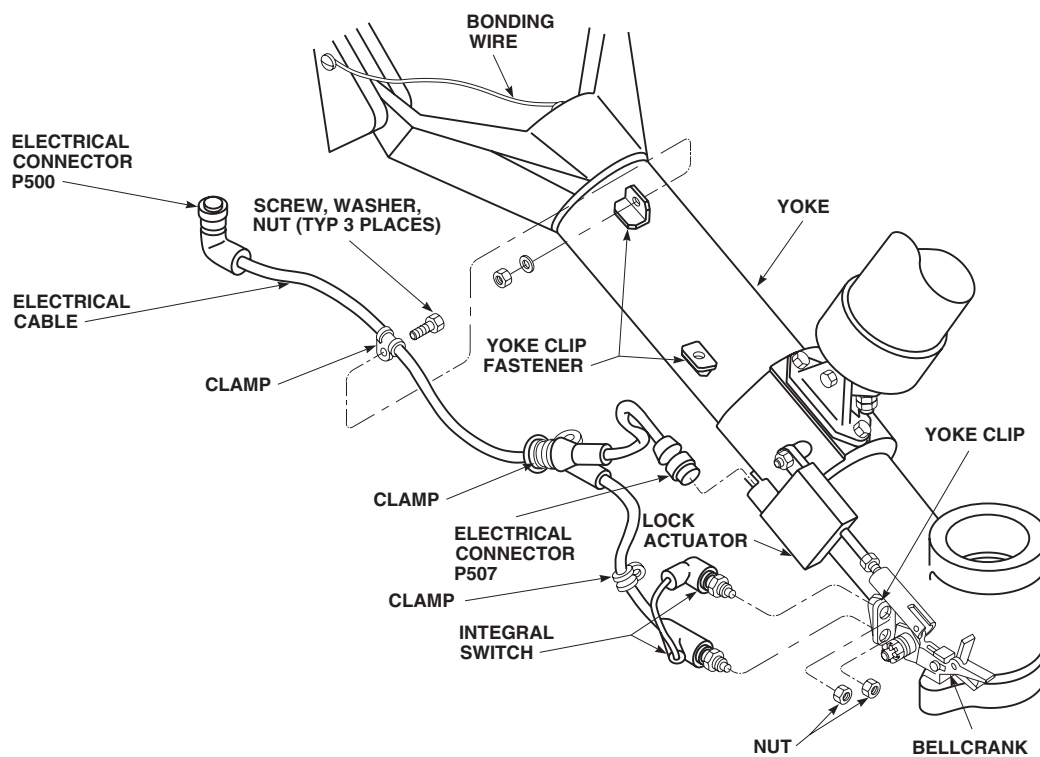
AA8937A
SA

Figure 1. Replace Tail Landing Gear Electrical Cable.

8. Turn outer hexnuts counterclockwise until you hear nitrogen hissing from valve.
9. When hissing stops install cap.
10. Remove tailwheel and tire (WP 0465 00).
11. Remove fork (WP 0463 00).
12. Disconnect and remove electrical cable with integral switches, by removing nuts on yoke clip at bellcrank (Figure 1).
13. Disconnect electrical connector, P507, from lock actuator.
14. Disconnect electrical connector, P500, from J500.
15. Remove electrical cable from yoke clips.
16. Place wood blocking, or other support, under yoke. Rest yoke on support when disconnected from shock strut.
17. Remove bolt, washers, and nut holding yoke to shock strut (Figure 2).
18. Disconnect bonding wire at yoke cover (Figure 1).
19. Remove bolt, washers, and nut holding yoke to fuselage. Remove yoke (Figure 2).
20. Remove lock actuator, locking pin, and bellcrank (WP 0464 00).
21. Remove shock strut attachment fitting and shims. Refer to, REPLACE TAIL LANDING GEAR SHOCK STRUT ATTACHMENT FITTING, in this work package.

REPLACE TAIL LANDING GEAR YOKE ASSEMBLY - Continued

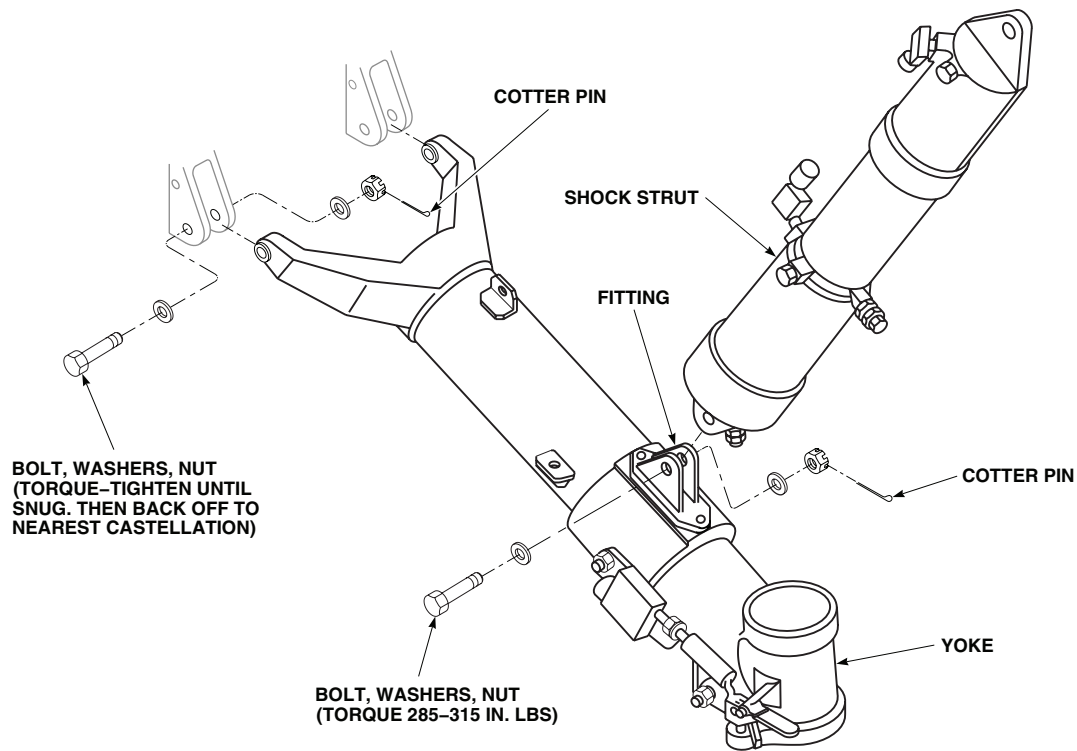
AK0668
SA

Figure 2. Replace Tail Landing Gear Yoke.

Inspect

1. Check Teflon coating on Teflon coated sleeve bearings and washers for wear. Replace Teflon coated washers, and/or sleeves. Refer to, REPLACE TAIL LANDING GEAR TEFLON COATED WASHERS and REPLACE TAIL LANDING GEAR TEFLON COATED SLEEVES, in this work package if:
 - a. **WEAR SURFACE LOOKS LIKE SMEARED BRONZE WITH NO DARK SPOTS.**
 - b. **20% OF SURFACE IS CORRODED OR SCORED.**
 - c. **TEFLON IS WORN FROM MORE THAN 1/3 OF BEARING SURFACE.**
 - d. **DISBONDING HAS OCCURRED OVER MORE THAN 25% OF TOTAL BONDED SURFACE.**
 - e. **THICKNESS OF TEFLON WASHERS, INCLUDING BOND LINE THICKNESS, IS LESS THAN 0.085-INCH.**
 - f. **INSIDE DIAMETER OF TEFLON COATED SLEEVE BEARING IS GREATER THAN 3.510-INCHES.**
2. Check lock actuator and locking pin bellcrank studs for:
 - a. Crossed, stripped, or flattened threads.
 - b. Looseness or bending.
 - c. Flange of stud or locking above surface of yoke.
3. **REPLACE YOKE IF STUDS ARE LOOSE OR DAMAGED.**

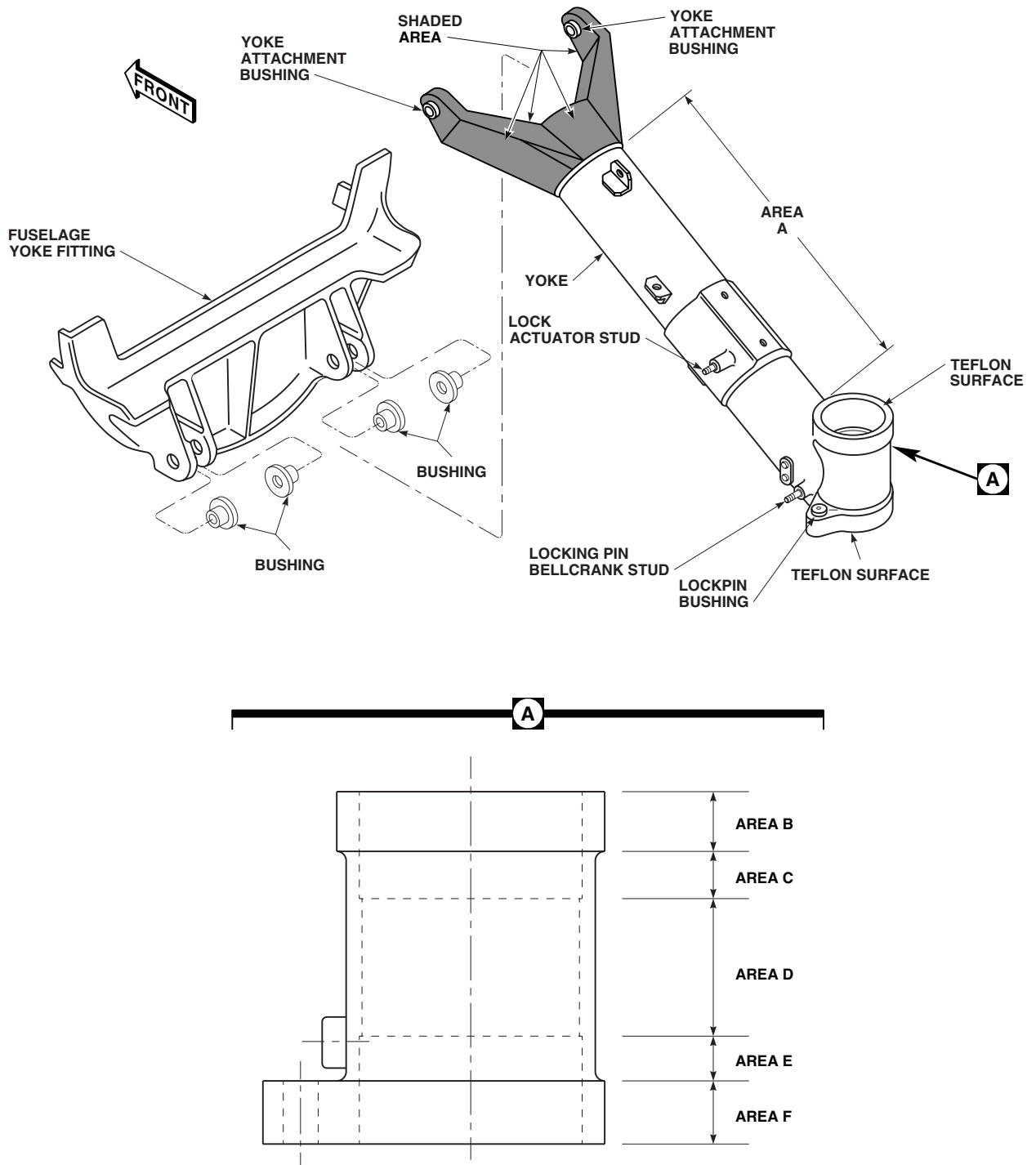
REPLACE TAIL LANDING GEAR YOKE ASSEMBLY - Continued

4. Measure lockpin bushing inside diameter. **LARGEST INSIDE DIAMETER SHALL BE NO MORE THAN 0.350-INCH.** Replace lockpin bushing if measurement is more than limit. Refer to, REPLACE TAIL LANDING GEAR YOKE LOCKPIN BUSHING, in this work package.
5. Measure yoke attachment bushings inside diameter. **LARGEST INSIDE DIAMETER SHALL BE NO MORE THAN 0.6365-INCH, WITH NO MORE THAN 0.010-INCH OUT-OF-ROUND PERMITTED.** Replace yoke attachment bushings if out of limits (WP 0474 00).
6. Measure fuselage yoke fitting bushings inside diameter. **INSIDE DIAMETER SHALL BE NO MORE THAN 0.6365-INCH, WITH NO MORE THAN 0.010-INCH OUT-OF-ROUND PERMITTED (Figure 3).** Replace all four bushings if one is found out of tolerance (WP 0474 00).
7. Inspect and repair fuselage yoke fitting (WP 0415 00).
8. Remove tail landing gear yoke dust cover and inspect inside of yoke for corrosion. Clean and treat any corrosion found. Install O-ring on dust cover. **TORQUE THE RETAINING SCREWS 20 - 25 INCH-POUNDS. USING NONELECTRICAL WIRE, Item 357, WP 1803 00 LOCKWIRE RETAINING SCREWS.** Seal around each retaining screw with sealing compound, Item 273, WP 1803 00.

Repair

1. Repair shaded outline areas as follows:
 - a. Blend out nicks, dents, scratches, and gouges with a 180-grit abrasive cloth, Item 77, WP 1803 00. **YOKE ARM TOTAL CUMULATIVE REPAIR DEPTH (INCLUDING ANY PREVIOUS BLENDING) AT ANY ONE LOCATION SHALL BE NOT MORE THAN 0.0312-INCH. REPAIR MUST NOT EXCEED 0.0312-INCH MAXIMUM.**
 - b. Replace yoke if damage is more than limit after blending (Figure 3).
2. Repair AREA A as follows:
 - a. **BLEND OUT NICKS, DENTS, SCRATCHES, OR GOUGES UP TO 0.010-INCH DEEP.**
 - b. Measure length of repair. **REPAIR AFTER BLENDING SHALL NOT BE MORE THAN 1.125-INCHES LONG.**
 - c. Replace yoke if damage, after repairs, is more than limits.
3. Repair AREA B (Figure 3) as follows:
 - a. **ON OUTER SURFACE, BLEND UP TO 0.020-INCH DEEP (MINIMUM OD IS 4.404-INCHES). ON INNER SURFACE, BLEND UP TO 0.010-INCH DEEP (MAXIMUM ID IS 3.6972-INCHES). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.3384-INCHES. BLENDING NOT TO EXCEED 1.375-INCHES LONG AT ANY CROSS SECTION.**
 - b. Replace yoke if damage is more than limit, after repair.
4. Repair AREA C as follows:
 - a. **ON OUTER SURFACE, BLEND UP TO 0.062-INCH DEEP (MINIMUM OD IS 4 INCHES). ON INNER SURFACE, BLEND UP TO 0.010-INCH DEEP (MAXIMUM ID IS 3.6972-INCHES). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.1204-INCH. BLENDING NOT TO EXCEED 1.375-INCHES LONG AT ANY CROSS SECTION.**
 - b. Replace yoke if damage is more than limit after repair.
5. Repair AREA D as follows:
 - a. **ON OUTER SURFACE, BLEND UP TO 0.062-INCH DEEP (MINIMUM OD IS 4 INCHES). ON INNER**

REPLACE TAIL LANDING GEAR YOKE ASSEMBLY - Continued



AK0669B
SA

Figure 3. Inspect Tail Landing Gear Yoke.

REPLACE TAIL LANDING GEAR YOKE ASSEMBLY - Continued

SURFACE, BLEND UP TO 0.010-INCH DEEP (MAXIMUM ID IS 3.6972-INCHES). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.179-INCH. BLENDING NOT TO EXCEED 1.375-INCHES LONG AT ANY CROSS SECTION.

- b. Replace yoke if damage is more than limit after repair.
6. Repair AREA E as follows:
 - a. **ON OUTER SURFACE, BLEND UP TO 0.062-INCH DEEP (MINIMUM OD IS 4 INCHES). ON INNER SURFACE, BLEND UP TO 0.030-INCH DEEP (MAXIMUM ID IS 3.7172-INCHES). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.1204-INCH. BLENDING NOT TO EXCEED 1.375-INCHES LONG AT ANY CROSS SECTION.**
 - b. Replace yoke if damage is more than limit after repair.
7. Repair AREA F as follows:
 - a. **ON OUTER SURFACE, BLEND UP TO 0.030-INCH DEEP (MINIMUM OUTSIDE RADIUS IS 2.182-INCHES). ON INNER SURFACE, BLEND UP TO 0.030-INCH DEEP (MAXIMUM ID IS 3.7172-INCHES). MINIMUM THICKNESS AT ANY CROSS SECTION AFTER BLENDING IS 0.3048-INCH. BLENDING NOT TO EXCEED 1.375-INCHES LONG AT ANY CROSS SECTION.**
 - b. Replace yoke if damage is more than limit after repair.
8. Fluorescent penetrant inspect reworked areas for cracks, as necessary, using fluorescent penetrant inspection kit (TM 1-1520-265-23).
9. Touch up reworked aluminum surfaces with corrosion resistant coating, Item 110, WP 1803 00. Apply primer and paint (WP 0381 00 and TM 55-1500-345-23).

Install

1. Turn off all electrical power.
2. Install shock strut attachment fitting on yoke. Refer to, REPLACE TAIL LANDING GEAR SHOCK STRUT ATTACHMENT FITTING, in this work package.
3. Install tailwheel lock actuator locking pin, and bellcrank assembly (WP 0464 00).
4. Position and support yoke for attachment to helicopter.
5. **QA** Install yoke to fuselage attaching bolts, washers, and nuts. **TIGHTEN NUTS UNTIL SNUG, THEN BACK OFF TO NEAREST CASTELLATION.** Install cotter pins (Figure 2).
6. Check lateral movement of yoke assembly. **IF LATERAL MOVEMENT OF YOKE EXCEEDS 0.047-INCH, REPLACE BUSHINGS** (WP 0474 00).
7. **QA** Attach bonding wire to yoke with screw (Figure 1).
8. Position and support yoke for attachment to shock strut.
9. **QA** Bolt shock strut to fitting with bolt, washers, and nut. Make sure thin washer is under bolthead and thick washer is under nut. **TORQUE NUTS TO 285 - 315 INCH-POUNDS.** Install cotter pin (Figure 2).
10. **QA** Install electrical cable with integral switches into yoke clip at bellcrank with nuts (Figure 1).
11. Inspect and treat electrical connector (WP 1703 00).
12. **QA** Connect electrical connector, P507, to tailwheel lock actuator.
13. Adjust locked and unlocked switches. Refer to, ADJUST TAIL LANDING GEAR TAILWHEEL LOCKED/ UNLOCKED SWITCHES, in this work package.
14. Inspect and treat electrical connector (WP 1703 00).

REPLACE TAIL LANDING GEAR YOKE ASSEMBLY - Continued

15. **QA** Connect electrical connector, P500 to J500.
16. **QA** Fasten cable to yoke clips with screws, washers, and nuts.
17. Apply sealing compound, Item 273, WP 1803 00, to all pockets, joints, and entrants on yoke cable clips.
18. Install fork assembly (WP 0463 00).
19. Install tailwheel and tire (WP 0465 00).
20. Lower tail section of helicopter.
21. Service shock strut upper and lower stages (WP 1599 00).
22. **WSPS** Install tail landing gear deflector and clamp (WP 0278 00 and WP 0279 00). ■
23. Install tail landing gear fairing (WP 0291 00).
24. Do an operational check of tailwheel lock (WP 0071 00).

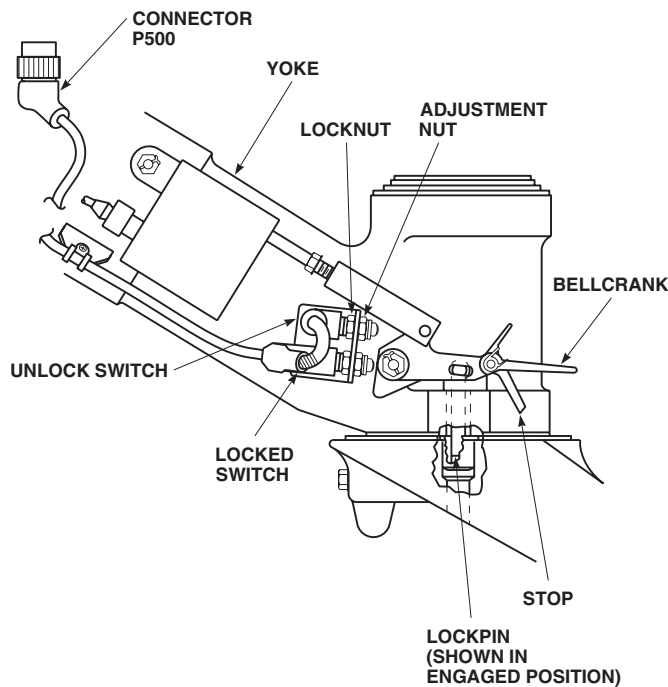
ADJUST TAIL LANDING GEAR TAILWHEEL LOCKED/UNLOCKED SWITCHES**Adjust**

1. Park helicopter on level surface.
2. Turn off all electrical power.
3. Disconnect electrical connector, P500 from J500 (Figure 4).
4. Manually unlock tailwheel.
5. Connect multimeter to pin 4 and pin 6 of electrical connector, P500.
6. Starting with the plunger in the free position, rotate the adjustment nut on the unlock switch clockwise until there is continuity between pin 4 and pin 6 of P500. Continue clockwise rotation of nut 1/3 of an additional turn.
7. **QA** With the adjustment nut held stationary, tighten the locking nut. Using nonelectrical wire, Item 354, WP 1803 00, lockwire locking nut.
8. Disconnect multimeter.
9. Manually lock tailwheel and make sure that the lockpin is fully engaged.
10. Connect multimeter to pin 1 and pin 3 of electrical connector, P500.
11. Starting with the plunger in the free position, rotate the adjustment nut clockwise until there is continuity between pin 1 and pin 3 of P500. Continue clockwise rotation of nut 2 to 3 additional turns.
12. **QA** With the adjustment nut held stationary, tighten the locking nut. Using nonelectrical wire, Item 354, WP 1803 00, lockwire locking nut.
13. Disconnect multimeter.
14. Connect electrical connector, P500 to J500.
15. Do an operational check of tailwheel lock (WP 0071 00).

REPLACE TAIL LANDING GEAR SHOCK STRUT ATTACHMENT FITTING**Remove**

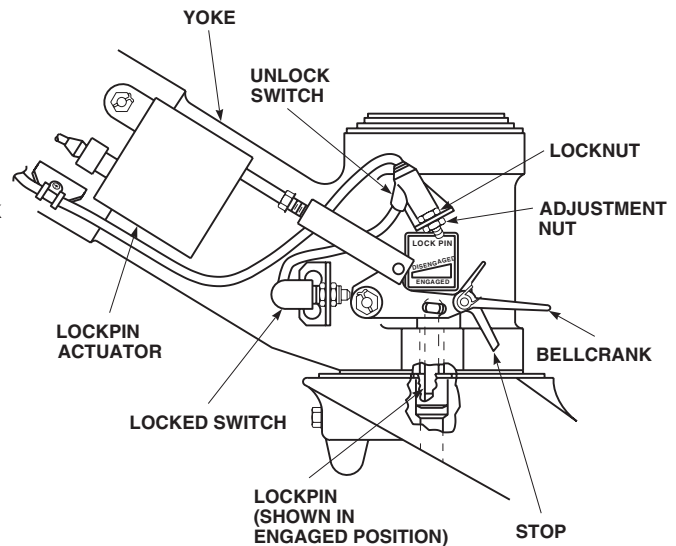
1. Turn off all electrical power.
2. Set parking brake.

REPLACE TAIL LANDING GEAR SHOCK STRUT ATTACHMENT FITTING - Continued



EFFECTIVITY

UH-60A 77-22724 - 77-22728, 78-22961 - SUBQ



EFFECTIVITY

UH-60A 77-22714 - 77-22723, 78-22960

AA3389
SA

Figure 4. Adjust Tail Landing Gear Tailwheel Locked/Unlocked Switches.

3. Engage tailwheel lock.
4. Chock main landing gear wheels.
5. Remove tail landing gear fairing (WP 0291 00).
6. Jack tail landing gear (WP 1675 00).
7. Remove bolt, washers, and nut holding shock strut to yoke (Figure 2).

CAUTION

Damage to fitting will occur if shim plates are removed. Shim plates are bonded to fitting. Do not remove shim plates from fitting.

8. Remove bolts, washers and shim washers. Remove fitting and shims (Figure 5, Detail A).

Install

1. **QA** Install fitting and shims. Insert bolts with thin washer next to bolthead and two 1/4-inch thick shim washers (Figure 5, Detail A).
2. Shim yoke as follows:
 - a. Install fitting on tail yoke.
 - b. Install yoke assembly with fuselage attachment bolts. Refer to REPLACE TAIL LANDING GEAR YOKE ASSEMBLY, in this work package.
 - c. Install fork and lock in trailing position (WP 0463 00).

REPLACE TAIL LANDING GEAR SHOCK STRUT ATTACHMENT FITTING - Continued

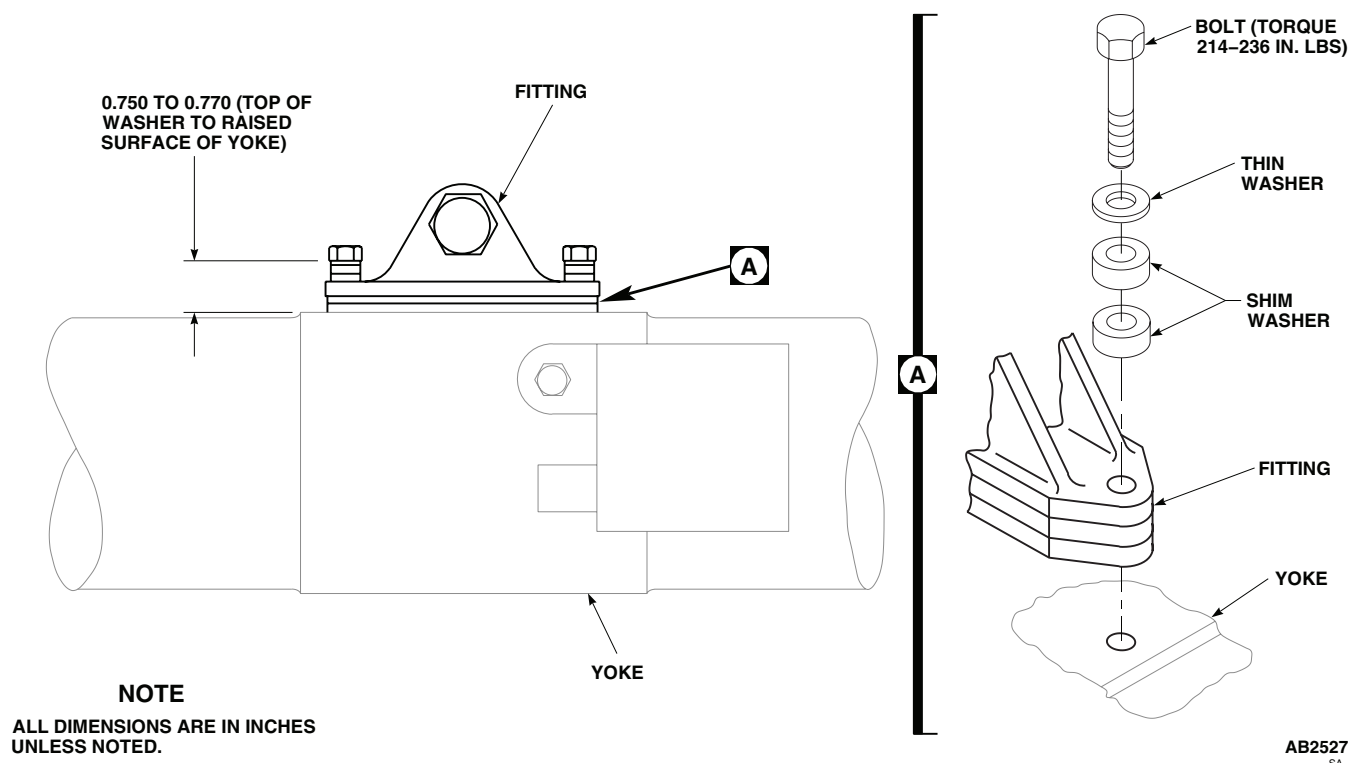


Figure 5. Replace Tail Landing Gear Shock Strut Attachment Fitting.

- d. Compress and restrict strut to length of 27.43-inches, from bolthole to bolthole (full kneeled length) and install in upper attachment fitting.
- e. Place jack under tailcone forward of jacking pad (STA 605.0) and jack tail until a dimension between center of yoke attachment bolts and floor or jacking surface (this duplicates WL 184.0) is 26.5-inches (this duplicates WL 210.5).
- f. Place jack under tail fork and raise to specific measurement of 21.94-inches (from ground or jacking surface) to center of axle (this will duplicate WL 205.94). Position strut to its lower attachment point.
- g. Adjust bonded shim on fitting assembly until strut attachment bolt can be installed. Adjust shimming under fitting attachment bolts until 0.750 to 0.770- inch is achieved.

CAUTION

Damage to threaded inserts may occur, if bolts are not shimmed properly. Shim bolts to maintain distance of 0.750 to 0.770-inch. Above fitting.

3. Measure distance from top of washer to flat yoke mounting surfaces (Figure 5). **DISTANCE SHALL MEASURE 0.750 TO 0.770-INCH.** Peel shim washer, as necessary, to get distance.
4. **QA** Install bolts, washers, and shim washers in fitting. **TORQUE BOLTS TO 214 - 236 INCH - POUNDS (Figure 5, Detail A).**
5. **QA** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts. Slide electrical insulation sleeving, Item 168, WP 1803 00, over nonelectrical wire between bolts to prevent chafing.

REPLACE TAIL LANDING GEAR SHOCK STRUT ATTACHMENT FITTING - Continued

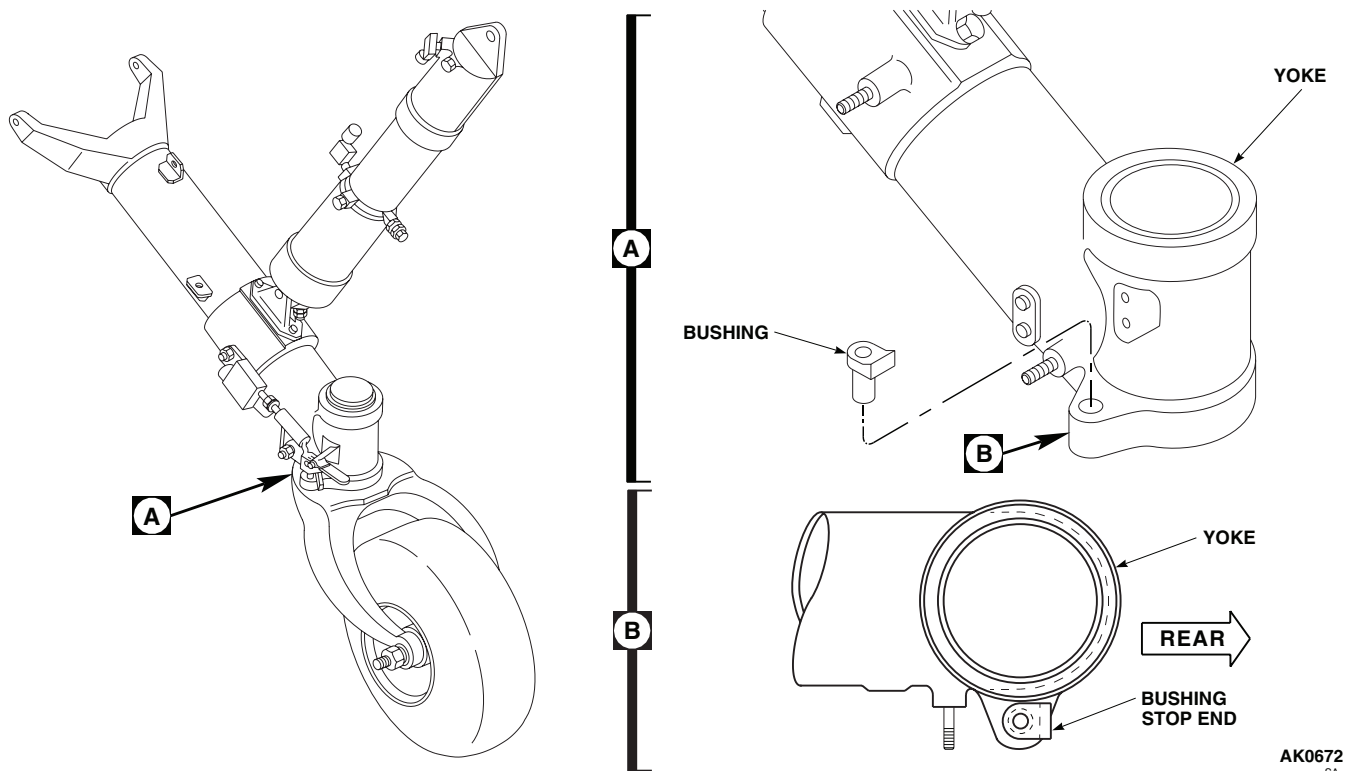


Figure 6. Replace Tail Landing Gear Yoke Lockpin Bushing.

6. **QA** Bolt yoke to shock strut with bolt, washers, and nut. Make sure thin washer is under bolthead and thick washer is under nut. **TORQUE NUT TO 285 - 315 INCH-POUNDS.** Install cotter pin (Figure 2).
7. Lower tail landing gear (WP 1675 00).
8. Service shock strut upper and lower stages (WP 1599 00).
9. Install tail landing gear fairing (WP 0291 00).

REPLACE TAIL LANDING GEAR YOKE LOCKPIN BUSHING

Remove

1. Turn off all electrical power.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

2. Using a nonmetallic scraper and technical acetone, Item 5, WP 1803 00, remove old sealing compound from bushing.
3. Drive lockpin bushing from yoke bore using a nonmetallic drift and nonmetallic hammer (Figure 6, Detail A).

REPLACE TAIL LANDING GEAR YOKE LOCKPIN BUSHING - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

4. Remove old sealing compound from bushing bore with technical acetone, Item 5, WP 1803 00, and abrasive mat, Item 2, WP 1803 00. Wipe bore and exterior surfaces clean with cheesecloth cloth, Item 85, WP 1803 00.

Inspect

1. Turn off all electrical power.
2. Visually inspect bushing bore for corrosion.
3. Remove corrosion and blend out pits, using abrasive cloth (Item 81, WP 1803 00). Diameter of lockpin bushing bore shall be 0.5625 to 0.5631-inch. Replace yoke if not within limits.

Install

1. Turn off all electrical power.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

2. Wipe new bushing clean with technical acetone, Item 5, WP 1803 00, and clean cheesecloth cloth, Item 85, WP 1803 00. Wipe dry.
3. Apply a coat of sealing compound, Item 273, WP 1803 00, to bushing outside surfaces.
4. Line up new lockpin bushing over yoke bore with square raised portion facing rear (Figure 6, Detail B).
5. **QA** Press new bushing into yoke while sealing compound is wet. Check that bushing shoulder is bottomed on yoke (Figure 6, Detail A).
6. Wipe extra sealing compound out of bushing bore.

REPLACE TAIL LANDING GEAR TEFLON COATED WASHERS**Remove**

1. Remove tail landing gear fork from yoke (WP 0463 00).

CAUTION

Damage to yoke will occur if a metallic drift is used to remove washers. Metal will scratch yoke. Use a nonmetallic drift to remove washers.

2. Using a nonmetallic drift, remove Teflon coated washer(s) from yoke (Figure 7).
3. Using a nonmetallic scraper, remove sealing compound from yoke.
4. If necessary, remove any remaining sealing compound with a clean cheesecloth cloth, Item 85, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.

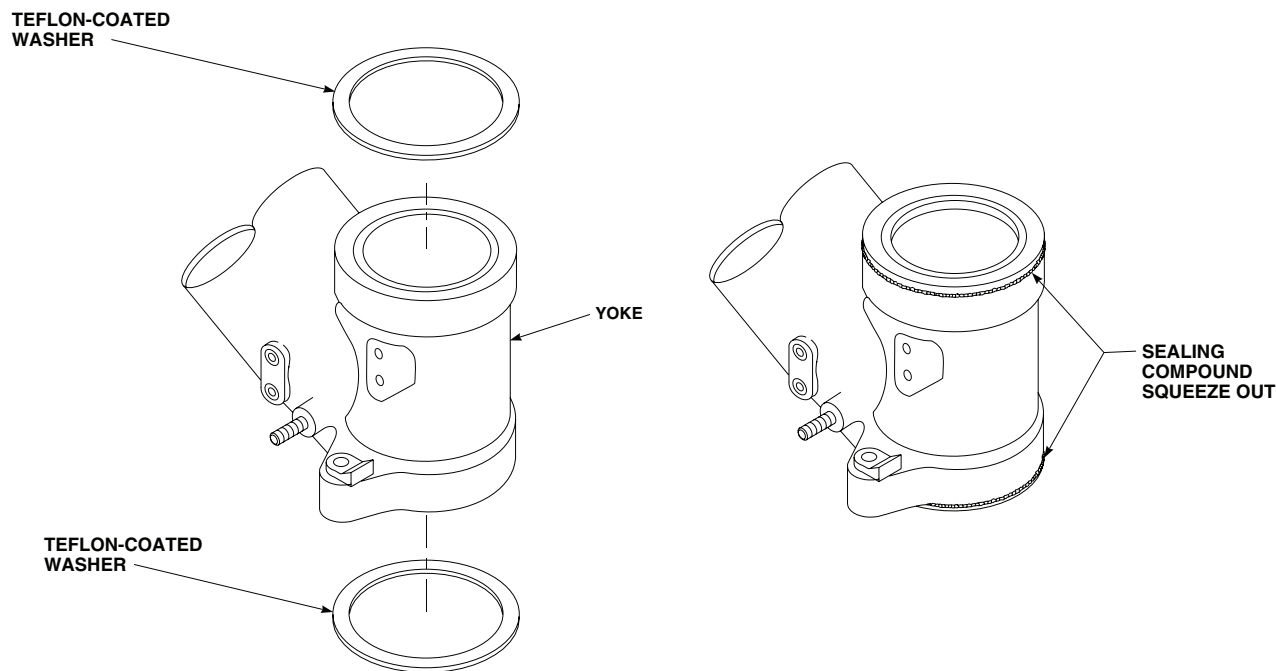
REPLACE TAIL LANDING GEAR TEFLON COATED WASHERS - ContinuedAK0673A
SA

Figure 7. Replace Tail Landing Gear Teflon Coated Washers.

Install

1. Using paint brush, Item 61, WP 1803 00, apply uniform coat, approximately 0.010 thick, of sealing compound, Item 11, WP 1803 00, to metallic surface of washer(s).
2. **QA** Install washer(s) on yoke. Apply pressure to Teflon surface of washer(s). Make sure that sealing compound has squeezed out uniformly around entire circumference of washer(s) (Figure 7).
3. If a continuous squeeze out is not achieved, remove washer(s) and repeat step 1. and step 2.
4. If necessary, make sealing compound flush with Teflon surface of washer(s) using a nonmetallic scraper. Allow sealing compound to cure for a minimum of 24 hours.
5. Install tail landing gear fork on yoke (WP 0463 00).

REPLACE TAIL LANDING GEAR TEFLON COATED SLEEVES**Remove****CAUTION**

Damage to yoke will occur if a metallic drift is used to remove sleeves. Metal will scratch yoke. Use a nonmetallic drift to remove sleeves.

1. Remove sleeves with a nonmetallic drift (Figure 8).
2. Clean yoke bore using clean cheesecloth cloth, Item 85, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00. Remove all grime.

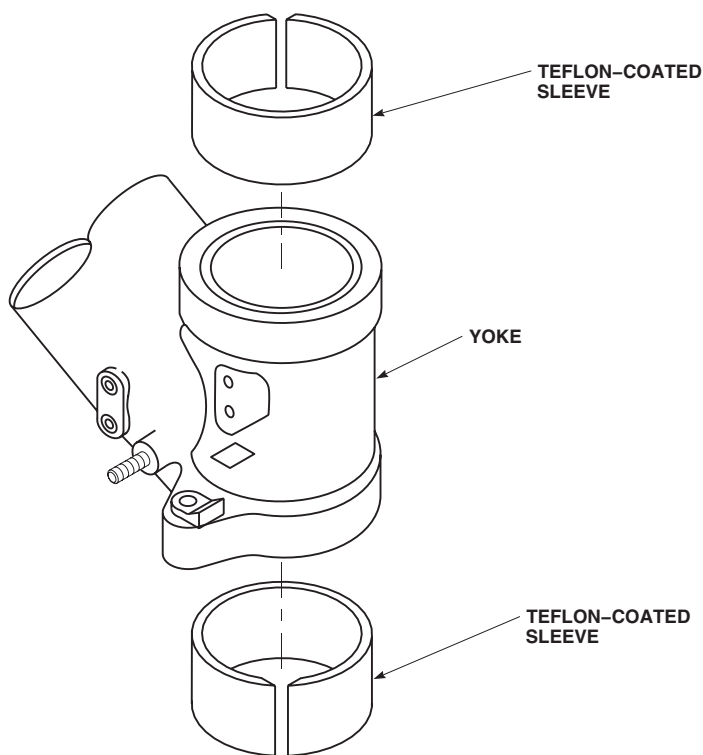
REPLACE TAIL LANDING GEAR TEFLON COATED SLEEVES - ContinuedAA9542
SA

Figure 8. Replace Tail Landing Gear Teflon Coated Sleeves.

Inspect

1. Inspect yoke bore for corrosion.
2. Repair yoke bore. Refer to, REPAIR, in this work package.

Install**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

1. Wipe yoke bore and sleeve mating surfaces with clean cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00. Wipe dry with clean cheesecloth cloth, Item 85, WP 1803 00. Continue cleaning until no more soil shows on a clean cloth. Do not allow solvent to dry on parts. Wipe parts dry.
2. Apply a coat of epoxy primer coating kit, Item 120, WP 1803 00, to sleeve OD and yoke bore.
3. **QA** Hand-press sleeve into yoke while primer is wet. Check that sleeve is bottomed in yoke bore (Figure 8).

REPLACE TAIL LANDING GEAR TEFLON COATED SLEEVES - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

4. Wipe primer off Teflon coated surfaces with a clean cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.

REPLACE TAIL LANDING GEAR YOKE FASTENERS.**Remove**

1. Remove electrical connectors. Refer to, REPLACE TAIL LANDING GEAR YOKE ASSEMBLY, REPLACE, in this work package.
2. If necessary, remove actuator (WP 0464 00).
3. Clean sealing compound from brackets using nonmetallic scraper.
4. Drill out rivets and remove bracket.

Install

1. Clean remaining sealing compound using machinery wiping towel, Item 344, WP 1803 00, dampened with cleaning compound solvent, Item 71, WP 1803 00.
2. Install new brackets using rivets (TM 1-1500-204-23).
3. Seal all pockets and joints from moisture with sealing compound, Item 273, WP 1803 00.
4. If necessary, install actuator (WP 0464 00).
5. Inspect and treat electrical connector (WP 1703 00).
6. Install electrical connectors. Refer to, REPLACE TAIL LANDING GEAR YOKE ASSEMBLY, REPLACE, in this work package.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR**TAIL LANDING GEAR FORK**

This WP supersedes WP 0463 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Gloves
 Goggles/Face Shield
 Fluorescent Penetrant Inspection Kit, XMA10
 Hand Reamer Set, Adjustable, GGG-R-180
 Heat Gun, LFT500
 Medium-Cut File
 Micrometer, 0 to 1 Inch
 Micrometer, 1 to 2 Inch
 Nonmetallic Drift
 Nonmetallic Scraper
 Tail Landing Gear Lockpin Bushing Drift, Locally-Made, Figure H-204, WP 1805 00
 Tail Landing Gear Oversized Axle Bushing, Locally-Made, Figure H-210, WP 1805 00
 Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Carbon Dioxide, Item 64, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 81, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Cloth, Cheesecloth, Item 85, WP 1803 00
 Corrosion Preventive Compound, Item 104, WP 1803 00

Corrosion Preventive Compound, Item 105, WP 1803 00
 Corrosion Resistant Coating, Item 110, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Insulation Sleeving, Electrical, Item 160, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Steering Assembly, Tail Gear, 70700-20374-041
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Assistant - (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-344-23
 TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0071 00
 WP 0381 00
 WP 0465 00
 WP 0475 00
 WP 1675 00
 WP 1803 00
 WP 1805 00

REPLACE TAIL LANDING GEAR FORK**Remove**

1. Turn off all electrical power.
2. Set parking brake.
3. Jack tail landing gear (WP 1675 00).
4. Remove wheel and tire (WP 0465 00).
5. Place lockpin in DISENGAGED position.

REPLACE TAIL LANDING GEAR FORK - Continued

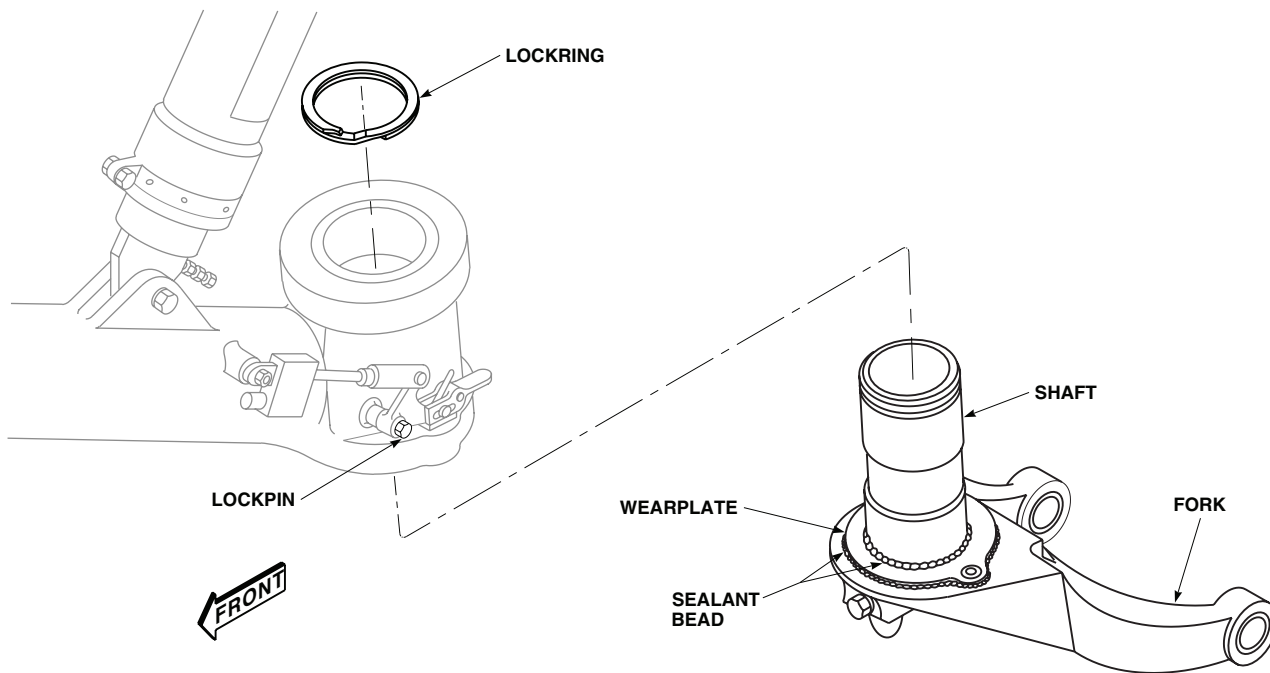
AK0678
SA

Figure 1. Replace Tail Landing Gear Fork.

CAUTION

Damage to equipment will occur if fork is not supported prior to removing locking.
Fork will drop from yoke. Support fork prior to removing locking.

6. Remove insulation sleeving from yoke if installed.
7. Remove locking and fork from yoke (Figure 1).

Inspect/Repair

1. Visually inspect locking for corrosion (TM 1-1500-344-23). Corrosion is indicated by red film or pitting.
2. Remove corrosion and blend out pits, using abrasive cloth, Item 81, WP 1803 00. **LOCKRING SHALL NOT BE LESS THAN 0.108-INCH THICK OR LESS THAN 0.264-INCH WIDE AFTER REPAIR.** Replace locking if not within limits.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves, goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

3. Clean locking and fork shaft using machinery wiping towel, Item 344, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00. Wipe dry using clean machinery towel, Item 344, WP 1803 00.

REPLACE TAIL LANDING GEAR FORK - Continued

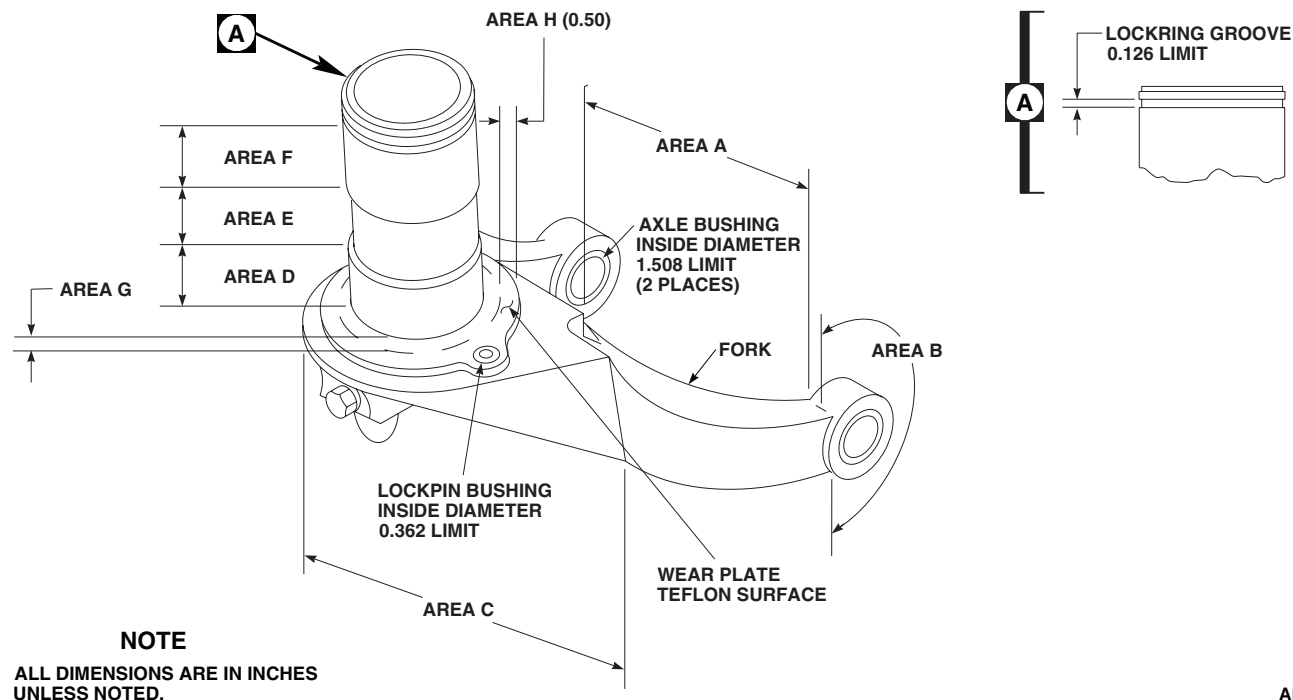


Figure 2. Repair Tail Landing Gear Fork.

4. Inspect Teflon coated wear plate as follows:

- Measure wear from Area H of Teflon coated wear plate to Area G of wear plate (Figure 2). **MAXIMUM WEAR SHALL NOT EXCEED 0.010-INCH.**
- Check Teflon coated wear plate Area G and Area H for scoring and corrosion. **SCORING OR CORROSION SHALL NOT EXCEED 20% OF TEFLON SURFACE.**

NOTE

- Absence of Teflon is acceptable if surface is sprayed with corrosion preventive compound, Item 104, WP 1803 00, every 30 days to prevent corrosion.
 - Disbonding, up to a total peripheral length of 50% with no indication of corrosion or wear plate movement must be repaired to the next 30 hour/43 day inspection. Inspect for corrosion on a daily basis until wear plate is replaced.
- Check Teflon coated wear plate for disbonding. **DISBONDING OF TEFLON WEAR PLATE SHALL NOT EXCEED A TOTAL OF 50% OF TOTAL PERIPHERAL LENGTH. NO CORROSION OR MOVEMENT OF PLATE IS ALLOWED.**
 - Check difference between thickness at inner diameter (Area G) and outer diameter (Area H). **DIFFERENCE SHALL NOT EXCEED 0.010-INCH.**
 - Replace Teflon coated wear plate if damage is more than limits. Refer to, REPLACE TAIL LANDING GEAR FORK LOCKPIN BUSHING AND JACKPAD, in this work package.

5. Inspect aluminum/bronze wear plate as follows:

REPLACE TAIL LANDING GEAR FORK - Continued

- a. Measure wear from Area H of aluminum/bronze wear plate to Area G of wear plate (Figure 2). **MAXIMUM WEAR SHALL NOT EXCEED 0.040-INCH.**
 - b. Check aluminum/bronze wear plate Area G and Area H for scoring and corrosion. **SCORING OR CORROSION SHALL NOT EXCEED 20% OF AREA G AND 30% OF AREA H ON WEAR PLATE SURFACE** (Figure 2).
 - c. Check aluminum/bronze wear plate for disbonding. **DISBONDING OF ALUMINUM/BRONZE WEAR PLATE SHALL NOT EXCEED A TOTAL OF 50% OF TOTAL PERIPHERAL LENGTH. NO CORROSION OR MOVEMENT OF PLATE IS ALLOWED.**
 - d. Replace aluminum/bronze wear plate if damage is more than limits. Refer to, REPLACE TAIL LANDING GEAR FORK WEARPLATE, in this work package.
6. Measure lockring groove for wear (Figure 2). **GROOVE WIDTH SHALL NOT BE MORE THAN 0.126-INCH.** Replace fork if groove width is more than limit.
 7. Measure axle bushings inside diameter (Figure 2). **LARGEST INSIDE DIAMETER SHALL BE NO MORE THAN 1.508-INCHES.** Replace axle bushings if measurement is more than limit. Refer to, REPLACE TAIL LANDING GEAR FORK AXLE BUSHINGS, in this work package.
 8. Measure lockpin bushing inside diameter. **LARGEST INSIDE DIAMETER SHALL BE NO MORE THAN 0.362-INCH.** Replace bushing if measurement is more than limit. Refer to, REPLACE TAIL LANDING GEAR FORK LOCKPIN BUSHING AND JACKPAD, in this work package.

NOTE

In the following steps, do all blending with an abrasive wheel, Item 350, WP 1803 00, abrasive cloth, Item 82, WP 1803 00, or a medium-cut file, as necessary.

9. Repair Area A (Figure 2) as follows:
 - a. **BLEND OUT NICKS, DENTS, SCRATCHES, AND GOUGES UP TO 0.062-INCH DEEP.** Replace fork if damage is more than limit after blending.
 - b. Measure length of repair. Measure length of four sides of fork leg at repair. **LENGTH OF REPAIR IN DIRECTION AROUND FORK LEG SHALL NOT BE MORE THAN 10% OF TOTAL LENGTH AROUND LEG.** Replace fork if damage is more than limit after repairs.
10. Repair Area B (Figure 2) as follows:
 - a. **BLEND OUT NICKS, DENTS, SCRATCHES, AND GOUGES UP TO 0.031-INCH DEEP AND 1/4-INCH IN DIAMETER.**
 - b. Replace fork if damage is more than limit after repair.
11. Repair Area C (Figure 2) as follows:
 - a. **BLEND OUT NICKS, DENTS, SCRATCHES, AND GOUGES UP TO 0.094-INCH DEEP AND 1/2-INCH IN DIAMETER.**
 - b. Replace fork if damage is more than limit after repair.
12. Repair Area D and Area F (Figure 2) as follows:
 - a. **CHECK CHROME-PLATING ON JOURNALS FOR CHIPPING, FLAKING, SCORING, OR WEAR THROUGH TO BASE METAL.** Replace fork if this damage is found.
 - b. Measure diameters of journals. **DIAMETERS SHALL BE NOT LESS THAN 3.497-INCHES.** Replace fork if journals measure below limit.

REPLACE TAIL LANDING GEAR FORK - Continued

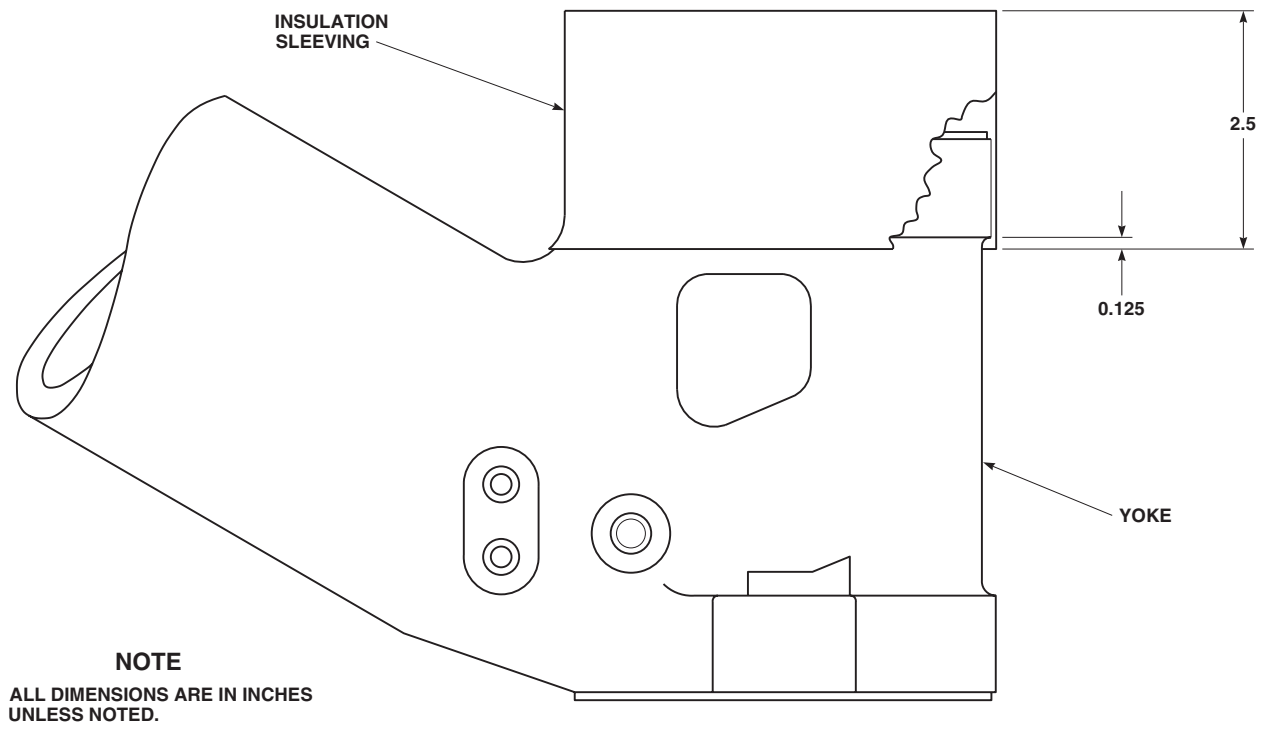


Figure 3. Replace Tail Landing Gear Fork Insulation Sleeving.

13. Repair Area E (Figure 2) as follows:

- a. **BLEND OUT NICKS, DENTS, SCRATCHES, AND GOUGES UP TO 0.010-INCH DEEP.**
- b. Measure length of repair. **REPAIR AFTER BLENDING, SHALL NOT BE MORE THAN 1.062-INCHES LONG.** Replace fork if measurement is more than limits.

14. Fluorescent penetrant inspect reworked areas for cracks by fluorescent penetrant inspection kit (TM 1-1520-265-23). **NONE ALLOWED.** Replace cracked fork.

15. Apply corrosion resistant coating, Item 110, WP 1803 00, to reworked areas. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).

Install

1. Turn off all electrical power.
2. Apply a bead of sealing compound, Item 273, WP 1803 00, around fork shaft between fork and wear plate and outer edge of wear plate (Figure 1). Using a nonmetallic scraper, make sealing compound flush with top surface of wear plate.
3. **QA** Apply corrosion preventive compound, Item 105, WP 1803 00 or Item 110, WP 1803 00, to exposed shaft and lockring.
4. Cut a piece of electrical insulation sleeving, Item 160, WP 1803 00, to a length of 2.5-inches.
5. **QA** Install insulation sleeving over top of yoke assembly until it extends 0.125-inch below shoulder of yoke (Figure 3).
6. Shrink insulation sleeving around body of yoke and over top of fork using heat gun.

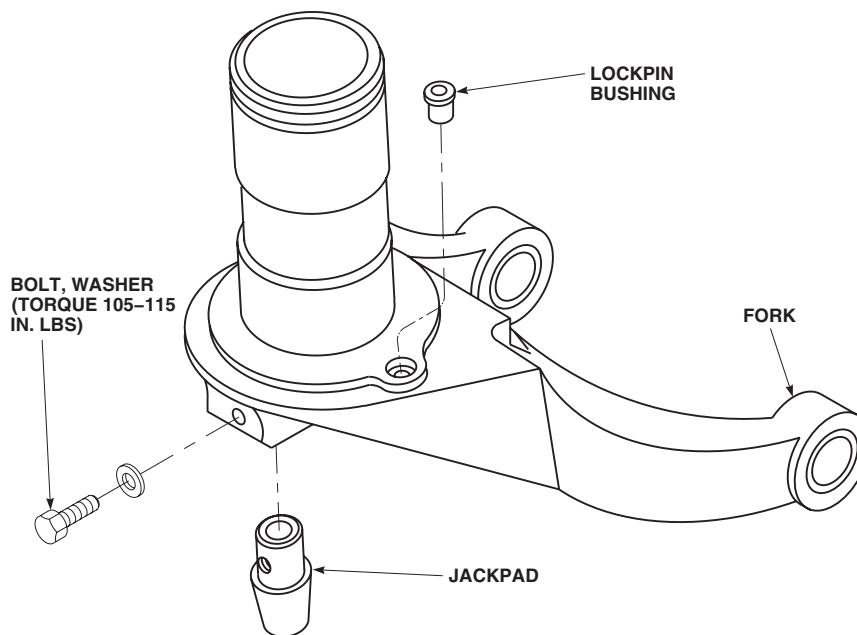
REPLACE TAIL LANDING GEAR FORK - ContinuedAK0681
SA

Figure 4. Replace Tail Landing Gear Fork Lockpin Bushing and Jackpad.

7. Place fork in trail position. Place lockpin in ENGAGED position.
8. Install wheel and tire (WP 0465 00).
9. Lower tail landing gear and remove jack (WP 1675 00).

REPLACE TAIL LANDING GEAR FORK LOCKPIN BUSHING AND JACKPAD**Remove**

1. Locally make tail landing gear lockpin bushing drift (WP 1805 00).
2. Install tail gear steering assembly 70700-20374-041.
3. Disengage lockpin.
4. Have assistant operate brakes.
5. Move chocks 8 to 12 inches away from front and rear of main gear tires.
6. Release parking brake.
7. Have assistant push on the right side of tail cone while turning tailwheel 20 to 30 degrees to right so lockpin bushing is moved clear of yoke-lockpin boss.
8. Chock main wheels. Set parking brake.
9. Working from the underside of fork, press or drive lockpin bushing up out of fork using tail landing gear lockpin bushing drift (Figure 4).
10. If removing jackpad, remove bolt and washer and remove from fork.

REPLACE TAIL LANDING GEAR FORK LOCKPIN BUSHING AND JACKPAD - Continued**Inspect**

Check lockpin bushing hole in fork for wear and elongation. Repair worn or elongated lockpin bushing hole (WP 0475 00).

Install

1. Put on a light coat of sealing compound, Item 273, WP 1803 00, around new bushing.
2. **QA** Press bushing into fork.
3. **QA** If jackpad was removed, install it in fork and line up hole in jackpad with hole in fork.
4. **QA** Install bolt and washer. **TORQUE BOLT TO 105 - 115 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolt to fork.
5. Have assistant operate brakes.
6. Release parking brake.
7. Move chocks 8 to 12 inches away from front and rear of main gear tires.
8. Have assistant push on left side of tailcone while turning tailwheel to trail position. Place lockpin in ENGAGED position.
9. Lock and unlock tailwheel a few times, to check operation of lockpin.
10. Remove tail gear steering assembly 70700-20374-041.
11. Do an operational check of tail wheel locking system (WP 0071 00).

REPLACE TAIL LANDING GEAR FORK WEARPLATE**Remove**

1. Remove fork. Refer to, REPLACE TAIL LANDING GEAR FORK, in this work package.

CAUTION

Damage to fork will occur if a metallic drift is used to remove wearplate. Metal will scratch fork. Use a nonmetallic drift to remove wearplate.

2. Using a nonmetallic drift, remove wearplate from fork (Figure 5).
3. Using a nonmetallic scraper, remove sealing compound from fork.
4. If necessary, remove any remaining sealing compound with a clean cheesecloth cloth, Item 85, WP 1803 00 dampened with cleaning compound solvent, Item 71, WP 1803 00.

Install

1. Using a paint brush, apply uniform coat, approximately 0.010-inch thick, of sealing compound, Item 273, WP 1803 00, to wearplate (side with chamfer on inside diameter).
2. **QA** Install wearplate on fork with adhesive coated surface down (Figure 5). Apply pressure to Teflon surface of wearplate. Make sure that sealing compound has squeezed out uniformly around entire circumference of wearplate.
3. If a continuous squeeze out is not achieved, remove wearplate and repeat step 1. and step 2.
4. Install lockpin bushing. Refer to, REPLACE TAIL LANDING GEAR FORK LOCKPIN BUSHING AND JACKPAD, in this work package.

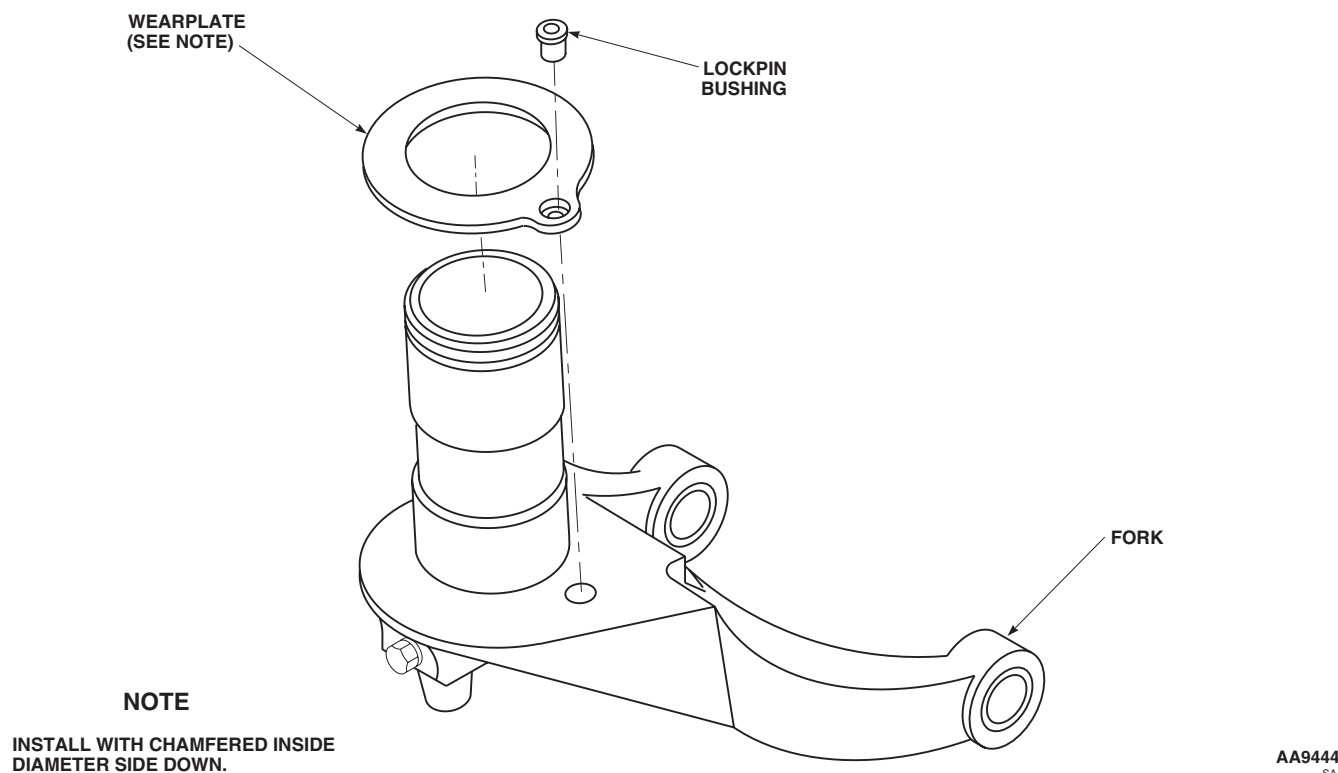
REPLACE TAIL LANDING GEAR FORK WEARPLATE - Continued

Figure 5. Replace Tail Landing Gear Fork Wearplate.

5. Apply a bead of sealing compound, Item 273, WP 1803 00, around fork shaft between fork and wearplate and outer edge of wearplate (Figure 6).
6. If necessary, make sealing compound flush with Teflon surface of wearplate using a nonmetallic scraper. Allow sealing compound to cure for a minimum of 72 hours.
7. Install fork. Refer to, REPLACE TAIL LANDING GEAR FORK, in this work package.

REPLACE TAIL LANDING GEAR FORK AXLE BUSHINGS**Remove**

1. Place fork in arbor press and carefully remove bushings (Figure 7).
2. Inspect fork inside diameter for damage. Nicks and scratches up to 0.031-inch deep may be repaired by reaming. Lug may be reamed up to 1.6875-inches in diameter.

Install

1. Locally made oversized axle bushing (WP 1805 00) must be used for reworked lugs. Interference fit between bushing and lug shall be 0.001 to 0.0035-inch.

WARNING

Injury to personnel will occur if protective gloves are not worn when handling carbon dioxide or chilled parts. Wear gloves when handling carbon dioxide or chilled parts.

2. Chill bushing in carbon dioxide, Item 64, WP 1803 00, to ease installation.

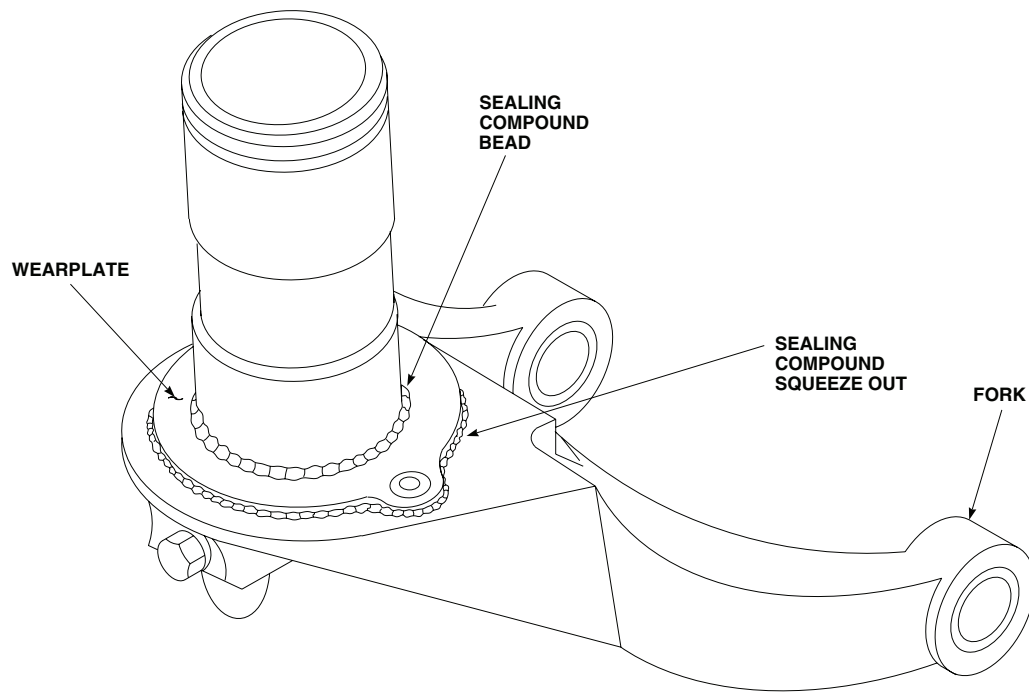
REPLACE TAIL LANDING GEAR FORK AXLE BUSHINGS - ContinuedAK0683A
SA

Figure 6. Tail Landing Gear Fork Seal Wearplate.

3. Remove bushing from carbon dioxide, coat with epoxy primer coating kit, Item 120, WP 1803 00.
4. **QA** Install bushing while primer is still wet, using arbor press (Figure 7).
5. Allow bushings to return to room temperature. Line-ream both bushings to 1.502 to 1.503-inches, using hand reamer set.
6. Clean ID of bushings using machinery wiping towel, Item 344, WP 1803 00.

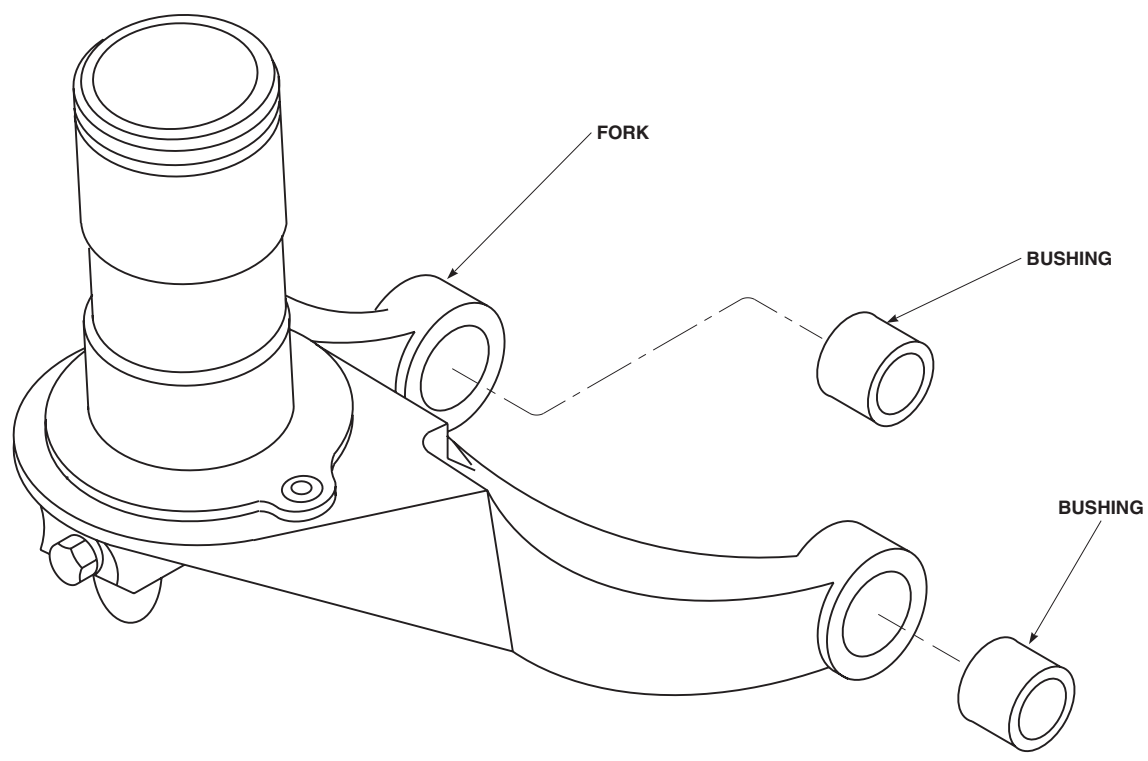
REPLACE TAIL LANDING GEAR FORK AXLE BUSHINGS - Continued

Figure 7. Replace Tail Landing Gear Fork Axle Bushings.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR**TAIL LANDING GEAR LOCK MECHANISM**

This WP supersedes WP 0464 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Gloves
 Goggles/Face Shield
 Tail Landing Gear Lock Mechanism Bellcrank Bushing
 Drift, Locally-Made, Figure H-203, WP 1805 00
 Torque Wrench, 30 - 200 in. lbs.

Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0071 00
 WP 1668 00
 WP 1703 00
 WP 1803 00
 WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Cheesecloth, Item 85, WP 1803 00
 Corrosion Preventive Compound, Item 104,
 WP 1803 00

REPLACE TAIL LANDING GEAR LOCK MECHANISM LOCK ACTUATOR**Remove**

1. Turn off all electrical power.
2. Disconnect electrical connector, P507, from actuator (Figure 1).
3. Unpin actuator and housing from bellcrank.
4. Remove nut and washers, and remove actuator from yoke.
5. Loosen jamnut on actuator shaft and unscrew adapter, with housing, from shaft (Figure 1, Detail C).
6. Remove lockwasher and jamnut from actuator shaft. Keep lockwasher and jamnut for reinstallation.
7. If damaged, remove snapring, bushing, collars, spring, and housing from adapter (Figure 1, Detail A).

Disassemble

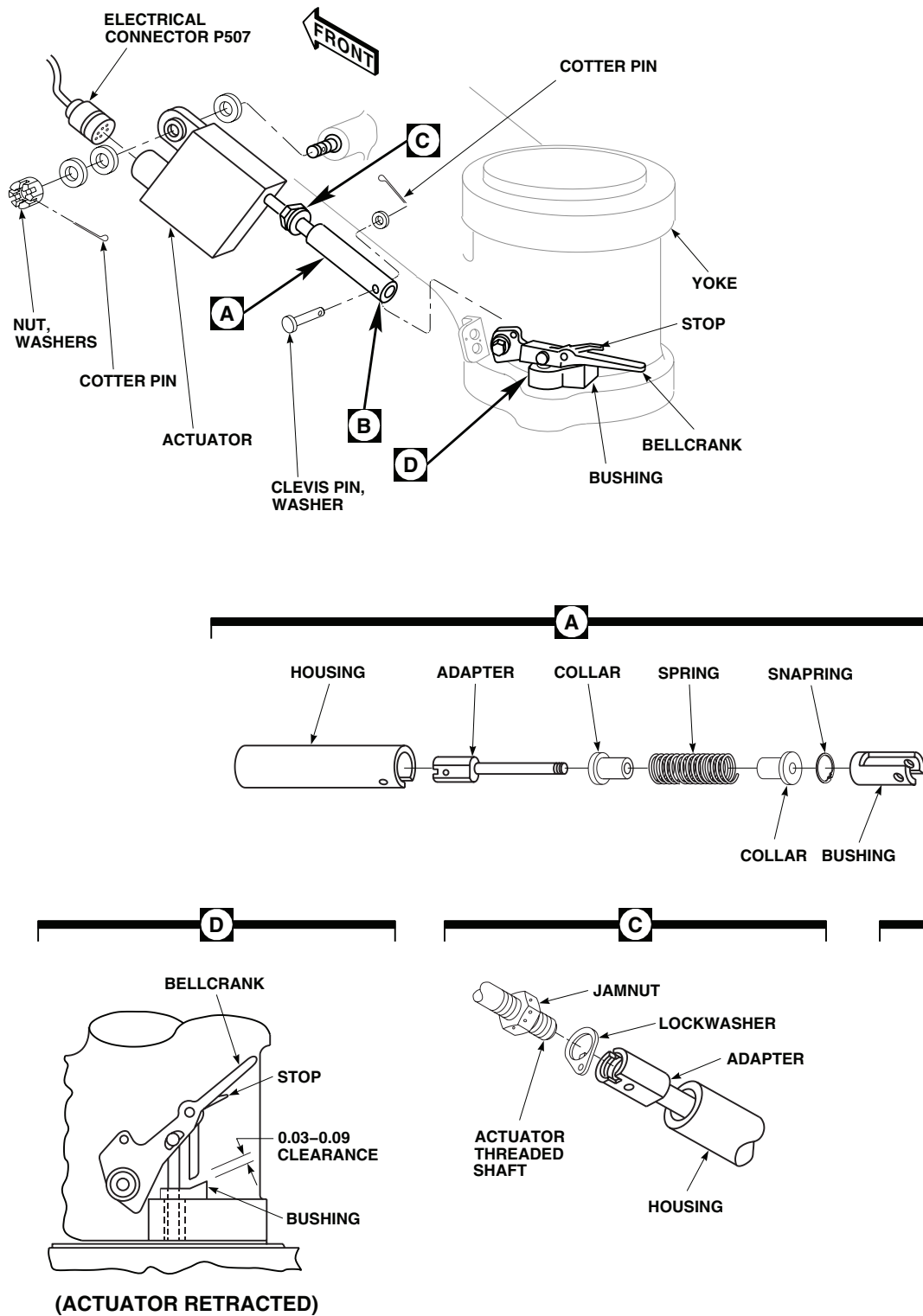
1. Remove cotter pin, washer, and clevis pin from housing. Remove washer from clevis pin (Figure 1).
2. Remove bushing.
3. Slide out adapter, with parts attached, from housing (Figure 1, Detail A).

WARNING

Injury to personnel and damage to equipment will result if care is not taken when removing collar. Spring is under pressure. Take care when removing collar.

4. Remove snapring from adapter groove while restraining collar, to avoid sudden release of spring pressure.
5. Remove two collars and spring.

REPLACE TAIL LANDING GEAR LOCK MECHANISM LOCK ACTUATOR - Continued

**NOTE**

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

AB2530
SA

Figure 1. Replace Tail Landing Gear Lock Mechanism Lock Actuator.

REPLACE TAIL LANDING GEAR LOCK MECHANISM LOCK ACTUATOR - Continued

6. Clean all parts with cleaning compound solvent, Item 71, WP 1803 00.

Inspect

1. Inspect parts for corrosion, damage, cracks, or deformation. **NONE ALLOWED.**
2. Replace damaged parts.

Assemble

1. Apply corrosion preventive compound, Item 104, WP 1803 00, to spring and to inside surfaces of housing.
2. **QA** Install adapter in housing (Figure 1, Detail A).
3. **QA** Coat inside and outside of bushing with corrosion preventive compound, Item 104, WP 1803 00, and install in housing with holes aligned.
4. **QA** Install clevis pin, washer, and cotter pin (Figure 1).

Install

1. Turn off all electrical power.
2. Lubricate tail wheel lock pin (WP 1668 00).
3. If housing is disassembled, put corrosion preventive compound, Item 104, WP 1803 00, on housing inside surfaces.
4. **QA** If housing is disassembled install housing, spring, collars, snapring, and bushing on adapter (Figure 1, Detail A).
5. **QA** Thread jamnut, lockwasher, and adapter with housing on actuator shaft (Figure 1, Detail C).
6. **QA** Pin housing and bushing to bellcrank with clevis pin. Install washer and cotter pin (Figure 1).
7. **QA** Install actuator on yoke with washers and nut. Install cotter pin.
8. Inspect and treat electrical connector (WP 1703 00).
9. **QA** Connect electrical connector, P507, to actuator.
10. Adjust throw of actuator as follows:
 - a. Turn on electrical power and retract actuator.
 - b. **ADJUST ADAPTER ON ACTUATOR SHAFT, SO THAT WHEN STOP ON BELLCRANK IS IN ENGAGED POSITION ON TOP OF BUSHING ON YOKE, THERE IS A 0.03 TO 0.09-INCH CLEARANCE BETWEEN STOP AND BUSHING (FIGURE 1, DETAIL D).**
 - c. Tighten jamnut against lockwasher and adapter. **TIGHTEN 1/6 TO 1/3 TURN BEYOND POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire jamnut to lockwasher (Figure 1, Detail C).
 - d. Squeeze and release stop to make sure it clears bushing on yoke. If stop hangs up on bushing, readjust adapter.
11. Do an operational check of tailwheel lock system (WP 0071 00).

REPLACE TAIL LANDING GEAR LOCK MECHANISM**Remove**

1. Turn off all electrical power.
2. Remove clevis pin from actuator shaft housing and disconnect actuator from bellcrank (Figure 2).
3. Unlock tail wheel.

REPLACE TAIL LANDING GEAR LOCK MECHANISM - Continued

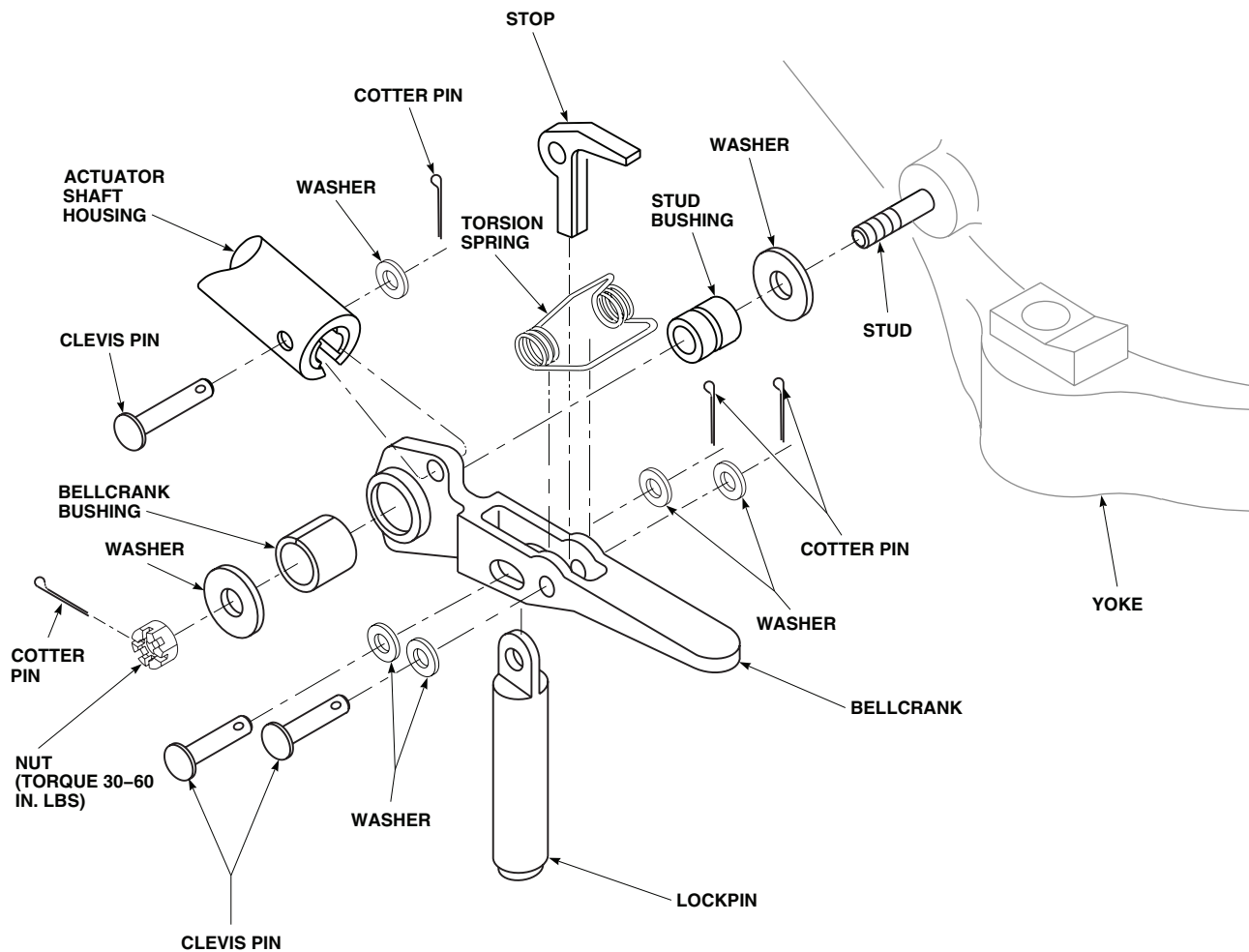


Figure 2. Replace Tail Landing Gear Lock Mechanism.

4. Remove clevis pin, stop, and torsion spring from bellcrank.
5. Remove clevis pin from bellcrank and lockpin.
6. Remove nut, washers, bellcrank, and stud bushing from stud.
7. Remove lockpin from yoke.

Inspect/Repair

1. Check stud bushing, lockpin, bellcrank, bellcrank bushing, clevis pins, stop, and torsion spring for cracks, wear, and corrosion. Repair or replace worn or corroded parts (Figure 2).
2. Locally make tail landing gear lock mechanism bellcrank bushing drift (WP 1805 00).
3. Remove worn bellcrank bushing using drift.

REPLACE TAIL LANDING GEAR LOCK MECHANISM - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Wipe replacement bushing and bellcrank mating surfaces clean using clean cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry.
5. Wet bushing and bellcrank mating surface with epoxy primer coating kit, Item 120, WP 1803 00.
6. Press bushing into bellcrank while primer is wet. Wipe primer out of bushing bore.

Install

1. Turn off all electrical power.
2. **QA** Install lockpin in yoke (Figure 2).
3. **QA** Install washer (with Teflon face outboard) on stud.
4. **QA** Install stud bushing and bellcrank on stud. Install washer (with Teflon face inboard) and nut. **TORQUE NUT TO 30 - 60 INCH-POUNDS.** Install cotter pin.
5. **QA** Fasten lockpin to bellcrank with clevis pin, washers, and cotter pin.
6. **QA** Install stop and torsion spring in bellcrank. Install clevis pin, washer, and cotter pin.
7. **QA** Install actuator shaft housing on bellcrank. Install clevis pin, washer, and cotter pin.
8. Do an operational check of lock mechanism system (WP 0071 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR**TAIL LANDING GEAR AXLE, WHEEL AND TIRE**

This WP supersedes WP 0465 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 High Pressure Nitrogen Servicing Cart, PD-EIG-90-2
 Tire Bead Breaker, 5372
 Tire Cage, NSN 4920-01-545-0763 SK2000TC-6
 Tire Inflator Assembly, M85352/1or M008348-1-1
 Torque Wrench, 30 - 200 in. lbs.

Nitrogen, Item 199, WP 1803 00
 Sealing Compound, Item 273, WP 1803 00
 Tags
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Alcohol, Denatured, Item 48, WP 1803 00
 Antiseize Compound, Item 51, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 79, WP 1803 00
 Grease, Aircraft, Item 130, WP 1803 00
 Grease, Aircraft, Item 131, WP 1803 00
 Lubricant, Inner Tube, Item 179, WP 1803 00

References

TM 55-2620-200-24
 WP 0476 00
 WP 1600 00
 WP 1675 00
 WP 1803 00

REPLACE TAIL LANDING GEAR AXLE, WHEEL, AND TIRE**Remove**

1. Park helicopter on level surface. Chock main landing gear wheels.
2. Jack tail landing gear (WP 1675 00).
3. Set parking brake. Leave brake on until wheel is replaced.

WARNING

Injury to personnel will result if valve core from valve stem is removed, before the tire is completely deflated. Release pressure from tire before removing valve core from valve stem.

4. Remove valve cap from valve stem. Completely deflate tire, and remove valve core from valve stem. Reinstall valve core. Attach deflation tag.

REPLACE TAIL LANDING GEAR AXLE, WHEEL, AND TIRE - Continued

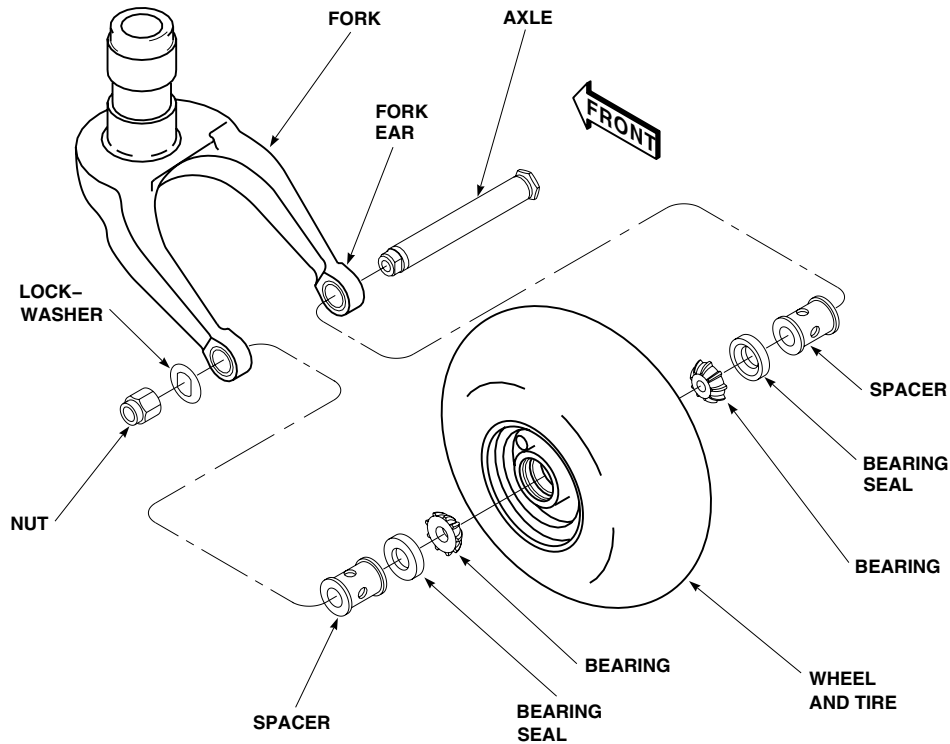
AK0687A
SA

Figure 1. Replace Tail Landing Gear Axle, Wheel, and Tire.

CAUTION

Damage to spacers, bearing seals, and bearings will result if dropped when removing axle. Spacers, bearing seals, and bearings will drop free when axle is removed. Catch bearings as axle is removed to avoid damaging cones.

NOTE

If wheel is being replaced, keep bearings and bearing seals with wheel and forward to intermediate level for repair.

5. Straighten tab on lockwasher. Remove nut, lockwasher, axle, wheel and tire and spacers from fork (Figure 1).

Inspect**WARNING**

Injury to personnel and damage to equipment will result if tire pressure is more than 20 psi prior to installation. Make sure tire pressure is 20 psi or less prior to installation.

1. Inspect wheel for obvious damage (TM 55-2620-200-24).
2. Check bearing cups in wheel halves for looseness, scratches, pitting, corrosion, and signs of overheating. **NONE ALLOWED.** Replace damaged cup (WP 0476 00).

REPLACE TAIL LANDING GEAR AXLE, WHEEL, AND TIRE - Continued**CAUTION**

Damage to bearings will result if they are allowed to spin while drying. Secure inner and outer race of bearings while drying.

3. Clean bearings with cleaning compound solvent, Item 71, WP 1803 00. Dry bearings with compressed air, if available, or air-dry them.
4. Inspect bearings for scratches, water stains, heat discoloration, worn rollers, or with a cracked or distorted bearing cage. **NONE ALLOWED.** Replace bearing if any of these conditions are found.
5. If bearing is determined serviceable, repack it with aircraft grease, Item 131, WP 1803 00.
6. Blend out nicks, scratches, or gouges on axle with abrasive wheel, Item 350, WP 1803 00, abrasive cloth, Item 79, WP 1803 00, or a medium cut file. **DEEPEST PART OF REPAIRED AREA MUST NOT BE OVER 0.005-INCH AFTER BLENDING. SCRATCHES AT RIGHT ANGLES TO CENTERLINE MUST NOT BE MORE THAN 1 INCH IN LENGTH.**

NOTE

Clamp-up forces will increase if length of spacer is less than 2.064-inches.

7. Measure length of spacer. If length of spacer is less than 2.064-inches, replace spacer.

Install

1. Lubricate bearing cups in wheel, and bearing seals with aircraft grease, Item 131, WP 1803 00.
2. **QA** Carefully install bearings and bearing seals in wheel (Figure 1).
3. Lubricate axle with aircraft grease, Item 131, WP 1803 00.
4. **QA** Place tailwheel and tire between fork ears while making sure bearings and seals do not fall out of wheel.
5. **QA** Place both spacers between fork ear and bearing seal, slide axle through fork ear, first spacer, wheel, second spacer and fork ear.

NOTE

Reuse of lockwasher is permitted for not more than three times, provided it is not cracked, corroded, or discolored, and new lockwasher is not available. If lockwasher is reused, bend shall not overlap or cross a previous bend.

6. **QA** Install lockwasher and nut on axle. Spin wheel and tighten nut until wheel can no longer be rotated by hand. Loosen nut until wheel rotates freely with no binding. Bend lockwasher over flat on nut. **NO END PLAY ALLOWED IN WHEEL ASSEMBLY.**

REPLACE TAIL LANDING GEAR AXLE, WHEEL, AND TIRE - Continued**WARNING**

- Injury to personnel or damage to equipment will result if nitrogen source is not vented at 140 psi maximum. Make sure to check tire inflator assembly nitrogen source is vented at 140 psi maximum.
 - Injury to personnel or damage to equipment will result if tire is inflated beyond recommended pressure. Do not inflate tire beyond its rated inflation pressure.
 - Injury to personnel or damage to equipment will result if an unregulated pressure source is used. Use an regulated pressure source to service tire.
 - Injury to personnel will result if the tire sidewall is facing personnel. Make sure personnel face tire tread and not tire sidewall when inflating tire.
 - Injury to personnel, or damage to equipment will result, if a tire inflation safety cage is not used while servicing tires. Use a tire inflation safety cage when inflating tire.
7. **INFLATE TIRE TO 115 TO 125 PSI** with nitrogen, Item 199, WP 1803 00 (TM 55-2620-200-24). Use a regulated nitrogen pressure source connected to tire inflator assembly.
 8. After a 5 to 10 minute waiting period, check wheel for leakage as follows:
 - a. Moisten air valve and around tire beads.
 - b. If bubbles appear, deflate, disassemble, and correct at point of leakage. Refer to, **REPLACE TAIL LANDING GEAR TIRE/TUBE**, in this work package.
 9. Install valve cap on valve stem.
 10. Lower tail landing gear and remove jack (WP 1675 00).
 11. Seal around nut and axle using sealing compound, Item 273, WP 1803 00.

REPLACE TAIL LANDING GEAR TIRE/TUBE**Remove****WARNING**

Injury to personnel and damage to equipment will occur when air pressure is not completely released from tire before removing tire from wheel. Wheel could explode. Deflate tire completely before removing valve core from valve stem and tire from wheel.

1. If not already done, completely deflate tire and remove valve core.

CAUTION

- Damage to component's protective finish will result if the bead breaker is not used on a clean flat surface. Disassemble and assemble wheel on a clean flat surface to protect finish.
 - Damage to components will occur if hand tools are used to pry between tire bead and wheel. Do not pry between tire bead and wheel with hand tools.
2. Break tire beads away from wheel flanges by applying even pressure to tire sides using tire bead breaker.

REPLACE TAIL LANDING GEAR TIRE/TUBE - Continued

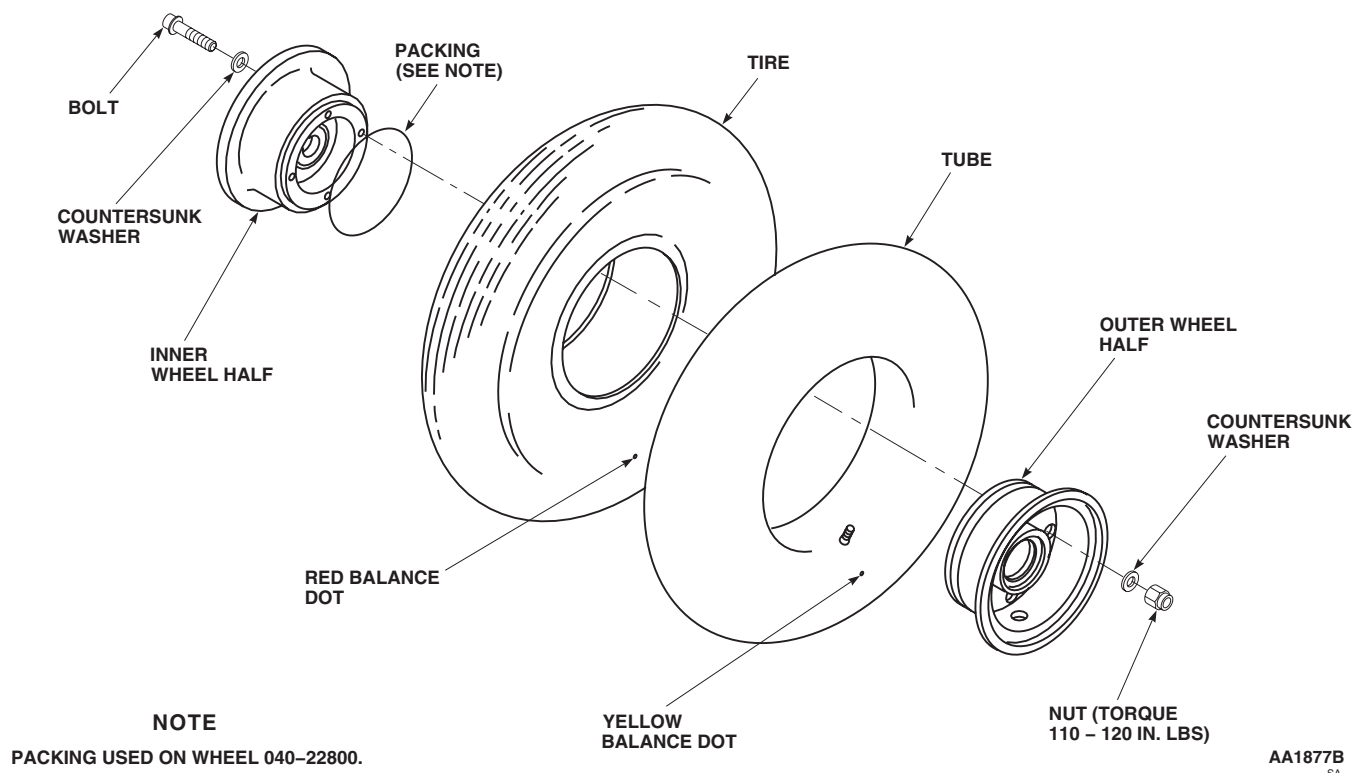


Figure 2. Replace Tail Landing Gear Tire/Tube.

CAUTION

Damage to wheel, wheel bolts, and nuts will result if impact or power wrenches are used to remove wheel bolt nuts. Do not use impact or power wrenches to remove wheel bolt nuts.

3. Remove bolts, countersunk washers, and nuts. Separate wheel halves (Figure 2).
4. On wheel, 040-22800, remove and discard packing on inner wheel half.
5. Remove tire from wheel half.
6. Remove tube from tire.

WARNING

Injury to personnel or damage to equipment may occur if wheel halves are not inspected whenever wheel is disassembled. Inspect wheel halves every time wheel is disassembled.

7. Inspect and repair wheel (WP 0476 00 00).

Inspect

1. Replace damaged tire.
2. Inspect new tire beads for damage (TM 55-2620-200-24). **NONE ALLOWED.**
3. Inspect wheel bolts for thread damage and cracks under bolt heads. Replace any bolt found damaged.

REPLACE TAIL LANDING GEAR TIRE/TUBE - Continued

4. Inspect nuts for obvious damage. Thread bolts into nuts by hand. Replace any nut that allows bolt to thread past top of nut.

Install

1. Before inserting inner tube in tire, dust tire generously with inner tube lubricant, Item 179, WP 1803 00. Clean wheel flanges and tire beads and remove any compound that settles on beads, using machinery wiping towel, Item 344, WP 1803 00, dampened with denatured alcohol Item 48, WP 1803 00.
2. **QA** Install valve core in stem.
3. Insert inner tube in tire, aligning red dot on tire with valve stem.
4. Using a regulated source of nitrogen, Item 199, WP 1803 00, connected to tire inflator assembly, inflate tire slowly until beads start to spread. Probe between tube and tire with fingers, to release possible trapped air.
5. **QA** On wheel, 040-22800, lubricate packing using aircraft grease, Item 130, WP 1803 00, and install packing on inner wheel half.
6. **QA** Carefully place tire on inner wheel half.
7. **QA** Place other wheel half in tire and line up boltholes in both wheel halves. Put antiseize compound, Item 51, WP 1803 00, on bolt and nut threads, face of nuts, under boltheads, and both sides of washers.

CAUTION

Damage to wheel, wheel bolts, and nuts will result if impact or power wrenches are used to remove wheel bolt nuts. Do not use impact or power wrenches to tighten or torque wheel bolts or nuts.

8. **QA** Bolt wheel halves together, with boltheads on side of wheel opposite valve stem. Install countersunk washers so countersunk sides are toward boltheads and nut faces, respectively. **TIGHTEN NUTS IN A CRISSCROSS PATTERN IN EVEN UNITS OF 30 INCH-POUNDS, UNTIL A FINAL TORQUE OF 110 - 120 INCH-POUNDS IS OBTAINED.**
9. Place wheel in safety cage (TM 55-2620-200-24).

WARNING

- Injury to personnel or damage to equipment will result if nitrogen source is not vented at 140 psi maximum. Make sure to check tire inflator assembly nitrogen source is vented at 140 psi maximum.
 - Injury to personnel or damage to equipment will result if tire is inflated beyond recommended pressure. Do not inflate tire beyond its rated inflation pressure.
 - Injury to personnel or damage to equipment will result if an unregulated pressure source is used. Use an regulated pressure source to service tire.
 - Injury to personnel will result if the tire sidewall is facing personnel. Make sure personnel face tire tread and not tire sidewall when inflating tire.
 - Injury to personnel, or damage to equipment will result, if a tire inflation safety cage is not used while servicing tires. Use a tire inflation safety cage when inflating tire.
10. Using a regulated source of nitrogen, Item 199, WP 1803 00, connected to tire inflator assembly, inflate tire with enough pressure to seat tire beads. **DO NOT EXCEED 110 PSI.**

REPLACE TAIL LANDING GEAR TIRE/TUBE - Continued**WARNING**

Failure to use proper procedures during servicing of main or tail landing gear tires may result in serious injury or death.

11. **ONCE TIRE IS CORRECTLY SEATED, INFLATE IT TO 115 TO 125 PSI.** Use primary or alternate methods for inflation (WP 1600 00).
12. After a 5 to 10 minute waiting period, check wheel for leakage as follows:
 - a. Moisten air valve and around tire beads.
 - b. If air bubbles appear, deflate, disassemble, and correct at point of leakage.
13. **REDUCE TIRE PRESSURE TO 20 PSI.** Tag wheel, stating psi in tire. Tire will be inflated to its rated inflation pressure after wheel is installed on helicopter.
14. Install valve cap on stem.
15. Remove wheel from safety cage.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR
MAIN LANDING GEAR SHOCK STRUT (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Caliper, Micrometer, Set, GGG-C-105
 Container, 1-Gallon
 Hydraulic Repairer's Toolkit, SC 5180-99-A03
 Hydraulic Shutoff Valve, M1400S
 Hydraulic Test Stand, MIL-T-7612D
 Inside Micrometer, Set, GGG-C-105
 Mallet, Rawhide
 Pail, Utility, RR-P-35
 Plastic/Wooden Pick
 Puller Kit, Mechanical, GGG-P-781
 Pump, Inflating, Manual, 2357
 Torque Wrench, 30 - 200 in. lbs.

Leak Detection Fluid, Item 175, WP 1803 00
 Nitrogen, Item 199, WP 1803 00
 Petrolatum, Item 216, WP 1803 00
 Screw, MS35275-247
 Tags
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wire, Nonelectrical, Item 354, WP 1803 00
 Wire, Nonelectrical, Item 356, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Pneudraulics Repairer MOS 15H (1)
 UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Hydraulic Fluid, Fire Resistant, Item 144, WP 1803 00
 Hydraulic Fluid, Petroleum Base, Item 145,
 WP 1803 00

References

TM 1-1520-265-23
 WP 0445 00
 WP 0446 00
 WP 0467 00
 WP 1597 00
 WP 1803 00

DISASSEMBLE

1. Clean strut exterior using cleaning compound solvent, Item 71, WP 1803 00, and machinery wiping towel, Item 344, WP 1803 00.

NOTE

Work area and tools must be clean. Use a surface covered with clean machinery wiping towel, Item 344, WP 1803 00, for disassembled parts.

2. Remove upper and lower stage air valve caps. Loosen outer hexnut and depressurize strut upper and lower stages. Leave valves open (Figure 1, Sheet 1, Detail A).

WARNING

Injury to personnel and damage to equipment will occur if shock strut is disassembled without removing upper and lower bleeder plugs. Residual pressure will cause shock strut piston to blow out during disassembly. Make sure hydraulic bleeder plugs are slowly removed to release possible residual pressure resulting from leakage from air side to hydraulic side of shock strut.

3. Remove upper and lower stage bleeder plugs. Discard packings. Drain hydraulic fluid from strut into 1 gallon container. Discard used fluid.

DISASSEMBLE - Continued

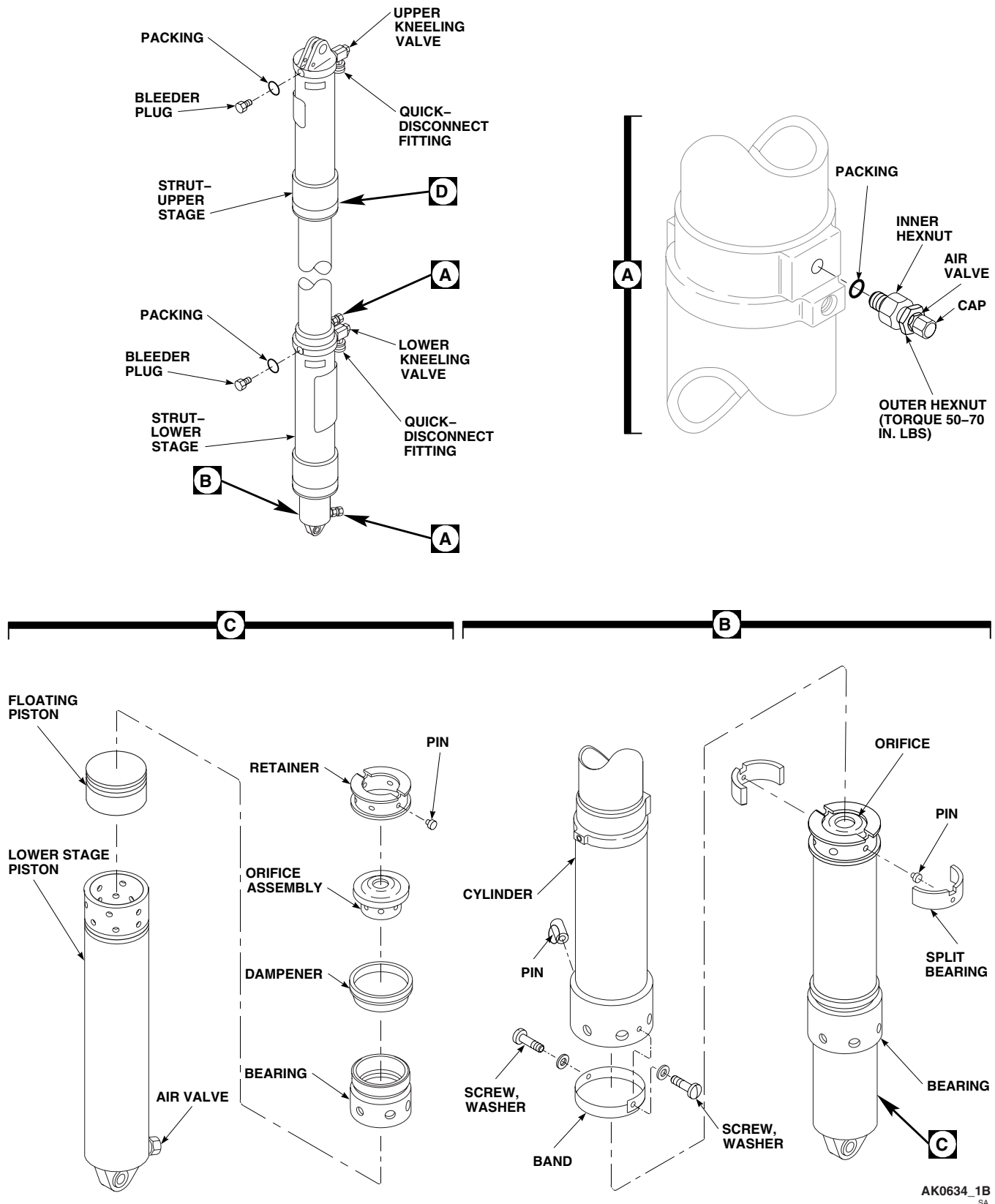


Figure 1. Disassemble Main Landing Gear Shock Strut (AVIM). (Sheet 1 of 2)

DISASSEMBLE - Continued

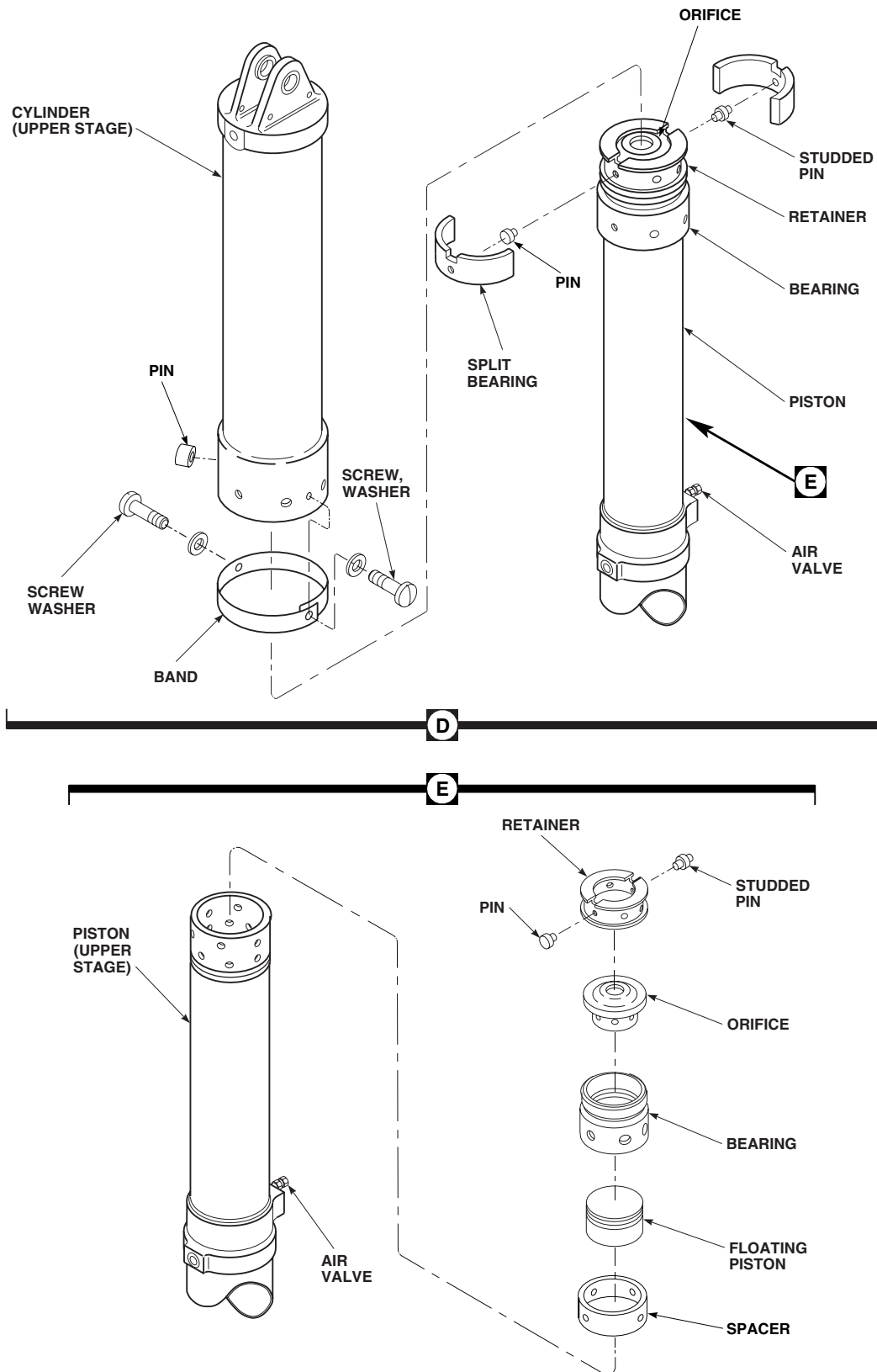
AK0634_2
SA

Figure 1. Disassemble Main Landing Gear Shock Strut (AVIM). (Sheet 2 of 2)

DISASSEMBLE - Continued

4. Remove upper and lower kneeling valves. Discard packings.
5. Lay strut down. Remove screws, washers, and band. Remove 6 pins using screw, MS35275-247, and puller (Figure 1, Sheet 1, Detail B).

NOTE

Split bearing halves may fall free as piston clears cylinder.

6. Separate piston from cylinder. Catch split bearing halves as piston is removed from cylinder.
7. Remove 6 pins, retainer, orifice assembly, dampener, and bearing (Figure 1, Sheet 1, Detail C). Do not disassemble orifice. Tag orifice for reassembly.

WARNING

Injury to personnel and damage to equipment will result if excessive air pressure is used to remove floating piston. Slowly operate hand pump at low pressure when removing floating piston.

8. Attach a manual air pump to lower air valve. Apply 2 to 5 psi pressure. Catch floating piston in a pail filled with machinery wiping towel, Item 344, WP 1803 00. Remove manual air pump.
9. Remove lower stage air valve. Discard packing.
10. Remove screws, washers, and band (Figure 1, Sheet 2, Detail D).
11. Remove 6 pins using screw, MS35275-247, and puller.

NOTE

Split bearing halves may fall free as piston clears cylinder.

12. Separate upper piston from cylinder. Catch split bearing halves as piston is removed from cylinder.
13. Remove 6 pins, retainer, orifice, and bearing (Figure 1, Sheet 2, Detail E). Do not disassemble orifice.

WARNING

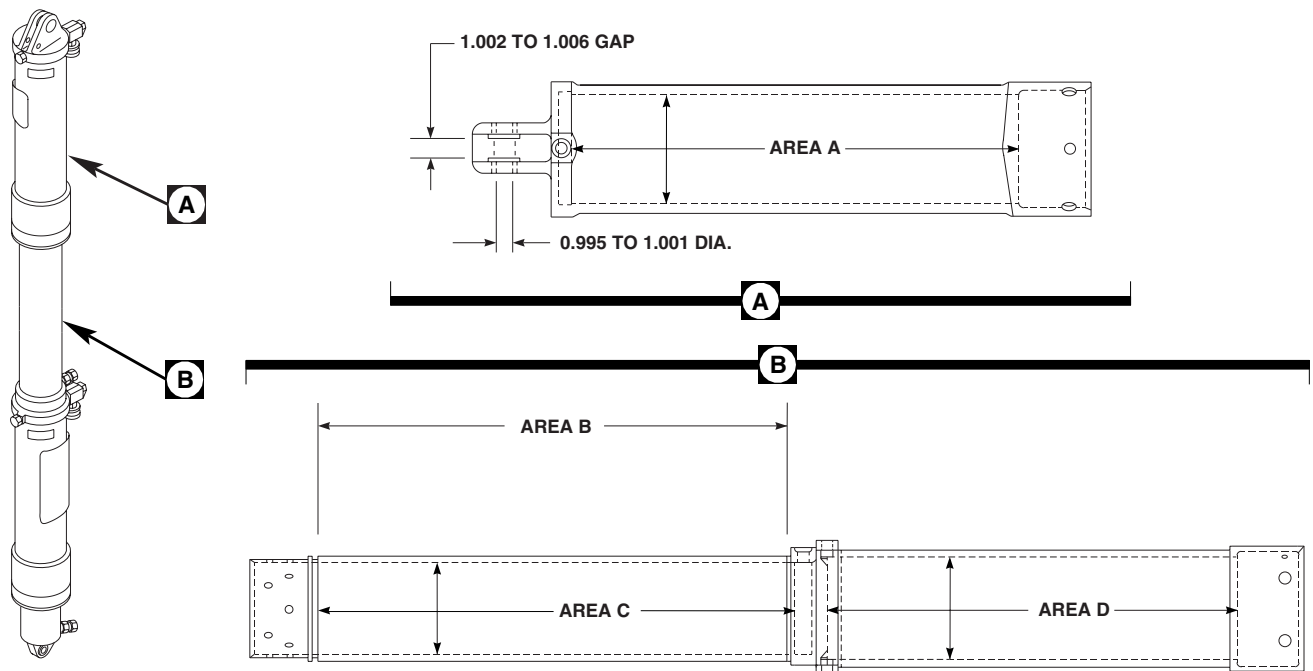
Injury to personnel and damage to equipment will result if excessive air pressure is used to remove floating piston. Slowly operate hand pump at low pressure when removing floating piston.

14. Attach manual air pump to upper air valve. Apply 2 to 5 psi pressure. Catch floating piston and spacer in a pail filled with machinery wiping towel, Item 344, WP 1803 00.
15. Remove upper air valve (Figure 1, Sheet 1, Detail A). Discard packing.
16. Remove all packings, seals, and scrapers using a plastic or wooden pick. Discard used packings, seals, and scrapers.
17. Clean parts with cleaning compound solvent, Item 71, WP 1803 00. Blow dry with dry compressed air.

INSPECT

1. Examine strut parts for cracks, scoring, scratches, and corrosion. Use metal flaw detection procedures as necessary to verify suspected cracks or fractures (TM 1-1520-265-23). **NO SCORING, CRACKS, OR CORROSION ALLOWED.** Replace damaged parts.
2. Inspect upper cylinder mounting lug bushings for wear. **BUSHING ID SHALL BE 0.995 TO 1.001-INCHES. DISTANCE BETWEEN BUSHINGS SHALL BE 1.002 TO 1.006-INCHES (FIGURE 2, DETAIL A).** Replace cylinders having worn bushings.

INSPECT - Continued

**NOTE**

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

AB2531
SA

Figure 2. Inspect Main Landing Gear Shock Strut Upper Stage (AVIM).

3. Check piston assembly chrome plated surface for surface blemishes or discolorations. Buff out surface blemishes or discolorations using abrasive cloth, Item 82, WP 1803 00. **PISTON OD SHALL NOT BE LESS THAN 3.495-INCHES AFTER BUFFING.** Slight discoloration after buffing is acceptable providing minimum dimensions and surface finish is maintained.
4. Check piston and cylinder chrome plated surfaces for wear. Take measurements along several points, 90 degrees apart as follows:
 - a. **AREA A - UPPER STAGE CYLINDER ID SHALL NOT BE MORE THAN 4.003-INCHES (FIGURE 2, DETAIL A).** Replace cylinder if out of tolerance.
 - b. **AREA B - UPPER STAGE PISTON OD SHALL NOT BE LESS THAN 3.495-INCHES (FIGURE 2, DETAIL B).** Replace piston/cylinder assembly if out of tolerance.
 - c. **AREA C - UPPER STAGE PISTON BORE ID SHALL NOT BE MORE THAN 3.336-INCHES (FIGURE 2, DETAIL B).** Replace piston/cylinder assembly if out of tolerance.
 - d. **AREA D - LOWER STAGE CYLINDER ID SHALL NOT BE MORE THAN 4.002-INCHES (FIGURE 2, DETAIL B).** Replace cylinder assembly if out of tolerance.
 - e. **AREA E - LOWER STAGE PISTON BORE ID SHALL NOT BE MORE THAN 3.336-INCHES (FIGURE 3).** Replace piston/cylinder assembly if out of tolerance.
 - f. **AREA F - LOWER STAGE PISTON OD SHALL NOT BE LESS THAN 3.495-INCHES (FIGURE 3).** Replace piston if out of tolerance.
5. Check split bearing halves for wear at several points around circumference. **SPLIT BEARING THICKNESS SHALL NOT BE LESS THAN 0.092-INCH (FIGURE 4, DETAIL A).** Replace bearings if out of tolerance.

INSPECT - Continued

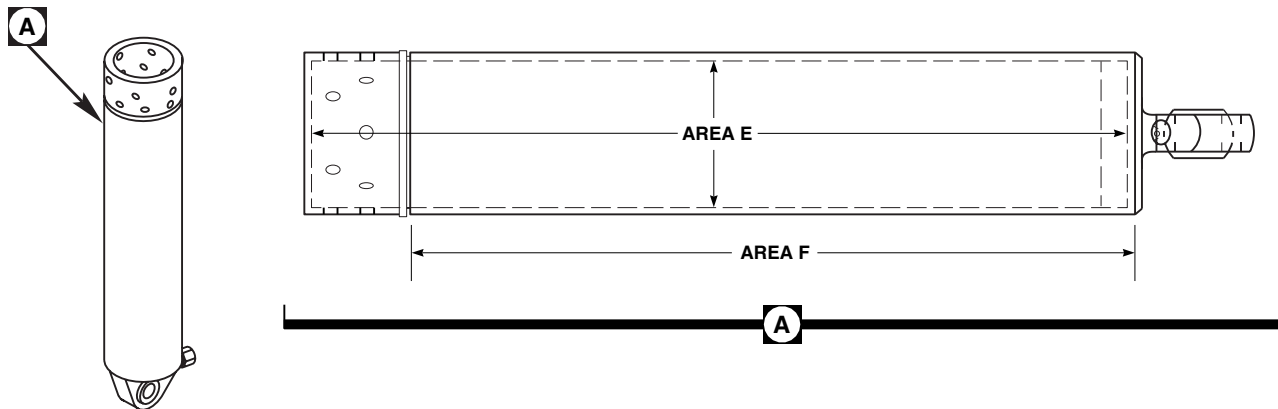
FB3320
SA

Figure 3. Inspect Main Landing Gear Shock Strut Lower Stage (AVIM).

6. Check retainer for wear at several points around OD. **ON STRUT, 70250-12051-042, UPPER AND LOWER STAGE RETAINERS SHALL MEASURE NOT LESS THAN 3.993-INCHES. ON STRUTS, 70250-12051-043 THROUGH 70250-12051-045, UPPER AND LOWER STAGE RETAINERS SHALL MEASURE NOT LESS THAN 3.973-INCHES (FIGURE 4, DETAIL B).** Replace retainer if out of tolerance.
7. Check bearings for wear at several points around ID. **ON STRUT, 70250-12051-042, UPPER AND LOWER STAGE BEARING ID SHALL MEASURE NOT MORE THAN 3.503-INCHES. ON STRUTS, 70250-12051-043 THROUGH 70250-12051-045, UPPER AND LOWER STAGE BEARING ID SHALL MEASURE NOT MORE THAN 3.506-INCHES (FIGURE 4, DETAIL C).** Replace out of tolerance bearings.
8. Check floating pistons for wear at several points around OD. **UPPER AND LOWER STAGE FLOATING PISTONS SHALL MEASURE NOT LESS THAN 3.326-INCHES (FIGURE 4, DETAIL D).** Replace floating pistons if out of tolerance.
9. Check lower stage piston bearing for maximum radial play. **LOWER STAGE PISTON BEARING MAY HAVE UP TO 0.010-INCH-MAXIMUM RADIAL PLAY AND 0.030-INCH-MAXIMUM AXIAL PLAY.** Replace bearing if out of tolerance (WP 0467 00).
10. Check Teflon liner for disbonding. **NO DISBONDING ALLOWED.** Replace bearing if liner is disbonded (WP 0467 00).

ASSEMBLE

1. Make sure all parts are clean.

NOTE

- If strut is to be stored, use fire resistant hydraulic fluid, Item 144, WP 1803 00, as an assembly and servicing fluid.
- If strut is scheduled for immediate return to service, use petroleum base hydraulic fluid, Item 145, WP 1803 00, as an assembly and servicing fluid.

2. Soak packings, retainers, G-T rings, and seals in hydraulic fluid before installation.
3. Coat all chrome-plated sliding and threaded parts with hydraulic fluid before assembly.

ASSEMBLE - Continued

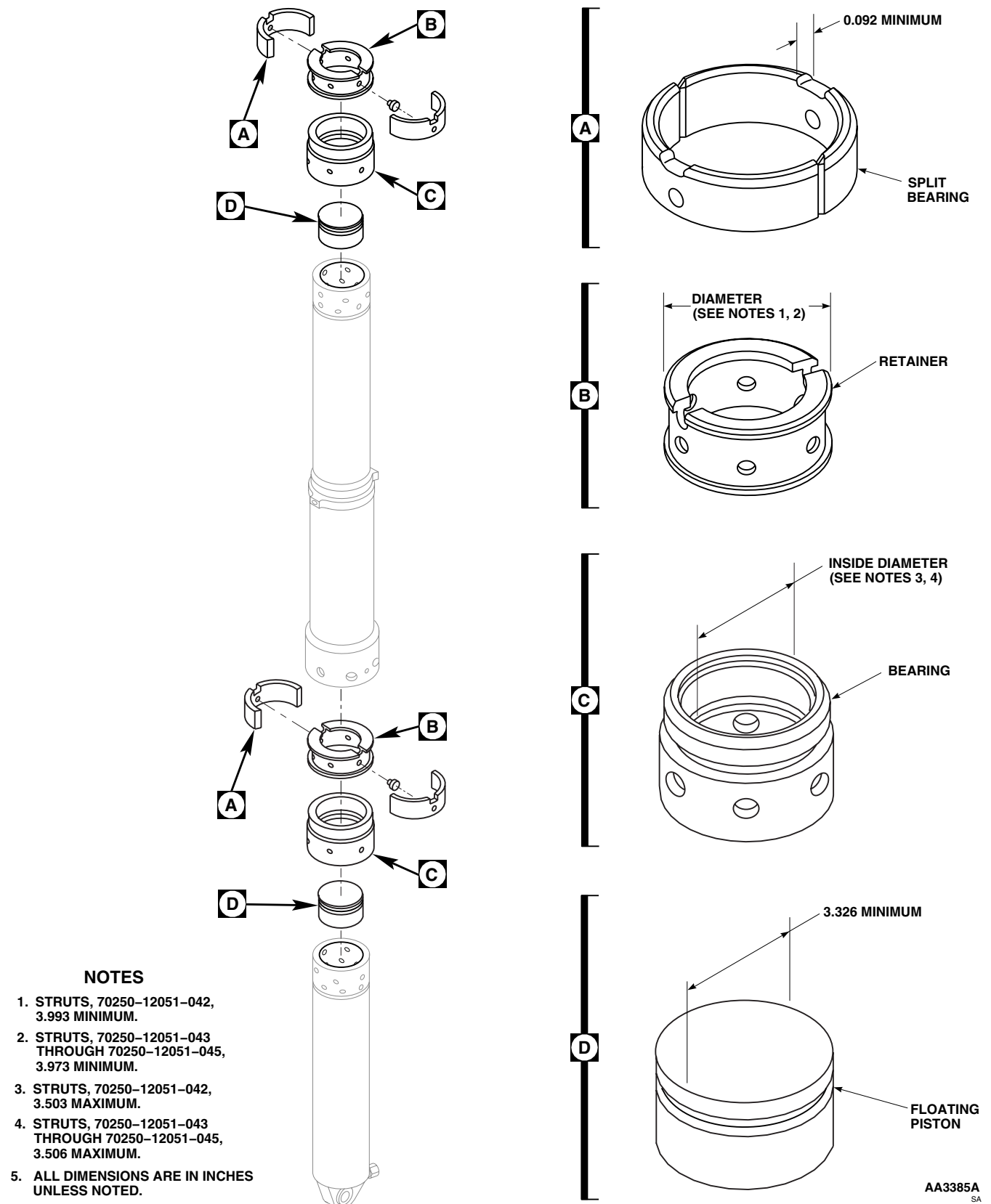


Figure 4. Inspect Main Landing Gear Shock Strut Upper And Lower Stage Components (AVIM).

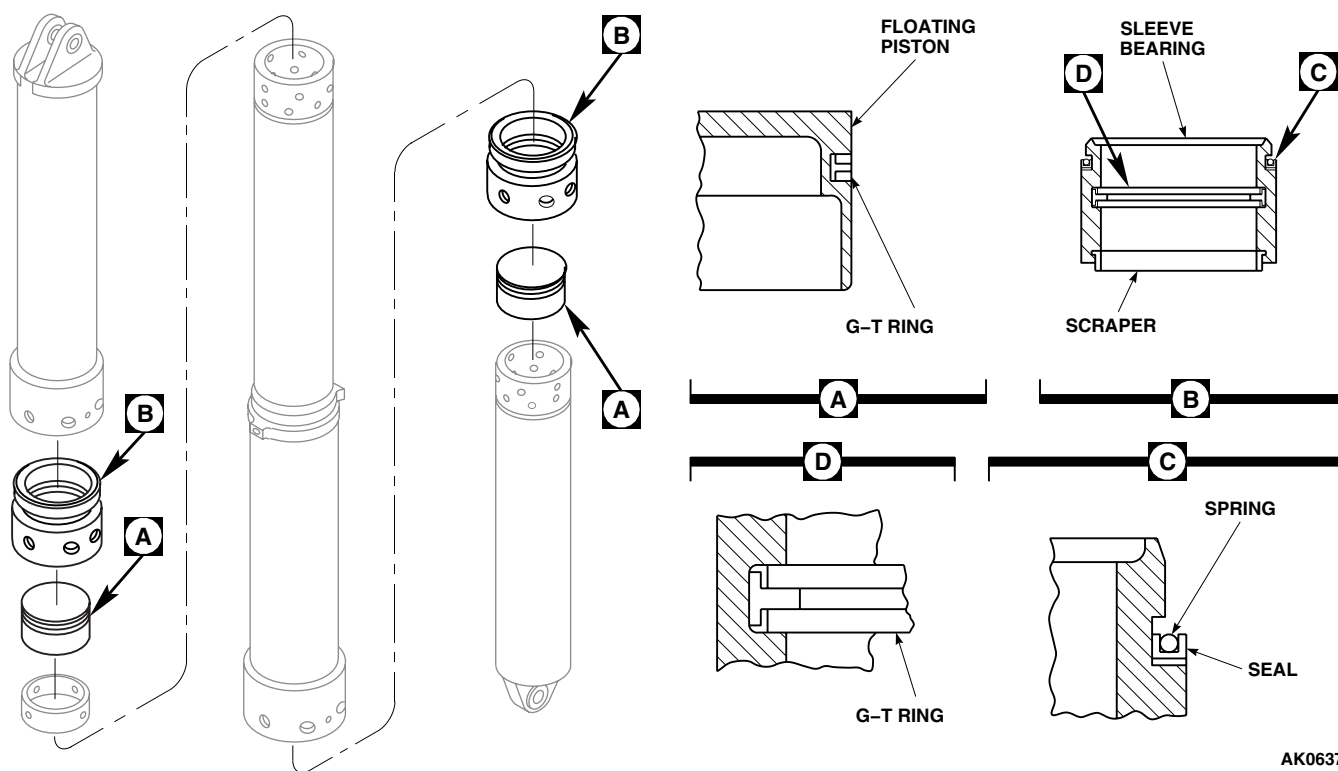
ASSEMBLE - Continued

4. **QA** Install G-T rings on upper and lower stage floating pistons, with radius of two Teflon rings facing center packing (Figure 5, Detail A).
5. **QA** Install G-T rings inside upper and lower bearings (Figure 5, Detail D).
6. **QA** Install seals on OD of upper and lower bearings, with springs facing tops of sleeves (Figure 5, Detail C).
7. **QA** Install scrapers, inside lower edge of bearings (Figure 5, Detail B).
8. **QA** Install spacer in upper stage piston with holes closest to spacer edge toward bottom. Line up one spacer hole with air valve port (Figure 6, Sheet 1, Detail A and Detail B).
9. **QA** Install floating pistons, open ends down in upper and lower stage piston bores. Push pistons to bottom of bore (Figure 6, Sheet 1, Detail A and Figure 7, Detail A).
10. **QA** Slip bearings, scraper ends first, over upper and lower stage pistons.

NOTE

Upper stage orifice assembly has a single hole in the center of the poppet valve and no coil spring.

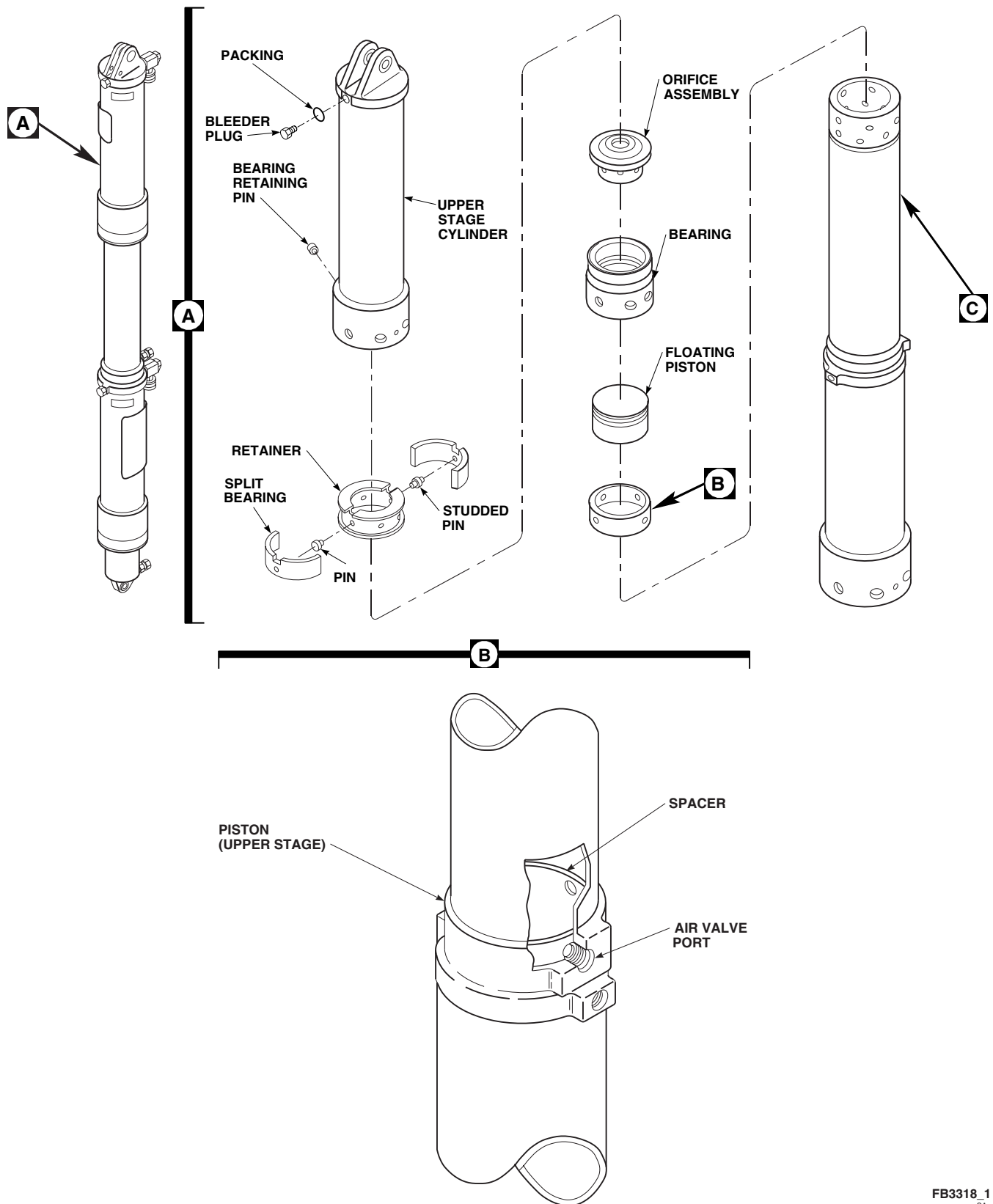
11. Remove identification tag and insert upper stage orifice in piston (Figure 6, Sheet 1, Detail A).
12. **QA** Install upper stage retainer. Line up slots with kneeling and air valve ports. Line up orifice retaining pin through retainer and piston wall in line with retainer notch, kneeling, and air valve ports (Figure 6, Sheet 2, Detail C).



AK0637
SA

Figure 5. Install Main Landing Gear Shock Strut Floating Piston/Sleeve Bearing (AVIM).

ASSEMBLE - Continued



FB3318_1
SA

Figure 6. Assemble Main Landing Gear Shock Strut Upper Stage (AVIM). (Sheet 1 of 2)

ASSEMBLE - Continued

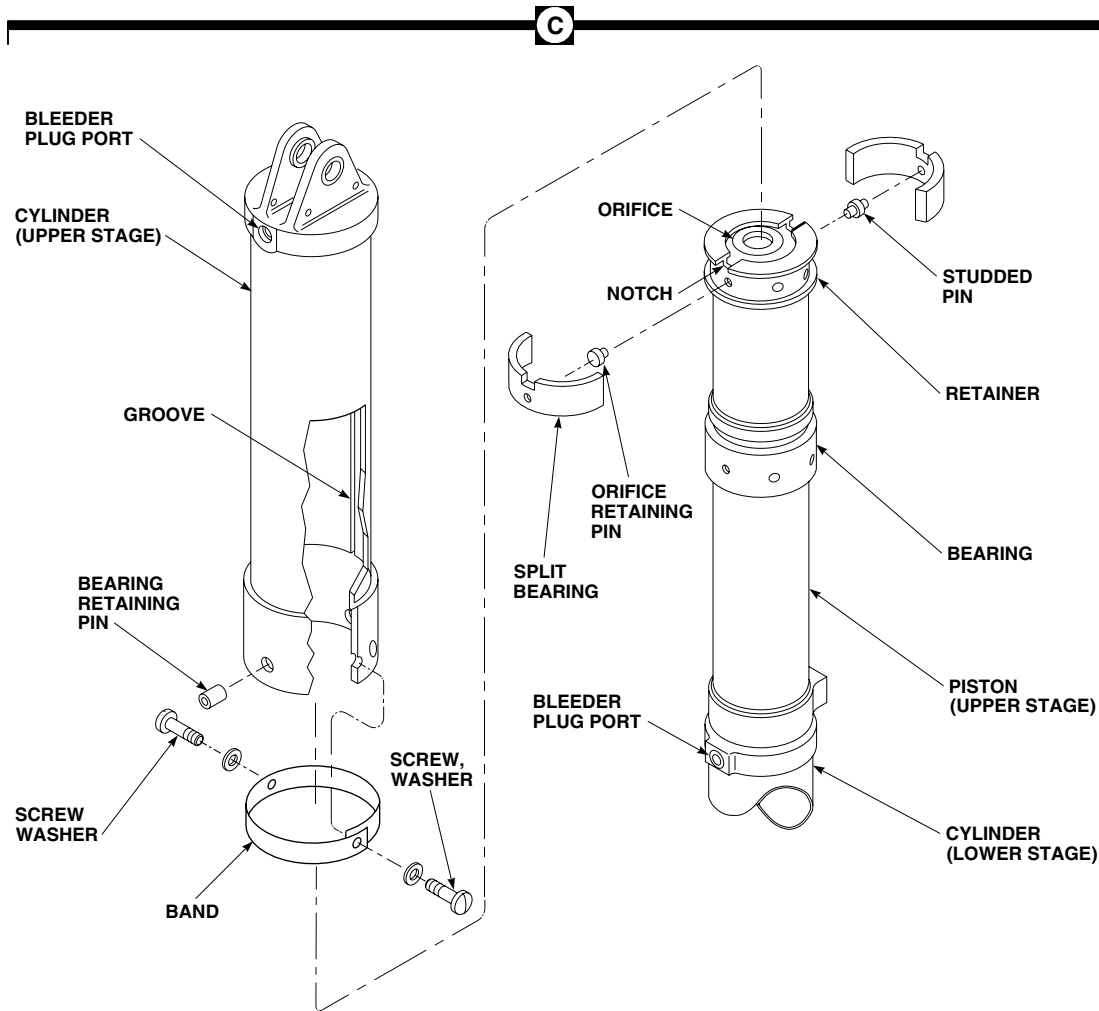
AK0638 2
SA

Figure 6. Assemble Main Landing Gear Shock Strut Upper Stage (AVIM). (Sheet 2 of 2)

NOTE

Installation of studded pin in any other pin hole will cause misalignment of strut stages.

13. **QA** Install 5 remaining orifice retaining pins.
14. **QA** Install split bearing halves on retainer. Line up upper stage cylinder groove with studded orifice retaining pin. Slide cylinder onto piston (Figure 6, Sheet 2, Detail C).
15. **QA** Slide bearing into cylinder. Line up bearing retaining pin holes and install 6 pins. Seat retaining pins with a rawhide mallet as required.

ASSEMBLE - Continued

16. **QA** Install band over bearing pins with screws and washers. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.
17. **QA** Slide bearing assembly, scraper end first, onto lower piston (Figure 7, Detail A).
18. **QA** Install dampener on piston, with dampener flange facing open end of piston.

NOTE

Lower stage orifice assembly has more than one hole in poppet valve and has a coil spring installed.

19. Remove identification tag and insert lower stage orifice into piston.
20. **QA** Install retainer. Line up holes and install 6 orifice retaining pins.
21. **QA** Install split bearing halves on retainer. Slide piston into cylinder.
22. **QA** Slide bearing into cylinder. Line up bearing retaining pin holes. Install 6 pins. Seat retaining pins with a rawhide mallet as required.
23. **QA** Install band over bearing pins with screws and washers. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.
24. Test and service strut (WP 1597 00). Refer to, TEST, in this work package.

TEST

1. Install upper and lower kneeling valves (WP 0445 00).

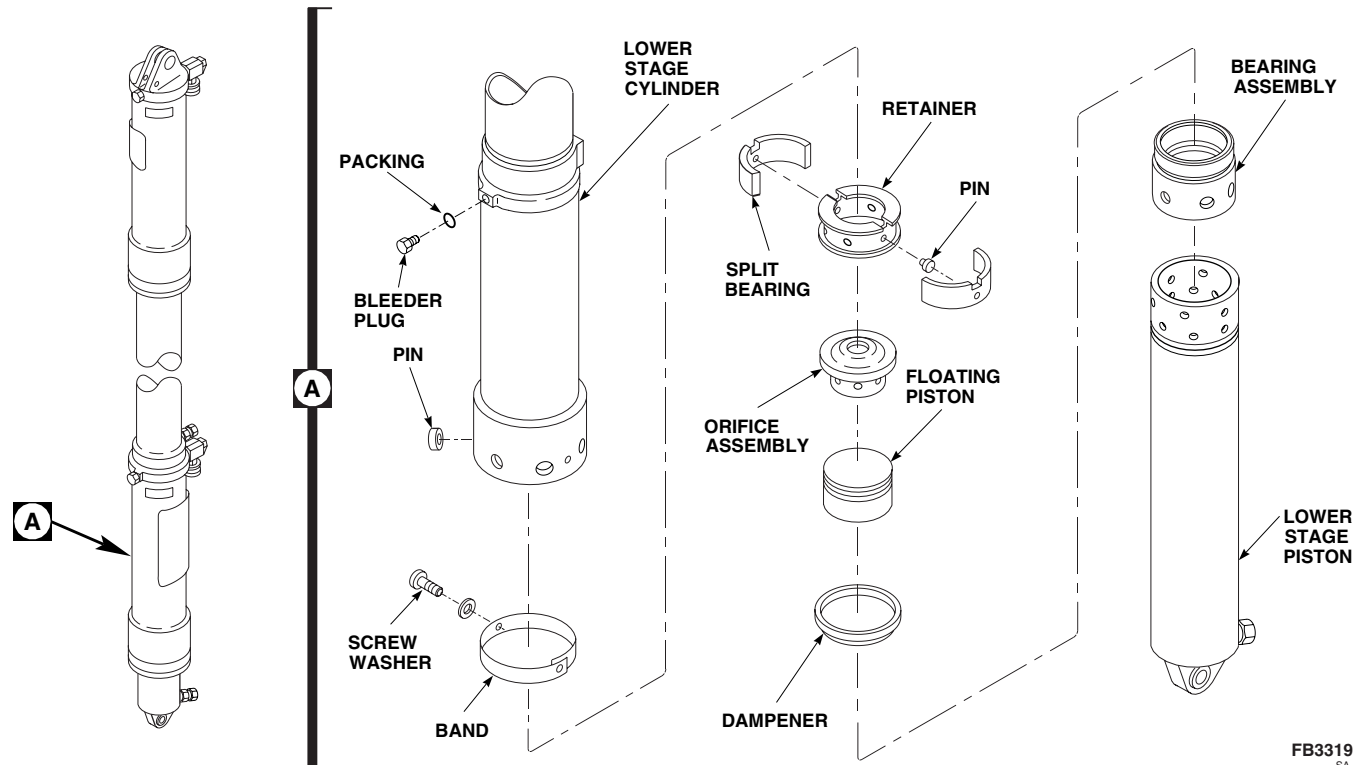


Figure 7. Assemble Main Landing Gear Shock Strut Lower Stage (AVIM).

TEST - Continued

2. Remove upper stage coupling dust cover. Install coupling half and drain hose on kneeling valve.
3. Support strut vertical. Route drain hose into a 1 gallon container.

CAUTION

Damage to strut will result if clearance does not exist prior to testing and servicing. Testing and servicing will extend strut to its maximum length of 72.94-inches measured between center of upper and lower attachment boltholes. Make sure enough clearance exists above strut.

4. Connect a shutoff valve and pressure line from hydraulic test stand to upper stage bleeder port. Start test stand. Open pressure line valve.
5. Apply 100 psi hydraulic pressure, at minimum flow, to fill strut upper stage and to bottom piston. Slowly close kneeling valve after fluid has run from drain line for 20 to 30 seconds.
6. Reduce hydraulic pressure to 0 psi.
7. Apply minimum test stand pressure and flow to strut. Open kneeling valve and maintain fluid flow until a steady air-free stream is obtained.
8. Close kneeling valve. Close pressure line shutoff valve. Reduce test stand pressure to 0 psi.
9. Allow a few minutes for trapped air to rise to top of strut. Open kneeling valve and fully compress strut. Close kneeling valve.
10. Apply 100 psi hydraulic pressure, at minimum flow, to fill strut upper stage.
11. Reduce hydraulic pressure and flow to minimum. Open kneeling valve. Check draining fluid for air.
12. Repeat preceding steps until hydraulic fluid is completely air-free.
13. Close pressure line shutoff valve. Compress strut upper stage, if extended.
14. Close kneeling valve. Reduce test stand pressure to 0 psi.
15. Disconnect hydraulic line at bleeder port.
16. Lubricate packing with petrolatum, Item 216, WP 1803 00.
17. Install packing on bleeder plug. Install bleeder plug in upper stage bleeder port. **TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
18. Disconnect drain line from upper kneeling valve. Install valve dust cap. Using nonelectrical wire, Item 356, WP 1803 00, lockwire kneeling valve knob.
19. Connect drain line to lower kneeling valve.
20. Connect pressure line to lower stage bleeder port.
21. Bleed strut lower stage, using same steps given for upper stage.
22. Install upper and lower stage air valves (WP 0446 00).
23. Connect pressure regulated nitrogen, Item 199, WP 1803 00, to upper stage valve. Slowly pressurize upper stage to 740 psi. Close air valve.
24. Connect nitrogen to lower stage air valve and pressurize lower stage to 210 psi. Close air valve. **TORQUE UPPER AND LOWER AIR VALVE OUTER HEXNUTS TO 50 - 70 INCH-POUNDS.**
25. Check upper and lower stage air valves for leakage using leak detection fluid, Item 175, WP 1803 00. **NO LEAKAGE ALLOWED.** Replace leaking air valve (WP 0446 00).

TEST - Continued

26. Check strut upper and lower stages for evidence of nitrogen and or hydraulic fluid leakage. **NO LEAKAGE ALLOWED.** Repair or replace a leaking strut.
27. Install air valve dust caps. Using nonelectrical wire, Item 357, WP 1803 00, lockwire lower hexnuts.
28. Using nonelectrical wire, Item 357, WP 1803 00, lockwire upper and lower stage bleed plugs.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****MAIN LANDING GEAR SHOCK STRUT LOWER STAGE SPHERICAL BEARING (AVIM)****This WP supersedes WP 0467 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Bearing Installation and Removal Set, 5180-00-980-2908
Bearing Staking Set, 5180-00-980-2925
Gloves
Goggles/Face Shield
Nonmetallic Scraper

Petrolatum, Item 216, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 0444 00
WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Corrosion Preventive Compound, Item 106, WP 1803 00

REMOVE**CAUTION**

Damage to fitting chamfer will occur if staked lip of bearing is improperly removed.
Be careful not to damage fitting chamfer when machining off staked lip of bearing.

1. Clamp fitting to baseplate of drill press and using bearing installation and removal set, machine off that part of bearing outer race swaged over chamfer on fitting (Figure 1, Detail B).
2. Press out bearing using arbor press and anvil/driver combination (TM 1-1500-204-23).

REPAIR**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

1. Clean primer residue from bore and chamfer, using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.
2. Check bearing bore lug for galling or scoring. **NO MORE THAN 0.010-INCH IF 25% OF CONTACT AREA IS AFFECTED.**

REPAIR - Continued

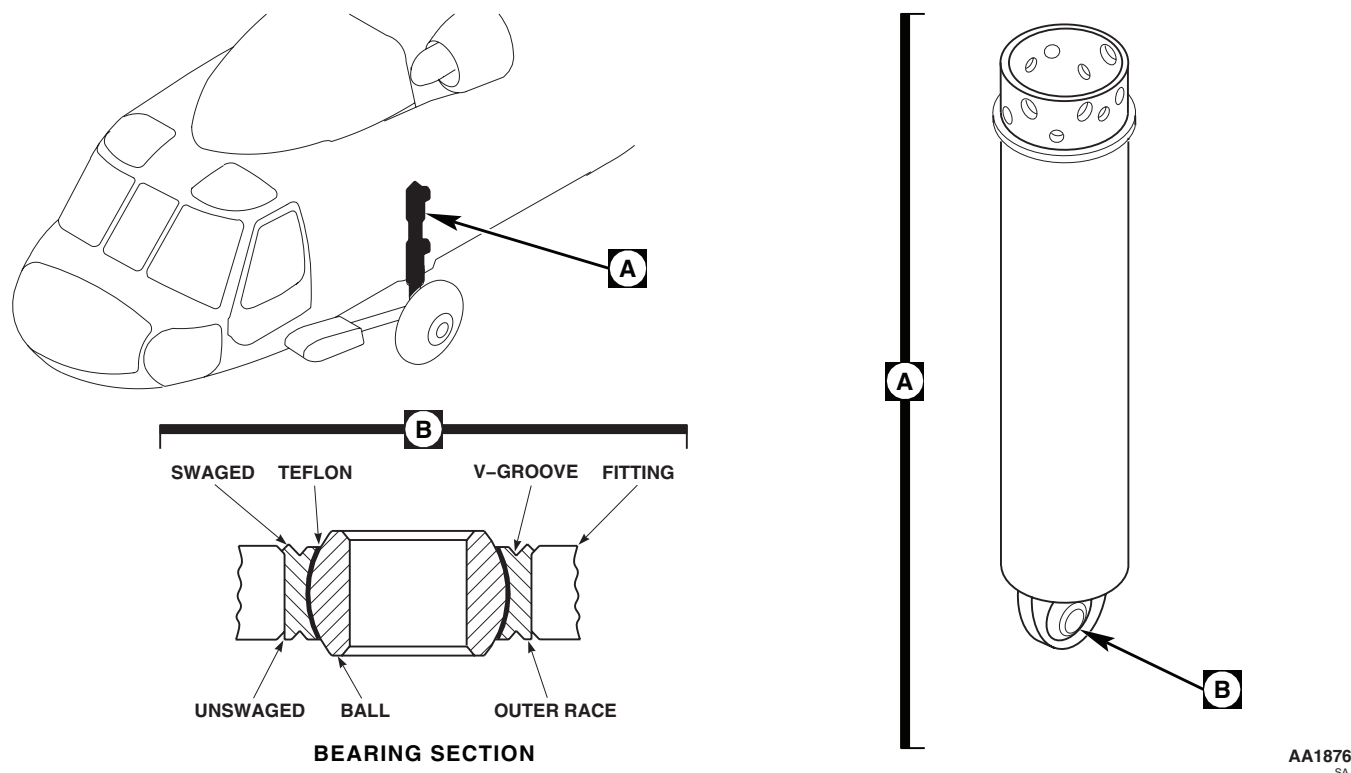


Figure 1. Replace Main Landing Gear Shock Strut Lower Stage Spherical Bearing (AVIM).

3. Blend out galling or scoring in bearing bore lug using abrasive cloth, Item 82, WP 1803 00.
4. Magnetic particle inspect bearing bore lug (TM 1-1520-265-23). Replace main landing gear shock strut if cracks are found (WP 0444 00).

INSTALL

1. Make sure that fitting bore, chamfers, and adjacent surfaces are clean and dry.
2. Apply a light coat of primer coating, Item 243, WP 1803 00, to inside of fitting bore and to outside surface of outer race of bearing (Figure 1, Detail B).
3. **QA** Press bearing into fitting, using arbor press with anvil and driver (TM 1-1500-204-23).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

4. Clean primer, if any, from bearing V-grooves, using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry.
5. Fix fitting with bearing on drill press base plate, with bearing staking set and anvil.

INSTALL - Continued**CAUTION**

Damage to bearing will result if petrolatum comes in contact on the ball of the bearing, or on the exposed edge of the fabric liner. If petrolatum gets on these areas, remove petrolatum immediately.

6. Lubricate V-groove of bearing and bearing staking set with petrolatum, Item 216, WP 1803 00.

CAUTION

Damage to bearing will result if bearing staking set makes contact with ball side of groove. Do not allow bearing staking set to make contact with ball side of groove as binding may result.

7. **QA** Swage V-groove lip over chamfer in fitting bore (TM 1-1500-204-23).
8. **QA** Turn fitting over and repeat swaging process on other side.
9. Clean petrolatum from V-groove with cheesecloth cloth, Item 85, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00.
10. Touch up bearing surfaces with corrosion preventive compound, Item 106, WP 1803 00.
11. Touch up paint (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR**MAIN LANDING GEAR WHEEL (AVIM)**

This WP supersedes WP 0468 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Mallet
 Nonmetallic Scraper
 Phenolic Drift Pin

Grease, Aircraft, Item 131, WP 1803 00
 Polyurethane Coating, Item 227, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Remover, Paint, Item 250, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Carbon Dioxide, Item 64, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Corrosion Resistant Coating, Item 110, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00

References

TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0381 00
 WP 0447 00
 WP 1803 00

REPAIR MAIN LANDING GEAR WHEEL (AVIM)**Repair****WARNING**

Injury to personnel and damage to equipment may occur if wheel halves are not inspected whenever wheel is disassembled. Make sure wheel halves are inspected whenever wheel is disassembled.

1. Remove tire from wheel (WP 0447 00).
2. Check seals for damaged reinforcing ring, visible damage, or deformation (Figure 1, Sheet 1). Replace any seal found unserviceable.

CAUTION

Damage to bearings will result if they are allowed to spin while drying. Secure inner and outer race of bearings while drying.

3. Clean bearings with cleaning compound solvent, Item 71, WP 1803 00. Dry bearings with compressed air, if available, or let air-dry.
4. Replace any bearing found worn, corroded, galled, scratched, pitted, water stained, heat-discolored, or with a cracked or distorted bearing cage.
5. If bearing is determined serviceable, repack it with aircraft grease, Item 131, WP 1803 00.

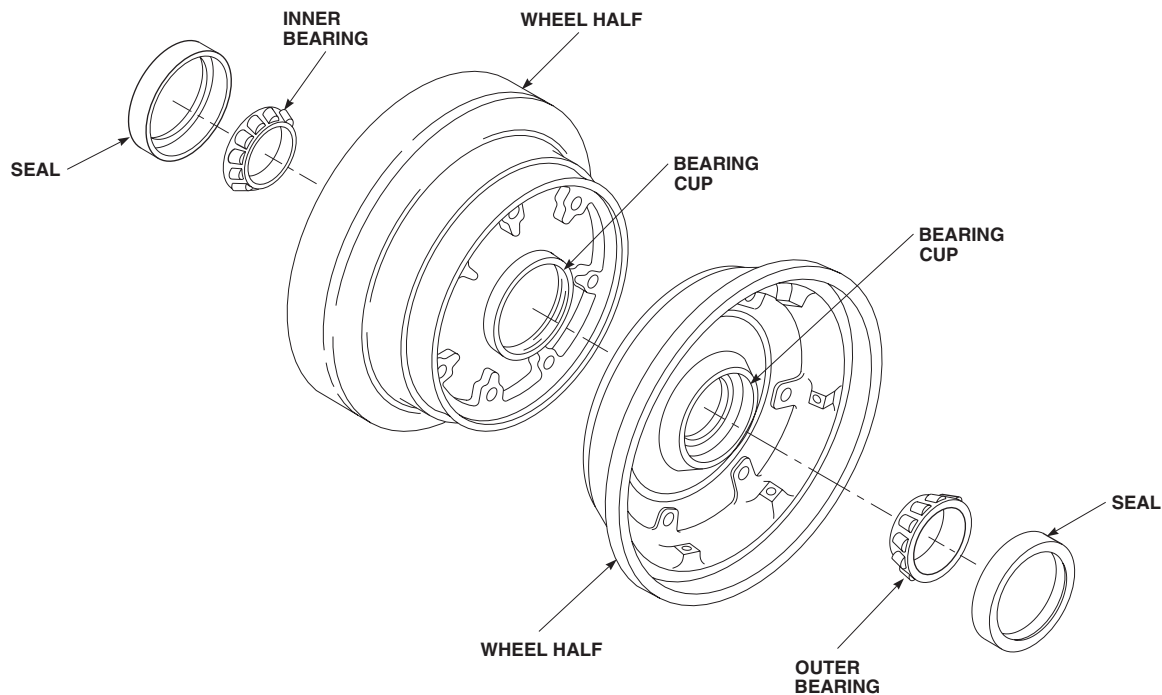
REPAIR MAIN LANDING GEAR WHEEL (AVIM) - ContinuedAA3383_1
SA

Figure 1. Repair Main Landing Gear Wheel (AVIM). (Sheet 1 of 3)

NOTE

New bearings packaged in preservative do not need degreasing before being packed with aircraft grease.

6. Inspect bearing cups in wheel halves for looseness, scratches, pitting, corrosion, and evidence of overheating. If cups are damaged, replace cups. Refer to, REPLACE BEARING CUPS (AVIM), in this work package.

WARNING

Injury to personnel or damage to equipment may occur if hardware is reused on wheel, 5003889. Replace all bolts and nuts on wheel, 5003889.

7. Check valve for stripped or crossed threads. Replace valve if damaged (WP 0447 00).
8. Replace instruction plate on wheel half if loose or not readable.

NOTE

Fluorescent penetrant inspection is not required on wheel halves, 040-25300.

9. Remove paint from wheel halves using paint remover, Item 250, WP 1803 00, and nonmetallic scraper.

REPAIR MAIN LANDING GEAR WHEEL (AVIM) - Continued

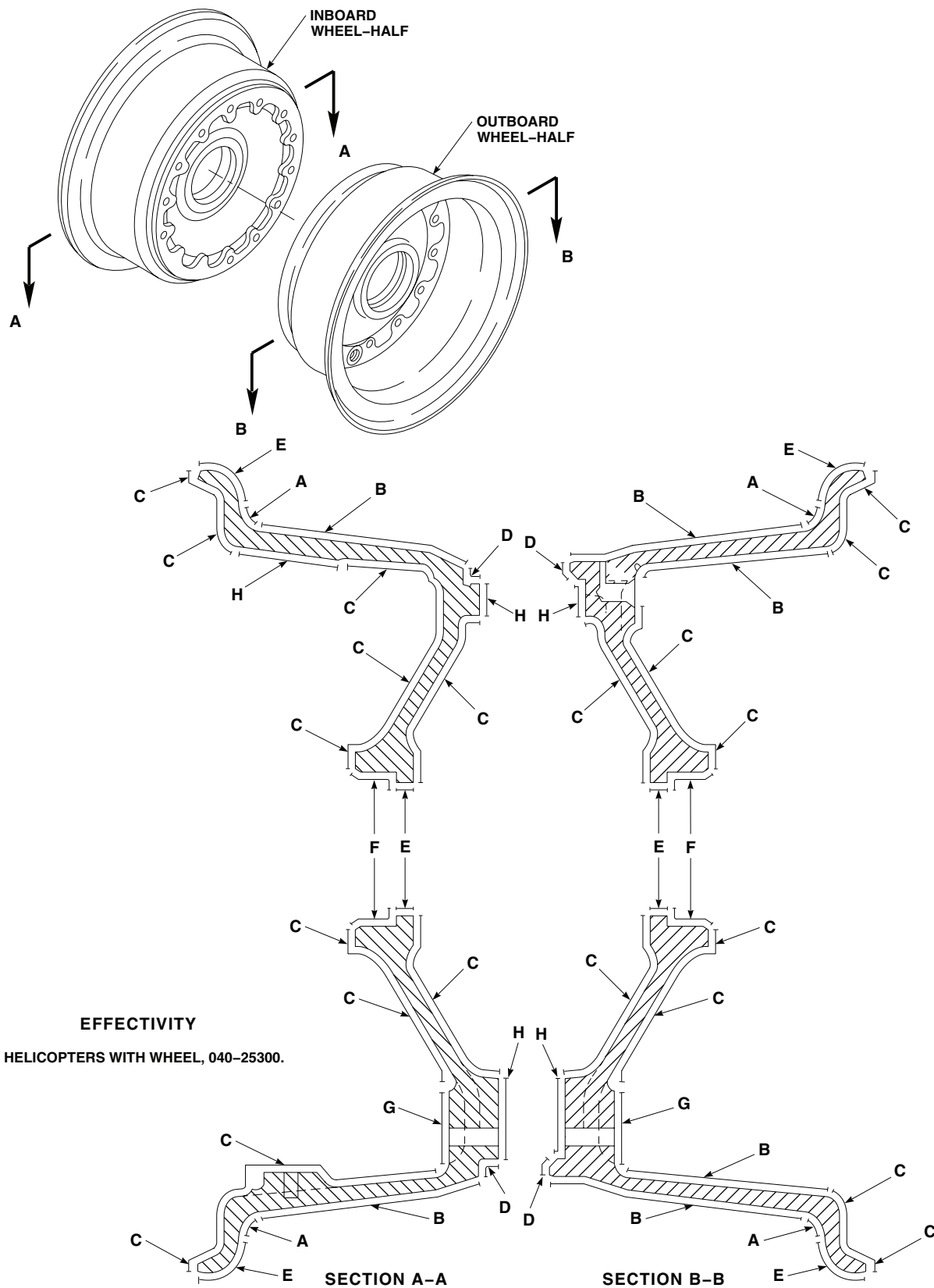


Figure 1. Repair Main Landing Gear Wheel (AVIM). (Sheet 3 of 3)

REPAIR MAIN LANDING GEAR WHEEL (AVIM) - Continued**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

10. Remove paint residue using machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.
11. Wipe dry using clean machinery wiping towel, Item 344, WP 1803 00.
12. Fluorescent penetrant inspect wheel halves, 5003889, for cracks and structural damage using fluorescent penetrant inspection kit (TM 1-1520-265-23). Pay particular attention to bead seat, bolt boss, male/female mating surfaces, and valve hole areas. **REPLACE WHEEL IF ANY CRACK IS PRESENT.**

NOTE

- Do all blending and polishing of wheel halves with abrasive cloth, Item 82, WP 1803 00.
 - Inspection areas are different for wheel, 5003889 and wheel, 040-25300. Refer to Figure 1, Sheet 2, for wheel, 5003889 and Figure 1, Sheet 3, for wheel, 040-25300.
13. Inspect wheel halves for scratches, nicks, or pits and repair as follows:
 - a. Repair AREA A as follows:
 - (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.015-INCH DEEP AND 1/2-INCH LONG.**
 - (2) Replace wheel if damage or repairs are more than limits.
 - b. Repair AREA B as follows:
 - (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.020-INCH DEEP AND 1.00 SQUARE INCH IN AREA.**
 - (2) **DAMAGE SHALL NOT BE DIRECTLY OPPOSITE OTHER DAMAGE OR A PREVIOUSLY REPAIRED AREA.**
 - (3) Replace wheel if damage or repairs are more than limits.
 - c. Repair AREA C as follows:
 - (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.020-INCH DEEP AND 1.00 SQUARE INCH IN AREA.**
 - (2) Replace wheel if damage or repairs are more than limits.
 - d. Repair AREA D as follows:
 - (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP.**
 - (2) **REPAIR AFTER BLENDING AND POLISHING, SHALL NOT INTERFERE WITH SEALING QUALITIES IN MALE/FEMALE MATING SURFACE AREAS.**
 - (3) Replace wheel if damage or repairs are more than limits, or if sealing qualities in male/female mating surface area is not maintained.
 - e. Repair AREA E as follows:

REPAIR MAIN LANDING GEAR WHEEL (AVIM) - Continued

- (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.030-INCH DEEP, 1/2 SQUARE INCH IN AREA, IN A MAXIMUM OF 2 PLACES.**

- (2) Replace wheel if damage or repairs are more than limits.

f. Repair AREA F as follows:

- (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS.**

- (2) **REPAIR AFTER BLENDING AND POLISHING SHALL NOT EFFECT BEARING CUP RETENTION.**

- (3) Replace wheel if damage or repairs effect bearing cup retention.

g. Repair AREA G as follows:

- (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP ON FACE OF EACH BOLT BOSS.**

- (2) Replace wheel if damage or repairs are more than limits.

h. Repair AREA H as follows:

- (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP AND 1/2 SQUARE INCH ON EACH INTERFACE BOSS.**

- (2) Replace wheel if damage or repairs are more than limits.

14. Touch up any blended areas with corrosion resistant coating, Item 110, WP 1803 00.

15. Prime wheel with epoxy primer coating kit, Item 120, WP 1803 00.

NOTE

Male/female mating surfaces, and bolt bosses need primer only.

16. Paint wheel with polyurethane coating, Item 227, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).

17. Install tire on wheel (WP 0447 00).

REPLACE BEARING CUPS (AVIM)**Remove****CAUTION**

Damage to wheel half will result if overheated. Overheating will effect service life of wheel. Do not overheat wheel half or heat longer than required.

1. Heat wheel half in an oven not over 250°F (121°C), or place in boiling water for 30 minutes.

CAUTION

- Damage to hub bore will result if bearing cups are removed improperly. Be careful when removing bearing cups to avoid raising burrs in hub bore.
- Damage to wheel half will result if wheel half is not supported properly. Support wheel half on hub and not on flange.

2. Remove wheel half from oven and immediately remove cup from bore (Figure 2). After heating wheel, bearing cup should be loose enough to fall out of bore when wheel is turned upside down. If cup does not fall out, tap cup from bore evenly using a phenolic drift pin and a mallet.

REPLACE BEARING CUPS (AVIM) - Continued

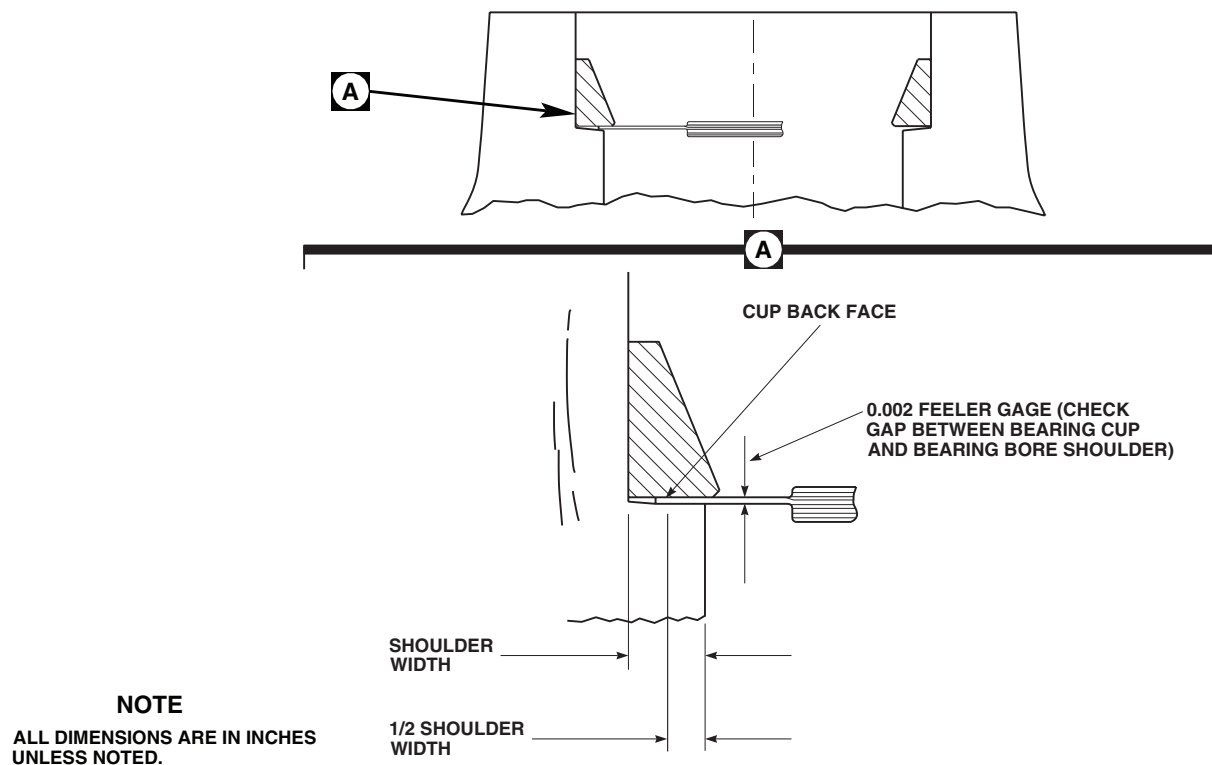


Figure 2. Replace Main Landing Gear Wheel Bearing Cups (AVIM).

- Carefully remove any primer or other foreign matter from hub bore. Make sure bearing bore and shoulder are free of burrs.

Install**WARNING**

Injury to personnel will result if carbon dioxide or chilled parts come in contact with skin. Use protective gloves when handling carbon dioxide or chilled parts.

- Place replacement bearing cup in a container of carbon dioxide, Item 64, WP 1803 00.

CAUTION

Damage to wheel half will result if overheated. Overheating will affect service life of wheel. Do not overheat wheel half or heat longer than required.

- Heat wheel half in an oven not over 250°F (121°C), or place in boiling water for 30 minutes.
- Remove wheel half from heat source.
- Remove bearing cup from carbon dioxide. Make sure cup is thoroughly dry.
- Coat wheel contacting surface of bearing cup with primer coating, Item 243, WP 1803 00, and install cup into hub bore while primer is still wet. Use a phenolic drift pin and a mallet to tap cup into bore. Make sure cup is against shoulder and avoid cocking cup during installation.

REPLACE BEARING CUPS (AVIM) - Continued**CAUTION**

Damage to wheel half will result if wheel half is not supported properly. Support wheel half on hub and not on flange.

6. **QA** Make sure cup is correctly seated by checking for any gap between cup back face and shoulder in wheel hub. Use a 0.002-inch feeler gage and try to insert it between cup and shoulder. **A 0.002-INCH-MAXIMUM GAP BETWEEN BACK FACE AND SHOULDER IS ALLOWED, PROVIDED THAT FEELER GAGE CANNOT BE INSERTED MORE THAN 1/2 THE WIDTH OF SHOULDER AT ANY LOCATION AROUND BACK FACE OF CUP (FIGURE 2, DETAIL A).**
7. Allow wheel to cool to room temperature. Apply a light coat of aircraft grease, Item 131, WP 1803 00, to bearing cup to prevent rust and corrosion.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****MAIN LANDING GEAR WHEEL BRAKE (AVIM)****This WP supersedes WP 0469 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Caliper Set, GGG-C-105
Dial Indicator, MIL-I-18422
Fixture, Test Stand, MIL-T-25456
Fluorescent Penetrant Inspection Kit, XMA101
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Test Stand, Hydraulic, D5B
Torque Wrench, 0 - 30 in. lbs.
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Alcohol, Denatured, Item 48, WP 1803 00
Antiseize Compound, Item 51, WP 1803 00
Brush, Acid Swabbing, Item 57, WP 1803 00
Brush, Paint, Item 61, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00

Cloth, Abrasive, Item 81, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Polyurethane Coating, Item 233, WP 1803 00
Remover, Paint, Item 250, WP 1803 00
Tape, Pressure Sensitive, Adhesive, Item 334,
WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Wheel, Abrasive, Item 350, WP 1803 00
Wire, Nonelectrical, Item 353, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 1803 00

CAUTION

Damage to brake will result if impact or power wrenches are used to remove brake bolts or nuts. Do not use impact or power wrenches to remove brake bolts or nuts.

DISASSEMBLE

1. Place brake assembly, housing down, on a clean flat surface.
2. Remove eight self-locking nuts. Do not remove bolts now (Figure 1).
3. Remove backing plate.
4. Remove six wear pads from backing plate.
5. Remove outboard rotating disc.
6. Remove stationary disc.
7. Remove inboard rotating disc.
8. Remove torque tube from housing.

DISASSEMBLE - Continued

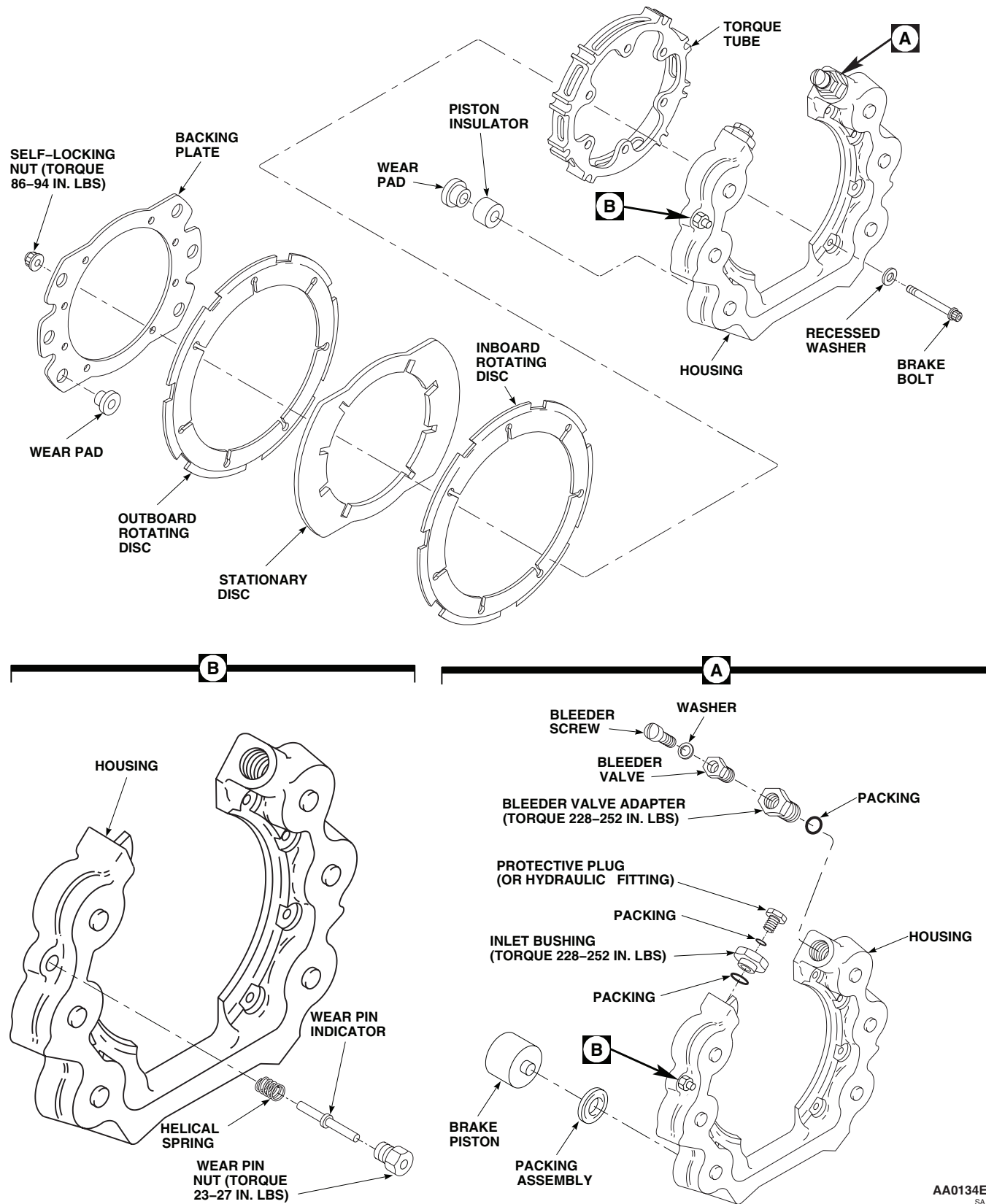
AA0134E
SA

Figure 1. Repair Wheel Main Landing Gear Wheel Brake (AVIM).

DISASSEMBLE - Continued

9. Remove eight brake bolts and recessed washers.
10. Remove six wear pads and six piston insulators from housing.
11. Remove six brake pistons from housing (Figure 1, Detail A).
12. Remove six packing assemblies from housing.
13. Remove bleeder screw and washer.
14. Remove bleeder valve.
15. Remove bleeder valve adapter.
16. Remove packing from housing.
17. Remove protective plug, or hydraulic fitting, from inlet in housing.
18. Remove packing from bushing.
19. Remove inlet bushing.
20. Remove packing.
21. Remove two wear pin nuts (Figure 1, Detail B).
22. Remove two wear pin indicators and helical springs from housing.

CLEAN

1. Clean two rotating and one stationary disc with compressed air and acid swabbing brush, Item 57, WP 1803 00.
2. Clean all other metal parts with cleaning compound solvent, Item 71, WP 1803 00. Use paint brush, Item 61, WP 1803 00, if needed.
3. Clean all piston insulators with denatured alcohol, Item 48, WP 1803 00. Dry with machinery wiping towel, Item 344, WP 1803 00.
4. Dry all metal parts thoroughly after cleaning, using filtered and dried compressed air.

INSPECT

1. Remove all foreign material from cavities, threaded holes, pockets or other entrapment areas in metal parts.
2. Visually inspect all parts for cracks, wear, structural damage, corrosion, and damaged threads. Replace all parts that show evidence of cracks, wear, corrosion, structural or thread damage.
3. Check backing plate for cracks around wear pad and mounting bolt holes. Replace cracked backing plate.
4. Check backing plate for dishing and distortion. **REPLACE BACKING PLATE IF WARPED 0.015-INCH OR MORE.**
5. Replace bolts that are bent, cracked, or have thread damage.
6. Check rotating discs for corrosion at relief slot terminal holes. Remove corrosion using abrasive cloth, Item 81, WP 1803 00, or abrasive wheel, Item 350, WP 1803 00. **CORROSION REMOVAL SHALL NOT EXCEED 0.015-INCH FOR RELIEF SLOTS AND 0.281-INCH DIAMETER FOR TERMINAL HOLES.**
7. Check rotating discs specifically for cracks around six relief slot terminal holes and seven drive slot radii. **NO CRACKS ALLOWED.** Replace any cracked discs.

INSPECT - Continued

8. Check rotating discs specifically for wear. **REPLACE DISC WHEN THICKNESS ACROSS UNCHIPPED MIX MEASURES 1/8-INCH OR LESS, OR IF MIX IS WORN TO 0.010-INCH THICK AT ANY POINT ON EITHER FACE OF DISC. AS MUCH AS 20% OF MIX MAY BE LOST THROUGH CHIPPING IF MIX LOSS IS SCATTERED. NO MORE THAN 25% LOSS OF MIX IS ALLOWED ON ANY SEGMENT.**
9. Check rotating discs for shrinkage and distortion. **REPLACE DISCS THAT ARE DISHED 0.015-INCH OR MORE, OR ARE 1/8-INCH OUT-OF-ROUND, OR HAVE SHRUNK TO AN INSIDE DIAMETER OF 6.940-INCHES OR LESS AT ANY POINT.**
10. Check rotating disc drive slots for wear and battering. **REPLACE DISC IF ONE OR MORE TANGS ARE BENT, OR IF ANY SLOT IS 0.780-INCH OR MORE IN WIDTH.**
11. Check stationary disc specifically for cracks around key slot corners. **NO CRACKS ALLOWED.** Replace cracked discs.
12. Check stationary disc for thickness. **REPLACE DISC IF WORN TO 0.140-INCH OR LESS IN THICKNESS AT ANY POINT ON DISC.**
13. Check stationary disc key slots for wear and battering. **REPLACE DISC IF WIDTH OF ANY KEY SLOT IS LESS THAN 0.515-INCH OR GREATER THAN 0.560-INCH.**
14. Check stationary disc for shrinkage and distortion. **REPLACE DISC IF DISHED 0.015-INCH OR MORE, OR IS 1/8-INCH OUT-OF-ROUND, OR HAS SHRUNK TO AN ID OF 6.320-INCHES OR LESS WHEN MEASURED BETWEEN ENDS OF KEY SLOT TANGS.**
15. Check ID of stationary disc for size. Replace if disc cannot be installed on torque tube without interference.
16. Check torque tube for cracks. Replace cracked tube.
17. Check width of torque tube key. **REPLACE TUBE IF KEY WIDTH IS 0.470-INCH OR LESS AT ANY POINT ALONG LENGTH OF ANY KEY.**
18. Check pistons for wear. Measure diameter of each piston at three places around it. **REPLACE PISTON IF DIAMETER MEASURES 1.240-INCHES OR LESS AT ANY POINT. REPLACE ANY PISTONS WITH BURRS, SCRATCHES, OR NICKS MORE THAN 0.003-INCH DEEP ON SEAL CONTACTING SURFACES.**
19. **REPLACE PISTON IF WEAR AT BOTTOM OF PISTON EXCEEDS 0.015-INCH.**
20. Inspect piston insulators for deterioration or wear. **REPLACE PISTON INSULATORS WORN TO 0.635-INCH OR LESS IN LENGTH, OR WITH BLISTERS AND RAISED AREAS, OR WITH END FACES NOT PARALLEL.**
21. Remove paint from housing using paint remover, Item 250, WP 1803 00, and nonmetallic scraper.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

22. Remove paint residue using machinery wiping towel, Item 344, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00.
23. Wipe dry using clean machinery wiping towel, Item 344, WP 1803 00.
24. Fluorescent penetrant inspect housing subassembly for cracks using fluorescent penetrant kit (TM 1-1520-265-23). Replace cracked housing.

INSPECT - Continued

25. Mask piston bores, fitting bosses, and wear pin holes using pressure sensitive adhesive tape, Item 334, WP 1803 00.
26. Prime housing with one coat of epoxy primer coating kit, Item 120, WP 1803 00.
27. Paint housing with polyurethane coating, Item 233, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).
28. Check thread size and condition in housing inlet and bleeder ports. Replace housing if threads are damaged.

CAUTION

Damage to brake will result if new wear pads are mixed with used pads. If any wear pad on piston or on backing plate is worn beyond limits, replace that set of six wear pads.

29. Check thickness of backplate wear pads. **REPLACE ALL SIX IF ONE OR MORE ARE WORN TO 0.380-INCH THICK OR LESS.**
30. Check thickness of piston wear pads. **REPLACE ALL SIX IF ANY ONE PAD IS WORN TO 0.380-INCH THICK OR LESS.**
31. Fluorescent penetrant inspect brake bolts specifically for cracks at radius under bolt head and within threads next to bolt shank using fluorescent penetrant inspection kit (TM 1-1520-265-23). Replace cracked brake bolts.

ASSEMBLE**CAUTION**

- Damage to packing will result if packing is not handled properly. Do not stretch or twist packings during assembly.
 - Damage to brake will result if impact or power wrenches are used to install brake bolts or nuts. Do not use impact or power wrenches to install brake bolts or nuts.
 - Damage to brake will result if the brake is not assembled on a clean flat surface. Assemble brake on clean, flat surface, free of foreign objects.
1. **QA** Install helical spring, wear indicator pin, and wear pin nut in housing (Figure 1, Detail B). **TORQUE WEAR PIN NUTS TO 23 - 27 INCH-POUNDS. Using nonelectrical wire, Item 353, WP 1803 00, lockwire wear pin nut to housing.**
 2. **QA** Lubricate packing with petroleum base hydraulic fluid, Item 145, WP 1803 00. Install packing and inlet bushing in inlet port of brake housing. **TORQUE INLET BUSHING TO 228 - 252 INCH-POUNDS (FIGURE 1, DETAIL A).**
 3. **QA** Lubricate packing with petroleum base hydraulic fluid, Item 145, WP 1803 00. Install packing and protective plug (or hydraulic fitting) in inlet bushing.
 4. **QA** Lubricate packing with petroleum base hydraulic fluid, Item 145, WP 1803 00. Install packing and bleeder valve adapter in bleeder port of brake housing. **TORQUE BLEEDER VALVE ADAPTER TO 228 - 252 INCH-POUNDS.**
 5. **QA** Install bleeder valve into bleeder valve adapter.
 6. **QA** Install bleeder washer and bleeder screw.
 7. **QA** Lubricate, with petroleum base hydraulic fluid, Item 145, WP 1803 00, six packing assemblies, and six housing cavities in areas which contact the piston surfaces. Install packing assemblies in piston cavities of housing.
 8. **QA** Lubricate six pistons with petroleum base hydraulic fluid, Item 145, WP 1803 00. Install six pistons in housing cavities, without damaging them or moving packings.

ASSEMBLE - Continued

9. Bottom pistons in housing, using finger pressure.
10. Coat threads of eight brake bolts and nuts, and bearing surfaces of boltheads, washers, and self-locking nuts with antiseize compound, Item 51, WP 1803 00 (Figure 1).
11. **QA** Install washers on eight brake bolts, with washer countersunk side against bolthead. Install bolts through bolt-holes in brake housing.
12. Place brake housing, with bolts in place, on a clean, flat surface, with pistons up.
13. **QA** Place torque tube over brake bolts, with torque tube scallops facing brake housing.
14. **QA** Insert wear pad into chamfered end of piston insulator (Figure 1, Detail A).
15. **QA** Install six piston insulators and six wear pads in housing (Figure 1, Detail A).
16. **QA** Install inboard rotating disc, over torque tube, add stationary disc and outboard rotating disc, over tube.
17. **QA** Install backplate and six wear pads, with flat face down, against torque tube, with bolts extending through back-plate.
18. **QA** Install self-locking nuts on eight brake bolts. **TORQUE EIGHT NUTS EVENLY TO 86 - 94 INCH-POUNDS.**

TEST

1. Connect repaired brake assembly, in vise, to hydraulic test stand with petroleum base hydraulic fluid, Item 145, WP 1803 00, able to apply 800 psi hydraulic pressure to brake.
2. Apply pressure and bleed brake.
3. Apply 794 psi pressure, then release pressure. Repeat several times. Rotating discs should be free to rotate as evidence of proper brake release function.
4. Apply 5.5 psi pressure to brake. Brake clearance must be zero and rotating discs must be movable by hand.
5. Apply 794 psi pressure. Hold pressure steady for 15 to 30 seconds. **NO LEAKS ARE PERMITTED.**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****MASTER BRAKE CYLINDER (AVIM)**

This WP supersedes WP 0470 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Hand Pump
Hydraulic Repairer's Toolkit, SC 5180-99-A03
Hydraulic Test Stand, MIL-T-7612D
Test Fixture, T-2355-1
Test Fixture, T-3430

Wire, Nonelectrical, Item 352, WP 1803 00

Wire, Nonelectrical, Item 353, WP 1803 00

Personnel Required

Aircraft Pnedraulics Repairer MOS 15H (1)

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00

References

TM 1-1500-204-23

TM 1-1500-344-23

WP 0452 00

WP 1803 00

DISASSEMBLE

1. Remove master brake cylinder (WP 0452 00).
2. Clean cylinder exterior surfaces with cleaning compound solvent, Item 71, WP 1803 00.
3. Remove cap screws and end cap (Figure 1).
4. Drain hydraulic fluid.
5. Remove piston metering valve assembly from cylinder body.
6. Compress return spring to expose spring clip and retaining pin (Figure 1, Detail A).
7. Remove spring clip, retaining pin, piston retainer, spring retainer and seat.
8. Remove piston head and guide.
9. Remove relief cap, poppet spring, nylon lock, poppet, and pressure sensing piston from piston assembly.
10. Remove filler cap and fluid level screw (Figure 1).
11. Remove rod end, washer, and locknut.
12. Remove piston rod from cylinder body.
13. Remove and discard all old packings.

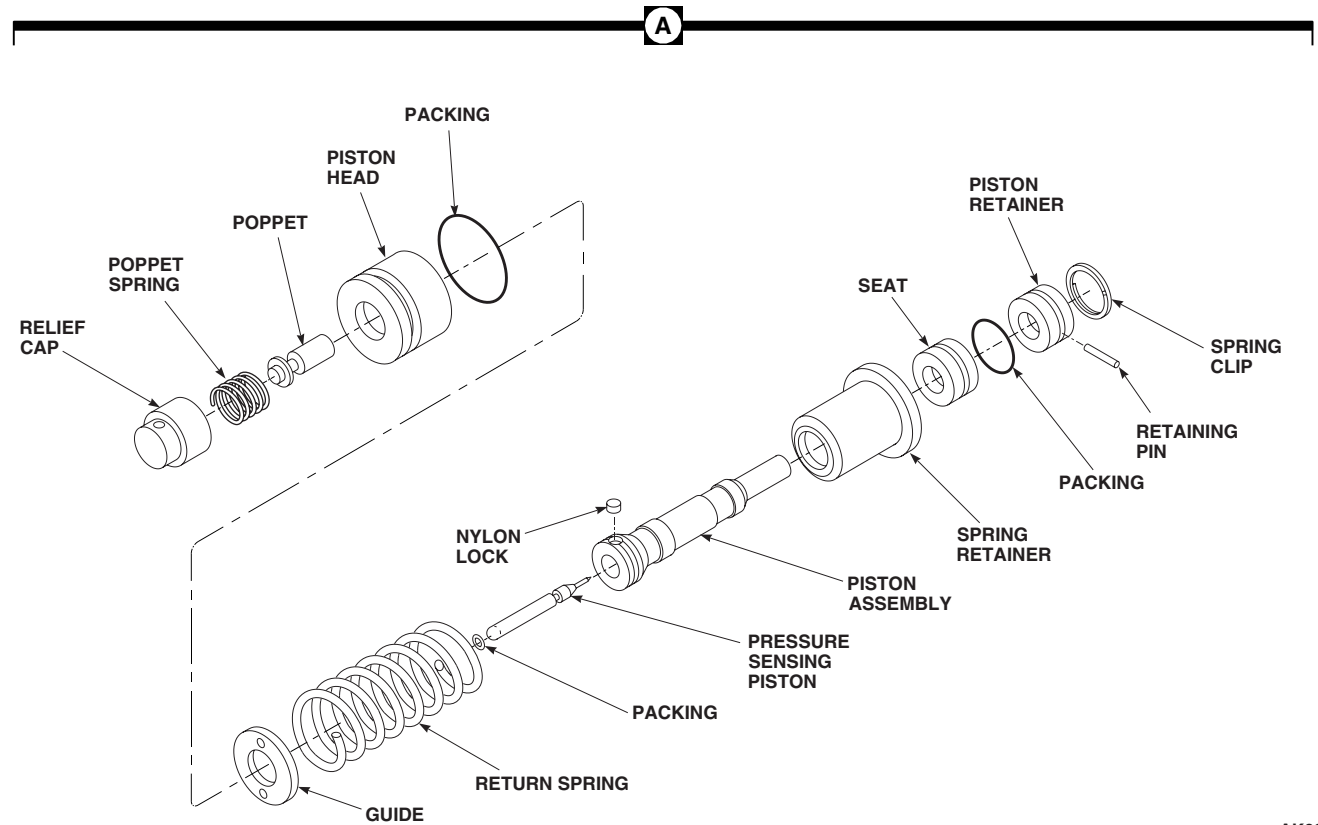
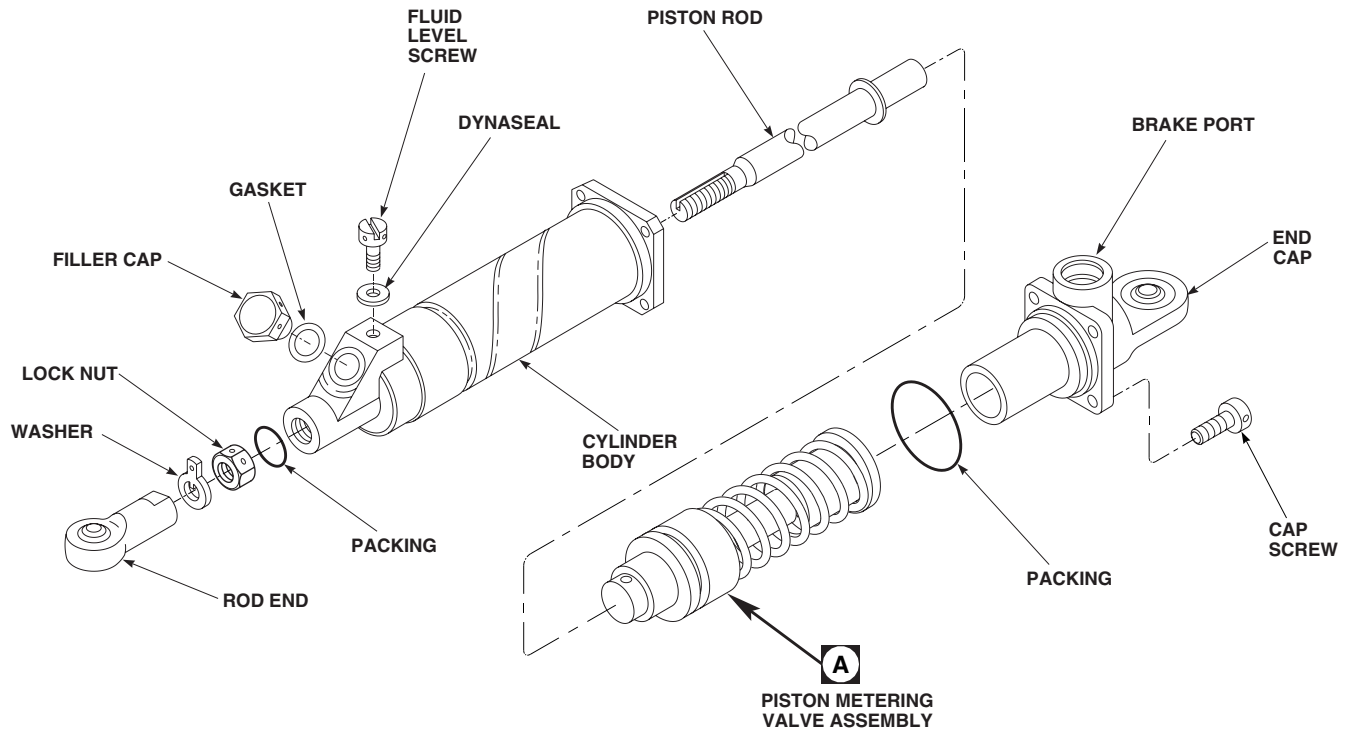
CLEAN

Wash all parts in clean cleaning compound solvent, Item 71, WP 1803 00. Blow parts dry with filtered dry compressed air.

INSPECT

1. Check all chrome plated wear surfaces for scratches, dents, gouges, corrosion or pitting (TM 1-1500-344-23). **NO DAMAGE ALLOWED.** Replace damaged parts.

INSPECT - Continued



AK0691
SA

Figure 1. Repair Master Brake Cylinder (AVIM).

INSPECT - Continued

2. Check threads for damage (TM 1-1500-204-23). Repair or replace damaged threaded parts.

ASSEMBLE**NOTE**

Submerge new packings and parts in petroleum base hydraulic fluid, Item 145, WP 1803 00, before assembly.

1. **QA** Install packing in cylinder body piston rod bore (Figure 1).
2. Position piston rod into cylinder body. Install locknut, washer, and rod end on piston rod. **DO NOT TIGHTEN ROD END.**
3. **QA** Install packing on piston head groove (Figure 1, Detail A).
4. **QA** Install packing on seat groove.
5. **QA** Install packing on pressure sensing piston.
6. **QA** Install pressure sensing piston, poppet, and nylon lock in piston assembly.
7. **QA** Position poppet spring in relief cap. Screw cap and spring onto piston assembly until bottomed.
8. **QA** Install piston head on piston assembly with open end away from relief cap.
9. Position guide and return spring inside piston head.
10. Stack seat and piston retainer in spring retainer.
11. **QA** Position spring retainer inside return spring. Compress spring. Line up retaining pin holes. Install retaining pin through piston retainer and piston assembly. Install spring clip over retaining pin.

WARNING

Injury to personnel will result if care is not taken when testing piston metering valve assembly. Hydraulic fluid is under pressure. Wear protective goggles and clothing when testing piston metering valve assembly.

12. Perform piston metering valve assembly operational test as follows:
 - a. Install appropriate fitting in test fixture, T-3430 (Figure 2).
 - b. Connect piston metering valve assembly in test fixture. Connect hydraulic test stand to test fixture port.
 - c. Using a hand pump, slowly increase inlet port pressure to the fixture past the cracking point of the piston metering valve allowing hydraulic fluid to flow from the exposed cap. Reduce inlet port pressure to zero.
 - d. Starting at 0 psi inlet port pressure, slowly increase pressure to the fixture until the piston valve cracks. **CRACKING PRESSURE SHALL BE 65 TO 85 PSI.**
 - e. If pressure is not within range, adjust cracking pressure by removing piston metering valve assembly from test fixture and either tighten cap to increase cracking pressure or loosen cap to decrease cracking pressure. Repeat step b. through step d.
 - f. Apply inlet port pressure to the fixture to the cracking point of the piston metering valve assembly. Block off inlet port pressure and observe for one minute. **GAGE PRESSURE AT THE INLET PORT SHALL NOT DECREASE BELOW 55 PSI MINIMUM.** Repair or replace piston metering valve if pressure decrease is below limits.
 - g. Repeat step b. through step f. to verify proper operation of the piston metering valve assembly.
13. **QA** Install piston metering valve assembly in cylinder body (Figure 1).

ASSEMBLE - Continued

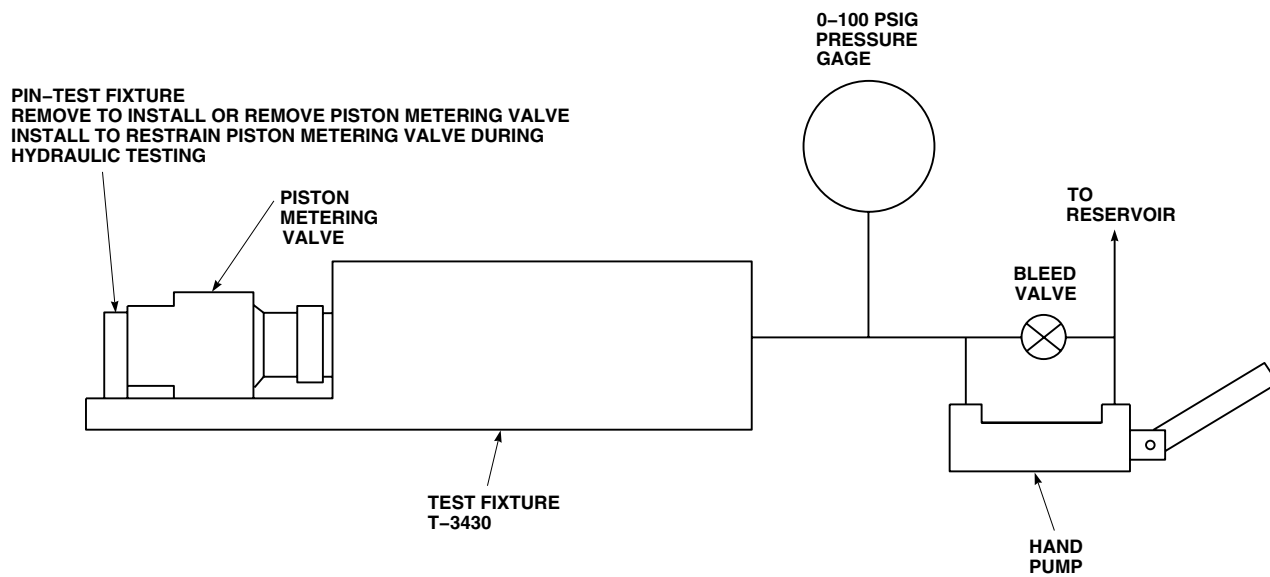
AA9400A
SA

Figure 2. Test Master Brake Cylinder Piston Metering Valve Assembly (AVIM).

14. **QA** Install packing on end cap groove.
15. **QA** Install end cap to cylinder body with cap screws. Using nonelectrical wire, Item 352, WP 1803 00, lockwire screws.
16. Fill cylinder with petroleum base hydraulic fluid, Item 145, WP 1803 00.
17. **QA** Install dynaseal on fluid level screw.
18. **QA** Install new gasket on filler cap.
19. **QA** Install fluid level screw and filler cap on cylinder. Using nonelectrical wire, Item 353, WP 1803 00, lockwire filler cap and screw.
20. Test master brake cylinder. Refer to, TEST MASTER BRAKE CYLINDER (AVIM), in this work package.

ASSEMBLE - Continued

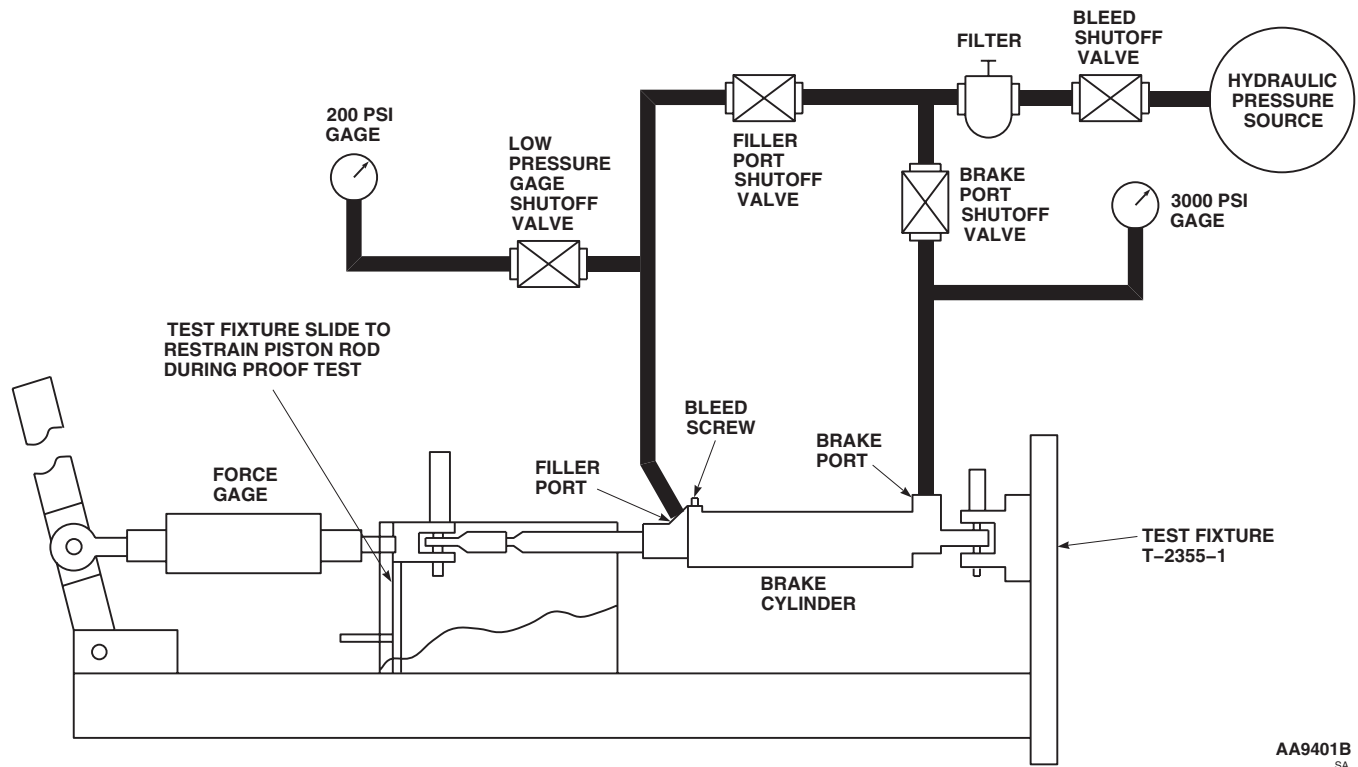


Figure 3. Test Master Brake Cylinder (AVIM).

TEST

WARNING

- Injury to personnel will result if care is not taken when testing master brake cylinder. Hydraulic fluid is under pressure. Wear protective goggles and clothing when testing master brake cylinder.
- Injury to personnel and damage to equipment will result if low pressure (200 psi) gage valve is not closed, when 2250 psi pressure is applied in the following steps. Close the valve for the low pressure (200 psi) gage.

1. Install master brake cylinder in test fixture, T-2355-1. Adjust test fixture to compress master cylinder piston rod 1/8-inch (Figure 3).
2. Attach hydraulic lines to **FILLER** and **BRAKE** ports.
3. Open **BRAKE**, **FILLER**, and bleeder line shutoff valves.
4. Pump hydraulic fluid through lines until air is removed. Refill hydraulic reservoir as required.
5. Close **FILLER** port shutoff valve. Open **BRAKE** port and bleeder line shutoff valves.
6. Close bleed line shutoff valves.
7. Close low pressure gage shutoff valve.
8. Apply 2250 psi to **BRAKE** port. Hold pressure for five minutes. Check exterior for leakage. **NO LEAKAGE ALLOWED.** Repair or replace leaking master brake cylinder.

TEST - Continued

9. Open **BRAKE** port shutoff valve and reduce pressure to 0 psi. **THE PISTON ROD SHALL RETRACT RAPIDLY AND SMOOTHLY AND THE PRESSURE SHALL IMMEDIATELY DROP TO 0 PSI.**
10. Close **BRAKE** port shutoff valve.
11. Open low pressure gage shutoff valve.
12. Open **FILLER** port shutoff valve. Apply 150 psi to **FILLER** port. Hold pressure for five minutes. Examine cylinder for external leakage. **NO LEAKAGE ALLOWED.** Reduce pressure to 0 psi and then close **FILLER** port shutoff valve.

NOTE

Staging pressure readings must be taken from the **FILLER** port gage.

13. Open **BRAKE** port shutoff valve. Apply gradually increasing force to **BRAKE** port until a change in brake pressure occurs (staging pressure). **PRESSURE SHALL BE 65 TO 85 PSI. PRESSURE FLUCTUATIONS FROM 0 PSI TO STAGING PRESSURE SHALL NOT EXCEED 6 PSI.**
14. Open **FILLER** port shutoff valve and apply 10 psi to piston rod. Close **FILLER** port shutoff valve and check pressure over 2 minute period for any drop in pressure. Pressure drop indicates internal leakage. **NO PRESSURE DROP ALLOWED.** Repair or replace leaking master brake cylinder.

NOTE

Pressure readings must be taken from the **FILLER** port gage.

15. Open **BRAKE** port shutoff valve and apply 65 to 85 psi to **BRAKE** port. Close **BRAKE** port shutoff valve and check pressure over two minute period for any drop in pressure. Pressure drop indicates internal leakage. **NO PRESSURE DROP ALLOWED.** Repair or replace leaking master brake cylinder.
16. Close low pressure gage shutoff valve.
17. Increase piston rod force to 265 to 285 psi and observe brake pressure. Insure that piston rod has not bottomed-out after two minute waiting period. **BRAKE PRESSURE SHALL BE 800 PSI MINIMUM AT 265 - 285 LBS ROD FORCE. THE PISTON SHALL NOT HAVE BOTTOMED-OUT AT THE END OF A TWO MINUTE PERIOD.**
18. Manually actuate piston rod as fast as possible without taking pressure gage needle over 1100 psi. **PRESSURE SHALL BE 850 PSI MINIMUM AT END OF STROKE.**
19. Remove brake cylinder from test fixture.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****PARKING BRAKE VALVE (AVIM)**

INITIAL SETUP:**Tools and Special Tools**

Air Regulator/Separator, Oil and Water, MIL-S-12928
Container, 2-Gallon
Depth Gage, Micrometer, 0 to 1 Inch
Filter, Hydraulic Fluid, MS28720-4
Fluorescent Penetrant Inspection Kit, XMA101
Hydraulic Repairer's Toolkit, SC 5180-99-A03
Pressure Gage, Hydraulic, 0 to 3000 psi, 1082A
Spring Scale, Dial Indicating, AAA-S133
Stopwatch
Tester, Pressure Gage, Type, MP-1
Torque Wrench, 30 - 200 in. lbs.
Valve Test Fixture, Locally-Made, Figure 202,
WP 1805 00

Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00
Lubricating Oil, General Purpose, Item 186,
WP 1803 00
Petrolatum, Item 216, WP 1803 00
Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Pneudraulics Repairer MOS 15H (1)

References

TM 1-1520-265-23
WP 1803 00
WP 1805 00

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00

DISASSEMBLE**WARNING**

Injury to personnel, or damage to components will result if caution is not used when removing can covers. Covers compress powerful springs. Use caution when removing can covers.

1. Remove bolts. Remove accumulator can covers (Figure 1, Sheet 2, Detail D).
2. Remove accumulator cans and accumulator springs.
3. Remove cylinders.
4. Remove nuts from ends of piston valves. Remove pistons from valve body.
5. Remove packings from piston valves. Discard packing.
6. Remove control lever from parking brake cam (Figure 1, Sheet 1).
7. Remove fitting ends from valve body.
8. Remove packing from fitting end. Discard packing.
9. Remove poppet spring from valve body.
10. Remove poppet valve assembly from valve body.
11. Remove plugs.
12. Remove packings from plugs. Discard packing.

DISASSEMBLE - Continued

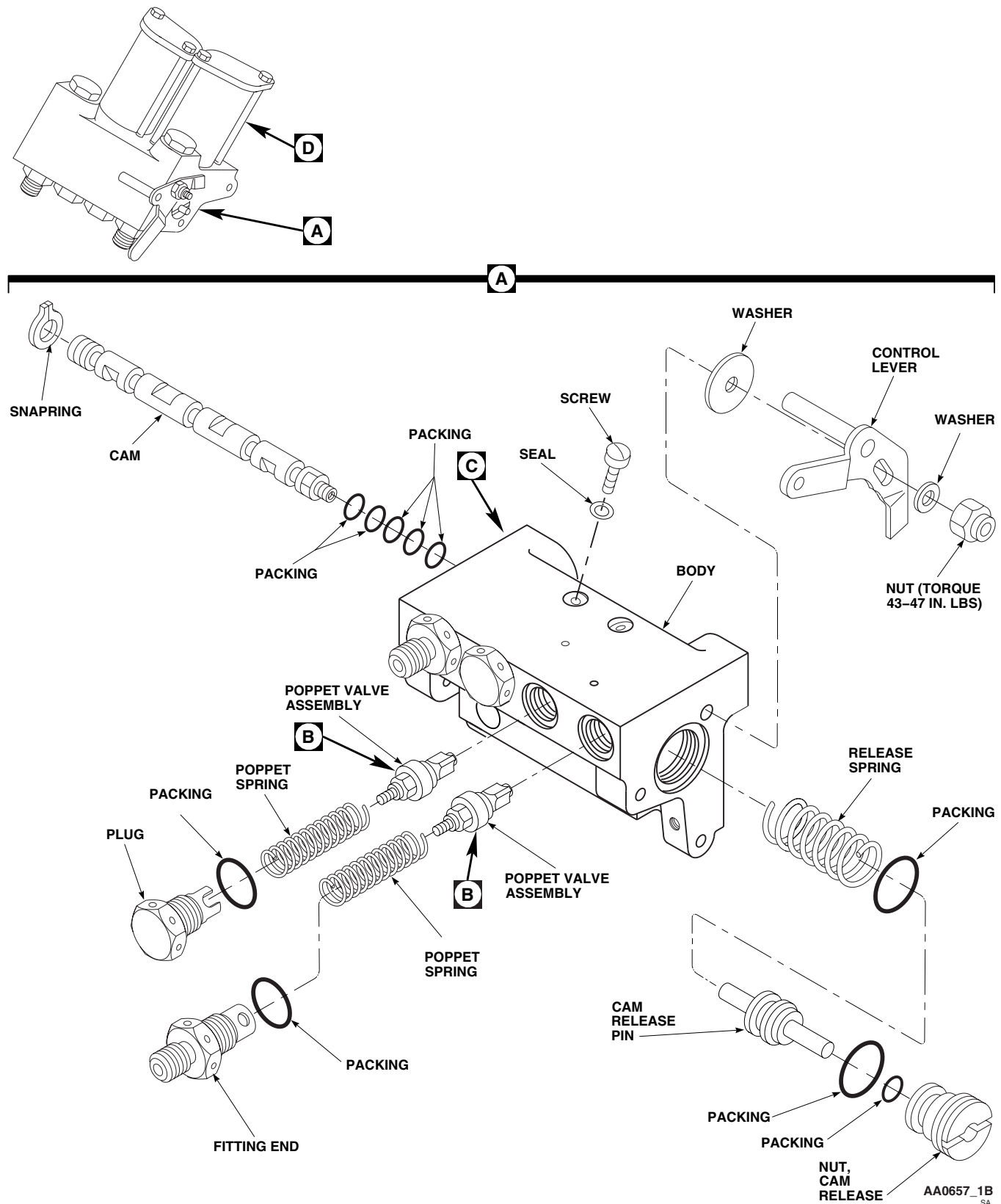


Figure 1. Repair Parking Brake Valve (AVIM). (Sheet 1 of 2)

DISASSEMBLE - Continued

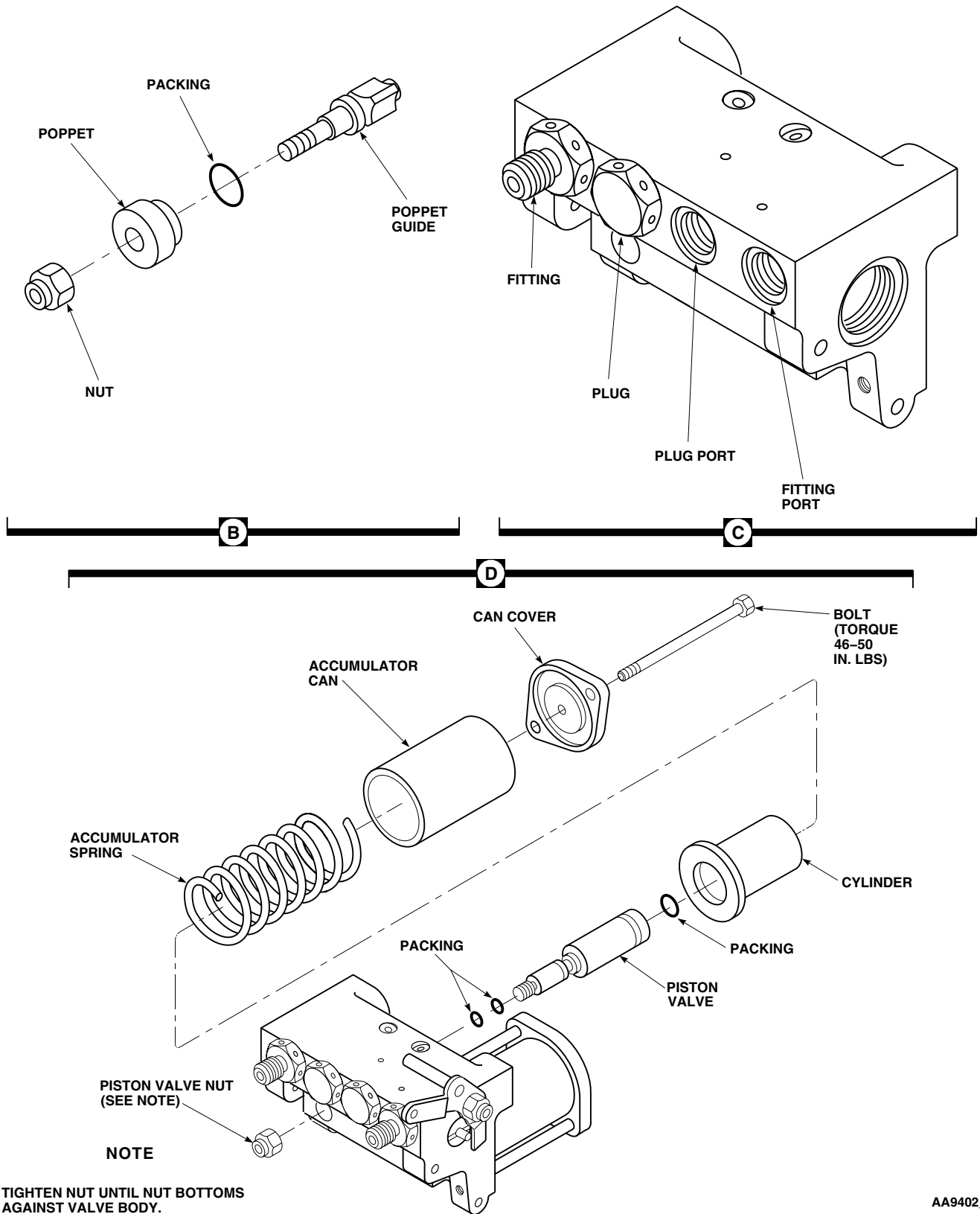


Figure 1. Repair Parking Brake Valve (AVIM). (Sheet 2 of 2)

DISASSEMBLE - Continued

13. Remove springs from valve body.
14. Remove poppet valve assemblies from valve body.
15. Slide cam from valve body.
16. Remove packings from cam. Discard packing.

NOTE

Cam release pin nut is spring loaded.

17. Remove cam release nut from valve body.
18. Remove packings from cam release nut. Discard packing.
19. Remove cam release pin from valve body.
20. Remove packing from cam release pin. Discard packing.
21. Remove release spring from valve body.
22. Remove screws from valve body.
23. Remove seals from screws.
24. Remove poppet valve nuts, poppets, and packings from poppet guides (Figure 1, Sheet 2, Detail B).

CLEAN**CAUTION**

Damage to valve parts will result if dirty solvent is used for cleaning. Make sure clean solvent is used.

Clean valve body and components with cleaning compound solvent, Item 71, WP 1803 00. Use filtered, dry compressed air to dry valve body and parts. Check that valve passages are dry and clean.

INSPECT

1. Check valve parts for scratches, wear, cracks, and corrosion. Replace corroded, cracked or scratched parts.
2. Check valve body and cam release pin for crack using fluorescent penetrant inspection kit (TM 1-1520-265-23).
NO CRACKS ALLOWED. Replace valve body or cam release pin, if cracks are found.
3. Check accumulator cans for dents interfering with spring action. Replace dented accumulator cans.
4. Check accumulator cylinder bores for foreign matter, corrosion, or scoring. Replace damaged accumulator cylinders.
5. Check threads for damage. Replace parts having damaged threads.
6. Replace all packings and springs.

ASSEMBLE

1. Immerse all packings in petroleum base hydraulic fluid, Item 145, WP 1803 00, before installing.
2. Lubricate threads with petroleum base hydraulic fluid, Item 145, WP 1803 00, or petrolatum, Item 216, WP 1803 00, before installation.
3. **QA** Place seals on screws and install screws in valve body. Using nonelectrical wire, Item 357, WP 1803 00, lock-wire screws (Figure 1, Sheet 1).

ASSEMBLE - Continued

4. **QA** Assemble packings, poppets and nuts onto guides. **TIGHTEN NUTS SO PACKINGS PROTRUDE 0.010-INCH (FIGURE 1, SHEET 2, DETAIL B).**
5. **QA** Install poppet valve assemblies with new springs, in valve body (Figure 1, Sheet 1).
6. **QA** Install packings on plugs.
7. **QA** Compress spring on poppet valve assemblies and install plugs. Using nonelectrical wire, Item 357, WP 1803 00, lockwire plugs.
8. **QA** Install packings on fitting ends.
9. **QA** Install fitting ends in valve body. Using nonelectrical wire, Item 357, WP 1803 00, lockwire fitting end.
10. **QA** Install new release spring in valve body.
11. **QA** Install packing on cam release pin.
12. **QA** Install cam release pin in valve body.
13. **QA** Install packing on outer groove and packing on inside groove of nut.
14. **QA** Install cam release nut in valve body. **TIGHTEN SO THAT NUT IS FLUSH TO 0.010-INCH BELOW VALVE BODY SURFACE.**
15. **QA** Install packings on cam.
16. **QA** Lubricate cam with petroleum base hydraulic fluid, Item 145, WP 1803 00, and while observing orientation of cam as shown, install in valve body.
17. **QA** Install 1 inch diameter flat washer over cam threaded end, followed by control lever, washer and nut. **TORQUE NUT TO 43 - 47 INCH-POUNDS.**
18. **QA** Install packings on small end of piston valves (Figure 1, Sheet 2, Detail D).
19. **QA** Install packings on large end of piston valves.
20. **QA** Lubricate piston valves with petroleum base hydraulic fluid, Item 145, WP 1803 00, and install in valve body.
21. **QA** Lubricate threads of piston valve nuts with general purpose lubricating oil, Item 186, WP 1803 00, and install. **TIGHTEN PISTON VALVE NUTS UNTIL NUTS BOTTOM AGAINST VALVE BODY. DO NOT OVERTIGHTEN.**
22. **QA** Install cylinders over piston valves.
23. Test parking brake valve. Refer to, TEST PARKING BRAKE VALVE (AVIM), in this work package.
24. **QA** Install accumulator springs over cylinders.
25. **QA** Install accumulator cans over accumulator springs.
26. **QA** Install accumulator can covers and bolts. **TORQUE BOLTS TO 46 - 50 INCH-POUNDS.** Using nonelectrical wire, Item 357, WP 1803 00, lockwire bolts.

TEST**NOTE**

Assemble valve to point described in ASSEMBLE, in this work package. Before completing assembly, test valve seals for fluid leakage.

1. Locally make valve test fixture (WP 1805 00).
2. Install parking brake valve in test fixture.

TEST - Continued

3. Make hydraulic test pressure and discharge manifolds (Figure 2).
4. Service gage tester with petroleum base hydraulic fluid, Item 145, WP 1803 00.
5. Install hydraulic fluid filter on outlet port of gage tester.
6. Attach pressure manifold to filter outlet port. Open both pressure manifold shut-off valves.
7. Pump gage tester until a steady stream of air free hydraulic fluid flows from hose.
8. Attach pressure manifold to parking brake valve cylinder ports.
9. Attach discharge manifold to parking brake valve wheel brake ports. Route hose end into a 2 gallon container.
10. Refill gage tester reservoir with new petroleum base hydraulic fluid, Item 145, WP 1803 00, as required.
11. Put control level in BRAKE-ON position.
12. Open all four manifold shut-off valves.
13. Pump all remaining air out of hoses and valves.
14. Close discharge manifold shut-off valves.
15. Raise pressure to 1940 to 2140 psi. Using stop watch, hold pressure for two minutes.
16. Check brake valve for external leakage or distortion. **NO LEAKAGE OR DISTORTION ALLOWED.** Repair or replace valve parts as required. Refer to, REPAIR PARKING BRAKE VALVE (AVIM), in this work package.
17. Open discharge manifold valves. Reduce pressure to zero.
18. Put control lever in BRAKE-OFF position.
19. Close discharge manifold shut-off valves.
20. Apply 1000 psi at cylinder ports. Move valve control lever to BRAKE-ON position.
21. Open gage tester return valve and reduce pressure to zero at brake valve cylinder ports.
22. Check lockpin extension. Failure of lockpin to extend may be caused by binding of release pin in lockpin and piston bore. Inspect and repair as necessary. Refer to, REPAIR PARKING BRAKE VALVE (AVIM), in this work package.
23. Using stopwatch, check for valve pressure loss over five minutes. Any pressure loss indicates brake poppet valve or accumulator packing seal leakage. **NO LEAKAGE ALLOWED.** Inspect and repair as necessary. Refer to, REPAIR PARKING BRAKE VALVE, in this work package.
24. Close gage tester return valve.
25. Close pressure manifold shut-off valves.
26. Apply 100 to 130 psi maximum with gage tester pump.
27. Attach a dial indicating spring scale to end of control lever. Apply 7 1/2-pounds tension while pulling lever toward BRAKE-OFF position.
28. Open left hand pressure manifold shut-off valve. **CONTROL LEVER LOCK-PIN SHALL RETRACT WITHOUT HESITATION** allowing control lever to be rotated toward BRAKE-OFF position. Inspect and repair faulty pin retraction or binding cam lever action. Refer to, REPAIR PARKING BRAKE VALVE (AVIM), in this work package.
29. Open right hand pressure manifold shut-off valve.
30. Apply 556 psi to valve cylinder ports.
31. Close discharge manifold shut-off valves.

TEST - Continued

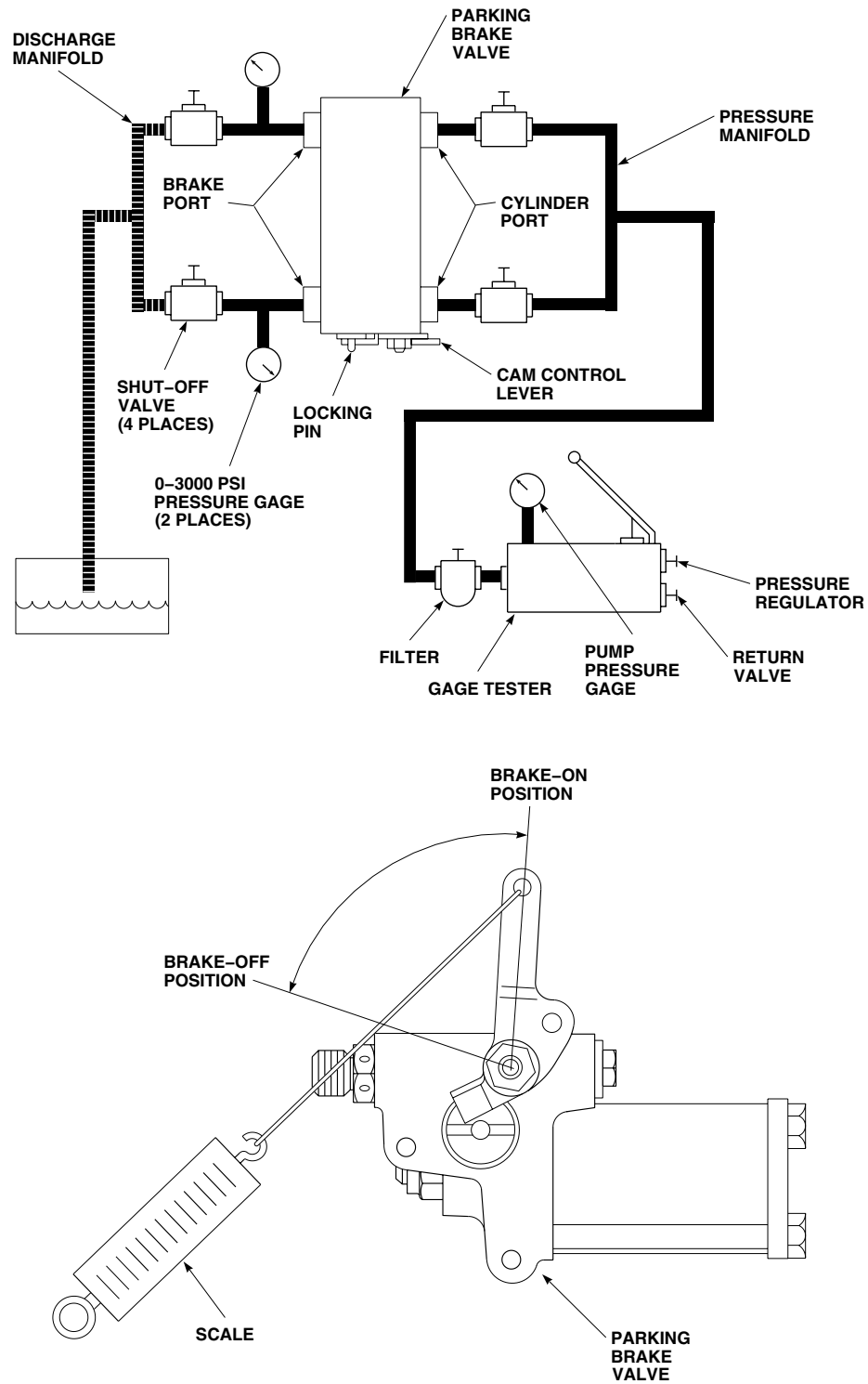


Figure 2. Test Parking Brake Valve (AVIM).

TEST - Continued

32. Put control lever in BRAKE-ON position to energize accumulators.
33. Open gage tester return valve to relieve pressure at valve cylinder ports.
34. Note time. Leave valve in brake-on position for eight hours. **INDICATED PRESSURE SHALL BE 556 TO 334 PSI AT THE END OF EIGHT HOURS.** A pressure drop below 334 psi indicates unacceptable leakage at poppet valves and accumulators. Disassemble, inspect, and repair as necessary. Refer to, REPAIR PARKING BRAKE VALVE (AVIM), in this work package.
35. Retest parking brake valve using the preceding test sequence. **FAILURE TO MEET ANY ONE OF THE TEST REQUIREMENTS SHALL BE CAUSE FOR REJECTION.**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****SLAVE MIXER VALVE (AVIM)**

INITIAL SETUP:**Tools and Special Tools**

Air Regulator/Separator, Oil and Water, MIL-S-12928
Dial Indicator, MIL-I-18422
Flow Meter, 2-1110-6MS
Fluorescent Penetrant Inspection Kit, XMA101
Hydraulic Repairer's Toolkit, SC 5180-99-A03
Pressure Gage, Hydraulic, 0 to 200 psi, 1082A
Pressure Gage, Hydraulic, 0 to 3000 psi
Test Stand, Hydraulic, MIL-T-7612

Hydraulic Fluid, Petroleum Base, Item 145,
WP 1803 00

Wire, Nonelectrical, Item 352, WP 1803 00

Personnel Required

Aircraft Pneudraulics Repairer MOS 15H (1)

References

TM 1-1520-265-23
WP 1803 00

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00

REPAIR SLAVE MIXER VALVE (AVIM)**Disassemble****WARNING**

Injury to personnel and damage to equipment will occur if care is not taken when removing slave mixer valve cap. Cap is spring loaded. Take care when removing cap.

1. Remove screws and cap from valve body (Figure 1).
2. Remove packing retainer, packing, and spring. Discard packing.
3. Remove piston assembly from valve body.
4. Remove valve cap, poppet, and poppet spring from piston (Figure 1, Detail A).
5. Remove nylon lock and packing from piston. Discard packing.
6. Remove plug and packing from valve body. Discard packing.

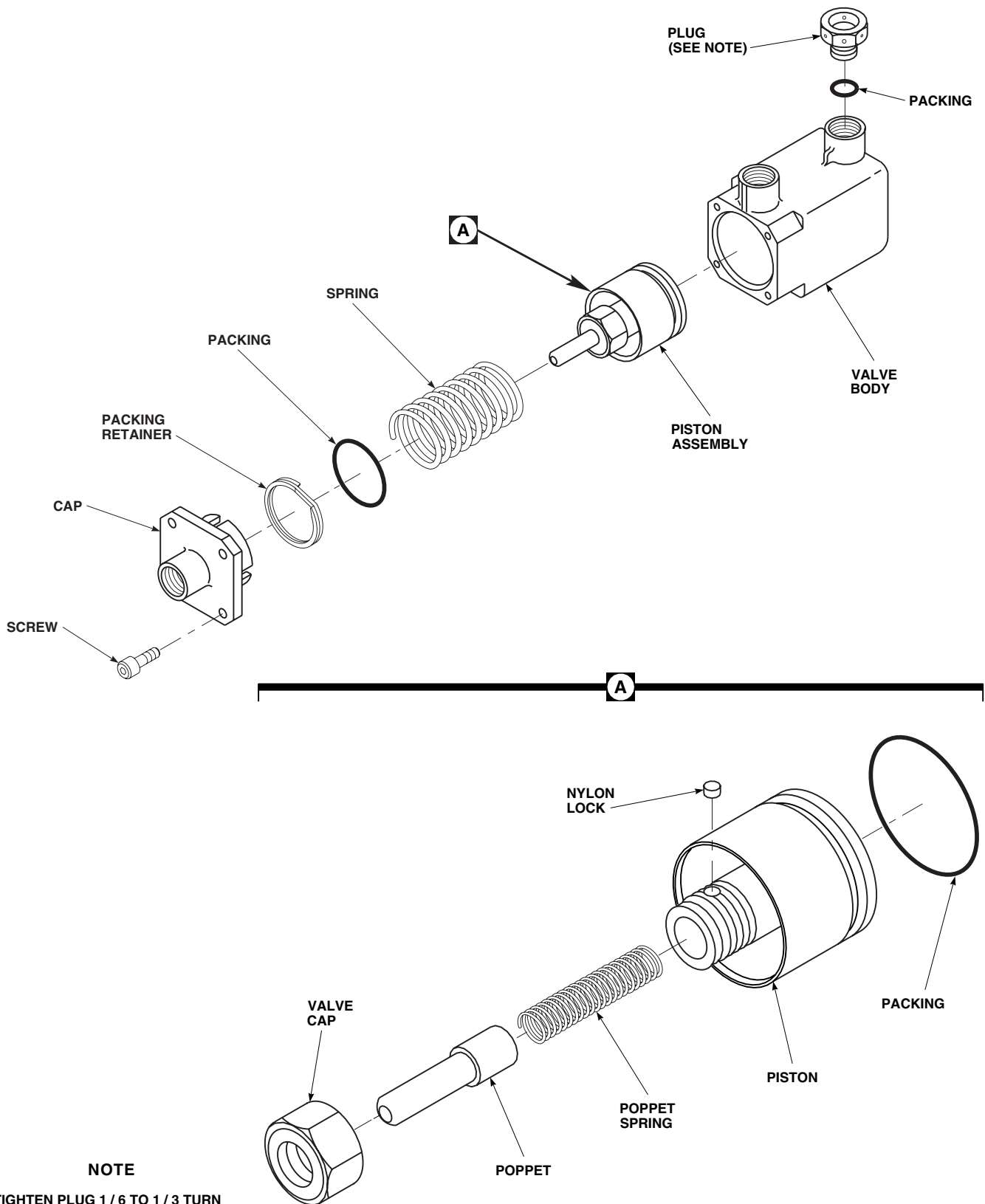
Clean

Clean parts using cleaning compound solvent, Item 71, WP 1803 00. Use filtered, dry compressed air to dry parts.

Inspect

1. Inspect valve parts for scratches, wear, cracks, and corrosion. **NONE ALLOWED.** Replace damaged or worn parts.
2. Inspect valve body and piston assembly for crack using Fluorescent Penetrant Inspection Kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace valve body or piston assembly, if cracks are found.
3. Inspect valve body bore for scratches, wear, corrosion, or scoring. **NONE ALLOWED.** Replace damaged or worn valve body.
4. Inspect threads for damage. **NO DAMAGE ALLOWED.** Replace parts having damaged threads.

REPAIR SLAVE MIXER VALVE (AVIM) - Continued



AK0693
SA

Figure 1. Repair Slave Mixer Valve (AVIM).

REPAIR SLAVE MIXER VALVE (AVIM) - Continued**Assemble**

1. Lubricate all packings using petroleum base hydraulic fluid, Item 145, WP 1803 00.
2. **QA** Install poppet spring and poppet into piston and secure with valve cap and nylon lock (Figure 1, Detail A).
3. **QA** Install packing on piston. Install piston assembly into valve body.
4. **QA** Install packing retainer and packing on cover (Figure 1).
5. **QA** Install spring and cover assembly into valve body. Secure cover with screws. Using nonelectrical wire, Item 352, WP 1803 00, lockwire screws.
6. Test slave mixer valve. Refer to, TEST SLAVE MIXER VALVE (AVIM), in this work package.
7. **QA** Install plug and packing into valve body. **TIGHTEN PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Using nonelectrical wire, Item 352, WP 1803 00, lockwire plug.

TEST SLAVE MIXER VALVE (AVIM)**WARNING**

Injury to personnel will result if care is not taken when testing slave mixer valve. Hydraulic fluid is under high pressure. Wear protective goggles and clothing when testing slave mixer valve.

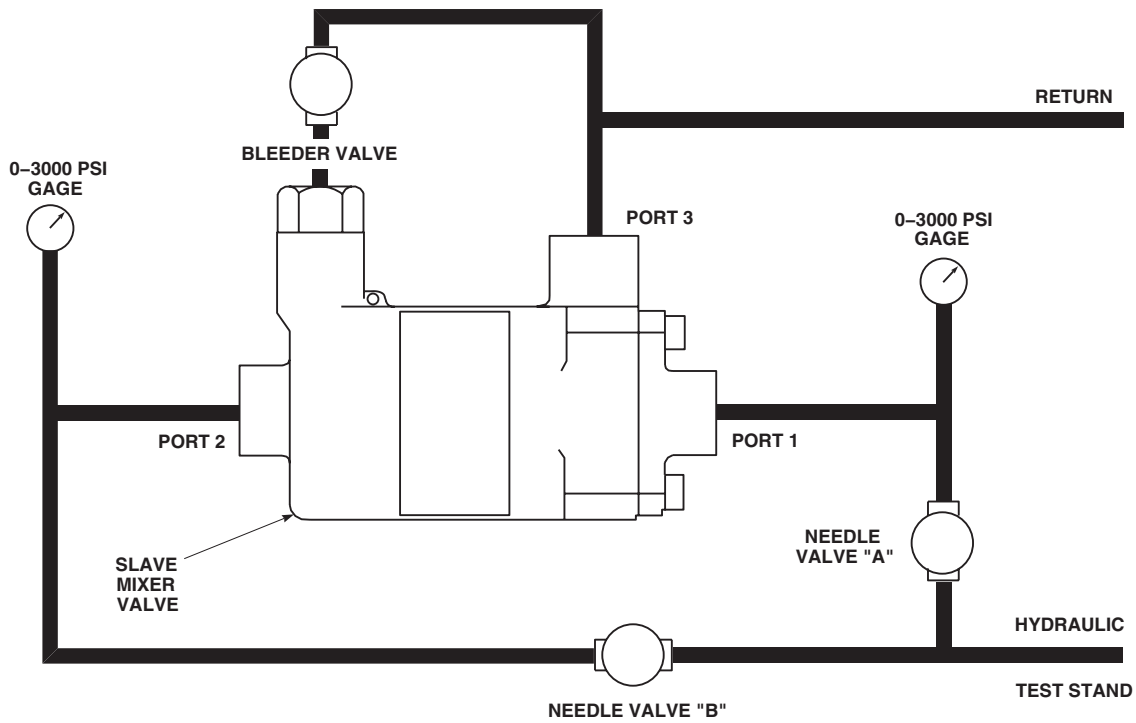
Proof Pressure Test

1. Set up slave mixer valve for test (Figure 2, Sheet 1).
2. Close needle valve B and open needle valve A.
3. Slowly apply hydraulic pressure to port 1 until air-free flow is observed from port 3. Close valve A.
4. Cap port 3 and adjust valve A to apply 2040 psi to port 1. Hold pressure for 2 minutes.
5. Check slave mixer valve for external or internal leakage from port 2. **NO LEAKAGE OR DEFORMATION ALLOWED.** Repair or replace slave mixer valve if leakage or deformation is noted.
6. Close valve A. Reduce hydraulic pressure to 100 psi.
7. Open bleeder valve and valve B. Apply low pressure to port 2 to purge air from slave mixer valve.
8. Close bleeder valve. Reduce test stand pressure to 0 psi. Open valve A and return line valve to vent port 1 and port 3. Close valve A and return line valve.
9. Adjust valve B to apply 2040 psi to port 2. Hold pressure for 2 minutes.
10. Check slave mixer valve for external or internal leaks from port 1 and port 3. **NO LEAKAGE OR DEFORMATION ALLOWED.** Repair or replace slave mixer valve if leakage or deformation is noted.
11. Reduce test stand pressure to 0 psi and close valve B.

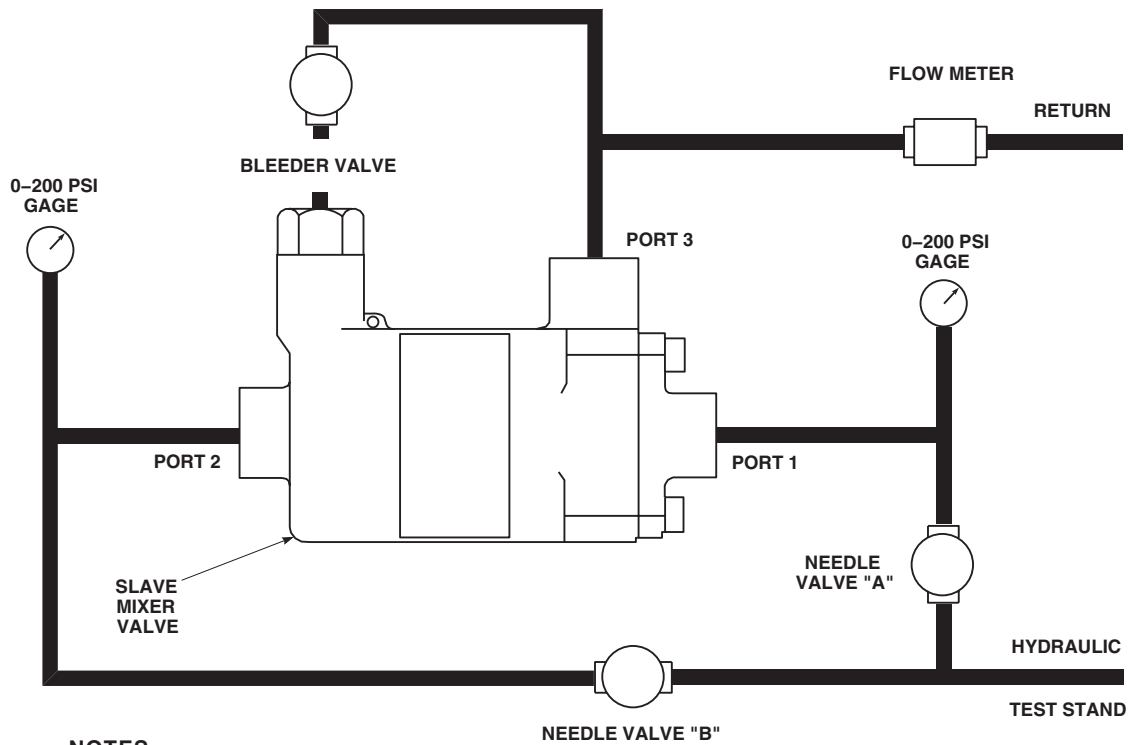
Check Valve Relief Pressure Test

1. Replace 0 to 3000 psi gages with 0 to 200 psi gages (Figure 2, Sheet 1).
2. Install flow meter along return line from port 3.
3. Close valve A. Open valve B and apply 100 psi to port 2. Open valve A and slowly apply pressure to port 1 until a flow of 20 cc/min is recorded out port 3.
4. Record pressure at port 1. **PORT 1 PRESSURE SHALL BE 30 PSI MAXIMUM.** Repair or replace slave mixer valve if port 1 pressure is more than limits.

TEST SLAVE MIXER VALVE (AVIM) - Continued



PROOF PRESSURE TEST



NOTES

1. LOW PRESSURE LEAKAGE TEST USE 0-200 PSI GAGES.
2. HIGH PRESSURE LEAKAGE TEST USE 0-3000 PSI GAGES.

CHECK VALVE RELIEF PRESSURE TEST

AK0694_1A
SA

Figure 2. Test Slave Mixer Valve (AVIM). (Sheet 1 of 2)

TEST SLAVE MIXER VALVE (AVIM) - Continued

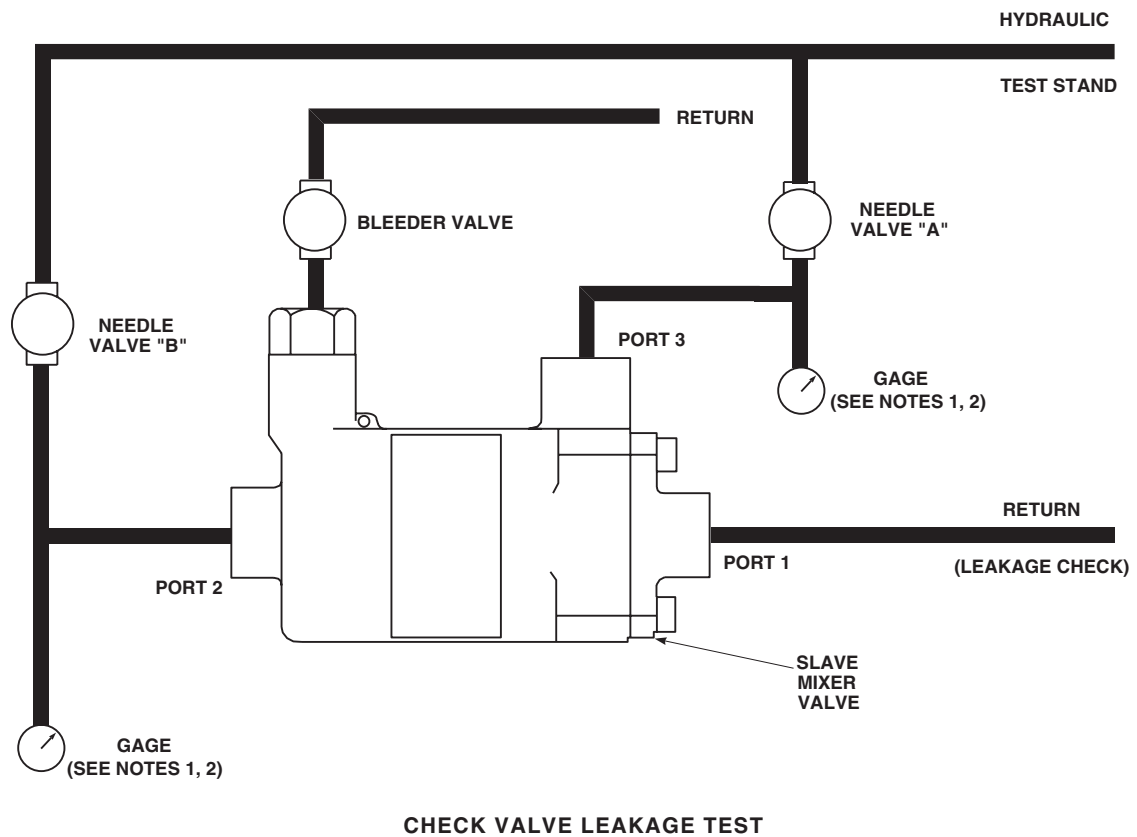
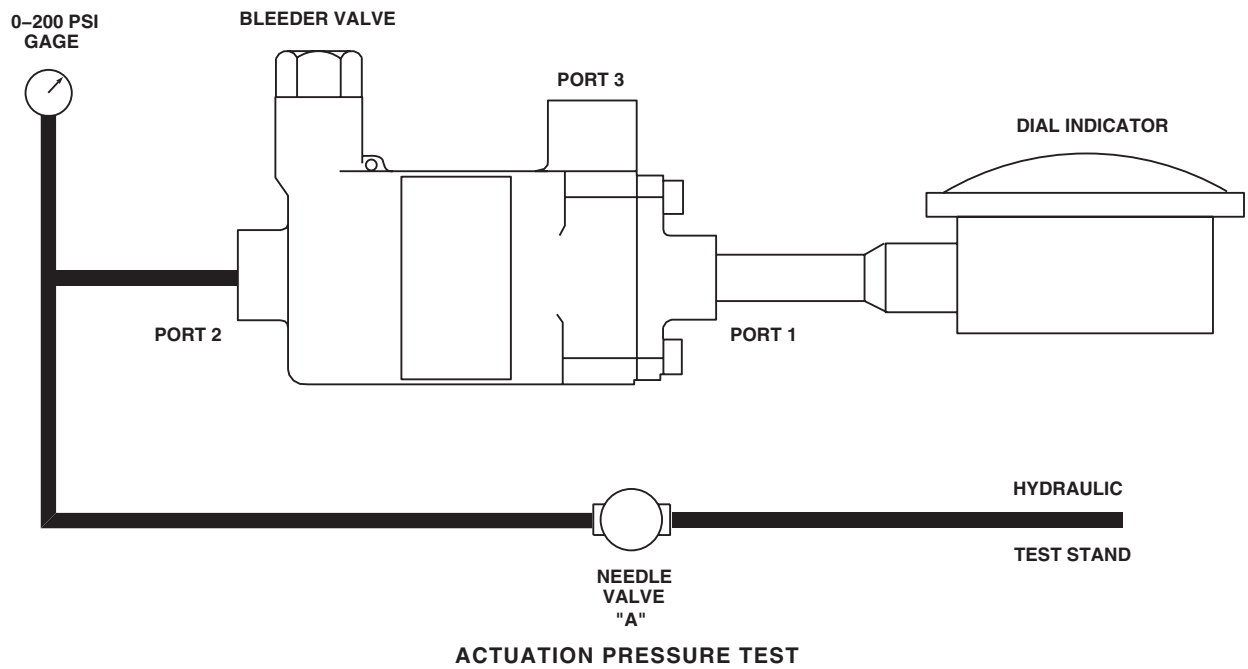
AK0694_2A
SA

Figure 2. Test Slave Mixer Valve (AVIM). (Sheet 2 of 2)

TEST SLAVE MIXER VALVE (AVIM) - Continued**Check Valve Leakage Test**

1. Perform low pressure leakage test as follows:
 - a. Set up slave mixer valve for test (Figure 2, Sheet 2). Use 0 to 200 psi gages.
 - b. Adjust valve B to apply 40 to 45 psi to port 2. Adjust valve A to apply 20 psi to port 3. Close valve A.
 - c. Hold pressure for 2 minutes and check port 1 for leakage. **PORT 1 LEAKAGE SHALL NOT EXCEED 1 DROP PER MINUTE.** Repair or replace slave mixer valve if leakage is more than limits.
2. Perform high pressure leakage test as follows:
 - a. Replace 0 to 200 psi gages with 0 to 3000 psi gages.
 - b. Adjust valve B to apply 520 to 525 psi to port 2. Adjust valve A to apply 500 psi to port 3. Close valve A.
 - c. Hold pressure for 2 minutes and check port 1 for leakage. **PORT 1 LEAKAGE SHALL NOT EXCEED 1 DROP PER MINUTE.** Repair or replace slave mixer valve if leakage is more than limits.
 - d. Adjust valve B to apply 1025 to 1030 psi to port 2. Adjust valve A to apply 1000 psi to port 3. Close valve A.
 - e. Hold pressure for 2 minutes and check port 1 for leakage. **PORT 1 LEAKAGE SHALL NOT EXCEED 1 DROP PER MINUTE.** Repair or replace slave mixer valve if leakage is more than limits.

Actuation Pressure Test

1. Set up slave mixer valve for test in a padded vise (Figure 2, Sheet 2).
2. Install plug and packing. Refer to, REPAIR SLAVE MIXER VALVE (AVIM), in this work package.
3. Insert dial indicator into port 1 until contact is made with poppet and reading is noted on indicator.
4. Adjust valve A until dial indicator movement is noted. **PRESSURE SHALL NOT EXCEED 12 PSI.** Repair or replace slave mixer valve if pressure is more than limits.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

LANDING GEAR

TAIL LANDING GEAR SHOCK STRUT (AVIM)

This WP supersedes WP 0473 00, dated 21 March 2007.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC-5180-99-B01
 Caliper, Micrometer, Set, GGG-C-105
 Container, 1-Gallon
 Coupler, 3205-4
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Fly Cutter, RST1013
 Hydraulic Repairer's Toolkit, SC 5180-99-A03
 Hydraulic Shutoff Valve, M1400S
 Hydraulic Test Stand, MIL-T-7612D
 Inside Micrometer, Set, GGG-C-105
 Magnifier, 06175
 Mallet, Rawhide
 Nonmetallic Scraper
 Pail, Utility, RR-P-35
 Plastic/Wooden Pick
 Puller Kit, Mechanical, GGG-P-781
 Pump, Inflating, Manual, 2357
 Spinning Tool, STD-1850-458-017
 Tags
 Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Cleaning Compound, Solvent, Item 71, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00

Cloth, Cheesecloth, Item 85, WP 1803 00
 Hydraulic Fluid, Fire Resistant, Item 144, WP 1803 00
 Hydraulic Fluid, Petroleum Base, Item 145, WP 1803 00
 Leak Detection Fluid, Item 175, WP 1803 00
 Nitrogen, Item 199, WP 1803 00
 Petrolatum, Item 216, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Screw, MS35275-247
 Towel, Machinery Wiping, Item 344, WP 1803 00
 Wire, Nonelectrical, Item 354, WP 1803 00
 Wire, Nonelectrical, Item 356, WP 1803 00
 Wire, Nonelectrical, Item 357, WP 1803 00

Personnel Required

Aircraft Pneudraulics Repairer MOS 15H (1)
 Assistant - (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
 TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0381 00
 WP 0459 00
 WP 0460 00
 WP 0461 00
 WP 1803 00

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM)

Disassemble

NOTE

Tail landing gear shock strut assemblies must be upgraded to latest configuration P/N 70250-13101-044, unless otherwise specified

1. Clean strut exterior using cleaning compound solvent, Item 71, WP 1803 00, and machinery wiping towel, Item 344, WP 1803 00.

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued**NOTE**

Work area and tools must be clean. Use a surface covered with clean machinery wiping towel, Item 344, WP 1803 00, for disassembled parts.

2. Remove upper and lower stage air valve caps. Loosen outer hexnut and depressurize strut upper and lower stages. Leave valves open (Figure 1, Detail B).

WARNING

Injury to personnel and damage to equipment will occur if shock strut is disassembled without removing upper and lower bleeder plugs. Residual pressure will cause shock strut piston to blow out during disassembly. Make sure hydraulic bleeder plugs are slowly removed to release possible residual pressure resulting from leakage from air side to hydraulic side of shock strut.

3. Remove upper and lower stage bleeder plugs. Discard packings. Drain hydraulic fluid from strut into 1 gallon container. Discard used fluid (Figure 1, Detail A).
4. Remove upper and lower kneeling valves. Discard packings.
5. Remove screws, washers, and band from piston and cylinder assembly. Remove pins using screw, MS35275-247, and puller.

NOTE

Split bearing halves may fall free as piston clears cylinder.

6. Separate lower stage piston assembly from piston and cylinder assembly. Catch split bearing halves as piston is removed from cylinder.
7. Remove pins, retainer, orifice assembly, spacer, dampener, and bearing. Do not disassemble orifice assembly.

WARNING

Injury to personnel and damage to equipment will result if excessive air pressure is used to remove floating piston. Slowly operate hand pump at low pressure when removing floating piston.

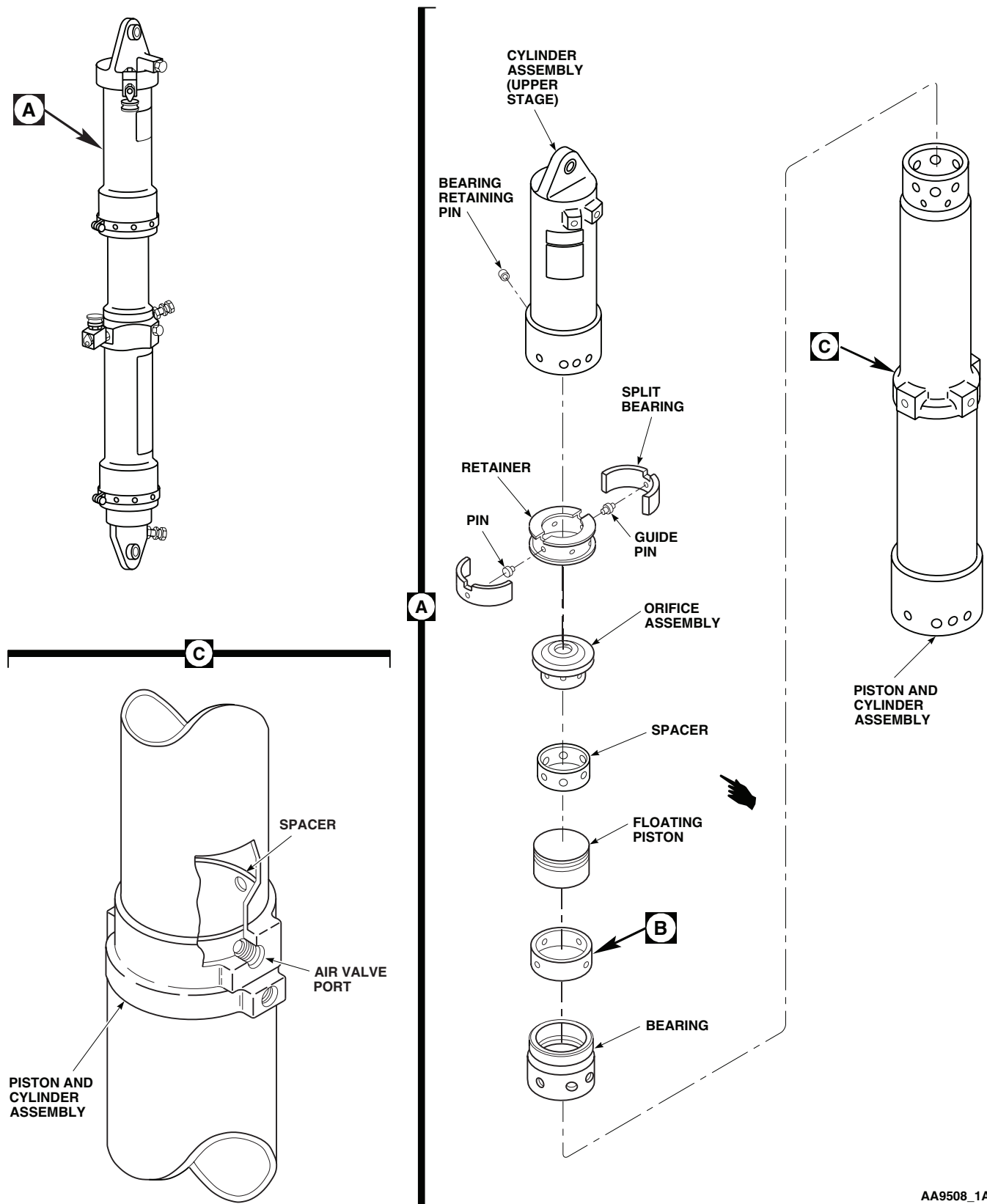
8. Attach a manual air pump to lower air valve. Apply 2 to 5 psi pressure. Catch floating piston in a pail filled with machinery wiping towel, Item 344, WP 1803 00. Remove manual air pump.
9. Remove lower stage air valve. Discard packing.
10. Remove screws, washers, and band from cylinder assembly.
11. Remove pins using screw, MS35275-247, and puller.

NOTE

Split bearing halves may fall free as piston clears cylinder.

12. Separate piston and cylinder assembly from upper stage cylinder assembly. Catch split bearing halves as piston is removed from cylinder.
13. Remove pins, retainer, orifice assembly, spacer, and bearing. Do not disassemble orifice.

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued



AA9508_1A
SA

Figure 1. Disassemble Tail Landing Gear Shock Strut (AVIM).

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued**WARNING**

Injury to personnel and damage to equipment will result if excessive air pressure is used to remove floating piston. Slowly operate hand pump at low pressure when removing floating piston.

14. Attach manual air pump to upper air valve. Apply 2 to 5 psi pressure. Catch floating piston in a pail filled with machinery wiping towel, Item 344, WP 1803 00.
15. Remove spacer from inside diameter of upper stage cylinder assembly.
16. Remove upper air valve. Discard packing.
17. Remove packings, seals, and scrapers using a plastic or wooden pick. Discard used packings, seals, and scrapers.
18. Clean parts with cleaning compound solvent, Item 71, WP 1803 00. Blow dry with dry compressed air.

Inspect

1. Examine strut parts for cracks, scoring, scratches, and corrosion. Use metal flaw detection procedures as necessary to verify suspected cracks or fractures (TM 1-1520-265-23). **NO SCORING, CRACKS, OR CORROSION ALLOWED.** Replace damaged parts.
2. Check piston assembly, piston and cylinder assembly, and cylinder assembly chrome plated surfaces for wear (Figure 2). Take measurements along several points, 90 degrees apart as follows:
 - a. **AREA A - CYLINDER ID SHALL NOT BE MORE THAN 3.503-INCHES.** Replace cylinder assembly if out of tolerance.
 - b. **AREA B - PISTON OD SHALL NOT BE LESS THAN 2.9935-INCHES.** Replace piston and cylinder assembly if out of tolerance.
 - c. **AREA C - PISTON BORE ID SHALL NOT BE MORE THAN 2.813-INCHES.** Replace piston and cylinder assembly if out of tolerance.
 - d. **AREA D - CYLINDER ID SHALL NOT BE MORE THAN 3.503-INCHES.** Replace piston and cylinder assembly if out of tolerance.
 - e. **AREA E - PISTON BORE ID SHALL NOT BE MORE THAN 2.812-INCHES.** Replace piston assembly if out of tolerance.
 - f. **AREA F - LOWER STAGE PISTON OD SHALL NOT BE LESS THAN 2.995-INCHES.** Replace piston if out of tolerance.
 - g. **AREA G - LOWER STAGE PISTON PIN ORIFICE BORE DIAMETER SHALL NOT BE MORE THAN 0.4385-INCH.** Replace piston if out of tolerance.
 - h. **AREA H - UPPER STAGE PISTON PIN ORIFICE BORE DIAMETER SHALL NOT BE MORE THAN 0.438-INCH AND GUIDE BORE DIAMETER SHALL NOT BE MORE THAN 0.3755-INCH.** Replace piston and cylinder assembly if out of tolerance.
3. Check split bearing halves for wear at several points around circumference. **SPLIT BEARING THICKNESS SHALL NOT BE LESS THAN 0.063-INCH (Figure 3, Detail A).** Replace bearings if out of tolerance.
4. Check retainers for wear at several points around OD. **RETAINER SHALL NOT BE LESS THAN 3.486-INCHES (Figure 3, Detail B).** Replace retainer if out of tolerance.
5. Check bearings for wear at several points around ID. **BEARING ID SHALL NOT BE MORE THAN 3.0035-INCHES (Figure 3, Detail C).** Replace out of tolerance bearings.
6. Check floating pistons for wear at several points around OD. **FLOATING PISTON SHALL NOT BE LESS THAN 2.8035-INCHES (Figure 3, Detail D).** Replace floating pistons if out of tolerance.

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued

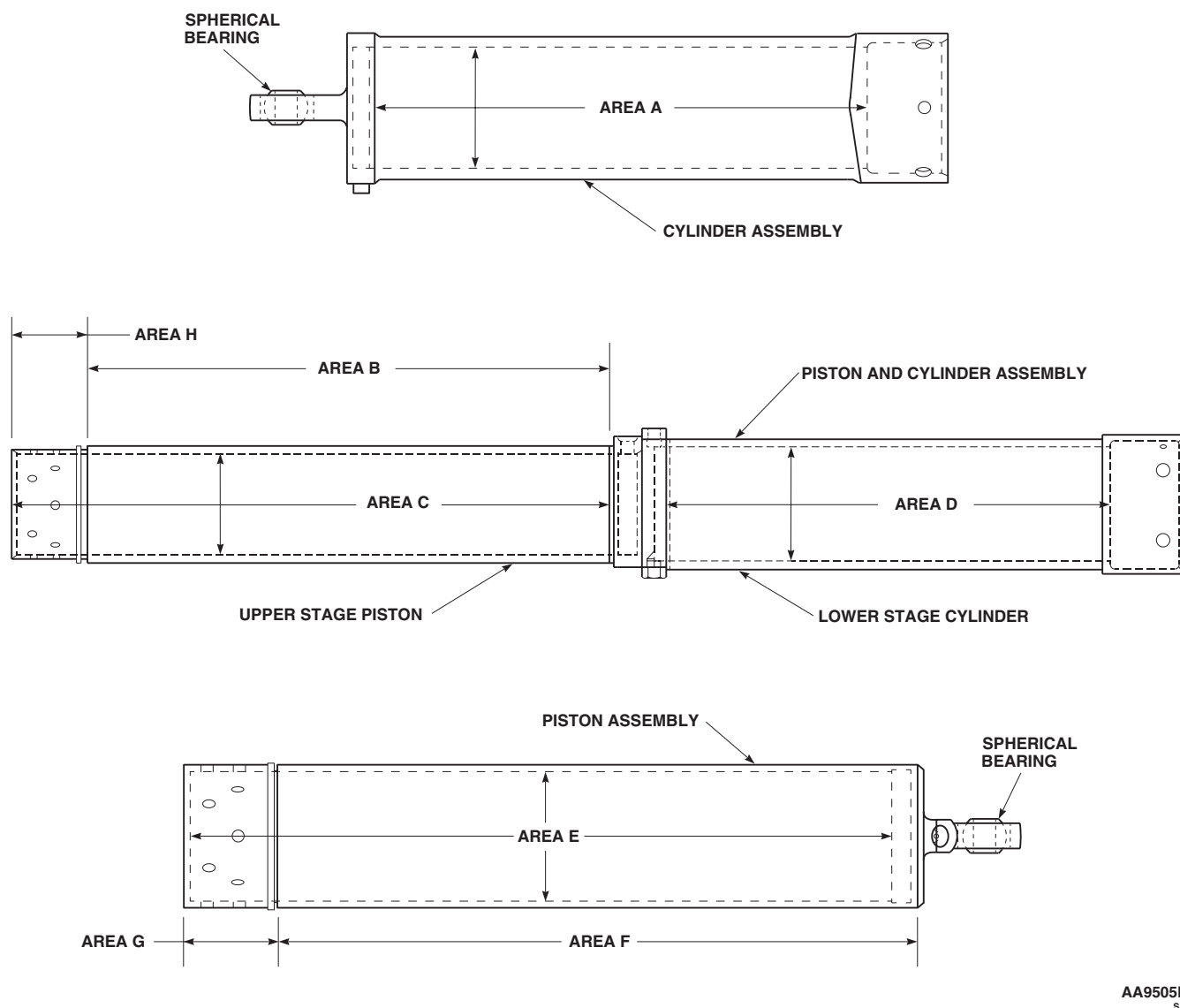
AA9505B
SA

Figure 2. Inspect Tail Landing Gear Shock Strut Cylinder, Piston and Cylinder, and Piston Assemblies (AVIM).

7. Check piston assembly and cylinder assembly spherical bearings for radial and axial play. **BEARINGS MAY HAVE 0.010-INCH MAXIMUM RADIAL PLAY AND 0.030-INCH MAXIMUM AXIAL PLAY.** Replace bearing if out of tolerance. Refer to, REPLACE TAIL LANDING GEAR SHOCK STRUT SPHERICAL BEARINGS (AVIM), in this work package.
8. Check Teflon liners for disbonding. **NO DISBONDING ALLOWED.** Replace bearing if liner is disbonded. Refer to, REPLACE TAIL LANDING GEAR SHOCK STRUT SPHERICAL BEARINGS (AVIM), in this work package.

Assemble

1. Make sure all parts are clean.

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued

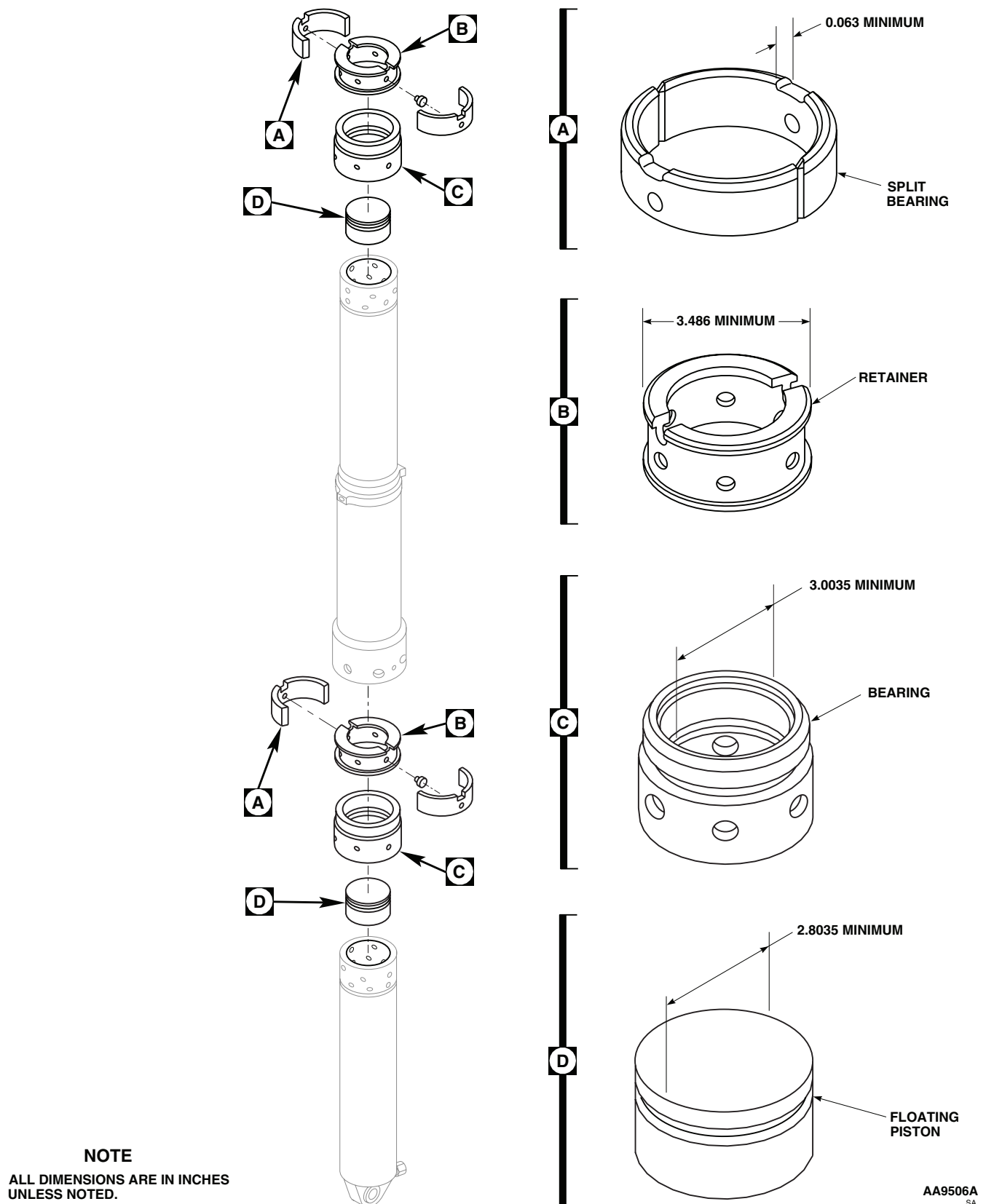


Figure 3. Inspect Tail Landing Gear Shock Strut Bearings, Retainer, and Floating Piston (AVIM).

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued

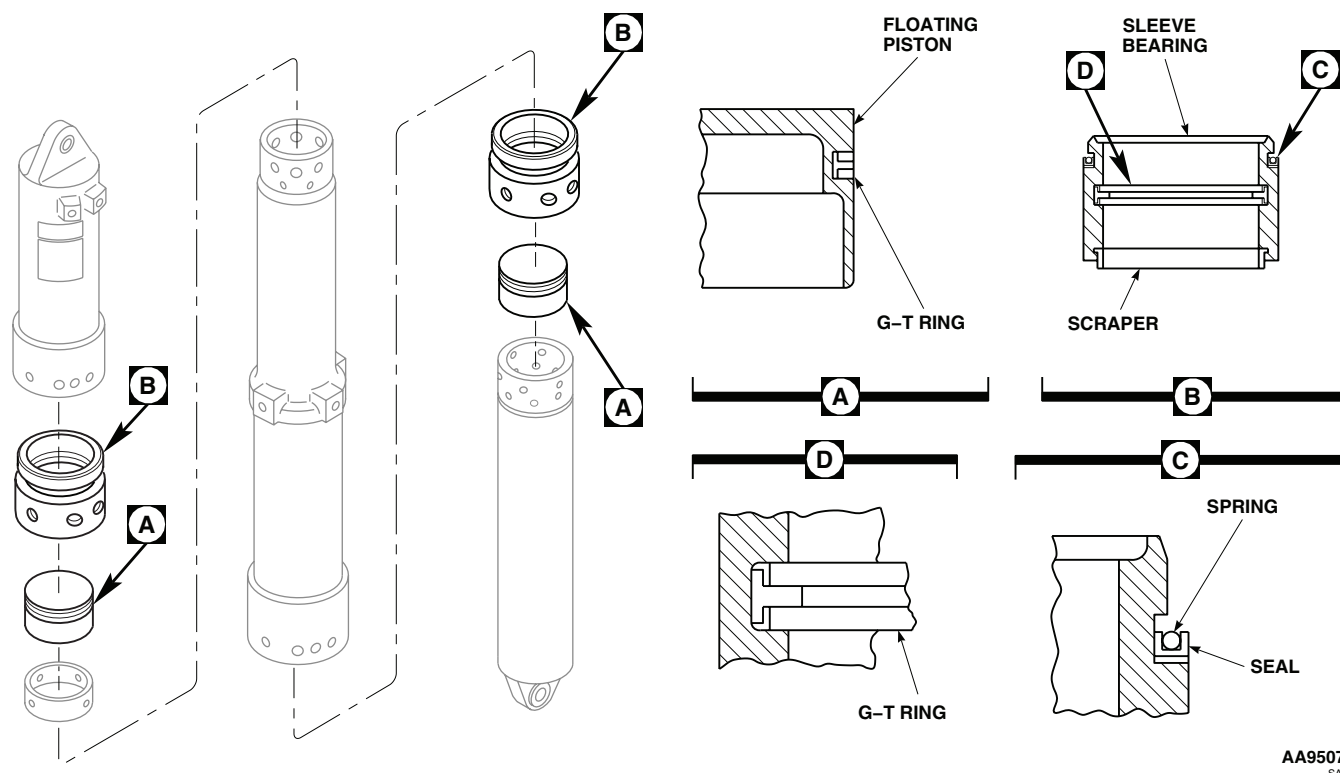
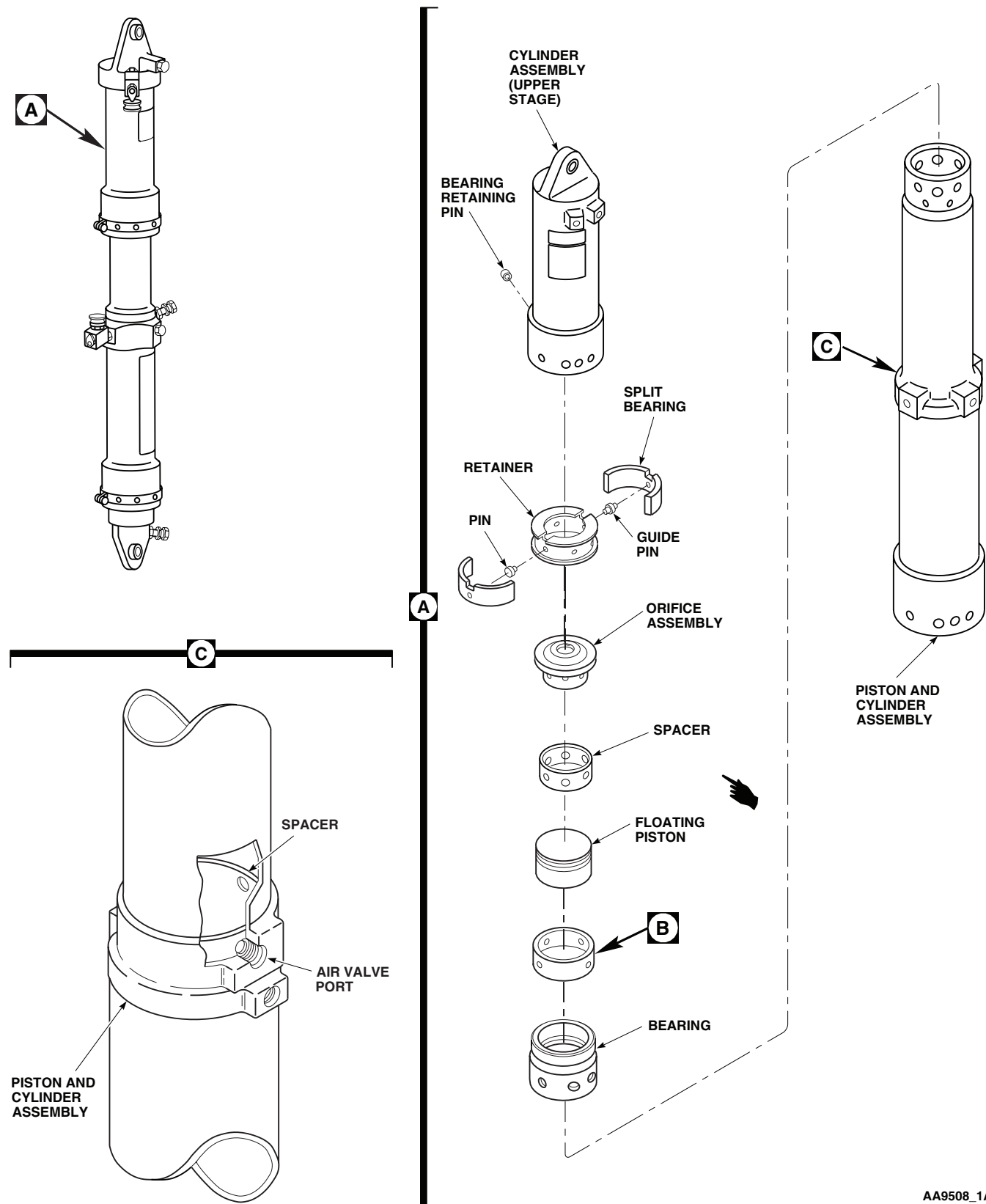


Figure 4. Install Tail Landing Gear Shock Strut Floating Piston/Sleeve Bearing (AVIM).

NOTE

- If strut is to be stored, use fire resistant hydraulic fluid, Item 144, WP 1803 00, as an assembly and servicing fluid.
 - If strut is scheduled for immediate return to service, use petroleum base hydraulic fluid, Item 145, WP 1803 00, as an assembly and servicing fluid.
 - Tail landing gear shock strut assemblies must be upgraded to latest configuration P/N 70250-13101-044, unless otherwise specified.
2. Soak packings, retainers, G-T rings, and seals in fire resistant hydraulic fluid, Item 144, WP 1803 00, or petroleum base hydraulic fluid, Item 145, WP 1803 00.
 3. Coat chrome-plated sliding and threaded parts with fire resistant hydraulic fluid, Item 144 or petroleum base hydraulic fluid, Item 145, WP 1803 00.
 4. **QA** Install G-T rings on upper and lower stage floating pistons, with radius of Teflon rings facing center packing (Figure 4, Detail A).
 5. **QA** Install G-T rings inside upper and lower bearings (Figure 4, Detail D).
 6. **QA** Install seals on OD of upper and lower bearings, with springs facing tops of sleeves (Figure 4, Detail C).
 7. **QA** Install scrapers, inside lower edge of bearings (Figure 4, Detail B).
 8. **QA** Insert and bottom spacer in upper stage piston.

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued



AA9508_1A
SA

Figure 5. Assemble Tail Landing Gear Shock Strut Upper Stage (AVIM). (Sheet 1 of 2)

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued

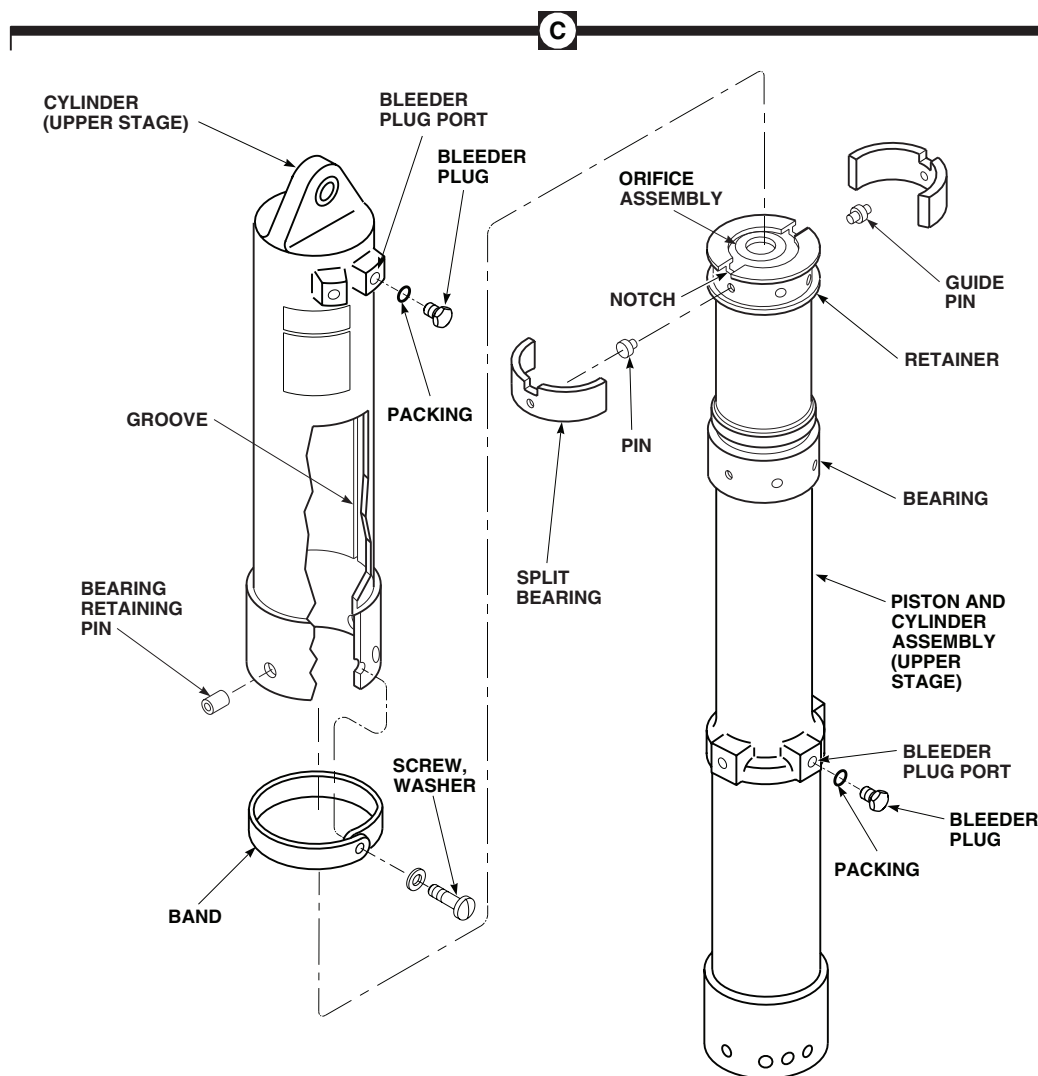
AA9508_2
SA

Figure 5. Assemble Tail Landing Gear Shock Strut Upper Stage (AVIM). (Sheet 2 of 2)

9. **QA** Install floating pistons, open ends down in upper and lower stage piston bores. Push pistons to bottom of bore (Figure 5, Sheet 1, Detail A, and Figure 6).
10. **QA** Install spacer in upper and lower stage piston with holes closest to spacer edge toward bottom. Line up one spacer hole with air vent port. (Figure 5, Sheet 1, Detail B,).
11. **QA** Slip bearings, scraper ends first, over upper and lower stage pistons.
12. **QA** Insert upper stage orifice assembly in piston (Figure 5, Sheet 1, Detail A).
13. **QA** Install upper stage retainer. Line up slots with kneeling and air valve ports. Line up guide pin through retainer and piston wall in line with cylinder assembly notch, kneeling, and air valve ports (Figure 5, Sheet 2, Detail C).

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued

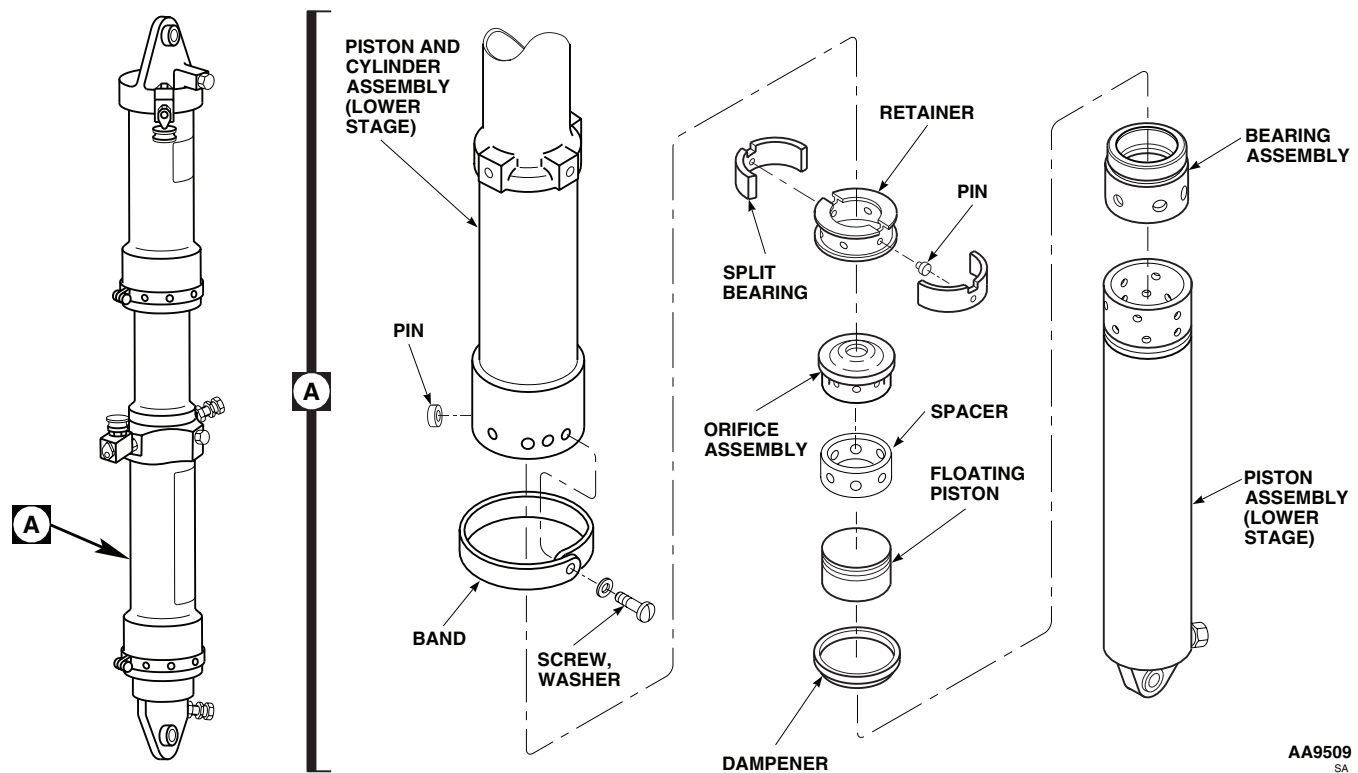


Figure 6. Assemble Tail Landing Gear Shock Strut Lower Stage (AVIM).

NOTE

Installation of guide pin in any other pin hole will prevent further strut assembly.

14. **QA** Install remaining orifice assembly retaining pins.
15. **QA** Install split bearing halves on retainer. Line up upper stage cylinder groove with studded pin. Slide cylinder onto piston.
16. **QA** Slide bearing into cylinder. Line up bearing retaining pin holes and install pins. Seat retaining pins with a rawhide mallet as required.
17. **QA** Install band over bearing pins with screws and washers. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.
18. **QA** Slide bearing assembly, scraper end first, onto lower piston (Figure 6).
19. **QA** Install dampener on piston, with dampener flange facing open end of piston.
20. **QA** Insert lower stage orifice assembly into piston.
21. **QA** Install retainer. Line up holes and install orifice assembly retaining pins.
22. **QA** Install split bearing halves on retainer. Slide piston into cylinder.
23. **QA** Slide bearing into cylinder. Line up bearing retaining pin holes. Install pins. Seat retaining pins with a rawhide mallet as required.
24. **QA** Install band over bearing pins with screws and washers. Using nonelectrical wire, Item 354, WP 1803 00, lockwire screws.

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued

25. Test shock strut. Refer to, TEST, in this work package.

Test

1. Install upper and lower kneeling valves (WP 0460 00).
2. Remove upper stage coupling dust cover. Install coupling half and drain hose on kneeling valve.
3. Route drain hose into a 1 gallon container.

CAUTION

Damage to strut will result if clearance does not exist prior to testing and servicing. Testing and servicing will extend strut to its maximum length of 39.54-inches. Make sure enough clearance exists above strut.

4. Connect a shutoff valve and pressure line from hydraulic test stand to upper stage bleeder port. Start test stand. Open pressure line valve.
5. Slowly apply hydraulic pressure to a maximum of 100 psig, at minimum flow, to fill strut upper stage and to bottom floating piston. Slowly close kneeling valve after fluid has run from drain line for 20 to 30 seconds.
6. Reduce hydraulic pressure to 0 psi.
7. Apply minimum test stand pressure and flow to strut. Open kneeling valve and maintain fluid flow until a steady air-free stream is obtained.
8. Close kneeling valve. Close pressure line shutoff valve. Reduce test stand pressure to 0 psi.
9. Allow a few minutes for trapped air to rise to top of strut. Open kneeling valve and slowly compress strut fully. Close kneeling valve.
10. Slowly apply hydraulic pressure to a maximum of 100 psig, at minimum flow, to extend and fill strut upper stage.
11. Reduce hydraulic pressure and flow to minimum. Open kneeling valve. Check draining fluid for air.
12. Repeat step 5. through step 11. until hydraulic fluid is completely air-free.
13. Close pressure line shutoff valve. Reduce test stand pressure to 0 psi. Compress strut upper stage, if extended.
14. Close kneeling valve.
15. Disconnect hydraulic line at bleeder port.
16. Lubricate packing with petrolatum, Item 216, WP 1803 00.
17. Install packing on bleeder plug. Install bleeder plug in upper stage bleeder port. **TIGHTEN BLEEDER PLUG 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
18. Disconnect drain line from upper kneeling valve. Install valve dust cap. Using nonelectrical wire, Item 356, WP 1803 00, lockwire kneeling valve knob.
19. Connect drain line to lower kneeling valve.
20. Connect pressure line to lower stage bleeder port.
21. Bleed strut lower stage, using step 2. through step 20. for upper stage.
22. Install upper and lower stage air valves (WP 0461 00).
23. Connect pressure regulated nitrogen, Item 199, WP 1803 00, to upper stage air valve. Slowly pressurize upper stage to 900 pounds line pressure on shock strut, 70250-13101-041, or 1025 pounds line pressure on shock struts, 70250-13101-042, 70250-13101-043 or 70250-13101-044. Close air valve, allow pressure to stabilize for 10 to 15 minutes, and check for leaks.

REPAIR TAIL LANDING GEAR SHOCK STRUT (AVIM) - Continued

24. Connect nitrogen to lower stage air valve and slowly pressurize lower stage to 275 pounds line pressure. Close air valve, allow pressure to stabilize for 10 to 15 minutes, and check for leaks. **TORQUE UPPER AND LOWER AIR VALVE OUTER HEXNUTS TO 50 - 70 INCH POUNDS.**
25. Check upper and lower stage air valves for leakage using leak detection fluid, Item 175, WP 1803 00. **NO LEAKAGE ALLOWED.** Replace leaking air valve (WP 0461 00).
26. Position strut assembly at a 45 degree angle and observe for a 24 hour period. **NO AIR LEAKAGE ALLOWED AFTER ONE HOUR FROM UPPER AND LOWER STAGES. MAXIMUM HYDRAULIC FLUID LEAKAGE AFTER 24 HOURS SHALL NOT EXCEED 2 DROPS PER STAGE.** Repair or replace leaking strut.
27. Remove air lines, open upper and lower air shutoff valves, and reduce pressure to 0 psi.
28. Make sure upper and lower kneeling valves are closed. If installed, remove coupling from upper and lower stage kneeling valves.
29. Connect hydraulic drain lines to upper and lower stage kneeling valves and vent lines to drain tray.
30. Open upper and lower stage kneeling valves and completely compress strut assembly. Close upper and lower stage kneeling valves.
31. Remove all hydraulic lines. Install coupling upper and lower stage kneeling valves. Install coupling dust cap.
32. Install air valve dust caps. Using nonelectrical wire, Item 357, WP 1803 00, lockwire lower hexnuts.
33. Using nonelectrical wire, Item 357, WP 1803 00, lockwire upper and lower stage bleed plugs.

REPLACE TAIL LANDING GEAR SHOCK STRUT SPHERICAL BEARINGS (AVIM)**Remove****CAUTION**

Damage to fitting chamfer will occur if staked lip of bearing is improperly removed.
Be careful not to damage fitting chamfer when machining off staked lip of bearing.

1. Clamp fitting to baseplate of drill press and using fly cutter and pilot, machine off that part of bearing outer race swaged over chamfer on fitting (Figure 7, Detail B).
2. Press out bearing using arbor press, retainer, bushing, washer, and nut (TM 1-1500-204-23).

Repair**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

1. Clean primer residue from bore and chamfer, using cheesecloth cloth, Item 85, WP 1803 00, wet with technical acetone, Item 5, WP 1803 00, and nonmetallic scraper.
2. Check bearing bore lugs for galling or scoring. Blend out galling or scoring in bearing bore lug using abrasive cloth, Item 82, WP 1803 00. **GALLING OR SCORING NO DEEPER THAN 0.010-INCH MAY BE BLENDED PROVIDED NOT MORE THAN 25% OF CONTACT AREA IS AFFECTED.**
3. Inspect bearing bore lugs using eddy current (TM 1-1520-265-23). Replace tail landing gear shock strut if cracks are found (WP 0459 00).

REPLACE TAIL LANDING GEAR SHOCK STRUT SPHERICAL BEARINGS (AVIM) - Continued

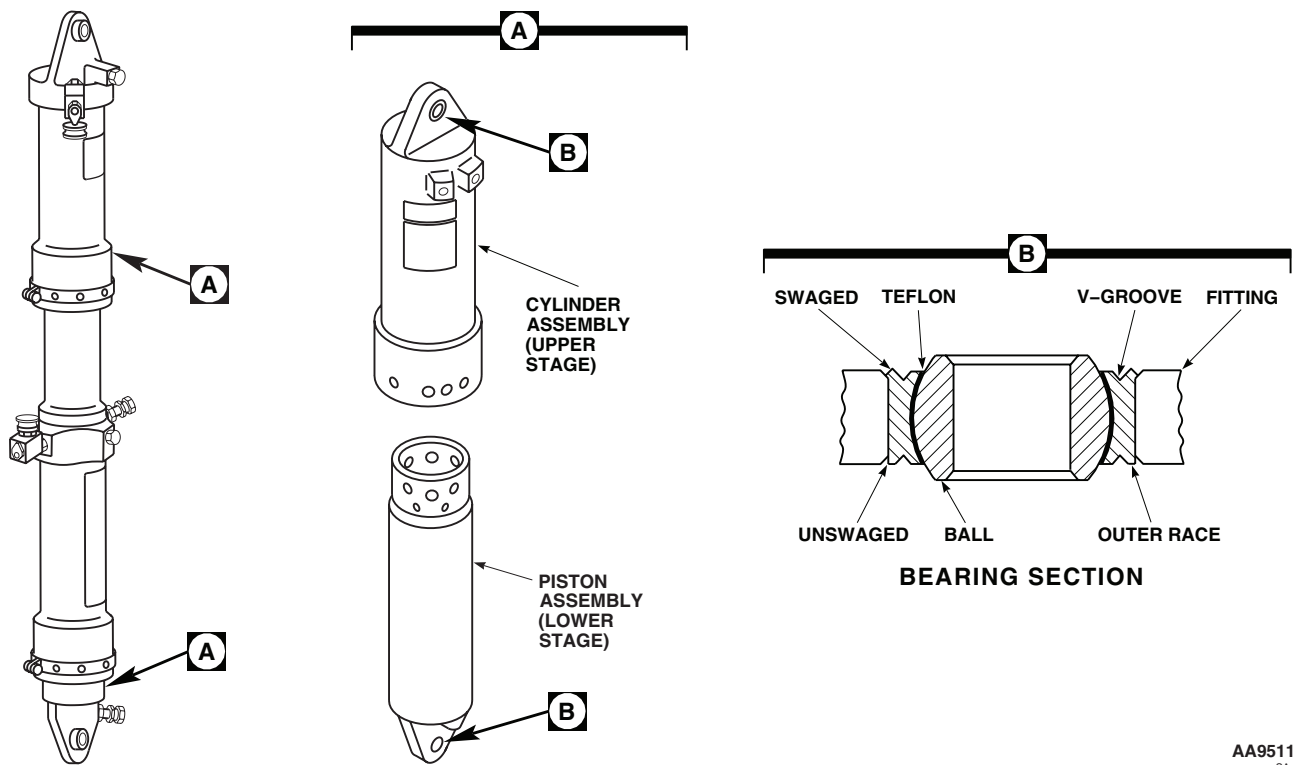
AA9511
SA

Figure 7. Replace Tail Landing Gear Shock Strut Spherical Bearings.

Install

1. Make sure that fitting bore, chamfers, and adjacent surfaces are clean and dry.
2. Apply a light coat of primer coating, Item 243, WP 1803 00, to inside of fitting bore and to outside surface of outer race of bearing.
3. Press bearing into fitting, using arbor press with anvil and driver (TM 1-1500-204-23).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

4. Clean primer, if any, from bearing V-grooves, using cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Wipe dry.
5. Fix fitting with bearing on drill press base plate, with proper spinning tool and anvil.

REPLACE TAIL LANDING GEAR SHOCK STRUT SPHERICAL BEARINGS (AVIM) - Continued**CAUTION**

Damage to bearing will result if petrolatum comes in contact on the ball of the bearing, or on the exposed edge of the fabric liner. If petrolatum gets on these areas, remove petrolatum immediately.

6. Lubricate V-groove of bearing and spinning tool with petrolatum, Item 216, WP 1803 00.

CAUTION

Damage to bearing will result if spinning tool makes contact with ball side of groove. Do not allow spinning tool to make contact with ball side of groove as binding may result.

7. Swage V-groove lip over chamfer in fitting bore (TM 1-1500-204-23).
8. Turn fitting over and repeat swaging process on other side.
9. Clean petrolatum from V-groove with cheesecloth cloth, Item 85, WP 1803 00, wet with cleaning compound solvent, Item 71, WP 1803 00.
10. Touch up bearing surfaces with corrosion removing compound, Item 108, WP 1803 00.
11. Touch up paint (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****TAIL LANDING GEAR YOKE - REPAIR (AVIM)****This WP supersedes WP 0474 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Slide Hammer, GGG-P-781
Tool Kit, Stud Removal, MIL-T-43113

Tape, Pressure Sensitive, Adhesive, Item 328,
WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0381 00
WP 0462 00
WP 1803 00

Material/Parts

Abrasive Mat, Item 2, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Brush, Paint, Item 61, WP 1803 00
Cloth, Cheesecloth, Item 85, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00

REPLACE TAIL LANDING GEAR YOKE ATTACHMENT BUSHINGS (AVIM)**Remove**

1. Remove tail landing gear yoke (WP 0462 00).
2. Place yoke assembly on sand bags or similar protective surface.
3. Using slide hammer, remove yoke attachment bushings (Figure 1 and TM 1-1500-204-23). Discard bushings.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

4. Remove old primer from yoke attachment bushing holes with nylon abrasive mat, Item 2, WP 1803 00, and technical acetone, Item 5, WP 1803 00. Wipe clean and dry with cheesecloth cloth, Item 85, WP 1803 00.

Install

1. Coat yoke attachment bushing bore and bushing outside surface with epoxy primer coating kit, Item 120, WP 1803 00, using paint brush, Item 61, WP 1803 00.
2. Press yoke attachment bushing halves into bore while primer is wet using an arbor press or equivalent (Figure 1, Detail A).
3. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
4. Install tail landing gear yoke (WP 0462 00).

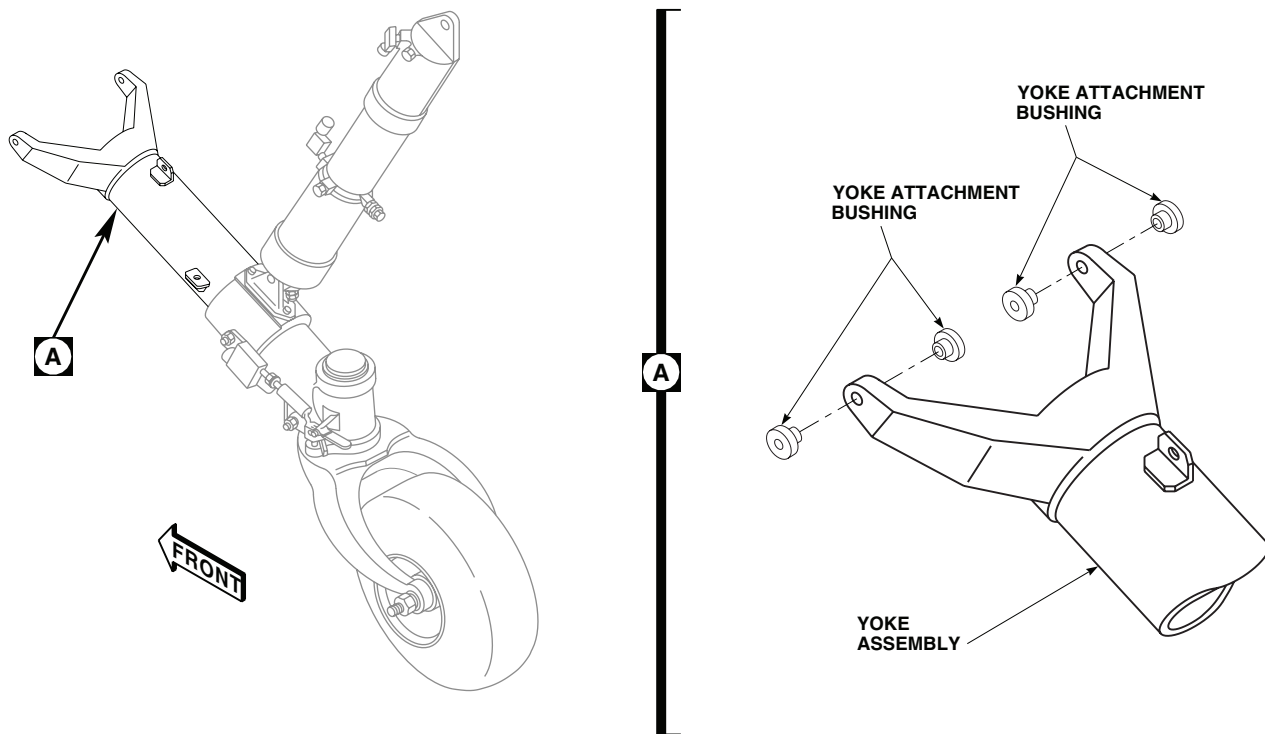
REPLACE TAIL LANDING GEAR YOKE ATTACHMENT BUSHINGS (AVIM) - ContinuedAK0675
SA

Figure 1. Replace Tail Landing Gear Yoke Attachment Bushings (AVIM).

REPLACE TAIL LANDING GEAR YOKE FUSELAGE/YOKE FITTING BUSHINGS (AVIM)**Remove**

1. Remove tail landing gear yoke (WP 0462 00).
2. Remove yoke fitting bushing halves using slide hammer. Discard bushings (Figure 2, Detail A).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

3. Remove old primer from fuselage fitting holes with technical acetone, Item 5, WP 1803 00, and abrasive mat, Item 2, WP 1803 00. Wipe clean and dry with cheesecloth cloth, Item 85, WP 1803 00.

Install

1. Make draw bolt (Figure 2, Detail B). Wrap draw bolt threads with pressure sensitive adhesive tape, Item 328, WP 1803 00, or equivalent.
2. Coat bushing bore and new yoke fitting bushing outside surface with epoxy primer coating kit, Item 120, WP 1803 00, using a paint brush, Item 61, WP 1803 00.

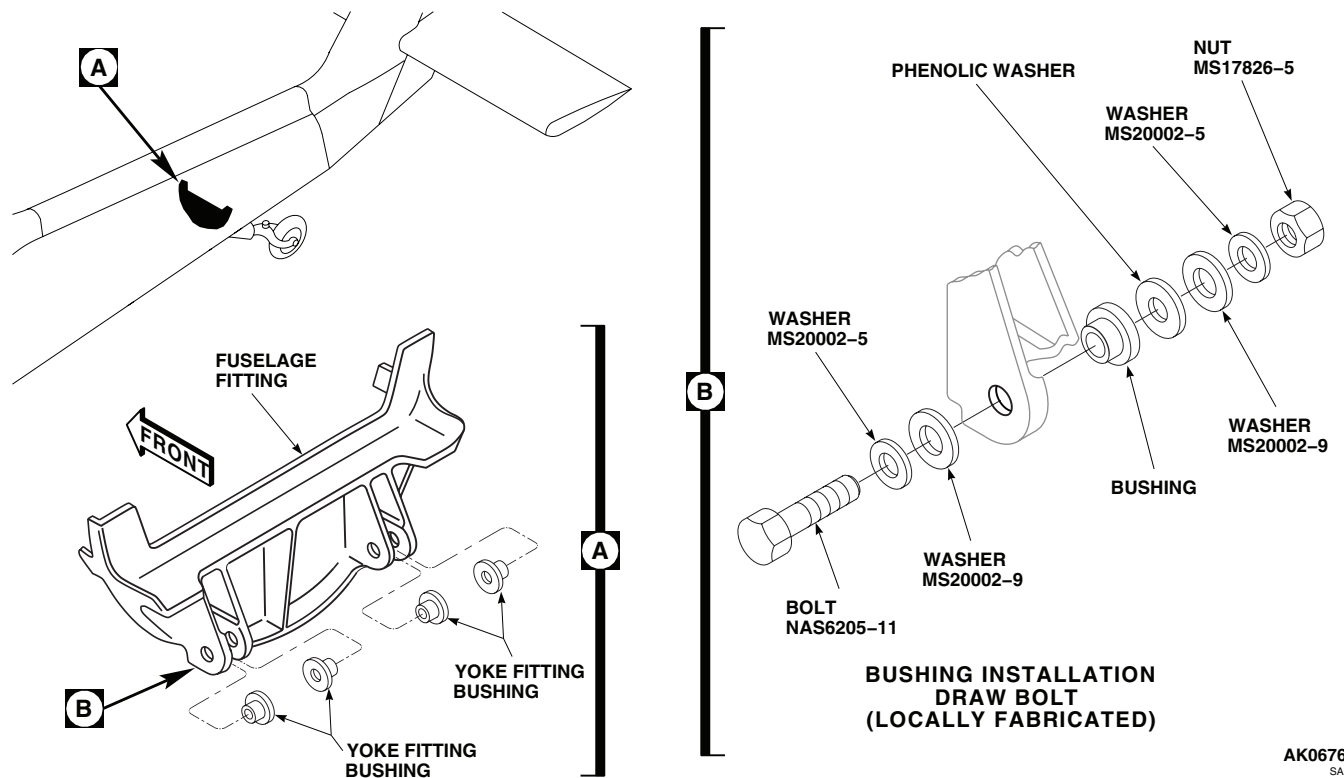
REPLACE TAIL LANDING GEAR YOKE FUSELAGE/YOKE FITTING BUSHINGS (AVIM) - Continued

Figure 2. Replace Tail Landing Gear Yoke Fuselage/Yoke Fitting Bushings (AVIM).

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

3. **QA** With yoke fitting bushing flange inside fuselage fitting boss, press yoke fitting bushing into hole until bushing flange is bottomed on fitting. Wipe extra primer out of bushing bore with cheesecloth cloth, Item 85, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
4. Touch up paint as required (WP 0381 00 and TM 55-1500-345-23).
5. Install tail landing gear yoke (WP 0462 00).

REPLACE TAIL LANDING GEAR YOKE INSERTS, STUDS, AND LOCKRINGS (AVIM)**Remove**

Remove lockring, stud, or insert from tail landing gear yoke using stud removal insert kit (Figure 3).

Install

Install lockring, stud, or insert in tail landing gear yoke using stud removal insert kit (Figure 3).

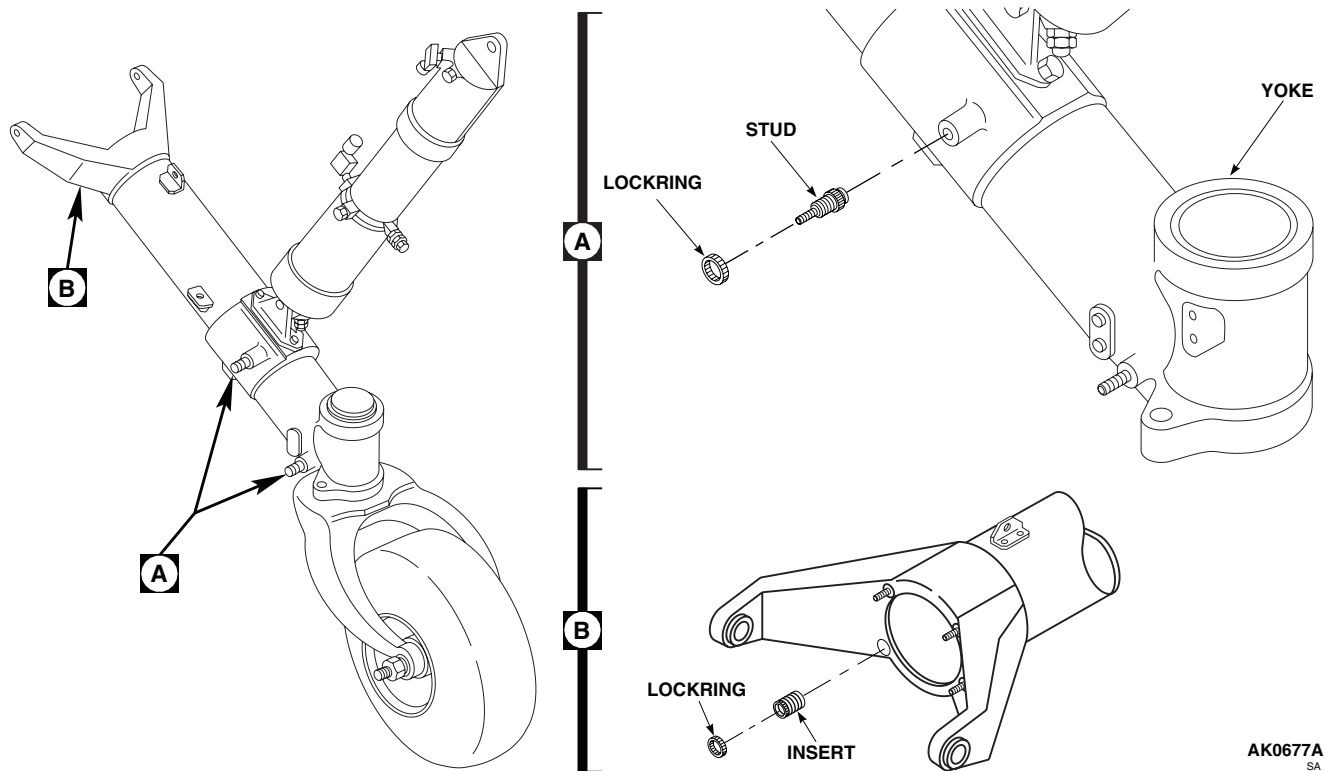
REPLACE TAIL LANDING GEAR YOKE INSERTS, STUDS, AND LOCKRINGS (AVIM) - Continued

Figure 3. Replace Tail Landing Gear Yoke Inserts, Studs, and Lockring (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**LANDING GEAR****TAIL LANDING GEAR FORK - REPAIR (AVIM)**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Hand Reamer Set, Adjustable, GGG-R-180
Tail Landing Gear Lockpin Oversize Bushing, Locally-Made, Figure 221, WP 1805 00

Material/Parts

Epoxy Primer Coating Kit, Item 120, WP 1803 00

Personnel Required

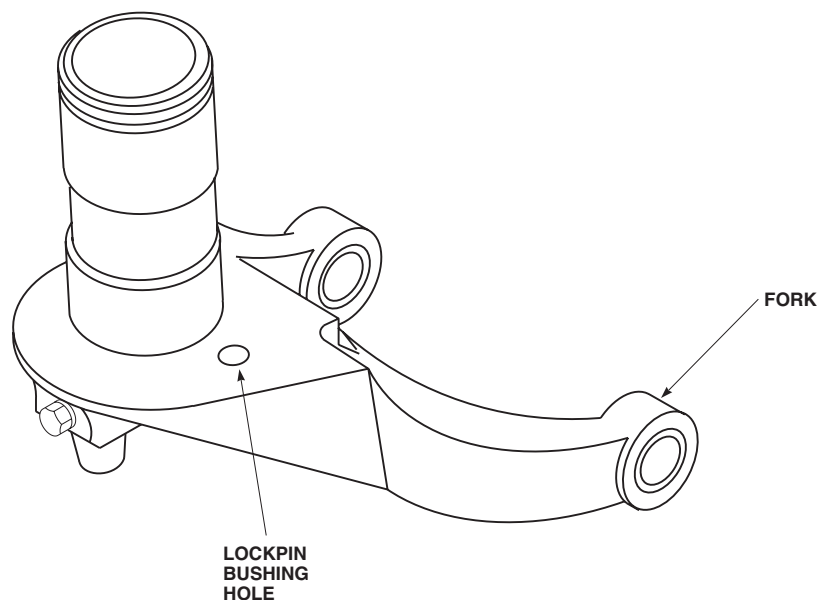
UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0463 00
WP 1803 00
WP 1805 00

REPAIR

1. Remove wearplate (WP 0461 00).
2. Locally make lockpin hole oversize bushing (WP 1805 00).
3. Clean up hole in tail wheel fork by drilling and using adjustable hand reamer set, reaming hole to 0.870 to 0.875-inch diameter (Figure 1).



AA9445
SA

Figure 1. Repair Tail Landing Gear Fork Lockpin Bushing Hole (AVIM).

REPAIR - Continued

4. Coat lockpin bushing hole with epoxy primer coating kit, Item 120, WP 1803 00.
5. Install bushing while primer is still wet, using arbor press.
6. Install wearplate (WP 0461 00).
7. Install lockpin bushing (WP 0461 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
LANDING GEAR
TAIL LANDING GEAR WHEEL - REPAIR (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101
Mallet
Phenolic Drift Pin

Polyurethane Coating, Item 227, WP 1803 00
Primer Coating, Item 243, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Carbon Dioxide, Item 64, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Corrosion Resistant Coating, Item 110, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Grease, Aircraft, Item 131, WP 1803 00

References

TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 0465 00
WP 1803 00

REPAIR TAIL LANDING GEAR WHEEL (AVIM)**Disassemble****NOTE**

Inspect and repair wheel whenever wheel is disassembled per TM 1-1520-265-23.

Remove tire from wheel (WP 0465 00).

Inspect/Repair

1. Fluorescent penetrant inspect wheel halves for cracks and structural damage using fluorescent penetrant inspection kit (TM 1-1520-265-23). Pay particular attention to bead seat, bolt boss, male/female mating surfaces, and valve hole areas. **NO CRACKS ALLOWED.** Replace wheel if any crack is present.
2. Inspect wheel halves for scratches, nicks, or pits (Figure 1, Sheet 2), and repair using abrasive cloth, Item 82, WP 1803 00, as follows:
 - a. Repair AREA A as follows:
 - (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP.**
 - (2) Replace wheel if damage or repairs are more than limits.
 - b. Repair AREA B as follows:
 - (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.010-INCH DEEP AND 1.00 SQUARE INCH IN AREA.**
 - (2) Replace wheel if damage or repairs are more than limits.
 - c. Repair AREA C as follows:

REPAIR TAIL LANDING GEAR WHEEL (AVIM) - Continued

- (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.031-INCH DEEP AND 1.00 INCH LONG.**

- (2) Replace wheel if damage or repairs are more than limits.

d. Repair AREA D as follows:

- (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.062-INCH DEEP AND 1.00 SQUARE INCH IN AREA.**

- (2) Replace wheel if damage or repairs are more than limits.

e. Repair AREA E as follows:

- (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS UP TO 0.015-INCH DEEP AND 1.00 INCH LONG.**

- (2) Replace wheel if damage or repairs are more than limits.

f. Repair AREA F as follows:

- (1) **BLEND AND POLISH NICKS, SCRATCHES, OR PITS.**

- (2) **REPAIR AFTER BLENDING AND POLISHING SHALL NOT EFFECT BEARING CUP RETENTION.**

- (3) Replace wheel if damage or repairs effect bearing cup retention.

3. Touch up blended area with corrosion resistant coating, Item 110, WP 1803 00.

4. Prime wheel with epoxy primer coating kit, Item 120, WP 1803 00.

NOTE

Male/female mating surfaces and bolt bosses need primer only.

5. Paint wheel with polyurethane coating, Item 227, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).
6. Check bearing cups for scratches, pitting, corrosion, and signs of overheating (Figure 1, Sheet 1). **NONE ALLOWED.** Replace damaged cup. Refer to, REPLACE TAIL LANDING GEAR WHEEL BEARING CUP (AVIM), in this work package.

CAUTION

Damage to bearings will result if they are allowed to spin while drying. Secure inner and outter race of bearings while drying.

7. Clean bearings with cleaning compound solvent, Item 71, WP 1803 00. Dry bearings with compressed air, if available, or let air-dry.
8. Inspect bearings for scratches, water stains, heat discoloration, worn rollers, or cracked or distorted bearing cage. **NONE ALLOWED.** Replace bearing if any of these conditions are found.
9. If bearing is determined serviceable, repack it with aircraft grease, Item 131, WP 1803 00.

NOTE

New bearings packaged in preservative do not need degreasing before being packed with aircraft grease.

10. Inspect bearing seals for damaged reinforcing rings, visible damage, or deformation. Replace any seal found unserviceable.

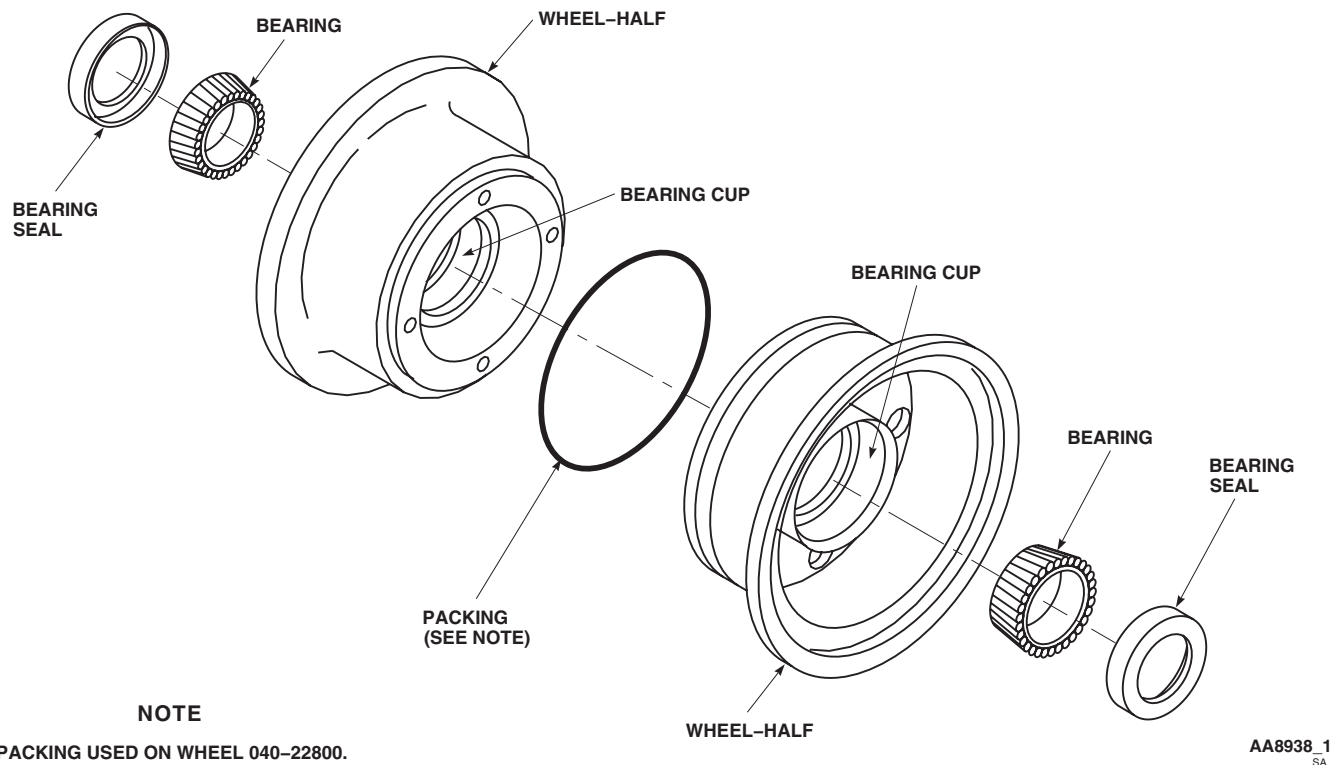
REPAIR TAIL LANDING GEAR WHEEL (AVIM) - Continued

Figure 1. Repair Tail Landing Gear Wheel (AVIM). (Sheet 1 of 2)

11. Inspect wheel bolts for thread damage and cracks under bolt heads. Magnetic particle inspect all wheel bolts (TM 1-1520-265-23). Replace any bolt found damaged.
12. Inspect nuts for obvious damage. Thread bolts into nuts by hand. Replace any nut that allows bolt to thread past top of nut.

Assemble

Install tire on wheel (WP 0465 00).

REPLACE TAIL LANDING GEAR WHEEL BEARING CUP (AVIM)**Remove****CAUTION**

Damage to wheel half will result if overheated. Overheating will affect service life of wheel. Do not overheat wheel half or heat longer than required.

1. Heat wheel half in an oven not over 300°F (149°C), or place in boiling water for 30 minutes.

REPLACE TAIL LANDING GEAR WHEEL BEARING CUP (AVIM) - Continued

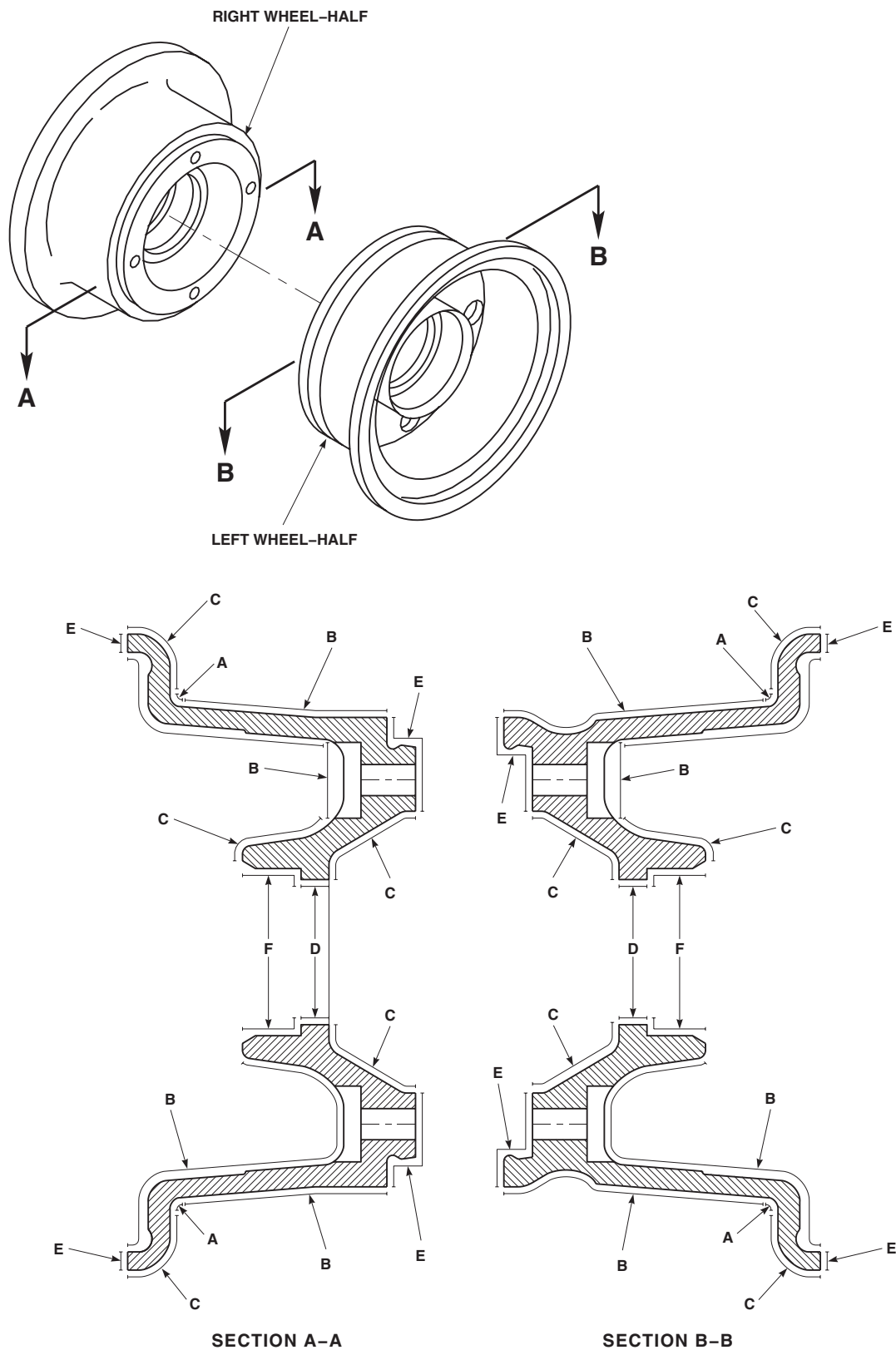
AK0689_2
SA

Figure 1. Repair Tail Landing Gear Wheel (AVIM). (Sheet 2 of 2)

REPLACE TAIL LANDING GEAR WHEEL BEARING CUP (AVIM) - Continued**CAUTION**

- Damage to hub bore will result if bearing cups are removed improperly. Be careful when removing bearing cups to avoid raising burrs in hub bore.
 - Damage to wheel half will result if wheel half is not supported properly. Support wheel half on hub and not on flange.
2. Remove wheel half from oven and immediately remove cup from bore (Figure 2). After heating wheel, bearing cup should be loose enough to fall out of bore when wheel is turned upside down. If cup does not fall out, tap cup from bore evenly using a phenolic drift pin and a mallet.
 3. Carefully remove any primer or other foreign matter from hub bore. Make sure bearing bore and shoulder are free of burrs.

Install**WARNING**

Injury to personnel will result if carbon dioxide or chilled parts are handled without protective gloves. Use protective gloves when handling carbon dioxide or chilled parts.

1. Place replacement bearing cup in a container of carbon dioxide, Item 64, WP 1803 00.

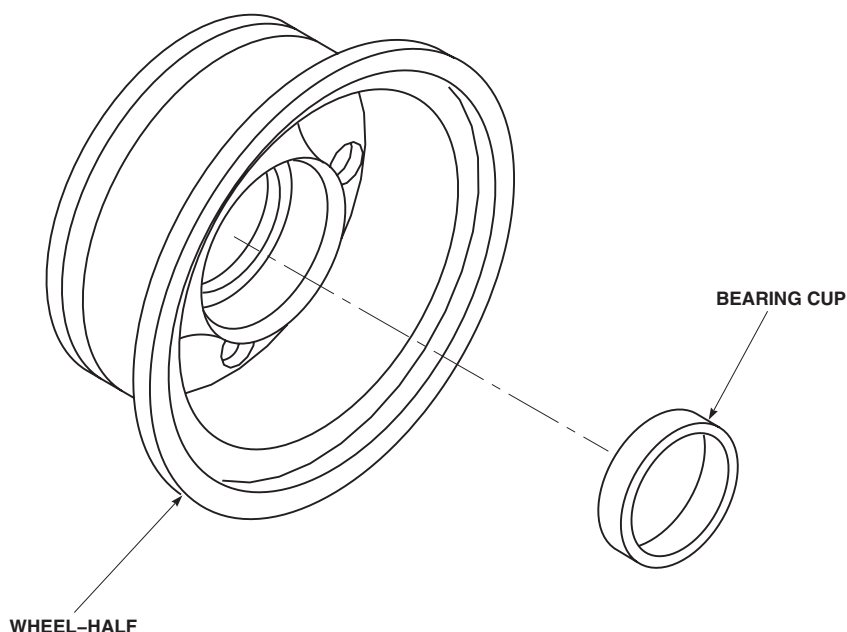
AK0690
SA

Figure 2. Replace Tail Landing Gear Wheel Bearing Cup (AVIM).

REPLACE TAIL LANDING GEAR WHEEL BEARING CUP (AVIM) - Continued**CAUTION**

Damage to wheel half will result if overheated. Overheating will effect service life of wheel. Do not overheat wheel half or heat longer than required.

2. Heat wheel half in an oven not over 300°F(149°C), or place in boiling water for 30 minutes.
3. Remove wheel half from heat source.
4. Remove bearing cup from carbon dioxide. Make sure cup is thoroughly dry.

CAUTION

Damage to wheel half will result if supported on flange. Support wheel half on hub and not on flange.

5. **QA** Coat wheel contacting surface of bearing cup with primer coating, Item 243, WP 1803 00, and install cup into hub bore while primer is still wet. Use a phenolic drift pin and a mallet to tap cup into bore. Make sure cup is against shoulder and avoid cocking cup during installation.
6. Allow wheel to cool to room temperature. Apply a light coat of aircraft grease, Item 131, WP 1803 00, to bearing cup to prevent rust and corrosion.

END OF WORK PACKAGE

UNIT LEVEL
ENGINE SYSTEM
ENGINE EXHAUST MODULE INSPECTION

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Fluorescent Penetrant Inspection Kit, XMA101

References

TM 1-1520-265-23
 WP 0520 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECTION

1. Inspect exhaust module component, Table 1.

Table 1. Exhaust Module Damage and Repair.

COMPONENT	DAMAGE	REPAIR
Exhaust Fairings (Titanium)	Cracks, maximum 4, each not more than 3 inches long, with distance between any 2 cracks not less than length of longer crack.	Weld per WP 0520 00.
Exhaust Ejector (Stainless Steel)	Dents, 2 inch maximum diameter or 0.250-inch maximum depth, except on welded seams, mating surfaces, and where cracks can occur.	If ejector has not been removed: no repair required. If ejector has been removed: use plastic faced hammer with support to remove dents.
	Dents, more than 2 inches in diameter or 0.250-inch deep, on welded seams, mating surfaces, or capable of causing cracks.	Repair at appropriate sheet metal maintenance facility.
	Crack, single, 3 inches maximum length.	Stop-drill, using 1/8-inch bit.
	Crack, single, more than 3 inches long.	Weld per WP 0520 00.
	Crack, continuous between 2 bolt holes or nutplates.	Weld per WP 0520 00.
Exhaust Ejector (Stainless Steel)	Cracks, maximum 4, each crack not more than 3 inches long and distance between 2 cracks more than 3 inches.	Weld per WP 0520 00.

INSPECTION - Continued**Table 1. Exhaust Module Damage and Repair. - Continued**

COMPONENT	DAMAGE	REPAIR
Attaching Angle (Stainless Steel)	Cracks, maximum 2, each crack from 6 to 12 inches long maximum, with 12 inches minimum distance between cracks.	Weld per WP 0520 00.
	Cracks, maximum 4, each crack 6 inches long maximum, with distance between any 2 cracks equal to length of longer crack.	Weld per WP 0520 00.

2. Repair exhaust module, Table 1 and WP 0520 00.

CAUTION

Do not repair cracks in ejector or ejector attaching angle with patches, because rivets can expand and cause more cracking.

3. Visually inspect exhaust ejector for dents and cracks. Determine length of crack by fluorescent penetrant inspection (TM 1-1520-265-23). Repair limits and methods are given in Table 1. **IF DAMAGE EXCEEDS LIMITS, REPLACE COMPONENT.**
4. Visually inspect attaching angle for cracks. Determine length of crack by fluorescent penetrant inspection (TM 1-1520-265-23). Repair limits and methods are given in Table 1. **IF DAMAGE EXCEEDS LIMITS, REPLACE COMPONENT.**

END OF WORK PACKAGE

UNIT LEVEL

ENGINE SYSTEM

ENGINE CONTROL CABLE INSPECTION

This WP supersedes WP 0478 00, dated 17 April 2006.

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0508 00

WP 0509 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0506 00

WP 0507 00

INSPECTION

NOTE

Visual inspection of cable spiral casing separation applies to load-demand, engine and fuselage power-available, and fuel selector valve push/pull cables.

1. Visually inspect push/pull cable for separation of spiral casing from end fitting. **NONE ALLOWED.** Replace cable (WP 0506 00, WP 0507 00, WP 0508 00, or WP 0509 00).
2. Visually inspect all push/pull cables for surface damage. Nicks or scratches that do not penetrate down to metal windings are acceptable and do not require repair. **SURFACE CUTS OR TEARS PENETRATING COVERING MUST BE REPAIRED (WP 0506 00).**
3. Inspect load-demand cable rod end for surface corrosion. **NONE ALLOWED.** Repair rod end (WP 0506 00).
4. Inspect fuselage power-available push/pull cable as follows:
 - a. Inspect rod end for radial and axial play. **NO PLAY ALLOWED.** Replace rod end if play is found (WP 0508 00).
 - b. Check stopnut self-locking feature by installing it on adapter. **REPLACE STOPNUT IF IT CAN BE ROTATED BY HAND.**
5. Inspect nut connecting push/pull cables to rotary inputs on HMU for cracks. **NONE ALLOWED.** Replace cable if nut is cracked.

END OF WORK PACKAGE

UNIT LEVEL**ENGINE SYSTEM**

ENGINE LOAD-DEMAND PUSH/PULL FRICTION CHECK

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powerplant Repairer's Toolkit, SC 5180-99-B07
Spring Scale, AAA-S-133
Torque Wrench, 0 - 30 in. lbs.

References

TM 1-2840-248-23
WP 0498 00
WP 0506 00
WP 1803 00

Material/Parts

Strap, Tiedown, Electrical, Item 306, WP 1803 00

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
UH-60 Helicopter Repairer MOS 15T (1)

FRICTION CHECK**NOTE**

The following procedures apply to both left and right side load-demand push/pull cables.

1. Open engine cowling.
2. Open main rotor pylon sliding cover.
3. Remove self-retaining bolt, nut, and washer holding LDS cable rod end to mixer lever (WP 0506 00).
4. Install electrical tiedown strap, Item 306, WP 1803 00, through LDS cable rod end for spring scale attachment point.
5. Attach spring scale to LDS cable rod end using tiedown strap. Measure cable friction by pulling the cable and measuring force at breakaway.
6. If friction force is less than 45 pounds, go to step 12..
7. If friction force is greater than 45 pounds, disconnect load-demand system rotary input assembly from HMU (WP 0506 00).
8. Attach spring scale to LDS cable rod end. Measure cable friction by pulling the cable and measuring force at break-away.
9. If friction force is less than 33 pounds, do this:
 - a. Do torque check on engine HMU spindle. **TORQUE SHALL BE LESS THAN 15 INCH-POUNDS.**
 - b. If torque is greater than 15 inch-pounds, replace HMU (TM 1-2840-248-23).
10. If friction force is greater than 33 pounds, do this:
 - a. Remove load-demand rotary input (WP 0498 00).

FRICTION CHECK - Continued

- b. Attach spring scale to LDS cable rod end. Measure cable friction by pulling the cable and measuring force at breakaway.
 - c. If friction force is less than 3 pounds, replace rotary input (WP 0498 00).
 - d. If friction force is greater than 3 pounds, replace engine load-demand push/pull cable (WP 0506 00).
- 11. Repeat step 5. through step 10. d. if load-demand push/pull cable, rotary input or HMU were replaced during friction check.
 - 12. Install load-demand push/pull cable on mixer lever (WP 0506 00).
 - 13. Close main rotor pylon sliding cover.
 - 14. Close engine cowling and latch.

END OF WORK PACKAGE

UNIT LEVEL
ENGINE SYSTEM
INSPECT PNEUMATIC TUBE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT PNEUMATIC TUBE

1. Inspect tubes for dents. **DENTS UP TO 25% OF TUBE DIAMETER ARE ACCEPTABLE.** Replace tube if dents exceed limit.
2. Inspect all welds on tubes for cracks. **NONE ALLOWED.** Replace tubes with any cracks at weld joints.
3. Inspect tube couplings for chafing, damage, stiffness, and tears. **NONE ALLOWED.** Replace tube couplings if too stiff to slide over tube, or if chafed or torn.
4. Check brackets for cracks, bending, and security.
5. Inspect bleed-air tubing installation for breaks and tears. Repair or replace if damaged.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM**ENGINE**

This WP supersedes WP 0481 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Adapter, 21C7082G01
 Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Hardware Kit, 70000-30100-043
 Hardware Kit, 70000-30100-044
 Hoist, to Support Weight of Engine
 Lifting Sling, 21C7081G02
 Maintenance Crane, 70700-20300-042
 Powerplant Repairer's Toolkit, SC 5180-99-B07
 Powertrain Repairer's Toolkit, SC 5180-99-B13
 Rubber Mallet, 25714
 Spring Scale, AAA-S-133
 Strap Wrench, GGG-W-651
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.
 Wood Block, 2 x 3 x 6 in. (5 x 8 x 15 cm)

Material/Parts

Abrasive Cloth, Assortment, Item 1, WP 1803 00
 Cloth, Abrasive, Item 82, WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Protective Caps and Plugs
 Sealing Compound, Item 285, WP 1803 00
 Washer, AN960C616
 Washer, AN960C616L

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
 UH-60 Helicopter Repairer MOS 15T (1)
 Aircraft Powertrain Repairer MOS 15D (1)

References

TM 1-1520-237-MTF
 TM 1-2840-248-23
 TM 1-6625-724-13&P
 TM 55-1500-345-23
 WP 0379 00
 WP 0381 00
 WP 0482 00
 WP 0484 00
 WP 0485 00
 WP 0486 00
 WP 0487 00
 WP 0619 00
 WP 0653 00
 WP 0655 00
 WP 0702 00
 WP 1590 00
 WP 1601 00
 WP 1688 00
 WP 1703 00
 WP 1720 00
 WP 1803 00

REMOVE**NOTE**

The following procedures apply to both No. 1 and No. 2 engines.

1. Turn off all electrical power.
2. Preserve engine before removal (TM 1-2840-248-23).
3. Open engine cowling.
4. Remove engine air inlet (WP 0484 00).
5. If maintenance crane will be used, assemble and set up crane (WP 1590 00).
6. Disconnect fuel hose at fuel pump elbow. Cap or plug hose and elbow (Figure 1, Sheet 1, Detail A).

REMOVE - Continued

7. Disconnect engine drain hoses from elbows on rear engine compartment deck. Cap or plug hoses and elbows.
8. **UH-60A 77-22727, 77-22728 UH-60A 78-22963 - SUBQ EH-60A UH-60L** Remove clamps holding common drain to engine compartment shield. ■
9. **UH-60A 77-22727, 77-22728 UH-60A 78-22963 - SUBQ EH-60A UH-60L** On No. 1 engine only, remove clamp holding swirl frame drain to airframe. ■
10. Disconnect electrical connector, P815 and P825, No. 1 engine, or P814 and P824, No. 2 engine from firewall receptacles. Cap receptacles.
11. Remove electrical harness clamp and bonding jumper from firewall if installed.
12. Disconnect power-available rotary input and load-demand rotary input from hydromechanical unit on engine.
13. Remove starter bleed-air tube from starter and upper firewall tube. Keep bleed-air tube, coupling, and clamps with helicopter.
14. **UH-60A** Remove and discard coupling holding cross bleed-air tube at firewall (WP 0702 00) (Figure 1, Sheet 1, Detail B). ■
15. **UH-60L** Loosen coupling on cross bleed-air tube at firewall, and slide sleeve and coupling onto cross bleed-air tube (Figure 1, Sheet 1, Detail B). ■
16. On No. 2 engine only, remove screw, washer, and spacer attaching engine control cables to right side IPS blower duct (Figure 1, Sheet 2, Detail C).
17. Disconnect coupling and remove inlet particle separator (IPS) blower duct from IPS blower. Remove duct.
18. **W/O HIRSS** Unclamp loop clamp from deswirl duct and engine power turbine flange and place deswirl duct and loop clamp inside of exhaust fairing. ■
19. **HIRSS** Unclamp loop clamp from deswirl duct and engine power turbine flange. Position loop clamp on deswirl duct. Leave deswirl duct and loop clamp installed in HIRSS module. ■
20. Install lifting sling on engine front and rear lifting lugs with quick-disconnect pins (Figure 2, Detail A and Detail B).
21. Pin lifting eye in No. 3 hole on lifting sling.

WARNING

To prevent injury to personnel or damage to equipment, do not use crane or hoist as a source of excessive force to free engine from helicopter. Also, make sure hoist hook does not touch pulley when lifting engine to avoid damage to the cable.

22. Attach sling to maintenance crane or a hoist with enough capacity to lift engine. Raise hoist until weight of engine is supported by sling.
23. Disconnect forward support mount as follows:
 - a. Hold output shaft from turning, using strap wrench (Figure 3, Sheet 1, Detail A).

REMOVE - Continued

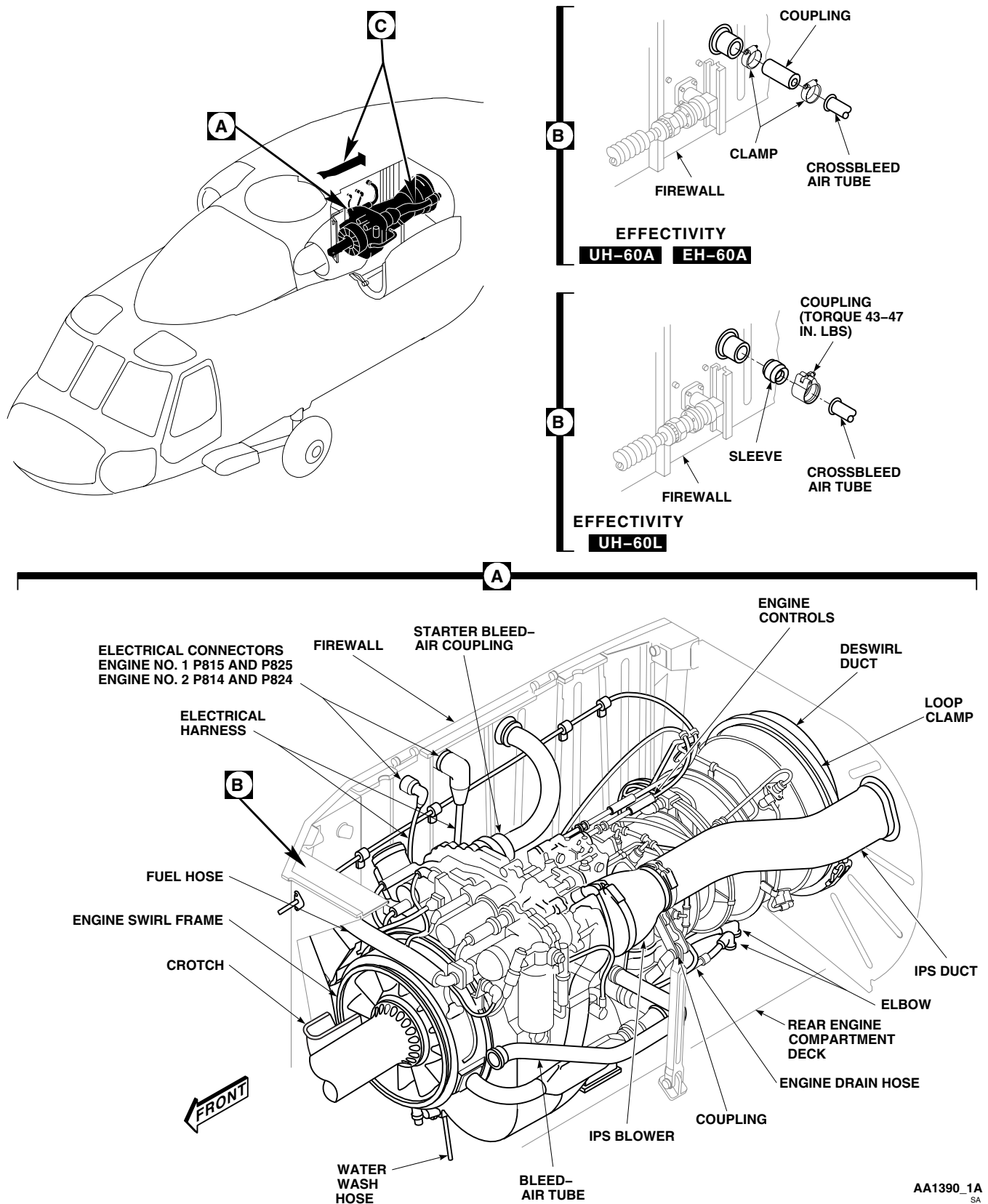


Figure 1. Disconnect Engine/Airframe Connections. (Sheet 1 of 2)

REMOVE - Continued

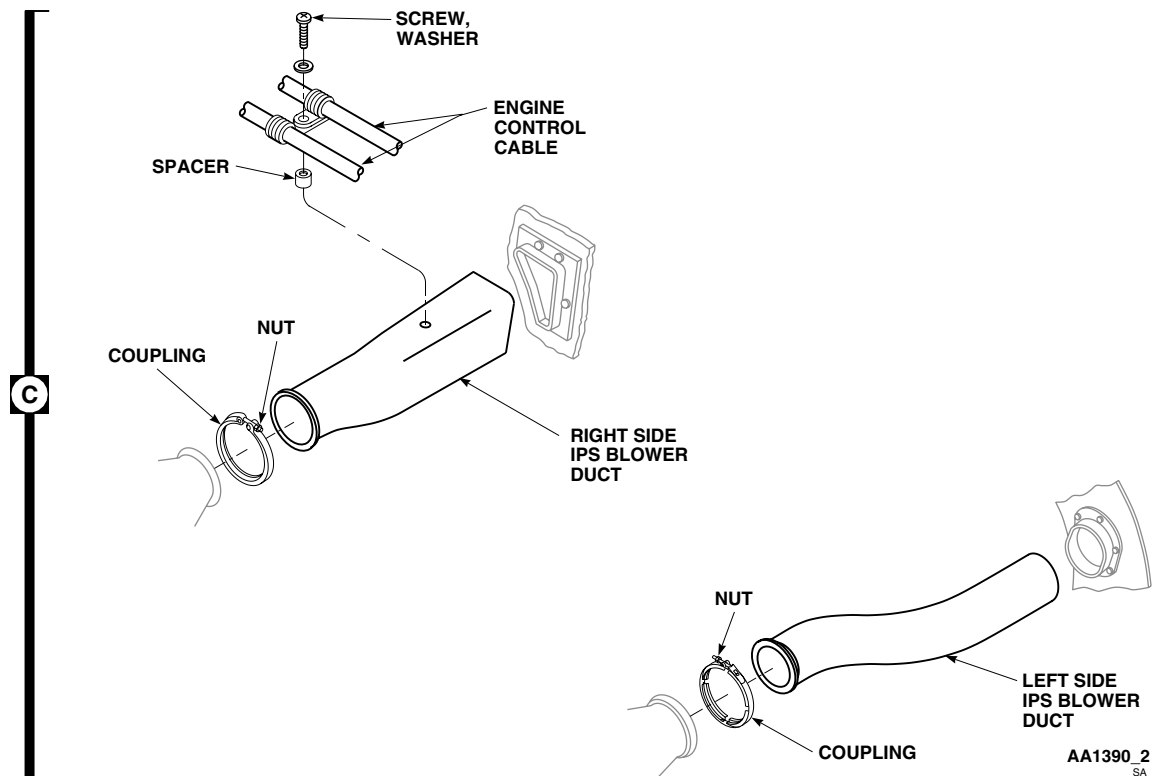


Figure 1. Disconnect Engine/Airframe Connections. (Sheet 2 of 2)

CAUTION

Excessive vibration will result if two or more nuts connecting engine output shaft to disc pack coupling are loosened or removed at any time. Engine output shaft, disc pack coupling and connecting hardware is a balanced assembly. Only one nut connecting engine output shaft to disc pack coupling may be loosened or removed at any one time. Loose or removed nuts must be torqued to their proper value before another nut is loosened or removed.

NOTE

- Helicopters with input module flange/coupling, 70351-08255-102, have nut-plates on forward side of flange/coupling to hold drive shaft.
 - Helicopters with input module flange, 70351-08206-102, use a self locking nut on forward side of flange to hold disc pack coupling to flange.
 - Helicopters with input module flange, 70351-08206-041, have nutplate on forward side of flange to hold disc pack coupling to flange.
- b. On helicopters with engine output drive shafts 70361-08004-042, or 70361-08004-043, remove bolts and washers and nuts (if installed) holding disc pack coupling to input module flange.
 - c. On helicopters with engine output drive shafts 70361-08014-041, remove bolts holding flange/coupling to drive-shaft.
 - d. Remove bolts and washers and nuts holding disc pack coupling to input module flange.

REMOVE - Continued

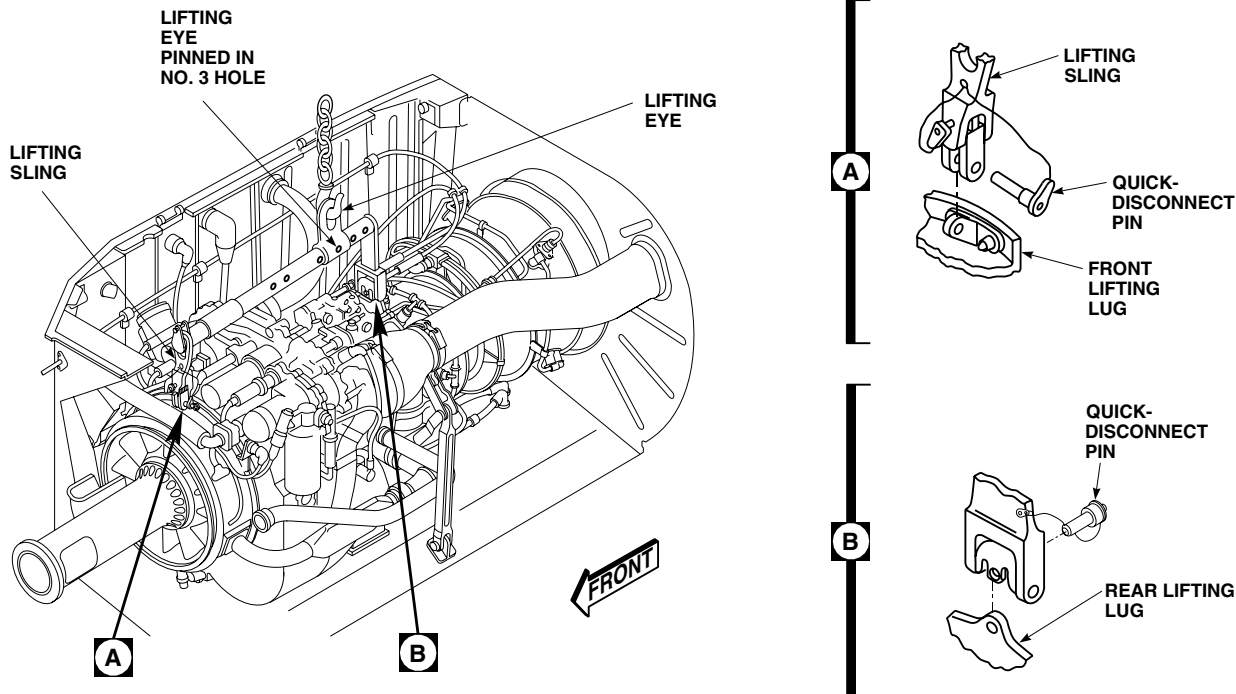
AK1239A
SA

Figure 2. Attach Lifting Sling.

- e. Remove bolts and washers holding forward support tube to gimbal.

NOTE

Replace any broken bolts on gimbal 70361-08001-102 (WP 0653 00).

- f. Keep bolts and washers with gimbal for installing engine.
- g. On helicopters with input module flange, 70351-08206-041, check input module flange nutplate rivets for proper installation, looseness, and fretting on heads. Replace input flange if rivets are not flush with flange surface, are loose, or fretting is found on heads (WP 0655 00).
- h. Do engine output shaft inspection after disconnect (WP 1720 00).
24. Disconnect support mounts as follows:

NOTE

Keep attaching hardware and shim washers with mount strut.

- a. Remove bolts, washers, and nuts holding inner and outer mount struts to mount links (Figure 3, Sheet 2, Detail D).

NOTE

Keep attaching hardware and shim washers with mount support.

- b. Remove bolts, washers, and nuts, holding center mount links to mount support under engine combustor section on engine compartment deck.

REMOVE - Continued

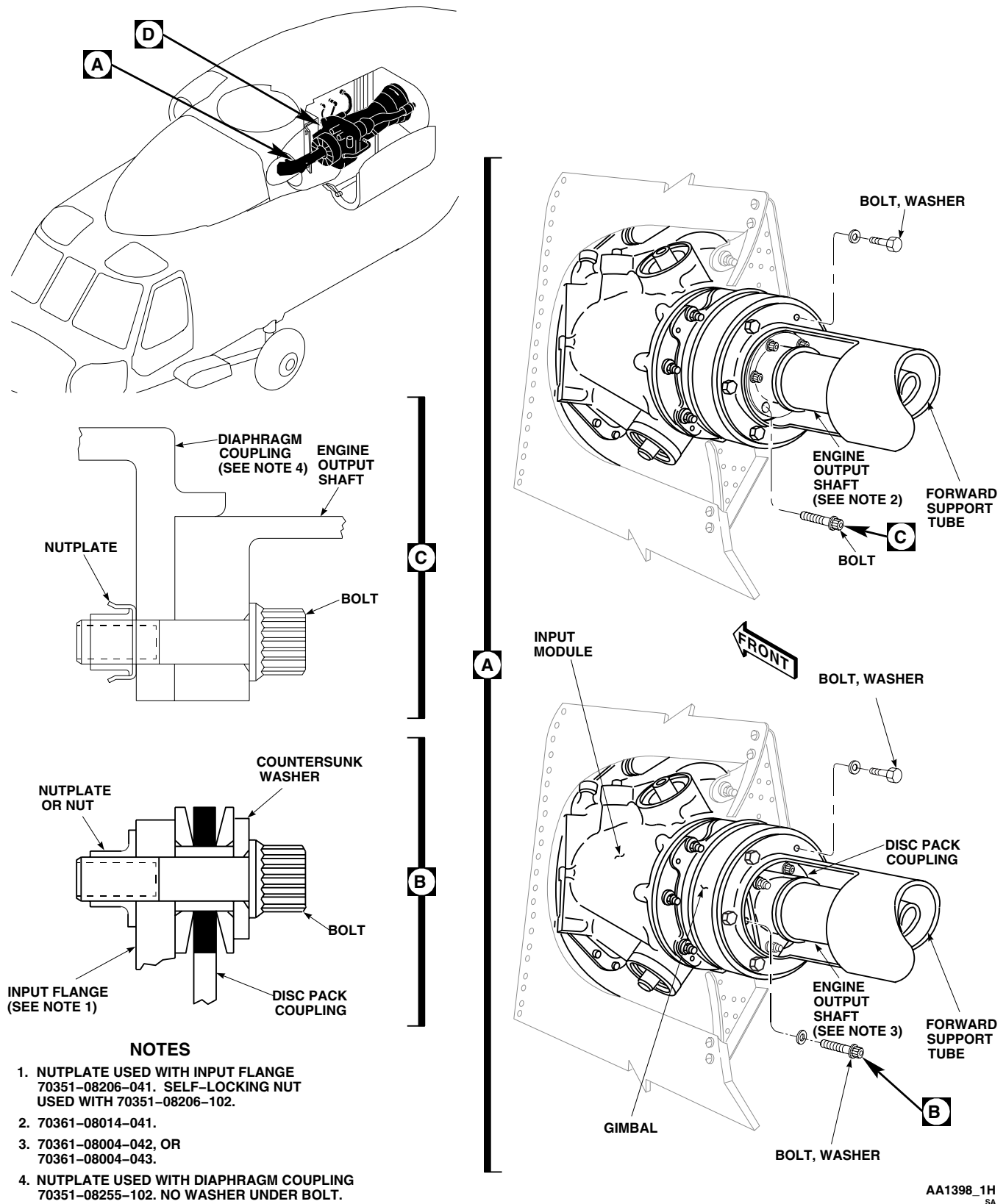


Figure 3. Disconnect Engine Mounts. (Sheet 1 of 2)

REMOVE - Continued

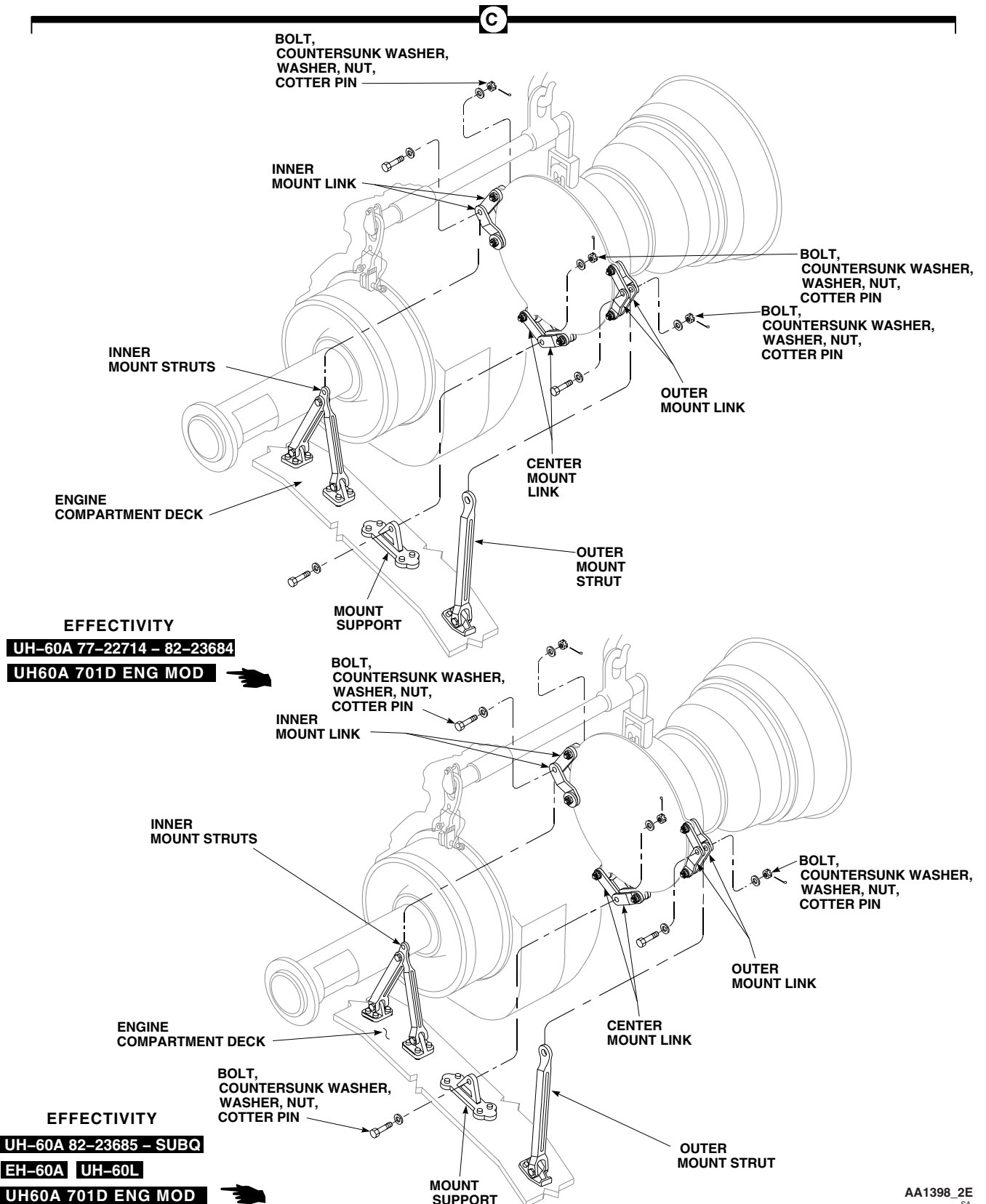


Figure 3. Disconnect Engine Mounts. (Sheet 2 of 2)

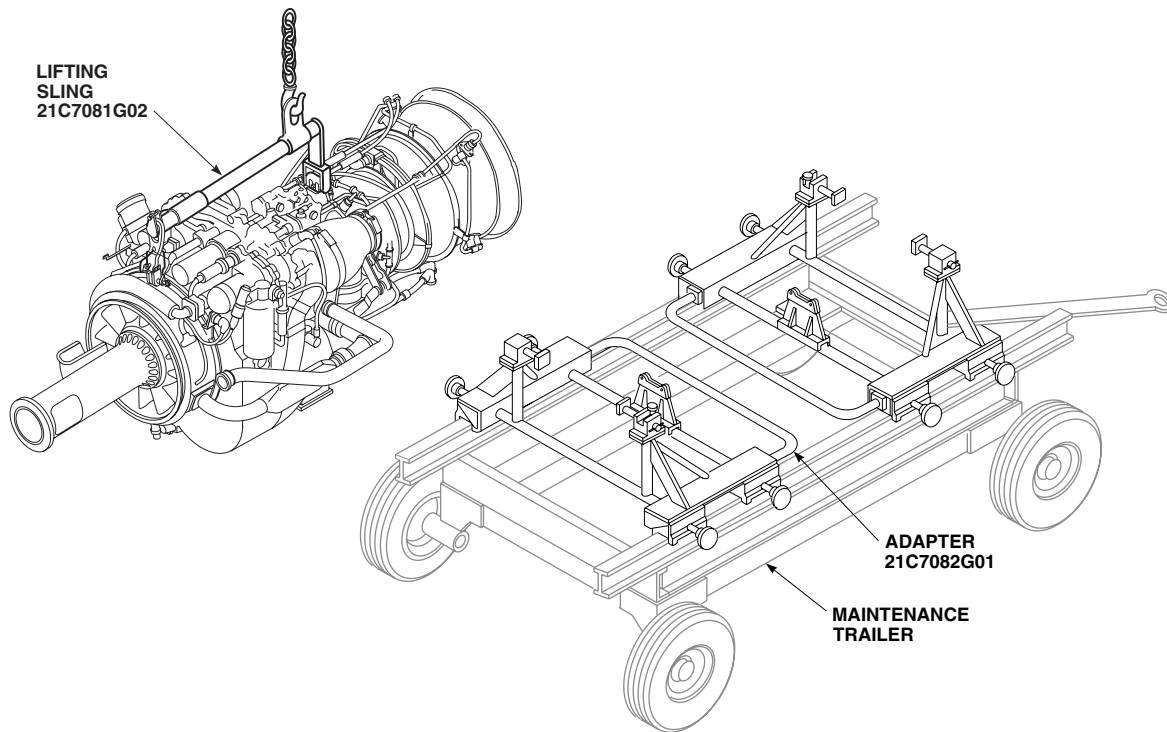
REMOVE - ContinuedAA1399
SA

Figure 4. Replace Engine.

CAUTION

- Damage to engine and helicopter will result if engine is allowed to swing while removing from helicopter. Carefully move engine when removing from helicopter.
- Damage to drive shaft and input flange/coupling will result on helicopters with drive shaft 70361-08014-041, if drive shaft is not completely separated from input flange/coupling.

25. On helicopters with drive shaft 70361-08014-041, move engine slightly aft to separate drive shaft from input flange/coupling.
26. Raise engine high enough to clear engine cowling.
27. Remove bolts, washers, and nuts holding aft mount links to flange under engine combustor section. Keep links and attaching parts with demountable power package.
28. Swing engine outward and lower it onto adapter (Figure 4).
29. Bolt engine to adapter (TM 1-2840-248-23).
30. Remove lifting sling from engine.

REMOVE - Continued**NOTE**

Do not tear down demountable power package if quick engine change (QEC) is available.

31. If engine is being replaced, tear down demountable power package and install components on new engine (WP 0482 00).
32. Inspect input pinion axial play (WP 0619 00).
33. On helicopters with engine output drive shaft 70361-08004-042, or 70361-08004-043, inspect engine output shaft and disc pack coupling (WP 0486 00).
34. On helicopters with engine output drive shaft 70361-08014-041, inspect engine output shaft (WP 0486 00).
35. Inspect engine output shaft and disc pack coupling (WP 0486 00).
36. Inspect input module flange (WP 0655 00).
37. Inspect elastomeric gimbal (WP 0619 00).
38. Perform an engine output shaft inspection after disconnect (WP 1720 00).
39. Inspect engine aft mounts (WP 0485 00).

INSTALL

1. Inspect deswirl duct as follows:
 - a. Check deswirl duct for cracks. **REPLACE DESWIRL DUCT IF CRACKED.**
 - b. Check for nicks, scratches, and gouges as follows:
 - (1) Clamp Flange: Any number, 1/32-inch deep with no high metal are acceptable. **CONTOUR AS NEEDED IF BEYOND LIMITS.**
 - (2) Vanes and Doublers: Any number, 0.005-inch deep with no high metal or sharp corners are acceptable. **CONTOUR AS NEEDED IF BEYOND LIMITS.**
 - (3) All Other Areas: Any number, 0.015-inch deep with no high metal are acceptable. **CONTOUR AS NEEDED IF BEYOND LIMITS.**
 - c. Check cowl for distortion. Cowl must be visually circular and concentric with the case. **COLD WORK IF BEYOND LIMITS.**
 - d. Check vanes for distortion. Vanes must be visually straight and parallel with flow path. **REPLACE DESWIRL DUCT IF BEYOND LIMITS.**
 - e. Check deswirl duct for missing vanes. A total of 6 vanes may be missing or removed provided that no two adjacent vanes are missing. **REPLACE DESWIRL DUCT IF BEYOND LIMITS.**
 - f. Inspect for cracks in vanes, specifically in the leading edge area of vanes. **CRACKS SHALL NOT EXCEED 1 INCH.** Remove vane if cracks of one inch or more are found as follows:

NOTE

Make sure that removal of a vane will not cause deswirl duct to be beyond limit.

- (1) Remove deswirl duct from engine.
- (2) Cut along base of vane using a high speed cutting wheel.
- (3) Grind smooth any rough edges.

INSTALL - Continued

- g. If cracks less than 1 inch are found, deswirl duct should be inspected every 30 hours thereafter.
2. Inspect deswirl duct loop clamp for cracks near T-bolt attachment point. **REPLACE DESWIRL DUCT LOOP CLAMP IF CRACKED.**
 3. **QA** **W/O HIRSS** Place exhaust deswirl duct and loop clamp inside of exhaust fairing before installing engine in compartment. ■
 4. **QA** **HIRSS** Install deswirl duct and loop clamp on HIRSS module by sliding duct inside finger seals of HIRSS core. ■
 5. If IPS duct was removed from No. 2 engine compartment, slide duct into exhaust fairing opening.
 6. Install lifting sling on engine front and rear lifting lugs with quick-disconnect pins (Figure 2, Detail A and Figure 2, Detail B).
 7. Pin lifting eye in No. 3 hole on lifting sling.
 8. Attach hoist to sling. Raise hoist until weight of engine is supported by sling.

CAUTION

Damage to engine and maintenance trailer will result if engine is allowed to swing while removing from maintenance trailer. Carefully move engine when removing from maintenance trailer.

9. Remove bolts and washers holding engine to adapter.

CAUTION

Due to possible damage to engine Cold Section, boltheads in Figure 5 connecting mount links to the engine must face aft. Boltheads connecting mount links to inner/outer mount struts and mount support may face in either direction (forward or aft).

NOTE

- Shim with washers if required, to install cotter pin when connecting center mount links to mounting flange.
 - Mount links should have open fork ends (widest end) attached to mount support or mount strut.
 - For the inner and outer mount links, the upper mount link should have the offset leg of fork facing aft and lower link should have the offset leg of fork facing forward.
 - When looking aft at the inner and outer mount links the bottom link should be forward (in front) of the upper link on both engines.
 - When looking aft at the center mount link, the right hand link is always forward (in front) of the left hand link on both engines.
 - Forks should be in front of adjoining upper mount link fork with mount or strut between these forks.
10. **QA** Bolt center mount links to engine with bolts, countersunk washers, washers and nuts. Make sure countersunk side of washers face bolthead and boltheads are facing aft. **TIGHTEN EACH NUT FINGERTIGHT, ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin (Figure 5, Sheet 1).

INSTALL - Continued**CAUTION**

- Disc pack coupling will be damaged if not connected to correct gimbal.
- Make sure to use the correct gimbal with input flange/coupling being used. Elastomeric gimbal 70361-08015-102 must be used with input flange/coupling 70351-08255-102 and engine output shaft 70361-08014-041. Elastomeric gimbal 70361-08001-101, 70361-08001-102, or 70361-08001-103, must be used with input flange 70351-08206-102 and engine output shaft 70361-08004-042, or 70361-08004-043 and disc pack 70361-08002-041.

NOTE

- On helicopters with input module flange 70351-08206-102, or 70351-08260-041, hardware used to hold disc pack coupling to input module flange is a balanced set and should not be replaced individually. If any individual pieces of hardware needs to be replaced, replace all hardware with balanced hardware kit, 70000-30100-043.
 - On helicopters with input module flange/coupling 70351-08255-102, hardware used to hold drive shaft to input module flange/coupling is a balanced set and should not be replaced individually. If any individual pieces of hardware needs to be replaced, replace all hardware with balanced hardware kit, 70000-30100-044.
11. On helicopters with engine output drive shaft 70361-08004-042, or 70361-08004-043, inspect engine output shaft installation bolts as follows:
 - a. Visually inspect bolts for signs of burrs or evidence of binding. **NONE ALLOWED.** Replace all bolts and washers with correct hardware kit if burrs or evidence of binding is found.
 - b. Inspect bolts for binding in disc pack coupling by inserting bolts into disc pack only and checking for freedom of rotation. **BOLTS MUST ROTATE FREELY WHEN TURNED BY HAND.** Replace all bolts and washers with correct hardware kit if bolts do not rotate freely. If replacement bolts still do not rotate freely, replace engine output shaft assembly and input module flange (WP 0486 00 and WP 0655 00).
 12. On helicopters with engine output drive shaft 70361-08014-041, inspect engine output drive shaft installation bolts for signs of burrs or evidence of binding. **NONE ALLOWED.** Replace all bolts with hardware kit 70000-30100-044, if burrs or evidence of binding is found.
 13. On helicopters with engine output drive shaft 70361-08004-042, or 70361-08004-043, do a torque check of nuts attaching disc pack coupling to engine output shaft as follows:
 - a. Use strap wrench on output shaft to prevent shaft from turning. Check disc pack coupling to engine output shaft nuts in tightening direction while holding bolts. **TORQUE SHALL NOT BE LESS THAN 90 INCH-POUNDS. REPLACE ENGINE OUTPUT SHAFT ASSEMBLY IF ANY NUT IS LESS THAN 90 INCH-POUNDS.**
 - b. Reset torque wrench to 130 inch-pounds and torque check disc pack coupling to engine output shaft nuts in tightening direction while holding bolts. **IF ANY NUT TURNS BELOW 130 INCH-POUNDS, RETORQUE NUT TO 140 INCH-POUNDS.**
 14. Slide engine output shaft into engine as far as it will go.
 15. Do torque check on input module flange nut. **PERFORM TORQUE CHECK OF THE INPUT MODULE FLANGE RETAINING NUT. CHECK TORQUE IN THE TIGHTENING DIRECTION ONLY. MINIMUM TORQUE IS 646 INCH-POUNDS, RETORQUE TO 646 - 714 INCH-POUNDS.**
 16. Remove bolts, washers, and nuts from input module flange and retain for later use.

INSTALL - Continued

17. Install bolts in input module flange, 70351-08206-041. Make note if bolts can be fully threaded into nutplate by hand for later use (Figure 5, Sheet 3, Detail B).
18. If input flange, 70351-08206-102, is installed, install bolts and input module flange nuts. If any one of the three bolts can be fully threaded by hand into the nut, replace the bolts, washers, and nuts as a matched set using hardware kit, 70000-30100-043.

CAUTION

On helicopters with engine output drive shaft 70361-08004-042, or 70361-08004-043, damage to input module flange will result if output shaft coupling is not properly aligned with input module flange. Bolts holding coupling to shaft may scrape against input module flange. Make sure coupling to flange bolt holes are properly aligned before placing engine in engine compartment.

19. On helicopters with engine output drive shaft 70361-08004-042, or 70361-08004-043, rotate engine output shaft to align 3 mounting holes in shaft coupling with 3 holes in input module flange.
20. On helicopters with engine output drive shaft 70361-08014-041, rotate engine output shaft to align 6 mounting holes in drive shaft with 6 holes in input module flange/coupling.

CAUTION

Damage to engine and helicopter will result if engine is allowed to swing while installing to helicopter. Carefully move engine when installing to helicopter.

21. **QA** Carefully raise and place engine in engine compartment making sure forward support tube is flush on input module gimble.

NOTE

Damage to mount struts, mount links and mounting bolt will result if force is used to install bolt. The bolt connecting mount struts to mount links is a close tolerance assembly. If bolt does not slide through strut and links, replace bolt.

22. **QA** **UH-60A 77-22714 - 82-23684** Connect support mounts as follows (Figure 5, Sheet 1):
 - a. **QA** Bolt inner and outer mount struts to mount links with bolts, washers, and nuts. **TIGHTEN NUTS FINGER-TIGHT, ADVANCE TO NEXT CASTELLATION.** Install cotter pins.
 - b. **QA** Bolt center mount links to mount support with bolts, washers, and nuts. **TIGHTEN NUT FINGERTIGHT, ADVANCE TO NEXT CASTELLATION.** Install cotter pin.
 - c. If shimming is required to allow cotter pin to be inserted in nut properly when connecting mount links to mount struts or support use washers, AN960C616 or AN960C616L, under bolthead or nut.
 - d. If required, additional engine mount instructions and torques are found in WP 0485 00.

NOTE

Damage to mount struts, mount links and mounting bolt will result if force is used to install bolt. The bolt connecting mount struts to mount links is a close tolerance assembly. If bolt does not slide through strut and links, replace bolt.

23. **QA** **UH-60A 82-23685 - SUBQ EH-60A UH-60L** Connect support mounts as follows (Figure 5, Sheet 1):
 - a. **QA** Bolt inner and outer mount struts to mount links with bolts, washers, and nuts. **TORQUE NUTS TO 181 - 199 INCH-POUNDS.** Install cotter pins.
 - b. **QA** Bolt center mount links to mount support with bolt, countersunk washer, washer and nut. **TIGHTEN NUT**

INSTALL - Continued

BOLT, COUNTERSUNK WASHER, WASHER, NUT
(FINGERTIGHTEN, ADVANCE TO NEXT CASTELLATION).
COTTER PIN

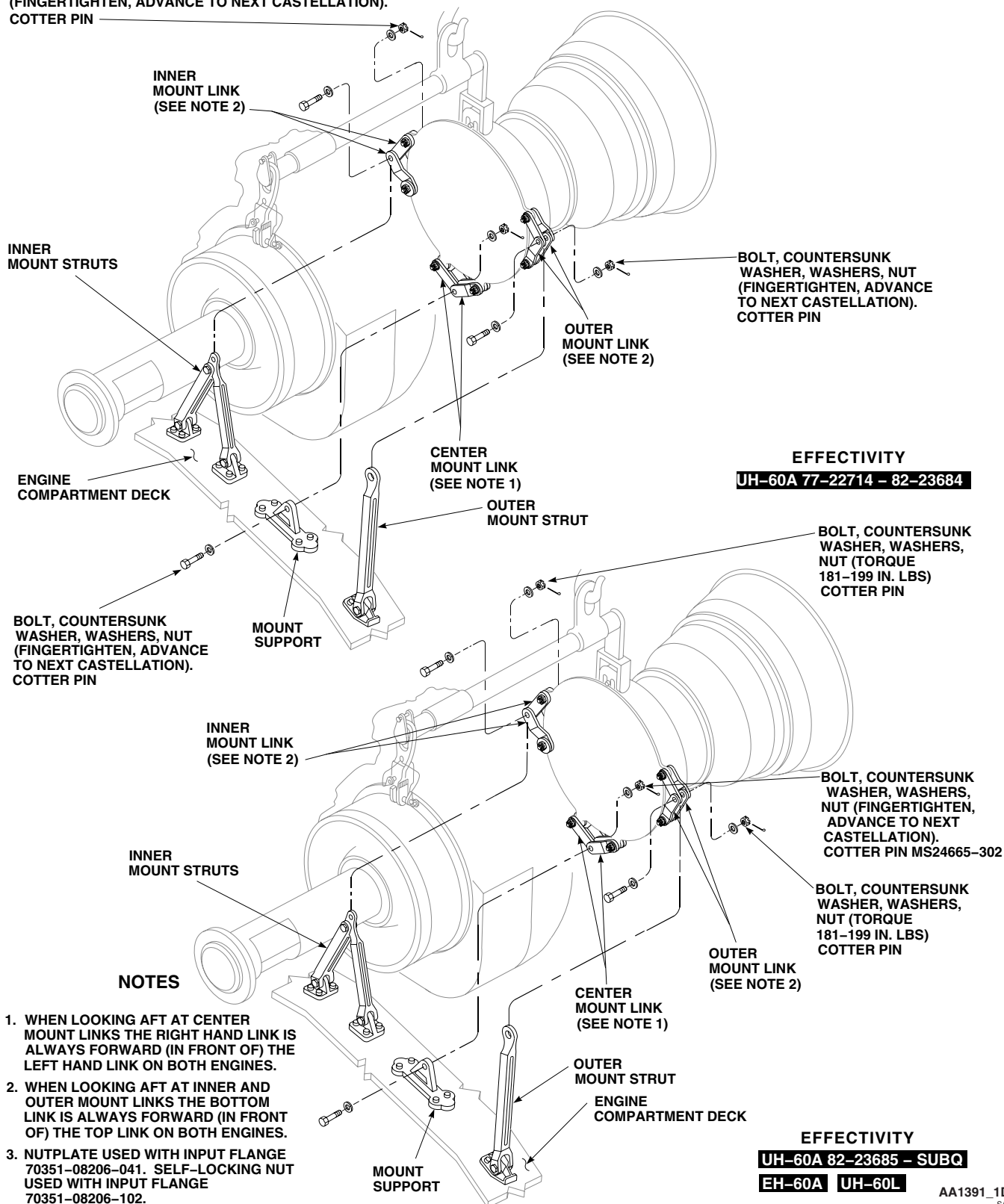
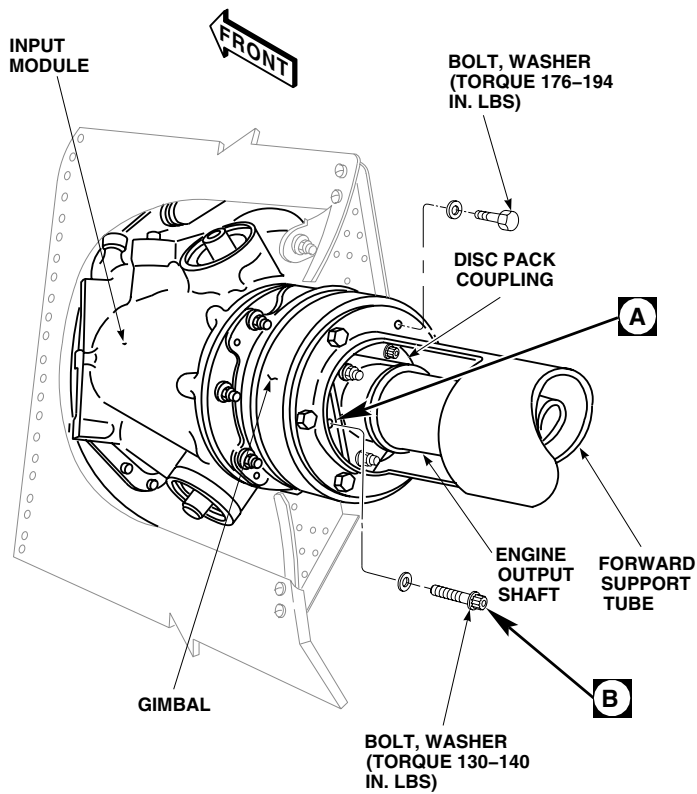
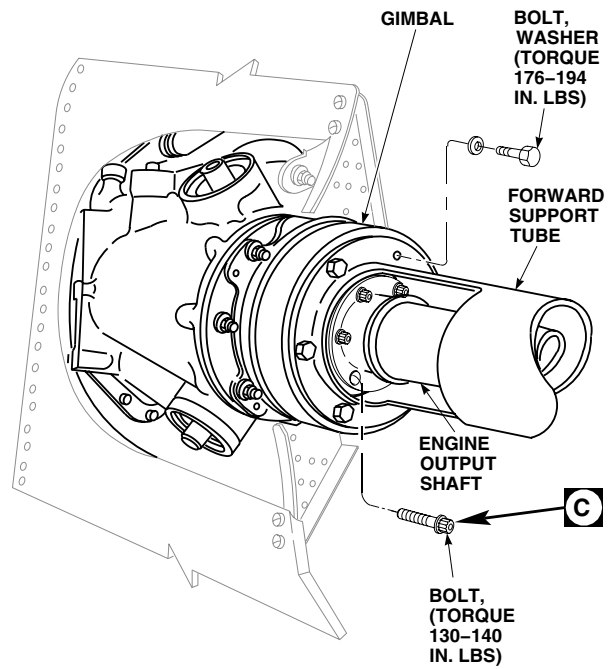


Figure 5. Connect Engine Mounts. (Sheet 1 of 3)

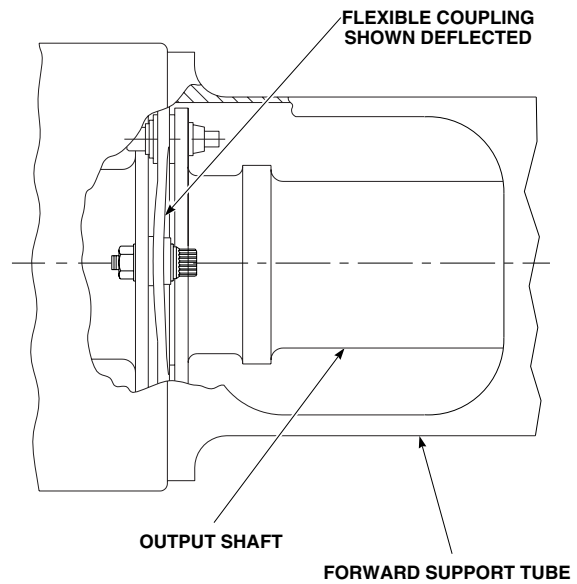
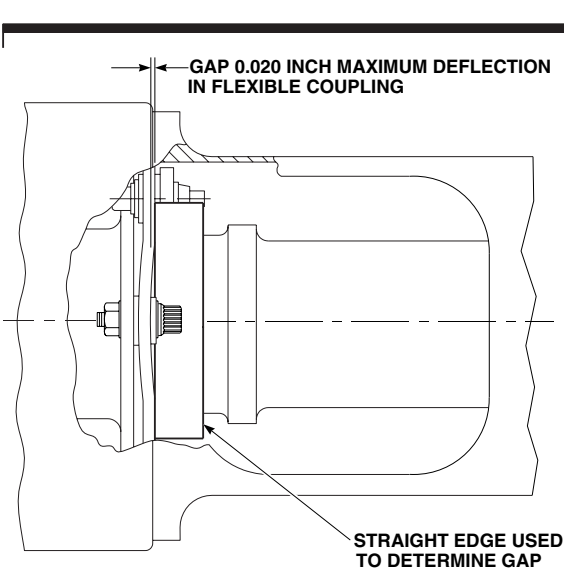
INSTALL - Continued



EFFECTIVITY
ENGINE OUTPUT DRIVE SHAFT
70361-08004-042, OR 70361-08004-043.



EFFECTIVITY
ENGINE OUTPUT DRIVE SHAFT
70361-08014-041.



AA1391_2G
SA

Figure 5. Connect Engine Mounts. (Sheet 2 of 3)

INSTALL - Continued

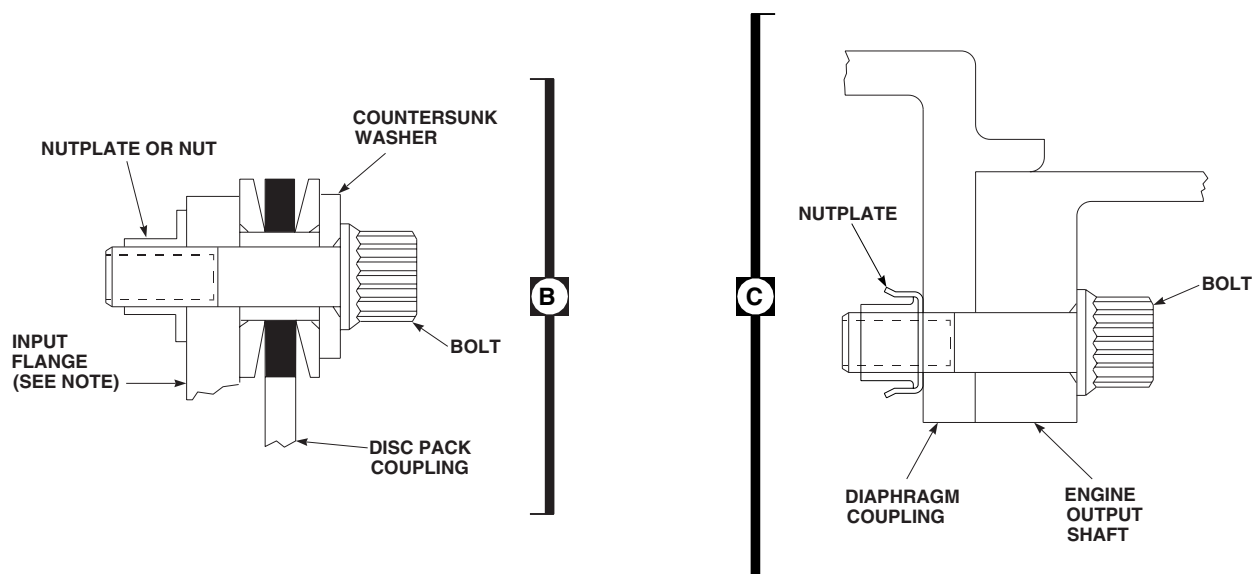
AA1391_3D
SA

Figure 5. Connect Engine Mounts. (Sheet 3 of 3)

UNTIL FIRMLY SEATED, THEN BACK OFF UNTIL WASHER UNDER NUT CAN BE TURNED BY HAND. RETIGHTEN NUT JUST ENOUGH SO THAT WASHER CAN NOT BE TURNED BY HAND. TIGHTEN NUT TO NEXT CASTELLATION. Install cotter pin. Do not tighten nut more than one castellation. Shim, if required, to install cotter pin.

- c. If required, additional engine mount instructions and torques are found in WP 0485 00.
- 24. Connect forward engine mount as follows:
 - a. **QA** Position forward support tube on gimbal (Figure 5, Sheet 2).

CAUTION

Damage to gimbal and support tube will result if bolts are used to draw support tube to gimbal. Position support tube flush against gimbal before installing bolts.

- b. **QA** Install bolts and washer attaching forward support tube to gimbal. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**

INSTALL - Continued**WARNING****CSI**

Verification of minimum torque of bolts is a critical characteristic.

NOTE

- Helicopters with input module flange, 70351-08206-102, use a self locking nut on forward side of flange to hold disc pack coupling to flange.
- Helicopters with input module flange, 70351-08206-041, have nutplate on forward side of flange to hold disc pack coupling to flange.
- The following procedure is the same for the input module flange utilizing the individual self locking nuts, except hold the nuts with a wrench and torque bolts the same as in the procedure.

- c. **QA** Connect output shaft disc pack coupling to input module flange with bolts, washers, and nuts (if required) as follows (Figure 5, Sheet 3, Detail B):
- (1) Tighten each bolt in equal steps of two turns to evenly draw output shaft into position until coupling is flush against input module flange. Do not exceed 40 inch-pounds of torque when tightening bolts (Figure 5, Sheet 2, Detail A):
 - (2) Place a 3 inch straight edge across coupling face and measure for deflection.
 - (3) If deflection is more than 0.020-inch, lightly tap shaft against balance ring in a forward direction using a 2 x 3 x 6 in. (5 x 8 x 15 cm) block of wood and a rubber mallet, until coupling deflection is less than 0.020-inch.
 - (4) **IF GAP IS LESS THAN 0.020-INCH, TORQUE BOLTS TO 130 - 140 INCH-POUNDS.**
 - (5) Check DA FORM 2408-13-1 to make sure torque check of bolts connecting disc pack coupling to input module flange is due after initial ground run and after 1 to 3 flight hours, and that torque check of bolts connecting disc pack coupling to engine output shaft is due after initial ground run and after 1 to 3 flight hours (WP 1688 00).
 - (6) If input module flange, 70351-08206-041, is used, and bolts were fully threaded into the nutplate by hand, then after torque has stabilized apply sealing compound, Item 285, WP 1803 00, to the end of the bolt on the threads extending past the nut. Insure the sealing compound is applied to the thread near the nut to allow it to seep into the nut through capillary action.
- d. Remove lifting sling from engine, remove maintenance crane and move hoist out of way. Remove lifting sling from crane or hoist (WP 1590 00).
25. On helicopters with engine output drive shaft 70361-08014-041, connect forward support engine mount as follows:
- a. Position forward support tube on gimbal (Figure 6, Sheet 1, Detail A).

CAUTION

Damage to gimbal and support tube will result if bolts are used to draw support tube to gimbal. Position support tube flush against gimbal before installing bolts.

- b. Install bolts and washers attaching forward support tube to gimbal. **TORQUE BOLTS TO 176 - 194 INCH-POUNDS.**
- c. Connect engine output drive shaft to input module flange/coupling with bolts as follows:
 - (1) Rotate engine output drive shaft to align 6 mounting holds with flange/coupling mounting holes.

INSTALL - Continued

- (2) Tighten each bolt in equal steps of two turns to evenly draw output shaft into position until drive shaft is flush against input module flange/coupling. Do not exceed 40 inch-pounds of torque when tightening bolts (Figure 6, Sheet 1, Detail A).
- (3) **TIGHTEN BOLTS UNTIL STABILIZED AT 130 - 140 INCH-POUNDS. RETIGHTEN BOLTS UNTIL NO MOTION IS FELT AT FINAL TORQUE VALUE. DO NOT EXCEED 140 INCH-POUNDS. DO NOT LOOSEN AND TIGHTEN BOLTS.**
- (4) Remove lifting sling from engine, remove maintenance crane and move hoist out of way. Remove lifting sling from crane or hoist (WP 1590 00).
- (5) Check DA FORM 2408-13-1 to make sure torque check of bolts connecting engine output drive shaft to input module flange/coupling is due after 3 to 13 flight hours, and meets minimum torque of 130 inch-pounds (WP 1688 00).

NOTE

Minimum gap between pneumatic tubes coupling connection is 0.120-inch. If necessary, metal may be removed from tube end using file to achieve this gap. Do not remove any metal from bead on tubes.

26. **QA UH-60A EH-60A** Connect cross bleed-air tube to firewall with new coupling (Figure 6, Sheet 1, Detail A). Place clamps next to beads on tubes. **TORQUE CLAMPS TO 35 - 45 INCH-POUNDS.** ■
27. **QA UH-60L** Connect cross bleed-air tube to firewall tube by sliding sleeve and coupling over cross bleed-air tube and firewall tube. **TORQUE NUT TO 43 - 47 INCH-POUNDS (Figure 6, Sheet 1, Detail A).** ■

NOTE

Minimum gap between pneumatic tubes coupling connection is 0.120-inch. If necessary, metal may be removed from tube end using file to achieve this gap. Do not remove any metal from bead on tubes.

28. **QA** Connect starter bleed-air tube to starter with clamp. **TORQUE CLAMP TO 35 - 45 INCH-POUNDS (Figure 6, Sheet 1, Detail A).**
29. **QA** Connect starter bleed-air tube to upper firewall tube with coupling and clamps. **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS (Figure 6, Sheet 1, Detail A).**

CAUTION

Make sure that rotary input alignment pin engages slot on engine hydromechanical unit. Handtighten retaining nut until it bottoms in support bracket. Slight axial/radial play is normal.

30. **QA** Attach power-available rotary input and load-demand rotary input to hydromechanical unit on engine (Figure 6, Sheet 1, Detail C).
31. Inspect and treat electrical connectors (WP 1703 00).
32. **QA** Uncap and connect electrical connectors to firewall mounted receptacles, P815 and P825, engine No. 1, or P814 and P824, engine No. 2 (Figure 6, Sheet 1, Detail B).
33. Remove caps and plugs from engine drain hoses and elbows on rear engine compartment deck.
34. **QA** Connect engine combustion drain hose to elbow on rear engine compartment deck. **TORQUE HOSE TO 135 - 190 INCH-POUNDS (Figure 6, Sheet 1, Detail D).**
35. **QA** Connect engine manifold drain hose to elbow on rear engine compartment deck. **TORQUE HOSE TO 215 - 280 INCH-POUNDS.**

INSTALL - Continued

36. **QA** **UH-60A 77-22727, 77-22728 UH-60A 78-22963 - SUBQ EH-60A UH-60L** Clamp common drain to engine compartment shield. ■
37. **QA** **UH-60A 77-22727, 77-22728 UH-60A 78-22963 - SUBQ EH-60A UH-60L** On No. 1 engine only, clamp engine swirl frame drain to airframe. ■

NOTE

Fuel hose for No. 1 engine should be positioned to avoid engine oil filler port and right front angled section of firewall. Additional contact between fuel hose and firewall is acceptable for both engines.

38. **QA** Uncap and connect fuel hose to fuel pump elbow.
39. **QA** Check for 0.750-inch clearance between fuel hose and fire extinguisher nozzle. If clearance is not correct, do this:
- a. Loosen bolts on fuel boost pump swivel flange (Figure 6, Sheet 2, Detail H).
 - b. Carefully move fuel boost pump elbow until at least 0.750-inch clearance is obtained between fuel hose and fire extinguisher nozzle.
 - c. **TORQUE BOLTS ON BOOST PUMP SWIVEL FLANGE TO 45 - 50 INCH-POUNDS.**
40. **TORQUE NUT ON FUEL HOSE TO 460 - 500 INCH-POUNDS.**
41. Inspect inlet particle separator (IPS) duct for cracks. If repairs are needed, refer to WP 0379 00.

CAUTION

Damage to engine cowling will result if coupling T-bolt holding IPS duct to IPS blower is installed incorrectly. Make sure coupling is positioned with T-bolt nut facing-inboard of aircraft and top of clamp is parallel to ground.

NOTE

When installing new IPS duct, refer to WP 0487 00.

42. **QA** Connect IPS duct to IPS blower with coupling (Figure 6, Sheet 2, Detail G). **TORQUE COUPLING NUT TO 60 - 70 INCH-POUNDS.** To seat clamp, tap it lightly with rubber mallet. Retorque clamp nut. Continue tapping clamp until two successive trials show no loss in torque.
43. **QA** On No. 2 engine only, clamp engine control cables to right side IPS duct with screw, washer, and spacer (Figure 1, Sheet 2, Detail C).
44. **QA** Clamp deswirl duct flange to engine power turbine flange as follows (Figure 6, Sheet 2, Detail E and Detail F):

NOTE

Left and right deswirl ducts are identical.

- a. Deswirl ducts have two alignment holes located on deswirl duct flange. One alignment hole shall be located top center while the other alignment hole shall point inboard when installed on the HIRSS module (Figure 6, Sheet 2).
- b. Make sure alignment hole in deswirl duct flange mates with alignment hole of engine power turbine flange.
- c. Visually inspect deswirl duct to make sure deswirl duct flange mates with engine power turbine flange through 360 degrees.

INSTALL - Continued

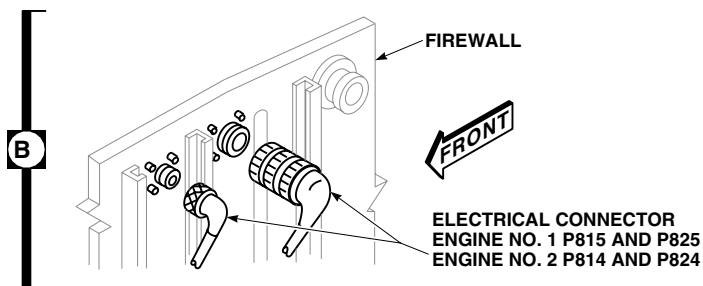
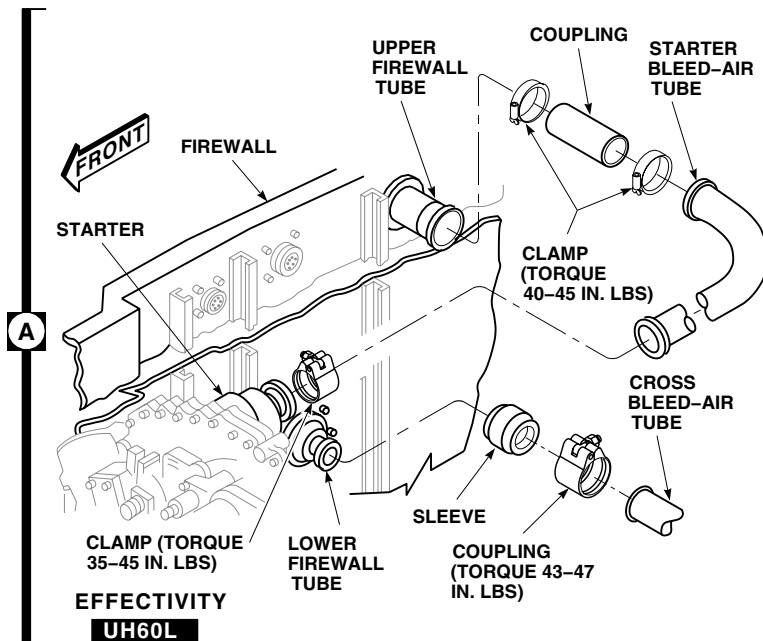
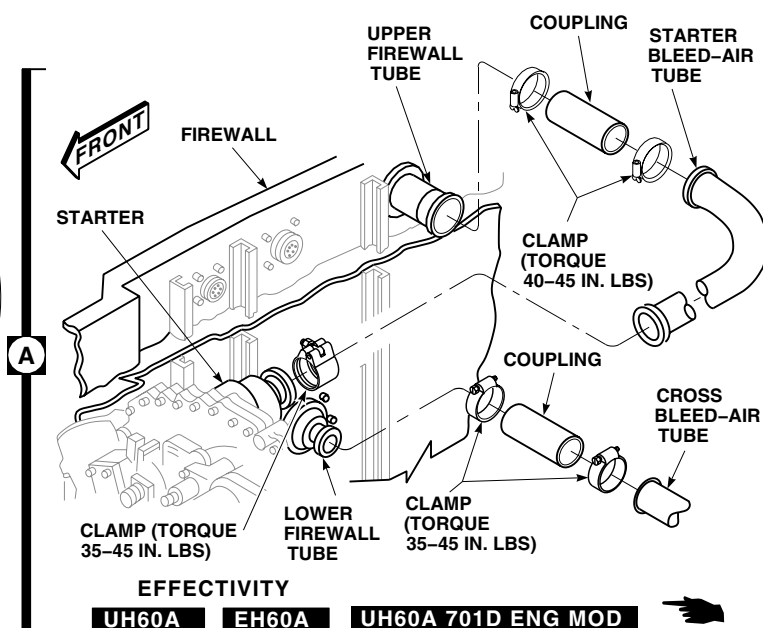
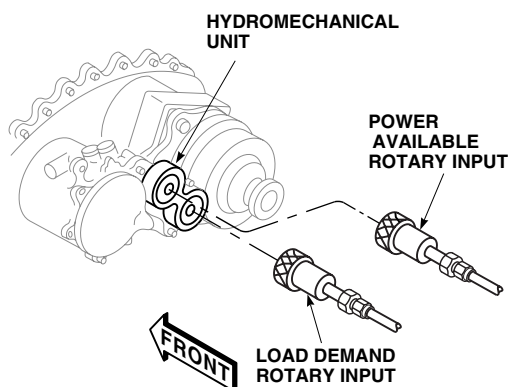
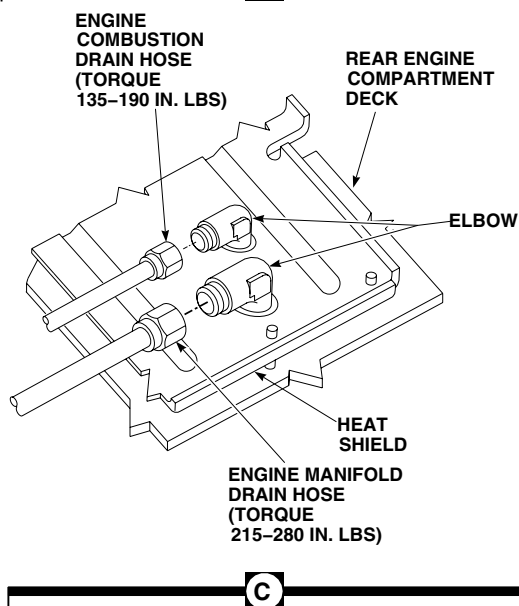
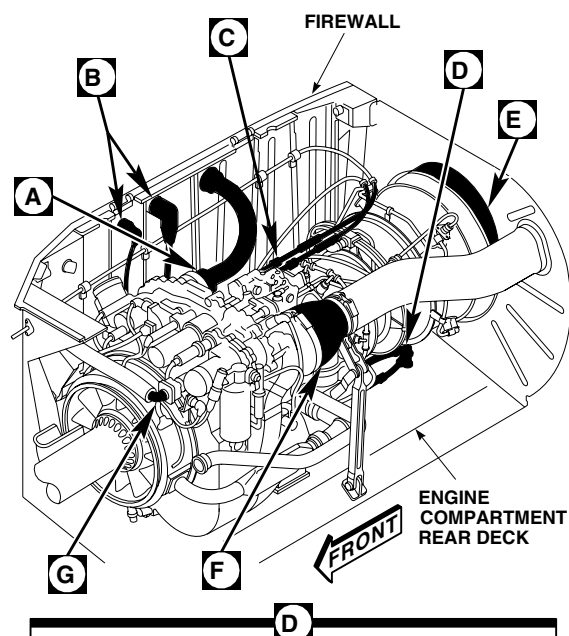
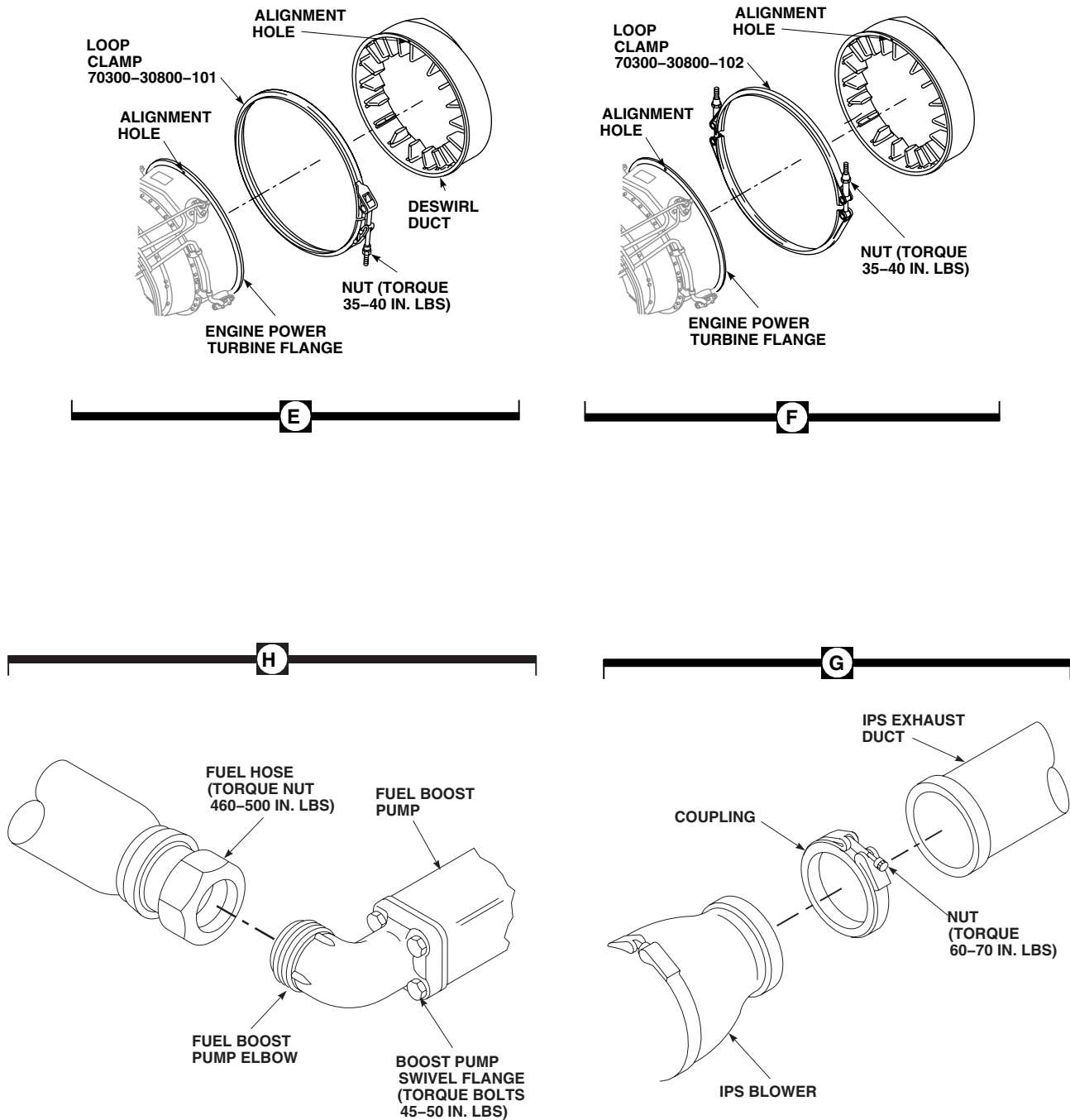
AA0136_1E
SA

Figure 6. Connect Engine/Airframe Connections. (Sheet 1 of 2)

INSTALL - Continued



AA0136_2D
SA

Figure 6. Connect Engine/Airframe Connections. (Sheet 2 of 2)

INSTALL - Continued

- d. If complete mating of flanges is not achieved, remove deswirl duct.
- e. Using a spring scale to measure load, apply no more than 15 pounds of load to remove warpage.
- f. If warpage cannot be removed using 15 pounds of load, replace deswirl duct.
- g. Lubricate pressure surfaces of loop clamp and T-bolt with aircraft turbine lubricating oil, Item 182, WP 1803 00.
- h. Install loop clamp over lips of deswirl duct and engine power turbine flange.

CAUTION

Damage to engine cowling will result if coupling T-bolt holding IPS duct to IPS blower is installed incorrectly. Make sure coupling is positioned with T-bolt nut facing inboard of aircraft and top of clamp is parallel to ground.

- i. Position loop clamp, 70300-30800-101, with T-bolt at 3 o'clock for left engine and 9 o'clock for right engine with nut facing down.

NOTE

Loop clamp is not a rigging tool. Deswirl duct must be supported until clamp is torqued.

- j. Position loop clamp, 70300-30800-102, with T-bolts at 3 o'clock for left engine and 9 o'clock for right engine with nuts facing up.

NOTE

When torquing loop clamp, 70300-30800-102, alternate from nut to nut to maintain even loading on clamp.

k. TORQUE LOOP CLAMP NUTS TO 35 - 40 INCH-POUNDS.

- l. To seat loop clamp, tap lightly with rubber mallet. Retorque loop clamp nuts. Continue tapping until two successive trials show no loss in torque.
 - m. Loosen loop clamp nuts slightly. **RETORQUE LOOP CLAMP NUTS TO 35 - 40 INCH-POUNDS.**
 - n. Visually inspect both latches to make sure no interference has occurred.
 - o. Visually inspect inside deswirl duct to make sure deswirl duct flange mates with engine power turbine flange through 360 degrees.
- 45. Install engine air inlet (WP 0484 00).
 - 46. Make sure area is clean and free of all foreign material.
 - 47. Check engine oil level and service as required (WP 1601 00).
 - 48. Close engine cowling and latch.
 - 49. Perform engine output shaft torque check after initial ground run and again after 1-3 flight hours (WP 0486 00).
 - 50. Perform an engine output shaft balance (TM 1-6625-724-13&P).
 - 51. If input module flange, 70351-08206-041 is installed, after torque has stabilized, place a drop of sealing compound, Item 285, WP 1803 00, on input module flange to disc pack coupling bolts adjacent to nutplates. Make sure sealing compound is allowed to seep into nutplates.
 - 52. Perform the following:

INSTALL - Continued

- a. Run up engine for at least 10 minutes. Let engine cool down for one hour minimum.
- b. Perform a torque check of the deswirl duct loop clamp nuts. **TORQUE SHOULD BE 35 - 40 INCH-POUNDS.** If torque is below limit, repeat step 44.1.

NOTE

Torque stripes to be used only on loop clamp, 70300-30800-102.

- c. Apply torque stripes to clamp nuts (WP 0381 00 and TM 55-1500-345-23).
53. Do engine checkout procedure as part of maintenance operation check/maintenance test flight (TM 1-2840-248-23).
54. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM**DEMOUNTABLE POWER PACKAGE**

This WP supersedes WP 0482 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Output Drive Shaft Puller, Locally-Made, Figure 216,
 WP 1805 00
 Powerplant Repairer's Toolkit, SC 5180-99-B07
 Rubber Mallet, 25714
 Spring Scale, AAA-S-133
 Torque Wrench, 30 - 200 in. lbs.
 Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Antiseize Compound, Item 49, WP 1803 00
 Lubricating Oil, Aircraft Turbine, Item 182,
 WP 1803 00
 Petrolatum, Item 216, WP 1803 00
 Protective Caps and Plugs
 Towel, Machinery Wiping, Item 344, WP 1803 00

Tubing, Plastic Spiral Wrap, Item 347, WP 1803 00
 Washer, AN960-10L

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-2840-248-23&P
 TM 55-1500-323-24
 WP 0079 00
 WP 0694 00
 WP 0695 00
 WP 1703 00
 WP 1803 00
 WP 1805 00

TEAR DOWN POWER PACKAGE**Remove Components**

1. Remove engine starter (WP 0694 00 or WP 0695 00).
2. Remove electrical control unit (ECU) or digital electronic control (DEC) and hydromechanical control unit (HMU) (TM 1-2840-248-23&P).
3. Remove engine controls support from HMU, if HMU is not to be installed on new engine.
4. Loosen loop clamp, remove deswirl duct from engine power turbine flange.
5. **HIRSS** Loosen captive nuts and remove center body cone from engine exhaust (Figure 1, Detail A). ■
6. **W/O HIRSS** Remove bolts, washers and cover plate from center tailpipe body. Remove nuts, washers, and center tailpipe body from engine exhaust (Figure 1, Detail A). ■

NOTE

Center mount links on flange under engine combustor section are removed when engine is installed on adapter, for teardown.

7. Remove bolts, countersunk washers, washers, and nuts, holding inner and outer mount links to engine rear mounting flange (Figure 1). Remove mount links.
8. Remove bolt, washer, and crotch from forward support tube. Slide crotch from forward support tube.

TEAR DOWN POWER PACKAGE - Continued

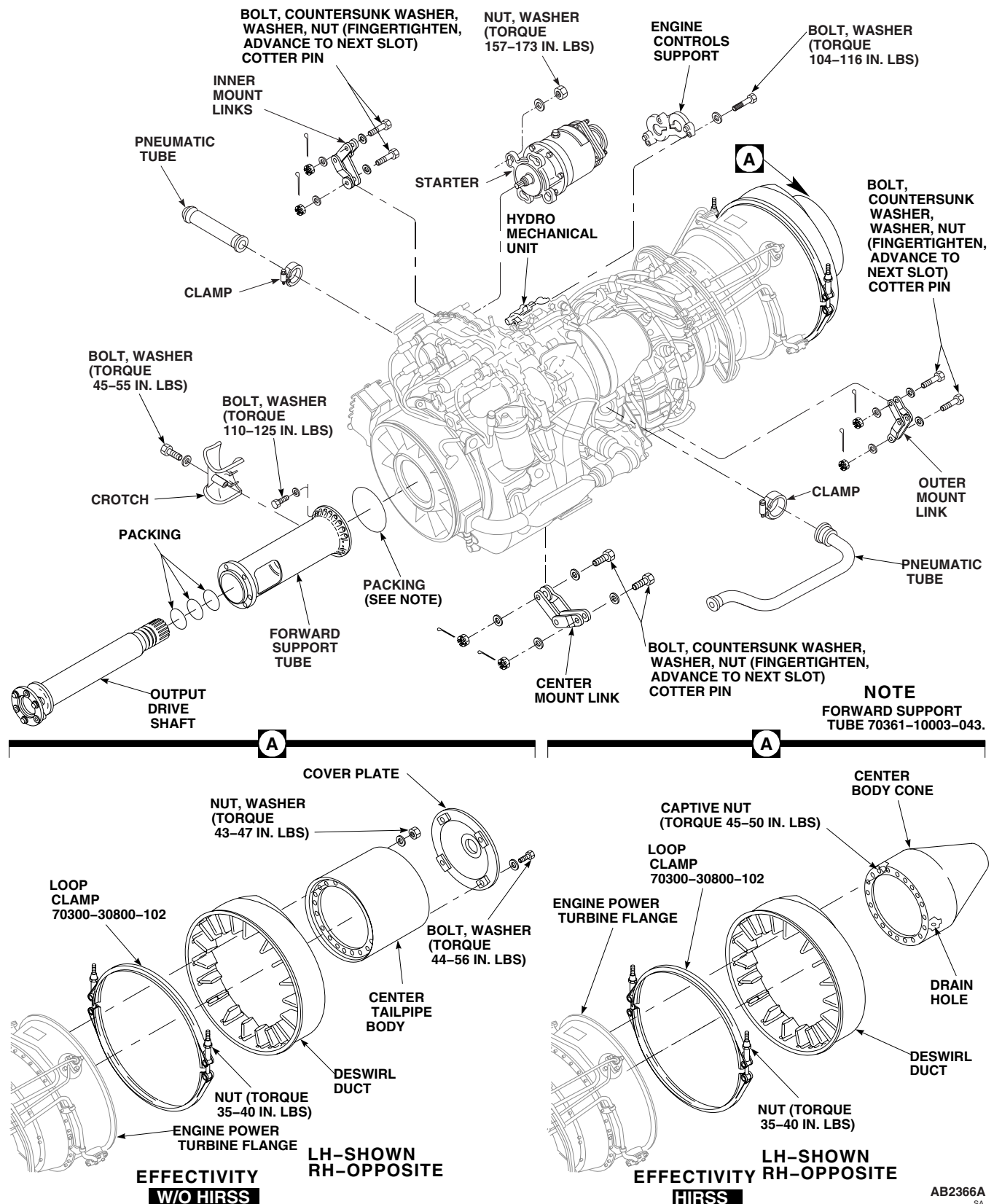


Figure 1. Remove/Install Components.

TEAR DOWN POWER PACKAGE - Continued

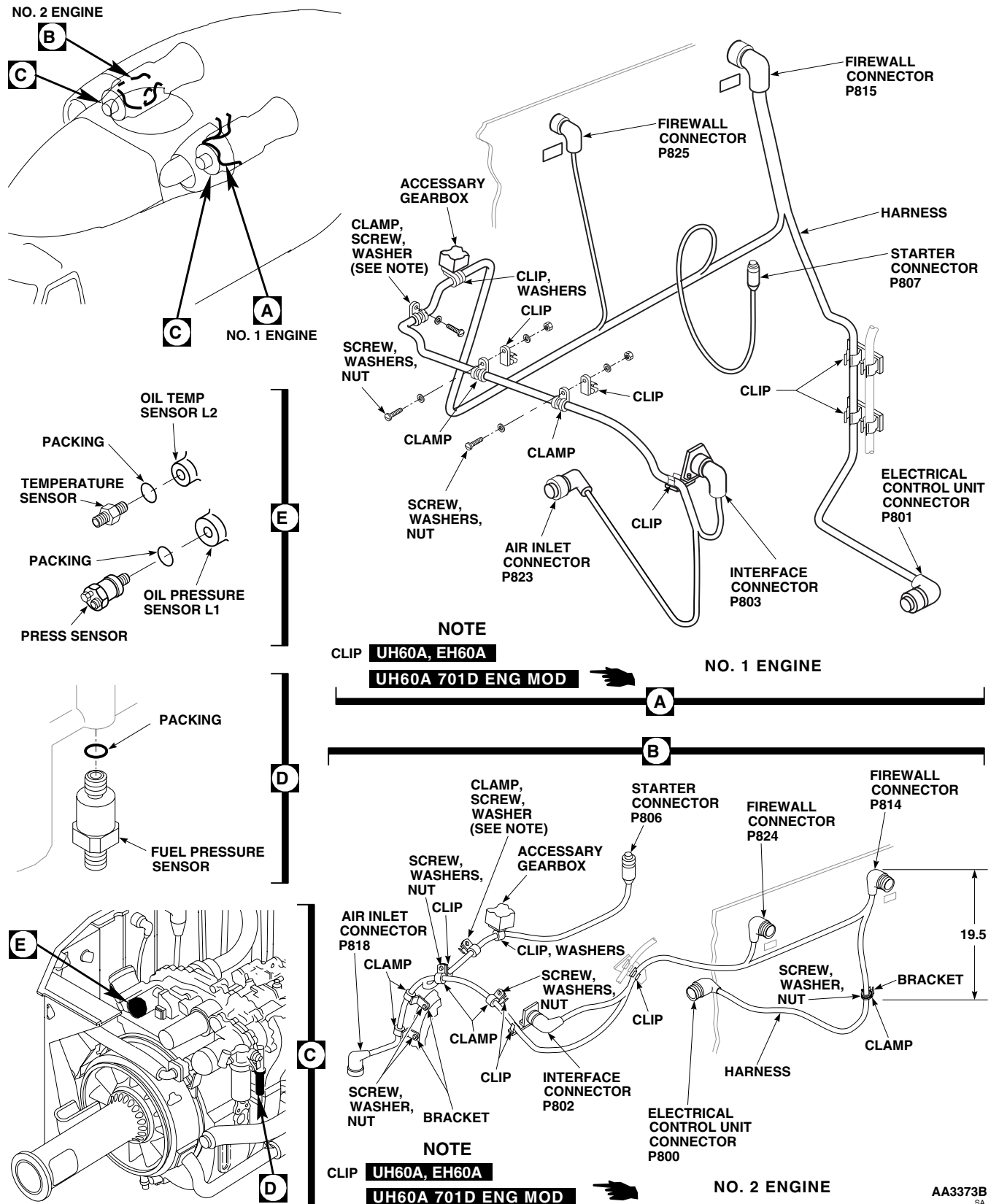


Figure 2. Remove/Install Wiring Harness And Sensing Components.

TEAR DOWN POWER PACKAGE - Continued

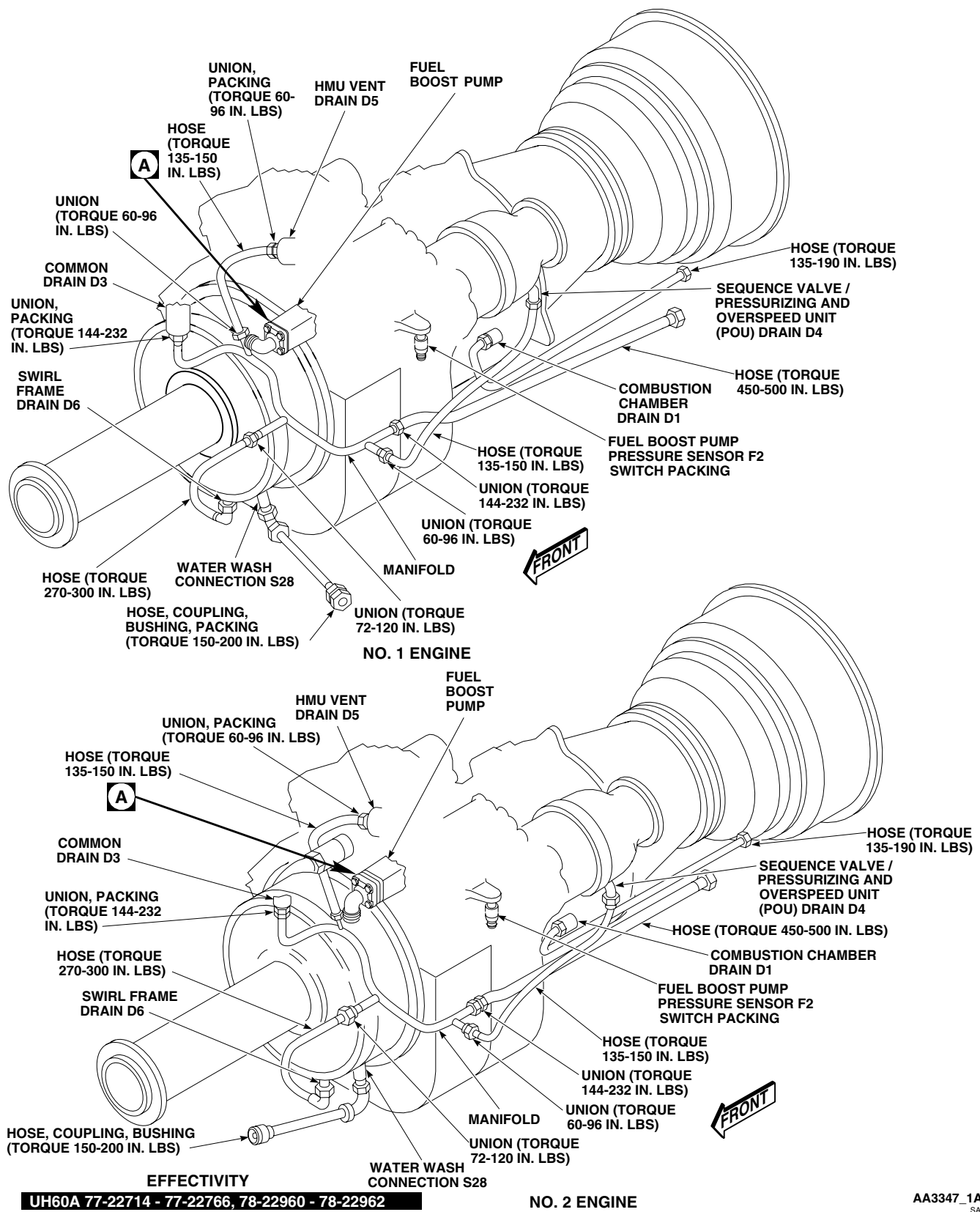
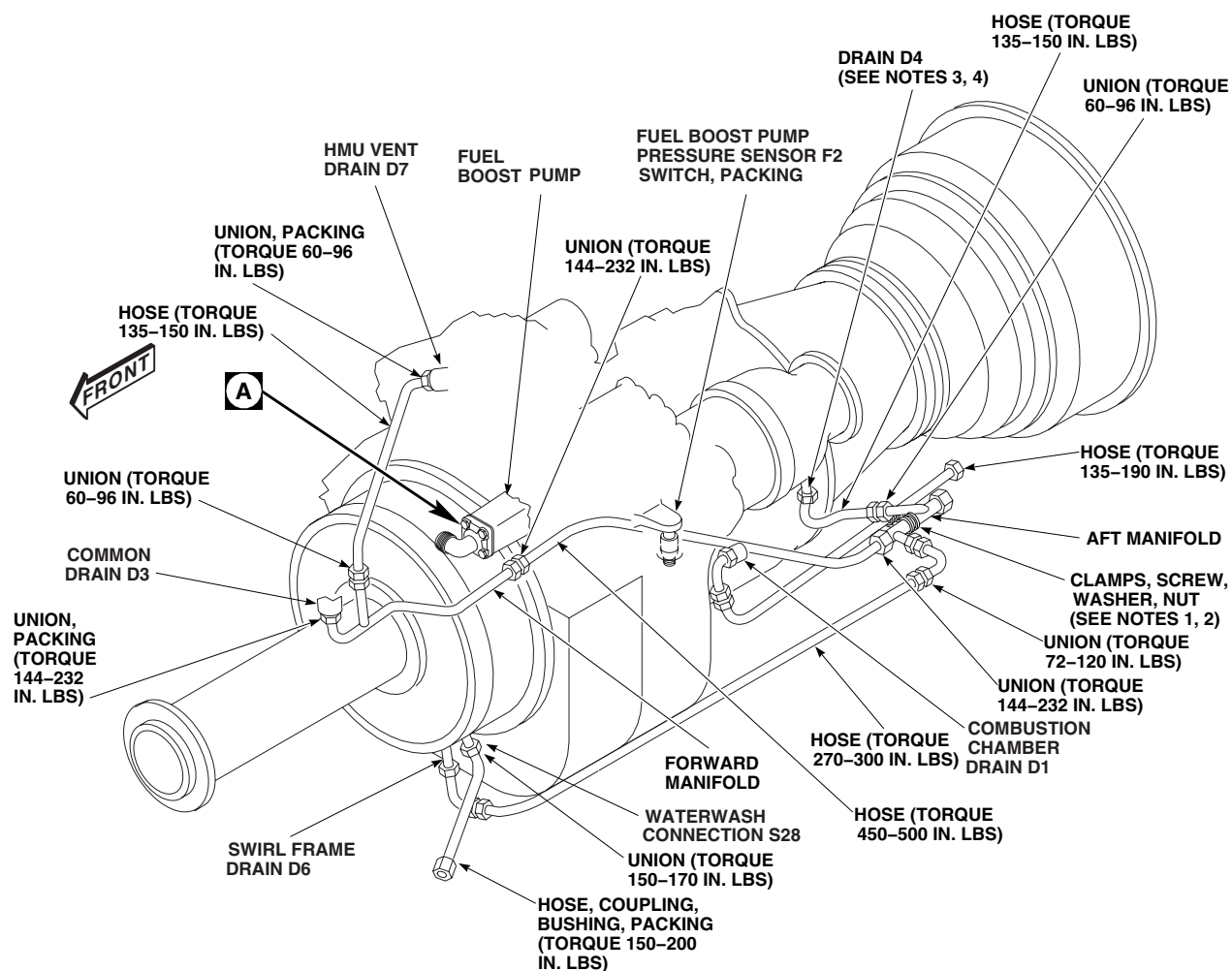


Figure 3. Remove/Install Fuel and Drain Lines. (Sheet 1 of 3)

TEAR DOWN POWER PACKAGE - Continued



NO. 1 ENGINE

EFFECTIVITY

UH60A 77-22727, 77-22728, 78-22963 – SUBQ

EH60A	UH60L
--------------	--------------

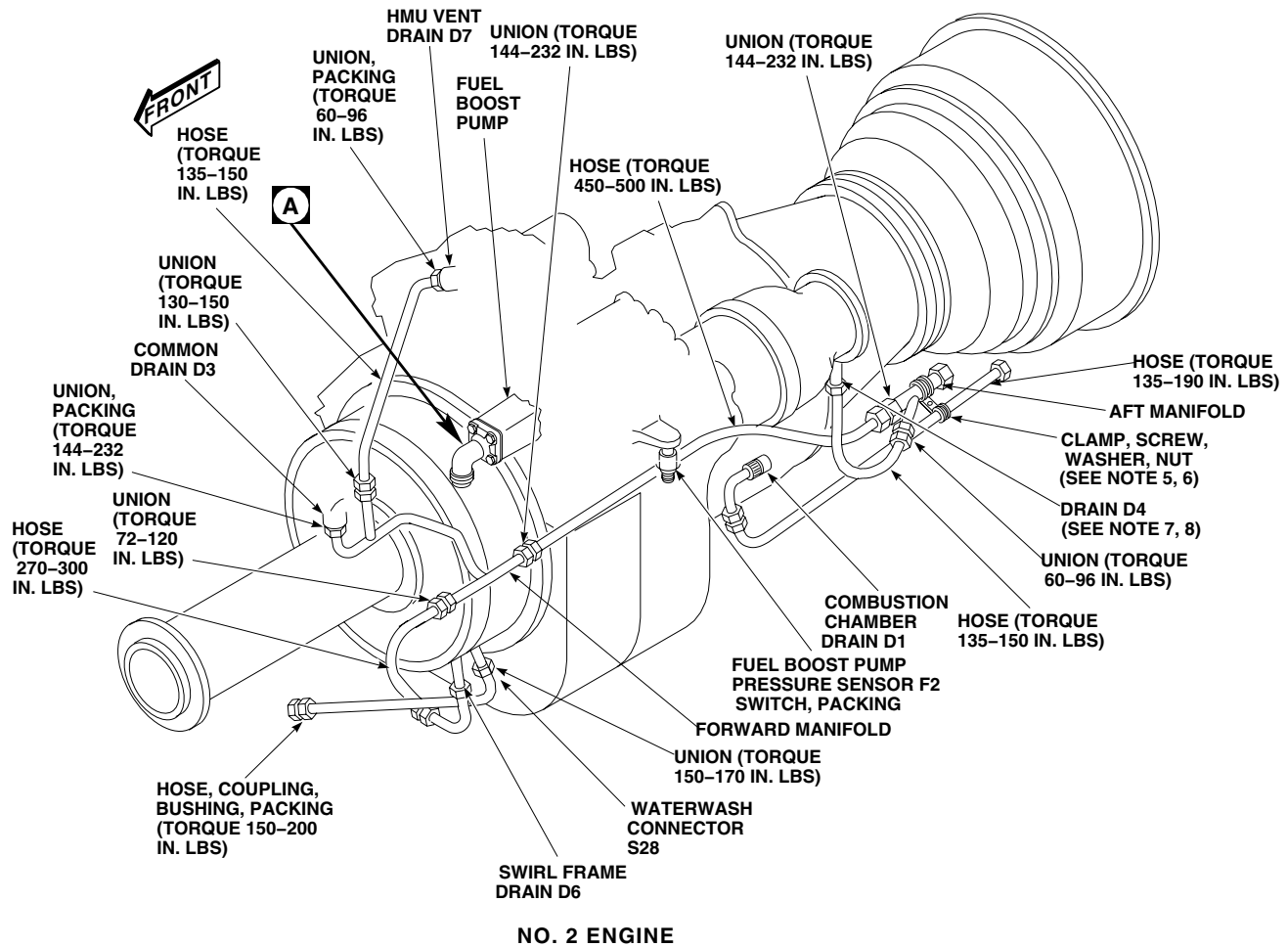
NOTES

1. **UH60A 81-23547 – SUBQ**
2. **EH60A**
3. **UH60A 77-22727, 77-22728, 78-22963 – SUBQ**
SEQUENCE VALVE/PRESSURIZING
AND OVERSPEED UNIT (POU).
4. **UH60L**
OVERSPEED AND DRAIN VALVE (ODV).

AA3347_2A
SA

Figure 3. Remove/Install Fuel and Drain Lines. (Sheet 2 of 3)

TEAR DOWN POWER PACKAGE - Continued



EFFECTIVITY

UH60A 77-22727, 77-22728, 78-22963 - SUBQ

EH60A UH60L

NOTES

5. UH60A 81-23547 - SUBQ

6. EH60A

7. UH60A 77-22727, 77-22728, 78-22963 - SUBQ

SEQUENCE VALVE/PRESSURIZING AND OVERSPEED UNIT (POU).

8. UH60L

OVERSPEED AND DRAIN VALVE (ODV).

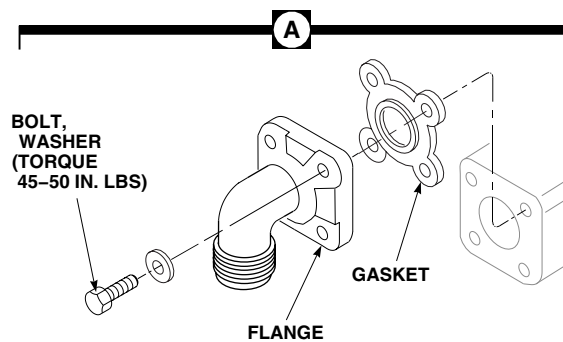
AA3347_3B
SA

Figure 3. Remove/Install Fuel and Drain Lines. (Sheet 3 of 3)

TEAR DOWN POWER PACKAGE - Continued

9. Remove bolts and washers and pull forward support tube and output drive shaft from engine. If output drive shaft fails to release from engine, do the following:
 - a. Locally make output drive shaft puller (WP 1805 00).
 - b. If necessary, loosen knurled nut on output drive shaft puller to release tension on donuts.
 - c. Clean any oil from inside of output drive shaft with a machinery wiping towel, Item 344, WP 1803 00.
 - d. Insert donut end of output drive shaft puller 6 inches (15 cm) into output drive shaft.
 - e. Tighten knurled nut to expand the donuts inside of output drive shaft.
 - f. Move slider against knurled nut and pull back sharply to pull output drive shaft from engine.
 - g. Loosen knurled nut and remove output drive shaft puller from output drive shaft.

NOTE

Packing is installed on forward support tube, 70361-10003-043, only.

10. Remove packings from shaft and forward support tube. Discard packings.
11. Remove clamps and pneumatic tubes from engine bleed ports.

Remove Wiring Harness and Sensing Components

1. Disconnect wiring harness electrical connector from interface connector, P803, engine No. 1 or P802, engine No. 2 (Figure 2, Detail A and Detail B).
2. Check electrical connectors for damage (TM 55-1500-323-24).
3. Unclamp and remove wiring harness from engine (Figure 2, Detail A and Detail B).
4. Disconnect engine green electrical cable connectors and remove oil temperature, oil pressure, and fuel pressure sensors from engine (Figure 2, Detail C, Detail D, and Detail E) (TM 1-2840-248-23&P).
5. Plug or cap all engine electrical connectors.

Remove Fuel and Drain Lines

1. Remove bolts, washers, elbow, flange and gasket from fuel boost pump on engine (Figure 3, Sheet 3, Detail A).
2. Disconnect drain hose from combustion chamber drain, D1 (Figure 3, Sheet 1, Detail A, Figure 3, Sheet 2, Detail A, or Figure 3, Sheet 3, Detail A).
3. Disconnect drain hose from union at common drain, D3. Remove union and packing from common drain. Remove packing from union. Discard packings.
4. Disconnect drain hose at swirl frame drain, D6.
5. Disconnect drain hose from union at HMU vent drain, D7. Remove union and packing from HMU vent drain. Discard packing.
6. Disconnect drain hose from sequence drain valve/pressurizing and overspeed unit or overspeed and drain valve, D4. Remove manifold and drain hoses from engine.
7. Remove hose from waterwash port.
8. Disassemble drain hoses and manifolds and remove unions.
9. Plug or cap engine drain lines and open ports.

BUILD UP POWER PACKAGE**Install Fuel and Drain Lines****CAUTION**

Damage to drain manifold will result if manifold chafes against engine main frame, swirl frame, or scroll case. If there is chafing, loosen nuts on hoses, reposition hoses, and retighten nuts. If chafing still exists, install clamps as required to maintain hose separation.

NOTE

Hose lengths and installation for No. 1 and No. 2 engine are different. Use correct figure.

1. Prevent waterwash line from chafing work platform fitting by installing plastic spiral wrap tubing, Item 347, WP 1803 00, over entire length of hose and routing hose over anti-ice duct on No. 1 engine and over sensor tube on No. 2 engine.
2. **QA** Install union in waterwash port, S28. **TORQUE UNION TO 150 - 170 INCH-POUNDS.**
3. **QA** Install waterwash hose on union. **TORQUE HOSE NUT TO 150 - 200 INCH-POUNDS.**
4. **QA** Install bushing, coupling half and packing onto waterwash hose. **TORQUE HOSE NUT TO 150 - 200 INCH-POUNDS.**
5. Lubricate all packings with petrolatum, Item 216, WP 1803 00.

CAUTION

Damage to engine will result if antiseize compound enters fuel lines. Do not allow antiseize compound to enter fuel lines.

6. Coat all male threads with antiseize compound, Item 49, WP 1803 00.
7. **QA** Lubricate gasket with petrolatum, Item 216, WP 1803 00. Install gasket, flange and elbow on fuel boost pump with bolts and washers. Position elbow for No. 1 or No. 2 engine installation. **TORQUE BOLTS TO 45 - 50 INCH-POUNDS (Figure 3, Sheet 3, Detail A).**
8. **QA** Install union and packing in common drain, D3. **TORQUE UNION TO 144 - 232 INCH-POUNDS.**
9. **QA** Install union and packing in HMU vent drain, D7. **TORQUE UNION TO 60 - 96 INCH-POUNDS.**
10. **QA** Install forward manifold to union in common drain, D3. **TORQUE UNION TO 144 - 232 INCH-POUNDS.**
11. **QA** Install union into forward manifold. **TORQUE UNION TO 60 - 96 INCH-POUNDS.**
12. **QA** Install hose between forward manifold and HMU vent drain, D7. **TORQUE HOSE NUT TO 135 - 150 INCH-POUNDS.**
13. **QA** **UH-60A 77-22714 - 77-22766 UH-60A 78-22960 - 78-22962** On No. 1 and No. 2. engine, install union into manifold. Install hose between swirl drain frame, D6, and manifold. **TORQUE HOSE NUT TO 270 - 300 INCH-POUNDS.** ■
14. **QA** **UH-60A 78-22963 - SUBQ UH-60A 77-22727, 77-22728 EH-60A UH-60L** On No. 1 engine, install union into aft manifold. **TORQUE UNION TO 72 - 120 INCH-POUNDS.** Install hose between swirl frame, D6, and aft manifold. **TORQUE HOSE NUT TO 270 - 300 INCH-POUNDS.** ■

BUILD UP POWER PACKAGE - Continued

15. **QA** **UH-60A 78-22963 - SUBQ UH-60A 77-22727, 77-22728 EH-60A UH-60L** On No. 2 engine only, install union into forward manifold. **TORQUE UNION TO 72 - 120 INCH-POUNDS.** Install hose between swirl drain frame, D6, and forward manifold. **TORQUE HOSE NUT TO 270 - 300 INCH-POUNDS.** ■
16. **QA** **UH-60A 78-22963 - SUBQ UH-60A 77-22727, 77-22728 EH-60A UH-60L** Install unions into the forward and aft manifolds. **TORQUE UNIONS TO 144 - 232 INCH-POUNDS.** ■
17. **QA** **UH-60A 78-22963 - SUBQ UH-60A 77-22727, 77-22728 EH-60A UH-60L** Install hose between forward manifold union and aft manifold union. **TORQUE HOSE NUTS TO 450 - 500 INCH-POUNDS.** ■
18. **QA** **UH-60A 7722714 - 77-22766 UH-60A 78-22960 - 78-22962** Install union into manifold. **TORQUE UNION TO 60 - 96 INCH-POUNDS.** Install hose between drain valve, D4, and manifold. **TORQUE HOSE NUTS TO 135 - 150 INCH-POUNDS.** ■
19. **QA** **UH-60A 78-22963 - SUBQ UH-60A 77-22727, 77-22728 EH-60A UH-60L** Install union into aft manifold. **TORQUE UNION TO 60 - 96 INCH-POUNDS.** Install hose between drain valve, D4, and aft manifold. **TORQUE HOSE NUTS TO 135 - 150 INCH-POUNDS.** ■
20. **QA** **UH-60A 7722714 - 77-22766 UH-60A 78-22960 - 78-22962** Install union into manifold. **TORQUE UNION TO 144 - 232 INCH-POUNDS.** Install hose on union. **TORQUE HOSE NUT TO 450 - 500 INCH-POUNDS.** ■
21. **QA** Install hose on combustor chamber drain, D1. **TORQUE HOSE NUT TO 135 - 190 INCH-POUNDS.** Do not clamp combustor chamber hose at this time, clamping occurs at installation.
22. **QA** Clamp hose assemblies every 18 inches to avoid chaffing.

Install Wiring Harness and Sensing Components**CAUTION**

Do not install white engine harness on the A-Axis nozzle stud. Failure to observe this caution may result in ejection of the oil nozzle and rapid loss of engine oil pressure.

NOTE

- If any cables are replaced perform a maintenance operational check on the new cable (WP 0079 00).
 - Harnesses for No. 1 and No. 2 engines are different. See Figure 2.
1. Check electrical connectors for damage (TM 55-1500-323-24).
 2. Inspect and treat electrical connectors (WP 1703 00).
 3. Position harness on engine to allow connection of three engine connectors.
 4. **QA** Connect electrical connector, P801, No. 1 engine or P800, No. 2 engine, to ECU or DEC (Figure 2, Detail A or Detail B).
 5. **QA** Connect electrical connector, P803, No. 1 engine or P802, No. 2 engine, to interface connector.
 6. **QA** Connect electrical connector, P807, No. 1 engine or P806, No. 2 engine, to engine starter.
 7. **QA** Clamp wiring harness to engine as shown.
 8. **QA** Install white engine/airframe harness to clip mounted on accessory gearbox stud.
 9. **QA** **UH-60A EH-60A** On No. 1 engine only, if required, remove existing single clip securing existing engine harness forward of engine oil jet. Install double clip using same hardware and install engine harness and existing harness. ■

BUILD UP POWER PACKAGE - Continued

10. **QA UH-60L** On No. 1 engine only, clamp engine harness to existing nutplate forward of engine oil jet using clamp, screw, and washer. ■
11. **QA UH-60A EH-60A** On No. 2 engine only, install clip to forward right hand side of engine using second stud on swirl frame counting down from top center line. Install engine harness and existing engine harness to clip. ■
12. **QA UH-60L** On No. 2 engine only, install engine harness to existing nutplate on forward right hand side of engine using clamp, screw, and washer. ■
13. **QA** Install clip to existing engine harness 1/2-inch right of center on upper front of engine. Clamp engine harness to clip using clamp, screw, washers, and nut.
14. **QA** Install clip to existing engine harness 3 inches left of enter on the upper front of engine. Clamp engine harness to clip using clamp, screw, washers, and nut.
15. **QA** Install clip to electrical connector bracket, P803, No. 1 engine or P802, No. 2 engine, using existing aft mounting screw. Attach engine harness to clip.
16. **QA** On No. 1 engine only, install 2 clips to engine harness and existing engine harness forward of the rear right hand side of engine.
17. **QA** On No. 2 engine only, install clip to engine harness and existing engine harness forward of existing harness clip on left hand side of engine.
18. **QA** On No. 2 engine only, install two brackets to forward right hand side of engine using the fourth and seventh stud on swirl frame counting down for top center line. Clamp engine harness to these brackets using clamps, screws, washers, and nuts.
19. **QA** On No. 2 engine only, install bracket on left hand side of engine using existing bolt on anti-ice valve. Clamp engine harness to bracket using clamp, screw, washer, and nut, do not tight now. Measure and allow 19.5 inches between clamp and electrical connector, P814. Tighten clamp.
20. Lubricate packing with petrolatum, Item 216, WP 1803 00. Install packing on oil temperature sensor (Figure 2, Detail E) (TM 1-2840-248-23&P).
21. **QA** Coat threads on oil temperature sensor with antiseize compound, Item 49, WP 1803 00. Install oil temperature sensor in engine accessory gear box (TM 1-2840-248-23&P).
22. **QA** Lubricate packing with petrolatum, Item 216, WP 1803 00. Install packing on oil pressure sensor (Figure 2, Detail E) (TM 1-2840-248-23&P).
23. **QA** Coat threads on oil pressure sensor with antiseize compound, Item 49, WP 1803 00. Install oil pressure sensor in engine accessory gear box (TM 1-2840-248-23&P).
24. **QA** Lubricate packing with petrolatum, Item 216, WP 1803 00. Install packing on fuel pressure sensor (Figure 2, Detail D) (TM 1-2840-248-23&P).
25. **QA** Coat threads on fuel pressure sensor with antiseize compound, Item 49, WP 1803 00. Install fuel pressure sensor in engine accessory gear box (TM 1-2840-248-23&P).
26. Inspect and treat electrical connectors (WP 1703 00).
27. **QA** Connect engine green electrical cable connectors to oil temperature, oil pressure and fuel pressure sensors (Figure 2, Detail C, Figure 2, Detail D, and Figure 2, Detail E).

BUILD UP POWER PACKAGE - Continued**Install Components****WARNING**

Injury to personnel and damage to equipment will result if pneumatic tube clamps interfere with engine variable geometry linkage. Make sure clamps are installed with clamp nuts and bolts straight up and down when connecting pneumatic tubes to engine bleed ports.

1. **QA** Clamp pneumatic tubes to engine bleed ports (Figure 1). Position clamps with nuts forward and up. Do not tighten clamps on anti-ice pneumatic tube until installation in helicopter.
2. **QA** Lubricate packings with petrolatum, Item 216, WP 1803 00. Install packings in grooves on engine output shaft and on forward support tube.

NOTE

Packing is installed on support tube, 70361-10003-043 only.

3. **QA** Carefully slide engine output shaft into front of engine. Place forward support tube with new packing against engine swirl frame so that access window is outboard and shaft can be seen. Bolt forward support tube to swirl frame. **TORQUE BOLTS TO 110 - 125 INCH-POUNDS.**

NOTE

Crotch bolt should be aligned in lower hole of support tube for both No. 1 and No. 2 engines.

4. **QA** Install crotch on forward support tube engaging alignment pins. Bolt crotch to forward support tube. Shim between crotch and support tube with washers, AN960-10L, as required to mate crotch to engine swirl frame. **TORQUE BOLTS TO 45 - 55 INCH-POUNDS.**

NOTE

Center mount links from flange under combustor section are to be installed after engine is removed from adapter, before installing engine in helicopter.

5. **QA** Install inner mount links to engine rear mounting flange, using countersunk washers under bolthead and washers as required under nut. Make sure countersunk side of washer faces bolthead and boltheads are facing aft. **TIGHTEN EACH NUT FINGERTIGHT. ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.
6. **QA** Install outer mount links to engine rear mounting flange, using countersunk washers under bolthead and washers as required under nut. Make sure countersunk side of washer faces bolthead and boltheads are facing aft. **TIGHTEN EACH NUT FINGERTIGHT. ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.
7. **HIRSS** Inspect center body cone as follows: 
 - a. Inspect center body cone for cracks. **NO CRACKS ALLOWED.** Replace center body cone if cracked.
 - b. Inspect center body cone for dents. **NO MORE THAN FOUR DENTS OF 1/4-INCH DEPTH CONTAINING NO SHARP CORNERS ARE ALLOWED.** If possible, cold work any dents beyond limits back to normal contours and within limits.
 - c. Inspect captive nuts for damage. **NO DAMAGE ALLOWED.** Replace center body cone if any captive nuts are damaged.

BUILD UP POWER PACKAGE - Continued

8. **QA HIRSS** On engine power turbine module, coat threads of center body cone attachment studs with aircraft turbine lubricating oil, Item 182, WP 1803 00. Install center body cone, with drain hole on bottom, on engine power turbine module with captive nuts. **TORQUE CAPTIVE NUTS TO 45 - 55 INCH-POUNDS (Figure 1, Detail A).** ■
9. **QA W/O HIRSS** Install center tailpipe body, with drain hole on bottom, on engine power turbine module, using washers and nuts. **TORQUE NUTS TO 43 - 47 INCH-POUNDS (Figure 1, Detail A).** ■
10. **QA W/O HIRSS** Bolt cover plate to center tailpipe body. **TORQUE BOLTS TO 44 - 56 INCH-POUNDS.** ■
11. **QA** Temporarily install deswirl duct by loosely installing loop clamp attaching deswirl duct flange to engine power turbine flange and secure loop clamp as follows (Figure 1, Detail A):

NOTE

Left and right deswirl ducts are identical.

- a. Deswirl ducts have two alignment holes located on deswirl duct flange. One alignment hole shall be located top center while the other alignment hole shall point inboard when installed on the HIRSS module(Figure 6).
- b. Make sure alignment hole in deswirl duct flange mates with alignment hole of engine power turbine flange.
- c. Visually inspect deswirl duct to make sure deswirl duct flange mates with engine power turbine flange through 360 degrees.
- d. If complete mating of flanges is not achieved, remove deswirl duct.
- e. Using a spring scale to measure load, apply no more than 15 pounds of load to remove warpage.
- f. If warpage cannot be removed using 15 pounds of load, replace deswirl duct.
- g. Lubricate pressure surfaces of loop clamp and T-bolt with aircraft turbine lubricating oil, Item 182, WP 1803 00.
- h. Install loop clamp over lips of deswirl duct and engine power turbine flange.

NOTE

Loop clamp is not a rigging tool. Deswirl duct must be supported until clamp is torqued.

- i. Position loop clamp, 70300-30800-102, with T-bolts at 3 o'clock and 9 o'clock positions with nuts facing up.

NOTE

When torquing loop clamp, 70300-30800-102, alternate from nut to nut to maintain even loading on clamp.

- j. **TORQUE LOOP CLAMP NUT TO 35 - 40 INCH-POUNDS.**
 - k. To seat loop clamp, tap lightly with rubber mallet. Retorque loop clamp nuts. Continue tapping until two successive trials show no loss in torque.
 - l. Loosen loop clamp nut slightly. **RETORQUE LOOP CLAMP NUTS TO 35 - 40 INCH-POUNDS.**
 - m. Visually inspect both latches to make sure no interference has occurred.
 - n. Visually inspect inside deswirl duct to make sure deswirl duct flange mates with engine power turbine flange through 360 degrees.
12. Install ECU or DEC and HMU (TM 1-2840-248-23&P).

BUILD UP POWER PACKAGE - Continued

13. **QA** If engine controls support is not already installed, install it on HMU with washers and bolts. **TORQUE BOLTS TO 104 - 116 INCH-POUNDS.**

14. Install engine starter (WP 0694 00 or WP 0695 00).

QUICK ENGINE CHANGE (QEC)**Tear Down QEC**

Remove starter from old engine (WP 0694 00 or WP 0695 00).

Build Up QEC

Install starter in new engine (WP 0694 00 or WP 0695 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

ENGINE LRU'S

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powerplant Repairer's Toolkit, SC 5180-99-B07

References

TM 1-2840-248-23
WP 0479 00
WP 0481 00
WP 0482 00

Material/Parts

Protective Caps and Plugs

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
UH-60 Helicopter Repairer MOS 15T (1)

NOTE

The following procedures apply to both No. 1 and No. 2 engine line replaceable units except as noted.

GENERAL

1. Turn off all electrical power.
2. Open engine cowling.
3. Remove and install engine line replaceable units (TM 1-2840-248-23). For other items such as lines, tubes, hoses, IPS, access panels, and engine control units, use additional instructions following each LRU listed.

REMOVE ACCESSORY SECTION MODULE LRU'S

Particle Separator Blower. After removing pneumatic tube from starter and firewall (No. 2 engine only) and IPS duct (WP 0481 00).

Remove Fuel System LRU's

1. Fuel Boost Pump. After disconnecting fuel hose from elbow, and removing elbow from pump (WP 0482 00).
2. Hydromechanical Control Unit (HMU). After removing engine control power-available and load-demand rotary input assemblies and support from HMU.

Remove Electrical System LRU's

1. History Recorder/History Counter on No. 1 engine firewall, remove access panel. Remove history recorder/counter lower attachment bolts.
2. Thermocouple Assembly. After removing IPS duct (WP 0481 00).

Remove Oil System LRU's

Oil and Scavenge Pump. After removing left/right access panels in firewall. On No. 1 engine, disconnect main fuel line at elbow on fuel pump. Cap line and elbow.

REMOVE ACCESSORY SECTION MODULE LRU'S - Continued**Remove Air System LRU's**

Anti-Icing Bleed and Start Valve on No. 2 engine, remove IPS blower and duct.

INSTALL ACCESSORY SECTION MODULE LRU'S

Particle Separator Blower. Install IPS duct (WP 0481 00) and pneumatic tube to starter and firewall. **QA check**

Install Fuel System LRU's

1. Fuel Boost Pump. Install elbow on pump and connect fuel hose to elbow (WP 0482 00).
2. **QA** Hydromechanical Control Unit (HMU). Install support and engine control power-available and load-demand rotary input assemblies on HMU.
3. Perform an engine load-demand push/pull cable friction check (WP 0479 00).

Install Electrical System LRU's

1. **QA** History Recorder. Install lower attachment bolts through opening in No. 1 engine firewall and install access panel on firewall in oil cooler compartment.
2. Thermocouple Assembly. Install IPS duct (WP 0481 00).

Install Oil System LRU's

Oil and Scavenge Pump. Install left/right access panels in firewall. On No. 1 engine, uncap and connect main fuel line at elbow on fuel pump. **QA check**

Install Air System LRU's

Anti-Icing Bleed and Start Valve on No. 2 engine, install IPS blower and duct. **QA check**

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****ENGINE AIR INLET**

This WP supersedes WP 0484 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Rubber Mallet, 25714
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

References

TM 55-1500-345-23
WP 0381 00
WP 1216 00
WP 1703 00
WP 1803 00

Material/Parts

Petrolatum, Item 216, WP 1803 00
Tags

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

Disassembly of engine air inlet is identical for left and right hand sides except where noted.

1. Turn off all electrical power.
2. If required, remove left hand shroud assembly as follows (Figure 1, Sheet 4, Detail K):
 - a. Remove screws and washers securing shroud assembly to engine air inlet and support assemblies.

NOTE

Tag support assemblies as upper and lower prior to removal from engine air inlet.

- b. Remove screws and washers securing support assembly to engine air inlet.

NOTE

HOIST BL-29900-30 Right side lower shroud not installed on helicopters using rescue hoist. ■

3. If required, remove right hand shroud assemblies by removing screws and washers securing shrouds to engine air inlet (Figure 1, Sheet 4, Detail L).
4. Open engine cowling.
5. Disconnect electrical connector, P823, No. 1 engine or P818, No. 2 engine, from air inlet receptacle (Figure 1, Sheet 1).
6. **UH-60A EH-60A** Loosen clamps and disconnect flexible coupling from air inlet and bleed-air tube. ■

REMOVE - Continued

7. **UH-60L** Loosen coupling on air inlet tube and bleed air tube, and slide compression sleeve and coupling on to bleed air tube (Figure 1, Sheet 1, Detail D). ■
8. Inspect coupling and flexible coupling for cracks, tears, hardening, or deterioration. **NONE ALLOWED.** Replace flexible coupling if required.
9. Unclamp waterwash hose from air inlet (Figure 1, Sheet 1).
10. Disconnect fire extinguisher discharge tube from fitting of firewall (Figure 1, Sheet 1, Detail A).
11. Remove screws and washers from clamps holding discharge tube to air inlet (Figure 1, Sheet 1, Detail B).
12. If deicing kit is installed on right inlet, disconnect ice detector electrical connector, P819, from receptacle on firewall.
13. Loosen both nuts equally. Disconnect coupling holding air inlet to engine swirl frame.
14. Remove bolts, screws, and washers from fairing. Remove fairing.
15. Slide air inlet forward and remove from helicopter.

DISASSEMBLE**NOTE**

Disassembly of engine air inlet is identical for left and right hand sides except where noted.

1. If required, remove left hand shroud assembly as follows (Figure 1, Sheet 4, Detail K):
 - a. Remove screws and washers securing shroud assembly to engine air inlet and support assemblies.

NOTE

Tag support assemblies as upper and lower prior to removal from engine air inlet.

- b. Remove screws and washers securing support assembly to engine air inlet.
2. If required, remove right hand shroud assemblies by removing screws and washers securing shrouds to engine air inlet (Figure 1, Sheet 4, Detail L).
3. Remove screws, washers, and plate from anti-ice valve cover (Figure 1, Sheet 2, Detail E).
4. Remove screws, washers and anti-ice valve cover from engine air inlet.
5. Remove anti-ice valve grommet from anti-ice valve cover as required.
6. If air inlet is to be replaced, remove engine air inlet anti-ice valve (WP 1216 00). Keep valve and attaching hardware with helicopter.

INSPECT/REPAIR

1. Check bleed-air exhaust slots for cracks or sharp corners (Figure 1, Sheet 2, Detail E and Figure 1, Sheet 3, Detail G) as follows:
 - a. Remove paint from ends of slots (WP 0381 00 and TM 55-1500-345-23).

INSPECT/REPAIR - Continued

NOTE

To prevent damaging inner fiberglass wall, do not insert file or drill more than 1/8-inch into slot.

- b. Round off ends of exhaust slots, using a 1/16-inch rotary file or drill bit. Remove only enough material to eliminate sharp corners.
 - c. A group of no more than three adjacent slots may have one crack between a slot and the next adjacent slot. Multiple groups of three slots or less may have cracks provided that there are four slots between each group of slots having cracks. **IF CRACKS ARE BEYOND LIMITS, REPLACE ENGINE AIR INLET.**
 - d. **MORE THAN ONE CRACK FROM A SLOT TO AN ADJACENT SLOT IS NOT ALLOWED. REPLACE AIR INLET.**
 - e. **CRACKS ORIGINATING FROM SEVERAL SLOTS CREATING A BOXED AREA IS NOT ALLOWED. REPLACE AIR INLET.**
 - f. Stop-drill any cracks within limits with a 1/16-inch drill bit.
 - g. Touch up reworked area (WP 0381 00 and TM 55-1500-345-23).
2. Check engine inlet for chafing next to alignment pin receptacle bracket.
- a. File corner of bracket if chafing is present. **MINIMUM THICKNESS BETWEEN EDGE OF BRACKET AND ALIGNMENT HOLE IS 0.20-INCH (Figure 1, Sheet 3, Detail J).**
 - b. Touch up reworked area (WP 0381 00 and TM 55-1500-345-23).

ASSEMBLE

NOTE

Assembly of engine air inlet is identical for left and right hand sides except where noted.

- 1. If removed, install engine air inlet anti-ice valve (WP 1216 00).
- 2. **QA** Install anti-ice valve grommet on anti-ice valve cover, as required (Figure 1, Sheet 2, Detail E).
- 3. **QA** Install anti-ice valve cover on engine air inlet with screws and washers.
- 4. **QA** Install plate on anti-ice valve cover with screws and washers.
- 5. **QA** If removed, secure left hand shroud assembly as follows (Figure 1, Sheet 4, Detail K):

NOTE

Make sure support assemblies are secured to the proper orientation (upper and lower) on the engine air inlet.

- a. Using screws and washers secure support assemblies to engine air inlet. Remove tags.
- b. Using screws and washers secure shroud assembly to engine air inlet and support assemblies.

NOTE

HOIST BL-29900-30 Right side lower shroud not installed on helicopters using rescue hoist. ■

- 6. **QA** If removed, use screws and washers to secure right hand shroud assemblies to engine air inlet (Figure 1, Sheet 4, Detail L).

ASSEMBLE - Continued

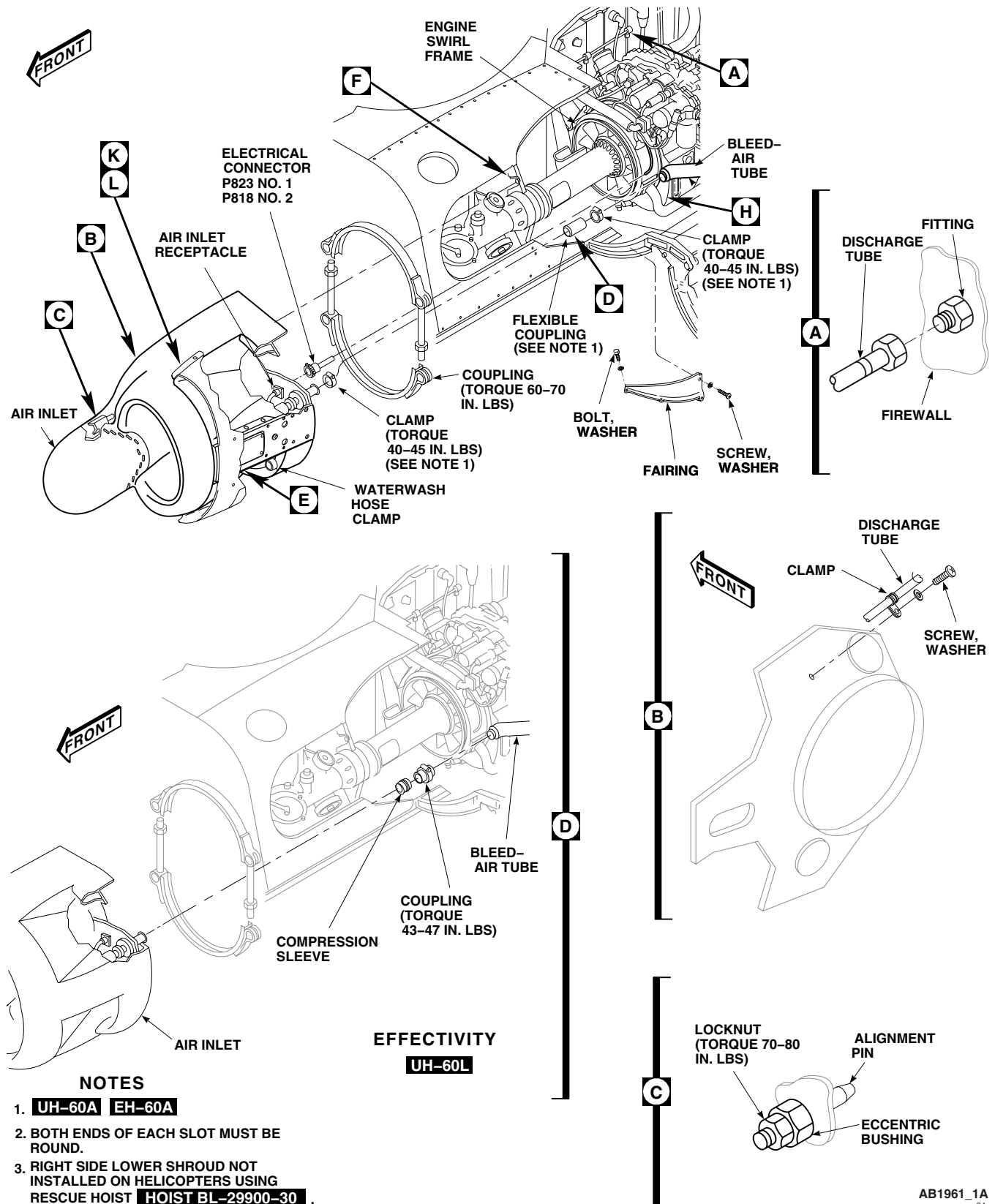


Figure 1. Replace Engine Air Inlet. (Sheet 1 of 4)

ASSEMBLE - Continued

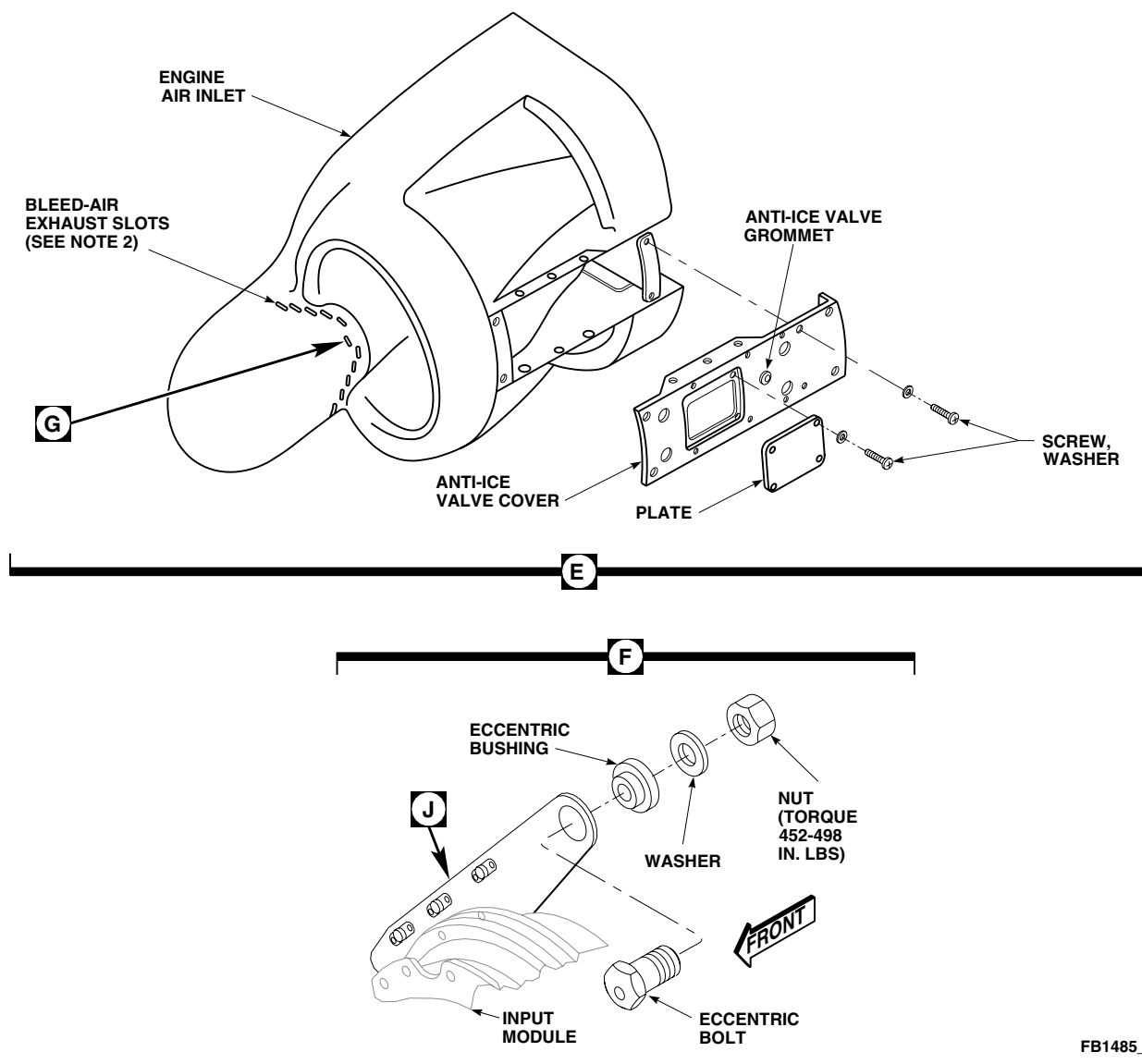
FB1485_2A
SA

Figure 1. Replace Engine Air Inlet. (Sheet 2 of 4)

INSTALL

NOTE

Assembly of engine air inlet is identical for left and right hand sides except where noted.

1. Turn off all electrical power.
2. **QA** Position air inlet on helicopter, making sure alignment pin is inserted in mating receptacle (Figure 1, Sheet 2, Detail F).
3. If required, adjust alignment pin on inlet and adjust alignment receptacle on input module by doing this:

INSTALL - Continued

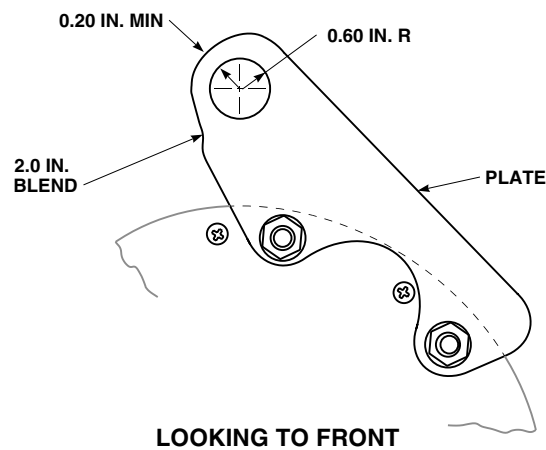
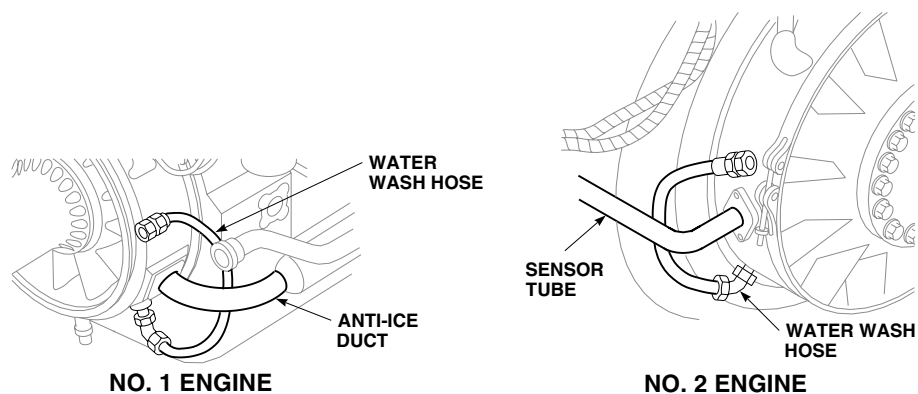
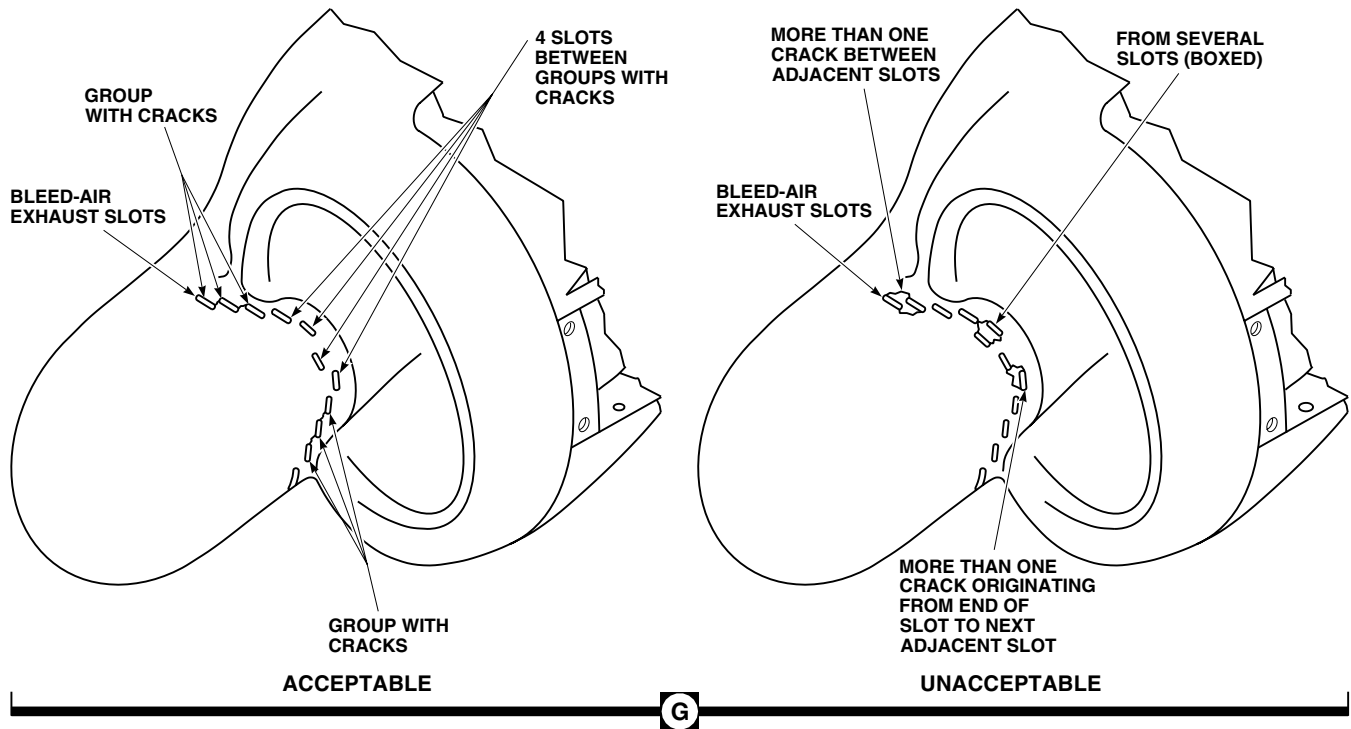
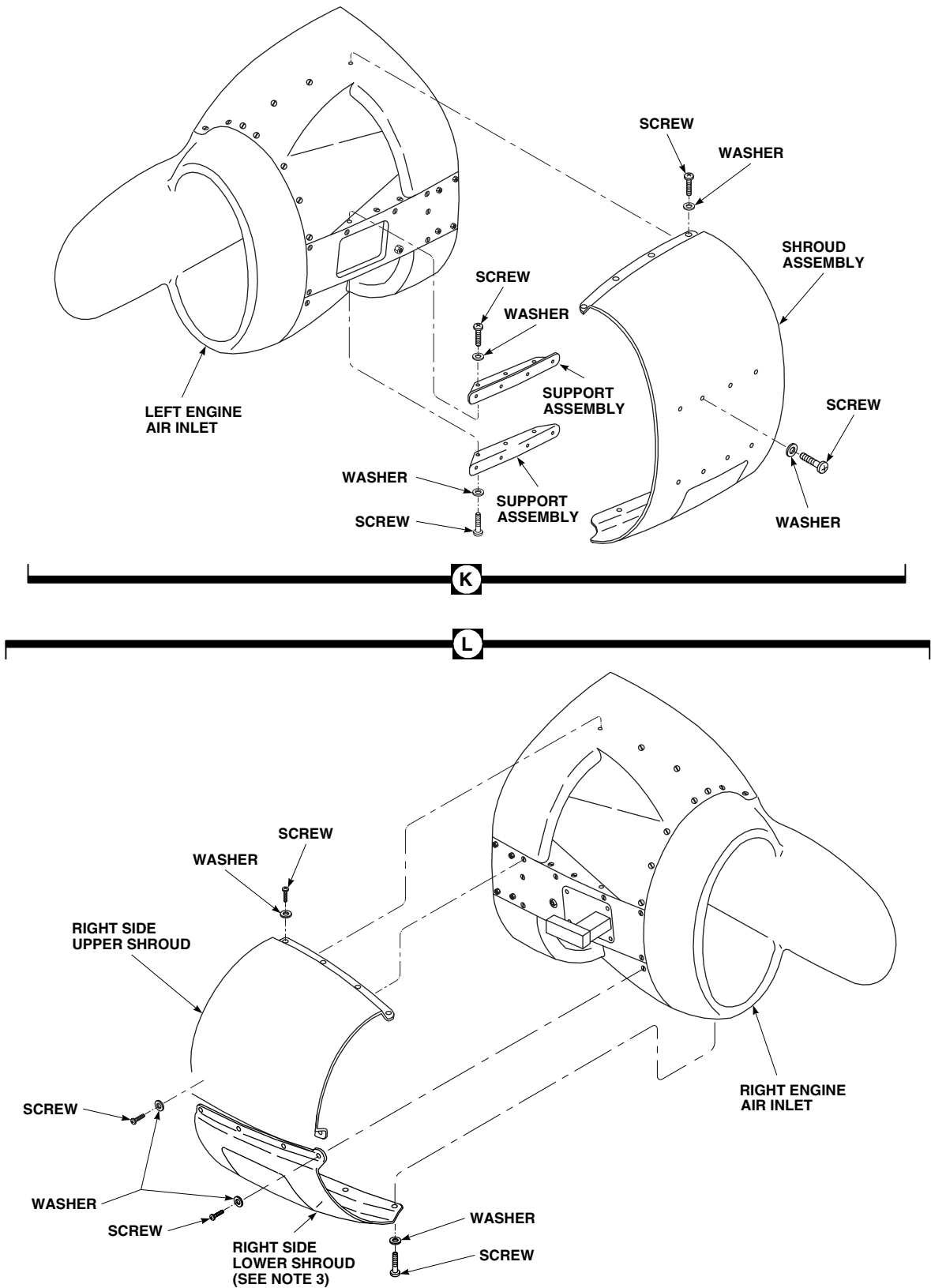


Figure 1. Replace Engine Air Inlet. (Sheet 3 of 4)

FB1485_3
SA

INSTALL - Continued



AB1961_4A
SA

Figure 1. Replace Engine Air Inlet. (Sheet 4 of 4)

INSTALL - Continued

- a. Loosen nut and center alignment pin receptacle by turning eccentric bolt and bushing. **TORQUE NUT TO 452 - 498 INCH-POUNDS (Figure 1, Sheet 2, Detail F).**
- b. Loosen alignment pin locknut (Figure 1, Sheet 1, Detail C).
- c. Slide inlet rearward until it contacts front face of engine swirl frame.
- d. Adjust eccentric bushing on alignment pin until circumference of inlet fairing matches swirl frame.
- e. **TORQUE ALIGNMENT PIN LOCKNUT TO 70 - 80 INCH-POUNDS.**
- f. If additional adjustment is required, remove inlet duct and realign receptacle.

NOTE

Be sure flange on air inlet and swirl frame on engine are clean.

4. Apply a light coat of petrolatum, Item 216, WP 1803 00, to mating surfaces of air inlet and engine swirl frame.
5. **QA** Slide air inlet back against engine swirl frame (Figure 1, Sheet 1).

NOTE

Locate T-bolts at the 3 o'clock and 9 o'clock positions to prevent moisture accumulation and corrosion problems.

6. **QA** Install coupling so that inboard nut of clamp is facing up (outboard nut faces down).
7. Tighten each nut a little at a time. Lightly tap all around clamp with a rubber mallet, while tightening, to correctly seat clamp. **ONCE CLAMP IS CORRECTLY SEATED, TORQUE NUTS TO 60 - 70 INCH-POUNDS.** Continue tapping the clamp until two successive trials show no loss of torque.
8. **QA** Install fairing beneath air inlet with screws and washers, and bolt and washer. Make sure rubber seal/anti-chafe is to the outside of fairing to keep air inlet and inlet fairing from chafing one another.
9. **QA** Connect fire extinguisher discharge tube to fitting on firewall (Figure 1, Sheet 1, Detail A).
10. **QA** Install clamps, screws, and washers attaching discharge tube to air inlet (Figure 1, Sheet 1, Detail B).
11. Inspect and treat electrical connector (WP 1703 00).
12. **QA** If deicing kit is installed on right inlet, connect ice detector electrical connector, P819, to electrical connector, J819, on firewall.

NOTE

Damage to water wash line will result if allowed to chafe against work platform. Position hose to avoid contact with work platform.

13. **QA** Route water wash line hose over anti-ice duct on No. 1 engine and over sensor tube on No. 2 engine (Figure 1, Sheet 3, Detail H).
14. **QA** Connect waterwash hose to air inlet using clamp, bolt, washer, and spacer.
15. **QA** Connect electrical connector, P823, No. 1 engine, or P818, No. 2 engine, to air inlet receptacle (Figure 1, Sheet 1).

INSTALL - Continued**NOTE**

Minimum gap between pneumatic tubes coupling connection is 0.120-inch. If necessary, metal may be removed from tube end using file to achieve this gap. Do not remove any metal from bead on tubes.

16. **QA UH-60A EH-60A** Connect flexible coupling to air inlet and bleed-air tube with clamps. **TORQUE CLAMPS TO 40 - 45 INCH-POUNDS.** ■
17. **QA UH-60L** Install compression sleeve and coupling over air inlet tube and bleed-air tube. **TORQUE NUT TO 43 - 47 INCH-POUNDS.** ■
18. Make sure area is clean and free of foreign material.
19. Close engine cowl and latch.
20. If removed, secure left hand shroud assembly as follows (Figure 1, Sheet 4, Detail K):

NOTE

Make sure support assemblies are secured in the proper orientation (upper and lower) on the engine air inlet.

- a. Using screws and washers secure support assemblies to engine air inlet. Remove tags.
 - b. Using screws and washers secure shroud assembly to engine air inlet and support assemblies.
21. If removed, use screws and washers to secure right hand shroud assemblies to engine air inlet (Figure 1, Sheet 4, Detail L).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM**AFT ENGINE MOUNTS**

This WP supersedes WP 0485 00, dated 15 December 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Airframe Repairer's Toolkit, SC 5180-99-B02
 Dial Indicator, MIL-I-18422
 Fluorescent Penetrant Inspection Kit, XMA101
 Gloves
 Goggles/Face Shield
 Hoist, to Support Weight of Engine
 Lifting Sling, 21C7081G02
 Maintenance Crane, 70700-20300-042
 Nonmetallic Scraper
 Torque Wrench, 30 - 200 in. lbs.

Washer, AN960C516
 Washer, AN960C516L
 Washer, AN960C616
 Washer, AN960C616L
 Washer, AN960C816

Personnel Required

UH-60 Helicopter Repairer MOS 15T (One)
 Aircraft Structural Repairer MOS 15G (One)

References

TM 1-1500-204-23
 TM 1-1520-265-23
 WP 0258 00
 WP 0391 00
 WP 0481 00
 WP 0515 00
 WP 0516 00
 WP 0517 00
 WP 0518 00
 WP 1590 00
 WP 1803 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
 Adhesive, Item 9, WP 1803 00
 Cloth, Abrasive, Item 75, WP 1803 00
 Corrosion Removing Compound, Item 108,
 WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Pad, Scouring, Item 205, WP 1803 00
 Sealing Compound, Item 274, WP 1803 00
 Towel, Machinery Wiping, Item 344, WP 1803 00

NOTE

The following procedures apply to both No. 1 and No. 2 engines.

REMOVE

1. Open engine cowl.
2. Disconnect coupling and remove inlet particle separator (IPS) duct from IPS blower.
3. Install lifting sling, maintenance crane (WP 1590 00), or hoist (WP 0481 00).

CAUTION

Damage to airframe, engine mounts, and engine may occur if shims are removed or disturbed. Do not attempt to remove or otherwise disturb the engine mount to airframe shims.

4. Remove sealing compound from edges of deck mounted fittings using nonmetallic scraper.

REMOVE - Continued

5. Remove bolts, countersunk washers, washers, and nuts holding outer mount strut to engine compartment deck mount fitting and upper links to engine. Remove outer mount from helicopter (Figure 1, Sheet 1 or Figure 1, Sheet 2).
6. Remove bolt, countersunk washer, washer, and nut holding upper links to strut. Remove links from strut.
7. Remove bolts, countersunk washers, washers, and nuts holding inner mount struts to engine compartment deck mount fittings and upper links to engine. Remove inner mount from helicopter.
8. Remove bolts, countersunk washers, washers, and nuts holding upper links to inner mount struts and bolt, washer, and nut holding struts together. Remove links from struts and separate struts.
9. Remove bolt, countersunk washer, washer, and nut holding lower links to support fitting under engine combustor section on engine compartment deck.
10. Remove bolts, countersunk washers, washers, and nuts holding links to engine. Remove links.
11. Remove soundproofing panels (WP 0258 00).
12. Remove bolts, washers, pins, collars, and nuts, holding mount fittings to engine compartment deck and remove mount fittings.

INSPECT/REPAIR

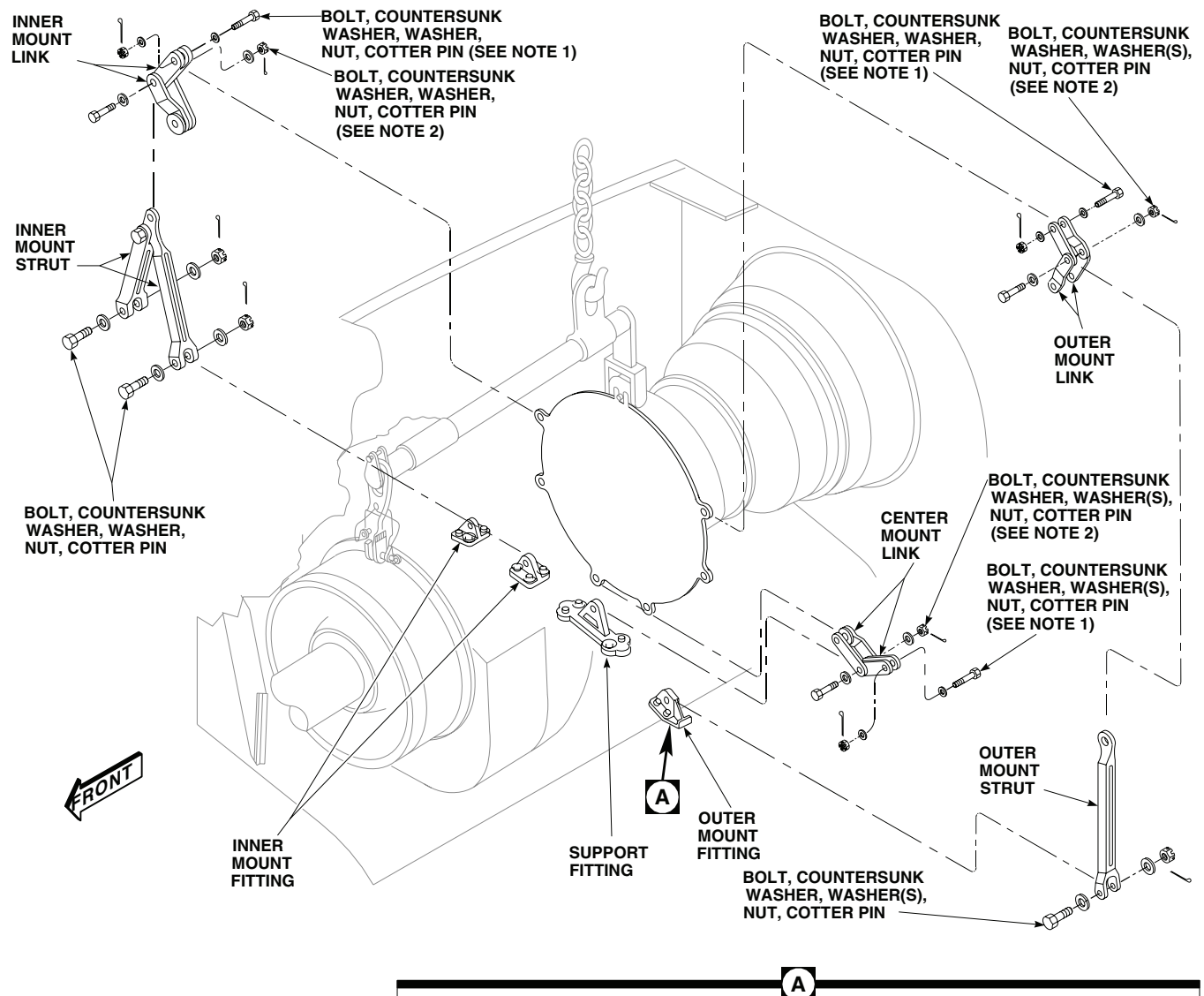
1. Check bearing radial play by doing this:
 - a. Install bolt, nut in bearing.
 - b. Clamp bolt in vise.
 - c. Clamp dial indicator to vise, place tip of indicator on fitting or strut.
2. **IF RADIAL PLAY IS GREATER THAN 0.005-INCH, REPLACE BEARING (WP 0516 00).**
3. Check bearing axial play by doing this:
 - a. Install bolt, nut in bearing.

CAUTION

Damage to strut or fitting will result if clamped in vise unprotected. Use padding to protect fitting or strut when clamping in vise.

- b. Clamp fitting or strut in vise.
 - c. Clamp dial indicator on vise, place tip of indicator on bolt.
4. **IF AXIAL PLAY IS GREATER THAN 0.020-INCH, REPLACE BEARING (WP 0516 00).**
5. Check unbushed hole in inner mount strut for wear. **IF WEAR IS FOUND OR SUSPECTED, DO A DETAILED INSPECTION (WP 0515 00).**
6. Inspect mount struts for corrosion, cracks, nicks, gouges, and scratches. If damage is found, **DETERMINE IF DAMAGE IS IN THE NICKEL PLATING OR THE PARENT MATERIAL** using the following procedure:
 - a. Replace strut if damage due to a nick, gouge, scratch, or pitting is found to be greater than 0.250-inches long or 0.032-inches deep.
 - b. Remove nickel plating in the immediate suspect area using abrasive cloth, Item 75, WP 1803 00. **DO NOT REMOVE PARENT MATERIAL.**

INSPECT/REPAIR - Continued



NOTES

1. DUE TO POSSIBLE DAMAGE TO ENGINE COLD SECTION, BOLTHEADS CONNECTING MOUNT LINKS TO ENGINE MUST FACE AFT.
2. BOLTHEADS CONNECTING MOUNT LINKS TO INNER / OUTER MOUNTS AND / OR MOUNT SUPPORT MAY FACE IN EITHER DIRECTION (FORWARD OR AFT).

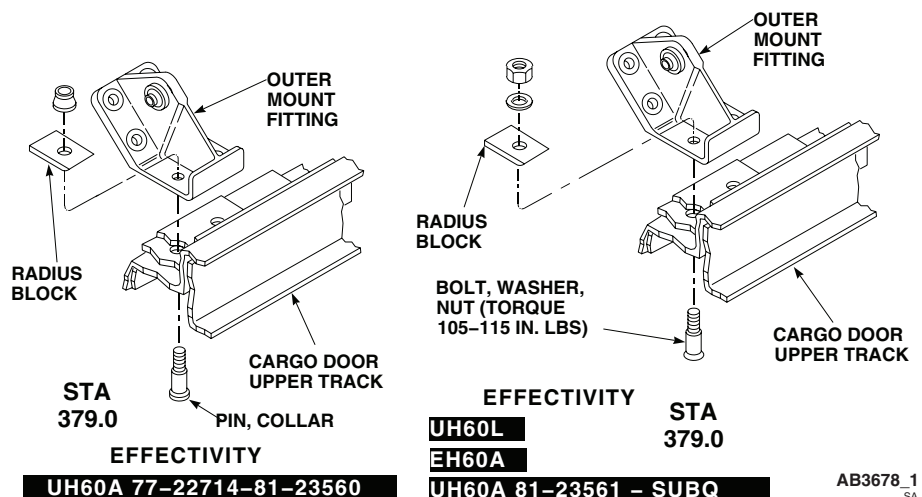


Figure 1. Replace Aft Engine Mounts. (Sheet 1 of 2)

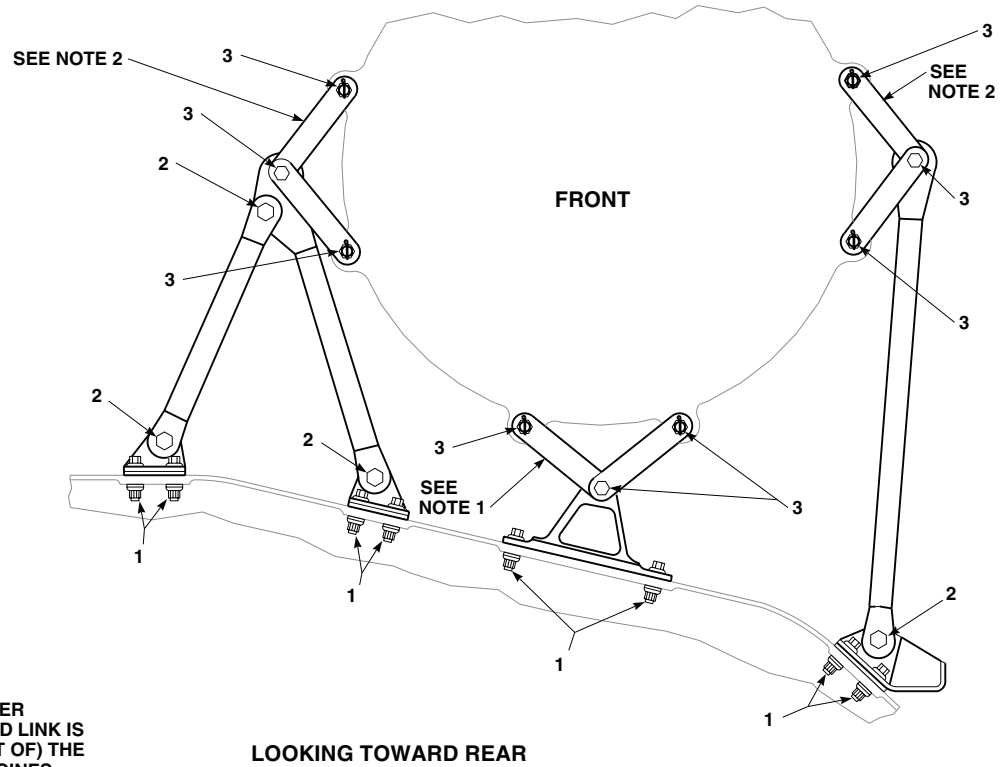
INSPECT/REPAIR - Continued

EFFECTIVITY

77-22714 - 82-23684

NOTES

1. WHEN LOOKING AFT AT CENTER MOUNT LINKS THE RIGHT HAND LINK IS ALWAYS FORWARD (IN FRONT OF) THE LEFT HAND LINK ON BOTH ENGINES.
2. WHEN LOOKING AFT AT THE INNER AND OUTER MOUNT LINKS THE BOTTOM LINK IS ALWAYS FORWARD (IN FRONT OF) THE TOP LINK ON BOTH ENGINES.



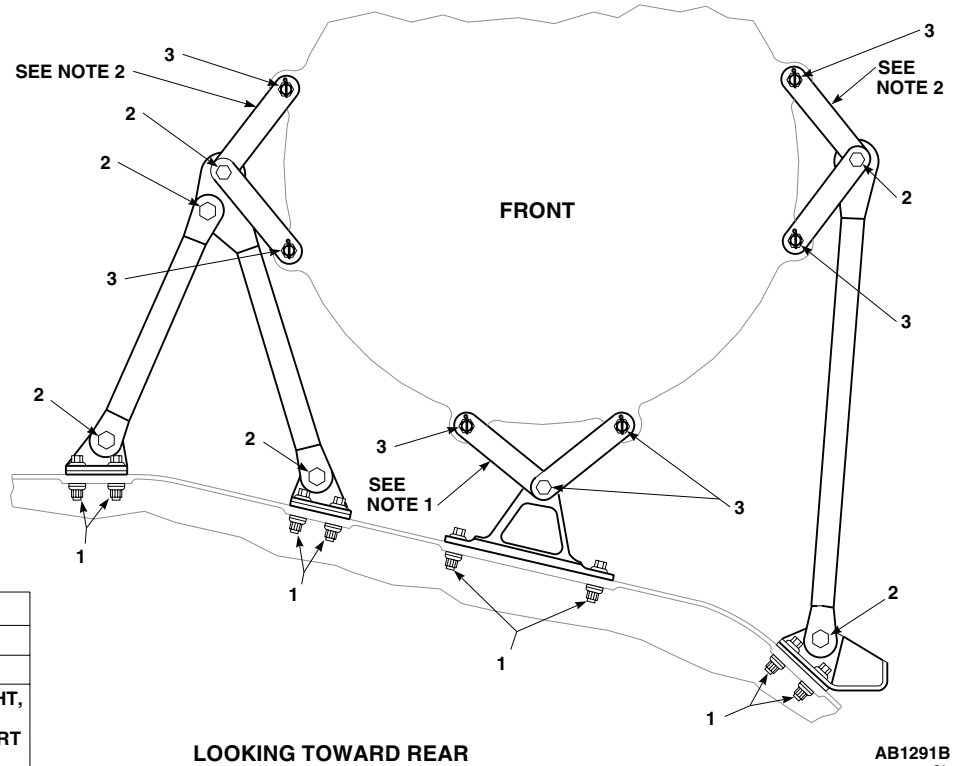
EFFECTIVITY

82-23685 - SUBQ

EH-60A UH-60L

TORQUE TABLE

LOCATION	TORQUE
1	119-131 IN. LBS
2	181-199 IN. LBS
3	TIGHTEN NUT FINGERTIGHT, ADVANCE TO NEXT CASTELLATION AND INSERT COTTER PIN.



AB1291B
SA

Figure 1. Replace Aft Engine Mounts. (Sheet 2 of 2)

INSPECT/REPAIR - Continued

- c. Determine if the damage is in the parent material. Inspect suspect area for cracks using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO DAMAGE ALLOWED IN THE PARENT MATERIAL.** Replace strut if any damage is detected.
- d. Perform a detailed inspection of the remainder of the strut per WP 0515 00.
- e. Clean surfaces where plating was removed and/or is missing with corrosion removing compound, Item 108, WP 1803 00.
- f. Apply two coats of epoxy primer coating kit, Item 120, WP 1803 00, to surfaces where plating was removed and/or is missing.
- g. Apply sealing compound, Item 274, WP 1803 00, to primed area.
- 7. Inspect mount fittings for corrosion, cracks, nicks, gouges, and scratches. **IF DAMAGE IS FOUND OR SUSPECTED, DO A DETAILED INSPECTION (WP 0516 00).**
- 8. Inspect mount links for corrosion, cracks, nicks, gouges, and scratches. **IF DAMAGE IS FOUND OR SUSPECTED, DO A DETAILED INSPECTION (WP 0517 00).**
- 9. Inspect support fitting for corrosion, cracks, nicks, gouges, and scratches. **IF DAMAGE IS FOUND OR SUSPECTED, DO A DETAILED INSPECTION (WP 0518 00).**
- 10. Inspect engine mount bolts for pitting. **NONE ALLOWED.** Replace pitted engine mount bolts.
- 11. Inspect engine mount bolts for corrosion. Remove minor corrosion from engine mount bolt using scouring pad, Item 205, WP 1803 00.
- 12. Measure mount strut to mount fitting mounting bolts and inner mount strut to strut mount bolts at mating points on shank. **MINIMUM SHANK DIAMETER OF BOLTS, NAS464P5-17 AND NAS464P5-12, IS 0.3088-INCH. MINIMUM SHANK DIAMETER OF BOLTS, NAS6705DU14 AND NAS6705DU18, IS 0.3095-INCH.** Replace bolts if shank diameter is less than limits.
- 13. Measure mount strut to mount link mounting bolts at mating points on shank. **MINIMUM SHANK DIAMETER OF BOLTS, NAS464P6-24, IS 0.3713-INCH. MINIMUM SHANK DIAMETER OF BOLTS, NAS6706DU26, IS 0.3720-INCH.** Replace bolts if shank diameter is less than limits.
- 14. Measure mount link to engine mount bolts at mating points on shank. **MINIMUM SHANK DIAMETER OF BOLTS, 74178-8-19, IS 0.4962-INCH.** Replace bolts if shank diameter is less than limits.
- 15. Measure ID of inner and outer mount strut bushing and mount fitting bearing. **MAXIMUM ID WHEN USING BOLTS, NAS464P5-17 AND NAS464P5-12, IS 0.3138-INCH. MAXIMUM ID WHEN USING BOLTS, NAS6705DU14, IS 0.3145-INCH.** If ID is more than limits, do this:
 - a. Subtract mount bolt shank diameter from bushing or bearing ID. **MAXIMUM GAP IS 0.005-INCH.**
 - b. Replace mount strut bushing or mount fitting bearing if gap is not within limits (WP 0515 00 or WP 0516 00).
- 16. Measure ID of inner and outer mount strut bearing and mount link bores which attach to mount struts. **MAXIMUM ID WHEN USING BOLTS, NAS464P6-24, IS 0.3763-INCH. MAXIMUM ID WHEN USING BOLTS, NAS6706DU26, IS 0.3770-INCH.** If ID is more than limits, do this:
 - a. Subtract mount bolt shank diameter from bearing or mount link bore ID. **MAXIMUM GAP IS 0.005-INCH.**
 - b. Replace mount strut bearing if gap is more than limits (WP 0515 00).
 - c. Replace mount link if gap is more than limits.
- 17. Measure ID of mount link bores which attach to engine mounts. **MAXIMUM ID IS 0.5012-INCH.** If ID is more than limits, do this:
 - a. Subtract mount bolt shank diameter from mount link bore ID. **MAXIMUM GAP IS 0.005-INCH.**

INSPECT/REPAIR - Continued

- b. Replace mount link if gap is more than limits.

INSTALL**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

NOTE

Mounting bolt, NAS624H10, may be substituted for mounting bolt, NAS624H8, if a longer bolt is required due to shim thickness.

1. Clean bottom and edges of mount fittings and area on deck where mount fittings are installed using a nonmetallic scraper and a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Do not allow solvent to dry on deck. Keep turning towel after each wipe.
2. Wipe area dry with a clean machinery towel.
3. Inspect mount fitting shims on deck for damage and security. If shims are loose or damaged, do the following:

NOTE

Up to three shims, totaling 0.270-inch, may be installed at any one location. Count the number of shims during removal so same amount may be reinstalled.

- a. Remove rivets or pins and collars securing shim(s) to top deck.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

- b. Clean deck area using a nonmetallic scraper and a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00.
- c. Position shim(s) and install rivets or pins and collars (TM 1-1500-204-23).
4. **QA** Install inner mount fittings and support fitting on deck with bolts, countersunk washers, washers and nuts. Make sure countersunk side of washer faces bolthead. **TORQUE NUTS TO 119 - 131 INCH-POUNDS (Figure 1, Sheet 2).**

NOTE

Outer engine mount fitting (STA 379.0) also supports upper cargo door track. During installation make sure radius block installation is properly oriented (WP 0391 00).

5. **QA** Install outer mount fittings on deck with bolts, washers, and nuts. **TORQUE NUTS TO 119 - 131 INCH-POUNDS (Figure 1, Sheet 2).**

INSTALL - Continued

6. **QA UH-60L EH-60A UH-60A 81-23561 - SUBQ** At STA 379.0 install outer mount fittings on cargo door upper track with bolts, washers, radius block, and nuts. **TORQUE NUTS TO 105 - 115 INCH-POUNDS (Figure 1, Sheet 1).** ■
7. **UH-60A 77-22714 - 81-23560** At STA 379.0 install outer mount fittings on cargo door upper track with pin, collar, and radius block (TM 1-1520-204-23) (Figure 1, Sheet 1). ■
8. Apply adhesive, Item 9, WP 1803 00, to all mating edges of engine mount shims, engine mount fitting, helicopter skin, and all boltheads.

CAUTION

Due to possible damage to engine Cold Section, boltheads in WP 0481 00, Figure 5, connecting mount links to the engine must face aft. Boltheads connecting mount links to inner/outer mount struts and mount support may face in either direction (forward or aft).

NOTE

- Mount links should have open fork ends (widest end) attached to mount support or mount strut.
 - For the inner and outer mount links, the upper mount link should have the offset leg of fork facing aft and lower link should have the offset leg of fork facing forward.
 - When looking aft at the inner and outer mount links the bottom link should be forward (in front) of the upper link on both engines.
 - When looking aft at the center mount link, the right hand link is always forward (in front) of the left hand link on both engines.
 - Forks should be in front of adjoining upper mount link fork with mount or strut between these forks.
 - Shim nut or bolthead with washers, AN960C816, if required, to install cotter pin when connecting center mount links to mounting flange.
9. **QA** Bolt center mount links to mounting flange under combustor on bottom of engine with bolts, countersunk washers, washers and nuts. Make sure countersunk side of washers face bolthead and boltheads are facing aft (Figure 1, Sheet 1). **TIGHTEN EACH NUT FINGERTIGHT, ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.
 10. Complete installation as follows:

NOTE

Shim nut or bolthead with washers, AN960C616 and AN960C616L, if required, to install cotter pin when connecting center mount links to support fitting.

- a. **QA** Bolt center mount links to support fitting under engine combustor section on engine compartment deck with bolt, countersunk washer, washers, and nut (Figure 1, Sheet 1). **TIGHTEN NUT UNTIL FIRMLY SEATED, THEN BACK OFF UNTIL WASHER UNDER NUT CAN BE TURNED BY HAND. RETIGHTEN NUT JUST ENOUGH SO THAT WASHER CANNOT BE TURNED BY HAND, THEN TIGHTEN NUT TO NEXT CASTELLATION. DO NOT TIGHTEN NUT MORE THAN ONE CASTELLATION.** Install cotter pin.

INSTALL - Continued**NOTE**

Shim nut or bolthead with washers, AN960C616 and AN960C616L, if required, to install cotter pin when connecting center mount links to support fitting.

- b. **QA** Bolt inner mount struts together with bolt, countersunk washers, washers and nut. Make sure countersunk side of washer faces bolthead (Figure 1, Sheet 1).
- (1) **UH-60A 77-22714 - 82-23684** **TIGHTEN NUT HANDTIGHT, ADVANCE TO NEXT CASTELLATION.** Install cotter pin. ■
- (2) **UH-60L EH-60A UH-60A 82-23685 - SUBQ** **TORQUE NUT TO 181 - 199 INCH POUNDS.** Install cotter pin. ■

NOTE

Shim nut or bolthead with washers, AN960C616 and AN960C616L, if required, to install cotter pin when connecting center mount links to support fitting.

- c. **QA** **UH-60A 77-22714 - 82-23684** Bolt mount links to inner mount strut with bolt, washers, and nut (Figure 1, Sheet 1). **TIGHTEN NUTS FINGERTIGHT, ADVANCE TO NEXT CASTELLATION.** Install cotter pin. ■
- d. **QA** **UH-60A 82-23685 - SUBQ UH-60L EH-60A** Bolt mount links to inner mount strut with bolt, washers, and nut (Figure 1, Sheet 1). **TORQUE NUT TO 181 - 199 INCH-POUNDS.** Install cotter pin. ■

NOTE

Shim nut or bolthead with washers, AN960C816, if required, to install cotter pin when connecting center mount links to support fitting.

- e. **QA** Bolt inner mount links to engine with bolts, countersunk washers, washers and nuts. Make sure countersunk side of washer faces bolthead and boltheads are facing aft (Figure 1, Sheet 1). **TIGHTEN EACH NUT FINGERTIGHT, ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.

NOTE

Shim nut or bolthead with washers, AN960C616 and AN960C616L, if required, to install cotter pin when connecting center mount links to support fitting.

- f. **QA** Bolt inner mount struts to mount fittings with bolts, countersunk washers, washers and nuts. Make sure countersunk side of washers face bolthead (Figure 1, Sheet 1). **TORQUE NUTS TO 181 - 199 INCH-POUNDS.** Install cotter pins.
- g. **QA** Bolt outer mount links to strut with bolts, washers, and nut (Figure 1, Sheet 1). **TORQUE NUT TO 181 - 199 INCH-POUNDS.** Install cotter pin.

NOTE

Shim nut or bolthead with washers, AN960C816, if required, to install cotter pin when connecting outer mount links to engine.

- h. **QA** Bolt outer mount links to engine with bolts, countersunk washers, washers and nuts. Make sure countersunk side of washers face bolthead and boltheads are facing aft (Figure 1, Sheet 1). **TIGHTEN EACH NUT FINGERTIGHT, ADVANCE NUT TO NEXT CASTELLATION.** Install cotter pin.

INSTALL - Continued**NOTE**

Shim nut or bolthead with washers, AN960C516 or AN960C516L, if required, to install cotter pin when connecting outer mount strut to mount fitting.

- i. **QA** Bolt outer mount strut to mount fitting with bolt, countersunk washers, washers and nut. Make sure countersunk side of washer faces bolthead (Figure 1, Sheet 1). **TORQUE NUT TO 181 - 199 INCH-POUNDS.** Install cotter pin.
11. Install soundproofing panels (WP 0258 00).
12. Remove lifting sling from engine, remove maintenance crane or move hoist out of the way. Remove lifting sling from crane or hoist (WP 1590 00).
13. **QA** If removed, install and connect inlet particle separator (IPS) duct to IPS blower with coupling. **TORQUE COUPLING BOLT TO 60 - 70 INCH-POUNDS.**
14. Make sure area is clean and free of foreign material.
15. Close engine cowlings and latch.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****FORWARD SUPPORT TUBE/ENGINE OUTPUT SHAFT**

This WP supersedes WP 0486 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101
Nonmetallic Scraper
Output Drive Shaft Puller (A), Locally-Made,
Figure 216, WP 1805 00
Output Drive Shaft Puller (B), Locally-Made,
Figure 265, WP 1805 00
Powerplant Repairer's Toolkit, SC 5180-99-B07
Torque Wrench, 30 - 200 in. lbs.
Torque Wrench, 150 - 1000 in. lbs.

Material/Parts

Adhesive, Item 12, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 82, WP 1803 00
Cloth, Abrasive, Item 78, WP 1803 00
Collar, HL87W-6
Corrosion Resistant Coating, Item 110, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Hi-Lok Fastener, HL64PB6-5
Petrolatum, Item 216, WP 1803 00
Rivet, MS90354-0604
Sealing Compound, Item 273, WP 1803 00

Tape, Insulation, Item 315, WP 1803 00
Towel, Machinery Wiping, Item 344, WP 1803 00
Washer, AN960-10L

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1500-344-23
TM 1-1520-237-MTF
TM 1-1520-265-23
TM 1-2840-248-23
TM 1-6625-724-13&P
TM 55-1500-345-23
WP 0381 00
WP 0481 00
WP 0484 00
WP 0519 00
WP 0619 00
WP 1688 00
WP 1803 00
WP 1805 00

NOTE

The following procedures apply to both No. 1 and No. 2 engines.

REMOVE

1. Open engine cowl.
2. Disconnect engine/airframe connections (WP 0481 00).
3. Remove engine air inlet (WP 0484 00).
4. Attach lifting sling to engine (WP 0481 00).
5. Disconnect engine mounts (WP 0481 00).

REMOVE - Continued**CAUTION**

- On helicopters with engine output drive shaft 70361-08004-042 or 70361-08004-043, damage may occur if the three (3) connecting bolts attaching the high speed shaft to the input module flange are not loosened one complete turn prior to removing each of the three (3) connecting bolts.
 - On helicopters with engine output drive shaft 70361-08004-041, damage may occur if the six (6) bolts attaching the high speed shaft to the input module flange are not loosened one complete turn prior to removing each of the six (6) connecting bolts.
 - Damage to drive shaft and input flange/coupling will result on helicopters with drive shaft 70361-08014-041, if drive shaft is not completely separated from input flange/coupling.
6. On helicopters with drive shaft 70361-08014-041, move engine slightly aft to separate drive shaft from input flange/coupling.
 7. Raise engine high enough off engine compartment deck to clear gear box input module.
 8. Remove bolts, washers and crotch from forward support tube. Slide crotch from forward support tube (Figure 2, Sheet 1, Detail A).
 9. Pull output drive shaft from engine. If output drive shaft fails to release from engine, do the following:
 - a. Locally make output drive shaft puller (A) or (B) (WP 1805 00).
 - b. If necessary, loosen knurled nut on output drive shaft puller to release tension on donuts.
 - c. Clean any oil from inside of output drive shaft with a machinery wiping towel, Item 344, WP 1803 00.
 - d. Insert donut end of output drive shaft puller 6 inches (15 cm) into output drive shaft.
 - e. Tighten knurled nut to expand the donuts inside of output drive shaft.
 - f. Move slider against knurled nut and pull back sharply to pull output drive shaft from engine.
 - g. Loosen knurled nut and remove output drive shaft puller from output drive shaft.
 10. Remove packings from engine output shaft. Discard packings.
 11. Remove bolts, washers, and forward support tube from engine swirl frame and remove from engine.
 12. On helicopters with forward support tube, 70361-10003-043, remove packing (Figure 2, Sheet 2, Detail B). Discard packing.
 13. Inspect input pinion for axial play (WP 0619 00).

INSPECT FORWARD SUPPORT TUBE

WARNING

The following procedure may involve asbestos containing part. Improper handling can result in release of asbestos fibers that may cause lung cancer or other related diseases. Do not grind, sand, drill or saw an asbestos containing part without consulting site safety, industrial hygiene and environmental procedures for these operations.

NOTE

Some forward support tubes may have an improper asbestos tape installed on the restrainer rings instead of the proper tape. Asbestos tape can be identified by its white color, cloth-like texture, and 0.125-inch thickness.

1. Check restrainer rings for asbestos tape. If asbestos tape is present, replace with Rulon tape (Refer to, RULON TAPE REPLACEMENT, in this work package) or replace forward support tube. If replacement forward support tube or insulation tape is not available, remove asbestos tape from retainer rings. Forward support tube may remain in service until a replacement tube or insulation tape is available and maintenance is scheduled.
2. Check forward support tube restrainer rings for disbonding of insulation tape. If disbonding is found, do this:
 - a. Using feeler gage, measure depth and extent of separation between restrainer ring and insulation tape. If access for use of a feeler gage is inadequate, a visual inspection is allowed (Figure 2, Sheet 2, Detail C).
 - b. **MAXIMUM ALLOWABLE DISBOND OF INSULATION TAPE IS 20% OF TOTAL BOND SURFACE, BUT NOT MORE THAN 50% IN A 45 DEGREE SEGMENT.**
 - c. **NO DISBONDING ALLOWED AT SLOT IN RESTRAINER RING AND 1/2-INCH ADJACENT TO SLOT.**
 - d. Replace tape or replace forward support tube if insulation tape is disbonded beyond limits. Refer to, RULON TAPE REPLACEMENT, in this work package. If replacement forward support tube or insulation tape is not available, remove tape from retainer rings. Forward support tube may remain in service until a replacement tube or insulation tape is available and maintenance is scheduled.
3. Check forward support tube for nicks, scratches, gouges, or damage. Repair or replace support tube, as required.

INSPECT ENGINE OUTPUT SHAFT

1. On helicopters with engine output drive shaft 70361-08004-042 or 70361-08004-043, inspect engine output shaft as follows:

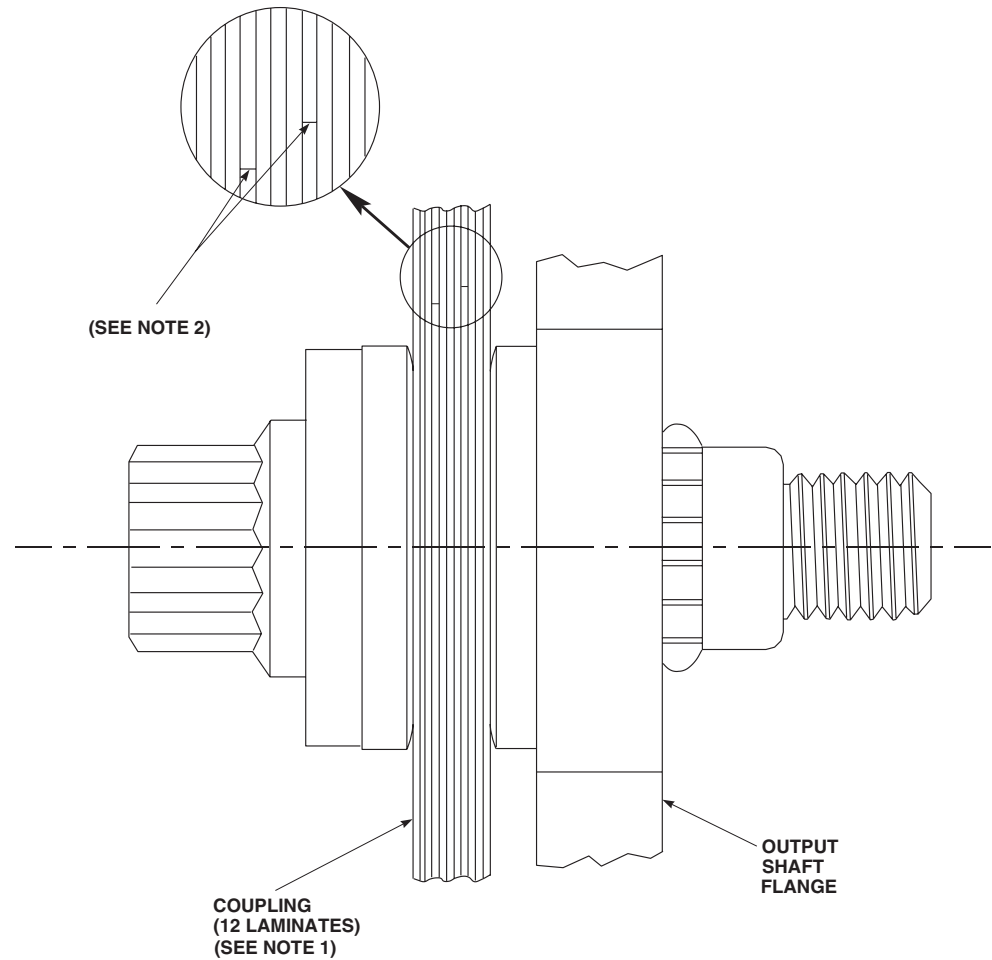
NOTE

- When inspecting engine output shaft and disc pack coupling assembly, one attaching nut may be removed and replaced if corroded or to aid inspection and repair.
 - Discs which make up disc pack coupling are a balanced set and cannot be replaced individually. If any disc is damaged beyond limits, engine output shaft assembly must be replaced.
- a. Check engine output shaft for scratches, nicks, gouges and corrosion (Table 1).
 - b. Check that repaired damage will not be more than limits given in Table 1. Replace engine output shaft and disc pack assembly if repair is more than limit given.
 - c. Check disc pack coupling for damage. If damage is more than limits given in Table 1, replace engine output shaft and disc pack coupling assembly (Figure 1).

INSPECT ENGINE OUTPUT SHAFT - Continued

- d. Inspect disc pack coupling for laminate cracks (Figure 1). **NO CRACKS ALLOWED.** Replace engine output shaft and disc pack coupling assembly if cracks are detected.

INSPECT ENGINE OUTPUT SHAFT - Continued



NOTES

1. COUPLING CRACKS MAY INITIATE IN OUTER LAMINATES, WHICH CAN BE EASILY SEEN ON THE SURFACE, OR INNER LAMINATES WHICH ONLY APPEAR ON THE OUTSIDE DIAMETER OF THE LAMINATE.
2. INNER LAMINATE CRACKS APPEAR AS THIN BLACK LINES ON THE OUTSIDE DIAMETER OF THE LAMINATE.

AB1108A
SA

Figure 1. Coupling Inspection.

INSPECT ENGINE OUTPUT SHAFT - Continued

2. On helicopters with engine output drive shaft, 70361-08014-041, inspect engine output shaft for scratches, nicks, gouges and corrosion (Table 1).
3. On helicopters with engine output drive shaft, 70361-08014-041, check that repaired damage on engine output shaft will not be more than limits given in Table 1. Replace engine output shaft if repair is more than limits given.

INSPECT CROTCH ASSEMBLY

1. Check for loose or missing pins. **NONE ALLOWED (Figure 2, Sheet 2, Detail D).** Replace any loose or missing pins (WP 0519 00).

NOTE

Backing strip is installed only when seal, 70302-10013-137, is used.

2. Check heat shield, firewall plate, backing strip and segment ring for bends and distortion. **NONE ALLOWED (Figure 2, Sheet 2, Detail E).** Replace bent or distorted parts as required (WP 0519 00).
3. Check crotch for bends or distortion. **NONE ALLOWED.** Replace crotch if bent or distorted (WP 0519 00).
4. Check seal for cracks and tears. **NONE ALLOWED.** Replace seal if cracked or torn (WP 0519 00).

NOTE

The crotch should be fluorescent penetrant inspected whenever a repair to the crotch has been performed, when damage is suspected, or when another periodic or special inspection specifically requires fluorescent penetrant inspection of the crotch.

5. Visually inspect crotch. If cracks are detected or suspected, fluorescent penetrant inspect, Type 1, Method C, crotch and segment ring for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace crotch or segment ring if cracked.
6. Inspect crotch assembly as follows:
 - a. Inspect crotch assembly for wear or damage (Figure 2, Sheet 3, Detail F).

CAUTION

Anti-ice or engine performance may be adversely affected if the entire surface is machined. It is not allowed to machine the entire surface.

- b. Scratches, gouges, wear grooves or rub marks shall be blended smooth, with as little flange removed as possible, using abrasive cloth, Item 82, WP 1803 00.
 - c. After repair perform local fluorescent penetrant inspection (TM 1-1520-265-23). **NO CRACKS ALLOWED.**
 - d. Brush corrosion resistant coating, Item 110, WP 1803 00, on repaired area and allow to dry.
 - e. Apply epoxy primer coating kit, Item 120, WP 1803 00, on repaired area.
7. Check crotch assembly for loose or missing rivets. **NONE ALLOWED.** Replace loose or missing rivets (WP 0519 00).

INSPECT CROTCH ASSEMBLY - Continued**NOTE**

Make sure repair of nick, scratches and corrosion will not cause part to be beyond limits.

8. Inspect heat shield, firewall plate, and backing strip for nicks and scratches. **DAMAGE SHALL NOT BE MORE THAN 0.010-INCH DEEP OR MORE THAN 25% OF TOTAL AREA OF PART.** Blend out nicks and scratches using abrasive cloth, Item 82, WP 1803 00. If damage is greater than limits, replace damaged part(s) (WP 0519 00).
9. Inspect firewall plate for corrosion. **CORROSION SHALL NOT BE MORE THAN 0.005-INCH DEEP OR MORE THAN 25% OF TOTAL AREA OF PART.** Blend out corrosion using abrasive cloth, Item 82, WP 1803 00. If corrosion is greater than limits, replace firewall plate (WP 0519 00).

REPAIR ENGINE OUTPUT SHAFT**WARNING****CSI**

Verification of 130 inch-pounds minimum torque and verification of balance repeatability within 0.80 gram - inch in any position are critical characteristics.

CAUTION

On helicopters with engine output drive shaft 70361-08004-042 or 70361-08004-043, excessive vibration will result if two or more nuts connecting engine output shaft to disc pack coupling are loosened or removed at any time. Engine output shaft, disc pack coupling and connecting hardware is a balanced assembly. Only one nut connecting engine output shaft to disc pack coupling may be loosened or removed at any one time. Loose or removed nuts must be torqued to their proper value before another nut is loosened or removed.

1. On helicopters with engine output drive shaft 70361-08004-042 or 70361-08004-043, if required, one nut at a time may be removed to facilitate repairs to the engine shaft flange.
2. Any scratches, through cadmium plating only, may be blended smooth using abrasive cloth, Item 82, WP 1803 00.
3. Blend damaged area that is through cadmium plating into base metal, removing least amount of material, using abrasive cloth, Item 78, WP 1803 00.
4. Polish blended area to remove abrasive marks and match surface smoothness using abrasive cloth, Item 82, WP 1803 00.
5. Clean reworked area with cleaning compound solvent, Item 71, WP 1803 00.
6. Paint the reworked area with epoxy primer coating kit, Item 120, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).

WARNING**CSI**

Verification of 130 inch-pounds minimum torque is a critical characteristic.

7. If removed, install nut on engine output shaft flange. **TORQUE NUT TO 130 - 140 INCH-POUNDS.**

REPAIR FORWARD SUPPORT TUBE

1. Repair support tube with scratches, dents, and gouges less than 0.015-inch deep x 0.030-inch wide x 0.500-inch long.
2. Replace tube if damage is more than limits given.

NOTE

Make sure polished-out damage will not be greater than allowable limits.

3. Blend damaged area to at least 0.500-inch radius, removing least amount of material, using abrasive cloth, Item 78, WP 1803 00.
4. Polish blended area to remove abrasive marks and match surface smoothness using abrasive cloth, Item 82, WP 1803 00.
5. Clean reworked area with solvent cleaning compound, Item 71, WP 1803 00.
6. Brush corrosion resistant coating, Item 110, WP 1803 00, on reworked area (TM 1-1500-344-23).
7. Touch up paint on reworked area with epoxy primer coating kit, Item 120, WP 1803 00 (WP 0381 00 and TM 55-1500-345-23).

NOTE

Make sure polished-out damage will not be greater than allowable limits.

8. Check forward support tube for loose or broken rivets holding restrainer rings. **NONE ALLOWED.** Replace loose or broken rivets as follows:

CAUTION

Damage to forward support tube will result if rivets are not replaced one at a time. Removing all rivets will cause loss of alignment. Replace rivets one at a time. If restrainer ring requires replacement see RULON TAPE REPLACEMENT

- a. Remove loose or broken rivet (TM 1-1500-204-23).
- b. Remove sealant from area using nonmetallic scraper.
- c. Measure hole diameter.
- d. **INSTALL RIVET, MS90354-0604, IF HOLE DIAMETER IS LESS THAN 0.202-INCH AS FOLLOWS:**
 - (1) Fluorescent penetrant inspect hole for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace forward support tube if cracks are found.
 - (2) Apply corrosion resistant coating, Item 110, WP 1803 00, to hole (TM 1-1500-344-23).
 - (3) Coat rivet with epoxy primer coating kit, Item 120, WP 1803 00. Install rivet while primer is still wet (TM 1-1500-204-23).

NOTE

A maximum gap of 0.020-inch is allowed between restrainer ring and support tube after a rivet or Hi-Lok has been installed.

- e. **INSTALL HI-LOK FASTENER, HL64PB6-5, AND COLLAR, HL87W-6, IF HOLE DIAMETER IS MORE THAN 0.202-INCH AS FOLLOWS:**
 - (1) Redrill hole using 13/64-inch drill bit. Deburr hole.

REPAIR FORWARD SUPPORT TUBE - Continued

- (2) Fluorescent penetrant inspect hole for cracks using fluorescent penetrant inspection kit (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace forward support tube if cracks are found.
- (3) Apply corrosion resistant coating, Item 110, WP 1803 00, to hole (TM 1-1500-344-23).
- (4) Coat Hi-Lok fastener with epoxy primer coating kit, Item 120, WP 1803 00. Install fastener and collar while primer is still is still wet (TM 1-1500-204-23).

NOTE

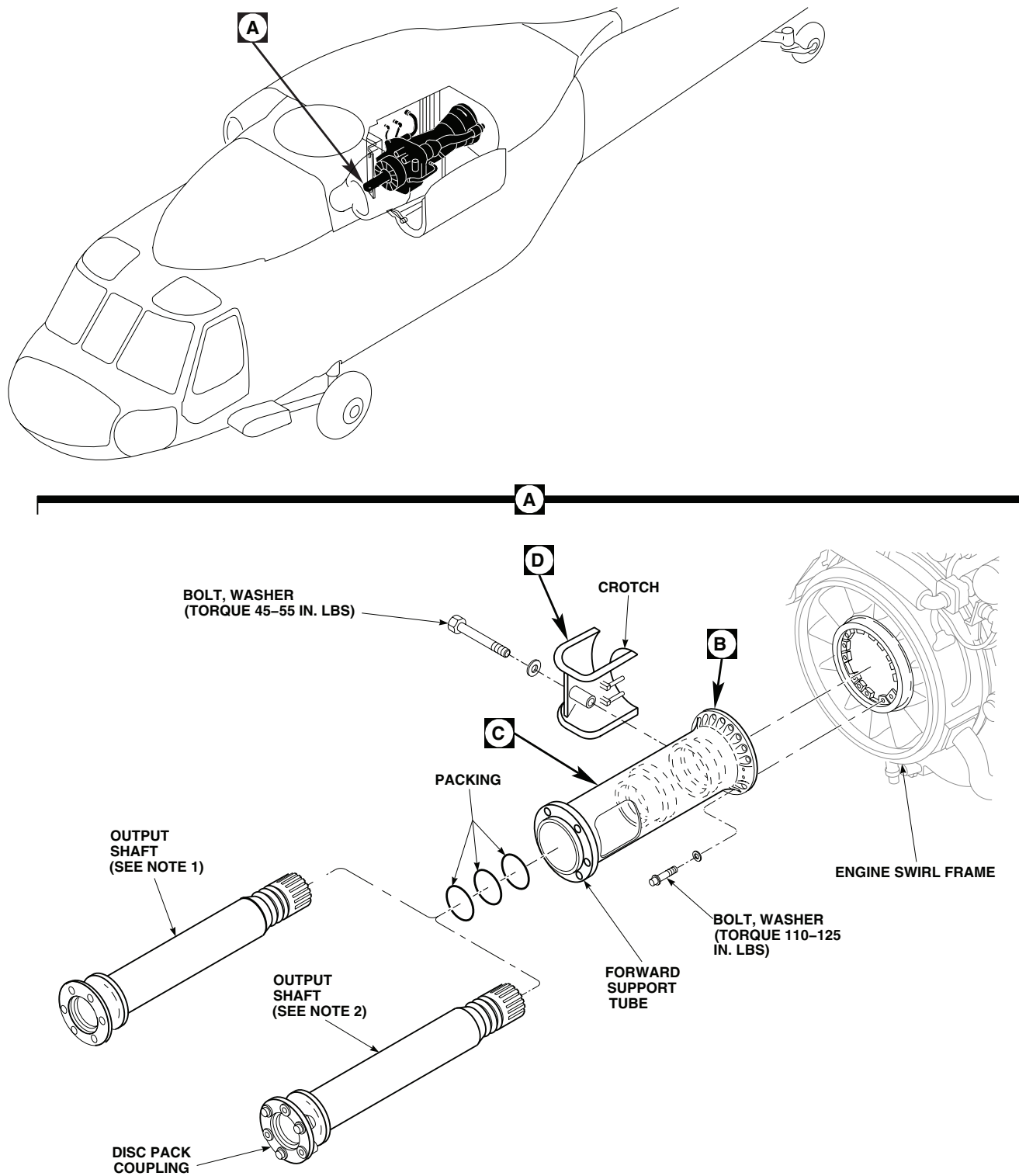
A maximum gap of 0.020-inch is allowed between restrainer ring and support tube after a rivet or Hi-Lok has been installed.

- f. Touch-up finish as required (WP 0381 00 and TM 55-1500-345-23).
- g. Apply sealing compound, Item 273, WP 1803 00, around head of rivet or Hi-Lok fastener.

REPAIR FORWARD SUPPORT TUBE FLANGE

1. Repair support tube flange with scratches, dents or gouges less than 0.015-inch deep and 0.030-inch wide. There cannot be more than 4 damaged areas in a 2 inch arc on flange.
2. Replace support tube if damage is greater than limits given.

REPAIR FORWARD SUPPORT TUBE FLANGE - Continued



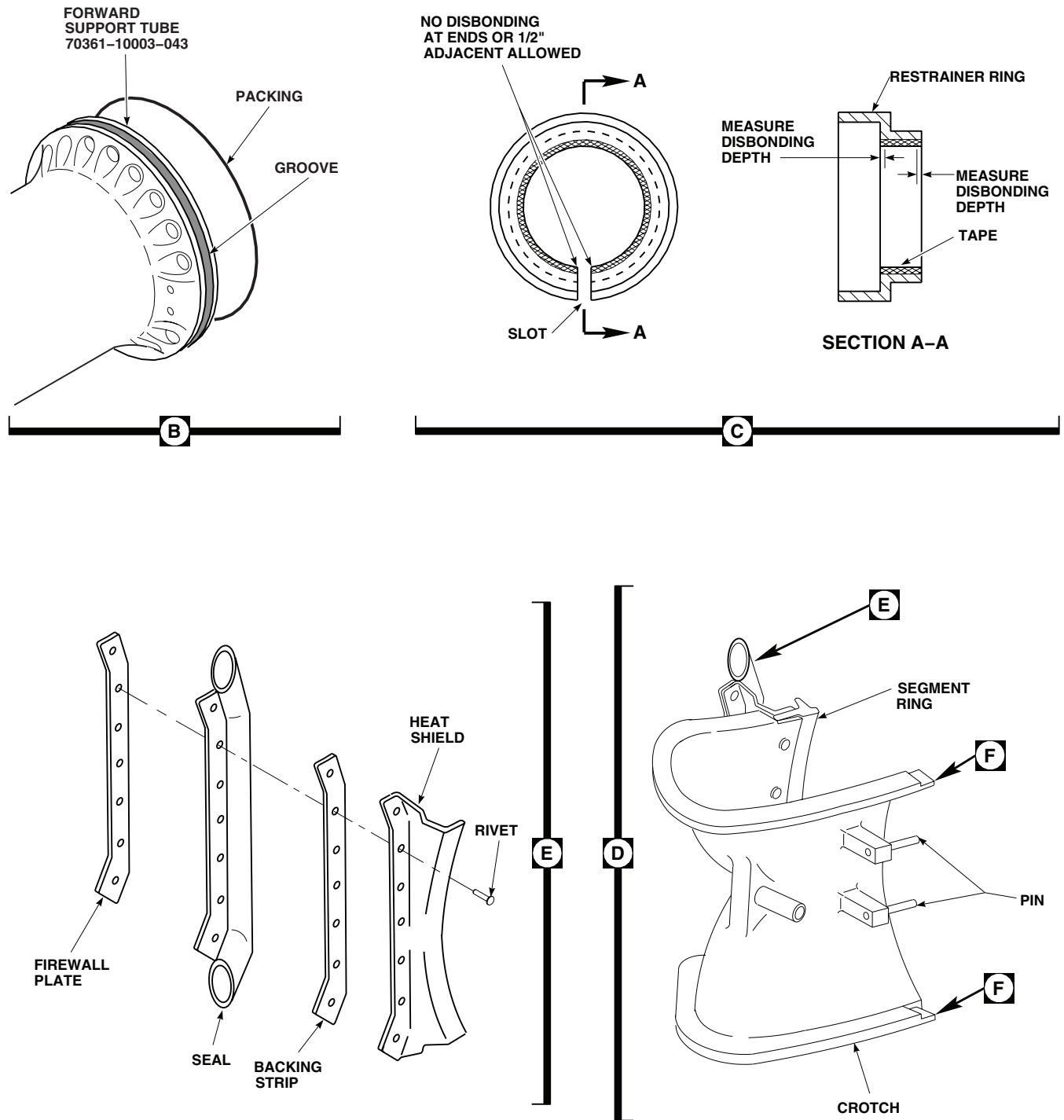
NOTES

1. 70361-08014-041.
2. 70361-08004-042, OR
70361-08004-043.

AB3351_1B
SA

Figure 2. Replace Forward Support Tube/Engine Output Shaft. (Sheet 1 of 3)

REPAIR FORWARD SUPPORT TUBE FLANGE - Continued



AB3351_2
SA

Figure 2. Replace Forward Support Tube/Engine Output Shaft. (Sheet 2 of 3)

REPAIR FORWARD SUPPORT TUBE FLANGE - Continued

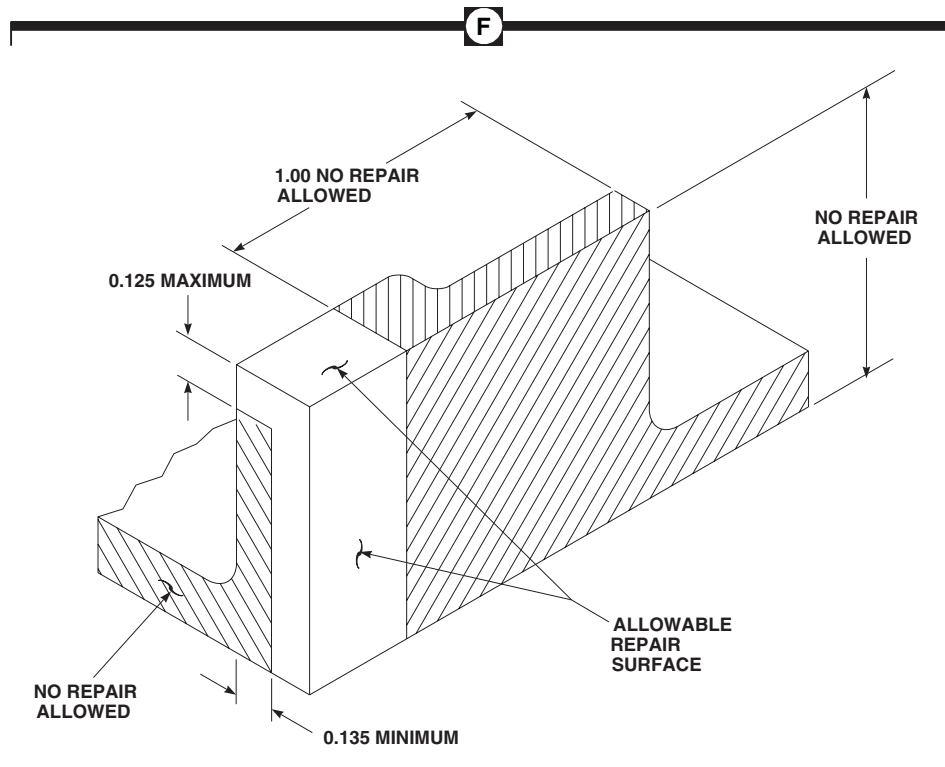


Figure 2. Replace Forward Support Tube/Engine Output Shaft. (Sheet 3 of 3)

NOTE

Make sure polished-out damage will not be greater than allowable limits.

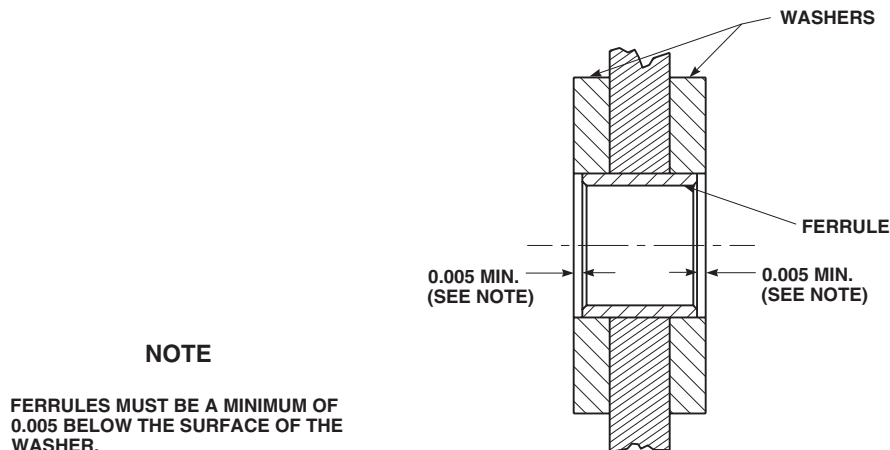
3. Blend damaged area to at least 0.060-inch radius, removing least amount of material, using abrasive cloth, Item 78, WP 1803 00.
4. Polish blended area to remove abrasive marks and match surface smoothness using abrasive cloth, Item 82, WP 1803 00.
5. Clean reworked area with solvent cleaning compound, Item 71, WP 1803 00.
6. Brush corrosion resistant coating, Item 110, WP 1803 00, on reworked area (TM 1-1500-344-23).

INSTALL**NOTE**

Packing is installed on support tube, 70361-10003-043, only. **QA Check**

1. On helicopters with support tube, 70361-10003-043, lubricate packing with petrolatum, Item 216, WP 1803 00, and install in groove on forward support tube (Figure 2, Sheet 2, Detail B).
2. **QA** Place forward support tube, with new packing, against engine swirl frame so that cutout is outboard and shaft can be seen. Bolt forward support tube to engine swirl frame. **TORQUE BOLTS TO 110 - 125 INCH-POUNDS (Figure 2, Sheet 1, Detail A).**
3. Lubricate packings with petrolatum, Item 216, WP 1803 00.

INSTALL - Continued

**NOTE**

FERRULES MUST BE A MINIMUM OF 0.005 BELOW THE SURFACE OF THE WASHER.

AB1711
SA

Figure 3. Disc Pack Assembly Coupling for Engine Driveshaft (Washers and Ferrules).

4. **QA** Install packings in grooves on output shaft.
5. **QA** Carefully slide shaft into forward support tube and engine.

NOTE

If crotch is new or has been repaired at another facility or maintenance level, perform inspection of crotch assembly. Refer to, INSPECT CROTCH ASSEMBLY, in this work package before installation.

6. **QA** Install crotch on forward support tube engaging alignment pins. Bolt crotch to forward support tube. Shim between crotch and support tube with washers, AN960-10L, as required to mate crotch to engine swirl frame.
TORQUE BOLT TO 45 - 55 INCH-POUNDS.
7. Do torque check on input module flange nut. **TORQUE CHECK NUT IN TIGHTENING DIRECTION ONLY TO 646 - 714 INCH-POUNDS.**
8. Inspect input pinion axial play and elastomeric gimbal (WP 0619 00).
9. Inspect work area to make sure it is clean and free of all tools and foreign matter.
10. Lower engine into engine compartment. Connect engine mounts (WP 0481 00).
11. Connect engine/airframe connections (WP 0481 00).
12. Do engine output shaft vibration level check (TM 1-6625-724-13&P).
13. Install engine air inlet (WP 0484 00).
14. Make sure area is clean and free of foreign material.
15. Close engine cowl and latch.
16. Do an engine checkout procedure (TM 1-2840-248-23) as part of maintenance test flight (TM 1-1520-237-MTF).

INSTALL - Continued**Table 1. Inspect Engine Output Shaft.**

DAMAGE	DEPTH	WIDTH	LENGTH	NUMBER OF DISCS
Scratches on shaft	0.005-inch	0.005-inch		
Nicks, gouges on shaft	0.010-inch	0.010-inch		
Nicks, gouges on flange	0.015-inch		0.125-inch	
Corrosion	0.001-inch			
The following inspections apply to drive shaft 70361-08004-042 and 70361-08004-043 only.				
Disc Pack Coupling dents and nicks on outer edge of disc	0.020-inch			no more than half
Disc Pack Coupling lamination spread		0.030-inch		no limit
Disc Pack Coupling dents and scratches	0.005-inch			
Disc Pack Coupling cracks	none allowed (See Figure 1)			
Inspect to insure ferrules are seated below the surface of the washers	(See Figure 3)			
Disc pack coupling ferrule looseness (none allowed)				

INSTALL - Continued**Table 1. Inspect Engine Output Shaft. - Continued**

DAMAGE	DEPTH	WIDTH	LENGTH	NUMBER OF DISCS
Corrosion and fretting on surface of disc pack coupling washers	0.0001 inch all over			

17. On helicopters with engine output drive shaft 70361-08004-042 or 70361-08004-043, maintain records by checking DA Form 2408-13-1 to make sure torque check of three bolts connecting disc pack coupling to input module flange is due after initial ground run and after 1 to 3 flight hours (WP 1688 00). Also, check DA FORM 2408-13-1 to make sure torque check of three bolts connecting disc pack coupling to engine output shaft is due after initial ground run and after 1 to 3 flight hours (WP 1688 00) if they were found to be loose during torque check.
18. On helicopters with engine output drive shaft 70361-08014-041, maintain records by checking DA Form 2408-13-1 to make sure torque check of six bolts connecting engine output drive shaft to input module flange/coupling is due after 3 to 13 flight hours and meets minimum torque of 130 inch-pounds (WP 1688 00).
19. Do a maintenance test flight (TM 1-1520-237-MTF).

RULON TAPE REPLACEMENT**WARNING**

The following procedure may involve an asbestos containing part. Improper handling can result in the release of asbestos fibers that may cause lung cancer or other related diseases. Do not grind, sand, drill, or saw an asbestos containing part without consulting site safety, industrial hygiene, and environmental procedures for these operations.

NOTE

Some forward support tubes may have an improper asbestos tape installed on the retainer rings instead of the proper tape. Asbestos tape can be identified by its white color, cloth-like texture, and 0.125-inch thickness.

1. Remove one restrainer ring at a time. Refer to, REPAIR FORWARD SUPPORT TUBE, in this work package. Make a note of slot position for reinstallation. When removing rivets start with one adjacent to slot and work around to other side of slot.
2. Remove tape and adhesive. Clean surface for bonding.
3. Cut insulation tape, Item 315, WP 1803 00, to 9.02-inches.
4. Apply adhesive, Item 12, WP 1803 00, to restrainer, and position tape.
5. Apply pressure to hold tape into position (1 to 15 PSI).
6. Allow to cure at 190° F (88° C) to 210° F (99° C) for one hour.
7. Inspect. Refer to, INSPECT FORWARD SUPPORT TUBE, in this work package.

RULON TAPE REPLACEMENT - Continued

8. Inspect holes in support tube and reinstall retractor per REPAIR FORWARD SUPPORT TUBE, in this work package. Make sure to position slot in same location, when installing rivets. When installing rivets start with one adjacent to slot and work around to other side of slot.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****ENGINE EXHAUST MODULE**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

References

WP 0477 00
WP 0481 00
WP 0521 00

Personnel Required

Assistant (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

The following procedure applies to both left and right engine exhaust modules.

1. Turn off all electrical power.
2. Remove screws, washers, and cover from upper rear cargo door track (Figure 1, Detail A).
3. Open engine cowl.
4. Remove screws, washers, and hinge cover from engine cowl rear hinge.
5. Remove starter air tube on right side engine (WP 0481 00).
6. On right side Inlet Particle Separator (IPS) duct only, remove screw, washer and spacer holding engine control cables to duct (Figure 1, Detail B).
7. Remove clamp securing IPS blower duct to IPS blower. Remove duct.
8. Remove screws and washers that secure exhaust module to helicopter (Figure 1, Detail A). Remove module.
9. If IPS flange is to be replaced, remove bolts and washers that secure IPS flange to exhaust module. Remove flange.
10. Inspect and repair exhaust module component (WP 0477 00).

INSTALL

1. Turn off all electrical power.
2. **QA** If IPS flange is being replaced, install IPS blower duct flange on exhaust module with bolts and washers (Figure 1, Detail B).
3. **QA** Install replacement exhaust module on helicopter with screws and washers (Figure 1, Detail A).
4. **QA** Install IPS blower duct on engine with clamp. **TORQUE CLAMP TO 60 - 70 INCH-POUNDS.** To seat clamp, tap it lightly with a plastic hammer. Retorque clamp nut. Continue tapping clamp until two successive trials show no loss in torque.
5. **QA** On right side only, clamp engine control cables to top of IPS duct with screw, washer, and spacer (Figure 1, Detail B).
6. Install starter air tube on right side engine (WP 0481 00).

INSTALL - Continued

7. **QA** If installing new IPS duct, do this:
 - a. **QA** Trial-fit IPS duct in flange.
 - b. If duct will not fit in flange, or is distorted by flange, cut slits as wide as necessary in duct.
 - c. Draw slits together, so that end of duct will slip fit in flange.
 - d. Repair slit end.
8. Rework duct using procedures in WP 0521 00.
9. **QA** Install hinge cover over rear hinge of engine cowling with screws and washers (Figure 1, Detail A).
10. **QA** Install cover over upper rear cargo door track with screws and washers.
11. Make sure area is clean and free of foreign material.
12. Close engine cowling and latch.

INSTALL - Continued

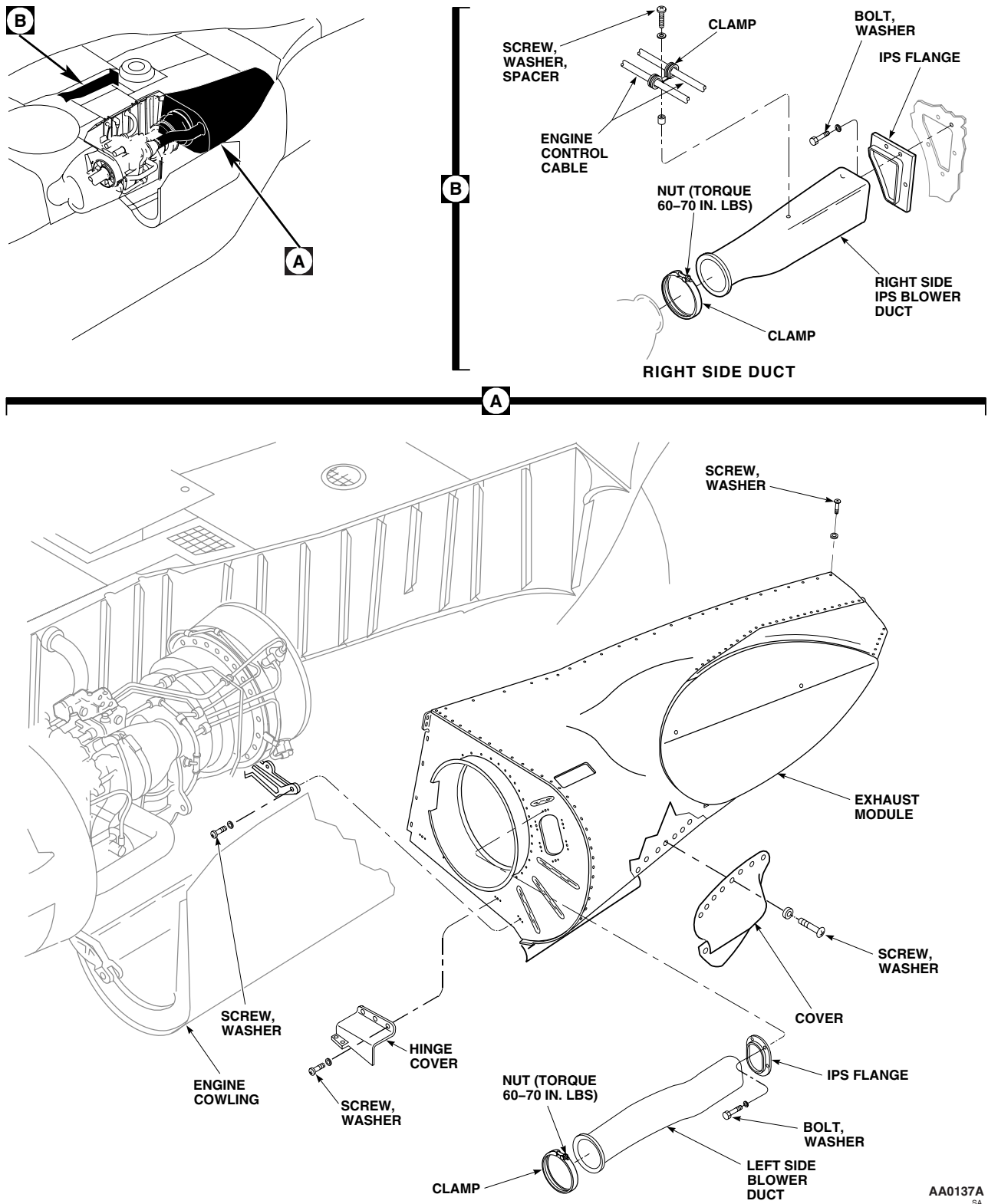


Figure 1. Replace Engine Exhaust Module.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE EXHAUST MODULE EJECTOR

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Personnel Required

Assistant - (1)

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0477 00

WP 0487 00

WP 0520 00

REMOVE

1. Turn off all electrical power.
2. Open engine cowl.
3. Remove engine exhaust module (WP 0487 00).
4. Remove screws, washers, and nuts securing top edge of exhaust ejector to exhaust module (Figure 1).
5. Remove screws and washers securing lower edge of exhaust ejector to lower edge of exhaust module.
6. Remove screw and washer securing exhaust ejector angle to exhaust module bracket on inside of module.
7. Remove exhaust ejector.
8. Inspect and repair exhaust ejector (WP 0477 00 and WP 0520 00).

INSTALL

1. Place replacement exhaust ejector in exhaust module and slide ejector over braided seal on exhaust module firewall (Figure 1).
2. **QA** Install screw and washer securing exhaust ejector angle to exhaust module bracket on inside of module.
3. **QA** Install screws and washers securing lower edge of exhaust ejector to exhaust module.
4. **QA** Install screws, washers, and nuts securing top edge of exhaust ejector to exhaust module.
5. Install engine exhaust module (WP 0487 00).
6. Make sure area is clean and free of foreign material.
7. Close engine cowl and latch.

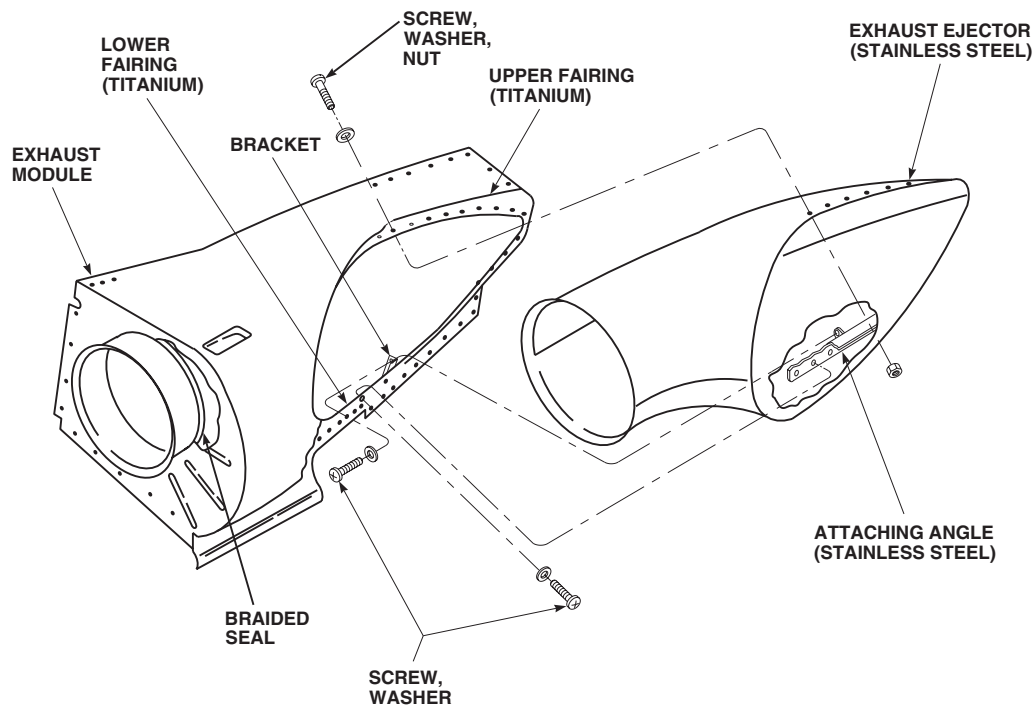
INSTALL - ContinuedAA0138A
SA

Figure 1. Replace Engine Exhaust Module Ejector.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****HIRSS**

This WP supersedes WP 0489 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Adapter, 70700-20424-041
Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101
Hoist Adapters, 70700-20468-041
Hoisting Sling, 70700-20320-042
Maintenance Crane, 70700-20300-042
Rubber Mallet, 25714
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Sealing Compound, Item 273, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)

UH-60 Helicopter Repairer MOS 15T (1)

References

DD Form 365C
FM 38-700
TM 1-1520-237-10
TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 0481 00
WP 0482 00
WP 0487 00
WP 0490 00
WP 0491 00
WP 0495 00
WP 1590 00
WP 1803 00

NOTE

The following procedure applies to both right and left sides except where noted.
Identify removed items with attaching hardware, and keep together for reinstallation.

REMOVE

1. Position main rotor head to allow overhead clearance.
2. Open engine cowling.
3. Position maintenance crane, hoist, or lifting device to lift HIRSS (WP 1590 00).
4. On right engine only, remove screw, washer, spacer and clamp holding engine control cables to IPS blower duct (Figure 1, Sheet 2, Detail B).
5. Remove right engine starter air tube (WP 0481 00).
6. Disconnect coupling and remove IPS blower duct from engine IPS blower (Figure 1, Sheet 2, Detail B).
7. Remove HIRSS extender (WP 0495 00).
8. Remove rear fairing and rear panel (WP 0490 00).
9. Remove screws, washers, and splice strap from top of bulkhead and oil cooler support beam (Figure 1, Sheet 1, Detail A).

REMOVE - Continued

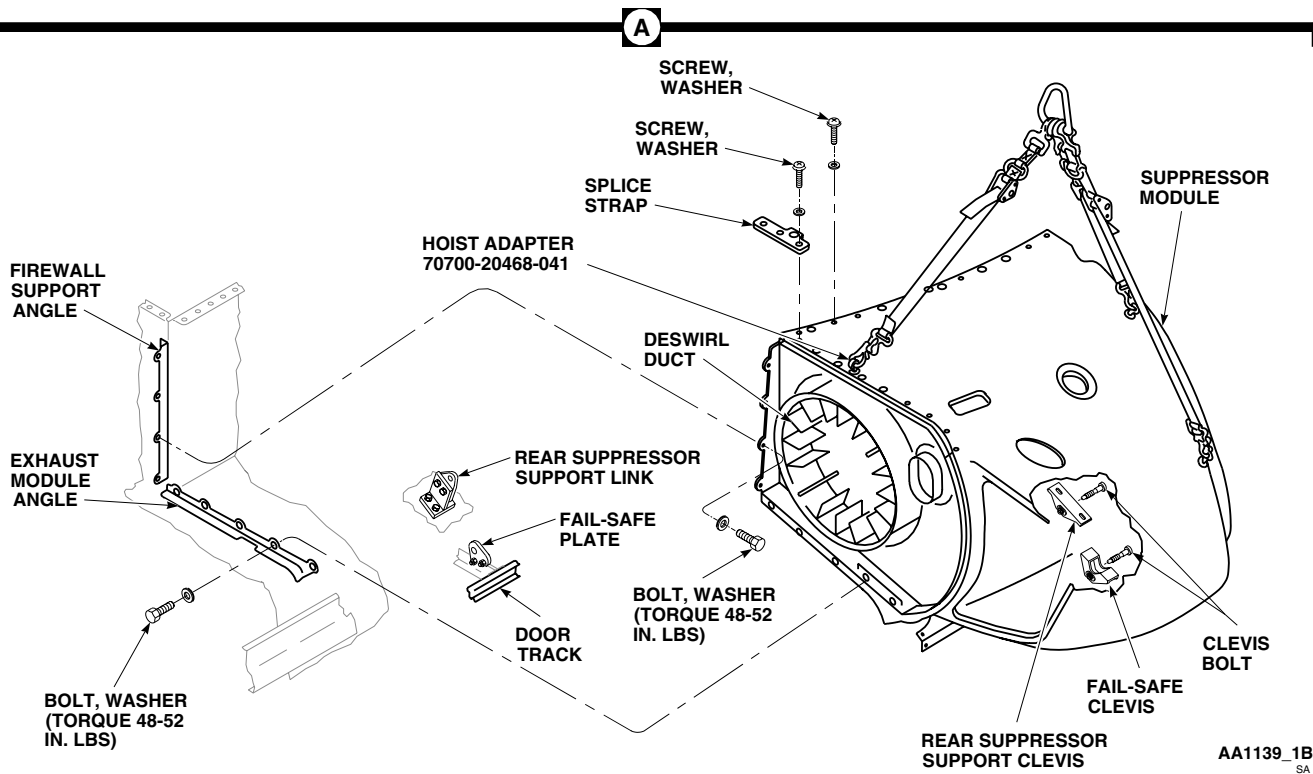
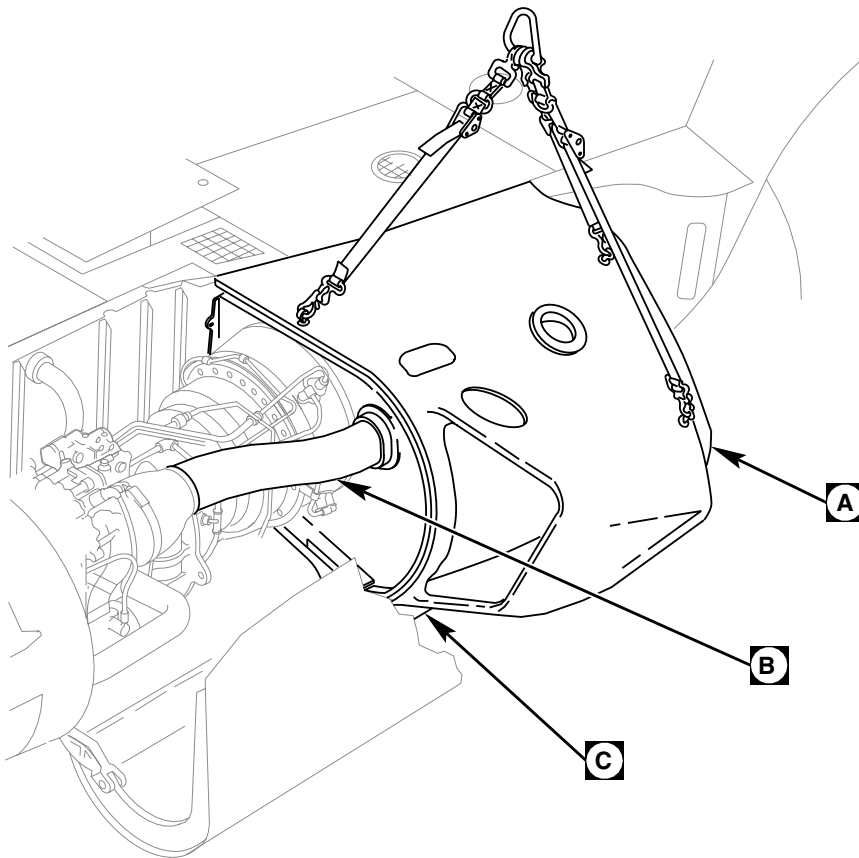


Figure 1. Remove HIRSS Module. (Sheet 1 of 2)

REMOVE - Continued

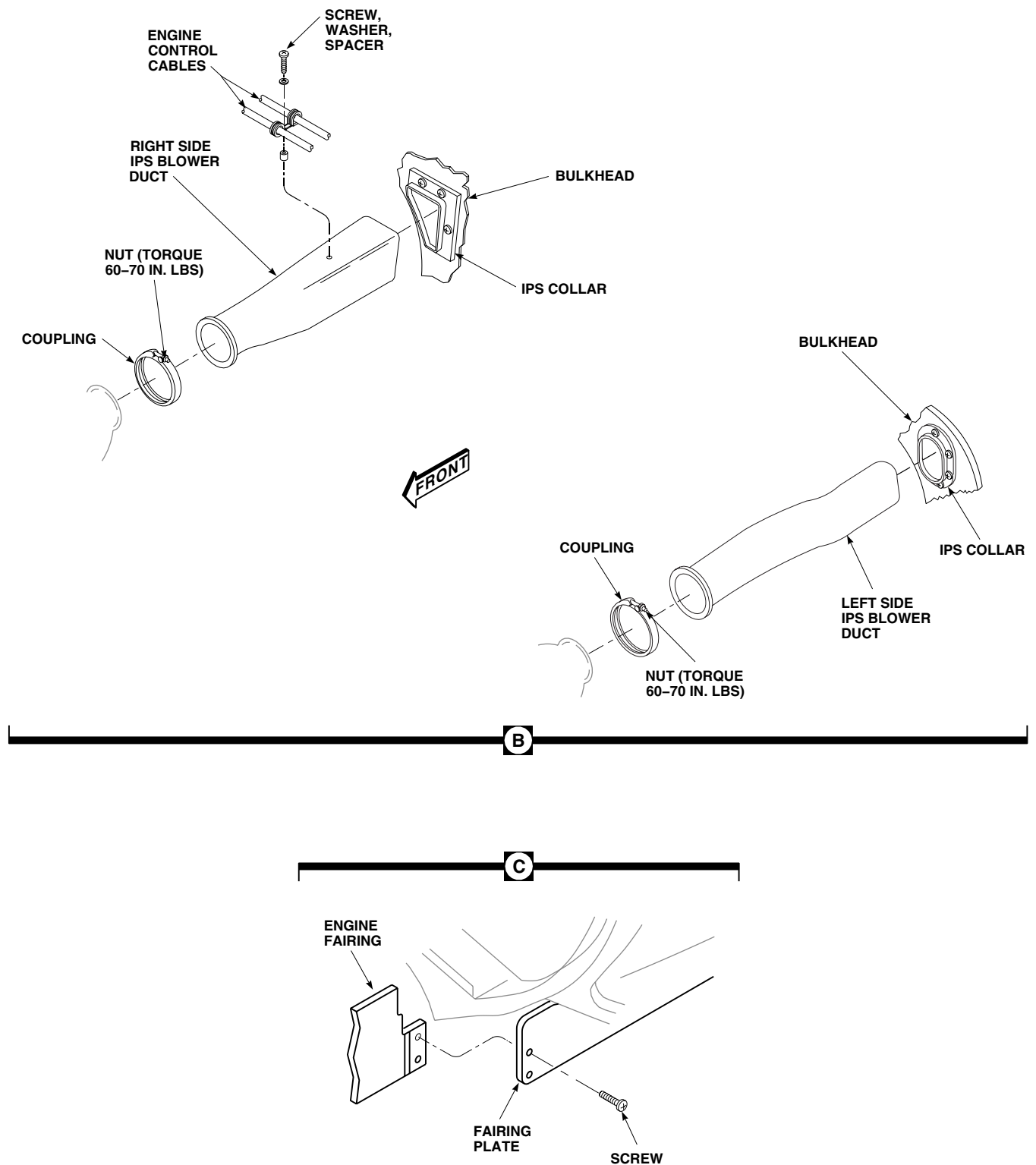
AA1139_2B
SA

Figure 1. Remove HIRSS Module. (Sheet 2 of 2)

REMOVE - Continued**WARNING**

- To prevent injury to personnel or damage to equipment, before lifting suppressor module, make sure sling is installed properly.
- Hoist cable may break, causing injury to personnel or damage to equipment if hoist is used to free a sticking component.
- Hoist cable may break, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

10. Install hoist adapters, 70700-20468-041, on module hoist points (Figure 1, Sheet 1, Detail A).
11. Attach hoisting sling legs between hoist adapters and ring.
12. Attach ring to maintenance crane, hoist, or lifting device. Adjust leg length of sling as needed.
13. Raise hoist enough to take slack from sling.
14. Loosen loop clamp holding deswirl duct to engine power turbine flange. Slide loop clamp over flanges onto deswirl duct (Figure 2, Detail A or Detail B).
15. Remove bolts and washers holding lower support angle to HIRSS bulkhead (Figure 3, Detail A).
16. Remove fail-safe clevis bolt from fail-safe plate (Figure 3, Detail C).
17. Remove clevis bolt holding rear suppressor support link to rear suppressor support clevis on HIRSS core (Figure 3, Detail B).
18. Remove bolts and washers holding suppressor module bulkhead to exhaust module angle and firewall support angle (Figure 1, Sheet 1, Detail A).
19. Remove bolts and washers holding rear panel to suppressor module.
20. Remove screws holding fairing plate to engine fairing (Figure 1, Sheet 2, Detail C).
21. Remove screws and washers holding nacelle fairing to firewall (Figure 1, Sheet 1, Detail A).

NOTE

Deswirl duct remains installed on HIRSS module during HIRSS removal.

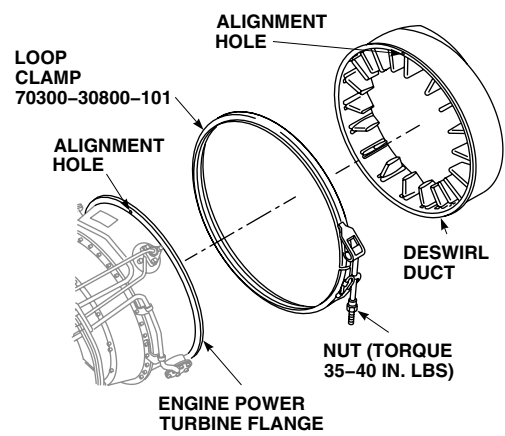
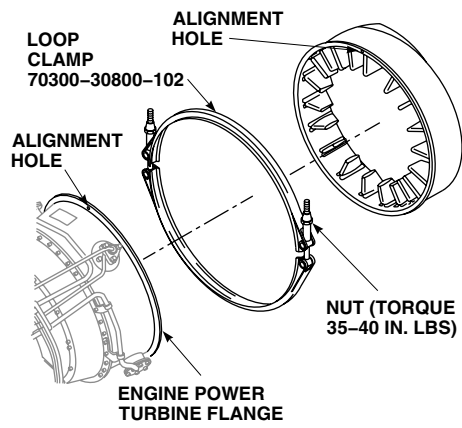
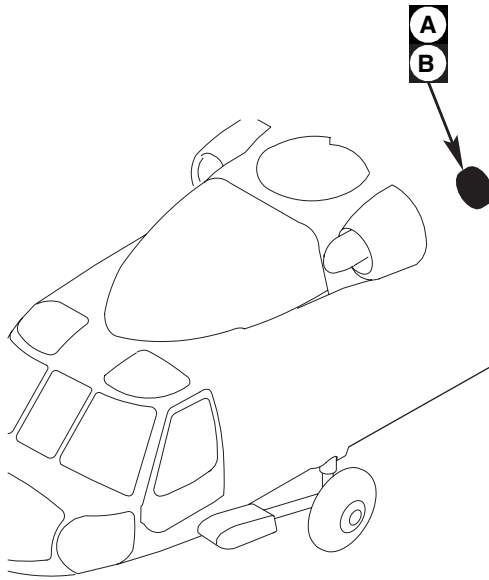
22. Slide suppressor module and deswirl duct rearward to clear center body cone and lift suppressor module off helicopter.
23. Remove deswirl duct and loop clamp from suppressor module.

NOTE

Do not lay module on bulkhead lower flange.

24. Lower module onto adapter, 70700-20424-041, and secure module with straps (Figure 4).
25. Remove sling and hoist adapters.
26. Inspect suppressor support mounts (WP 0491 00).
27. If engine exhaust module is to be installed in place of HIRSS, do this:
 - a. Remove bolts, washers, and nuts holding fail-safe plate to cargo door track. Remove fail-safe plate (Figure 3, Detail C).

REMOVE - Continued



AA8932A
SA

Figure 2. Remove Deswirl Duct.

REMOVE - Continued

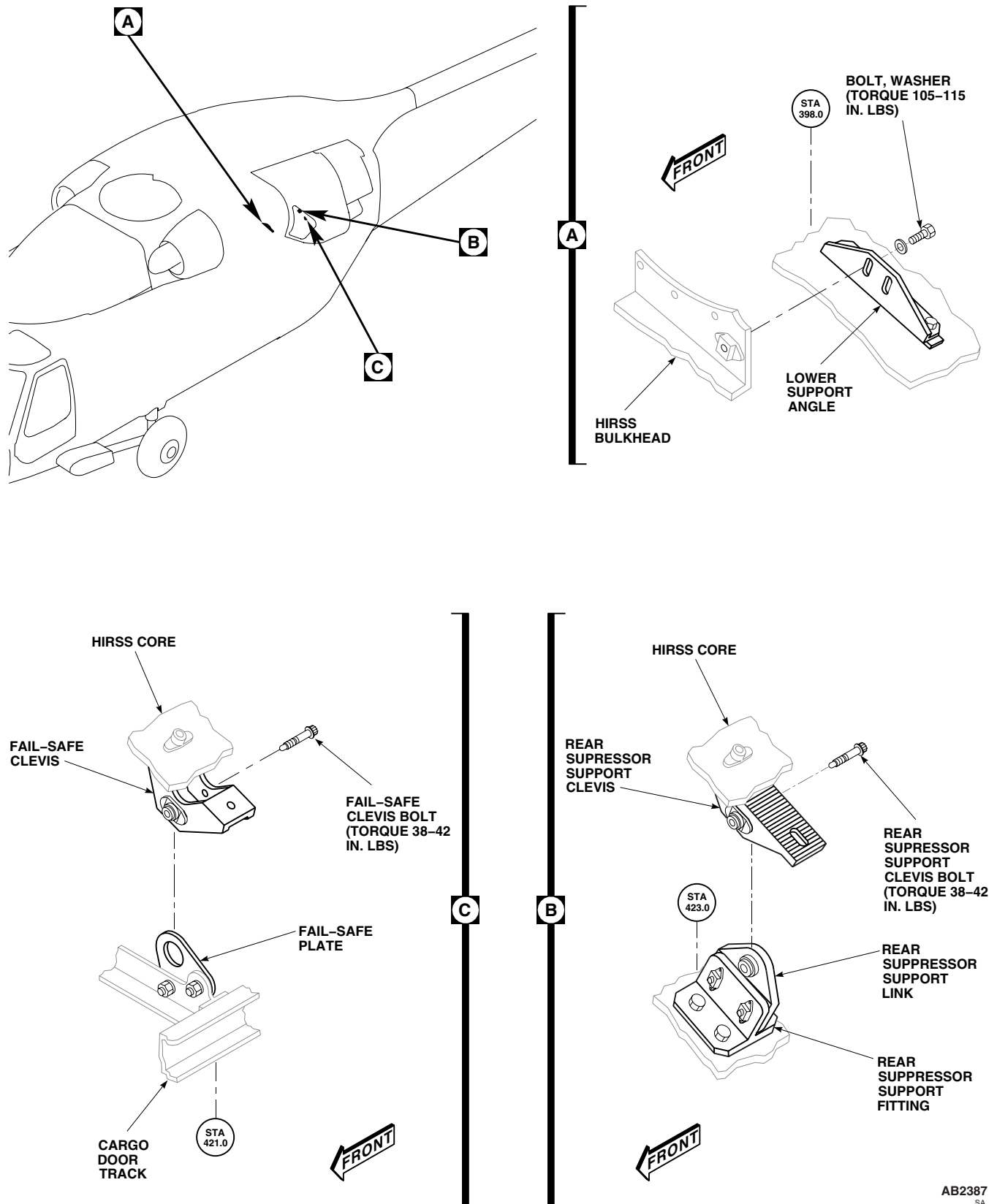


Figure 3. Disconnect HIRSS Mounts.

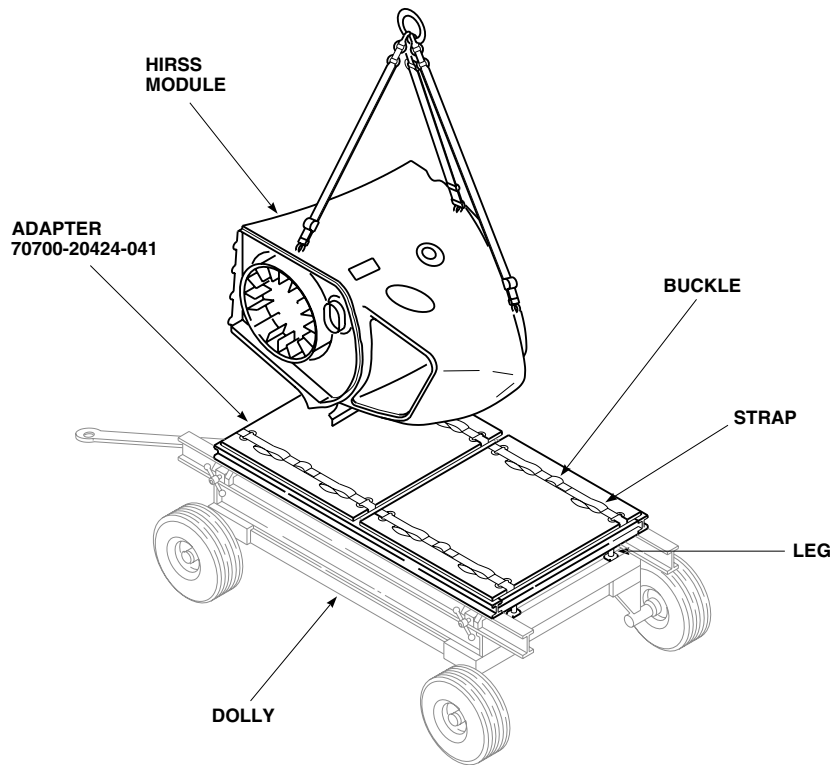
REMOVE - ContinuedAK1259A
SA

Figure 4. HIRSS Module Adapter.

- b. Remove rear suppressor support fitting, bolts and washers from fitting. Remove fitting from airframe (Figure 3, Detail B).
 - c. Apply sealing compound, Item 273, WP 1803 00, under bolt heads. Install bolts and washers in rear suppressor support fitting nutplates.
 - d. Remove bolts, washers, shims, and lower support angle from airframe.
 - e. Apply sealing compound, Item 273, WP 1803 00, under bolt heads. Install bolts and washers in lower support angle nutplates (Figure 3, Detail A).
 - f. Stow HIRSS components and attaching hardware to protect them from damage and have them ready for reinstallation. Repair any damage before stowing (FM 38-700).
 - g. Remove center body cone (WP 0482 00).
 - h. Remove maintenance crane, hoist or lift (WP 1590 00).
 - i. Install original center tailpipe body and cover (WP 0482 00).
 - j. Install engine exhaust module (WP 0487 00).
28. Make sure area is clean and free of foreign material.
 29. Close engine cowl and latch.

INSPECT SUPPRESSOR AND NACELLE FAIRING SUPPORT MOUNTS

1. Inspect suppressor support mount as follows:

INSPECT SUPPRESSOR AND NACELLE FAIRING SUPPORT MOUNTS - Continued

- a. Visually check spherical bearing in rear support link for looseness or damage. **NO LOOSENESS OR DAMAGE ALLOWED.** Replace parts that have loose or damaged bearings.
 - b. Visually inspect rear suppressor support fitting, rear support link, and rear suppressor support clevis for flats or damage to adjustment ridges. **NO DAMAGE ALLOWED.** Replace rear suppressor support clevis if adjustment ridges are damaged.
 - c. Inspect fail-safe clevis, fail-safe plate, rear suppressor support clevis, rear suppressor support fitting and link using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cracked parts.
 - d. Check fail-safe and rear suppressor support clevis bolts for wear. **DIAMETER OF CLEVIS BOLT SHANK IN CONTACT AREA MUST NOT BE LESS THAN 0.281-INCH.** Replace worn bolt.
2. Inspect nacelle fairing support mount as follows:
- a. Visually check spherical bearings in upper nacelle support fitting and lower nacelle support link for looseness or damage. **NO LOOSENESS OR DAMAGE ALLOWED.** Replace parts that have loose or damaged bearings.
 - b. Check upper support clevis for flats or damage to adjustment ridges. **NO DAMAGE ALLOWED.** Replace upper support clevis if adjustment ridges are damaged.
 - c. Inspect lower nacelle support fitting and link using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace lower nacelle support fitting and link if cracks are found.
 - d. Inspect upper nacelle support fitting and clevis using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace upper nacelle support fitting and clevis if cracks are found.
 - e. Check upper and lower clevis bolts for wear. **DIAMETER OF CLEVIS BOLT SHANK IN CONTACT AREA MUST NOT BE LESS THAN 0.281-INCH.** Replace worn bolt.

INSTALL**NOTE**

Initial installation of the HIRSS requires the following entry on helicopter basic weight and balance record (DD Form 365C): WEIGHT CHANGE +172.4, MOMENT ARM 420.1, MOMENT/1000 +72,431.

1. If exhaust module is installed, do this:
 - a. Remove exhaust module (WP 0487 00).
 - b. Remove engine center tailpipe (WP 0482 00).

NOTE

If HIRSS has never been installed on helicopter, firewall support angles holes must be drilled.

- c. Position firewall support angles on firewall at STA 402.0 (Figure 5, Detail A). Mark 4 bolt holes on firewall support angle.
- d. Drill 4 holes in firewall support angle using No. 40 drill bit.
- e. Redrill holes using 1/4-inch diameter drill bit. Deburr holes.
- f. Attach firewall support angles to firewall at STA 402.0, with bolts and washers.

INSTALL - Continued

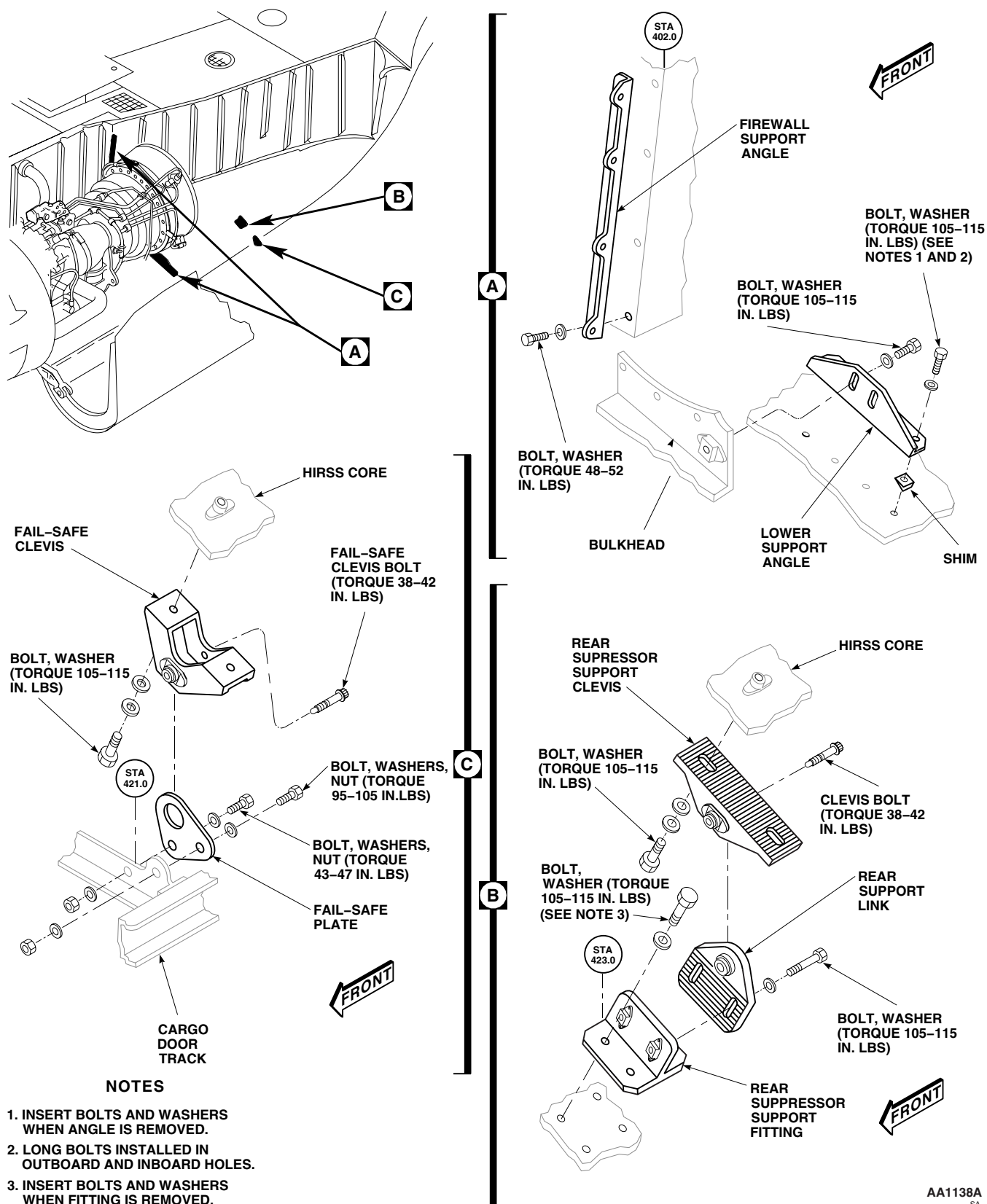


Figure 5. HIRSS Suppressor Support Mounts.

INSTALL - Continued

- g. Apply sealing compound, Item 273, WP 1803 00, under boltheads and around edges of rear suppressor support fitting.
 - h. Attach rear suppressor support fitting to airframe with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS (Figure 5, Detail B).**
 - i. Attach fail-safe plates to airframe above cargo door track with bolt, washers, and nut. **TORQUE NUT TO 95 - 105 INCH-POUNDS (Figure 5, Detail C).**
 - j. Install bolt, washer, and nut to fail-safe plate. **TORQUE NUT TO 43 - 47 INCH-POUNDS (Figure 5, Detail C).**
 - k. Install center body cone (WP 0482 00).
2. Position main rotor head to allow overhead clearance.

NOTE

The following procedure applies to right and left sides except where indicated.

- 3. Position maintenance crane (WP 1590 00), hoist, or lifting device to lift HIRSS.
- 4. Inspect deswirl duct (WP 0481 00).

NOTE

Deswirl duct must be temporarily installed on HIRSS module before module is installed on helicopter.

5. **QA** Mate deswirl duct to HIRSS module by doing this:

NOTE

Left and right deswirl ducts are identical.

- a. Deswirl ducts have two alignment holes located on deswirl duct flange. One alignment hole shall be located top center while the other alignment hole shall point inboard when installed in the HIRSS modules (Figure 2).
 - b. Temporarily install deswirl duct on HIRSS module by sliding duct inside finger seals on core.
 - c. Position loop clamp on deswirl duct beyond flange.
6. Attach three hoist adapters, 70700-20468-041, to HIRSS module hoist points (Figure 1, Sheet 1, Detail A).
7. Attach hoisting legs between hoist adapters and sling apex. Adjust leg lengths to position module for installation.
8. Attach ring to hoist or lifting device and raise sling enough to remove slack.

INSTALL - Continued**WARNING**

- To prevent injury to personnel or damage to equipment, before lifting suppressor module, make sure sling is installed properly.
- Hoist cable may break, causing injury to personnel or damage to equipment if hoist is used to free a sticking component.
- Hoist cable may break, causing injury to personnel or damage to equipment, if hook is allowed to contact pulley when hoisting components. Do not allow hook assembly to contact pulley.

CAUTION

Damage to center body cone and/or deswirl vanes will occur if HIRSS module is not carefully hoisted in place. Use care when hoisting and positioning HIRSS module on helicopter.

9. Lift HIRSS module into position (Figure 1, Sheet 1, Detail A).
10. **QA** Line up and loosely install HIRSS module hardware by doing this:
 - a. Install screws and washers holding nacelle fairing to firewall.

NOTE

Suppressor support link should be centered between sides of suppressor support clevis after torque is applied.

- b. Attach rear suppressor support link to rear suppressor support clevis on HIRSS core with clevis bolt (Figure 3, Detail B).
 - c. Attach fail-safe plate to fail-safe clevis on HIRSS core with fail-safe clevis bolt (Figure 3, Detail C).
 - d. Install bolts and washers holding bulkhead to exhaust module angle (Figure 1, Sheet 1, Detail A).
 - e. Attach bulkhead to firewall support angle with bolts and washers (Figure 1, Sheet 1, Detail A).
 - f. Attach lower support angle to HIRSS bulkhead with bolts and washers (Figure 3, Detail A).
 - g. Install screws and washers holding fairing plate to engine fairing (Figure 1, Sheet 2, Detail C).
 - h. Attach splice strap to top of bulkhead and oil cooler support beam with screws and washers.
11. Install deswirl duct (WP 0481 00).
 12. To torque all hardware, do this:
 - a. Tighten all screws holding nacelle fairing to firewall.

NOTE

Rear suppressor support link must seat with the rear of suppressor support fitting after the bolts are torqued.

- b. **TORQUE REAR SUPPRESSOR SUPPORT FITTING TO REAR SUPPRESSOR SUPPORT LINK BOLTS TO 105 - 115 INCH-POUNDS (Figure 3, Detail B).**

INSTALL - Continued**NOTE**

Suppressor support link should be centered between sides of suppressor support clevis after torque is applied.

- c. **TORQUE REAR SUPPRESSOR SUPPORT CLEVIS BOLT TO 38 - 42 INCH-POUNDS (Figure 3, Detail B).**
- d. **IF ADJUSTMENT WAS MADE, TORQUE REAR SUPPRESSOR SUPPORT CLEVIS TO HIRSS CORE BOLTS TO 105 - 115 INCH-POUNDS (Figure 3, Detail B).**

NOTE

After torquing fail-safe clevis bolt, it is acceptable for a gap to exist between the underside of bolthead and the fail-safe clevis.

- e. **TORQUE FAIL-SAFE CLEVIS BOLT TO 38 - 42 INCH-POUNDS (Figure 3, Detail C).**
 - f. **TORQUE LOWER SUPPORT ANGLE TO AIRFRAME BOLTS TO 105 - 115 INCH-POUNDS (Figure 3, Detail A).**
 - g. Tighten screws holding splice strap to oil cooler support beam and bulkhead.
 - h. **TORQUE BOLTS HOLDING SUPPRESSOR MODULE TO FIREWALL SUPPORT ANGLE TO 48 - 52 INCH-POUNDS (Figure 1, Sheet 1, Detail A).**
 - i. **TORQUE BOLTS HOLDING SUPPRESSOR MODULE BULKHEAD TO EXHAUST MODULE ANGLE TO 48 - 52 INCH-POUNDS (Figure 1, Sheet 1, Detail A).**
 - j. **TORQUE BOLTS HOLDING LOWER SUPPORT ANGLE TO HIRSS BULKHEAD TO 105 - 115 INCH-POUNDS (Figure 3, Detail A).**
 - k. Install support to lower outboard side of bulkhead and upper cargo door track fairing with screws.
 - l. Tighten screws holding fairing plate to engine fairing (Figure 1, Sheet 2, Detail C).
- 13. **QA** Install IPS blower duct in collar on bulkhead (Figure 1, Sheet 2, Detail B).
 - 14. **QA** Clamp forward end of blower ducts to engine with coupling. **TORQUE COUPLING NUTS TO 60 - 70 INCH-POUNDS.**
 - 15. Seat coupling around circumference by tapping with a plastic mallet and retorquing coupling nut.
 - 16. Continue tapping coupling until two successive trials show no loss of torque.
 - 17. **QA** On right engine only, clamp engine control cables to top of IPS blower duct with screw, spacer, and washer (Figure 1, Sheet 2, Detail B).
 - 18. Install right engine starter air tube (WP 0481 00).
 - 19. Install panel and rear fairing (WP 0490 00).
 - 20. Remove hoisting sling from nacelle.
 - 21. Remove hoist adapters from HIRSS module and install screws.
 - 22. Install HIRSS extender (WP 0495 00).
 - 23. Make sure area is clean and free of foreign material.
 - 24. Close engine cowling and latch.
 - 25. Touch up paint on exposed fasteners (WP 0381 00 and TM 55-1500-345-23).
 - 26. Operate engine for a minimum of 10 minutes (TM 1-1520-237-10).

INSTALL - Continued

27. Allow one hour cool down time. **TORQUE CHECK DESWIRL DUCT LOOP CLAMP TO 35 - 40 INCH-POUNDS.** Retorque loop clamp if below limit (WP 0481 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

HIRSS REAR FAIRINGS AND PANELS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 55-1500-345-23
WP 0379 00
WP 0381 00
WP 0493 00
WP 0495 00

REPLACE HIRSS REAR FAIRINGS AND PANELS**Remove****NOTE**

If panel and fairing are to be removed as one piece, do not remove vertical row of bolts holding panel to rear fairing.

1. Remove HIRSS extender (WP 0495 00).
2. On right side only, remove bolts and washers holding suppressor module IPS blower duct to panel (WP 0493 00).
3. Remove screws and washers securing rear fairing and panel to firewall and nacelle. Remove rear fairing and panel (Figure 1, Detail A or Detail B).
4. Disconnect panels from rear fairings by removing vertical row of bolts and washers (Figure 1, Detail A or Detail B).

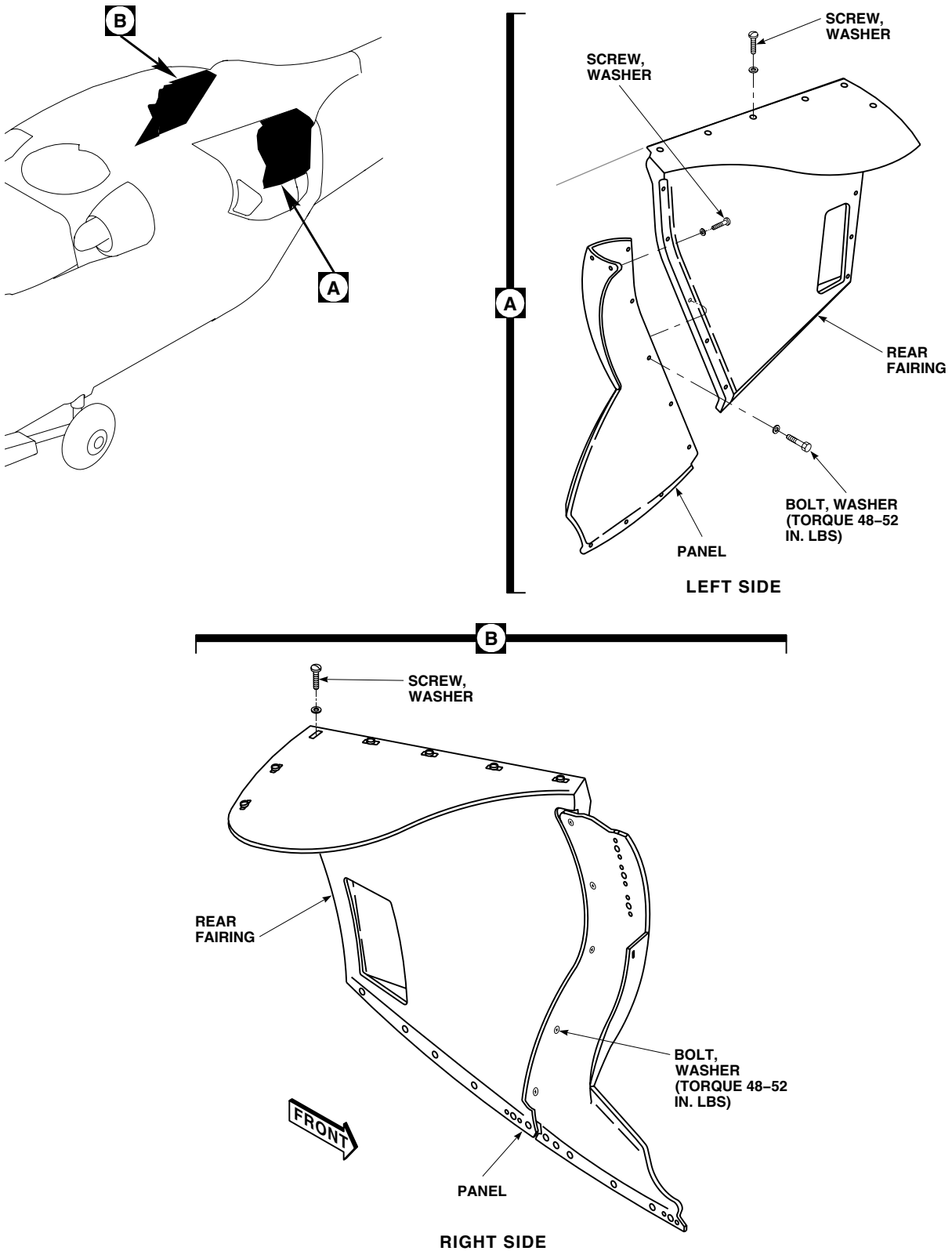
Inspect

1. Check panels and rear fairings for cracks in fiberglass. **ANY NUMBER OF CRACKS UP TO 1 INCH LONG ARE ALLOWED.** Stop-drill cracks using No. 52 drill bit. Repair fairings (WP 0379 00).
2. Check panels and rear fairings for loose, broken, or missing fasteners. **UP TO 15% OF FASTENERS MAY BE DAMAGED OR MISSING, WITH AT LEAST TWO GOOD FASTENERS BETWEEN EACH DAMAGED OR MISSING FASTENER.** Replace fasteners as required (TM 1-1500-204-23).

Install

1. **QA** If panels were disconnected from fairings, install panels to rear fairings with bolts and washers. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS (Figure 1, Detail A or Detail B).**
2. **QA** Install panel and rear fairing on firewall and nacelle with screws and washers.
3. **QA** On right side only, attach rear panel to suppressor module IPS blower duct with bolts and washers (WP 0493 00).
4. Install HIRSS extender (WP 0495 00).
5. Touch up paint on exposed screws as required (WP 0381 00 and TM 55-1500-345-23).

REPLACE HIRSS REAR FAIRINGS AND PANELS - Continued



AA2172
SA

Figure 1. Replace HIRSS Rear Fairings and Panels.

REPLACE HIRSS REAR FAIRINGS AND PANELS - Continued

6. Make sure area is clean and free of foreign material.

REPAIR HIRSS REAR FAIRINGS AND PANELS**Remove**

Remove panel from rear fairing. Refer to, REPLACE HIRSS FAIRINGS AND PANELS, in this work package.

Repair**NOTE**

Rear panels, fairings, and outlet panels are constructed of fiberglass skin.

1. Evaluate damage to fiberglass components (WP 0379 00). Refer to, REPLACE HIRSS FAIRINGS AND PANELS, in this work package.
2. Repair damaged areas of panels, fairings, or outlet panels (WP 0379 00).
3. Evaluate limit of missing, broken, or loose nutplates. Refer to, REPLACE HIRSS FAIRINGS AND PANELS, in this work package.
4. Replace nutplates as required (TM 1-1500-204-23) (Figure 2, Detail A or Figure 2, Detail B).
5. Replace stiffener on outlet panel as required (TM 1-1500-204-23) (Figure 2, Detail C).

Install

1. Attach rear panel to rear fairing. Refer to, REPLACE HIRSS FAIRINGS AND PANELS, in this work package.
2. Touch up paint in repaired areas as required (WP 0381 00 and TM 55-1500-345-23).

The diagrams illustrate the assembly of a rear panel in several steps:

- Diagram 1 (Top Left):** Shows a side profile of a vehicle with two areas marked B and A.
- Diagram 2 (Top Right, labeled A):** Shows the rear panel left side being attached to the rear fairing using rivets and nutplates. Labels include: NUTPLATE, RIVET, REAR FAIRING, REAR PANEL LEFT SIDE, and FRONT.
- Diagram 3 (Bottom Left, labeled C):** Shows the rear panel right side being attached to the rear fairing using rivets and nutplates. Labels include: STIFFENER, REAR FAIRING, OUTLET PANEL, RIVET, and NUTPLATE.
- Diagram 4 (Bottom Right, labeled B):** Shows the rear panel right side being attached to the rear fairing using rivets and nutplates. Labels include: RIVET, NUTPLATE, REAR PANEL, and FRONT.

The final assembly is labeled **REAR PANEL RIGHT SIDE**.

AA1142A

Figure 2. Repair HIRSS Rear Fairings and Panels.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
HIRSS SUPPRESSOR SUPPORT MOUNTS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

References

WP 0489 00
WP 1803 00

Material/Parts

Sealing Compound, Item 273, WP 1803 00

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

It is not necessary to remove module if mounts are replaced one at a time.

1. Remove HIRSS module (WP 0489 00).
2. Remove bolts, washers and rear support link from rear suppressor support fitting (Figure 1, Detail B).
3. Remove bolts, washers and rear suppressor support fitting from helicopter.
4. Remove bolts, washers and rear suppressor support clevis from HIRSS core.
5. Remove bolts, washers and fail-safe clevis from HIRSS core (Figure 1, Detail C).
6. Remove bolts, washers, nuts, and fail-safe plate from helicopter.
7. Remove bolts and washers, and remove lower support angle and shim from helicopter (Figure 1, Detail A).

INSTALL

1. **QA** Install fail-safe plate on helicopter with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS (Figure 1, Detail C).**
2. **QA** Install fail-safe clevis to HIRSS core with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
3. **QA** Attach rear suppressor support clevis to HIRSS core with bolts and washers. Line up serrations on core with serrations on clevis to center bolts in clevis slots (Figure 1, Detail B). Do not torque bolts now.
4. Assemble rear suppressor support fitting to rear support link with bolts and washers. Do not torque bolts now.
5. Apply sealing compound, Item 273, WP 1803 00, under boltheads and around edges of rear suppressor support fitting.
6. **QA** Install rear suppressor support fitting and rear support link to helicopter. **TORQUE REAR SUPPRESSOR SUPPORT FITTING BOLTS TO 105 - 115 INCH-POUNDS (Figure 1, Detail B).**
7. Apply sealing compound, Item 273, WP 1803 00, under boltheads and around edges of lower support angle.

INSTALL - Continued

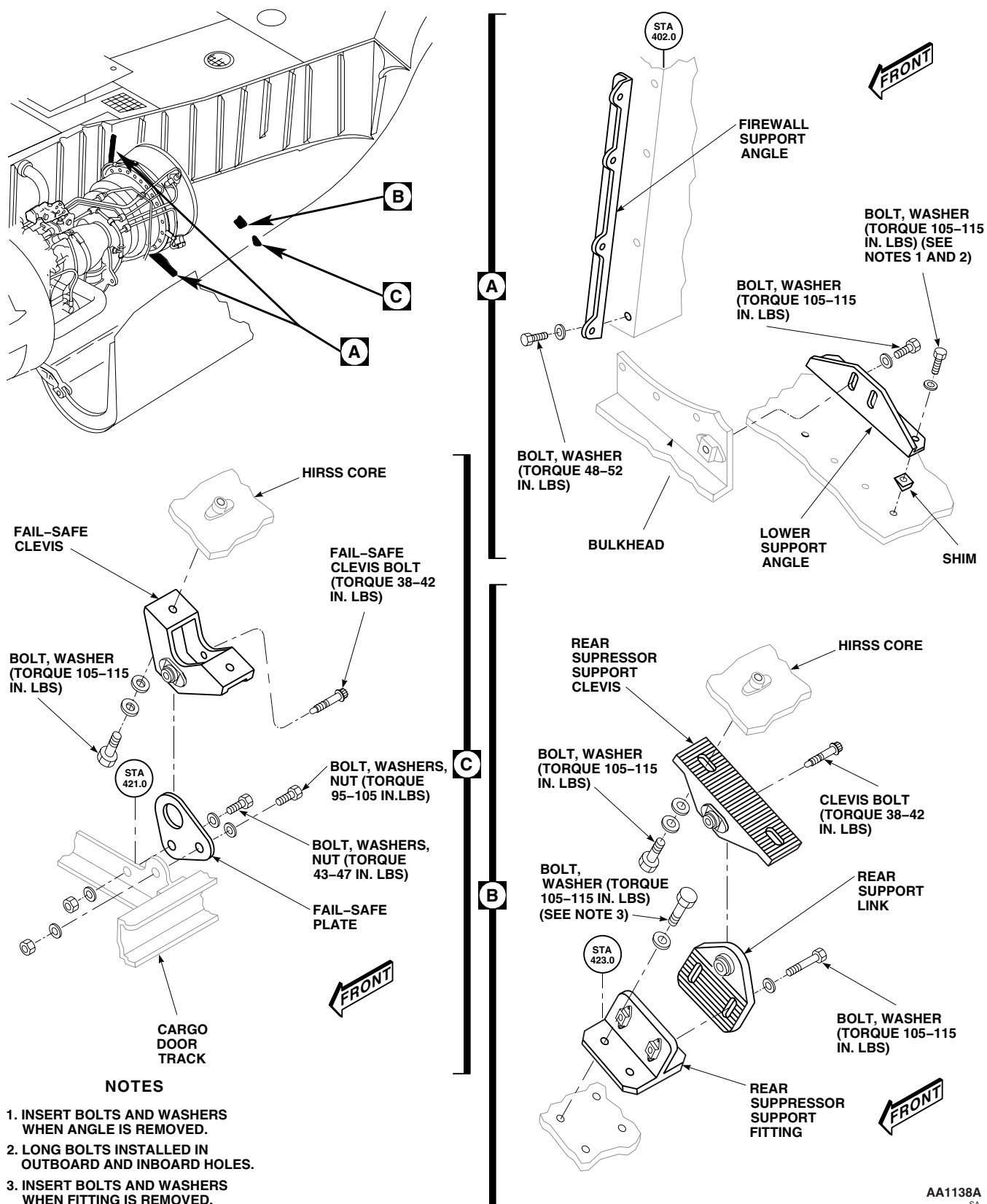


Figure 1. Replace HIRSS Suppressor Support Mounts.

INSTALL - Continued

8. **QA** Position lower support angle and shim on airframe and install bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS (Figure 1, Detail A).**
9. Install HIRSS module (WP 0489 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
HIRSS NACELLE FAIRING SUPPORT MOUNTS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Torque Wrench, 30 - 200 in. lbs.

References

WP 0493 00

Personnel Required

Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

It is not necessary to remove nacelle fairing if mounts are replaced one at a time.

1. Remove HIRSS nacelle fairing (WP 0493 00).
2. Remove bolts, washers, lower nacelle support fitting and shim(s) from lower nacelle (Figure 1, Detail B).
3. Remove bolts, washers, nuts and lower nacelle support link from HIRSS core.
4. Remove bolts, washers, nuts, upper nacelle support fitting, and shim(s) from upper nacelle fairing (Figure 1, Detail A).
5. Remove bolts, washers, nuts, and upper support clevis from HIRSS core.

INSTALL**NOTE**

- A maximum of 3 shims may be used under upper nacelle support fitting.
- Two different shims are available, a solid and a laminated one. The laminated shim may appear to be two shims but should be counted as one. One shim is approximately 0.180-inch thick.

1. **QA** Install upper nacelle support fitting and shim(s) with bolts, washers, and nuts. **TORQUE NUTS TO 95 - 105 INCH-POUNDS (Figure 1, Detail A).**

NOTE

Gap under head of bolt is acceptable.

2. **QA** Install lower nacelle support fitting and shim(s) with bolts and washers. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS (Figure 1, Detail B).**

INSTALL - Continued

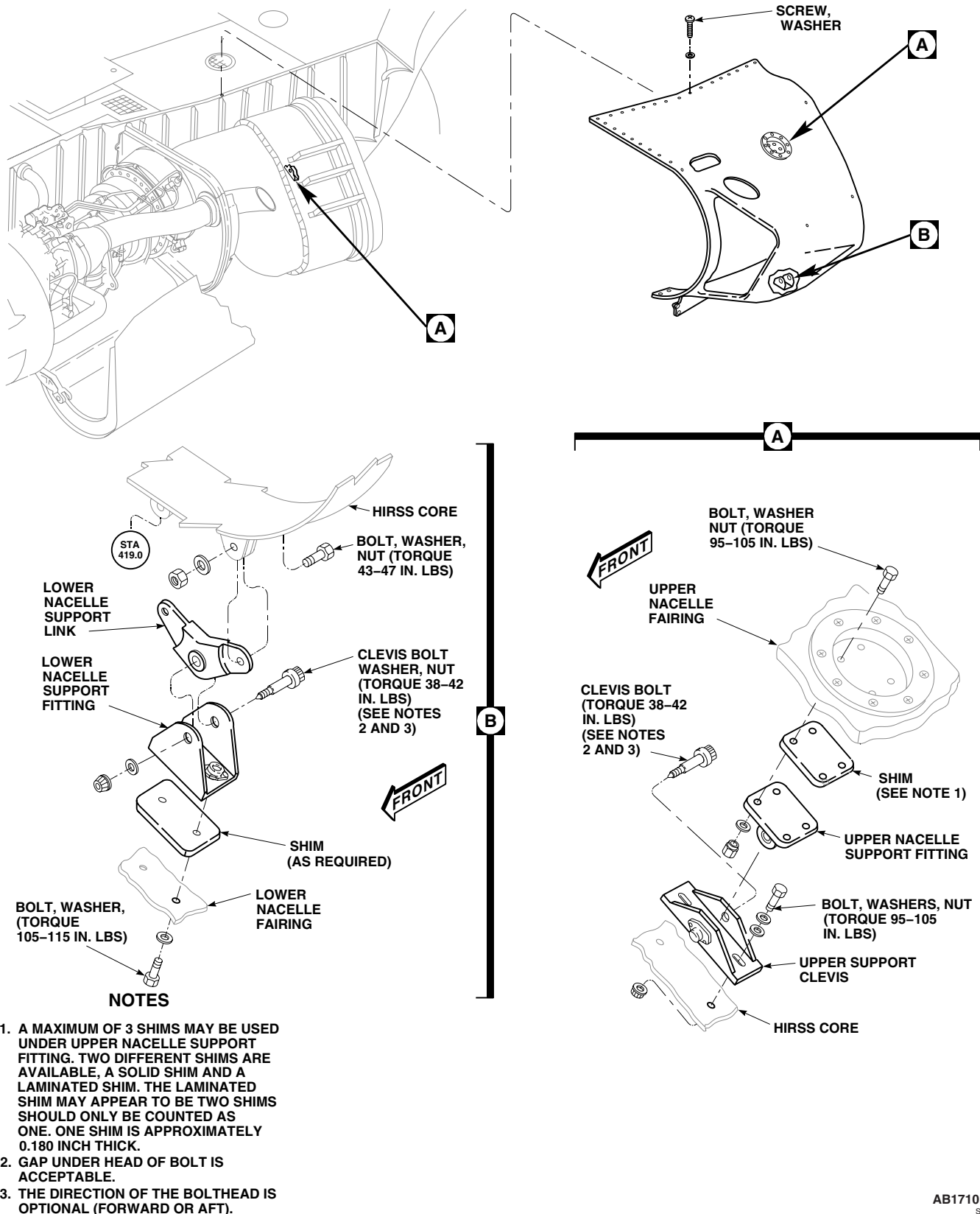


Figure 1. Replace HIRSS Nacelle Support Mounts.

INSTALL - Continued**NOTE**

Gap under head of bolt is acceptable.

3. **QA** Install upper support clevis to HIRSS core with bolts, washers, and nuts (Figure 1, Detail A). **TORQUE NUTS TO 95 - 105 INCH-POUNDS.**
4. **QA** Install lower nacelle support link to HIRSS core with bolts, washers, and nuts (Figure 1, Detail B). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
5. Install HIRSS nacelle fairing (WP 0493 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

HIRSS NACELLE FAIRING

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

WP 0377 00

WP 0379 00

WP 0381 00

WP 0489 00

WP 0492 00

WP 0495 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 55-1500-345-23

REPLACE HIRSS NACELLE FAIRING**Remove****NOTE**

Removal of left and right nacelle fairing and attaching components is the same except as noted.

1. Remove HIRSS extender (WP 0495 00).
2. Remove HIRSS module (WP 0489 00).
3. Remove nacelle fairing by doing this:
 - a. Remove screws and washer holding nacelle fairing to bulkhead (Figure 1, Sheet 1).
 - b. Remove clevis bolt from upper nacelle support fitting (Figure 1, Sheet 1, Detail A).
 - c. Remove clevis bolt, washer, and nut from lower nacelle support fitting (Figure 1, Sheet 1, Detail B).
 - d. Carefully lift nacelle fairing from suppressor core.
 - e. Remove bolts and washers from IPS collar. Remove collar and suppressor IPS blower duct from bulkhead (Figure 1, Sheet 2, Detail C).
 - f. On right side only, remove bolts and washers holding suppressor, module IPS blower duct to rear panel.

Inspect

1. Check nacelles for cracks. **NO CRACKS ALLOWED WITHIN 2 INCHES OF CORE SUPPORT OR NACELLE TO AIRFRAME ATTACHMENTS. ANY NUMBER OF CRACKS UP TO 1 INCH LONG ARE ALLOWED IN ALL OTHER AREAS.** Repair fiberglass (WP 0379 00).
2. Check nacelle fairing for loose, broken, or missing hoist point nutplates. **NONE ALLOWED.** Replace nutplates as required (TM 1-1500-204-23).

REPLACE HIRSS NACELLE FAIRING - Continued

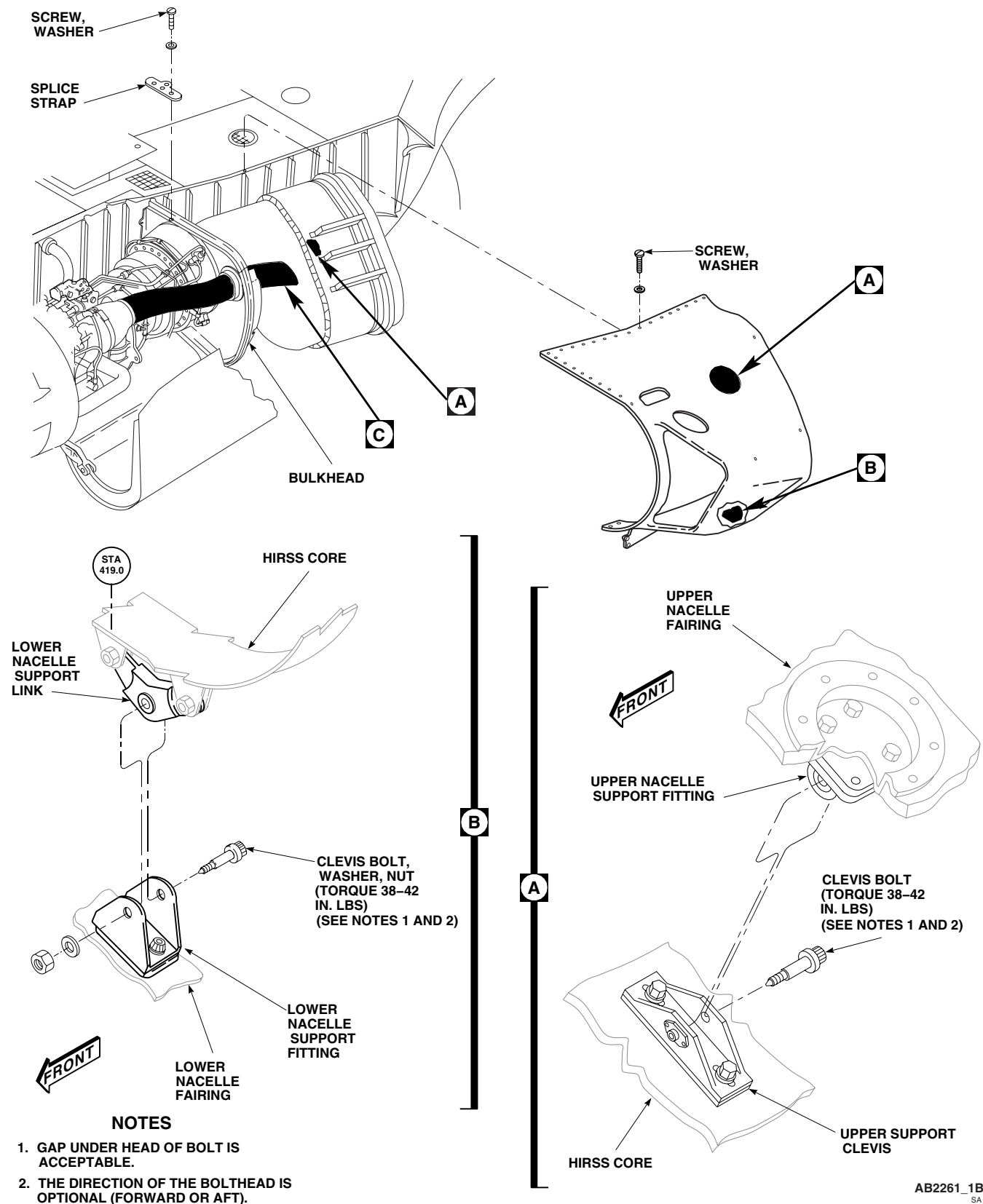


Figure 1. Repair HIRSS Nacelle Fairing. (Sheet 1 of 2)

REPLACE HIRSS NACELLE FAIRING - Continued

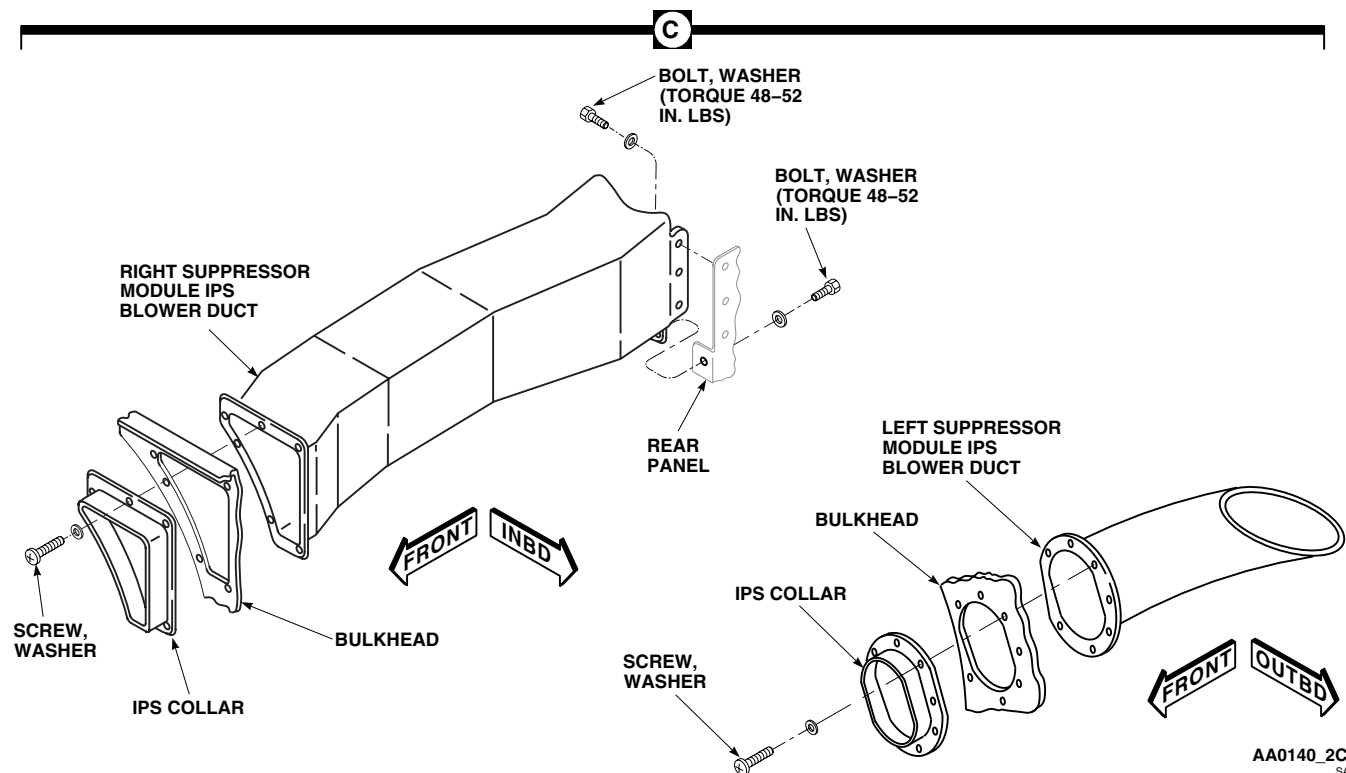


Figure 1. Repair HIRSS Nacelle Fairing. (Sheet 2 of 2)

3. Check nutplates on firewall and bulkhead attachment areas. **TWO MISSING, LOOSE, OR BROKEN NUTPLATES ARE ALLOWED WITH AT LEAST TWO GOOD NUTPLATES BETWEEN DAMAGED NUTPLATES.** Replace damaged nutplates (TM 1-1500-204-23).

Install**NOTE**

Installation of left and right nacelle and attaching components is the same except where noted.

1. Install IPS collar and suppressor module IPS blower duct on bulkhead with screws and washers (Figure 1, Sheet 2, Detail C).
2. On right side only install bolts and washers holding suppressor module IPS blower duct to rear panel. **TORQUE BOLTS TO 48 - 52 INCH-POUNDS.**
3. Install nacelle fairing by doing this:
 - a. With aid of assistant, carefully lift nacelle fairing into place on suppressor core and loosely install screws and washer holding nacelle fairing to bulkhead (Figure 1, Sheet 1).

NOTE

Gap under head of bolt is acceptable.

- b. Install clevis bolt in upper support clevis and upper nacelle support fitting (Figure 1, Sheet 1, Detail A). **TORQUE BOLT TO 38 - 42 INCH-POUNDS.**

REPLACE HIRSS NACELLE FAIRING - Continued**NOTE**

Gap under head of bolt is acceptable.

- c. Install clevis bolt, washer, and nut in lower nacelle support link and lower nacelle support fitting (Figure 1, Sheet 1, Detail B). **TORQUE BOLT TO 38 - 42 INCH-POUNDS.**
- d. Tighten screws holding nacelle fairing to bulkhead.
- 4. Install HIRSS module (WP 0489 00).
- 5. Install HIRSS extender (WP 0495 00).

REPAIR HIRSS NACELLE FAIRING**Disassemble**

- 1. Remove attaching components from nacelle fairing by doing this:
 - a. Remove upper and lower nacelle support fittings and shims (WP 0492 00).
 - b. Remove suppressor module IPS blower duct components. Refer to REPLACE HIRSS NACELLE FAIRING, in this work package.
 - c. Remove screws, washers, nuts and support panel from nacelle fairing (Figure 2, Detail A).
 - d. Remove handhold by drilling out rivets (TM 1-1500-204-23).
 - e. Remove screws and rear panel support angle.
 - f. Remove hoist point nutplates by drilling out rivets (TM 1-1500-204-23) (Figure 2, Detail B).
 - g. Remove screws securing fairing plate to nacelle fairing (Figure 2, Detail C) (TM 1-1500-204-23).
 - h. On left side only, remove screws, washers, nuts, and rear panel support angle from nacelle fairing (Figure 2, Detail A).

Repair**NOTE**

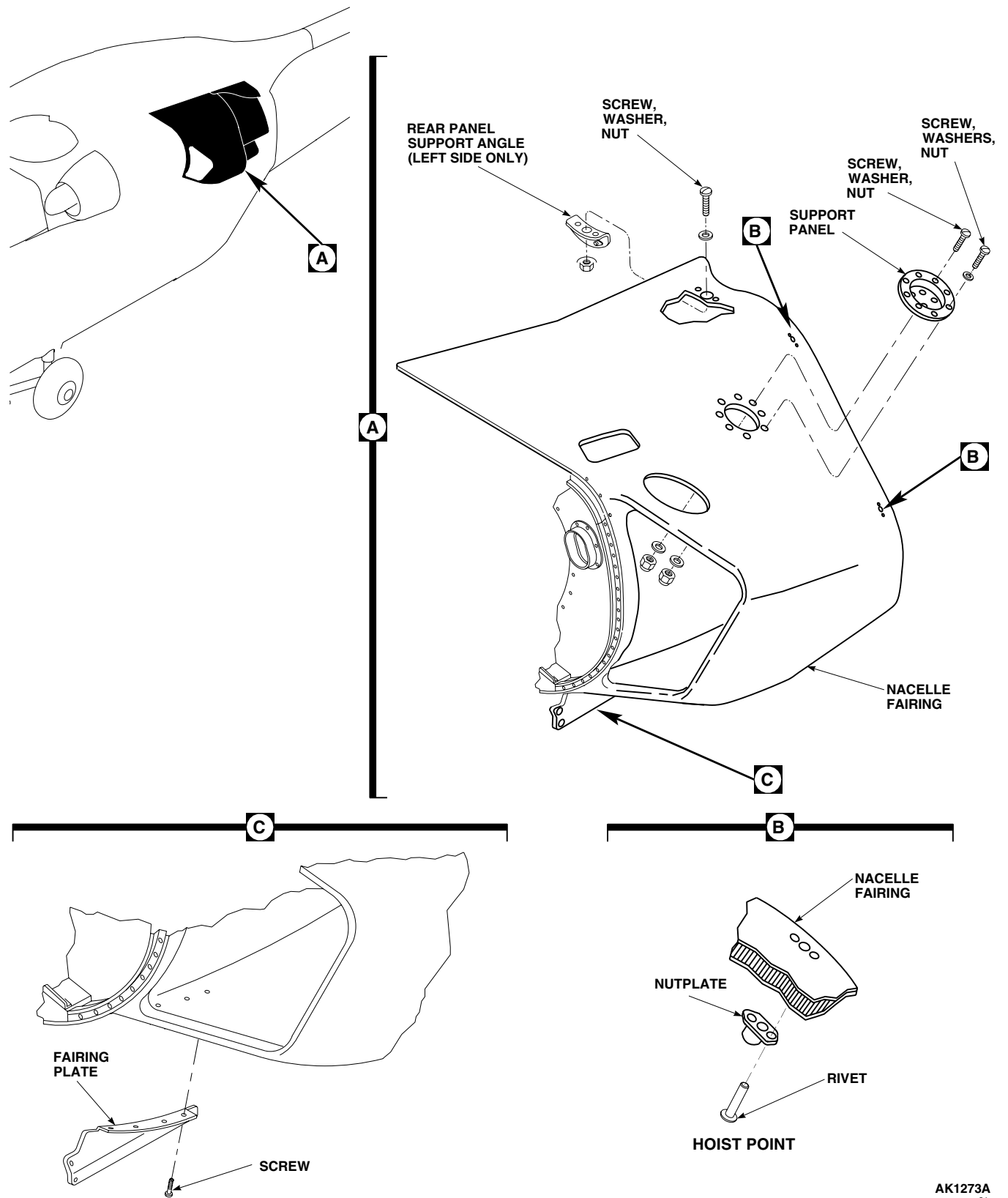
Nacelle fairing is constructed of a fiberglass outer skin and an inner core of aluminum alloy 2024/F40 honeycomb.

- 1. Evaluate damage to fairing skin and core. Refer to REPLACE HIRSS NACELLE FAIRING, in this work package. (WP 0377 00).
- 2. Repair nacelle fairing outer skin (WP 0379 00).

Assemble

- 1. **QA** Install fairing plate on nacelle fairing with screws (Figure 2, Detail C) (TM 1-1500-204-23).
- 2. **QA** Install nutplates for hoist points with rivets (Figure 2, Detail B) (TM 1-1500-204-23).
- 3. Install suppressor module IPS ducting. Refer to, REPLACE HIRSS NACELLE FAIRING, in this work package.
- 4. **QA** Install rear panel support angle on nacelle fairing with screws, washers, and nuts (Figure 2, Detail A).
- 5. **QA** Install support panel in opening on top of nacelle fairing with screws, washers, and nuts (Figure 2, Detail A).
- 6. **QA** Install upper and lower nacelle support fittings and shims (WP 0492 00).

REPAIR HIRSS NACELLE FAIRING - Continued



AK1273A
SA

Figure 2. Repair HIRSS Nacelle Fairing.

REPAIR HIRSS NACELLE FAIRING - Continued

7. Touch up paint in repaired areas as required (WP 0381 00 and TM 55-1500-345-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****HIRSS SUPPRESSOR CORE AND BAFFLE**

This WP supersedes WP 0494 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101
Torque Wrench, 30 - 200 in. lbs.

Material/Parts

Antiseize Compound, Item 49, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-265-23
WP 0489 00
WP 0493 00
WP 0497 00
WP 0522 00
WP 1803 00

REMOVE

1. Remove HIRSS module (WP 0489 00).
2. Remove HIRSS nacelle fairing (WP 0493 00).
3. Remove bolts, washers, nuts, and slide baffle out of suppressor core (Figure 1).
4. Remove bolts and washers from IPS collar. Remove collar and IPS blower duct from right bulkhead (WP 0493 00).
5. Remove screws, washers and nuts and remove bulkhead from forward support flange on suppressor core (Figure 1).

INSPECT

1. Table 1 defines allowable damage and wear to suppressor core (Figure 2, Sheet 1 and Figure 2, Sheet 2).
2. Table 2 defines allowable damage and wear to baffle (Figure 3).

INSPECT - Continued

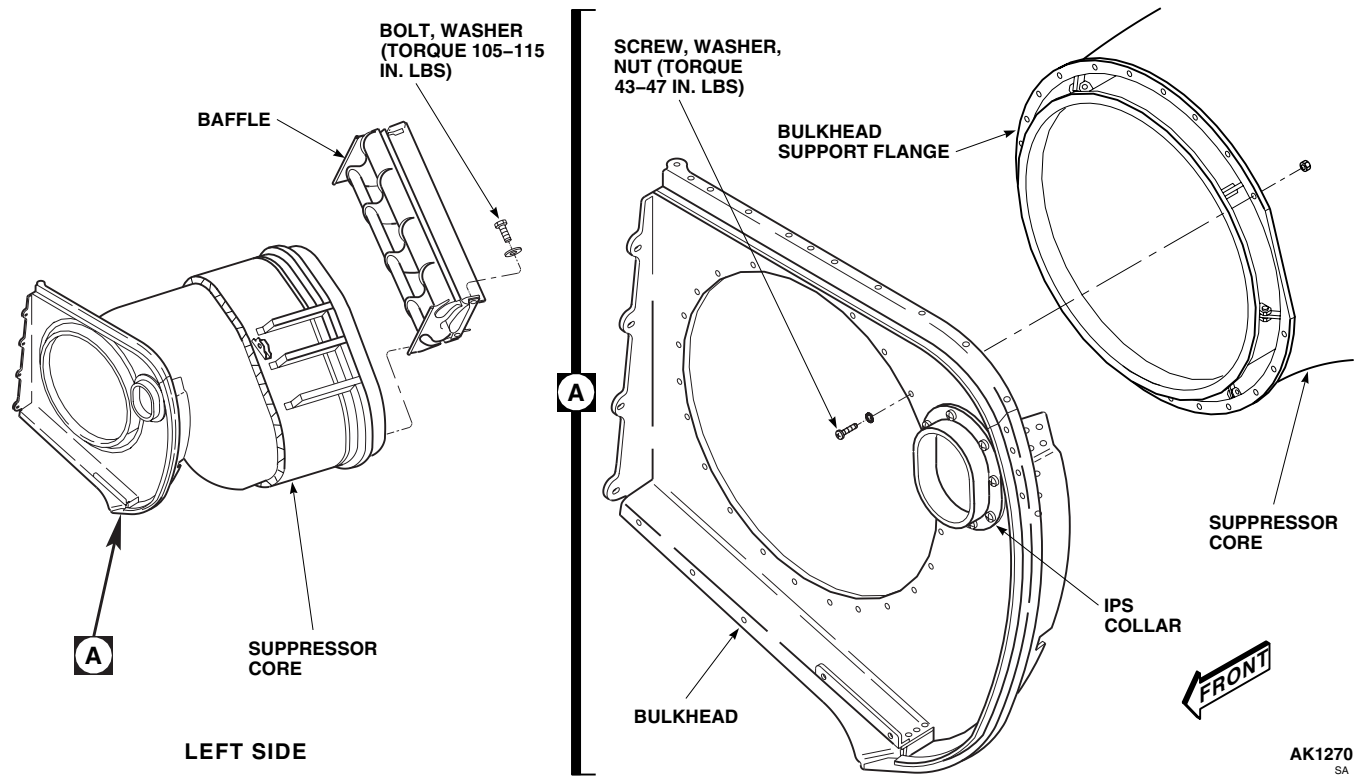
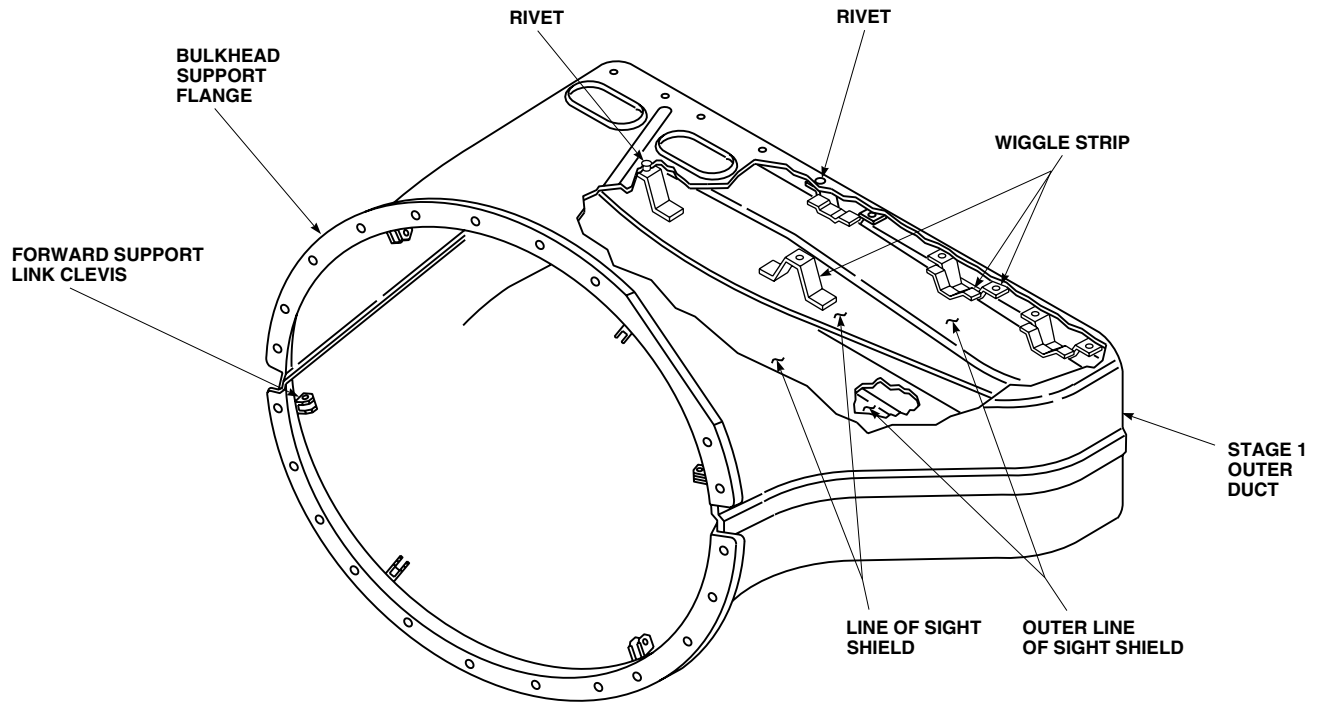
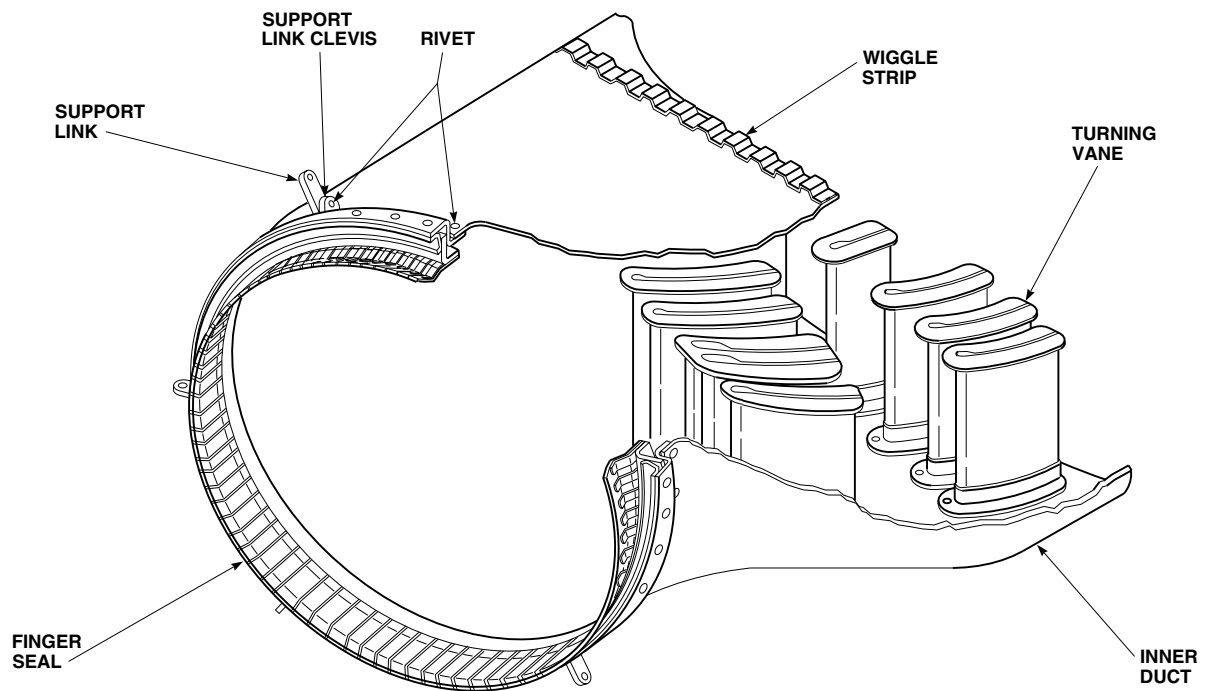


Figure 1. Replace HIRSS Suppressor Core and Baffle.

INSPECT - Continued



STAGE 1 OUTER DUCT AND
LINE OF SIGHT SHIELDS

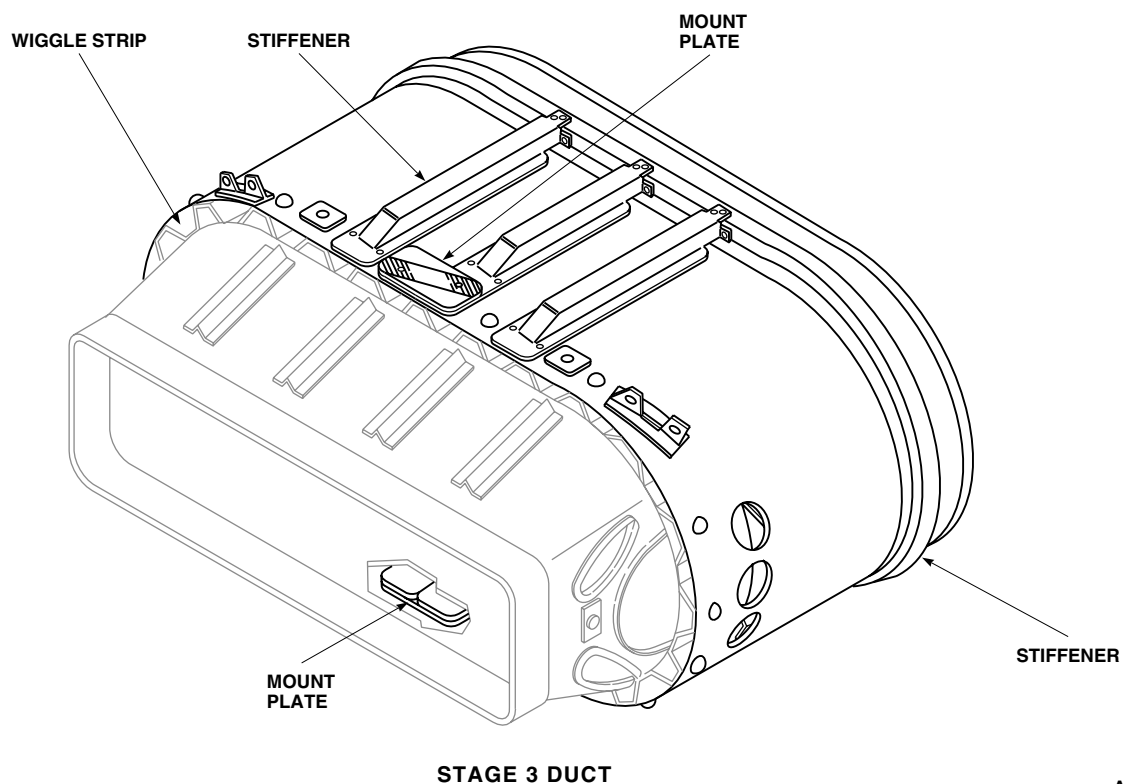
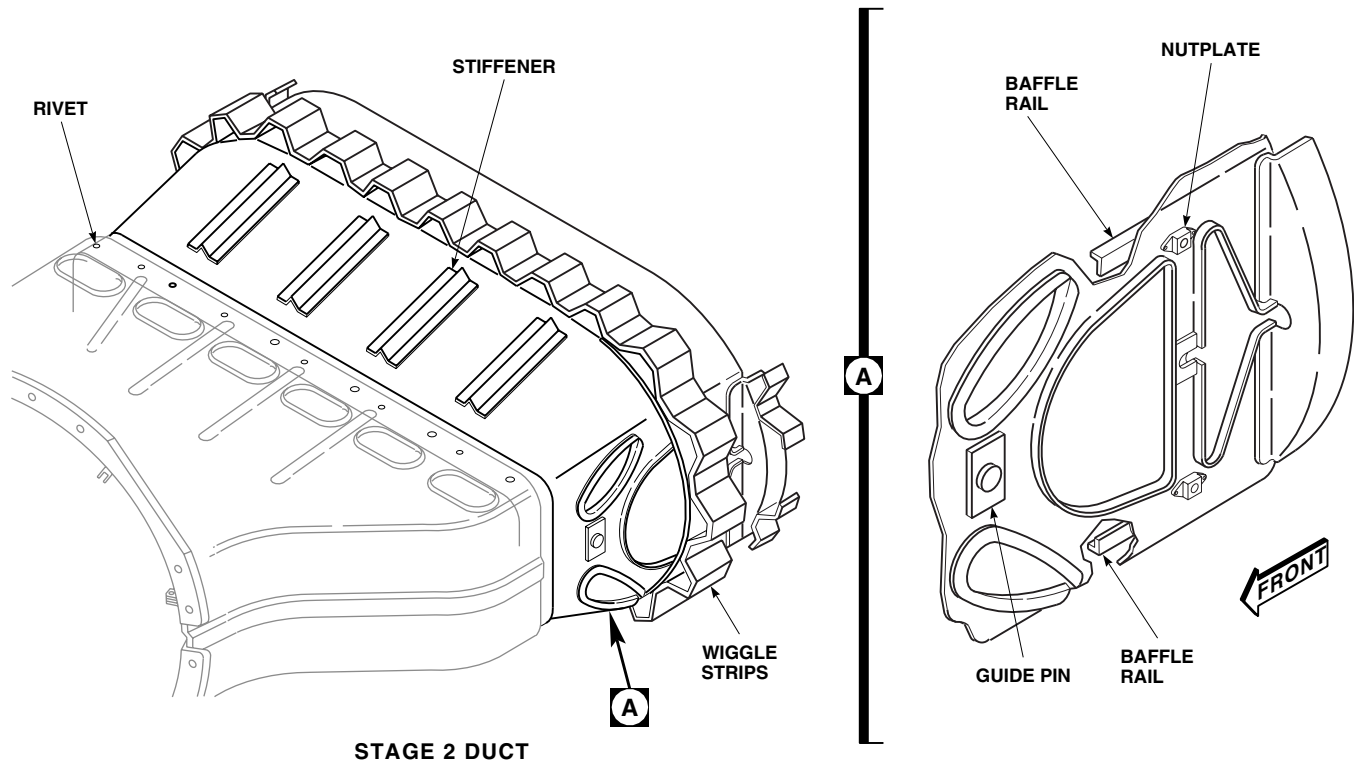


STAGE 1 INNER DUCT

AK1268_1
SA

Figure 2. Inspect HIRSS Suppressor Core. (Sheet 1 of 2)

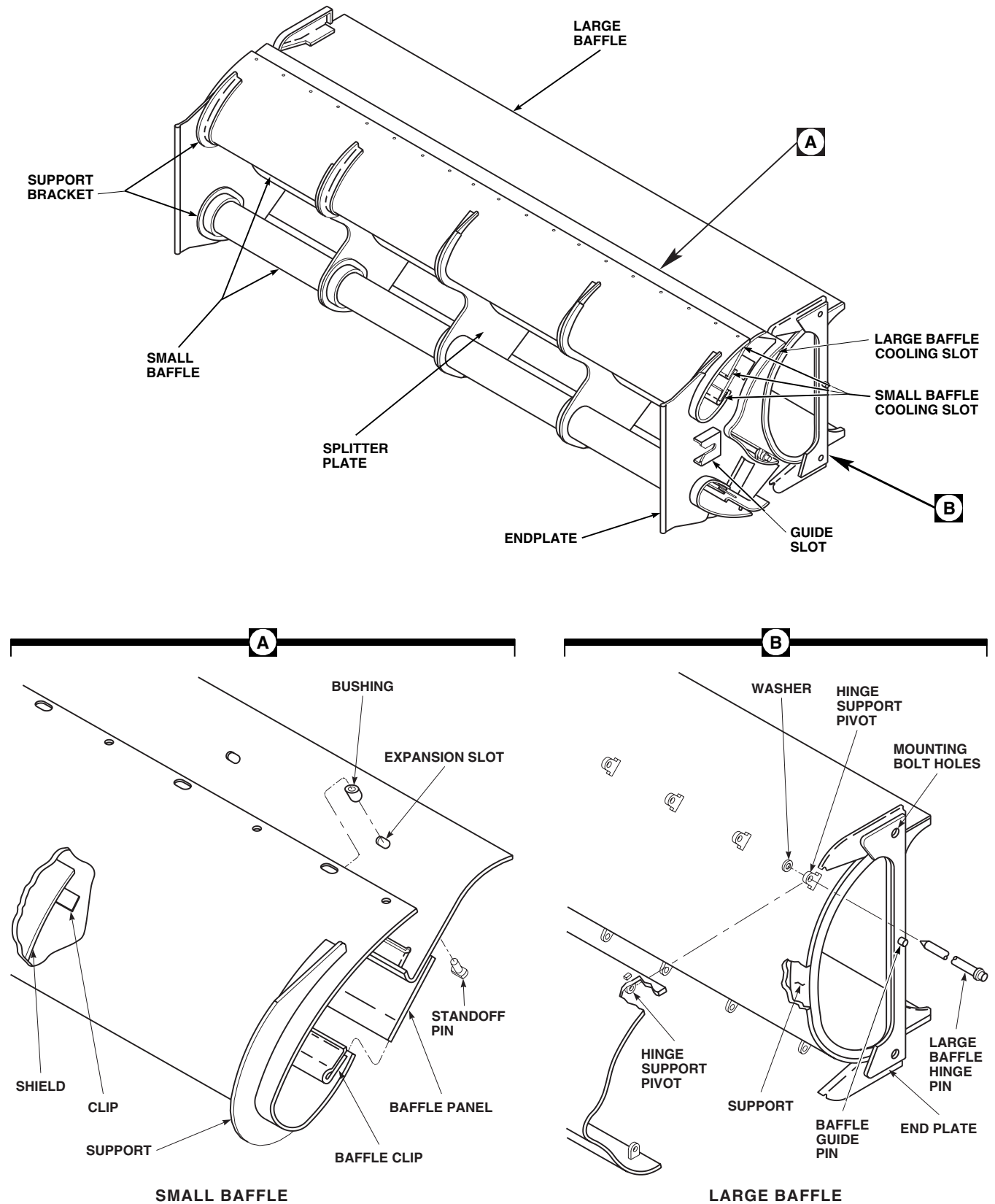
INSPECT - Continued



AK1268_2
SA

Figure 2. Inspect HIRSS Suppressor Core. (Sheet 2 of 2)

INSPECT - Continued



AK1269
SA

Figure 3. Inspect HIRSS Baffle.

INSPECT - Continued

Table 1. HIRSS Suppressor Core Inspection.

COMPONENT	TYPE AND SIZE	REPAIR ACTION
Overall Assy	Discoloration	
	- - Any amount.	No repair required.
	Nicks	
	- - Any amount.	No repair required.
	Scratches	
	- - Any amount.	No repair required.
	Dents	
	- - Up to 2 inches in diameter and 1/4-inch deep.	No repair required. Touchup emissive coating, as required (WP 0497 00).
	- - Dents larger than above limits.	Straighten dents by lightly tapping with plastic mallet and if necessary, touch up emissive coating (WP 0497 00). If dents cannot be straightened, replace core.
	Cracks (see note)	
	- - Within 2 inches of mounts or stage-to-stage attachment locations, except Stage 2 duct.	Unacceptable. Replace core.
	Corrosion	
	- - Any amount that does not penetrate component.	No repair required.
	- - Any corrosion which causes holes or breaks in component.	Unacceptable. Replace core.
	Warping	
	- - With baffle installed, view core from rear to see if any first stage inner duct surface is visible around line-of-sight shields.	Unacceptable. Replace core.
	- - With strong light source placed in first stage duct cavity and baffle installed behind light, view core from rear to see if any direct light is visible from first stage inner duct.	Unacceptable. Replace core.

INSPECT - Continued

Table 1. HIRSS Suppressor Core Inspection. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION
Bulkhead	Missing or damaged emissive coating	
	- - Up to 2 square inches.	No repair required.
	- - More than 2 square inches.	Touch up emissive coating (WP 0497 00).
	Loose rivets	
	- - Up to 10% of rivets in any joint other than nutplates.	No repair required.
	- - Greater than above limit or any nut-plate rivets.	Replace rivets.
	Missing rivets	
	- - Any rivets.	Replace rivets.
	Cracks (see note)	
	- - Within 2 inches of bulkhead where it attaches to support flange on core.	Unacceptable. Replace bulkhead.
Finger Seals	- - In all other areas, any number of cracks up to 1 inch long which do not intersect or will not cause a piece to break off bulkhead.	No repair required.
	Cracks (see note)	
	- - Breaking into rivet holes in AMS 5510 bracket or bracket holder.	Unacceptable. Replace Inlet Seal (WP 0522 00).
	- - In all other areas of bracket, any number of cracks up to 0.50-inch long which do not intersect or will not cause a piece to break off bracket.	No repair required.
	- - Any crack greater than 0.25-inch long in INCO 718 finger seal and brace.	Unacceptable. Replace Inlet Seal (WP 0522 00).
	- - Cracks in above component not more than 0.25-inch long.	No repair required.
	Bends or deformation	

INSPECT - Continued

Table 1. HIRSS Suppressor Core Inspection. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION
Stage 1 inner duct assy	- - Fingers bent or deformed out of plane ± 0.218 -inch.	Bend back into shape. Inspect for cracks. If cracked, replace Inlet Seal (WP 0522 00).
	- - Greater than above limits.	Unacceptable. Replace Inlet Seal (WP 0522 00).
	Missing seal fingers	Up to 3 fingers may be missing provided spacing between missing fingers is greater than 60 degrees.
	Cracks (see note)	
	- - In turning vanes, three cracks per vane up to 0.50-inch long not in same axial plane and do not intersect.	No repair required.
	- - In any other area of duct skin or turning vane supports, any number of cracks up to 0.50-inch long which do not intersect or will not cause a piece to break off duct.	No repair required.
	- - Greater than above limits or breaking into forward link supports.	Unacceptable. Replace core.
	Vane bending or distortion	
	- - Up to 0.50-inch as long as front/rear and/or up/down movement of front vanes is not more than 0.25-inch or bending/twisting is not enough to disconnect vanes.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace core.
	Inner duct support link	
	- - Check rivet holes for elongation.	None allowed. Replace support link.
	- - Check rivet hole diameters 0.158 to 0.160-inches.	Replace support link if outside diameter limits.
	Loose rivets	
	- - Check for gaps at rivet and between mating surfaces.	None allowed. Replace rivet or restrike cold formed rivet.
Stage 1 outer duct assy	Cracks (see note)	

INSPECT - Continued

Table 1. HIRSS Suppressor Core Inspection. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION
Stage 2 duct	- - In duct skin or line-of-sight shields, any number of cracks up to 1 inch long that do not intersect or will not cause a piece to break off component.	No repair required.
	- - In wiggle strips, two cracks per leg up to 0.25-inch long which do not intersect.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace core.
	- - Within 2 inches of bulkhead support flange or breaking into forward link supports or rivet holes.	Unacceptable. Replace core.
	Cracks (see note)	
	- - Breaking into baffle guide pin or slot.	Unacceptable. Replace core.
	- - Within 2 inches of mounts or stage-to-stage attachment locations at 8 or more locations with a length of 2 inches each or longer, within 1 1/4-inches of another crack with individual cracks having no more than one connecting perpendicular segment not more than 1 1/2-inches long.	Unacceptable. Replace core.
	- - In all other areas, any number of cracks up to 1 inch long which do not intersect or will not cause a piece to break off duct.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace core.
	Missing or damaged emissive coating	
Baffle guide pin	- - Up to 6 square inches.	No repair required.
	- - More than 6 square inches.	Touch up emissive coating (WP 0497 00).
	Wear	
	- - Diameter of pin not less than 0.218-inch.	No repair required.
	- - Diameter of pin less than 0.218-inch.	Unacceptable. Replace core.

INSPECT - Continued**Table 1. HIRSS Suppressor Core Inspection. - Continued**

COMPONENT	TYPE AND SIZE	REPAIR ACTION
Baffle rails	Missing or loose	Unacceptable. Replace core.
	Bends or distortion	Unacceptable. Replace core.
Second/Third stage wiggle strips	Cracks (see note)	
	- - Two per leg, not more than 0.25-inch long, which do not intersect.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace core.
	Distortion	
Stage 3 duct	- - Any amount as long as gap dimension between second and third stage remains same.	No repair required.
	- - Any distortion which reduces gap.	Unacceptable. Replace core.
	Cracks (see note)	
	- - Any size against or breaking into mount plates or brackets.	Unacceptable. Replace core.
	- - In all other areas, any number of cracks up to 1 inch long that do not intersect or will not cause a piece to break off duct.	No repair required.
NOTE:		
Fluorescent penetrant inspect, Type I, Method C (TM 1-1520-265-23).		

Table 2. HIRSS Baffle Inspection.

COMPONENT	TYPE AND SIZE	REPAIR ACTION
Baffle guide pin	Wear	
	- - Diameter of pin not less than 0.218-inch.	No repair required.
	- - Diameter of pin less than 0.218-inch.	Unacceptable. Replace baffle.
	Missing or loose	Unacceptable. Replace baffle.

INSPECT - Continued

Table 2. HIRSS Baffle Inspection. - Continued

COMPONENT	TYPE AND SIZE	REPAIR ACTION
Baffle guide slot	Cracks (see note) or bends	Unacceptable. Replace baffle.
	Wear	
	- - Width of slot not more than 0.395-inch.	No repair required.
	- - Greater than above limit.	Unacceptable. Replace baffle.
Endplates	Cracks (see note)	
	- - Breaking into mounting bolt holes.	Unacceptable. Replace baffle.
	- - All other areas, any number of cracks not more than 0.50-inch long.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace baffle.
	Bending or twisting	
	- - Any amount as long as baffle fits properly in rails, guide pin slot, and forward fish mouth seal on stage 2 duct.	No repair required.
Splitter plates and support brackets	- - Greater than above limits.	Unacceptable. Replace baffle.
	Cracks (see note)	
	- - In parent metal, any number of cracks up to 0.50-inch long which do not intersect or will not cause a piece to break off plates or brackets.	No repair required.
	- - In braze, any cracks not more than 1 inch long.	No repair required.
Baffle cooling slots (six in small baffles and two in large baffles)	- - Greater than above limits.	Unacceptable. Replace baffle.
	Distortion	
	- - Any amount as long as gap dimension of slot remains same.	No repair required.

INSPECT - Continued**Table 2. HIRSS Baffle Inspection. - Continued**

COMPONENT	TYPE AND SIZE	REPAIR ACTION
	- - Any distortion which reduces gap.	Unacceptable. Replace baffle.
Small baffle cooling slot standoff pins	Loose, bent, missing	Unacceptable. Replace baffle.
Baffles	Cracks (see note)	
	- - Breaking into small baffle standoff pins or expansion slots.	Unacceptable. Replace baffle.
	- - Breaking into large baffle hinge support pivots.	Unacceptable. Replace baffle.
	- - All other areas on baffles, any number of cracks up to 1 inch long which do not intersect or will not cause a piece to break off baffles.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace baffle.
Large baffle hinge support pivots	Braze	
	- - Up to 20% of any joint may be damaged or broken.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace baffle.
Overall baffle assy	Distortion	
	- - Up to 0.25-inch in any area beyond original baffle dimensions.	No repair required.
	- - Greater than above limits.	Unacceptable. Replace baffle.
NOTE:		
Fluorescent penetrant inspect, Type I, Method C (TM 1-1520-265-23).		

INSTALL

1. **QA** Install bulkhead on suppressor core bulkhead support flange with screws, washers, and nuts (Figure 1, Detail A). **TORQUE NUTS TO 43 - 47 INCH-POUNDS.**
2. **QA** Install IPS collar and IPS blower duct on right bulkhead, using bolts and washers.

INSTALL - Continued**NOTE**

If helicopter is not operated in a hostile environment, it is not necessary to install baffle.

3. Apply a light coat of antiseize compound, Item 49, WP 1803 00, to threads of baffle installation bolts.
4. **QA** Slide baffle into core. Install bolts and washers (Figure 1). **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
5. Install HIRSS nacelle fairing (WP 0493 00).
6. Install HIRSS module (WP 0489 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****HIRSS EXTENDER**

This WP supersedes WP 0495 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101
Torque Wrench, 30 - 200 in. lbs.

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1520-265-23
WP 1803 00

Material/Parts

Sealing Compound, Item 274, WP 1803 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)

REPLACE HIRSS EXTENDER**Remove**

1. Remove bolts and washers holding inboard support struts to rear panel and outboard support strut to fuselage (Figure 1, Sheet 2, Detail B and Detail C).
2. With an assistant supporting the extender, remove bolts, washers, and spacers, holding the extenders two upper support channels and one lower support channel to nacelle fairing. Remove extender (Figure 1, Sheet 1, Detail A).
3. If required, remove bolts, washers, nuts, and remove ends of struts from angles on lower side of extender (Figure 1, Sheet 2, Detail B and Detail C).

Inspect

Table 1 defines allowable damage and wear to extender.

REPLACE HIRSS EXTENDER - Continued

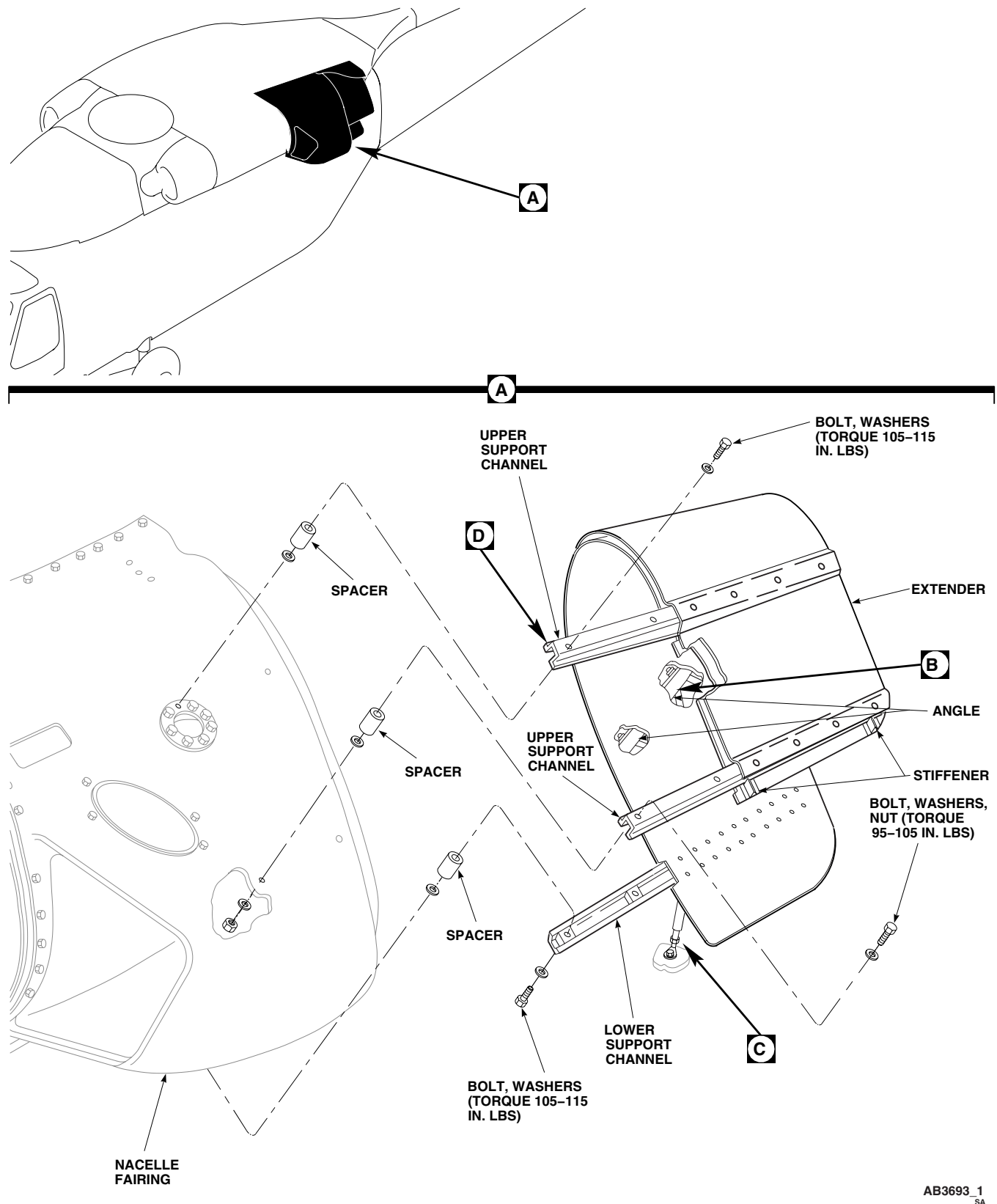


Figure 1. Replace HIRSS Extender. (Sheet 1 of 2)

REPLACE HIRSS EXTENDER - Continued

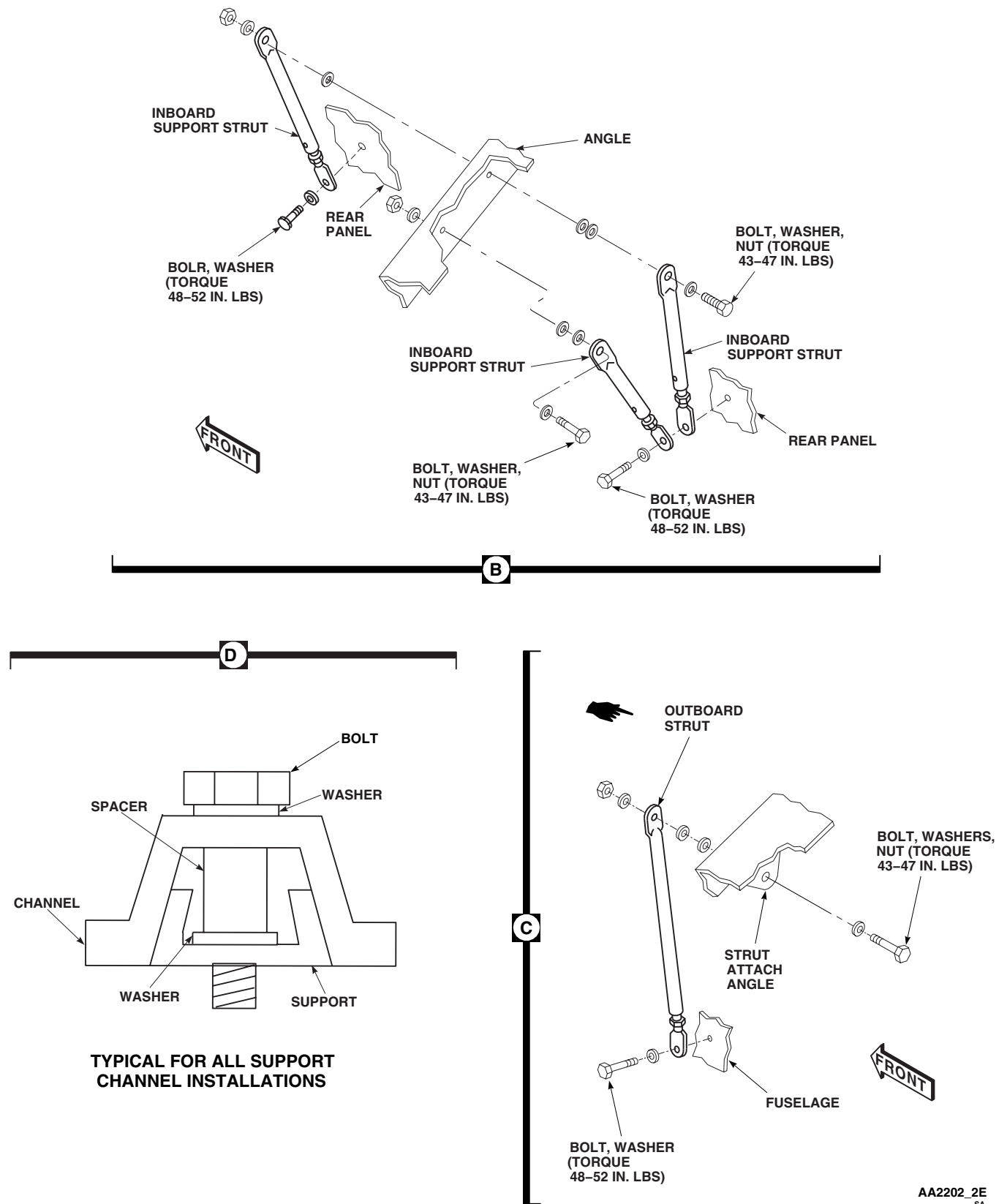


Figure 1. Replace HIRSS Extender. (Sheet 2 of 2)

REPLACE HIRSS EXTENDER - Continued

Table 1. HIRSS Extender Inspection.

COMPONENT	TYPE AND SIZE	REPAIR ACTION
Extender	Cracks (see note)	
	- - In strut attach angles.	Repair strut attach angles TM 1-1500-204-23. Refer to, REPAIR HIRSS EXTENDER, in this work package.
	- - In forward lower surface at lower support channel attachment point less than 1/2 the width of reinforcing doubler.	Install reinforcing doubler. Refer to, REPAIR HIRSS EXTENDER, in this work package.
	- - More than 1/2 the width of reinforcing doubler.	Unacceptable. Replace extender.
	- - In support beams, struts, within 3 inches of support beams, strut attach angle, or any lower surface angle or channel.	Unacceptable. Replace extender.
	- - All other locations any amount not over 1 inch in length, that does not intersect or will not cause a piece of extender to fall off.	Repair weld (see note).
	Loose, broken, or missing fasteners	
	- - None allowed in support beams or struts.	Unacceptable. Replace extender.
	- - All other areas up to 15% of fasteners damaged, with at least two good fasteners between damaged fasteners.	Replace fastener.
NOTE:		
Fluorescent penetrant inspect, Type I, Method C (TM 1-1520-265-23).		

Install

1. **QA** Slide exhaust extender over nacelle fairing and loosely attach upper and lower support channels with bolts, washers, spacers, and nut (Figure 1, Sheet 1, Detail A).

NOTE

One washer may be installed between strut assemblies and the rear panels/fuselage to prevent chafing between the HIRSS fairing and strut.

2. **QA** Connect lower end of support strut assemblies to rear panel and fuselage with bolts, and washers. Do not tighten now (Figure 1, Sheet 2, Detail B and Detail C).

REPLACE HIRSS EXTENDER - Continued

3. **QA** Connect upper end of support strut assemblies with bolts, washers, and nuts. Do not tighten now (Figure 1, Sheet 2, Detail B and Detail C).
4. **TORQUE UPPER AND LOWER SUPPORT CHANNEL BOLTS TO 105 - 115 INCH-POUNDS (Figure 1, Sheet 1, Detail A).**
5. **TORQUE UPPER SUPPORT CHANNEL NUT TO 95 - 105 INCH-POUNDS.**

NOTE

- Adjust strut length using adjustable rod ends as necessary to allow strut installation.
 - Inboard strut, 70308-03611-050, may not install properly to the inboard support strut angle due to the radius block being in the way. The inboard strut may be trimmed a maximum of 3/16-inch in order to obtain proper installation. Trim the minimum amount necessary to obtain proper installation.
6. Adjust length of inboard and outboard support struts to allow for proper installation of struts. **TORQUE STRUT JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Make sure there is clearance between exhaust extender and suppressor assembly (Figure 2).
 7. **TORQUE UPPER SUPPORT STRUT ASSEMBLY FITTING NUTS TO 43 - 47 INCH-POUNDS (Figure 1, Sheet 2, Detail B and Detail C).**
 8. **TORQUE LOWER SUPPORT STRUT ASSEMBLY FITTING BOLTS TO 48 - 52 INCH-POUND (Figure 1, Sheet 2, Detail B and Detail C).**
 9. Seal bolts, washers and struts at fuselage attachment point with sealing compound, Item 274, WP 1803 00.
 10. Make sure area is clean and free of foreign material.

REPAIR HIRSS EXTENDER**Remove**

Remove extender from HIRSS. Refer to, REPLACE HIRSS EXTENDER, in this work package.

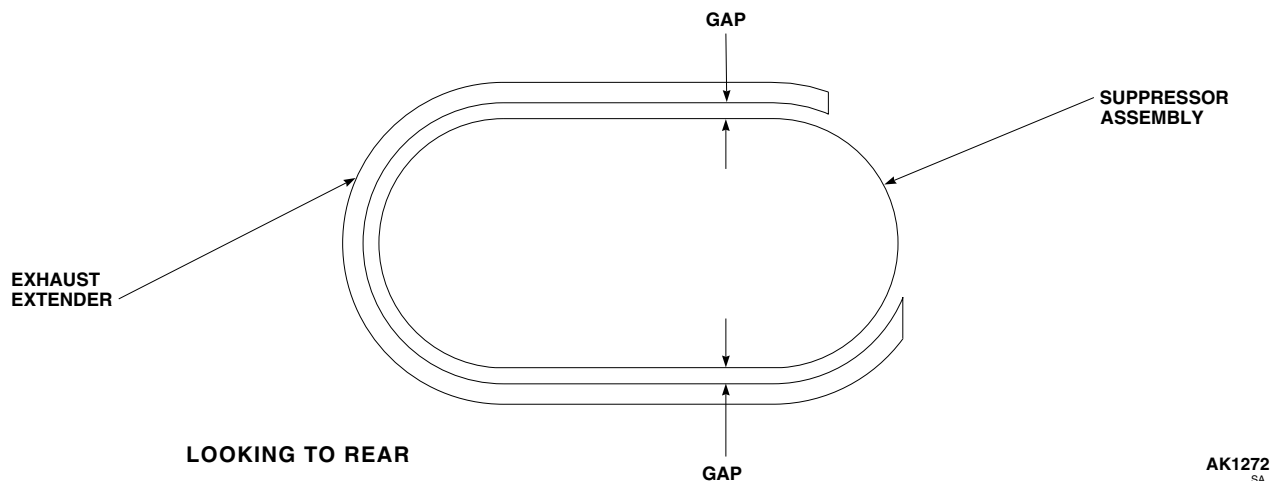


Figure 2. Adjust Exhaust Extender.

REPAIR HIRSS EXTENDER - Continued**Repair****NOTE**

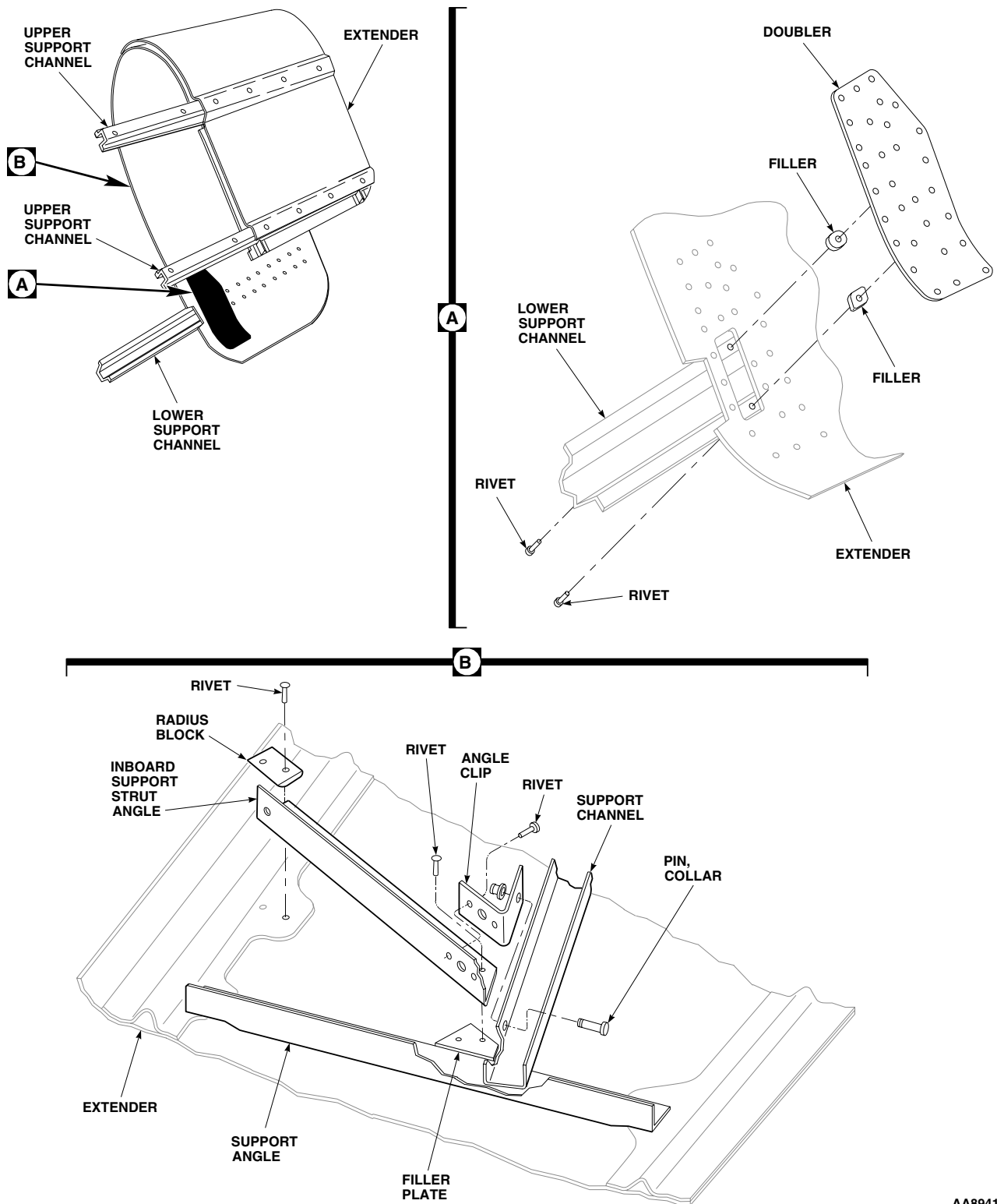
Repair of extender is limited to replacement of the inboard support strut angle and the addition of a reinforcing doubler for the lower support channel.

1. Evaluate damage to extender. Refer to, REPLACE HIRSS EXTENDER, in this work package.
2. Replace the inboard support strut angle as follows:
 - a. Drill out rivets attaching inboard support strut angle to extender (TM 1-1500-204-23).
 - b. Install replacement inboard support strut angle, radius block, filler plate, and angle clip and secure to extender with pins, collars, and rivets (TM 1-1500-204-23) (Figure 3, Detail B).
3. Add reinforcing doubler as follows:
 - a. Drill out rivets attaching forward end of lower support angle to extender (TM 1-1500-204-23) (Figure 3, Detail A).
 - b. Install doubler and fillers on extender and secure with rivets (TM 1-1500-204-23).

Install

Install extender on HIRSS. Refer to, REPLACE HIRSS EXTENDER, in this work package.

REPAIR HIRSS EXTENDER - Continued



AA8941
SA

Figure 3. Repair HIRSS Extender.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

HIRSS BULKHEAD REPAIR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02

References

TM 1-1500-204-23

WP 0489 00

WP 0494 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

REMOVE

Remove firewall support angle from bulkhead (WP 0489 00).

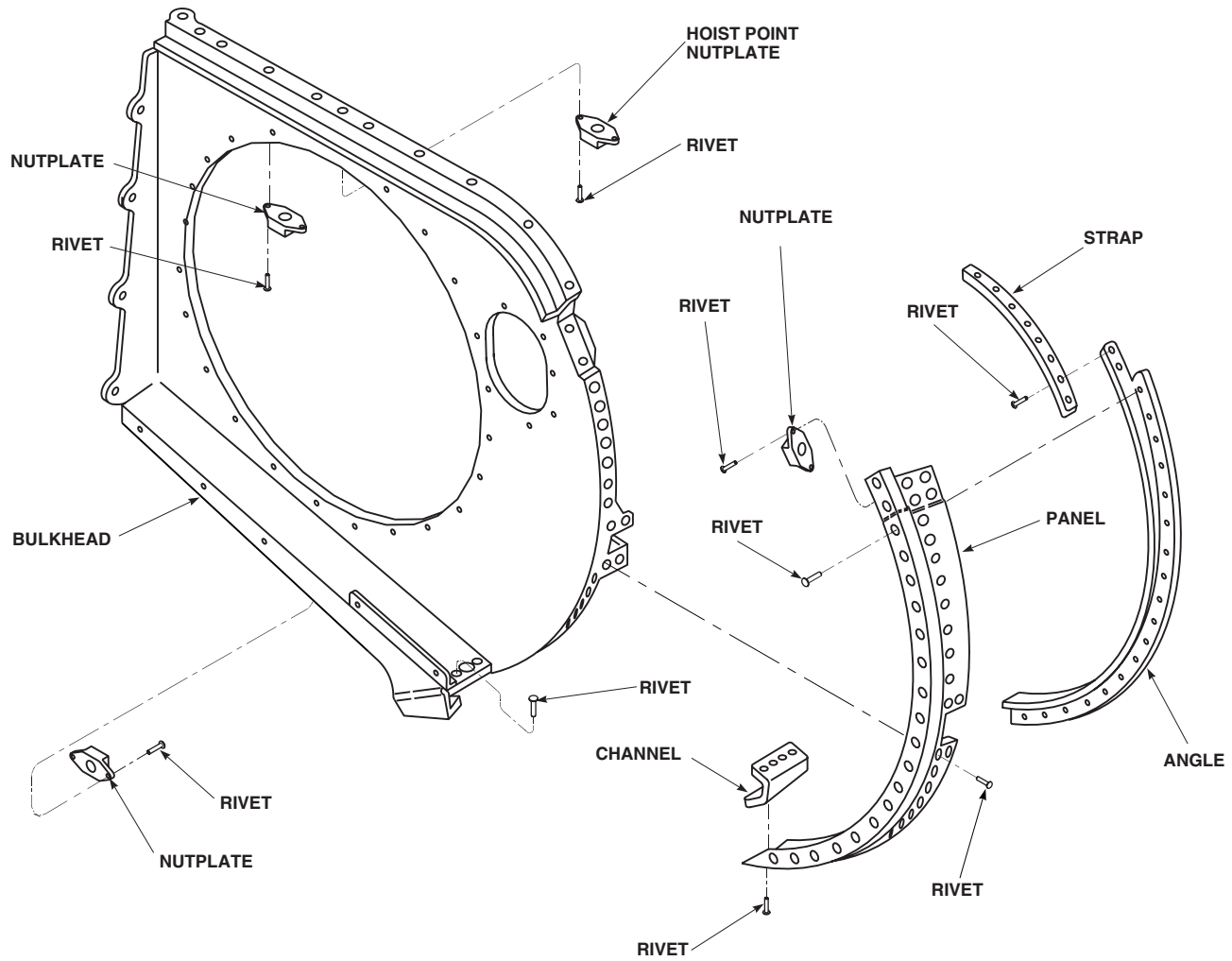
REPAIR**NOTE**

Bulkhead repair is limited to replacement of nutplates and riveted components.

1. Evaluate damage to bulkhead (WP 0494 00).
2. Repair or replace components as required (TM 1-1500-204-23) (Figure 1).
3. Replace loose, broken, or missing rivets as required (TM 1-1500-204-23).

INSTALLInstall firewall support angle on bulkhead (WP 0489 00). **QA Check**

INSTALL - Continued



AK1275
SA

Figure 1. Repair HIRSS Bulkhead.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

HIRSS EMISSIVE COATING REPAIR

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Heat Gun, LFT500

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

Material/Parts

Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 78, WP 1803 00
Paint, Heat Resisting, Item 206, WP 1803 00

References

WP 0494 00
WP 1803 00

REPAIR**NOTE**

- Coating repair should be last item done after all other maintenance or repairs are done to suppressor, either on or off helicopter.
- Inspection areas and damage limits for emissive coating repairs are listed in Table 1, WP 0494 00.

1. Examine coating for areas of erosion, looseness, cracking, or flaking.
2. Remove damaged coating by wire brushing or careful scraping. Using abrasive cloth, Item 78, WP 1803 00, clean area down to parent metal, and feather edges around area to be recoated.

CAUTION

Only cleaning compound solvent, Item 71, WP 1803 00, may be used. Solvent shall be new and clean. Do not use other cleaning compounds.

NOTE

- Allow at least 30 minutes for solvent to evaporate.
 - Be sure that all oily residue is removed by solvent wash. Repeated solvent wash may be required.
3. Clean areas to be coated by washing with cleaning compound solvent, Item 71, WP 1803 00.
 4. Mix heat resisting paint, Item 206, WP 1803 00, per instructions on heat resisting paint kit. Heat resisting paint must be used within 24 hours after mixing.
 5. Brush on a smooth, wet coat using broad, even strokes.
 6. Allow heat resisting paint to air-dry until it becomes matte gray.
 7. Cure heat resisting paint using heat gun at 600°F(316°C) for at least 30 minutes.

REPAIR - Continued**NOTE**

Complete curing of heat resisting paint can be done by running engine at idle speed for 5 minutes.

8. Inspect heat resisting paint after curing. Repeat application procedure if heat resisting paint is blistering, cracking, flaking, or if coverage is incomplete.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****LOAD-DEMAND ROTARY INPUT**

This WP supersedes WP 0498 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Rod, 0.125-inch diameter
Torque Wrench, 30 - 200 in. lbs.

References

TM 1-1520-237-MTF
WP 0479 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

NOTE

The following procedures apply to both No. 1 and No. 2 engine load-demand rotary inputs.

REMOVE

1. Open engine cowl.
2. Open main rotor pylon sliding cover.
3. Loosen coupling nut holding engine load-demand push/pull cable to adapter (Figure 1, Detail C).
4. Disconnect load-demand rotary input from engine hydromechanical unit (Figure 1, Detail B).

WARNING

Injury to personnel and damage to equipment will result if engine load-demand rotary input spring is not compressed and released carefully. Slowly compress and release spring to avoid sudden release of spring tension.

5. Compress spring on engine load-demand rotary input by inserting a 3/8-inch hex wrench through splines in front end of rotary barrel. Push against front of input drive until rear of input drive extends beyond adapter.
6. Unhook rear of input drive from engine load-demand push/pull cable rod end (Figure 1, Detail C). Slowly release pressure on hex wrench until input drive is in full front position.
7. Remove retaining nut from rotary input.

INSTALL

1. **QA** Install retaining nut on rotary input.

INSTALL - Continued

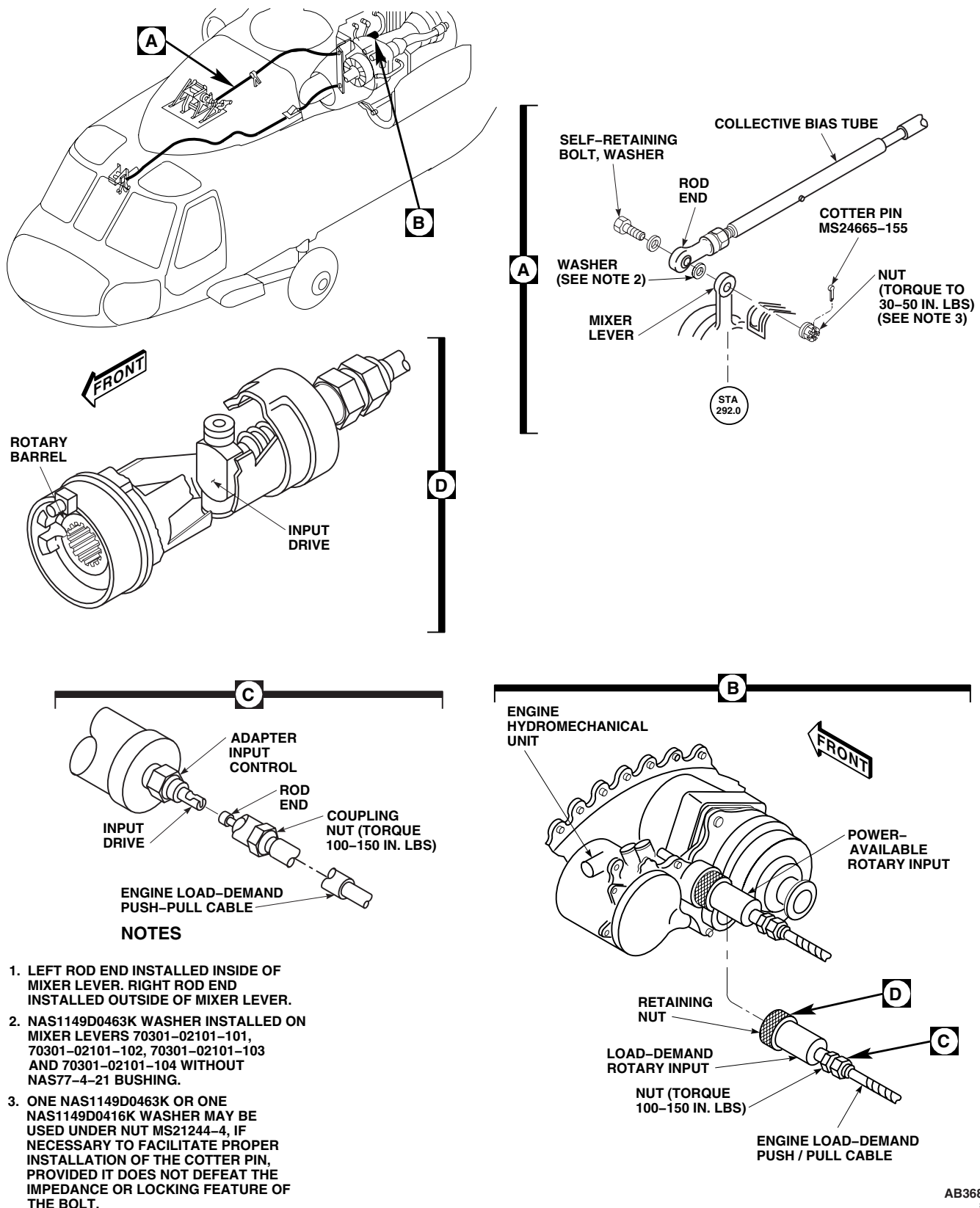
AB3687
SA

Figure 1. Replace Load-Demand Rotary Input.

INSTALL - Continued**WARNING**

Injury to personnel and damage to equipment will result if engine load-demand rotary input spring is not compressed and released carefully. Slowly compress and release spring to avoid sudden release of spring tension.

2. Compress spring of engine load-demand rotary input by inserting a 3/8-inch hex wrench through splines in front end of rotary barrel. Push against front of input drive until rear of input drive extends beyond adapter.

NOTE

When connecting push/pull cable to rotary input, make sure internal connection is made.

3. **QA** Hook engine load-demand push/pull cable rod end on exposed end of input drive (Figure 1, Detail C). Slowly release pressure on hex wrench, and remove hex wrench from rotary adapter.
4. **QA** Thread coupling nut on adapter. Do not torque nut connecting engine load-demand push/pull cable to rotary input now.
5. Connect external electrical or hydraulic power.

CAUTION

Damage to collective bias tube spring pin will result if collective stick is moved with push/pull cable attached to mixer lever and drill rod installed in rotary input. Do not move collective stick with cable attached and drill rod installed.

6. Turn on backup hydraulic pump and hold collective stick in low stick position.
7. Make sure 0.125-inch diameter drill rod in rotary input slides freely in and out of rigging hole. If required, readjust collective bias tube rod end by doing this:
 - a. Loosen jamnut on collective bias tube rod end (Figure 1, Detail A).
 - b. Remove bolt, washers, and nut holding collective bias tube rod end to mixer lever.
 - c. Turn collective bias tube rod end a half-turn at a time until 0.125-inch diameter drill rod slides freely in and out of rigging hole.
 - d. **QA** Reinstall collective bias tube rod end on mixer with self-retaining bolt, washer(s), and nut (Figure 1, Detail A).
 - e. **QA** Tighten jamnut on collective bias tube rod end. **TIGHTEN JAMNUT 1/6 - 1/3 TURN BEYOND POINT WHERE SHARP RISE IN TORQUE IS FELT.**
 - f. Recheck that 0.125-inch diameter drill rod slides freely in and out of rigging hole when collective stick is in low position.

NOTE

Do the next step only if collective bias tube rod end has been removed from mixer lever.

8. **QA TORQUE NUT THAT SECURES COLLECTIVE BIAS TUBE ROD END TO MIXER LEVER TO 30 - 50 INCH-POUNDS.** Install cotter pin (Figure 1, Detail A).
9. Remove 0.125-inch diameter drill rod from load-demand rotary input.
10. **QA** Connect load-demand rotary input to engine hydromechanical unit (Figure 1, Detail B).
11. **QA** Make sure that rotary input alignment pin engages slot on engine hydromechanical unit. Tighten retaining nut

INSTALL - Continued

until it bottoms in support bracket. Slight axial/radial play is normal. **TORQUE NUT CONNECTING LOAD-DEMAND PUSH/PULL CABLE TO ROTARY INPUT TO 100 - 150 INCH-POUNDS.**

12. Do an engine load-demand push/pull cable friction test (WP 0479 00).
13. Disconnect external electrical or hydraulic power.
14. Make sure area is clean and free of foreign material.
15. Close main rotor pylon sliding cover and latch.
16. Close engine cowling and latch.
17. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
POWER-AVAILABLE ROTARY INPUT

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Rod, 0.125-inch diameter
Torque Wrench, 30 - 200 in. lbs.

References

TM 1-1520-237-MTF
WP 0512 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

NOTE

The following procedures apply to both No. 1 and No. 2 engine power-available rotary inputs.

REMOVE

1. Open engine cowling.
2. Place quadrant ENG POWER CONT lever OFF.
3. Loosen retaining nut and disconnect power-available rotary input from engine hydromechanical unit (Figure 1, Detail A).
4. Loosen nut and disconnect power-available rotary input from engine power-available push/pull cable. Remove retaining nut from rotary input.

INSTALL**NOTE**

When connecting push/pull cable to rotary input, make sure internal connection is made.

1. **QA** Install retaining nut on rotary input and connect rotary input to engine power-available push/pull cable. Do not torque nut connecting power-available push/pull cable to rotary input now (Figure 1, Detail A).
2. Place ENG POWER CONT lever on quadrant in IDLE detent.
3. **QA** Insert 0.125-inch diameter drill rod in rotary input. If rod can be inserted, remove rod and attach rotary input to engine. **TORQUE NUT CONNECTING POWER-AVAILABLE PUSH/PULL CABLE TO ROTARY INPUT TO 100 - 150 INCH-POUNDS.**
4. If rod cannot be inserted, rig engine power-available system (WP 0512 00).
5. When all rigging has been completed, place quadrant ENG POWER CONT lever OFF.
6. Make sure area is clean and free of foreign material.
7. Close engine cowling and latch.

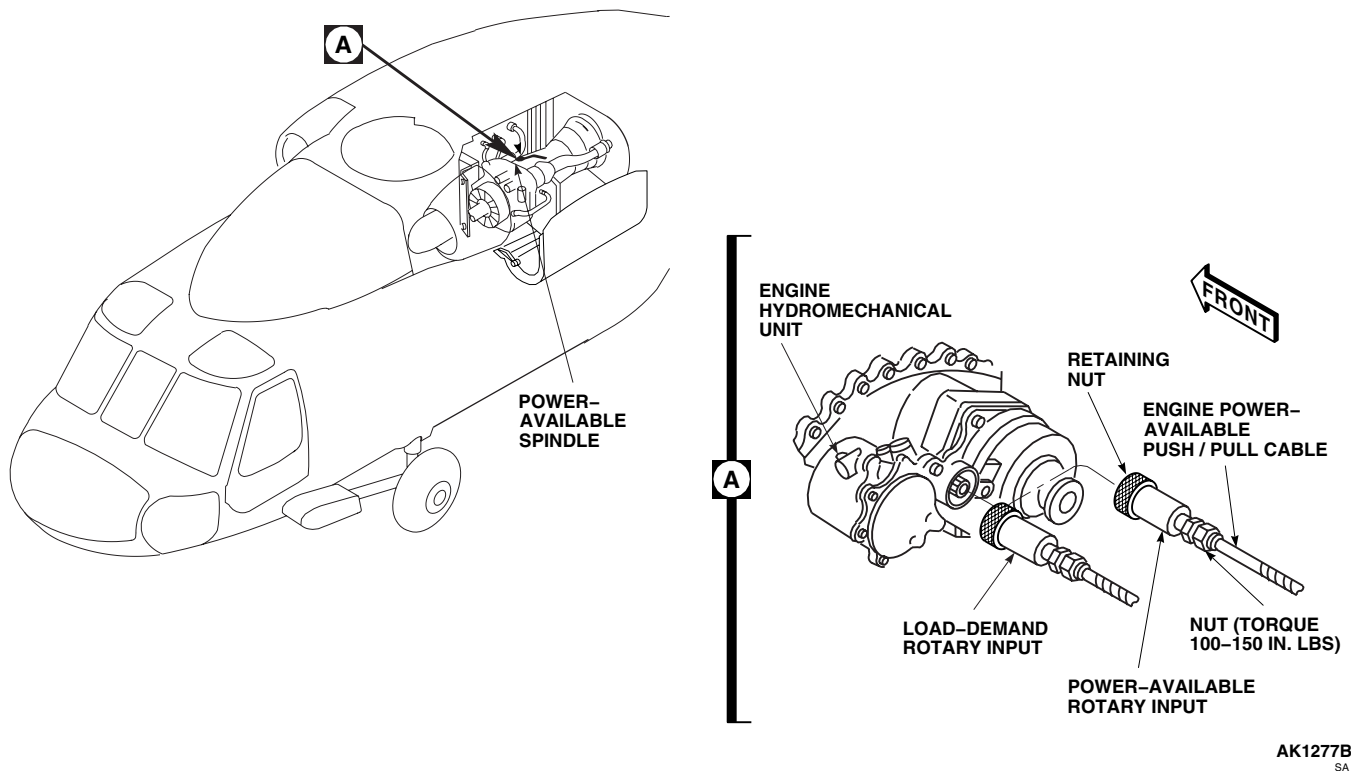
INSTALL - Continued

Figure 1. Replace Power-Available Rotary Input.

8. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE SPEED CONTROL POTENTIOMETER

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

WP 0076 00

WP 0258 00

WP 1703 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Remove soundproofing panels as required to access engine speed control potentiometer (WP 0258 00).
3. Disconnect electrical connector, P901, from potentiometer (Figure 1, Detail A).
4. Remove screws, washers and lockwashers securing potentiometer to supports. Remove potentiometer from supports.

INSTALL

1. Turn off all electrical power.
2. **QA** Secure potentiometer to supports using screws, washers, and lockwashers (Figure 1, Detail A).
3. Inspect and treat electrical connector (WP 1703 00).
4. **QA** Connect electrical connector, P901, to potentiometer.
5. Make sure area is clean and free of foreign material.
6. Install soundproofing panel in overhead cabin (WP 0258 00).
7. Do an operational check of engine speed trim system (WP 0076 00).

INSTALL - Continued

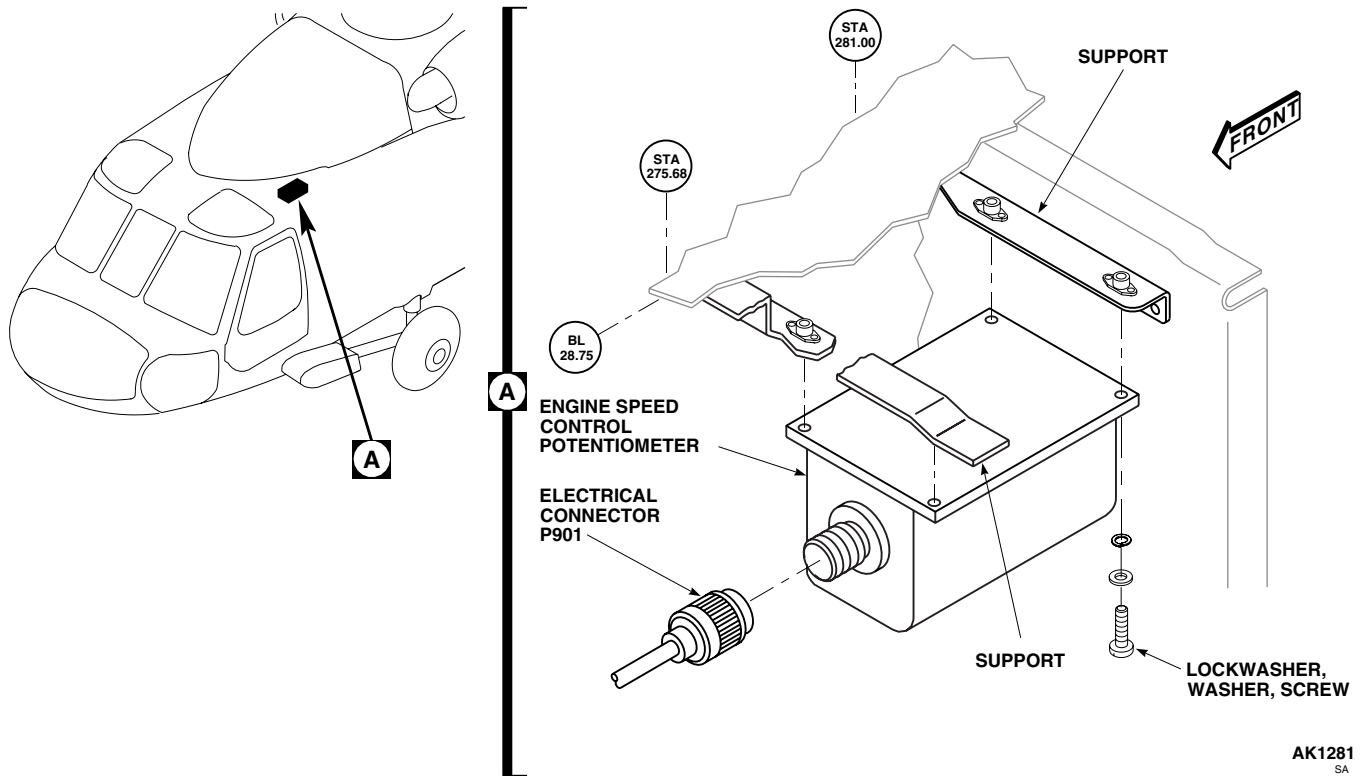


Figure 1. Replace Engine Speed Control Potentiometer.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****ENGINE CONTROL QUADRANT**

This WP supersedes WP 0501 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Rod, 0.125-inch diameter
Torque Wrench, 30 - 200 in. lbs.

References

WP 0512 00
WP 0526 00
WP 1703 00
WP 1803 00

Material/Parts

Sealing Compound, Item 273, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

The following procedures apply to both No. 1 quadrant and No. 2 quadrant.

1. Turn off all electrical and hydraulic power.
2. Move fuel selector levers forward.
3. Unlock fastener studs holding upper console to cockpit overhead, and hinge down upper console.
4. Remove screws and bracket at rear of quadrant (Figure 1, Detail A).
5. Remove screws and washers from center cover. Remove cover before disconnecting electrical connector from receptacle on right side cover.
6. Remove screws and washers from left side cover and lower cover. Disconnect electrical connector, P394. Remove stopbolt.
7. Remove screws and washers from right side cover and lower cover. Disconnect electrical connectors, P395 and P396. Remove stopbolt.
8. Remove bolts, washers and nuts holding power-available push/pull cable and fuel selector valve push/pull cables to quadrant levers (Figure 1, Detail B).
9. Remove clamp and disconnect electrical connector, P105 (No. 1 quadrant) or P106 (No. 2 quadrant).
10. Remove bolt and washers holding quadrant to cockpit overhead. Remove quadrant.

INSPECT

1. Check all threaded inserts for proper engagement of support and broken or cracked threads.
2. **IF INSERT IS LOOSE, BROKEN OR CRACKED, REPLACE INSERT (WP 0526 00).**

INSPECT - Continued

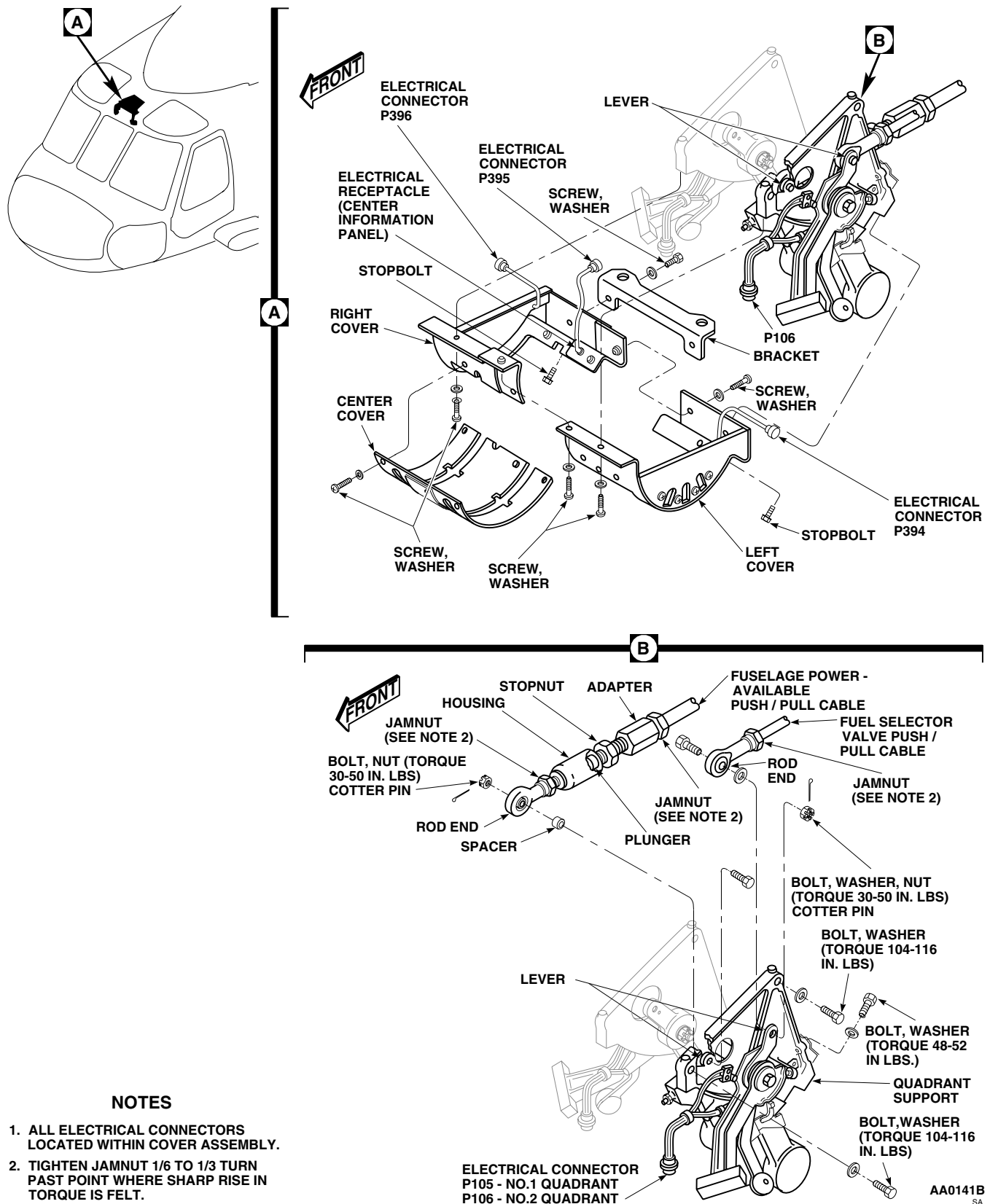


Figure 1. Replace Engine Control Quadrant.

INSTALL

CAUTION

Damage to starter switch actuators will result if ENG POWER CONT lever is moved from FLY or IDLE to OFF, while pressing starter button. Do not press starter button while moving ENG POWER CONT lever.

1. Turn off all electrical and hydraulic power.
2. **QA** Position quadrant on cockpit overhead and install bolts and washers. **TORQUE OUTBOARD FACING BOLTS TO 104 - 116 INCH-POUNDS. TORQUE REAR FACING BOLTS TO 48 - 52 INCH-POUNDS (Figure 1, Detail A).**
3. Inspect and treat electrical connectors (WP 1703 00).
4. **QA** Connect electrical connector, P105 (No. 1 quadrant), or P106 (No. 2 quadrant) (Figure 1, Detail A). Install clamp around connector and secure with screw, washer, and nut.
5. Move ENG POWER CONT lever so rigging holes in lever and quadrant support line up. Insert 0.125-inch diameter drill rod through rigging holes in lever and quadrant support.
6. **QA** Adjust rod end on power-available push/pull control cable until rod end bearing and quadrant ENG POWER CONT lever holes line up. **TIGHTEN JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.** Bolt quadrant lever to control cable rod end. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.
7. **QA** Move fuel selector lever forward. Install fuel selector valve push/pull cable to lever with bolt, washer and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin (Figure 1, Detail B).
8. Rig engine power-available system (WP 0512 00).
9. Make sure area is clean and free of foreign material.
10. Inspect and treat electrical connectors (WP 1703 00).
11. **QA** Connect electrical connectors, J395 and J396. Install right side cover on quadrant with screws and washers. Install stopbolt.
12. Inspect and treat electrical connectors (WP 1703 00).
13. **QA** Connect electrical connector, J394. Install left side cover on quadrant with screws and washers. Install stopbolt.
14. **QA** Insert electrical connector on center cover into receptacle on right side cover, and secure center cover with screws and washers.
15. Adjust secondary stop blocks. Refer to, ADJUST, in this work package.
16. Seal edges of center cover with sealing compound, Item 273, WP 1803 00.
17. **QA** Install bracket at rear of quadrant with screws and washers.
18. **QA** Hinge up upper console and lock fastener studs holding upper console to cockpit overhead.
19. Move fuel selector levers back to the off position.

ADJUST

1. Turn off all electrical and hydraulic power.

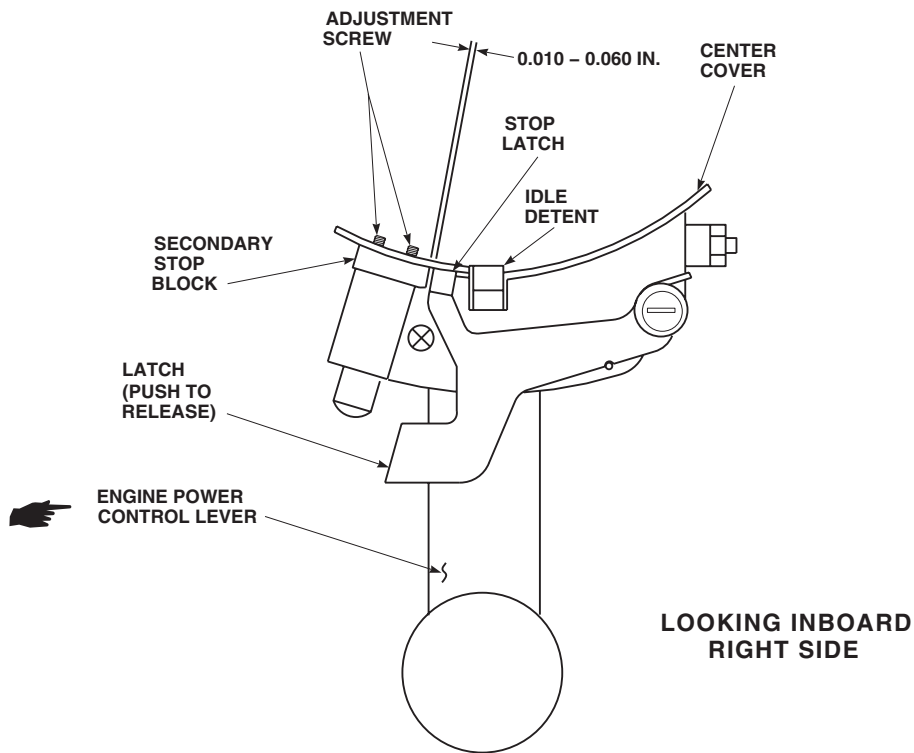
ADJUST - ContinuedAB0978B
SA

Figure 2. Adjust Secondary Stop Blocks.

NOTE

The following adjustment is for both No. 1 and No. 2 secondary stop blocks.

2. With engine power control lever in the IDLE position and fully engaged in the ground idle position detent located in the center cover assembly, using feeler gage, measure the gap between the stop latch and the secondary stop block located on the center access cover (Figure 2).
3. Adjust gap between 0.010 to 0.060-inch by loosening the two adjustment screws on the stop block.
4. Tighten adjustment screws and place engine power control lever in the OFF position.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE CONTROL QUADRANT CANOPY PLUNGER
This WP supersedes WP 0502 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Spring Scale, AAA-S-133

References

WP 0501 00
WP 0526 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Move fuel selector levers forward.
3. Unlock fastener studs holding upper console to cockpit overhead, and hinge down upper console.
4. Remove quadrant covers (WP 0501 00).
5. Remove jamnut and washer from canopy plunger (Figure 1).
6. Remove canopy plunger from fitting on support.

INSPECT

1. Check all threaded inserts for proper engagement of support and broken or cracked threads.
2. **IF INSERT IS LOOSE, BROKEN OR CRACKED, REPLACE INSERT (WP 0526 00).**

INSTALL

1. Position fire extinguisher T-handle so detent is lined up with hole in fitting.
2. **QA** Thread canopy plunger into fitting until plunger is lightly touching detent in handle.
3. **QA** Install jamnut and washer on canopy plunger. Do not tighten jamnut now (Figure 1).
4. Using a spring scale attached to T intersection of handle, move handle through full range of travel while adjusting canopy plunger. **ADJUST PLUNGER UNTIL A FORCE OF 5 - 7 POUNDS IS OBTAINED TO MOVE PLUNGER FROM DETENTS.**
5. **QA TIGHTEN JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
6. Install quadrant covers (WP 0501 00).
7. Hinge up upper console and lock fastener studs holding upper console to cockpit overhead.
8. Move fuel selector levers back to the off position.

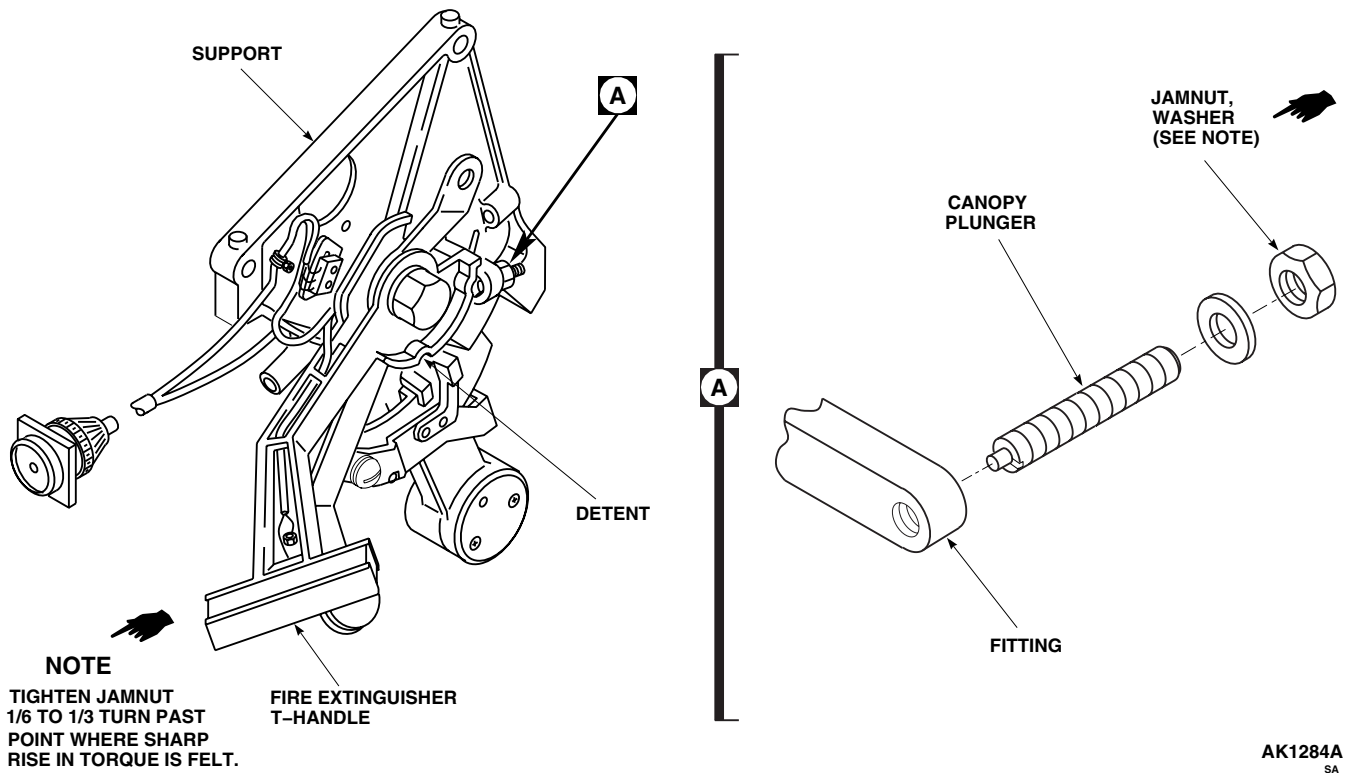
INSTALL - Continued

Figure 1. Replace/Adjust Engine Control Quadrant Canopy Plunger.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE STARTER SWITCH ADJUSTMENT

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

WP 0079 00

WP 0501 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

ADJUSTMENT**CAUTION**

Damage to starter switch actuator will result if ENG POWER CONT lever is moved from FLY or IDLE to OFF, while pressing starter button. Do not press starter button while moving ENG POWER CONT lever.

NOTE

Do not adjust starter switch on bench. Adjust switch after quadrant is installed in helicopter, and power-available push/pull cable is rigged.

1. Turn off all electrical power.
2. Move fuel selector levers forward.
3. Unlock fastener studs holding upper console to cockpit overhead, and hinge down upper console.
4. Remove quadrant covers (WP 0501 00).
5. Place ENG POWER CONT lever on quadrant against off stop.
6. Loosen screws holding starter switch to quadrant support, just enough to move switch (Figure 1, Detail A).
7. **QA** With starter button fully depressed, move starter switch until arm is against starter button, and switch clicks. Tighten screws to hold switch in place.
8. Move ENG POWER CONT lever back and forth several times, then place lever against off stop again.
9. Make sure that starter switch cannot be actuated by moving lever against off stop.
10. Press starter button and check for click of switch. Readjust as necessary.
11. Make sure area is clean and free of foreign material.
12. Install quadrant covers (WP 0501 00).
13. **QA** Hinge up and fasten upper console.
14. Move fuel selector levers back to the off position.
15. Do an operational check of engine start and ignition system (WP 0079 00).

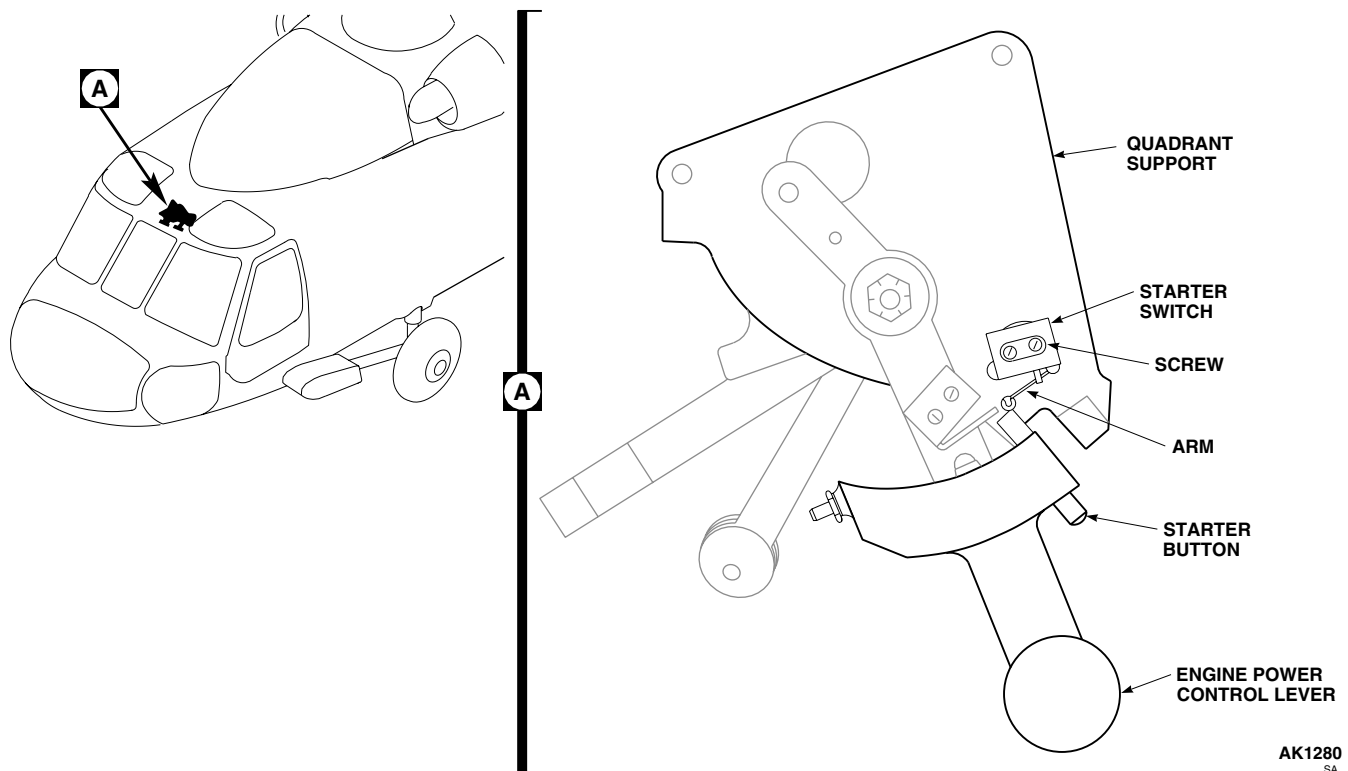
ADJUSTMENT - Continued

Figure 1. Adjust Engine Starter Switch.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE STARTER ABORT SWITCH ADJUSTMENT

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

WP 0080 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

ADJUSTMENT

1. Turn quadrant adjusting nut fully clockwise (Figure 1).
2. Pull ENG POWER CONT lever handle down as far as it will go and hold. Using a pencil, mark a line on lever 0.060-inch above top of handle.
3. **QA** With handle down as far as it will go, turn quadrant adjusting nut counterclockwise until switch clicks. Release handle.
4. While pulling ENG POWER CONT lever handle down, look at lever to see where top of handle is when switch clicks.
5. If top of handle is then below pencil marking, no further adjustment is required.
6. If there is no click with handle fully down, turn quadrant adjusting nut counterclockwise in quarter-turn units until a click is heard.
7. If top of handle is covering pencil marking, turn quadrant adjusting nut clockwise in quarter-turn units until switch clicks within 0.060-inch spread range.
8. Perform operational check of engine controls quadrant (WP 0080 00).

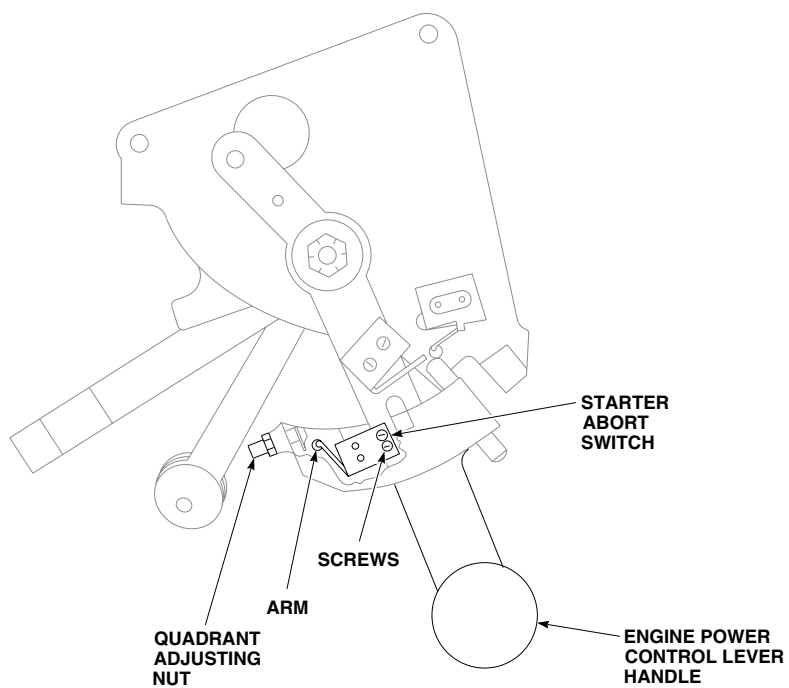
ADJUSTMENT - ContinuedAA9699A
SA

Figure 1. Engine Starter Abort Switch Adjustment.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE STARTER OVERRIDE SWITCH ADJUSTMENT

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

WP 0080 00

WP 0501 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

ADJUSTMENT

1. Remove screws, washers, and center cover of quadrant (WP 0501 00).
2. Loosen screws holding starter override switch to ENG POWER CONT lever, just enough to move switch (Figure 1).
3. **QA** With starter button fully pressed, move starter override switch until arm is against starter button, and switch clicks. Hand tighten screws holding switch to ENG POWER CONT lever bracket (Figure 1).
4. Make sure area is clean and free of foreign material.
5. Install center quadrant cover with screws and washers (WP 0501 00).
6. Perform operational check of engine controls quadrant (WP 0080 00).

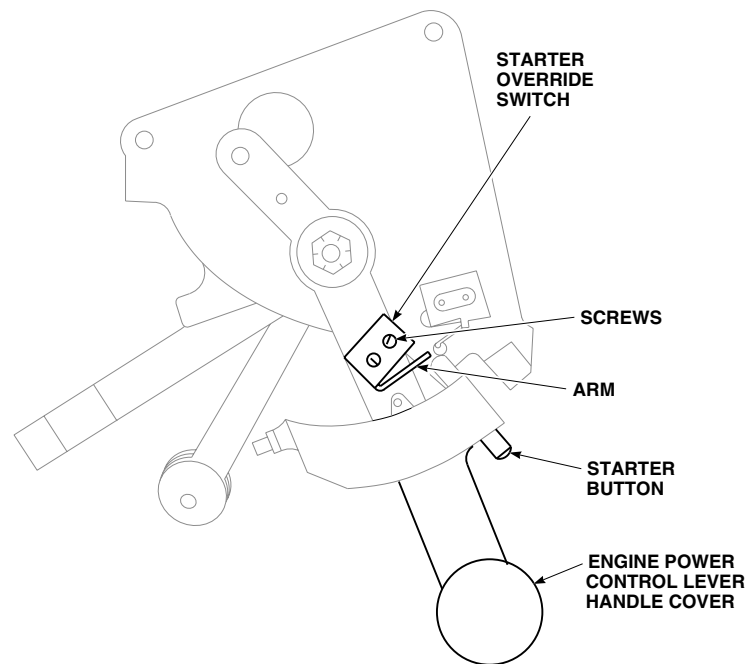
ADJUSTMENT - ContinuedAA9698
SA

Figure 1. Engine Starter Override Switch Adjustment.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

ENGINE LOAD-DEMAND PUSH/PULL CABLE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Rods, 0.125-inch diameter
Powerplant Repairer's Toolkit, SC 5180-99-B07
Torque Wrench, 30 - 200 in. lbs.

Sealing Compound, Item 273, WP 1803 00
Spring Pin, 094-0562LDP
Strap, Tiedown, Electrical, Item 302, WP 1803 00
Tubing, Plastic Spiral Wrap, Item 347, WP 1803 00
Wire, Nonelectrical, Item 354, WP 1803 00

Material/Parts

Adhesive, Item 9, WP 1803 00
Adhesive, Item 37, WP 1803 00
Corrosion Removing Compound, Item 108,
WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Insulation Sleeving, Item 161, WP 1803 00
Lubricating Oil, Aircraft Turbine, Item 182,
WP 1803 00
Pad, Scouring, Item 205, WP 1803 00

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-237-MTF
WP 0479 00
WP 0484 00
WP 1803 00

NOTE

The following procedures apply to both left and right side load-demand push/pull cables.

REMOVE

1. Open engine cowling.
2. Remove engine air inlet (WP 0484 00).
3. Open main rotor pylon sliding cover.
4. Connect external electrical power.
5. Turn on backup hydraulic power.
6. Move collective stick in cockpit to low stick position.
7. Disconnect load-demand rotary input from engine hydromechanical unit (HMU) (Figure 1, Detail A). Insert 0.125-inch diameter drill rod through rigging hole in outer housing and inner rotary driver of rotary input to hold spring compressed.

REMOVE - Continued

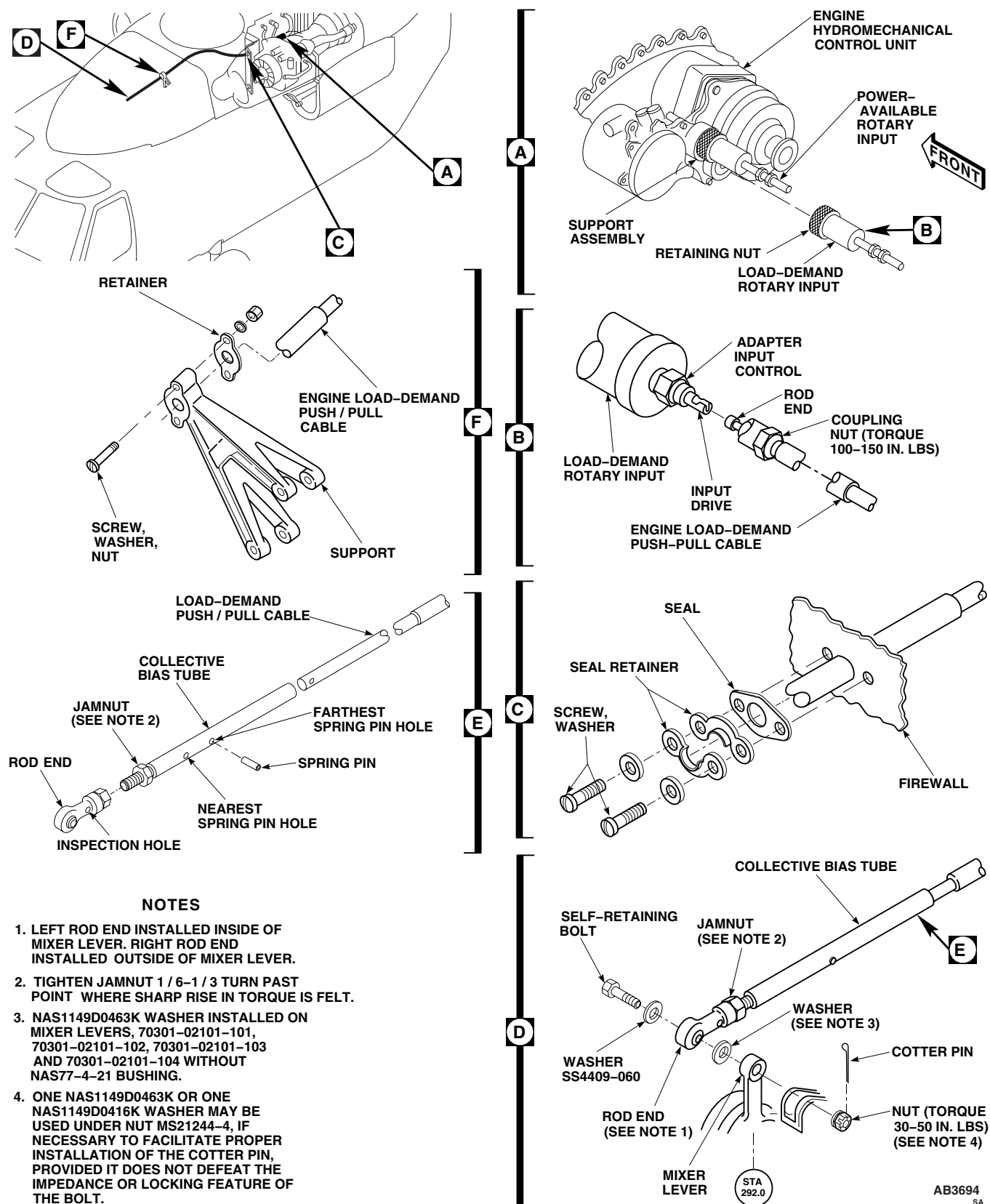


Figure 1. Replace Engine Load-Demand Push/Pull Cable.

REMOVE - Continued**CAUTION**

Shearing of spring pin in collective bias tube will result if collective stick is moved while drill rod is installed. Do not move collective stick when drill rod is installed.

8. Disconnect engine load-demand push/pull cable from load-demand rotary input. Make sure retaining nut stays on load-demand rotary input (Figure 1, Detail B).
9. Turn off backup hydraulic power.
10. Unclamp engine load-demand push/pull cable from firewall.
11. Remove screws, seal retainers, and seal from firewall and cable (Figure 1, Detail C).
12. Remove tie down straps and remove spiral wrap from cable.
13. Remove bolt, washer, and nut holding collective bias tube rod end to mixer lever on main rotor pylon deck. Remove rod (Figure 1, Detail D).
14. Remove rod end and nut from collective bias tube. Inspect rod end for radial and axial play. **NO PLAY ALLOWED.** Replace rod end if play is found (Figure 1, Detail E).
15. Remove screws, washers, and nuts and slide retainer away from support (Figure 1, Detail F).
16. Remove load-demand push/pull cable from support and helicopter. Remove retainer from cable.
17. If collective bias tube and load-demand push/pull cable need to be separated, do this:
 - a. Remove spring pin from collective bias tube and load-demand push/pull cable (Figure 1, Detail E).
 - b. Slide load-demand push/pull cable from collective bias tube.
 - c. Connect new load-demand push/pull cable to collective bias tube with spring pin.

NOTE

- There are two spring pin holes in collective bias tube. Either hole can be used now until final adjustment at installation.
 - The spring pin is part of a shear section. Only use one wrap (loop) when lockwiring this pin.
- d. Using only one wrap (loop), lockwire spring pin to collective bias tube using nonelectrical wire, Item 354, WP 1803 00.

INSPECT

Visually inspect engine load-demand push/pull cable for surface corrosion and/or damage. Nicks or scratches that do not penetrate down to metal windings are acceptable and do not require repair. **SURFACE CORROSION AND/OR CUTS OR TEARS PENETRATING COVERING MUST BE REPAIRED.**

REPAIR

1. Turn off all electrical power.
2. Remove surface corrosion from LDS cable rod end using scouring pad, Item 205, WP 1803 00.
3. Clean surface with corrosion removing compound, Item 108, WP 1803 00, and apply 2 coats of epoxy primer coating kit, Item 120, WP 1803 00, to surfaces where plating was disturbed.
4. Repair cuts or tears up to 1 inch in length in engine load-demand push/pull cable by applying adhesive, Item 9, WP 1803 00, to damaged area and blending out.

REPAIR - Continued

5. To repair cuts or tears greater than 1 inch in length in cable do the following:
 - a. Cut insulation sleeving, Item 161, WP 1803 00, 1 1/2-inches longer than damaged area.
 - b. Slide insulation sleeving, Item 161, WP 1803 00, over cable and center over damaged area.
 - c. Apply adhesive, Item 37, WP 1803 00, to both ends of insulation sleeving.

INSTALL

1. **QA** Insert cable and collective bias tube through support (Figure 1, Detail F). Install nut and rod end on collective bias tube (Figure 1, Detail E).
2. **QA** Route load-demand push/pull cable from mixer lever at STA 292.0 on upper forward deck through engine compartment firewall.
3. Add plastic spiral tubing wrap, Item 347, WP 1803 00, to cable from STA 325.0 through STA 330.0, and from STA 353.0 through STA 368.0. Hold spiral wrap in place with electrical tiedown straps, Item 302, WP 1803 00.
4. **QA** Install retainer on support with screws, washers, and nuts (Figure 1, Detail F).

NOTE

Make sure retaining nut is installed on load-demand rotary input before connecting to engine load-demand push/pull cable.

5. Compress spring of load-demand rotary input and insert a 0.125-inch diameter drill rod.

NOTE

- When connecting push/pull cable to rotary input, make sure internal connection is made.
 - There should be no play at connection point between push/pull cable and rotary input.
6. Connect engine load-demand push/pull cable to load-demand rotary input and loosely install coupling nut. Do not torque coupling nut now (Figure 1, Detail B).
 7. **QA** Install seal and seal retainers around push/pull cable and install on firewall with screws (Figure 1, Detail C).
 8. **QA** Clamp load-demand push/pull cable to every rib along firewall.
 9. Check for a minimum clearance of 0.250-inch between LDS cable and engine fuel filter head. Reposition cable or add clamps as necessary.

NOTE

Cyclic stick position has no effect on rigging of the load-demand push/pull cable.

10. Turn on backup hydraulic power. Hold collective stick down against stop and adjust rod end bearing to line up with hole in mixer lever at STA 292.0 on upper forward deck as follows (Figure 1, Detail D):
 - a. Thread rod end on or off collective bias tube to line up holes.
 - b. If rod end bottoms on collective bias tube before boltholes line up, remove spring pin, line up hole in load-demand push/pull cable with hole in collective bias tube nearest to rod end, and install spring pin. Using only one wrap (loop), lockwire spring pin to tube using nonelectrical wire, Item 354, WP 1803 00. Adjust rod end again.
 - c. If boltholes line up, check that nonelectrical wire, Item 354, WP 1803 00, cannot pass through inspection hole in rod end. If it does, remove spring pin, line up hole in load-demand push/pull cable with hole in collective bias

INSTALL - Continued

tube farthest from rod end, and install spring pin. Using only one wrap (loop), lockwire spring pin to tube using nonelectrical wire, Item 354, WP 1803 00. Adjust rod end again.

- d. **QA TIGHTEN JAMNUT 1/6 TO 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**

CAUTION

Shearing of spring pin in collective bias tube will result if collective stick is moved while drill rod is installed. Do not move collective stick with push/pull cable attached to mixer lever and drill rod installed in rotary input.

NOTE

- Right side load-demand push/pull cable rod end is connected to the outside of mixer lever. Left side load-demand push/pull cable rod end is connected to the inside of mixer lever.
 - Make sure that washer, SS4409-060, or equivalent is installed between bolt head and collective bias tube rod end. Install washer, AN960KD416, between mixer lever and collective bias tube rod end only if mixer lever, 70301-02101-101, 70301-02101-102, 70301-02101-103, or 70301-02101-104 without bushing installed.
 - Adjust the rod end as required to allow the 0.125-inch diameter drill rod in rotary input to slide freely before rod end is connected at mixer level.
11. **QA** Connect rod end to mixer lever with bolt, washer(s), and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin (Figure 1, Detail D).
12. Be sure 0.125-inch diameter drill rod in rotary input slides freely in and out of rigging hole after installing bolt in rod end and mixer lever.

NOTE

Slight axial and radial play from the load-demand rotary input is acceptable when retaining nut is correctly installed against support assembly.

13. Remove 0.125-inch diameter drill rod from load-demand rotary input.
14. Turn off backup hydraulic power.
15. **QA** Connect load-demand rotary input to engine hydromechanical unit. Make sure that rotary input alignment pin engages in slot on engine hydromechanical unit. Tighten retaining nut until it bottoms in support assembly (Figure 1, Detail A).
16. **QA** Coat threads of LDS rod end with aircraft turbine lubricating oil, Item 182, WP 1803 00. **TORQUE NUT CONNECTING LOAD-DEMAND PUSH/PULL CABLE TO ROTARY INPUT TO 100 - 150 INCH-POUNDS.**
17. Do an engine load-demand push/pull cable friction check (WP 0479 00).
18. After installing LDS cable rod end, seal threads extending past nut with sealing compound, Item 273, WP 1803 00.
19. Close main rotor pylon sliding cover.
20. Install engine air inlet (WP 0484 00).
21. Close engine cowling and latch.
22. Disconnect external electrical power.

INSTALL - Continued

23. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE POWER-AVAILABLE PUSH/PULL CABLE

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powerplant Repairer's Toolkit, SC 5180-99-B07
Torque Wrench, 30 - 200 in. lbs.

References

TM 1-1520-237-MTF
WP 0484 00
WP 0512 00
WP 1803 00

Material/Parts

Strap, Tiedown, Electrical, Item 302, WP 1803 00
Tubing, Plastic Spiral Wrap, Item 347, WP 1803 00

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
UH-60 Helicopter Repairer MOS 15T (1)

NOTE

The following procedures apply to both No. 1 and No. 2 engine power-available push/pull cables.

REMOVE

1. Open engine cowl.
2. Remove engine air inlet (WP 0484 00).
3. Place quadrant ENG POWER CONT lever OFF.
4. Disconnect power-available rotary input from engine hydromechanical unit (Figure 1, Detail B).
5. Disconnect fuselage power-available push/pull cable from engine power-available push/pull cable at firewall (Figure 1, Detail A). Disconnect engine power-available push/pull cable from firewall. Keep nut for installing new cable.
6. Unclamp engine power-available push/pull cable from firewall. Remove nut from cable. Keep nut for installing new cable.
7. Remove tiedown straps and remove spiral wrap from cable.
8. Disconnect engine power-available push/pull cable from power-available rotary input. Make sure retaining nut stays on power-available rotary input (Figure 1, Detail B).

REMOVE - Continued

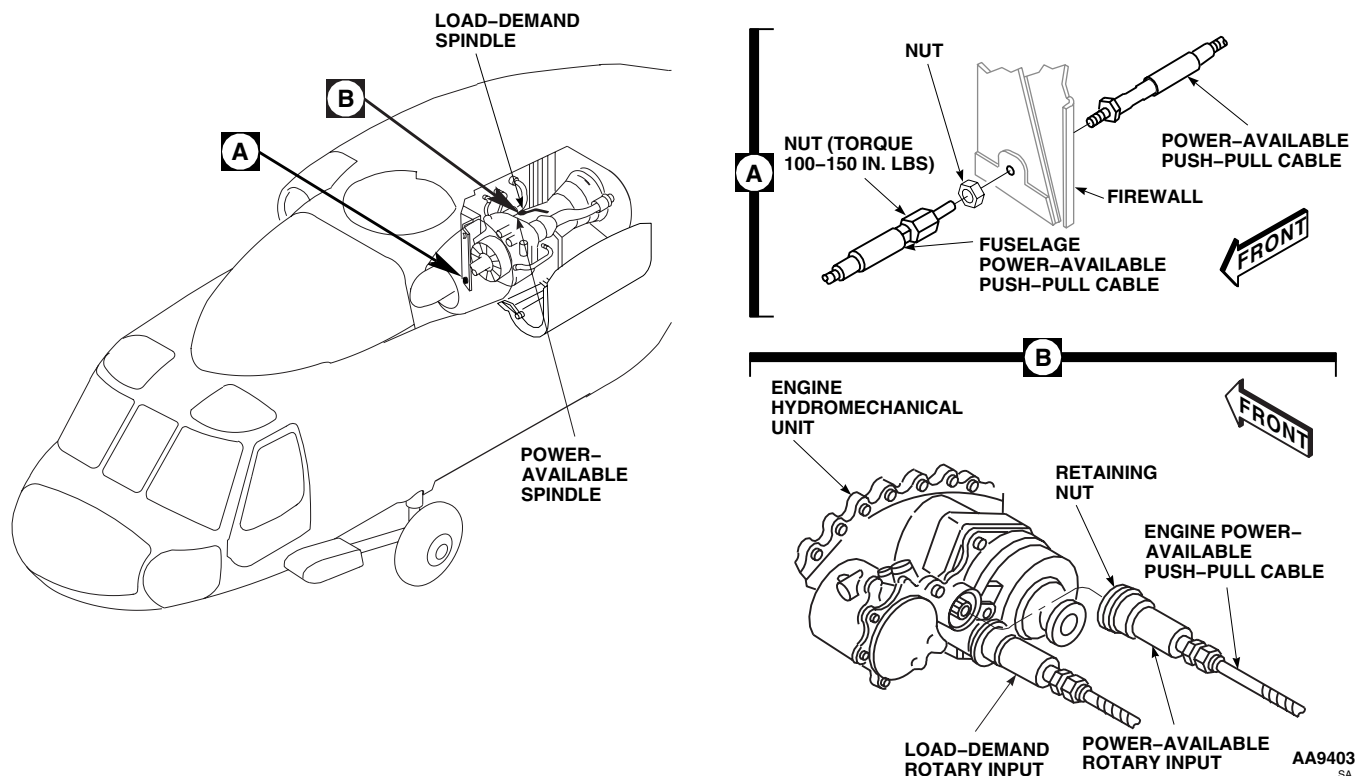


Figure 1. Replace Engine Power-Available Push/Pull Cable.

INSTALL

NOTE

- Make sure retaining nut is installed on power-available rotary input before connecting to engine power-available push/pull cable.
- When connecting push/pull cable to rotary input, make sure internal connection is made.

1. **QA** Connect engine power-available push/pull cable to power-available rotary input. Do not torque nut now (Figure 1, Detail B).
2. **QA** Install nut on end of engine power-available push/pull cable.
3. Add plastic spiral wrap tubing, Item 347, WP 1803 00, as necessary from firewall to power-available rotary input in areas where cable may chafe. Hold spiral wrap in place with electrical tiedown straps, Item 302, WP 1803 00.
4. **QA** Place end of engine power-available push/pull cable through firewall. Install nut holding cable to firewall (Figure 1, Detail A).
5. **QA** Clamp engine power-available push/pull cable to firewall.

INSTALL - Continued**NOTE**

When connecting push/pull cables, make sure internal connection is made.

6. **QA** Connect fuselage power-available push/pull cable to engine power-available push/pull cable and clamp to every rib along firewall. **TORQUE NUT CONNECTING PUSH/PULL CABLES TO 100 - 150 INCH-POUNDS.**
7. Rig engine power-available system (WP 0512 00).
8. Install engine air inlet (WP 0484 00).
9. Close engine cowl and latch.
10. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****FUSELAGE POWER-AVAILABLE PUSH/PULL CABLE****This WP supersedes WP 0508 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Powerplant Repairer's Toolkit, SC 5180-99-B07
Torque Wrench, 30 - 200 in. lbs.

Towel, Machinery Wiping, Item 344, WP 1803 00
Tubing, Plastic Spiral Wrap, Item 347, WP 1803 00

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Abrasive Cloth, Assortment, Item 1, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 7, WP 1803 00
Sealing Compound, Item 273, WP 1803 00
Strap, Tiedown, Electrical, Item 302, WP 1803 00

References

TM 1-1520-237-MTF
WP 0258 00
WP 0484 00
WP 0501 00
WP 0512 00
WP 1803 00

REMOVE

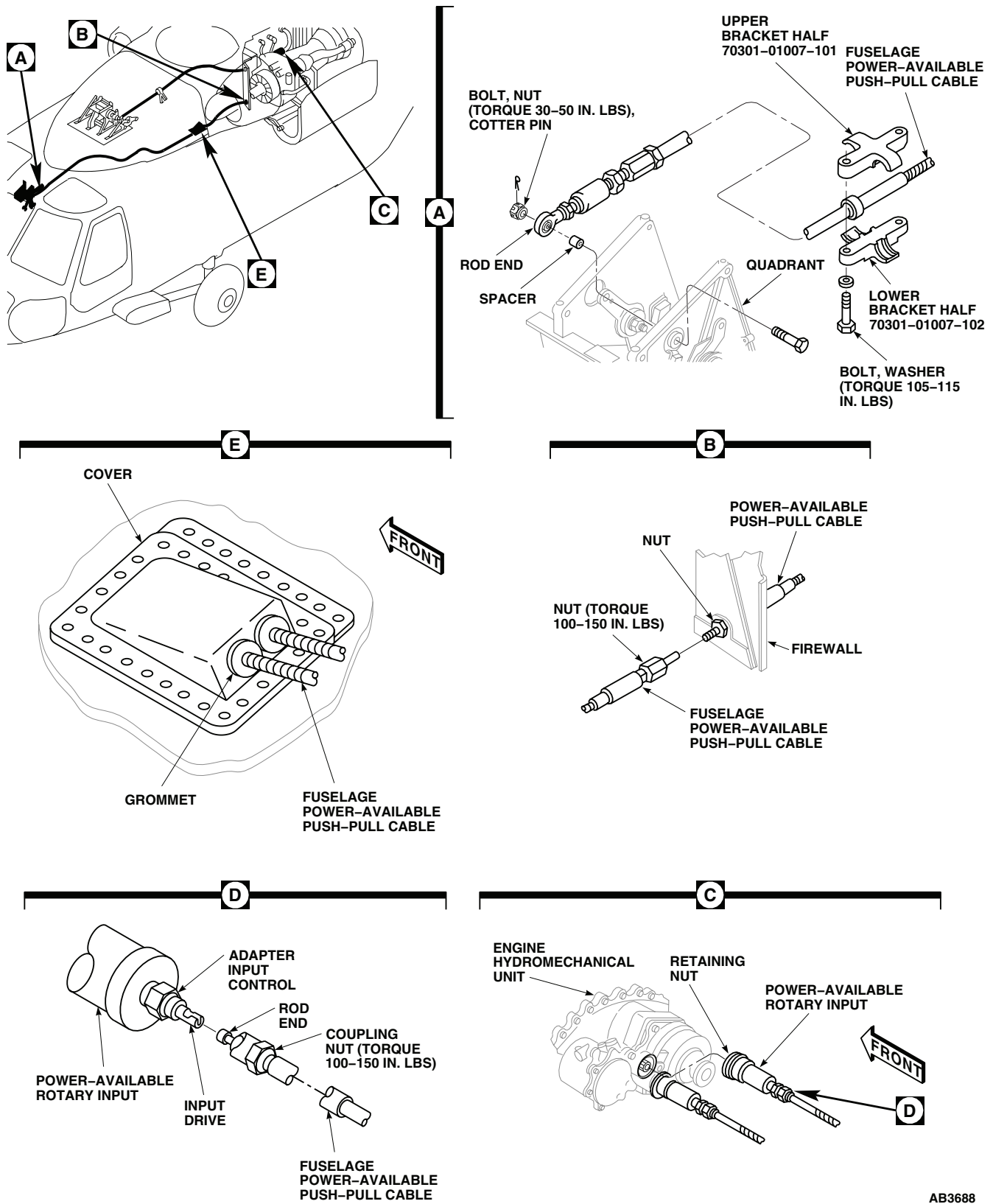
1. Turn off all electrical power.
2. Unlock fastener studs holding upper console to cockpit overhead, and hinge down.
3. Remove quadrant covers (WP 0501 00).
4. Move quadrant ENG POWER CONT lever in IDLE detent.
5. Open engine cowling.
6. Disconnect power-available rotary input from engine hydromechanical unit (Figure 1, Detail C).

NOTE

The following procedures apply to both left and right side fuselage power-available push/pull cables.

7. Remove bolt, spacer, and nut holding fuselage power-available push/pull cable rod end to quadrant ENG POWER CONT lever (Figure 1, Detail A).
8. Remove bolt, washer, and bracket halves holding fuselage power-available push/pull cable and fuel selector valve push/pull cable to cockpit ceiling (Figure 1, Detail A).
9. Remove engine air inlet (WP 0484 00).
10. Remove soundproofing from cabin ceiling (WP 0258 00).
11. Disconnect fuselage power-available push/pull cable from engine power-available push/pull cable at firewall (Figure 1, Detail B).
12. Unclamp fuselage power-available push/pull cable from helicopter.

REMOVE - Continued



AB3688
SA

Figure 1. Replace Fuselage Power-Available Push/Pull Cable.

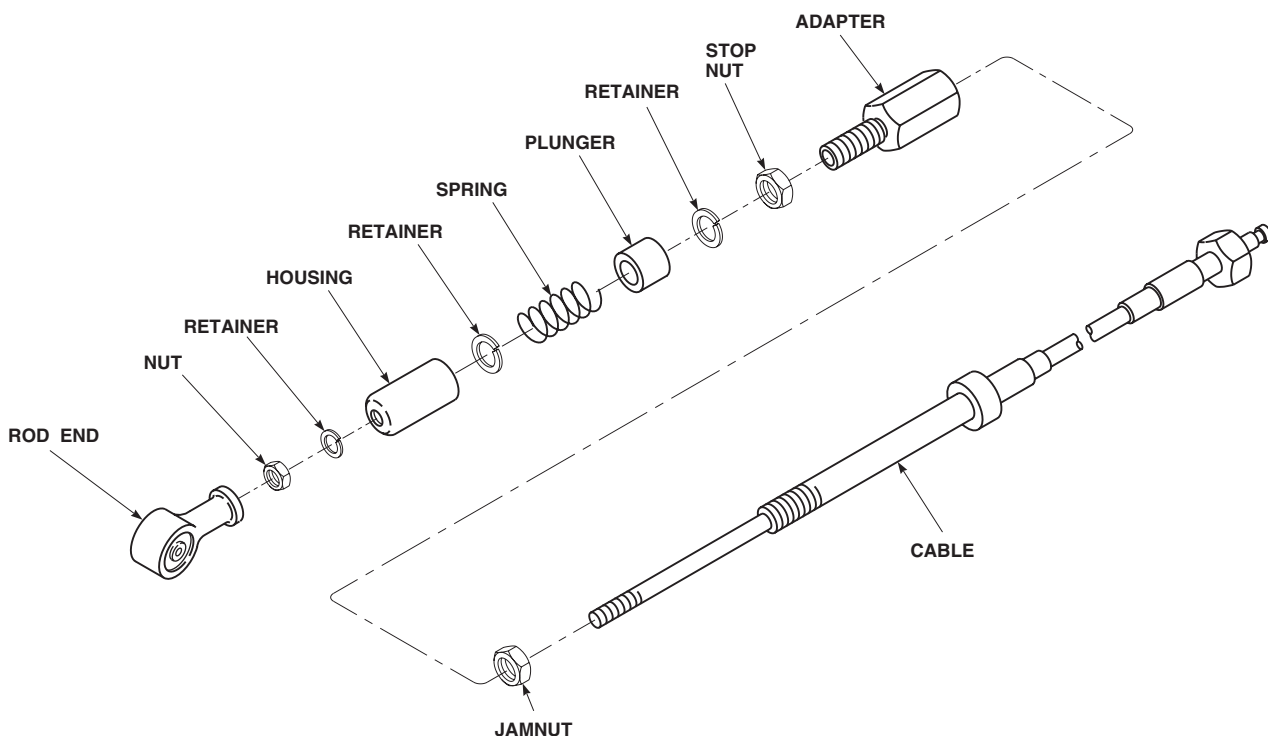
REMOVE - ContinuedAK1289
SA

Figure 2. Install Fuselage Power-Available Push/Pull Cable.

13. Remove electrical component tiedown straps and spiral wrap tubing from cable.
14. Carefully pull cable from grommet and remove fuselage power-available push/pull cable from helicopter (Figure 1, Detail E).
15. Remove rod end, nut, retainer, housing, retainer, spring, plunger, retainer, stopnut, adapter, jamnut, and cable (Figure 2).

INSTALL

1. Inspect grommet in cover at STA 338.0, if grommet is loose or worn through, do this (Figure 1, Detail E):
 - a. Make a 45 degree cut through edge of replacement grommet.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/ industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

- b. Remove old grommet and clean adhesive from cover using a nonmetallic scraper and a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Do not allow solvent to dry on cover. Keep turning cloth after each wipe.
- c. Lightly scuff inside of outer groove of grommet with assortment abrasive cloth, Item 1, WP 1803 00.

INSTALL - Continued

- d. Apply a light coat of adhesive, Item 7, WP 1803 00, to groove of grommet and edge of hole in cover. Allow adhesive to air-dry for 10 minutes or until tacky.
- e. **QA** Install grommet in hole in cover. For best results, allow adhesive to cure for 24 hours.
2. **QA** Install jamnut, adapter, stopnut, retainer, plunger, spring, retainer, housing, retainer, nut, and rod end on replacement cable (Figure 2).
3. **QA** Route fuselage power-available push/pull cable from quadrant ENG POWER CONT lever in cockpit ceiling to engine compartment firewall.
4. **QA** Carefully slide fuselage power-available push/pull cable through grommet.
5. Apply sealing compound, Item 273, WP 1803 00, to edge of grommet and cover, at 45 degrees slice and around cable.
6. Add plastic spiral wrap tubing, Item 347, WP 1803 00, 5 inches long, where cable passes through each fuselage frame. Hold spiral wrap in place with electrical tiedown straps, Item 302, WP 1803 00.

NOTE

When connecting push/pull cable, make sure internal connection is made.

7. **QA** Connect fuselage power-available push/pull cable to engine power-available push/pull cable at firewall (Figure 1, Detail B). **TORQUE NUT TO 100 - 150 INCH-POUNDS.**
8. **QA** Clamp fuselage power-available push/pull cable to fuel selector valve push/pull cable every 6 inches from ENG POWER CONT lever to STA 340.0.

CAUTION

Damage to equipment will result if bracket halves are not installed properly. Bracket halves are not interchangeable. Bracket half, 70301-01007-101, is upper bracket half, 70301-01007-102, is lower bracket half.

NOTE

Some aircraft may have been built with two upper bracket halves, 70301-01007-101, instead of one of each.

9. **QA** Bolt fuselage power-available push/pull cable and fuel selector valve push/pull cable to cockpit overhead with bolts, washers, and bracket halves (Figure 1, Detail A). **TORQUE BOLTS TO 105 - 115 INCH-POUNDS.**
10. **QA** Clamp fuselage power-available push/pull cable to structure.
11. Rig engine power-available system (WP 0512 00).
12. Install engine air inlet (WP 0484 00).
13. Install soundproofing in cabin ceiling (WP 0258 00).
14. Close engine cowlings and latches.
15. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****FUEL SELECTOR VALVE PUSH/PULL CABLE**

This WP supersedes WP 0509 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Gloves
Goggles/Face Shield
Nonmetallic Scraper
Powerplant Repairer's Toolkit, SC 5180-99-B07
Torque Wrench, 30 - 200 in. lbs.

Towel, Machinery Wiping, Item 344, WP 1803 00
Tubing, Plastic Spiral Wrap, Item 347, WP 1803 00

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Abrasive Cloth, Assortment, Item 1, WP 1803 00
Acetone, Technical, Item 5, WP 1803 00
Adhesive, Item 7, WP 1803 00
Sealing Compound, Item 273, WP 1803 00

References

WP 0258 00
WP 0484 00
WP 0501 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Move ENG FUEL SYS lever OFF.
3. Unlock fastener studs holding upper console to cockpit overhead, and hinge down.
4. Remove quadrant covers (WP 0501 00).

NOTE

The following procedures apply to both left and right side fuel selector valve push/pull cables.

5. Remove bolt, washer, and nut holding fuel selector valve push/pull cable rod end to quadrant fuel selector lever. Remove rod end and nut from cable (Figure 1, Detail A).
6. Inspect rod end for axial and radial play. **NONE ALLOWED.**
7. Remove bolt, washer, and bracket halves holding fuel selector valve push/pull cable and fuselage power-available push/pull cable to cockpit overhead.
8. Open engine cowling.
9. Open oil cooler compartment access doors.
10. Remove engine air inlet (WP 0484 00).
11. Remove soundproofing from cabin ceiling (WP 0258 00).
12. Disconnect fuel selector valve push/pull cable from fuel selector valve control box (Figure 1, Detail B).
13. Unclamp fuel selector valve push/pull cable from firewall and structure.

REMOVE - Continued

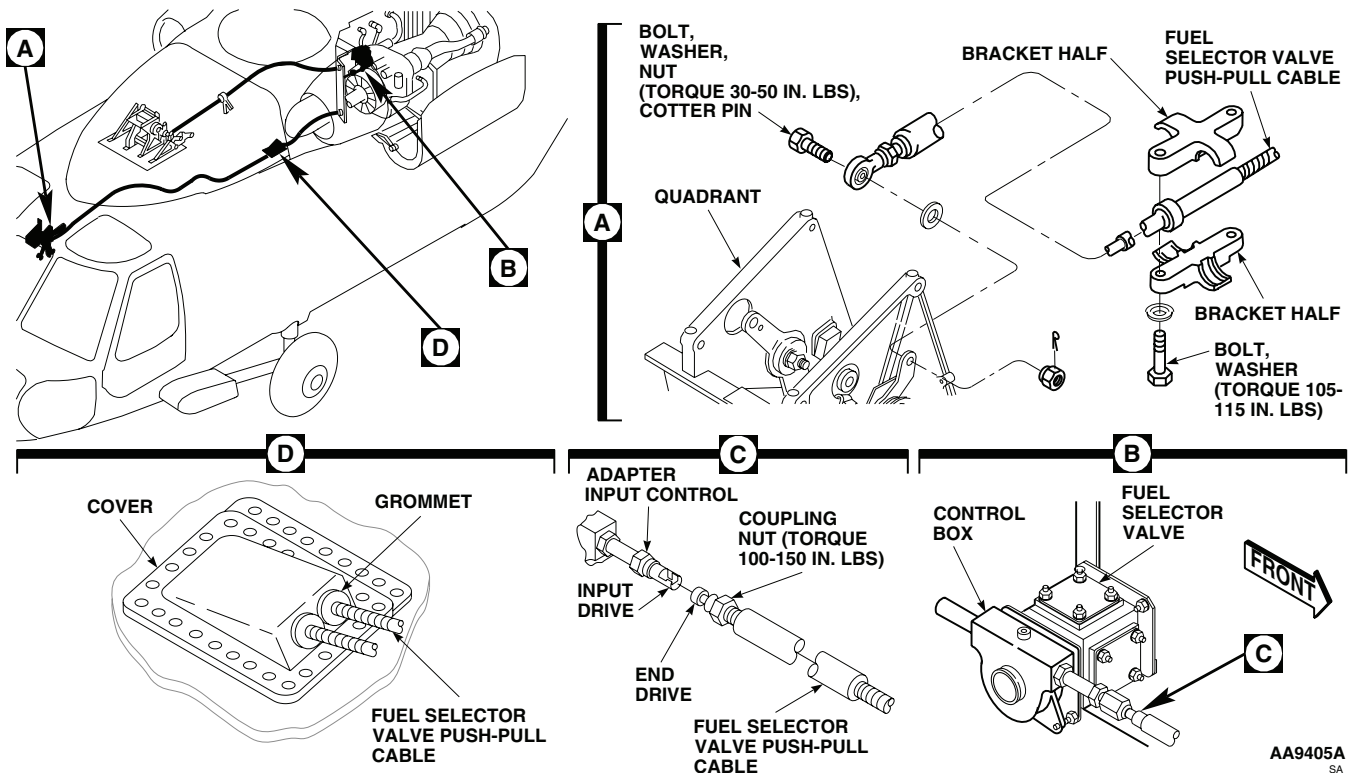


Figure 1. Replace Fuel Selector Valve Push/Pull Cable.

14. Remove tiedown straps and remove spiral wrap from cable.
15. Carefully pull cable from grommet and remove cable from helicopter.

INSTALL

1. Inspect grommet at STA 338.0, if grommet is loose or worn through, do this (Figure 1, Detail D):
 - a. Make a 45 degree cut through edge of replacement grommet.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

- b. Remove old grommet. Clean adhesive from cover using a nonmetallic scraper and a machinery wiping towel, Item 344, WP 1803 00, dampened with technical acetone, Item 5, WP 1803 00. Do not allow solvent to dry on airframe. Keep turning towel after each wipe.
- c. Lightly scuff insides of outer groove of grommet with abrasive cloth assortment, Item 1, WP 1803 00.
- d. Apply a light coat of adhesive, Item 7, WP 1803 00, to groove of grommet and edge of hole in cover. Allow adhesive to air-dry for 10 minutes or until tacky.
- e. **QA** Install grommet in hole in cover. For best results, allow adhesive to cure for 24 hours.

INSTALL - Continued

2. **QA** Route fuel selector valve push/pull cable from quadrant fuel selector lever to oil cooler compartment firewall.
3. **QA** Carefully slide fuel selector valve push/pull cable through grommet.
4. Apply sealing compound, Item 273, WP 1803 00, to edge of grommet and cover at 45 degree slice and around cable.
5. **QA** Connect fuel selector valve push/pull cable to fuel selector valve control box. **TORQUE NUT TO 100 - 150 INCH-POUNDS (Figure 1, Detail C).**
6. Add plastic spiral wrap tubing, Item 347, WP 1803 00, to fuel selector valve push/pull cable at these locations:
 - a. From STA 245.0 to STA 253.0.
 - b. From STA 257.0 to STA 263.0.
 - c. From STA 270.0 to STA 275.0.
 - d. From STA 278.0 to STA 283.0 at BL 5.0.
 - e. From BL 8.0 to BL 12.0 at STA 283.0.
 - f. From STA 290.0 to STA 294.0.
 - g. From STA 297.0 to STA 303.0.
 - h. From STA 313.0 to STA 318.0.
 - i. From STA 325.0 to STA 330.0.
 - j. From STA 350.0 to STA 360.0.
7. **QA** Clamp fuel selector valve push/pull cable to fuselage power-available push/pull cable every 6 inches from ENG POWER CONT lever to STA 340.0.

NOTE

Keyway on fuel selector valve and control box should be OFF.

8. **QA** Bolt fuel selector valve push/pull cable and fuselage power-available push/pull cable to cockpit overhead with bolts, washers, and bracket halves. **TORQUE BOLTS TO 105 - 115 INCH-POUNDS (Figure 1, Detail A).**
9. **QA** Clamp fuel selector valve push/pull cable to firewall and structure.

NOTE

Make sure ENG FUEL SYS lever is OFF.

10. **QA** Thread jamnut and rod end onto fuel selector valve push/pull cable until rod end bearing and fuel selector lever holes line up. **TIGHTEN JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
11. **QA** Bolt fuel selector valve push/pull cable rod end to quadrant fuel selector lever.
12. Move fuel selector lever through DIRECT and X FEED positions and check by feel for fuel selector control box detenting in each position.
13. **QA TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.
14. Install quadrant covers (WP 0501 00).
15. Move fuel selector levers back to the off position.
16. **QA** Hinge up and lock upper console to cockpit overhead.

INSTALL - Continued

17. Install soundproofing in cabin ceiling (WP 0258 00).
18. Close oil cooler compartment access doors.
19. **QA** Install engine air inlet (WP 0484 00).
20. Close engine cowling and latch.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
FUEL SELECTOR VALVE CONTROL BOX

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Powerplant Repairer's Toolkit, SC 5180-99-B07
Torque Wrench, 30 - 200 in. lbs.

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
UH-60 Helicopter Repairer MOS 15T (1)

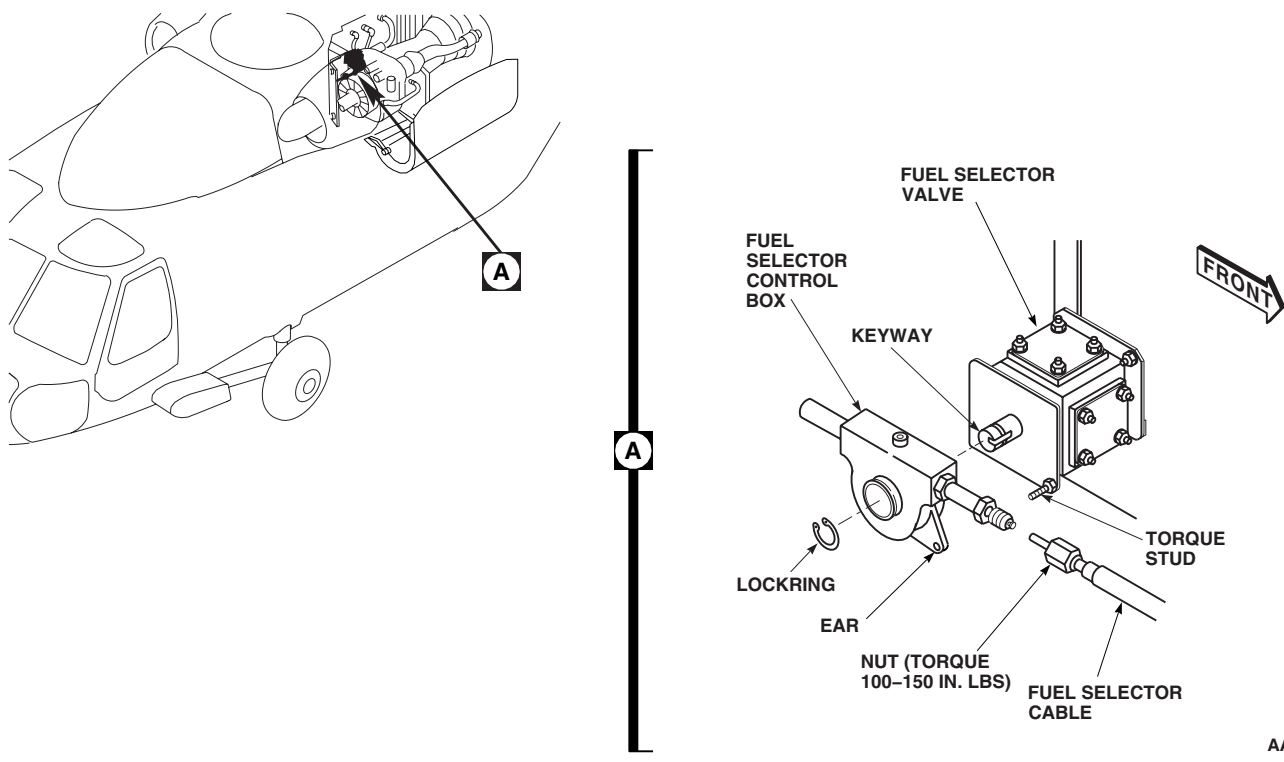
REMOVE

1. Place ENG FUEL SYS lever OFF.
2. Open engine cowling.
3. Open oil cooler compartment access doors.

NOTE

The following procedures apply to both left and right side fuel selector control boxes.

4. Disconnect fuel selector valve push/pull cable from control box (Figure 1).



AA9406
SA

Figure 1. Replace Fuel Selector Valve Control Box.

REMOVE - Continued

5. Remove lockring and control box from fuel selector valve.

INSTALL

1. **QA** Line up keyway on fuel selector valve and control box at OFF. Install control box on fuel selector valve with slot in ear over torque stud. Install lockring (Figure 1).
2. **QA** Connect fuel selector valve push/pull cable to control box. **TORQUE NUT TO 100 - 150 INCH-POUNDS.**
3. Move ENG FUEL SYS lever to DIR, XFD, and OFF while an assistant checks to be sure shaft of fuel selector valve turns.
4. Make sure area is clean and free of foreign material.
5. Close oil cooler compartment access doors.
6. Close engine cowling and latch.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE LOAD-DEMAND PUSH/PULL CABLE SUPPORTS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Fluorescent Penetrant Inspection Kit, XMA101
Powerplant Repairer's Toolkit, SC 5180-99-B07
Torque Wrench, 30 - 200 in. lbs.

References

TM 1-1520-265-23
WP 0479 00
WP 0506 00
WP 0533 00

Personnel Required

Aircraft Powerplant Repairer MOS 15B (1)
UH-60 Helicopter Repairer MOS 15T (1)

NOTE

Replacement procedures for left and right cable supports are the same except as noted.

REMOVE

1. Open main rotor pylon sliding cover.
2. Disconnect engine load-demand push/pull cable from cable support (WP 0506 00).
3. Remove bolts, washers, and nuts holding cable support to accessory module (Figure 1, Detail A). Remove cable support.
4. On left hand cable support, 70301-02119-101, remove bolt, washers, nut, and clip from cable support (Figure 1).

INSPECT

1. Inspect clip (WP 0533 00).
2. Inspect cable support using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace cable support if cracks are found.

INSTALL

1. **QA** Position clip in left hand cable support, 70301-02119-101, and install washers between clip and lugs of support. Install bolt, washers and nut attaching clip to cable support (Figure 1). **TORQUE NUT TO 105 - 115 INCH-POUNDS.**
2. **QA** Position cable support on accessory module. Install washers between lugs or clip as necessary to fill gaps. Install bolts, washers, and nuts attaching cable support to accessory module (Figure 1, Detail A). **TORQUE NUTS TO 105 - 115 INCH-POUNDS.**
3. Connect engine load-demand push/pull cable to cable support (WP 0506 00).
4. Do an engine load-demand push/pull cable friction check (WP 0479 00).
5. Make sure area is clean and free of foreign material.

INSTALL - Continued

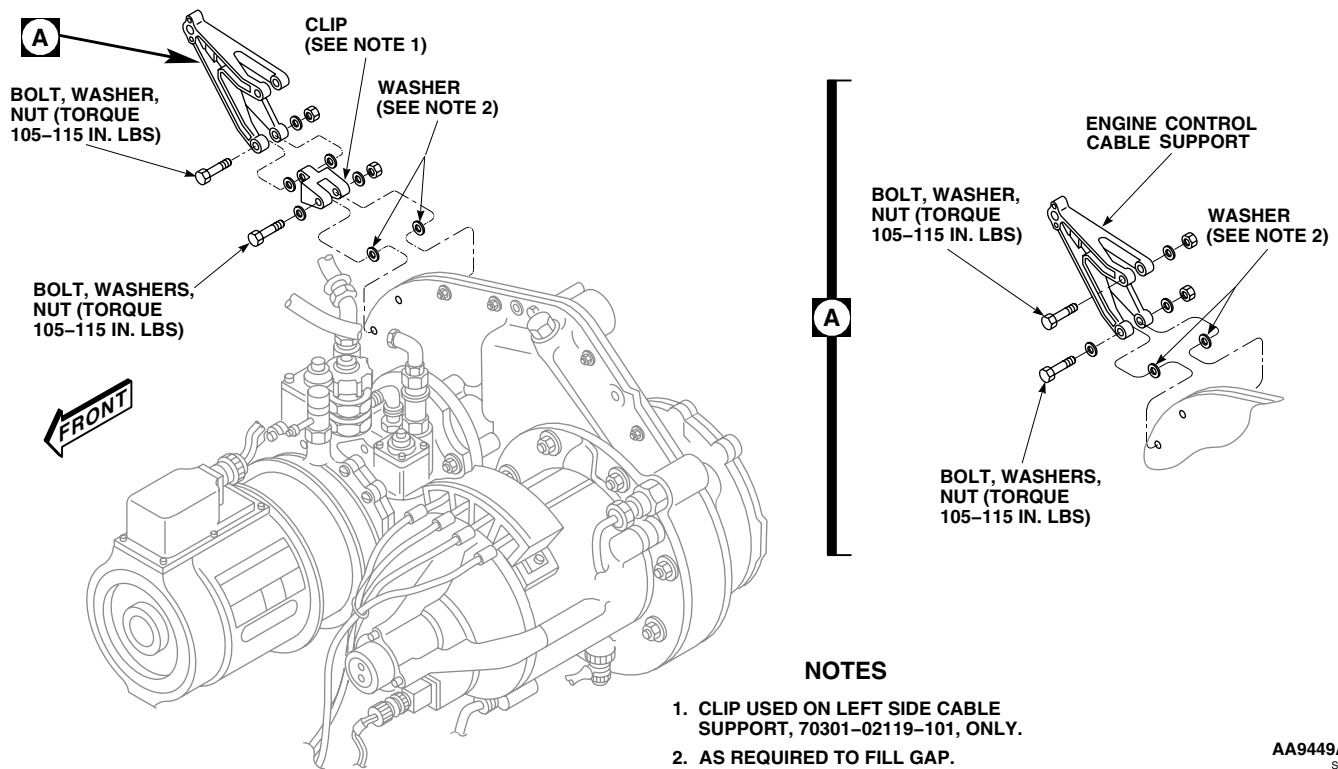


Figure 1. Replace Engine Load-Demand Push/Pull Cable Supports.

6. Slide main rotor pylon sliding cover closed and latch.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

ENGINE POWER-AVAILABLE SYSTEM RIGGING AND ADJUSTMENT

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Drill Rod, 0.125-inch diameter
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 30 - 200 in. lbs.

WP 0501 00

WP 0503 00

WP 0504 00

WP 0505 00

WP 0507 00

WP 0508 00

WP 0530 00

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-237-MTF
WP 0499 00

RIGGING AND ADJUSTMENT**NOTE**

The following procedures apply to both No. 1 and No. 2 systems.

1. Turn off all electrical power.
2. Move fuel selector levers forward.
3. Unlock fastener studs holding upper console to cockpit overhead, and hinge down.
4. Remove quadrant covers (WP 0501 00).
5. Move ENG POWER CONT lever to OFF.
6. Open engine cowling.
7. Disconnect power-available rotary input from engine hydromechanical unit (Figure 1, Detail A).
8. Move both quadrant ENG POWER CONT levers to IDLE detent and insert 0.125-inch diameter drill rod through rigging holes in lever and quadrant support.
9. Position rotary input to approximate installed position. Insert 0.125-inch diameter drill rod through rigging hole in outer housing and inner rotary driver of rotary input.
10. If drill rod slides in and out freely, or with slight friction, proceed to step 12.
11. If drill rod cannot be inserted in rotary input, do this:
 - a. Remove bolt, spacer, and nut holding power-available push/pull cable to ENG POWER CONT lever (Figure 2, Detail A).
 - b. Position rotary input barrel to align holes and insert drill rod.
 - c. Pull rod end bearing forward to remove slack in cable. Loosen jamnut and adjust rod end until rod end bearing and quadrant POWER CONT lever holes line up.

RIGGING AND ADJUSTMENT - Continued

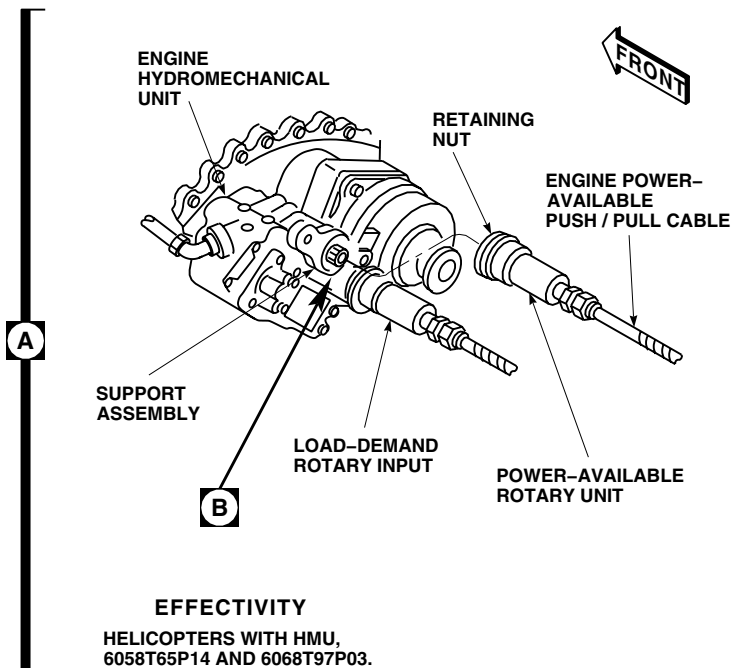
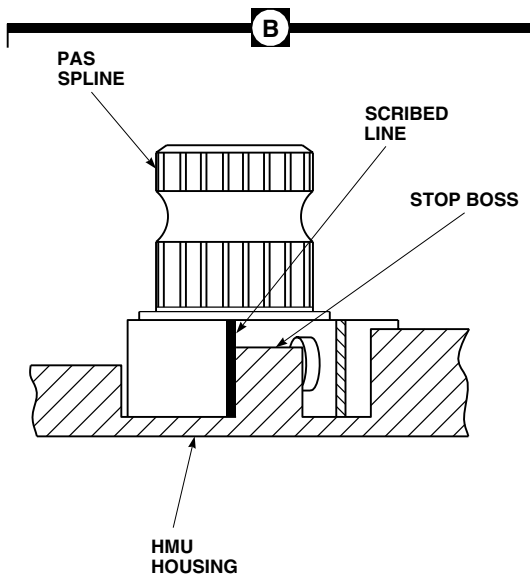
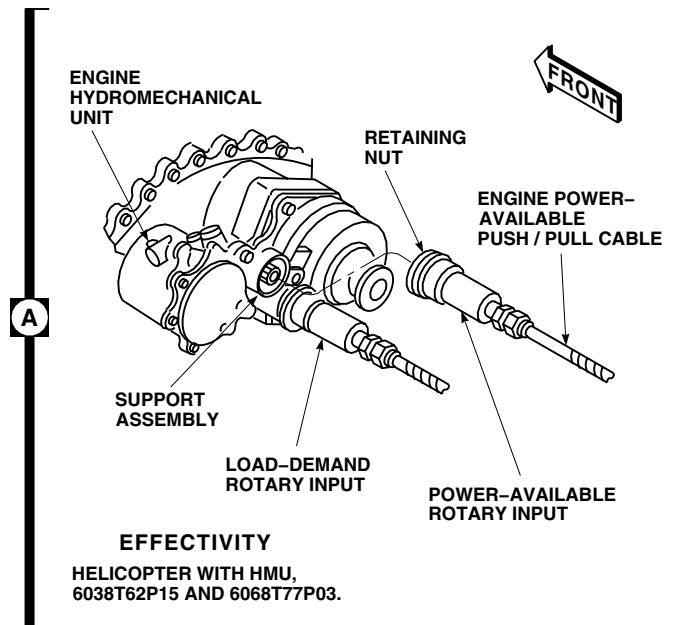
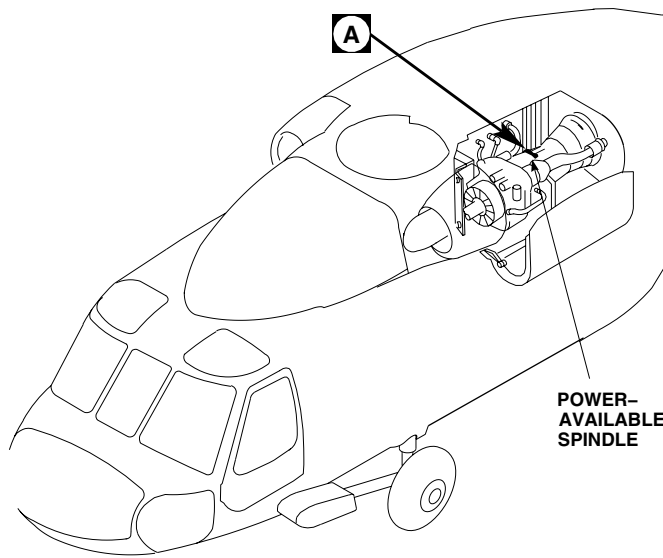


Figure 1. Rig Engine Power-Available System.

RIGGING AND ADJUSTMENT - Continued

- d. **QA TIGHTEN JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT.**
 - e. **QA** Install fuselage power-available push/pull cable rod on quadrant ENG POWER CONT lever, with bolt, washer, spacer, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.
- 12. Remove drill rods from rigging holes in quadrant support, lever, and rotary input.
 - 13. Position rotary input at HMU mounting location and check for any twisting or rotation required to make connection. If twisting or rotation is required, remove it by adjusting connection of rotary input and power-available cable (WP 0499 00).

NOTE

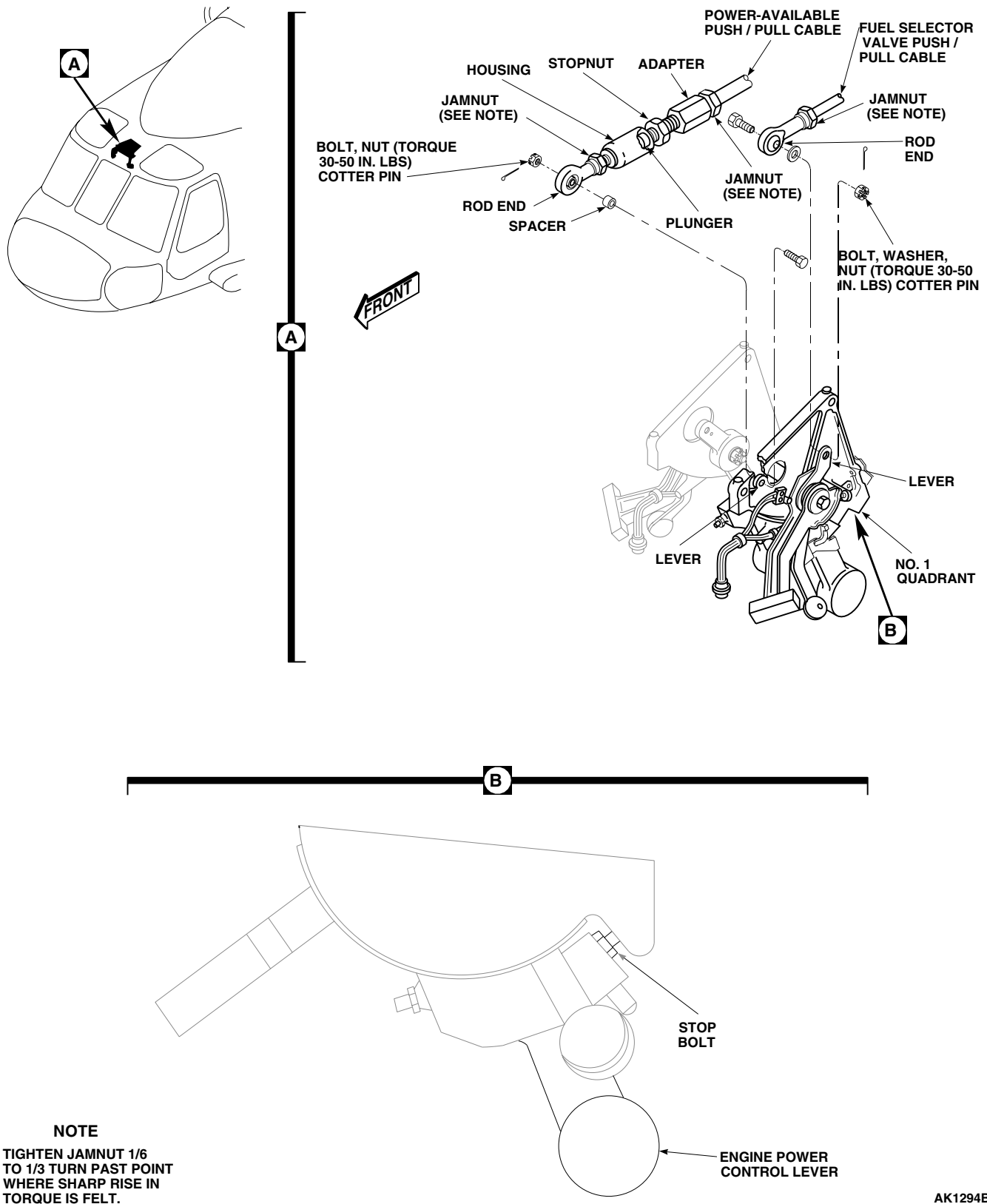
Slight axial and radial play from the power-available rotary input is acceptable when retaining nut is correctly installed against support assembly.

- 14. **QA** Connect power-available rotary input to engine hydromechanical unit (Figure 1, Detail A). Make sure that rotary input alignment pin engages slot on engine HMU. Tighten retaining nut until it bottoms in support assembly.
- 15. Install quadrant covers (WP 0501 00).
- 16. Move quadrant ENG POWER CONT lever to FLY detent. If stopnut contacts housing before adapter contacts plunger, rotate stopnut away from housing.
- 17. **QA** Adjust adapter on fuselage power-available push/pull cable to contact face of plunger. Turn adapter one complete turn against plunger face. **TIGHTEN JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT (Figure 2, Detail A).**

NOTE

- There are two types of engine hydromechanical units (HMU). HMU's are identified by information on data plate and external appearance.
 - HMU, 6038T62P15 and 6068T77P03, have an audible click when DEC lockout is reached.
 - HMU, 6058T65P14 and 6068T97P03, do not have a click. DEC lockout is indicated by alignment of scribed line on PAS spindle with stopboss on HMU housing.
- 18. Inspect data plate on HMU to determine which HMU is installed.
 - 19. On HMU, 6038T62P15 and 6068T77P03, rig for DEC lockout as follows:
 - a. Move quadrant ENG POWER CONT lever from FLY detent towards LOCKOUT and listen at engine HMU for a click to indicate DEC lockout has occurred. Stop moving lever as soon as click is heard.
 - b. Adjust stopnut against housing. Turn stopnut back two full turns.
 - c. Move ENG POWER CONT lever to IDLE to reset DEC.
 - d. Move ENG POWER CONT lever to LOCKOUT and listen at HMU for click. Click must be heard before stopnut touches housing.
 - 20. On HMU, 6058T65P14 and 6068T97P03, rig for DEC lockout as follows:
 - a. Move quadrant ENG POWER CONT lever from FLY detent towards LOCKOUT. Stop moving lever when scribed line on PAS spindle is aligned with stop boss on HMU housing (Figure 1, Detail B). Scribed line must be completely visible.
 - b. Adjust stopnut in quadrant against housing. Turn stopnut back two full turns.

RIGGING AND ADJUSTMENT - Continued



AK1294B
SA

Figure 2. Rig Engine Power-Available System.

RIGGING AND ADJUSTMENT - Continued

- c. Move quadrant ENG POWER CONT lever to IDLE to reset DEC.
 - d. Move quadrant ENG POWER CONT lever towards LOCK OUT and check that scribed line on PAS spindle is aligned with stop boss on HMU housing and is completely visible. Alignment must occur before stopnut touches housing.
- 21. If click is not heard before housing touches stopnut, or if housing touches stopnut before alignment occurs, repeat step 18. through step 19.
 - 22. If attempt to rig is still unsuccessful, repeat step 8. through step 19.
 - 23. If attempt to rig is still unsuccessful, replace engine power-available push/pull cable and fuselage power-available push/pull cable (WP 0507 00 and WP 0508 00).
 - 24. Adjust OFF stop as follows:
 - a. Turn bolt in so push/pull cable internal stop will limit travel when quadrant ENG POWER CONT lever is moved to rear (Figure 2, Detail B).
 - b. Turn bolt out until it contacts face of quadrant ENG POWER CONT lever when lever limits have been reached.
 - c. Release quadrant ENG POWER CONT lever and turn bolt out one full turn to make sure OFF stop is reached before push/pull cable internal stop or engine stop.
 - 25. Adjust quadrant ENG FUEL SYS lever as follows:
 - a. Place quadrant ENG FUEL SYS lever in line with OFF on quadrant side cover.
 - b. Adjust rod end on fuel selector valve push/pull cable so rod end bearing and lever holes line up. **TIGHTEN JAMNUT 1/6 - 1/3 TURN PAST POINT WHERE SHARP RISE IN TORQUE IS FELT (Figure 2, Detail A).**
 - c. Install fuel selector valve push/pull cable to quadrant ENG FUEL SYS lever with bolt, washer, and nut. **TORQUE NUT TO 30 - 50 INCH-POUNDS.** Install cotter pin.
 - 26. Adjust engine starter switch (WP 0503 00).
 - 27. Adjust starter override switch (WP 0505 00).
 - 28. Adjust starter abort switch (WP 0504 00).
 - 29. Adjust fire extinguisher arming switch (WP 0530 00).
 - 30. Adjust secondary stop blocks (WP 0501 00).
 - 31. Make sure area is clean and free of foreign material.
 - 32. Hinge up and lock upper console to cockpit overhead.
 - 33. Move fuel selector levers back to OFF.
 - 34. Close engine cowling and latch.
 - 35. Do a maintenance test flight (TM 1-1520-237-MTF).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

ENGINE MOUNT RESTRAINER RINGS

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101

Personnel Required

Airframe Structural Repairer MOS 15G (1)

Material/Parts

Cloth, Abrasive, Item 82, WP 1803 00
Collar, HL1087W-6
Corrosion Resistant Coating, Item 110, WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Rivet, MS90354-0604
Sealing Compound, Item 273, WP 1803 00

References

TM 1-1500-204-23
TM 1-1520-265-23
TM 55-1500-345-23
WP 0381 00
WP 1803 00

NOTE

- This procedure applies to forward support tube, 70361-10003-043, only.
- Replacement of forward and rear restrainer ring is the same.

REMOVE

1. Remove 4 rivets securing restrainer ring to forward support tube using number 4 drill. Do not allow rivet to spin in hole. Remove ring from tube (TM 1-1500-204-23) (Figure 1).
2. Remove any remaining primer from area around holes using abrasive cloth, Item 82, WP 1803 00.

INSTALL**NOTE**

- Slots in restrainer rings shall be installed opposite (180 degrees) from each other.
- Restrainer rings shall be installed with boss facing towards ends of forward support tube.

1. Remove sealing compound from area around removed fastener holes.
2. If installing new restrainer ring, do the following:
 - a. Position restrainer ring in installed position.
 - b. Mark hole in forward support tube that aligns with pilot hole in restrainer ring.
 - c. Remove restrainer ring.
 - d. Measure marked fastener hole in forward support tube.
 - e. If marked fastener hole is greater than 0.202-inch, redrill hole using 13/64-inch drill bit. Deburr hole.

INSTALL - Continued

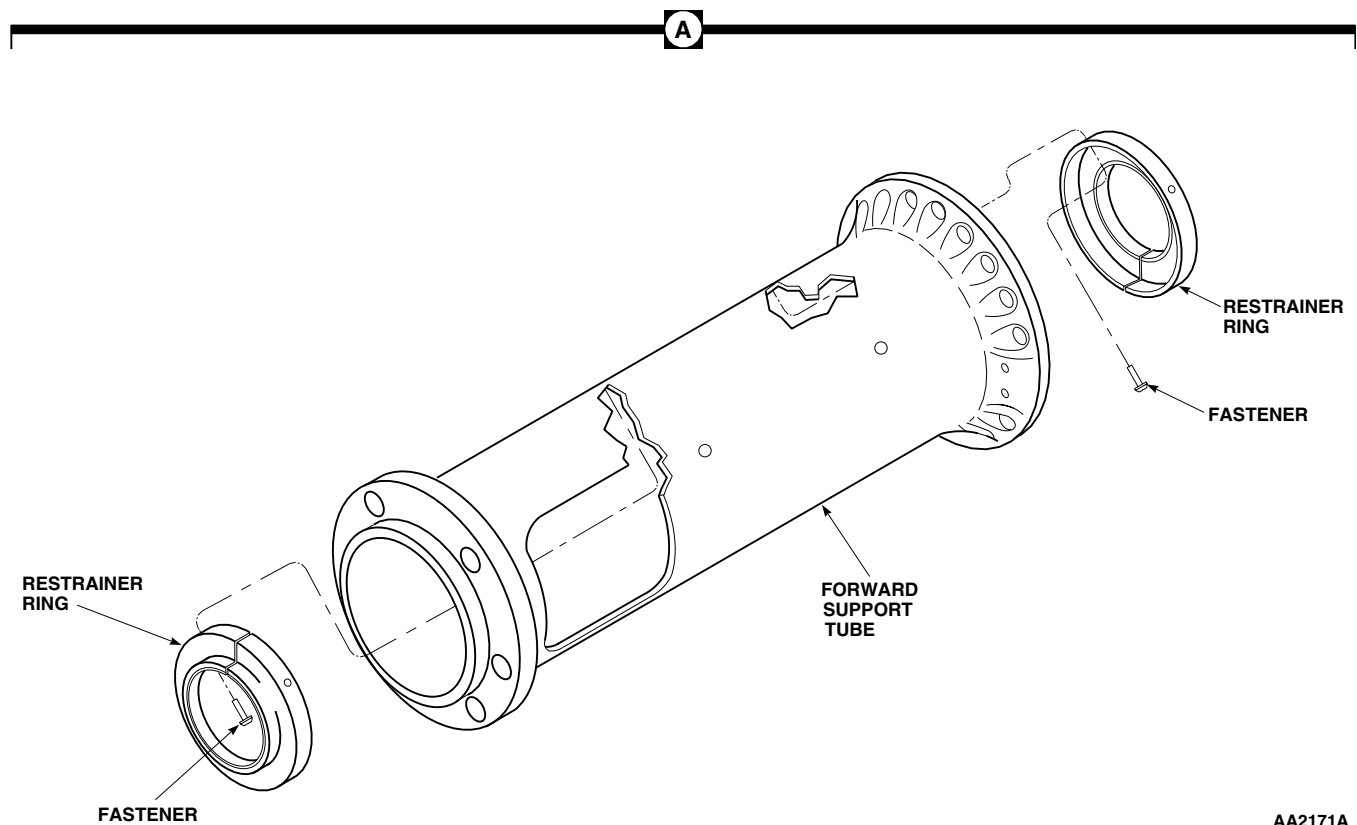
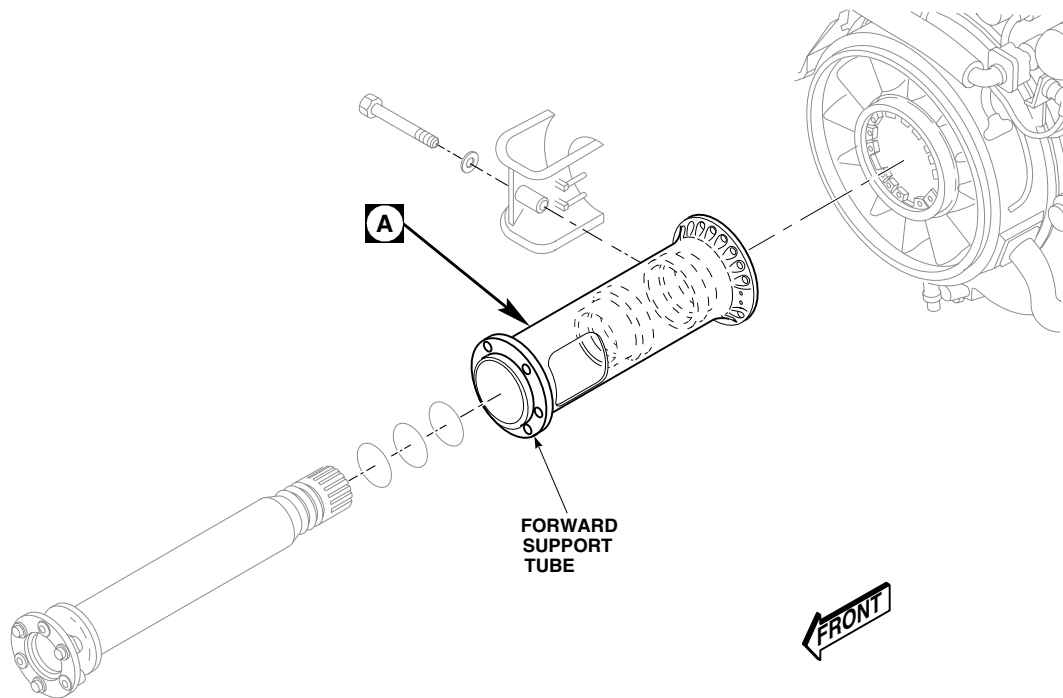
AA2171A
SA

Figure 1. Replace Engine Mount Restrainer Rings.

INSTALL - Continued

- f. Drill out pilot hole in restrainer ring to diameter of marked hole in forward support tube. Deburr hole.
- g. Measure remaining fastener holes in forward support tube.
- h. Redrill any of the remaining fastener holes using 13/64-inch drill if fastener hole is greater than 0.020-inch. Deburr holes.
- i. Check fastener holes for cracks using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace forward tube if cracks are found.
- j. Position restrainer ring in forward support tube aligning hole in restrainer ring with marked hole in forward support tube. Install cleco.

NOTE

A nonmetallic wedge in slot of restrainer may be used to make sure restrainer ring is making contact with inner surface of forward support tube.

- k. While making sure restrainer ring is in contact with inner surface of forward support tube, drill same diameter hole in restrainer ring as hole in forward support tube. Install cleco.
- l. Remove clecos and restrainer ring. Deburr holes.
- m. **QA** Position restrainer ring in forward support tube and secure with clecos.

NOTE

Clecos shall be removed one at a time to install fasteners.

- n. Remove cleco. Apply corrosion resistant coating, Item 110, WP 1803 00, to hole (TM 1-1500-344-23).

NOTE

- Install Hi-Lok fastener, HL-64PB6-5, and collar, HL87W-6, if hole was drilled to 13/64-inch. Install rivet, MS 90354-0604, in other holes.
 - A maximum gap of 0.020-inch is allowed between restrainer ring and forward support tube after rivet or Hi-Lok has been installed.
- o. Coat rivet or Hi-Lok with epoxy primer coating kit, Item 120, WP 1803 00. Install rivet or Hi-Lok while primer is still wet (TM 1-1500-204-23).
 - p. Touch-up finish as required (WP 0381 00 and TM 55-1500-345-23).
 - q. Apply sealing compound, Item 273, WP 1803 00, around head of rivet or Hi-Lok fasteners.
3. If installing a previously installed restraining ring, do the following:
- a. Measure fastener holes in forward support tube.
 - b. If fastener hole(s) is greater than 0.020-inch, redrill hole using 13/64-inch drill bit. Deburr hole(s).
 - c. Check fastener holes for cracks using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace forward tube if cracks are found.
 - d. Position restrainer ring in forward support tube. Install clecos.
 - e. If fastener hole in forward support tube was drilled to 13/64-inch, drill hole in restrainer ring to 13/64-inch.
 - f. Remove restrainer ring and deburr holes.
 - g. **QA** Position restrainer ring in forward support tube and secure with clecos.

INSTALL - Continued**NOTE**

Clecos shall be removed one at a time to install fasteners.

- h. Remove cleco. Apply corrosion resistant coating, Item 110, WP 1803 00, to hole.

NOTE

- Install Hi-Lok fastener, HL-64PB6-5, and collar, HL87W-6, if hole was drilled to 13/64-inch. Install rivet, MS 90354-0604, in other holes.
 - A maximum gap of 0.020-inch is allowed between restrainer ring and forward support tube after rivet or Hi-Lok has been installed.
- i. Coat rivet or Hi-Lok with epoxy primer coating kit, Item 120, WP 1803 00. Install rivet or Hi-Lok while primer is still wet (TM 1-1500-204-23).
- j. Touch-up finish as required (WP 0381 00 and TM 55-1500-345-23).
- k. Apply sealing compound, Item 273, WP 1803 00, around head of rivet or Hi-Lok fasteners.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENGINE SYSTEM

ENGINE CONTROL QUADRANT FRONT COVER INFORMATION PLATE

INITIAL SETUP:

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

WP 0106 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Remove screws and washers and disconnect front information plate from quadrant center cover (Figure 1).

INSTALL

1. Turn off all electrical power.
2. **QA** Gently position front information plate on quadrant center cover and push into electrical connector.

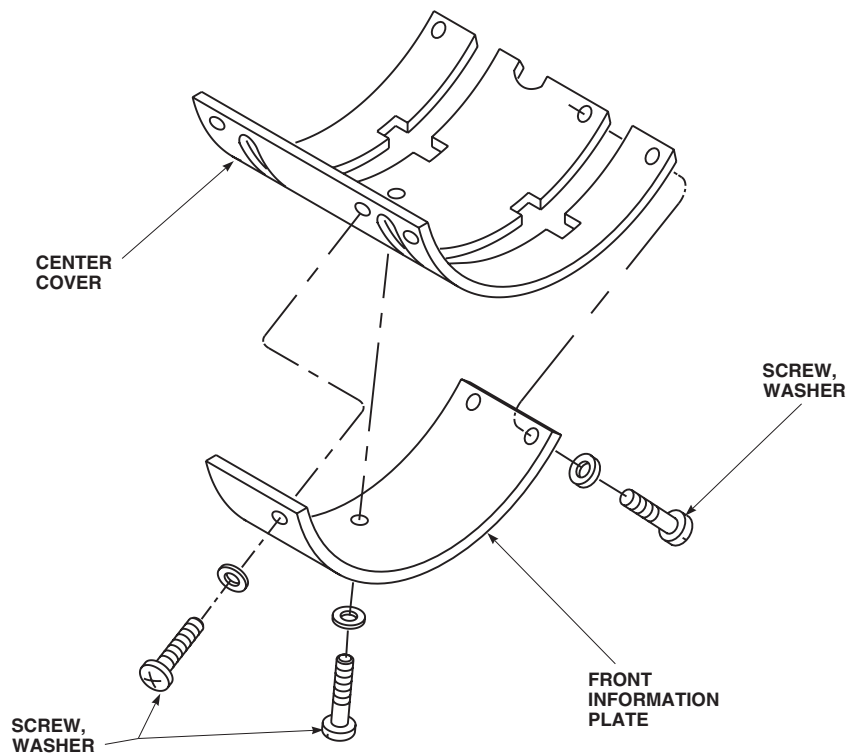
AA7713
SA

Figure 1. Replace Engine Control Quadrant Front Cover Information Plate.

INSTALL - Continued

3. **QA** Install front information plate to center cover with screws and washers (Figure 1).
4. Do an operational check of upper console lights (WP 0106 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
AFT ENGINE MOUNT STRUTS (AVIM)

INITIAL SETUP:
Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Bushing Installation and Removal Kit, 65700-20002-041
 Fluorescent Penetrant Inspection Kit, XMA101
 Roller Staking Tool, RST 2154

Pad, Scouring, Item 205, WP 1803 00
 Primer Coating, Item 243, WP 1803 00
 Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Adhesive, Item 26, WP 1803 00
 Corrosion Removing Compound, Item 108, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00

References

TM 1-1520-265-23
 TM 55-1500-322-24
 WP 1803 00

NOTE

Strut repair consists of replacing bearings and bushings, inspecting, and refinishing.

REMOVE BEARING/BUSHINGS

1. Remove bearing, using instructions in TM 55-1500-322-24 (Figure 1).
2. Remove bushing(s) from strut, using bushing installation and removal kit and arbor press.

INSPECT

1. Inspect bearing for corrosion and damage. **REPLACE BEARING IF PITS ARE VISIBLE IN A 0.250-INCH DIAMETER OR IF PITS EXCEED 0.010-INCH DEEP.**
2. Inspect bearing for axial play. **AXIAL PLAY SHALL NOT BE MORE THAN 0.020-INCH.** Replace bearing if axial play is beyond limits.
3. Inspect bearing for radial play. **RADIAL PLAY SHALL NOT BE MORE THAN 0.005-INCH.** Replace bearing if radial play is beyond limits.
4. Inspect bearing bore for scratches. **REPLACE BEARING IF SCRATCHES ARE GREATER THAN 0.500-INCH LONG AND 0.005-INCH DEEP.**
5. Remove minor corrosion and scratches from bearings using scouring pad, Item 205, WP 1803 00.
6. Inspect strut for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace strut if cracks are found.
7. Inspect strut for nicks, gouges, scratches, and pitting. **REPLACE STRUT IF DAMAGE IS GREATER THAN 0.250-INCH LONG AND 0.032-INCH DEEP.**
8. Measure bearing hole wear per applicable dimension in Table 1.
9. Measure bushing hole wear per applicable dimension in Table 1.
10. Measure inner mount strut to strut mount bolt hole wear per applicable dimension in Table 1.

INSPECT - Continued

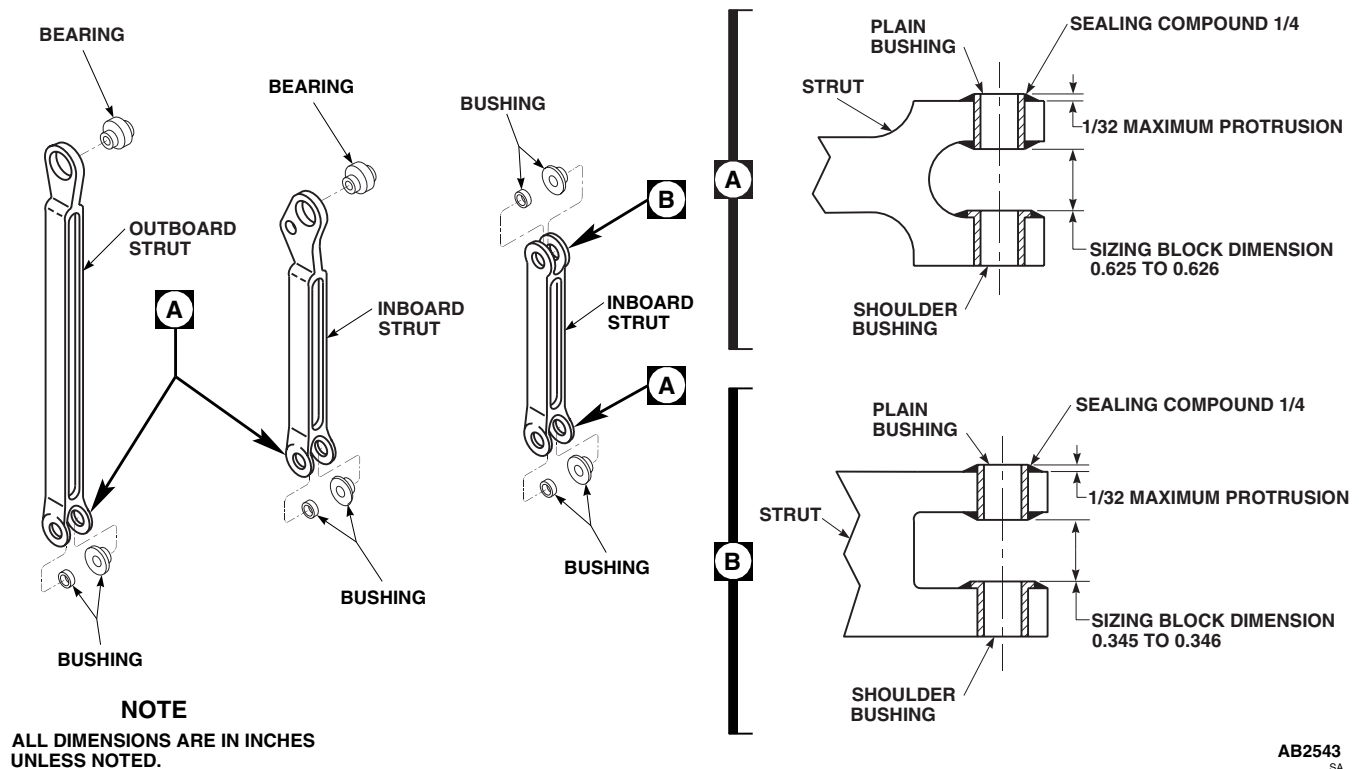


Figure 1. Repair Aft Engine Mount Struts (AVIM).

11. Blend out nicks, gouges, scratches, and pitting using 1/2-inch radius abrasive wheel, Item 350, WP 1803 00. **BLEND MUST NOT BE MORE THAN 0.250-INCH LONG AND 0.032-INCH DEEP.** Replace strut if repair is beyond limits.
12. Clean surface with corrosion removing compound, Item 108, WP 1803 00. Apply 2 coats of epoxy primer coating kit, Item 120, WP 1803 00, to surfaces where plating was disturbed.

Table 1. Maximum Allowable Bore Dimensions For Engine Strut, Support, Links, and Bushings.

Basic Assembly Part Number/Bore	Applicable Bushing Dash Number	
	-101	-102
70305-02105/bushing	0.4376	0.4376
70305-02106/bushing	0.4376	0.4376
70305-02119/bushing	0.4376	0.4376
Basic Bushing Part Number/ Diameter	Applicable Bushing Dash Number	
	-101	-102

INSPECT - Continued

Table 1. Maximum Allowable Bore Dimensions For Engine Strut, Support, Links, and Bushings. - Continued

70305-22112/Inside	(see notes)	(see notes)
70305-22112/Outside	0.4396	0.4396
70305-02112/Inside	(see notes)	(see notes)
70305-02112/Outside	0.4396	0.4396
Basic Assembly Part Number/Bore		Maximum Diameter
70305-02105/unbushed	(see notes)	
70305-02105/bearing	0.9063	
70305-02106/bearing	0.9063	
70305-02110/bearing	0.6878	
70305-02111/bearing	0.6878	
70305-02115/unbushed	0.6370	
Link Assembly Part Number	Large Bore	Small Bore
70305-02107-101	0.5040	0.3757
70305-02107-102	0.5035	0.3772
NOTES:		
0.3158 WHEN USING BOLTS, NAS464P5-17 AND NAS464P5-12.		
0.3165 WHEN USING BOLTS, NAS6705DU14 AND NAS6705DU18.		

INSTALL BEARING/BUSHING

1. Install bearing, using instructions in TM 55-1500-322-24 (Figure 1).
2. **QA** Stake bearing in housing, using roller staking tool.
3. Coat bushings and strut bushing holes with primer coating, Item 243, WP 1803 00 (Figure 1, Detail A or Detail B).
4. **QA** Press bushing(s) into strut, using installation and removal kit and arbor press. Remove extra primer.
5. Line ream bushings as follows:
 - a. Line ream both lower bushings on struts, 70305-02105-043 and 70305-02106-043 as follows:
 - (1) **FOR 70305-02105-043 STRUTS, LINE REAM 0.3122 TO 0.3132-INCH CENTERED WITHIN 0.020-INCH AND PERPENDICULAR WITHIN 0.005-INCH.**

INSTALL BEARING/BUSHING - Continued

- (2) **FOR 70305-02106-043 STRUTS, LINE REAM 0.3122 TO 0.3132-INCH CENTERED WITHIN 0.005-INCH AND PERPENDICULAR WITHIN 0.003-INCH.**
- b. Line ream both upper and both lower bushings on strut, 70305-02119-042. **LINE REAM 0.3122 TO 0.3132-INCH CENTERED WITHIN 0.005-INCH AND PERPENDICULAR WITHIN 0.003-INCH.**
- 6. Apply adhesive, Item 26, WP 1803 00, around bushings.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****AFT ENGINE MOUNT FITTINGS (AVIM)**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Fluorescent Penetrant Inspection Kit, XMA101
Staking Tool, RST 2459

Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

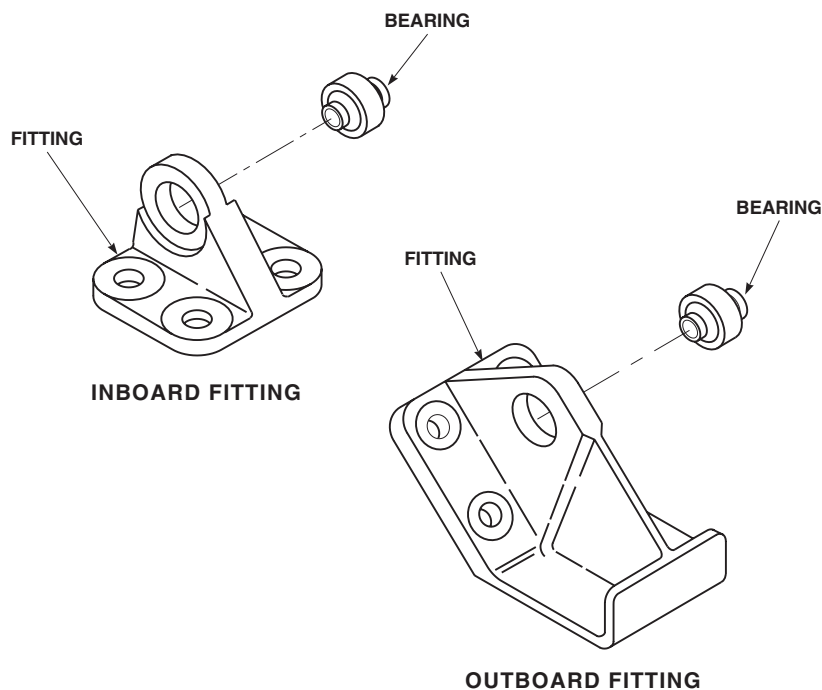
Corrosion Removing Compound, Item 108,
WP 1803 00
Epoxy Primer Coating Kit, Item 120, WP 1803 00
Pad, Scouring, Item 205, WP 1803 00
Remover, Paint, Item 250, WP 1803 00

References

TM 1-1500-344-24
TM 1-1520-265-23
TM 55-1500-322-24
WP 1803 00

REMOVE BEARING

Remove bearing per instructions in TM 55-1500-322-24 (Figure 1).



AK1253
SA

Figure 1. Repair Aft Engine Mount Fittings (AVIM).

REMOVE BEARING - Continued**INSPECT**

1. Inspect bearing for corrosion and damage. **REPLACE BEARING IF PITS ARE VISIBLE IN A 0.250-INCH DIAMETER OR IF PITS EXCEED 0.010-INCH DEEP.**
2. Inspect bearing for axial play. **AXIAL PLAY SHALL NOT BE MORE THAN 0.020-INCH.** Replace bearing if axial play is beyond limits.
3. Inspect bearing for radial play. **RADIAL PLAY SHALL NOT BE MORE THAN 0.005-INCH.** Replace bearing if radial play is beyond limits.
4. Inspect bearing bore for scratches. **SCRATCHES MUST NOT BE MORE THAN 0.500-INCH LONG AND 0.005-INCH DEEP.** Replace bearing if scratches are beyond limits.
5. Remove minor corrosion and scratches using scouring pad, Item 205, WP 1803 00.
6. Remove paint from fitting using paint remover, Item 250, WP 1803 00.
7. Inspect aft engine mount fitting for cracks (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace aft engine mount fitting if cracks are found.
8. Inspect fitting for nicks, gouges, scratches, and pitting. **DAMAGE MUST NOT BE MORE THAN 0.250-INCH LONG AND 0.032-INCH DEEP.** Replace fitting if damage is beyond limits.
9. Blend out nicks, gouges, scratches, and pitting on inboard fitting using 1/2-inch radius abrasive wheel, Item 350, WP 1803 00. **BLENDED AREA MUST NOT BE MORE THAN 0.250-INCH LONG AND 0.032-INCH DEEP.** Replace fitting if repair is beyond limits.
10. Blend out nicks, gouges, scratches, and pitting on outboard fitting using 1/2-inch radius abrasive wheel, Item 350, WP 1803 00. **BLENDED AREA MUST NOT BE MORE THAN 0.500-INCH LONG AND 0.032-INCH DEEP.** Replace fitting if repair is beyond limits.
11. Blend out pits on inboard and outboard fitting using scouring pad, Item 205, WP 1803 00, then apply corrosion removing compound, Item 108, WP 1803 00.
12. Clean surface and apply epoxy primer coating kit, Item 120, WP 1803 00, to surfaces where plating was disturbed per instructions in TM 1-1500-344-23.

INSTALL BEARING

1. Install bearing per instructions in TM 55-1500-322-24 (Figure 1).
2. **QA** Stake bearing in housing using staking tool.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****AFT ENGINE MOUNT LINKS (AVIM)**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Fluorescent Penetrant Inspection Kit, XMA101

References

TM 1-1520-265-23
WP 0515 00
WP 1803 00

Material/Parts

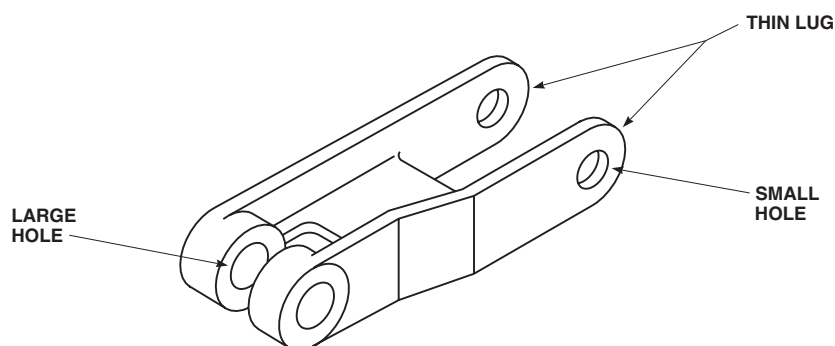
Wheel, Abrasive, Item 350, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

INSPECT

1. Measure for wear in holes at both ends of link per Table 1, (refer to WP 0515 00).
2. Inspect link in area of thin lugs for nicks, gouges, scratches, and pitting. **DAMAGE MUST NOT BE MORE THAN 0.250-INCH LONG AND 0.020-INCH DEEP.** Replace link if damage is beyond limits (Figure 1).



AK1254
SA

Figure 1. Repair Aft Engine Mount Links (AVIM).

INSPECT - Continued

3. Inspect remaining area of link for nicks, gouges, scratches, and pitting. **DAMAGE MUST NOT BE MORE THAN 0.500-INCH LONG AND 0.032-INCH DEEP.** Replace link if damage is beyond limits.
4. Inspect link for cracks, using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace link if cracks are found.
5. Blend out nicks, gouges, scratches, and pitting in area of thin lugs using 1/2-inch radius abrasive wheel, Item 350, WP 1803 00. **BLENDED AREA MUST NOT BE MORE THAN 0.250-INCH LONG AND 0.020-INCH DEEP.** Replace link if repair is beyond limits.
6. Blend out nicks, gouges, scratches, and pitting in remaining area of link 1/2-inch radius abrasive wheel, Item 350, WP 1803 00. **BLENDED AREA MUST NOT BE MORE THAN 0.500-INCH LONG AND 0.032-INCH DEEP.** Replace link if repair is beyond limits.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM**AFT ENGINE MOUNT SUPPORTS (AVIM)**

This WP supersedes WP 0518 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Fluorescent Penetrant Inspection Kit, XMA101

References

TM 1-1520-265-23
 WP 0515 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REPAIR

1. On aft mount supports, 70305-02115-042 and 70305-02115-043 only, inspect bolt attachment hole for damage to asbestos tape. Remove and discard tape if loose, missing or worn through to support. Do not replace tape (Figure 1).
2. Inspect aft mount support fitting inspection area (Figure 1) for nicks, gouges, scratches, and pitting. **DAMAGE MUST NOT BE MORE THAN 0.032-INCH DEEP AND THE MINIMUM LUG THICKNESS OF 0.43-INCH**

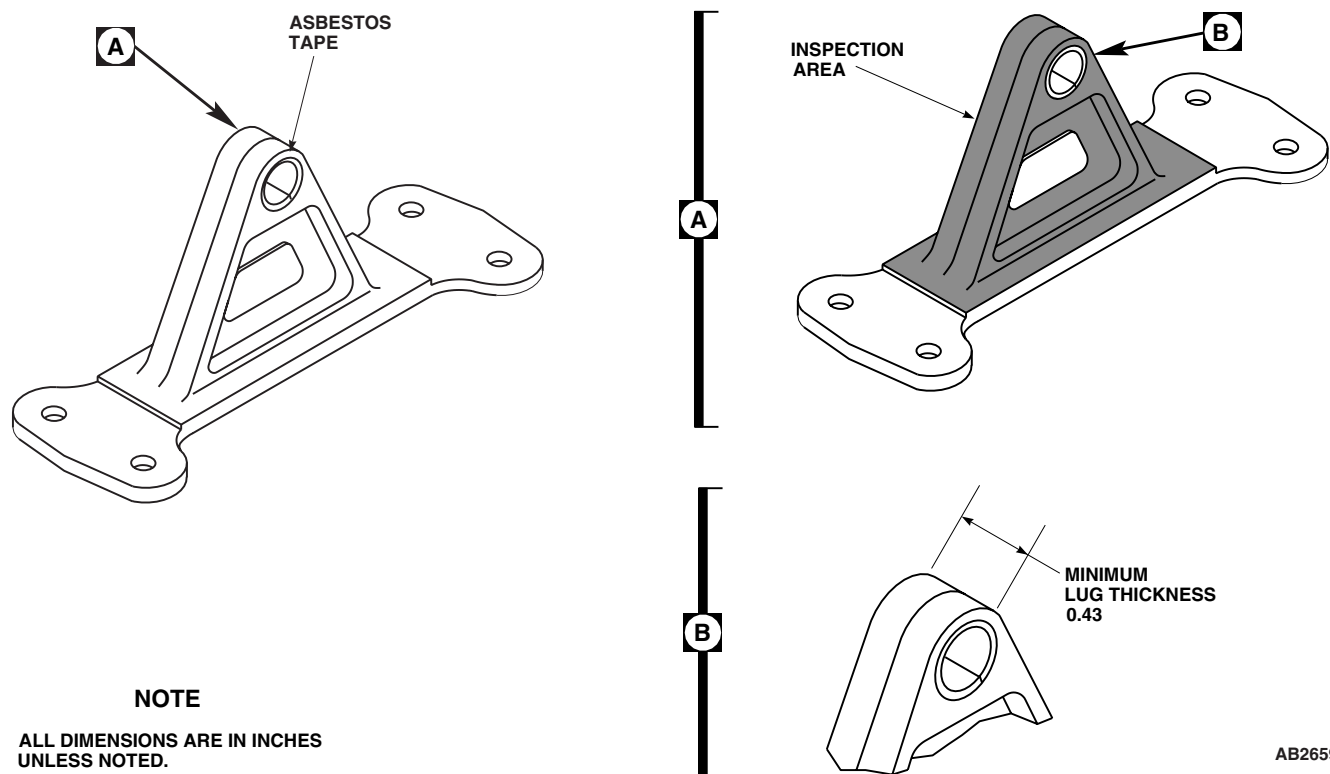


Figure 1. Repair Aft Engine Mount Supports (AVIM).

REPAIR - Continued

MUST BE MAINTAINED. Inspect all other areas (Figure 1) of aft mount support for nicks, gouges, scratches, and pitting. **DAMAGE MUST NOT BE MORE THAN 0.032-INCH DEEP OVER THE ENTIRE SHADED REGION ON BOTH THE FORWARD AND AFT SURFACES OF THE FITTING.** Replace aft mount support if damage exceeds limits.

3. Inspect aft mount support mount bolt bore wear per applicable dimension in Table 1, refer to WP 0515 00.
4. Inspect support for cracks using fluorescent penetrant inspection method (TM 1-1520-265-23). **NO CRACKS ALLOWED.** Replace support if cracks are found.
5. Blend out nicks, gouges, scratches, and pitting on aft mount support fitting inspection area (Figure 1). **BLENDED AREA MUST NOT BE MORE THAN 0.032-INCH DEEP AND THE MINIMUM LUG THICKNESS OF 0.43-INCH MUST BE MAINTAINED AFTER BLENDING.** Blend out nicks, gouges, scratches, and pitting on all other areas of aft mount support fitting (Figure 1). **BLENDED DAMAGE MUST NOT BE MORE THAN 0.032-INCH DEEP OVER THE ENTIRE SHADED REGION ON BOTH THE FORWARD AND AFT SURFACES OF THE FITTING.** Replace aft mount support if repair exceeds limits.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL

ENGINE SYSTEM

ENGINE MOUNT CROTCH ASSEMBLY (AVIM)

INITIAL SETUP:

Tools and Special Tools

Airframe Repairer's Toolkit, SC 5180-99-B02
 Backing Strip, Locally-Made, Figure 130, WP 1805 00
 Engine Inlet Firewall Seal, Locally-Made, Figure 65,
 WP 1805 00

Material/Parts

Carbon Dioxide, Item 64, WP 1803 00
 Sealing Compound, Item 292, WP 1803 00

Personnel Required

Airframe Structural Repairer MOS 15G (1)

References

TM 1-1500-204-23
 WP 0486 00
 WP 1803 00
 WP 1805 00

REPLACE PINS

1. Remove loose or damaged pins (TM 1-1500-204-23) (Figure 1).
2. Inspect pin bores for wear. **PIN BORES SHALL NOT EXCEED 0.1275-INCH.** Replace crotch assembly if either pin bore is greater than limit.

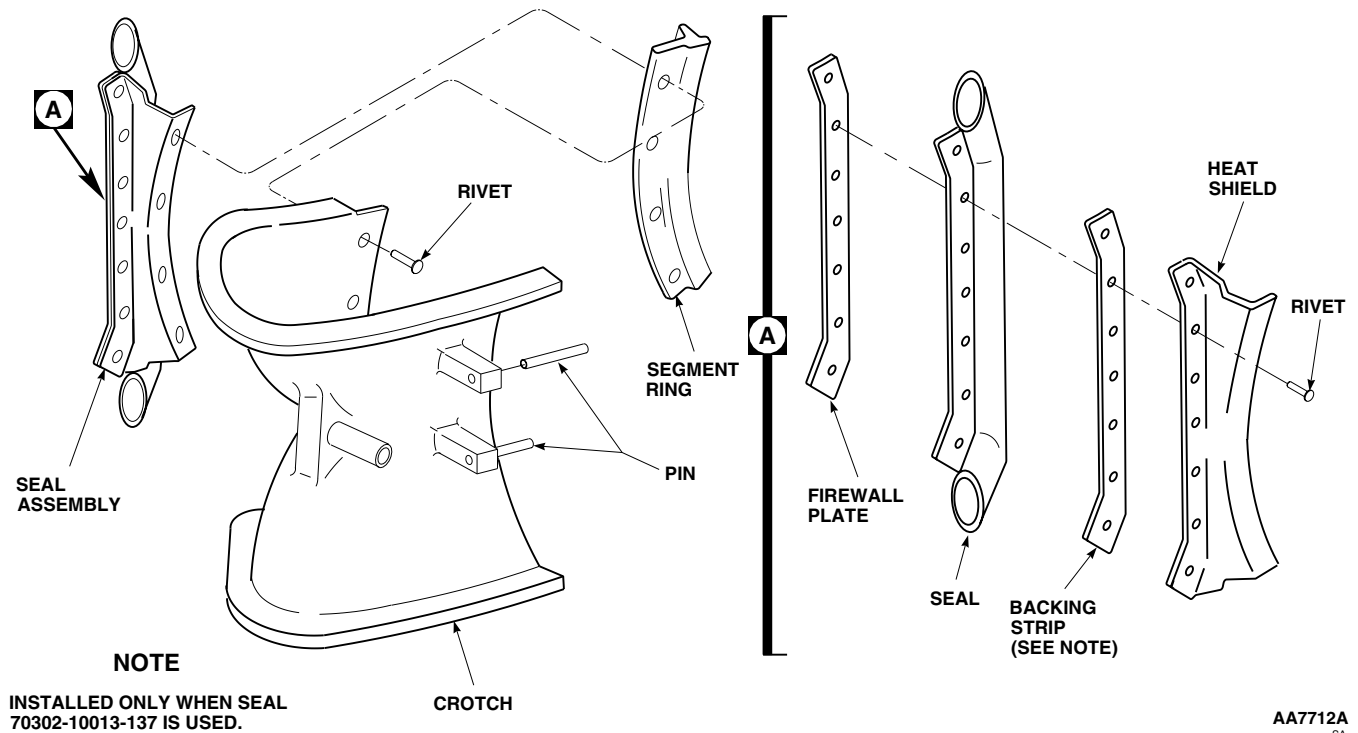


Figure 1. Repair Engine Mount Crotch Assembly (AVIM).

REPLACE PINS - Continued

3. Place replacement pin(s) in carbon dioxide, Item 64, WP 1803 00, for a minimum of 30 minutes.
4. Coat pin bores with sealing compound, Item 292, WP 1803 00. Insert pins in bores and allow to dry for 1 hour.

REPLACE CROTCH COMPONENTS**NOTE**

Individual parts of seal assembly may be replaced after seal assembly is removed from crotch and disassembled.

1. Remove rivets securing seal assembly and segment ring to crotch (TM 1-1500-204-23).

NOTE

Backing strip is installed only when seal, 70302-10013-137, is used.

2. **QA** Remove rivets securing heat shield, backing strip, seal, and firewall plate. Replace parts as required (Figure 1).
3. Check crotch for missing or damaged anodized finish. **NONE ALLOWED.** Replace crotch if anodized finish is missing or damaged (WP 0486 00).
4. If replacement seal is required, locally make engine inlet firewall seal (WP 1805 00).
5. If replacement backing strip is required, locally make backing strip (WP 1805 00).
6. Assemble heat shield, backing strip (if installed), seal, and firewall plate and secure with rivets (TM 1-1500-204-23).
7. **QA** Attach seal assembly and segment ring to crotch with rivets (TM 1-1500-204-23).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE EXHAUST EJECTOR AND FAIRINGS (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Fluorescent Penetrant Inspection Kit, XMA101

Material/Parts

Filler Wire, AMS 4951
Filler Wire, 347 MIL-R-5031, Class 5A

References

AMS 2689
MIL-W-8611
TM 1-1520-265-23
WP 0488 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

REPAIR

1. Remove exhaust ejector from exhaust module (WP 0488 00).
2. Weld stainless steel ejector and attaching angle by gas tungsten (GTA) method per MIL-W-8611, using 347 filler wire, MIL-R-5031, Class 5A (Figure 1).
3. Weld titanium upper and lower fairings per AMS 2689, using filler wire, AMS 4951.

INSPECT

1. Inspect repaired area using fluorescent penetrant method per TM 1-1520-265-23 with the following exceptions:
 - a. Inspector shall be dark-conditioned for a minimum of 5 minutes each time dark area is entered from white light area.
 - b. Inspector shall be dark-conditioned for a minimum of 5 minutes after looking directly into beam of black light or after using white light in inspection booth.
 - c. White light source shall provide a minimum of 200 foot-candles of illumination, at part surface, for inspection of fluorescent penetrant indications.
2. Permanently mark repaired component "PS-004". Marking shall be in same style and location as part number.

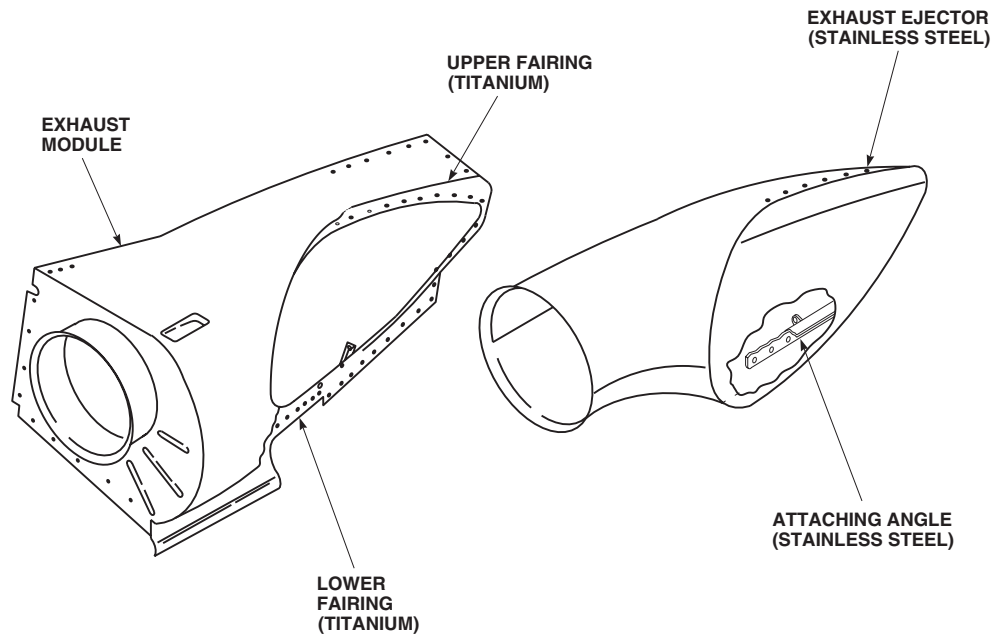
INSPECT - ContinuedAA0139A
SA

Figure 1. Repair Engine Exhaust Ejector and Fairings (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE EXHAUST MODULE DUCT ASSEMBLY (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Airframe Repairer's Toolkit, SC 5180-99-B02
Exhaust Fairing Bulkhead Seal, Locally-Made, Figure
115, WP 1805 00

References

TM 1-1500-204-23
WP 1805 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

REMOVE**NOTE**

Removal of left and right duct assemblies is the same.

1. Remove bulkhead from exhaust module by drilling out rivets on outside edge of bulkhead (TM 1-1500-204-23) (Figure 1).
2. Drill out rivets holding duct assembly to bulkhead and remove duct assembly (TM 1-1500-204-23) (Figure 1, Detail A).

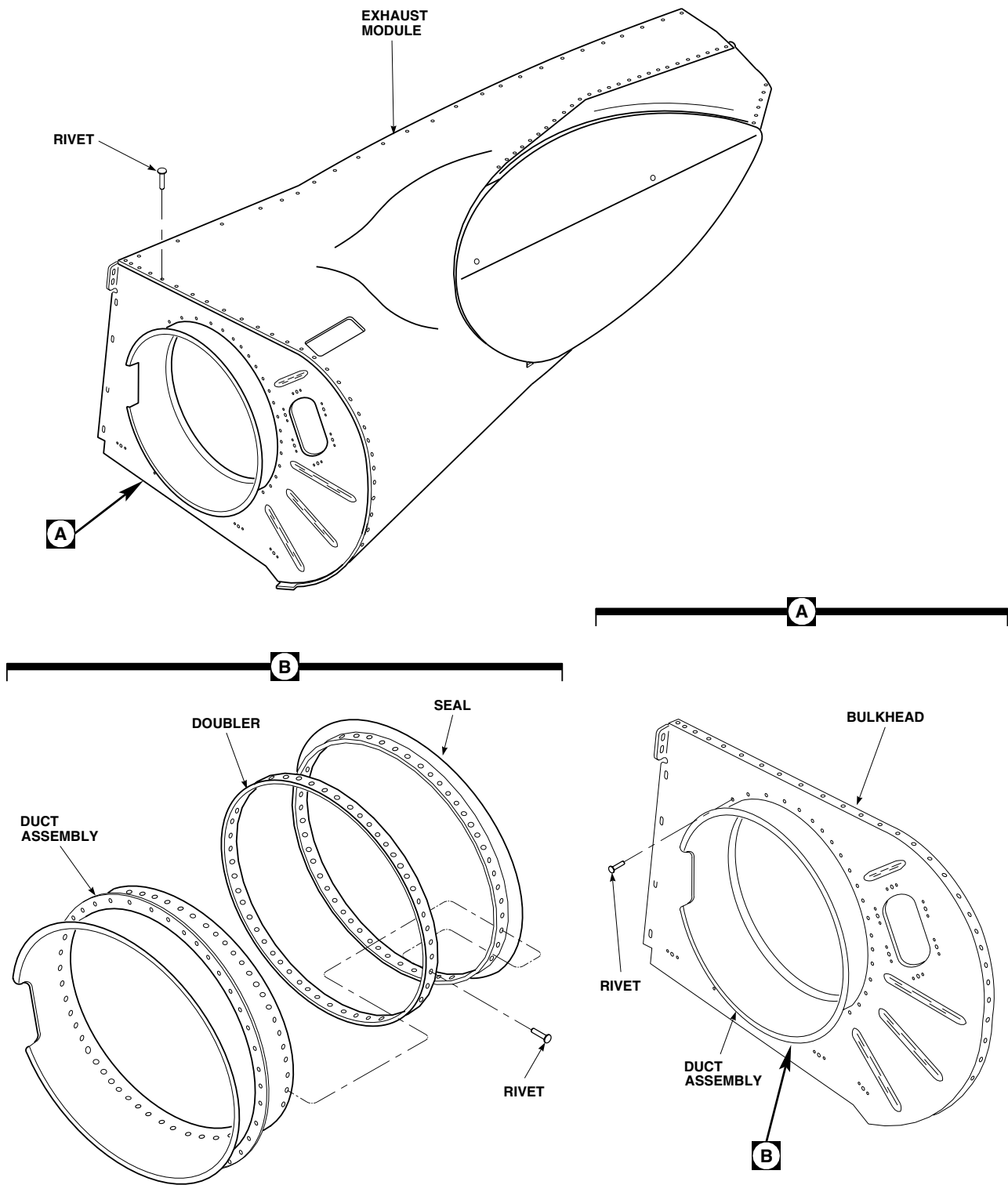
REPLACE SEAL

1. Drill out rivets holding doubler and seal to duct (TM 1-1500-204-23) (Figure 1, Detail B).
2. Locally make new exhaust fairing bulkhead seal (WP 1805 00).
3. Assemble seal and doubler on duct and secure with rivets (TM 1-1500-204-23).

INSTALL

1. **QA** Install duct assembly on bulkhead and secure with rivets (TM 1-1500-204-23).
2. **QA** Install bulkhead on exhaust module and secure outside edge with rivets (TM 1-1500-204-23).

INSTALL - Continued



AA7711
SA

Figure 1. Replace Engine Exhaust Module Duct Assembly (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

HIRSS INLET SEAL (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Torque Wrench, 30 - 200 in. lbs.

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
WP 0489 00
WP 0493 00

Personnel Required

Aircraft Structural Repairer MOS 15G (1)
Assistant - (1)

REMOVE**NOTE**

Removal of left and right inlet seals is the same.

1. Remove HIRSS module (WP 0489 00).
2. Remove nacelle fairing (WP 0493 00).
3. Remove bulkhead from bulkhead support flange by removing screws, washers and nuts (Figure 1, Detail A).
4. Drill out rivets securing inlet seal to stage 1 inner duct (Figure 1, Detail B) (TM 1-1500-204-23).
5. Remove inlet seal from stage 1 inner duct of suppressor core.

INSTALL

1. **QA** Install new inlet seal into stage 1 inner duct of suppressor core (Figure 1, Detail B and Figure 1, Detail C).
2. Position bracket of inlet seal 0.963 to 0.883-inch from stage 1 outer duct flange (Figure 1, Detail C).
3. **QA** Install inlet seal to stage 1 inner duct with rivets (TM 1-1500-204-23).
4. **QA** Install bulkhead on bulkhead support flange of suppressor core and attach with screws, washers and nuts.
TORQUE NUTS TO 43 - 47 INCH-POUNDS (Figure 1, Detail A).
5. Install nacelle fairing (WP 0493 00).
6. Install HIRSS module (WP 0489 00).

INSTALL - Continued

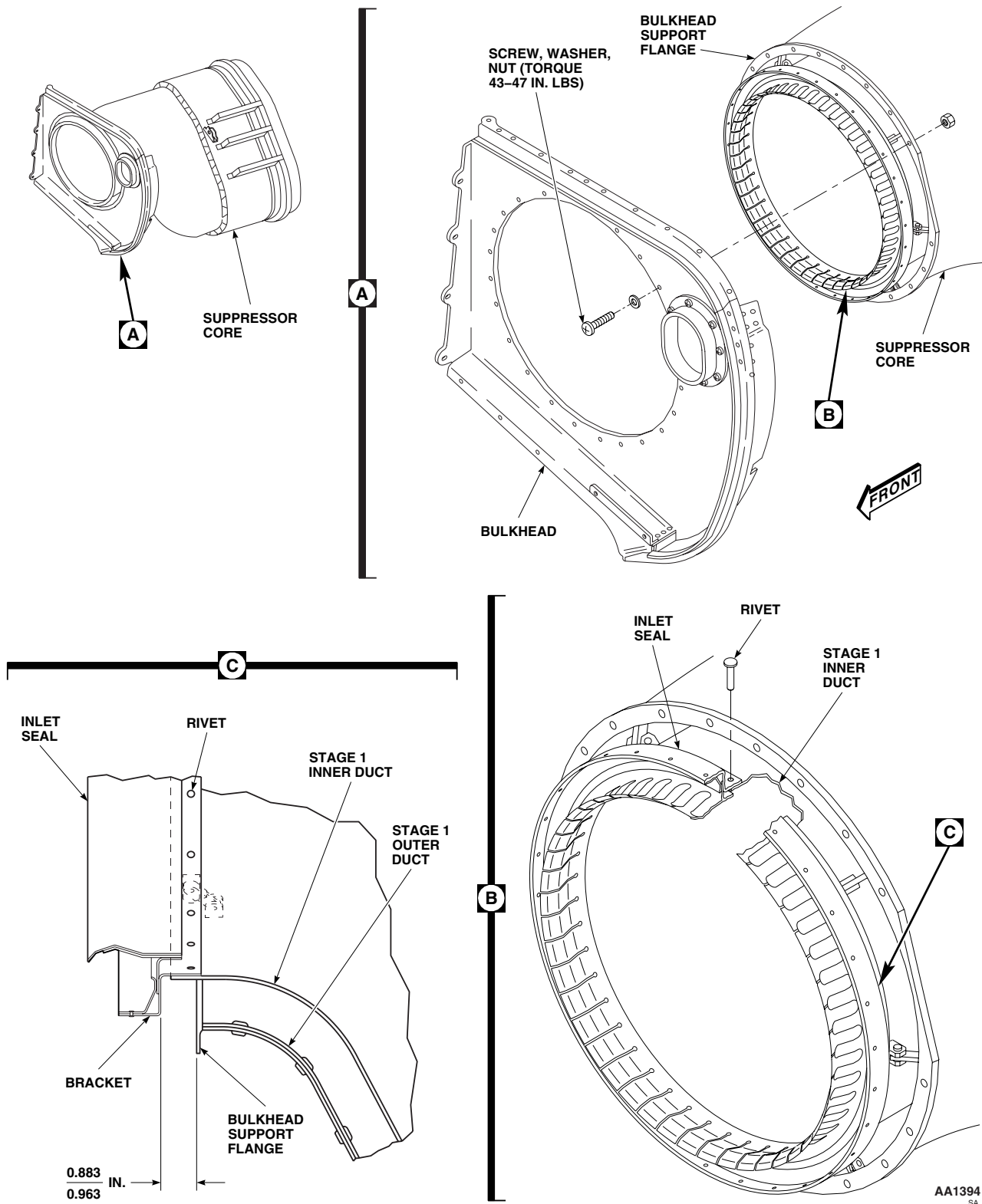


Figure 1. Replace HIRSS Inlet Seal (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****ROTARY INPUT (AVIM)**

This WP supersedes WP 0523 00, dated 21 March 2007.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Fluorescent Penetrant Inspection Kit, XMA101
Gloves
Goggles/Face Shield
Rotary Input Bearing Removal Tube, Locally-Made,
Figure 211, WP 1805 00
Spring Tester, PB4-50D
Torque Wrench, 30 - 200 in. lbs.

Cloth, Cleaning, Item 86, WP 1803 00
Grease, Aircraft, Item 132, WP 1803 00
Primer Coating, Item 243, WP 1803 00
Sealing Compound, Item 281, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1500-204-23
TM 1-1520-265-23
TM 55-1500-322-24
WP 1803 00
WP 1805 00

Material/Parts

Acetone, Technical, Item 5, WP 1803 00
Cleaning Compound, Solvent, Item 71, WP 1803 00
Cloth, Abrasive, Item 81, WP 1803 00

NOTE

The following procedure applies to both the power-available and load-demand rotary inputs. When maintaining these rotary inputs, keep parts separated.

DISASSEMBLE**NOTE**

Heating of rotary input may be necessary in order to weaken Loctite bond between adapter and housing.

1. Place rotary input in a 250°F(121°C) preheated oven for 15 to 20 minutes.
2. Place rotary input in padded vise.
3. Remove adapter from housing (Figure 1 and Figure 2). Repeat step 1 and step 2 if necessary.
4. Remove spring from adapter (only on load-demand input) (Figure 2).
5. Remove large retaining ring from housing (Figure 1 and Figure 2).
6. Remove small retaining ring from end of rotary barrel.
7. Remove front bearing from housing. If bearing is seated tightly in housing, do the following:
 - a. Locally make rotary input bearing removal tube (WP 1805 00).
 - b. Place tube in rear of housing over drive assembly.
 - c. Lightly tap tube to unseat bearing.
8. Remove rotary barrel and drive assembly from housing.
9. Remove drive assembly from rotary barrel.

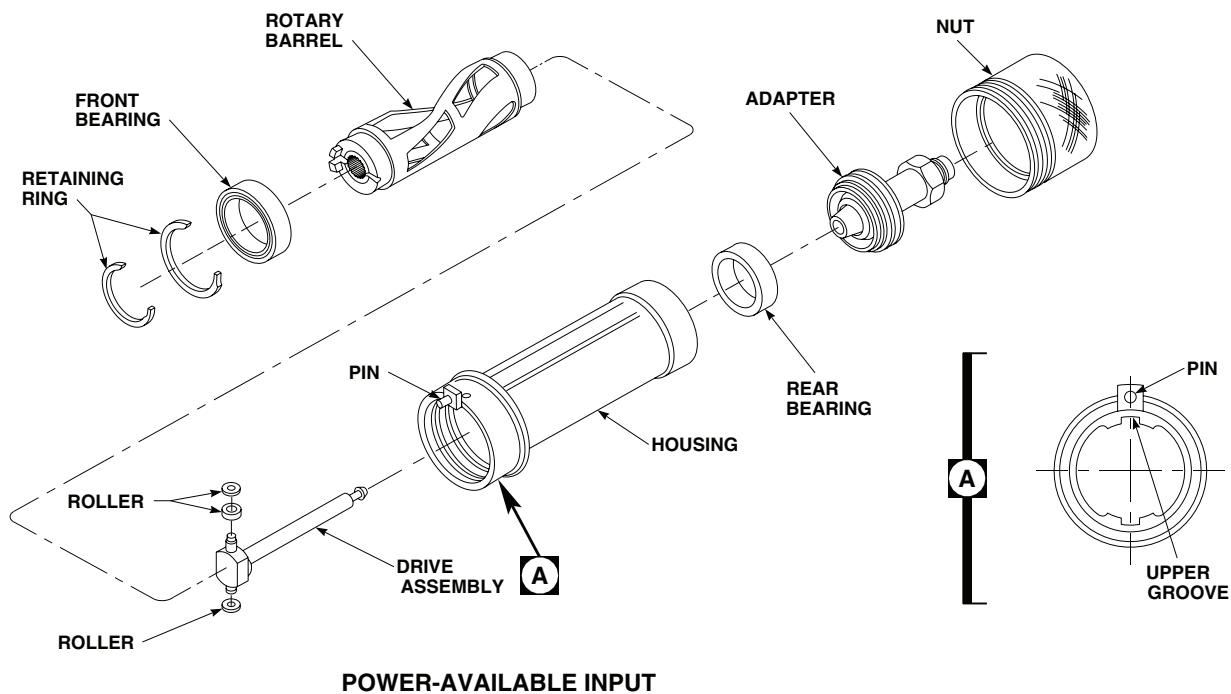
DISASSEMBLE - Continued

Figure 1. Power-Available Input Repair (AVIM).

10. Remove rollers from drive assembly.
11. Remove rear bearing from housing. If necessary, lightly tap bearing with phenolic drift to unseat it from housing.

INSPECT**WARNING**

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

CAUTION

Damage to bearings will result if contact with solvents occur. Do not allow solvents to contact bearings.

1. Clean housing bearing bores with cleaning cloth, Item 86, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00 (Figure 1 and Figure 2).
2. Clean remaining metal parts with cleaning compound solvent, Item 71, WP 1803 00.
3. Inspect bearings (TM 55-1500-322-24).

INSPECT - Continued

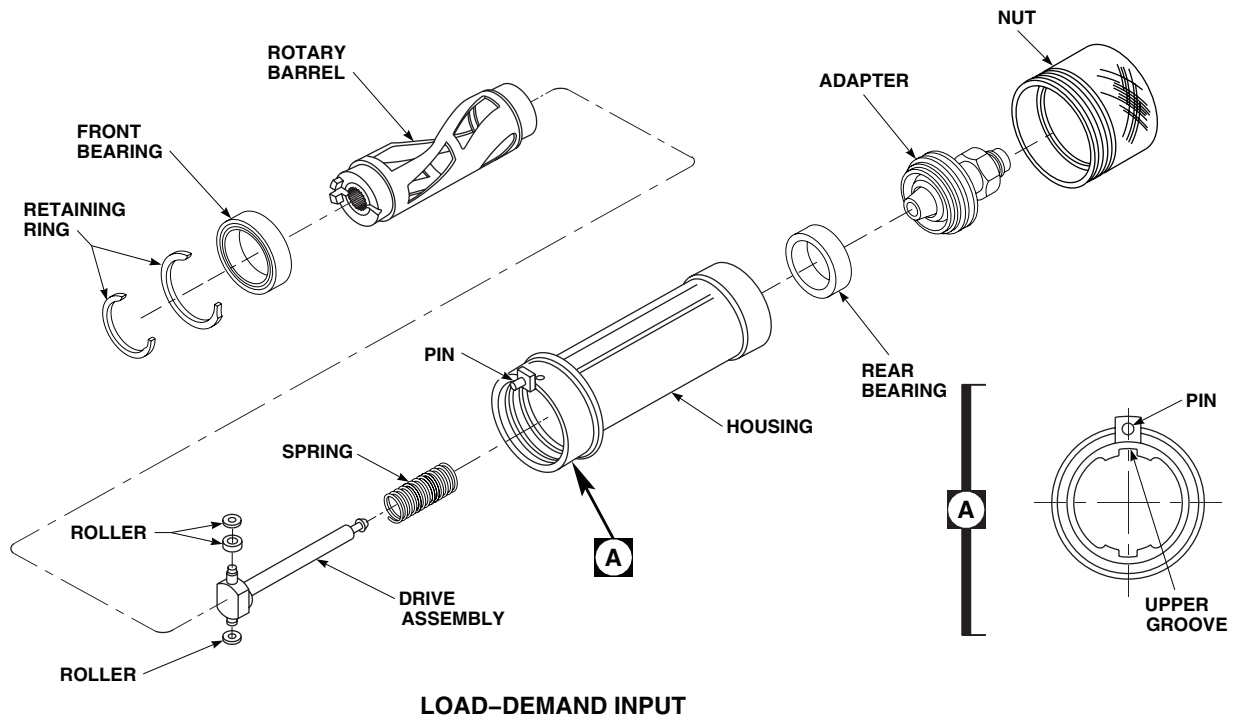
AA2096B
SA

Figure 2. Load-Demand Input Repair (AVIM).

4. Check load-demand input spring with spring tester. **COMPRESS LOAD-DEMAND INPUT SPRING TO 1.250-INCHES. SPRING PRESSURE MUST BE AT LEAST 24 POUNDS, BUT NOT MORE THAN 26 POUNDS.** Replace spring if not within limits.
5. Check retaining rings for damage. **NONE ALLOWED.** Replace retaining rings if damaged.
6. Visually inspect rotary barrel for nicks, scratches, and corrosion. **NONE ALLOWED.** Replace rotary barrel if nicks, scratches or corrosion are found.
7. Fluorescent penetrant inspect, Type 1, Method C, rotary barrel for cracks and fractures (TM 1-1520-265-23). **NONE ALLOWED.** Replace rotary barrel if cracked or fractured.
8. Inspect slots of rotary barrel for wear. **WIDTH OF NARROW SLOT SHALL NOT EXCEED 0.2530-INCH. OD OF SLOTTED END SHALL NOT BE LESS 1.0615-INCHES. OD OF NONSLOTTED END SHALL NOT BE LESS THAN 0.9995-INCH. ID OF NONSLOTTED END SHALL NOT EXCEED 0.870-INCH.** Replace rotary barrel if not within limits.
9. Inspect drive assembly for nicks, scratches, and corrosion. **NONE ALLOWED.** Replace drive assembly if damage is found.
10. Fluorescent penetrant inspect, Type 1, Method C, drive assembly for cracks and fractures (TM 1-1520-265-23). **NONE ALLOWED.** Replace drive assembly if cracked or fractured.
11. Inspect rollers for nicks, scratches, and corrosion. **NONE ALLOWED.** Replace roller if damage is found.
12. Measure rollers for wear. **OD OF ROLLERS 70301-02117-101 AND -102 SHALL NOT BE LESS THAN 0.249-INCH. ID OF ROLLER 70301-02117-101 SHALL NOT BE MORE THAN 0.153-INCH. ID OF ROLLER 70301-02117-102 SHALL NOT BE MORE THAN 0.125-INCH.** Replace roller if not within limits.

INSPECT - Continued

13. Inspect adapter as follows:
 - a. Inspect all external surfaces of adapter except threads for nicks, scratches, and corrosion. **DEPTH OF EXTERNAL DAMAGE SHALL NOT EXCEED 0.010-INCH DEEP.** Replace adapter if damage is beyond limits. Clean up external surface corrosion within limits using abrasive cloth, Item 81, WP 1803 00.
 - b. Inspect internal surfaces and all threads of adapter for damage. **NONE ALLOWED.** Replace adapter if any internal surface or thread damage is found.
14. Fluorescent penetrant inspect, Type 1, Method C, adapter for cracks and fractures (TM 1-1520-265-23). **NONE ALLOWED.** Replace adapter if cracked or fractured.
15. Inspect pin as follows:
 - a. Check pin at rear of housing for looseness. **NONE ALLOWED.** Replace pin if loose or missing.
 - b. If required, carefully remove damaged pin.
 - c. Inspect pin bore. **BORE MUST BE 0.1555 TO 0.1559 INCHES. IF BORE EXCEEDS 0.1559 INCHES, REPLACE HOUSING.**
 - d. Press new pin into housing, pin must protrude 0.230 +/- 0.010 inches from housing.
16. Inspect housing as follows:
 - a. Inspect all external surfaces of housing except threads for nicks, scratches, and corrosion. **DEPTH OF EXTERNAL DAMAGE SHALL NOT EXCEED 0.010-INCH DEEP.** Replace housing if damage is beyond limits. Clean up external surface corrosion within limits using abrasive cloth, Item 81, WP 1803 00.
 - b. Inspect internal surfaces and all threads of housing for damage. **NONE ALLOWED.** Replace housing if any internal surface or thread damage is found.
17. Fluorescent penetrant inspect, Type 1, Method C, housing for cracks and fractures (TM 1-1500-265-23). **NONE ALLOWED.** Replace housing if cracked or fractured.
18. Inspect nut as follows:
 - a. Inspect all external surfaces of nut except threads for nicks, scratches, and corrosion. **DEPTH OF EXTERNAL DAMAGE SHALL NOT EXCEED 0.010-INCH DEEP.** Replace nut if damage is beyond limits. Clean up external surface corrosion within limits using abrasive cloth, Item 81, WP 1803 00.
 - b. Inspect internal surfaces and all threads of nut for damage. **NONE ALLOWED.** Replace nut if any internal surface or thread damage is found.

NOTE

False indications of cracking are possible due to knurling on nut. If cracks are detected in knurled area, re-inspect to verify crack.

19. Fluorescent penetrant inspect, Type 1, Method C, nut for cracks and fractures (TM 1-1520-265-23). **NONE ALLOWED.** Replace nut if cracked or fractured.
20. Inspect front bearing bore of housing for wear. **ID OF FRONT BEARING BORE SHALL NOT EXCEED 1.501-INCH.** Replace housing if front bearing bore is not within limit.
21. Inspect rear bearing bore of housing for wear. **ID OF REAR BEARING BORE SHALL NOT EXCEED 1.2505-INCH.** Replace housing if rear bearing bore is not within limit.

ASSEMBLE

1. Apply a thin coat of primer coating, Item 243, WP 1803 00, to ID of front bearing and OD of rear bearing.

ASSEMBLE - Continued

2. **QA** Install rear bearing in housing while primer is wet (TM 1-1500-204-23).
3. **QA** Install front bearing onto rotary barrel and secure with small retaining ring (TM 1-1500-204-23).
4. Lubricate all moving parts with aircraft grease, Item 132, WP 1803 00.
5. **QA** Install rollers on drive assembly (Figure 1 and Figure 2).
6. **QA** Install drive assembly in rotary barrel. The side of drive assembly with two rollers fits into narrow slot in rotary barrel.
7. Apply a thin coat of primer coating, Item 243, WP 1803 00, to OD of front bearing.
8. **QA** Install rotary barrel and drive assembly into housing. Make sure that side of drive assembly with 2 rollers is in upper groove (Figure 1, Detail A and Figure 2, Detail A).
9. If necessary, reseal rear bearing by lightly tapping with mallet.

WARNING

Acetone is extremely flammable and toxic to the eyes, skin and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces, or other sources of ignition.

10. Clean retaining ring bore in housing with cleaning cloth, Item 86, WP 1803 00, moistened with technical acetone, Item 5, WP 1803 00.
11. **QA** Install large retaining ring in groove in housing assembly.
12. **QA** On load-demand rotary input only, push drive assembly to rear of housing assembly. Place spring onto drive assembly (Figure 2).
13. Apply sealing compound, Item 281, WP 1803 00, to threads of adapter and housing.
14. **QA** Place adapter over end of drive assembly. Thread adapter into housing. **TORQUE ADAPTER TO 95 - 105 INCH-POUNDS (Figure 1 and Figure 2).**
15. Check assembled rotary input for smooth operation.

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE CONTROL QUADRANT POWER LEVER (AVIM)
This WP supersedes WP 0524 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Material/Parts

Washer, AN960-616

References

TM 1-1500-204-23

WP 0528 00

WP 0529 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

The following procedures apply to both No. 1 quadrant and No. 2 quadrant. When maintaining No. 1 quadrant and No. 2 quadrant, keep parts separate.

1. Remove override and abort switches from engine power control lever (WP 0529 00 and WP 0528 00).
2. Remove nut and washers from bolt (Figure 1).
3. Slide lever from bolt.
4. Remove bushing and laminated shim from lever.

DISASSEMBLE

1. **UH-60A 86-24516 - SUBQ EH-60A 85-24469 - SUBQ UH-60L MWO 50-39** Remove nut, washers, bolt, spacers, torsion spring, and latch stop from control lever (Figure 1). 

NOTE

There are two small compressed springs and a latch in the lever assembly. Be careful when removing cover to avoid losing parts.

2. Remove screws and cover from lever (Figure 2).
3. Remove lever from handle.
4. Remove heavy spring, light spring, and latch from lever.
5. Remove abort switch screw and spring from handle.
6. Remove pin from starter button and remove button and spring from handle.
7. Remove plunger from handle.

INSPECT/REPAIR

1. **CHECK WEAR OF BUSHING FOR MAXIMUM ID OF 0.375-INCH.** If ID is worn beyond tolerance, replace bushing (Figure 1).
2. Inspect bearing for damage and wear in lever (TM 1-1500-204-23).
3. If required, press bearing from lever, using an arbor press.

ASSEMBLE

1. **QA** If bearing was removed, install new bearing in lever, using an arbor press (Figure 2).
2. Position plunger in handle.
3. **QA** Install starter button and spring on plunger, using pin.
4. Position abort switch screw and spring in handle.
5. **QA** Install nut and washer on abort switch screw.

NOTE

- Light spring refers to C0240-022-0810M.
- Heavy spring refers to C0240-038-1000M.

6. **QA** Place latch in light spring and install latch and spring, as an assembly, in upper slot of lever assembly.
7. **QA** Install heavy spring in lower slot of lever.

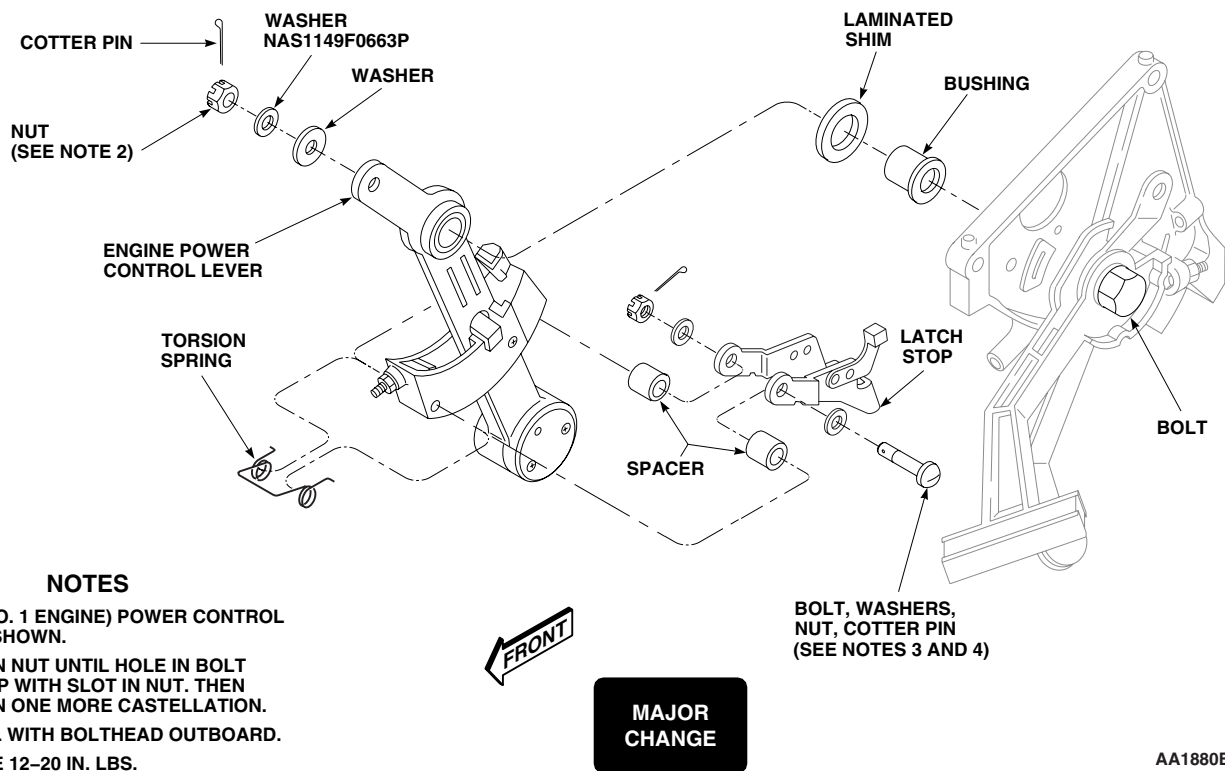


Figure 1. Replace Engine Control Quadrant Power Lever (AVIM).

ASSEMBLE - Continued

8. Position lever in handle.
9. **QA** Install cover on lever, using screws.
10. **UH-60A 86-24516 - SUBQ EH-60A 85-24469 - SUBQ UH-60L MWO 50-39** Install latch stop assembly as follows:
 - a. Position torsion spring on lever assembly (Figure 1).
 - b. Install latch stop on lever assembly then engage torsion spring.
 - c. **QA** Install bolt, washers, spacers, and nut, holding latch stop assembly in place.
 - d. **QA** Install cotter pin in nut.

INSTALL**NOTE**

If a new laminated shim is being installed, peel laminations as required to obtain minimum clearance between parts, with control lever moving freely, without drag.

1. **QA** Install laminated shim, with stainless steel side facing engine power control lever, on bushing (Figure 1).
2. **QA** Install bushing and laminated shim in engine power control lever.
3. **QA** Install engine power control lever on bolt with washers and nut.

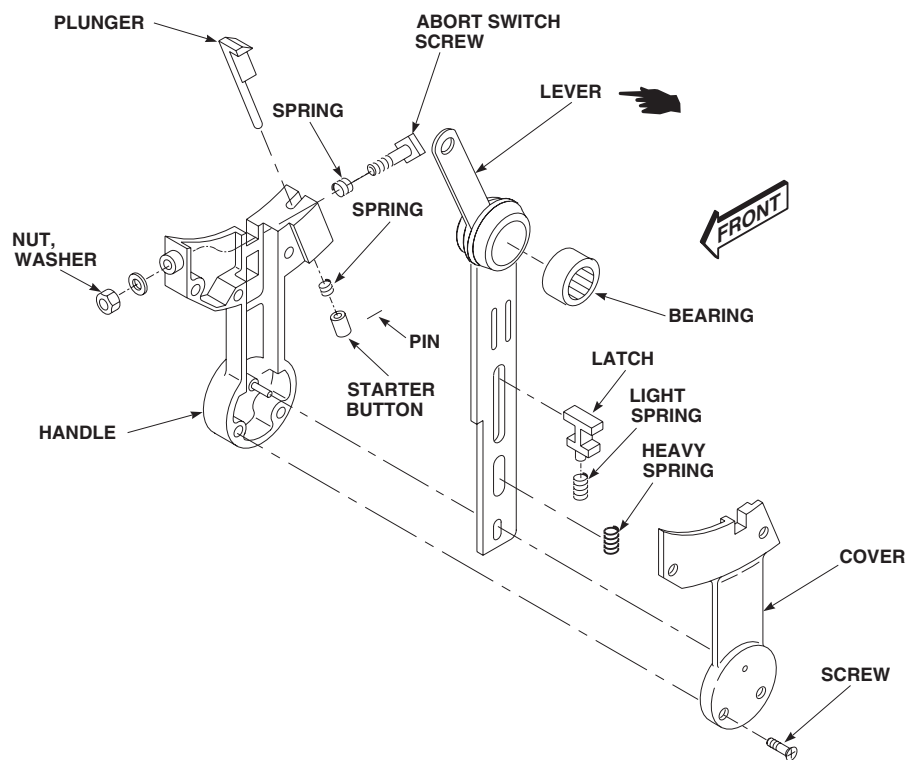
AA1881A
SA

Figure 2. Repair Engine Control Quadrant Power Lever (AVIM).

INSTALL - Continued**NOTE**

Shim with washers, AN960-616, as required to properly install cotter pin.

4. **TIGHTEN NUT UNTIL HOLE IN BOLT LINES UP WITH SLOT IN NUT. THEN TIGHTEN ONE MORE CASTELLATION.**
5. Move engine power control lever, fire extinguisher T-handle, and fuel control lever, and check for binding or looseness.
6. If handles or levers bind, or are too loose add, or remove shims as required.
7. **QA** Install cotter pin in nut.
8. Install override and abort switches (WP 0529 00 and WP 0528 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

ENGINE CONTROL QUADRANT T-HANDLE AND FUEL LEVER (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

WP 0524 00

WP 0527 00

WP 0530 00

Material/Parts

Tags

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

The following procedures apply to both No. 1 quadrant and No. 2 quadrant.

1. Remove engine starter and fire extinguisher arming switches (WP 0527 00 and WP 0530 00).
2. Remove engine power control lever (WP 0524 00).
3. Slide bolt, fire extinguisher T-handle, and fuel control lever, as an assembly, from quadrant support (Figure 1).
4. Remove bolt and washer from T-handle and fuel control lever.
5. Remove bushing from T-handle and fuel control lever.
6. **UH60A 80-23500 - SUBQ EH60A UH60L** Remove laminated shim from bushing. ■
7. Remove fuel control lever, washer, and laminated shim from fire extinguisher T-handle. Tag shim for assembly.
8. Remove electrical connector from electrical wires.
9. Carefully remove electrical tubing insulation from wire bundle.
10. Remove T-handle wire from wire bundle.

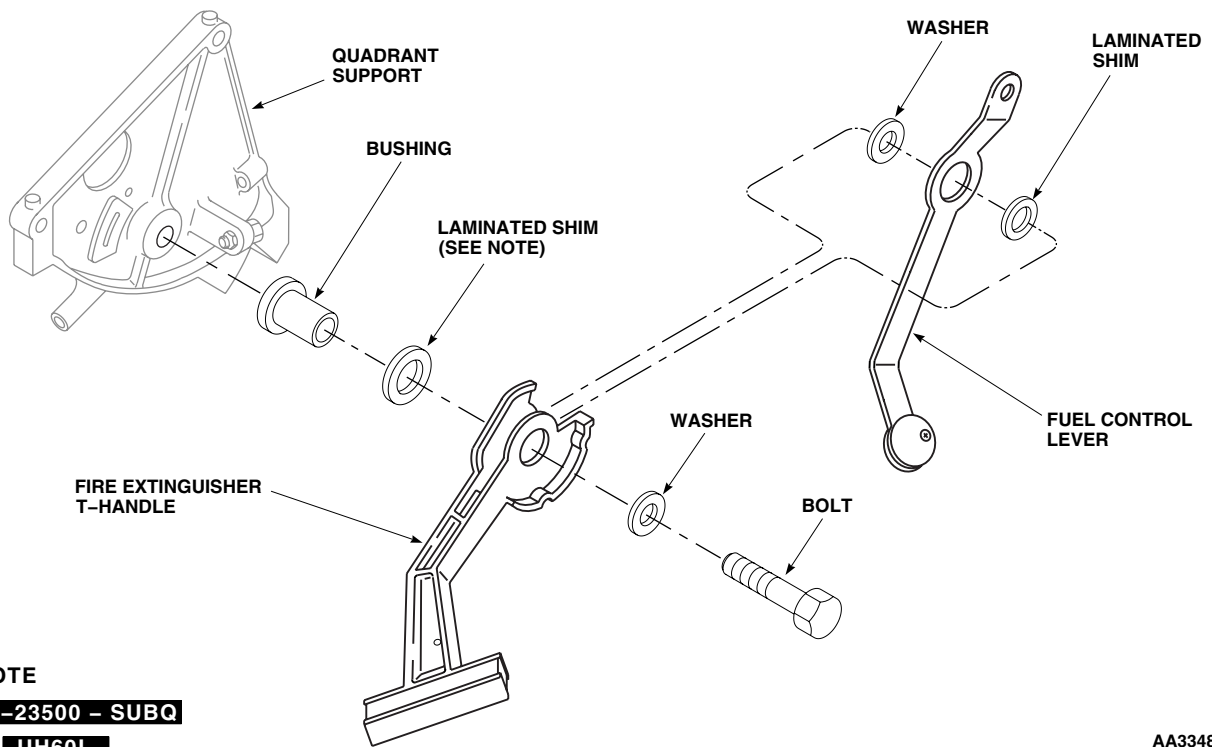
INSPECT/REPAIR

CHECK ID OF FIRE EXTINGUISHER T-HANDLE BUSHING FOR MAXIMUM WEAR OF 0.3765-INCH. If wear is beyond tolerance, replace bushing.

INSTALL

1. **QA** Position wire from fire extinguisher T-handle in wire bundle.
2. Measure and cut electrical wire to correct length.
3. Install electrical tubing insulation on wire bundle.
4. **QA** Connect electrical wire to electrical connector.
5. Remove tag from laminated shim.

INSTALL - Continued



NOTE

UH60A 80-23500 - SUBQ

EH60A UH60L

AA3348
SA

Figure 1. Replace Engine Control Quadrant T-Handle and Fuel Lever (AVIM).

6. If a new laminated shim is being installed, peel laminations as required to obtain minimum clearance between parts.
7. Position fuel control lever, washer, and laminated shim, with stainless steel side of shim against lever, between ears of T-handle (Figure 1).
8. **UH60A 80-23500 - SUBQ EH60A UH60L** Install laminated shim with stainless steel side facing fire extinguisher T-handle on bushing.
9. **QA** Install bushing through T-handle, washer, fuel control lever, and shim.
10. **QA** Install bolt and washer through T-handle and fuel control lever. Install shim on bolt.
11. **QA** Slide bolt, T-handle, and fuel control lever, as an assembly, in quadrant support.
12. Install engine power control lever (WP 0524 00).
13. Install fire extinguisher and starter switches (WP 0527 00 and WP 0530 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

ENGINE CONTROL QUADRANT THREADED INSERTS (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Insert, Screw Thread, SR162L
Insert, Screw Thread, SR192L
Insert, Screw Thread, SR258L
Tool Kit, Screw Thread Insert, K4-162-162L

Tool Kit, Screw Thread Insert, K4-192-192L

Tool Kit, Screw Thread Insert, K4-258-258L

Personnel Required

Aircraft Structural Repairer MOS 15G (1)

UH-60 Helicopter Repairer MOS 15T (1)

REMOVE**NOTE**

Removal and installation of all threaded inserts is the same except as noted. Use screw thread insert tool kit, K4-162-162L, for screw thread insert, SR162L; screw thread insert tool kit, K4-192-192L, for screw thread insert, SR192L; and screw thread insert tool kit, K4-258-258L, for screw thread insert, SR258L.

1. Install required removal bit from tool kit in drill press.
2. Using drill press and removal bit, drill to a depth of 0.090-inch, for insert, SR162L.
3. Using drill press and removal bit, drill to a depth of 0.107-inch for inserts, SR192L and SR258L.
4. Using installation wrench, turn insert counterclockwise and remove insert (Figure 1).
5. Use clean, dry compressed air to remove any loose chips from hole.

INSTALL

1. Using installation wrench, thread screw thread insert, SR162L, into hole until top of screw thread insert is 0.010 to 0.020-inch below boss surface of support (Figure 1).
2. Using installation wrench, thread screw thread insert, SR192L or SR258L, into hole until top of screw thread insert is 0.015 to 0.025-inch below boss surface of support.
3. **QA** Insert swage tool in insert and apply downward pressure until shoulder of tool is bottomed against boss surface. Make sure shoulder is bottomed against boss surface.

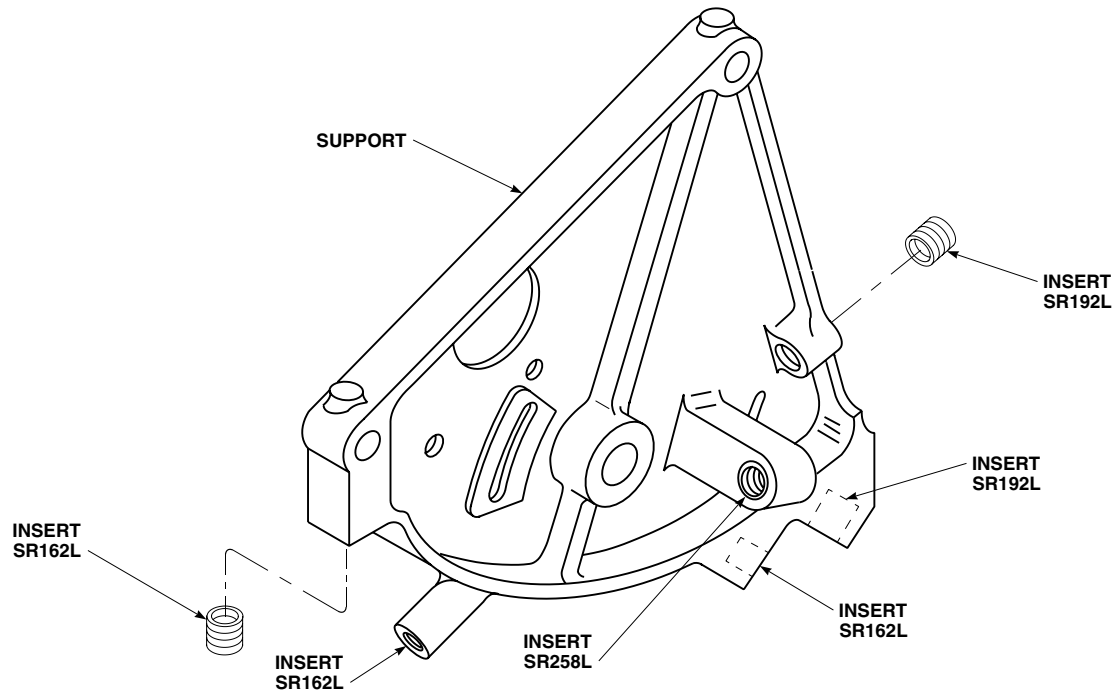
INSTALL - ContinuedAA9702
SA

Figure 1. Replace Engine Control Quadrant Threaded Inserts (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

ENGINE CONTROL QUADRANT ENGINE STARTER SWITCH (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Adhesive, Item 9, WP 1803 00
Solder, Tin Alloy, Item 299, WP 1803 00
Tags

References

WP 0501 00
WP 0503 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Lower overhead console.
3. Remove quadrant covers (WP 0501 00).
4. Remove screws, lockwashers, nutplate, and spacer plate holding starter switch to quadrant support (Figure 1, Detail A).
5. Place identification tags on starter switch wires.
6. Remove sealing compound to expose solder connections on engine starter switch.
7. Unsolder wires from starter switch.

INSTALL

1. Turn off all electrical power.
2. **QA** Identify and solder wires to starter switch using tin alloy solder, Item 299, WP 1803 00. Remove tags.
3. Apply adhesive, Item 9, WP 1803 00, to cover soldered connections.
4. **QA** Install starter switch on quadrant support with screws, lockwashers, nutplate, and spacer plate. Do not tighten screws now (Figure 1, Detail A).
5. Adjust engine starter switch (WP 0503 00).
6. **QA** Install quadrant covers (WP 0501 00).
7. **QA** Hinge up and secure upper console.

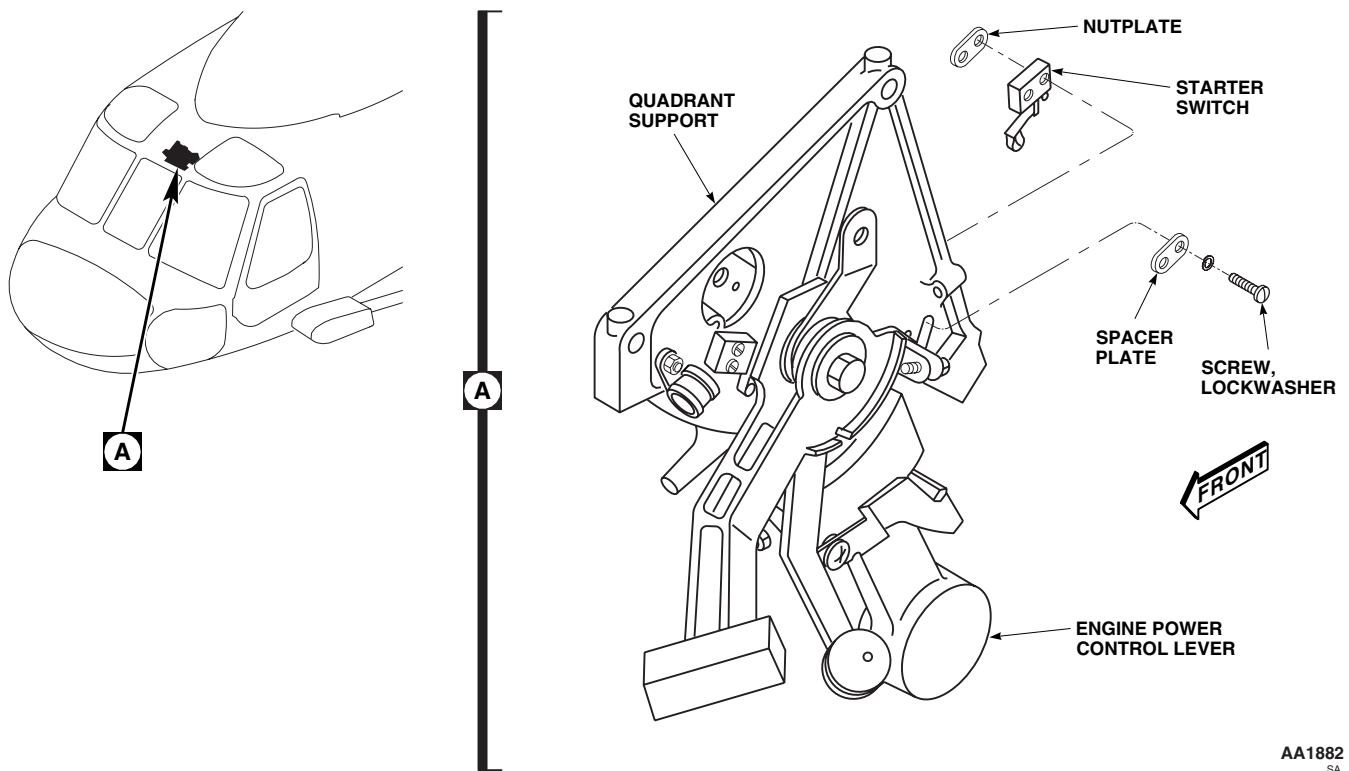
INSTALL - Continued

Figure 1. Replace Engine Control Quadrant Engine Starter Switch (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****ENGINE CONTROL QUADRANT STARTER ABORT SWITCH (AVIM)****This WP supersedes WP 0528 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06
Torque Wrench, 0 - 30 in. lbs.

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Adhesive, Item 9, WP 1803 00
Solder, Tin Alloy, Item 299, WP 1803 00
Tags
Tape, Lacing and Tying, Item 321, WP 1803 00

References

WP 0079 00
WP 0501 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Lower overhead console.
3. Remove quadrant covers (WP 0501 00).
4. **UH-60A 86-24516 - SUBQ EH-60A 85-24469 - SUBQ UH-60L MWO 50-39** Remove nut, washers, bolt, spacers, torsion spring, and latch stop from control lever (Figure 1).

NOTE

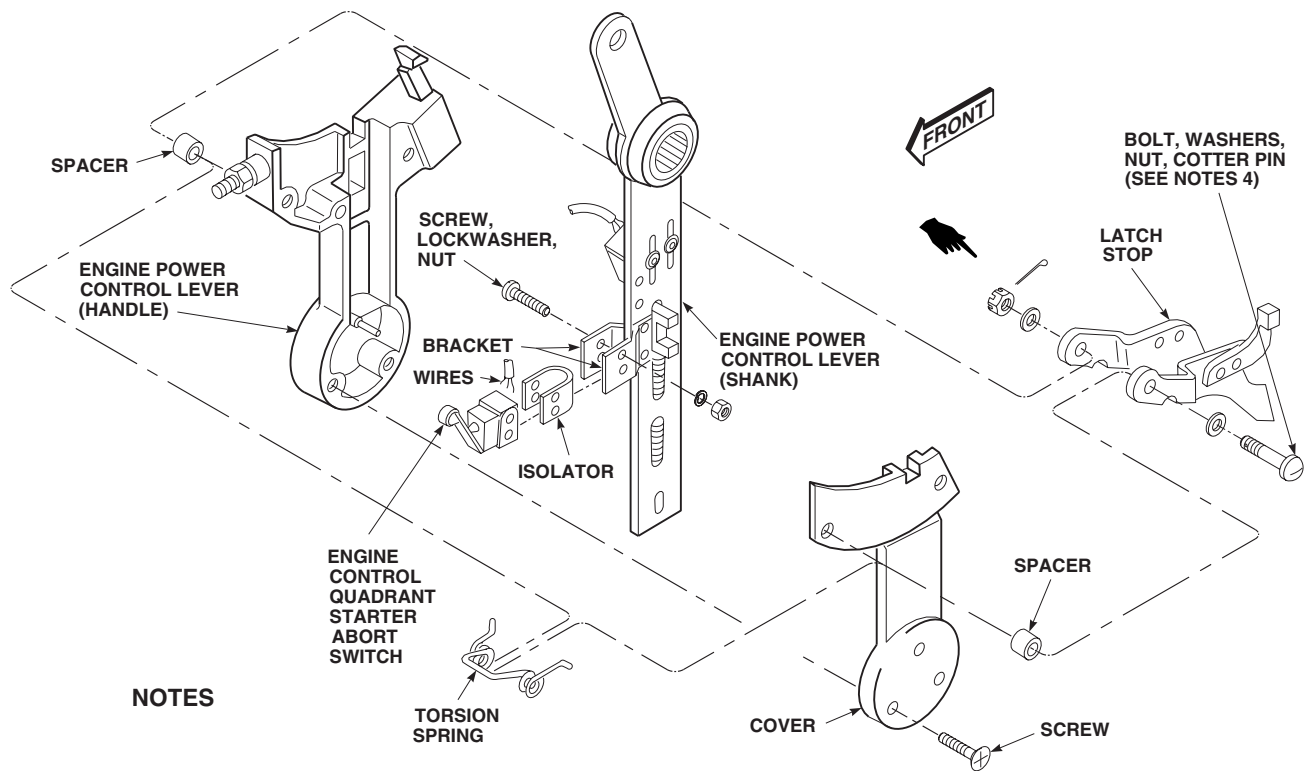
There are two small compressed springs and a latch in the engine power control lever assembly. Be careful when removing cover to avoid losing parts.

5. Remove screws and cover from handle on engine power control lever.
6. Remove screws, lockwashers, nuts, and insulator holding switch to brackets on engine power control lever.
7. Place identification tags on starter abort switch wires.
8. Remove sealing compound to expose solder connections on starter abort switch.
9. Unsolder wires from starter abort switch.

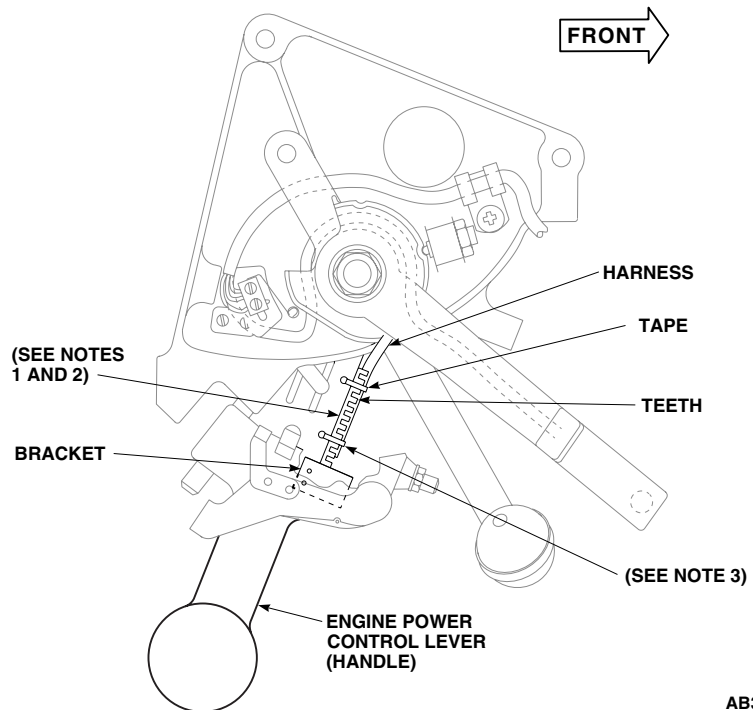
INSTALL

1. Turn off all electrical power.
2. **QA** Identify and solder wires to starter abort switch using tin alloy solder, Item 299, WP 1803 00. Remove tags.
3. Apply adhesive, Item 9, WP 1803 00, to cover soldered connections.
4. **QA** Attach starter abort switch with insulator between brackets on engine power control lever and secure with screws, lockwashers, and nuts (Figure 1).

INSTALL - Continued



NOTES



NOTES

1. WHEN FABRICATING GROMMET, MAKE SURE THERE ARE 11 FULL TEETH SQUARE AND FLUSH AT EACH END.
2. POSITION GROMMET AS CLOSE AS POSSIBLE TO BRACKET.
3. WHEN SECURING GROMMET AT LOWER HOLE, MAKE SURE TAPE IS SECURED BETWEEN THIRD AND FOURTH TOOTH FROM BOTTOM.
4. INSTALL WITH BOLTHEAD OUTBOARD.

AB3689A
SA

Figure 1. Replace Engine Control Quadrant Starter Abort Switch (AVIM).

INSTALL - Continued**NOTE**

When placing grommet against shank of PCL lever, the switch lead dressing may not allow grommet to reach starter abort switch bracket. Install grommet as close to switch bracket as possible.

5. Place grommet against shank of PCL lever directly above starter abort switch bracket. Place abort switch harness between grommet teeth.

NOTE

When securing harness/grommet, make sure tape is positioned between teeth prior to tightening.

6. Using two No. 40 holes, secure harness/grommet with lacing and tying tape, Item 321, WP 1803 00, in two places (Figure 1, Detail A). Thoroughly clean work area of all FOD.

CAUTION

Damage to wiring will result if caution is not used when installing covers on engine power control lever. Carefully position wires and install covers to avoid pinching and chaffing conditions.

7. **QA** Install cover on handle of engine power control lever with screws.
8. **UH-60A 86-24516 - SUBQ EH-60A 85-24469 - SUBQ UH-60L MWO 50-39** Install latch stop as follows:
 - a. Position torsion spring on engine power control lever.
 - b. Install latch stop on lever assembly and engage torsion spring.
 - c. **QA** Install bolt, washers, spacers, and nut holding latch stop in place. **TORQUE NUT TO 12 - 20 INCH-POUNDS.**
 - d. **QA** Install cotter pin in nut.
9. Install quadrant covers (WP 0501 00).
10. **QA** Hinge up and secure upper console.

ADJUST

1. Turn quadrant adjusting nut fully clockwise (Figure 2).
2. Pull ENG POWER CONT lever handle down as far as it will go and hold. Using a pencil, mark a line on lever 0.060-inch above top of handle.
3. With handle down as far as it will go, turn adjusting nut counterclockwise until switch clicks. Release handle.
4. While pulling ENG POWER CONT lever handle down, look at lever to see where top of handle is when switch clicks.
5. If top of handle is then below pencil marking, no further adjustment is required.
6. If there is no click with handle fully down, turn adjusting nut counterclockwise in quarter-turn units until a click is heard.
7. If top of handle is covering pencil marking, turn adjusting nut clockwise in quarter-turn units until switch clicks within 0.060-inch spread range.

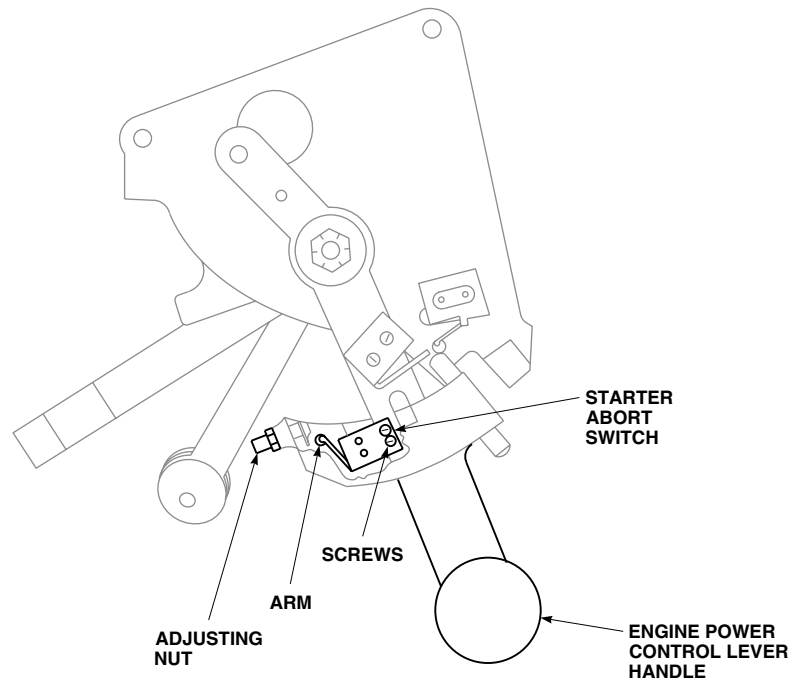
ADJUST - ContinuedAB1417
SA

Figure 2. Adjust Engine Control Quadrant Starter Abort Switch (AVIM).

8. Perform operational check of engine start and ignition system (WP 0079 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM****ENGINE CONTROL QUADRANT STARTER OVERRIDE SWITCH (AVIM)****This WP supersedes WP 0529 00, dated 17 April 2006.**

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer's Toolkit, SC 5180-99-B06

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Adhesive, Item 9, WP 1803 00
Solder, Tin Alloy, Item 299, WP 1803 00
Tags

References

WP 0079 00
WP 0501 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Lower overhead console.
3. Remove quadrant covers (WP 0501 00).
4. Remove screws, lockwashers, and nutplate holding starter override switch to engine power control lever (Figure 1).
5. Place identification tags on starter override switch wires.
6. Remove sealing compound to expose solder connections on starter override switch.
7. Unsolder wires from starter override switch.

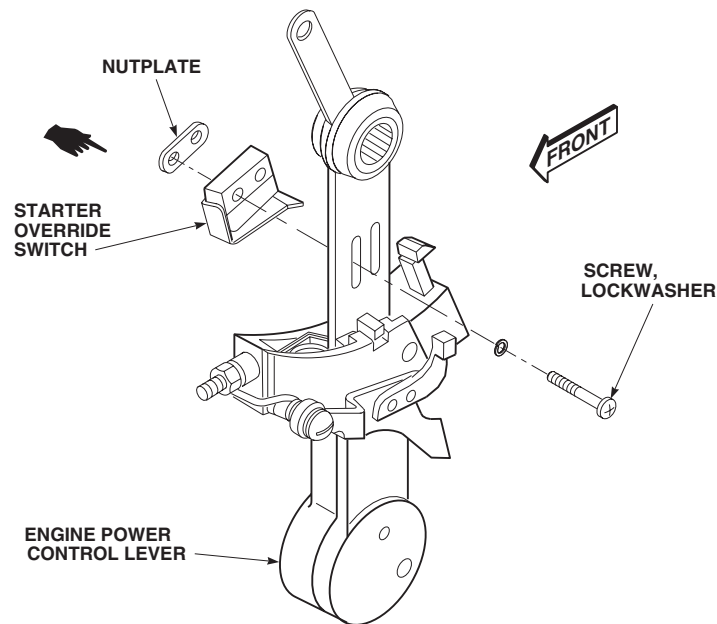
INSTALL

1. Turn off all electrical power.
2. **QA** Identify and solder wires to starter override switch using tin alloy solder, Item 299, WP 1803 00. Remove tags.
3. Apply adhesive, Item 9, WP 1803 00, to cover soldered connections.

NOTE

Starter override switch is mounted to inboard side of both NO. 1 and NO. 2 engine power control levers.

4. **QA** Install starter override switch on engine power control lever with screws, lockwashers, and nutplate. Do not tighten screws now (Figure 1).
5. Adjust starter override switch. Refer to, ADJUST, in this work package.
6. Install quadrant covers (WP 0501 00).
7. **QA** Hinge up and secure upper console.

INSTALL - Continued**NOTE**

ENGINE POWER CONTROL LEVER
SHOWN REMOVED FOR CLARITY.

AB3690A
SA

Figure 1. Replace Engine Control Quadrant Starter Override Switch (AVIM).

ADJUST

1. Remove screws, washers, and center cover of quadrant (WP 0501 00).
2. Loosen screws holding starter override switch to ENG POWER CONT lever, just enough to move switch (Figure 1).
3. With starter button fully pressed, move starter override switch until arm is against starter button, and switch clicks. Tighten screws holding switch to ENG POWER CONT lever bracket (Figure 2).
4. Make sure area is clean and free of foreign material.
5. Install center quadrant cover with screws and washers (WP 0501 00).
6. Perform operational check of engine start and ignition system (WP 0079 00).

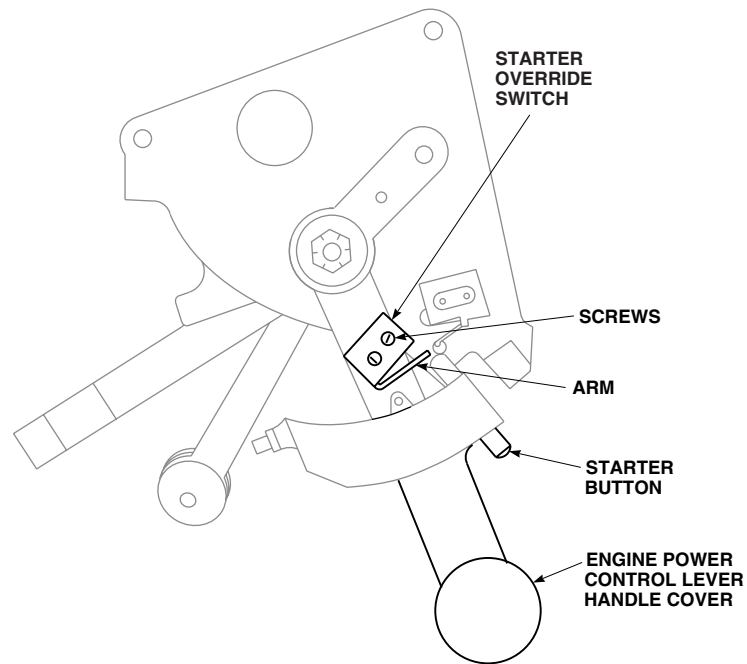
ADJUST - ContinuedAB1416
SA

Figure 2. Adjust Engine Control Quadrant Starter Override Switch (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL**ENGINE SYSTEM**

ENGINE CONTROL QUADRANT FIRE EXTINGUISHER ARMING SWITCH (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Airframe Repairer's Toolkit, SC 5180-99-B02
Electrical Repairer's Toolkit, SC 5180-99-B06
Multimeter, 60741-310
Spring Scale, AAA-S-133

Tags

Personnel Required

Aircraft Electrician MOS 15F (1)
Aircraft Structural Repairer MOS 15G (1)
UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Adhesive, Item 29, WP 1803 00
Solder, Tin Alloy, Item 299, WP 1803 00

References

WP 0501 00
WP 1803 00

REMOVE

1. Turn off all electrical power.
2. Lower overhead console.
3. Remove quadrant covers (WP 0501 00).
4. Remove screws, lockwashers, and nutplate holding fire extinguisher arming switch to quadrant support (Figure 1, Sheet 1).
5. Place identification tags on fire extinguisher arming switch wires.
6. Remove sealing compound to expose solder connections on fire extinguisher arming switch.
7. Unsolder wires from fire extinguisher arming switch.

INSTALL

1. Turn off all electrical power.
2. **QA** Identify and solder wires to fire extinguisher arming switch using tin alloy solder, Item 299, WP 1803 00. Remove tags.
3. Apply adhesive, Item 29, WP 1803 00, to cover soldered connections.
4. **QA** Install fire extinguisher arming switch on quadrant support with screws, lockwashers, and nutplate. Do not tighten screws now (Figure 1, Sheet 1).
5. Adjust fire extinguisher arming switch. Refer to, ADJUST, in this work package.

CAUTION

To prevent chafing to left side fire extinguisher arming switch wiring make sure loop does not protrude past 1.25-inch diameter hole.

6. If required, on left side only, loosen clamp and adjust fire extinguisher arming switch wiring loop so that loop does not protrude past 1.25-inch diameter hole (Figure 1, Sheet 2).

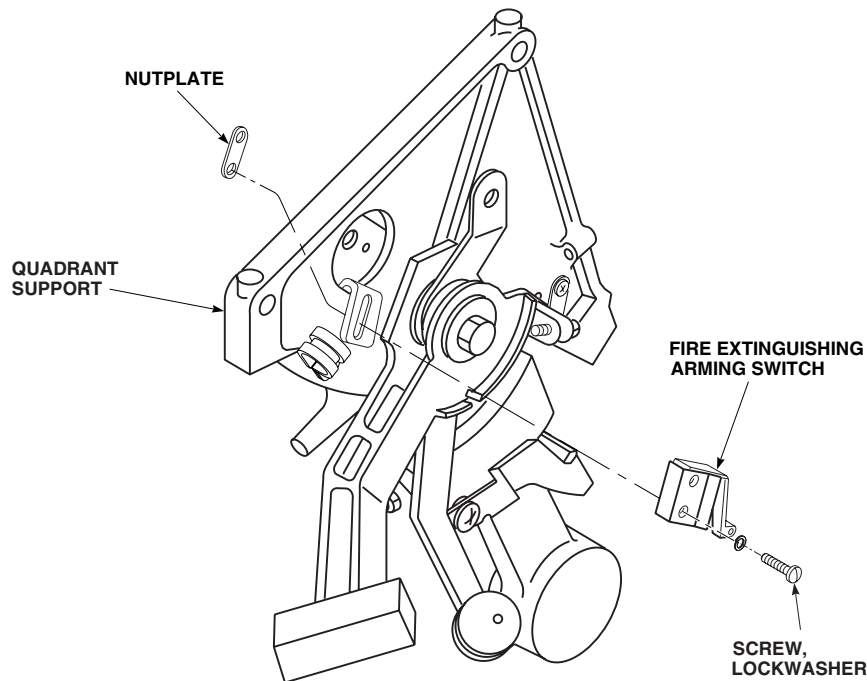
INSTALL - ContinuedAB3691_1
SA

Figure 1. Replace Engine Control Quadrant Fire Extinguisher Arming Switch (AVIM). (Sheet 1 of 2)

7. Install quadrant covers (WP 0501 00).
8. **QA** Hinge up and secure upper console.

ADJUST

1. Move fire extinguisher T-handle forward until plunger engages in stowed detent (Figure 2).
2. Attach spring scale, to apex of T-handle.
3. Using spring scale, adjust plunger to release from detent position when breakout force of 5 to 7 pounds is applied at T intersection of lever assembly.
4. Loosen screws holding fire extinguisher arming switch to support just enough to move switch (Figure 1).
5. Move fire extinguisher arming switch until arm is against fire extinguisher lever and switch clicks. Tighten screws to hold switch in place.
6. Move fire extinguisher T-handle aft as far as travel permits. Check that switch is off.
7. Move fire extinguisher T-handle from stowed position detent to aft several times and check for proper switch operation.
8. Verify switch operation by using multimeter.

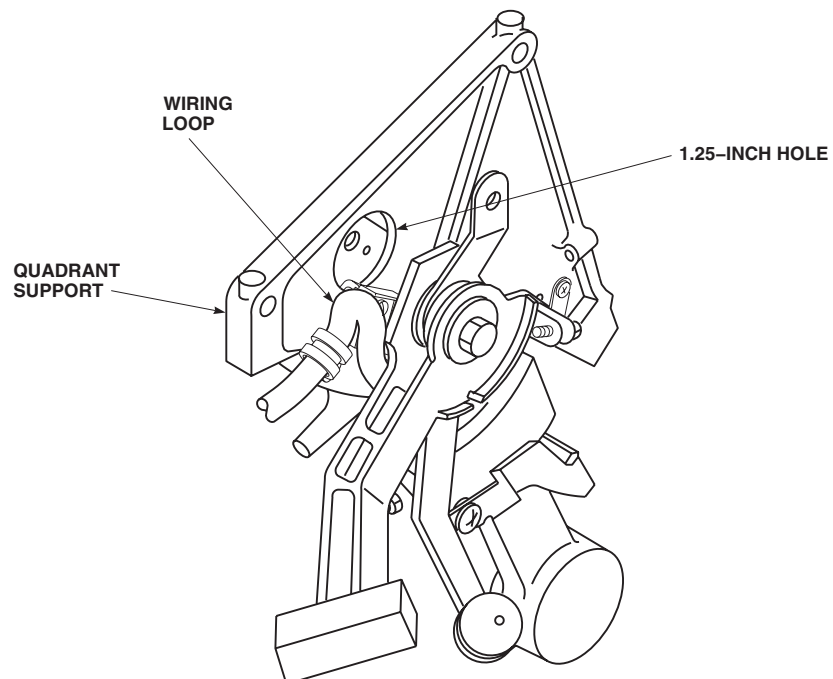
ADJUST - ContinuedAA1943.2
SA

Figure 1. Replace Engine Control Quadrant Fire Extinguisher Arming Switch (AVIM). (Sheet 2 of 2)

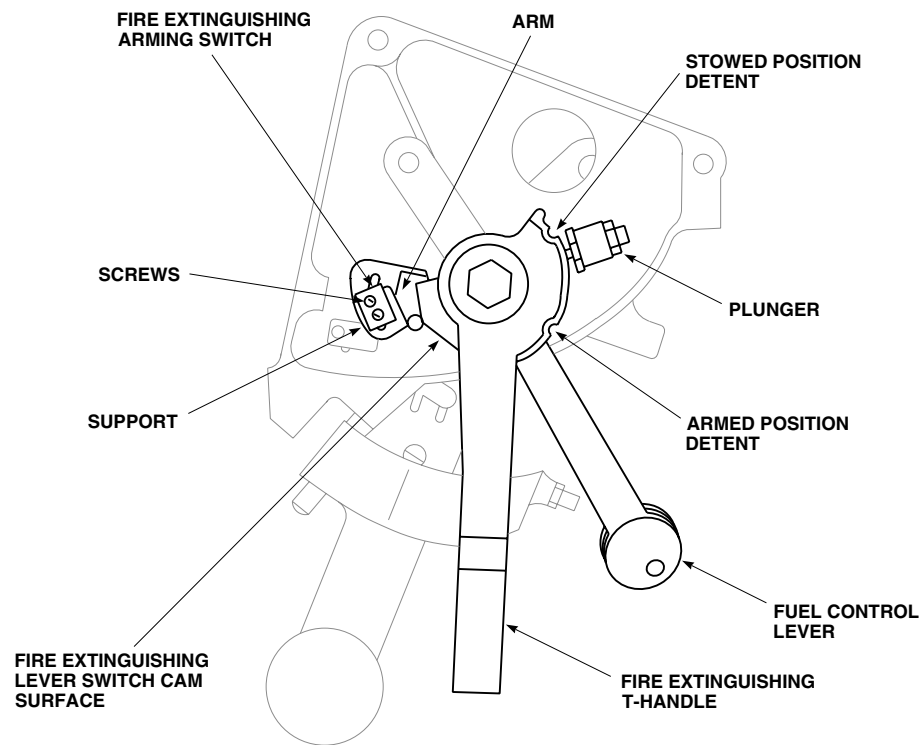
ADJUST - ContinuedAA9700
SA

Figure 2. Adjust Engine Control Quadrant Fire Extinguisher Arming Switch (AVIM).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM

ENGINE CONTROL QUADRANT LEFT AND RIGHT COVER INFORMATION PLATE (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Crimp Tool, M22520/2-01
 Electrical Repairer's Toolkit, SC 5180-99-B06
 Heat Gun, LFT500
 Insertion-Extraction Tool, GGG-E-936

Material/Parts

Insulation Sleeving, Electrical, Item 155, WP 1803 00
 Insulation Sleeving, Electrical, Item 168, WP 1803 00

Personnel Required

Aircraft Electrician MOS 15F (1)
 UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0106 00
 WP 0501 00
 WP 1803 00

REMOVE**NOTE**

Replacement of the left and right information plate is the same except as noted.

1. Remove engine quadrant cover (WP 0501 00).
2. Remove screws holding information plate to side cover (Figure 1).
3. Remove wires and pins from connector using insertion-extraction tool. Save connector for replacement information plate.

INSTALL

1. Install electrical insulation sleeving, Item 168, WP 1803 00, and electrical insulation sleeving, Item 155, WP 1803 00, and slide back. Do not heat shrink now.
2. **QA** Install pins on information plate electrical wires using crimp tool.
3. Feed wires through backshell and insert pins into connector using insertion-extraction tool as follows:

WIRE NO.	CONNECTOR PIN NO.
A-20	1
B-20N	2

4. Slide previously installed electrical insulation sleeving, Item 155, WP 1803 00 over backshell and using heat gun apply heat to both electrical insulation sleeveings to shrink fit. Do not overheat.
5. **QA** Install information plate to inside of side cover with screws (Figure 1).
6. Install quadrant cover assembly (WP 0501 00).

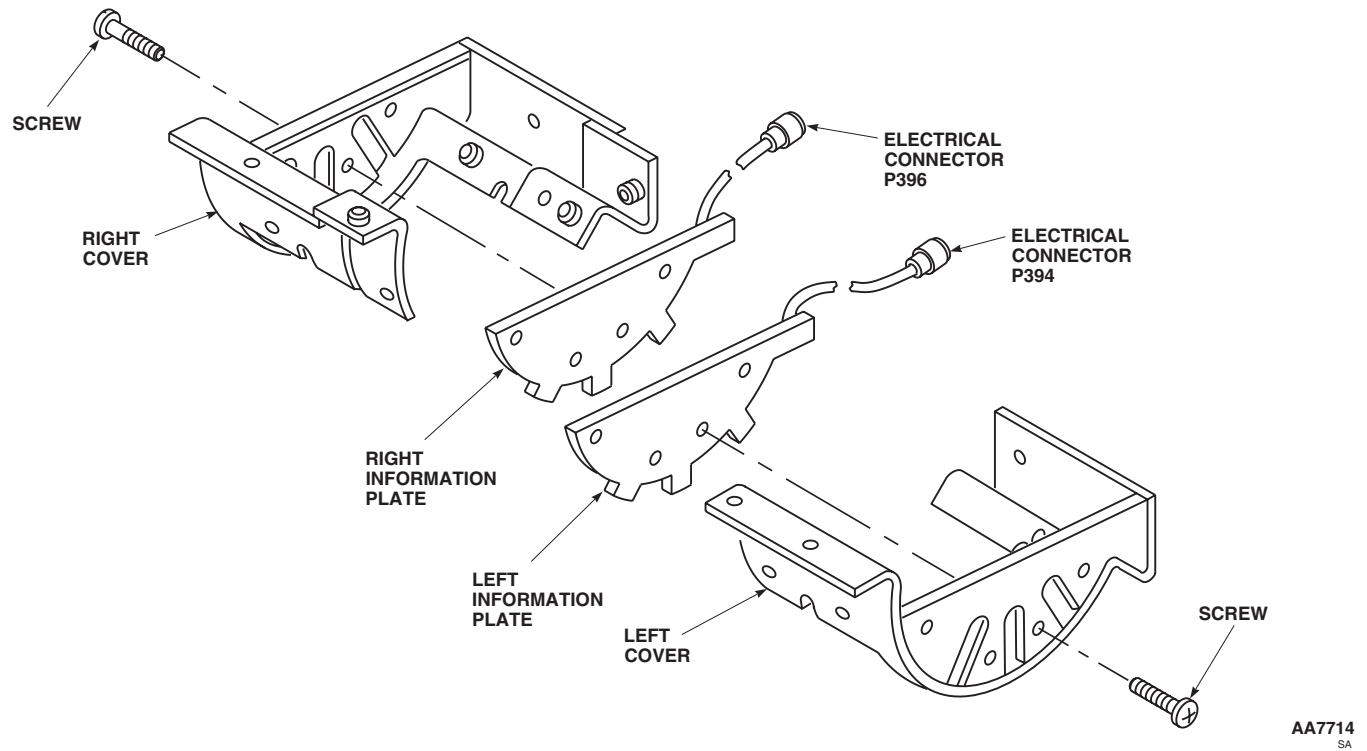
INSTALL - Continued

Figure 1. Replace Engine Control Quadrant Left and Right Cover Information Plates (AVIM).

7. Do operational check of upper console lights (WP 0106 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM

ENGINE CONTROL QUADRANT FRONT COVER INFORMATION PLATE WIRING HARNESS (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Electrical Repairer's Toolkit, SC 5180-99-B06

References

WP 0106 00
 WP 0501 00
 WP 1703 00
 WP 1803 00

Material/Parts

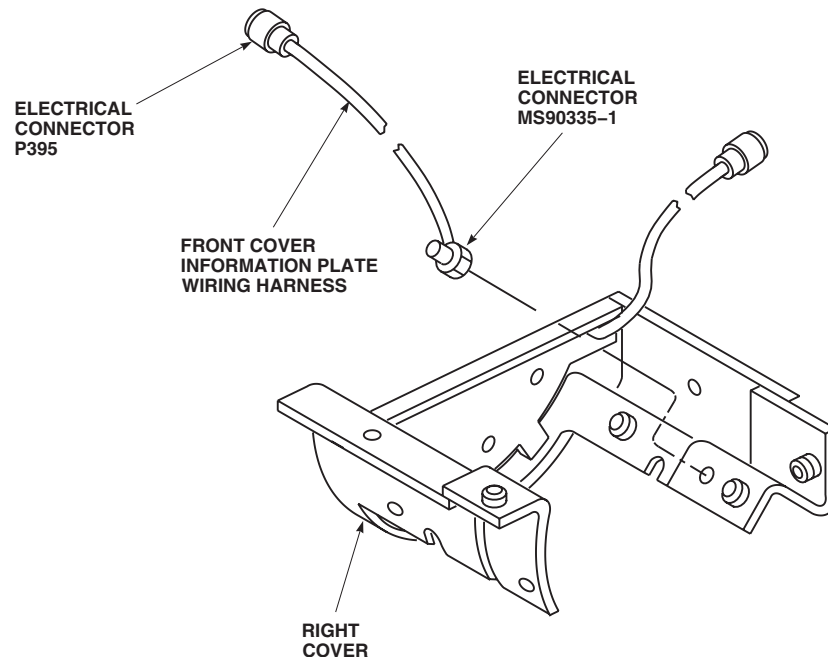
Solder, Tin Alloy, Item 299, WP 1803 00
 Tags

Personnel Required

Aircraft Electrician MOS 15F (1)
 UH-60 Helicopter Repairer MOS 15T (1)

REMOVE

1. Turn off all electrical power.
2. Remove center and right side covers from quadrant assembly (WP 0501 00).



AA7715
 SA

Figure 1. Replace Engine Control Quadrant Front Cover Information Plate Wiring Harness (AVIM).

REMOVE - Continued

3. Tag and unsolder wires from connector on quadrant right side cover assembly and remove front cover information wiring harness (Figure 1).
4. Remove nut, washer, solder lug, and electrical connector.

INSTALL

1. Turn off all electrical power.
2. Inspect electrical connector (WP 1703 00).
3. **QA** Install electrical connector on quadrant cover assembly (Figure 1).
4. **QA** Solder wiring harness to terminals using tin alloy solder, Item 299, WP 1803 00. Untag wires.
5. Install right side and center covers on quadrant assembly (WP 0501 00).
6. Do an operational check of upper console lights (WP 0106 00).

END OF WORK PACKAGE

AVIATION UNIT AND INTERMEDIATE LEVEL
ENGINE SYSTEM
ENGINE LOAD-DEMAND CABLE SUPPORT CLIP (AVIM)

INITIAL SETUP:
Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
 Bushing Pusher
 Fluorescent Penetrant Inspection Kit, XMA101
 Nonmetallic Scraper
 Sizing Block

Sealing Compound, Item 273, WP 1803 00
 Tape, Pressure Sensitive, Adhesive, Item 321,
 WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

Material/Parts

Cloth, Abrasive, Item 82, WP 1803 00
 Epoxy Primer Coating Kit, Item 120, WP 1803 00
 Polyurethane, Coating, Item 227, WP 1803 00
 Remover, Paint, Item 251, WP 1803 00

References

TM 1-1520-265-23
 TM 55-1500-345-23
 WP 0381 00
 WP 1803 00

REMOVE BUSHINGS
NOTE

Replace bushings as a complete set of two per support clip (Figure 1).

Remove bushings from support clip using bushing pusher and arbor press.

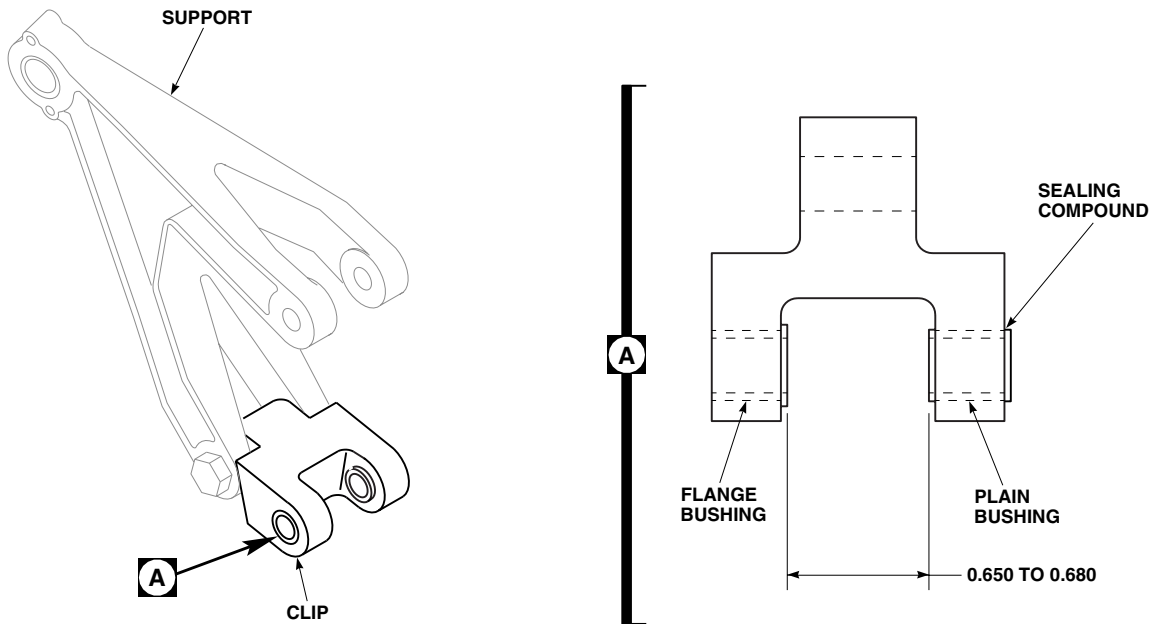
INSPECT/REPAIR

1. Remove sealant from bores in support clip with nonmetallic scraper.
2. Remove paint from support clip bores using paint remover, Item 251, WP 1803 00.
3. Fluorescent penetrant inspect support clip for cracks (TM 1-1520-265-23). Replace support clip if cracks are found.
4. Check for corrosion, galling, or scoring in support clip bushing bores. Smooth out any corrosion, galling, or scoring in support clip bushing bores using abrasive cloth, Item 82, WP 1803 00. **REPLACE CLIP IF DAMAGE IS DEEPER THAN 0.010-INCH AFTER BLENDING, OR IF BLENDED AREA IS GREATER THAN 25% OF BORE AREA.**
5. Measure clip bore after blending. **REPLACE CLIP IF EITHER BORE IS LARGER THAN 0.3761-INCH.**
6. Paint support clip as follows (WP 0381 00 and TM 55-1500-345-23):
 - a. Mask bushing bores with pressure sensitive adhesive tape, Item 321, WP 1803 00.
 - b. Apply one coat of epoxy primer coating kit, Item 120, WP 1803 00, to surface.
 - c. Apply two coats of coating polyurethane, Item 227, WP 1803 00.
 - d. Remove pressure sensitive adhesive tape from bushing bores.

INSTALL BUSHINGS

1. Manufacture a sizing block 0.650 to 0.680-inch.

INSTALL BUSHINGS - Continued



NOTE

ALL DIMENSIONS ARE IN INCHES
UNLESS NOTED.

AA9701A
SA

Figure 1. Repair Engine Load-Demand Cable Support Clip (AVIM).

2. Coat bushings and support clip bushing bores with epoxy primer coating kit, Item 120, WP 1803 00.
3. **QA** Using arbor press, press bushings into clip bores while primer is still wet. Remove any excess primer. Check for 0.650 to 0.680-inch block dimension (Figure 1).
4. Seal bushings with sealing compound, Item 273, WP 1803 00.

END OF WORK PACKAGE

By Order of the Secretary of the Army:

Official:

A handwritten signature in black ink, reading "Joyce E. Morrow". The signature is fluid and cursive, with the first name "Joyce" and last name "Morrow" clearly legible.

JOYCE E. MORROW
*Administrative Assistant to the
Secretary of the Army*
0608617

GEORGE W. CASEY, JR.
*General, United States Army
Chief of Staff*

DISTRIBUTION:

To be distributed in accordance with Initial Distribution Number (IDN) 311289, requirements for TM 1-1520-237-23-6.

These are the instructions for sending an electronic 2028

The following format must be used if submitting an electronic 2028. The subject line must be exactly the same and all fields must be included; however only the following fields are mandatory: 1, 3, 4, 5, 6, 7, 8, 9, 10, 13, 15, 16, 17, and 27.

From: "Whomever" <whomever@wherever.army.mil>

To: 2028@redstone.army.mil

Subject: DA Form 2028

1. From: Joe Smith
2. Unit: home
3. Address: 4300 Park
4. City: Hometown
5. St: MO
6. Zip: 77777
7. Date Sent: 19--OCT--93
8. Pub no: 55--2840--229--23
9. Pub Title: TM
10. Publication Date: 04--JUL--85
11. Change Number: 7
12. Submitter Rank: MSG
13. Submitter FName: Joe
14. Submitter MName: T
15. Submitter LName: Smith
16. Submitter Phone: 123-123-1234
17. Problem: 1
18. Page: 2
19. Paragraph: 3
20. Line: 4
21. NSN: 5
22. Reference: 6
23. Figure: 7
24. Table: 8
25. Item: 9
26. Total: 123
27. Text:

This is the text for the problem below line 27.

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25-30; the proponent agency is ODISC4.						Use Par reverse) for Repair Parts and Spe- cial Tool Lists (RPSTL) and Supply Catalogs/ Supply Manuals (SC/SM)	DATE 8/30/02
TO: (Forward to proponent of publication or form)(Include ZIP Code) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, 35898						FROM: (Activity and location)(Include ZIP Code) MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565	
PART 1 - ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS							
PUBLICATION/FORM NUMBER TM 9-1005-433-24						DATE 16 Sep 2002	TITLE Organizational, Direct Support, And General Support Maintenance Manual for Machine Gun, .50 Caliber M3P and M3P Machine Gun Electrical Test Set Used On Avenger Air Defense Weapon System
ITEM NO.	PAGE NO.	PARA- GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON	
1	W PG 3		2			Test or Corrective Action column should identify a different WP number.	
EXAMPLE							
* Reference to line numbers within the paragraph or subparagraph.							
TYPED NAME, GRADE OR TITLE MSG, Jane Q. Doe, SFC						TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTEN- SION 788--1234	SIGNATURE

TO: (Forward direct to addressee listed in publication) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, 35898				FROM: (Activity and location) (Include ZIP Code) MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565			DATE 8/30/02	
PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS								
PUBLICATION NUMBER				DATE		TITLE		
PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION
PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)								
EXAMPLE								
TYPED NAME, GRADE OR TITLE MSG, Jane Q. Doe, SFC				TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION 788-1234			SIGNATURE	

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25-30; the proponent agency is ODISC4.						Use Par reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/Supply Manuals (SC/SM)	DATE
TO: (Forward to proponent of publication or form)(Include ZIP Code) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898 - 5000						FROM: (Activity and location)(Include ZIP Code)	
PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS							
PUBLICATION/FORM NUMBER						DATE	TITLE
ITEM NO.	PAGE NO.	PARA- GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON	
* Reference to line numbers within the paragraph or subparagraph.							
TYPED NAME, GRADE OR TITLE					TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTENSION		SIGNATURE

TO: (Forward direct to addressee listed in publication) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898				FROM: (Activity and location) (Include ZIP Code)			DATE	
PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS								
PUBLICATION NUMBER				DATE		TITLE		
PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION
PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)								
TYPED NAME, GRADE OR TITLE				TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION			SIGNATURE	

The Metric System and Equivalents

Linear Measure

1 centimeter = 10 millimeters = .39 inch
 1 decimeter = 10 centimeters = 3.94 inches
 1 meter = 10 decimeters = 39.37 inches
 1 dekameter = 10 meters = 32.8 feet
 1 hectometer = 10 dekameters = 328.08 feet
 1 kilometer = 10 hectometers = 3,280.8 feet

Weights

1 centigram = 10 milligrams = .15 grain
 1 decigram = 10 centigrams = 1.54 grains
 1 gram = 10 decigrams = .035 ounce
 1 decagram = 10 grams = .35 ounce
 1 hectogram = 10 decagrams = 3.52 ounces
 1 kilogram = 10 hectograms = 2.2 pounds
 1 quintal = 100 kilograms = 220.46 pounds
 1 metric ton = 10 quintals = 1.1 short tons

Liquid Measure

1 centiliter = 10 milliliters = .34 fl. ounce
 1 deciliter = 10 centiliters = 3.38 fl. ounces
 1 liter = 10 deciliters = 33.81 fl. ounces
 1 dekaliter = 10 liters = 2.64 gallons
 1 hectoliter = 10 dekaliters = 26.42 gallons
 1 kiloliter = 10 hectoliters = 264.18 gallons

Square Measure

1 sq. centimeter = 100 sq. millimeters = .155 sq. inch
 1 sq. decimeter = 100 sq. centimeters = 15.5 sq. inches
 1 sq. meter (centare) = 100 sq. decimeters = 10.76 sq. feet
 1 sq. dekameter (are) = 100 sq. meters = 1,076.4 sq. feet
 1 sq. hectometer (hectare) = 100 sq. dekameters = 2.47 acres
 1 sq. kilometer = 100 sq. hectometers = .386 sq. mile

Cubic Measure

1 cu. centimeter = 1000 cu. millimeters = .06 cu. inch
 1 cu. decimeter = 1000 cu. centimeters = 61.02 cu. inches
 1 cu. meter = 1000 cu. decimeters = 35.31 cu. feet

Approximate Conversion Factors

<i>To change</i>	<i>To</i>	<i>Multiply by</i>	<i>To change</i>	<i>To</i>	<i>Multiply by</i>
inches	centimeters	2.540	ounce-inches	Newton-meters	.007062
feet	meters	.305	centimeters	inches	.394
yards	meters	.914	meters	feet	3.280
miles	kilometers	1.609	meters	yards	1.094
square inches	square centimeters	6.451	kilometers	miles	.621
square feet	square meters	.093	square centimeters	square inches	.155
square yards	square meters	.836	square meters	square feet	10.764
square miles	square kilometers	2.590	square meters	square yards	1.196
acres	square hectometers	.405	square kilometers	square miles	.386
cubic feet	cubic meters	.028	square hectometers	acres	2.471
cubic yards	cubic meters	.765	cubic meters	cubic feet	35.315
fluid ounces	milliliters	29.573	cubic meters	cubic yards	1.308
pints	liters	.473	milliliters	fluid ounces	.034
quarts	liters	.946	liters	pints	2.113
gallons	liters	3.785	liters	quarts	1.057
ounces	grams	28.349	liters	gallons	.264
pounds	kilograms	.454	grams	ounces	.035
short tons	metric tons	.907	kilograms	pounds	2.205
pound-feet	Newton-meters	1.356	metric tons	short tons	1.102
pound-inches	Newton-meters	.11296			

Temperature (Exact)

F	Fahrenheit	5/9 (after	Celsius	°C
	temperature	subtracting 32)	temperature	

